AURIC ARTISAN · USER DOCUMENTATION

Website tools

Every tool on auricartisan.com, grouped by what it helps you do, with its address and a link to its documentation.

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Contents

- Website tools
  - Accessibility and vision tools
    - Accessibility Checklist
      - Accessibility Checklist documentation
      - Getting started with the Accessibility Checklist
      - Attach evidence
      - Export your results
      - Laws by jurisdiction
      - The 50 checks
      - Work through the checklist
      - Accessibility Checklist reference
      - Troubleshooting the Accessibility Checklist
      - Accessibility Checklist FAQ
      - Accessibility Checklist: supporting material
      - Accessibility Checklist keyboard shortcuts
      - Accessibility Checklist glossary
      - Limits and accuracy
      - Accessibility Checklist privacy
      - Related tools and reading
    - Accessibility Compliance
      - Accessibility Compliance documentation
      - Getting started
      - Accessibility Compliance FAQ
      - Accessibility Compliance: supporting material
    - Analyzer
      - Analyzer documentation
      - Getting started with the Analyzer
      - The accessibility engine (A11y+)
      - Accessibility rules
      - Arrange the workspace
      - Choose what to scan
      - Colour, contrast and vision
      - Export and share
      - Fix and preview
      - Performance, SEO and page health
      - Read the Overview and scores
      - Run an analysis
      - Save, reopen and compare
      - Scan a whole site
      - Analyzer reference
      - Troubleshooting the Analyzer
      - Analyzer FAQ
      - Analyzer: supporting material
      - Analyzer keyboard shortcuts

- Analyzer glossary
- Limits and accuracy
- Analyzer privacy
- Analyzer release notes
- Related tools and reading

Contrast Checker

- Contrast Checker documentation
- Getting started with the Contrast Checker
- Check a colour pair
- Make a pair pass
- Share a pair
- Contrast Checker reference
- Troubleshooting the Contrast Checker
- Contrast Checker FAQ
- Contrast Checker: supporting material
- Contrast Checker glossary
- Limits and accuracy
- Contrast Checker privacy
- Related tools and reading

Ishihara Plate Generator

- Ishihara Plate Generator documentation
- Getting started with the Ishihara Plate Generator
- Build the standard 38
- Export plates
- Generate plates
- Read the diagnostics
- Ishihara Plate Generator reference
- Troubleshooting the Ishihara Plate Generator
- Ishihara Plate Generator FAQ
- Ishihara Plate Generator: supporting material
- Ishihara Plate Generator keyboard shortcuts
- Ishihara Plate Generator glossary
- Limits and accuracy
- Ishihara Plate Generator privacy
- Ishihara Plate Generator release notes
- Related tools and reading

Vision Lab

- Vision Lab documentation
- Getting started with the Vision Lab
- Audit a palette
- Browse the Catalogue and panels
- Compare conditions
- Explore Analytics
- Links, saving and exports
- The 78 conditions
- Use the Workbench
- Vision Lab reference

- Troubleshooting the Vision Lab
- Vision Lab FAQ
- Vision Lab: supporting material
- Vision Lab keyboard shortcuts
- Vision Lab glossary
- Limits and accuracy
- Vision Lab privacy
- Vision Lab release notes
- Related tools and reading

Colour workspace

- ASCII Art Generator
  - ASCII Art Generator documentation
  - Getting started with the ASCII Art Generator
  - Charts, benchmark and Compare
  - Copy, export and batch
  - Fine tuning
  - Looks, colour and detail
  - Render modes and full screen
  - Choose a source
  - ASCII Art Generator reference
  - Troubleshooting
  - Frequently asked questions
  - Supporting material
  - Shortcuts
  - Glossary
  - Limits and accuracy
  - Privacy
  - Related tools and reading
- Basic Color Tools
  - Basic Color Tools documentation
  - Getting started with Basic Color Tools
  - Import, export and the Library
  - Workspace and panels
  - Basic Color Tools reference
  - Troubleshooting Basic Color Tools
  - Basic Color Tools FAQ
  - Colour Tools workbench
  - Accessibility Lab
  - Animation Contrast
  - CMYK Soft-Proof
  - Color Formulation
  - Color Inspector
  - Color Library
  - Color Name Finder
  - Color Psychology
  - Color Space Converter
  - Contrast System

- Gradient Library
- Harmony Studio
- Open the Colour Tools and manage their panels
- Pantone & Named Lookup
- Type Readability Sim
- Panel tools
- Collage Maker (panel)
- Color Battle
- Color Picker
- Gradient Maker
- Palette Remix
- Palette Studio
- Quick Converter
- Supporting material
- Shortcuts
- Glossary
- Limits and accuracy
- Privacy
- Related tools and reading

Color Science Lab

- Color Science Lab documentation
- Getting started with the Color Science Lab
- Arrange, link, save and export panels
- The Catalogue and its panels
- Catalogue Compare and Analytics
- Compare two colours
- Gamut audit
- Illuminants
- The 45 tools
- The Workbench
- Color Science Lab reference
- Troubleshooting the Color Science Lab
- Color Science Lab FAQ
- Supporting material
- Shortcuts
- Glossary
- Limits and accuracy
- Privacy
- Related tools and reading

Design System Generator

- Design System Generator documentation
- Getting started with the Design System Generator
- Compare styles and industries
- Use the component playground
- Check contrast with the audit
- Export tokens
- Read the system

- Set the brief
- Share, save and connect
- Design System Generator reference
- Troubleshooting the Design System Generator
- Design System Generator FAQ
- Supporting material
- Glossary
- Limits and accuracy
- Privacy
- Related tools and reading

Font Library

- Font Library documentation
- Getting started with the Font Library
- Browse, search and filter
- Collections and the recommender
- Develop and embed
- Family details
- Your kit, compare and download
- The specimen panel
- Font Library reference
- Troubleshooting
- Frequently asked questions
- Supporting material
- Shortcuts
- Glossary
- Limits and accuracy
- Privacy
- Related tools and reading

Icon Library

- Icon Library documentation
- Getting started with the Icon Library
- Browse, search and filter
- The icon panel
- Your kit and export
- Sets, collections and Find your icon
- Icon Library reference
- Troubleshooting
- Frequently asked questions
- Supporting material
- Shortcuts
- Glossary
- Limits and accuracy
- Privacy
- Related tools and reading

Personalization Generator

- Personalization Generator documentation
- Getting started with the Personalization Generator

- Brand Studio
- Build your colour pool
- Design tokens
- Export, save and share
- Presets
- Palettes, gradients, typography, UI pairs and posters
- Run a sweep and read the results
- Tune the sweep
- Personalization Generator reference
- Troubleshooting
- Frequently asked questions
- Supporting material
- Glossary
- Limits and accuracy
- Privacy
- Related tools and reading

Colorimetry and rendering tools

- Chromatic Adaptation
  - Chromatic Adaptation documentation
  - Getting started with Chromatic Adaptation
  - Adapt a list of colours
  - Adapt an image to a new light
  - Compare transforms and copy a matrix
  - Explore the adaptation time course
  - Chromatic Adaptation reference
  - Troubleshooting Chromatic Adaptation
  - Chromatic Adaptation FAQ
  - Chromatic Adaptation: supporting pages
  - Chromatic Adaptation shortcuts
  - Chromatic Adaptation glossary
  - Chromatic Adaptation limits and accuracy
  - Chromatic Adaptation privacy
  - Chromatic Adaptation release notes
  - Related to Chromatic Adaptation
- Gamut Lab
  - Gamut Lab documentation
  - Getting started with the Gamut Lab
  - Compare colour spaces
  - Define a custom colour space
  - Inspect transfer functions
  - Gamut Lab reference
  - Troubleshooting the Gamut Lab
  - Gamut Lab FAQ
  - Gamut Lab: supporting pages
  - Gamut Lab shortcuts
  - Gamut Lab glossary
  - Gamut Lab limits and accuracy

- Gamut Lab privacy
- Gamut Lab release notes
- Related to the Gamut Lab

Gamut Mapping

- Gamut Mapping documentation
- Getting started with Gamut Mapping
- Compare spaces and methods
- Map a list of colours
- Map a colour into a smaller gamut
- Gamut Mapping reference
- Troubleshooting Gamut Mapping
- Gamut Mapping FAQ
- Gamut Mapping: supporting pages
- Gamut Mapping shortcuts
- Gamut Mapping glossary
- Gamut Mapping limits and accuracy
- Gamut Mapping privacy
- Gamut Mapping release notes
- Related to Gamut Mapping

HDR Gamut Plot

- HDR Gamut Plot documentation
- Getting started with the HDR Gamut Plot
- Compare broadcast gamuts
- Explore ICtCp and Jzazbz planes
- Read HDR transfer curves
- HDR Gamut Plot reference
- Troubleshooting the HDR Gamut Plot
- HDR Gamut Plot FAQ
- HDR Gamut Plot: supporting pages
- HDR Gamut Plot shortcuts
- HDR Gamut Plot glossary
- HDR Gamut Plot limits and accuracy
- HDR Gamut Plot privacy
- HDR Gamut Plot release notes
- Related to the HDR Gamut Plot

Illuminants

- Illuminants documentation
- Getting started with Illuminants
- Adapt colours between white points
- Compare two light sources
- Estimate measurement uncertainty
- Query many illuminants at once
- Illuminants reference
- Troubleshooting Illuminants
- Illuminants FAQ
- Illuminants: supporting pages
- Illuminants shortcuts

- Illuminants glossary
- Illuminants limits and accuracy
- Illuminants privacy
- Illuminants release notes
- Related to Illuminants
- Tonal Steps
  - Tonal Steps documentation
  - Getting started with Tonal Steps
  - Build a ramp
  - Check contrast
  - Export tokens
  - Make tints, shades, tones and key sets
  - Tonal Steps reference
  - Troubleshooting Tonal Steps
  - Tonal Steps FAQ
  - Tonal Steps: supporting pages
  - Tonal Steps shortcuts
  - Tonal Steps glossary
  - Tonal Steps limits and accuracy
  - Tonal Steps privacy
  - Tonal Steps release notes
  - Related to Tonal Steps
- Tone Mapping
  - Tone Mapping documentation
  - Getting started with Tone Mapping
  - Compare operators
  - Export a LUT
  - Map a frame
  - Read the scopes
  - Tone Mapping reference
  - Troubleshooting Tone Mapping
  - Tone Mapping FAQ
  - Tone Mapping: supporting pages
  - Tone Mapping shortcuts
  - Tone Mapping glossary
  - Tone Mapping limits and accuracy
  - Tone Mapping privacy
  - Tone Mapping release notes
  - Related to Tone Mapping
- Perception and spectral tools
  - Industrial Dye Chemistry (Dye in Solution)
    - Using Dye in Solution
    - Getting started with Dye in Solution
    - Build a bath
    - Compare lights
    - Export and share
    - Run a titration

- Run a tolerance sweep
- Use a measured spectrum
- Dye in Solution reference
- Troubleshooting Dye in Solution
- Dye in Solution FAQ
- Dye in Solution background
- Dye in Solution shortcuts
- Dye in Solution glossary
- Limits and accuracy
- Privacy
- Release notes
- Related tools and articles

Material Lighting (Material Under Light)

- Using Material Under Light
- Getting started with Material Under Light
- Compare with the reference light
- Export and share
- Run a sensitivity sweep
- Set up a scene
- Use a measured reflectance
- Material Under Light reference
- Troubleshooting Material Under Light
- Material Under Light FAQ
- Material Under Light background
- Material Under Light shortcuts
- Material Under Light glossary
- Limits and accuracy
- Privacy
- Release notes
- Related tools and articles

Neuroscience (Visual Pathway)

- Using the Visual Pathway
- Getting started with the Visual Pathway
- Cones, opponent channels and the cone table
- Explore the five stages
- Export and share
- Visual Pathway reference
- Troubleshooting the Visual Pathway
- Visual Pathway FAQ
- Visual Pathway background
- Visual Pathway shortcuts
- Visual Pathway glossary
- Limits and accuracy
- Privacy
- Release notes
- Related tools and articles

Perception and Vision

- Using Perception and Vision
- Getting started with Perception and Vision
- Check a palette and export
- Read the checks
- Simulate a design or image
- Perception and Vision reference
- Troubleshooting Perception and Vision
- Perception and Vision FAQ
- Perception and Vision background
- Perception and Vision shortcuts
- Perception and Vision glossary
- Limits and accuracy
- Privacy
- Release notes
- Related tools and articles
- Psychophysical Experiment Engine
  - Using the Psychophysics Bench
  - Getting started with the Psychophysics Bench
  - Design an experiment
  - Export your data
  - Read the results
  - Run a session
  - Psychophysics Bench reference
  - Troubleshooting the Psychophysics Bench
  - Psychophysics Bench FAQ
  - Psychophysics Bench background
  - Psychophysics Bench shortcuts
  - Psychophysics Bench glossary
  - Limits and accuracy
  - Privacy
  - Release notes
  - Related tools and articles
- Image and file tools
  - Collage Maker
    - Collage Maker documentation
    - Getting started with the Collage Maker
    - Arrange pictures in a grid
    - Build and edit a collage
    - Export, copy and save sessions
    - Read the measurements
    - Collage Maker reference
    - Troubleshooting the Collage Maker
    - Collage Maker FAQ
    - Collage Maker background material
    - Collage Maker shortcuts
    - Collage Maker glossary
    - Limits and accuracy

- Privacy
- Release notes
- Related tools and reading

ICC Profile Parser

- ICC Profile Parser documentation
- Getting started with the ICC Profile Parser
- Check conformance
- Export a reading
- Identify a profile and read its gamut
- Understand ICC profiles
- ICC Profile Parser reference
- Troubleshooting the ICC Profile Parser
- ICC Profile Parser FAQ
- ICC Profile Parser background material
- ICC Profile Parser glossary
- Limits and accuracy
- Privacy
- Release notes
- Related tools and reading

Image Picker

- Image Picker documentation
- Getting started with the Image Picker
- Analyse a list of hex colours
- Chart the palette
- Export and share
- Extract a palette
- Pick exact colours
- Read contrast pairs, tone bands and harmony
- Image Picker reference
- Troubleshooting the Image Picker
- Image Picker FAQ
- Image Picker background material
- Image Picker shortcuts
- Image Picker glossary
- Limits and accuracy
- Privacy
- Release notes
- Related tools and reading

LUT Lab

- LUT Lab documentation
- Getting started with LUT Lab
- Apply a look
- Load a .cube file
- LUT concepts
- Read what a transform moved
- Write a .cube file
- LUT Lab reference

- Troubleshooting LUT Lab
- LUT Lab FAQ
- LUT Lab background material
- LUT Lab glossary
- Limits and accuracy
- Privacy
- Release notes
- Related tools and reading

M-Gradient Palette

- M-Gradient Palette documentation
- Getting started with the M-Gradient Palette
- Build a ramp
- Check contrast and colour vision
- Compare routes and easings
- Export a ramp
- Interpolation and gamut
- M-Gradient Palette reference
- Troubleshooting the M-Gradient Palette
- M-Gradient Palette FAQ
- M-Gradient Palette background material
- M-Gradient Palette glossary
- Limits and accuracy
- Privacy
- Release notes
- Related tools and reading

SVG Recolour

- SVG Recolour documentation
- Getting started with SVG Recolour
- Check contrast and colour vision
- Export the result
- Recolour an SVG
- Recolour from a harmony
- SVG Recolour reference
- Troubleshooting SVG Recolour
- SVG Recolour FAQ
- SVG Recolour background material
- SVG Recolour glossary
- Limits and accuracy
- Privacy
- Release notes
- Related tools and reading

WEBSITE › TOOLS

Website tools

Every tool on auricartisan.com, grouped by what it helps you do, with its address and a link to its documentation.

Auric Artisan's tools run in your browser at https://auricartisan.com. Most need no account: what you make is kept in your browser, and you can export it as files. Each tool below has its own folder of documentation with a quick start, task guides, a complete reference, troubleshooting and a glossary.

The tools fall into five groups. If you are not sure which one you need, start with Which tool for which job.

Accessibility and vision

Check colour contrast, audit whole websites, work through WCAG, and see colour the way people with different vision do. Group overview: Accessibility and vision.

Tool	Address	What it does
Analyzer	/tool/analyzer/	Audits a website's colours, contrast and accessibility, and keeps the reports.
Contrast Checker	/tool/contrast-checker/	Grades a text and background pair with WCAG 2.x ratios and APCA.
Accessibility Checklist	/resources/accessibility-checklist/	50 WCAG 2.2 checks with the law by jurisdiction and an inspector.
Accessibility Compliance	/accessibility-compliance/	Where this address takes you and what it offers.
Vision Lab	/tool/vision-simulation/	Shows a colour, palette or image through 78 vision conditions.
Ishihara Plate Generator	/tool/ishihara-plate-generation/	Makes, checks and exports Ishihara-style plates.

Colour workspace

Build palettes and design systems, convert and study colours, and choose fonts and icons. Group overview: Colour workspace.

Tool	Address	What it does
Basic Color Tools	/tool/basic-tools/	A panel workspace of colour tools, plus the site-wide Colour Tools.
Color Science Lab	/tool/color-science-lab/	Workbench, Compare, Gamut audit and Illuminants views with 45 panel tools.
Personalization Generator	/tool/personalization-gen/	Turns a colour pool and a brief into palettes, type, UI pairs, tokens and a brand studio.
Design System Generator	/tool/design-engine/	Generates, checks and exports a complete design system.
ASCII Art Generator	/tool/ascii/	Turns a picture or a word into character art.
Font Library	/tool/font/	Browse, compare and embed fonts from the Auric font CDN.
Icon Library	/tool/icon/	Browse, customise and export icons from the Auric icon CDN.

Colorimetry and rendering

Light sources, adaptation, gamuts, HDR and tone. Group overview: Colorimetry and rendering.

Tool	Address	What it does
Illuminants	/tool/general/colorimetry/illuminants/	Standard illuminants, their spectra, white points and colour temperature.

Tool	Address	What it does
Chromatic Adaptation	/tool/general/colorimetry/chromatic-adaptation-dynamic/	Adapts colours and images from one white point to another.
Gamut Mapping	/tool/general/colorimetry/gamut-map/	Brings out-of-gamut colours into a target space.
Gamut Lab	/tool/general/gamut-and-rendering/gamut/	Compares colour spaces on chromaticity diagrams and slices.
HDR Gamut Plot	/tool/general/gamut-and-rendering/hdr-gamut-plot/	Plots HDR and wide-gamut colour.
Tonal Steps	/tool/general/gamut-and-rendering/tonal-steps-tints-shades/	Builds tint, tone and shade ramps.
Tone Mapping	/tool/general/gamut-and-rendering/tone-mapping/	Compares tone-mapping operators on HDR frames.

Perception and spectral

How the eye and brain see colour, psychophysics experiments, and colour from dyes and materials. Group overview: Perception and spectral.

Tool	Address	What it does
Visual Pathway	/tool/general/perception-and-neuroscience/neuroscience/	Follows a stimulus from the cones to the visual cortex.
Perception and Vision	/tool/general/perception-and-neuroscience/perception-vision/	Colour-vision simulation over your own image.
Psychophysical Experiment Engine	/tool/general/perception-and-neuroscience/psychophysical-experiment-engine/	Measures thresholds with psychophysical methods.
Dye in Solution	/tool/general/spectral-tools/industrial-dye-chemistry/	Dye colour against pH, concentration and path length.
Material Under Light	/tool/general/spectral-tools/material-lighting/	Renders materials under chosen light sources.

Image and file tools

Work with pictures, colour files and vector art. Group overview: Image and file tools.

Tool	Address	What it does
Collage Maker	/tool/general/tools/collage-maker/	Lays out pictures in a collage and exports it.
ICC Profile Parser	/tool/general/tools/icc-profile-parser/	Reads and explains ICC colour profiles.
Image Picker	/tool/general/tools/image-picker/	Takes colours and palettes from a picture.
LUT Lab	/tool/general/tools/lut-lab/	Previews looks and reads and writes .cube LUT files.
M-Gradient Palette	/tool/general/tools/m-gradient-palette/	Builds perceptual gradient ramps with accessibility checks.
SVG Recolour	/tool/general/tools/svg-recolor/	Maps and replaces the colours in an SVG.

Which tool for which job

You want to…	Use
Check that text is readable on its background	Contrast Checker
Audit a whole website for colour and accessibility	Analyzer
Work through WCAG 2.2 success criteria	Accessibility Checklist
See your design as people with colour-vision deficiencies do	Vision Lab
Build a palette, shades or a gradient quickly	Basic Color Tools
Generate a whole design system with tokens	Design System Generator
Take colours from a photograph	Image Picker
Make a perceptually even ramp	M-Gradient Palette or Tonal Steps
Understand a colour profile file	ICC Profile Parser
Find a font or icon for a project	Font Library or Icon Library
Learn how light and white points change colour	Illuminants and Chromatic Adaptation

The floating kits — a calculator, unit converter, notes and more — and the colour libraries are documented in their own sections.

In this folder

- accessibility-and-vision/
- colour-workspace/
- colorimetry-and-rendering/
- perception-and-spectral/
- image-and-file-tools/

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Accessibility and vision tools

What the accessibility and vision tools on auricartisan.com are for, and which one to open for each job.

This group covers the tools on auricartisan.com that help you make a design or a website usable by more people. Some of them measure a live page against the Web Content Accessibility Guidelines (WCAG), the W3C standard that most accessibility laws point at. Others show you how colour looks to people with colour-vision deficiencies or low vision, so you can catch problems that a number alone does not reveal.

The tools range from a one-question checker to a full audit console. Use the small tools when you already know what you are asking, and the Analyzer when you want to find the problems you do not know about yet.

Every tool here is free to use. A few actions that run on Auric Artisan's servers, such as rendering someone else's website or crawling a whole site, need an account and are metered in tokens. Each tool's own docs say exactly which actions those are.

The tools

Tool	Address	Use it to
Analyzer	https://auricartisan.com/tool/analyzer/	Audit a whole web page: accessibility (126 rules), contrast, performance, SEO, security and more
Contrast Checker	https://auricartisan.com/tool/contrast-checker/	Grade one text colour against one background with the WCAG 2.1 ratio
Accessibility Compliance	https://auricartisan.com/accessibility-compliance/	Old address; it now forwards you to the Accessibility Checklist
Accessibility Checklist	https://auricartisan.com/resources/accessibility-checklist/	Work through 50 WCAG 2.2 checks, record results with evidence, and read the law by country
Vision Lab	https://auricartisan.com/tool/vision-simulation/	Simulate colour-vision deficiencies and other vision conditions on colours and images
Ishihara Plate Generator	https://auricartisan.com/tool/ishihara-plate-generation/	Generate Ishihara-style dot plates for teaching and demonstrations

Which tool should I open?

- You have two colours and want a yes or no: open the Contrast Checker.
- You have a URL and want every problem on the page: open the Analyzer.
- You are preparing a manual audit or an accessibility statement: open the Accessibility Checklist.
- You want to see a palette or an image the way someone with a colour-vision deficiency sees it: open the Vision Lab.
- You are teaching how colour-vision tests work: open the Ishihara Plate Generator.

Important: None of these tools gives a legal certification or a medical diagnosis. Automated testing finds real problems quickly, but a conformance claim needs manual review as well, and a vision simulation is an approximation, not an eye test.

In this folder

- analyzer/ — the URL and website Analyzer

- contrast-checker/ — the WCAG colour-pair checker
- accessibility-compliance/ — the old compliance address and where it goes now
- accessibility-checklist/ — the checklist, the Inspector and the laws by jurisdiction
- vision-simulation/ — the Vision Lab
- ishihara-plate-generator/ — the Ishihara plate generator

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Accessibility Checklist

What the Accessibility Checklist is, who it is for, and how to start working through its 50 WCAG 2.2 checks.

The Accessibility Checklist is a free working checklist of 50 accessibility checks, grouped into nine areas, from semantics and keyboard focus to forms, media, colour, motion, ARIA, zoom and testing. Each check is explained in full: what it means, who it affects, how to test it, what the fix looks like in code, and what usually breaks it.

You work through it in the **Inspector**: a list of every check beside the selected check in full. Mark each one **Pass**, **Fail** or **N/A**, attach annotated screenshots as evidence, and watch a meter count what is left. When you are done, print the findings or copy them as Markdown into your issue tracker. Everything stays in your browser; there is no account and nothing is uploaded.

The page also carries a **Reference** of what the WCAG levels mean and what the law asks for in 15 jurisdictions, India first, with links to the primary sources.

Who it is for

- Front-end developers, designers and accessibility testers preparing or running a manual review.
- Teams who want evidence of a process to go with an automated audit.
- Anyone who needs to know which accessibility law applies where, and to which WCAG version.

Where to find it

- Address: https://auricartisan.com/resources/accessibility-checklist/
- From any page: open the **Learn** menu in the site header and select **Accessibility Compliance** ("WCAG checks & guidance").
- The old addresses https://auricartisan.com/accessibility-compliance/ and https://auricartisan.com/compliance forward here. See Accessibility Compliance.
- A single check: add `#check-` and the check's ID, for example `https://auricartisan.com/resources/accessibility-checklist/#check-k2`.

Quick start

1. Open https://auricartisan.com/resources/accessibility-checklist/. The first check, "Landmarks wrap every region of the page", is selected.
2. Optionally run the automated pass first: select **Run the automated pass first** → to open the Analyzer on your page.
3. Read the check in the pane on the right (on a phone, tap a row to open it).
4. Test your page as the **How to test it** block describes.
5. Select **Pass**, **Fail** or **N/A**. With focus in the list or the pane, you can press `Space`, `F` or `N`.
6. Press `J` to move to the next check, and repeat.
7. For a failure, select **Attach screenshot**, add a screenshot and mark it up.
8. Open the **Export** tab and select **Copy as Markdown** or **Export the report**.

What you can do

- Read 50 checks, each tied to a WCAG 2.2 success criterion or marked as good practice, with a code example for most.
- See which checks the Analyzer can decide for you (**AUTO**), narrow down (**PARTLY**), or cannot judge (**BY HAND**).

- Mark each check **Pass**, **Fail** or **N/A**, and clear a mark by pressing it again.
- Filter to **Required** (Level A and AA), **AAA** or **Practice** checks, or hide passed checks with **Open only**.
- Attach any number of screenshots to a check and annotate them with a pen, highlighter, arrows, boxes, redaction and numbered callouts.
- Print the flagged findings, or copy them as Markdown for Jira, Linear or GitHub.
- Look up the accessibility law for India, the United States, the European Union, the United Kingdom, Canada, Australia, Japan and eight more countries.
- Work entirely from the keyboard.

In this folder

- docs/ — how to use the checklist
  - Getting started — the screen tour and your first check
  - Work through the checklist — selecting, marking, filtering and the meter
  - Attach evidence — screenshots and the annotation pen
  - Export your results — print, Markdown and reset
  - The 50 checks — every check by area, with its criterion and who can decide it
  - Laws by jurisdiction — the Reference tab
  - Reference — every control, label and message
  - Troubleshooting
  - FAQ
- others/ — supporting material
  - Glossary
  - Keyboard shortcuts
  - Limits and accuracy
  - Related tools and reading
  - Privacy

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Accessibility Checklist documentation

The list of Accessibility Checklist guides with one line on each.

These guides explain how to use the checklist at https://auricartisan.com/resources/accessibility-checklist/. Start with Getting started; the reference lists every control.

Guide	What it covers
Getting started	The masthead, the tab rail and meter, the Inspector, and your first check
Work through the checklist	Selecting checks, marking them, filters, folding areas, the meter, and the phone layout
Attach evidence	Adding screenshots to a check and annotating them
Export your results	The Export tab: printing, copying Markdown, and clearing your work
The 50 checks	Every check, area by area, with its criterion and how much the Analyzer can decide
Laws by jurisdiction	The Reference tab: WCAG levels and the law in 15 jurisdictions
Reference	Every control, label, message and stored item
Troubleshooting	Common problems and what to do
FAQ	Short answers to common questions

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Getting started with the Accessibility Checklist

Open the Accessibility Checklist, learn the screen, and mark your first check.

Open it

Go to https://auricartisan.com/resources/accessibility-checklist/, or open the **Learn** menu in the site header and select **Accessibility Compliance**. No account is needed.

The screen, top to bottom

Masthead

The kicker "Auric Artisan · Accessibility", the title **The accessibility checklist**, and a line: "Fifty checks across the whole of WCAG 2.2, each one explained: what it means, who it affects, how to test it, and what the fix looks like in code."

A note under it says: "The analyzer decides 11 of these outright and narrows 23 more. The rest needs a person, which is what the explanations are for." Two links follow:

- **Run the automated pass first** → opens the Analyzer.
- **Wire it into CI** → opens the public API documentation.

Tab rail and meter

Three tabs: **Checklist**, **Export** and **Reference**. With a tab focused, `Left` and `Right` move between tabs (wrapping round), and `Home` and `End` jump to the first or last.

Beside the tabs is the **meter**: "0 passed · 0 flagged · 50 left", a two-colour bar (passed and flagged), and "of 50". The meter always counts all 50 checks, whatever filter is on. When a filter is on, a small chip beside it names the filter and how many rows it shows, for example "REQUIRED · 35".

On wide screens the tab rail and meter stay pinned under the site header as you scroll.

The Checklist tab: the Inspector

On screens at least 1080 pixels wide, the Checklist tab is two columns:

- **The index**, on the left: filter buttons, then nine numbered areas, each with a header showing how many of its checks you have decided (for example "0 / 6"), then one row per check. Each row shows a state symbol, the check title and its tag (for example "A 1.3.1" or "PRACTICE"). Required checks show their tag in gold.
- **The pane**, on the right: the selected check in full. It stays in view while the index scrolls.

The first check is selected when you arrive, unless the address names another one.

On narrower screens you see the index only. Tapping a row opens that check full-screen, with an **All checks** button to go back. See Work through the checklist.

What the pane shows

From top to bottom:

1. A bar with the check's position ("3 of 50") and **Previous check** and **Next check** buttons.
2. The area, for example "01 Semantics & structure".
3. The check title.
4. Chips: the criterion tag, **Required** for Level A and AA checks, and **AUTO**, **PARTLY** or **BY HAND** with a sentence saying how much the Analyzer can decide.
5. A one-sentence summary of the check.
6. The verdict buttons: **Pass**, **Fail**, **N/A** and **Attach screenshot**.
7. **Evidence**, once you have attached screenshots.
8. The explanation: **What it means**, **Who it affects**, **How to test it**, **What the fix looks like** (on most checks), **Usually breaks when**, and links such as "Understanding 1.3.1" to the W3C.

Mark your first check

1. Make sure the first check, "Landmarks wrap every region of the page", is selected in the pane.
2. Open the page you are testing in another tab.
3. Follow **How to test it**. For this check, you run a short snippet in the browser console and look for exactly one `main` landmark and a label on every repeated `nav`.
4. Back in the checklist, select **Pass** if the page meets it, **Fail** if it does not, or **N/A** if it does not apply.
5. Look at the index row: its symbol changes, and the meter updates.
6. Select **Next check** (or press `J` with focus in the list or pane) to move on.

The result: your mark is saved in this browser straight away. Close the tab and come back later, and it is still there.

Next steps

- Work through the checklist for filters, keys and the phone layout.
- Attach evidence to back up a failure with an annotated screenshot.

- Export your results when you are done.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Attach evidence

Add screenshots to a check and annotate them with the pen, highlighter, arrows, boxes, redaction and numbered callouts.

A finding is easier to fix, and harder to dispute, when it carries a screenshot. Every check can hold any number of screenshots, each with its own annotations.

Add a screenshot

1. Select the check in the index.
2. Select **Attach screenshot** in the pane. On narrow phones this is an icon-only button with the same name. The **Screenshot evidence** dialog opens.
3. Add one or more images in any of three ways:
   - drag and drop image files anywhere on the dialog;
   - paste an image with `Ctrl` + `V`;
   - select **Choose a file** and pick one or more images.
4. The dialog announces how many screenshots were attached.

Only image files are accepted; other files are ignored. Screenshots are saved in your browser as soon as they are added.

Annotate it

The toolbar on the left of the dialog ("Annotation tools") holds six tools. Each has a key while the dialog is open.

Tool	Key	What it draws
Pen	`P`	A freehand line
Highlighter	`H`	A wide, translucent freehand stroke
Arrow	`A`	A straight arrow. This is the default tool.
Box	`B`	A rectangle outline
Redact	`R`	A solid filled rectangle in the chosen colour
Callout	`C`	A numbered circle (1, 2, 3 and so on) with one click

- **Colours:** Red (default), Blue, Green and Black.
- **Line widths:** Fine, Medium (default) and Broad.

Drag across the screenshot to draw. Every mark is saved as you make it.

The **Marks** panel lists the marks on the current screenshot, newest first, each with a remove button ("Remove this mark"). Before you draw anything it says "Nothing marked yet. Pick a tool and drag across the screenshot."

Manage screenshots

The footer of the dialog has:

- a strip of thumbnails for this check's screenshots, each showing its mark count, with a tile to add another;

- a status line such as "2 screenshots · 5 marks · no limit";
- **Undo**, which removes the last mark;
- **Clear marks**, which removes every mark on the current screenshot at once, without asking;
- **Remove**, which deletes the current screenshot. The first press changes the label to **Remove this one?** for four seconds; press again to confirm. The dialog announces "Screenshot removed."

Select **Done**, or press `Esc`, to close the dialog. Focus returns to **Attach screenshot**.

In the pane

Once a check has screenshots, the pane shows an **Evidence** block with the number pinned and numbered thumbnails ("Screenshot 1" and so on). Select a thumbnail to reopen it in the dialog, or the **Add** tile to add another. The caption reads: "No cap on images or on marks, and the pen never touches the picture underneath."

What the marks do to the image

Marks are stored separately from the original image, which is never changed. When you close the dialog, a copy of each screenshot is made with the marks baked in; that copy is what the thumbnails and the printed report show.

> **Important: Redact** covers an area only in that marked-up copy. The original, unredacted screenshot stays stored in your browser. Do not rely on Redact to remove sensitive information from what is stored on your device.

Limits

- The page sets no limit on screenshots or marks; your browser's own storage limits still apply.
- Screenshots need your browser's database storage. In some private windows it is unavailable; marks on checks still work, but screenshots cannot be attached.
- Screenshots are not included in the Markdown export, which only says how many are attached. The printed report shows up to four small thumbnails per check.
- The checklist's single-key shortcuts are switched off while the dialog is open, and the site's own right-click menu gives way to your browser's.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Export your results

Use the Export tab to print your findings, copy them as Markdown for an issue tracker, or clear your work.

Open the **Export** tab. Its heading is "Turn the flags into a report".

What the tab shows

- **What the report will contain** — four tiles with your current counts: **Flagged**, **Passed**, **Not applicable** and **Untouched** (which starts at 50).
- **Decided, by level** — a bar for each of **A** (out of 22), **AA** (out of 13), **AAA** (out of 2) and **Practice** (out of 13). Here "decided" means Passed or Flagged; N/A does not count.
- **Evidence** — "No screenshots attached", or how many screenshots are attached across how many checks.
- **Preview — the first lines, as they will read** — the first lines of the Markdown export, exactly as it will be copied.

- **Three ways out** — the three buttons below.
- A privacy note: "Your work stays on this device."
- **Before you send it** — how many checks are untouched, and a reminder that "A worked checklist is evidence of a process, not a conformance claim."

Copy as Markdown

"Straight into the tracker." Best for Jira, Linear, GitHub and similar tools.

1. Select **Copy as Markdown**.
2. The page says "The findings are on your clipboard, as Markdown."
3. Paste into your tracker.

The Markdown always covers all 50 checks, whatever filter is on. It has this shape:

```
# Accessibility review

Worked 24 September 2026 · WCAG 2.2 · 15 of 50 checks decided

- Flagged: 3
- Passed: 10
- Not applicable: 2
- Never checked: 35

Source: https://auricartisan.com/resources/accessibility-checklist/

## Flagged (3)

### 2.4.7 Focus is always visible, and never removed without a replacement
Level AA · Keyboard & focus · 2 screenshots attached
outline: none with nothing in its place is the single most common accessibility bug in the wild.
> (the "What it means" paragraph)

## Never checked (35)
These were not looked at. They are not passes.
- 1.3.1 Landmarks wrap every region of the page

A worked checklist is evidence of a process, not a conformance claim.
```

Details worth knowing:

- Each flagged check has its criterion and title as a heading, then its level (or "Practice"), area and screenshot count, its one-line summary, and its "What it means" paragraph as a quote. Practice checks have no criterion number.
- Passed and not-applicable checks appear only as counts.
- Screenshots are never included; only their number.
- "Worked" is the date you copy, not the date you did the work.
- The "checks decided" count includes N/A.
- If nothing is flagged, the lists are left out and the Markdown says either that nothing is flagged but some checks were never looked at, or simply "Nothing is flagged."

If the browser refuses the clipboard, the page says "Could not reach the clipboard — the browser refused it."

Export the report (print)

The button is described as "One printable document". In the current version it opens your browser's **print dialog** for the checklist; it does not download a file. To get a PDF, choose **Save as PDF** in the print dialog.

When you press it, the page announces how many findings it is printing, or "Nothing is flagged — printing the list as it stands."

The printout contains the list of checks:

- **Flagged** checks with their full explanation and a heavy left border;
- **Untouched** checks collapsed to their title, tags and summary;
- up to four small thumbnails of the marked-up screenshots per check.

Passed and **N/A** checks are left out of the printout, and the tab rail, meter, index and pane are not printed. There is no summary of counts on paper; use **Copy as Markdown** for those.

> **Tip:** Select **All 50** and switch off **Open only** before printing, so that a filter does not limit what prints. Pressing `Ctrl + P` gives the same printout as the button.

Reset

"Clear this device." This removes every mark and every screenshot from every check.

> **Important:** Although the button's description says it asks first, in the current version **Reset clears everything immediately, with no confirmation and no undo**. There is no server copy to fall back on. Copy your Markdown or print first if you might need your work.

After a reset the page says "Cleared — marks and screenshots." Filters and folded areas are not changed.

What is not available

- No CSV, JSON or file download.
- No import or restore, so progress cannot be moved to another browser or device.
- No notes field: only marks and screenshots are kept.

The "What a claim actually needs" link

The link at the foot of the Export tab, **What a claim actually needs** →, is meant to take you to the Reference tab. If it does not switch tabs, select **Reference** in the tab rail. See Laws by jurisdiction.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Laws by jurisdiction

The Reference tab of the Accessibility Checklist: what the WCAG levels mean and what the law asks for in 15 jurisdictions.

The **Reference** tab is headed "What the levels mean, and what the law asks for". It explains the four levels used on the page and summarises the accessibility law in 15 jurisdictions, with links to the primary sources.

> **Important:** The page states this is "Orientation, not legal advice." Dates move: the page notes that two of its dates moved in the eighteen months before its last review. It was last reviewed on **11 September 2026**, covering 29 instruments in 15 jurisdictions. Check the primary source before you rely on any date. The summaries below repeat the page's content as of that review.

The four levels on this page

Level	Checks	What the page says
A	22	The floor. Nearly every instrument below names it, or something built on it.
AA	13	What the statutes and procurement documents actually cite. A + AA is the legal line.
AAA	2	Stricter than any law here requires. Worth doing; never demanded wholesale.
Practice	13	No criterion covers it, and every audit finds it anyway.

The page's summary: WCAG 2.1 Level AA is the common denominator almost everywhere. The UK and Australia have already moved to WCAG 2.2 AA, and EN 301 549 V4 takes the European standard there too. Build to WCAG 2.2 AA and you are above every line on the page.

Finding your way

A side rail headed **Jurisdictions** jumps to each card: India (5 instruments), United States (4), European Union (3), United Kingdom (2), Canada (3), Australia (2), Japan (2) and **Elsewhere** (8). On narrow screens it becomes a row of buttons you scroll sideways.

Each card shows the jurisdiction, its standard, a table of instrument, what it does and when, and who it binds. Instrument names link to their primary sources. Dates that are current are highlighted.

A second rail, **Also on this site**, links to the Learn library ("The science behind each rule"), the Analyzer ("What the analyzer checks") and WCAG 2.2 at the W3C.

India

Standard: WCAG 2.1 AA via GIGW 3.0. Binds government, establishments and service providers, public and private.

Instrument	When	What it does
Rights of Persons with Disabilities Act (Act 49 of 2016, ss. 40, 42, 46)	2016	Empowers accessibility standards, requires accessible ICT, and places the duty on service providers
Rajive Raturi v. Union of India, Supreme Court of India	8 Nov 2024	Held merely recommendatory guidelines ultra vires; gave the Union three months to set a mandatory floor
GIGW 3.0, Guidelines for Indian Government Websites (MeitY / NIC)	2023, revised 2025	88 mandatory checkpoints; names WCAG 2.1 Level AA; certified by STQC
IS 17802 (Parts 1 and 2), Bureau of Indian Standards	2021	India's own ICT accessibility standard, written to be citable in procurement

Instrument	When	What it does
Draft ICT Accessibility Rules (consultation, after Raturi)	In progress	The mandatory floor the Court ordered; check its current status

Non-compliance with the Act is an offence under section 89, according to the page.

United States

Standard: WCAG 2.1 AA, with Section 508 still at WCAG 2.0 AA. Binds federal, state and local government, and private business through Title III litigation.

Instrument	When	What it does
ADA Title II, DOJ final rule (28 CFR Part 35, April 2024)	2027 / 2028	State and local government web content and apps to WCAG 2.1 AA
Section 508, Rehabilitation Act (the 2017 Refresh)	2018	Federal agencies and what they buy, to WCAG 2.0 AA by reference
Section 504 (HHS and Department of Education rules)	2024 onwards	Recipients of federal funding, such as hospitals and universities, to WCAG 2.1 AA
ADA Title III (public accommodations)	1990	No technical standard; where most private litigation happens, with WCAG as the practical measure

The page notes that in April 2026 the DOJ pushed both Title II deadlines back a year: large entities to April 2027, and smaller entities and special districts to April 2028. The underlying duty not to discriminate never paused.

European Union

Standard: EN 301 549 V3.2.1, which points at WCAG 2.1 AA. Binds the public sector and, since 2025, much of the private sector across 27 member states.

Instrument	When	What it does
European Accessibility Act (Directive (EU) 2019/882)	28 Jun 2025	Reaches the private sector: e-commerce, banking, e-books, communications, ticketing, ATMs
Web Accessibility Directive (Directive (EU) 2016/2102)	2019 / 2021	Public sector sites and apps, with an accessibility statement and a feedback route
EN 301 549 (V3.2.1, March 2021)	V3.2.1	The harmonised standard; clause 9 is WCAG 2.1 AA, and V4.1.1 moves it to WCAG 2.2

Service microenterprises are exempt from the European Accessibility Act.

United Kingdom

Standard: WCAG 2.2 AA. Binds the public sector by regulation and everyone else by duty.

Instrument	When	What it does
Public Sector Bodies Accessibility Regulations (SI 2018/952)	2.2 since 2024	Public sector to WCAG 2.2 AA, monitored against 2.2 since October 2024, plus a statement

Instrument	When	What it does
Equality Act 2010 (s. 20, reasonable adjustments)	2010	Outcome-based duty with no named standard; applies to private service providers too

Canada

Standard: CAN/ASC-EN 301 549:2024. Binds federally regulated bodies; provinces add their own rules.

Instrument	When	What it does
Accessible Canada Act (S.C. 2019, c. 10) and Regulations	2019	Federally regulated organisations publish accessibility plans, take feedback and report
CAN/ASC-EN 301 549:2024, Accessibility Standards Canada	31 May 2024	Adopts the European standard: WCAG 2.1 AA plus extra ICT requirements
AODA, Ontario Integrated Accessibility Standards (s. 14)	2021	WCAG 2.0 AA, without live captions and audio description, for large organisations and the public sector

Australia

Standard: WCAG 2.2 AA. Binds government directly and business through the Disability Discrimination Act.

Instrument	When	What it does
Disability Discrimination Act 1992 (and Maguire v SOCOG, 2000)	1992	Names no standard; the Sydney Olympics case is why WCAG is the measure
Government digital standards (with AS EN 301 549:2020 for procurement)	2.2 AA	WCAG 2.2 AA formally adopted as the minimum, up from 2.0

Japan

Standard: JIS X 8341-3:2016, which carries WCAG 2.0. Binds public bodies by standard and private business by duty since 2024.

Instrument	When	What it does
JIS X 8341-3:2016, Japanese Industrial Standard	2016	The success criteria of WCAG 2.0, with its own conformance and testing procedure
Act on Eliminating Discrimination against Persons with Disabilities (2021 amendment)	Apr 2024	Reasonable accommodation required of private business, not merely encouraged

Elsewhere: eight more worth knowing

These rows have no links on the page.

Country	What the page says
Israel	IS 5568, pointing at WCAG 2.0 AA. Binds most private sites, which is unusually broad.
Brazil	LBI 13.146/2015 article 63; eMAG for government.
South Korea	KWCAG 2.1, with a national certification mark.

Country	What the page says
Norway	WCAG 2.1 AA, and it reaches private sites aimed at the public.
France	RGAA 4.1; Law 2005-102 article 47 carries real fines and a public declaration.
Germany	BITV 2.0 for public bodies; the BFSG implements the European Accessibility Act.
Italy	Legge Stanca, extended to large private companies.
New Zealand	Web Accessibility Standard 1.1, pointing at WCAG 2.1 AA for government.

What this is not

The page closes with two statements, quoted here:

- "**Orientation, not legal advice.** Every row points at a primary source. Dates move — two of the dates here moved in the eighteen months before this review — so check the source before you rely on one."
- "**A worked checklist is not a conformance claim.** It is evidence of a process. A claim needs a full evaluation against the whole standard, and the people who use assistive technology every day."

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

The 50 checks

Every check in the Accessibility Checklist, area by area, with its WCAG criterion, level and how much the Analyzer can decide.

The checklist has 50 checks in nine areas. This page lists them all, with the one-line summary the page shows for each (a few are slightly shortened to fit the table). Open a check on the page for its full explanation, test and fix.

How to read the tables

- **ID** is the check's identifier. Open it directly with `#check-` plus the ID, for example `https://auricartisan.com/resources/accessibility-checklist/#check-k2`.
- **Tag** is the level and WCAG success criterion, as the page shows it. **Practice** means good practice with no criterion of its own.
- Checks at Level A or AA are **Required**; AAA and Practice checks are optional.
- **Decided by** says how much the Analyzer can do:
  - **AUTO** — "The analyzer decides this one on its own"
  - **PARTLY** — "The analyzer narrows this down; a person confirms it"
  - **BY HAND** — "Only a person can decide this one"

The checklist itself does not import Analyzer results. Run the Analyzer and mark the checks yourself.

Totals	Count
Level A	22
Level AA	13
Level AAA	2
Practice	13
Required (A and AA)	35
AUTO / PARTLY / BY HAND	11 / 23 / 16

01 Semantics & structure

"What a screen reader builds its map from. Nothing below works if this is wrong."

ID	Check	Tag	Decided by	Summary
d1	Landmarks wrap every region of the page	A 1.3.1	PARTLY	header, nav, main, footer — one main per page, and any repeated nav named.
d2	Headings describe structure, not size	A 1.3.1	AUTO	One h1, no skipped levels. Style with CSS; never pick a heading level to get a font size.
d3	html lang is set, and switches on foreign passages	A 3.1.1	AUTO	It picks the speech synthesiser. A French quote inside English needs its own lang.
d4	Every route has a unique, front-loaded title	A 2.4.2	PARTLY	In a SPA the title is yours to set on navigation — the browser will not do it.
d5	Lists, tables and figures use real elements	A 1.3.1	PARTLY	A table needs th with scope and a caption. A div grid announces nothing.
d6	The reading order matches the visual order	Practice	BY HAND	CSS order, grid placement and absolute positioning all move the eye without moving the DOM.

02 Keyboard & focus

"The whole product, without a mouse. This is where most single-page apps fail."

ID	Check	Tag	Decided by	Summary
k1	Every interactive element is reachable and operable by keyboard	A 2.1.1	PARTLY	A div with onClick is not. Use a button, or add role, tabindex and both key handlers.
k2	No keyboard trap — focus can always get back out	A 2.1.2	BY HAND	Modals, embedded players and custom editors are the usual offenders.
k3	Focus is always visible, and never removed without a replacement	AA 2.4.7	PARTLY	outline: none with nothing in its place is the most common accessibility bug in the wild.
k4	Focus order follows meaning	A 2.4.3	PARTLY	No positive tabindex. tabindex="-1" for programmatic targets, "0" to add to the order.
k5	Focus is managed across route changes and dialogs	Practice	BY HAND	On open, move focus in; on close, return it to the trigger. On navigation, move it to the new h1.
k6	A skip link comes first and actually moves focus	A 2.4.1	PARTLY	It must be the first tabbable thing, visible on focus, and target a focusable container.
k7	Pointer targets are at least 24x24 CSS pixels	AA 2.5.8	AUTO	New in WCAG 2.2. Spacing counts toward it — small icons need room around them.

03 Forms & errors

"Where an inaccessible product stops being annoying and starts costing money."

ID	Check	Tag	Decided by	Summary
f1	Every control has a programmatic label	A 3.3.2	AUTO	A placeholder is not a label. It vanishes on input and many screen readers skip it.
f2	Errors are identified in text and tied to the field	A 3.3.1	PARTLY	aria-describedby links the message; aria-invalid marks the state. Colour alone says nothing.
f3	Common fields declare autocomplete	AA 1.3.5	AUTO	name, email, tel, address — it is a requirement, and it halves the typing for everyone.
f4	Related controls are grouped with fieldset and legend	A 1.3.1	PARTLY	Radio sets and checkbox groups are meaningless read one at a time.
f5	Validation is announced, not just painted	Practice	BY HAND	A live region or a focus move — otherwise a screen reader user submits into silence.
f6	Legal and financial submissions are reversible or confirmed	A 3.3.4	BY HAND	Reversible, checked, or confirmed. Pick one before taking the payment.

04 Images & media

"Alt text is a decision, not a field to fill in."

ID	Check	Tag	Decided by	Summary
m1	Every image is either described or explicitly decorative	A 1.1.1	PARTLY	alt="" and aria-hidden for decoration. A filename in the alt is worse than nothing.
m2	Text in an image is repeated in real text	A 1.4.5	BY HAND	It cannot be resized, restyled, translated or searched — screenshots of code are the worst case.
m3	Video has captions; audio has a transcript	A 1.2.2	BY HAND	Auto-captions are a draft. Someone reads them before they ship.
m4	Video has audio description where it carries visual information	AA 1.2.5	BY HAND	If the picture says something the soundtrack does not, it needs describing.
m5	Nothing plays automatically for more than three seconds	A 1.4.2	PARTLY	Or there is a control to stop it, reachable before everything else.
m6	Icons carry an accessible name or are hidden	Practice	PARTLY	An icon-only button needs aria-label; an icon beside a label needs aria-hidden.

05 Colour & contrast

"The part an automated tool can genuinely finish for you."

ID	Check	Tag	Decided by	Summary
c1	Body text reaches 4.5:1, large text 3:1	AA 1.4.3	AUTO	Against the composited background — the colour after every translucent layer, not the token.
c2	Controls and meaningful graphics reach 3:1	AA 1.4.11	AUTO	Input borders, toggles, chart series, focus rings.
c3	Colour is never the only carrier of meaning	A 1.4.1	PARTLY	Links underlined, errors iconned, chart series labelled or patterned.

ID	Check	Tag	Decided by	Summary
c4	Hover, focus, active and selected states all pass too	AA 1.4.11	PARTLY	A rule that wins at rest can lose on hover — measure the state, not the resting colour.
c5	Dark mode and forced-colors are checked separately	Practice	PARTLY	An inverted palette almost never passes, and Windows High Contrast drops your backgrounds.
c6	Simulated for protanopia, deuteranopia and tritanopia	Practice	AUTO	On real screens, not only on the swatch sheet.

06 Motion & preferences

"The settings the operating system already knows about."

ID	Check	Tag	Decided by	Summary
p1	prefers-reduced-motion is honoured	AAA 2.3.3	PARTLY	Not only decorative animation — parallax, autoplaying carousels and scroll-jacking.
p2	Nothing flashes more than three times a second	A 2.3.1	BY HAND	This one can cause seizures. It is not a preference.
p3	Moving or auto-updating content can be paused	A 2.2.2	PARTLY	Carousels, tickers, live feeds — pause, stop or hide.
p4	Timeouts are announced, or there are none	AAA 2.2.6	BY HAND	A session that dies silently loses the form someone spent twenty minutes on.

07 ARIA & custom widgets

"No ARIA beats bad ARIA. Reach for a native element first, every time."

ID	Check	Tag	Decided by	Summary
a1	Custom widgets expose name, role and value	A 4.1.2	PARTLY	And keep them in sync as state changes. A silent aria-expanded is a broken disclosure.
a2	The widget follows its APG keyboard pattern	Practice	BY HAND	Tabs, menus, comboboxes and trees each have an expected key map. Users know it already.
a3	Status messages reach a live region	AA 4.1.3	PARTLY	polite for confirmations, assertive only for genuine interruptions. Never both.
a4	No aria-hidden on a focusable element	Practice	AUTO	It creates a control a screen reader cannot see but can still land on.
a5	Link text makes sense out of context	A 2.4.4	PARTLY	"Read more" repeated eleven times is a list of eleven identical links.

08 Zoom, reflow & spacing

"The same page in the conditions people actually browse it."

ID	Check	Tag	Decided by	Summary
z1	No horizontal scroll at 320 CSS pixels wide	AA 1.4.10	AUTO	Equivalent to 400% zoom on a desktop. Two-dimensional scrolling is the failure.
z2	Text resizes to 200% without loss	AA 1.4.4	PARTLY	Fixed-height containers with overflow: hidden are where the text goes.
z3	User text-spacing overrides do not break the layout	AA 1.4.12	PARTLY	Line height 1.5, paragraph 2em, letter 0.12em, word 0.16em — nothing may be clipped.
z4	Both orientations work	AA 1.3.4	PARTLY	No locking to portrait unless the content genuinely requires it.
z5	Focused elements are not hidden behind sticky chrome	AA 2.4.11	BY HAND	New in 2.2, and a sticky header plus scroll-into-view is exactly how it happens.

09 Testing & proof

"How it stays fixed after the sprint that fixed it." All five are Practice checks, and each links to the W3C's guidance on evaluation.

ID	Check	Tag	Decided by	Summary
x1	An automated pass runs in CI on every pull request	Practice	AUTO	It catches about a third. That third should never reach a human reviewer again.
x2	A keyboard-only pass is part of review	Practice	BY HAND	Unplug the mouse and do the primary task. Ten minutes, most of the remaining bugs.
x3	A screen reader pass on one real combination	Practice	BY HAND	NVDA with Firefox, or VoiceOver with Safari. Not the browser plugin that reads the DOM.
x4	Findings carry evidence and a criterion	Practice	BY HAND	A screenshot and a number turn "this feels wrong" into something a developer can close.
x5	People with disabilities are in the testing loop	Practice	BY HAND	Every checklist including this one is a proxy for that, never a replacement.

Links inside the checks

Every check links to the W3C Understanding document for its criterion ("Understanding 1.3.1" and so on); area 09 links to "W3C on evaluation". Some checks add a tool link:

Check	Link	Opens
d1	Check the landmarks	The Analyzer
k3	Check the focus ring	The Analyzer
c1	Check a pair	The tools collection page
c2	Audit the page	The Analyzer
c3	Simulate it	The Vision Lab
c6	Run the simulator	The Vision Lab
x1	Automate with the API	The API documentation

To check a single colour pair for c1, use the Contrast Checker.

Checks with no **What the fix looks like** block: d6, f6, m2, m3, m4, m5, c2, c3, c4, c6, p2, p3, p4, a2, z1, z4, x2, x3, x4 and x5.

For notes on the level tags of m2 and f6, see Limits and accuracy.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Work through the checklist

Select checks, mark them Pass, Fail or N/A, filter the list, fold areas, read the meter, and use the one-check view on a phone.

Select a check

- **With a mouse or touch:** select a row in the index.
- **With the keyboard:** focus a row, then press `Down` or `Up` to move to the next or previous visible check, and `Home` or `End` for the first or last. Only one row is in the `Tab` order at a time, so `Tab` moves out of the list rather than through every row.
- **From the pane:** select **Previous check** or **Next check**, or press `K` or `J`.
- **From the address:** open `#check-` plus the check's ID, for example `#check-f2`.

The pane's bar shows where the check sits in the full list of 50, for example "12 of 50", whatever filter is on. **Previous check** and **Next check** are disabled at either end of the visible list.

Mark a check

Button	Key	Meaning	State word
Pass	`Space`	The page meets this check	Passed
Fail	`F`	The page does not meet it: it is flagged	Failed
N/A	`N`	The check does not apply to this page	Not applicable

- Selecting the active verdict again clears it back to **Not checked yet**.
- `Space` marks **Pass** only when focus is on a row, on the pane, or on the check title. Elsewhere `Space` presses the focused button as usual.
- A screen reader hears the new state word after every mark.

Each row shows its state as a shape as well as a colour:

State	Symbol
Not checked yet	An empty ring
Passed	A gold disc with a tick; the row title is dimmed
Failed	A red disc with an exclamation mark
Not applicable	A dashed ring with a dash

Marks are saved in this browser the moment you make them.

Where the keys work

The single-key shortcuts (`J`, `K`, `Space`, `F`, `N`, `E`) only respond while focus is inside the index or the pane. They are ignored in text fields, when `Ctrl`, `Alt` or `Cmd` is held, and while the screenshot pen is open. This is deliberate: WCAG 2.1.4 asks that

single-character shortcuts do not fire by accident, and the page follows its own advice.

Key	Action
`E`	Open the check: on a phone, the one-check view; on a wide screen, focus moves to the pane title
`Esc` in the pane	On a phone, back to the list; on a wide screen, focus returns to the check's row
`J` from the pane	Next check; focus stays in the pane and the new title is announced

See Keyboard shortcuts for the full list.

Filter the list

The filter buttons at the top of the index:

Button	Shows	Count
All 50	Every check (default)	50
Required	Level A and AA checks	35
AAA	Level AAA checks	2
Practice	Good-practice checks with no WCAG criterion of their own	13
Open only	A switch that hides checks marked Passed; it combines with any of the buttons above	—

- **All 50**, **Required**, **AAA** and **Practice** are mutually exclusive.
- **Open only** keeps flagged, not-applicable and untouched checks.
- If the selected check is filtered out, the selection moves to the first visible check.
- `J`, `K`, the arrow keys and **Previous check** / **Next check** skip hidden rows.
- If nothing matches, the index says "Nothing matches these filters." Select **All 50** and switch off **Open only**.
- Filters are not remembered: a reload returns to **All 50**.

There is no text search, no filter by area, and no separate A or AA filter.

Fold an area

Select an area header (for example "02 Keyboard & focus 0 / 7") to fold or unfold its rows. Folding is not a filter and does not change the meter, but keyboard navigation skips folded rows. Folds are not remembered; every area is open after a reload.

The header's tally is decided out of total, where decided means any mark, including N/A.

Read the meter

The meter beside the tabs reads, for example, "12 passed · 3 flagged · 33 left … of 50".

- **left** is 50 minus passed, flagged and not-applicable checks. N/A has no bar segment of its own.
- The meter always counts all 50 checks. Filters never change it; the chip beside it says what the list is showing.
- Screen readers also hear a fuller sentence, for example "12 passed, 3 flagged, 33 not yet checked of 50. 2 of the flagged are A or AA.", or, with nothing flagged, how many **required** checks are left.

On a phone: the one-check view

Below 1080 pixels wide, the index is the page and the pane is hidden.

1. Tap a row. The check opens full-screen, the address changes to `#check-` plus its ID, and focus moves to the title.
2. Read the check and mark it.
3. Use **Previous check** and **Next check** to move through checks without adding to your browser history.
4. Tap **All checks**, or use your browser's Back, to return to the list. The list scrolls to the row you came from and focuses it, and keeps it highlighted.

Turning the phone or widening the window past 1080 pixels closes the one-check view.

Suggested order of work

1. Run the Analyzer on the page first, so the **AUTO** checks are already decided.
2. Filter to **Required** and work area by area.
3. Flag failures with **Fail** and attach a screenshot to each (see Attach evidence).
4. Switch on **Open only** to see what is still to do.
5. Finish with the **Practice** checks, especially area 09, **Testing & proof**.
6. Export (see Export your results).

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Accessibility Checklist reference

Every control, label, message and stored item of the Accessibility Checklist.

Page

Item	Detail
Address	https://auricartisan.com/resources/accessibility-checklist/
Old addresses	`/accessibility-compliance/` and `/compliance` forward here
Menu entry	**Learn** › **Accessibility Compliance** ("WCAG checks & guidance")
Deep link to a check	`#check-` plus the ID, for example `#check-k2`
Masthead links	**Run the automated pass first** → (Analyzer), **Wire it into CI** → (API documentation)

Tabs and meter

Control	What it does
Checklist	The Inspector: index and pane
Export	Counts, a preview of the Markdown, and the three export actions
Reference	WCAG levels and the law in 15 jurisdictions
Meter	"N passed · N flagged · N left … of 50", with a two-part bar. Always counts all 50.
Scope chip	Shown when a filter is on: the filter's name and how many rows are visible

Index

Control	What it does
All 50	Show every check (default)
Required	Show Level A and AA checks (35)
AAA	Show Level AAA checks (2)
Practice	Show good-practice checks (13)
Open only	Hide checks marked Passed; combines with the filters above
Area header	Fold or unfold the area; shows decided out of total
Row	Select the check; shows state, title and tag

Pane

Control	What it does
All checks	Back to the list (phone layout only)
Position	"N of 50", the check's place in the full list
Previous check / **Next check**	Move through the visible checks (`K` / `J`)
Chips	Criterion tag, **Required**, and **AUTO**, **PARTLY** or **BY HAND**
Pass	Mark as passed (`Space`); select again to clear
Fail	Mark as flagged (`F`); select again to clear
N/A	Mark as not applicable (`N`); select again to clear
Attach screenshot	Open the screenshot evidence dialog
Evidence	Thumbnails of attached screenshots, and an **Add** tile
Explanation	What it means, who it affects, how to test it, what the fix looks like, what usually breaks it, links

Screenshot evidence dialog

Control	What it does	Default
Choose a file	Pick one or more images	—
Drop or paste	Drop image files, or paste with `Ctrl` + `V`	—
Pen, Highlighter, Arrow, Box, Redact, Callout	Drawing tools (`P` , `H` , `A` , `B` , `R` , `C`)	Arrow
Colour	Red, Blue, Green, Black	Red
Line width	Fine, Medium, Broad	Medium
Marks list	Every mark on this screenshot, with a remove button each	—
Thumbnail strip	Switch screenshot, or add another	—
Undo	Remove the last mark	—
Clear marks	Remove every mark on this screenshot, without asking	—

Control	What it does	Default
Remove	Delete this screenshot; confirm with **Remove this one?** within four seconds	—
Done	Close the dialog (`Esc` does the same)	—

Export tab

Control	What it does
Export the report	Opens the browser's print dialog with flagged and untouched checks
Copy as Markdown	Copies a Markdown report of all 50 checks to the clipboard
Reset	Clears every mark and screenshot at once, without asking

Messages

Message	When
"Nothing matches these filters."	No check matches the active filters
"N screenshot(s) attached."	Screenshots were added
"Screenshot removed."	A screenshot was deleted
"Printing N finding(s)."	**Export the report** with flagged checks
"Nothing is flagged — printing the list as it stands."	**Export the report** with nothing flagged
"The findings are on your clipboard, as Markdown."	**Copy as Markdown** worked
"Could not reach the clipboard — the browser refused it."	**Copy as Markdown** failed
"Cleared — marks and screenshots."	After **Reset**

Stored in this browser

Item	Kept
Your marks (Pass, Fail, N/A) for each check	Yes
Screenshots, their marks, and the marked-up copies	Yes
Filters, folded areas, the selected check	No (a `#check-` address selects a check)
Notes or reasons	Not available

Nothing is sent to a server. See Privacy.

Layout

Width	Layout
1080 px and wider	Index and pane side by side; the pane stays in view; the tab rail and meter stay under the header
Under 1080 px	Index only; a row opens the one-check view
620 px and narrower	Key hints on the verdict buttons are hidden; **Attach screenshot** becomes an icon button

Without JavaScript, the page shows the plain list of all 50 checks with their explanations, and no filters, meter or marking.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Troubleshooting the Accessibility Checklist

Common problems with the Accessibility Checklist, why they happen, and what to do.

My progress has gone

- **Why:** marks and screenshots live only in this browser on this device. They are lost if you cleared this site's data, used a private window, switched browser, profile or device, or selected **Reset** (which does not ask first).
- **What to do:** there is no server copy to restore from. In future, copy the Markdown or print before clearing anything.

Reset cleared everything without asking

- **Why:** in the current version, **Reset** clears all marks and screenshots immediately, even though its description says it asks first.
- **What to do:** nothing can be recovered. Export before you reset.

Screenshots will not attach

- **Why:** screenshots need your browser's database storage, which some private windows block. Only image files are accepted.
- **What to do:** use a normal window, and attach PNG, JPEG or other image files.

The single-key shortcuts do nothing

- **Why:** they only work while focus is in the index or the pane, not with `Ctrl`, `Alt` or `Cmd` held, and not while the screenshot dialog is open. `Space` marks **Pass** only from a row, the pane or the check title. During the site's first-visit guide tour, the tour may take the arrow keys.
- **What to do:** select a row first, then press the key.

I cannot find a check

- **Why:** a filter is on, **Open only** is hiding passed checks, or the area is folded.
- **What to do:** look at the chip beside the meter; select **All 50**, switch off **Open only**, and unfold the area.

Previous check or Next check is greyed out

- **Why:** you are at the start or end of the visible list.
- **What to do:** change the filter or unfold areas to reach more checks.

The printout is missing checks

- **Why:** passed and not-applicable checks are left out of the printout by design. An active filter may also limit it.
- **What to do:** select **All 50** and switch off **Open only** before printing. For counts and a full list of untouched checks, use **Copy as Markdown**.

Export the report did not download a file

- **Why:** it opens the browser's print dialog instead.
- **What to do:** choose **Save as PDF** in the print dialog.

Copy as Markdown failed

- **Why:** the browser refused clipboard access. The message says "Could not reach the clipboard — the browser refused it."
- **What to do:** select the button again after clicking on the page, and check your browser's clipboard permission for auricartisan.com.

"What a claim actually needs →" does nothing

- **Why:** the link does not switch tabs in the current version.
- **What to do:** select the **Reference** tab.

Remove says "Remove this one?"

- **Why:** deleting a screenshot needs a second press within four seconds.
- **What to do:** press **Remove this one?** to confirm, or wait for it to reset.

On a phone, the check fills the screen

- **Why:** below 1080 pixels wide the checklist shows one check at a time.
- **What to do:** tap **All checks**, or use your browser's Back.

Redacted areas are still visible in stored screenshots

- **Why:** redaction applies only to the marked-up copy used for thumbnails and printing; the original is kept unchanged.
- **What to do:** crop or blur sensitive information before attaching the screenshot.

"Check a pair" does not open a contrast checker

- **Why:** the link on check c1 opens the tools collection page.
- **What to do:** open the Contrast Checker directly.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Accessibility Checklist FAQ

Short answers to common questions about the Accessibility Checklist.

Is it free? Do I need an account?

It is free, and there is no account. Nothing to install either: it works in any modern browser.

Is a finished checklist a WCAG certification?

No. The page says it plainly: "A worked checklist is evidence of a process, not a conformance claim." A claim needs a full evaluation against the whole standard, and the involvement of people who use assistive technology every day.

Does it cover all of WCAG 2.2?

It covers the ground most audits hit, in 50 checks tied to 33 WCAG criteria, plus 13 good-practice checks. It does not have a check of its own for every criterion; for example, the WCAG 2.2 additions Dragging Movements (2.5.7), Consistent Help (3.2.6), Redundant Entry (3.3.7) and Accessible Authentication (3.3.8) have no dedicated check. The Analyzer has automated rules for three of those four.

What do AUTO, PARTLY and BY HAND mean?

How much the Analyzer can decide: **AUTO** checks it decides on its own, **PARTLY** checks it narrows down for a person to confirm, and **BY HAND** checks only a person can judge. The checklist does not read Analyzer results; you mark every check yourself.

Which checks are required?

Level A and AA checks, 35 in all, marked **Required**. They are what laws and procurement documents cite. AAA and Practice checks are optional.

How do I undo a mark?

Select the same verdict again. The check goes back to **Not checked yet**.

Where is my work saved?

In this browser, on this device. Nothing is uploaded. It survives closing the tab, but not clearing site data, and it does not follow you to another browser or device.

Can I move my progress to another computer, or share it with a colleague?

Not as a file you can import. Share the result instead: **Copy as Markdown** for your issue tracker, or **Export the report** and save it as a PDF.

Can I add notes to a check?

There is no notes field. Use screenshots with numbered callouts, or add notes in your tracker after copying the Markdown.

Are screenshots included in the export?

The printed report shows up to four small marked-up thumbnails per check. The Markdown says how many screenshots are attached but does not include them.

Is the legal information up to date?

It was last reviewed on 11 September 2026, and the page notes that dates move. It is orientation, not legal advice; check the primary source linked from each instrument.

Why is the menu entry called Accessibility Compliance?

The site header's **Learn** menu lists this page as **Accessibility Compliance**, and the old `/accessibility-compliance/` address forwards here. The page itself is titled The Accessibility Checklist.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Accessibility Checklist: supporting material

The checklist's glossary, keyboard shortcuts, limits, related tools and privacy notes.

This folder supports the checklist guides with the terms they use, the full list of keys, an honest account of what the checklist covers, related tools, and where your work is kept. The public changelog has no entries for the checklist, so there is no release-notes page.

File	What it covers
Glossary	Terms used on the page and in these docs
Keyboard shortcuts	Every key the checklist responds to
Limits and accuracy	What the checklist covers, known differences between its wording and its behaviour, and what it is not
Related tools and reading	Tools and articles that pair with the checklist
Privacy	What stays in your browser

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Accessibility Checklist keyboard shortcuts

Every key the Accessibility Checklist responds to, and where each one works.

The single-letter keys work only while focus is inside the index (including its filter buttons and area headers) or the pane. They are ignored in text fields, when `Ctrl`, `Alt` or `Cmd` is held, and while the screenshot dialog is open.

Moving between checks

Keys	Where	Action
`Down` / `Up`	On a row	Next or previous visible check; focus moves with it
`Home` / `End`	On a row	First or last visible check
`J` / `K`	Index or pane	Next or previous check. From the pane, focus stays and the new title is announced.
`E`	Index or pane	Open the check: the one-check view on a phone, the pane title on a wide screen
`Esc`	Pane	On a phone, back to the list; on a wide screen, back to the check's row

Marking

Key	Action
`Space`	**Pass** (only on a row, the pane or the check title)
`F`	**Fail**
`N`	**N/A**

Pressing the key for the current verdict again clears it.

Tabs

Keys	Action
`Left` / `Right`	Previous or next tab, wrapping round
`Home` / `End`	First or last tab

In the screenshot evidence dialog

Keys	Action
`P`	Pen
`H`	Highlighter
`A`	Arrow
`B`	Box
`R`	Redact
`C`	Callout
`Ctrl` + `V`	Paste a screenshot
`Esc`	Close the dialog, like **Done**

Printing

`Ctrl` + `P` prints the same report as **Export the report**.

For shortcuts that work across the whole site, see Keyboard shortcuts.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Accessibility Checklist glossary

Plain-language definitions of the terms used by the Accessibility Checklist.

APG — The W3C ARIA Authoring Practices Guide, which sets out the expected keyboard behaviour for widgets such as tabs and menus.

Area — One of the nine numbered groups of checks, 01 to 09.

AUTO, PARTLY, BY HAND — How much the Analyzer can decide a check: on its own, by narrowing it down for a person, or not at all.

Check — One of the 50 items on the checklist.

Decided — Marked. On the Export tab's level bars this means Passed or Flagged; in the Markdown export it also includes N/A.

Evidence — Screenshots attached to a check, with their annotations.

Flagged — A check marked **Fail**. The button says Fail, the state word says Failed, and the meter and exports say flagged.

Inspector — The two-column working view: the index of checks beside the selected check in full.

Index — The list of checks on the left of the Inspector.

Level A, AA, AAA — WCAG conformance levels. A and AA are what laws cite; AAA is stricter and optional.

Meter — The counter beside the tabs: passed, flagged and left, out of 50.

One-check view — The phone layout that shows one check at a time.

Pane — The selected check in full, on the right of the Inspector.

Pen — The screenshot evidence dialog and its annotation tools.

Practice — A good-practice check with no WCAG success criterion of its own.

Required — A check at Level A or AA.

Scope chip — The label beside the meter that names the active filter and how many rows it shows.

Success criterion — One testable requirement in WCAG, such as 2.4.7 Focus Visible.

Untouched — A check that is not marked. Also shown as "Not checked yet", "Never checked" or "left". An untouched check is not a pass.

Verdict — A mark on a check: Pass, Fail or N/A.

VPAT — Voluntary Product Accessibility Template, the report format procurement often asks for.

WCAG — The Web Content Accessibility Guidelines published by the W3C. The checklist is written against WCAG 2.2.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Limits and accuracy

What the Accessibility Checklist covers, where its wording and behaviour differ, and what it is not.

What it covers

- 50 checks: 37 tied to 33 distinct WCAG success criteria at Levels A, AA and AAA, and 13 good-practice checks.
- The masthead describes the checks as spanning "the whole of WCAG 2.2". In practice some criteria have no check of their own, including four added in WCAG 2.2: Dragging Movements (2.5.7), Consistent Help (3.2.6), Redundant Entry (3.3.7) and Accessible Authentication (Minimum) (3.3.8). The Analyzer checks 2.5.7, 3.2.6 and 3.3.8 automatically.
- Two level tags differ from WCAG: check m2 (text in images, 1.4.5) and check f6 (legal and financial submissions, 3.3.4) are tagged Level A, but WCAG places both criteria at Level AA. Both are still counted as **Required**, so your work is unaffected; only the A and AA counts shift by two.
- Check d3 also asks for `lang` on foreign-language passages, which WCAG covers under 3.1.2 Language of Parts (AA), although the check cites only 3.1.1.

AUTO, PARTLY and BY HAND

These labels describe what the Analyzer can decide. The checklist does not import Analyzer results. Two AUTO labels are really judgement calls: x1 (an automated pass in continuous integration) is a process check, and c6 (simulating colour-vision deficiencies) needs a person to look at the result.

Where the wording and the behaviour differ

The current page describes a few things differently from how they work. The docs describe the behaviour.

The page says	What happens
Export the report: "Self-contained HTML with the screenshots inline"	It opens the browser's print dialog; choose **Save as PDF** for a file
Reset: "Asks first, and says how much it is about to discard"	It clears everything at once, with no confirmation
Passed: "Listed by name" in the report	The Markdown gives a count; the printout leaves passed checks out
Not applicable: "Carried as you marked them"	The Markdown gives a count; the printout leaves them out
"The checks, your reasons and every screenshot"	There is no field for reasons; marks and screenshots are kept
What a claim actually needs →	The link does not switch to the Reference tab; select **Reference** yourself
Check c1's **Check a pair** link	Opens the tools collection page, not a contrast checker

The Evidence count reads "1 screenshots pinned" when there is one screenshot; this is a wording slip, not an error.

Legal information

The Reference tab is orientation, not legal advice. It was last reviewed on 11 September 2026. Dates in law and regulation move; always check the primary source.

What it is not

- Not a conformance claim or certification. A worked checklist is evidence of a process.
- Not a replacement for testing with assistive technology and with disabled people; check x5 says so itself.
- Not shared or synced: it is one browser's record of one person's work.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Accessibility Checklist privacy

Where the Accessibility Checklist keeps your marks and screenshots, and how to delete them.

The page puts it simply: "Your work stays on this device."

What is kept, and where

Item	Where
Your marks on each check	Your browser's local storage
Screenshots, the original image and its marks, and a marked-up copy	Your browser's database storage

- There is no account, no sync and no server copy. The checklist sends none of your marks or screenshots anywhere.
- Your work is per browser, per profile and per device.
- In a private window, or when storage is blocked, the page still works but forgets when you close it. Without database storage, screenshots cannot be attached, but marks still work.

What leaves your device

Only what you choose to take out:

- **Copy as Markdown** puts the findings on your clipboard. Screenshots are never included.

- **Export the report** prints the page, including small marked-up screenshot thumbnails.

Redaction

The **Redact** tool covers an area only in the marked-up copy used for thumbnails and printing. The original screenshot stays stored in your browser, unredacted. Crop or blur sensitive details before you attach a screenshot.

Working in two tabs

Each open tab keeps its own copy of your marks and saves it as a whole. Working in two tabs at once may overwrite marks made in the other. Use one tab at a time.

Delete your data

- **Export** › **Reset** removes every mark and every screenshot at once, without asking.
- Clearing this site's data in your browser settings removes everything too.

The rest of the page loads the site's normal scripts, including advertising, under the site-wide consent settings. See Company and legal for the site's privacy policy.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Related tools and reading

Tools and articles that pair well with the Accessibility Checklist.

Where	Why
Analyzer	Run the automated pass first; it decides many AUTO checks and narrows the PARTLY ones
Contrast Checker	Settle a single colour pair for checks c1, c2 and c4
Vision Lab	Simulate colour-vision deficiencies for checks c3 and c6
Public API	Automate an audit in continuous integration, as check x1 suggests
Learn	The science behind each rule, in plain language
Accessibility Compliance	The old address for this page

External

- WCAG 2.2 quick reference at the W3C: https://www.w3.org/WAI/WCAG22/quickref/
- W3C Understanding documents, linked from every check: https://www.w3.org/WAI/WCAG22/Understanding/
- W3C guidance on evaluating accessibility, linked from area 09: https://www.w3.org/WAI/test-evaluate/

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Accessibility Compliance

What the Accessibility Compliance address offers a visitor today, and where it takes you.

https://auricartisan.com/accessibility-compliance/ is the address of Auric Artisan's accessibility compliance resource. It no longer has a page of its own: opening it forwards you straight to the Accessibility Checklist at https://auricartisan.com/resources/accessibility-checklist/, which is where the compliance material now lives.

What a visitor gets, therefore, is the checklist: 50 WCAG 2.2 checks explained and ready to mark, the tools to record evidence and export findings, and a **Reference** tab on what the law asks for in 15 jurisdictions. The site header still calls this destination **Accessibility Compliance** in its **Learn** menu.

Who it is for

Anyone following an old link, a bookmark, or the **Accessibility Compliance** menu entry, who wants to know what WCAG and the law require and how to check a site against them.

Where to find it

Address	What happens
https://auricartisan.com/accessibility-compliance/	Forwards immediately to the Accessibility Checklist
https://auricartisan.com/compliance/	Redirects permanently to `/accessibility-compliance/`, which then forwards to the checklist
https://auricartisan.com/compliance	Shows the `/accessibility-compliance/` page, which forwards to the checklist
Site header: **Learn** › **Accessibility Compliance** ("WCAG checks & guidance")	Opens the checklist directly

Quick start

1. Open https://auricartisan.com/accessibility-compliance/.
2. Your browser moves to the Accessibility Checklist at once. If it does not, select the **Accessibility Checklist** link on the short "Redirecting" page.
3. Select the **Reference** tab to read what the law asks for where you work.
4. Select the **Checklist** tab to work through the checks.
5. Select **Run the automated pass first** → to audit your page with the Analyzer.
6. Use the **Export** tab to print or copy your findings.

What you can do

Everything the Accessibility Checklist offers:

- read 50 WCAG 2.2 checks, each explained, with a test and a fix;
- mark checks and attach annotated screenshots;
- export your findings;
- read the accessibility law for India, the United States, the European Union, the United Kingdom, Canada, Australia, Japan and eight more countries.

In this folder

- docs/ — what happens at this address
  - Getting started — the forward, and where to go next
  - FAQ
- others/ — supporting material and links

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Accessibility Compliance documentation

The guides for the Accessibility Compliance address.

The Accessibility Compliance address forwards to the Accessibility Checklist, so these guides are short. For everything the checklist does, see the Accessibility Checklist docs.

Guide	What it covers
Getting started	What happens when you open the address, and what to do next
FAQ	Why the address forwards, and what happened to the old page

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Getting started

What happens when you open the Accessibility Compliance address, and how to use the checklist it takes you to.

What happens when you open the address

1. You open https://auricartisan.com/accessibility-compliance/ (or `/compliance/`).
2. A very short page titled "Redirecting to Accessibility Checklist" loads, with the text "Redirecting to Accessibility Checklist..." and a link.
3. Your browser moves on at once to https://auricartisan.com/resources/accessibility-checklist/. The forward replaces the old address in your history, so your browser's Back button skips it.

If your browser blocks the automatic forward, select the **Accessibility Checklist** link on that page.

The result: you are on the Accessibility Checklist, with its **Checklist**, **Export** and **Reference** tabs.

What to do next

If you want to	Go to
Know which law applies to you, and which WCAG level it cites	The **Reference** tab. See Laws by jurisdiction.
Check your site against WCAG 2.2 by hand	The **Checklist** tab. See Work through the checklist.
Find the problems a tool can find automatically	**Run the automated pass first** →, which opens the Analyzer
Hand findings to your team	The **Export** tab. See Export your results.

Update your links

If you link to the compliance page from your own site or documents, link to https://auricartisan.com/resources/accessibility-checklist/ directly. The old address still works, but the direct address saves a step.

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Accessibility Compliance FAQ

Short answers about the Accessibility Compliance address and where its content went.

Why does /accessibility-compliance/ open the checklist?

The compliance resource was folded into the Accessibility Checklist, which now carries both the checks and the legal reference. The old address forwards there so existing links keep working.

Is there still a separate compliance page?

No. Everything a visitor needs is on the checklist page: the 50 checks, evidence and export, and the **Reference** tab on the law in 15 jurisdictions.

Does the forward keep anything I had on the old page?

The address has no content of its own to keep. Work you do on the checklist is saved in your browser, as described in the checklist's Privacy page.

Does the checklist make my site compliant?

No. It is a structured way to check and record your work. The page says a worked checklist is evidence of a process, not a conformance claim, and its legal summaries are orientation, not legal advice.

Where is the menu entry?

In the site header, **Learn** › **Accessibility Compliance** ("WCAG checks & guidance"). It opens the checklist.

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Accessibility Compliance: supporting material

Where to find the glossary, limits, privacy notes and related material for the compliance resource.

The Accessibility Compliance address has no page or behaviour of its own beyond forwarding to the Accessibility Checklist, so its supporting material is the checklist's. There is nothing stored at this address, no shortcuts, and no separate release history.

Topic	Where
Terms such as Required, Practice and WCAG levels	Checklist glossary
Keyboard shortcuts	Checklist shortcuts
What the checklist covers and what it is not	Checklist limits and accuracy
Where your work is kept	Checklist privacy
Related tools	Checklist related tools
The automated audit	Analyzer

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Analyzer

What the Auric Artisan Analyzer audits, who it is for, and how to run your first audit.

The Analyzer audits a web page from its URL. It reads the page, measures what it can in a real browser, and reports back across accessibility, colour contrast, palette, performance, SEO and metadata, media, responsive behaviour, security and reliability. The page it lives on introduces it as "One URL, audited seven ways." Every finding names the element that caused it, explains why it matters, and, where it can, gives you the code to fix it.

At its centre is an accessibility engine of 126 rules mapped to WCAG 2.0, 2.1 and 2.2 success criteria, Section 508 and EN 301 549. Around it sit a contrast audit with suggested replacement colours, a colour-vision simulation of the page's own palette, Core Web Vitals, a raw-versus-rendered HTML comparison, a site map built from robots.txt and the sitemap, a security-header check, and exports in JSON, HTML, PDF, CSS, CSV and Markdown.

The Analyzer has two modes. **Simple** shows the sections most people need and runs a fixed set of checks. **Advanced** opens all 22 sections as a workspace of panels you can arrange, lets you choose which checks run, and adds whole-site scanning, live editing and custom rules. Every run is filed in your browser so you can reopen it and compare it with later runs of the same site.

Who it is for

- **Accessibility specialists** who want a rule-by-rule WCAG audit, an accessibility tree, a screen-reader transcript and a VPAT-shaped conformance table.
- **Front-end and full-stack developers** who want the failing selectors, a paste-ready CSS patch and a comparison of raw and rendered HTML.
- **Designers** who want to see every failing colour pair, a passing replacement that keeps the hue, and the palette under colour-vision filters.
- **SEO and content teams** who want metadata, headings, structured data, robots rules and sitemap discovery in one place.
- **Site owners** who want a score to track over time and a short list of what to fix first.

Where to find it

- Address: https://auricartisan.com/tool/analyzer/
- From any page: open the **Tools** menu in the site header and select **Analyze URLs**.
- A link of the form `https://auricartisan.com/tool/analyzer/?url=https://example.com` opens the Analyzer and starts an audit of that address straight away.

The Analyzer runs in any modern browser with JavaScript enabled. Auditing Auric Artisan's own pages needs no account. Auditing another website needs you to be signed in, because that page is rendered on Auric Artisan's servers (see Limits and accuracy).

Quick start

1. Open https://auricartisan.com/tool/analyzer/.
2. Type or paste a full address into the URL field, for example `https://example.com`.
3. Select **Analyze**, or press `Enter` in the field.
4. Watch the progress card in the Overview panel. It lists each check as it runs.
5. Read the **Overall Audit Score** and the **Top 3 Things To Improve Score** in the Overview panel.
6. Open the **Contrast** panel to see failing colour pairs, and the **Fixes** panel for the code to change.

7. Open the **Report** panel and select **Export HTML** or **Export JSON** to keep a copy.
8. Run the same URL again after you fix something, then open **Compare** to see what changed.

What you can do

- Audit one page, or crawl up to 24 pages of a site in one run (Artisan plan and above).
- Choose which of 13 checks run, and see the token cost before you start (Advanced).
- Read an accessibility grade, a WCAG 2.2 conformance verdict and a success-criteria scorecard.
- Filter accessibility findings by impact, area and WCAG level, and copy the selector or the fix for each one.
- Get a ready-to-paste CSS patch for contrast, focus and target-size failures.
- See every text colour pair with its WCAG ratio and a passing replacement colour.
- Preview the page through colour-vision and low-vision filters.
- Compare raw HTML with the rendered DOM to find what JavaScript adds or changes.
- Measure Core Web Vitals and see a loading waterfall and performance budgets.
- Check metadata, headings, schema, robots.txt and every sitemap.
- Write your own selector rules and run them on every audit.
- Arrange sections into a saved workspace layout (Advanced).
- Reopen past runs, compare two versions of the same site, and keep a copy in your account.
- Export JSON, HTML, PDF, CSS, CSV and Markdown reports, or share a link that re-runs the audit.

In this folder

- docs/ — how to use the Analyzer
  - Getting started — open it, the screen tour, your first audit
  - Run an analysis — what happens when you select Analyze, and how pages are reached
  - Choose what to scan — the 13 checks, the scan picker and what a run costs
  - Scan a whole site — site scan settings, progress and paged results
  - Arrange the workspace — panels, sections, layouts and Focus
  - Read the Overview and scores — the score, grades, weights and issue cards
  - The accessibility engine — every view of the A11y+ panel
  - Accessibility rules — all 126 rules, category by category
  - Colour, contrast and vision — Contrast, Palette, Simulation, Before / After
  - Performance, SEO and page health — Performance, SEO, Media, Deep, Devices, Security, Project, Site Map, Insights
  - Fix and preview — Fixes, Playground, Rules and CSP Setup
  - Save, reopen and compare — the library, Workspace, Compare and your account
  - Export and share — every export format and share link
  - Reference — every control, option and output
  - Troubleshooting — symptom, cause and what to do
  - FAQ
- others/ — supporting material
  - Glossary
  - Keyboard shortcuts
  - Limits and accuracy
  - Related tools and reading
  - Release notes

- Privacy

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Analyzer documentation

The list of Analyzer guides, from a first audit to exports, with one line on each.

These guides explain how to use the Analyzer at https://auricartisan.com/tool/analyzer/. Start with Getting started, then open the guide for the job in front of you. The reference page lists every control and output in one place.

Guide	What it covers
Getting started	Open the Analyzer, learn the screen, and run your first audit
Run an analysis	What happens after you select Analyze, how a page is reached, and the progress card
Choose what to scan	The 13 checks, the What to scan picker in Advanced, and how a run is priced
Scan a whole site	Site scan settings, the live progress console and paged per-page results
Arrange the workspace	Panels, the Sections picker, width, Focus, Collapse, and saved Layouts
Read the Overview and scores	The overall score, grade bands, category weights and issue cards
The accessibility engine	Every view of the A11y+ panel: findings, fixes, widgets, tree, screen reader, conformance, export
Accessibility rules	All 126 accessibility rules, grouped by category, with impact and WCAG mapping
Colour, contrast and vision	The Contrast, Palette, Simulation and Before / After panels, WCAG ratio and APCA Lc
Performance, SEO and page health	Performance, SEO, Media, Deep, Devices, Security, Project, Site Map and Insights
Fix and preview	The Fixes, Playground, Rules and CSP Setup panels
Save, reopen and compare	How runs are kept, the Workspace panel, Compare, and keeping a copy in your account
Export and share	Every export, its format and file name, and the share links
Reference	Every control, option, input and output
Troubleshooting	Common problems, why they happen, and what to do
FAQ	Short answers to common questions

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Getting started with the Analyzer

Open the Analyzer, learn what each part of the screen does, and run your first audit.

This guide takes you from opening the Analyzer to reading your first result. It takes about five minutes.

Open the Analyzer

Go to https://auricartisan.com/tool/analyzer/, or open the **Tools** menu in the site header and select **Analyze URLs**.

You do not need an account to audit a page on auricartisan.com. To audit another website you need to be signed in, because that page is rendered on Auric Artisan's servers. See Run an analysis for the details.

The screen, top to bottom

The masthead

The top of the page names the tool: the kicker "Auric Artisan · Analyzer", the headline "One URL, audited seven ways.", and a line explaining that accessibility, metadata, palette, contrast, performance, reliability and security are all checked against the page's own markup. Under it, a thin rail lists **WCAG 2.1 / 2.2** and **LCP · CLS · INP**, with a link, **Read the API reference**, to the public API documentation.

The API hint

A one-line note may appear under the masthead: "Automate this — 12 API credits per call to `POST /v1/analyzer/inspect`." It tells developers that the same kind of audit is available from the REST API, and that API credits are a separate balance from your tool tokens. Select **API docs** to open that endpoint's documentation, or the **×** button to dismiss the note. Dismissing it hides the hint on every tool page in this browser.

The strip

The dark strip is where you work. From left to right:

Control	What it does
URL field	Where you type or paste the address to audit. The placeholder reads "Enter a URL to analyze…" in Simple mode.
Analyze	Starts the audit. Pressing `Enter` in the URL field does the same.
Cancel	Appears only while an audit or site scan is running. Stops it.
Share	Copies a link that opens the Analyzer on the same URL and re-runs the audit.
Sections and **Layouts**	Advanced mode only. Open the section picker and the saved-layout menu.
Simple / **Advanced**	Switches mode. Your choice is remembered in this browser.

In Advanced mode two folding rows appear under the strip: **What to scan**, which lets you choose the checks and shows their cost, and **Site scan settings**, which holds the whole-site scan controls.

The run line

Under the strip is one line that shows where the run is and what it costs:

- The **progress rail** has five steps: **1 Analyze**, **2 Issues**, **3 Fix**, **4 Preview** and **5 Export**. It lights up as a run moves on. Each step is also a button: **Fix** shows the Fixes section, **Preview** shows Before / After, **Export** shows Report, and **Analyze** and **Issues** show the Overview.
- The **token meter** shows **This run** (the cost of the next run in tokens) and **Remaining** (your balance). When you are signed out it says "Sign in to claim 100 free tokens." with a **Sign in · 100 free tokens** button. Select the meter to open a sheet that explains the price. See Choose what to scan.

The search bar

After the first run, a **Search results** bar appears with the placeholder "Filter issues, fixes, files, media, rules…". Typing filters the cards and rows in the open sections. **Clear** empties it.

The workspace

Below everything is the workspace: a grid of panels, one per section of the report. Before your first run it holds only the **Overview** panel, which lists what a run will give you:

- Accessibility, against WCAG 2.1 and 2.2, with the rule each finding breaks
- Every text pair that fails contrast, and a passing colour to replace it
- How the page looks with colour-vision and low-vision filters on
- The palette, the type, and every image, video and audio file
- Rendered DOM against raw HTML: framework, SEO signals, what JavaScript wrote
- Core Web Vitals, security headers, broken links, and a report you can export

Each panel has a title bar with a grip (drag it to move the panel) and four buttons: **Width**, **Focus**, **Collapse** and **Close**. See Arrange the workspace.

Simple or Advanced?

	Simple	**Advanced**
Sections you can open	8: Overview, Project, Insights, Site Map, Contrast, Fixes, Compare, Report	All 22
Checks that run	A fixed set (see below)	Your choice of 13, from the **What to scan** picker
Opening more sections	`Ctrl` + `P`, or the progress-rail steps	The **Sections** button, or `Ctrl` + `P`
Saved layouts	No menu (your arrangement is still remembered)	**Layouts** menu with ready-made and named layouts
Whole-site scan	Not available	**Site scan settings**
Custom rules, Playground, CSP Setup	Not available	Available

Simple runs these checks: page structure (always), Project model, Colour contrast, SEO and metadata, Site map, Accessibility and Responsive behaviour. Some of these feed the score without having a section in Simple mode.

Switching mode does not re-run anything. Each mode remembers its own panel arrangement.

Run your first audit

1. Select **Simple** in the strip if it is not already selected.
2. Select the URL field and type `https://auricartisan.com/tool/contrast-checker/`. Any auricartisan.com page works without an account.
3. Select **Analyze**.
4. Watch the Overview panel. A card titled **Auditing** shows a live line ("Opening the page…", "Reading the page…", "Checking the page against the accessibility rules…") and a list of the steps this run is doing, each ticked off as it finishes.
5. When the run ends, the progress rail moves to **2 Issues** and the workspace opens the sections your checks filled. In Simple mode that is **Overview** across the full width, then **Project**, **Contrast** and **Site Map**.
6. In the Overview panel, read the large score and the **Overall Audit Score** grade.
7. Open **Why this score?** under the score to see which categories cost points.
8. Scroll to **Top 3 Things To Improve Score**. Each card gives the estimated points it is worth, why it matters, and the fix.
9. Press `Ctrl` + `P`, select **Fixes**, and read the **Fix This First** list.

The result: you have a score, a ranked list of what to fix, and the code for each fix. The run is also filed in your browser, so it appears in **Compare** and in the library the next time you open the Analyzer.

Where to go next

- Read the Overview and scores to understand the numbers.
- The accessibility engine to work through accessibility findings in Advanced mode.
- Export and share to keep a copy or send the result to someone.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

The accessibility engine (A11y+)

Read and act on every view of the A11y+ panel, from the grade and findings to the accessibility tree, screen-reader transcript, conformance table and exports.

The **A11y+** panel holds the Analyzer's accessibility engine: 126 rules checked against the page as your browser actually renders it, plus widget-pattern checks, an accessibility tree, a screen-reader transcript, a colour-vision table, a focus and target map, a WCAG 2.2 conformance table and five exports. It is available in Advanced mode, and runs when the **Accessibility** check is on.

In Simple mode the same engine still runs and feeds the **Accessibility** category of the overall score; you just do not see its panel.

How the panel is organised

A row of four groups sits at the top of the panel. Selecting a group opens its first view; groups with more than one view show a second row of view buttons.

Group	Views
Overview	Overview
Findings (with the issue count)	**Issues**, **Fixes**
Inspect	**Widgets**, **Annotations**, **A11y tree**, **Screen reader**, **Vision**, **Focus & targets**
Report	**Conformance**, **Export**

Most blocks inside a view are folding cards: select the card's title to open or close it.

Key terms

- **Rule** — one automated check, such as "Buttons must have discernible text". Each rule applies to some elements and passes, fails, or needs review on each.
- **Impact** — how badly a failure affects someone using the page. The engine uses four levels, each with a one-line meaning in the panel:

Impact	Meaning shown
critical	Blocks people from using the page at all.
serious	Makes a task very difficult or unreliable.
moderate	Causes friction and confusion.
minor	Polish — worth fixing, rarely blocking.

- **Needs review** — the rule ran and could not decide. Text over a photograph or a gradient is the usual cause, because there is no single background colour to measure. A person has to look.
- **Not tested** — no rule can judge that WCAG criterion automatically. It is never counted as passing.
- **Accessible name** — the name assistive technology announces for an element, worked out from its text, its label, `aria-label`, `aria-labelledby` or `alt`.
- **Accessibility tree** — the simplified version of the page that browsers give to screen readers and other assistive technology.

Overview view

The top of the view is the engine's headline:

- **Accessibility** with a score from 0 to 100 and a grade.
- A conformance line, for example "Does not conform to WCAG 2.2 AA." followed by how many criteria need manual review before a conformance claim can be made.
- A bar and key: how many rules **passed**, **failed** and **need review**, out of the rules that ran.

The engine's grade scale

This scale is stricter than the one used for the overall score:

Grade	Label	Score
A+	Excellent	95–100
A	Strong	90–94
B	Good	80–89
C	Needs work	70–79
D	Poor	55–69
F	Critical	0–54

How the engine scores

Every failing rule adds a penalty. The penalty is larger for higher impact (critical counts ten times as much as minor) and grows with the number of elements affected, but only slowly, so one component repeated forty times does not swamp an otherwise sound page. Rules that need review add a much smaller penalty. The score then falls smoothly as the penalty grows: it never jumps to zero, and a single critical failure is always visible.

Two kinds of rule run and are reported but are left out of the score, because no Level AA claim depends on them: **Level AAA** rules and **best-practice** rules (those with no WCAG criterion).

The cards

Card	What it shows
Severity summary	"N issues on N failing rules", with a button and a count for each impact level
Coverage limits for this run	Shown when computed styles or element geometry were unavailable; affected rules report needs review
Where the problems are	One bar per category with its issue count, for example Pointer & touch, Colour & contrast, Landmarks & regions

Card	What it shows
Quick wins — highest value per unit of effort	Up to eight low-effort, high-impact fixes, each with its node count and fix
Reading level (WCAG 3.1.5)	Flesch reading ease, reading grade, sentence length and long sentences, over the page's prose
Full remediation backlog	Every failing rule as a task, ranked by score gain per unit of effort

Impact levels with no issues are listed together, for example "Nothing critical, minor." The reading-level card measures paragraphs, list items and quotes only; a page that is mostly interface copy says a reading grade would not mean anything. The backlog says what score fixing everything would reach, and its table has these columns: **Fix** (with category and WCAG criterion), **Impact**, **Nodes**, **Effort**, **Gain** and **Value**.

Findings: Issues view

Every failing rule and every rule that needs review, as one folding row each. The row shows the rule's title, a badge for its impact (or **needs review**), its level (**WCAG A**, **WCAG AA**, **WCAG AAA** or **best practice**) and the number of elements affected.

Filter the list

Filter	Options
Impact	**All**, critical, serious, moderate, minor
Area	**All**, plus one button per category that has findings
Level	**WCAG A**, **WCAG AA**, **WCAG AAA**, **Best practice**
Search	"Search rules, elements, selectors…" matches rule names, IDs, descriptions and categories

Inside a rule

Open a row to see:

- a plain-language description of the rule;
- the rule ID (for example `color-contrast`), the WCAG criterion with its level as a link (for example "1.4.3 Contrast (Minimum) (AA)"), and badges for **Section 508** and **EN 301 549** when the rule maps to them;
- **How to meet this ↗**, a link to the W3C Understanding page for that criterion;
- each affected element, with:
  - its impact and the exact problem, for example "Contrast 1.19:1 is below the required 4.5:1 for normal text (12px). #ffffff on #efebe0.";
  - **Where**, the element's CSS selector;
  - for contrast, a live specimen of the text as it is ("Now") and with the recommended colour ("Recommended"), each with its ratio;
  - for other rules, the element's code as it is now and as recommended;
  - **Fix**, the concrete change;
  - **Copy selector** and **Copy fix** buttons.

A rule with many elements shows the first few and says "Showing N of M affected elements."

Findings: Fixes view

Every failure the engine could turn into a concrete patch, grouped so that a component repeated many times is one job. Groups are ordered by impact, then by how many places they affect.

The summary lists **Fix groups**, **Auto-fixable** (for example "31 of 40"), **High confidence** and **Need human wording**.

Ready-to-paste CSS patch is one stylesheet with every CSS change the engine proposes, each labelled with its rule. Select **Copy the whole stylesheet** to copy it. Review it against your design system before you ship it.

Each fix group shows:

- a one-line summary, such as "#d3af37 on #e5e0d1 is 1.6:1 — below the 4.5:1 minimum. #74601e reaches 4.62:1.";
- badges for impact, confidence, **auto-fixable** and the number of places;
- the rule and criterion;
- the code before and after;
- a note, for example "Keeps the original hue and only moves lightness, so the design stays recognisable.";
- **Where (N)**, a list of every selector with its own **Copy** button;
- **Copy fix** and **Copy CSS**.

Confidence

Confidence	Meaning shown
high	Derived from the element's own markup or computed colours — safe to apply nearly verbatim.
medium	Inferred from a filename, icon class or URL — the shape is right, read the wording.
review	The defect is certain, the wording is not — a person has to supply the text.

For example, a missing `alt` inside a link that already has a name is proposed as `alt=""`, and an icon link's name is taken from where it points rather than from its icon. Always read medium and review suggestions before you use them.

Inspect: Widgets view

Per-element rules ask "is this attribute legal?". This view asks "is this widget wired up correctly?", checked against the W3C ARIA Authoring Practices. It recognises eight widget patterns:

- Tabs
- Disclosure / Accordion
- Combobox
- Dialog (modal)
- Menu / Menubar
- Tree view
- Slider
- Carousel

For each, it checks structure and state: whether a tab actually controls a panel, whether `aria-expanded` matches what is really on screen, whether an auto-rotating carousel can be paused. The summary shows **Widgets detected**, **Fully conformant**, **Not conformant** and **Pattern findings**, and each widget gets a card with its name, pattern, status and any findings.

Closed dialogs and collapsed menus are hidden from the accessibility tree and are deliberately not judged. If no widgets are found, the view says "No composite widgets detected".

Inspect: Annotations view

A pass over the whole page in the style of a visual annotation tool, grouped and counted. Six tiles sit at the top; select one to isolate that category:

Tile	Contains
Errors	Accessibility failures that will block users
Contrast errors	Text that does not meet the contrast minimum
Alerts	Likely problems that need a human decision
Features	Accessibility features found on the page
Structural elements	Headings, landmarks and lists that shape the page
ARIA	ARIA roles, states and properties in use

Each group lists its items with a count and code; open **Show elements** to see every selector.

Inspect: A11y tree view

What assistive technology actually sees. Generic wrappers are transparent and `aria-hidden` subtrees are removed, the way a browser builds the tree. Each node shows its role, its accessible name (or "(no accessible name)") and its selector.

The summary lists **Exposed nodes**, **Landmarks**, **Interactive widgets**, **Widgets/landmarks with no name** and **Subtrees hidden from AT**. Very large pages are cut off, with a note such as "Tree truncated at N nodes."

Inspect: Screen reader view

The linear announcement stream a screen reader would produce, followed by the lists screen-reader users navigate by.

- ▶ **Speak transcript** reads the stream aloud with your browser's built-in speech. ■ **Stop** stops it. **Copy transcript** copies the text.
- **Announcement stream** lists each line in order, such as "Site header, banner landmark" or "Company, button, collapsed, has pop up menu". The intro says how many lines announce a bare role with no name.
- The rotor lists: **Headings** (with level), **Landmarks**, **Links** and **Form controls & widgets**. Items with no name are flagged.
- **Heading outline problems** appears when the heading order has gaps.

The speech is your browser's voice, not a real screen reader. Test with a real screen reader before you rely on the result.

Inspect: Vision view

Every text colour pair on the page, re-rendered through four colour-vision deficiencies, with the WCAG contrast ratio for each and the APCA Lc value for reference.

- A summary line per condition: **Protanopia**, **Deuteranopia**, **Tritanopia** and **Achromatopsia**, each with how many pairs drop below AA and how many lose contrast.
- **Palette under colour-vision deficiency** — a table with the original pair and its contrast (ratio and Lc), the ratio under each condition, and how many times the pair is used.
- **Colours that collide** — palette colours that become indistinguishable from each other under a simulated deficiency. If none do, it says "No colour collisions".

Up to 400 text elements are sampled. See Colour, contrast and vision for what the ratio and Lc mean.

Inspect: Focus & targets view

Keyboard focus and touch target sizes.

- The summary lists **Elements in the tab order**, **Positive tabindex**, **Focusable with no name**, **Focusable off-screen**, **Targets below 24×24px** and **Targets below 44×44px**.
- **Tab order** lists every stop in the order the `Tab` key reaches it, with its name, role, size and selector.
- **Touch target sizes** is a table with **Control**, **Role**, **Size**, **2.5.8 (24px)** and **2.5.5 (44px)**. Each size column reads **Pass** or **Below**.

WCAG 2.5.8 (Level AA) asks for targets of at least 24 by 24 CSS pixels, or enough clear space around them; 2.5.5 (Level AAA) asks for 44 by 44.

Report: Conformance view

The WCAG 2.2 conformance picture, separate from the score.

- **Automated verdict** — the highest level the page meets with no failures (**A**, **AA** or **AAA**), or **Does not conform**, plus how many criteria need manual review. Conformance is all or nothing: one failure at a level denies that level.
- A note that automated testing can only confirm a subset of WCAG, and that criteria shown as untested need human evaluation and are never counted as passing.
- Three level cards, **WCAG 2.2 A**, **WCAG 2.2 AA** and **WCAG 2.2 AAA**, each with passes out of the total and counts of fail, review and untested.
- **Success criteria scorecard** — all 60 criteria the engine knows, each marked ✓ pass, ✗ fail, ? review or – untested.
- **Accessibility Conformance Report (VPAT-style)** — one row per Level A and AA criterion, with **Criterion**, **Level**, **Conformance** and **Remarks**.

Conformance	Remark shown
Supports	Automated checks found no failures.
Partially Supports	N item(s) require manual verification.
Does Not Support	N instance(s) found, with the failing rule
Not Evaluated	Not covered by automated testing — manual evaluation required.

A VPAT (Voluntary Product Accessibility Template) is the format most procurement processes ask for. This table is evidence to support a claim, not the claim itself.

Report: Export view

"Everything the engine produced, in the formats the rest of your toolchain expects." File names start with `a11y-` followed by a short form of the audited address.

Export	Contents	File
Results JSON (axe format)	Violations, passes and incomplete results in the shape continuous-integration tools already read	`a11y-<site>.json`
Issues CSV	One row per affected element: rule, impact, WCAG, selector, problem, fix, suggested patch, confidence	`a11y-<site>-issues.csv`

Export	Contents	File
Remediation brief (Markdown)	A developer write-up: score, conformance, every violation with selectors and fixes, and the suggested patches	`a11y-<site>-remediation.md`
Contrast fix CSS	A stylesheet patch that raises every failing text pair to the AA minimum	`a11y-<site>-contrast.css`
Conformance report CSV	VPAT-shaped Section 508 and EN 301 549 rows with conformance level and remarks. Marked **Specialist**.	`a11y-<site>-conformance-report.csv`

The conformance report CSV is a paid perk of the Specialist plan. The other four exports are free.

Run metadata at the foot of the view lists the **Engine** ("Auric Accessibility Engine"), the **Rule set** ("WCAG 2.2 (A/AA/AAA) · Section 508 · EN 301 549 · ARIA 1.2 · APCA preview"), **Rules available**, **Rules evaluated**, **Run at**, **URL**, and whether **Computed styles** and **Element geometry** were available, with how many **Stylesheets** were read. The card's badge is how long the engine took.

Below the engine: Advanced Accessibility

Under the engine block, the panel continues with a summary card headed **Advanced Accessibility**: the same score, a **Score Breakdown**, **Conformance Levels** (Level A and Level AA, with how many were not tested), and **Voice Flow** (steps a screen reader visits, landmarks announced, issues in the flow, and **▶ Play** / **■ Stop** buttons that speak the flow).

What comes after depends on your plan:

- **Paid plans** see **Top Accessibility Priorities**, detailed cards comparing raw and rendered accessibility (keyboard, ARIA and contrast findings, and a WCAG mapping), and, for a site scan, **A11y+ · Page-by-Page**.
- **Free and signed-out visitors** see a card headed **The full accessibility report** in their place. It says how many findings are shown, lists what the paid report adds, and offers **See plans** and **Sign in**. It also states that every other tool on the site stays free and that this is the one report that is not.

A practical order of work

1. In **Overview**, read the grade and the conformance line, then open **Quick wins**.
2. In **Issues**, filter **Impact** to critical, then serious. Fix what blocks a task first: controls with no name, fields with no label, keyboard traps.
3. In **Fixes**, copy the CSS patch for contrast and target-size failures, and review the medium and review suggestions by hand.
4. In **Inspect**, check **Focus & targets** by pressing `Tab` through the real page, and read the **Screen reader** transcript for unnamed controls.
5. Work through **needs review** items yourself: text over images, whether animation can be paused, whether focus order still makes sense visually.
6. Re-run the audit and open **Compare** to confirm what changed.

For every rule the engine checks, see Accessibility rules.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Accessibility rules

Every one of the 126 accessibility rules the Analyzer checks, grouped by category, with its impact, WCAG mapping and what it looks for.

The accessibility engine in the **A11y+** panel checks 126 rules. This page lists every one, grouped into the 15 categories the panel uses for its **Area** filter and its **Where the problems are** bars. For how to read the results, see The accessibility engine.

How to read the tables

- **Rule** is the title the Issues view shows.
- **ID** is the rule's identifier, shown in each rule's details and used in the exports.
- **Impact** is how badly a failure affects people: critical, serious, moderate or minor.
- **WCAG** lists the success criteria the rule maps to. "Best practice" means the rule maps to no criterion: it is good advice, but not part of conformance.
- The **Level** filter in the Issues view uses the lowest level among a rule's criteria. A rule mapped to a Level A criterion is filtered as **WCAG A**.

Rules that map to WCAG 2.0 Level A or AA criteria also carry a **Section 508** badge, and rules that map to WCAG 2.0 or 2.1 Level A or AA criteria carry an **EN 301 549** badge.

Across the catalogue, 18 rules are critical, 61 serious, 30 moderate and 17 minor. By level, 81 are filtered as Level A, 20 as Level AA, 5 as Level AAA and 20 as best practice.

What every rule has in common

- Rules only judge what is actually exposed. Elements hidden from assistive technology are skipped, apart from a few rules that exist to check hidden content, such as `aria-hidden` misuse, `<meta>` tags and image-map areas.
- Rules that need the page's layout or computed styles (contrast, reflow, target size, focus order) report **needs review** instead of guessing when those are unavailable.
- Content inside open shadow roots (web components) is checked too.

Text alternatives (12 rules)

Images and other non-text content need a text equivalent that a screen reader can announce.

Rule	ID	Impact	WCAG	Looks for
Images must have an alt attribute	`image-alt`	critical	1.1.1	`<img>` with no `alt`, no ARIA name and no presentational role
Alt text must not be a file name	`image-alt-filename`	serious	1.1.1	Alt text such as `hero_banner.jpg`
Alt text must be meaningful	`image-alt-placeholder`	serious	1.1.1	Placeholder alt such as "image", "photo", "logo" or "icon"
Alt text should not duplicate adjacent text	`image-alt-redundant`	minor	Best practice	Alt that repeats the text of its link or figure, so it is announced twice
Alt text should be concise	`image-alt-long`	minor	Best practice	Alt text longer than about 150 characters
Image buttons must have alternative text	`input-image-alt`	critical	1.1.1, 4.1.2	`<input type="image">` with no text
Image map areas must have alternative text	`area-alt`	critical	1.1.1, 2.4.4	Clickable image-map regions with no name
Embedded objects must have alternative text	`object-alt`	serious	1.1.1	`<object>` with no text alternative
Elements with role="img" must have an accessible name	`role-img-alt`	serious	1.1.1	`role="img"` with no name

Rule	ID	Impact	WCAG	Looks for
Informative SVGs must have an accessible name	`svg-img-alt`	serious	1.1.1	Meaningful inline SVG with no `<title>`, `aria-label` or `aria-labelledby`
Canvas elements need a text alternative	`canvas-fallback`	serious	1.1.1	`<canvas>` with no fallback content or ARIA name
Server-side image maps must not be used	`server-side-image-map`	minor	1.1.1, 2.1.1	Image maps that need a mouse

Audio & video (4 rules)

People who cannot hear or see media need captions, descriptions and transcripts, and nobody should be surprised by sound.

Rule	ID	Impact	WCAG	Looks for
Video must have captions	`video-caption`	critical	1.2.2	Prerecorded video with no captions track
Video should provide audio description	`video-description`	serious	1.2.3, 1.2.5	Video with no description track or text alternative
Audio needs a transcript	`audio-transcript`	serious	1.2.1	Audio-only content with no transcript
Audio must not autoplay for more than 3 seconds without a control	`no-autoplay-audio`	serious	1.4.2	Sound that starts by itself with no pause control

Structure & semantics (12 rules)

Screen-reader users move through a page by its headings, lists and frames. The structure has to be real markup, not just styling.

Rule	ID	Impact	WCAG	Looks for
Heading levels must not skip	`heading-order`	moderate	1.3.1, 2.4.6	A jump such as `h2` straight to `h4`
Headings must not be empty	`empty-heading`	moderate	1.3.1, 2.4.6	Headings with no text
Page must contain a level-one heading	`page-has-heading-one`	moderate	Best practice	A page with no `h1`
Lists may only contain list items	`list-structure`	serious	1.3.1	Other content placed directly inside `<ul>` or `<ol>`
List items must be inside a list	`listitem-parent`	serious	1.3.1	An `<li>` outside a list
Definition lists must be structured correctly	`definition-list`	serious	1.3.1	A `<dl>` with anything other than `dt` / `dd` pairs
dt and dd must be inside a definition list	`dlitem-parent`	serious	1.3.1	`<dt>` or `<dd>` outside a `<dl>`
Styled paragraphs must not be used as headings	`p-as-heading`	moderate	1.3.1	Bold, oversized paragraphs that look like headings
Frames must have a title	`frame-title`	serious	2.4.1, 4.1.2	Frames announced only as "frame"
Frame titles must be unique	`frame-title-unique`	serious	4.1.2	Two frames with the same title
IDs on interactive elements must be unique	`duplicate-id-active`	serious	4.1.2	Duplicate IDs that break labels and ARIA references

Rule	ID	Impact	WCAG	Looks for
Hidden content should be reviewed	`hidden-content`	minor	Best practice	Hidden content that scripts may reveal later; check it is exposed properly when shown

Tables (9 rules)

Data tables must tell screen-reader users which header each cell belongs to.

Rule	ID	Impact	WCAG	Looks for
Table headers must reference data cells	`th-has-data-cells`	serious	1.3.1	Header cells with no data cells, often a layout table marked as data
Data tables must have header cells	`table-headers`	serious	1.3.1	Data tables with no `<th>`
scope attribute must be valid and on a header cell	`scope-attr-valid`	moderate	1.3.1	Invalid or misplaced `scope` values
headers attribute must reference cells in the same table	`td-headers-attr`	serious	1.3.1	Broken `headers` references
Data tables should have a caption	`table-caption`	minor	Best practice	Data tables with no `<caption>`
Table header cells must not be empty	`empty-table-header`	minor	Best practice	Header cells with no name
Do not fake a caption with a full-width first-row cell	`table-fake-caption`	serious	1.3.1	A spanning first row used as a visual title instead of `<caption>`
Table caption and aria-label must not be identical	`table-duplicate-name`	minor	Best practice	A caption repeated in `aria-label` or `title`
Data cells in large tables must be associated with a header	`td-has-header`	serious	1.3.1	In tables of 3 or more rows and columns, data cells no header reaches

Forms & labels (15 rules)

Every field and button needs a name a screen reader can announce and a speech-input user can say.

Rule	ID	Impact	WCAG	Looks for
Radio and checkbox groups need a group label	`fieldset-legend`	moderate	1.3.1, 3.3.2	Related radios or checkboxes with no `<fieldset>` and `<legend>`
Form fields must not have multiple label elements	`form-field-multiple-labels`	moderate	1.3.1, 4.1.2	A field with more than one `<label>`
autocomplete attribute must use valid tokens	`autocomplete-valid`	serious	1.3.5	Invalid `autocomplete` values
Fields collecting user data should declare their purpose	`identify-input-purpose`	moderate	1.3.5	Personal-data fields (name, email, phone, address) with no `autocomplete`
Accessible name must contain the visible label	`label-in-name`	serious	2.5.3	An `aria-label` that replaces the visible text with something different
Changing a control must not automatically change context	`no-onchange-submit`	serious	3.2.2	A select that navigates as soon as its value changes

Rule	ID	Impact	WCAG	Looks for
autofocus should not be used	`no-autofocus`	moderate	Best practice	`autofocus`, which drops users into the middle of the page
Form elements must have labels	`form-field-label`	critical	1.3.1, 3.3.2, 4.1.2	Fields with no programmatic label
Placeholder text must not be the only label	`label-placeholder-only`	serious	3.3.2	Fields labelled only by placeholder text
Required fields must be programmatically marked	`required-field-indicated`	moderate	3.3.2	Fields marked required visually but not with `required` or `aria-required`
Invalid fields must describe the error in text	`error-identification`	serious	3.3.1, 3.3.3	`aria-invalid` with no linked error message
Authentication must not depend on a cognitive test	`accessible-authentication`	serious	3.3.8	Login flows that block paste or require memorising or transcribing
Forms need an explicit submit control	`form-submit-control`	moderate	3.2.2	Forms with no submit button
Buttons must have discernible text	`button-name`	critical	4.1.2	Buttons announced only as "button"
Select elements must have an accessible name	`select-name`	critical	4.1.2	Selects announced only as a combo box

Zoom, reflow & motion (6 rules)

People who zoom, enlarge text or change spacing must still be able to use the page.

Rule	ID	Impact	WCAG	Looks for
Content must not be locked to one orientation	`css-orientation-lock`	serious	1.3.4	Media-query transforms that lock portrait or landscape
Zooming and scaling must not be disabled	`meta-viewport`	critical	1.4.4	`user-scalable=no` or a maximum scale below 2
Text spacing must be overridable	`avoid-inline-spacing`	serious	1.4.12	Inline `!important` spacing that user styles cannot override
Content must reflow at 320 CSS pixels	`reflow-fixed-width`	serious	1.4.10	Declared fixed widths that force two-way scrolling at 320px
Hover content must be dismissible, hoverable and persistent	`content-on-hover`	moderate	1.4.13	Native `title` tooltips, which fail all three
Users should be able to zoom to at least 500%	`meta-viewport-large`	minor	1.4.4	A maximum scale below 5

Colour & contrast (4 rules)

Text and controls must stand out from what is behind them, and colour must not be the only cue.

Rule	ID	Impact	WCAG	Looks for
Text must meet the minimum contrast ratio	`color-contrast`	serious	1.4.3	Body text under 4.5:1, or large text under 3:1
Text should meet the enhanced contrast ratio	`color-contrast-enhanced`	moderate	1.4.6	Body text under 7:1, or large text under 4.5:1 (Level AAA)

Rule	ID	Impact	WCAG	Looks for
Links in body text must not rely on colour alone	`link-in-text-block`	serious	1.4.1	Links in prose with no underline and under 3:1 against the surrounding text
UI component boundaries need 3:1 contrast	`non-text-contrast`	serious	1.4.11	Controls whose visible boundary is under 3:1, or that have no boundary at all

Contrast is measured from what the page actually paints. Semi-transparent colours are blended with what sits behind them first. Text over a gradient, photograph or video has no single background colour, so it is reported as needs review. Each failing pair comes with a suggested colour that keeps the hue and adjusts only the lightness. See Colour, contrast and vision.

Keyboard & focus (12 rules)

Everything must work with a keyboard alone, and the user must always see where focus is.

Rule	ID	Impact	WCAG	Looks for
Click handlers must be on focusable, named elements	`keyboard-handler-focusable`	critical	2.1.1, 4.1.2	A `div` or `span` with a click handler that keyboard users cannot reach
Interactive controls must not be nested	`nested-interactive`	serious	4.1.2	A button inside a link, or the reverse
accesskey values must be unique	`accesskeys-unique`	minor	Best practice	Duplicate `accesskey` values
Single-character shortcuts must be remappable or scoped	`character-key-shortcuts`	serious	2.1.4	Single-key shortcuts that fire while someone is dictating
Positive tabindex must not be used	`tabindex-positive`	serious	2.4.3	`tabindex` above 0
Tab order must follow the visual reading order	`focus-order-visual`	serious	2.4.3	CSS reordering that makes the tab sequence jump around
Focused elements must show a visible focus indicator	`focus-visible`	serious	2.4.7	Outlines removed with nothing to replace them
Focus indicator must be large enough and contrast enough	`focus-appearance`	moderate	2.4.13, 1.4.11	Focus indicators thinner than 2 CSS pixels or under 3:1
Focused elements must not be hidden behind sticky UI	`focus-not-obscured`	serious	2.4.11	Sticky headers, cookie bars or chat widgets covering the focused element
Scrollable regions must be keyboard-accessible	`scrollable-region-focusable`	serious	2.1.1	Scrolling areas that neither take focus nor contain anything focusable
Frames removed from the tab order must not contain focusable content	`frame-focusable-content`	serious	2.1.1	An `<iframe>` with `tabindex="-1"`; usually needs a manual check
Focusable elements should convey a role	`focus-order-semantics`	moderate	Best practice	Elements put in the tab order with no role

Timing & animation (3 rules)

Moving, blinking and refreshing content must be under the user's control.

Rule	ID	Impact	WCAG	Looks for
Pages must not auto-refresh or redirect on a timer	`meta-refresh`	critical	2.2.1	A timed `<meta>` refresh or redirect

Rule	ID	Impact	WCAG	Looks for
Blinking and scrolling content must be stoppable	`blink-marquee`	serious	2.2.2	`<blink>` and `<marquee>`
Long or infinite animation must respect reduced-motion	`animation-pausable`	serious	2.2.2, 2.3.3	Animation over five seconds, or motion that ignores the reduced-motion setting

Navigation (9 rules)

People need to skip repeated blocks, know where a link goes, and find pages more than one way.

Rule	ID	Impact	WCAG	Looks for
Page must offer a way to skip repeated blocks	`bypass`	serious	2.4.1	No skip link, `<main>` landmark or heading structure
Page must have a descriptive title	`document-title`	serious	2.4.2	A missing or empty page title
Links must have discernible text	`link-name`	critical	2.4.4, 4.1.2	Links announced only as "link", often icon-only links
Link text must describe the destination	`link-name-generic`	moderate	2.4.4, 2.4.9	Generic text such as "click here", "read more" or "learn more"
Links with the same name must go to the same place	`identical-links-same-purpose`	minor	2.4.9	Two links with the same name and different destinations
Site should offer more than one way to find pages	`multiple-ways`	moderate	2.4.5	No search, sitemap or index alongside the navigation
Links opening a new window should say so	`link-new-window`	minor	Best practice	Links that open a new tab without warning
Help mechanisms must appear in a consistent place	`consistent-help`	minor	3.2.6	Help and contact links that move between pages
Skip links must point at an existing target	`skip-link`	moderate	Best practice	A skip link whose target ID is not on the page

Safe linking (1 rule)

Rule	ID	Impact	WCAG	Looks for
target="_blank" should carry rel="noopener"	`target-blank-noopener`	minor	Best practice	New-tab links that let the opened page control the original tab

Pointer & touch (4 rules)

Targets must be big enough to hit, and gestures need simple alternatives.

Rule	ID	Impact	WCAG	Looks for
Touch targets must be at least 24×24 CSS pixels	`target-size-minimum`	serious	2.5.8	Targets under 24×24px that are not inline and lack 24px of clear space
Touch targets should be at least 44×44 CSS pixels	`target-size-enhanced`	moderate	2.5.5	Targets under 44×44px (Level AAA)

Rule	ID	Impact	WCAG	Looks for
Path-based and multipoint gestures need a single-pointer alternative	`pointer-gestures`	moderate	2.5.1	Swipe, pinch or drag with no tap or click equivalent
Drag interactions need a non-dragging alternative	`dragging-movements`	moderate	2.5.7	Drag-and-drop with no click alternative

Language & readability (5 rules)

The page's language drives the screen reader's voice, and plain prose helps everyone.

Rule	ID	Impact	WCAG	Looks for
`<html>` must have a lang attribute	`html-has-lang`	serious	3.1.1	A page with no language
lang attribute must be a valid BCP 47 tag	`html-lang-valid`	serious	3.1.1	An invalid language tag
lang attributes on page parts must be valid	`valid-lang`	serious	3.1.2	Invalid `lang` on passages in another language
Text should be readable at lower-secondary level	`reading-level`	minor	3.1.5	Prose harder than lower-secondary level (Level AAA)
Abbreviations should be expanded	`abbr-expansion`	minor	3.1.4	Abbreviations with no expansion (Level AAA)

ARIA correctness (24 rules)

ARIA adds roles, states and names for assistive technology. Used wrongly, it silently breaks them.

Rule	ID	Impact	WCAG	Looks for
role attribute must contain a valid, non-abstract role	`aria-role-valid`	critical	4.1.2	Unknown or abstract roles, which browsers drop
role must be allowed on the element it is placed on	`aria-allowed-role`	moderate	4.1.2	A role the element may not carry
ARIA attributes must be spelled correctly	`aria-attr-valid`	critical	4.1.2	Misspelt `aria-*` attributes, such as `aria-labeledby`
ARIA attribute values must be valid	`aria-attr-value-valid`	critical	4.1.2	Out-of-range tokens, non-numbers and broken ID references
ARIA attributes must be allowed on the element's role	`aria-allowed-attr`	serious	4.1.2	States and properties the role does not support
Roles must carry their required ARIA attributes	`aria-required-attr`	critical	4.1.2	For example `role="checkbox"` with no `aria-checked`
Composite roles must own the right child roles	`aria-required-children`	critical	1.3.1, 4.1.2	For example a `tablist` that does not own `tab` elements
Roles must sit inside their required parent role	`aria-required-parent`	critical	1.3.1, 4.1.2	For example an `option` outside a `listbox`
aria-hidden elements must not contain focusable content	`aria-hidden-focus`	serious	1.3.1, 4.1.2	Focusable elements inside `aria-hidden`
aria-hidden must not be on `<body>`	`aria-hidden-body`	critical	4.1.2	The whole page hidden from assistive technology
Naming attributes must not be used on roles that prohibit names	`aria-prohibited-attr`	serious	4.1.2	`aria-label` on roles such as generic, paragraph or code

Rule	ID	Impact	WCAG	Looks for
Deprecated ARIA attributes should be removed	`aria-deprecated-attr`	minor	Best practice	ARIA drag-and-drop attributes, which nothing supports
Dialogs must have an accessible name	`aria-dialog-name`	serious	4.1.2	Dialogs announced only as "dialog"
ARIA commands must have an accessible name	`aria-command-name`	serious	4.1.2	Elements with a widget role and no name
ARIA input fields must have an accessible name	`aria-input-field-name`	serious	4.1.2	Custom input widgets with no name
Meters and progress bars must have an accessible name	`aria-meter-progress-name`	serious	4.1.2	Meters and progress bars announced as a bare number
Tooltips must have an accessible name	`aria-tooltip-name`	serious	4.1.2	Empty tooltips
role=presentation must not be combined with ARIA or focus	`presentation-role-conflict`	moderate	4.1.2	Presentational elements that are focusable or carry global ARIA
ARIA ID references must resolve	`aria-reference-valid`	serious	1.3.1, 4.1.2	`aria-labelledby` and similar pointing at missing IDs
Status messages must be exposed to assistive technology	`status-messages`	moderate	4.1.3	Toasts, cart counts or errors that appear with no live region
Live regions must be configured correctly	`aria-live-valid`	moderate	4.1.3	Misconfigured or over-assertive live regions
Conditional ARIA state must match the element's native state	`aria-conditional-attr`	serious	4.1.2	An `aria-checked` that disagrees with a native checkbox
aria-roledescription must be on an element with a semantic role	`aria-roledescription`	serious	4.1.2	`aria-roledescription` on an element with no role
role="text" must not contain focusable descendants	`aria-text`	serious	4.1.2	Links or fields inside `role="text"`, which disappear from the tree

Landmarks & regions (6 rules)

Landmarks let people jump straight to the main content, the navigation or the footer.

Rule	ID	Impact	WCAG	Looks for
Page must have exactly one main landmark	`landmark-one-main`	moderate	Best practice	No `<main>`, or more than one
banner and contentinfo landmarks must be unique	`landmark-no-duplicate`	moderate	Best practice	More than one top-level header or footer landmark
Landmarks of the same type must have distinct names	`landmark-unique`	moderate	Best practice	Two `<nav>` landmarks both announced as "navigation"
Top-level landmarks must not be nested inside another landmark	`landmark-top-level`	moderate	Best practice	A banner, main or contentinfo landmark inside another landmark
All page content should be inside a landmark	`region`	moderate	Best practice	Content outside every landmark
Form landmarks should be named	`form-landmark-name`	minor	Best practice	Several unnamed form landmarks

ARIA widget patterns

In addition to the 126 rules, the **Widgets** view checks eight composite widget patterns: Tabs, Disclosure / Accordion, Combobox, Dialog (modal), Menu / Menubar, Tree view, Slider and Carousel. Its findings are reported under the category **ARIA widget patterns** and count in the score like any other rule. When a widget finding repeats a rule that has already failed (for example an unnamed dialog), it is reported once, not twice.

WCAG criteria the engine knows

The Conformance view lists 60 WCAG success criteria. Eleven of them are never checked by any rule, because software cannot judge them. They always appear as untested and need a person:

Criterion	Name	Level
1.3.2	Meaningful Sequence	A
1.3.3	Sensory Characteristics	A
1.4.5	Images of Text	AA
2.1.2	No Keyboard Trap	A
2.3.1	Three Flashes or Below Threshold	A
2.5.2	Pointer Cancellation	A
2.5.4	Motion Actuation	A
3.2.1	On Focus	A
3.2.3	Consistent Navigation	AA
3.2.4	Consistent Identification	AA
3.3.7	Redundant Entry	A

A criterion can also show as untested on a particular page when none of its rules found anything to check, for example the captions criteria on a page with no video.

The other 49 criteria are covered by at least one rule: 1.1.1, 1.2.1, 1.2.2, 1.2.3, 1.2.5, 1.3.1, 1.3.4, 1.3.5, 1.4.1, 1.4.2, 1.4.3, 1.4.4, 1.4.6, 1.4.10, 1.4.11, 1.4.12, 1.4.13, 2.1.1, 2.1.4, 2.2.1, 2.2.2, 2.3.3, 2.4.1, 2.4.2, 2.4.3, 2.4.4, 2.4.5, 2.4.6, 2.4.7, 2.4.9, 2.4.11, 2.4.13, 2.5.1, 2.5.3, 2.5.5, 2.5.7, 2.5.8, 3.1.1, 3.1.2, 3.1.4, 3.1.5, 3.2.2, 3.2.6, 3.3.1, 3.3.2, 3.3.3, 3.3.8, 4.1.2 and 4.1.3.

To check the untested criteria yourself, work through the Accessibility Checklist.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Arrange the workspace

Open, move, resize, focus and close report panels, find sections with the Sections picker, and save layouts.

The Analyzer shows its report as a workspace: each section is a panel in a grid, and you choose which panels are on screen and where. This lets you read a finding and the thing it is about side by side, for example the accessibility findings next to the contrast pairs and the fixes.

Both modes use the workspace. Simple mode offers 8 sections and no **Layouts** menu; Advanced offers all 22 sections and saved layouts.

What opens by itself

- **Before your first run**, the workspace holds only the **Overview** panel, across the full width.
- **After a run**, the section of every check that ran is opened, if your mode can show it. The **Devices** section is never opened automatically.
- **The first time you use Advanced**, the default arrangement is **Overview** and **Contrast** side by side, with **Fixes** across the full width below. It replaces the single Overview panel when your first run finishes, but only if you have not already changed the arrangement.

Sections you closed during a run are not reopened for you.

The panel title bar

Every panel has a title bar with a grip and four buttons:

Control	What it does
Grip (the section icon and name)	Drag to move the panel. With the keyboard, focus it and press `Left` or `Right` to move it one place.
Width	Cycles the panel through a third of the row, half the row and the full row.
Focus	Shows only this panel, full width. Select it again ("Show the other panels again") to bring your arrangement back.
Collapse / **Expand**	Folds the panel to its title bar and back. The panel keeps its place.
Close	Removes the panel from the workspace. Its data is kept; open it again from the Sections picker.

Panels are as tall as their content, and the grid packs them so a short panel beside a long one does not leave a gap.

Open sections with the Sections picker

In Advanced mode select **Sections** in the strip. In either mode you can press `Ctrl` + `P` while the Analyzer is open.

> **Note:** While the Analyzer page is open, `Ctrl` + `P` opens the Sections picker instead of your browser's print dialog. To get the report on paper, use the exports in the **Report** section.

The picker is a dialog with a filter field ("Type to filter…") and every section your mode offers, in three groups:

Group	Contains
On screen	Sections that are open now. Their tick box is ticked.
Ready to open	Sections the last run filled, or that are always available
Not scanned	Sections whose check was switched off for the last run. Their count reads "scan".

Each row shows the section's finding count, taken from the same badge the section uses. A dash means the section has no count.

To use it:

- Select a row to open or close that section. The picker stays open, so you can open several in one visit.
- Type in the filter field to narrow the list.

- Press `Up` and `Down` to move between rows, following the rows as they are laid out on screen; `Left` and `Right` move across columns when the filter field is empty.
- Press `Enter` to open the highlighted section (or scroll to it if it is already open) and close the picker.
- Press `Space` to open or close the highlighted section and keep the picker open.
- Press `Esc`, or click outside the dialog, to close it.

The footer shows these keys and how many sections are on screen.

Choosing a section from **Not scanned** turns its check on. If there is already a report, the Analyzer runs the audit again so the section has something to show. In Simple mode this also switches you to Advanced, because Simple has no scan picker.

Open a section from the progress rail

The steps in the run line also open sections: **3 Fix** shows Fixes, **4 Preview** shows Before / After, **5 Export** shows Report, and **1 Analyze** and **2 Issues** show the Overview. A section opened this way appears at the top of the results area, above the panels, rather than as a panel of its own.

Layouts (Advanced)

Select **Layouts** in the strip to open the layout menu.

Ready-made layouts

Layout	What it puts on screen
Accessibility review	The engine's findings, the contrast that fails, and the fixes: A11y+ full width, then Contrast and Fixes side by side
Colour and vision	Every failing pair beside its palette, both under a filter: Contrast, Palette, Simulation, Before / After
Speed and weight	Vitals, what JavaScript wrote, and the media that costs the most: Performance full width, then Media and Deep
Content and search	Metadata and headings against the shape of the whole site: SEO and Project, then Site Map full width
Everything, thin	One narrow column of each, for a first look: Overview, Contrast, A11y+, SEO, Performance, Site Map

The arrangement currently on screen is marked **on screen**.

Save your own layout

1. Arrange the panels the way you want them.
2. Open **Layouts**.
3. Type a name in the **Name this arrangement** field (up to 40 characters).
4. Select **Save**.

Your layout now appears in the menu. Select it to apply it. Select the × beside a saved layout to forget it.

Nothing is lost when you switch

When you apply a layout or select **Reset to the default layout**, the arrangement you are leaving is kept, unless it is already saved under a name. It appears under **Kept automatically** as the date and time, for example "12 Sep 14:05". Only the three newest automatic entries are kept; named layouts are never removed for you.

What is remembered

- Each mode remembers its own arrangement, in this browser, between visits.
- Saved layouts are stored in this browser.
- A **Focus** view is temporary and is not remembered.

If the workspace cannot load in your browser, the Analyzer falls back to a row of tabs, one section at a time.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Choose what to scan

Pick which of the 13 checks run in Advanced mode, understand which sections each check fills, and read what a run costs in tokens.

A full audit runs every check the Analyzer has. You rarely need all of them. In Advanced mode you can switch checks off, which makes a run faster and can make it cheaper. In Simple mode the set of checks is fixed.

The checks

One check, **Page structure**, always runs: it reads the page and its resources, and everything else is built on it. The other 13 can be switched on and off.

Check	What it does	Fills the section	Notes
Project model	One deduplicated model of the page: resources, colours, media, schema, dependencies	Project	from Page structure
Colour contrast	Every text pair measured against WCAG, with a passing colour to replace it	Contrast	
Before / After	The page as it is beside the page with the contrast fixes applied	Before / After	from Colour contrast
Accessibility	The full rule engine: WCAG 2.1 and 2.2, the a11y tree, keyboard and screen-reader order	A11y+	slow
Palette	Dominant colours and generated accessible palettes	Palette	
Vision simulation	The palette through colour-vision and low-vision filters	Simulation	from Palette
SEO and metadata	Title, description, headings, schema, robots, canonical, keyword density	SEO	
Media	Every image, video and audio file: size, dimensions, alt text, captions, loading	Media	
Performance	Core Web Vitals measured in your browser, bundle weight, network simulation	Performance	slow
Responsive behaviour	Viewport, breakpoints, tap-target sizes, fixed widths that will not fit a phone	Devices	
Deep DOM and framework	Rendered DOM against raw HTML, framework detection, what JavaScript wrote	Deep	slow
Security and reliability	Security headers, broken links, server stability	Security	
Site map	Discovers the rest of the site from robots.txt and the whole sitemap tree	Site Map	slow

- **slow** marks the checks that take noticeably longer.
- **from …** marks a check that adds no measurement of its own and reads from another one. Ticking it turns its source on as well. Unticking the source turns off everything that reads from it.

Eight sections are never switched on or off, because they present what the checks found or what you saved: **Insights**, **Fixes**, **Playground**, **Rules**, **Workspace**, **Report**, **CSP Setup** and **Compare**. The **Overview** always has something, because Page structure always runs.

What Simple mode runs

Simple mode has no picker. It always runs Page structure plus: Project model, Colour contrast, SEO and metadata, Site map, Accessibility and Responsive behaviour. Responsive behaviour and Accessibility count towards the overall score even though Simple has no Devices or A11y+ section to show them.

Use the picker (Advanced)

1. Select **Advanced** in the strip.
2. Open the row under the strip that starts **What to scan**. Its heading shows how many checks are on and what they cost, for example "What to scan · 13 of 13 · 4 tokens".
3. Tick or untick checks. Ticking a check also opens its section in the workspace; unticking it closes that section.
4. Or use a shortcut button at the foot of the list:
   - **Everything** turns every check on.
   - **The fast three** keeps Page structure, Colour contrast and SEO and metadata, and turns the rest off.
5. Select **Analyze**.

Your choice is remembered in this browser for the next visit.

The notes at the foot of the picker repeat the rules: "4 checks to a token, rounded up. A blocked page adds the cost of rendering it on a server.", which section always runs, and which sections are always available.

Sections whose check did not run

If you open a section whose check was switched off for the last run, it says that the check "was switched off for this run, so there is nothing measured to show here", with a **Turn it on and run again** button. The Analyzer never shows "no issues found" for a check that did not run.

Choosing such a section from the **Sections** picker has the same effect: its check is turned on and, if there is already a report, the audit runs again.

Categories whose check was switched off are also left out of the overall score, rather than counted as zero or as 100. The Overview says which categories were excluded.

What a run costs

Tokens pay for work done on Auric Artisan's servers, such as rendering a page from another website in a real browser, or crawling a site. The token meter in the run line quotes the next run before you start it.

The quote

- The chosen checks are priced at **four checks to a token**, rounded up, with a minimum of one token. Page structure is not counted.
- If the page blocks embedding and your checks need a live, rendered page, rendering it on a server adds **2 tokens**. The meter shows this as "+2 more if the page blocks embedding".

- Checks that only read the page's HTML never need a server render. A run made only of Project model, SEO and metadata, Responsive behaviour, Security and reliability and Site map costs the same whatever the page does. The meter's note then says so.

Choice	Checks counted	Quote
Everything	13	4 tokens, +2 if the page blocks embedding
Simple mode's fixed set	6	2 tokens, +2 if the page blocks embedding
The fast three	2	1 token, +2 if the page blocks embedding

The meter's **Remaining** figure is your balance. Signed-out visitors see "Sign in to claim 100 free tokens." and a **Sign in · 100 free tokens** button; signed-in accounts that have not claimed them see **Claim 100 free tokens**; **See plans** links to the pricing page.

The token sheet

Select the token meter, or press `Enter` or `Space` while it has focus, to open a sheet that answers what the meter cannot. It has these parts:

- **Where you stand** — your balance and what this run would leave.
- **Why** followed by the cost — the lines that make up the price, such as "Reading the page, once" for one page, or "Finding the pages" plus a per-page line for a site scan. It notes that contrast, palette, media, accessibility and SEO run on your own machine and cost nothing, and that a token only ever buys work on a server. For a site scan it adds that tokens are "Held, not spent: you are charged for the pages that come back and refunded the rest."
- **What would fit** — what your balance still covers: another single page, a rendered snapshot, or a whole-site scan of a chosen size. A **Pages** stepper and a **Pages that block embedding** switch let you price a job before you run it. Pricing a job here does not change the run the **Analyze** button is about to make.
- **Lately** — your recent token activity, such as free tokens claimed or a monthly allowance added.

Monthly analyses

Server-side URL analyses are also counted against a monthly allowance. The free Apprentice plan includes 20 URL analyses a month; the pricing page at https://auricartisan.com/company/pricing/ lists the allowance for each paid plan. When the allowance is used up, the Analyzer shows **You have used this month's scans**.

> **Note:** Auditing a page on auricartisan.com in your own browser does not need the server, so it works without an account. Vision simulations elsewhere on the site are not metered at all.

For plans, allowances and top-ups, see the account and billing docs in Account and billing.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Colour, contrast and vision

Read the Contrast, Palette, Simulation and Before / After panels, and understand the WCAG contrast ratio and the APCA Lc value.

Four sections deal with colour: **Contrast** measures every text colour pair on the page, **Palette** extracts the page's colours, **Simulation** shows the page through colour-vision filters, and **Before / After** previews the contrast fixes. The **Vision** view of the A11y+ panel adds a table of every pair under four colour-vision deficiencies (see The accessibility engine).

Two ways to measure contrast

The WCAG contrast ratio

WCAG 2.x measures contrast as a ratio between two colours' **relative luminance**, which is how much light a colour gives off on a scale from 0 (black) to 1 (white). Green counts most towards it and blue least, as in human vision. The ratio is the lighter luminance plus 0.05, divided by the darker plus 0.05. It runs from 1:1 (the same colour twice) to 21:1 (black on white), and it does not matter which colour is the text.

What	Needs at least	Level
Normal text	4.5:1	AA
Large text	3:1	AA
The visible edge of a control, focus indicators, meaningful graphics	3:1	AA
Normal text, enhanced	7:1	AAA
Large text, enhanced	4.5:1	AAA

Large text means at least 24 CSS pixels, or at least 18.66 CSS pixels in bold. It is a size threshold, not a heading rule.

The APCA Lc value

APCA, the Accessible Perceptual Contrast Algorithm, is a newer contrast model being developed for a future version of WCAG. It gives a **lightness contrast** value, written **Lc**, that tracks how people actually perceive contrast more closely than the ratio, especially for light text on dark backgrounds and thin type.

- Lc runs from 0 (no contrast) to a little over 100 for black on white.
- It has a direction: dark text on a light background gives a positive value, light text on a dark background a negative one. Compare the size of the number, not its sign.
- A larger number is needed for smaller or thinner text.

The Analyzer shows Lc beside the ratio for reference, for example "needs 4.5:1 · 12px · APCA Lc 22.9 (needs 90)". The "needs" value comes from this size-and-weight guide:

Text size	Normal weight	Bold (700 and above)
24px and above	Lc 60	Lc 45
18px to under 24px	Lc 75	Lc 60
16px to under 18px	Lc 90	Lc 75
Under 16px	Lc 90	Lc 90

Note: APCA is advisory. It is not part of any current conformance requirement, so it never affects a pass, a fail or a score. Use it for judgement calls, and use the WCAG ratio for compliance.

The Contrast section

The Contrast section lists every text colour pair the page actually paints, measured against its effective background. Semi-transparent colours are blended with what is behind them first, because that blend is what a reader sees.

Blocks in the section

Block	What it shows
What is contrast?	A short explanation of the ratio and the AA, AAA and FAIL labels
Statistics (unique pairs)	Counts of **Failing**, **AA Pass** and **AAA Pass** pairs
Pair Count Breakdown	Raw, duplicate, unique, shared and same-colour pair counts, and total fail occurrences
Impact-Based Contrast Score	The section's score, with text, UI and decorative failures counted separately
Rendered Scene Contrast	When canvas or rendered layers were sampled: scene samples, low-contrast regions, contrast range and penalty
Detected Brand Colors	The page's dominant colours. Suggested fixes keep these where they can and adjust the text or UI colour instead.
Contrast Pairs	Every pair, grouped as **Failing Combinations**, **AA Passes** and **AAA Passes**
Live page preview	The analysed page, with reload, open in a new tab, zoom and full-screen controls

The **Pair Count Breakdown** lists **Raw candidate pairs**, **Duplicate pairs removed**, **Unique pairs retained**, **Shared across ≥2 pages**, **Same-colour (no contrast)** and **Total fail occurrences**. A pair from a shared stylesheet is counted once, with the pages it appears on, not once per page.

How the section is scored

The Contrast score starts at 100 and loses points for each failing pair, weighted by what it is: text failures count most (×12), then interface components (×9), then decorative colour (×3). Critical failures and low contrast in the rendered scene cost extra. The **Why this score?** box shows the breakdown.

Read a pair

Each pair shows:

- the ratio and its grade, for example "4.6:1 AA", and what it needs ("needs 4.5:1", or "needs 3:1 · large" for large text);
- the element type and a sample of its text;
- a sentence such as "Text colour #15803d is rendered on its effective background #ecf8f1 (e.g. `<span>`). Ratio 4.6:1 meets WCAG AA for normal text (AA needs 4.5:1)."

A failing pair also offers a suggested replacement colour. Select it ("Click to copy fix") to copy the corrected colour. The suggestion keeps your hue and changes only the lightness, so the result still looks like your brand.

What the section does not measure

- **Text over gradients, photographs and video.** There is no single background colour, so no honest ratio exists. The accessibility engine reports these as needs review.
- **Text nobody can see.** Empty layout boxes, decorative bars and hidden controls are skipped.

The Palette section

The Palette section extracts the colours the page uses and generates accessible alternatives.

Block	What it shows
What is this?	A short explanation. Select any swatch to copy its hex code.
Extracted Dominant Colors	The most-used colours, with a count
Theme Mode Detection	Whether light, dark or both modes were detected, with the signals and colour references counted

Block	What it shows
All Text Colors and **All Background Colors**	Every text and background colour found
Fonts	The font families the page uses
Export as CSS and **Export as JSON**	Download the extracted palette as `palette.css` or `palette.json`
CVD Previews	The palette under deuteranopia, protanopia, tritanopia and achromatopsia
Color Usage Frequency	Each colour with how many times it is used, split into computed styles and colours written in the HTML
Generated Accessible Palettes	Twelve AA-tuned palettes: Complementary, Triadic, Split-Comp and Analogous, three of each

Select **Save** on a generated palette to keep it in the **Workspace** section's **Saved Palettes** list.

If you open the Analyzer with a palette sent from your library, the Palette section shows that palette's contrast audit even before a page has been analysed.

The Simulation section

The Simulation section shows the analysed page through colour-vision and low-vision filters, so you can see whether information carried by colour survives.

The panel explains that about 8% of males and 0.5% of females have some form of colour blindness. Choose a **Vision Type**:

Vision Type	What it shows
Normal Vision	The page unfiltered
Deuteranopia	No working green-sensitive cones: reds and greens are hard to tell apart
Protanopia	No working red-sensitive cones: reds look darker and merge with greens
Tritanopia	No working blue-sensitive cones: blues and yellows are confused
Achromatopsia	No colour at all: the page in greyscale
Low Vision	A slight blur with reduced saturation

The live preview has buttons to reload, open the page in a new tab, zoom out, zoom in and go full screen. The active filter is named under the preview.

A strip at the top says how the preview is running, for example "Live preview — This page is on our origin, so it runs here exactly as it does for a visitor."

When the page will not embed

Many sites send headers that forbid other sites from showing them in a frame. The preview then shows **Couldn't embed this page** and explains that the site is blocking embedding with X-Frame-Options or a Content-Security-Policy frame-ancestors rule. You have four choices:

- **Run in browser** opens a real pop-up window with the page as it was captured, with the active filter applied. It works whatever the site's framing policy. Allow pop-ups for auricartisan.com if nothing opens.
- **Open in new tab** opens the live site, without a filter.
- **Try again** reloads the preview.
- **View CSP setup** opens the **CSP Setup** section, which explains how a site you own can allow Auric Artisan to embed it. See Fix and preview.

These filters are quick previews. For the full range of conditions, severities and models, use the Vision Lab.

The Before / After section

Before / After shows the page as it is beside the page with the Analyzer's contrast fixes applied, so you can judge the fixes before you change your CSS.

- **BEFORE (Original)** and **AFTER (Fixed)** previews sit side by side under **Original vs. Fixed**.
- **Applied Fixes** lists every colour change, with **Locate** to find the element and **Highlight all** to mark them all.
- **Score Comparison** shows the score now and the estimated score after the fixes.
- **Regression Check (after applying fixes)** is a list to tick through by hand after you apply the fixes to your real site: the page still loads without errors, critical user paths still work, keyboard navigation still works, the mobile breakpoint is still responsive, performance has not regressed, the visual layout looks correct, and the score went up or stayed the same. **Mark all validated** ticks them all.

If every pair already passes, the section says "No fixes needed" and "All contrast pairs pass WCAG requirements. Your colors are accessible!"

A practical workflow for colour

1. Run an audit with **Colour contrast**, **Palette** and **Vision simulation** switched on, or apply the **Colour and vision** layout in Advanced mode.
2. In **Contrast**, open **Failing Combinations** and note which components the failing pairs belong to.
3. Copy a suggested colour, or take it to the Contrast Checker to adjust it by hand.
4. Check **Before / After** to see the fixes in place.
5. Look at the page under **Protanopia**, **Deuteranopia** and **Achromatopsia** in **Simulation**, and make sure no information is carried by colour alone.
6. Check both your light and dark themes: a colour that passes on one often fails on the other.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Export and share

Every way to take a result out of the Analyzer — reports, data files, stylesheets and share links — with formats and file names.

This guide lists every export the Analyzer offers, what each file contains, and the ways to share a result. Most exports are in the **Report** section; others live in the section they come from.

The Report section

Report opens with **Full Analysis Report**, the date it was generated and the audited address, the score and the issue count. Then comes the **Export** row.

Button	What you get	File
Export JSON	The full report data, for automation, regression checks and comparing fields over time	`analysis.json`
Export HTML	A self-contained HTML report that opens in any browser and can be printed	`auric-analyzer-report.html`
Export PDF	A PDF of the report (paid plans; see below)	`auric-report-<number>.pdf`, or your browser's print dialog

Button	What you get	File
Export Fix CSS	One clean stylesheet of the generated fixes	`auric-analyzer-fixes.css`
Developer Report	A Markdown report written for developers	`auric-developer-report.md`
Client Report	A Markdown report written for clients and stakeholders	`auric-client-report.md`
Copy Report	The developer report, copied to the clipboard as Markdown	—
Share Link	A link to re-run this audit (see below)	—

Export PDF

PDF is a Pro export format, included with paid plans. On the free plan, selecting it opens an upgrade prompt that explains the audit stays free and that **Export HTML**, **Export JSON**, the fix CSS and both Markdown reports carry the same data.

When you are entitled, the button shows its progress ("Building PDF…"). If a server PDF cannot be built, the Analyzer opens the HTML report in a new window with your browser's print dialog ("Opening print view…"): choose **Save as PDF** there. If your browser blocks that window, the HTML report is downloaded instead, with a message telling you to open it and print it to PDF.

Copy Report

When the copy works, a message says "Report copied to the clipboard as Markdown." If the browser refuses clipboard access, the message says nothing was copied and suggests **Developer Report** to download the same Markdown instead.

What the Report section shows

Below the buttons, **Details** repeats the whole audit in one long page:

- **Category Scores**, with a button per category;
- **Narrative Insights** (human summary, confidence, estimated score after top fixes, recommended next steps);
- **Score Weighting Transparency**;
- **Meta & SEO**, **Accessibility**, **Contrast**, **Color Palette**, **Fonts**, **DOM Structure**, **Performance**, **Metadata Intelligence**, **Responsive Devices**, **Security & Reliability** and **Custom Rules** summaries;
- **All Issues**, each with its fix;
- **How Your Score Was Calculated**, a table of **Category**, **Score**, **Weight**, **Contribution** and **Points lost**, with the final score. Excluded categories show why, for example "excluded (13% unused — no value produced)". The button **How this score is combined, and what it cannot tell you** explains the method;
- a footer naming the engine version, the rule sets used ("WCAG 2.1 AA + 2.2 + WCAG 3 (preview) + Core Web Vitals 2024 + HTTP Archive 2024"), the data sources, the run time, and the limitations ("Cross-origin pages cannot be deeply inspected; some metrics are lab estimates and not real-user data (RUM).").

Exports in other sections

Section	Button	File
A11y+ › Export	**Results JSON (axe format)**	`a11y-<site>.json`
A11y+ › Export	**Issues CSV**	`a11y-<site>-issues.csv`
A11y+ › Export	**Remediation brief (Markdown)**	`a11y-<site>-remediation.md`
A11y+ › Export	**Contrast fix CSS**	`a11y-<site>-contrast.css`

Section	Button	File
A11y+ › Export	**Conformance report CSV** (Specialist plan)	`a11y-<site>-conformance-report.csv`
A11y+ › Fixes	**Copy the whole stylesheet**	Clipboard
Fixes	**Export Fix CSS**	`auric-analyzer-fixes.css`
Fixes	**Copy Batch CSS**	Clipboard
Palette	**Export as CSS** / **Export as JSON**	`palette.css` / `palette.json`
Project	**Export project model JSON**	`project-model-<date>.json`
Site Map	**Export map JSON**	`site-map-<date>.json`
Playground	**Export CSS Fixes** / **Export Palette**	`accessibility-fixes.css` / `fixed-palette.css`
Playground › Design Lab	**Download .css**, **Theme tokens (JSON)**, **Runtime JS**, **Palette JSON**	`auric-design-lab.css`, `auric-design-tokens.json`, `auric-design-lab.js`, `auric-palette.json`
Devices	Live custom CSS **Export**	`device-custom.css`
Rules	**Export rules**	`analyzer-custom-rules.json`
Workspace	**Export Workspace**	`workspace.json`
Compare	**Download it**	`<host>-<date>.json`

In these names, `<site>` is a short form of the audited address and `<date>` is the date in year-month-day form.

Share a result

There are four ways to share, and none of them sends the full report. Each link makes the recipient's Analyzer run the audit again on the same address, so they see the page as it is when they open the link.

Where	Control	What the recipient gets
Strip	**Share**	A link that re-runs the audit on the same URL; the original score travels as context only
Report	**Share Link**	A copied link carrying the score, issue count and URL; the scanned pages stay on your machine
Insights	**Copy share link**	A link that re-runs the audit on the same address
Insights	**Email**	Opens your email app with a message titled "Auric Artisan audit:" and the URL, containing the score and the headlines

The strip's **Share** opens the site-wide share dialog. The Report's **Share Link** copies its link straight away and shows it in a **Share this report** sheet with **Copy again** and **Done**.

You can also build a link by hand: `https://auricartisan.com/tool/analyzer/?url=https://example.com` opens the Analyzer and starts an audit of that address.

> **Note:** Because a shared link re-runs the audit, the recipient needs whatever the audit needs. Auditing another website requires them to be signed in, and the run uses their own tokens and monthly allowance.

For the site-wide share dialog, see Shareable links.

Which format should I use?

You want to	Use
Hand results to a developer	**Developer Report**, or the A11y+ **Remediation brief** and **Issues CSV**
Send a summary to a client or manager	**Client Report**, or **Export HTML**
Keep a record for an audit trail	**Export HTML** or **Export PDF**, plus **Export JSON**
Feed a pipeline or compare runs programmatically	**Export JSON**, or the A11y+ **Results JSON (axe format)**
Paste fixes straight into a stylesheet	**Export Fix CSS**, or the A11y+ **Contrast fix CSS**
Support a procurement or VPAT request	The A11y+ **Conformance report CSV** (Specialist), reviewed by a person
Show a colleague the live result	**Share**

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Fix and preview

Turn findings into code with the Fixes panel, try changes live in the Playground, add your own checks in Rules, and let your site be previewed with CSP Setup.

Four sections help you act on what the audit found. **Fixes** gives you the code. **Playground** lets you try colour changes on the page and watch the score move. **Rules** adds your own checks to every audit. **CSP Setup** explains how to let the Analyzer preview a site you own.

Fixes is available in both modes; the other three are Advanced only. For the accessibility engine's own patches (the ready-to-paste CSS and per-element suggestions), see the Fixes view in The accessibility engine.

Fixes

The developer hand-off: every actionable fix for the page, grouped by type.

Parts of the section

Part	What it does
How to use this	Explains that colour fixes give exact CSS values and structural fixes describe the HTML to change
Total	The number of actionable fixes
Fix This First	The top fixes, numbered, each with its reason and an impact label such as "Low impact"
Fix Mode	**Safe fixes** or **Aggressive fixes** (below)
Heuristic Auto-Fix	Four buttons that apply or export the generated fix stylesheet (below)
Fix Details	**Color Fixes (High Priority)** with a **Copy fix** button on each, and **Structural & Semantic Fixes**

Fix Mode

- **Safe fixes** favours brand-safe contrast changes and structural recommendations. This is the default.
- **Aggressive fixes** also includes broader CSS normalisation.

Changing the mode rebuilds the generated stylesheet. Your choice is remembered in this browser.

Heuristic Auto-Fix

The generated stylesheet contains safe visual fixes: focus rings, media sizing, motion reduction and generated contrast colours.

Button	What it does
Apply in Playground	Applies the stylesheet to the Playground preview only, and updates its live score. Nothing on your site changes.
Copy Batch CSS	Copies the whole stylesheet to the clipboard
Diff View	Shows the affected rules as they are now beside the fixed version
Export Fix CSS	Downloads the stylesheet as `auric-analyzer-fixes.css`

If the Playground preview is not open yet, **Apply in Playground** opens it first. If it still cannot apply, it says "Open the Playground preview first, then apply heuristic fixes."

A safe way to apply fixes

1. Copy a small group of fixes, not all of them.
2. Apply them in your code.
3. Re-run the audit and check **Before / After** and **Compare**.
4. Repeat, then export the final report.

Playground

An editor over the analysed page. Select any element in the preview, change its colours, and watch the score change live. It also holds a palette generator and a design lab that map a palette onto the page's elements.

Palette Generator

Open the **Palette Generator** bar to show it.

Control	What it does	Range or options	Default
Undo, redo, save, reverse	Step through palette history, save a version, reverse the order	Buttons	—
Colour slots	Each colour has a picker, a lock (kept when regenerating) and a remove button; **+** adds a slot	—	5 slots
Size	Number of colours	2 to 12	5
Base hue	Starting hue in degrees	0 to 360	220°
Seed	Text that makes seeded palettes repeatable	Any text	`auric`
Generation mode	How the next palette is built	Ten modes (below)	—

The generation modes are **Random**, **Seeded**, **Harmony**, **Gradient**, **Contrast**, **WCAG-friendly**, **CVD-safe**, **Accessible**, **Compliant** and **Comparable**.

Score Comparison

Shows **Original Score** and **Live Score** side by side, or "No changes yet" before you edit anything.

Filter & Rules

- An element filter: **All Elements**, **Images (img)**, **Buttons**, **Links (a)**, **Headings (h1-h6)**, **Forms & Inputs**, **Text Elements**, **Interactive**.
- A rule set for the live score: **WCAG 2.1 AA**, **WCAG 2.2**, **Core Web Vitals**, **WCAG 3 (Preview)**, **All Rules**.

Element Editor

Select any text or element in the **Live Preview**. The editor then lets you change its text colour and background and shows accessible alternatives straight away. Every change is logged and counted in the live score.

Button	What it does
Export CSS Fixes	Downloads your element edits as `accessibility-fixes.css`
Export Palette	Downloads the fixed palette as `fixed-palette.css`
Reset All	Removes every edit

Design Lab Editor

Maps each kind of element to a slot in your palette, and applies the result live.

- **Element Mapping** has one menu each for **Backgrounds**, **Text**, **Headings**, **Buttons**, **Button text**, **Cards**, **Sections**, **Borders**, **Accent / Links**, **Shadows** and **Form fields**. Each can be set to a palette slot or **(unset)**.
- **Live Audit Warnings** flags pairs that are hard to read or not safe for colour-vision deficiencies. When all is well it says "No accessibility warnings — palette looks readable and CVD-friendly."
- **Detected Site Colors** → **Suggestions** lists the page's own colours with a suggested replacement and an **Apply** button.
- **Generated CSS** shows the stylesheet for the mapping, with these buttons:

Button	File
Copy CSS	Copies it to the clipboard
Download .css	`auric-design-lab.css`
Theme tokens (JSON)	`auric-design-tokens.json`
Runtime JS	`auric-design-lab.js`, a small script that applies the palette to a live page
Palette JSON	`auric-palette.json`

Optional Runtime JS shows the script before you download it.

When the page will not load in the preview

The Playground needs to load the page in a frame. If the site blocks embedding, the preview shows **Couldn't embed this page** with **Run in browser**, **Open in new tab**, **Try again** and **View CSP setup**. See Colour, contrast and vision for what each does.

Rules

Create your own selector-based checks without writing JSON. Each rule becomes part of the next analysis and can check that something exists, count it, or require an attribute. Custom rule results appear in the report under **Custom Rules**.

Pre-built templates

Select a template to fill in the form below, then adjust and save it.

- **SEO** (10): SEO title exists; Meta description exists; Canonical link present; Open Graph title; Open Graph image; H1 heading present; Single H1 per page; HTML lang attribute; No accidental noindex; Viewport meta tag.
- **A11Y** (9): Buttons have names; Images have alt text; Inputs have labels; Links have descriptive text; Skip navigation link; Main landmark present; Nav landmark present; No outline:none on focused elements; Tables have header cells.
- **UX** (5): No hidden priority content; 404 page has search or nav; No positive tabindex values; Print stylesheet present; No links pointing to #.
- **Performance** (6): Lazy-load non-critical images; Scripts defer or async; Preconnect to font origins; No `<style>` blocks in body; Images declare width/height; No autoplay videos without muted.
- **Security** (5): No inline handlers; External links use rel noopener; No mixed-content resources; Meta charset declared; Password fields allow autocomplete.

Add a rule

1. Enter a **Rule name**, for example "Primary CTA has label".
2. Enter a **CSS selector**, for example `.cta-primary, #signup-btn`.
3. Choose a **Check type**: **Element exists**, **Element must not exist**, **Minimum count**, **Maximum count** or **Attribute required**.
4. Fill in **Min**, **Max** or **Attribute** (for example `aria-label`) if the check type needs it.
5. Choose a **Severity**: **Medium** (default), **Critical** or **Minor**.
6. Write an **Issue message** and a **Fix suggestion**.
7. Select **Add Rule**.
8. Select **Save and re-run** to audit the page again with your rule.

The result: your rule appears under **Rule Results** and in **Latest Results** as passed or failed, with the number of matches.

Manage rules

Control	What it does
Filter Rules	Show **All Checks**, or only **Critical**, **Medium** or **Minor** rules
Advanced JSON editor	Edit every rule as JSON directly
Save and re-run	Saves the rules and runs the audit again
Reset defaults	Restores the two starting rules
Export rules	Downloads your rules as `analyzer-custom-rules.json`
Disable / **Remove**	On each rule in **Rule Results**

The Analyzer starts with two rules: **Main landmark** (a `main` element or `role="main"` must exist, Medium) and **Primary heading** (at least one `h1`, Critical). Your rules are stored in this browser.

CSP Setup

CSP & Access Setup explains how to configure a server you own so the Analyzer can load its pages in the **Simulation**, **Device Preview**, **Before / After** and **Playground** sections. It is a guide to your own server's headers; it does not change anything by itself.

What it covers

- **Why pages are blocked**: `X-Frame-Options: DENY` or `SAMEORIGIN`, CSP `frame-ancestors 'none'`, and CORS restrictions, with the solution: add the Auric Artisan origin to `frame-ancestors` and remove `X-Frame-Options`.
- **Minimum required**: the header to add to every page you want to preview (shown below), and removing or updating `X-Frame-Options`, which overrides CSP in some browsers.
- **Current page detection**: the analysed URL, its origin, and the CSP it sent.
- **Server Config Snippets** with a **Copy** button each, for **Nginx**, **Apache (.htaccess)**, **Express / Node.js**, **Cloudflare (Transform Rules)**, **Netlify (_headers file)**, **Vercel (vercel.json)**, **HTML Meta Tag (limited)** and **CORS Headers (full set)**.
- **Verification Checklist**: check the headers in your browser's developer tools, confirm `X-Frame-Options` is gone, re-analyse your URL, check the console for CSP errors, and review the **Security** section. → **Open Security Tab** takes you there.
- **How to Remove / Delete These Rules**: snippets that undo the change for each platform, and **Removal Verification Steps**.

The minimum header is:

```
Content-Security-Policy: frame-ancestors 'self' https://auricartisan.com https://www.auricartisan.com
```

Important: In some of the platform snippets the header value repeats the header name, as in `"Content-Security-Policy: frame-ancestors …"` inside a `Content-Security-Policy` header. Before you deploy, make sure the value starts with `frame-ancestors`, exactly as in the **Minimum required** line above. A duplicated name makes the header invalid.

Note: `frame-ancestors` cannot be set with a `<meta>` tag. It only works as an HTTP response header.

Only allow embedding on sites you control, and remove the rule when you no longer need it.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Performance, SEO and page health

What the Performance, SEO, Media, Deep, Devices, Security, Project, Site Map and Insights sections measure, and how to read them.

Beyond accessibility and colour, the Analyzer checks how fast the page is, how search engines see it, what JavaScript changes, how it copes with small screens, which security headers it sends, and how it fits into the rest of the site. This guide covers each of those sections. Most are available in Advanced mode only; **Project**, **Site Map** and **Insights** are also in Simple mode.

Every scored section has a **Why this score?** box, built like the one in the Overview, that explains its formula, what moved the score and the top recommendations. See Read the Overview and scores.

Performance

A Lighthouse-style audit built from the page's HTML, CSS, JavaScript, assets, DOM size and layout stability, and from real browser timing when it is available.

What it shows

Block	What it shows
Score and one-line verdict	For example "Fast, stable, and well within most budgets."

Block	What it shows
Data Source	Whether scores use real browser timing or estimates
Performance Context	Critical performance issues, acceptable architecture patterns, and the real-world load time
Quick Wins	The first changes to make
Sub-scores	**Loading**, **Interactivity**, **Visual Stability**, **Asset Weight**, **Maintainability**
Browser Timing	DOM content loaded, load event, transfer size and number of resources
Top Performance Priorities	Ranked issues with impact and fix
Page totals	Weights, scripts, stylesheets, CSS rules, DOM nodes and depth, images, iframes, third-party origins
File-Level Assets	Every script and stylesheet: blocking or not, origin, format, size, impact and fix
Performance Budgets	Pass or fail against fixed budgets (below)
Performance Findings	Each finding with why and fix
Loading Timeline / Waterfall	An estimated load order marking TTFB, FCP, LCP and page complete; red bars block rendering
Asset Dependency Graph	Every asset, grouped by type and origin
Structural Analysis	Repeated templates, template inconsistencies, z-index conflicts and layout-break risks
Generated Performance CSS	A small stylesheet of safe improvements, such as a visible focus style and responsive images

Performance budgets

Budget	Limit
HTML weight	250 KB
CSS weight	160 KB
Inline JS	120 KB
DOM size	1,500 nodes
DOM depth	32 levels
Blocking scripts	2
Image dimensions	0 missing
Third-party origins	8

Core Web Vitals

Core Web Vitals are Google's measures of loading, visual stability and responsiveness. The Analyzer reports six metrics. They are shown in the **Insights** section, in a card headed **Core Web Vitals** followed by where the numbers came from: "(measured · lab)" when your browser measured them, "(measured · backend lab)" when a server browser did, or "(lab estimates)" when they were estimated from the page's size and structure.

Metric	Good	Needs improvement	Poor
Largest Contentful Paint (LCP)	up to 2,500 ms	up to 4,000 ms	over 4,000 ms
Cumulative Layout Shift (CLS)	up to 0.10	up to 0.25	over 0.25
Interaction to Next Paint (INP)	up to 200 ms	up to 500 ms	over 500 ms
First Contentful Paint (FCP)	up to 1,800 ms	up to 3,000 ms	over 3,000 ms

Metric	Good	Needs improvement	Poor
Time to First Byte (TTFB)	up to 800 ms	up to 1,800 ms	over 1,800 ms
Total Blocking Time (TBT)	up to 200 ms	up to 600 ms	over 600 ms

The card's score is the share of metrics in the good range. Where it can, it names the element responsible (for example the LCP image) and the likely causes. When vitals are measured, the Performance score is capped by them, so a page cannot score 100 while its vitals are poor.

Network Simulation lets you re-estimate the numbers for **Fast 4G**, **Slow 4G**, **Fast 3G**, **Slow 3G** or **Offline**. It changes estimated values only, not measured ones, and your choice is remembered in this browser.

All of these are lab numbers from one load, not data from real visitors.

SEO

Search-engine readiness: meta tags, headings, structured data, indexing, links, content depth and keyword distribution.

Block	What it shows
Score	100 minus weighted deductions per failed check (critical −14, medium −7, minor −3)
Indexability	Robots meta, whether the page is indexable and followable, and the canonical URL
Content	Word count, title length and meta description length
Deep SEO Analysis — Raw + Rendered	Raw and rendered SEO scores, **SEO Delta Score**, **Crawler Confidence**, **Dynamic SEO Risk**
Dynamic SEO Detection and **Dynamic SEO Flags**	Whether the title, description, canonical, robots, Open Graph tags, JSON-LD or H1 only appear after JavaScript runs
Raw vs Rendered SEO	Each field before and after rendering, marked stable or changed
Raw/Rendered Check Matrix	Around 27 checks, each pass or fail in the raw HTML and the rendered page
SEO Checks	15 pass-or-fail checks, listed below the table
Top SEO Priorities	Ranked fixes
Sitemap & Robots Discovery (hyper-detailed)	robots.txt rules, every sitemap found, and **All discovered URLs**; **Re-probe** checks again
Structured Data (JSON-LD)	Blocks found, valid and invalid, and schema types
Top Keywords (content density)	The most frequent terms with counts and share
SEO Findings	Each finding with its fix
SEO vs Competitor Archetypes	Your SEO, performance and accessibility scores beside reference averages by site type
Technical SEO	Title, meta description, canonical, viewport, language, charset, robots
Asset SEO	Images, missing alt, decorative images, bad file names, heavy formats, below-the-fold images not lazy-loaded
URL Quality	Slug, slug length, lowercase, hyphens, path segments, query parameters
Content Quality	Word count, sentences, average words per sentence
Schema (JSON-LD) Validation	Types found, errors, and recommended types that are missing
Crawl Quality	Internal and external links, broken anchors, malformed links, H1 count, URL depth, duplicate headings, indexability

The 15 **SEO Checks** are: title present, title length 30–65, meta description, canonical URL, lang attribute, viewport meta, Open Graph title, Open Graph image, robots not noindex, single H1, heading order, structured data (JSON-LD), internal links, content depth (at least 300 words) and descriptive link anchors.

Sitemap discovery reads robots.txt (user agents, allow and disallow rules, crawl delay and sitemap references) and every sitemap and sitemap index it finds, then lists every URL with a filter.

The competitor archetypes are fixed reference figures that the panel describes as industry averages based on HTTP Archive data. They are for orientation, not a live ranking.

Media

Element-level detail for every image, video and audio file.

- Counts of **Images**, **Videos** and **Audio**.
- **Media Interpretation**: priority and LCP candidates, decorative images, and the alt-text model (missing alt is a real problem; empty alt is treated as decorative when the image is not content).
- **Media Checks**: images have alt decisions, images reserve layout space, autoplay control, visible controls, captions for video, audio transcripts.
- For each item: source URL, file size, dimensions, loading behaviour, alt text or captions, selector, parent, DOM path and layout impact.
- **Top Media Priorities**.

A page with no media says "No media assets detected" and is not scored for media.

Deep

The **Deep Analyzer** compares the raw HTML the server sends with the DOM after JavaScript has run. Use it to find content, metadata or links that only exist after rendering, which some crawlers and assistive technologies may miss.

Block	What it shows
Headline	Score, rendering classification (for example "Static HTML / progressively enhanced") and detected framework
Rendering	Mode, confidence, framework, DOM delta, whether content and SEO are in the raw HTML
Page Size	Raw HTML against rendered DOM: size, nodes, words, scripts and media
SEO Generation	How metadata is produced, and the canonical
Accessibility Signals	Live regions, dialogs, canvases with no fallback, hidden focusable elements
Raw HTML Summary (Pre-JS)	Metadata, social tags, schema, semantics, forms, media, scripts, fonts and hydration signals
Completeness checks	13 raw SEO, 11 raw accessibility and 20 deep HTML quality checks, each with a fix
Rendered DOM Analysis (Post-JS)	Rendered SEO fields, headings, links, controls, media, hydrated content and a text sample
Framework Detection	The detected framework with evidence and confidence
Detected Script Tools	Analytics, tag managers, ads, chat, replay, A/B testing, consent, bundles, routers and more
Execution Risks	Large bundles, inline event handlers, suspicious script URLs, mixed-content scripts
Third-Party Hosts and **Script Loading Order**	Where scripts come from, and loading-order warnings

Devices

The **Responsive Design Analyzer** checks viewport metadata, fixed-width risk, image dimensions and interactive density across common device profiles. This section never opens by itself after a run; open it from the Sections picker.

Device profiles

Each profile gets a score and a risk level. Select one to test it in the Device Lab.

Profile	Size (CSS pixels)	Profile	Size (CSS pixels)
Small phone	320 × 568	iPad landscape	1024 × 768
Android small	360 × 640	iPad Pro portrait	1024 × 1366
iPhone SE	375 × 667	iPad Air landscape	1180 × 820
iPhone 12/13	390 × 844	HD laptop	1280 × 720
Android large	412 × 915	Laptop	1366 × 768
Phone Max	428 × 926	Desktop	1440 × 900
Fold compact	540 × 720	Wide desktop	1536 × 864
Small tablet	600 × 960	MacBook Pro	1728 × 1117
iPad portrait	768 × 1024	Full HD	1920 × 1080
iPad Air portrait	820 × 1180	QHD monitor	2560 × 1440

The headline names the most broken device, for example "Most broken device: Small phone at 320 x 568. Risk: Low."

Device Lab

The Device Lab shows the page at the chosen size, with **Auto**, **Live**, **Screenshot**, **Rotate** and **Fullscreen** buttons, and the usual preview controls. If the site blocks embedding, the same **Couldn't embed this page** overlay appears as in Simulation.

Live custom CSS lets you type CSS that is applied to the preview as you type, with no reload. Use **Import page CSS**, **Format**, **Clear**, **Copy** and **Export** (downloads `device-custom.css`). The text is stored locally for the session, and the last edit is re-applied.

Device Screenshot Wall

Capture all devices renders the page in a real headless browser on Auric Artisan's servers at every device size, with full-page captures and measured overflow, tap-target and text-size checks. It works even when the site blocks embedding. Capturing another site needs you to be signed in, and server captures are metered in tokens.

> **Note:** On a page that blocks embedding, the Device Lab starts server-rendered captures by itself a few seconds after it opens. That is why the Devices section is never opened for you automatically.

Responsive Signals sums up: viewport meta, touch target inventory, fixed-width risk, image dimensions and render-blocking scripts.

Security

The **Security Policy** section checks HTTPS, the Content-Security-Policy, framing, referrer and permissions policies, MIME sniffing, link reliability and server timing.

It shows a **Security** score, a **Reliability** score and the number of **Broken Links**, then these checks, each ticked or crossed with a fix:

Check	What passes
HTTPS	The page is served over HTTPS
Strict Transport Security	An HSTS header is sent
Content Security Policy	A CSP header is sent
CSP default source	The CSP includes `default-src`
CSP object restriction	The CSP includes `object-src 'none'`
Frame protection	CSP `frame-ancestors` or `X-Frame-Options` is set
Referrer policy	A referrer policy is set
Permissions policy	A permissions policy is set
MIME sniffing protection	`X-Content-Type-Options: nosniff` is sent
Cross-origin isolation policy	COOP, CORP or COEP is set where compatible
No mixed content	No insecure resources on a secure page
Safe new-tab links	New-tab links carry `rel="noopener"`
Secure forms	Forms submit securely
Inline script surface	Inline script use is limited

Environment-Aware Scoring notes when the page is on a local development server; HTTPS and deployment-only headers are then scored more gently, and the raw score is shown beside the adjusted one. **Code-Level vs Server-Level** separates findings you fix in your code from those you fix in server configuration. **Reliability & Stability** lists links checked, internal and external links, broken links and wall time. **Detected Policy** shows the Content-Security-Policy, Permissions-Policy and Referrer-Policy the page actually sent.

The security score checks whether protective headers are present, not how good they are.

Project

The **Unified Project Model** is one shared model for every section: pages, links and the sitemap are discovered once and reused, so totals stay consistent and shared files are counted once.

- **Project**: URL, origin, source, pages in the model and when it was scanned. **Export project model JSON** downloads it as `project-model-<date>.json`.
- **Shared Entities**: tiles for pages, stylesheets, scripts, fonts, images, media items, colours, contrast pairs, schema objects, classes, IDs and dependencies. Select a tile to open the section that owns it.
- **Page Discovery vs Scan**: discovered, scanned (completed), failed, not yet scanned, and duplicate URLs removed.
- **Findings by Module** and **Unified Findings (cross-linked)**: every finding with the section it belongs to.

Site Map

The **Full Site Map** crawls the site, builds a project tree, and shows which CSS and JavaScript files connect to which pages, with each shared file listed once and a "used on N pages" count.

Block	What it shows
Crawl & Map	**Crawl Full Site**, **Cancel**, **Pages** (1 to 500, default 60) and **Workers** (1 to 8, default 4)
Resources (deduped)	Unique stylesheets, scripts, font files, images and external files; duplicate references collapsed
Design tokens (deduped)	Unique colours, font families, classes, and IDs reused across pages
Export map JSON	Downloads the map as `site-map-<date>.json`
Project Tree and **Page Map**	Files and pages as folders
Dependency Graph	An interactive graph of pages and files (controls below the table)
Page → CSS & JS Connections	Each page with its stylesheets and scripts
Dependency Chains	Full CSS `@import` and JavaScript import trees, filled in by a crawl
Stylesheets → Pages & imports, **Scripts → Pages & imports**, **Images → Pages**	Each file with the pages that use it
Site-wide Findings (merged)	Findings that repeat across pages, with the pages they appear on

The **Dependency Graph** has three layouts, **Radial**, **Force** and **Grouped**; check boxes to show or hide **Pages**, **CSS**, **JS**, **Fonts** and **All labels**; a search field ("Find a file or page…"); and zoom in, zoom out, **Fit**, **Reset** and full-screen buttons. Select a node to highlight what connects to it, drag nodes to rearrange them, and scroll to zoom.

Crawl Full Site is available when the audited page is on auricartisan.com, or when the run used Auric Artisan's servers. For another website otherwise, the button is disabled with the hint "Cross-origin page — use Scan Site (backend) instead". The token meter prices a site-map crawl at one token for the whole job, not per page.

Insights

Smart Insights is an overview of the whole report written for decision-makers: a summary, prioritised fixes, a fix simulator, a benchmark percentile and progress over time.

Block	What it shows
Health	A health score, a tier, a percentile ("Better than N% of audited sites") and a UX score
Headlines	Two or three plain sentences about the result
Benchmark Breakdown	Your percentile per category against a reference median
vs. Competitor Archetypes	Reference scores by site type, each marked "you beat this" or "below this"
Share Report	**Copy share link** and **Email**
Top 5 Things to Fix First	Each with **Plain language**, **Developer view**, **Suggested fix** and an estimated score gain
Priority Engine	A ranked table: issue, severity, impact, effort, business impact, UX and revenue
Core Web Vitals	The six vitals (see Performance above) and **Network Simulation**
Fix Simulator	Tick fixes to see the projected score change as you go

Block	What it shows
Progress Over Time	Your past runs of this URL

The benchmark and archetype figures are fixed reference values for orientation. They are not a live comparison with other sites.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Read the Overview and scores

Understand the overall score, the grade bands, how categories are weighted, and how to work through the issue cards.

The **Overview** panel is the summary of a run: one score, the categories behind it, and the issues ranked by how much fixing them is worth. This guide explains each part and how the numbers are made.

The score

The large number at the top is the **Overall Audit Score**, from 0 to 100, with a letter grade beside it. Under it, a line names the categories it was weighted from. If some checks were switched off, the line says so, for example "Scored on 4 of 10 categories — performance, reliability, security, media, colour and palette, rendered DOM were not scanned for this run and were excluded rather than counted as zero."

A page that could not be measured (a bot-challenge page or an almost empty page) gets no score at all, rather than a misleading one. See Run an analysis.

Grade bands

The overall score and each section's own score use the same bands:

Grade	Tier	Range	Meaning shown
A+	Excellent	90–100	Production-ready; meets or exceeds best practices.
A	Strong	80–89	Solid foundations with only minor refinements left.
B	Good	70–79	Generally healthy, with a few clear gaps to close.
C	Fair	55–69	Usable, but several material issues should be addressed.
D	Weak	40–54	Significant issues are affecting real users.
F	Critical	0–39	Serious, widespread problems need urgent attention.

The accessibility engine in the **A11y+** panel uses its own, stricter letter scale. See The accessibility engine.

Why this score?

Open **Why this score?** under the score. It shows:

- the tier, its meaning, and how many points you need to reach the next tier, for example "To reach Excellent (A+), gain about 6 more points.";
- **How it's calculated** — a one-line formula;
- **Category weights & impact** — each category's share and how many points it cost;
- **Checks** — how many failed, how many items and pages were analysed, and a **Confidence** level;
- **What moved the score** — the categories that cost the most;
- **Top recommendations** — the first fixes to make.

Every section's panel has its own **Why this score?** box built the same way.

How the overall score is made

The overall score is a weighted average of up to ten category scores. The **Report** panel's table **How Your Score Was Calculated** shows the exact weights, contributions and points lost for your run.

Category	Base weight	Comes from
Accessibility	20%	The accessibility engine's score, when it ran
Colour contrast	16%	The Contrast audit
Performance	13%	The Performance audit, capped by measured Core Web Vitals
Deep analysis	10%	The Deep DOM comparison
Metadata	9%	The SEO and metadata check
Responsive	8%	The Responsive behaviour check
Visual quality	8%	What is painted on screen, sampled from the rendered page
Reliability	7%	Whether the links that were checked resolved
Security	7%	Whether the expected protective headers are declared
Media	2%	Images, video and audio alternatives

Two rules keep the number honest:

- **A category with nothing to measure is excluded**, and the remaining weights are scaled up to fill the gap. A page with no images is not penalised for media, and a check you switched off does not count as zero or as 100. Excluded rows show **Not scored**.
- **SEO and A11y+ are shown but not counted** in the overall score. The SEO score is reported for its own sake; the A11y+ score is the same engine result that already feeds Accessibility.

A score can go down after you improve a page. Fixing a component so that content becomes visible can expose text that was never checked before. That is the report becoming more accurate, not the page getting worse.

The rest of the Overview

Block	What it shows
What is this?	A short explanation of the score for first-time readers
Human Summary	Score, strong and weak areas, confidence, the previous run, and the estimated score after top fixes
Category Scores	Every category's score, including SEO and A11yPlus
Score Weight Transparency	The categories that cost points, with their weight and point impact
Analysis Context	**Environment**, **Performance context** and **Template inference** (shared header and footer)
Canvas / GPU / Non-DOM Layer (hyper-detail)	Canvas layers, pixels sampled, low-contrast regions, unreadable and unnamed canvases
Statistics	Counts of Critical, Medium and Minor issues, colours, contrast pairs and elements
Summary	A plain count, for example "Your site has 8 issues." with a line per severity

Block	What it shows
Issue Details	The issue cards, described below

For a whole-site scan, the Overview adds **Full-Site Scan** and **Page-By-Page Breakdown**. See Scan a whole site.

Work through the issue cards

Top 3 Things To Improve Score

The first block under **Issue Details** is the three fixes worth the most. Each shows the estimated points it would add, the category, **Why** it matters, the **Fix**, and a **Fix first** label.

Issue Classification

A row of badges separates **Real issues** from **Branding-related** and **Dev/temporary** ones, so that, for example, a finding caused by a development server is not mixed with real defects.

The issue list

Issues are grouped by severity: **Critical**, **Important** and **Low Impact**. Each card shows:

- the issue title and its current status;
- tags such as the metric it affects (for example LCP) and its area;
- **Fix: Safe** or **Medium**, which says how risky the change is, and an estimated score gain;
- **Cause chain**, a disclosure that explains what **caused by**, **blocked by** and **affects** — the chain from the defect to the user;
- **Why** and **Fix** lines;
- four status buttons.

Button	What it does
Open	The default status: still to do
Fixed	Marks the issue as fixed
Ignore	Marks the issue as accepted and out of scope
Recheck	Marks the issue for checking and re-runs the audit straight away

Statuses are remembered in this browser. **Recheck** starts a new run, which is metered like any other run.

If nothing was found, the list says "No issues detected — your page passed all checks."

Use the score well

- Watch it as a trend on the same page, not as a target. Hiding an element from assistive technology can remove a finding and make the page worse.
- Count causes, not findings. One button component used in twelve places produces twelve findings and needs one fix.
- Use **Compare** after a fix to see exactly what changed. See Save, reopen and compare.
- A good score is not a legal conformance claim. See Limits and accuracy.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Run an analysis

What happens when you select Analyze, which addresses are accepted, how the page is reached, and what the progress card tells you.

This guide explains a single-page audit from the moment you select **Analyze** to the moment results appear. For a whole-site crawl, see Scan a whole site.

Start a run

1. Type or paste an address into the URL field.
2. Select **Analyze**, or press `Enter` in the field.
3. To stop a run part-way, select **Cancel**. The Overview panel then says "Analysis cancelled".

There are three other ways to start a run:

- Open a link of the form `https://auricartisan.com/tool/analyzer/?url=<address>`. The Analyzer fills the field and starts at once.
- Open a link someone made with **Share**. It re-runs the audit on the same address.
- Select **Re-analyze** next to a past run in the **Workspace** panel, or **Recheck** on an issue card.

Which addresses are accepted

You type	What the Analyzer uses
`https://example.com/page`	Exactly that address
`example.com` or `example.com/page`	The same with `https://` added in front
`/tool/contrast-checker/`	That path on auricartisan.com
Nothing at all	The Analyzer's own page
`ftp:`, `mailto:`, `tel:`, `javascript:` or a malformed address	Refused with "That URL doesn't look valid" and a hint to try a full address such as `https://example.com`

When the Analyzer tidies what you typed, it writes the tidied address back into the field so you can see what was actually audited.

How the page is reached

A web page can only be measured once the Analyzer has a copy of it. The Analyzer tries several ways, in order, and uses the first that works. The live line in the progress card tells you which one it is on.

Order	Live line	When it is used
1	"Rendering the page on our servers…"	You are signed in. The page is rendered in a real browser on Auric Artisan's servers.
2	"Loading the page in a frame…"	The page loads in a hidden frame in your browser. Works for auricartisan.com pages.
3	"The page would not embed — reading its HTML instead…"	Your browser reads the page's HTML; needs the site's permission or **Public proxy**.
4	"Using the copy saved from a previous run…"	Everything else failed, and this browser holds a copy of the page from a run in the last seven days.

In step 1 the rendered page comes back to your browser for the rest of the analysis. Before step 2 the Analyzer checks the page's headers and skips the frame at once if they refuse embedding. Step 3 only works when the other site allows cross-origin requests, or when you have turned on **Public proxy** in the site scan settings.

If the server render in step 1 is refused, for example because this month's analyses are used up, the Analyzer still tries steps 2 to 4. Pages on auricartisan.com therefore still work.

> **Note:** When a page is read from its HTML (step 3 or 4), its scripts are removed before it is displayed, apart from JSON-LD structured data. Anything the page builds with JavaScript may then be missing from the audit. The **Deep** panel compares raw HTML with the rendered page so you can see how much that matters.

When you are not signed in

Auditing another website without an account ends with a panel headed **Sign in to scan another site**, with **Sign in** and **See plans** buttons. It explains that analysing an auricartisan.com page needs no account, and that scanning another site runs on Auric Artisan's servers, which the account meters. It is not an error in the Analyzer.

Two related panels appear for signed-in accounts:

- **Scanning another site needs a plan** — your plan does not include URL analysis. Select **See plans**.
- **You have used this month's scans** — your monthly allowance of URL analyses is spent. Select **See plans**, or wait for the next month.

The progress card

While a run is working, the Overview panel shows a card headed **Auditing**. It has:

- a live line saying what is happening now;
- a bar that fills as steps finish;
- a numbered list of the steps this run will take, in the order they really happen. Each step is marked waiting, active or done.

The steps are named after the checks you chose:

Step	Check
Reading the page	Page structure (always runs)
Building the project model	Project model
Measuring contrast	Colour contrast
Previewing the fixes	Before / After
Running the accessibility rules	Accessibility
Extracting the palette	Palette
Simulating colour vision	Vision simulation
Checking metadata	SEO and metadata
Listing media	Media
Measuring performance	Performance
Checking responsive behaviour	Responsive behaviour
Comparing rendered and raw HTML	Deep DOM and framework

Step	Check
Checking headers and links	Security and reliability
Discovering the site map	Site map

If you switched some checks off, a line under the list says how many, for example "3 checks switched off for this run."

Two steps take noticeably longer than the rest:

- **Measuring performance** loads the page a second time and waits for it to settle ("Measuring how fast the page loads…").
- **Running the accessibility rules** runs 126 rules over the whole page ("Checking the page against the accessibility rules…").

The site map step keeps working after the results appear, because it reads robots.txt and every sitemap. The **Site Map** and **SEO** panels fill in when it finishes.

The checks for a run are fixed when you select **Analyze**. Ticking or unticking a check part-way through does not change what that run measures.

When the run finishes

- The progress rail moves to **2 Issues**.
- The workspace opens the section of every check that ran and that your mode can show. The **Devices** section is the exception: it never opens by itself, because opening it on a page that blocks embedding can start paid screenshot captures.
- The run is filed in your browser's library as a new version of that site, and added to the history and monitoring lists in the **Workspace** panel. See Save, reopen and compare.
- The tab badges update with finding counts: **Contrast**, **SEO**, **A11y+** and **Fixes**.

Pages that cannot be scored

The Analyzer refuses to give a number when the page it received is not the real page:

- **A bot-challenge page.** If the page title looks like a challenge ("Just a moment", "Attention required", "Access denied", "Checking your browser", "Verify you are human", a captcha and similar), the report says the host answered with a bot-challenge page instead of the real content, and the score is left empty.
- **An almost empty page.** If the page has under 40 characters of text and under 10 elements, the report says it returned almost no content, which is likely an error page, a redirect stub or a challenge.

Rate limits on the target site

If the site you are auditing answers "too many requests" (HTTP 429), the Analyzer waits and retries by itself. The live line reads "Target rate-limited (HTTP 429) — waiting N s then retrying…". If it keeps happening, the run ends with a panel headed **Rate-limited by target (HTTP 429)** and advice to wait 30–60 seconds, lower the scan workers, or analyse one URL at a time.

Re-running

Select **Analyze** again with the same address. The progress rail returns to **1 Analyze** for the length of the run. Every re-run of the same address becomes another version of the same saved project, so you can compare them later.

Save, reopen and compare

How finished runs are kept in your browser, how to reopen them, how to compare two versions of a site, and how to keep a copy in your account.

You do not have to save a run by hand. Every finished audit is filed in your browser as a version of the site it audited, so you can reopen it later without running it again, and compare it with the next run to see what your changes did.

How runs are kept

- **One project per site.** A single-page audit is filed as a project named "Audit · " followed by the host and path. A whole-site scan is its own project, named "Audit · " followed by the host.
- **Every run is a version.** Running the same address again adds a new version to the same project instead of creating a new entry. Versions are never deleted to make room; after the 25 newest, older versions keep their scores and dates in the list and their full reports stay stored.
- **One collection per host.** Projects for the same host are grouped in a library collection named "URL Analyzer (" plus the host and ")", together with the site's palette.
- **The palette is filed too.** The colours the page is built from are saved as a palette named "Palette · " plus the host, so colour tools in your library can open it.
- **The full report is stored.** Every section of the report is stored in your browser, so reopening a version redraws every panel exactly as it was. The site map, which finishes after the rest, is added to the version when it arrives.

Runs are kept in your browser's storage on this device. They do not follow you to another browser or computer unless you keep a copy in your account (below), and clearing this site's data in your browser removes them.

Filing a run counts as a saved project in your library. The free plan allows 20 saved projects; if your library is full, a new project is not filed. See the Library Kit docs in Library Kit for how the library works.

The **Save this run** button fixed at the bottom right of the page is the site-wide library button. It saves the current output to your library, the same as on other tools.

Reopen a past run

There are three ways:

- In the **Workspace** section, under **Saved projects**, select **Open** beside a project.
- In the **Compare** section, pick the version on the right and select **Open this version**.
- Open the project from your library.

The Analyzer loads the stored report and redraws every section. Nothing is re-run and no tokens are used.

The Workspace section (Advanced)

"History, saved palettes, monitoring, and workspace management."

Block	What it holds
Saved projects	Every saved site, with its latest score, the number of versions, and **Open**
Also on this machine	The older history list: URL, score and date for up to 50 recent runs, each with **Re-analyze** and a remove button
Workspace Actions	**Export Workspace**, **Import Workspace** and **Clear All**

Block	What it holds
Saved Palettes	Palettes you saved from the **Palette** section's generated palettes
Continuous Monitoring	Each analysed URL with its last score, date, trend and checks; **Re-check** and remove buttons

Re-check current URL, at the top of **Continuous Monitoring**, runs the current address again.

Export, import and clear

- **Export Workspace** downloads `workspace.json`: the history list, saved palettes, notes and pinned findings.
- **Import Workspace** reads a `workspace.json` file back in. A file that is not analyzer data is refused with "That file could not be read as analyzer data."
- **Clear All** asks "Clear all workspace data?" and explains that saved palettes, notes and pinned findings are deleted from this browser, that analysis reports are not affected, and that this cannot be undone. Select **Clear workspace** to confirm.

Monitoring is not automatic. The Analyzer does not run in the background; it records each run you make, and **Re-check** runs it again when you ask.

Compare two versions

The **Compare** section sets two saved runs of the same site side by side: what was fixed, what appeared, and what stayed. It is available in both modes, and it works without a new run because it reads your saved versions.

1. Open **Compare**.
2. Choose the **Site** from the first menu.
3. Choose the earlier version under **Was** and the later one under **Now**. By default the newest run is compared with the one before it.
4. Read the result (below).

If nothing has been saved yet, the section says "Nothing has been saved to compare yet." with an **Enter a URL** button. If the site has one version, it says a comparison needs two and offers a button to analyse the site again.

Reading a comparison

Part	What it shows
Header	The **Was** and **Now** scores and the change: **better**, **worse**, **no change** or **not scored**
Scope note	Appears when the two runs did not measure the same things, naming the checks that ran only then or only now
By category	Each category whose score moved, with a bar, the change, and the scores before and after
Section by section	Each section's finding count before and after, and the change
What changed	A section's detail: **Gone** (fixed), **New** (appeared) and how many stayed in both

A section that ran in only one of the two versions is marked **only measured once** instead of showing a change, because a check that did not run finds nothing, which is not the same as finding nothing wrong. Categories measured in only one run are left out of **By category** for the same reason.

Contrast differences list each pair as two swatches with its ratio. Palette differences list colour swatches. Other sections list the finding and where it was found.

Buttons for the Now version

Button	What it does
Open this version	Loads that version's full report into the Analyzer
Download it	Downloads that version's report as JSON, for example `example.com-2026-09-24.json`
Keep in my account	Uploads that version to your account (below). Shows **In your account** ✓ once kept.

If a version's stored report has been removed from your browser, these buttons say "That version's results are no longer stored." Analysing the site again rebuilds it.

Keep a copy in your account

Browser storage is one browser on one machine. To open a report elsewhere, keep it in your account.

1. Sign in.
2. Open **Compare** and choose the version as **Now**.
3. Select **Keep in my account**.

The button changes to **In your account** ✓, and a message confirms "Kept in your account." Its tooltip shows the stored size. Select it again to remove the account copy; the message says "Removed from your account. This browser still has it."

- Nothing is uploaded unless you select the button. A whole-site report can be large, so the Analyzer never spends your connection or storage without being asked.
- The button only appears when you are signed in and report storage is available for your account.
- Kept reports stay until you remove them, or until your account is deleted.

Good habits

- Run the same page before and after a change, then compare. The **Gone** list is your proof that a fix worked.
- Keep the same checks switched on between runs you want to compare, so the scope note does not appear.
- Keep important versions in your account before you clear your browser's data.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Scan a whole site

Audit several pages of one site in a single run, follow the live progress, and page through the per-page results.

A site scan finds the pages of a site from its links and sitemap, audits each one in a real browser on Auric Artisan's servers, and combines the results into one report with a page-by-page breakdown. Use it to find the shared components that fail on every page, or to check a set of pages before a release.

Before you start

- You need to be **signed in**. Without an account the scan stops with "Sign in to scan a site. Analyzing this page still works without an account."
- Whole-site scanning is part of the **Artisan plan and above**. On other plans the scanner answers "Whole-site crawling is part of the Artisan plan and above."
- A site scan costs tokens. The pricing page describes it as one token per page analysed, so a 24-page crawl costs 24. The token sheet shows the exact quote, with a line for finding the pages and a line for the pages themselves. Tokens are held

when the scan starts; you are charged for the pages that come back and refunded the rest.
- One run audits **up to 24 pages**.

The site scan settings

Switch to **Advanced** and open **Site scan settings** under the strip.

Setting	What it does	Values	Default
Scan Site	Starts the site scan for the address in the URL field	Button	—
Robots	Follow robots.txt rules while scanning	On or off	On
Subdomains	Include subdomains in the scan	On or off	Off
Pages	How many pages to audit, with a time estimate beside it	1 to 24	10
Show	How many detailed results to show per page of the results list	25 or 50	25
Workers	Concurrent scan workers: higher is faster, lower is gentler on the site	1 to 20	3
Public proxy	Fetch cross-origin pages through third-party public CORS proxies	On or off	Off

Pages limits only how many pages are audited. It does not limit the site map: discovery still reads robots.txt and the whole sitemap tree, so the **Site Map** section covers the entire site however few pages you audit. The estimate beside the field, for example "≈ 23s", updates as you type.

With **Public proxy** on, the address being analysed is shared with those third-party services. It is off unless you turn it on.

> **Note:** On auricartisan.com the crawl runs on Auric Artisan's servers, and the scanner receives the address and the **Pages** number. In the current version the **Robots**, **Subdomains** and **Workers** settings do not change a site scan on the live site. **Public proxy** only affects single-page runs that fall back to reading a page's HTML in your browser.

Run a site scan

1. Type the site's address in the URL field, for example `https://example.com/`.
2. Select **Advanced**, then open **Site scan settings**.
3. Set **Pages** to the number of pages you want audited, up to 24.
4. Check the token meter, or open the token sheet to see the quote.
5. Select **Scan Site**.
6. Follow the live console in the Overview panel (below).
7. When it ends, the live line reads "Scan complete: N pages scanned" and the report opens.

To stop part-way, select **Cancel** in the strip.

The live console

While a site scan runs, the Overview panel shows a console headed **Live site audit**. The large title names the current phase, starting with "Discovering pages", then "Found N pages — auditing…". Below it, five folding groups count pages as they move:

Group	What it lists
In flight	Pages being audited right now
Completed pages	Pages audited, with their score, title and issue count
Failed pages	Pages that could not be audited, and why
Skipped	Pages that were found but not audited
Discovery & budget	How pages were found and how the page budget was spent

The progress rail's bar tracks the share of pages finished.

Read the results

A site scan produces one combined report. Every section reads the combined result, and the Overview adds two cards:

- **Full-Site Scan** — counts of **Completed**, **Failed**, **Skipped** and **Found** pages (shown as "Found (sitemap)" when a sitemap was used).
- **Page-By-Page Breakdown** — one row per audited page, each with its overall score, accessibility, SEO, performance and security scores, issue count and its findings.

The breakdown list has its own controls:

Control	What it does
Sort menu	**Lowest score** (default), **Highest score**, **Most issues**, **Title A-Z**, **URL A-Z**, **Scan order**
25 / page, **50 / page**	Rows per page of the list
First, **Prev**, **Next**, **Last**	Move through the list
Expand visible, **Collapse visible**	Open or close every row on screen

A line above the list says which rows are showing, for example "Showing 1-25 of 24 pages". If full details for a page of rows are still loading, compact rows are shown meanwhile.

The **A11y+** panel adds an **A11y+ · Page-by-Page** card for the same pages, and the **Site Map** panel merges findings across pages under **Site-wide Findings (merged)**.

How the combined score is worked out

- Each page is scored on its own, then the scores are combined into the site's result.
- For accessibility, each page's score leans on the in-depth audit of that page's rendered markup, and the combined accessibility score can only be lowered by it, never raised.
- A page the scanner could not reach is listed under **Failed pages** rather than scored.

Pages the scanner cannot audit

The scanner runs in a real browser on Auric Artisan's servers. It cannot audit:

- pages behind a login, a bot challenge or a strict firewall. The page is reported as refused, with the HTTP status, and the advice to run the audit from a browser that is already on the page;
- private and local addresses such as `localhost`, `127.0.0.1`, `10.x.x.x`, `192.168.x.x`, `172.16.x.x` to `172.31.x.x` and `169.254.x.x`. These are refused with "That address is not reachable from the scanner.";

- anything other than `http` and `https` addresses ("Only http and https URLs can be scanned.").

After the scan

The site scan is filed in your library as its own project, named "Audit · " followed by the host, separate from single-page audits of the same site. Re-scanning the same site adds a new version you can compare with the last one. See Save, reopen and compare.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Analyzer reference

Every control, option, section, input and output of the Analyzer in one place.

A complete list of the Analyzer's controls and outputs. Each entry links to the guide that explains it in more depth.

Page elements

Element	What it does
Masthead	The headline "One URL, audited seven ways.", a lede, and a rail with **Read the API reference**
API hint	The REST API equivalent and its price in API credits; dismissing it hides it on every tool page
Save this run	The site-wide library button, fixed at the bottom right

The strip

Control	What it does	Values	Default
URL field	The address to audit	Any http or https address, a bare host name, or a path on auricartisan.com	Empty
Analyze	Starts a single-page audit. `Enter` in the URL field does the same.	Button	—
Cancel	Stops the running audit or site scan	Shown only while running	—
Share	Opens the share dialog with a link that re-runs the audit	Button	—
Sections	Opens the Sections picker. Shows `Ctrl` + `P`.	Advanced only	—
Layouts	Opens the layout menu	Advanced only	—
Simple / **Advanced**	Switches mode; remembered in this browser	Simple, Advanced	Simple

URL field placeholder: "Enter a URL to analyze…" (Simple) or "Enter a URL for deep DOM, SEO, contrast & accessibility audit…" (Advanced). See Run an analysis.

What to scan (Advanced)

The row's heading reads "What to scan · N of 13 · N tokens". See Choose what to scan.

Control	What it does	Default
13 check boxes	One per switchable check (listed below)	All on
Everything	Turns every check on	—
The fast three	Keeps only Page structure, Colour contrast and SEO and metadata	—

The 13 checks are Project model, Colour contrast, Before / After, Accessibility, Palette, Vision simulation, SEO and metadata, Media, Performance, Responsive behaviour, Deep DOM and framework, Security and reliability, and Site map.

Page structure always runs. Simple mode always runs Page structure, Project model, Colour contrast, SEO and metadata, Site map, Accessibility and Responsive behaviour.

Site scan settings (Advanced)

See Scan a whole site.

Control	What it does	Values	Default
Scan Site	Starts a whole-site scan	Button	—
Robots	Follow robots.txt rules	On, off	On
Subdomains	Include subdomains	On, off	Off
Pages	Pages to audit, with a time estimate	1–24	10
Show	Rows per page of detailed results	25, 50	25
Workers	Concurrent scan workers	1–20	3
Public proxy	Fetch cross-origin pages through third-party public CORS proxies	On, off	Off

The run line

Element	What it does
1 Analyze	Progress step; shows the Overview
2 Issues	Progress step; shows the Overview
3 Fix	Progress step; shows Fixes
4 Preview	Progress step; shows Before / After
5 Export	Progress step; shows Report
This run	The quoted cost of the next run, in tokens, with "+2 more if the page blocks embedding" when a render may be needed
Remaining	Your token balance
Sign in · 100 free tokens / **Claim 100 free tokens** / **See plans**	Account actions shown when they apply
Token meter (select it)	Opens the token sheet: **Where you stand**, **Why**, **What would fit**, **Lately**

Search results

Control	What it does
Search results field	Filters cards and rows in open sections; shown after the first run
Clear	Empties the field

Workspace controls

See Arrange the workspace.

Control	What it does
Panel grip	Drag to move; `Left` / `Right` when focused
Width	Third, half or full row
Focus	Show only this panel; select again to restore
Collapse / **Expand**	Fold to the title bar
Close	Remove the panel
Sections picker	A filter field and three groups: **On screen**, **Ready to open**, **Not scanned**
Layouts menu	Ready-made, saved and kept layouts; **Save** a name; **Reset to the default layout**

Ready-made layouts: **Accessibility review**, **Colour and vision**, **Speed and weight**, **Content and search**, **Everything, thin**.

Sections

Section	Simple	Advanced	Filled by	Contents
Overview	Yes	Yes	Page structure (always)	Score, grade, categories, statistics, issue cards
Project	Yes	Yes	Project model	Shared entities, discovery, findings by module
Palette	No	Yes	Palette	Extracted colours, CVD previews, generated palettes
Contrast	Yes	Yes	Colour contrast	Every text pair, ratios, suggested colours
Performance	No	Yes	Performance	Sub-scores, budgets, assets, waterfall
SEO	No	Yes	SEO and metadata	Meta, raw versus rendered, sitemap, schema, keywords
A11y+	No	Yes	Accessibility	The 126-rule engine and its views
Insights	Yes	Yes	Always available	Summary, benchmarks, priority engine, Core Web Vitals, fix simulator
Media	No	Yes	Media	Images, video and audio in detail
Deep	No	Yes	Deep DOM and framework	Raw HTML against rendered DOM
Site Map	Yes	Yes	Site map	Crawl, project tree, dependency graph
Simulation	No	Yes	Vision simulation	Page under colour-vision and low-vision filters
Devices	No	Yes	Responsive behaviour	Device profiles, Device Lab, screenshot wall
Security	No	Yes	Security and reliability	Header checks, links, detected policy
Playground	No	Yes	Always available	Element editor, palette generator, design lab

Section	Simple	Advanced	Filled by	Contents
Fixes	Yes	Yes	Always available	Fix list, fix mode, heuristic auto-fix
Before / After	No	Yes	Before / After	Original and fixed previews
Rules	No	Yes	Always available	Custom selector rules and templates
Workspace	No	Yes	Always available	Saved projects, history, palettes, monitoring
Compare	Yes	Yes	Always available	Two saved versions side by side
CSP Setup	No	Yes	Always available	Server headers to allow embedding
Report	Yes	Yes	Always available	Full report and exports

Section badges show finding counts on **Contrast**, **SEO**, **A11y+** and **Fixes**.

A11y+ views

Group	View	Contents
Overview	Overview	Grade, conformance line, severity, categories, quick wins, reading level, backlog
Findings	Issues	Every failing or needs-review rule, with filters **Impact**, **Area**, **Level** and search
Findings	Fixes	Fix groups, confidence, the ready-to-paste CSS patch
Inspect	Widgets	Eight ARIA widget patterns checked
Inspect	Annotations	Errors, Contrast errors, Alerts, Features, Structural elements, ARIA
Inspect	A11y tree	The accessibility tree
Inspect	Screen reader	Transcript, rotor lists, speech
Inspect	Vision	Pairs under four colour-vision deficiencies, with APCA Lc
Inspect	Focus & targets	Tab order and target sizes
Report	Conformance	WCAG 2.2 verdict, scorecard, VPAT-style table
Report	Export	Five exports and run metadata

See The accessibility engine and Accessibility rules.

Issue card statuses

Status	Effect
Open	Default; still to do
Fixed	Marked as fixed
Ignore	Marked as accepted
Recheck	Marked, and the audit runs again at once

Scores and thresholds

Item	Value
Overall and section grade bands	A+ 90–100, A 80–89, B 70–79, C 55–69, D 40–54, F 0–39
A11y+ engine grade bands	A+ 95–100, A 90–94, B 80–89, C 70–79, D 55–69, F 0–54
Category weights, first five	Accessibility 20%, contrast 16%, performance 13%, deep analysis 10%, metadata 9%
Category weights, last five	Responsive 8%, visual quality 8%, reliability 7%, security 7%, media 2%
Contrast thresholds	4.5:1 normal text, 3:1 large text and UI, 7:1 and 4.5:1 enhanced
Large text	24px, or 18.66px bold
Target size	24 × 24 CSS px (AA), 44 × 44 CSS px (AAA)
Core Web Vitals, good	LCP ≤ 2,500 ms, CLS ≤ 0.10, INP ≤ 200 ms, FCP ≤ 1,800 ms, TTFB ≤ 800 ms, TBT ≤ 200 ms

Limits

Limit	Value
Pages per site scan	24
Site Map crawl pages	1–500 (default 60), 1–8 workers (default 4)
Saved copy of a page for fallback	Kept 7 days in this browser
History list in Workspace	50 entries
Free plan URL analyses	20 a month
Free plan saved projects	20
Layout names	40 characters
Automatic kept layouts	The newest 3

Links you can build

Link	Effect
`https://auricartisan.com/tool/analyzer/?url=<address>`	Opens the Analyzer and audits the address
A **Share** or **Share Link** link	Opens the Analyzer and re-runs the audit of the shared address

Remembered in this browser

- Simple or Advanced mode, and each mode's panel arrangement
- Saved layouts and the three newest automatic ones
- Which checks are ticked in **What to scan**
- **Fix Mode** (Safe or Aggressive)
- The **Public proxy** choice
- The Network Simulation profile
- Issue statuses (Open, Fixed, Ignore, Recheck)

- Custom rules
- Every run's report, versions, palettes and history (see Save, reopen and compare)
- A copy of each audited page for up to 7 days, used as a last-resort fallback
- Whether you dismissed the API hint

For exports and file names, see Export and share.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Troubleshooting the Analyzer

What each Analyzer error or odd result means, why it happens, and what to do about it.

Each entry gives the symptom as you see it, the cause, and what to do.

Starting a run

"That URL doesn't look valid"

- **Why:** the address could not be turned into a web address, or it uses a scheme the Analyzer cannot fetch, such as `mailto:`, `tel:`, `ftp:` or `javascript:`.
- **What to do:** type a full address including `https://`, for example `https://example.com`.

"Sign in to scan another site"

- **Why:** auditing a website other than auricartisan.com runs on Auric Artisan's servers, which needs an account. Nothing has failed.
- **What to do:** select **Sign in**, then run the audit again. Pages on auricartisan.com can be audited without an account.

> **Note:** The tips under this and other error panels may mention a "This Page" button. That button has been removed from the strip. To audit a page on auricartisan.com, type or paste its address into the URL field.

"Scanning another site needs a plan"

- **Why:** your account's plan does not include URL analysis, or the server refused the run for a plan reason. The message under the heading gives the exact reason. If it says "This run costs N tokens and your balance is M", your token balance is too low.
- **What to do:** select **See plans**. On a paid plan the monthly token allowance renews; the free plan's starter balance does not.

"You have used this month's scans"

- **Why:** you have used your monthly allowance of server-side URL analyses (20 a month on the free plan).
- **What to do:** wait for the next month, or select **See plans**. Pages on auricartisan.com can still be audited in your browser.

"The token meter could not be read, so this run cannot be metered"

- **Why:** a server-side run cannot start while your token balance is unavailable.
- **What to do:** wait a moment and try again.

Reaching the page

"Couldn't reach the target" or "You are offline"

The panel checks three things and shows each as a row:

Row	What it tells you
Your connection	Whether your browser reports it is online
Server-side scanning	That scanning another site runs on Auric Artisan's servers via Scan Site, which needs an account
Target origin	Whether the address is on auricartisan.com or on a site your browser cannot fetch

- **Why:** you are offline, the address has a typo or a private host name, the site is down, or the site is on another origin and you are not signed in.
- **What to do:** check your connection and the address; sign in to scan another site; check any VPN or firewall on your network.

"Could not fetch this cross-origin page from the browser"

- **Why:** the page would not embed, and your browser is not allowed to read another site's HTML directly.
- **What to do:** sign in, so the page is rendered on Auric Artisan's servers. Or turn on **Public proxy** in **Site scan settings** (Advanced), which routes the fetch through a third-party public proxy. Only do this if you are happy for that service to see the address.

"Could not analyze this page. Every method the analyzer has was tried"

- **Why:** every way of reaching the page failed. The numbered list names each method and why it failed: the frame, a rendered snapshot, a browser fetch and the server.
- **What to do:** read the last line (the server) first. It is usually the one that explains the problem, such as a site that refused the scanner.

"That site refused the scanner (HTTP 403)" (or 401 or 429)

- **Why:** the site declined to serve an automated browser. Sites behind a bot challenge, a login or a strict firewall cannot be audited remotely.
- **What to do:** audit a page that is publicly reachable. If it is your own site, allow the scanner or test a staging copy that is publicly reachable.

"That address is not reachable from the scanner."

- **Why:** the address is private or local, such as `localhost`, `127.0.0.1`, `10.x.x.x`, `192.168.x.x` or a link-local address. The scanner never visits private networks.
- **What to do:** audit a publicly reachable copy of the page.

"Request timed out"

- **Why:** the site did not respond in time. It may be slow, blocking automated visitors, or behind a captcha.
- **What to do:** try again later, or audit a lighter page first.

"Rate-limited by target (HTTP 429)"

- **Why:** the site you are auditing (or a proxy in front of it) asked the Analyzer to slow down. The Analyzer already waits and retries.
- **What to do:** wait 30–60 seconds and select **Analyze** again. If you were scanning a whole site, lower **Workers** and **Pages**, or audit one URL at a time. A recent copy of the page, if this browser has one, may be used instead.

Results

The score is empty and the report says the page returned a bot-challenge page

- **Why:** the site answered with a challenge page ("Just a moment", "Verify you are human" and similar) instead of its content. Scoring it would describe the challenge, not your page.
- **What to do:** audit a page the site serves to automated browsers, or allow the scanner on your own site.

The score is empty and the report says the page returned almost no content

- **Why:** the page had under 40 characters of text and under 10 elements. It is probably an error page, a redirect stub or a challenge.
- **What to do:** check the address. If the page builds its content with JavaScript, sign in so it is rendered in a real browser.

A section says it "was switched off for this run"

- **Why:** the check that fills that section did not run.
- **What to do:** select **Turn it on and run again**.

"Accessibility engine did not run"

- **Why:** the engine needs the page's rendered structure, and this run did not produce one.
- **What to do:** sign in so the page is rendered on the server, or audit a page on auricartisan.com.

"Coverage limits for this run" appears in A11y+

- **Why:** computed styles or element geometry were not available, so contrast, reflow, target-size and focus-order rules could not measure.
- **What to do:** those rules report **needs review** instead of guessing. Check them by hand, or re-run with the page rendered in a browser.

Some content is missing from the audit

- **Why:** the page was read from its HTML rather than rendered, so anything it builds with JavaScript is absent.
- **What to do:** sign in so the page is rendered in a real browser. Compare **Raw HTML Summary** and **Rendered DOM Analysis** in **Deep** to see what depends on JavaScript.

My score dropped and I did not change anything

- **Why:** the page changed underneath it (new content, an advert, a third-party embed or an A/B test), or a fix made hidden content visible so it was checked for the first time.
- **What to do:** open **Compare** and read the **New** lists rather than the score.

Performance numbers change from run to run

- **Why:** they come from a single lab load. Network, server load and caching change between runs, and estimated values depend on the **Network Simulation** profile.
- **What to do:** compare several runs, and treat single numbers as indicative.

The Analyzer's result differs from another tool

See the FAQ. Compare individual findings, not scores.

Previews

"Couldn't embed this page" in Simulation, Playground, Devices or Before / After

- **Why:** the site sends `X-Frame-Options` or a Content-Security-Policy `frame-ancestors` rule that stops other sites from showing it in a frame.
- **What to do:** select **Run in browser** to open the captured page in a pop-up window with the filter applied. Allow pop-ups for auricartisan.com if nothing opens. If it is your own site, follow **CSP Setup**.

Devices started capturing screenshots by itself

- **Why:** on a page that blocks embedding, the Device Lab starts server-rendered captures a few seconds after it opens. Captures are metered.
- **What to do:** close the **Devices** panel if you do not want captures.

Crawl Full Site is greyed out

- **Why:** the audited page is on another site and the run did not use Auric Artisan's servers, so the crawl cannot read it. The button's hint says "Cross-origin page — use Scan Site (backend) instead".
- **What to do:** sign in and run the audit again, or use **Scan Site**.

Robots, Subdomains or Workers seem to have no effect

- **Why:** in the current version, a site scan on auricartisan.com uses only the address and the **Pages** number.
- **What to do:** nothing; set **Pages** to control the size of the scan.

Exports, saving and sharing

Export PDF shows an upgrade prompt

- **Why:** PDF is a Pro export format for paid plans.
- **What to do:** use **Export HTML** and print it to PDF from your browser. It carries the same data.

Export PDF opens a print dialog instead of downloading

- **Why:** a server PDF was not available, so the Analyzer opened the HTML report for printing.
- **What to do:** choose **Save as PDF** in the print dialog. If the window was blocked, the HTML report was downloaded instead; open it and print it.

"Copy failed — the browser refused the clipboard write"

- **Why:** your browser did not allow the page to write to the clipboard.

- **What to do:** use **Developer Report** to download the same Markdown.

`Ctrl` + `P` opens the Sections picker instead of printing

- **Why:** the Analyzer uses `Ctrl` + `P` for its section picker while the page is open.
- **What to do:** use the exports in **Report** to get a printable copy.

"That version's results are no longer stored."

- **Why:** the stored report for that version has been removed, usually because this site's data was cleared in your browser.
- **What to do:** analyse the site again. To avoid this, keep important versions in your account.

A run does not appear in Compare or the library

- **Why:** the library may be full (the free plan allows 20 saved projects), or your browser may be blocking storage, for example in some private windows.
- **What to do:** remove old projects from your library, or use a normal browser window.

The Keep in my account button is missing

- **Why:** you are signed out, or report storage is not available for your account.
- **What to do:** sign in. If it still does not appear, download the version with **Download it** instead.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Analyzer FAQ

Short answers to the questions people most often ask about the Analyzer.

Is the Analyzer free?

Yes. The tool, every section and most exports are free. Work that runs on Auric Artisan's servers is metered: auditing another website (rendered on a server), whole-site scans, and server screenshots. Signed-in accounts can claim 100 free tokens, and the free plan includes 20 server-side URL analyses a month. PDF export is a paid Pro export, and the VPAT-shaped conformance CSV is a Specialist-plan perk. See Choose what to scan.

Do I need an account?

Not to audit pages on auricartisan.com. To audit another website you need to be signed in, because your browser is not allowed to read another site's pages directly; the page is rendered on Auric Artisan's servers instead.

Can I audit a page on my computer, or on a company intranet?

Generally no. The server scanner refuses private and local addresses such as `localhost` and `192.168.x.x`, and your browser cannot read another origin's pages unless that server explicitly allows it. Put a copy of the page somewhere publicly reachable, such as a staging site.

What is the difference between Simple and Advanced?

Simple shows 8 sections and runs a fixed set of checks. Advanced shows all 22 sections, lets you choose which of 13 checks run, and adds site scanning, the Playground, custom rules, saved layouts and CSP Setup. The audit itself is the same quality in

both. See Getting started.

How many accessibility rules does it check?

126 rules, mapped to WCAG 2.0, 2.1 and 2.2 success criteria, Section 508 and EN 301 549, plus checks for eight ARIA widget patterns. See Accessibility rules.

Does a good score mean my site is accessible, or legally compliant?

No. Automated testing proves failures; it rarely proves success. Some WCAG criteria cannot be checked by software at all, and they are shown as untested rather than passed. A clean report is strong evidence that you have done the automatable work, not a certificate. For a conformance claim you need manual review as well; the Accessibility Checklist helps with that.

Why does another tool score my page 100 when the Analyzer does not?

The tools measure different things. The Analyzer checks more criteria than many automated tools, counts "needs review" results instead of treating them as passes, weights findings by severity and by how many elements they affect, and measures the page as rendered. Compare individual findings instead: when two tools disagree about one element and one rule, look at the element.

Why are there two different grade scales?

The overall score and each section use one set of bands (A+ from 90). The accessibility engine in A11y+ uses a stricter scale (A+ from 95, F below 55), because its score is meant to track WCAG conformance closely. See Read the Overview and scores.

What does "needs review" mean?

The rule ran and could not decide, usually because the answer depends on something only a person can judge, such as text over a photograph. It costs a little in the accessibility score, less than a failure, because "we could not tell" is not evidence that something is fine.

What is APCA Lc, and does it affect my score?

APCA is a newer model of perceived contrast being developed for a future version of WCAG. Lc is its lightness contrast value. The Analyzer shows it beside the WCAG ratio for reference only; it never affects a pass, a fail or a score. See Colour, contrast and vision.

Can I scan my whole site?

Yes, on the Artisan plan and above, up to 24 pages per run. See Scan a whole site.

Where are my results stored?

In your browser on this device. Every run is filed as a version of its site, with the full report, so you can reopen and compare it. Nothing is uploaded unless you choose **Keep in my account** in Compare. See Save, reopen and compare and Privacy.

Can I compare before and after a fix?

Yes. Run the page, fix something, run it again, then open **Compare**. It shows what was fixed (**Gone**), what appeared (**New**) and what stayed.

Can I share a report with someone who has no account?

A share link re-runs the audit for the recipient, so they need what the audit needs: an account if the page is on another website. To send the results themselves, export **Export HTML**, **Client Report** or **Export PDF** and send the file.

Why does `Ctrl` + `P` not print?

While the Analyzer is open, `Ctrl` + `P` opens its Sections picker. Use the exports in **Report** for a printable copy.

Can I automate audits?

Yes, through the REST API. The API hint on the page names the endpoint and its price in API credits, which are a separate balance from tool tokens. See the public API documentation at https://auricartisan.com/library/documentation/api/.

Why do some sections say the page could not be embedded?

The site forbids other sites from showing it in a frame. Use **Run in browser** to open a captured copy in a pop-up window, or, for your own site, follow **CSP Setup**. See Fix and preview.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Analyzer: supporting material

The Analyzer's glossary, shortcuts, limits, related tools, release notes and privacy notes.

This folder holds material that supports the Analyzer guides: the terms they use, the keyboard shortcuts, an honest account of what the Analyzer measures and where it stops, related tools and reading, the history of changes, and what happens to your data.

File	What it covers
Glossary	Every technical term used in the Analyzer docs, in one sentence each
Keyboard shortcuts	Keys for the URL field, the Sections picker, panels and the token meter
Limits and accuracy	What the Analyzer measures, how precisely, and what it is not
Related tools and reading	Other Auric Artisan tools, extensions and articles that pair with the Analyzer
Release notes	User-facing history of the Analyzer
Privacy	What stays in your browser, what goes to a server, and what you can delete

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Analyzer keyboard shortcuts

The keys the Analyzer responds to, in the URL field, the Sections picker, the workspace panels and the token meter.

The Analyzer can be used entirely from the keyboard. `Tab` and `Shift` + `Tab` move between controls as usual; the keys below are the Analyzer's own.

Anywhere on the Analyzer page

Keys	What they do
`Ctrl` + `P`	Open the Sections picker, in Simple or Advanced; replaces the print shortcut on this page

In the URL field

Keys	What they do
`Enter`	Start the audit, the same as **Analyze**

In the Sections picker

Keys	What they do
Type any text	Filter the sections by name
`Up` / `Down`	Move to the row above or below, as the rows are laid out
`Left` / `Right`	Move across columns, when the filter field is empty
`Enter`	Open the highlighted section (or scroll to it) and close the picker
`Space`	Open or close the highlighted section and keep the picker open

Keys	What they do
`Esc`	Close the picker

On a panel

Keys	What they do
`Left` / `Right` on the panel grip	Move the panel one place earlier or later in the workspace
`Enter` or `Space` on **Width**, **Focus**, **Collapse**, **Close**	Use that button

On the token meter

Keys	What they do
`Enter` or `Space`	Open the token sheet

Disclosures and cards

Folding cards and disclosures, such as **Why this score?**, **Cause chain** and the A11y+ cards, open and close with `Enter` or `Space`.

For shortcuts that work across the whole site, see Keyboard shortcuts.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Analyzer glossary

Plain-language definitions of the terms used in the Analyzer and its documentation.

Accessible name — The name assistive technology announces for an element, worked out from its visible text, its label, `alt`, `aria-label` or `aria-labelledby`.

Accessibility tree — The simplified version of a page that browsers give to screen readers and other assistive technology.

APCA — The Accessible Perceptual Contrast Algorithm, a newer model of perceived contrast being developed for a future version of WCAG. Advisory only in the Analyzer.

API credits — The balance that pays for calls to Auric Artisan's REST API. Separate from tool tokens; neither pays for the other.

ARIA — Accessible Rich Internet Applications: attributes such as `role`, `aria-label` and `aria-expanded` that add meaning for assistive technology.

ARIA widget pattern — A standard way to build a composite control, such as tabs or a menu, described in the W3C ARIA Authoring Practices.

Axe format — The JSON shape used by a widely used open-source accessibility testing library, which many continuous-integration tools can read. The A11y+ **Results JSON** export uses it.

Before / After — The section that shows the page as it is beside the page with the contrast fixes applied.

Bot challenge — A page a site shows automated visitors instead of its content ("Just a moment", "Verify you are human"). The Analyzer does not score it.

Canonical URL — The address a page tells search engines is its main version.

Check — One of the 13 switchable parts of a run, such as Colour contrast or Performance. Page structure always runs.

Conformance — Meeting every WCAG success criterion at a level. One failure at a level denies that level.

Core Web Vitals — Google's measures of loading (LCP), visual stability (CLS) and responsiveness (INP). The Analyzer also reports FCP, TTFB and TBT.

CLS (Cumulative Layout Shift) — How much the page's content jumps around while it loads. Good is 0.10 or less.

CORS (cross-origin resource sharing) — The browser rule that stops one site reading another site's pages unless that site allows it.

CSP (Content Security Policy) — A response header that tells browsers which sources a page may load and who may embed it.

CVD (colour-vision deficiency) — Reduced ability to tell some colours apart, often called colour blindness.

Deuteranopia, protanopia, tritanopia — Colour-vision deficiencies in which the green-, red- or blue-sensitive cones do not work.

Achromatopsia — Seeing no colour at all, only light and dark.

DOM, raw and rendered — The page's structure. Raw is the HTML the server sends; rendered is the structure after JavaScript has run.

EN 301 549 — The European accessibility standard for ICT, which references WCAG 2.1 Level A and AA.

FCP (First Contentful Paint) — When the first text or image appears. Good is 1,800 ms or less.

frame-ancestors — The CSP directive that says which sites may show a page in a frame.

Hydration — When JavaScript takes over a page the server already rendered and makes it interactive.

Impact — How badly a failure affects people: critical, serious, moderate or minor.

INP (Interaction to Next Paint) — How quickly the page responds to a click, tap or key press. Good is 200 ms or less.

JSON-LD — Structured data written as JSON inside a page, which search engines read.

Landmark — A region of a page, such as the header, navigation, main content or footer, that assistive technology can jump to.

Large text — Text of at least 24 CSS pixels, or at least 18.66 CSS pixels in bold. It needs less contrast than normal text.

LCP (Largest Contentful Paint) — When the largest image or text block appears. Good is 2,500 ms or less.

Lc — APCA's lightness contrast value, from 0 to a little over 100. Negative values mean light text on a dark background.

Level A, AA, AAA — WCAG's three conformance levels. A is the minimum, AA is what most laws require, AAA is the highest.

Needs review — A rule ran and could not decide; a person has to look.

Not tested — No rule can judge that WCAG criterion automatically. Never counted as a pass.

Panel — One section of the report shown in the workspace grid.

Relative luminance — How much light a colour gives off, from 0 for black to 1 for white, used by the WCAG contrast ratio.

Rule — One automated accessibility check, such as "Buttons must have discernible text".

Section 508 — The US federal accessibility standard, which references WCAG 2.0 Level A and AA.

Site scan — A run that finds and audits several pages of one site, up to 24.

Success criterion — One testable requirement in WCAG, such as 1.4.3 Contrast (Minimum).

TBT (Total Blocking Time) — How long the main thread was blocked during loading. Good is 200 ms or less.

Token — The unit that pays for work done on Auric Artisan's servers, such as rendering another site's page or crawling a site.

TTFB (Time to First Byte) — How long the server took to start answering. Good is 800 ms or less.

Version — One saved run of a site. Re-running the same address adds a version to the same project.

VPAT — Voluntary Product Accessibility Template, the report format most procurement processes ask for.

WCAG — The Web Content Accessibility Guidelines published by the W3C. The Analyzer maps its rules to versions 2.0, 2.1 and 2.2.

WCAG contrast ratio — The ratio between two colours' relative luminance, from 1:1 to 21:1.

Workspace — The grid of panels the Analyzer shows its report in. Also the name of the Advanced section that holds saved projects and history.

X-Frame-Options — An older response header that stops a page being shown in a frame on other sites.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Limits and accuracy

What the Analyzer measures, how precisely, which pages it can reach, and what its results are not.

The Analyzer gives you a short, specific, evidence-backed list of things that are wrong with a page. It does not certify a page, predict every visitor's experience, or replace a manual review. This page is honest about where the line is.

What it measures

- **One page per run**, or up to 24 pages in a whole-site scan.
- **The page as it was at one moment.** Content that loads later, changes with an A/B test, or appears only after a user action may not be seen. A page part-way through an animation measures differently from a page at rest.
- **Lab results, not visitor data.** Performance figures come from one load in one browser, not from real visitors on real devices and networks.

Which pages it can reach

Page	Can it be audited?
Any page on auricartisan.com	Yes, in your browser, with or without an account
A public page on another website	Yes, when you are signed in: it is rendered in a real browser on Auric Artisan's servers
Another website, signed out	Only if that site allows cross-origin requests, or with **Public proxy** turned on, and then only from its HTML
A page behind a login, a bot challenge or a strict firewall	No. The scanner is refused, or receives a challenge page and declines to score it.

Page	Can it be audited?
`localhost`, private network and link-local addresses	No. The scanner never visits private addresses.
Anything other than `http` and `https`	No

When a page is read from its HTML rather than rendered, its scripts are removed first (apart from JSON-LD). Anything built by JavaScript is then missing.

Accessibility

- **Automated coverage has a ceiling.** Of the 60 WCAG criteria the engine lists, 11 can never be checked automatically and always show as untested: Meaningful Sequence, Sensory Characteristics, Images of Text, No Keyboard Trap, Three Flashes, Pointer Cancellation, Motion Actuation, On Focus, Consistent Navigation, Consistent Identification and Redundant Entry.
- **A pass is not a good experience.** A link called "click here" has an accessible name; a heading order can be valid and meaningless.
- **Needs review is not a pass.** Text over images, gradients and video, and anything else the engine could not decide, is left for a person.
- **Level AAA and best-practice rules** run and are reported, but do not count in the accessibility score.
- **Hidden content is not judged**, apart from rules that exist to check hidden content. A closed dialog or collapsed menu is checked only when it is open.
- **Web components**: content inside open shadow roots is checked; content inside closed shadow roots cannot be seen.
- **Rules that need layout** (contrast, reflow, target size, focus order) report needs review when computed styles or geometry are not available, and a note under **Coverage limits for this run** says so.
- **The screen-reader transcript** is a model of what a screen reader announces, spoken with your browser's voice. It is not a real screen reader.
- **Suggested fixes** marked medium or review confidence need a person to check the wording.

Colour and contrast

- Contrast is measured from the colours the page actually paints, with semi-transparent colours blended over what is behind them.
- Text over gradients, photographs or video has no single background colour and is reported as needs review, not given a number.
- The **Vision** view samples up to 400 text elements.
- The **Simulation** filters and the colour-vision tables are approximations of how a typical person with each condition sees colour. Real vision varies from person to person.
- APCA Lc is shown for reference and never affects a result.
- Suggested colours keep the hue and change only lightness. They meet the target ratio but may not suit your brand; any colour that meets the ratio is acceptable.

Scores

- The overall score is a weighted summary of up to ten categories. Categories with nothing to measure, or whose check was switched off, are excluded rather than counted as zero or 100.
- Scores are designed to be hard to max out. A page with a couple of serious issues lands in the seventies or below.
- A score can fall after you improve a page, when a fix exposes content that was never checked before.

- The **Security** score checks whether protective headers are present, not how good they are. Scoring is softened for local development servers.
- **Reliability** reflects only the links that were checked. Read the count beside it.
- Benchmark, percentile and competitor-archetype figures in **Insights** and **SEO** are fixed reference values for orientation, not a live comparison with other sites.

Performance

- Core Web Vitals are measured when possible and estimated otherwise; the card says which. Estimates depend on the **Network Simulation** profile.
- The waterfall is an estimated load order, not a recorded network trace.
- Performance budgets are fixed general-purpose limits; your own budgets may differ.

Site scans

- Up to 24 pages per run, found from same-site links and the sitemap.
- In the current version, the **Robots**, **Subdomains** and **Workers** settings do not change a site scan on the live site.

What the Analyzer is not

- It is not a legal conformance certificate. Regulations reference WCAG conformance, which needs manual review too.
- It is not a replacement for testing with real assistive technology and real users.
- It is not a security scanner. It reads headers and links; it does not probe for vulnerabilities.
- It is not real-user monitoring.

Use the Analyzer to clear the automatable problems first, so that a manual review can spend its time on what only a person can judge. The Accessibility Checklist covers the manual side.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Analyzer privacy

What the Analyzer keeps in your browser, what it sends to Auric Artisan's servers or third parties, and how to delete it.

This page explains where your data goes when you use the Analyzer. For the site's full privacy policy, see the company and legal docs in Company and legal.

What happens in your browser

When you audit a page on auricartisan.com and you are not signed in, the whole audit runs in your browser. The page is loaded into a hidden frame on your device, measured there, and nothing about the audit is sent anywhere.

Contrast, palette, media and SEO analysis run on your own device, whichever way the page was reached. When a page is rendered on Auric Artisan's servers, the accessibility rules run there, in the server's browser, because they need the page as it was laid out.

What goes to Auric Artisan's servers

When	What is sent	Why
You are signed in and audit another website	The address you entered	The page is rendered on the server and the results come back to you
You run **Scan Site**	The address and the number of pages	The site is crawled and each page audited on the server
You use **Capture all devices** or the Device Lab captures screenshots	The address	Screenshots are rendered on the server
A run, scan or capture is metered	Your account and the token amount	Tokens and the monthly allowance are recorded against your account
You select **Keep in my account** in Compare	That version's full report	So you can open it on another device

The audited page is fetched by Auric Artisan's servers, so the site you audit sees a visit from them, not from you.

What goes to third parties

When	Who	What
You turn on **Public proxy** (off by default)	Third-party public CORS proxy services	The address being analysed, when your browser falls back to reading a page's HTML
You open a **How to meet this** link	The W3C website	A normal visit to that page
You select **Email** in Insights	Your own email app	A draft with the URL, score and headlines

What is kept in your browser

- Every run's full report, its versions, and the site's palette, in your library
- The history list, saved palettes and monitoring list in the **Workspace** section
- A copy of each audited page's HTML for up to 7 days, used only if a later run cannot reach the page
- Your preferences: mode, panel layouts, saved layouts, ticked checks, fix mode, **Public proxy**, network profile, issue statuses and custom rules
- Whether you dismissed the API hint

This data stays on this device and in this browser. It is not shared with other visitors.

What share links contain

- The strip's **Share** link carries the audited address and its score.
- The **Share Link** in Report carries the address, the score and the issue count.
- The **Copy share link** in Insights carries the address.

Anyone who has the link can read these values. None of the links carries the full report or the scanned pages.

Delete your data

To delete	Do this
The Workspace history, saved palettes, notes and pinned findings	**Workspace** › **Clear All**, then **Clear workspace**
One saved project or all projects	Remove them in your library
A report kept in your account	**Compare**, choose the version, select **In your account** ✓ to remove the account copy
Everything the Analyzer stored in this browser	Clear this site's data in your browser settings
Reports kept in your account, all at once	Deleting your account removes them

Clearing site data cannot be undone: versions that are not kept in your account are lost.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Analyzer release notes

User-facing history of the Analyzer, from the public changelog and the current page.

The Analyzer changes with the website. Dated entries below come from the public changelog at https://auricartisan.com/library/changelog/. The first section describes how the current page differs from the last changelog entry that mentions the Analyzer.

Since v0.9.11 (current page, September 2026)

These changes are visible on the page today:

- **126 accessibility rules**, up from the 112 announced in v0.9.10. The engine also checks eight ARIA widget patterns, and proposes concrete fixes with a confidence level in a new **Fixes** view.
- **A workspace instead of a tab row.** Sections open as panels you can move, resize, focus, collapse and close, in both modes. Advanced adds a **Sections** picker (`Ctrl` + `P`) and a **Layouts** menu with ready-made and saved layouts.
- **What to scan.** In Advanced mode you choose which of 13 checks run. The token meter prices the checks you chose, at four checks to a token.
- **Simple mode** now shows 8 sections, including **Project**, **Site Map** and **Compare**.
- **Runs are kept.** Every run is filed in your browser as a version of its site, with the full report, and can be reopened without running again.
- **Compare** sets two versions of a site side by side and lists what was fixed, what appeared and what stayed. A version can be kept in your account.
- **The progress card** lists the steps a run is really taking, based on the checks you chose.
- **This Page** has been removed from the strip. Type an auricartisan.com address to audit a page on this site.
- The page has a masthead, "One URL, audited seven ways.", and a one-line note on automating audits through the API.

v0.9.11 — 12 August 2026

- Tokens and API credits were separated and explained: tokens pay for server work started in a tool, and appear in the token bar on the Analyzer; API credits pay for REST API calls.

v0.9.10 — 27 July 2026

- The **A11y+** tab was rebuilt on a 112-rule accessibility engine covering ARIA 1.2, accessible-name computation and APCA contrast, with an accessibility tree view, a screen-reader transcript, colour-vision deficiency simulation and a VPAT export.
- The engine's rules load only when needed, so the Analyzer page stays fast.

v0.9.8 — 2 May 2026

- Improved the Analyzer's interface, workflow and feature behaviour for clearer workflows and more reliable interactions.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Related tools and reading

Other Auric Artisan tools, extensions and articles that work well with the Analyzer.

Tools in this group

Tool	Use it with the Analyzer to
Contrast Checker	Settle one colour pair by hand after the Analyzer finds it failing
Accessibility Checklist	Cover the WCAG criteria that automated testing cannot check, and read the law by country
Vision Lab	Simulate many more vision conditions, at chosen severities, on colours and images
Ishihara Plate Generator	Show colleagues how colour-vision deficiencies change what people see

Other parts of Auric Artisan

Where	Why
Basic colour tools	Build a palette or a family of tints after the Analyzer has told you a colour must move
Library Kit	Find, organise and reopen saved Analyzer projects and palettes
Shareable links	How the site-wide share dialog behind the **Share** button works
Account and billing	Tokens, plans, the monthly allowance and keeping reports in your account
Browser extension	Inspect colour and accessibility on any page from your own browser
Auric Artisan Studio: Accessibility	Catch accessibility problems in your code editor before they reach a page
Public API	Automate audits through the REST API

Articles on auricartisan.com

- **Understanding Your Accessibility Report** — how to read an accessibility score, severities, conformance and contrast results: https://auricartisan.com/library/documentation/articles/2026-08-20-understanding-your-accessibility-report
- **URL Analyzer UI/UX Guide** — an earlier guide to the Analyzer's interface. Some details, such as the **This Page** button and the Simple-mode section list, have changed since it was written: https://auricartisan.com/library/documentation/articles/2026-05-24-url-analyzer-ui-ux-guide
- The public API reference: https://auricartisan.com/library/documentation/api/
- The Learn library, with plain-language articles on contrast, colour vision and readable type: see Learn.

Standards

- WCAG 2.2 and the W3C Understanding documents, linked from every rule as **How to meet this**: https://www.w3.org/WAI/WCAG22/Understanding/
- W3C ARIA Authoring Practices, used by the **Widgets** view: https://www.w3.org/WAI/ARIA/apg/patterns/

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Contrast Checker

What the Contrast Checker does, who it is for, and how to grade a colour pair in under a minute.

The Contrast Checker answers one question quickly: does this text colour work on this background? You give it two hex colours. It gives back the exact WCAG 2.1 contrast ratio to two decimal places, a **Pass** or **Fail** badge against five thresholds, and a preview panel painted in the pair you are testing.

Two buttons do the rest. **Make it pass AA** appears when body text fails and moves the text colour to the nearest value that reaches 4.5:1. **Copy link** copies the page address, which always carries both colours, so a pair can be pasted into a ticket or a review and reopened exactly as you left it.

The tool is deliberately small. All the maths runs in your browser: nothing is uploaded, nothing is stored, and no account is needed. It tests one pair at a time. To find every failing pair on a whole page, use the Analyzer.

Who it is for

- Designers choosing text and background colours for a component.
- Developers and reviewers who need a number, not an opinion, in a code or design review.
- Accessibility specialists checking a single pair reported by another tool.
- Anyone learning what the WCAG contrast levels mean.

Where to find it

- Address: https://auricartisan.com/tool/contrast-checker/
- With a pair already loaded: `https://auricartisan.com/tool/contrast-checker/?fg=1A1400&bg=F7F3E6` (`fg` is the text colour, `bg` the background, as hex without the `#`).
- Site search opens it with a colour or a pair filled in when your search is about a colour.

Quick start

1. Open https://auricartisan.com/tool/contrast-checker/. It opens with a default pair already graded.
2. Type or paste your text colour into **Text colour**, for example `#6B7280`.
3. Type or paste your background colour into **Background colour**, for example `#FFFFFF`.
4. Read the large ratio under **Contrast ratio**.
5. Read the five badges. **Body text · AA** at 4.5:1 is the one most laws and contracts point at.
6. If body text fails, select **Make it pass AA** to get a passing text colour.
7. Look at the preview panel to judge the pair at real reading sizes.
8. Select **Copy link** to share the result.

What you can do

- Grade any pair of hex colours against WCAG 2.1 AA and AAA for body text, large text, and interface components and graphics.
- See the pair as large text, body text and an interface element.
- Swap the two colours with one button.
- Nudge the text colour to the nearest value that passes AA.

- Share or bookmark a pair as a link.
- Use it entirely from the keyboard and with a screen reader.

In this folder

- docs/ — how to use the Contrast Checker
  - Getting started — the screen, part by part, and your first check
  - Check a colour pair — entering colours and reading the ratio and grades
  - Make a pair pass — using Make it pass AA well
  - Share a pair — links, bookmarks and ways in
  - Reference — every control, threshold and output
  - Troubleshooting
  - FAQ
- others/ — supporting material
  - Glossary
  - Limits and accuracy
  - Related tools and reading
  - Privacy

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Contrast Checker documentation

The list of Contrast Checker guides with one line on each.

These guides explain how to use the Contrast Checker at https://auricartisan.com/tool/contrast-checker/. Start with Getting started; the reference lists every control and threshold.

Guide	What it covers
Getting started	The screen, part by part, and your first check
Check a colour pair	What the colour fields accept, how the ratio is worked out, and which grade applies to you
Make a pair pass	What Make it pass AA does, what it does not do, and how to keep your brand colour
Share a pair	How the address carries the pair, Copy link, and the ways into the tool
Reference	Every control, threshold, output and keyboard behaviour
Troubleshooting	Odd results and what causes them
FAQ	Short answers to common questions

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Getting started with the Contrast Checker

Open the Contrast Checker, learn each part of the screen, and grade your first colour pair.

Open it

Go to https://auricartisan.com/tool/contrast-checker/. The page loads with a default pair, text `#1A1400` on background `#F7F3E6`, and grades it straight away, so there is no empty state to get past.

The screen, part by part

On a wide screen the tool is two columns over a full-width strip: controls on the left, the result on the right, and the preview underneath. Below 720 pixels wide it becomes one column in the order controls, result, preview, and the swap button turns so its arrows point up and down. Nothing is hidden on a narrow screen.

Area	What it holds
Breadcrumb	**Home**, **Tools** (links to the tools collection) and **Contrast Checker**
Heading	The kicker "WCAG 2.1 · Contrast", the title **Color Contrast Checker**, and a line saying nothing is uploaded
API hint	"Automate this — 1 API credit per call to `POST /v1/accessibility/check`", with **API docs** and a dismiss button
Controls	**Text colour** and **Background colour**, each a colour swatch beside a hex field, with the swap button between them
Result	The ratio (**Contrast ratio**), five grade rows, **Make it pass AA**, **Copy link** and a status line
Preview	A panel painted in your pair, with a line of large text, a paragraph of body text and an "Interface element" chip
Below	**What the numbers mean**, four questions and answers, and three cards linking to other tools

The API hint is for developers who want to check pairs from code; API credits are a separate balance from tool tokens. Dismissing it hides it on every tool page in this browser.

Check your first pair

1. Select the **Text colour** hex field and replace the value with `#6B7280`.
2. Select the **Background colour** hex field and replace the value with `#FFFFFF`.
3. Read the ratio. It updates as you type, and reads **4.83:1**.
4. Read the five rows. **Body text · AA** reads **Pass**; **Body text · AAA** reads **Fail**.
5. Look at the preview. Judge whether the paragraph is comfortable to read at length, not just whether it passes.

The result: you know the exact ratio and which WCAG levels the pair meets.

Try the other controls

- Select the swap button (between the two colours) to exchange them. The ratio stays the same; the preview turns inside out.
- Type a failing pair, such as `#D3AF37` on `#F7F3E6`. A **Make it pass AA** button appears. Select it, and the text colour moves to the nearest value that reaches 4.5:1. The status line says what it chose.
- Select **Copy link**. The status line says "Share link copied." Paste the link anywhere to reopen this exact pair.

Next steps

- Check a colour pair explains the ratio, the five grades and what counts as large text.
- Make a pair pass explains how the fix chooses its colour.
- To check every pair on a live page at once, use the Analyzer.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Check a colour pair

Enter a text and background colour, read the WCAG 2.1 ratio, and work out which of the five grades applies to what you are colouring.

Enter the colours

Each colour has two controls that stay in step: a colour swatch and a hex field. Use the swatch to explore and the field to paste a value from a stylesheet or a design file.

Control	What it does	Starts at
Text colour swatch	Opens your system's colour picker. Moving it writes the hex into the field and re-grades as you go.	`#1a1400`
Text colour hex field	Accepts a typed or pasted hex value and re-grades on every keystroke	`#1A1400`
Swap button	Exchanges the two colours and re-grades	—
Background colour swatch	The same picker for the background	`#f7f3e6`
Background colour hex field	The same typed entry for the background	`#F7F3E6`

What the hex fields accept

Hex only, in three or six digits, with or without a leading `#`. These are all the same colour to the checker: `#1A1400`, `1A1400`. Three digits are expanded, so `#f00` and `f00` both mean `#FF0000`.

When you leave a field, a value it understood is rewritten in full: uppercase, six digits, with a `#`. Anything else is not accepted: `rgb()`, `hsl()`, colour names such as `rebeccapurple`, and eight-digit hex with an alpha channel.

> **Important:** While a field holds something it cannot read, such as a half-typed `#1A`, the checker grades pure black in its place (for the text) or pure white (for the background). The ratio you see then is not yours. Finish typing, or press `Tab` to leave the field, before you trust the number.

Read the ratio

The large number under **Contrast ratio** is the WCAG 2.1 contrast ratio, to two decimal places, for example **4.53:1** or **21.00:1**.

- **Range:** from 1:1 (two identical colours) to 21:1 (black on white).
- **Updates:** on every keystroke and every movement of a swatch. There is no Calculate button.
- **Direction:** none. Dark on light and light on dark give the same number, so swapping never changes it.

How the ratio is worked out

WCAG measures each colour's **relative luminance**, which is how much light it gives off on a scale from 0 (black) to 1 (white), with green counting most and blue least, as in human vision. The ratio is the lighter luminance plus 0.05, divided by the darker luminance plus 0.05. This is the sRGB formula in WCAG 2.1, and it is the number an auditor will quote.

Rounding at the threshold

The badges compare the exact, unrounded ratio with each threshold, but the number on screen is rounded to two decimals. A pair measuring 4.4996 prints **4.50:1** beside a **Fail** badge for body text AA. The badge is right. If a pair lands that close, treat it as failing and give yourself some margin.

A ratio exactly equal to a threshold passes: a true 4.5 reads **Pass**.

Read the five grades

Every pair is graded against five thresholds at once, so you never have to choose a mode first.

Row	Threshold	Covers
Body text · AA	4.5:1	Ordinary running text at Level AA, the level most laws and contracts point at
Body text · AAA	7:1	Body text at Level AAA, worth aiming for on long-form reading
Large text · AA	3:1	Text at or above the large-text size, at Level AA
Large text · AAA	4.5:1	Large text at Level AAA
UI & graphics	3:1	Interface components, focus indicators and meaningful graphics, against what sits next to them

Which row applies to you

Choose the row by what you are colouring, not by which badge is greenest.

- **Paragraphs, labels, table cells, help text:** body text. Use the 4.5:1 row, with 7:1 as a stretch goal.
- **Headings and display type:** large text, but only if the size qualifies (below).
- **Input borders, toggle states, icon buttons, focus rings, the meaningful parts of a chart:** the **UI & graphics** row at 3:1. Measure against the colour actually next to the component, which may not be the page background.

What counts as large text

At least 18.66 pixels in bold, or at least 24 pixels otherwise. It is a size threshold, not a heading rule: a 20-pixel heading in a light weight is body text as far as WCAG is concerned, and must reach 4.5:1.

What is out of scope

WCAG 2.1 exempts inactive (disabled) controls, pure decoration, text that is part of a logo, and text inside a picture of something else. The checker grades whatever pair you give it, so a **Fail** on a disabled button's label is not automatically a defect. Everything a person has to read or operate is in scope.

Use the preview

The panel under the controls is painted in your pair and holds:

- a large line, "Large text looks like this", at the large-text size;
- a body-text paragraph at a realistic line length, so you can judge the pair over several lines;
- an "Interface element" chip with a border in the text colour, so you can see the pair as a control outline.

Look at it. The ratio is a floor, not a verdict. The WCAG 2.1 formula ignores font weight, ignores size beyond the large-text threshold, and does not model the way light text on a dark background appears to spread. A thin weight at 4.6:1 can be tiring to read; a heavy weight at 4.3:1 can look crisp and still fail an audit.

The preview always shows your two colours, in both light and dark site themes.

A check in five steps

1. Enter the text colour.
2. Enter the background colour.
3. Read the ratio.
4. Find the row that matches what you are colouring, and read its badge.
5. If it fails, see Make a pair pass. Either way, share the pair if someone else needs to see it.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Make a pair pass

Use Make it pass AA to find a passing text colour, understand how it chooses, and keep your brand hue.

Make it pass AA is the main button in the result area. It only appears while the pair fails body text AA (under 4.5:1), so seeing it is itself a signal that body text is failing.

Use it

1. Enter a failing pair, for example text `#D3AF37` on background `#F7F3E6` (1.90:1).
2. Select **Make it pass AA**.
3. Read the status line. For this pair it says "Adjusted text to #836C22 (4.58:1)."

The result: the new colour is in the **Text colour** field and swatch, the pair is re-graded, the button disappears, and the address updates a moment later with the new pair.

How it chooses

1. It looks at the background. If the background's relative luminance is above 0.5 it moves the text towards black; otherwise towards white.
2. It walks the text colour along a straight line towards that end, in about fifty small steps, checking the ratio at each step.
3. It stops at the first step that reaches 4.5:1. If none does, it uses pure black or pure white.

What it will not do

- **It only aims at 4.5:1.** There is no option for 3:1 or 7:1. For AAA, use the button first, then keep darkening (or lightening) by hand while you watch the ratio.
- **It never changes the background.** Only the text colour moves. If the background is the colour you are allowed to change, select the swap button, use the fix, then swap back.
- **It does not keep your hue or saturation.** The path is a straight line towards black or white, so the result is usually a greyer relative of your colour. Treat it as a target, not a colour to ship.
- **It is close to minimal, not exactly minimal.** It takes the first of about fifty samples that passes.

The mid-grey surprise

The direction is chosen by relative luminance, not by how light a colour looks. Mid grey `#808080` has a relative luminance of about 0.22, which counts as dark, so text on it is pushed towards white even though the grey looks like the middle of the range. That is correct WCAG behaviour. If you wanted the other direction, move the text colour towards black yourself and watch the ratio.

Keep your brand colour

1. Select **Make it pass AA** to see how far the pair has to move, and note the colour and ratio it reports.
2. Go back to your original colour.
3. Change it a few digits at a time towards the suggested colour, keeping the hue, and watch the ratio.
4. Stop as soon as **Body text · AA** reads **Pass**, and leave some margin.

You end up with a colour that is still yours and that passes.

For suggestions that keep the hue automatically across a whole page, the Analyzer offers a replacement for every failing pair that adjusts only lightness.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Share a pair

Share or bookmark a colour pair as a link, and the ways other parts of the site open the checker.

The Contrast Checker keeps no history. The page address is the only place a pair is kept, and it is always up to date.

The address is the state

Each time the checker re-grades, it rewrites the address with the current pair, without the `#`:

```
https://auricartisan.com/tool/contrast-checker/?fg=1A1400&bg=F7F3E6
```

`fg` is the text colour and `bg` the background. Open that address and the checker restores both colours and grades them at once. Bookmark it, or paste it into a pull request, a bug report or a design review.

Copy link

1. Set up the pair you want to share.
2. Select **Copy link**.
3. The status line says "Share link copied." Paste the link where you need it.

If your browser does not allow clipboard access (for example on an insecure connection, or after you declined the permission), the full address is printed in the status line instead, so you can select and copy it by hand.

Partial links

Only values that are valid hex are used. A link with `?fg=D3AF37` and nothing else sets the text colour and leaves the background at the default. Site search uses exactly this when its answer is about a single colour, and sends both values when its answer is about a pair.

Two things the address does not do

- **It does not add history entries.** The address is replaced, not added to, so your browser's Back button will not step through the colours you tried. Copy a link at the time if you want a pair back later.
- **It does not reload the page.** The update is silent and happens a moment after you stop typing.

Ways into the checker

From	What it opens
Site search colour answers	The checker with the colour, or the pair, already filled in
The homepage colour demo	The checker on its default pair, from **Open the full checker**
Your dashboard	The **Contrast** entry in the launcher row
Learn articles about contrast	The checker from the article's call to action

What is kept

Nothing. There is no saved history, no library of pairs and no account state. If a result matters, copy its link before you move on. There is nothing to clear, and nothing leaves your device.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Contrast Checker reference

Every control, input, threshold, output, message and keyboard behaviour of the Contrast Checker.

Controls

Control	What it does	Values	Default
Text colour swatch	Opens the system colour picker; writes uppercase hex into the field	Any colour	`#1a1400`
Text colour hex field	Typed or pasted text colour	3- or 6-digit hex, with or without `#`	`#1A1400`
Swap button	Exchanges text and background	—	—
Background colour swatch	Opens the system colour picker	Any colour	`#f7f3e6`
Background colour hex field	Typed or pasted background colour	3- or 6-digit hex, with or without `#`	`#F7F3E6`
Make it pass AA	Moves the text colour to the nearest value that reaches 4.5:1	Shown only when body text fails AA	—
Copy link	Copies the address, which carries both colours	—	—

The swatches are your browser's own colour input, not the site's custom colour picker, so the swatch always shows the value that is being graded.

Outputs

Output	What it shows
Contrast ratio	The WCAG 2.1 ratio to two decimals, from 1.00:1 to 21.00:1. Shows a dash until the first grading.
Grade badges	**Pass** or **Fail** for each of the five thresholds below
Status line	The result of **Make it pass AA** or **Copy link**
Preview	Your pair as large text, a body-text paragraph and an "Interface element" chip

Thresholds

Row	Threshold
Body text · AA	4.5:1
Body text · AAA	7:1
Large text · AA	3:1
Large text · AAA	4.5:1
UI & graphics	3:1

A ratio equal to a threshold passes. Badges use the exact ratio; the display is rounded to two decimals. Large text means at least 18.66px bold or 24px regular.

Status messages

Message	When
"Adjusted text to #RRGGBB (N.NN:1)."	After **Make it pass AA**

Message	When
"Share link copied."	After **Copy link**, when the clipboard accepted it
The full address	After **Copy link**, when the clipboard is unavailable

Address parameters

Parameter	Meaning	Example
`fg`	Text colour, hex without `#`	`fg=1A1400`
`bg`	Background colour, hex without `#`	`bg=F7F3E6`

Invalid values are ignored and the default is used for that colour. The address updates shortly after each change, replacing the current history entry.

Keyboard and screen readers

- `Tab` order: text swatch, text hex field, swap button, background swatch, background hex field, then the action buttons.
- The focus ring is drawn around the whole hex row, so you can always see which field you are in.
- The ratio is a polite live region: a screen reader reads the new ratio after a change without interrupting.
- The status line is a status region, so the outcome of both buttons is announced.
- The swap button is labelled "Swap text and background colours".
- The questions under **Questions people actually ask** are native disclosures: press `Enter` or `Space` to open one.

There are no custom keyboard shortcuts on this page.

Layout

- Wide screens: controls and result side by side, preview across the full width below.
- Under 720 pixels: one column (controls, result, preview), with the swap button turned a quarter turn.
- The page follows the site's light or dark theme. The preview always shows your own two colours.

Content below the tool

Section	Contents
What the numbers mean	4.5:1, 3:1, 7:1 and the definition of large text
Questions people actually ask	Four questions: the ratio you need, large text, readability, and disabled controls and logos
Next	**Check a whole page** (the Analyzer), **Build a palette** (Basic tools) and **Learn the rules** (the Learn library)

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Troubleshooting the Contrast Checker

Odd results in the Contrast Checker, why they happen, and what to do.

The ratio looks wrong while I am typing

- **Why:** a half-typed value such as `#1A` cannot be read, so the checker grades pure black (for the text) or pure white (for the background) in its place.
- **What to do:** finish typing all three or six digits, or press `Tab` to leave the field. The field rewrites a valid value in full, and the ratio is then yours.

My colour is not accepted

- **Why:** the fields take hex only. `rgb()`, `hsl()`, colour names and eight-digit hex with transparency are not accepted.
- **What to do:** convert the colour to six-digit hex first. For a semi-transparent colour, work out the colour it actually shows over its background, and test that.

The ratio says 4.50:1 but body text AA says Fail

- **Why:** the display is rounded to two decimals; the badge uses the exact value, which is just under 4.5.
- **What to do:** treat it as failing and move the colour a little further.

The ratio did not change when I swapped the colours

- **Why:** the ratio does not depend on which colour is the text. Only the preview, and which colour **Make it pass AA** moves, change.
- **What to do:** nothing; this is correct.

Make it pass AA made my colour grey

- **Why:** it moves the text colour in a straight line towards black or white and stops at the first passing step, which removes saturation.
- **What to do:** use its result as a target and adjust your original colour towards it by hand. See Make a pair pass.

Make it pass AA went towards white on a mid-grey background

- **Why:** the direction follows relative luminance. Mid greys count as dark.
- **What to do:** darken the text yourself if you want the other direction.

The Make it pass AA button is missing

- **Why:** it only appears when body text fails AA. If the pair already reaches 4.5:1 there is nothing to fix.
- **What to do:** for AAA (7:1), darken or lighten the text by hand.

Copy link printed the address instead of copying it

- **Why:** your browser did not allow clipboard access, for example on an insecure connection or because you declined the permission.
- **What to do:** select the address in the status line and copy it by hand.

Back does not return to my previous colours

- **Why:** the address is replaced, not added to history.
- **What to do:** copy a link whenever you want to keep a pair.

The UI & graphics row passes but my control is still hard to see

- **Why:** the row compares the two colours you entered. A control's edge is judged against the colour actually next to it, which may not be the background you typed.
- **What to do:** enter the control's border colour and the colour directly behind the control.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Contrast Checker FAQ

Short answers to common questions about the Contrast Checker and colour contrast.

What contrast ratio do I actually need?

4.5:1 for body text and 3:1 for large text is WCAG 2.1 Level AA, the level most laws and contracts reference. AAA asks for 7:1 and 4.5:1. Interface components and meaningful graphics need 3:1 against what sits next to them.

What counts as large text?

At least 18.66 pixels in bold, or 24 pixels otherwise. A large heading in a light weight can still fall under the body-text requirement.

Does passing the ratio mean the text is readable?

No. The WCAG 2.1 formula ignores font weight, size beyond the large-text threshold, and the way light text on dark backgrounds appears to spread. A pass is a floor, not a verdict. Check the preview as well as the number.

Do disabled controls and logos have to pass?

No. WCAG 2.1 exempts inactive controls, pure decoration, text that is part of a logo, and text in a picture of something else. Everything a person has to read or operate is in scope.

Does the checker show APCA?

No. The page's title mentions APCA, but the tool itself calculates only the WCAG 2.1 ratio. For APCA Lc values beside the ratio, audit a page with the Analyzer; its A11y+ panel shows them. The REST API endpoint named in the page's API hint also covers APCA.

Can I check a colour with transparency?

Not directly. Work out the solid colour it shows over its background and test that.

Can I check many pairs at once?

Not here. It checks one pair at a time. The Analyzer checks every pair on a live page.

Is anything uploaded or saved?

No. The maths runs in your browser and nothing is stored. The pair lives only in the page address.

Do I need an account?

No. The tool is free and needs no sign-in.

Can I use it with a keyboard and a screen reader?

Yes. Every control is reachable with `Tab`, the ratio is announced when it changes, and the result of each button is announced too. See the Reference.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Contrast Checker: supporting material

The Contrast Checker's glossary, limits, related tools and privacy notes.

This folder holds the material that supports the Contrast Checker guides. The tool has no keyboard shortcuts of its own (keyboard use is covered in the Reference), and the public changelog has no entries for it, so there are no shortcut or release-note pages.

File	What it covers
Glossary	The terms used in the Contrast Checker docs
Limits and accuracy	What the ratio measures, how precisely, and what it is not
Related tools and reading	Tools and articles that pair with the checker
Privacy	What stays in your browser

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Contrast Checker glossary

Plain-language definitions of the terms used by the Contrast Checker.

AA and AAA — WCAG conformance levels. AA is what most laws and contracts require; AAA is stricter and not required everywhere.

APCA — The Accessible Perceptual Contrast Algorithm, a newer contrast model being developed for a future version of WCAG. The Contrast Checker does not calculate it.

Contrast ratio — The WCAG measure of how different two colours are in lightness, from 1:1 (identical) to 21:1 (black on white).

Hex colour — A colour written as a `#` and six hexadecimal digits for red, green and blue, such as `#1A1400`, or three digits as shorthand, such as `#F00`.

Large text — Text of at least 18.66 pixels in bold or 24 pixels otherwise. It needs less contrast than body text.

Relative luminance — How much light a colour gives off, from 0 for black to 1 for white, weighted towards green as in human vision.

sRGB — The standard colour space of the web, which the ratio formula assumes.

UI & graphics — Interface components, focus indicators and meaningful graphics, which need 3:1 against the colour next to them.

WCAG — The Web Content Accessibility Guidelines published by the W3C. The checker uses the WCAG 2.1 contrast formula and thresholds.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Limits and accuracy

What the Contrast Checker measures, how precisely, and where it stops.

What it measures

The WCAG 2.1 contrast ratio between two solid sRGB colours, using the relative-luminance formula from the WCAG specification. The number is exact for the two colours you enter; it is displayed to two decimal places, and the pass or fail decisions use the unrounded value.

What it does not measure

- **APCA, WCAG 3 drafts or colour difference.** Despite the page title, there is no APCA or Lc score, no colour-difference metric and no colour-vision simulation in this tool.
- **Transparency or layering.** Hex values only, with no alpha. Text at partial opacity, or over an image or a gradient, has to be reduced to its effective solid colour first.
- **Font weight and spread.** The formula ignores weight, ignores size beyond the large-text threshold, and does not model how light text on a dark background appears to spread. A pass is a floor, not a verdict; use the preview.
- **Context.** The **UI & graphics** row compares the two colours you typed. In a real design, a component's edge is judged against whatever is next to it.
- **Scope decisions.** Disabled controls, decoration, logos and text in pictures are exempt from WCAG, but the checker grades any pair you give it. Deciding what is in scope is up to you.

Make it pass AA

- Aims at 4.5:1 only.
- Changes only the text colour.
- Moves in a straight line towards black or white, so it does not preserve hue or saturation.
- Takes the first of about fifty steps that passes, so the result is close to minimal, not exactly minimal.
- Chooses its direction by relative luminance, so mid greys push text towards white.

One pair at a time

There is no batch mode, no palette import and no matrix of every colour against every other colour. For every pair on a live page, use the Analyzer.

Not a certification

A passing ratio shows that one pair meets one WCAG threshold. It does not make a page or a product accessible or compliant.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Contrast Checker privacy

What the Contrast Checker does with the colours you enter.

- **Everything runs in your browser.** The ratio, the grades and the fix are calculated on your device. The colours you enter are not sent anywhere.
- **Nothing is stored.** There is no history, no saved pairs and no account state. The only record of a pair is the page address, which your browser keeps in its own history like any other address.
- **Shared links contain the colours.** A copied link carries both colours in plain text in the address. Anyone with the link can see them.

- **The API hint.** If you dismiss the "Automate this" note, that choice is remembered in this browser so the note stays hidden on every tool page.

The site-wide privacy policy is covered in Company and legal.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Related tools and reading

Tools and articles that pair well with the Contrast Checker.

Where	Why
Analyzer	Find every failing pair on a live page, with the elements named and a replacement that keeps your hue
Vision Lab	See how your colours look with colour-vision deficiencies
Accessibility Checklist	Work through the rest of WCAG, including the contrast checks, by hand
Basic colour tools	Build ramps, tints and shades, or pull colours from an image, once you know a colour must move
Learn	Plain-language articles on contrast, colour vision and readable type
Public API	Check pairs from code with `POST /v1/accessibility/check` (1 API credit per call)

Articles on auricartisan.com

- **Color Contrast Checker User Guide** — the site's own long-form guide to this tool: https://auricartisan.com/library/documentation/articles/2026-09-03-contrast-checker-user-guide
- **Understanding Your Accessibility Report** — how contrast results fit into a full accessibility audit: https://auricartisan.com/library/documentation/articles/2026-08-20-understanding-your-accessibility-report

Standard

- WCAG 2.1 Understanding document for success criterion 1.4.3, Contrast (Minimum): https://www.w3.org/WAI/WCAG21/Understanding/contrast-minimum.html

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Ishihara Plate Generator

What the Ishihara Plate Generator makes, who it is for, and how to generate, check and export your first plates.

The Ishihara Plate Generator makes dot plates in the style of the Ishihara colour-vision test: a circle of coloured dots in which a figure (a number, a letter, a symbol, a shape or a path to trace) stands out to typical colour vision and blends into the background for a particular colour-vision deficiency. The page describes them as "Dot plates that carry a figure to normal colour vision and hide it from a specific deficiency — generated, measured and scored on this device."

Every plate is scored as it is drawn. The generator measures how far apart the figure and background colours are for typical vision, then recomputes that distance through simulated protanopia, deuteranopia and tritanopia, and combines four measurements into a quality score. You can preview plates under each deficiency, build a standard set of 38 plates, and export plates as PNG or SVG with a seed that rebuilds any plate exactly.

Everything runs in your browser. There is no account, and nothing is uploaded.

> **Important:** The page states: "Not a colour-vision test. These plates are illustrative. A screen is not colour-calibrated and a real Ishihara examination is printed, lit and administered under controlled conditions. For diagnosis, see an eye-care professional."

Who it is for

- Educators and students learning how colour-vision tests work.
- Designers and accessibility specialists demonstrating colour-vision deficiency to colleagues.
- Researchers and developers who need reproducible, seeded plates as figures or test material.

Where to find it

- Address: https://auricartisan.com/tool/ishihara-plate-generation/

Quick start

1. Open https://auricartisan.com/tool/ishihara-plate-generation/. Six starter plates are generated for you.
2. Choose a **Plate Type**, a **Color Palette** and a **Difficulty**.
3. Select **+ Generate**, or **×3** or **×6** for several plates at once.
4. On a plate card, switch between **Normal**, **Protan**, **Deutan** and **Tritan** to see how it looks under each deficiency.
5. Double-click a plate to open it full size.
6. Open the **Diagnostics** tab to see each plate's score taken apart.
7. Download a plate as PNG or SVG from its card, or open the **Export** tab to export every plate.

What you can do

- Generate 7 plate types in 9 colour palettes (or at random), at 4 difficulty tiers.
- Make a custom plate with your own number, letter, symbol or shape, and an optional seed.
- Fine-tune dot count, plate shape, placement, dot style, card material, colour separation, edge blend, dot scale and texture grain.

- Preview plates under protanopia, deuteranopia and tritanopia.
- Read a quality score for each plate and its four components.
- Build the standard 38-plate set from a fixed seed.
- Export PNG (up to 4096 pixels), SVG (one circle per dot), a ZIP of every plate, and a seeds file to rebuild the batch.

In this folder

- docs/ — how to use the generator
  - Getting started — the screen and your first plates
  - Generate plates — the Lab: types, palettes, difficulty, custom plates, fine tuning and plate cards
  - Read the diagnostics — the score and its four components
  - Build the standard 38 — the fixed set
  - Export plates — PNG, SVG, ZIP and seeds
  - Reference — every control and output
  - Troubleshooting
  - FAQ
- others/ — supporting material
  - Glossary
  - Keyboard shortcuts
  - Limits and accuracy
  - Related tools and reading
  - Release notes
  - Privacy

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Ishihara Plate Generator documentation

The list of Ishihara Plate Generator guides with one line on each.

These guides explain how to use the generator at https://auricartisan.com/tool/ishihara-plate-generation/. Start with Getting started; the reference lists every control.

Guide	What it covers
Getting started	The masthead, the four tabs, and your first plates
Generate plates	Plate types, palettes, difficulty, batches, custom plates, fine tuning, cards and the full-size view
Read the diagnostics	The quality score, its four components, and the ΔE through each deficiency
Build the standard 38	The fixed 38-plate set and what it contains
Export plates	PNG, SVG, ZIP and seed files, and reproducing a plate
Reference	Every control, option, default and output
Troubleshooting	Common problems and what to do
FAQ	Short answers to common questions

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Getting started with the Ishihara Plate Generator

Open the Ishihara Plate Generator, learn the screen, and generate and inspect your first plates.

Open it

Go to https://auricartisan.com/tool/ishihara-plate-generation/. No account is needed. When the page loads, it generates six starter plates and says "Welcome! 6 starter plates generated."

The screen, top to bottom

Masthead

The kicker "Auric Artisan · Vision", the title **Ishihara Plate Generation**, and a one-line description. The description depends on the site-wide Simple or Advanced preference; in Simple mode it reads "Dot plates that carry a figure to normal colour vision and hide it from a specific deficiency — generated, measured and scored on this device."

A row of facts follows: 7 plate types, 9 colour palettes, 4 difficulty tiers, 3 CVD simulations, OKLab colour science, ΔE validation, PNG / SVG export, On-device, and a **Developer reference** → link.

API hint

"Automate this — 3 API credits per call to `POST /v1/ishihara/plate`." It tells developers that plates can also be generated from the REST API, and that API credits are a separate balance from tool tokens. **API docs** opens the endpoint's documentation; **×** hides the hint on every tool page.

The notice

"Not a colour-vision test. These plates are illustrative. A screen is not colour-calibrated and a real Ishihara examination is printed, lit and administered under controlled conditions. For diagnosis, see an eye-care professional."

The four tabs

Tab	What it is for
Lab	Generate plates and see them on the bench
Diagnostics	Each plate's score, taken apart
Set 38	Build the standard 38-plate set
Export	Export every plate as PNG, SVG, a ZIP or a seeds file

Your first plates

1. On the **Lab** tab, open **Plate Type** and choose **Number**.
2. Open **Color Palette** and choose **Red vs Green**.
3. Open **Difficulty** and choose **Easy**.
4. Select **+ Generate**. A new card appears on the bench, and the page says "Plate #7 generated —" followed by the number drawn.
5. On the new card, select **Protan** and then **Deutan**. The plate is redrawn as it might look to someone with protanopia or deuteranopia; the number should become harder to see.
6. Select **Normal** to return to the original.
7. Double-click the plate to open it full size. Press `Right` and `Left` to move between plates, and `Esc` to close.
8. Open the **Diagnostics** tab and find your plate: its score, whether it passed, and its four components.

The result: a scored plate you can download from its card as PNG or SVG.

> **Note:** Plates are not saved between visits. Export anything you want to keep, or keep its seed so you can rebuild it.

Next steps

- Generate plates for every Lab control.
- Read the diagnostics to understand the score.
- Export plates to keep your plates.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Build the standard 38

Generate the fixed 38-plate set, the composition a printed plate book uses, from a single seed.

The **Set 38** tab builds "The standard thirty-eight": "Twenty-four numbers, six letters, four symbols, two shapes and two traceable paths — the composition a printed book uses, built from a single seed so it is the same every time."

Build it

1. Open the **Set 38** tab.
2. Select **Generate the set**. The button reads "Building…" while it works.
3. When it finishes, the tab shows every plate in a grid, and a summary with **Plates** (38), **Seed** (42) and **Built** (how long it took, a few seconds).

The Lab is left alone: "The set is built and shown here; nothing on the bench is cleared or replaced. Thirty-eight plates take a moment and a quarter of a million dots."

You can also start the set from the empty Lab with **Or build the standard 38**.

What the set contains

Kind	Count	Plates (content and difficulty)
Number	24	0 to 23; each block of eight runs easy, easy, medium, medium, medium, hard, hard, extreme
Letter	6	A and B (easy), C, D and E (medium), F (hard)
Symbol	4	star (hard), arrow (extreme), triangle (easy), heart (easy)
Shape	2	circle and square (medium)
Path	2	two paths, medium and hard

Because the set is built from one seed (42) in a fixed composition, "the same call gives the same book every time".

Use the set

- Show it to a class to explain how plates grade from easy to extreme.
- Compare how each plate behaves under the three deficiencies in the Lab, using **CVD Preview**.
- Keep the seed: 42 always rebuilds the same set.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Export plates

Download plates as PNG or SVG, export every plate as a ZIP, save a seeds file, and rebuild a plate from its seed.

Every file is built in your browser and handed straight to the download. Nothing leaves your device.

One plate, from its card

Button	What you get	File name
Download PNG	A raster image, 4,096 pixels square, on the chosen card material	`ishihara-plate-<number>-<content>.png`
Download SVG	A vector image with one circle per dot, which scales to any print size	`ishihara-plate-<number>-<content>.svg`
Copy seed	The plate's seed, on your clipboard	—

The page says "Rendering high-res PNG…" while it builds the PNG. For example, plate 1 showing 93 downloads as `ishihara-plate-1-93.png`.

Every plate, from the Export tab

The **Export** tab has four actions. They cover every plate currently in the Lab.

Action	What you get
Everything at once › **Export every plate**	One ZIP, `ishihara-plates.zip`, with a PNG (2,048 pixels), an SVG and a JSON description for each plate
PNG (Raster) › **Download every .png**	The same ZIP of every plate
SVG (Vector) › **Download every .svg**	One SVG file per plate, downloaded one after another
Seeds (Reproducible) › **Download the seeds**	A text file, `ishihara-seeds.txt`, with the seed, palette and settings of every plate

↓ **Export All** in the Lab does the same as **Export every plate**.

Inside the ZIP, files are named by plate number, type and content, for example `plate-001-number-93.png`, `plate-001-number-93.svg` and `plate-001-number-93.json`. The JSON records the plate's ID, seed, type, content, palette, difficulty, plate shape, dot count, fine-tuning settings and its quality report.

The seeds file is a table with the columns content, seed, type, palette, difficulty, dots and quality, one row per plate.

> **Note:** Your browser may ask permission before it saves several files in a row, as **Download every .svg** does. Allow it for auricartisan.com.

Reproduce a plate

"A seed rebuilds a plate exactly. Keep it with the file and the same settings will produce the same dots on any machine."

1. Copy the plate's seed with **Copy seed**, or read it from the seeds file.
2. Select **Custom** in the Lab.
3. Choose the same type, content, palette and difficulty, and use the same fine-tuning settings.
4. Paste the seed into **Seed (optional)** and select **Generate**.

The result: the same plate, dot for dot.

Over the REST API

The Export tab ends with "Or ask for one over HTTP": "Three API credits a call. A separate balance from the tool's tokens." It shows an example request to `POST /v1/ishihara/plate` with a type, content, palette, difficulty and seed. See the public API documentation at https://auricartisan.com/library/documentation/api/.

Messages

Message	Meaning
"PNG downloaded" / "SVG downloaded"	The file was saved
"Loading ZIP library…"	The ZIP builder is loading, the first time you export
"Exporting all plates…" then "N plates exported as ZIP"	The ZIP is ready

Message	Meaning
"No plates to export"	The bench is empty
"ZIP library unavailable — downloading first plate as PNG"	The ZIP builder could not load, so only the first plate was saved

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Generate plates

Use the Lab to choose a plate type, palette and difficulty, generate single plates or batches, make custom plates, fine-tune the dots, and read the plate cards.

Everything in this guide happens on the **Lab** tab.

Choose what to generate

Plate Type

Type	What the figure is
Number	A number to read
Letter	A letter to read
Symbol	A symbol, such as a star or an arrow
Number (Gradient)	A number drawn with a colour gradient
Shape	A shape, such as a ring
Multi-layer (Shape+Number)	A shape and a number together
Hidden Path	A winding path to trace

Color Palette

The colour families the figure and background are drawn from. The names say which colours are set against each other.

- **Auto (Random)**, a palette chosen at random for each plate (default)
- **Red vs Green**
- **Green vs Brown**
- **Blue vs Purple**
- **Orange vs Green**
- **Teal vs Magenta**
- **Yellow vs Blue**
- **Sepia Mono**
- **Mixed Pastel**

Difficulty

Easy, **Medium** (default), **Hard** or **Extreme**. Difficulty "Controls colour similarity between figure and background": the harder the plate, the closer the figure's colours are to the background's.

Generate

Button	What it does
+ Generate	Draws one plate with the current settings and adds it to the bench
×1, **×3**, **×6**	Generates 1, 3 or 6 plates in a row
Custom	Opens **Create Custom Plate** (below)
Clear All	Removes every plate from the bench ("All plates cleared")
↓ **Export All**	Downloads every plate as a ZIP (see Export plates)

Each new plate is announced, for example "Plate #8 generated — 5", and scored as it lands.

When the bench is empty it says "No plates on the bench" and offers **Generate a plate** and **Or build the standard 38**.

Create a custom plate

Select **Custom** to choose the figure yourself.

1. Choose a **Plate Type**.
2. Type the **Content**: a number, a letter, or a symbol or shape name. The dialog lists the names it knows:
   - symbols: star, arrow, triangle, heart, diamond, circle, square, moon, sun, check, x, target, squareDot, dot, leaf, wave, shield;
   - shapes: pentagon, octagon, rounded-square, annulus, oval.
3. Choose a **Color Palette** and a **Difficulty**.
4. Optionally type a **Seed**. "Same seed + parameters = identical plate."
5. Select **Generate**. The page says "Custom plate #N generated".

Select **Cancel** to close the dialog without generating.

Plate size

Plate Size sets how large plates are drawn on the bench: **S** (210 pixels), **M** (290 pixels, default), **L** (440 pixels) or **XL** (900 pixels). The size is remembered in this browser. It changes the view only; exports have their own sizes.

"Double-click a plate to open it full size."

Fine tuning

Select **Fine tuning** to show more controls. They apply to plates you generate next.

Control	What it does	Range or options	Default
Dot Count	Number of dots to render. More = finer texture.	2,000–15,000	7,000
Plate Shape	Overall shape of the plate	**Circle**, **Square**	Circle
CVD Preview	Redraws every plate as it might appear with a deficiency	**Off**, **Protanopia**, **Deuteranopia**, **Tritanopia**	Off
Placement Engine	How dots are packed	**Balanced**, **Organic**, **Precision**, **Dense**	Balanced
Dot Style	The look of each dot	**Mixed**, **Round**, **Soft Edge**, **Rings**	Mixed

Control	What it does	Range or options	Default
Material	The card the plate is printed on	**Paper**, **Warm Card**, **Cool Card**, **Transparent**	Paper
Color separation	How far apart figure and background colours are pushed	0–100	55
Edge blend	How softly the figure's edge merges into the background	0–100	42
Dot scale	Size of the dots, as a percentage	70–145	100
Texture grain	Grain of the card surface	0–100	35

This batch

A summary above the bench counts the plates you have now:

Figure	Meaning
Plates	Plates on the bench
Rendered	Plates drawn in this session
Avg dots	Average dots per plate
Color space	"P3 HDR" when your display supports wide-gamut colour, otherwise "sRGB"
Avg QA	Average quality score

The plate card

Each plate appears on a card with:

- its number and content, for example "#1 - 45";
- tags for its type, palette, difficulty, dot count, placement engine and dot style;
- **QA PASS** or a review mark, and the quality score as a percentage;
- a per-plate preview switch: **Normal**, **Protan**, **Deutan**, **Tritan**;
- figures: **Normal ΔE** (figure against background for typical vision), **CVD min** (the smallest of those distances through the three deficiencies), **Density**, **Figure** (the share of dots in the figure), **Seed** and **Material**;
- buttons: **Copy seed** ("Seed copied"), **Download PNG** and **Download SVG**.

See Read the diagnostics for what the figures mean.

Open a plate full size

Double-click a plate, or focus it and press `Enter` or `Space`. The full-size view shows the content as its title, a line with type, palette, difficulty and dot count, and a footer with the plate number, QA score, **Separation ΔE**, **Concealment ΔE**, seed, and its position (for example "1 of 10"). It has the same **Normal**, **Protan**, **Deutan** and **Tritan** switch.

Press `Right` or `Left` for the next or previous plate, and `Esc` to close.

Remember

Plates live only on this page. Reloading or leaving the page loses them. Keep a plate by exporting it or saving its seed.

Read the diagnostics

Understand each plate's quality score, its four weighted components, and how well it hides from each colour-vision deficiency.

A good plate does two things at once: its figure is easy to see with typical colour vision, and it disappears for the deficiency it is meant to catch. The **Diagnostics** tab shows how well each plate does both.

"One number hides four measurements, so the score is shown taken apart. A plate fails when it is either invisible to normal vision or still legible to the deficiency it is meant to catch."

How colour distance is measured

The generator works in **OKLab**, a colour space designed so that equal distances look roughly equally different. **ΔE** here is the straight-line distance between two colours in OKLab. On this scale, 0 means identical and a few hundredths is already a visible difference; figures such as 0.150 are clearly different colours.

- **Separation** is the ΔE between the average figure colour and the average background colour: how far apart they are for someone with typical colour vision.
- **Concealment** is that same distance recomputed after simulating protanopia, deuteranopia and tritanopia. The smaller it gets, the better the plate hides.

The summary

At the top, **This batch** gives **Plates**, **Passing** (for example "9 of 10"), **Mean score** and **Weakest** (the lowest-scoring plate and its score).

Each plate's card

- The plate's content, its type, palette and difficulty, a thumbnail, its **quality score** out of 100, and **Pass** or **Review**.
- Four component bars, each with its weight and its measured value:

Component	Weight	What it rewards
Dot density	.24	A density close to the ideal of about 0.36 (the share of the plate covered by dots)
Figure ratio	.20	A figure that takes up about 28% of the dots
Separation	.32	A clear ΔE between figure and background for typical vision
Concealment	.24	A figure that nearly vanishes through the most confusing deficiency, relative to its separation

A bar is highlighted when its component scores under 60%.

- **Through each deficiency — ΔE from the background**: the figure's distance from the background under **Protanopia**, **Deuteranopia** and **Tritanopia**. Here the small number is the good one; values under 0.03 are marked as hidden.
- A sentence such as "Hides best from deuteranopia; still most legible under tritanopia."

When a plate passes

A plate passes when all of these hold:

- dot density is between 0.08 and 0.95;
- the figure takes between 5% and 65% of the dots;

- separation is above 0.01;
- the quality score is 45 or more.

Otherwise it is marked **Review**.

Using the diagnostics

- A plate with low **Separation** may be invisible even to typical vision. Lower the difficulty, raise **Color separation** in **Fine tuning**, or try another palette.
- A plate with low **Concealment** is still legible through the deficiency. Choose a palette aimed at that deficiency, such as **Red vs Green** for red-green deficiencies, or raise the difficulty.
- The deficiency named in "Hides best from …" is the one the plate is best suited to demonstrate.

If no plates exist yet, the tab says "Nothing measured yet" with a **Go to the Lab** button.

> **Note:** These scores measure the plate as drawn on screen with simulated deficiencies. They do not make it a validated clinical test. See Limits and accuracy.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Ishihara Plate Generator reference

Every control, option, default and output of the Ishihara Plate Generator.

Page

Item	Detail
Address	https://auricartisan.com/tool/ishihara-plate-generation/
Tabs	**Lab**, **Diagnostics**, **Set 38**, **Export**
API hint	3 API credits per call to `POST /v1/ishihara/plate`
On load	Six starter plates are generated
Notice	"Not a colour-vision test. These plates are illustrative…"

Lab: main controls

Control	Options	Default
Plate Type	Number, Letter, Symbol, Number (Gradient), Shape, Multi-layer (Shape+Number), Hidden Path	Number
Color Palette	Auto (Random) or one of eight named palettes (see Generate plates)	Auto (Random)
Difficulty	Easy, Medium, Hard, Extreme	Medium
+ Generate	One plate	—
×1, **×3**, **×6**	Several plates	—
Custom	Opens **Create Custom Plate**	—
Clear All	Removes every plate	—

Control	Options	Default
↓ **Export All**	ZIP of every plate	—
Plate Size	S (210 px), M (290 px), L (440 px), XL (900 px)	M

Create Custom Plate

Field	Values
Plate Type	As above
Content	A number, a letter, or a symbol or shape name
Color Palette	As above
Difficulty	As above
Seed (optional)	Any whole number; the same seed and settings give an identical plate
Cancel / **Generate**	—

Symbol names: star, arrow, triangle, heart, diamond, circle, square, moon, sun, check, x, target, squareDot, dot, leaf, wave, shield. Shape names: pentagon, octagon, rounded-square, annulus, oval.

Fine tuning

Control	Range or options	Default
Dot Count	2,000–15,000	7,000
Plate Shape	Circle, Square	Circle
CVD Preview	Off, Protanopia, Deuteranopia, Tritanopia	Off
Placement Engine	Balanced, Organic, Precision, Dense	Balanced
Dot Style	Mixed, Round, Soft Edge, Rings	Mixed
Material	Paper, Warm Card, Cool Card, Transparent	Paper
Color separation	0–100	55
Edge blend	0–100	42
Dot scale	70–145	100
Texture grain	0–100	35

Plate card

Item	Meaning
Heading	Plate number and content, for example "#1 - 45"
Tags	Type, palette, difficulty, dot count, placement engine, dot style
QA PASS and percentage	Pass or review, and the quality score
Preview switch	**Normal**, **Protan**, **Deutan**, **Tritan**
Normal ΔE	Figure against background for typical vision (OKLab)

Item	Meaning
CVD min	The smallest figure-to-background distance through the three deficiencies
Density	Share of the plate covered by dots
Figure	Share of dots in the figure
Seed	The number that rebuilds the plate
Material	The card stock
Copy seed, **Download PNG**, **Download SVG**	Actions

Full-size view

Opened by double-clicking a plate, or `Enter` or `Space` on it. Shows content, type, palette, difficulty, dots, plate number, QA score, **Separation ΔE**, **Concealment ΔE**, seed and position; the preview switch; `Left` / `Right` to move; `Esc` to close.

This batch (Lab)

Plates, **Rendered**, **Avg dots**, **Color space** ("P3 HDR" or "sRGB"), **Avg QA**.

Diagnostics

Item	Detail
Summary	**Plates**, **Passing**, **Mean score**, **Weakest**
Components	**Dot density** (.24), **Figure ratio** (.20), **Separation** (.32), **Concealment** (.24)
Through each deficiency	ΔE from the background under Protanopia, Deuteranopia, Tritanopia
Pass rule	Density 0.08–0.95, figure 5–65% of dots, separation above 0.01, score 45 or more

Set 38

Item	Detail
Generate the set	Builds 38 plates without touching the Lab
Composition	24 numbers, 6 letters, 4 symbols, 2 shapes, 2 paths
Seed	42

Export

Action	Output
Card **Download PNG**	`ishihara-plate-<number>-<content>.png`, 4,096 px
Card **Download SVG**	`ishihara-plate-<number>-<content>.svg`
Export every plate, **Download every .png**, ↓ **Export All**	`ishihara-plates.zip`: PNG (2,048 px), SVG and JSON per plate
Download every .svg	One SVG per plate

Action	Output
Download the seeds	`ishihara-seeds.txt`: content, seed, type, palette, difficulty, dots, quality

Remembered in this browser

Only the **Plate Size**. Plates themselves are not kept between visits.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Troubleshooting the Ishihara Plate Generator

Common problems with the Ishihara Plate Generator, why they happen, and what to do.

My plates disappeared

- **Why:** plates are kept only while the page is open. Reloading or leaving the page loses them.
- **What to do:** export plates you want to keep, or download the seeds file so you can rebuild them.

I can see the figure clearly under Protan, Deutan or Tritan

- **Why:** the plate's colours still differ for that deficiency. Palettes and difficulty decide how well a figure hides.
- **What to do:** check **Concealment** in **Diagnostics**, choose a palette that targets that deficiency, raise the difficulty, or regenerate.

I cannot see the figure even with normal preview

- **Why:** at **Hard** and **Extreme** the figure's colours are very close to the background's, and your screen, brightness and room light all affect what you see.
- **What to do:** lower the difficulty, raise **Color separation** in **Fine tuning**, or check the plate's **Separation** in **Diagnostics**.

A plate is marked Review instead of Pass

- **Why:** its quality score is under 45, or its density, figure share or separation is out of range.
- **What to do:** see Read the diagnostics and regenerate with adjusted settings.

"No plates to export"

- **Why:** the bench is empty.
- **What to do:** generate plates first. **Set 38** plates are shown in their own tab and are not on the bench.

"ZIP library unavailable — downloading first plate as PNG"

- **Why:** the ZIP builder could not load, often because of a network problem or a content blocker.
- **What to do:** reload and try again, or download plates one at a time from their cards.

Download every .svg saved only one file, or asked for permission

- **Why:** browsers limit or ask before a page saves several files in a row.
- **What to do:** allow multiple downloads for auricartisan.com, or use **Export every plate** for a single ZIP.

"Copy failed"

- **Why:** the browser refused clipboard access.

- **What to do:** read the seed from the card, or download the seeds file.

A rebuilt plate does not match the original

- **Why:** the seed only reproduces a plate with exactly the same type, content, palette, difficulty and fine-tuning settings.
- **What to do:** check every setting against the seeds file or the plate's JSON in the export ZIP.

Colour space shows sRGB, not P3 HDR

- **Why:** your display or browser does not report wide-gamut support.
- **What to do:** nothing; plates are still generated and scored the same way.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Ishihara Plate Generator FAQ

Short answers to common questions about the Ishihara Plate Generator.

Can I use these plates to test someone's colour vision?

No. The page says: "Not a colour-vision test. These plates are illustrative. A screen is not colour-calibrated and a real Ishihara examination is printed, lit and administered under controlled conditions. For diagnosis, see an eye-care professional."

Is it free? Do I need an account?

It is free and needs no account. Plates are generated in your browser. Generating plates from code through the REST API costs 3 API credits per call.

What does the quality score mean?

It combines four measurements: dot density, figure ratio, separation for typical vision, and concealment through the simulated deficiencies. A plate passes when it is visible to typical vision and hidden from the deficiency it targets. See Read the diagnostics.

Which colour-vision deficiencies are simulated?

Protanopia, deuteranopia and tritanopia, using the Brettel, Viénot and Mollon (1997) model.

Why can I not keep my plates after closing the page?

Plates are not stored between visits. Export them, or keep their seeds.

Can I make a plate with my own number or word?

Use **Custom** and type a number, a letter, or one of the listed symbol or shape names. See Generate plates.

What is a seed?

A number that fixes every random choice in a plate. The same seed with the same settings produces the same plate, dot for dot, on any machine.

Which format should I export?

PNG for slides and documents; SVG for printing at any size, since every dot is a separate circle; the ZIP for everything at once, with a JSON description of each plate.

What is the standard 38?

A fixed set of 38 plates (24 numbers, 6 letters, 4 symbols, 2 shapes and 2 paths), built from seed 42 so it is the same every time. See Build the standard 38.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Ishihara Plate Generator: supporting material

The Ishihara Plate Generator's glossary, shortcuts, limits, related tools, release notes and privacy notes.

This folder supports the generator's guides with the terms they use, the keys it responds to, an honest account of what the plates are and are not, related tools, its history, and what happens to your data.

File	What it covers
Glossary	Terms used in the generator's docs
Keyboard shortcuts	Keys for plates and the full-size view
Limits and accuracy	What the scores measure, and why this is not a vision test
Related tools and reading	Tools and articles that pair with the generator
Release notes	User-facing history
Privacy	What stays in your browser

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Ishihara Plate Generator keyboard shortcuts

The keys the Ishihara Plate Generator responds to.

Keys	Where	Action
`Enter` or `Space`	A focused plate	Open it full size
Double-click	A plate	Open it full size
`Right`	Full-size view	Next plate
`Left`	Full-size view	Previous plate
`Esc`	Full-size view	Close it

All other controls are standard buttons and menus: move between them with `Tab` and `Shift` + `Tab`, and use them with `Enter` or `Space`.

For shortcuts that work across the whole site, see Keyboard shortcuts.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Ishihara Plate Generator glossary

Plain-language definitions of the terms used by the Ishihara Plate Generator.

Concealment — How well a figure hides under a simulated deficiency: its distance from the background there, compared with its distance for typical vision. Lower distance means better concealment.

CVD (colour-vision deficiency) — Reduced ability to tell some colours apart, often called colour blindness.

Deuteranopia, protanopia, tritanopia — Colour-vision deficiencies in which the green-, red- or blue-sensitive cones do not work. The generator simulates all three.

Dot density — The share of the plate's area covered by dots.

ΔE (delta E) — A measure of colour difference. The generator measures it in OKLab, where 0 is identical and a few hundredths is a visible difference.

Difficulty — How similar the figure's colours are to the background's: Easy, Medium, Hard or Extreme.

Figure — What the plate shows: a number, letter, symbol, shape or path.

Figure ratio — The share of dots that belong to the figure.

Ishihara plate — A circle of coloured dots containing a figure, used in colour-vision screening. The originals were published by Shinobu Ishihara.

OKLab — A colour space designed so that equal distances look roughly equally different.

P3 HDR / sRGB — Colour spaces. sRGB is the standard web colour space; Display P3 is a wider gamut some screens can show.

Placement engine — The method used to scatter and pack dots: Balanced, Organic, Precision or Dense.

Quality score (QA) — A 0–100 score combining dot density, figure ratio, separation and concealment.

Seed — A number that fixes every random choice in a plate, so the same seed and settings rebuild it exactly.

Separation — The colour distance between the average figure colour and the average background colour, for typical vision.

Standard 38 — A fixed set of 38 plates in the composition a printed plate book uses.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Limits and accuracy

What the Ishihara Plate Generator's scores measure, how far to trust them, and why the plates are not a colour-vision test.

Not a colour-vision test

The page is explicit: "Not a colour-vision test. These plates are illustrative. A screen is not colour-calibrated and a real Ishihara examination is printed, lit and administered under controlled conditions. For diagnosis, see an eye-care professional."

In particular:

- **Screens differ.** Colour, brightness, gamut and night modes change what a plate looks like. The same file looks different on two screens.
- **Printing differs.** Paper, ink and light change colours again. A printed plate from this tool is not a calibrated plate.
- **Examinations are controlled.** A real test controls lighting, viewing distance and timing, and uses validated plates. These plates are not validated against people with colour-vision deficiencies.

What the scores measure

- Colour distances are calculated in **OKLab** between the average figure colour and the average background colour.
- Deficiencies are simulated with the **Brettel, Viénot and Mollon (1997)** model for protanopia, deuteranopia and tritanopia. A simulation models a typical, complete (dichromatic) deficiency; milder forms and individual differences are not modelled.
- The quality score weighs four components (density .24, figure ratio .20, separation .32, concealment .24). It describes how well a plate is built, not whether it would detect a deficiency in a real person.
- A plate passes when density, figure share and separation are in range and the score is at least 45.

Previews

The **CVD Preview** and the per-plate **Protan**, **Deutan** and **Tritan** switches show the same simulation. They are an illustration of how a plate might look, not what any particular person sees.

Reproducibility

A seed reproduces a plate exactly only with the same type, content, palette, difficulty and fine-tuning settings. The export ZIP includes a JSON description of every plate for this reason.

Nothing is kept

Plates exist only while the page is open. Export or keep seeds for anything you need later.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Ishihara Plate Generator privacy

What the Ishihara Plate Generator does with your plates and settings.

- **Everything happens in your browser.** Plates are generated, scored and exported on your device. The Export tab says: "Nothing leaves the browser — each file is built here and handed straight to the download."
- **Nothing about your plates is stored.** Plates exist only while the page is open. The one thing remembered in this browser is your **Plate Size** choice.
- **Custom content stays local.** A number or word you type into **Custom** is used only to draw the plate.
- **The ZIP builder.** The first time you export a ZIP, the page loads its ZIP library from auricartisan.com. No plate data is sent.
- **The API hint.** If you dismiss the "Automate this" note, that choice is remembered in this browser so the note stays hidden on every tool page.

The site-wide privacy policy is covered in Company and legal.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Ishihara Plate Generator release notes

User-facing history of the Ishihara Plate Generator.

Dated entries come from the public changelog at https://auricartisan.com/library/changelog/. The first section describes the page as it is now.

Current page (September 2026)

- Four tabs: **Lab**, **Diagnostics**, **Set 38** and **Export**.
- **Diagnostics** shows each plate's quality score taken apart into its four components, with the ΔE through each simulated deficiency.
- **Set 38** builds the standard 38-plate set from seed 42 without touching the plates in the Lab.
- **Export** offers a ZIP of every plate, individual SVGs, and a seeds file.
- **×1**, **×3** and **×6** generate several plates at once; **Plate Size** offers S, M, L and XL.
- Each card has a per-plate **Normal**, **Protan**, **Deutan** and **Tritan** preview, and double-clicking opens a plate full size.

v0.9.75 — 19 April 2026

- The Ishihara plate generator was added to the tools collection.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Related tools and reading

Tools and articles that pair well with the Ishihara Plate Generator.

Where	Why
Vision Lab	See colours, palettes and images through 78 vision conditions, with measurements
Contrast Checker	Check that text colours are readable, a different question from colour confusion
Analyzer	See a web page's colours under colour-vision filters
Learn	Plain-language articles on colour vision
Public API	Generate plates from code with `POST /v1/ishihara/plate` (3 API credits per call)

Articles on auricartisan.com

- **Ishihara Plate Generation** — an earlier article about the generator. The page has been redesigned since, so some names and layouts differ; this documentation describes the current page: https://auricartisan.com/library/documentation/articles/2026-04-18-ishihara-plate-generation

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Vision Lab

What the Vision Lab simulates, who it is for, and how to see a colour, a palette or an image through 78 vision conditions.

The Vision Lab (page title: Vision Deficiency Lab) shows how colours, palettes and images look to people with 78 different vision conditions, from the common colour-vision deficiencies such as deuteranopia to cataract, glaucoma, macular degeneration and dozens of rarer retinal, optic-nerve and corneal conditions. Its masthead reads "Seventy-eight ways of seeing, simulated."

Seventeen of the conditions are **per-colour transforms**: every colour has an exact answer, computed in linear light with the published colour-vision models of Brettel (1997), Viénot (1999) and Machado (2009). The other sixty-one are **field effects**: they change where and how clearly you see (blur, lost regions, glare, distortion) rather than changing each colour. Every condition runs the same ten reference swatches, so two conditions can be compared fairly.

The lab has five views. The **Workbench** takes one condition apart on a colour, a palette or your own image. The **Catalogue** is every condition as a card that opens a detailed floating panel. **Compare** lays one source out under many conditions side by side. **Palette audit** checks every pair in your palette under all 17 colour conditions and suggests fixes. **Analytics** maps the whole collection by category and severity. Everything runs in your browser; images you add are never uploaded.

> **Important:** The page states: "Educational visualisation only. Not a medical simulation, a diagnostic tool, or a substitute for professional eye care — and screen colours are not calibrated."

Who it is for

- Designers checking that a palette, chart or interface still works for people with colour-vision deficiencies.
- Accessibility specialists demonstrating the effect of low vision and field loss.
- Educators and students learning how different eye conditions affect sight.
- Colour scientists comparing simulation models.

Where to find it

- Address: https://auricartisan.com/tool/vision-simulation/
- A single condition in the Workbench: `https://auricartisan.com/tool/vision-simulation/?vw=deuteranopia:100:brettel` (condition, severity 0–100, model).

Quick start

1. Open https://auricartisan.com/tool/vision-simulation/. It opens on the **Workbench**, with Deuteranopia selected.
2. Under **Specimen**, choose **Interface**, **Palette** or **Image**.
3. Type your brand colour into **Base colour**, for example `#D3AF37`, and read what it becomes under the condition.
4. Pick another condition from the list on the right, for example Protanopia or Tritanopia.
5. Read **What collapses**: pairs of reference colours that stop being distinguishable.
6. Select **Compare this** to see the same source under several conditions at once.
7. Open **Palette audit** and enter your palette's colours to find every pair that merges.
8. Select **Copy link to this view** to share what you are looking at.

What you can do

- Simulate 78 conditions in 11 categories, each with an estimated severity score and a prevalence note.
- Set severity from 0 to 100% and choose the Brettel, Viénot or Machado model for colour-vision conditions.
- See interface mock-ups, a ten-colour palette, or your own image under a condition, side by side with the original.
- Measure colour change with ΔE76 and CIEDE2000, and count reference pairs that stop being distinguishable.
- Open up to five detailed floating panels, with colour-appearance settings, chromaticity diagrams, metrics, snapshots and exports.
- Feed one image to every open panel, from your own file or 15 built-in samples and 4 test charts.
- Compare many conditions on one contact sheet.
- Audit a palette of up to 24 colours under 17 colour conditions, with suggested lightness fixes.
- Export the catalogue as JSON or CSV, panels as images, and share any view as a link.

In this folder

- docs/ — how to use the Vision Lab
  - Getting started — the screen, the five views, and a first simulation
  - Use the Workbench — one condition, taken apart
  - Browse the Catalogue and panels — cards, filters, favourites, floating panels and the source image
  - Compare conditions — the contact sheet
  - Audit a palette — pairs that merge, and fixes
  - Explore Analytics — the matrix, map and statistics
  - The 78 conditions — every condition, by category
  - Links, saving and exports
  - Reference — every control and output
  - Troubleshooting
  - FAQ
- others/ — supporting material
  - Glossary
  - Keyboard shortcuts
  - Limits and accuracy
  - Related tools and reading
  - Release notes
  - Privacy

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Vision Lab documentation

The list of Vision Lab guides with one line on each.

These guides explain how to use the Vision Lab at https://auricartisan.com/tool/vision-simulation/. Start with Getting started; the reference lists every control.

Guide	What it covers
Getting started	The masthead, the five views, the notice, and a first simulation
Use the Workbench	Specimens, base colour, severity, model, stage, what collapses, and the measurements
Browse the Catalogue and panels	Cards, search, filters, favourites, floating panels, Vision Control and the source image
Compare conditions	The contact sheet: tiles, order, columns and severity
Audit a palette	Which of your colour pairs merge under 17 conditions, and how to fix them
Explore Analytics	The severity matrix, the map, statistics and exports
The 78 conditions	Every condition with its category, kind, severity score and prevalence
Links, saving and exports	Sharing views, what is remembered, and every export
Reference	Every control, option and output
Troubleshooting	Common problems and what to do
FAQ	Short answers to common questions

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Getting started with the Vision Lab

Open the Vision Lab, learn the five views, and run your first simulation.

Open it

Go to https://auricartisan.com/tool/vision-simulation/. No account is needed to use it.

The screen, top to bottom

Masthead

The kicker "Auric Artisan · Vision", the headline "Seventy-eight ways of seeing, simulated.", and a line explaining that every condition runs the same ten reference swatches through the published models, in linear light, so two cards differ because the conditions differ. A rail of facts follows: **Conditions 78**, **Categories 11**, **Brettel 1997 · Viénot 1999 · Machado 2009**, and **Read the API reference** →.

API hint

"Automate this — 5 API credits per call to `POST /v1/vision/audit`." It tells developers that the palette audit is also available from the REST API, and that API credits are a separate balance from tool tokens. Select **API docs** to read about the endpoint, or **×** to dismiss the hint on every tool page.

The view bar

Five buttons switch between the lab's views. The lab remembers which one you last used.

View	What it is for
Workbench	One condition taken apart: a specimen, a severity dial, the model, and measurements
Catalogue	Every condition as a card; a card opens a detailed floating panel
Compare	One source shown under many conditions at once, as a contact sheet
Palette audit	Every pair in your palette checked under all 17 colour conditions
Analytics	The whole collection mapped by category and severity

The notice

At the foot of the lab: "Educational visualisation only. Not a medical simulation, a diagnostic tool, or a substitute for professional eye care — and screen colours are not calibrated."

Two kinds of condition

- **Per-colour transforms (17).** Colour-vision deficiencies, cone isolation and colour tints. Every colour has an exact answer: the lab computes what each colour becomes.
- **Field effects (61).** Cataract, glaucoma, retinal and optic-nerve conditions and others. They change where and how clearly you see (blur, dark or lost regions, glare, distortion). On a card, the reference swatches are left as they are and the loss is painted over them, "because colour survives glaucoma and the field does not."

Your first simulation

1. On the **Workbench**, look at the stage in the middle. With the **Interface** specimen and **Split** stage, it shows a small interface mock-up as it normally looks and as it looks with deuteranopia.
2. Select the **Base colour** field on the left and type your brand colour, for example `#D3AF37`. Next to it, "under condition →" shows the colour it becomes.
3. In the list on the right, select **Protanopia**. The stage, the base colour and the measurements update.
4. Read **What collapses** under the stage: reference colour pairs that were clearly different and now are not.
5. Read **Measured · base colour**: how far your colour moved (ΔE76 and ΔE2000), how many of the 45 reference pairs were lost, and the contrast of the mock-up's button label on its fill.
6. Drag the **Severity** slider down to see a milder form of the deficiency.

The result: you know what your colour becomes for someone with protanopia, and which colours they can no longer tell apart.

Next steps

- Use the Workbench for every Workbench control.
- Audit a palette to check your whole palette in one go.
- The 78 conditions to see what each condition is.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Audit a palette

Check every pair of colours in your palette under all 17 colour conditions, see which pairs merge, and get a lightness fix for each.

Simulating one colour tells you what it becomes. It does not tell you that two of your colours became the same thing, which is what breaks a chart legend, a status badge or a map. **Palette audit** checks every pair in your palette, under all 17 colour conditions, on the same engine the panels use.

Enter your palette

1. Select **Palette audit** in the view bar. It opens with a five-colour sample palette.
2. For each colour, use its swatch ("Change colour N") or type into its hex field.
3. Select **Add colour** to add another, up to 24 colours.
4. Select **Remove** beside a colour to take it out.

Each colour shows how many collapses it is involved in, for example "15 collapses". Select a swatch to see only where that colour collapses. The palette is remembered in this browser.

Set the threshold

Merge below, measured in CIEDE2000, is how close two simulated colours must be to count as merged.

Setting	Range	Default
ΔE threshold	1 ("identical") to 15 ("loose"), in steps of 0.5	5

A lower threshold only flags pairs that become almost identical; a higher one also flags pairs that become merely similar.

Read the result

The middle column shows one condition at a time:

- a verdict, for example "4 of 10 pairs merge under Achromatopsia";
- **Your palette** above the same colours **Under** that condition, with the ΔE of each merged pair now and what it was ("ΔE 3.0 was 67");
- **What collapses**, naming the condition, its severity score and prevalence, and listing each merged pair with its distance before and after (for example "16.6 → 1.59");
- **Measured · this condition**: pairs merged, the closest ΔE after simulation, how many of the 17 conditions fail, and how many of your colours are safe in every condition.

If nothing merges, the verdict says "Every pair survives all 17 colour conditions."

Choose the condition

The right column lists the 17 colour conditions, worst first. Each shows how many pairs merge (or "clean"), its severity score and prevalence.

- **Search 17 conditions** filters the list.
- **All 17** and **Only failing** switch between every condition and those with merges.

Select a condition to show it in the middle column.

Fix a merged pair

Select **Suggest a fix** beside a merged pair. The audit finds the smallest change in **lightness** to one of the two colours that pulls the pair back apart under that condition, and shows the new colour, whether it is lighter or darker, and the distance it clears at. Lightness is used rather than hue because hue is exactly what these conditions remove.

If no change of lightness under 80% separates the pair, it says "No lightness change under 80% separates these." Consider adding a pattern, label or shape instead of relying on colour.

Take it further

- **Open this condition in the workbench** opens the selected condition in the Workbench.
- **Copy the palette** copies your palette's colours.

The footer notes that the audit uses CIEDE2000 on the simulated colours, in your browser, and names the REST API endpoint that does the same audit (`POST /v1/vision/audit`).

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Browse the Catalogue and panels

Find conditions in the Catalogue, open them as floating panels, arrange panels with Vision Control, and feed them an image.

The **Catalogue** shows every condition as a card. Each card runs the same ten reference swatches, so the differences between two cards are real differences between two conditions. Selecting a card opens that condition as a floating panel, where you can run your own image through it and use every advanced control.

The Catalogue bar

Control	What it does
Search simulations…	Filters cards by name. `/` jumps to it; `Esc` clears it. **Clear** empties it.
Scope button	Shows the current category and its count, for example "All conditions 17"; opens a menu of categories
Filter	Opens the **Kind** and **Level** filters, with **Done**; a badge counts active filters
Count	How many conditions are showing, for example "17 of 78"
About this catalogue	Explains how the cards are built
Copy link	Copies a link to the whole current workspace (see Links, saving and exports)
Daily tally	On the free plan, how many simulations you have opened today out of 25

Kind offers **All**, **Per-colour** and **Field**; **Level** offers **All**, **Basic** and **Advanced**.

Why do I see 17 of 78?

The **Level** filter follows the site-wide Simple and Advanced preference. **Basic** shows the 17 per-colour conditions, **Advanced** shows the 61 field conditions, and **All** shows all 78. On a first visit the Catalogue usually opens at the basic level. Select **Filter**, then **Level** › **All**, to see everything.

Categories

A row of tabs narrows the cards: **All conditions**, **Favourites**, the 11 categories, **Source image** and **Open panels**. Each shows its count at the current filter. The lab remembers the last category you browsed.

If nothing matches, the grid says "Nothing matches that. No condition matches every filter you have on." Select **Clear the filters**.

The cards

Each card shows:

- the severity score (1 to 10), an estimated visual-impact score, not a clinical grade;
- the category and the condition's name;
- a short description, such as "Green-cone absent";
- its kind (**Per-colour** or **Field**) and level (**Basic** or **Advanced**);
- its prevalence, such as "~1% males" or "Educational";
- the ten reference swatches as the condition affects them;
- a star to add it to **Favourites**.

Select a card, or focus it and press `Enter` or `Space`, to open its panel. `Shift` + click or `Ctrl` + click opens a second copy of a panel that is already open.

Daily limit on the free plan

On the free plan you can open 25 simulations a day from the Catalogue. The tally in the bar ("0 / 25", read as "0 of 25 simulations used today") counts them. After the 25th, opening another card shows an upgrade prompt: "You've run your 25 free simulations for today. Upgrade for unlimited runs — every simulation stays free, this just lifts the daily cap." Paid plans have no daily cap. The tally counts simulation cards you open from the Catalogue; working in the Workbench is not counted.

Floating panels

A panel is a window over the page with the full simulation for one condition. You can move it and open up to five at once.

Title bar

Button	What it does
Link panel	Syncs controls with other linked panels
Duplicate	Opens another instance of this simulation
Pin panel	Prevents the panel being replaced when you go over the panel limit
Minimize	Sends the panel to the dock of minimised panels
Maximize panel	Maximises or restores the panel
Close panel	Closes it

When five panels are open, opening a sixth replaces the oldest panel that is not pinned. If all five are pinned, nothing more opens until you close one.

Panel controls (colour-vision panels)

Controls vary by condition. A colour-vision panel, such as Deuteranopia, has:

Control	What it does	Range or options	Default
Base color	The colour to simulate: a picker, a hex field and **Random**	Any colour	`#D3AF37`
Severity	How strongly the condition is expressed	0–100%	100%
Perceptual severity curve	Applies a perceptual curve so mid severities feel natural	On, off	Off
Multi-pass deep	Runs two passes of the simulation, which exaggerates the effect	On, off	Off
Model	The simulation model	**Brettel (1997)**, **Viénot (1999)**, **Machado-like**	Brettel (1997)
Workflow preset	Applies recommended parameters and colour-appearance settings	Custom (no preset), Default, Beginner mode, Material testing, Color matching, Print proof, Photography (D65)	Custom
Copy HEX / **Copy CSS**	Copies the reference hex, or CSS variables for the reference and simulated colours	—	—
Show advanced controls	Reveals cone-weight controls and exporters	On, off	Off
Cone channel weights	Scale the L, M and S cone signals	0 to 2 each	1
Export all CVD	Downloads a JSON file of the base colour under each colour-vision type, at several severities	—	—
Illuminant	The standard illuminant used for the chromaticity maths	Several	D65

The **Color appearance (CAM)** section models how the colour looks in context:

Control	Options or range	Default
CAM model	CAT16 adaptation, CAM16-UCS perceptual, CIECAM02 surround, Hunt-Stevens brightness	CAT16 adaptation
White point	D65, D50, A, F2, F7, F11, and warm, neutral and cool LED	D65 — Daylight (6500 K)
Correlated color temperature	1,667–25,000 K	6517 K
Perceived brightness	5–400%	100%
Observer	2° (small or close field), 10° (large field)	2°
Surround	Average (typical room), Dim (low-light), Dark (theatre)	Average
Viewing distance	Near (50 cm or less), Normal (50–80 cm), Far (80 cm or more)	Normal

Snapshots save the panel's whole state under a name ("Name (e.g. Version A)", then **Save**), kept in this browser. **Recent changes** lists your edits, with **Undo last** and **Clear**.

Panel readouts

- **Reference** and the simulated colour, each with HEX, RGB and CIE xy values, and the **ΔE2000** and **ΔE76** between them.

- Tabs:
  - **Chromaticity** — a CIE 1931 xy diagram with the reference, the simulation and the drift between them, plus figures for xy drift, saturation collapse, lightness change and angle.
  - **Metrics** — RGB drift, the LMS cone response, drift over time, and reference and simulated HSL.
  - **Image** — "Run your own image through it": **Choose image** and **Clear**, with **Original** and **Simulated** side by side. Every control applies to the image just as it does to the colour.
  - **Mindmap** — how the condition relates to others.
  - **Model audit** — the steps from input to output (linear-light conversion, disease transform, colour appearance, validation) and what each model covers.

Field-condition panels have their own controls for the condition's effect.

Vision Control

When panels are open, a dock titled **Vision Control** appears. It has:

- arrangement buttons: **Tile in a grid**, **Split left and right**, **Split top and bottom**, **Cascade with an offset** and **Stack them centred**;
- one tab per open panel, to bring it forward;
- a **More panel actions** menu: **Sync all panels**, **Minimize all**, **Restore all**, **Export a comparison PNG**, **Open the verification panel**, and **Close all but pinned**.

Minimised panels wait in a separate dock ("Minimized panels"); select one to restore it, or its × to close it. Press `T` to tile all panels, and `1` to `5` to bring a panel forward.

The source image

One image can feed every panel. Open the **Source image** tab (or the **Image** button in the category bar). The card is titled **Universal Source Image** and says "One image feeds every simulation. A default sample is loaded for you."

Control	What it does	Options	Default
Drop area	Drop, paste, or click to upload an image	Any image file	A built-in reference chart
Apply to all panels	Sends the image to every open panel	—	—
Quality	The working size the image is processed at	Fast · 1024px, Balanced · 1600px, Full · 4096px	Fast · 1024px
Auto-apply to new panels	New panels receive the image automatically	On, off	On
Pause auto / **Clear**	Pause auto-apply, or remove the image	—	—

What this image loses shows the image's size and working size, and ranks the conditions that change this image most, with **Open most-affected** →.

Samples offers 15 images: Forest scene, Skin tones, Eye, Farmers market, Coastal village, Rainy city, India · Himalayas, India · Kerala, India · Rajasthan, India · Tea hills, China · Guilin, China · Zhangjiajie, China · Yunnan, Thrown paint and Abstract drawing. **Test charts** offers Color bars, Spectrum, Red–Green and Chromatic grid.

Images you add are processed in your browser and are not uploaded.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Compare conditions

Lay one source out under many vision conditions at once on the Compare contact sheet, and find the condition that costs the most.

Compare is a contact sheet: the same source shown under several conditions side by side, each tile measured the same way. Use it to answer "which of these conditions hurts my design most?"

Open the sheet

- Select **Compare** in the view bar.
- From the Workbench, select **Compare this** to add the current condition.
- Press `C` (outside text fields) to jump to Compare.

The tiles

The first tile is **Normal vision**, marked "control", so you always have the original to compare against. Every other tile shows:

- the source under that condition;
- "N pairs lost of 45", the number of reference colour pairs that stop being distinguishable;
- "CTA" and a contrast ratio: the contrast of the mock-up's call-to-action button label on its fill, under the condition;
- the condition's severity score and name;
- a grip to reorder it ("Reorder … — drag, or use the arrow keys") and **×** to remove it.

Select **Add a condition** (it says how many are available to choose from) to add another tile.

The controls

Control	What it does	Options or range	Default
Severity for every tile	One severity for all tiles	0–100%	100%
Order	How tiles are sorted	**By impact**, **By severity**, **As arranged**	—
Columns	Tiles per row	**3**, **4**, **5**	—
Copy link to this sheet	Copies a link that reopens this sheet	—	—
Send the worst to the workbench	Opens the most damaging condition in the Workbench	—	—

Order and **Columns** are under **Sheet options**. A heading names the set, for example "Colour vision · 8 conditions", and a sentence names the condition that costs the most, for example "Achromatopsia costs the most: 23 of 45 reference pairs stop being tellable apart."

Reorder with the keyboard

Focus a tile's grip, then press `Left` or `Up` to move it one place earlier, `Right` or `Down` to move it later, and `Home` or `End` to send it to the start or end.

A way to use it

1. Put your design's colours in the Workbench as the base colour, or load your screenshot as the Image specimen.

2. Open **Compare** and set **Order** to **By impact**.
3. Read the first few tiles: these are the conditions to design for.
4. Select **Send the worst to the workbench** to study the worst one in detail.
5. Select **Copy link to this sheet** to share the comparison.

The sheet is remembered in this browser between visits.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Explore Analytics

See the whole collection of vision conditions by category and severity, and export the catalogue as JSON or CSV.

Analytics maps the collection of conditions: which categories they fall into, how severe they are estimated to be, and which are the most severe. It reflects the conditions currently showing, so it follows the Catalogue's **Level** and **Kind** filters.

Views

Choose a view under **View**:

View	What it shows
Matrix	A grid of categories against severity. Each cell counts how many conditions land there.
Map	A map of how the conditions relate to each other
Statistics	Figures for the collection

The matrix

- Rows are categories and columns are severity scores, running left to right. Each row ends with its count and mean severity, and the last row totals the columns.
- Colour shows how **many** conditions are in a cell, not how bad they are: the column already says that. A key shows the range of counts.
- Select a cell to list the conditions in it.

A note explains: "Severity is an estimated visual-impact score, not a clinical grade. It orders the collection; it does not rank anyone's sight."

The collection

A summary of what is showing: how many conditions, how many categories, the mean severity, and how many are per-colour transforms and field effects. It notes that the 17 per-colour conditions are exactly the 17 the Catalogue calls basic.

Most severe lists the top conditions by severity score.

Actions

Button	What it does
Open a random condition	Opens a random condition as a panel (`R` does the same)
Export JSON	Downloads the catalogue as `vd-catalog-<date>.json`

Button	What it does
Export CSV	Downloads severity data as `vd-severity-<date>.csv` (columns listed below)
Keyboard shortcuts	Shows the lab's keyboard shortcuts (`?` does the same)

The CSV columns are slug, name, category, complexity, severity_score, description and prevalence.

A footer line says how many conditions are shown and their severity range, for example "17 of 78 shown · severity 3–9".

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Links, saving and exports

Share a Vision Lab view as a link, see what the lab remembers in your browser, and export results.

Share a view as a link

From	Button	What the link restores
Workbench	**Copy link to this view**	The Workbench, on the same condition, severity and model
Catalogue bar	**Copy link**	The level, category, open panels (linked or pinned) and source-image settings
Compare	**Copy link to this sheet**	The Compare sheet

The Catalogue link restores the whole workspace: the level filter, the active category, the open panels (and which are linked or pinned), and the source-image settings. The Analytics CSV has the columns slug, name, category, complexity, severity_score, description and prevalence. **Export all CVD** records the base colour at several severities for each colour-vision type.

The Workbench link

A Workbench link has the form:

```
https://auricartisan.com/tool/vision-simulation/?vw=deuteranopia:60:vienot
```

The value is the condition's short name, then optionally the severity from 0 to 100, then optionally the model (`brettel`, `vienot` or `machado`). Opening such a link always opens the Workbench, whatever view you used last.

The workspace link

The Catalogue's link adds `#v=` and an encoded value to the address. The lab keeps that part of the address up to date as you work, without adding history entries, so the address bar always holds a link to your current workspace. Built-in samples and test charts travel with the link; an image of your own does not, because it never leaves your device.

What the lab remembers in your browser

- the view you last used;
- the Workbench's condition, severity, model, specimen and base colour;
- the Catalogue category you last browsed, and your **Favourites**;
- the Compare sheet;
- the Palette audit palette;
- panel snapshots you saved, and the panels you had open;
- where you left the **Vision Control** dock;

- on the free plan, today's simulation count.

All of it stays in this browser. Clearing the site's data resets it.

Exports

Where	Control	What you get
Analytics	**Export JSON**	The catalogue of conditions, as `vd-catalog-<date>.json`
Analytics	**Export CSV**	Severity data, as `vd-severity-<date>.csv`
Vision Control › More panel actions	**Export a comparison PNG**	An image of the open panels side by side, as `vision-comparison-<date and time>.png`
A colour-vision panel	**Export all CVD**	A JSON file of the base colour under each colour-vision type, ending `-all-cvds.json`
A panel	**Copy HEX** / **Copy CSS**	The reference hex, or CSS variables for the reference and simulated colours, on your clipboard
Palette audit	**Copy the palette**	Your palette's colours, on your clipboard

Some condition panels have exports of their own. For example, the Metameric Failure panel can download a comparison CSV, a JSON of all conditions and a PNG grid.

Save to your library

The site-wide **Save** button at the bottom right of the page saves the current output to your Auric Artisan library. See Library Kit.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

The 78 conditions

Every condition the Vision Lab simulates, by category, with its kind, severity score, summary and prevalence as the cards show them.

The Vision Lab simulates 78 conditions in 11 categories. This page lists them as the cards show them.

How to read the tables

- **Severity** is an estimated visual-impact score from 1 to 10. It orders the collection; it is not a clinical grade and does not rank anyone's sight.
- **Kind** is **Per-colour** (a colour-in, colour-out transform: every colour has an exact answer) or **Field** (an effect on the visual field, such as blur, lost regions or glare). The 17 per-colour conditions are the Catalogue's **Basic** level; the 61 field conditions are **Advanced**.
- **Summary** and **Prevalence** are the short texts on each card, with abbreviations such as "/yr" and "HTN" spelled out here. Prevalence figures are approximate, and some cards say "Educational" or "N/A" for conditions that are teaching aids rather than diagnoses.

Color Vision (9)

Condition	Severity	Kind	Summary	Prevalence
Protanopia	6	Per-colour	Red-cone absent	~1% males
Protanomaly	4	Per-colour	Red-cone weakened	~1% males
Deuteranopia	6	Per-colour	Green-cone absent	~1% males
Deuteranomaly	3	Per-colour	Green-cone weakened	~5% males
Tritanopia	5	Per-colour	Blue-cone absent	under 0.01%
Tritanomaly	3	Per-colour	Blue-cone weakened	under 0.01%
Achromatopsia	9	Per-colour	Total color absence	~1 in 33,000
Color Inversion	7	Per-colour	All channels inverted	N/A
Grayscale	7	Per-colour	Luminance only	N/A

Cone Isolation (4)

Condition	Severity	Kind	Summary	Prevalence
Red-Cone Monochromacy	7	Per-colour	Red channel isolated	Educational
Green-Cone Monochromacy	7	Per-colour	Green channel isolated	Educational
Blue-Cone Monochromacy	8	Per-colour	Blue channel isolated	Educational
Rod Monochromacy	8	Per-colour	Scotopic (rod) vision	Educational

Ocular Conditions (5)

Condition	Severity	Kind	Summary	Prevalence
Cataract	5	Field	Lens opacity	Very common (aging)
Glaucoma	7	Field	Optic nerve damage	~2.5% over 40
Diabetic Retinopathy	7	Field	Vascular retinal damage	~40% of diabetics
Macular Degeneration	8	Field	Macular damage	~8% over 60
Post-LASIK Halos	4	Field	Halos and glare	Post-surgical

Retinal Dystrophies (20)

Condition	Severity	Kind	Summary	Prevalence
Retinitis Pigmentosa	8	Field	Peripheral degeneration	~1:4,000
Stargardt Disease	7	Field	Macular dystrophy	~1:10,000
Cone Dystrophy	7	Field	Cone cell loss	under 1:30,000
Cone-Rod Dystrophy	8	Field	Cone + rod loss	Rare
Choroideremia	8	Field	Choroidal atrophy	~1:50,000

Condition	Severity	Kind	Summary	Prevalence
Best Vitelliform Macular Dystrophy	6	Field	Macular deposits	Rare
Pattern Dystrophy	5	Field	Macular patterns	Rare
Fundus Albipunctatus	6	Field	Night blindness	Very rare
Oguchi Disease	6	Field	Stationary night blindness	Very rare
Goldmann-Favre Syndrome	7	Field	Vitreoretinal	Very rare
X-Linked Retinoschisis	7	Field	Schisis layers	~1:5,000 males
Leber Congenital Amaurosis	10	Field	Severe early onset	~1:30,000
Retinopathy of Prematurity	7	Field	Preterm vascular	~1:3,000 births
Central Serous Chorioretinopathy	6	Field	Fluid under retina	~1:10,000
Macular Hole	7	Field	Central hole	~3:1,000 over 60
Epiretinal Membrane	6	Field	Macular puckering	~7% over 60
Myopic Macular Degeneration	7	Field	High-myopia degeneration	High myopes
Acute Macular Neuroretinopathy	6	Field	Paracentral loss	Very rare
Autoimmune Retinopathy	8	Field	Immune-mediated	Very rare
Paraneoplastic Retinopathy	8	Field	Cancer-associated	Very rare

Optic Nerve Disorders (9)

Condition	Severity	Kind	Summary	Prevalence
Optic Neuritis	8	Field	Optic nerve inflammation	~1:30,000 a year
Papilledema	8	Field	Disc swelling (ICP)	Secondary condition
Ischemic Optic Neuropathy	8	Field	Vascular optic loss	~2:100,000 a year
Toxic Optic Neuropathy	7	Field	Toxin-induced	Rare
Leber Hereditary Optic Neuropathy	9	Field	Mitochondrial	~1:50,000
Homonymous Hemianopia	8	Field	Half visual field lost	Stroke related
Bitemporal Hemianopia	8	Field	Temporal fields lost	Rare
Scotoma	7	Field	Blind spot	Various causes
Cortical Visual Impairment	9	Field	Brain-based vision loss	Leading cause under 2 years

Corneal & Anterior Segment (10)

Condition	Severity	Kind	Summary	Prevalence
Keratoconus	6	Field	Corneal thinning	~1:2,000
Corneal Edema	6	Field	Corneal swelling	Various
Bullous Keratopathy	7	Field	Blister formation	Various
Fuchs' Endothelial Dystrophy	6	Field	Endothelial loss	~4% over 40

Condition	Severity	Kind	Summary	Prevalence
Pterygium	4	Field	Tissue overgrowth	Tropical common
Uveitis	6	Field	Eye inflammation	~1:4,500 a year
Iridocyclitis	5	Field	Iris/ciliary inflammation	Various
Scleritis	5	Field	Scleral inflammation	Rare
Endophthalmitis	9	Field	Intraocular infection	Post-surgical risk
Sympathetic Ophthalmia	9	Field	Bilateral uveitis	Very rare

Perceptual Effects (6)

Condition	Severity	Kind	Summary	Prevalence
Age-Related Shift	3	Per-colour	Blue-yellow lens shift	Universal aging
Chloropsia	5	Per-colour	Green tint	Medication/rare
Cyanopsia	5	Per-colour	Blue tint	Post-cataract surgery
Erythropsia	5	Per-colour	Red tint	Snow/glare exposure
Neurological Distortion	7	Field	Shape distortion	Neurological
Metameric Failure	4	Field	Metamerism breakdown	Lighting context

Vascular & Retinal (4)

Condition	Severity	Kind	Summary	Prevalence
Branch Retinal Vein Occlusion	7	Field	Branch vessel occlusion	~1:1,000 over 40
Central Retinal Vein Occlusion	8	Field	Central vessel occlusion	~1:1,000 over 60
Central Retinal Artery Occlusion	10	Field	Arterial occlusion	~1:10,000 a year
Hypertensive Retinopathy	7	Field	Hypertension damage	Common with high blood pressure

Vitreous & Structural (4)

Condition	Severity	Kind	Summary	Prevalence
Retinal Detachment	9	Field	Retina separation	~1:10,000 a year
Posterior Vitreous Detachment	6	Field	Vitreous separation	Very common over 60
Vitreous Haemorrhage	8	Field	Blood in vitreous	~7:100,000 a year
Pigment Dispersion Syndrome	5	Field	Pigment release	~1:100

Developmental & Congenital (5)

Condition	Severity	Kind	Summary	Prevalence
Amblyopia	7	Field	Lazy eye	~2–3% of children

Condition	Severity	Kind	Summary	Prevalence
Aniridia	8	Field	Iris absence	~1:64,000
Coloboma	7	Field	Ocular gap defect	~1:10,000
Congenital Cataract	7	Field	Lens opacity at birth	~1:250
Ocular Albinism	7	Field	Pigmentation absent	~1:17,000

Functional Aging (2)

Condition	Severity	Kind	Summary	Prevalence
Presbyopia	4	Field	Near focus loss	Universal over 40
AMD (Advanced)	9	Field	Severe AMD	~2% over 60

The 17 colour conditions used by Palette audit

Palette audit checks every per-colour condition: Protanopia, Protanomaly, Deuteranopia, Deuteranomaly, Tritanopia, Tritanomaly, Achromatopsia, Color Inversion, Grayscale, the four cone-isolation conditions, Age-Related Shift, Chloropsia, Cyanopsia and Erythropsia.

> **Important:** These simulations are educational visualisations, not medical simulations or diagnostic tools. Real experience of any condition varies from person to person.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Use the Workbench

Take one condition apart in the Workbench: choose a specimen, set the base colour, severity and model, and read what collapses.

The Workbench shows one condition at a time, in three columns: the source on the left, the stage in the middle, and the list of conditions on the right. It opens with **Deuteranopia** selected, or with the condition named in the page address.

The source column

Specimen

What the stage shows:

Specimen	What you see
Interface	A small interface mock-up: a header with a **Get started** button, and a "Palette health" card
Palette	The ten reference swatches as a strip
Image	Your own image: drop it on the stage, or select **Choose an image**

With the **Image** specimen, the stage says "Drop an image on the stage" and "Whatever you drop stays on your machine — the pipeline runs in this tab and nothing is uploaded." To open a condition's floating panel, see Browse the Catalogue and panels.

Base colour

Type a hex colour into the field (up to seven characters, such as `#E53E3E`). Next to it, "under condition →" shows the colour it becomes. Under the field, ten swatches (the reference strip) each have a **Use** button that makes that swatch the base colour.

Severity

A slider from 0 to 100%, with marks for **Normal**, the condition's clinical default (for example **Clinical default 100%**) and **Full**. The figure above it says how strongly the condition is expressed, for example "100% · full expression".

Selecting a condition resets the slider to that condition's clinical default. As the page puts it: "Severity is a dial on the model, not a milder diagnosis."

Model

For the colour-vision conditions, choose how the simulation is calculated: **Brettel**, **Viénot** or **Machado**. The note beside the heading describes the method, for example "projection in LMS" (LMS being the three cone types). See Limits and accuracy for how the models differ.

Stage

- **Split** shows the reference and the simulation side by side.
- **Simulated only** shows just the simulation.

The stage

The heading names what you are looking at, for example "COLOUR TRANSFORM · Deuteranopia · 100%". Under the stage are two readouts.

What collapses

Pairs of reference colours that were comfortably apart (CIEDE2000 of 12 or more) and land within 8 under this condition, ranked by how far they fell. The heading says how many, for example "2 of 45 reference pairs fall under ΔE2000 8", and each pair shows its distance before and after, such as "24 → 1.5".

The ten reference swatches make 45 pairs. A pair that collapses is two colours that a person with this condition can no longer tell apart.

Measured · base colour

Measure	Meaning
ΔE76	How far the base colour moved, as a simple distance in Lab colour space
ΔE2000	How far it moved, with the CIEDE2000 formula, which follows perceived difference more closely
pairs lost / 45	How many reference pairs collapsed
button label / fill	The contrast ratio of the mock-up's button label against its fill, under the condition

ΔE (delta E) is a measure of colour difference: around 1 is barely noticeable, and larger numbers mean more obvious differences.

The conditions column

- **Search 78 conditions** filters the list by name.
- Filter buttons: **All 78**, **Colour** (the 17 per-colour transforms), **Field** (the 61 field effects) and **Basic only**.
- The selected condition is described at the top: its severity score in a ring, its name and category, a one-line description (for example "Green-cone absent. A per-colour transform: every pixel has an answer."), and chips for its kind, its prevalence and its level.
- Below, every condition is listed by category, with its severity score. Field conditions are marked **FIELD**.

Select a condition to put it on the stage.

The action rail

Button	What it does
Open the full panel	Opens this condition as a floating panel with every advanced control
Compare this	Sends the condition to the **Compare** sheet
Copy link to this view	Copies a link that reopens the Workbench on this condition, severity and model

The rail also notes "78 conditions · 11 categories" and "Brettel 1997 · Viénot 1999 · Machado 2009, in linear light".

What is remembered

The Workbench remembers your condition, severity, model, specimen and base colour in this browser, and a link with `?vw=` restores a condition directly. See Links, saving and exports.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Vision Lab reference

Every control, option and output of the Vision Lab in one place.

Page

Item	Detail
Address	https://auricartisan.com/tool/vision-simulation/
Masthead facts	Conditions 78, Categories 11, Brettel 1997 · Viénot 1999 · Machado 2009
API hint	5 API credits per call to `POST /v1/vision/audit`
Views	**Workbench**, **Catalogue**, **Compare**, **Palette audit**, **Analytics**
Notice	Educational visualisation only; not a medical simulation or diagnostic tool; screens are not calibrated

Workbench

Control	What it does	Values	Default
Specimen	What the stage shows	**Interface**, **Palette**, **Image**	Interface
Base colour	The colour to simulate	Hex, up to 7 characters	`#E53E3E`

Control	What it does	Values	Default
Reference strip	Ten swatches, each with a **Use** button	—	—
Severity	Strength of the condition	0–100%	The condition's clinical default
Model	Colour-vision model	**Brettel**, **Viénot**, **Machado**	Brettel
Stage	Layout of the stage	**Split**, **Simulated only**	Split
Search 78 conditions	Filters the condition list	Text	—
List filters	Narrow the list	**All 78**, **Colour**, **Field**, **Basic only**	All 78
Open the full panel	Opens the condition as a floating panel	—	—
Compare this	Adds the condition to Compare	—	—
Copy link to this view	Copies a `?vw=` link	—	—

Readouts: **What collapses** (reference pairs that fall under ΔE2000 8 from 12 or more) and **Measured · base colour** (**ΔE76**, **ΔE2000**, **pairs lost / 45**, **button label / fill**).

Catalogue

Control	What it does	Values
Search simulations…	Filters cards	Text; `/` to focus, `Esc` to clear
Scope button	Chooses a category	All conditions, Favourites, 11 categories, Source image, Open panels
Filter › **Kind**	Per-colour or field conditions	**All**, **Per-colour**, **Field**
Filter › **Level**	Basic or advanced conditions (linked to the site-wide preference)	**All**, **Basic**, **Advanced**
About this catalogue	Explains the cards	—
Copy link	Copies a `#v=` workspace link	—
Daily tally	Simulations opened today (free plan)	Up to 25
Card	Opens a floating panel; star adds to Favourites	`Enter` or `Space`; `Shift` or `Ctrl` + click duplicates

Floating panel

Control	Values	Default
Title bar	**Link panel**, **Duplicate**, **Pin panel**, **Minimize**, **Maximize panel**, **Close panel**	—
Base color	Picker, hex, **Random**	`#D3AF37`
Severity	0–100%	100%
Perceptual severity curve, **Multi-pass deep**	On, off	Off
Model	Brettel (1997), Viénot (1999), Machado-like	Brettel (1997)
Workflow preset	Custom (no preset), Default, Beginner mode, Material testing, Color matching, Print proof, Photography (D65); **Apply**	Custom

Control	Values	Default
Copy HEX, **Copy CSS**	—	—
Show advanced controls	Cone channel weights L, M, S (0–2); **Export all CVD**	Off; weights 1
Illuminant	Standard illuminants	D65
CAM model	CAT16 adaptation, CAM16-UCS perceptual, CIECAM02 surround, Hunt-Stevens brightness	CAT16 adaptation
White point	D65, D50, A, F2, F7, F11, LED warm, neutral, cool	D65
Correlated color temperature	1,667–25,000 K	6517 K
Perceived brightness	5–400%	100%
Observer	2°, 10°	2°
Surround	Average, Dim, Dark	Average
Viewing distance	Near, Normal, Far	Normal
Snapshots	Name and **Save**	—
Recent changes	**Undo last**, **Clear**	—
Readout tabs	**Chromaticity**, **Metrics**, **Image**, **Mindmap**, **Model audit**	Chromaticity

Up to 5 panels can be open. A sixth replaces the oldest unpinned panel.

Vision Control

Control	What it does
Arrange	Tile in a grid, split left and right, split top and bottom, cascade, or stack centred
Panel tabs	Bring a panel forward
More panel actions	Sync, minimise or restore all; export a comparison PNG; verification panel; close all but pinned
Minimised dock	Restore or close a minimised panel

Universal Source Image

Control	Values	Default
Drop area	Drop, paste or click to upload an image	A built-in reference chart
Apply to all panels	—	—
Quality	Fast · 1024px, Balanced · 1600px, Full · 4096px	Fast · 1024px
Auto-apply to new panels	On, off	On
Pause auto, **Clear**	—	—
What this image loses	Conditions ranked by effect on the image; **Open most-affected →**	—
Samples	15 images	—
Test charts	Color bars, Spectrum, Red–Green, Chromatic grid	—

Compare

Control	Values	Default
Tiles	Normal vision control, plus one per condition; reorder by dragging or with arrow keys; × removes	—
Add a condition	Any of the 78	—
Severity for every tile	0–100%	100%
Order	**By impact**, **By severity**, **As arranged**	—
Columns	**3**, **4**, **5**	—
Copy link to this sheet	—	—
Send the worst to the workbench	—	—

Palette audit

Control	Values	Default
Palette	Up to 24 colours: swatch, hex field, **Remove**; **Add colour**	A 5-colour sample
Merge below (CIEDE2000 ΔE)	1–15, steps of 0.5	5
Search 17 conditions	Text	—
Condition filter	**All 17**, **Only failing**	All 17
Suggest a fix	Smallest lightness change that separates a merged pair	—
Open this condition in the workbench, **Copy the palette**	—	—

Analytics

Control	Values
View	**Matrix**, **Map**, **Statistics**
Matrix cell	Lists the conditions in it
Open a random condition	—
Export JSON, **Export CSV**	`vd-catalog-<date>.json`, `vd-severity-<date>.csv`
Keyboard shortcuts	Opens the shortcuts overlay

Numbers used throughout

Item	Value
Conditions	78 (17 per-colour, 61 field)
Categories	11
Reference swatches	10, making 45 pairs
Collapse rule (Workbench, Compare)	A pair 12 or more apart (CIEDE2000) that falls within 8
Severity score	1 to 10, an estimated visual-impact score

Item	Value
Open panels	Up to 5
Free plan	25 simulations a day from the Catalogue

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Troubleshooting the Vision Lab

Common problems in the Vision Lab, why they happen, and what to do.

The Catalogue shows only 17 of 78 conditions

- **Why:** the **Level** filter is set to **Basic**, which shows only the per-colour conditions. It follows the site-wide Simple/Advanced preference.
- **What to do:** select **Filter**, then **Level** › **All**.

"Nothing matches that."

- **Why:** the search and filters together exclude every condition.
- **What to do:** select **Clear the filters**, or clear the search.

Opening a card shows an upgrade prompt

- **Why:** on the free plan you can open 25 simulations a day from the Catalogue, and today's are used.
- **What to do:** use the **Workbench**, which is not counted, or come back tomorrow. Paid plans have no daily cap.

A panel I had open disappeared when I opened another

- **Why:** up to five panels can be open; a sixth replaces the oldest panel that is not pinned.
- **What to do:** select **Pin panel** on panels you want to keep, or close panels you no longer need.

Nothing opens when I select a card

- **Why:** five panels are open and all five are pinned, or today's free simulations are used.
- **What to do:** unpin or close a panel, or check the daily tally.

My image does not appear in the panels

- **Why: Auto-apply to new panels** may be off or paused, or the image was loaded before the panels opened.
- **What to do:** open **Source image** and select **Apply to all panels**.

My image is processed slowly

- **Why: Quality** is set to Balanced or Full, which processes more pixels.
- **What to do:** choose **Fast · 1024px** while exploring.

The model buttons change nothing

- **Why:** the model applies to the colour-vision conditions. Field conditions, such as glaucoma, are not colour transforms.
- **What to do:** choose a per-colour condition to compare models.

The simulation does not match what someone with the condition sees

- **Why:** simulations model a typical case on an uncalibrated screen. Real vision varies from person to person, and your display, its settings and the room light all change how colours appear.
- **What to do:** treat the result as a guide, not a measurement. See Limits and accuracy.

A shared link opened a different view

- **Why:** a `?vw=` link always opens the Workbench; a `#v=` link restores the Catalogue workspace.
- **What to do:** use **Copy link to this view** in the Workbench, **Copy link** in the Catalogue, or **Copy link to this sheet** in Compare, depending on what you want to share.

My own image is missing from a shared link

- **Why:** your images never leave your device, so they cannot travel in a link. Built-in samples and test charts do.
- **What to do:** send the image separately, or share a built-in sample.

My settings are gone

- **Why:** the lab remembers your work in this browser only. A different browser, a private window, or clearing site data starts afresh.
- **What to do:** share important views as links, and export what you need.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Vision Lab FAQ

Short answers to common questions about the Vision Lab.

Is it free?

Yes. Every simulation is free on every plan. The free plan counts up to 25 simulations a day opened from the Catalogue; paid plans lift that cap. The Workbench, Compare, Palette audit and Analytics are free to use.

Is it a medical tool?

No. The page says: "Educational visualisation only. Not a medical simulation, a diagnostic tool, or a substitute for professional eye care — and screen colours are not calibrated." It cannot diagnose or test anyone's vision.

Which models does it use?

For the colour-vision conditions, the published models of Brettel (1997), Viénot (1999) and Machado (2009), calculated in linear light. You can switch between them in the Workbench and in each panel.

What is the difference between per-colour and field conditions?

A per-colour condition changes each colour in a predictable way, so every colour has an exact answer. A field condition changes how and where you see, such as blur, dark patches or a lost half of the visual field; colour may survive while the field does not.

What does the severity score mean?

It is an estimated visual-impact score from 1 to 10. It orders the collection; it is not a clinical grade and does not rank anyone's sight.

What does the severity slider do?

It sets how strongly the model is applied, from 0 (normal vision) to 100% (full expression). As the page puts it, severity is a dial on the model, not a milder diagnosis.

How do I check a whole palette?

Use **Palette audit**. It checks every pair of up to 24 colours under all 17 colour conditions and suggests lightness fixes for pairs that merge. See Audit a palette.

Are my images uploaded?

No. The page says: "Whatever you drop stays on your machine — the pipeline runs in this tab and nothing is uploaded."

Can I share what I am looking at?

Yes: **Copy link to this view** (Workbench), **Copy link** (Catalogue) or **Copy link to this sheet** (Compare). See Links, saving and exports.

How is this different from the Analyzer's Simulation section?

The Analyzer previews a whole web page under six filters. The Vision Lab goes much deeper: 78 conditions, severities, models, measurements, a palette audit and image input.

Can I automate it?

The palette audit is available from the REST API as `POST /v1/vision/audit`, at 5 API credits per call. See the public API documentation at https://auricartisan.com/library/documentation/api/.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Vision Lab: supporting material

The Vision Lab's glossary, shortcuts, limits, related tools, release notes and privacy notes.

This folder supports the Vision Lab guides with the terms they use, the keyboard shortcuts, an honest account of what the simulations are and are not, related tools, the history of changes, and what happens to your images.

File	What it covers
Glossary	Terms used in the Vision Lab docs
Keyboard shortcuts	Every key the lab responds to
Limits and accuracy	The models, what they approximate, and what the lab is not
Related tools and reading	Tools and articles that pair with the lab
Release notes	User-facing history from the public changelog
Privacy	What stays in your browser

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Vision Lab keyboard shortcuts

Every key the Vision Lab responds to.

The lab's single-key shortcuts work anywhere on the page except while you are typing in a field, and not with `Ctrl`, `Alt` or `Cmd` held. Press `?` to see them on screen.

Navigation

Key	Action
`/`	Focus the search field
`Esc`	Clear the search (while it has focus)
`?`	Show or hide the keyboard shortcuts overlay
`F`	Favourites
`A`	Analytics
`C`	Compare

Panels

Keys	Action
`R`	Open a random simulation
`T`	Tile all panels
`1` to `5`	Bring panel 1 to 5 forward
`Shift` + click, or `Ctrl` + click, on a card	Duplicate a panel that is already open

Cards and tiles

Keys	Where	Action
`Enter` or `Space`	A focused Catalogue card	Open its panel
`Left` / `Up`	A Compare tile's grip	Move the tile one place earlier
`Right` / `Down`	A Compare tile's grip	Move the tile one place later
`Home` / `End`	A Compare tile's grip	Move the tile to the start or end

For shortcuts that work across the whole site, see Keyboard shortcuts.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Vision Lab glossary

Plain-language definitions of the terms used by the Vision Lab.

Achromatopsia — Seeing no colour at all, only light and dark.

CAM (colour appearance model) — A model of how a colour looks in context: under a given light, surround and brightness. The panels offer CAT16, CAM16-UCS, CIECAM02 and Hunt-Stevens options.

CIE xy — A two-number description of a colour's hue and saturation (chromaticity), plotted on the CIE 1931 diagram.

CIEDE2000 (ΔE2000) — A formula for colour difference that follows human perception closely. Around 1 is barely noticeable.

Clinical default — The severity at which a condition is shown by default.

Collapse — Two colours that were clearly different becoming indistinguishable under a condition. The Workbench counts a reference pair as collapsed when it was 12 or more apart and falls within 8 (CIEDE2000).

Cone — A light-sensitive cell in the eye. L, M and S cones respond to long (red), medium (green) and short (blue) wavelengths.

Correlated colour temperature (CCT) — The colour of a white light, in kelvin: lower is warmer, higher is cooler.

CVD (colour-vision deficiency) — Reduced ability to tell some colours apart, often called colour blindness.

Deuteranopia, protanopia, tritanopia — Colour-vision deficiencies in which the green-, red- or blue-sensitive cones are absent.

Deuteranomaly, protanomaly, tritanomaly — Milder forms, in which those cones are present but weakened.

ΔE (delta E) — A measure of colour difference. **ΔE76** is a simple distance in Lab colour space; **ΔE2000** is the more perceptual CIEDE2000.

Field effect — A condition that changes how or where you see (blur, lost regions, glare, distortion) rather than changing each colour.

Illuminant — A standard light source used in colour calculations, such as D65 (average daylight).

Linear light — Colour values converted from screen encoding to values proportional to light, which the models need to work correctly.

LMS — The three cone responses (long, medium, short). Colour-vision models work by adjusting colours in LMS space.

Observer (2° or 10°) — The standard field of view assumed by colour maths: 2° for small or close things, 10° for large areas.

Per-colour transform — A condition in which each colour becomes another colour in a predictable way.

Prevalence — Roughly how common a condition is.

Severity — On the slider, how strongly the model is applied (0–100%). As a score, an estimated visual-impact rating from 1 to 10.

Specimen — What the Workbench stage shows: the Interface mock-up, the Palette strip or your Image.

White point — The colour treated as white under a given light.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Limits and accuracy

How the Vision Lab's simulations are made, how far to trust them, and what the lab is not.

What the simulations are

- **Educational visualisations.** The page states they are "not a medical simulation, a diagnostic tool, or a substitute for professional eye care". They show how a typical case of a condition might affect what you see, to help design and teaching.
- **Per-colour conditions** use published colour-vision models, calculated in linear light (colours are converted from screen values to values proportional to light before any maths):
  - **Brettel (1997)** projects colours onto the plane a dichromat can see, in LMS cone space.
  - **Viénot (1999)** is a simplified, single-plane version of the same idea.
  - **Machado (2009)** models reduced cone sensitivity, which suits the milder "-anomaly" forms and partial severities. The floating panels label their version "Machado-like". The models agree on the broad picture and differ in detail. Try more than one.
- **Field conditions** are illustrative effects (blur, dimming, lost regions, glare, distortion) drawn over the image. They show the kind of loss, not a measured map of any real patient's field.
- **Severity** is a dial on the model, not a milder diagnosis. The **severity score** (1 to 10) is an estimated visual-impact score that orders the collection; it is not a clinical grade.

What changes the result you see

- **Your screen.** Screen colours are not calibrated. Brightness, colour profile, night modes and panel type all change what you see.
- **The room.** Ambient light changes how colours look.
- **Your own vision.** If you have a colour-vision deficiency yourself, a simulation of another deficiency is filtered through your own.
- **Real people vary.** Two people with the same diagnosis can see differently.

Measurements

- **ΔE76** and **ΔE2000** are calculated between the original and simulated colours. ΔE2000 follows perception more closely.
- **Collapse** in the Workbench and Compare means a reference pair that was at least 12 apart (ΔE2000) falls within 8. Palette audit uses your chosen threshold, 5 by default.
- **Button label / fill** and **CTA** figures are WCAG contrast ratios, calculated on the simulated colours of the mock-up.
- **Palette audit** covers the 17 per-colour conditions only; field conditions are not colour transforms.

Images

- Images are processed at a working size set by **Quality** (1024, 1600 or 4096 pixels), not always at full resolution.
- Very large images take longer at the higher quality settings.

What the lab is not

- Not a vision test or a diagnosis.
- Not a guarantee that a design is accessible. Pair it with contrast checks and with testing by people who have these conditions.
- Not a substitute for professional eye care.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Vision Lab privacy

What the Vision Lab does with your colours and images, and what it keeps in your browser.

Your images stay on your device

The page says it plainly: "Whatever you drop stays on your machine — the pipeline runs in this tab and nothing is uploaded." Every simulation, measurement and audit runs in your browser.

What is kept in your browser

- the view you last used, and the Workbench's condition, severity, model, specimen and base colour;
- the Catalogue category, your favourites, the Compare sheet and the Palette audit palette;
- the panels you had open, their snapshots, and the position of the Vision Control dock.

Clearing this site's data in your browser removes all of it.

The daily count

On the free plan, the lab counts how many simulations you open from the Catalogue each day, and the count is also checked by Auric Artisan. Only the count is involved; your colours and images are not sent.

Links

A workspace link (`#v=`) and a Workbench link (`?vw=`) carry your settings in the address, readable by anyone who has the link. They never contain your own images.

The site-wide privacy policy is covered in Company and legal.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Vision Lab release notes

User-facing history of the Vision Lab, from the public changelog and the current page.

Dated entries come from the public changelog at https://auricartisan.com/library/changelog/. The first section describes the page as it is now.

Current page (September 2026)

- The lab opens on a **Workbench** that takes one condition apart, with **Interface**, **Palette** and **Image** specimens, a severity dial, a choice of Brettel, Viénot or Machado, and measurements of what collapses.
- Five views sit behind one bar: **Workbench**, **Catalogue**, **Compare**, **Palette audit** and **Analytics**.
- **Compare** is a contact sheet with a normal-vision control tile, ordering by impact or severity, and a choice of 3, 4 or 5 columns.
- **Palette audit** checks every pair of up to 24 colours under 17 colour conditions and suggests lightness fixes.
- **Analytics** shows the collection as a matrix of categories against severity, with JSON and CSV exports.
- Floating panels are arranged from a compact **Vision Control** dock.
- The medical notice appears once, at the foot of the lab.

v0.9.8 — 2 May 2026

- Vision Simulation moved into the Unified Simulation system as a panel-based workflow.

v0.9.6 — 13 April 2026

- Improved protanopia, deuteranopia and tritanopia simulation models, with revised coefficients validated against published datasets.
- A multi-model comparison view for side-by-side analysis.
- Expanded exports for reproducible experiments, with CSV and JSON outputs.

WEBSITE › TOOLS › ACCESSIBILITY AND VISION

Related tools and reading

Tools and articles that pair well with the Vision Lab.

Where	Why
Analyzer	Preview a whole web page under colour-vision filters, and see every text pair under four deficiencies in its A11y+ panel
Contrast Checker	Check the contrast of a pair once you have chosen new colours
Ishihara Plate Generator	Generate dot plates that show how colour-vision tests work
Accessibility Checklist	Checks c3 and c6 ask for exactly what the Vision Lab shows
Perception and vision tools	Other perception tools on the site
Learn	Plain-language articles on colour vision
Public API	Run the palette audit from code with `POST /v1/vision/audit`

Articles on auricartisan.com

- **Vision Simulation User Guide** — an earlier guide to the lab. The page has been redesigned since, so some names and layouts differ; this documentation describes the current page: https://auricartisan.com/library/documentation/articles/2026-05-24-vision-simulation-user-guide

WEBSITE › TOOLS › COLOUR WORKSPACE

Colour workspace

The Auric Artisan tools for building palettes and design systems, converting and studying colours, and choosing fonts and icons.

The colour workspace is the group of Auric Artisan website tools you use to make things with colour: palettes, gradients, harmonies and whole design systems, plus the fonts and icons that go with them. It also includes the site-wide Colour Tools, fourteen floating colour tools you can open over any page.

Every tool runs in your browser. Most work without an account; saving to your Library and some account features need you to sign in. Each tool has its own folder with an overview, task guides, a full reference, troubleshooting, a glossary and notes on accuracy and privacy.

Tools in this group

Tool	Address	What it is for	Docs
Basic Color Tools	https://auricartisan.com/tool/basic-tools/	A panel workspace with seven everyday tools (Palette Studio, Palette Remix, Gradient Maker, Color Picker, Color Battle, Collage Maker, Quick Converter) and the home of the fourteen site-wide Colour Tools.	basic-colour-tools/
Color Science Lab	https://auricartisan.com/tool/color-science-lab/	Colour-science models and analyses in floating panels: colour spaces, conversions, colour difference and more.	colour-science-lab/
Personalization Generator	https://auricartisan.com/tool/personalization-gen/	Sweeps thousands of candidate palettes built from your own colour pool and a brief, then derives gradients, type pairings, UI pairs, posters, design tokens and a sixteen-mockup brand Studio.	personalisation-generator/
Design System Generator	https://auricartisan.com/tool/design-engine/	Builds a complete design system (colour, type, icons, spacing, components, tokens) from an industry, a style and an accessibility target, and exports it.	design-engine/
ASCII Art Generator	https://auricartisan.com/tool/ascii/	Turns pictures and text into character art, with five looks, fine tuning, colour, and TXT, PNG, SVG, HTML, JSON, CSV and ANSI exports.	ascii-art-generator/
Font Library	https://auricartisan.com/tool/font/	Browse, test, compare and embed 2,213 font families from the Auric Font CDN, with suitability scores, pairings, licences and self-hosting kits.	font-library/
Icon Library	https://auricartisan.com/tool/icon/	Search 12,896 icons from seven open-licence sets, restyle them, and copy them as SVG, JSX, a data URI or a CDN tag, or export a ZIP or sprite.	icon-library/

Which tool should I use?

- **I need a palette quickly**: Palette Studio in Basic Color Tools.
- **I need a whole design system with tokens**: Design System Generator or Personalization Generator.
- **I need to check contrast while reading another page**: the Colour Tools, opened with the right-click menu.
- **I need colour science detail**: Color Science Lab, or Color Space Converter for one colour.
- **I need a typeface or icons**: Font Library and Icon Library.

Related

- Accessibility and vision tools
- Image and file tools
- Colour libraries
- Font CDN and Icon CDN

WEBSITE › TOOLS › COLOUR WORKSPACE

ASCII Art Generator

What the ASCII Art Generator is, who it is for, where to find it, and how to turn a picture or a word into character art and take it away as text, image or code.

The ASCII Art Generator turns a picture into character art: rows of letters, symbols, blocks or Braille dots whose density follows the light and dark of the image. You drop in a photo, a screenshot or a logo, or type a word, and the art appears straight away and updates as you change anything.

Five ready-made looks cover most needs. Behind **Fine tuning** sit the controls that decide how the picture is read: seven character sets or your own ramp, the cell shape, brightness model, gamma and contrast, three sampling methods, Floyd–Steinberg and Bayer dithering, edge-aware glyph choice, the output font, filters such as flip, rotate and frame, and a seed. The art can be shown as plain text, colour HTML, a canvas image or SVG.

Everything happens in your browser. Pictures are read and converted on your device and are not uploaded. You can copy the art, or download it as TXT, PNG, SVG, HTML, JSON, CSV or a colour ANSI file for terminals. An **Analysis** tab and a **Compare** tab measure the character sets and the render for anyone who wants to know why the art looks the way it does.

Who it is for

- Developers who want a banner for a README, a terminal or a commit message.
- Designers and artists making text-based artwork or retro graphics.
- Teachers and students exploring how images are sampled and dithered.
- Anyone who wants a picture of themselves, a pet or a logo in characters.

Where to find it

- Page: https://auricartisan.com/tool/ascii/ (the short address https://auricartisan.com/ascii also opens it).
- No sign-in is needed.
- The masthead links to the **Developer reference**, the technical write-up of the engine.

Quick start

1. Open https://auricartisan.com/tool/ascii/.
2. Select **Choose a file** and pick a photo, or drop one onto the tool, or paste a screenshot with `Ctrl` + `V` (`Cmd` + `V` on a Mac). No picture to hand? Select **Try an example**.
3. Under **Look**, try **Classic**, **Blocks**, **Fine**, **Bold** and **Line art**.
4. Move the **Detail** slider to set how many characters wide the art is.
5. Under **Colour**, choose **From image** to colour each character from the picture.
6. Select **Copy** under the art, or open the **Export** tab and download the format you need.

What you can do

- Convert an image (PNG, JPEG, GIF, WebP, BMP, SVG or TIFF, up to 20 MB) or a line of text. The page also has a webcam source, but the camera cannot currently be used on this site; see Choose a source.
- Pick a look, or fine-tune character set, shape, tone, sampling, dithering, edges, type and filters.
- Show the result as text, colour HTML, a canvas image or SVG, and view it full screen.
- Measure the character sets, the edge field, character frequencies and render times.

- Render a list of words in one go and time each one.
- Export as TXT, PNG, SVG, HTML, JSON, CSV and ANSI.

In this folder

- docs/ — how to use the generator.
  - Getting started
  - Choose a source
  - Looks, colour and detail
  - Fine tuning
  - Render modes and full screen
  - Charts, benchmark and Compare
  - Copy, export and batch
  - Reference
  - Troubleshooting
  - FAQ
- others/ — supporting material.
  - Glossary
  - Shortcuts
  - Limits and accuracy
  - Related
  - Privacy

WEBSITE › TOOLS › COLOUR WORKSPACE

ASCII Art Generator documentation

The list of guides for the ASCII Art Generator.

File	What it covers
getting-started.md	Open the page, learn the layout, make your first piece.
sources.md	Image, Text and Webcam sources; dropping, pasting and the example.
looks-colour-and-detail.md	The five looks, mono or colour, ink and paper, detail and invert.
fine-tuning.md	Character set, shape, tone, sampling, edges, type, filters and advanced settings.
render-modes.md	Text, HTML, Canvas and SVG renders; the readouts; full screen.
charts-and-compare.md	The Analysis tab's charts and benchmark, and the Compare tab's character set table.
export.md	Copy, every download format, Export all, and batch rendering.
reference.md	Every control, range, default and message.
troubleshooting.md	Symptoms, causes and fixes.
faq.md	Short answers to common questions.

Related

- ASCII Art Generator overview
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Getting started with the ASCII Art Generator

Open the ASCII Art Generator, learn its layout, and make and copy your first piece of character art.

Open the page

Go to https://auricartisan.com/tool/ascii/. The masthead reads **ASCII Art Generator**, *Drop in a picture and get character art back. Nothing is uploaded — every pixel is read and converted in this browser.*, with a row of labels and **Developer reference** →.

A short strip under it reads **Drop a picture**, **Pick a look**, **Take it away** and *It converts as you go.* Its **×** hides it for good on this browser.

The layout

Four tabs: **Lab**, **Analysis**, **Compare** and **Export**. You do almost everything in **Lab**.

Lab has two parts:

- **The control rail** on the left:
  - **Source**: **Image**, **Text** or **Webcam**, and the controls for the one you pick;
  - **Look**: five named looks, a strip showing the characters in use from **Light** to **Dark**, and their count, such as **10 glyphs**;
  - **Colour**: **Mono** or **From image**, and in Mono the **Ink** and **Paper** colours;

  - **Detail**: how many characters wide the art is, and **Invert for dark paper**;
  - **Fine tuning**: a button that opens every other setting, in folded groups.
- **The stage** on the right: **Render** chips (**Text**, **HTML**, **Canvas**, **SVG**) and the render time, the art itself, and a footer with **Grid**, **Chars**, **Render**, **Copy** and **Full screen**.

Select any group heading in the rail to fold it away or open it again.

On screens narrower than about 1,180 pixels the stage comes first and the controls follow below it.

The empty stage

Before you give it anything, the stage reads **Drop an image on the stage** and offers four buttons: **Choose a file**, **Try an example**, **Type some text** and **Use the webcam**. Under them, *Your ramp, right now* previews how the current character set draws a gradient.

Your first piece

1. Select **Try an example**. The tool draws a lit sphere and converts it.
2. Under **Look**, select **Blocks**, then **Line art**, and watch the art change.
3. Drag **Detail** to about 160.
4. Under **Colour**, select **From image**. The render switches to **HTML** and each character takes its colour from the picture.
5. Select **Full screen** to see it large, then close it with **×**.
6. Under **Colour**, select **Mono** again, and select **Copy**. The message **Copied!** confirms the text is on your clipboard.

Result: a piece of plain-text art ready to paste into a document, a chat or a code comment.

There is no Render button. The art converts as soon as you give it a source and again shortly after each change.

Next steps

- Choose a source
- Looks, colour and detail
- Copy, export and batch

Related

- ASCII Art Generator overview
- Reference

WEBSITE › TOOLS › COLOUR WORKSPACE

Charts, benchmark and Compare

The measurements, charts and benchmark in the Analysis tab, and the character set comparison in the Compare tab.

The **Analysis** and **Compare** tabs explain why the art looks the way it does. You do not need them to make art; they help when a result bands, looks flat or converts slowly.

The Analysis tab

This render

The left column repeats the facts of the last conversion:

- **Grid**, **Characters** and **Render** (time in milliseconds);
- **Character set**, and **Ramp** (the number of glyphs, for example **10 glyphs**);
- **Sampler** and **Dithering**.

Benchmark

Run benchmark converts the current source ten times with the current settings and reports the average, for example **Avg: 18.4 ms (10 runs)**, and the **Min**, **Max** and **Spread**. Use it to see which setting makes conversion slow. With the Image source and no picture loaded, it reads **Load an image first**.

The charts

Six charts. Each has a ⛶ button, and you can also double-click a chart, to open it full screen. Close the full-screen chart with ×, `Esc`, or by selecting the dark background.

Chart	What it shows	What to look for
Glyph density calibration	The ink of each character in the current set, measured by drawing it at 32 × 32 pixels, sorted light to dark.	A smooth rise. Big jumps or flat runs mean some tones have no matching character.
Character frequency	How often each character appears in the latest output.	A spike at one end means gamma or contrast is crushing the mid-tones.
Top character distribution	The twelve most frequent characters as a ring, with the rest grouped as one slice.	Whether a few characters dominate the art.
Sobel edge field	The strength of edges in the source picture; bright means a strong edge.	Where edge-aware glyph selection can act. Where it is dark, **Edge weight** has nothing to work on.
Sampling comparison	The same source through area averaging, bilinear and point sampling.	Point sampling breaking fine detail into noise.
Render timeline	Conversion time for the last hundred conversions.	Where live updates stop keeping up with the sliders.

The charts redraw after every conversion and when you open the tab.

The Compare tab

Measure all sets draws every character of all seven sets at 32 × 32 pixels, measures how much ink each one has, and fills a table:

Column	Meaning
Charset	The set: standard, detailed, blocks, minimal, symbols, lines or braille.
Glyphs	The number of characters, including the space.
Min density	The ink of the lightest character, from 0 to 1.
Max density	The ink of the darkest character.
Range	Max minus min. A wider range carries more contrast.

Column	Meaning
Monotonic	✓ if every character has at least as much ink as the one before it, in the set's own order; ✗ if not.

A set that is not monotonic has at least one character darker than the one after it, and a smooth gradient drawn with it shows bands. More glyphs carry more detail; a wider range carries more contrast. They are not the same thing.

The measurement uses the fonts on your device, so results can differ slightly between computers. Your custom ramp is not in the table; when it is in use, its densities show in the **Glyph density calibration** chart.

Related

- Fine tuning
- Limits and accuracy
- Glossary

WEBSITE › TOOLS › COLOUR WORKSPACE

Copy, export and batch

Copy the art, download it as TXT, PNG, SVG, HTML, JSON, CSV or ANSI, export several files at once, and render a list of words in one go.

Every file is built in your browser and handed straight to your downloads. Nothing is uploaded.

Copy

- **Copy** under the art copies the plain characters. The message reads **Copied!**
- **Copy for terminal** in the **ANSI** card copies the art with colour codes, for pasting into a terminal. The message reads **Copied with colour — paste into a terminal**, or just **Copied** if colour is off.

Paste plain text into a place that uses a monospaced font (a code block, a terminal, a text editor), or the columns will not line up.

Download one format

Open the **Export** tab. Each card has a download button.

Card	Button	File	What it contains
TXT	**Download .txt**	ascii-art.txt	The characters, one line per row, with filters applied.
PNG	**Download .png**	ascii-art.png	A picture of the art.
SVG	**Download .svg**	ascii-art.svg	Vector art with selectable characters.
HTML	**Download .html**	ascii-art.html	A standalone web page. The only download that keeps colour as coloured text.
JSON	**Download .json**	ascii-research.json	The characters and every setting that produced them.
CSV	**Download .csv**	ascii-analysis.csv	How often each character appears.
ANSI	**Download .ans**	ascii-art.ans	The characters with 24-bit colour codes for terminals.

A message confirms each download, for example **TXT exported**. If nothing has been converted yet, it reads **Nothing to export**.

Match the download to the render

The PNG, SVG and HTML downloads are built from a render, so choose the matching **Render** chip on the **Lab** tab first:

For	Show this render first	Otherwise
PNG	**Canvas**	The PNG is drawn from the last HTML render if there has been one, or from the text using the output font, **Ink** and **Paper**.
SVG	**SVG**	The SVG reuses the last SVG render or SVG download if there has been one, or wraps the last HTML render, or draws the text in black on white.
HTML	**HTML**	The page contains the last HTML render if there has been one, or the plain text in black on white.

Without that step, a PNG, SVG or HTML download can show an earlier version of the art. See Troubleshooting.

The HTML file

With an HTML render, the file is a page on a black background holding the coloured characters on your paper colour. It opens in any browser.

The JSON file

It records the time, the source (image, text or webcam), the render mode, the characters in the ramp, every option, the text of the art, the timings of the last twenty conversions and the glyph densities. It does not include the source picture. The generator cannot load a JSON file back in; keep it as a record of the settings, or use it in your own code.

The CSV file

Three columns, **Character**, **Count** and **Frequency** (per cent), sorted from the most common character down. Line breaks are not counted. The card on the page also mentions density per glyph, but the file holds frequencies only; the densities are in the JSON file.

The ANSI file

A text file whose colour runs are 24-bit terminal colour codes, with a reset at the end of every line. Print it in a terminal that supports true colour (for example with `cat ascii-art.ans` on macOS or Linux) and the picture appears in colour.

It needs **From image** colour. With **Mono**, **Download .ans** shows **Turn colour on first — ANSI carries the image's colours** and saves nothing. After a successful download the message reads **Saved ascii-art.ans — cat it in a terminal**.

Export several files at once

Export all six, in the left column of **Export**, downloads five files one after another: the TXT, PNG, SVG, JSON and CSV files. It does not include the HTML or ANSI files; download those from their cards. Your browser may ask whether to allow several downloads from the page.

Batch: render a list of words

The **Batch** box takes one line of text per row, for example:

```
AURIC
ARTISAN
ASCII
```

Render each line converts each line as **Text** source, with the current text font and conversion settings, and fills **Batch results** with a table of **Input** (the first 30 characters), **Cols**, **Rows**, **Chars** and **Time (ms)**. With nothing in the box it reads **No inputs**.

Batch measures; it does not produce files or change the art on the stage. It does leave the last line of the list in the **Text** box on the **Lab** tab.

Save to your Library

On this page, the Library's **Save to Library** stores the current art and your settings in your Auric Artisan Library, and opening the item later puts them back. See the Library Kit.

Related

- Render modes and full screen
- Troubleshooting
- Privacy

WEBSITE › TOOLS › COLOUR WORKSPACE

Fine tuning

Every setting behind the Fine tuning button — character set and custom ramp, shape, tone, sampling and dithering, edges, type, filters and advanced options.

Select **Fine tuning** at the bottom of the rail to open eight more groups. Each starts folded; select a group's heading to open it. Every change re-converts the art a moment later.

Changing anything here clears the highlight on the **Look** buttons, since the art no longer matches a named look.

Character set

A menu of seven sets, each listed light to dark:

Set	Characters	Count
Standard	space `.:-=+*#%@`	10
Detailed	space and 69 punctuation marks, letters and digits	70
Blocks	space ` `	5
Minimal	space `.:#@`	5
Symbols	space `·••◉●`	6
Lines	space and box-drawing characters `─│┌┘└` and more	23
Braille	space and all 255 Braille patterns	256

Custom ramp lets you type your own characters, lightest first and darkest last, up to 200. It needs at least two characters; with fewer, the set chosen in the menu is used. Start with a space so the lightest areas stay empty, for example `.oO@`.

The **Look** strip in the rail always shows the set actually in use, custom ramp included.

Shape

Character cells are taller than they are wide, so without correction the art comes out stretched.

Control	Range	Default	What it does
Match the output font	on or off	on	Measures the output font and sets **Cell aspect** to match it.
Cell aspect	0.2 to 1.2	set by the font	Width of a cell divided by its height. Lower values give fewer rows.
Lock row count	on or off	off	Fixes the number of rows instead of working it out from the picture.
Rows	10 to 300	50	The fixed row count, when locked.

Moving **Cell aspect** by hand works best with **Match the output font** off; while it is on, choosing another output font measures it again.

Tone

Control	Range	Default	What it does
Brightness model	**Luminosity (Rec.601)**, **Average**, **Lightness (HSL)**	Luminosity	How a pixel's colour becomes a single brightness. Luminosity weights green most, as the eye does.
Gamma	0.2 to 3.0	1.0	Above 1 lifts the mid-tones (lighter); below 1 deepens them.
Contrast	0 to 1	0	Stretches the picture's contrast before conversion.
Saturation	0 to 2	0	Boosts colour before conversion. Mainly affects **From image** colour.

Sampling

Control	Options	Default	What it does
Sampling method	**Area average (SAT)**, **Bilinear**, **Point / Nearest**	Area average	How the pixels under each cell become one value. Area averaging uses every pixel and is the most faithful; point sampling is fastest but can shimmer on fine detail.
Dithering	**None**, **Floyd-Steinberg**, **Ordered (Bayer)**	None	Spreads the rounding error between neighbouring cells so smooth gradients show as texture instead of bands.
Bayer matrix	**Bayer 4×4**, **Bayer 8×8**	8×8	The pattern size for ordered dithering.

Floyd–Steinberg gives an organic, grainy texture; ordered dithering gives a regular cross-hatch.

Edges

Control	Range	Default	What it does
Sobel glyph selection	on or off	on	Finds edges in the picture and, at strong edges, swaps in a thin line character of similar brightness.
Edge weight	0 to 1	0.65	How strongly edges steer the choice of character.

The line characters it can use are the backtick, `-`, `/`, `|` and the backslash, and only those present in the current set. So it has most effect with **Detailed**, a little with **Standard** (which has `-`), and none with Blocks, Minimal, Symbols, Lines or Braille. The character is chosen by brightness, not by the direction of the edge.

The **Sobel edge field** chart in **Analysis** shows where the edges are.

Type

Control	Range	Default	What it does
Output font	monospace, Courier New, Consolas, Monaco	monospace	Font of the HTML, Canvas and SVG renders and of downloads.
Size	6 to 32 pixels	12	Character size for the Canvas and SVG renders and downloads.
Line height	0.8 to 2.0	1.0	Row spacing.
Kerning	−2 to 4	0	Extra space between characters, in pixels.
Render mode	**Text (plain)**, **HTML (color spans)**, **Canvas (rasterised)**, **SVG (vector)**	Text	The same choice as the **Render** chips above the art.

On screen, the Text and HTML renders are scaled to fit the stage whatever size you choose.

Filters

Eight check boxes, applied to the finished characters:

Filter	Effect
Flip X	Mirrors left to right.
Flip Y	Mirrors top to bottom.
Rotate 90°	Turns the art a quarter turn clockwise.
Border	Surrounds the art with `#`.
Frame	Surrounds the art with a double-line box.
Brighten	Moves each Standard-set character one step towards `@`.
Darken	Moves each Standard-set character one step towards a space.
Noise	Replaces a share of the characters with random Standard-set characters.

Noise amount (0 to 1, default 0.1) sets that share.

Filters work on the characters, not the picture, so:

- **Brighten**, **Darken** and **Noise** use the Standard characters; with other sets, Brighten and Darken leave characters they do not know unchanged, and Noise mixes Standard characters in;
- the **HTML** render does not show filters at all, although **Copy** and the TXT download include them;
- with **From image** colour, shape-changing filters (**Rotate 90°**, **Border**, **Frame**) make the Canvas and SVG renders fall back to one colour, and the flips can leave colours in their original places.

Advanced

Control	Range	Default	What it does
Thin out	0 to 0.95	0	Drops characters from runs of the same character. Lines become shorter, so the picture skews; use it for effect, not to save space.

Control	Range	Default	What it does
Seed	a whole number	2654435769	Fixes the random choices of **Noise** and the position of the ordered-dither pattern, so the same seed gives the same result.

Related

- Looks, colour and detail
- Charts, benchmark and Compare
- Reference

WEBSITE › TOOLS › COLOUR WORKSPACE

Looks, colour and detail

The five named looks, mono or colour art, ink and paper, the Detail slider and Invert for dark paper.

The three groups under **Source** are the ones most people need: **Look**, **Colour** and **Detail**.

Look

Five buttons. Each sets a character set and a few tone and edge settings at once, then re-converts the art.

Look	Character set	What it sets
Classic	Standard	Gamma 1, contrast 0.15, edge-aware on, edge weight 0.65
Blocks	Blocks	Gamma 1, contrast 0.2, edge-aware off
Fine	Detailed	Gamma 1, contrast 0.1, edge-aware on, edge weight 0.5
Bold	Minimal	Gamma 0.9, contrast 0.45, edge-aware off
Line art	Standard	Gamma 1.4, contrast 0.6, edge-aware on, edge weight 1

Classic is highlighted when the page opens, but its contrast is not applied until you select it. If you change anything in **Fine tuning**, the highlight clears, because the art no longer matches a named look.

Under the buttons, a strip shows the characters in use from **Light** to **Dark**, and the heading counts them, for example **10 glyphs**. Sets longer than 24 characters are shown as an even sample of 24.

Choosing a look

- **Classic** suits most photos.
- **Blocks** gives solid, graphic shading and works well at small sizes.
- **Fine** uses a long ramp of 70 characters and keeps the most tonal detail at large widths.
- **Bold** uses only four marks and a space, for strong, poster-like art.
- **Line art** pushes contrast and edges, for logos, drawings and outlines.

Colour

Two buttons:

- **Mono**: every character in one colour. The **Ink** and **Paper** colour wells appear. The render switches to **Text**.

- **From image**: each character takes the colour of the part of the picture it stands for. **Ink** and **Paper** are hidden, and the render switches to **HTML**, which can show colour.

After choosing **From image** you can also switch to the **Canvas** or **SVG** render; both keep the colour. See Render modes and full screen.

Ink and paper

Ink defaults to #f0f0f0 (near white) and **Paper** to #0f0e0c (near black).

- **Paper** is the background of the **HTML**, **Canvas** and **SVG** renders, and of their downloads.
- **Ink** is the character colour of the **Canvas** and **SVG** renders in Mono.
- A PNG download made while the **Text** render is showing uses both.
- The **Text** render on screen ignores them and uses the page's own colours, and a Mono **HTML** render uses the page's text colour on your paper.

With **From image**, the paper colour is still used as the background.

Detail

The **Detail** slider sets the width of the art in characters, from 20 to 300 (default 100); the heading shows it, for example **100 wide**. The number of rows follows from the picture's proportions and the cell shape, unless you lock the row count in **Fine tuning** (see Fine tuning).

- 60 to 120 columns suit a README, a chat message or a terminal window.
- 150 to 300 columns keep more detail but need a small font or a wide screen to look right.

On screen the art is scaled to fit the stage, so a wider setting shows smaller characters, not a larger picture.

Invert for dark paper

The character sets run from the least ink (a space) to the most. On light paper, a dark part of the picture should get the most ink; on dark paper it should get the least. **Invert for dark paper** flips the mapping.

The tool sets it for you: it turns on while **Paper** is dark and off while it is light. Once you select the box yourself, it stays as you set it.

Related

- Fine tuning
- Render modes and full screen
- Reference

WEBSITE › TOOLS › COLOUR WORKSPACE

Render modes and full screen

The Text, HTML, Canvas and SVG renders, the readouts under the art, and the full-screen view.

The **Render** chips above the art choose how it is shown. The characters are the same in every mode; what changes is how they are drawn and what they can carry.

The four renders

Chip	Shows the art as	Colour	Filters	Best for
Text	Plain characters in the page's text colour	No	Yes	Copying and pasting
HTML	Characters in coloured runs on your paper colour	Yes	No	Colour art on screen and the HTML download
Canvas	A picture drawn at the output font and size	Yes	Yes	A pixel-exact PNG
SVG	Vector text, one line per row	Yes	Yes	Scalable art and the SVG download

- **Colour** means the render can show **From image** colour. See Looks, colour and detail.
- **Filters** means the render shows the filters from **Fine tuning**. The HTML render always shows the art before filters.
- Choosing **From image** switches to **HTML**, and choosing **Mono** switches back to **Text**. You can pick another chip afterwards.

The Text and HTML renders are scaled to fit the stage. The Canvas and SVG renders are drawn at the **Size** set under **Type** in **Fine tuning**, and shrink to fit the stage width if they are wider.

The readouts

The footer under the art shows:

- **Grid**: columns × rows, for example **100 × 42**;
- **Chars**: the number of characters;
- **Render**: how long the last conversion took, in milliseconds.

The render time also appears beside the **Render** chips. The **Analysis** tab repeats these figures with the character set, glyph count, sampler and dithering. See Charts, benchmark and Compare.

Copy

Copy in the footer copies the plain characters to your clipboard, with filters applied, whichever render is showing. The message **Copied!** confirms it. To copy with colour for a terminal, use **Copy for terminal** in **Export**. See Copy, export and batch.

Full screen

Full screen opens the art over the whole window on a dark background.

- For the Canvas render, it shows the canvas picture.
- For the other renders, it shows the plain characters in gold, without colour.

Close it with **×** in the top-right corner.

Related

- Copy, export and batch
- Fine tuning
- Limits and accuracy

WEBSITE › TOOLS › COLOUR WORKSPACE

Choose a source

Convert an image, a line of text or a webcam frame, and every way to get a picture into the ASCII Art Generator.

The **Source** group at the top of the rail has three buttons: **Image**, **Text** and **Webcam**. Switching source converts the new source at once if it has something to show, or returns to the empty stage if it has not.

Image

Use a photo, screenshot, logo or drawing.

Ways to load one:

- select the drop area (*Drop an image, or click to browse*) or **Choose a file** on the empty stage, and pick a file;
- drag a file anywhere onto the tool; the stage highlights while you drag;
- copy an image and press `Ctrl` + `V` (`Cmd` + `V` on a Mac) anywhere on the page, as long as you are not typing in a field;
- select **Try an example** to load a built-in picture of a lit sphere on a dark gradient.

Accepted types are PNG, JPEG, GIF, WebP, BMP, SVG and TIFF, up to 20 MB. Whether TIFF opens depends on your browser.

After loading, the rail shows the picture's size, such as **1200 × 800 px**. Pictures larger than 4,000 pixels on a side, or larger than 12 megapixels, are scaled down first, and a note says so, for example **Downscaled from 6000×4000 to 4000×2667**. The art never needs that much detail, so you will not normally see a difference.

Clear removes the picture and returns the stage to its empty state.

An animated GIF is converted as a single still frame.

Text

Select **Text** (or **Type some text** on the empty stage) to turn a word or short line into art.

Control	What it does	Default
Text box	The text to draw.	ASCII
Font	monospace, Courier New, Consolas, Monaco, Fira Code or Source Code Pro.	monospace
Size	8 to 128 pixels.	48
Weight	**Normal**, **Bold** or **Black**.	Bold
White on black	Draws light letters on a dark ground instead of dark on light.	off

The text is first drawn as a picture in the chosen font, then converted like an image. The font must be installed on your device; if it is not, your browser substitutes another. Bigger sizes give smoother letter shapes; set **Detail** wide enough to hold them.

If the box is empty, the stage returns to its empty state.

Webcam

Select **Webcam** (or **Use the webcam** on the empty stage) to see the camera controls.

At the time of writing, the camera cannot be used on auricartisan.com. The site tells browsers not to give any of its pages camera access, so **Start** does not turn the camera on, and a message under the buttons says that camera permission was

denied. Changing the camera permission in your browser does not help. To convert a picture of yourself, take a photo with your camera app and load it with **Image**.

For reference, the controls are:

Control	What it is for
Start	Ask for the camera and show a small live preview.
Capture	Convert the current frame.
Keep converting every frame	Convert continuously, several times a second.
Smooth between frames	Blend each frame a little with the one before, to reduce flicker.
Stop	Turn the camera off.

Related

- Looks, colour and detail
- Troubleshooting
- Privacy

WEBSITE › TOOLS › COLOUR WORKSPACE

ASCII Art Generator reference

Every control, range, default, file and message in the ASCII Art Generator in one place.

Address

- https://auricartisan.com/tool/ascii/
- https://auricartisan.com/ascii opens the same page.

Tabs

Tab	Purpose
Lab	Source, look, colour, detail, fine tuning, and the art.
Analysis	Readouts, benchmark and six charts.
Compare	Ink measurements for all seven character sets.
Export	Seven download formats, Export all, and batch rendering.

Source

Control	Values	Default
Source	**Image**, **Text**, **Webcam**	Image
Image file	PNG, JPEG, GIF, WebP, BMP, SVG, TIFF; up to 20 MB	none
Large images	Scaled to at most 4,000 px a side and 12 megapixels	always
Text	any text	ASCII

Control	Values	Default
Text font	monospace, Courier New, Consolas, Monaco, Fira Code, Source Code Pro	monospace
Text size	8 to 128 px	48
Text weight	Normal, Bold, Black	Bold
White on black	on or off	off
Keep converting every frame	on or off	off
Smooth between frames	on or off	off

Look, colour and detail

Control	Values	Default
Look	**Classic**, **Blocks**, **Fine**, **Bold**, **Line art**	Classic highlighted
Colour	**Mono**, **From image**	Mono
Ink	any colour	#f0f0f0
Paper	any colour	#0f0e0c
Detail	20 to 300 columns	100
Invert for dark paper	on or off	follows the paper

Fine tuning

Group	Control	Values	Default
Character set	Set	Standard, Detailed, Blocks, Minimal, Symbols, Lines, Braille	Standard
Character set	**Custom ramp**	2 to 200 characters, light to dark	off
Shape	**Match the output font**	on or off	on
Shape	**Cell aspect**	0.2 to 1.2	set by the font
Shape	**Lock row count**	on or off, 10 to 300 rows	off, 50
Tone	Brightness model	Luminosity (Rec.601), Average, Lightness (HSL)	Luminosity
Tone	**Gamma**	0.2 to 3.0	1.0
Tone	**Contrast**	0 to 1	0
Tone	**Saturation**	0 to 2	0
Sampling	Method	Area average (SAT), Bilinear, Point / Nearest	Area average
Sampling	Dithering	None, Floyd-Steinberg, Ordered (Bayer)	None
Sampling	Bayer matrix	4×4, 8×8	8×8
Edges	**Sobel glyph selection**	on or off	on
Edges	**Edge weight**	0 to 1	0.65
Type	Output font	monospace, Courier New, Consolas, Monaco	monospace
Type	Size	6 to 32 px	12

Group	Control	Values	Default
Type	**Line height**	0.8 to 2.0	1.0
Type	**Kerning**	−2 to 4 px	0
Type	Render mode	Text, HTML, Canvas, SVG	Text
Filters	Check boxes	Flip X, Flip Y, Rotate 90°, Border, Frame, Brighten, Darken, Noise	all off
Filters	**Noise amount**	0 to 1	0.1
Advanced	**Thin out**	0 to 0.95	0
Advanced	**Seed**	whole number	2654435769

Character sets

Set	Characters (light to dark)	Count
Standard	space `.:-=+*#%@`	10
Detailed	space and 69 marks, letters and digits	70
Blocks	space ` `	5
Minimal	space `.:#@`	5
Symbols	space `·••◉⬤`	6
Lines	space and box-drawing characters	23
Braille	space and 255 Braille patterns	256

Stage

Item	What it does
Render chips	**Text**, **HTML**, **Canvas**, **SVG**
Grid	Columns × rows
Chars	Number of characters
Render	Time of the last conversion
Copy	Copies the plain characters
Full screen	Shows the art over the whole window

Files

File	From
ascii-art.txt	**Download .txt**, Export all
ascii-art.png	**Download .png**, Export all
ascii-art.svg	**Download .svg**, Export all
ascii-art.html	**Download .html**

File	From
ascii-research.json	**Download .json**, Export all
ascii-analysis.csv	**Download .csv**, Export all
ascii-art.ans	**Download .ans**

Messages

Message	Meaning
Rendering…	A conversion is in progress.
Unsupported file type: …	The file is not one of the accepted image types.
File too large (… MB). Max 20 MB.	The file is over 20 MB.
Failed to load image	The browser could not read the file.
Downscaled from … to …	A large picture was scaled down before conversion.
Copied!	The art is on the clipboard.
Nothing to copy	Nothing has been converted yet.
Nothing to export	Nothing has been converted yet.
Turn colour on first — ANSI carries the image's colours	ANSI download needs **From image**.
Load an image first	The benchmark needs a picture.
No inputs	The batch box is empty.
Camera permission denied. …	The camera could not start. See Troubleshooting.

Related

- Fine tuning
- Glossary
- Shortcuts

WEBSITE › TOOLS › COLOUR WORKSPACE

Troubleshooting

Fixes for common problems in the ASCII Art Generator, from files that will not load to downloads that do not match the art.

Nothing happens when I open the page

That is expected: the stage waits for a source. Choose a file, drop or paste a picture, select **Try an example**, or switch to **Text**. There is no Render button; conversion starts by itself.

My file will not load

- **Unsupported file type**: use PNG, JPEG, GIF, WebP, BMP, SVG or TIFF. Convert other formats, such as HEIC, first.
- **File too large**: the limit is 20 MB. Save a smaller copy or a JPEG.

- **Failed to load image**: the file may be damaged, or your browser cannot read that format. TIFF in particular opens only in some browsers.

Pasting a picture does nothing

Click on an empty part of the page first, so you are not typing in a field, then press `Ctrl` + `V` (`Cmd` + `V` on a Mac). The clipboard must hold an image, not a link to one.

The webcam will not start

At the time of writing, auricartisan.com tells browsers not to allow camera access on any of its pages. **Start** then shows a message that camera permission was denied. Changing permissions in your browser does not help. Take a photo with your camera app and load it with **Image** instead.

The art looks stretched or squashed

- Leave **Match the output font** on in **Fine tuning › Shape**, so the cell shape matches the font.
- If you pasted the art somewhere else, use a monospaced font there too; a proportional font breaks the columns.
- If **Lock row count** is on, the row count no longer follows the picture. Turn it off.

The art looks like a negative

The characters with the most ink should stand for the dark parts on light paper, and the light parts on dark paper. Check **Invert for dark paper**. Once you have changed it yourself, it no longer follows the paper colour automatically.

The art is too dark, too light or flat

- Try **Gamma**: above 1 lightens the mid-tones, below 1 darkens them.
- Raise **Contrast**.
- Look at **Character frequency** in **Analysis**: a spike at one end means the tones are crushed.
- Try the **Line art** or **Bold** look for strong shapes, or **Fine** for subtle tones.

Smooth areas show bands

Turn on **Floyd-Steinberg** or **Ordered (Bayer)** dithering, or choose a longer set such as **Detailed**. The **Compare** tab shows which sets are not ordered light to dark.

Colour does not show

- Under **Colour**, choose **From image**. Colour shows in the **HTML**, **Canvas** and **SVG** renders, not in **Text**.
- With **Rotate 90°**, **Border**, **Frame** or **Thin out** on, the Canvas and SVG renders fall back to one colour. Turn those off for colour art.

Filters do nothing

The **HTML** render shows the art before filters. Switch to **Text**, **Canvas** or **SVG**. **Brighten**, **Darken** and **Noise** work on Standard-set characters, so with other sets they change little or mix Standard characters in.

The PNG, SVG or HTML download does not match the art

Those downloads can reuse an earlier render. Before downloading, select the matching render on the **Lab** tab: **Canvas** for PNG, **SVG** for SVG, **HTML** for HTML. Then download again.

The ANSI download says to turn colour on

The ANSI file carries the picture's colours, so it needs **From image**. For plain characters, use the TXT download or **Copy**.

Export all gave me five files, not six

That is how it works at present: **Export all six** downloads TXT, PNG, SVG, JSON and CSV. Download the HTML and ANSI files from their own cards. If only one file arrived, allow the page to download several files when your browser asks.

Full screen shows no colour

Full screen shows the plain characters for every render except **Canvas**. Switch to **Canvas** first to see the colour art full screen. Close full screen with **×**.

The Analysis charts are empty

Convert something first; the charts describe the last conversion. If they are still blank, switch to another tab and back.

Conversion is slow or lags behind the sliders

Lower **Detail**, switch **Sampling** to **Bilinear** or **Point / Nearest**, turn off dithering, or use the **Text** render. The **Render timeline** chart and **Run benchmark** show the effect.

Related

- FAQ
- Limits and accuracy
- Help centre

WEBSITE › TOOLS › COLOUR WORKSPACE

Frequently asked questions

Short answers to common questions about the ASCII Art Generator.

Are my pictures uploaded?

No. Pictures are read and converted in your browser and never leave your device. Downloads are built on your device too. See Privacy.

Do I need an account?

No. The generator works without signing in.

Is it really ASCII?

Only the **Standard**, **Detailed** and **Minimal** sets use plain ASCII characters. **Blocks**, **Symbols**, **Lines** and **Braille** use Unicode characters, which need a font that has them and may not display in every terminal or app.

Which settings give the best result?

Start with a look: **Classic** for photos, **Line art** for logos and drawings, **Blocks** for small, bold art, **Fine** for large, detailed art. Then adjust **Detail** for the space you have. Pictures with a clear subject and strong contrast convert best.

How wide should the art be?

It depends on where it goes. A README or chat message fits 60 to 100 columns; a terminal is often 80 to 120 columns wide; a poster or a PNG can go up to 300.

Can I keep the colours?

Yes, with **From image**. Colour survives in the HTML, PNG (from the Canvas render), SVG and ANSI downloads. Plain text and **Copy** carry no colour; **Copy for terminal** does.

Can I make animated ASCII art?

Not on this page. An animated GIF is converted as one still frame, and the webcam source cannot currently be used on the site.

Can I convert text into big letters?

Yes. Choose **Text** as the source, type a word, and pick a font, size and weight. The word is drawn and then converted. **Batch** on the **Export** tab measures a whole list of words.

Why does my text look wrong when I paste it?

The place you pasted into is probably using a proportional font. Put the art in a code block or set a monospaced font there.

Can I load a JSON file back in?

No. The JSON records the settings and the art, but the page has no import. You can save the art and settings to your Library instead; see Copy, export and batch.

Does it work on a phone?

Yes. On narrow screens the art comes first and the controls follow. Pasting and dragging depend on the phone's browser; **Choose a file** always works.

Related

- Troubleshooting
- Getting started

WEBSITE › TOOLS › COLOUR WORKSPACE

Supporting material

The glossary, shortcuts, limits, related tools and privacy notes for the ASCII Art Generator.

File	What it covers
glossary.md	Terms used by the generator.
shortcuts.md	Keyboard and mouse shortcuts.
limits-and-accuracy.md	Known limits and what the measurements mean.
related.md	Tools and reading that go with the generator.
privacy.md	What stays in your browser and what is remembered.

Related

- ASCII Art Generator overview
- Documentation contents

WEBSITE › TOOLS › COLOUR WORKSPACE

Shortcuts

Keyboard and mouse shortcuts in the ASCII Art Generator.

Shortcut	Where	What it does
`Ctrl` + `V` (`Cmd` + `V` on a Mac)	Anywhere on the page, when not typing in a field	Loads an image from the clipboard.
Drag a file onto the tool	**Lab**	Loads the image.
`Enter` or `Space`	On a group heading in the rail	Folds or opens the group.
Double-click a chart	**Analysis**	Opens the chart full screen.
`Esc`	Full-screen chart	Closes it.

The full-screen view of the art closes with × only.

Related

- Getting started
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Glossary

Terms used in the ASCII Art Generator and its documentation.

Term	Meaning
ANSI colour codes	Escape sequences that terminals read as colour instructions. The ANSI download uses 24-bit (true colour) codes.
Area average	A sampling method that averages every pixel under a character cell. The page labels it SAT, after the summed-area table that makes it fast.

Term	Meaning
Bayer matrix	A fixed grid of thresholds used for ordered dithering, here 4 × 4 or 8 × 8.
Bilinear sampling	Reads a cell's value by blending the four nearest pixels.
Braille pattern	One of the 256 characters made of up to eight raised dots, used here as fine texture.
Cell aspect	A character cell's width divided by its height. Used to keep the art in proportion.
Character set, ramp	The characters the art is drawn with, ordered from least ink to most.
Density	How much of a character's cell is covered by ink, from 0 to 1.
Dithering	Spreading the rounding error between cells so that smooth tones appear as texture rather than bands.
Edge-aware selection	Swapping in a thin line character where the picture has a strong edge.
Floyd–Steinberg	An error-diffusion dithering method that passes each cell's error to its neighbours.
Gamma	A curve applied to brightness before conversion. Here, above 1 lightens the mid-tones.
Glyph	A single character as drawn by a font.
Luminosity (Rec.601)	A brightness formula that weights red, green and blue as the eye does, green most.
Monospaced font	A font in which every character has the same width, so columns line up.
Monotonic	Of a character set: each character has at least as much ink as the one before it.
Point sampling	Reads a cell's value from a single pixel. Fastest, but can shimmer on fine detail.
Sobel operator	A standard way to measure how quickly brightness changes, used to find edges.
True colour	24-bit colour, about 16.7 million colours.

Related

- Reference
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Limits and accuracy

Known limits of the ASCII Art Generator and what its measurements can and cannot tell you.

Size limits

Limit	Value
Image file	20 MB
Image size used	Up to 4,000 pixels a side and 12 megapixels; larger pictures are scaled down
Width of the art	20 to 300 characters
Locked row count	10 to 300 rows
Custom ramp	2 to 200 characters
Render timeline	The last 100 conversions

Sources

- The webcam source cannot currently be used on auricartisan.com, because the site does not allow camera access on its pages.
- An animated GIF is converted as a single still frame.
- TIFF opens only in browsers that can display TIFF images.
- The **Text** source draws with fonts installed on your device; a missing font is replaced by another.

Fonts decide how the art looks

Character art depends on a monospaced font. The same characters look different in different fonts, and wrong in a proportional one. **Match the output font** measures the chosen output font to keep proportions right, but the place you paste the art into uses its own font.

The glyph density measurements in **Analysis** and **Compare** are made by drawing characters with the fonts on your device, so they can differ slightly from one computer to another.

Colour

- **Text** and **Copy** carry no colour. Colour needs the HTML, Canvas or SVG render, or the ANSI file.
- The **HTML** render shows the art before filters and **Thin out**.
- With colour on, filters that change the shape (**Rotate 90°**, **Border**, **Frame**) and **Thin out** make the Canvas, SVG and ANSI output fall back to one colour, and **Flip X** and **Flip Y** do not move the colours with the characters.

Filters

Brighten, **Darken** and **Noise** are built around the Standard characters. With other sets, Brighten and Darken change only the characters they know, and Noise adds Standard characters. **Thin out** removes repeated characters, so rows become uneven and the picture skews.

Edge-aware selection

Edge-aware selection can only use the line characters in the current set, and chooses by brightness, not by the direction of the edge. It has no effect with the Blocks, Minimal, Symbols, Lines or Braille sets.

Downloads

- The PNG, SVG and HTML downloads can reuse an earlier render unless the matching render is showing. See Copy, export and batch.
- **Export all six** downloads five files; the HTML and ANSI files are separate.
- The SVG made from an HTML render embeds web content inside the SVG, which some design apps cannot open. The SVG made from the **SVG** render uses plain text elements.
- The JSON file does not contain the source picture and cannot be loaded back into the page.

The benchmark

Run benchmark times ten conversions in your browser. Timings depend on your device, your browser and whatever else it is doing, so compare settings on the same device in the same session rather than across devices.

Related

- Troubleshooting
- Fine tuning
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Privacy

What the ASCII Art Generator keeps in your browser and what uses the network.

Stays in your browser

Pictures you load, paste or drop are read and converted on your device. They are not uploaded. Text you type is drawn and converted the same way. Every download is built in your browser.

Remembered

- In your browser's local storage: whether you dismissed the **Drop a picture · Pick a look · Take it away** strip.
- In the page address: like other pages on the site, the generator keeps the values of its controls in the address after `#p=`, so a reload restores them. Your picture is not part of the address. If you share the address, the other person gets your settings, not your picture. See Shareable links.

Uses the network

- Loading the page and its scripts.
- The Library's **Save to Library**, if you use it, stores the art and settings in your Auric Artisan Library. See the Library Kit.
- Site-wide analytics and advertising follow your cookie choices. See the Help centre.

Related

- Limits and accuracy
- Copy, export and batch

WEBSITE › TOOLS › COLOUR WORKSPACE

Related tools and reading

Tools and reading that go with the ASCII Art Generator.

Tools

- Image and file tools: prepare, crop or recolour a picture before converting it.

- Basic Color Tools: pick ink and paper colours, or build a palette from a picture.
- Font Library: find a monospaced font for your art.

Site features

- Shareable links: how the page address keeps your settings, so a reload or a shared link restores them.
- Library Kit: find art you saved to your Library.

Reading

- ASCII Art Generator developer reference: the technical write-up linked from the page's masthead.

Related

- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Basic Color Tools

What Basic Color Tools is, who it is for, where to find it and how to start using its panel workspace and the site-wide Colour Tools.

Basic Color Tools is a workspace for everyday colour work on the Auric Artisan website. It brings seven small tools together on one page: **Palette Studio**, **Palette Remix**, **Gradient Maker**, **Color Picker**, **Color Battle**, **Collage Maker** and **Quick Converter**. Each tool opens as a floating panel with a controls sidebar on the left and a work area on the right. You can keep up to five panels open at once, move and resize them, dock them, and switch between them without leaving the page.

The same folder also documents the **Colour Tools** workbench: fourteen further floating tools (such as **Color Inspector**, **Contrast System** and **CMYK Soft-Proof**) that you can open on any page of the site from the right-click menu, with `Ctrl` + `Shift` + `C`, or by adding `?color=` to a page address. When a page cannot open those tools, the site sends you to Basic Color Tools instead, so the two belong together.

Everything runs in your browser. Palettes, gradients, colours and collages are computed on your device, images you add are read locally, and your preferences (favourites, level, last view) are remembered in your browser's local storage. You can save results to your Auric Artisan Library and reopen them in the right tool later.

Who it is for

- Designers who need a quick palette, gradient or colour conversion without opening a large application.
- Front-end developers who want CSS, SCSS, Tailwind or JSON output for a palette or gradient.
- Accessibility reviewers who want contrast ratios, WCAG pass levels and colour-vision previews next to the palette they are building.
- Students and colour enthusiasts who want to see a colour in many formats and colour spaces at once.

Where to find it

- Page: https://auricartisan.com/tool/basic-tools/ (the short address https://auricartisan.com/basic-tools also opens it).
- The Colour Tools workbench: right-click anywhere on a page of the site and use the **Colour tools** group, or press `Ctrl` + `Shift` + `C`.
- No sign-in is needed to use the tools. Saving to the Library uses your Library; see Import, export and the Library.

Quick start

1. Open https://auricartisan.com/tool/basic-tools/. The page opens on the **Workspace** view.
2. In the **Open a tool** list on the left, select **Palette Studio**. It opens as a panel on the stage to the right.
3. Select **Generate** (or press `Space` while the panel is focused) until you like the palette.
4. Hover over a colour and select the lock button to keep it, then generate again: locked colours stay put.
5. Open the **Contrast Matrix** or **WCAG Pairs** tab in the panel to check contrast between your colours.
6. In the **Export** section of the sidebar, select **CSS** to copy the palette as CSS custom properties, or **PNG** to download an image of it.
7. Open **Gradient Maker** from the list as well. Both panels stay open side by side; drag a title bar to arrange them.
8. Select **Save to Library**, the floating button near the bottom-right corner of the page, to keep the focused panel's result for later.

What you can do

- Generate palettes from thirteen harmony and accessibility modes, lock colours, edit them in OKLCH and extract colours from an image.
- Remix a set of seed colours into four styled variant palettes.
- Build linear, radial and conic gradients with any number of stops, perceptual interpolation and easing, then export CSS, SVG, PNG (1600 px or 4K), JSON or a Tailwind snippet.
- Pick one colour and read it in HEX, RGB, HSL, HSV, OKLCH, Lab, XYZ and CMYK, with harmonies, colour-vision previews and contrast against common backgrounds.
- Score two palettes against each other for contrast, perceptual separation and colour-vision resilience.
- Compose local images into one of 29 collage layouts and export a 4K PNG.
- Convert any colour between fourteen notations and compare two colours by ΔE and contrast.
- Open fourteen more colour tools over any page through the Colour Tools workbench.
- Save results to your Library and reopen them later in the matching tool.

In this folder

- docs/ — how to use the workspace and every tool.
  - Getting started — open the page, tour the screen, make a first palette.
  - Workspace and panels — the three views, the catalogue, search and level, favourites, panel windows.
  - Import, export and the Library — every export format, the share link, Save to Library and restoring.
  - Panel tools — one guide per workspace tool.
  - Colour Tools workbench — the fourteen site-wide colour tools and their launcher.
  - Reference — every control, option, range and format.
  - Troubleshooting — symptoms, causes and fixes.
  - FAQ — short answers to common questions.
- others/ — supporting material.
  - Glossary — the colour and accessibility terms used here.
  - Shortcuts — keys and mouse gestures.
  - Limits and accuracy — what the numbers mean and what they are not.
  - Related — related tools, libraries and Learn articles.
  - Privacy — what stays on your device and what is stored.

WEBSITE › TOOLS › COLOUR WORKSPACE

Basic Color Tools documentation

The list of every guide for Basic Color Tools and the Colour Tools workbench.

This folder explains how to use the Basic Color Tools page and everything on it: the three views, the panel windows, the seven workspace tools, and the fourteen site-wide Colour Tools. Start with the getting-started guide, then open the guide for the tool you need. The reference lists every control in one place.

Guides in this folder

File	What it covers
getting-started.md	Open the page, tour the screen, make and export a first palette.
workspace-and-panels.md	Workspace, Catalogue and Panels views; search, level and favourites; moving, docking and closing panels.
import-export-and-library.md	Every copy and download format, the palette share link, Save to Library and reopening saved items.
reference.md	Every control, option, range, default and output format of the workspace and its seven tools.
troubleshooting.md	Symptoms, why they happen and what to do.
faq.md	Short answers to common questions.

Subfolders

Folder	What it covers
panel-tools/	One guide per workspace tool: Palette Studio, Palette Remix, Gradient Maker, Color Picker, Color Battle, Collage Maker, Quick Converter.
colour-tools/	The Colour Tools workbench: how to open it on any page, its launcher and panels, and one guide for each of its fourteen tools.

Related

- Basic Color Tools overview
- Supporting material: glossary, shortcuts, limits, privacy

WEBSITE › TOOLS › COLOUR WORKSPACE

Getting started with Basic Color Tools

Open Basic Color Tools, learn the parts of the screen, and make, check and export your first palette.

This guide takes you from opening the page to exporting a checked palette. It takes about five minutes.

Open the page

1. Go to https://auricartisan.com/tool/basic-tools/ in a current desktop or mobile browser. JavaScript must be enabled.
2. Wait for the page to load. You see the title **Basic Color Tools**, a short description, and a row of labels: **4 Categories**, **7 Tools**, **Palette Remix · Color Battle**, **Sidebar + Panel Layout** and **In-page Floating Panels**.
3. Below the title is a row of three view buttons: **Workspace**, **Catalogue** and **Panels**. The page opens on **Workspace** the first time, and on the view you used last after that.

A tour of the screen

The view buttons

View	What it shows
Workspace	A launcher list on the left and a stage on the right. Tools you open appear as panels on the stage. This is where you work.
Catalogue	A searchable grid of tool cards with a category filter, a level switch, favourites and a copy-link button. Use it to browse and filter.
Panels	A list of the panels that are open, each with its state, and a reminder of what the panel title-bar buttons do.

The Workspace view

- **Open a tool** (left column) lists every tool the catalogue currently shows, grouped by category: **Generate**, **Pick**, **Compose** and **Convert**. Each row has a small colour band, the tool name, an **Advanced** label on advanced tools, and the word **open** when that tool already has a panel.
- **Panels** (left column, below the list) shows how many panels are open out of five, as a number and a bar. It tells you how much room is left ("Room for 3 more.") or that you have reached the ceiling. Every open panel is listed with its state: **Floating**, **Docked** or **Maximised**. Select one to bring it back to the front.
- **The stage** (right) is where panels open. Before you open anything it shows **Nothing open yet.** and a short explanation. Panels stay inside the stage while you work in this view. On screens narrower than about 1,100 pixels the launcher column sits above the stage instead of beside it.

A panel

Every tool opens in the same kind of window:

- A **title bar** with the tool's name and category, and four buttons: **Open another instance** (duplicate), **Minimize**, **Maximize or restore**, and **Close**.
- A **controls sidebar** on the left with the tool's inputs and export buttons.
- A **main area** on the right with the result, often split into tabs.
- A **resize handle** in the bottom-right corner.

On a narrow screen the sidebar and main area stack vertically inside the panel, and the panel body scrolls.

Save to Library

A floating **Save to Library** button sits near the bottom-right corner of the page. It saves the result of the focused panel to your Library. See Import, export and the Library.

Your first palette

1. In **Open a tool**, select **Palette Studio**. A panel opens on the stage with five colour strips in the main area.
2. Select **Generate** in the sidebar. Five new colours replace the old ones. You can also press `Space` while the panel is focused and your cursor is not in a text box or list.
3. Open the **Harmony** list and choose **Analogous**. The palette regenerates as neighbouring hues.
4. Hover over the second colour strip. Small buttons appear at its top. Select the lock button. The strip is marked as locked.
5. Select **Generate** a few more times. The locked colour stays; the others change.
6. Select **+ Add** to add a sixth colour. A palette holds between two and nine colours.
7. Select the main area's **Contrast Matrix** tab. A table shows the WCAG contrast ratio of every pair of colours, colour-coded from fail to AAA.

8. Open the **CVD Preview** list in the sidebar and choose **Deuteranopia**. The strips now show how the palette may look to someone with deuteranopia. Your real colour values are unchanged; set the list back to **Normal vision** when you are done.
9. In the **Export** section, select **CSS**. The palette is copied to your clipboard as CSS custom properties, and a short **CSS copied** message appears.

Result: a palette you generated, locked and checked, copied as code:

```
:root {
  --color-1: #3F6E8C;
  --color-2: #5B8A74;
  --color-3: #A3B86A;
  --color-4: #D9C27A;
  --color-5: #E89F6B;
  --color-6: #B5655C;
}
```

The colours above are only an example; yours will differ.

Work with two tools at once

1. With Palette Studio open, select **Gradient Maker** in **Open a tool**. A second panel opens, slightly offset from the first.
2. Drag the Gradient Maker title bar to move the panel. Drag the corner handle to make it smaller.
3. Select anywhere on the Palette Studio panel. It comes to the front.
4. Select **Minimize** on the Gradient Maker title bar. The panel is hidden from the stage, and the **Panels** list on the left shows it as **Docked**. Its work is kept.
5. Select **Gradient Maker** in the **Panels** list (or select **Gradient Maker** again in **Open a tool**). It returns at its previous size and position, with your stops as you left them.

Result: two tools on screen, and you know how to arrange, dock and restore them. For more, read Workspace and panels.

Next steps

- Learn each tool: Panel tools.
- Open the site-wide Colour Tools from any page: Colour Tools workbench.
- See every export format: Import, export and the Library.

Related

- Basic Color Tools overview
- Shortcuts
- Troubleshooting

WEBSITE › TOOLS › COLOUR WORKSPACE

Import, export and the Library

Every way to bring colours and images into Basic Color Tools, every copy and download format, the palette share link, and saving to and reopening from your Library.

Open panels in Basic Color Tools are not kept when you reload or leave the page. To keep your work, copy it, download it, or save it to your Auric Artisan Library. This guide lists every way in and out.

Bringing colours and images in

Into	How	Accepts
Palette Studio	**From Image**	Any image file your browser can open. Colours are extracted on your device.
Palette Studio	Select a HEX label on a strip	Any colour, through the site colour picker.
Palette Remix	**Seed colors** box	HEX, `rgb()`, `rgba()`, `hsl()`, `hsla()`, `hsv()`, `oklch()`, separated by lines, spaces, commas, semicolons or vertical bars.
Color Battle	**Team A** and **Team B** boxes	Same as Palette Remix, up to nine colours per team.
Gradient Maker	Stop HEX boxes and colour wells	Three- or six-digit HEX, or the browser's picker.
Color Picker	HEX box, colour well, **Pick** eyedropper	HEX; any pixel on screen where the eyedropper is supported.
Quick Converter	Input box, colour well	HEX, `rgb()`, `rgba()`, `hsl()`, `hsla()`, `hsv()`, `oklch()`.
Collage Maker	**+ Add Images**, or select a cell	Image files from your device.
Any tool	Your Library	Saved palettes, gradients and colours reopen in the matching tool. See below.

Colours you type in any of the text boxes are converted to sRGB HEX codes. Values outside the sRGB range are clipped.

Copy and download formats

Palette Studio

Button	Kind	Output
Share	Copy	A link with the HEX codes after `#palette=`
CSS	Copy	CSS custom properties `--color-1` …
SCSS	Copy	Sass variables `$color-1` …
TW	Copy	Tailwind `extend.colors` snippet with `palette-1` …
JSON	Copy	Array of colour objects with metrics
SVG	Download	`palette.svg`, 1200 × 240
PNG	Download	`palette.png`, 1200 × 240
ASE	Copy	Tab-separated text lines

Examples:

```
:root {
  --color-1: #D3AF37;
  --color-2: #1A1A2E;
}
```

```
$color-1: #D3AF37;
$color-2: #1A1A2E;
```

```
extend: { colors: {
    "palette-1": "#D3AF37",
    "palette-2": "#1A1A2E"
} }
```

```
[
  {
    "hex": "#D3AF37",
    "rgb": { "r": 211, "g": 175, "b": 55 },
    "oklch": { "L": 0.766, "C": 0.138, "H": 91.6 },
    "lab": { "L": 72.75, "a": 0.95, "b": 62.79 },
    "luminance": 0.448,
    "bestText": { "hex": "#000000", "ratio": 9.96, "label": "dark", "passes": true },
    "locked": false
  }
]
```

For role palettes each object also has `role`, `token`, `pairWith`, `target` and `note`. The real output carries more decimal places than shown here.

The **ASE** text looks like this, with tab characters between the parts:

```
Color 1  #D3AF37  RGB(211, 175, 55)
Color 2  #1A1A2E  RGB(26, 26, 46)
```

It is plain text, not a binary Adobe Swatch Exchange file.

Palette Remix

Button	Kind	Output
CSS	Copy	One `:root` block per set, `--remix-1-1` …
JSON	Copy	Array of the four sets with `name`, `note`, `hexes`
All HEX	Copy	Every HEX code, one per line
Copy (card)	Copy	One set's HEX codes, space-separated
Swatch	Copy	One HEX code

Gradient Maker

Button	Kind	Output
Copy CSS	Copy	`background: linear-gradient(…);` (or radial or conic)
SVG	Download	`gradient.svg`, 800 × 400
1600px	Download	`gradient.png`, 1600 × 800
4K	Download	`gradient.png`, 3840 × 1920
JSON	Copy	The JSON shown on the **Export** tab
Tailwind	Copy	`backgroundImage: { 'gradient': '…' }`

Open the **Export** tab after your last change before you copy JSON; see Gradient Maker.

Color Picker and Quick Converter

Every row on the **Formats** tab (Color Picker) and the **Convert** tab (Quick Converter) has its own copy button. Quick Converter's **CSS oklch()**, **CSS oklab()** and **CSS lab()** rows are ready for stylesheets.

Color Battle

Button	Kind	Output
Copy winner	Copy	The winning team's HEX codes, one per line (Team A in a draw)
Copy report	Copy	A short text report with the result and both scores
Swatch	Copy	One HEX code

Collage Maker

Button	Kind	Output
Export PNG (4K)	Download	`collage.png`, 3,840 pixels wide when the preview is at least 800 pixels wide

What "copied" looks like

Every copy button shows a short message such as **Copied**, **CSS copied** or **JSON copied** near the bottom of the page. If your browser blocks clipboard access, nothing is copied; see Troubleshooting.

The palette share link

Palette Studio's **Share** button copies a link like this:

```
https://auricartisan.com/tool/basic-tools/#palette=D3AF37-1A1A2E-F5DEB3-0F172A-FFFFFF
```

The HEX codes are listed after `#palette=`, without `#` signs and separated by hyphens. Opening the link opens Basic Color Tools, but the page does not rebuild the palette from it automatically. Use the link as a compact record: copy the codes and paste them into Palette Remix, Color Battle or Quick Converter.

The Catalogue's **Copy link** button copies the page address as it is in your browser.

Save to Library

The **Save to Library** button floats near the bottom-right corner of the page. On narrow screens it shows only its icon.

1. Focus the panel you want to save by selecting it.
2. Select **Save to Library**.
3. A save window opens listing what will be saved. For each item you can edit the **Name** and **Description**. You can also choose a **Collection** (or leave it **Unfiled**) and add **Tags (comma-separated)** for all items.
4. Select **Save to Library** in the window, or press `Ctrl` + `Enter` (`Cmd` + `Enter` on a Mac). Select **Cancel** to close without saving.

Result: a message such as **Saved "Palette · 5 colors (analogous)"** appears, and the item is in your Library.

What each tool saves:

Focused panel	Saved as
Palette Studio	One palette, with the harmony mode and CVD preview setting.
Palette Remix	Four palettes, one per remix set, named after the set.
Gradient Maker	One gradient, with its CSS, stops, type, angle, interpolation and easing.
Color Picker	One colour, with its formats and harmony colours.
Color Battle	Two palettes, **Team A** and **Team B**, each remembering both teams and the result.
Quick Converter	One colour, with every format.
Collage Maker	Nothing from the collage. Export the PNG instead.
No panel open, or only Collage Maker	A **Basic Tools workspace** bookmark that records your level setting.

Save acts on the focused panel. Docked (minimised) panels are not saved. If there is nothing to save, a message says **Nothing to save yet — try creating something first.**

Reopening saved items

You can reopen items from your Library in Basic Color Tools. The item opens in a fresh panel of the tool that suits it, replacing any panel of that tool that is already open:

Saved item	Reopens in
A palette saved from Palette Studio	Palette Studio (**Loaded N colors from library**)
A remix set	Palette Remix, as seed colours (**Loaded remix from library**)
A gradient	Gradient Maker (**Loaded gradient from library**)
A colour saved from Color Picker	Color Picker (**Loaded color from library**)
A colour saved from Quick Converter	Quick Converter (**Loaded color from library**)
A Color Battle team	Color Battle, with both teams (**Loaded teams from library**)
A palette, gradient or colour saved from another tool	Palette Studio, Gradient Maker or Color Picker, by type
A saved analysis run from the URL Analyzer	Palette Studio, with the colours of the audited page

Because a restore opens a fresh panel, any unsaved work in the replaced panel is lost.

For how to browse, organise and send items from the Library, see the Library Kit.

Related

- Basic Color Tools overview
- Panel tools
- Reference
- Privacy

WEBSITE › TOOLS › COLOUR WORKSPACE

Workspace and panels

How the Workspace, Catalogue and Panels views work, how to find tools, and how to move, resize, dock, duplicate and close panels.

Basic Color Tools is a panel workspace. You open tools as floating windows called panels, keep several on screen, and arrange them as you like. This guide explains the three views of the page, the catalogue's filters, and every way to handle a panel.

A **panel** is a movable, resizable window inside the page. It holds one tool. Panels are not browser windows or tabs, so they disappear when you reload or leave the page.

The three views

The row of buttons under the page title switches between three views. The page remembers the view you used last, in your browser's local storage, and opens on it next time. The first visit opens on **Workspace**.

View	Use it to
Workspace	Open tools from a short list and work with them on a stage.
Catalogue	Browse tool cards, search, filter by category and level, and mark favourites.
Panels	See which panels are open and in what state, and read what each title-bar button does.

Open panels are never closed by switching views. In the Workspace view they sit inside the stage. In the Catalogue and Panels views they float over the whole page. When you return to Workspace they move back into the stage.

The Workspace view

Open a tool

The **Open a tool** list on the left shows the tools, grouped under their category names. The number next to the heading is how many tools the list shows.

- Each row shows a small colour band that hints at what the tool makes, the tool's name, and an **Advanced** label for the three advanced tools.
- A row reads **open** when that tool already has a panel.
- Select a row to open the tool. If the tool is already open, its panel comes to the front instead, and a docked panel is restored.

The list mirrors what the Catalogue currently shows. If you searched or filtered the Catalogue (for example by level or by **Favorites**), the Workspace list shows the same subset. Clear the Catalogue filters to see all seven tools again.

Panels and capacity

The **Panels** section under the list shows how many panels are open, out of a maximum of five, as a number and a bar.

- While there is room it reads, for example, **Room for 3 more.**
- At five it reads **That is the ceiling. Close or dock one before opening another.**
- Each open panel is listed with its state: **Floating**, **Docked** or **Maximised**. Select an entry to bring that panel back to the front, restoring it if it is docked.

The stage

The stage is the large surface on the right. Panels open inside it and cannot be dragged outside it. **Maximize** makes a panel fill the stage. When nothing is open the stage shows **Nothing open yet.** with a short explanation.

On screens narrower than about 1,100 pixels the launcher column moves above the stage.

The Catalogue view

The Catalogue is a grid of tool cards under a single filter bar.

The filter bar

Control	What it does
Basic Tools	A link that reloads the Basic Color Tools page.
Search box	Filters cards as you type. It matches tool names, summaries and tags; every word you type must appear. Placeholder: *Search palette, remix, battle, gradient, picker, collage, converter...*
Clear	Empties the search box and shows all cards again.
Category list	Shows one category: **All Tools**, **Generate**, **Pick**, **Compose**, **Convert**, **Favorites** or **Open Panels**.
Level	Three buttons: **Basic**, **Advanced** and **Both**. See below.
Count	Shows how many cards are visible out of the total, for example **4 of 7**.
Copy link	Copies the page address to your clipboard. The label briefly changes to **Copied**.
Quick start button	The round information button. Shows or hides the quick start note.

The quick start note says: pick any tool card to open it as a movable floating panel; each panel has a controls sidebar on the left and an interactive workspace on the right; open up to five at once, drag to reposition, resize from the corner, maximize to full screen, or close with the × button. The × on the note hides it and the page remembers that choice.

Level: Basic, Advanced and Both

The level switch filters tools by depth. **Both** is the default.

Level	Tools shown
Basic	Palette Studio, Gradient Maker, Color Picker, Quick Converter
Advanced	Palette Remix, Color Battle, Collage Maker
Both	All seven

The page remembers your level in your browser's local storage.

Categories

Category	Tools
Generate	Palette Studio, Palette Remix, Gradient Maker
Pick	Color Picker, Color Battle
Compose	Collage Maker
Convert	Quick Converter

With **All Tools** or **Favorites** selected, cards are grouped under category headings, each with a count such as **3 tools**. The page remembers the category you chose.

Tool cards

Each card shows the category, the tool name, a colour band, a one-line summary and tags: the category, **Basic** or **Advanced**, up to three topic tags, and ● **Open** when the tool has a panel.

- Select a card to open the tool, or to bring its existing panel forward.
- Hold `Shift`, `Ctrl` or `Cmd` while you select a card to open another instance even if one is already open.
- Select the star on a card to add it to **Favorites** or remove it. Favourites are saved in your browser's local storage.

If no card matches your search, the grid shows **No basic tools match.** If **Favorites** is empty it shows **No favorites yet.** with a reminder to select the star on a card.

The Open Panels category

Choose **Open Panels** in the category list to see a card for each open panel. A card reads **Floating workspace is active** or **Minimized in dock**, and shows **Maximized** when the panel is maximised. Select a card to restore and focus that panel. With nothing open it shows **No panels open.**

The Panels view

The Panels view has two cards.

- **Open panels** lists every panel with its name, category and state (**floating**, **docked** or **maximised**) and a count such as **2 of 5**. With nothing open it says **No panels open. The Workspace tab is where you start one.**
- **What the bar does** explains the four title-bar buttons: **Duplicate** makes a second copy with its own state; **Minimise** parks the panel in the dock and keeps its work; **Maximise** gives it the whole page and back; **Close** frees one of the five slots.

This view is for reading. To bring a panel back, use the Workspace **Panels** list or the Catalogue's **Open Panels** category.

Working with panels

Parts of a panel

- **Title bar**: a grip, the tool's name and category, and four buttons.
- **Controls sidebar** on the left and **main area** on the right. When the panel is narrow they stack, and the panel body scrolls.
- **Resize handle** in the bottom-right corner.
- Tabs across the top of the main area in most tools. Select a tab to switch what the main area shows.

Title-bar buttons

Button	Tooltip	What it does
Duplicate	**Open another instance**	Opens a second panel of the same tool with its own, separate state.
–	**Minimize**	Hides the panel and docks it. Its work is kept. The Workspace **Panels** list shows it as docked.
□	**Maximize or restore**	Fills the stage (Workspace view) or the browser window (other views). Select again to return to the previous size.
×	**Close**	Closes the panel and discards its work.

Move, resize and focus

1. Drag the title bar to move a panel. Panels cannot be dragged while maximised.

2. Drag the bottom-right handle to resize. The smallest size is 480 by 380 pixels, or the available space if that is smaller.
3. Double-click the title bar to maximise or restore.
4. Select anywhere on a panel to bring it to the front. The front panel is the *focused* panel: keyboard shortcuts and **Save to Library** act on it.

Panels always stay inside the stage or the window with a small margin. If you make the browser window smaller, open panels are pulled back inside it.

The Escape key

When the keyboard focus is inside a panel, `Escape` restores a maximised panel to its previous size. If the panel is not maximised, `Escape` closes it and its work is lost. Select outside the panel first if you only want to leave a text box.

Opening the same tool twice

Selecting a tool that is already open brings the existing panel forward. To get a second copy, select the title bar's **Open another instance** button, or hold `Shift`, `Ctrl` or `Cmd` while you select the tool's card in the Catalogue. Each copy keeps its own colours and settings.

The five-panel limit

Up to five panels can be open at once. If you open a sixth, the panel that was opened earliest is closed automatically to make room, and its work is lost. Save anything you want to keep to your Library first, or close a panel yourself.

What is kept and what is not

- Kept in your browser's local storage: your last view, level, category, favourites, and whether you dismissed the quick start note.
- Not kept: open panels and their contents. Reloading or leaving the page closes them. To keep a result, copy or download it, or use **Save to Library**. See Import, export and the Library.

Related

- Getting started
- Panel tools
- Shortcuts
- Troubleshooting
- Basic Color Tools overview

WEBSITE › TOOLS › COLOUR WORKSPACE

Basic Color Tools reference

Every control, option, range, default and output of the Basic Color Tools page, its views and its seven panel tools, with pointers to the Colour Tools references.

This page lists every control on the Basic Color Tools page. Defaults are given only where the page sets one. For the fourteen Colour Tools, each tool's guide ends with its own full controls table; see Colour Tools at the end.

Page

Item	Value
Address	https://auricartisan.com/tool/basic-tools/ (also https://auricartisan.com/basic-tools)
Tools	7, in 4 categories
Maximum open panels	5
Sign-in	Not needed to use the tools
Works offline	The tools compute everything in the browser once the page has loaded

View buttons

Control	What it does	Values	Default
Workspace	Launcher, capacity meter and stage	—	Shown on first visit
Catalogue	Card grid with filters	—	—
Panels	List of open panels and a title-bar legend	—	—
Last view	Reopens on the view you used last	Stored in your browser	—

Workspace view

Control	What it does
Open a tool rows	Open a tool, or bring its open panel forward (restoring it if docked). Rows show **Advanced** and **open** labels.
Panels meter	Shows **N of 5 open**, a bar, and **Room for N more.** or the ceiling message.
Panels entries	One per panel with **Floating**, **Docked** or **Maximised**. Select to bring it forward.
Stage	Holds the panels in this view. Shows **Nothing open yet.** when empty.

Catalogue view

Control	What it does	Values or range	Default
Basic Tools link	Reloads the page	—	—
Search box	Filters cards by name, summary and tags; all words must match	Free text	Empty
Clear	Empties the search	—	—
Category list	Shows one category	All Tools, Generate, Pick, Compose, Convert, Favorites, Open Panels	All Tools (then remembered)
Level	Filters by depth	Basic, Advanced, Both	Both (then remembered)
Count	Visible cards out of the total	For example **7 of 7**	—
Copy link	Copies the page address	—	—
Quick start button	Shows or hides the quick start note	On or off	Off

Control	What it does	Values or range	Default
Quick start ×	Hides the note permanently	—	—
Tool card	Opens the tool or focuses its panel	—	—
`Shift`, `Ctrl` or `Cmd` + select a card	Opens another instance	—	—
Card star	Adds or removes a favourite	On or off	Off
Open Panels card	Restores and focuses a panel	—	—

Panels view

Item	What it shows
Open panels	Each panel's name, category and state, and **N of 5**
What the bar does	Duplicate, Minimise, Maximise and Close explained

Panel window

Control	What it does	Values or range
Title bar drag	Moves the panel	Stays within the stage or window
Title bar double-click	Maximises or restores	—
Open another instance	Opens a copy with separate state	—
Minimize	Hides and docks the panel, keeping its work	—
Maximize or restore	Fills the stage or window, and back	—
Close	Closes the panel and discards its work	—
Resize handle	Resizes	Minimum 480 × 380 pixels, or the available space
`Escape` (focus inside a panel)	Restores a maximised panel; otherwise closes the panel	—
Select a panel	Brings it to the front (focus)	—

Palette Studio

Control	What it does	Values or range	Default
Harmony	Generation mode; regenerates on change	Random, Analogous, Complementary, Triadic, Monochromatic, Split complementary, Tetradic, Compound, Shades, Tints, WCAG AA palette, WCAG AAA palette, Accessibility palette	Random
Generate	Replaces unlocked colours	—	—
`Space`	Generate (panel focused, not typing)	—	—
+ Add	Adds a random colour	Up to 9 colours	—
From Image	Extracts dominant colours from an image	As many colours as the palette has	—

Control	What it does	Values or range	Default
WCAG AA / **WCAG AAA** / **Accessibility**	Switch mode and generate	—	—
CVD Preview	Simulates colour-vision deficiency on the strips	Normal vision, Deuteranopia, Protanopia, Tritanopia, Achromatopsia	Normal vision
Strip **Lock**	Keeps the colour on Generate	On or off	Off
Strip **Copy hex**	Copies the HEX code	—	—
Strip **Adjust**	OKLCH sliders with **Apply** and **Cancel**	L 0–1, C 0–0.4, H 0–360	Current colour
Strip **Remove**	Removes the colour	At least 2 remain	—
Strip HEX label	Opens the site colour picker	—	—
Drag a strip	Reorders	—	—
Tabs	Palette, Contrast Matrix, WCAG Pairs, Curves, CVD Comparison, Color Info	—	Palette
Share	Copies a `#palette=` link	—	—
CSS, **SCSS**, **TW**, **JSON**, **ASE**	Copy	—	—
SVG, **PNG**	Download 1200 × 240 image	—	—

Palette Remix

Control	What it does	Values or range	Default
Seed colors	Colours to remix; applied when you leave the box	HEX, rgb(), rgba(), hsl(), hsla(), hsv(), oklch()	#D3AF37, #1A1A2E, #F5DEB3, #0F172A, #FFFFFF
Remix style	Recipe family	Editorial kit, Analog drift, Complement battle, Tonal system	Editorial kit
Colors per palette	Colours per set	3–9	5
Remix energy	Random variation	0–100	30
Remix	Re-rolls variation	—	—
New seed	Random seeds	—	—
Card **Copy** / **Use as seed**	Copy one set / move it into the seeds	—	—
CSS, **JSON**, **All HEX**	Copy all sets	—	—

Gradient Maker

Control	What it does	Values or range	Default
Type	Shape	Linear, Radial, Conic	Linear
Angle	Direction or start angle (hidden for Radial)	0–360	90

Control	What it does	Values or range	Default
Interpolation	Blend space	sRGB (native CSS), OKLCh (perceptual), HSL, CIE Lab	sRGB (native CSS)
Easing	Blend pace	Linear, Ease in, Ease out, Ease in-out, Sine	Linear
Reverse, **+ Stop**, **Even**, **Random**	Edit stops	—	—
Track	Add a stop / highlight a marker	—	—
Stop row	Colour well, HEX box, position slider, ✕	Position 0–100, step 0.5; at least 2 stops	#D3AF37 at 0, #1D2A3E at 100
Tabs	Design, Curves, Presets, Export	—	Design
Presets	Sunset, Ocean, Aurora, Coral, Mint, Royal, Auric, Spectrum, Mono dark	—	—
Copy CSS, **JSON**, **Tailwind**	Copy	—	—
SVG, **1600px**, **4K**	Download	800 × 400; 1600 × 800; 3840 × 1920	—

Color Picker

Control	What it does	Values or range	Default
Colour well, HEX box	Set the colour	HEX	—
Pick	System eyedropper	Needs browser support	—
Random	Random colour	—	—
Saturation-value plane	Saturation (x) and value (y)	0–100 %	—
Hue slider	Hue	0–360°	—
Tabs	Formats, Harmonies, CVD, Chromaticity, Info	—	Formats
Format copy buttons	HEX, RGB, HSL, HSV, OKLCH, Lab, XYZ, CMYK	—	—
Harmony rows	Complementary, Analogous, Triadic, Tetradic, Split-comp., Square, Shades (L ramp), Tints (C ramp)	Select a swatch to apply	—
Contrast tiles	White, Off-white, Black, Near-black, Dark slate, Auric gold	AAA, AA, Fail	—

Color Battle

Control	What it does	Values or range	Default
Team A, **Team B**	Palettes to compare	Up to 9 colours each	Five colours each
Dice buttons	Random team of five	—	—
Battle, **Swap**	Score; exchange and score	—	Scores on opening
Copy winner, **Copy report**	Copy	—	—
Score weights	Avg contrast 35, AA pairs 30, Avg ΔE2000 20, CVD ΔE avg 15	0–100 total	—

Collage Maker

Control	What it does	Values or range	Default
Layout	Arrangement	29 layouts	Grid
Cols, **Rows**	Grid size for grid-based layouts	1–6 each	3, 2
Gap, **Radius**	Spacing and rounding (px)	0–40 each	6, 8
Background	Grid colour	Any	#101010
Border width / color	Inner border	0–20 px; any colour	0; #FFFFFF
Drop shadow	Preview shadow	On or off	On
Global filter	Filter for all images	None, Grayscale, Sepia, Invert, Blur, Warm, Cool, Contrast, Vintage	None
Title overlay + colour	Title in the export	Text; any colour	Empty; #FFFFFF
+ Add Images, cell, shuffle, clear	Manage images	Image files	—
`Shift` + right-click a filled cell	Cycle that cell's filter	Filter list	—
Export PNG (4K)	Download `collage.png`	—	—

Quick Converter

Control	What it does	Values or range	Default
Input box	Sets the colour	HEX, rgb(), rgba(), hsl(), hsla(), hsv(), oklch()	#D3AF37
Colour well, dice	Set or randomise	—	—
Convert rows	HEX, RGB, RGB %, HSL, HSV, OKLCH, OKLab, Lab (D65), XYZ (D65), CIE xyY, CMYK, CSS oklch(), CSS oklab(), CSS lab()	Copy each	—
Compare ΔE	Color A and Color B with ΔE 76, ΔE 2000, WCAG ratio, ΔL, ΔC, ΔH, Same hue?	HEX	A #D3AF37, B #1F2A3E
Wide Gamut	Linear sRGB, Display P3, Rec.2020, Adobe RGB with In gamut or Out	—	—
Visualize	CIE 1931, hue wheel, shade strip, hue strip	—	—
Quick Info	Luminance, Nearest name	—	—

Save to Library

Control	What it does
Save to Library (floating button)	Captures the focused panel and opens the save window
Name, **Description**	Per item
Shared description, **Collection**, **Tags (comma-separated)**	For all items being saved
Save to Library (in the window) or `Ctrl` + `Enter`	Saves

Control	What it does
Cancel	Closes without saving

Colour Tools

The fourteen site-wide Colour Tools each have their own controls table at the end of their guide:

Tool	Group	Guide
Color Inspector	Essentials	colour-inspector.md
Accessibility Lab	Essentials	accessibility-lab.md
Contrast System	Essentials	contrast-system.md
Color Name Finder	Essentials	colour-name-finder.md
Animation Contrast	Essentials	animation-contrast.md
Type Readability Sim	Essentials	type-readability-sim.md
Color Library	Explore	colour-library.md
Color Psychology	Explore	colour-psychology.md
Pantone & Named Lookup	Explore	pantone-and-named-lookup.md
Harmony Studio	Design	harmony-studio.md
Gradient Library	Design	gradient-library.md
CMYK Soft-Proof	Design	cmyk-soft-proof.md
Color Formulation	Deep dive	colour-formulation.md
Color Space Converter	Deep dive	colour-space-converter.md

The launcher, the shared panel window and the `?color=` address parameters are covered in launcher-and-panels.md.

Related

- Basic Color Tools overview
- Shortcuts
- Glossary

WEBSITE › TOOLS › COLOUR WORKSPACE

Troubleshooting Basic Color Tools

Symptoms you may meet in Basic Color Tools, why they happen, and what to do.

Each entry gives the symptom, the reason, and what to do. For problems with the site-wide Colour Tools, see also Open the Colour Tools and manage their panels.

Panels

A panel closed by itself when I opened another tool

- **Why:** five panels is the limit. Opening a sixth closes the panel that was opened earliest.
- **What to do:** before opening a sixth tool, save or export what you need from the oldest panel, or close a panel you no longer need. The Workspace **Panels** meter shows how many slots are left.

A panel closed when I pressed Escape

- **Why:** `Escape` inside a panel closes it (or, if it is maximised, restores its size first).
- **What to do:** select outside the panel before pressing `Escape` for other purposes. Closed panels cannot be reopened with their work; save to your Library as you go.

I minimised a panel and now I cannot find it

- **Why:** a minimised panel is hidden completely and kept in a dock.
- **What to do:** in the Workspace view, select it in the **Panels** list, or select the tool again in **Open a tool**. In the Catalogue, choose **Open Panels** in the category list and select its card.

All my panels disappeared after a reload

- **Why:** open panels and their contents live only in the page. They are not saved when you reload or leave.
- **What to do:** copy, download or **Save to Library** before leaving. See Import, export and the Library.

A panel is partly off the screen or too small to use

- **Why:** panels are kept inside the stage or window, but a very small window leaves little room.
- **What to do:** make the browser window larger; panels are pulled back inside automatically. Maximise the panel with its **Maximize or restore** button or by double-clicking its title bar.

I cannot drag or resize a panel

- **Why:** maximised panels cannot be moved or resized.
- **What to do:** select **Maximize or restore** to return it to normal size first.

The Workspace list shows fewer than seven tools

- **Why: Open a tool** lists what the Catalogue currently shows. A search, a **Level** other than **Both**, or the **Favorites** or **Open Panels** category hides tools.
- **What to do:** open the Catalogue, select **Clear**, set **Level** to **Both** and choose **All Tools**.

Copying, downloading and sharing

Nothing is copied when I select a copy button

- **Why:** the browser can refuse clipboard access, for example in some private windows, embedded views or when the page is not focused.
- **What to do:** select the button again directly, and make sure the page has focus. For Gradient Maker, the **Export** tab shows the CSS, SVG and JSON in text boxes you can select and copy by hand.

Gradient Maker's JSON button copies nothing, or an old gradient

- **Why:** the **JSON** button copies the text in the **Export** tab's JSON box, which is only refreshed while the **Export** tab is open.
- **What to do:** open the **Export** tab after your last change, then select **JSON**.

The share link opens the page but not my palette

- **Why:** the Palette Studio share link records the HEX codes in the address, but the page does not rebuild the palette from it.
- **What to do:** copy the codes from the link (after `#palette=`) into Palette Remix or Color Battle, or save the palette to your Library to reopen it properly.

The collage PNG is smaller than 4K

- **Why:** the export is scaled from the preview. When the preview is narrower than 800 pixels, the image is under 3,840 pixels wide.
- **What to do:** widen or maximise the Collage Maker panel before selecting **Export PNG (4K)**.

The conic gradient PNG does not download or looks wrong

- **Why:** conic PNG export needs a browser that can draw conic gradients on a canvas.
- **What to do:** update your browser, or copy the CSS instead. Note that SVG export has no conic form and exports a straight gradient.

The ASE file does not open in my design app

- **Why: ASE** copies tab-separated text, not a binary Adobe Swatch Exchange file.
- **What to do:** paste the text into a document or spreadsheet, or use **CSS**, **JSON** or **SVG** instead.

Colours and images

Typing a colour in Quick Converter changes it before I finish

- **Why:** the input converts as soon as its contents are a valid colour and then rewrites itself as HEX. A partial value such as `#D3A` is already valid.
- **What to do:** paste complete values, or use the colour well.

Space does not generate a new palette

- **Why:** `Space` works only when the Palette Studio panel is the focused panel and the cursor is not in a text box or list.
- **What to do:** select an empty part of the Palette Studio panel, then press `Space`, or select **Generate**.

"Max 9" or "Min 2 swatches" appears

- **Why:** a Palette Studio palette holds between two and nine colours.
- **What to do:** remove a colour before adding one, or add one before removing.

"Eyedropper not supported" appears in Color Picker

- **Why:** the system eyedropper uses a browser feature available mainly in Chromium-based desktop browsers.
- **What to do:** use the colour well or type the HEX code, or switch to a browser that supports the eyedropper.

"Cannot extract (CORS)" appears when extracting from an image

- **Why:** the browser would not let the page read the image's pixels. This can happen with images that come from another website.
- **What to do:** save the image to your device first, then choose the local file with **From Image**.

Colours in a palette or ramp look duller than the numbers I set

- **Why:** OKLCH and Lab values can describe colours outside the sRGB range that screens and CSS HEX codes can show. The tools clip such colours to the nearest displayable one.
- **What to do:** lower the chroma, or accept the clipped colour. The HEX code always reflects what you see.

Dropping images onto the collage does nothing

- **Why:** the preview's hint mentions dropping files, but the current version adds images only through the file picker.
- **What to do:** select **+ Add Images** and choose several files at once.

Right-clicking a collage cell opens the site menu instead of changing the filter

- **Why:** the site's own right-click menu takes the plain right-click.
- **What to do:** hold `Shift` while you right-click the cell. Each `Shift` + right-click moves the cell to the next filter.

Saving to the Library

"Nothing to save yet — try creating something first."

- **Why:** the focused panel has nothing the Library can store yet.
- **What to do:** create a result in the panel, select the panel to focus it, and try again.

Saving from Collage Maker saved a workspace bookmark, not my collage

- **Why:** the Library cannot capture the collage preview. With only Collage Maker focused, it saves a Basic Tools workspace bookmark instead.
- **What to do:** use **Export PNG (4K)** to keep the collage.

Opening a saved item replaced my open panel

- **Why:** restoring an item opens a fresh panel of that tool and closes the existing one.
- **What to do:** save the open panel's work before you reopen a Library item in the same tool.

Related

- FAQ
- Workspace and panels
- Limits and accuracy
- Help centre

WEBSITE › TOOLS › COLOUR WORKSPACE

Basic Color Tools FAQ

Short answers to common questions about Basic Color Tools and the Colour Tools workbench.

Do I need an account?

No. All seven tools and the Colour Tools work without signing in. Saving to your Library uses the site's Library; see the Library Kit and Account and billing.

Does it cost anything or use tokens?

The seven tools compute everything in your browser and do not call the server for their results, so they do not spend tool tokens. See Account and billing for plans.

Are my images uploaded?

No. Images you add to Palette Studio (**From Image**) or Collage Maker are read by your browser on your device. See Privacy.

How many panels can I have open?

Five. Opening a sixth closes the oldest one. See Workspace and panels.

Why are some tools missing from the list?

The Catalogue's **Level** switch or search is filtering them, and the Workspace list follows the Catalogue. Set **Level** to **Both**, clear the search and choose **All Tools**.

Will my work still be there tomorrow?

Open panels are not saved. Your favourites, level, category and last view are remembered in your browser. To keep results, export them or use **Save to Library**.

Can I open the same tool twice?

Yes. Select **Open another instance** on the panel's title bar, or hold `Shift`, `Ctrl` or `Cmd` while selecting the tool's card in the Catalogue. Each copy has its own state.

What is the difference between Color Picker and Quick Converter?

Color Picker is built around choosing a colour by eye (plane, hue slider, eyedropper) and shows harmonies, colour-vision previews and contrast tiles. Quick Converter is built around typing or pasting a value; it shows fourteen notations, compares two colours and lists wide-gamut values.

What is the difference between Basic Color Tools and the Colour Tools?

Basic Color Tools is a page with seven panel tools. The Colour Tools are fourteen further floating tools you can open on any page of the site from the right-click menu, with `Ctrl` + `Shift` + `C`, or with a `?color=` address parameter. See Colour Tools.

Which contrast standard do the tools use?

The seven panel tools use the WCAG 2.x contrast ratio: 4.5:1 for AA normal text, 7:1 for AAA, 3:1 for large text and non-text interface parts. Some Colour Tools, such as Contrast System, also report APCA.

Is the ASE export an Adobe swatch file?

No. It copies tab-separated text. Use CSS, SCSS, Tailwind, JSON, SVG or PNG for other tools.

Where are the palette, shade and gradient collections?

Large browsable collections of palettes, shades, gradients, harmonies and colours live in the site's library area. See Colour libraries.

Related

- Troubleshooting
- Getting started
- Basic Color Tools overview

WEBSITE › TOOLS › COLOUR WORKSPACE

Colour Tools workbench

The fourteen floating Colour Tools you can open on any page of Auric Artisan, how to open them, and a guide for each.

The Colour Tools are fourteen floating colour tools that open over whatever page you are on: right-click and use the **Colour tools** group, press `Ctrl` + `Shift` + `C`, right-click a colour value, or add `?color=` to a page address. They share one launcher, one kind of panel window and a control bar for arranging panels. Basic Color Tools is their home page, and this folder documents them.

Start with launcher-and-panels.md, then open the guide for the tool you need.

Opening and managing the tools

File	What it covers
launcher-and-panels.md	Every way to open the tools, the `?color=` address parameters, the launcher, the panel window, linking and pinning, the control bar, and what all fourteen tools share.

Essentials

File	Tool	In one line
colour-inspector.md	Color Inspector	Everything about one colour: formats, contrast, vision, harmony, names, psychology, scales, science and code export.
accessibility-lab.md	Accessibility Lab	Two to six colours under WCAG 2.1, 2.2, APCA or ISO 9241-3, with a vision-filter grid.
contrast-system.md	Contrast System	Up to five foregrounds on one background: three measures, fixes, APCA size table, contrast curve and interface preview.
colour-name-finder.md	Color Name Finder	The nearest of 1,058 named colours and seventeen similar ones.
animation-contrast.md	Animation Contrast	Text contrast across 48 samples of an animated gradient or a video, with a safe text colour.
type-readability-sim.md	Type Readability Sim	Reading comfort from colours, font style, size, weight and viewing condition.

Explore

File	Tool	In one line
colour-library.md	Color Library	Browse 1,058 named colours by family with full detail views and swatch downloads.
colour-psychology.md	Color Psychology	Mood, emotions, traits, brand fit and cultural meanings for one to six colours.
pantone-and-named-lookup.md	Pantone & Named Lookup	Nearest matches in CSS, Pantone Solid C, RAL Classic, NCS and Crayola subsets.

Design

File	Tool	In one line
harmony-studio.md	Harmony Studio	Eight harmony schemes, optionally adjusted to a contrast target on your background.
gradient-library.md	Gradient Library	512 curated gradients with CSS, Tailwind, SVG and JSON export.
cmyk-soft-proof.md	CMYK Soft-Proof	A colour or image simulated on five print conditions, with CMYK values.

Deep dive

File	Tool	In one line
colour-formulation.md	Color Formulation	How a colour is built: channels, conversion pipeline, a reflectance sketch and a paint-mix suggestion.
colour-space-converter.md	Color Space Converter	One colour in 34 notations across eight groups, with diagrams and JSON export.

Related

- Basic Color Tools overview
- Documentation contents
- Shortcuts
- Limits and accuracy

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Accessibility Lab

Compare two to six colours under WCAG 2.1, WCAG 2.2, APCA or ISO 9241-3, preview text in them, and see every colour through seven vision filters.

Accessibility Lab checks a small set of colours, from two to six, as a group. The first colour is the background; the others are foreground colours such as text, buttons and icons. The lab shows a contrast matrix for every pair under the standard you choose, a live text sample in your first two colours, and a grid showing each colour through seven vision filters with its contrast against the background.

It supports four ways of measuring contrast:

- **WCAG 2.1** and **WCAG 2.2**: the luminance contrast ratio, from 1:1 to 21:1. Both versions use the same formula.
- **WCAG 3 (APCA)**: the Accessible Perceptual Contrast Algorithm, reported as **Lc** (lightness contrast). It takes into account which colour is the text and which is the background.
- **ISO 9241-3**: luminance modulation, a measure from an older ergonomics standard for display text.

Accessibility Lab is in the launcher's **Essentials** group.

Open it

Open the launcher and select **Accessibility Lab**, or pick it from the right-click menu's **Colour tools** group. See Open the Colour Tools. It starts with a dark navy background (#0F172A) and one foreground: the colour you opened it with, or gold (#D3AF37).

Screen tour

Sidebar

- **Standard**: WCAG 2.1, WCAG 2.2, WCAG 3 (APCA) or ISO 9241-3.
- **Large text mode**: a check box that applies the large-text thresholds.
- **Colors (first = background)**: one colour field per colour, labelled **Background**, **Color 1**, **Color 2** and so on, with **+ Color**, **- Color** and **Randomize** buttons.
- **Presets**: four ready-made colour sets.
- **Preview CVD filter**: a vision filter applied to the live preview only.
- **Standard info**: a short explanation of the chosen standard.

Main area

- **Contrast matrix**: every colour against every other colour, labelled with the standard's name.
- **Live preview**: a heading (**Read me clearly**), a paragraph, a line of small print and two buttons (**Outlined** and **Primary**), drawn in the background colour and **Color 1**.
- **Color-vision deficiency grid** (*All 7 filters* × *all colors*): one row per filter, one column per colour.

Tasks

Check a brand palette

1. Set **Background** to your page background colour.
2. Set **Color 1** to your body text colour. Select **+ Color** and add your link, button and accent colours (up to six colours in total).
3. Choose a **Standard**. **WCAG 2.1** is the usual legal and procurement baseline.
4. Read the matrix row or column for **BG**. Each cell shows the value, a bar and a level pill: green for a pass, amber for a borderline result, red for a fail.

Result: you can see which of your colours are readable on your background, and which pairs of foreground colours also contrast with each other.

Check large text or headings

1. Tick **Large text mode**.
2. The matrix levels switch to the large-text thresholds, and the live preview's text grows from 14 to 18 pixels.

Result: levels for text that is at least 18 point regular or 14 point bold (about 24 and 19 CSS pixels).

Compare standards

1. Keep the same colours and switch **Standard** between **WCAG 2.1** and **WCAG 3 (APCA)**.
2. Read **Standard info** for what each value means.

Result: you see where the two models agree and where APCA is stricter or more lenient, for example with dark backgrounds or very light text.

See the colours through vision filters

1. Read the **Color-vision deficiency grid**. Each row shows all your colours as simulated for one filter. Under each foreground swatch is its WCAG 2.1 contrast ratio against the simulated background, coloured green (AA or AAA), amber or red (fail).
2. To see real text, choose a filter in **Preview CVD filter**. The live preview is redrawn through it.

Result: you can spot colours that lose contrast or merge for people with a colour-vision deficiency.

Standards and levels

Standard	Value shown	Normal text	Large text
WCAG 2.1 / **WCAG 2.2**	Ratio, for example `7.21:1`	**AAA** at 7:1, **AA** at 4.5:1, otherwise **Fail**	**AAA** at 4.5:1, **AA** at 3:1, otherwise **Fail**
WCAG 3 (APCA)	`Lc` value, for example `Lc -86.4`	**AAA** at 90, **AA** at 75, **Body** at 60, otherwise **Fail**	**AAA** at 75, **AA** at 60, **Body** at 45, otherwise **Fail**
ISO 9241-3	Modulation, for example `M 0.912`	**Excellent** at 0.7, **Good** at 0.5, **Marginal** at 0.3, otherwise **Poor**	Same

APCA thresholds apply to the size of the Lc value, whether it is positive or negative. A negative Lc means light text on a dark background; a positive Lc means dark text on a light background. In the matrix, the row colour is treated as the background and the column colour as the text.

The **Standard info** text for each:

- **WCAG 2.1**: body text needs 4.5:1 (AA) or 7:1 (AAA); large text (18 pt regular or 14 pt bold and up) needs 3:1 for AA.
- **WCAG 2.2**: the same ratio as 2.1; version 2.2 adds criteria for focus visibility, target size and consistent help, which a colour tool cannot check.
- **WCAG 3 (APCA)**: Lc runs from about −108 to +106; body text needs an absolute Lc of 75 (AA) or 90 (AAA).
- **ISO 9241-3**: M = (Lmax − Lmin) / (Lmax + Lmin); at least 0.5 is recommended for legible display text under office lighting.

Vision filters

Filter	What it simulates
Normal vision	No change.
Deuteranopia	Missing green-sensitive cones (the most common red-green deficiency).
Protanopia	Missing red-sensitive cones; reds also look darker.
Tritanopia	Missing blue-sensitive cones (rare).
Achromatopsia	No colour vision; greyscale.
Low vision	Blur with reduced saturation. Live preview only.
Cataract haze	A yellowish haze with slight blur. Live preview only.

In the grid, the **Low vision** and **Cataract haze** rows show the colours unchanged, because those two filters are blur effects that apply to a picture, not to single colour values. Use **Preview CVD filter** to see them.

Presets

Preset	Colours
Auric dark	#0F172A, #FFFFFF, #D3AF37
Brand blues	#1E40AF, #F9FAFB, #F59E0B, #10B981
Alert states	#111827, #F3F4F6, #EF4444, #3B82F6, #10B981
High contrast	#000000, #FFFFFF, #FF3B30, #34C759, #007AFF, #FFD60A

The first colour of each preset becomes the background.

Controls

Control	What it does	Values or range	Default
Standard	Contrast measure for the matrix	WCAG 2.1, WCAG 2.2, WCAG 3 (APCA), ISO 9241-3	WCAG 2.1
Large text mode	Large-text thresholds; bigger preview text	On or off	Off
Colour fields	Background and foreground colours	2 to 6 colours; HEX or colour well; dice for random	#0F172A and #D3AF37
+ Color	Adds a random colour	Up to 6	—
- Color	Removes the last colour	At least 2 remain	—
Randomize	Keeps the count, resets the background to #0F172A and randomises the others	—	—
Preset buttons	Load a preset set	4 presets	—
Preview CVD filter	Filter for the live preview	Normal vision, Deuteranopia, Protanopia, Tritanopia, Achromatopsia, Low vision, Cataract haze	Normal vision

Outputs and exports

The lab is for checking; it has no export. To copy values, open a colour in the Color Inspector, or use Contrast System for copyable reports.

Accuracy and limits

- WCAG ratios use the standard sRGB relative-luminance formula.
- The APCA figure follows a simplified form of the published algorithm. APCA is still a draft for WCAG 3, and its thresholds may change.
- ISO 9241-3 values are computed from relative luminance, not measured screen luminance.
- The colour-vision filters use simple 3 × 3 matrices for complete deficiencies. Milder, more common forms change colours less. The low-vision and cataract filters are illustrations, not medical models.
- Contrast is only one part of accessibility. Font size, weight, spacing and layout matter too.

Related

- Colour Tools documentation
- Contrast System for deeper contrast work and readability
- Type Readability Sim
- Accessibility and vision tools
- Basic Color Tools

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Animation Contrast

Sample an animated background, from CSS keyframe colours or a video file, at 48 points, check text contrast at each one, find the worst frame and get a text colour that passes throughout.

Most contrast checks assume the background stays still. Hero sections with animated gradients, background videos and colour transitions do not: text that is readable at the start of a loop can become unreadable for part of it. Animation Contrast checks the whole timeline.

You give it a text colour and an animated background, either a list of keyframe colours or a video file. It takes 48 evenly spaced samples along the timeline, checks the text against each one with the standard you choose, shows a pass or fail strip, picks out the worst moment, and suggests a single text colour that would pass at every sample.

Animation Contrast is in the launcher's **Essentials** group.

Open it

Open the launcher and select **Animation Contrast**. See Open the Colour Tools. The text colour starts as the colour you opened the tool with, or white (#FFFFFF).

Screen tour

1. **Setup** (*Text + animated background*):
   - **Text color**: a colour field.
   - **Threshold**: the standard to test against.
   - **Source**: **CSS keyframes** or **Video frames**.
2. The background section:
   - For **CSS keyframes**: **Keyframe stops**, with a note such as *4 colors · interpolated in OKLab*, one numbered colour row per stop, and **+ Add stop**, **− Remove last** and **Reset**.
   - For **Video frames**: **Video source**, a drop area with **Choose file**, and once a video is loaded, a player, **Sample 48 frames** and **Clear video**.
3. The results:
   - **Timeline**: 48 ticks coloured with each sample's background, marked pass or fail, an axis from 0 % to 100 %, and two pills: *N / 48 pass (P%)* and *N fail*.
   - **Worst frame** (*Lowest contrast in timeline*): a preview of body text and an 11-pixel caption on the worst background, with its position (*At t = 62%*), its HEX code, the measured value, and its WCAG and APCA levels.
   - **Suggested safe text color**: a colour chip, the worst contrast it reaches, and **Use as text color**.

Thresholds

Option	Passes when
WCAG 2.x (4.5:1)	The contrast ratio is at least 4.5:1 (AA, normal text).
WCAG 2.x large (3:1)	The ratio is at least 3:1 (AA, large text).
WCAG 2.x AAA (7:1)	The ratio is at least 7:1.
APCA Lc 75 (body)	The APCA lightness contrast is at least 75 (either sign).
APCA Lc 60 (large)	The APCA lightness contrast is at least 60.

Tasks

Check text over a CSS gradient animation

1. Set **Text color** to your heading or body text colour.
2. Keep **Source** on **CSS keyframes**.
3. Enter the colours your background passes through, in order, as keyframe stops. Use **+ Add stop** for more (up to eight) and **− Remove last** to remove (at least two remain). **Reset** restores the starting set: #0F172A, #1E3A8A, #9333EA, #F97316.
4. Choose a **Threshold**.
5. Read the **Timeline**. Hover over a tick to see its position, value and pass or fail.

The tool blends between neighbouring stops in OKLab and spaces the stops evenly across the timeline, like a CSS animation with evenly spaced keyframes and linear timing.

Result: you see what share of the animation keeps your text readable, and where it fails.

Check text over a background video

1. Set **Source** to **Video frames**.
2. Drop a video file on the drop area, or select **Choose file**. The video appears with player controls, muted.
3. Wait until it has loaded, then select **Sample 48 frames**. A **Sampling frames…** message appears, then **Sampled 48 frames**.
4. Read the timeline. Select a tick to move the video to that moment.

For each sample the tool takes the average colour of the whole frame. Short clips (the drop area suggests about 30 seconds at most) sample fastest.

Result: a pass or fail reading across the whole video.

Get a text colour that always passes

1. After any check, read **Suggested safe text color**. Its note says **Passes every sampled frame** or **Best available — still fails somewhere**.
2. Select **Use as text color** to apply it. **Applied #HEX** appears and the timeline updates.

The suggestion keeps your text colour's hue and chroma and moves its OKLCH lightness up and down in small steps until the worst sample passes, then picks whichever direction needed the smaller change. If neither direction passes, it offers the colour that did best.

Result: one static text colour you can use over the whole animation, or a clear sign that the background needs changing (for example an overlay or a darker range).

Controls

Control	What it does	Values or range	Default
Text color	The text colour to test	HEX, colour well, dice	The colour you opened with, or #FFFFFF
Threshold	The pass standard	5 options (see Thresholds)	WCAG 2.x (4.5:1)
Source	Where background colours come from	CSS keyframes, Video frames	CSS keyframes
Keyframe stop	One background colour	Colour well or HEX	4 stops
+ Add stop	Adds a white stop at the end	Up to 8 stops	—
− Remove last	Removes the last stop	At least 2 stops	—
Reset	Restores the four default stops	—	—
Drop area / **Choose file**	Loads a video	Video files	—
Sample 48 frames	Samples the loaded video	48 samples	—
Clear video	Removes the video	—	—
Timeline tick	Shows details; in video mode, seeks the video	48 ticks	—
Use as text color	Applies the suggested colour	—	—

Outputs and exports

The tool has no file export. Copy the suggested colour's HEX code from the **Text color** field after applying it.

Accuracy and limits

- 48 samples catch most problems in a loop, but a very brief flash between two samples can be missed.
- Keyframe mode assumes evenly spaced keyframes and linear timing in OKLab. Real CSS animations may use different timing, blend in sRGB, or overlap several layers.
- Video mode uses the average colour of each frame. Text over a small bright area in an otherwise dark frame can fail even when the average passes. Check the area behind your text separately if the video has strong local contrast.
- The video is read in your browser and is not uploaded. Some video formats may not play in every browser.
- APCA figures use a simplified form of a draft method; see Contrast System.

Related

- Colour Tools documentation
- Contrast System
- Accessibility Lab
- Basic Color Tools

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CMYK Soft-Proof

Simulate a screen colour or an image on five press and paper conditions, see the CMYK separation after dot gain and ink limits, spot colours that will shift, and copy CMYK values.

Screens mix light; printing presses lay down ink on paper. Many bright screen colours cannot be printed, paper is never pure white, printed black is never pure black, and ink spreads as it soaks in. CMYK Soft-Proof gives you a quick feel for these effects. It takes a colour, or an image you drop in, and shows how it may look on one of five print conditions, the CMYK ink amounts involved, and which parts of an image will change most.

Soft-proofing means previewing on screen how something will print. **CMYK** is the four-ink process (cyan, magenta, yellow and black). **Dot gain** is the darkening of mid-tones as ink spreads on paper. **TAC** (total area coverage) is the maximum combined ink a press allows, in percent (C + M + Y + K, up to 400 %).

This tool is an intuition builder, not a colour-managed proof. It does not use ICC profiles.

CMYK Soft-Proof is in the launcher's **Design** group.

Open it

Open the launcher and select **CMYK Soft-Proof**. See Open the Colour Tools. It starts with the colour you opened with, or a process blue (#0085CA).

Screen tour

1. **Source** (*Soft-proof preview*): a colour field, the **Press condition** list, and a note describing the condition (its TAC limit, paper colour, black point and typical use).
2. **Side-by-side preview** (*Screen vs. simulated press*): **Screen (sRGB)** and **Press simulation** chips with their HEX codes, a ΔE pill with a label, and a TAC pill such as *TAC 173% / 330%* (the ink used against the limit). When the ink limit is exceeded, a note says how much cyan, magenta and yellow were reduced.
3. **CMYK separation** (*After dot gain & TAC*): bars for **Cyan**, **Magenta**, **Yellow** and **Black** with percentages, and three copy buttons: **CSS cmyk()**, **Channel values** and **Simulated hex**.
4. **Image soft-proof**: a drop area (*Drop an image here, or paste from clipboard*) with **Choose file**. Once an image is loaded, three versions side by side and **Clear**.

Press conditions

Condition	TAC limit	Paper	Black point	Note shown
ISO Coated v2 (FOGRA39)	330 %	#F4F2E8	#1B1814	European coated commercial offset. The most common print spec.
SWOP (US Web Coated)	300 %	#F1EFE3	#1F1B16	US web offset on coated stock.
ISO Uncoated (FOGRA52)	320 %	#EFEBD8	#252118	Premium uncoated stock; higher dot gain, warmer paper.
ISO Newsprint 26v5	240 %	#EAE3CE	#2B2719	Heavy dot gain, low TAC.
Fine art / cotton rag	360 %	#F8F3DF	#1C1610	Inkjet on cotton rag; closer to a monitor's range than offset.

Each condition also has its own dot-gain amount: lowest for fine art, highest for newsprint.

How the simulation works

1. The screen colour is split into CMYK with a simple formula.
2. Dot gain darkens the mid-tones of each ink.
3. If the total ink exceeds the TAC limit, cyan, magenta and yellow are reduced evenly and black is kept, which preserves contrast.
4. The inks are recombined over the condition's paper colour, with its practical black point, to give the simulated screen colour.
5. The difference between the original and the simulation is measured with ΔE2000.

ΔE2000	Label
Below 2	**Match**
2 to 5	**Close**
5 to 10	**Shift**
10 and above	**Lost**

Tasks

Check a brand colour for print

1. Enter the colour.
2. Choose the **Press condition** your printer uses (ask them; FOGRA39 is common in Europe, SWOP in the US).
3. Compare **Screen (sRGB)** and **Press simulation**, and read the ΔE label.
4. Check the TAC pill. Red means the colour needed more ink than the press allows.

Result: a warning if the colour will shift noticeably or is too ink-heavy, before you send files.

Copy CMYK values

Under **CMYK separation**:

- **CSS cmyk()** copies the separation as a CSS function. For the starting blue (#0085CA) on FOGRA39 it reads `cmyk(100%, 45%, 0%, 28%)`.
- **Channel values** copies the same numbers as `C100 M45 Y0 K28`.
- **Simulated hex** copies the HEX code of the press simulation.

These values are after dot gain and the ink limit. Use them as a starting point to discuss with your printer, not as final separations.

Soft-proof an image

1. Drop an image on the drop area, select **Choose file**, or paste an image while the panel has focus.
2. Three versions appear: **Original**, the **Press simulation**, and an **Out-of-gamut warning** view where pink marks pixels that change by more than ΔE 10.
3. Change **Press condition** to compare conditions.
4. Select **Clear** to remove the image.

The image is scaled to at most 360 pixels on its longest side for the preview. Large images (the drop area suggests about 2 megapixels at most) are processed fastest.

Result: you see which areas of the image, typically bright blues, greens and saturated oranges, will print noticeably differently.

Controls

Control	What it does	Values or range	Default
Colour field	Colour to simulate	HEX, colour well, dice	#0085CA, or the colour you opened with
Press condition	Paper, ink limit and dot gain	5 conditions	ISO Coated v2 (FOGRA39)
CSS cmyk()	Copies the separation as `cmyk()`	—	—
Channel values	Copies `C… M… Y… K…`	—	—
Simulated hex	Copies the simulated colour	—	—
Drop area / **Choose file** / paste	Loads an image	Image files	—
Clear	Removes the image	—	—

Accuracy and limits

- This is not colour-managed soft-proofing. The CMYK split is a naive formula, and the press conditions are simple models (paper colour, black point, dot gain and ink limit) named after common standards. They do not use the official characterisation data or ICC profiles.
- Your screen's own calibration affects what you see.
- Real separations depend on the printer's profile, rendering intent, black generation and more. Ask for a contract proof for colour-critical work.
- Images are processed in your browser and are not uploaded.

Related

- Colour Tools documentation
- Pantone & Named Lookup
- Color Formulation
- CMYK and the Four-Color Printing Process on Learn
- Basic Color Tools

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Color Formulation

Read one colour as channels, follow it through the conversion pipeline, see a reflectance sketch, and get a two-pigment paint mix that comes close to it.

Color Formulation takes one colour apart. It shows the numbers a screen stores for the colour, how those numbers are converted into the perceptual spaces designers use, a sketch of how a painted surface of that colour might reflect light, and which two artist pigments, mixed in which ratio, come closest to it.

Use it when you want to understand a colour rather than only copy its code: to teach or learn how sRGB, linear light and OKLab relate, to judge whether a colour is a light tint or a deep shade, or to get a starting point for mixing a physical paint that matches a screen colour.

Everything is calculated in your browser from the colour you enter. The paint recipe and the reflectance chart are simplified models, not measurements; see Accuracy and limits.

How to open it

Open Color Formulation from the **Deep dive** group of the Colour Tools launcher (tag **FORM**) or from the Color Inspector's **Open in** list, which passes the inspected colour in; see Open the Colour Tools and manage their panels.

If you open it with a colour, it starts with that colour. Otherwise it starts with `#D3AF37`.

Screen tour

The panel has a sidebar on the left and a tabbed main area on the right. When the panel is narrow, the sidebar moves above the main area.

Sidebar

- **Target color** — the colour field: a swatch that opens your browser's colour picker, a text box for a hex code, and a dice button (**Random color** tooltip) that sets a random colour.
- **The large chip** — a 100-pixel block filled with the colour and labelled with its hex code. The label switches between dark and light text so it stays readable.
- **Properties** — seven quick readings:

Reading	What it tells you
Lightness (OKL)	OKLCh lightness as a percentage (0% is black, 100% is white).
Chroma	OKLCh chroma, to three decimals. 0 is a neutral grey; screen colours rarely pass about 0.37.
Hue angle	OKLCh hue in degrees.
Saturation (HSL)	HSL saturation as a whole percentage.
Rel. luminance	WCAG relative luminance, from 0 (black) to 1 (white), to four decimals.
Tone character	**Light tint** above 72% OKLCh lightness, **Deep shade** below 35%, otherwise **Mid tone**.
Chroma character	**Vivid / saturated** above chroma 0.20, **Moderate chroma** above 0.09, otherwise **Muted / neutral**.

Channels tab

The tab that opens first. It draws a labelled bar for every channel in five groups, with the value at the end of each bar:

Group	Channels and values
sRGB (gamma-encoded, 0–255)	**R**, **G**, **B** as whole numbers from 0 to 255: the values in the hex code.
Linear-light sRGB (gamma decoded)	**R-lin**, **G-lin**, **B-lin** from 0 to 1, to four decimals.
HSL (hue–saturation–lightness)	**Hue** in degrees, **Sat** and **Lit** as percentages.
OKLab (perceptually uniform, Björn Ottosson)	**L**, **a** (marked "red–green") and **b** (marked "blue–yellow"), to four decimals.
OKLCh (cylindrical OKLab)	**L**, **C** (chroma) and **h°** (hue in degrees).

The OKLab **a** and **b** bars run from −0.4 on the left to +0.4 on the right, so a bar that stops half-way means zero. The OKLCh **C** bar runs from 0 to 0.4. The two hue bars are tinted with the hue they show.

Pipeline tab

Six numbered steps show how the colour travels between spaces, with this colour's own numbers filled in:

1. **Encoded sRGB** — the three whole numbers stored in the hex code, gamma-compressed for display.
2. **Linear-light sRGB** — the same colour after gamma decoding, proportional to the amount of light.
3. **CIE XYZ (D65)** — tristimulus values under the D65 daylight white; Y is the relative luminance.
4. **OKLab** — L, a and b; equal distances here look like roughly equal colour differences.
5. **OKLCh** — the cylindrical form of OKLab, where you can change chroma or turn the hue without changing lightness.
6. **Back to sRGB** — each step reversed, gamma-encoded again and clamped to 0–255.

Reflectance tab

- **Approximate reflectance** ("380–780 nm visible spectrum") — a chart of 36 bars across the visible spectrum, labelled every 50 nm from 380 nm to 780 nm. Taller bars mean more light reflected at that wavelength. A rainbow strip along the bottom marks the spectrum, and the whole chart is tinted with your colour. The note under the chart explains that this is a heuristic and that a spectroradiometer gives the real curve.
- **Peak reflectance data** — a table of the five wavelengths with the highest bars, with columns **Wavelength**, **Reflectance** (a percentage) and **Bar**.

Paint recipe tab

- **Best two-pigment mix** — a chip of the best mix the model found, labelled with its hex code and "Recreated — ΔE" and the difference from your colour. Beside it: "Mix *pigment A* and *pigment B* at *x*% / *y*% ratio", a swatch and hex code for each pigment, and a bar split in the two proportions.
- **Top 6 alternatives** — the next six pairs, each with the mixed colour, the two pigment names, the ratio (for example `45/55`) and the ΔE.
- A note under the cards: "Subtractive OKLab heuristic — best ΔE2000 match across 13 artist primaries × 21 ratios. Actual yield depends on pigment opacity, binder, and substrate."

The 13 pigments the model can combine, with the colours it uses for them:

Pigment	Colour used	Pigment	Colour used
Cadmium Red Light	`#E63A2E`	Phthalo Blue	`#0B3D7A`
Alizarin Crimson	`#A0182E`	Phthalo Green	`#0E7A4A`
Cadmium Yellow	`#FFD500`	Sap Green	`#3F6E2A`
Yellow Ochre	`#C5A050`	Dioxazine Purple	`#3B1B5C`
Burnt Sienna	`#8B3A1F`	Titanium White	`#FAF8F2`
Raw Umber	`#7B4F2E`	Mars Black	`#1A1A1A`
Ultramarine Blue	`#1E3A8A`		

The model tries every pair of these pigments, including a pigment with itself (91 pairs), at 21 ratios from 100/0 to 0/100 in steps of 5%.

Walkthroughs

Read how a colour is encoded

1. Open Color Formulation.

2. In the sidebar's hex box, type `D3AF37`.
3. Look at **Properties**. You see **Lightness (OKL)** 76.6%, **Chroma** 0.138, **Hue angle** 91.6°, **Saturation (HSL)** 64%, **Rel. luminance** 0.4478, **Tone character** Light tint and **Chroma character** Moderate chroma.
4. On the **Channels** tab, compare the **sRGB** bars (211, 175, 55) with the **Linear-light sRGB** bars. The blue channel drops from about a fifth of its range to under 4%, because gamma decoding makes dark values much smaller.
5. Select the **Pipeline** tab and read the six steps.

Result: you can see the same gold as stored numbers, as linear light, and as OKLab and OKLCh coordinates.

Find a two-pigment paint mix

1. Enter the colour you want to mix, for example `#D3AF37`.
2. Select the **Paint recipe** tab.
3. Read the **Best two-pigment mix** card. For `#D3AF37` it says "Mix **Cadmium Yellow** and **Yellow Ochre** at **45% / 55%** ratio", and the chip shows `#D1AA2E`, "Recreated — ΔE 1.52".
4. Look through **Top 6 alternatives** for pairs you already own. The ΔE under each card tells you how far that mix lands from your colour; lower is closer.

Result: a starting recipe and six fallbacks, each with the colour the model expects the mix to produce.

> **Tip:** Treat the recipe as a direction to start mixing in, then adjust by eye under the light where the work will be seen.

Look at the reflectance sketch

1. Enter a colour.
2. Select the **Reflectance** tab. The chart draws for the current colour.
3. To fill the **Peak reflectance data** table, change the colour while the tab is open (for example, pick a new colour, or change one digit of the hex code and change it back).

Result: a chart and a table of the five strongest wavelengths in the sketch. Read Accuracy and limits before drawing conclusions from them.

Controls

Control	What it does	Values or range	Default
Colour swatch (**Color picker**)	Opens your browser's colour picker and updates every reading as you pick.	Any sRGB colour.	The colour the tool was opened with, or `#D3AF37`.
Hex box (**Color hex**)	Type a hex code. The tool updates as soon as the text is a valid code; invalid text is ignored.	3, 4, 6 or 8 hex digits, with or without `#`. Transparency digits are ignored.	As above.
Dice button (**Random color**)	Sets a random colour.	Any sRGB colour.	—
Channels tab	Shows the channel bars.	—	Selected when the tool opens.
Pipeline tab	Shows the six conversion steps.	—	—
Reflectance tab	Shows the reflectance chart and the peak table.	—	—
Paint recipe tab	Shows the best mix and six alternatives.	—	—

Outputs and exports

Color Formulation has no copy or download buttons. Everything it shows is on screen: the readings, the channel values, the pipeline numbers, the reflectance sketch and the recipes. To copy the colour in a particular format, open it in Color Space Converter or the Color Inspector.

Accuracy and limits

- **The conversions are standard.** sRGB to linear light uses the sRGB transfer function, XYZ uses the D65 white, and OKLab and OKLCh follow Björn Ottosson's published definition. The numbers match what other colour tools give, within rounding.
- **The paint recipe is a rough model, not a pigment database.** It stands each pigment in as one sRGB colour and mixes pairs in OKLab with a small darkening step. It does not model pigment strength, opacity, the binder or the surface, which is why the tool's own note says actual yield varies.
- **In this model the ratio moves the hue and chroma but not the lightness.** The lightness of a mix is always the average of the two pigments' lightness, a little darker. A recipe that reads 100% / 0% therefore still depends on the second pigment: in the model, "Yellow Ochre + Titanium White at 100% / 0%" is lighter than Yellow Ochre alone. Read such a result as "this pigment, lightened (or darkened) towards the second one".
- **ΔE is the CIEDE2000 difference** between your colour and the modelled mix. Below about 2 most people see no difference side by side; above 5 the difference is clear.
- **The reflectance chart is a teaching sketch, not a spectrum.** It is drawn from the colour's HSL hue, saturation and lightness. The height of the chart follows the lightness, and the peak's position follows the hue angle, not the wavelengths a real surface of that colour reflects most. For example, a pure blue peaks near 643 nm in this chart, although a real blue surface reflects least there. Do not use the chart or the peak table as spectral data; measure the surface with a spectrophotometer instead.
- **Different surfaces can match on screen and differ in light.** Two paints that look the same under one light can separate under another (metamerism). No screen tool can predict that without real spectral data.

Related

- Colour Tools documentation — all fourteen tools
- Color Space Converter — the same colour in 34 notations, with gamut diagrams
- Color Inspector — every reading for one colour, and the **Open in** list
- CMYK Soft-Proof — how the colour might print on a press
- Perception and spectral tools — tools that work with real spectra, dyes and materials
- Basic Color Tools

WEBSITE › TOOLS › COLOUR WORKSPACE

Color Inspector

Use the Color Inspector to read one colour in every format, check it for contrast and colour-vision safety, find its name, build scales and harmonies, and export it as code.

The Color Inspector takes one colour and shows you everything the site can work out about it. A plain-language summary tells you how light, saturated, warm and readable the colour is. Around it sit ten tabs of detail: more than forty notations, contrast against real surfaces, how the colour looks with colour-vision deficiencies, harmonies, the nearest named colours and industry swatches, colour psychology, tint and shade scales, spectral science, and ready-to-paste code.

Use it when you meet a colour and want to know whether you can use it, and how. It is also the tool behind the site's `?color=` links: add `?color=` and a hex code to any page address and the Inspector opens over that page.

The Inspector is one of the six **Essentials** in the Colour Tools. It opens in a floating panel; for moving, resizing and closing panels, see Open the Colour Tools and manage their panels.

How to open it

- **Launcher:** right-click the page, choose **Colour tools** › **All colour tools…** (or press `Ctrl` + `Shift` + `C`), then select **Color Inspector** in the **Essentials** group.
- **Right-click a colour:** when you right-click a hex code or a colour swatch on the site, choose **Inspect in Colour Tools**. The Inspector opens with that colour.
- **A link:** add `?color=` and a colour to any page address, for example `https://auricartisan.com/?color=CEA1B5`. Add `&ctab=contrast` to open on a tab, or `&on=FFFFFF` to add a comparison colour. The full list of accepted forms is in Open the Colour Tools and manage their panels.

Opened without a colour, the Inspector starts on the site's gold, `#D3AF37`.

Screen tour

The panel has three areas: a strip of ten tabs across the top, a sidebar, and the main area that shows the selected tab. In a narrow panel the sidebar moves above the main area.

The sidebar

- **The hero swatch.** A large swatch of the colour with its hex code, its closest colour name, and four badges: the hue family, the correlated colour temperature in kelvin, the CIE L* lightness, and a readability badge (**AA on white**, **AA on black** or **Decorative**). Select the swatch to copy the hex code. When the name is only approximate, it starts with `≈`. If the colour came from a link with transparency, the swatch shows a checkerboard and an `α` tag; if the link held a fragment the site had to interpret, a tag shows the text it started from.
- **The colour field.** The colour swatch opens your browser's colour picker, the text box takes a hex code, and two icon buttons follow: the dice (**Random color** tooltip, "Jump to a perceptually balanced random color") picks a random colour of medium saturation and lightness, and the drop (**Screen picker** tooltip, "Sample any pixel on screen") lets you pick a colour from anywhere on your screen. The drop only appears in browsers that support picking colours from the screen.
- **Read-out.** Nine cells: **RGB**, **HSL**, **OKLCh**, **CIE L***, **Luminance**, **Chroma**, **Temp**, **On white** and **On black**. Select any cell to copy its value.
- **Compare.** Chips for comparison colours. They come from a link (every colour after the first, and any `on=` colours), and every colour you move away from in this panel is added too, most recent first, up to eight. Select a chip to inspect that colour instead. The section is hidden while it is empty.
- **Actions. Copy hex**, **Copy CSS**, **Share link**, **Save** and **Download swatch**. See Outputs and exports.
- **Open in.** Eight buttons that open another Colour Tool with this colour: **Contrast System**, **Harmony Studio**, **Space Converter**, **Pantone & Named**, **Accessibility Lab**, **Formulation**, **CMYK Soft-Proof** and **Psychology**.

Almost every swatch in the tabs is a button. Select one and the Inspector switches to that colour, and the colour you were on goes into **Compare**, so you can always step back.

The tab strip

The tabs are **Overview**, **Formats**, **Contrast**, **Vision**, **Harmony**, **Names**, **Psychology**, **Scales**, **Science** and **Export**. When the panel is too narrow to show them all, arrow buttons appear at each end of the strip to scroll it.

With a tab focused, `←` and `→` move to the previous or next tab (wrapping round), and `Home` and `End` jump to the first and last.

Overview

- **Summary.** A sentence or two in plain words, for example: "#D3AF37 — closest named color … is a light yellow, vivid. It is warm (3352 K) and carries body text on black but not on white." It describes lightness (from near-black to near-white), saturation (from achromatic to "at the edge of what sRGB can show"), temperature, which text use the colour can carry, and the colour's dominant wavelength.
- **Live preview.** The colour as a background with a headline, 14 px body copy, 11.5 px small print and a **Primary action** button, all in whichever of black or white reads better on it. Below it: **White text** and **Black text** ratios, and **Best ink**.
- **Channels.** Bars for R, G, B and alpha, and for hue, saturation, lightness and OKLCh chroma.
- **Quick numbers.** Hex, RGB, HSL, HSV, CMYK, CIELAB, CIELCh, OKLCh, luminance Y, hue family, temperature and total ink.
- **Shades → tints.** Eleven steps from dark to light with your colour in the middle, mixed in OKLCh. Select a step to inspect it.

Formats

A button, **Copy the full read-out (JSON)**, and 43 notations in five groups. Select any row to copy just that value.

Group	Notations
Web & CSS (19)	HEX, HEX (short), RGB, RGBA, RGB %, HSL, HSLA, HSV, CMYK, CIE XYZ (D65), CIELAB (D65), CIELCh (D65)
	OKLab, OKLCh, Linear sRGB, HEX + alpha, CSS modern rgb, HWB, HSI
Wide gamut (5)	Display P3, Adobe RGB, Rec. 2020 (as CSS `color()` values), ACEScg (linear), Linear sRGB
Perceptual & CIE (9)	CIE xyY, CIE u′v′, CIELAB (D50), CIELCh (D50), CIELUV, LCh(uv), Hunter Lab, IPT, JzAzBz
Video & broadcast (3)	YCbCr, YUV, YIQ
Code (7)	Decimal, Android (ARGB), Swift UIColor, SwiftUI, Flutter, OpenGL vec4, .NET ARGB

Contrast

Against real surfaces is a table that treats your colour as text on seven standard surfaces: White `#FFFFFF`, Paper `#F5F5F3`, Light gray `#E5E5E5`, Mid gray `#767676`, Charcoal `#333333`, Near black `#111111` and Black `#000000`. When the Inspector was opened by a link, two more rows show **This page's background** and **This page's text**, sampled from the page underneath. Every comparison colour adds a **Compared color** row.

Column	Meaning
Ratio	The WCAG 2.x contrast ratio.
Body / **Large**	The WCAG level for normal and for large text: AAA, AA or Fail.
UI	Pass at 3:1 or more, the WCAG threshold for icons, borders and other interface parts.
APCA Lc	The APCA lightness contrast of your colour as text on that surface.
APCA verdict	AAA (Lc 90 or more), AA (75), Body (60) or Fail.
ISO mod.	The ISO 9241-3 luminance modulation and its rating.
Preview	"Aa" in your colour on the surface.

A line under the table gives the reverse reading: white and paper text on your colour, in Lc.

Nearest accessible version offers four swatches, each your colour with its lightness changed and its hue kept, just enough to reach **AA body text on white** (4.5:1), **AAA body text on white** (7:1), **AA body text on black** and **AAA body text on black**.

Each shows the new hex and ratio, or "already passes". Select one to inspect it.

As a background shows your colour behind white text and behind black text, with the ratio and the body and large-text levels for each.

Vision

- **How this color is seen.** A table of typical vision and four colour-vision deficiencies: **Protanopia** (no long-wave, red-sensitive cones), **Deuteranopia** (no medium-wave, green-sensitive cones), **Tritanopia** (no short-wave, blue-sensitive cones) and **Achromatopsia** (no colour discrimination). For each: the simulated colour, its hex (select it to inspect), and the ΔE2000 shift from the original. A shift above ΔE 10 means the colour reads as a clearly different hue.
- **Filtered preview.** Five sample cards of your colour, one per vision type, drawn through live filters.
- **Grayscale survival.** Your colour next to its grey equivalent, with the grey's hex and L*. This is how it fares in black-and-white print, photocopies and e-ink.
- **Confusion risk.** Shown only when there are comparison colours. For each one: the ΔE between it and your colour for typical vision, the vision type under which they are closest, the ΔE there, and a verdict: **Distinct** (ΔE 10 or more), **Tight** (5 to 10) or **Confusable** (under 5).

Harmony

Every scheme is rotated in OKLCh, so companion colours keep the lightness and chroma of your colour instead of drifting the way HSL rotations do. The tab shows eight schemes, each swatch labelled with its hue and OKLCh lightness:

Scheme	Hue offsets
Complementary	0°, +180°
Split-Complementary	0°, +150°, +210°
Analogous	−30°, 0°, +30°
Triadic	0°, +120°, +240°
Tetradic	0°, +90°, +180°, +270°
Square	0°, +90°, +180°, +270°
Monochromatic	Six steps of lightness at your hue and chroma
Double Split	0°, +30°, −30°, +150°, +210°

Tetradic and Square use the same 90° spacing in this tab. **Accessible triad** then shows the triadic colours adjusted to reach 4.5:1 on white and, separately, on black, with each ratio.

Names

- **Closest named color.** Your colour beside the nearest of 1,058 named colours, the ΔE2000 between them, and what that distance means, from "indistinguishable" (under ΔE 1) to "the nearest entry, but visibly different" (ΔE 10 or more).
- **Industry systems.** The three closest entries in each of five systems: HTML/CSS (141 colours), Pantone Solid C (105), RAL Classic (148), NCS (56) and Crayola (111).
- **Similar named colors.** The next 17 names by ΔE2000.

This tab downloads its data the first time you open it, and shows "Loading names data…" until it arrives.

Psychology

- **Emotional read.** Emotion words for the colour's hue family at its lightness, and meters for **Warmth** (Warm, Neutral or Cool), **Energy** (High, Moderate or Calm), **Saturation pull** (OKLCh chroma) and **Lightness** (L*). Very light colours read

softer and calmer; very dark ones read more serious and grounded.
- **Personality.** Trait words.
- **Brand fit.** A sentence on where this hue family suits a brand.
- **Cultural meaning.** Positive and negative associations by region or tradition, for example China, India and Japan for red.

This tab also downloads its data the first time you open it.

Scales

- **Tints**, **Shades** and **Tones.** Ten steps each from your colour towards white, black and mid grey, mixed in OKLCh. Tones lower the chroma and keep the hue.
- **Design-token ramp.** Eleven steps named 50 to 950, each at a fixed OKLCh lightness, with chroma eased off towards the ends and limited to what sRGB can show at that lightness. Each step shows its hex and its contrast ratio on white and on black, and the step nearest your colour is marked with `◂`.
- **CSS custom properties.** The ramp as `--color-50` to `--color-950` inside `:root`, with a **Copy** button.

Science

- **Where this color sits in the spectrum.** Dominant wavelength (with its band: violet, blue, cyan, green, yellow, orange or red), excitation purity, CIE x, y, CIE u′, v′, tristimulus Y, correlated colour temperature and a temperature description from **Very warm** to **Very cool**. Purples and magentas have no single wavelength, so they show a complementary wavelength and "Line of purples" instead.
- **Approximate reflectance.** A spectrum strip from 380 to 780 nm, masked to show roughly which light a surface of this colour would send back.
- **Gamut headroom.** How much of the available chroma your colour uses at its lightness and hue in sRGB, the current and maximum chroma, and a 12-step ramp from grey to the most saturated version.
- **Coordinates in other gamuts.** The colour's R, G, B values in sRGB, Display P3, Adobe RGB and Rec. 2020.
- **On press.** A simple CMYK split, the total ink, and a pass mark for total ink up to 300% (a warning up to 320%). Most web presses cap total ink around 300%.

Export

- **Share this read-out.** A read-only box with the current page address plus `?color=`, and three buttons: **Copy link**, **Swatch file** and **Save to library**. A note reminds you that `?on=ffffff` adds a comparison colour and `&ctab=contrast` opens a tab.
- Six code blocks, each with a **Copy** button: **CSS custom properties**, **SCSS**, **Tailwind**, **Android colors.xml**, **SwiftUI** and **JSON — everything above**.

Walkthroughs

Check whether a brand colour can carry text

1. Open the Inspector with your colour, for example with `?color=D3AF37` on any page.
2. Read the summary on **Overview**. For `#D3AF37` it says the colour "carries body text on black but not on white".
3. Open the **Contrast** tab. The **White** row shows a ratio of 2.11 and **Fail** for body text; the **Black** row shows 9.96 and **AAA**.
4. Under **Nearest accessible version**, select **AA body text on white**.

The Inspector switches to a darker version of your gold that reaches 4.5:1 on white, and your original colour moves into **Compare** so you can go back to it.

Check that two colours stay distinct for colour-blind readers

1. Open the Inspector with both colours in one link, for example `?color=D3AF37,4F9D69`.
2. Open the **Vision** tab.
3. Read **Confusion risk**. It names the vision type under which the two colours are closest and gives a verdict.

If the verdict is **Confusable** or **Tight**, do not rely on the colour difference alone: add a label, an icon or a pattern.

Export a colour as design tokens

1. Set the colour you want in the colour field.
2. Open the **Export** tab.
3. Select **Copy** above the block you need, for example **Tailwind**.

The clipboard now holds a `tailwind.config.js` snippet with a `brand` scale from 50 to 950. The **CSS custom properties** block for `#D3AF37` starts like this:

```
:root {
  --brand: #d3af37;
  --brand-rgb: 211 175 55;
  --brand-oklch: oklch(0.7655 0.1385 91.55);
  --brand-ink: #000000;
  --brand-50: #fff5c2;
  --brand-100: #ffe998;
  /* … through --brand-950 */
}
```

Share exactly what you are looking at

1. Open the tab you want the other person to see, for example **Names**.
2. In the sidebar, select **Share link**.

A message confirms "Share link copied" with the address. The link opens the current page with the Inspector on the same colour and the same tab.

Controls

Control	What it does	Values or range	Default
Hero swatch	Copies the hex code.	—	—
Colour swatch (picker)	Opens the browser's colour picker.	Any sRGB colour	`#D3AF37`, or the colour you opened with
Hex text box	Sets the colour.	3, 4, 6 or 8 hex digits, `#` optional	Same as above
Dice button (**Random color**)	Picks a random colour of medium saturation and lightness.	—	—
Drop button (**Screen picker**)	Picks a colour from your screen.	Supported browsers only	—
Read-out cells	Copy one value each.	9 cells	—
Compare chips	Inspect a comparison colour.	Up to 8	Empty, or from the link
Copy hex	Copies the hex code.	—	—
Copy CSS	Copies `--color: #rrggbb;`.	—	—

Control	What it does	Values or range	Default
Share link	Copies a link to this page, colour and tab.	—	—
Save	Saves the colour to your Library.	—	—
Download swatch	Downloads a 512 × 512 px swatch.	SVG; hold `Shift` for PNG	SVG
Open in buttons	Open another Colour Tool with this colour.	8 tools	—
Tabs	Switch the main view.	10 tabs	**Overview**, or the `ctab` value
Tab scroll arrows	Scroll the tab strip.	Shown when tabs overflow	—
Swatches in tabs	Inspect that colour.	—	—
Formats rows	Copy one notation.	43 rows	—
Copy the full read-out (JSON)	Copies the JSON read-out.	—	—
Copy (Scales, Export)	Copies that code block.	7 blocks	—
Copy link / **Swatch file** / **Save to library**	Same as **Share link**, **Download swatch** and **Save**.	—	—

Outputs and exports

Output	Format	Notes
Copy hex	Text, for example `#D3AF37`	—
Copy CSS	`--color: #d3af37;`	—
Share link / **Copy link**	A URL with `color` and `ctab`	Opens the Inspector on the same page, colour and tab.
Download swatch / **Swatch file**	512 × 512 px SVG, or PNG with `Shift` held	Named after the hex code, for example `D3AF37.svg`.
Save / **Save to library**	A colour item in your Library	Named with the hex and closest name, tagged with the hue family.
Read-out cells, Formats rows	Text	One value each.
Copy the full read-out (JSON) and the JSON block	JSON	Hex, alpha, RGB, HSL, HSV, CMYK, CIELAB, CIELCh, OKLab, OKLCh, XYZ, CIE xy, luminance, contrast on white and on black, temperature, hue family, dominant wavelength, excitation purity, nearest name and total ink.
CSS custom properties	CSS	`--brand`, `--brand-rgb`, `--brand-oklch`, `--brand-ink`, `--brand-50` to `--brand-950`.
SCSS	SCSS variables	`$brand`, `$brand-ink`, `$brand-50` to `$brand-950`.
Tailwind	`tailwind.config.js` snippet	A `brand` colour scale 50 to 950.
Android colors.xml	XML resources	`brand_50` to `brand_950`.
SwiftUI	Swift `Color` extension	`brand50` to `brand950`.

After a copy, save or download, the button you used briefly shows a tick and a short message confirms what happened. If you select **Save** on a page where the Library is not available, the message says "The library isn't available on this page".

Accuracy and limits

- **Terms used on this page.** *WCAG contrast ratio*: the ratio between the relative luminance of two colours, from 1:1 to 21:1. WCAG 2 asks for 4.5:1 for normal text (AA), 3:1 for large text and interface parts, and 7:1 for AAA. *APCA Lc*: a newer measure of perceived lightness contrast, roughly −108 to +106. Positive values mean dark text on a lighter background, negative values light text on a darker one. *ΔE2000*: the CIEDE2000 colour difference; under about 1 two colours look the same side by side. *OKLCh*: a colour model with lightness, chroma and hue that changes evenly to the eye. *Colour-vision deficiency (CVD)*: reduced or missing sensitivity in one type of cone.
- **APCA verdicts are a guide.** WCAG 3, which may adopt APCA, is still a draft. The AAA, AA and Body labels are the site's reading of the published APCA guidance, not a conformance result. For a WCAG 2.2 audit, use the **Ratio** column.
- **ISO 9241-3 ratings.** The modulation is (Lmax − Lmin) / (Lmax + Lmin) from 0 to 1. The tool rates 0.7 and over Excellent, 0.5 Good, 0.3 Marginal and below that Poor.
- **Vision simulations are approximations.** They model full-strength dichromacy with simple fixed matrices. Real colour vision varies from person to person, and milder forms (anomalous trichromacy) are more common than the full forms shown. Use the results to find risks, not to certify a design.
- **Temperature.** Correlated colour temperature is calculated with McCamy's formula. It is meaningful for near-neutral colours and only indicative for saturated ones.
- **Reflectance is a heuristic.** The curve explains the colour; it is not a spectrophotometer reading and cannot be used to match pigments.
- **CMYK is arithmetic.** The **On press** split is a simple conversion, not an ICC profile conversion. Soft-proof in CMYK Soft-Proof before sending artwork to print.
- **Named-system matches are approximate.** Pantone, RAL and NCS entries are published sRGB approximations of printed swatches, so the tool always shows the ΔE distance rather than calling a match exact.
- **sRGB only.** Colours are handled as sRGB. Transparency from a link is shown and exported, but contrast is calculated on the solid colour.
- **Page sampling.** The "This page's" rows use the page's main background and text colour, not the colour behind a particular element.
- **Network.** The **Names** and **Psychology** tabs download their data from the site the first time you open them. If that fails, the tab says "That data set could not be loaded."

Related

- Colour Tools documentation
- Open the Colour Tools and manage their panels — `?color=` parameters, panels and the control bar
- Contrast System — deeper contrast work on several foregrounds
- Accessibility Lab — a whole palette against the standards and vision filters
- Color Name Finder — a quick name lookup
- Harmony Studio, Pantone & Named Lookup, Color Space Converter, CMYK Soft-Proof, Color Psychology
- Library Kit — where saved colours go

WEBSITE › TOOLS › COLOUR WORKSPACE

Color Library

Search and filter 1,058 named colours by family, open any colour's detail view with sixteen formats, a triadic harmony and similar colours, and download a swatch as SVG or PNG.

Color Library is a browsable catalogue of 1,058 named colours, such as *Crimson*, *Dusty Rose* or *Gunmetal*, grouped into 13 families. You can search by name or HEX code, filter by family, page through the results, and open any colour for a detail view

with every common format, its temperature, contrast against white and black, a triadic harmony and the twelve most similar named colours.

This in-panel library is different from the site's large colour collections (palettes, shades, gradients and the colour atlas) in the library area; for those, see Colour libraries.

Color Library is in the launcher's **Explore** group.

Open it

Open the launcher and select **Color Library**, or pick it from the right-click menu's **Colour tools** group. See Open the Colour Tools. If you open it with a colour (for example from another tool), it starts on that colour's detail view; otherwise it starts on the list.

Screen tour

Toolbar

- A search box (*Search by name or hex…*).
- A **Category** list: **All** and the 13 families.
- A count of matching colours, for example **146 colors**.

List view

- A grid of up to 60 colour cells per page, each with a swatch, name and HEX code.
- A pager: **‹ Prev**, **Page N of M** (the page number is a box you can type in) and **Next ›**.

Detail view

- **← Back to library**, **Copy HEX**, **Download SVG** and **Download PNG**.
- A large chip with the name and HEX code.
- A grid with **Hue category**, **HSL**, **OKLCh**, **Temperature** (an estimated colour temperature in kelvin and a label such as *Warm* or *Neutral cool*), **vs White** and **vs Black** (WCAG contrast ratios).
- **Formats** (*Click to copy*): sixteen rows.
- **Harmony — triadic**: three colours 120° apart in OKLCH hue.
- **Similar colors** (*by ΔE2000*): the twelve closest named colours with their ΔE.

Families

Family	Colours
Red	119
Pink	83
Orange	82
Yellow	86
Green	136
Blue	146
Purple	98

Family	Colours
Brown	92
Gray	79
White	47
Black	31
Metallic	33
Neon	26

Tasks

Find a named colour

1. Type part of a name (for example `sage`) or a HEX code (for example `#8a`) in the search box. The grid filters as you type and the page resets to 1.
2. Optionally choose a family in **Category**.
3. Use **Next ›**, **‹ Prev** or the page box to move through the results.

Result: the named colours that match, 60 per page.

Read a colour in full

1. Select a colour cell. The detail view opens.
2. Select any row under **Formats** to copy it. The rows are **HEX**, **HEX (short)**, **RGB**, **RGBA**, **RGB %**, **HSL**, **HSLA**, **HSV**, **CMYK**, **CIE XYZ (D65)**, **CIELAB (D65)**, **CIELCh (D65)**, **OKLab**, **OKLCh**, **Linear sRGB** and **CSS keyword** (the lower-case HEX code).
3. Select a harmony or similar colour to open its detail view.
4. Select ← **Back to library** to return to the list.

Result: the colour's values on your clipboard, and related colours to explore.

Download a swatch

In the detail view, select **Download SVG** or **Download PNG**. You get a 512 by 512 pixel square of the colour, named after the colour (for example `Dusty_Rose.svg`).

Send a colour to another tool

1. In the list, right-click a colour cell to get the site's colour menu (copy options, **Check its contrast**, **Build harmonies**, **Find its name**, **Inspect in Colour Tools**).
2. Hold `Shift` while you right-click for the Color Library's own menu: **Copy HEX**, **Download SVG**, **Download PNG**, **Open in Spaces**, **Find name**, **Open Accessibility** and **Open Psychology**.

Result: the colour opens as the starting colour of the tool you chose.

Controls

Control	What it does	Values or range	Default
Search box	Filters by name or HEX text	Free text	Empty

Control	What it does	Values or range	Default
Category	Filters by family	All and 13 families	All
Colour cell	Opens the detail view	—	—
‹ **Prev** / **Next** › / page box	Moves between pages	60 colours per page	Page 1
← **Back to library**	Returns to the list	—	—
Copy HEX	Copies the HEX code	—	—
Download SVG / **Download PNG**	Downloads a 512 × 512 swatch	—	—
Format row	Copies that format	16 rows	—
`Shift` + right-click a list cell	Opens the tool's colour menu	7 actions	—

Accuracy and limits

- The names are descriptive and come from a general list. Different sources use different names for the same colour.
- **Temperature** is estimated from the colour's chromaticity with a standard approximation for light sources. For strongly coloured or very dark colours the figure is rough, and some colours show *Indeterminate*.
- The search matches names and HEX text exactly as typed (not colour similarity). To find names by similarity, use Color Name Finder.
- The library is not a licensed colour system. For Pantone, RAL, NCS and similar matches, use Pantone & Named Lookup.

Related

- Colour Tools documentation
- Color Name Finder
- Pantone & Named Lookup
- Color Psychology
- Basic Color Tools

WEBSITE › TOOLS › COLOUR WORKSPACE

Color Name Finder

Enter a colour and find the closest of more than a thousand named colours, with seventeen similar colours and ten copyable formats.

Color Name Finder answers "what is this colour called?". You give it one colour and it finds the nearest match in a built-in list of 1,058 named colours, shows seventeen more close matches, and lists the colour in ten copyable formats. It is the quickest tool in the workbench: one input, three result sections.

The names come from a general-purpose colour-name list grouped into 13 families: red, pink, orange, yellow, green, blue, purple, brown, gray, white, black, metallic and neon. For matches against formal systems such as Pantone, RAL or NCS, use Pantone & Named Lookup.

Color Name Finder is in the launcher's **Essentials** group.

Open it

Open the launcher and select **Color Name Finder**, pick it from the right-click menu's **Colour tools** group, or right-click a colour value and select **Find its name**. See Open the Colour Tools. It starts with the colour you opened it with, or gold (#D3AF37).

Screen tour

1. **Source**: the colour field (a colour well, a HEX box and a dice button for a random colour).
2. **Nearest match** (with the note **ΔE2000**):
   - two large chips side by side: your colour with its HEX code and hue family, and the nearest named colour with its name, HEX code, **ΔE** distance and name family;
   - a small grid with **HSL**, **OKLCh**, **RGB** and **Hue family**.
3. **Similar colors**: a grid of the next seventeen closest named colours, each with a swatch, its name, HEX code and ΔE.
4. **Full color spaces** (with the note **Click to copy**): ten rows you can select to copy.

Tasks

Name a colour

1. Type a HEX code in the Source box, or use the colour well.
2. Read the second chip under **Nearest match**. Its label is the colour's nearest name.
3. Check the **ΔE** figure under it. Below about 2 the name is a close match; above about 5 it is only the nearest of the names available.

Result: a name you can use in conversation, a style guide or a token name.

Walk to a nearby named colour

1. Under **Similar colors**, select any cell.
2. The Source changes to that named colour and every section updates.

Result: you can step through related names until you find the one you want.

Copy a format

Select a row under **Full color spaces**. The value is copied and a short **Copied** message appears. The rows are:

Row	Example for #D3AF37
HEX	`#D3AF37`
HEX (short)	`#D3AF37` (a three-digit form when one exists, such as `#FA0` for #FFAA00)
RGB	`rgb(211, 175, 55)`
RGBA	`rgba(211, 175, 55, 1)`
RGB %	`rgb(82.7% 68.6% 21.6%)`
HSL	`hsl(46.2, 63.9%, 52.2%)`
HSLA	`hsla(46.2, 63.9%, 52.2%, 1)`
HSV	`hsv(46.2, 73.9%, 82.7%)`

Row	Example for #D3AF37
CMYK	`cmyk(0%, 17.1%, 73.9%, 17.3%)`
CIE XYZ (D65)	`xyz(0.4289, 0.4479, 0.1)`

How matching works

The tool converts your colour and every named colour to CIELAB and measures the difference with **ΔE2000**, the CIE's standard formula for how different two colours look. The list is sorted by that difference: the smallest is the nearest match, the next seventeen are the similar colours.

The **Hue family** is a plain-language label worked out from the colour's HSL values: Black, White or Gray for very dark, very light or unsaturated colours, otherwise Red, Orange, Yellow, Lime, Green, Teal, Cyan, Blue, Purple, Magenta or Pink by hue.

Controls

Control	What it does	Values or range	Default
Colour well	Sets the source colour with your browser's picker	Any sRGB colour	#D3AF37, or the colour you opened with
HEX box	Sets the source colour as you type	3, 4, 6 or 8 hex digits, with or without `#`	Same
Dice (**Random color**)	Picks a random colour	—	—
Similar colour cell	Makes that named colour the source	17 cells	—
Format row	Copies that value	10 rows	—

Outputs and exports

- Copy any of the ten formats from **Full color spaces**.
- The tool has no file export. For code snippets and scales, open the colour in the Color Inspector.

Accuracy and limits

- The name list is a general one. Names are descriptive, not standards; two lists can call the same colour different things.
- ΔE2000 is computed from sRGB values with a D65 white point. It predicts differences seen side by side under good viewing conditions.
- The nearest match is only as near as the list allows. Check the ΔE before relying on a name.
- A colour's transparency is ignored.

Related

- Colour Tools documentation
- Pantone & Named Lookup for Pantone, RAL, NCS, Crayola and CSS names
- Color Library to browse the same named colours by family
- Color Inspector
- Basic Color Tools

WEBSITE › TOOLS › COLOUR WORKSPACE

Color Psychology

Enter one to six colours and read their combined warmth and energy, emotion words, personality traits, brand-fit notes and cultural meanings by region.

Color Psychology gives a quick reading of what colours tend to suggest. For up to six colours it shows a combined palette mood (how warm and how energetic the set feels, with emotion words and personality traits), and for each colour its emotional associations, personality, the kinds of brand it tends to suit, and what it can mean in different cultures and traditions.

The readings are organised by hue family. The tool decides each colour's family (for example Red, Teal or Gray) from its hue, saturation and lightness, looks up that family's associations, and adjusts them for very light or very dark colours.

Color Psychology is in the launcher's **Explore** group.

Open it

Open the launcher and select **Color Psychology**. See Open the Colour Tools. It starts with one colour: the colour you opened with, or gold (#D3AF37).

Screen tour

1. A toolbar with **+ Color**, **− Color**, **Random** and a count such as **2 colors**.
2. **Colors** (*1–6*): one colour field per colour, labelled **Color 1**, **Color 2** and so on.
3. **Palette mood**:
   - a strip of your colours;
   - a **Warmth** meter from **Cool** to **Warm** and an **Energy** meter from **Calm** to **Stimulating**;
   - a sentence such as *The palette feels warm and inviting, with high energy.*;
   - up to eight emotion tags and up to twelve trait tags drawn from all the colours.
4. One reading per colour:
   - a large swatch, the heading *Color 1 — Red*, and an estimated colour temperature with a label;
   - **Emotion**, **Personality** and **Brand fit**;
   - **Cultural meanings**: a table of regions or traditions, each with a positive association (+) and, where the data has one, a negative one (−).

Hue families

The tool uses 14 families plus a fallback: **Red**, **Orange**, **Yellow**, **Lime**, **Green**, **Teal**, **Cyan**, **Blue**, **Purple**, **Magenta**, **Pink**, **Black**, **White**, **Gray**, and **Other** for anything that does not fit.

Very light colours (HSL lightness above 78 %) gain the emotions *Soft* and *Calming*, the traits *gentle* and *airy*, and lower energy. Very dark colours (lightness below 25 %) gain *Serious* and *Grounded*, the traits *intense* and *deep*, lower energy and a little less warmth.

How the mood is worked out

Each family has a warmth score and an energy score from −100 to +100. The palette mood averages them across your colours:

Average	Sentence
Warmth above 30	*warm and inviting*
Warmth below −30	*cool and considered*
Warmth in between	*balanced*
Energy above 30	*high energy*
Energy below −10	*low energy*
Energy in between	*moderate energy*

Cultural meanings

The cultural table covers regions and traditions such as Western, China, Japan, Korea, India, Hindu, Buddhism, Tibetan, Thailand, Egypt, Middle East, Islam, Africa, South Africa, Ireland and the Netherlands. Which ones appear depends on the colour family; the Western row appears for almost every family.

Tasks

Check the mood of a brand palette

1. Enter your main brand colour in **Color 1**.
2. Select **+ Color** to add each supporting colour (up to six).
3. Read **Palette mood**: the two meters, the sentence and the tags.

Result: a short description of how the palette is likely to come across, useful for a brief or a design review.

Check a colour for an international audience

1. Enter the colour.
2. Read **Cultural meanings** under its reading, and look for negative associations in the markets you care about.

Result: a list of associations to discuss with people who know those markets.

Explore random combinations

1. Select **Random**. Every colour is replaced with a new random colour and the readings update.

Result: quick inspiration for moods you might not have considered.

Controls

Control	What it does	Values or range	Default
+ Color	Adds a random colour	Up to 6 colours	—
− Color	Removes the last colour	At least 1 remains	—
Random	Replaces every colour with a random one	—	—
Colour fields	Set each colour	HEX, colour well, dice	#D3AF37, or the colour you opened with

Outputs and exports

The tool has no export. Copy colours from their fields, and quote the readings as you need.

Accuracy and limits

- Colour associations are cultural generalisations drawn from common design and marketing references. They vary between people, generations, industries and contexts, and they change over time.
- The same family reading applies to every colour in the family; two different blues get the same associations unless one is very light or very dark.
- Cultural meanings are brief summaries, not research. Check important decisions with people from the audience in question.
- The colour temperature figure is an estimate for light sources and can read *Indeterminate* for some colours.

Related

- Colour Tools documentation
- Color Library
- Harmony Studio
- Personalization Generator
- Basic Color Tools

WEBSITE › TOOLS › COLOUR WORKSPACE

Color Space Converter

Convert one colour into 34 notations across eight groups, from HEX and HSL to Lab, OKLCH, Display P3, Rec.2020, ACEScg, JzAzBz and YCbCr, with gamut badges, two diagrams, channel bars and a JSON export.

Color Space Converter writes one colour in every colour space you are likely to meet, from web notations to scientific and broadcast ones: 34 notations in eight groups. Each row has a copy button, wide-gamut rows show whether the colour fits the space, and two diagrams place the colour on the CIE 1931 chromaticity chart and on an OKLab slice. A third tab shows the main channels as bars.

A **colour space** is a system of numbers that describes colours. Different spaces suit different jobs: HEX and RGB for screens and code, CMYK for print, Lab and OKLab for measuring how different colours look, P3 and Rec.2020 for wide-gamut screens, YCbCr for video. A **gamut** is the range of colours a space or device can represent.

Color Space Converter is in the launcher's **Deep dive** group.

Open it

Open the launcher and select **Color Space Converter**, or pick it from the right-click menu's **Colour tools** group. See Open the Colour Tools. It starts with the colour you opened with, or gold (#D3AF37).

Screen tour

Sidebar

- **Color**: a colour field.

- A large chip showing the colour and its HEX code.
- Key figures: **Luminance** (WCAG relative luminance), **OKLCh L**, **Chroma**, **Hue°**, **Saturation** (HSL) and **CCT** (estimated colour temperature with a label).
- **Copy All (JSON)**.

Main area tabs

Tab	What it shows
All Spaces	A search box (*Search 30 spaces…*), a count, and the 34 notations in eight groups.
Diagrams	**CIE 1931 xy Chromaticity** and **OKLab a–b Gamut Plane**, plus six figures.
Channels	Bars for the channels of eight spaces.

The notations

Group	Rows
Web / Device-RGB	HEX, HEX (short), RGB, RGBA, RGB %
Cylindrical / Perceptual Device	HSL, HSV / HSB, HWB, HSI
Print / Subtractive	CMYK, Linear sRGB
CIE Standard	XYZ D65, XYZ D50, Lab (D65), LCh (D65), Lab (D50), LCh (D50)
Perceptual Uniform	OKLab, OKLCh, CIELuv, LChuv
Wide Gamut	Display P3, Linear P3, Adobe RGB 1998, Linear AdobeRGB, Rec.2020, Linear Rec.2020, ACEScg (AP1)
Scientific / Appearance	Hunter Lab, IPT, JzAzBz
Video / Broadcast	YCbCr BT.709, YUV, YIQ (NTSC)

Each group heading shows how many spaces it holds. The wide-gamut rows carry a badge: ✓ **In** when all three channels are between 0 and 1, **! Out** otherwise.

What some of these are:

- **HWB**: hue, whiteness and blackness, a CSS colour notation.
- **HSI**: hue, saturation and intensity, used in image processing.
- **Linear sRGB**: sRGB with the gamma curve removed, so values are proportional to light.
- **XYZ D65 / D50**: CIE tristimulus values for a daylight (D65) or print-standard (D50) white.
- **Lab and LCh (D65 and D50)**: CIELAB and its cylindrical form. CSS `lab()` and `lch()` are defined relative to D50, so use the D50 rows for CSS.
- **OKLab and OKLCh**: a modern perceptual space where equal steps look roughly equal; CSS supports `oklab()` and `oklch()`.
- **CIELuv and LChuv**: another CIE perceptual space, used for lighting and displays.
- **Display P3, Adobe RGB 1998, Rec.2020**: wider-gamut RGB spaces for modern screens, photography and HDR video. The encoded rows use CSS `color()` syntax, such as `color(display-p3 …)`.
- **ACEScg (AP1)**: a linear space used in film and visual-effects rendering.
- **Hunter Lab**: an older opponent-colour space still used in some industries.
- **IPT** and **JzAzBz**: perceptual spaces from colour science research; JzAzBz is designed to also cover high dynamic range.
- **YCbCr BT.709**, **YUV**, **YIQ (NTSC)**: luma and colour-difference encodings used in video.

Tasks

Copy a colour in a particular notation

1. Set **Color**.
2. On **All Spaces**, type part of a name or value in the search box, for example `p3` or `lab`. Only matching rows show, and the count reads *N matching spaces*.
3. Select the copy button on the row. **Copied** appears.

The copied text sometimes differs slightly from the displayed text: for example HSL is copied with commas (`hsl(46.2, 63.9%, 52.2%)`) and the degree signs shown in LCh rows are left out, so the copied value is valid in CSS or code.

Result: the notation you need on your clipboard.

Export every notation at once

Select **Copy All (JSON)**. A JSON object with one key per notation (for example `"hex"`, `"oklch"`, `"p3"`, `"rec2020"`) is copied, and **All spaces copied as JSON** appears.

```
{
  "hex": "#D3AF37",
  "hex-s": "#D3AF37",
  "rgb": "rgb(211, 175, 55)",
  "oklch": "oklch(0.7655 0.1385 91.6)",
  "p3": "color(display-p3 …)"
}
```

The real output has all 34 keys.

See where a colour sits

1. Open **Diagrams**.
2. **CIE 1931 xy Chromaticity** shows the horseshoe of visible colours, the white point (D65) as a small white dot, and four gamut triangles: **sRGB**, **P3**, **a98-RGB** and **Rec.2020**. Your colour is a ringed dot with guide lines.
3. **OKLab a–b Gamut Plane** shows a slice through OKLab at your colour's lightness (given as *L = N%*). The coloured area is the part of the slice that sRGB can show; the grey area is outside sRGB. Your colour is the ringed dot.
4. Under the diagrams, read **xy chromaticity**, **Y (luminance)**, **Rel. luminance**, **CCT**, **Temp. kind** and **Hue category**.

Result: you can see how close the colour is to the edge of sRGB, and how much room the wider gamuts leave.

Compare channels

Open **Channels** to see bars for **sRGB (gamma-encoded, 0–255)**, **Linear-light sRGB (0–1)**, **HSL**, **OKLab (perceptually uniform)**, **OKLCh (cylindrical)**, **CIE XYZ D65**, **Display P3 (encoded, 0–1)** and **YCbCr BT.709 (0–1)**. Each bar shows the value as a number and as a filled length.

Controls

Control	What it does	Values or range	Default
Color	The colour to convert	HEX, colour well, dice	#D3AF37, or the colour you opened with
Copy All (JSON)	Copies all notations as JSON	34 keys	—

Control	What it does	Values or range	Default
Tabs	Switch view	All Spaces, Diagrams, Channels	All Spaces
Search box	Filters rows by name or value	Free text	Empty
Row copy button	Copies one notation	—	—

Accuracy and limits

- Every conversion starts from an 8-bit sRGB colour, so values are only as precise as that input. A colour chosen here is always inside sRGB, so the wide-gamut badges read ✓ **In**; the badges are useful for colours near the edges and as a reminder of what each space can hold.
- D65 values use the sRGB white point directly. D50 values use a standard chromatic adaptation from D65.
- **CCT** uses a common approximation for light sources. It is meaningful for near-white and near-neutral colours and rough for saturated ones; some colours show *Indeterminate*.
- The chromaticity diagram's fill colours are approximate and clipped to what your screen can show.
- The scientific spaces (IPT, JzAzBz, Hunter Lab) follow their published formulas with standard assumptions (for example an absolute luminance for JzAzBz). Use a colour-science library for research-grade work.

Related

- Colour Tools documentation
- Quick Converter, the fourteen-notation converter in the workspace
- Color Inspector
- Color Science Lab
- Oklab and Oklch on Learn
- Basic Color Tools

WEBSITE › TOOLS › COLOUR WORKSPACE

Contrast System

Check up to five foreground colours on one background under WCAG 2.x, APCA and ISO 9241-3, read an APCA font-size table, sweep lightness on a contrast curve, fix colours to AA or AAA, and preview them in a realistic interface.

Contrast System is the workbench's in-depth contrast tool. You set one background and up to five foreground colours. For each pair it reports three measures side by side, suggests the nearest colour that passes, shows which font sizes and weights the pair can carry under APCA, and draws the pair in a realistic interface sample.

The three measures:

- **WCAG 2.x contrast ratio**: the luminance ratio used by WCAG 2.0, 2.1 and 2.2, from 1:1 to 21:1. 4.5:1 is AA for normal text, 7:1 is AAA, 3:1 is AA for large text.
- **APCA Lc** (shown as **WCAG 3 APCA**): the Accessible Perceptual Contrast Algorithm, proposed for WCAG 3. **Lc** (lightness contrast) runs from about −108 to +106. A negative value means light text on a dark background, a positive value dark text on a light background. Readability depends on its size and on the font's size and weight.
- **ISO 9241-3 modulation (M)**: (Lmax − Lmin) / (Lmax + Lmin), from 0 to 1, from the ergonomics standard for display text.

Contrast System is in the launcher's **Essentials** group.

Open it

Open the launcher and select **Contrast System**, pick it from the right-click menu's **Colour tools** group, or right-click a colour value and select **Check its contrast**. See Open the Colour Tools. It starts with a dark navy background (#0F172A) and one foreground: the colour you opened it with, or gold (#D3AF37).

Screen tour

Sidebar

- **Background**: one colour field.
- **Foreground colors**: one colour field per foreground, labelled **FG 1**, **FG 2** and so on, with **+ Add**, **− Remove** and **Swap BG↔FG1**.
- **Text size**: **Normal**, **Large** or **Small**. Changes the scale of the **Preview** tab.
- **Preview filter**: a vision filter for the **Preview** tab.
- **Quick presets**: five background-and-foreground sets.
- A summary with one line per foreground: its WCAG ratio (coloured green, amber or red) and its APCA Lc.

Main area tabs

Tab	What it shows
Analysis	One card per foreground with all three measures and fix buttons.
APCA Guide	The Lc value for one pair, a plain-language verdict, and a font size and weight table.
Curve	How the contrast ratio changes as the foreground gets lighter or darker, with suggested alternatives and the nearest AA and AAA colours.
Preview	A realistic interface drawn in your colours, with a stats card per foreground.

The Analysis tab

Each card shows:

- A header with the background swatch, a small **Aa — FG 1 on BG** sample, both HEX codes and the foreground swatch.
- **WCAG 2.x**: a bar, the ratio (for example `7.21:1`) and two level pills, **norm** and **large**.
- **WCAG 3 APCA**: a bar, the Lc value with its sign (for example `Lc -86.4`) and two level pills, **body** and **large**.
- **ISO 9241**: a bar, the modulation (for example `M 0.912`) and a level pill.
- Four buttons: copy the foreground HEX code, → **Fix to AA**, → **Fix to AAA** and **Curve**.

Pills are green for a pass, amber for a borderline result and red for a fail. The levels:

Measure	Levels
WCAG 2.x, normal	**AAA** 7:1, **AA** 4.5:1, else **Fail**
WCAG 2.x, large	**AAA** 4.5:1, **AA** 3:1, else **Fail**
APCA, body	**AAA** Lc 90, **AA** Lc 75, **Body** Lc 60, else **Fail**
APCA, large	**AAA** Lc 75, **AA** Lc 60, **Body** Lc 45, else **Fail**
ISO 9241-3	**Excellent** 0.7, **Good** 0.5, **Marginal** 0.3, else **Poor**

APCA levels use the size of Lc, whichever its sign.

Tasks

Check your text colours on a background

1. Set **Background** to your surface colour.
2. Set **FG 1** to your body text colour. Select **+ Add** for headings, links and muted text (up to five foregrounds).
3. Read each card on **Analysis**. Look at the **norm** pill for body text and the **large** pill for headings.

Result: every text colour rated under three measures at once.

Fix a colour that fails

1. On a failing card, select → **Fix to AA** (4.5:1) or → **Fix to AAA** (7:1).
2. The foreground changes to the nearest passing colour and **Fixed to AA** or **Fixed to AAA** appears.

The fix keeps the colour's hue and chroma and moves only its OKLCH lightness, in small steps: lighter on a dark background, darker on a light one. It stops at the first step that passes. If no lightness can reach the target (for example 7:1 on a mid-grey background), it stops at the lightest or darkest version it tried, which may still fail; the card shows the result.

Result: a foreground that looks like the original but meets the ratio.

Pick a size and weight with the APCA guide

1. Open **APCA Guide**. Choose a pair with the **FG** buttons next to **Pair:**.
2. Read the large Lc value and the verdict under it:

Lc (absolute)	Verdict shown
90 or more	AAA — suitable for all body text, small and large.
75 to 90	AA — passes body text at regular weights and sizes.
60 to 75	Fluent — readable for mid-size and larger text.
45 to 60	Spot — suitable for short labels and larger UI text only.
30 to 45	Non-text / large decorative elements only.
Under 30	Insufficient — fails all text readability thresholds.

3. Read the table. Rows are font sizes (10, 12, 14, 16, 18, 24, 32 and 48 pixels); columns are font weights (100 to 900). Each cell shows:
   - ✓: passes with room to spare (or the combination needs no minimum);
   - ✓~: passes with a small margin (less than 15 Lc above the minimum);
   - ✗: fails; hover to see the Lc needed;
   - —: that size and weight is not usable at any contrast.

Result: you know the smallest and thinnest type the pair can support, for example "16 px needs weight 400 or more".

Explore lighter and darker alternatives

1. Open **Curve**, or select **Curve** on an analysis card. Choose a pair next to **Pair:**.
2. Read the chart **WCAG contrast ratio curve — FG lightness sweep**. The line shows the contrast ratio (vertical axis, up to 21) for every OKLCH lightness of the foreground (horizontal axis, dark to light), keeping its hue and chroma. Dashed lines

mark **7:1 AAA**, **4.5:1 AA** and **3:1 Lg**. A ringed dot marks the current foreground.
3. Under the chart, read the three current figures (**WCAG 2.x**, **APCA Lc**, **ISO mod**) with their levels.
4. Under **Alternatives**, eight chips show the foreground at eight lightness levels with their ratio and level. Select one to use it.
5. Below, two cards show **Nearest AA (4.5:1)** and **Nearest AAA (7:1)**, each with a preview, its HEX code and ratio. Select **Use this color** to apply it.

Result: the same colour family at a lightness that works, chosen by eye.

Preview the pair in an interface

1. Open **Preview**.
2. The sample draws a small product interface in the background colour, with **FG 1** as the text colour and **FG 2** as the accent: a header (**Panel Preview**, the active filter and the FG 1 ratio), a heading (*The quick brown fox*), body text, 11-pixel small print, **Primary** and **Secondary** buttons, an **FG 2 link**, a **Badge**, a dashboard card with APCA Lc and ISO figures, a form with an email field and a status list, a table of levels (**Normal text**, **Large text**, **APCA body**) and three alert states.
3. A strip at the bottom shows *Aa* under **Normal**, **Deuteranopia**, **Protanopia**, **Tritanopia** and **Achromatopsia**.
4. Change **Text size** to **Small** or **Large** to scale the sample, and **Preview filter** to see it through a vision filter.
5. Under the sample, one stats card per foreground shows its ratio, APCA Lc and level, ISO modulation and level, and a WCAG pill.

Result: you judge the colours in context, not only as numbers.

Quick presets

Preset	Background	Foregrounds
Dark UI	#0F172A	#FFFFFF, #D3AF37
Light UI	#F9FAFB	#111827, #1D4ED8
High contrast	#000000	#FFFFFF, #FFFF00
Brand gold	#1A1A2E	#D3AF37, #E8E8E8
Alert red	#FEF2F2	#991B1B, #B91C1C

Controls

Control	What it does	Values or range	Default
Background	Background colour	HEX, colour well, dice	#0F172A
FG 1 … **FG 5**	Foreground colours	1 to 5	#D3AF37, or the colour you opened with
+ Add	Adds a random foreground	Up to 5	—
− Remove	Removes the last foreground	At least 1 remains	—
Swap BG↔FG1	Exchanges the background and FG 1	—	—
Text size	Scale of the Preview sample	Normal, Large, Small	Normal
Preview filter	Vision filter on the Preview sample	Normal vision, Deuteranopia, Protanopia, Tritanopia, Achromatopsia, Low vision	Normal vision

Control	What it does	Values or range	Default
Quick presets	Load a preset	5 presets	—
Copy HEX (card)	Copies the foreground HEX	—	—
→ Fix to AA / **→ Fix to AAA**	Adjusts lightness to 4.5:1 or 7:1	—	—
Curve (card)	Opens the Curve tab for that pair	—	—
Pair: buttons	Choose the foreground for APCA Guide and Curve	FG 1 to FG 5	FG 1
Alternative chips	Apply one of eight lightness variants	OKLCH L 0.05 to 0.95	—
Use this color	Apply the nearest AA or AAA colour	—	—

Outputs and exports

- Copy a foreground's HEX code from its analysis card.
- Fixed and suggested colours replace the foreground, so you can copy them from the colour field.
- There is no file export. For code output, open the colour in the Color Inspector.

Accuracy and limits

- WCAG ratios use the standard sRGB relative-luminance formula.
- The APCA value follows a simplified form of the published algorithm, and the size and weight table is based on the APCA bridge guidance for WCAG 3. APCA is still a draft; treat its results as guidance.
- ISO 9241-3 modulation is computed from relative luminance, not measured on a screen.
- The fix and the curve change only OKLCH lightness. Colours outside the sRGB range are clipped, which can shift them slightly.
- Vision filters are simple simulations; see Accessibility Lab for their limits.

Related

- Colour Tools documentation
- Accessibility Lab for a whole palette at once
- Type Readability Sim
- Animation Contrast for text over moving backgrounds
- Contrast Checker and other accessibility tools
- Basic Color Tools

WEBSITE › TOOLS › COLOUR WORKSPACE

Gradient Library

Browse and filter 512 curated linear gradients by colour scheme and number of stops, open any gradient's details, and copy it as CSS, a CSS variable, a Tailwind class, SVG or JSON, or download it as SVG or PNG.

Gradient Library is a collection of 512 ready-made linear gradients. Each has a name, an angle, two to four colour stops and a colour scheme it was built from. You can filter by scheme and number of stops, search by name, and open any gradient to see

its stops in detail and copy it in five code formats or download it as an image.

When you want to build your own gradient instead, use Gradient Maker in the Basic Color Tools workspace. For the site's full gradient collection, see Colour libraries.

Gradient Library is in the launcher's **Design** group.

Open it

Open the launcher and select **Gradient Library**, or pick it from the right-click menu's **Colour tools** group. See Open the Colour Tools. It shows **Loading gradients…** while it downloads the collection from the site the first time you open it.

Screen tour

Toolbar

- A search box (*Search gradients…*).
- A **Scheme** list.
- A **Stops** list: **Any stops**, **2 stops**, **3 stops** or **4 stops**.
- A count of matching gradients, for example **72 gradients**.

List view

A grid of up to 24 gradient cards per page. Each shows the gradient, its name, and its number of stops and scheme. A pager with **‹ Prev**, **Page N of M** and **Next ›** sits underneath.

Detail view

- **← Back to library**, **Download SVG** and **Download PNG**.
- A large preview, and **Name**, **Scheme**, **Angle** and **Stops**.
- **Stops**: one card per stop with its HEX code, OKLCH lightness, chroma and hue, and its position.
- **Export**: five rows, each with the code and a **Copy** button.

The collection

Scheme	Gradients
Analogous	76
Complement	76
Triadic	72
Tetradic	72
WarmCool	72
Sunset	72
Mono	72

By number of stops: 128 gradients have two stops, 128 have three and 256 have four. Names follow the pattern *Duo Analogous #001*.

Tasks

Find a gradient

1. Choose a **Scheme**, for example **Sunset**.
2. Choose a number of **Stops**.
3. Optionally type part of a name in the search box.
4. Page through the results.

Result: a grid of matching gradients, 24 at a time.

Copy a gradient into your project

1. Select a gradient card to open its detail view.
2. Under **Export**, select **Copy** beside the format you need:

Row	Example
CSS	`linear-gradient(15deg, #9A0076 0.0%, #FFBBCF 100.0%)`
CSS variable	`--gradient: linear-gradient(15deg, …);`
Tailwind arbitrary	`bg-[image:linear-gradient(15deg,_#9A0076_0.0%,_#FFBBCF_100.0%)]`
SVG	An SVG with a left-to-right linear gradient
JSON	The gradient's full record: angle, stops (position, HEX and OKLCH), name and scheme

Copied appears after each copy.

Result: the gradient in the format your code expects.

Download an image

In the detail view, select **Download SVG** or **Download PNG**. You get a 1280 by 320 pixel image named after the gradient (for example `Duo_Analogous__001.png`), drawn at the gradient's angle.

The SVG code under **Export** always runs left to right; the downloaded SVG follows the gradient's angle.

Work with a single stop colour

1. In the detail view, right-click a stop card for the site's colour menu (copy options, **Check its contrast**, **Build harmonies**, **Find its name**, **Inspect in Colour Tools**).
2. Hold `Shift` while you right-click for the Gradient Library's own menu: **Copy HEX**, **Open in Spaces** and **Find name**.

Controls

Control	What it does	Values or range	Default
Search box	Filters by name	Free text	Empty
Scheme	Filters by scheme	All, Analogous, Complement, Triadic, Tetradic, WarmCool, Sunset, Mono	All
Stops	Filters by number of stops	Any, 2, 3, 4	Any
Gradient card	Opens the detail view	—	—

Control	What it does	Values or range	Default
Pager	Moves between pages	24 per page	Page 1
← **Back to library**	Returns to the list	—	—
Copy (Export rows)	Copies that format	5 formats	—
Download SVG / **Download PNG**	Downloads a 1280 × 320 image	—	—
`Shift` + right-click a stop	Opens the tool's colour menu	3 actions	—

Accuracy and limits

- All gradients are linear and blend in sRGB, as CSS does by default.
- The collection needs a network connection the first time it loads in a session. If it cannot load, the panel shows a message that the gradient data could not be loaded.
- Gradients were generated from colour-scheme rules; they are starting points, not checked for text contrast. Check any text you place on a gradient with Contrast System or Animation Contrast.

Related

- Colour Tools documentation
- Gradient Maker
- Color Gradients and Interpolation on Learn
- Basic Color Tools

WEBSITE › TOOLS › COLOUR WORKSPACE

Harmony Studio

Generate eight kinds of colour harmony from one base colour, optionally adjust every swatch to meet a contrast target on your background, preview them and export HEX, CSS, JSON or PNG.

Harmony Studio builds a palette from one base colour using a classic colour-harmony pattern, such as complementary, triadic or square, and shows it on a colour wheel. It can then adjust each colour so that it reaches a contrast target against a background you choose, which turns a pretty palette into one you can use for text and interface elements.

A **colour harmony** is a set of hues chosen by their position on the colour wheel: opposite each other (complementary), evenly spaced (triadic, square), or close together (analogous). Harmony Studio rotates hue in OKLCH, a perceptual colour space, so the colours keep the base colour's lightness and chroma.

Harmony Studio is in the launcher's **Design** group.

Open it

Open the launcher and select **Harmony Studio**, pick it from the right-click menu's **Colour tools** group, or right-click a colour value and select **Build harmonies**. See Open the Colour Tools. The base colour starts as the colour you opened with, or gold (#D3AF37); the background starts as dark navy (#0F172A).

Screen tour

Sidebar

- **Base color** and **Background**: colour fields.
- **Scheme**: the harmony pattern.
- **Mode**: how the colours are adjusted for contrast.
- **Mono count**: a slider for the number of colours in the **Monochromatic** scheme.
- **Export**: **Copy HEX list**, **Copy CSS variables**, **Copy JSON** and **Download PNG**.

Main area

1. **Harmony wheel**, with a summary such as *3 colors · Standard (math)*: a hue ring with a dot for each colour joined by a dashed outline, the base colour ringed in gold and shown as a disc in the centre. Beside it, a legend lists each colour's HEX code, OKLCH hue, lightness and chroma, and its contrast ratio against the background; the first is marked **Base**.
2. **Palette**: a card per colour with a swatch, HEX code, a level pill (**AAA**, **AA**, **AA Lg** or **Fail**) for its contrast against the background, the ratio, and its OKLCH lightness and chroma.
3. **Preview on background**: the background with a **Sample Heading** in the first colour, body text in the second colour, up to five buttons (**Btn 1** to **Btn 5**) filled with the palette colours, and a tag for each colour.

Schemes

Scheme	Hue offsets from the base	Colours
Complementary	0°, 180°	2
Split-Complement	0°, 150°, 210°	3
Analogous	−30°, 0°, +30°	3
Triadic	0°, 120°, 240°	3
Tetradic	0°, 90°, 180°, 270°	4
Square	0°, 90°, 180°, 270°	4
Monochromatic	Same hue, lightness from dark to light	3 to 9 (**Mono count**)
Double Split	0°, +30°, −30°, 150°, 210°	5

Tetradic and **Square** produce the same four hues in this tool. In **Analogous** the list starts with the hue 30° below your base colour, so the first card, the gold-ringed dot and the **Base** label in the legend show that colour; your base colour is the second one.

Modes

Mode	What it does
Standard (math)	Uses the harmony colours as calculated.
WCAG 4.5:1 (AA)	Adjusts each colour until it reaches 4.5:1 against the background.
Soft 3.5:1	Adjusts each colour to 3.5:1, a gentler target that keeps more colour.
High-contrast 7:1 (AAA)	Adjusts each colour to 7:1.

The adjustment changes only lightness: each colour is made lighter if it is already lighter than the background, or darker if it is darker, in small steps until it meets the target. If a colour cannot reach the target within the steps allowed, it keeps the closest result, and its card shows the real level.

Tasks

Make an accessible accent set for a dark interface

1. Set **Background** to your dark surface colour.
2. Set **Base color** to your brand colour.
3. Choose a **Scheme**, for example **Split-Complement**.
4. Set **Mode** to **WCAG 4.5:1 (AA)**.
5. Check the **Palette** cards: each should show **AA** or **AAA**.
6. Look at **Preview on background** to judge the colours together.

Result: a set of related colours, each readable as text on your background.

Build a monochromatic ramp

1. Choose **Monochromatic**.
2. Move **Mono count** to the number of steps you need, from 3 to 9.

The ramp keeps the base colour's hue and chroma and spreads OKLCH lightness from dark (0.18) to light (0.88).

Result: a tonal ramp you can use for surfaces, borders and text.

Export the palette

- **Copy HEX list**: comma-separated codes in palette order, starting with the base colour. Message: **HEX list copied**.
- **Copy CSS variables**: one line per colour, `--color-1: #D3AF37;` and so on. Message: **CSS variables copied**.
- **Copy JSON**: the scheme, mode, base, background and palette. Message: **JSON copied**.
- **Download PNG**: downloads `harmony-palette.png`, a strip 80 pixels wide per colour and 100 pixels tall, with HEX labels.

Example JSON:

```
{
  "scheme": "triadic",
  "mode": "standard",
  "base": "#D3AF37",
  "background": "#0F172A",
  "palette": ["#D3AF37", "#…", "#…"]
}
```

Send a colour to another tool

1. Right-click a palette card. The site's colour menu opens with copy options and **Check its contrast**, **Build harmonies**, **Find its name** and **Inspect in Colour Tools**.
2. For Harmony Studio's own menu, hold `Shift` while you right-click the card. It offers **Copy HEX**, **Open in Color Spaces**, **Open in Accessibility**, **Find name** and **Open in Psychology**. `Esc` or a click elsewhere closes it.

Result: the colour opens in the chosen tool as its starting colour.

Controls

Control	What it does	Values or range	Default
Base color	The colour the harmony is built from	HEX, colour well, dice	#D3AF37, or the colour you opened with
Background	Colour for contrast checks and preview	HEX, colour well, dice	#0F172A
Scheme	Harmony pattern	8 schemes	Triadic
Mode	Contrast adjustment	Standard (math), WCAG 4.5:1 (AA), Soft 3.5:1, High-contrast 7:1 (AAA)	Standard (math)
Mono count	Colours in Monochromatic	3 to 9	5
Copy HEX list	Copies codes, comma-separated	—	—
Copy CSS variables	Copies `--color-N` lines	—	—
Copy JSON	Copies the full state	—	—
Download PNG	Downloads `harmony-palette.png`	—	—
Palette card, `Shift` + right-click	Opens the tool's colour menu	5 actions	—

Accuracy and limits

- Hues are rotated in OKLCH. The wheel places the dots by their HSL hue, so their positions on the ring can look slightly uneven even when the OKLCH offsets are exact.
- Rotated colours outside the sRGB range are clipped, which can change their lightness or chroma a little.
- Contrast uses the WCAG 2.x ratio. A colour adjusted to 4.5:1 is suitable for normal text on that background only.
- Harmony rules are conventions, not guarantees of a pleasing result.

Related

- Colour Tools documentation
- Palette Studio for generated palettes with locking and many exports
- Accessibility Lab
- Color Space Converter
- Color Theory and Harmony on Learn
- Basic Color Tools

WEBSITE › TOOLS › COLOUR WORKSPACE

Open the Colour Tools and manage their panels

Every way to open the 14 floating Colour Tools, how the launcher works, and how to move, arrange, pin and close the panels they open in.

The Colour Tools are 14 small colour tools that open in floating panels over the page you are on. You do not have to leave an article, a tool page or the home page to check a contrast ratio or look up a colour's name. The panel opens on top, you work

in it, and you close it when you are done.

This page covers everything the 14 tools share: the ways to open them, the **Color Tools** launcher, the floating panel window and the control bar that appears while panels are open. Each tool has its own page in this folder.

Where the Colour Tools are available

The Colour Tools load on almost every page of the site, a moment after the page finishes loading. The Basic Color Tools page (https://auricartisan.com/tool/basic-tools/) is their home page.

A few pages do not load them, for example the Analyzer. On those pages the right-click menu has no **Colour tools** group, and `Ctrl` + `Shift` + `C` takes you to the Basic Color Tools page instead of opening the launcher.

Ways to open a tool

From the right-click menu

Right-click anywhere on a page (outside a text field) to open the site's menu. The **Colour tools** row opens a submenu:

Item	What it opens
All colour tools...	The launcher, at the place you right-clicked. The row shows the shortcut `Ctrl` + `Shift` + `C`.
Contrast System	Contrast System
Accessibility Lab	Accessibility Lab
Harmony Studio	Harmony Studio
Color Space Converter	Color Space Converter
Color Name Finder	Color Name Finder
Gradient Library	Gradient Library
Color Library	Color Library

The other seven tools are always one step away through **All colour tools...**.

> **Tip:** While the menu is open you can type to filter it. Typing part of a tool's name finds it, even inside a submenu.

To use the browser's own menu instead, hold `Shift` while you right-click.

Right-click a colour value

When the thing you right-click is a colour, the menu header changes to **Colour** and shows the hex code and its RGB value. The menu recognises a colour when:

- you have selected text that is a hex code starting with `#` (3 or 6 digits), or six hex characters that include at least one digit, and you right-click on that selection;
- you right-click a swatch or chip that the site marks with its colour (the colour cards in many tools and libraries do this);
- you right-click a short piece of text that is only a hex code, such as `#D3AF37`.

The colour section then offers:

Item	What it does
Copy HEX	Copies the hex code in capitals.
Copy RGB	Copies `rgb(R G B)`.
Copy HSL	Copies the HSL value.
Copy OKLCH	Copies the OKLCH value.
Check its contrast	Opens Contrast System with this colour as the first foreground.
Build harmonies	Opens Harmony Studio with this colour.
Find its name	Opens Color Name Finder with this colour.
Inspect in Colour Tools	Opens the Color Inspector with this colour.

The **Colour tools** row is still there too, and every tool you open from it starts with the colour you right-clicked.

The keyboard shortcut

Press `Ctrl` + `Shift` + `C` on any page that loads the Colour Tools to open the launcher. It opens where you last right-clicked, or near the top-left corner if you have not right-clicked yet.

If you have text selected that is exactly a hex code (3 or 6 digits, with or without `#`), the tool you then pick in the launcher opens with that colour. Select `#1E40AF` in an article, press the shortcut, choose **Contrast System**, and the pair is ready to check.

A `?color=` link

Add `?color=` and a colour to the address of any page on the site, and the Color Inspector opens over that page with the full read-out for the colour. For example:

```
https://auricartisan.com/?color=CEA1B5
https://auricartisan.com/library/?color=1E40AF&on=FFFFFF&ctab=contrast
```

Parameter	What it does	Accepted values
`color`, `colour` or `hex`	The colour to inspect. If you list several, the first is inspected and the rest appear as comparison colours.	See the forms below. Separate several colours with a comma, a semicolon or a vertical bar.
`on`, `bg`, `against` or `vs`	One or more comparison colours. They appear under **Compare** and as extra rows in the Contrast and Vision tabs.	Same forms as `color`.
`ctab`	The Inspector tab to open on.	`overview`, `formats`, `contrast`, `vision`, `harmony`, `names`, `mind` (the Psychology tab), `scales`, `science`, `export`

The colour can be written as:

- a hex code with 3, 4, 6 or 8 digits: `CEA`, `CEA1B5`, `CEA1B5CC` (the 4- and 8-digit forms carry transparency);
- `rgb()` or `rgba()`, `hsl()` or `hsla()`, `hwb()`, for example `rgb(206 161 181)`;
- three numbers separated by spaces, read as red, green and blue: `206+161+181`;
- a CSS colour name such as `rebeccapurple` or `tomato`;
- a fragment of a hex code. One digit is read as a grey (`C` becomes `#CCCCCC`), two digits are repeated (`CE` becomes `#CECECE`), five or seven digits are padded with the last digit, and anything longer than eight digits keeps the first six. The Inspector then shows a small tag saying which text it started from.

> **Important:** Leave out the `#`, or write it as `%23`. In an address, `#` starts the page fragment, so `?color=#CEA1B5` reaches the site without a colour.

In an address bar, a space can be typed as `+` or `%20`. When a value contains brackets, such as `rgb(206 161 181)`, separate several colours with `;` or `|`, because a comma is then read as part of the colour.

When the Inspector was opened by a link, it keeps the address bar in step with what you see: change the colour or the tab and the `color` and `ctab` values update, so copying the page address shares the current read-out. If you open a second `?color=` link, or go back and forward between two, the open Inspector switches colour instead of opening another panel. An Inspector opened from the launcher leaves the address bar alone.

From other tools

- The Color Inspector's **Open in** list opens eight other Colour Tools with the colour you are inspecting. See Color Inspector.
- The Design Engine can open Colour Tools with your system's primary colour. See Design Engine.

The launcher

The launcher is a small dark card titled **Color Tools**, with "Auric Artisan · Tools" in small capitals above the title. It lists all 14 tools in four groups, each heading followed by the number of tools in it.

- **Essentials** (6): Color Inspector (INSP), Accessibility Lab (A11Y), Contrast System (CR), Color Name Finder (NAME), Animation Contrast (ANIM), Type Readability Sim (TYPE).
- **Explore** (3): Color Library (LIB), Color Psychology (PSY), Pantone & Named Lookup (PMS).
- **Design** (3): Harmony Studio (HARM), Gradient Library (GRAD), CMYK Soft-Proof (CMYK).
- **Deep dive** (2): Color Formulation (FORM), Color Space Converter (SPACE).

Each row shows an icon, the tool's name and its short tag. Hover over a row to see a tooltip with a one-line description of the tool.

Search

The search box at the top says "Search 14 tools…". Type to filter the list as you go. The search matches words in the tool names and in their descriptions, so `apca`, `video`, `pantone` or `print` all find the right tool. Groups with no match disappear. If nothing matches, the launcher says "No tools match your search."

Keyboard use

Key	What it does
`Tab` / `Shift` + `Tab`	Moves between the search box, the tool rows, the footer links and the close button.
`Enter` or `Space`	Opens the focused tool.
`Esc` in the search box	Clears the search. Press it again, with the box empty, to close the launcher.
`Esc` elsewhere	Closes the launcher.

Moving, resizing and closing

- **Move:** drag the launcher by its header.
- **Resize:** drag its bottom-right corner. On a narrow screen the tools stack in one column instead of two.

- **Close:** select the close button (×), press `Esc`, click anywhere outside the launcher, or open a tool. Opening a tool closes the launcher.

The footer has two links that leave the page: **View all tools** ("Browse the complete toolkit") and **Open dashboard** ("Your tools & recent activity").

The panel window

Every tool opens in the same kind of floating window.

The title bar

The title bar shows a grip, the tool's name and the label "Color Tools", then six icon buttons. Hover over one to see its tooltip.

Button (tooltip)	What it does
Chain icon ("Link — sync controls with other linked panels")	Turns linking on or off for this panel (see below).
Two squares ("Open another instance of this simulation")	Opens a second copy of this tool, offset slightly from the first.
Pin icon ("Pin — prevent auto-replacement when over the panel limit")	Keeps the panel open when the panel limit is reached.
Dash ("Minimize to dock")	Hides the panel. Bring it back from its tab on the control bar.
Square ("Maximize / restore")	Fills the browser window, leaving a small margin. Select it again to go back.
Cross ("Close")	Closes the panel.

Linking. When two or more panels are linked, changing a control in one changes the matching control in the others. This is most useful with two copies of the same tool, made with the two-squares button. Tools that share a control, such as a background colour field, follow each other too. A pinned panel is also skipped by **Close all but pinned** on the control bar.

Moving, resizing and focus

- Drag the title bar to move a panel. You cannot drag a maximised panel.
- Double-click the title bar to maximise or restore.
- Drag the corner handle at the bottom right to resize. On a computer a panel is at least 520 by 420 pixels, and it always stays inside the browser window.
- Click anywhere in a panel to bring it to the front.
- Each tool remembers the size and position you last gave it, and whether it was pinned, and reopens there next time.

While a tool loads, the panel shows a small card with the tool's name and a progress bar. If a tool cannot open, the panel says "Couldn't open" with the tool's name and a short reason.

Keyboard in a panel

Key	What it does
`Esc`	Leaves full-window view if the panel is maximised. Otherwise it closes the panel that has focus.

> **Note:** `Esc` closes the panel even when your cursor is in one of its fields. If you only meant to dismiss a dropdown list, click outside the list instead.

How many panels can be open

Up to five panels can be open at once. If you open a sixth, the oldest panel that is not pinned closes to make room. If all five are pinned, the new tool does not open until you unpin or close one.

Opening a tool that is already open brings its panel to the front (and restores it if it was minimised) instead of opening a second one. The panel keeps the colour it already has. Use the two-squares button in the title bar when you want two copies side by side.

Theme and small screens

The launcher, the panels, their dropdown lists and the control bar keep the same dark look whichever site theme you use, so the tools look the same on a light page and a dark page.

When a panel is narrow (under roughly 780 pixels wide), its sidebar moves above the main area. On a phone, a panel takes the full width of the screen and buttons grow to a comfortable touch size.

The control bar

While at least one panel is open, a slim bar appears at the bottom-right of the window. Drag it by its grip (the dotted handle at its left end) to move it; its position is remembered.

From left to right the bar holds five layout buttons, one tab per open panel, a count and a menu button.

Part (tooltip)	What it does
Grid icon ("Tile in a grid")	Tiles the panels to fill the window: 1 fills it, 2 side by side, 3–4 in 2 × 2, 5 in 3 × 2.
Split icon ("Split left and right")	Places the panels side by side at full height.
Split icon ("Split top and bottom")	Stacks the panels at full width.
Cascade icon ("Cascade with an offset")	Gives each panel the same size and staggers them diagonally.
Stack icon ("Stack them centred")	Centres every panel in the same place, the last one in front.
Panel tabs	Select a tab to bring that panel to the front and restore it if minimised.
Count	How many panels are open out of five, for example `2/5`.
Three dots ("More panel actions")	Opens the menu below.

The layout buttons act on panels that are neither minimised nor maximised. Small icons on each tab show whether that panel is pinned, synced, minimised or maximised, and its tooltip spells the state out.

The three-dots menu (**More panel actions**):

Item	What it does
Sync all panels / **Unsync all panels**	Links or unlinks every open panel at once. Needs at least two panels.
Minimize all	Minimises every panel.
Restore all	Brings every minimised panel back.
Export a comparison PNG	Downloads one image with a cell per open panel.
Open the verification panel	Opens a comparison overlay.
Close all but pinned	Asks "Close all unpinned floating panels?" and then closes every panel that is not pinned.

> **Note:** The same control bar serves the Vision Simulation panels. **Export a comparison PNG** and **Open the verification panel** are built for those: with Colour Tools panels, the PNG shows only each panel's name and a "no image data" placeholder.

On a phone the bar runs along the bottom edge and shows the panel tabs, the count and the three-dots button; the layout buttons are hidden.

What all 14 tools share

- **Colour fields.** Most tools use the same colour field: a colour swatch that opens your browser's colour picker, a text box for a hex code, and a dice button (**Random color** tooltip) that picks a random colour. The text box accepts 3, 4, 6 or 8 hex digits, with or without `#`. The tool updates as soon as the text is a valid code.
- **Live results.** There is no "calculate" button. Every change updates the results at once.
- **Copying.** When you copy a value, a short message such as "Copied: #D3AF37" appears at the bottom of the screen.
- **Dropdowns.** Select a dropdown to open its list, then select an option. `Esc`, a click outside, or scrolling closes the list.
- **Tooltips.** Many buttons and controls explain themselves when you hover over them.
- **Vision simulations.** The tools that preview colour-vision deficiency use the same set of filters, so a result in one tool matches the others. See Accessibility Lab for what each filter simulates and its limits.
- **A fresh start each time.** A tool starts from its default colours, or from the colour you opened it with. Colours and settings are not kept after you close a panel; only the panel's size, position and pin state are.
- **Everything runs in your browser.** Colours, images and videos you use in the tools are not uploaded.

Related

- Colour Tools documentation — the contents of this folder
- Basic Color Tools
- The Essentials tools: Color Inspector, Accessibility Lab, Contrast System, Color Name Finder, Animation Contrast, Type Readability Sim
- The other tools: Color Library, Color Psychology, Pantone & Named Lookup, Harmony Studio, Gradient Library, CMYK Soft-Proof, Color Formulation, Color Space Converter
- Site features — the right-click menu, settings and themes

WEBSITE › TOOLS › COLOUR WORKSPACE

Pantone & Named Lookup

Enter a colour and find its nearest matches in HTML/CSS named colours, a Pantone Solid Coated subset, RAL Classic, NCS and Crayola, ranked by ΔE2000 with a plain-language closeness label.

Pantone & Named Lookup finds the closest matches for a colour in five named colour systems at once. It shows the single closest match across all systems, the top matches in each system, and a small list of values to copy. Each match carries its ΔE2000 distance and a label that says in plain words how close it is.

Designers use named systems to talk to printers, paint suppliers and manufacturers. This tool helps you find a starting reference, for example "which Pantone is nearest to our brand blue?", before you check a physical swatch book.

Pantone & Named Lookup is in the launcher's **Explore** group.

Open it

Open the launcher and select **Pantone & Named Lookup**. See Open the Colour Tools. It starts with the colour you opened with, or #0F4C81 (a classic blue).

The systems

System	Entries	Example name
HTML/CSS	141	aliceblue
Pantone Solid C	105	PANTONE Yellow C
RAL Classic	148	RAL 1000 Green beige
NCS	56	NCS S 0500-N
Crayola	111	Crayola Red

The HTML/CSS list holds the named colours of CSS. The Pantone, RAL, NCS and Crayola lists are curated subsets, not complete systems, with widely published approximate screen (sRGB) renderings of each colour.

Screen tour

1. **Source color** (*ΔE2000 nearest-match*): a colour field and a **Filter** list (**All systems** or one system).
2. **Closest overall**, labelled with the winning system: your colour and the best match side by side, a pill such as *ΔE 1.84 · Very close*, and your colour's OKLCH, HSL and CMYK values.
3. One section per system (*N chips · top 5*):
   - the top match as a wide button with its swatch, name, HEX code, ΔE and closeness label;
   - the next matches as smaller cells with name, ΔE and label.
4. **Quick copy** (*Click to copy*): **HEX**, **RGB**, **CMYK (naive)** and **Nearest overall**.

Closeness labels

ΔE2000	Label
Below 1	**Imperceptible**
1 to 2	**Very close**
2 to 5	**Close**
5 to 10	**Noticeable**
10 to 20	**Distinct**
20 and above	**Far**

ΔE2000 is the CIE's standard measure of how different two colours look. Under about 1, most people cannot see a difference side by side.

Tasks

Find the nearest Pantone to a brand colour

1. Enter your brand colour in the colour field.
2. Set **Filter** to **Pantone Solid C**. The section now lists the top 18 Pantone matches instead of 5.
3. Read the top match's name and closeness label.
4. Select any match to make it the source colour; **Selected** and the name appear.

Result: a Pantone reference to look up in a physical guide.

Compare all systems at a glance

1. Leave **Filter** on **All systems**.
2. Read **Closest overall** and then the top five in each system.

Result: the nearest CSS name, Pantone, RAL, NCS and Crayola colour for the same input.

Copy a value

Select a row under **Quick copy**. **CMYK (naive)** is a simple formula conversion; **Nearest overall** copies the match's name followed by its HEX code in brackets.

Controls

Control	What it does	Values or range	Default
Colour field	The colour to match	HEX, colour well, dice	#0F4C81, or the colour you opened with
Filter	Shows all systems or one	All systems, HTML/CSS, Pantone Solid C, RAL Classic, NCS, Crayola	All systems
Match (top or cell)	Makes that match the source colour	—	—
Quick copy rows	Copy a value	4 rows	—

With **All systems**, each system shows its top 5 matches; with one system selected, it shows the top 18.

Accuracy and limits

- The Pantone, RAL and NCS entries are approximate sRGB renderings of printed or painted colours. Real swatches vary with paper, ink, finish and lighting, and a screen cannot show some of them accurately at all.
- The lists are subsets. A closer official colour may exist that this tool does not know about.
- Matching uses ΔE2000 from sRGB under a D65 white point. It predicts side-by-side differences under good viewing, not how a colour will print.
- Always confirm with the official physical guide before specifying a colour for production.

Related

- Colour Tools documentation

- Color Name Finder for general colour names
- CMYK Soft-Proof for a print preview
- Color Order Systems: Munsell, NCS, Pantone, and RAL on Learn
- Basic Color Tools

WEBSITE › TOOLS › COLOUR WORKSPACE

Type Readability Sim

Combine text and background colours with font style, size, weight, letter-spacing and a viewing condition to get an APCA reading, a comfort score, size and weight sweeps, and recommendations.

A contrast ratio on its own does not tell you whether text will be comfortable to read. The same colour pair can be easy at 24 pixels and bold, and hard at 12 pixels and light. Viewing conditions matter too: blur from low vision or glare on a screen eats into the margin.

Type Readability Sim puts these together. You choose a text colour, a background, a font style, size, weight and letter-spacing, and a viewing condition. It reports the APCA contrast and the minimum the APCA guidance asks for at that size and weight, a comfort score from 0 to 100, the WCAG ratio, a live preview, two sweeps across sizes and weights, and practical recommendations.

Type Readability Sim is in the launcher's **Essentials** group.

Open it

Open the launcher and select **Type Readability Sim**. See Open the Colour Tools. The text starts white (#FFFFFF). The background starts as the colour you opened the tool with, or a deep navy (#1A1A2E).

Screen tour

Sidebar

- **Text** and **Background**: colour fields.
- **Font**: one of five font styles.
- **Size** (shows the current value, for example **Size · 16px**), **Weight** and **Letter-spacing** sliders.
- **Viewing condition**.
- **Sample content**.

Main area

1. **Reading comfort** (*Composite estimate*): three meters.
   - **Comfort**: a score out of 100 with a bar and a label.
   - **APCA Lc**: the absolute Lc value, *Needs N for this size/weight*, and **Passes tier** or **Below tier**.
   - **WCAG 2.x**: the ratio, the levels for body and large text, and **Passes** or **Fails**.
2. **Live preview**: your sample text drawn with all your settings, under the chosen viewing condition. The heading shows the font style, size and weight.
3. **Size sweep**: *The quick brown fox* at 12, 14, 16, 18, 24 and 32 pixels in your weight, each marked **OK** or **fail**.
4. **Weight sweep**: the same text at your size in weights 300 to 800, each marked **OK** or **fail**.
5. **Recommendations** (*Strongest fix first*): up to four suggestions, or a note that the pair passes all checks.

Options

Font

Option	Font stack used
System sans (Inter-like)	Your system's interface sans-serif
Humanist (Georgia-like)	Georgia or a similar serif
Monospace (JetBrains-like)	JetBrains Mono or your system monospace
Display thin	Bodoni Moda, Didot or a similar high-contrast serif
Rounded (Quicksand-like)	Quicksand, Avenir Next, Nunito or your system sans

The preview uses fonts installed on your device; if a named font is not available, your browser falls back to a similar one. Each style also changes the comfort score slightly: the thin display style scores lower, the humanist style a little lower, and monospace a little higher.

Viewing condition

Option	Effect on the preview	Effect on the score
Normal vision · indoor	None	None
Low vision 20/40 (blur 0.6px)	0.6-pixel blur	−12
Low vision 20/80 (blur 1.8px)	1.8-pixel blur	−28
Bright glare (10% white veil)	A 10 % white veil	−8
Tired eyes (blur 0.3px + 5% veil)	0.3-pixel blur and a 5 % white veil	−4

Sample content

Option	Text
Body paragraph	The quick brown fox jumps over the lazy dog, letters and digits
UI label / button	Save changes, Continue, Cancel, Sign out
Long-form (4 sentences)	A short passage about readability, long enough to feel fatigue
Tabular / numeric	A row of revenue and percentage figures

How the numbers work

- **APCA Lc** is the Accessible Perceptual Contrast Algorithm's lightness contrast; only its size matters here.
- **Needs** is the minimum Lc from the APCA size and weight table (the same table as the Contrast System APCA Guide), blended between the nearest sizes and weights. When a size and weight is not usable at any contrast, it shows —.
- **Comfort** starts at 50, adds 0.9 points for every Lc above the requirement (and subtracts below it), adds a little for larger sizes and heavier weights, subtracts the viewing-condition penalty, adjusts for the font style, and is kept between 0 and 100.

Comfort score	Label
85 and above	**Effortless**

Comfort score	Label
65 to 84	**Comfortable**
50 to 64	**Legible**
35 to 49	**Strained**
Below 35	**Unreadable**

The comfort score is the tool's own heuristic, tuned to APCA. It is not a published standard.

Tasks

Check whether small text is comfortable

1. Set **Text** and **Background** to your colours.
2. Set **Size** to your caption size, for example 12 pixels, and **Weight** to its weight.
3. Read **Comfort** and the **APCA Lc** meter.
4. Look at **Size sweep** to see from which size the pair passes.

Result: you know whether the text is comfortable at that size, and the smallest size that passes.

Test under reduced vision

1. Change **Viewing condition** to **Low vision 20/40 (blur 0.6px)**, then **Low vision 20/80 (blur 1.8px)**.
2. Watch the preview blur and the **Comfort** score drop.
3. If the score falls to **Strained** or below, read **Recommendations**.

Result: an idea of how much safety margin the pair has.

Follow a recommendation

The recommendations can include:

- *Raise contrast by ~N Lc — current pair fails the size/weight tier.*
- *Bump size to N px — passes at current weight.* (the next table size that passes)
- *Bump weight to N — passes at current size.* (the next weight that passes)
- *Try text color #HEX — would pass current size/weight.* (the text colour's OKLCH lightness moved away from the background until it passes)
- *Under reduced vision, padding contrast above the minimum tier improves reading speed.* (shown for any viewing condition other than normal when the margin is under 15 Lc)

Apply a suggestion by changing the slider or typing the suggested colour into **Text**.

Result: the meters turn to **Passes tier** and the comfort label improves.

Controls

Control	What it does	Values or range	Default
Text	Text colour	HEX, colour well, dice	#FFFFFF
Background	Background colour	HEX, colour well, dice	The colour you opened with, or #1A1A2E

Control	What it does	Values or range	Default
Font	Font style	5 styles	System sans (Inter-like)
Size	Font size	10 to 48 px, step 1	16 px
Weight	Font weight	100 to 900, step 100	500
Letter-spacing	Tracking	−0.05 em to 0.20 em, step 0.01	0.00 em
Viewing condition	Simulated condition	5 options	Normal vision · indoor
Sample content	Preview text	4 options	Body paragraph

Letter-spacing changes the preview only; it does not change the scores.

Outputs and exports

The tool has no export. Copy colours from the colour fields.

Accuracy and limits

- The APCA values use a simplified form of a draft method, and the size and weight table is guidance, not a standard.
- The comfort score is a design heuristic, not a measurement of reading speed or a clinical test.
- Blur and veil effects are illustrations of low vision and glare; real vision loss varies a great deal.
- Fonts render with what is installed on your device, so the preview can differ from the font you plan to use.

Related

- Colour Tools documentation
- Contrast System
- Accessibility Lab
- Font Library to choose a real typeface
- Basic Color Tools

WEBSITE › TOOLS › COLOUR WORKSPACE

Panel tools

The seven tools that open as panels in the Basic Color Tools workspace, with a guide for each.

These are the seven tools you open from the Basic Color Tools workspace. Each opens as a floating panel with a controls sidebar and a main area. The level column shows whether the tool appears under **Basic** or **Advanced** in the Catalogue's level switch.

Guide	Category	Level	In one line
palette-studio.md	Generate	Basic	Live palette generator with harmony, WCAG and accessibility modes, locking, OKLCH editing, image extraction and eight exports.
palette-remix.md	Generate	Advanced	Turns seed colours into four styled variant palettes; copy CSS, JSON or all HEX codes.
gradient-maker.md	Generate	Basic	Linear, radial and conic gradients with any number of stops, perceptual interpolation, easing and CSS, SVG, PNG, JSON and Tailwind export.
colour-picker.md	Pick	Basic	One colour in eight formats, with harmonies, colour-vision previews, chromaticity and contrast notes.
colour-battle.md	Pick	Advanced	Scores two palettes against each other for contrast, separation and colour-vision resilience.
collage-maker.md	Compose	Advanced	Arranges your images in one of 29 layouts and exports a 4K PNG.
quick-converter.md	Convert	Basic	Converts one colour into fourteen notations and compares two colours.

For how panels open, move and close, see Workspace and panels. Every control is also listed in the Reference.

Related

- Basic Color Tools overview
- Colour Tools workbench

WEBSITE › TOOLS › COLOUR WORKSPACE

Collage Maker (panel)

Put your own images into one of 29 collage layouts, style the grid, apply filters, add a title and export a 4K PNG, all inside a Basic Color Tools panel.

The Collage Maker panel arranges images from your device into a layout: a plain grid, an editorial magazine page, a filmstrip, a pinboard and 25 more. You can set spacing, corner radius, background, borders and a drop shadow, apply a colour filter to all images or to single cells, add a title, and export the result as a large PNG. It is useful for mood boards, palette boards and quick visual references.

This is the compact collage tool inside Basic Color Tools. The site also has a separate, fuller Collage Maker page; see Image and file tools.

The panel's Collage Maker is in the **Compose** category at the **Advanced** level.

Open it

In the Workspace view, select **Collage Maker** in **Open a tool**. It opens with an empty three-by-two grid.

Screen tour

Sidebar

- **Layout**: 29 layouts.
- **Cols**, **Rows**, **Gap** and **Radius** number boxes.
- **Background** colour.
- **Border width / color**: a number and a colour.
- **Drop shadow** check box.
- **Global filter**.
- **Title overlay**: text and colour.
- **+ Add Images**, a shuffle button and a clear (bin) button.
- **Export PNG (4K)**.

Main area: Preview

A 4:3 preview of the collage. Empty cells read **+ Add**. Under it: *Click any cell to assign an image. Drag from desktop to drop multiple at once. Right-click a cell to apply a per-cell filter.* Dropping files is not supported in the current version, and a plain right-click opens the site's own menu; use the methods in the tasks below.

Layouts

Cols and **Rows** shape the layouts marked "your grid". The others have a fixed number of cells.

Layout	Cells	Notes
Grid	your grid	Equal cells, Cols by Rows.
Brick	your grid	Cols by Rows.
Mosaic	your grid	A finer grid with a repeating pattern of large and small cells.
Hero	your grid (at least 2)	The first cell is large, the rest fill around it.
Split	2	Two large halves.
Two-up	2	Two side by side.
Triptych	3	Three side by side.
Sidebar	4	One large image and three small ones.
Magazine	5	Editorial page with one large image.
Feature	4	A wide feature image above smaller ones.
Panorama	5	A wide image above four small ones.
Banner	4	One large image beside three tall, narrow ones.
Golden	4	One dominant image and diminishing ones.
Story	4	A tall format with a wide top image.
Poster	4	A large image above three small ones.
Frame	5	Images framing a centre image.
Quilt	10	Mixed sizes.

Layout	Cells	Notes
Tiles	9	Mixed sizes.
Checkerboard	16	Four by four; alternate cells are slightly faded.
Diagonal	7	Cells stepping diagonally.
Staircase	5	Rows that grow wider, like steps.
Masonry	12	Four columns of varied heights.
Columns	4	Four tall columns.
Rows	3	Three wide rows.
Contact sheet	20	Five by four small frames.
Pinboard	12	Four by three, each image slightly tilted.
Polaroid	Cols (at least 3)	One row, images tilted like prints.
Strip	Cols (at least 3)	One row.
Filmstrip	5	Five frames in a row.

Tasks

Make a mood board

1. Choose a **Layout**, for example **Magazine**.
2. Select **+ Add Images** and choose several images at once. They fill the empty cells in order. Images beyond the layout's number of cells are ignored.
3. To place or replace a single image, select that cell and choose one file.
4. Select the shuffle button to mix the order of the images.
5. Adjust **Gap**, **Radius**, **Background**, border and **Drop shadow** until it looks right.

Result: your images arranged in the chosen layout in the preview.

Apply filters

1. Choose a **Global filter** to apply one look to every image: **None**, **Grayscale**, **Sepia**, **Invert**, **Blur**, **Warm**, **Cool**, **Contrast** or **Vintage**.
2. To give one cell its own filter, hold `Shift` and right-click that cell. Each `Shift` + right-click moves the cell to the next filter in the list above, and a message such as **Filter: sepia** appears. A cell's own filter overrides the global filter.

If you have set the site's right-click menu to use the browser's menu, a plain right-click cycles the filter instead. See Site features.

Result: every image carries the look you chose, with individual exceptions.

Add a title and export

1. Type a **Title overlay** and choose its colour.
2. Select **Export PNG (4K)**.

The title does not appear in the preview. It is drawn only in the exported image: bold, centred near the top, with a soft shadow.

Result: `collage.png` downloads, with the preview's proportions, the background colour, rounded corners, borders, filters and title. When the preview is at least 800 pixels wide, the image is 3,840 pixels wide. When the preview is narrower, the image is 4.8 times the preview's size and so less than 3,840 pixels wide; widen or maximise the panel before exporting to get the full size.

Controls

Control	What it does	Values or range	Default
Layout	Chooses the arrangement.	29 layouts	Grid
Cols	Columns for grid-based layouts; images per row for Polaroid and Strip.	1 to 6	3
Rows	Rows for grid-based layouts.	1 to 6	2
Gap	Space between cells, in pixels.	0 to 40	6
Radius	Corner rounding of cells, in pixels.	0 to 40	8
Background	Colour behind the cells.	Any colour	#101010
Border width	Inner border on each cell, in pixels.	0 to 20	0
Border colour	Colour of that border.	Any colour	#FFFFFF
Drop shadow	Shadow under each cell in the preview.	On or off	On
Global filter	Filter for every image.	None, Grayscale, Sepia, Invert, Blur, Warm, Cool, Contrast, Vintage	None
Title overlay	Text drawn at the top of the export.	Any text	Empty
Title colour	Colour of the title.	Any colour	#FFFFFF
+ Add Images	Adds several images to empty cells.	Image files	—
Cell	Select to place one image in that cell.	One image file	—
`Shift` + right-click a filled cell	Cycles that cell's filter.	Same list as Global filter	—
Shuffle	Randomises the image order.	—	—
Clear (bin)	Removes all images.	—	—
Export PNG (4K)	Downloads the collage as `collage.png`.	3,840 pixels wide when the preview is 800 pixels wide or more	—

Accuracy and limits

- Images are read from your device in the browser and are not uploaded. Very large images or many images make the export slower.
- The export redraws the collage at high resolution from the preview's layout. The drop shadow and the slight fade of alternate **Checkerboard** cells appear only in the preview; the pinboard and polaroid tilts are not drawn in the export either.
- Images are cropped to fill their cells, keeping their centre.
- Open panels are not saved when you reload the page. Export before you leave.

Related

- Panel tools
- Collage Maker page and other image tools
- Troubleshooting

WEBSITE › TOOLS › COLOUR WORKSPACE

Color Battle

Enter two palettes, score each for contrast, perceptual separation and colour-vision resilience, compare them in a cross-palette matrix, and copy the winner or a report.

Color Battle compares two palettes, **Team A** and **Team B**, and gives each a score out of 100. The score rewards palettes whose colours contrast well with each other, look clearly different, and stay different for people with colour-vision deficiencies. It also shows a matrix of contrast between every colour in Team A and every colour in Team B, which helps when one palette will be used for text and the other for backgrounds.

Use it to choose between two candidate palettes, to test a palette against a competitor's, or to see how much a change improves things.

Color Battle is in the **Pick** category at the **Advanced** level.

Open it

In the Workspace view, select **Color Battle** in **Open a tool**. It opens with two five-colour teams already scored.

Screen tour

Sidebar

- **Team A**, with the hint *HEX/RGB/HSL/OKLCH, one per line.*, a dice button that fills the team with random colours, and a text box.
- **Team B**, with the hint *Competing palette.*, its own dice button and text box.
- **Battle** and **Swap**.
- **Copy winner** and **Copy report**.
- **Result**: **Team A wins**, **Team B wins** or **Draw**.

Main area: Score Results

- Two score cards side by side (stacked on narrow panels), one per team.
- A **Cross-palette contrast matrix** card below them.

What you can type

Each team box accepts the same colour forms as Palette Remix: HEX (three or six digits, with or without `#`), `rgb()` / `rgba()`, `hsl()` / `hsla()`, `hsv()` and `oklch()`. Put one colour per line, or separate them with spaces, commas, semicolons or vertical bars. When the battle runs, each box is tidied into HEX codes, one per line, duplicates are removed, and only the first nine colours are kept. If a box has no valid colour, it falls back to its default team.

How the score works

For each team the tool looks at every pair of colours inside that team. A five-colour team has 10 pairs; a nine-colour team has 36. It then combines four measures:

Measure	Shown as	Weight	Full marks at
Average WCAG contrast ratio of the pairs	**Avg contrast**	35 points	an average of 7:1 or more
Share of pairs at 4.5:1 or more	**AA pairs**	30 points	100 % of pairs
Average perceptual difference (ΔE2000) of the pairs	**Avg ΔE2000**	20 points	an average of 44 or more
Average ΔE2000 after simulating deuteranopia, protanopia, tritanopia and achromatopsia	**CVD ΔE avg**	15 points	an average of 34 or more

The total is rounded to a whole number from 0 to 100. The team with the higher total wins; equal totals are a **Draw**. A team with only one colour has no pairs and scores 0.

ΔE2000 (delta E 2000) is a standard measure of how different two colours look. Around 1 is barely noticeable; values in the tens mean clearly different colours.

Tasks

Compare two candidate palettes

1. Replace the text in **Team A** with your first palette, one colour per line.
2. Replace the text in **Team B** with your second palette.
3. Select **Battle**. (Leaving a text box also runs the battle.)
4. Read each score card: the large number is the **battle score**; under it are **N internal pair tests**, the swatches, the four measures and a bar showing the score.
5. Read **Result** in the sidebar.

Result: two scored palettes and a winner, with the measures that explain the difference.

Check one palette as text on the other

1. Put your text colours in **Team A** and your background colours in **Team B**.
2. Select **Battle**.
3. Read the **Cross-palette contrast matrix**. Rows are Team A colours, columns are Team B colours (the corner reads **A ↓ / B →**). Each cell is the contrast ratio, tinted green at 4.5:1 or more, amber from 3:1, and red below 3:1.

Result: you can see at a glance which text and background combinations are readable.

Stress-test with random teams

1. Select the dice button beside **Team A** or **Team B**. That team is replaced by five random colours spread around the hue circle, and the battle runs.
2. Select **Swap** to exchange the two teams and run again.

Result: quick comparisons against random palettes, useful for seeing what a good score looks like.

Share the outcome

- **Copy winner** copies the winning team's HEX codes, one per line. In a draw it copies Team A.

- **Copy report** copies a short text report:

```
Color Battle Report
Team A wins

Team A: 72/100
#D3AF37 #0F172A #F8E7A1 #FFFFFF #1A1A2E

Team B: 65/100
#3B82F6 #111827 #93C5FD #F9FAFB #EF4444
```

The scores above are only an example.

- Select any swatch on a score card to copy its HEX code.

Save to and restore from the Library

With Color Battle focused, **Save to Library** saves each team as a palette, named **Team A — N colors** and **Team B — N colors**, and records both teams and the result. Opening a saved team in Basic Color Tools reopens Color Battle with both teams and runs the battle; **Loaded teams from library** appears.

Controls

Control	What it does	Values or range	Default
Team A box	First palette.	Up to 9 colours kept	#D3AF37, #0F172A, #F8E7A1, #FFFFFF, #1A1A2E
Team B box	Second palette.	Up to 9 colours kept	#3B82F6, #111827, #93C5FD, #F9FAFB, #EF4444
Dice (per team)	Five random colours for that team, then battle.	—	—
Battle	Scores both teams.	—	Runs once on opening
Swap	Exchanges the teams, then battles.	—	—
Copy winner	Copies the winning team.	—	—
Copy report	Copies the text report.	—	—
Swatch	Copies one HEX code.	—	—

Accuracy and limits

- The score is a heuristic for comparing palettes, not a pass or fail test. A palette of colours meant to sit side by side (such as a chart palette) needs different things from a text-and-background palette, so read the measures, not only the total.
- Contrast uses the WCAG 2.x formula. The AA share counts pairs at 4.5:1, the normal-text threshold.
- Colour-vision simulation uses simple matrices and shows likely effects, not an exact view for any individual.

Related

- Panel tools
- Palette Studio
- Palette Remix
- Accessibility Lab in the Colour Tools workbench

- Limits and accuracy

WEBSITE › TOOLS › COLOUR WORKSPACE

Color Picker

Pick a colour with a saturation-value plane, hue slider, eyedropper or HEX code, then copy it in eight formats and see harmonies, colour-vision previews, chromaticity and contrast.

Color Picker is for one colour at a time. You choose the colour with a saturation-value plane and a hue slider, the browser's colour well, a HEX code, a random button or the system eyedropper. The main area then shows the colour in eight notations, eight harmony sets built from it, how it looks under colour-vision deficiencies, where it sits on the CIE 1931 chromaticity diagram, and how well it contrasts with common backgrounds.

The picker works in **HSV** (hue, saturation, value): the hue slider picks the hue, and the square plane picks saturation (left to right) and value, or brightness (bottom to top).

Color Picker is in the **Pick** category at the **Basic** level.

Open it

In the Workspace view, select **Color Picker** in **Open a tool**.

Screen tour

Sidebar

- A colour well and a HEX text box, side by side.
- **Pick** (tooltip **Use OS eyedropper**) and **Random**.
- The saturation-value plane, with a round marker.
- The hue slider, with a bar-shaped marker.

Main area tabs

Tab	What it shows
Formats	The colour in HEX, RGB, HSL, HSV, OKLCH, Lab, XYZ and CMYK, each with a copy button.
Harmonies	Eight rows of colours derived from the current hue.
CVD	The colour under normal vision and four colour-vision deficiencies, with the difference from the original.
Chromaticity	The colour on the CIE 1931 diagram and on a hue wheel.
Info	Key numbers, the closest named colour, and contrast against six backgrounds.

Ways to set the colour

Method	How
Plane	Press on the plane and drag. Right is more saturated, up is brighter.
Hue slider	Press on the slider and drag left or right.
Colour well	Select the well and choose a colour in your browser's picker.

Method	How
HEX box	Type a code such as `#1E90FF` and press `Enter` or select outside the box.
Pick	Opens the system eyedropper so you can sample any pixel on your screen.
Random	Chooses a random hue with fairly strong saturation and brightness.
Harmony swatch	On the **Harmonies** tab, select a swatch to make it the current colour.

The eyedropper uses the browser's EyeDropper feature, which is available in Chromium-based browsers such as Chrome and Edge on desktop. Where it is missing, **Eyedropper not supported** appears; use the colour well or the HEX box instead.

Tasks

Get a colour's CSS in several formats

1. Set the colour with any method above.
2. On the **Formats** tab, select the copy button beside the format you need.

Result: the value is on your clipboard and **Copied** appears. For the colour #D3AF37 the formats read:

Format	Example
HEX	`#D3AF37`
RGB	`rgb(211, 175, 55)`
HSL	`hsl(46, 64%, 52%)`
HSV	`hsv(46, 74%, 83%)`
OKLCH	`oklch(76.6% 0.138 92)`
Lab	`lab(72.8 0.9 62.8)`
XYZ	`0.429, 0.448, 0.100`
CMYK	`cmyk(0%, 17%, 74%, 17%)`

Lab is CIE Lab (CIELAB) with a D65 white point. XYZ values are on a 0 to 1 scale (Y = 1 is white). CMYK is a simple device-independent conversion, not a press profile.

Build a harmony set from one colour

1. Set your base colour.
2. Open **Harmonies**. Each row shows a set of swatches with HEX labels.
3. Select any swatch to make it the current colour. **Applied #HEX** appears and every tab updates.

The rows are computed in OKLCH, keeping the base colour's lightness and chroma and changing the hue:

Row	Hues or steps
Complementary	Base and base + 180°
Analogous	Base − 30°, base, base + 30°
Triadic	Base, + 120°, + 240°
Tetradic	Base, + 90°, + 180°, + 270°

Row	Hues or steps
Split-comp.	Base, + 150°, + 210°
Square	Base, + 90°, + 180°, + 270°
Shades (L ramp)	Five lightness steps (0.18, 0.32, 0.46, 0.60, 0.78) at the base hue and chroma
Tints (C ramp)	Five chroma steps (0.05 to 0.25) at the base hue and lightness

Tetradic and **Square** use the same four hues here. Colours outside the sRGB range are clipped to the nearest displayable colour.

Result: a related colour set you can walk through one swatch at a time.

Check a colour for accessibility

1. Open **Info**.
2. Read **Contrast against common backgrounds**: six tiles for **White** (#FFFFFF), **Off-white** (#F5F5F5), **Black** (#000000), **Near-black** (#1A1A1A), **Dark slate** (#0F172A) and **Auric gold** (#D3AF37). Each shows the WCAG contrast ratio and a level: **AAA** (7:1 or more), **AA** (4.5:1 or more) or **Fail**.
3. Open **CVD** to see the colour as simulated for **deuteranopia**, **protanopia**, **tritanopia** and **achromatopsia**. Each tile shows the simulated HEX code and **ΔE2000**, how different the simulated colour is from the original. Larger numbers mean a bigger change.

Result: you know which backgrounds the colour can carry text on and how much it shifts for colour-blind viewers.

The Info tab

Item	Meaning
HEX	The current colour.
Luminance	WCAG relative luminance, 0 (black) to 1 (white), four decimals.
Lab L	CIE Lab lightness, 0 to 100.
OKLCh L	OKLCH lightness as a percentage.
Chroma	OKLCH chroma.
Hue	OKLCH hue in degrees.
Warm/Cool	**Warm** for OKLCH hues below 60° or from 300° up, **Cool** from 180° to below 300°, **Neutral** in between. A rough guide only.
Text on	A **Sample** word in black or white, whichever the tool picks as readable on the colour.
Closest named color	The nearest of 64 built-in colour names, with its HEX code and ΔE2000 distance.

The Chromaticity tab

- **CIE 1931 chromaticity position (sRGB triangle = dashed gold)**: the horseshoe-shaped chart of all colours people can see, with the sRGB range drawn as a dashed gold triangle and your colour marked.
- **Hue wheel position**: your colour's OKLCH hue marked on a hue ring.

Save to and restore from the Library

With Color Picker focused, **Save to Library** saves the colour with its formats and its harmony colours. A saved colour reopens in Color Picker and **Loaded color from library** appears.

Controls

Control	What it does	Values or range	Default
Colour well	Sets the colour with the browser's picker.	Any sRGB colour	—
HEX box	Sets the colour by code on `Enter` or when you leave the box.	3- or 6-digit HEX, with or without `#`	—
Pick	System eyedropper.	Needs browser support	—
Random	Random colour.	Saturation 50–100 %, value 40–90 %	—
Saturation-value plane	Sets saturation and value.	0–100 % each	—
Hue slider	Sets hue.	0–360°	—
Format copy buttons	Copy one format.	8 formats	—
Harmony swatch	Applies that colour.	8 rows	—
Tabs	Switch the main area.	Formats, Harmonies, CVD, Chromaticity, Info	Formats

Accuracy and limits

- Values are computed from 8-bit sRGB, so conversions round to the displayed precision.
- Colour-vision tiles use simple simulation matrices. They show the likely direction and size of the change, not an exact view for any one person.
- The named-colour match uses a short built-in list of 64 names. For larger naming systems, use Color Name Finder or Pantone & Named Lookup.
- CMYK here is a formula without an ink or paper profile. For print previews, use CMYK Soft-Proof.

Related

- Panel tools
- Quick Converter for fourteen notations and two-colour comparison
- Color Inspector, the in-depth single-colour tool of the Colour Tools workbench
- Import, export and the Library

WEBSITE › TOOLS › COLOUR WORKSPACE

Gradient Maker

Build linear, radial and conic gradients with any number of stops, choose the colour space they blend in, add easing, and export CSS, SVG, PNG, JSON or Tailwind.

Gradient Maker builds CSS gradients. You set a type (linear, radial or conic), add colour stops along a track, and choose how the colours blend between stops. It shows a large live preview, charts of how lightness, chroma and hue change along the gradient, nine presets, and export options for code and images.

A **colour stop** is a colour pinned at a position along the gradient, from 0 % (the start) to 100 % (the end). **Interpolation** is the colour space the gradient blends in between stops: blending in sRGB (what CSS does by default) can pass through dull or dark middles, while blending in OKLCH keeps lightness and saturation more even. **Easing** changes the pace of the blend along the gradient.

Gradient Maker is in the **Generate** category at the **Basic** level.

Open it

In the Workspace view, select **Gradient Maker** in **Open a tool**. It opens with a two-stop linear gradient from gold (#D3AF37) to deep blue (#1D2A3E) at 90 degrees.

Screen tour

Sidebar

- **Type**: **Linear**, **Radial** or **Conic**.
- **Angle**: the direction in degrees. Hidden for radial gradients.
- **Interpolation**: **sRGB (native CSS)**, **OKLCh (perceptual)**, **HSL** or **CIE Lab**.
- **Easing**: **Linear**, **Ease in**, **Ease out**, **Ease in-out** or **Sine**.
- **Reverse**, **+ Stop**, **Even** and **Random**.
- **Export**: **Copy CSS**, **SVG**, **1600px**, **4K**, **JSON** and **Tailwind**.

Main area tabs

Tab	What it shows
Design	The preview, the stop track, and the **Stops** list with a count, for example **Stops (3)**.
Curves	Four charts of the gradient's colour along its length.
Presets	Nine ready-made gradients.
Export	Read-only boxes with the CSS, SVG and JSON code.

The stop track and list

The track under the preview shows the gradient left to right with a round marker for each stop.

- Select an empty spot on the track to add a stop there. Its colour is a straight sRGB blend of the two stops either side of that point.
- Select a marker to highlight it.

The **Stops** list has one row per stop:

- A colour well: select it to choose a colour.
- A HEX text box: type a six-digit or three-digit HEX code.
- A position slider from 0 to 100 in steps of 0.5, with the position shown as a percentage.
- A ✕ remove button, shown while there are more than two stops.

Tasks

Build a three-colour gradient

1. Select **+ Stop**. A new stop with a random colour appears at 50 %.
2. In its row, type a HEX code or use the colour well.
3. Drag its position slider to where you want it.
4. Select **Even** to space all stops evenly.
5. Change **Type** to **Radial** to see a circular version, or back to **Linear** and set **Angle** (0 to 360).

Result: the preview shows your gradient and updates with every change.

Make the blend perceptually smooth

1. Set **Interpolation** to **OKLCh (perceptual)**.
2. Open the **Curves** tab. Compare the **Lightness across gradient** chart with the sRGB result: the line becomes smoother.
3. Try an **Easing** such as **Ease in-out** to hold the end colours longer and move quickly through the middle.

Result: a gradient without a muddy middle. The CSS output now contains 20 generated stops that imitate the chosen blend, because CSS blends in sRGB unless told otherwise.

Start from a preset

1. Open the **Presets** tab.
2. Select a preset. It loads and the panel switches back to the **Design** tab.
3. Edit its stops as you like.

The presets are **Sunset**, **Ocean**, **Aurora**, **Coral**, **Mint**, **Royal**, **Auric**, **Spectrum** and **Mono dark**.

Result: a starting gradient you can adjust.

Export the gradient

1. Select **Copy CSS** to copy a declaration such as `background: linear-gradient(90deg, #D3AF37 0.0%, #1D2A3E 100.0%);`.
2. Or select **SVG**, **1600px** or **4K** to download a file.
3. For JSON, open the **Export** tab first, then select **JSON** (see the note below).

Result: code on your clipboard or an image in your downloads folder.

Other buttons

- **Reverse** mirrors every stop's position, so the gradient runs the other way.
- **Even** keeps the order of the stops and spaces them evenly from 0 % to 100 %.
- **Random** replaces the colours with new ones (keeping the number of stops), evenly spaced in hue around a random starting hue, and spreads the stops evenly.

Curves

The charts sample the gradient at 60 points, using the current interpolation and easing:

- **Lightness across gradient (samples 60 points)**: OKLCH lightness from 0 to 1.

- **Chroma across gradient**: OKLCH chroma from 0 to 0.4.
- **Hue rotation along gradient**: OKLCH hue from 0 to 360 degrees.
- **CIE 1931 path**: sample points on the chromaticity diagram, with the sRGB triangle drawn dashed.

Export

Button	Result
Copy CSS	Copies `background: <gradient>;`.
SVG	Downloads `gradient.svg`, 800 by 400 pixels.
1600px	Downloads `gradient.png`, 1600 by 800 pixels.
4K	Downloads `gradient.png`, 3840 by 1920 pixels.
JSON	Copies the JSON shown on the **Export** tab: type, angle, interpolation, easing, and each stop's HEX, position and OKLCH values.
Tailwind	Copies `backgroundImage: { 'gradient': '<gradient>' }` for a Tailwind theme.

The CSS forms are:

```
background: linear-gradient(90deg, #D3AF37 0.0%, #1D2A3E 100.0%);
background: radial-gradient(circle at center, #D3AF37 0.0%, #1D2A3E 100.0%);
background: conic-gradient(from 90deg at center, #D3AF37 0.0%, #1D2A3E 100.0%);
```

Notes on each format:

- **CSS**: with **sRGB (native CSS)** or **HSL** interpolation and **Linear** easing, the CSS lists your stops as they are, and the browser blends them in sRGB. With any other combination the CSS lists 20 generated stops that reproduce the chosen interpolation and easing.
- **SVG**: uses your stops as they are, blended in sRGB. SVG has no conic gradient, so a conic gradient is exported as a straight left-to-right gradient.
- **PNG**: with **sRGB (native CSS)** and **Linear** easing the image uses your stops directly; otherwise it uses 40 generated stops. Conic PNGs need a browser that supports conic gradients on a canvas.
- **JSON**: the **JSON** button copies the text in the **Export** tab's JSON box, which is refreshed only while that tab is open. Open the **Export** tab after your last change, then select **JSON**, to copy the current gradient.

Example of the JSON:

```
{
  "type": "linear",
  "angle": 90,
  "interp": "oklch",
  "easing": "linear",
  "stops": [
    { "hex": "#D3AF37", "pos": 0, "oklch": { "L": 0.766, "C": 0.138, "H": 91.6 } },
    { "hex": "#1D2A3E", "pos": 100, "oklch": { "L": 0.283, "C": 0.041, "H": 258.8 } }
  ]
}
```

The copied JSON is indented over several lines and the OKLCH numbers carry more decimal places than shown here.

Save to and restore from the Library

With Gradient Maker focused, **Save to Library** saves the gradient with its CSS, stops, type, angle, interpolation and easing. A saved gradient reopens in Gradient Maker with all of those settings, and **Loaded gradient from library** appears.

Controls

Control	What it does	Values or range	Default
Type	Gradient shape.	Linear, Radial, Conic	Linear
Angle	Direction for linear, starting angle for conic.	0 to 360 degrees	90
Interpolation	Colour space for blending between stops.	sRGB (native CSS), OKLCh (perceptual), HSL, CIE Lab	sRGB (native CSS)
Easing	Pace of the blend.	Linear, Ease in, Ease out, Ease in-out, Sine	Linear
Reverse	Mirrors stop positions.	—	—
+ Stop	Adds a random stop at 50 %.	No fixed maximum	—
Even	Spaces stops evenly.	—	—
Random	New colours, even spacing.	Keeps the stop count	—
Track	Select empty space to add a stop; select a marker to highlight it.	0 % to 100 %	—
Stop colour well	Sets a stop's colour.	Any colour	—
Stop HEX box	Sets a stop's colour by code.	3- or 6-digit HEX	—
Stop position slider	Moves a stop.	0 to 100, step 0.5	—
Stop ✕	Removes a stop.	At least 2 stops remain	—
Preset	Loads a preset.	9 presets	—
Export buttons	Copy or download.	See Export	—

Accuracy and limits

- The preview and the copied CSS are the same gradient, so what you see is what you copy.
- Non-sRGB blends and easing are approximated with generated stops. Twenty stops in CSS and forty in PNG are enough to look smooth in normal use, but the result is a close imitation, not an exact curve.
- Colours outside the sRGB range during OKLCH or Lab blending are clipped to the nearest displayable colour.
- The HSL blend takes the shorter way round the hue circle.

Related

- Panel tools
- Palette Studio
- Gradient Library in the Colour Tools workbench
- Import, export and the Library
- Troubleshooting

WEBSITE › TOOLS › COLOUR WORKSPACE

Palette Remix

Paste seed colours, choose a remix style, and get four controlled variant palettes you can copy as CSS, JSON or HEX.

Palette Remix takes colours you already have, such as a brand palette, and makes four variations of them in one step. Each variation follows a recipe: it shifts hue, lightness and chroma by set amounts, spreads the colours a little across the palette, and adds as much random variation as you ask for.

Use it when you need alternatives that still feel related to the original: an editorial set, a darker mood, a set of UI neutrals or a complementary accent family.

Palette Remix is in the **Generate** category at the **Advanced** level. With the Catalogue's level set to **Basic** it is hidden.

Open it

In the Workspace view, select **Palette Remix** in **Open a tool**. It opens with five seed colours already filled in and four remix sets shown.

Screen tour

Sidebar

- **Seed colors**: a text box, one colour per line to start with.
- **Remix style**: the recipe family.
- **Colors per palette**: how many colours each output palette has.
- **Remix energy**: a slider for how much random variation to add.
- **Remix** and **New seed** buttons.
- **Export**: **CSS**, **JSON** and **All HEX**.

Main area

- **Output Palettes**, with a summary such as **5 seed colors → 4 remix sets**.
- Four cards, numbered 1 to 4. Each shows the recipe name, a short note, a row of swatches with HEX labels, and two buttons: **Copy** and **Use as seed**.

Seed colours you can paste

The seed box accepts colours in these forms, in any mix:

- HEX with three or six digits, with or without `#`: `#D3AF37`, `d3af37`, `#FA0`.
- `rgb()` and `rgba()` with comma-separated numbers: `rgb(211, 175, 55)`.
- `hsl()` and `hsla()` with percentages: `hsl(46, 64%, 52%)`.
- `hsv()`: `hsv(46, 74%, 83%)`.
- `oklch()` with space-separated values; lightness may be written as 0 to 1 or as a percentage: `oklch(76% 0.13 87)`.

Separate colours with new lines, spaces, commas, semicolons or vertical bars. Duplicates are dropped and anything the tool cannot read is ignored. If nothing in the box is a valid colour, the tool falls back to its five default seeds.

When a palette needs more colours than you gave seeds, the seeds repeat in order.

Remix styles

Each style produces four named sets.

Style	Sets and their notes
Editorial kit	**Hero** (balanced brand-forward remix), **Ink** (deeper and moodier), **Glow** (soft highlight system), **Punch** (higher chroma accents)
Analog drift	**North** (cool-side analogous), **Core** (seed-respectful), **South** (warm-side analogous), **Mist** (low-chroma UI neutrals)
Complement battle	**Primary** (original contrast base), **Complement** (opposite hue family), **Split A** (split-complement left), **Split B** (split-complement right)
Tonal system	**Paper** (light surface tones), **Stone** (middle surfaces), **Metal** (dense interface tones), **Night** (dark-mode palette)

Every set is computed in OKLCH. Lightness is kept between 0.10 and 0.95 and chroma between 0.015 and 0.32, so results stay usable.

Tasks

Remix a brand palette

1. Select the text in **Seed colors** and paste your own colours, one per line.
2. Select outside the box. The output updates.
3. Choose a **Remix style**, for example **Tonal system**. The four cards update at once.
4. Set **Colors per palette** to the number you need, from 3 to 9.
5. Move **Remix energy** to the left for faithful variations or to the right for looser ones.
6. Select **Remix** to roll new random variation with the same settings.

Result: four named palettes built from your colours, each with a one-line description of its purpose.

Iterate on a result you like

1. On the card you like, select **Use as seed**. Its colours replace the seed box, the sets are rebuilt from them, and **Remix set moved into seed** appears.
2. Change the style or energy and keep going.

Result: a chain of remixes that moves steadily away from the original in the direction you choose.

Start from random colours

1. Select **New seed**. The seed box fills with random OKLCH colours, as many as **Colors per palette**, and the sets rebuild.

Result: four remix sets from a fresh random starting point.

Copy what you need

- Select a swatch to copy its HEX code.
- Select **Copy** on a card to copy that set's HEX codes, separated by spaces.
- Use the **Export** buttons for all four sets at once (below).

Export

Button	Result
CSS	Copies one `:root` block per set, with variables named `--remix-1-1`, `--remix-1-2` … for set 1, `--remix-2-1` … for set 2, and so on.
JSON	Copies an array of the four sets, each with `name`, `note` and `hexes`.
All HEX	Copies every HEX code from all four sets, one per line.

Example of **CSS** output for the first set (your colours will differ):

```
:root {
  --remix-1-1: #D8B23E;
  --remix-1-2: #1C1A31;
  --remix-1-3: #F2DDB6;
  --remix-1-4: #10182D;
  --remix-1-5: #FBF9F5;
}
```

Save to and restore from the Library

With Palette Remix focused, **Save to Library** saves each of the four sets as a separate palette, named after the set. A saved remix set reopens in Palette Remix with its colours in the seed box, and **Loaded remix from library** appears.

Controls

Control	What it does	Values or range	Default
Seed colors	The colours to remix. Applied when you leave the box.	Any mix of the accepted formats	#D3AF37, #1A1A2E, #F5DEB3, #0F172A, #FFFFFF
Remix style	Chooses the recipe family. Updates at once.	Editorial kit, Analog drift, Complement battle, Tonal system	Editorial kit
Colors per palette	Number of colours in each set. Updates at once.	3 to 9	5
Remix energy	Random variation in hue, lightness and chroma. Updates at once.	0 to 100	30
Remix	Rebuilds the sets with new random variation.	—	—
New seed	Fills the seed box with random colours and rebuilds.	—	—
Swatch	Copies one HEX code.	—	—
Copy (card)	Copies that set's HEX codes, space-separated.	—	—
Use as seed (card)	Moves that set into the seed box and rebuilds.	—	—
CSS, **JSON**, **All HEX**	Copy all four sets.	See Export	—

Accuracy and limits

- Remix energy adds random variation, so the same settings give different results each time. Copy or save a result you want to keep.
- At the highest energy the hue can move by up to about 35 degrees in either direction on top of the recipe's own shift.
- The tool does not check contrast. To check a set, paste it into Color Battle or open it in Palette Studio through the Library.

Related

- Panel tools
- Palette Studio
- Color Battle
- Import, export and the Library

WEBSITE › TOOLS › COLOUR WORKSPACE

Palette Studio

Generate palettes from harmony, WCAG and accessibility modes, lock and edit colours, extract colours from an image, check contrast and colour-vision behaviour, and export.

Palette Studio is a live palette generator. It shows your palette as tall colour strips that you can regenerate, lock, reorder, edit and remove. Around the strips are analysis tabs for contrast, lightness and chroma curves, colour-vision previews and colour details, and a set of export buttons.

It has thirteen generation modes. Ten are colour-harmony shapes. Three build **role palettes** designed for accessible interfaces: each colour has a job (page canvas, body ink, primary surface, text on the primary surface, focus ring and so on) and the tool picks lightness so the pairs meet a contrast target.

Palette Studio is in the **Generate** category at the **Basic** level.

Open it

In the Workspace view, select **Palette Studio** in **Open a tool**. In the Catalogue, select its card. It opens with five colours generated in **Random** mode.

Screen tour

Sidebar

- **Harmony**: the generation mode. Changing it regenerates the palette straight away.
- **Generate**, **+ Add** and **From Image**.
- **Accessibility Generate**: three shortcut buttons, **WCAG AA**, **WCAG AAA** and **Accessibility**, that switch the mode and generate in one step.
- **CVD Preview**: shows the strips as they may appear with a colour-vision deficiency.
- **Export**: **Share**, **CSS**, **SCSS**, **TW**, **JSON**, **SVG**, **PNG** and **ASE**.

Main area tabs

Tab	What it shows
Palette	The colour strips, and a hint line: *Space generate · Click options · Click hex edit · Drag reorder*.
Contrast Matrix	The WCAG contrast ratio of every pair of colours.
WCAG Pairs	Text-and-background pairs with their ratio, level and pass or fail.
Curves	OKLCH lightness and chroma across the palette, a hue wheel, and a CIE 1931 chromaticity diagram.
CVD Comparison	The palette under normal vision and four colour-vision deficiencies, one row each.
Color Info	Every colour's values in several colour spaces.

A colour strip

Each strip shows its HEX code near the bottom and, under it, a small detail: the colour's role in a role palette, the simulation name while a CVD preview is on, or **OKLCh** otherwise. Hover over a strip (or tap it on a touch screen) to show its buttons, stacked at the top:

Button	Tooltip	What it does
Padlock	**Lock** / **Unlock**	Keeps the colour when you generate again. Locked strips always show their buttons.
Copy	**Copy hex**	Copies the HEX code.
Gear	**Adjust**	Opens the **Adjust OKLCh** sliders for this colour.
×	**Remove**	Removes the colour. Only shown while the palette has more than two colours.

The strip under your pointer widens slightly so you can see it better.

Generation modes

All modes pick colours in OKLCH, a colour space where equal steps look roughly equally different, and add a little random variation so each press of **Generate** gives a fresh result. Each press starts from a new random base hue.

Mode	What you get
Random	Unrelated colours of medium lightness and moderate chroma.
Analogous	Neighbouring hues about 30 degrees apart, getting lighter across the palette.
Complementary	Colours alternating between a base hue and its opposite.
Triadic	Colours from three hues 120 degrees apart.
Monochromatic	One hue, from dark to light.
Split complementary	A base hue and the two hues either side of its opposite (150 and 210 degrees away).
Tetradic	Four hues 90 degrees apart.
Compound	A base hue, a neighbour 30 degrees away, and the opposites of both.
Shades	One hue, lightness rising from very dark, low chroma.
Tints	One hue, from medium to very light, with chroma fading as the colours get lighter.
WCAG AA palette	A role palette whose text pairs reach 4.5:1.

Mode	What you get
WCAG AAA palette	A stricter role palette whose text pairs reach 7:1.
Accessibility palette	A low-chroma canvas, near-black ink, and blue, orange, green and purple surfaces spread apart for colour-vision differences, each with a readable ink.

Role palettes

The two WCAG modes produce these roles, in this order: **Canvas**, **Ink**, **Primary**, **Primary ink**, **Focus**, **Secondary**, **Secondary ink**, **Success**, **Success ink**. The **Accessibility** mode produces **Canvas**, **Ink**, then **CVD blue**, **CVD blue ink**, **CVD orange**, **CVD orange ink**, **CVD green** and **CVD green ink**.

The palette keeps its current number of colours, so a five-colour palette gets the first five roles. To get every role, add colours with **+ Add** (up to nine) and generate again.

The targets the generator aims for:

- Text roles (the *ink* colours) aim for 4.5:1 against their surface in **WCAG AA** and **Accessibility** mode, and 7:1 in **WCAG AAA** mode.
- Coloured surfaces aim for at least 3:1 against the canvas (4.5:1 in **Accessibility** mode) so they stand out as UI shapes.
- **Focus** aims for 3:1, the WCAG 2.2 contrast for non-text focus indicators.

The generator searches for a lightness that meets the targets. The **WCAG Pairs** tab shows whether each pair actually passes, so check it rather than assuming.

Tasks

Generate a palette you like

1. Choose a mode in **Harmony**.
2. Select **Generate**, or press `Space` while the panel is focused and the cursor is not in a text box or list.
3. Lock the colours you want to keep. Generate again until the rest fit.
4. Drag a strip onto another strip's position to reorder the palette.

Result: a palette of two to nine colours in the order you want, with the locked colours unchanged.

Fine-tune one colour

1. Hover over the strip and select **Adjust**. The **Adjust OKLCh** box opens over the strip with three sliders: **L** (lightness, 0 to 1), **C** (chroma, 0 to 0.4) and **H** (hue, 0 to 360 degrees).
2. Move the sliders. The strip previews the change as you go.
3. Select **Apply** to keep it, or **Cancel** (or select anywhere outside the box) to discard it.

Alternatively, select the HEX code on the strip to open the site's colour picker, choose a colour, and confirm it. The strip updates as you pick.

Result: the colour changes to exactly the value you set; other colours are untouched.

Build an accessible role palette

1. Select **+ Add** until the palette has nine colours.
2. Select **WCAG AA** under **Accessibility Generate**. The strips show role names such as **Canvas** and **Primary ink**.

3. Open the **WCAG Pairs** tab. Read the four figures at the top: **Min pair** (the lowest ratio among the pairs), **AA normal** and **AAA normal** (how many pairs reach 4.5:1 and 7:1), and **CVD min dE** (the smallest colour difference between any two colours under simulated colour-vision deficiency).
4. Check the table: each row names a pair (for example **Primary ink on Primary**), shows both colours, the ratio, the WCAG level for normal text (**AAA**, **AA** or **Fail**) and whether the pair meets its own target (**Pass** or **Fail**).
5. If a pair fails, select **WCAG AA** again for a new attempt, or lock the colours that pass and regenerate.

Result: a nine-role palette with a table proving which text and surface pairs pass.

Extract a palette from an image

1. Select **From Image** and choose an image file from your device.
2. Wait a moment. The palette is replaced by the image's dominant colours, most common first, and a message such as **Extracted 5 colors** appears.

The tool extracts as many colours as the palette currently has (five if it is empty). To extract more, add colours first. The image is scaled down to at most 400 pixels on its longest side for analysis, transparent pixels are ignored, and the colours are found by clustering similar pixels (k-means). The result can differ slightly between runs. Extracted colours are unlocked.

Result: a palette taken from your picture, ready to lock, edit and export.

Analysis tabs in detail

Contrast Matrix

A table with your colours as both rows and columns. Each cell is the WCAG 2.x contrast ratio between the two colours; the diagonal shows a dash. Cells are tinted green for 4.5:1 and above, amber for 3:1 to 4.5:1, and red below 3:1. The note under the table lists the thresholds: 7.0 AAA, 4.5 AA, 3.0 AA Large, below 3.0 Fail.

WCAG Pairs

For role palettes the tool pairs each colour with the colour it is meant to sit on. Ink pairs use a 4.5:1 target (7:1 for AAA roles); surface and focus pairs use 3:1. For palettes without roles, each row is **Best text on Color N**: the tool picks whichever of four near-black and near-white inks (#111111, #000000, #FFFFFF, #F8FAFC) gives the highest contrast on that colour, with a 4.5:1 target. The footnote reads: *WCAG normal text targets use 4.5:1 for AA and 7:1 for AAA. Focus and non-text UI pairs use 3:1.*

Curves

- **Lightness (OKLCh L) across palette**: a line from the first colour to the last, scale 0 to 1.
- **Chroma (OKLCh C) across palette**: scale 0 to 0.4.
- **Hue distribution — Hue wheel**: each colour marked on a hue ring.
- **CIE 1931 chromaticity positions**: each colour on the chromaticity diagram, with the sRGB triangle drawn as a dashed gold line.

A smooth lightness line usually means an even ramp; a jagged one means some colours may be hard to tell apart in greyscale.

CVD Comparison

Five rows: **Normal vision**, then **deuteranopia**, **protanopia**, **tritanopia** and **achromatopsia**, each showing the whole palette as simulated with HEX labels. Use it to spot colours that merge for some viewers.

Color Info

One card per colour with **HEX**, **RGB**, **HSL**, **OKLCh** (lightness as 0 to 100, chroma, hue), **Lab**, **Luminance** (WCAG relative luminance, 0 to 1), and for role palettes **Role** and **Target**.

CVD Preview

The **CVD Preview** list recolours the strips to simulate **Deuteranopia**, **Protanopia**, **Tritanopia** or **Achromatopsia**. It changes only what you see: the HEX codes, the exports and the analysis tabs keep the real colours. Choose **Normal vision** to turn it off.

Export

Button	Tooltip	Result
Share	Copy share URL	Copies a link to Basic Color Tools that records the palette's HEX codes, for example `https://auricartisan.com/tool/basic-tools/#palette=D3AF37-1A1A2E-F5DEB3`.
CSS	Copy CSS vars	Copies `:root { --color-1: #…; … }`.
SCSS	Copy SCSS	Copies one `$color-1: #…;` line per colour.
TW	Copy Tailwind	Copies a Tailwind `extend` snippet with `palette-1`, `palette-2` … keys.
JSON	Copy JSON	Copies an array with each colour's HEX, role details, RGB, OKLCH, Lab, luminance, best text colour and lock state.
SVG	Download SVG	Downloads `palette.svg`, 1200 by 240 pixels, equal stripes with HEX labels.
PNG	Download PNG	Downloads `palette.png`, 1200 by 240 pixels, equal stripes with HEX labels.
ASE	Copy ASE-style txt	Copies plain text, one line per colour: `Color 1`, the HEX code and `RGB(r, g, b)`, separated by tabs.

The **ASE** button copies text in a tab-separated layout, not a binary Adobe Swatch Exchange file. Paste it into a text file or spreadsheet.

The share link stores the codes in the part of the address after `#`. Opening the link opens Basic Color Tools; the palette is not rebuilt from the link automatically, so treat it as a record of the codes you can copy back into a tool such as Palette Remix.

Examples of the copied code are in Import, export and the Library.

Save to and restore from the Library

With Palette Studio focused, select **Save to Library** to save the palette with its harmony mode and CVD preview setting. When you open a saved palette in Basic Color Tools later, it reopens in Palette Studio and a message such as **Loaded 5 colors from library** appears.

Controls

Control	What it does	Values or range	Default
Harmony	Sets the generation mode and regenerates.	13 modes (see above)	**Random**
Generate	Replaces every unlocked colour.	—	—

Control	What it does	Values or range	Default
`Space`	Same as **Generate** when the panel is focused.	—	—
+ Add	Adds a random colour at the end.	Up to 9 colours; message **Max 9**	—
From Image	Extracts the palette from an image file.	Any image type the browser can open	—
WCAG AA	Switches to **WCAG AA palette** and generates.	—	—
WCAG AAA	Switches to **WCAG AAA palette** and generates.	—	—
Accessibility	Switches to **Accessibility palette** and generates.	—	—
CVD Preview	Simulates colour-vision deficiency on the strips.	Normal vision, Deuteranopia, Protanopia, Tritanopia, Achromatopsia	Normal vision
Lock	Keeps a colour when generating.	On or off	Off
Copy hex	Copies one colour's HEX code.	—	—
Adjust	Opens OKLCH sliders for one colour.	L 0–1 (step 0.005), C 0–0.4 (step 0.005), H 0–360 (step 1)	The colour's current values
Remove	Removes one colour.	At least 2 colours remain; message **Min 2 swatches**	—
HEX label	Opens the site colour picker for that colour.	Any colour	—
Drag a strip	Reorders the palette.	—	—
Export buttons	Copy or download the palette.	See the Export table	—

Accuracy and limits

- Contrast ratios use the WCAG 2.x relative-luminance formula on sRGB values.
- Colour-vision previews use simple simulation matrices. They show the general effect, not exactly what any individual sees.
- The WCAG and accessibility modes aim for their targets but are not guaranteed to reach every one; the **WCAG Pairs** tab is the check.
- OKLCH values outside the sRGB range are clipped to the nearest displayable colour, so a requested chroma may not be reached.
- Image extraction samples a scaled-down copy of the image and uses random starting points, so results can vary slightly.

Related

- Panel tools
- Palette Remix for variations of a palette you already have
- Color Battle to compare two palettes
- Harmony Studio in the Colour Tools workbench
- Import, export and the Library

WEBSITE › TOOLS › COLOUR WORKSPACE

Quick Converter

Enter a colour in any common notation and copy it in fourteen formats; compare two colours by ΔE and contrast; check wide-gamut values; see the colour on charts.

Quick Converter is the fastest way to restate one colour in many notations. Paste a HEX, RGB, HSL, HSV or OKLCH value and it shows fourteen conversions, each with a copy button, plus the colour's luminance and nearest name. Three more tabs compare the colour with a second colour, show its coordinates in wide-gamut display spaces, and plot it on charts.

Quick Converter is in the **Convert** category at the **Basic** level.

Open it

In the Workspace view, select **Quick Converter** in **Open a tool**. It opens showing gold (#D3AF37).

Screen tour

Sidebar

- A large swatch showing the colour and its HEX code.
- An input box (placeholder *#D3AF37, rgb(…), hsl(…), oklch(…)*), a colour well and a dice button for a random colour.
- **Quick Info**: **Luminance** and **Nearest name**.

Main area tabs

Tab	What it shows
Convert	Fourteen notations, each with a copy button.
Compare ΔE	Two colours side by side with difference and contrast figures and a chart.
Wide Gamut	Linear coordinates in sRGB, Display P3, Rec.2020 and Adobe RGB, with in-gamut status.
Visualize	CIE 1931 position, hue wheel, a lightness ramp and a hue ramp.

Entering a colour

The input box understands:

- HEX with three or six digits, with or without `#`.
- `rgb(211, 175, 55)` and `rgba(…)` (the alpha value is ignored).
- `hsl(46, 64%, 52%)` and `hsla(…)`.
- `hsv(46, 74%, 83%)`.
- `oklch(76.6% 0.138 91.6)`, with lightness as a percentage or from 0 to 1.

The box converts as soon as its contents form a valid colour, and then rewrites itself as the HEX code. Because of this, typing a value character by character can convert too early (for example, `#D3A` is already a valid three-digit code). Paste complete values, or use the colour well.

The dice button picks a random colour from the whole RGB range.

The fourteen formats

For #D3AF37 the **Convert** tab shows:

Row	Example
HEX	`#D3AF37`
RGB	`rgb(211, 175, 55)`
RGB %	`rgb(82.7%, 68.6%, 21.6%)`
HSL	`hsl(46, 64%, 52%)`
HSV	`hsv(46, 74%, 83%)`
OKLCH	`76.55% / 0.1385 / 91.55°`
OKLab	`L 76.55% · a -0.0037 · b 0.1384`
Lab (D65)	`L 72.75 · a 0.95 · b 62.79`
XYZ (D65)	`X 0.4289 · Y 0.4479 · Z 0.1000`
CIE xyY	`x 0.4391 · y 0.4585 · Y 0.4479`
CMYK	`0, 17, 74, 17`
CSS oklch()	`oklch(76.6% 0.138 92)`
CSS oklab()	`oklab(76.6% -0.004 0.138)`
CSS lab()	`lab(72.8% 0.95 62.79)`

The three **CSS** rows are ready to paste into a stylesheet. The other rows are readable summaries. Select the copy button beside any row to copy it; **Copied** appears.

What the less familiar notations mean:

- **OKLCH** and **OKLab**: a perceptual colour space. L is lightness, C is chroma (colourfulness), H is hue in degrees; a and b are the green–red and blue–yellow axes.
- **Lab (D65)**: CIE Lab with a D65 (daylight) white point. L runs from 0 to 100.
- **XYZ (D65)**: the CIE 1931 tristimulus values on which the other spaces are built, on a 0 to 1 scale.
- **CIE xyY**: chromaticity (x, y) plus luminance (Y).
- **CMYK**: a simple percentage conversion without an ink or paper profile.

Tasks

Convert a colour for a stylesheet

1. Paste your colour into the input box.
2. On **Convert**, select the copy button beside **CSS oklch()**, **CSS oklab()** or **CSS lab()**.

Result: a CSS colour function on your clipboard.

Compare two colours

1. Open **Compare ΔE**.

2. Set **Color A** with its colour well or HEX box. Color A is the converter's main colour, so this also changes the sidebar and the **Convert** tab.
3. Set **Color B** with its colour well or HEX box. It starts as #1F2A3E.
4. Read the figures:

Figure	Meaning
ΔE 76	The simple straight-line distance in Lab.
ΔE 2000	The modern perceptual difference. Around 1 is just noticeable; above about 10 is clearly different.
WCAG ratio	The contrast ratio and level (**AAA**, **AA** or **Fail** for normal text).
ΔL (Lab)	Difference in Lab lightness, B minus A.
ΔL (OKLCh)	Difference in OKLCH lightness, as percentage points.
ΔC (OKLCh)	Difference in OKLCH chroma.
ΔH	Hue difference in degrees, the short way round.
Same hue?	**Yes** when the hue difference is under 10 degrees.

The chart **Both colors on CIE 1931 chromaticity** marks both colours.

Result: you know how different the two colours look and whether they work as text and background.

Check wide-gamut values

1. Open **Wide Gamut**.
2. Read the four tiles: **Linear sRGB**, **Display P3 (linear)**, **Rec.2020 (linear)** and **Adobe RGB (linear)**. Each shows three linear channel values and **In gamut** or **Out**.

Because the input is an sRGB colour, it is in gamut for the wider spaces. The tiles are useful for seeing how the same colour is written in each space's linear values. The tab notes that the preview is sRGB-coded.

Result: the colour's coordinates in four RGB spaces.

See the colour on charts

Open **Visualize**:

- **CIE 1931 chromaticity**: the colour's position.
- **Hue wheel (OKLCh hue)**: its hue.
- **All shades (L 0→1 at fixed C, H)**: a strip from black to white keeping the colour's chroma and hue.
- **All hues (H 0→360 at fixed L, C)**: a strip around the hue circle keeping the colour's lightness and chroma.

Colours that fall outside sRGB in the strips are clipped to the nearest displayable colour.

Quick Info

- **Luminance**: WCAG relative luminance, 0 to 1, four decimals.
- **Nearest name**: the closest of 64 built-in colour names, with its HEX code and ΔE2000 distance, for example `Auric Gold (#D3AF37) ΔE 0.00` for the starting colour, because Auric Gold is one of the built-in names.

Save to and restore from the Library

With Quick Converter focused, **Save to Library** saves the colour with all its formats. A saved converter item reopens in Quick Converter and **Loaded color from library** appears.

Controls

Control	What it does	Values or range	Default
Input box	Sets the colour from text.	HEX, rgb(), rgba(), hsl(), hsla(), hsv(), oklch()	#D3AF37
Colour well	Sets the colour with the browser's picker.	Any sRGB colour	#D3AF37
Dice	Random colour.	Any RGB value	—
Copy buttons (Convert)	Copy one notation.	14 rows	—
Color A well and HEX box	Sets the main colour from the Compare tab.	HEX	#D3AF37
Color B well and HEX box	Sets the comparison colour.	HEX	#1F2A3E
Tabs	Switch the main area.	Convert, Compare ΔE, Wide Gamut, Visualize	Convert

Accuracy and limits

- Conversions start from 8-bit sRGB and use the standard sRGB, CIE and OKLab formulas with a D65 white point.
- Values are rounded for display; copying gives the rounded text shown.
- The Lab conversion here does not adapt to a D50 white point, so its numbers can differ slightly from tools that report Lab relative to D50 (as CSS `lab()` formally does).

Related

- Panel tools
- Color Picker
- Color Space Converter, which covers more than thirty colour spaces
- Import, export and the Library

WEBSITE › TOOLS › COLOUR WORKSPACE

Supporting material

The glossary, shortcuts, limits, related links and privacy notes for Basic Color Tools and the Colour Tools workbench.

This folder holds the reference material that supports the Basic Color Tools guides: the terms they use, every shortcut, what the numbers mean and do not mean, where to go next, and what happens to your data.

File	What it covers
glossary.md	Colour, contrast and workspace terms used in these docs.
shortcuts.md	Keys and mouse gestures for the workspace, the panel tools and the Colour Tools.
limits-and-accuracy.md	How precise the conversions, contrast checks and simulations are, and what they are not.
related.md	Related tools, colour libraries, kits and Learn articles.
privacy.md	What stays on your device, what is stored in your browser, and what uses the network.

Related

- Basic Color Tools overview
- Documentation contents

WEBSITE › TOOLS › COLOUR WORKSPACE

Shortcuts

Every keyboard shortcut and mouse gesture in Basic Color Tools, its panel tools and the Colour Tools workbench.

`Cmd` applies on a Mac where the table lists it. Shortcuts that act on "the panel" act on the focused panel, the one you selected last.

Basic Color Tools page

Keys or gesture	Where	What it does
`Space`	Palette Studio, panel focused, not typing	Generates a new palette (locked colours stay).
`Escape`	Focus inside a panel	Restores a maximised panel; otherwise closes the panel.
Double-click the title bar	Any panel	Maximises or restores.
Drag the title bar	Any panel (not maximised)	Moves the panel.
Drag the bottom-right corner	Any panel (not maximised)	Resizes the panel.
`Shift`, `Ctrl` or `Cmd` + select a tool card	Catalogue	Opens another instance of the tool.
Drag a colour strip	Palette Studio	Reorders the palette.
`Shift` + right-click a filled cell	Collage Maker	Cycles that cell's filter.
`Enter` or leave the box	Color Picker HEX box; Quick Converter Compare HEX boxes	Applies the typed HEX code.
`Ctrl` + `Enter` (`Cmd` + `Enter`)	Save to Library window	Saves.

Colour Tools workbench

Keys or gesture	Where	What it does
`Ctrl` + `Shift` + `C`	Any page that loads the Colour Tools	Opens the Colour Tools launcher. If the selected text is a HEX code, the tool you pick opens with that colour.
Right-click	Any page	Opens the site menu with the **Colour tools** group.
`Shift` + right-click	Any page	Opens the browser's own menu instead.
Type while the right-click menu is open	Site menu	Filters the menu items.
`Tab`, `Shift` + `Tab`	Launcher	Moves between the search box, tool rows, links and the close button.
`Enter` or `Space`	Launcher	Opens the focused tool.
`Esc`	Launcher search box	Clears the search; press again to close the launcher.
`Esc`	Launcher, elsewhere	Closes the launcher.
`Esc`	Colour Tools panel	Leaves full-window view if maximised; otherwise closes the panel.
Double-click the title bar	Colour Tools panel	Maximises or restores.

Individual Colour Tools may have their own keys; each tool's guide lists them. See Colour Tools.

Site-wide shortcuts

The site's search, settings and other global shortcuts are documented in Site features.

Related

- Workspace and panels
- Open the Colour Tools and manage their panels
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Glossary

Plain-language definitions of the colour, contrast, print and workspace terms used in the Basic Color Tools and Colour Tools documentation.

Terms are in alphabetical order.

Term	Meaning
Achromatopsia	Complete absence of colour vision; everything appears in shades of grey.
ACEScg	A linear, wide-gamut RGB space used in film and visual-effects rendering.
Adobe RGB (1998)	A wider-gamut RGB space used in photography and print workflows, especially for greens and cyans.
APCA	Accessible Perceptual Contrast Algorithm, a contrast method proposed for WCAG 3 that accounts for which colour is text and which is background.
APCA Lc	APCA's lightness-contrast value, from about −108 to +106; its size, together with font size and weight, predicts readability.

Term	Meaning
Analogous	A harmony of hues next to each other on the colour wheel, typically about 30° apart.
Black point	The darkest colour a printing process can actually produce, which is never a pure black.
CCT	Correlated colour temperature, in kelvin; an estimate of how warm or cool a light-like colour appears.
Chroma	How colourful a colour is compared with a grey of the same lightness.
Chromaticity	The colour quality of a light apart from its brightness, usually given as CIE x and y.
CIE 1931 chromaticity diagram	A horseshoe-shaped chart of all colours people can see, used to compare gamuts.
CIELAB (Lab)	A CIE colour space with lightness (L) and two opponent axes (a green–red, b blue–yellow), designed for measuring colour difference.
CIELuv	A CIE perceptual space often used for lights and displays.
CMYK	The four process inks of printing: cyan, magenta, yellow and black (key).
Colour harmony	A set of hues chosen by their positions on the colour wheel, such as complementary or triadic.
Colour space	A system of numbers for describing colours, such as sRGB, Lab or OKLCH.
Colour stop	A colour fixed at a position along a gradient.
Complementary	Two hues opposite each other on the colour wheel.
Contrast ratio	The WCAG 2.x measure of luminance difference between two colours, from 1:1 to 21:1.
CVD	Colour-vision deficiency, often called colour blindness.
D50, D65	Standard white points: D65 is average daylight (used by sRGB and screens); D50 is used in print and for CSS lab() and lch().
Deuteranopia	A red–green colour-vision deficiency caused by missing green-sensitive cones; the most common type.
Display P3	A wide-gamut RGB space used by many modern phones and monitors.
Dock	Where a minimised panel is kept, out of view, until you restore it.
Dot gain	The darkening of mid-tones in print as ink spreads into the paper.
Easing	How the pace of a blend changes along a gradient, such as ease in or ease out.
ΔE (delta E)	A number for how different two colours look. ΔE 76 is the simple Lab distance; ΔE2000 (CIEDE2000) is the more accurate modern formula.
Focused panel	The panel you selected last; it sits in front and receives keyboard shortcuts and Save to Library.
Gamut	The range of colours a colour space, screen or printer can reproduce.
Gamut clipping	Replacing a colour that is outside a gamut with the nearest colour inside it.
HEX code	A six-digit (or three-digit) hexadecimal way of writing an sRGB colour, such as #D3AF37.
HSL, HSV, HWB	Cylindrical versions of RGB: hue with saturation and lightness, saturation and value, or whiteness and blackness.
Hue	The attribute that makes a colour red, yellow, green, blue and so on, given as an angle from 0° to 360°.
Interpolation	Calculating the colours between two stops; the colour space used changes how the middle looks.
JzAzBz	A perceptual colour space designed to work for high dynamic range as well as normal images.
k-means	A clustering method used to find the main colours of an image by grouping similar pixels.
Level (Basic, Advanced, Both)	The Catalogue filter that shows simpler tools, advanced tools, or all of them.

Term	Meaning
Linear RGB	RGB with the gamma curve removed, so the numbers are proportional to light energy.
Luminance (relative)	How bright a colour is on a 0 (black) to 1 (white) scale, used by WCAG contrast.
Metamerism	When two colours match under one light and differ under another.
Monochromatic	A set of colours with one hue and different lightness.
Named colour system	A catalogue of colours with names or codes, such as Pantone, RAL, NCS or the CSS named colours.
OKLab, OKLCH	A modern perceptual colour space (and its lightness, chroma, hue form) where equal steps look roughly equal; supported in CSS.
Panel	A movable, resizable window inside the page that holds one tool.
Protanopia	A red–green colour-vision deficiency caused by missing red-sensitive cones; reds also look darker.
Rec.2020	A very wide-gamut RGB space defined for ultra-high-definition and HDR video.
Role palette	A palette where each colour has a job, such as canvas, ink, primary or focus, chosen so pairs meet a contrast target.
Saturation	How intense a colour is relative to its brightness, in HSL or HSV terms.
Soft-proofing	Previewing on screen how a design is likely to look when printed.
sRGB	The standard RGB colour space of the web and most screens.
Stage	The surface in the Workspace view where panels open.
TAC	Total area coverage: the maximum combined percentage of CMYK inks a press allows.
Tetradic, square	Four hues spaced around the colour wheel; in these tools both use four hues 90° apart.
Triadic	Three hues 120° apart on the colour wheel.
Tritanopia	A rare blue–yellow colour-vision deficiency caused by missing blue-sensitive cones.
WCAG	Web Content Accessibility Guidelines. Their contrast levels: AA needs 4.5:1 for normal text and 3:1 for large text and interface parts; AAA needs 7:1 for normal text.
XYZ, xyY	CIE tristimulus values that other colour spaces are built on, and the chromaticity-plus-luminance form of them.
YCbCr, YUV, YIQ	Luma and colour-difference encodings used in video and broadcast.

Related

- Limits and accuracy
- Supporting material
- Help centre

WEBSITE › TOOLS › COLOUR WORKSPACE

Limits and accuracy

How the colour conversions, contrast checks, colour-vision simulations and scores in Basic Color Tools are calculated, how precise they are, and what they are not.

Basic Color Tools and the Colour Tools calculate everything with published formulas in your browser. The results are good for design decisions and checks. They are not certified measurements, and a few of them are deliberate simplifications. This page says which.

Colour values

- **Starting point.** Every panel tool works from 8-bit sRGB colours (the HEX codes you see). Values you type in other notations are converted to sRGB first.
- **Out-of-range colours are clipped.** OKLCH, OKLab and Lab can describe colours that sRGB cannot show. When a slider, ramp or harmony asks for one, the tools clip each channel to the displayable range. The HEX code shows the colour you actually get, which can be less saturated than the numbers suggest.
- **Rounding.** Readouts are rounded for display (for example, OKLCH chroma to three or four decimals). Copy buttons copy the rounded text.
- **White point.** Lab, XYZ and xyY use a D65 white point without adapting to D50. Tools that report Lab relative to D50 can show slightly different numbers.
- **CMYK.** The CMYK figures in Color Picker and Quick Converter come from a simple formula with no ink, paper or press profile. Use CMYK Soft-Proof for a press-aware preview, and a real proof for print decisions.
- **Wide gamut.** Quick Converter's **Wide Gamut** tab shows linear coordinates for Display P3, Rec.2020 and Adobe RGB. The input is always an sRGB colour, and the page cannot show colours beyond sRGB.

Contrast

- The seven panel tools use the WCAG 2.x contrast ratio: (lighter luminance + 0.05) divided by (darker luminance + 0.05), from 1:1 to 21:1.
- Levels: 4.5:1 is AA for normal text, 7:1 is AAA, 3:1 is AA for large text and for non-text parts such as focus indicators.
- A ratio says nothing about font size, weight or how the colours are used. A passing ratio is necessary for readable text, not a guarantee.
- Some Colour Tools also report APCA, a newer contrast model. APCA is not yet part of a published WCAG standard.

Colour-vision simulation

- The panel tools simulate **deuteranopia**, **protanopia**, **tritanopia** and **achromatopsia** with simple 3 × 3 colour matrices applied to sRGB values.
- These show the general direction and size of the change. They do not show exactly what any person sees, and they model complete (dichromatic) deficiencies rather than the milder forms that are more common.
- The Colour Tools use their own shared set of filters; see Accessibility Lab.

Colour difference

- **ΔE 76** is the straight distance in Lab. **ΔE 2000** (CIEDE2000) corrects for how people actually perceive differences and is the one to use. A ΔE 2000 around 1 is just noticeable side by side; above about 10 the colours are clearly different.
- Nearest-name matches in Color Picker and Quick Converter use a short built-in list of 64 names, so the "nearest" name can still be quite far away. Check the ΔE shown with it.

Generators and scores

- **Palette Studio** modes add random variation, so the same mode gives different palettes each time. The WCAG and accessibility modes search for colours that meet their targets but can fall short; the **WCAG Pairs** tab reports the real result.
- **Image extraction** samples a scaled-down copy of the image (400 pixels on the longest side) and groups colours with k-means from random starting points. Results can differ slightly between runs and will not include colours that cover only a few pixels.

- **Palette Remix** adds random variation controlled by **Remix energy**.
- **Color Battle** scores are a weighted heuristic (contrast 35, AA share 30, ΔE 20, colour-vision ΔE 15). Use them to compare palettes, not as a pass mark.
- **Gradient Maker** imitates OKLCH, HSL, Lab and eased blends with generated stops (20 in CSS, 40 in PNG). The imitation is close but not exact.

Images

- **Collage Maker** exports at 3,840 pixels wide only when its preview is at least 800 pixels wide. Drop shadows, the checkerboard fade and the tilt of pinboard and polaroid cells show in the preview but not in the export.

What these tools are not

- Not a colour-management system. The site shows sRGB colours as your screen and browser render them. An uncalibrated screen changes how colours look.
- Not an accessibility audit. Contrast and colour-vision checks cover colour only. For a full review, use the Accessibility and vision tools.
- Not a source of licensed colour-system data. Named-colour matches are approximations.

Related

- Glossary
- Troubleshooting
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Privacy

What Basic Color Tools and the Colour Tools keep on your device, what they store in your browser, and what uses the network.

Basic Color Tools and the Colour Tools do their work in your browser. This page lists what stays on your device, what is remembered, and what reaches the network.

Stays on your device

- **Colours you enter or generate.** Palettes, gradients, conversions and scores are calculated in the page.
- **Images.** Images you choose for Palette Studio's **From Image** or for Collage Maker are read by your browser from your device. They are not uploaded.
- **Images and videos in the Colour Tools** (for example CMYK Soft-Proof or Animation Contrast) are also processed in the browser.
- **The eyedropper.** Color Picker's **Pick** button uses your browser's eyedropper. It samples a single pixel only when you choose one, and the page receives only that colour.
- **Clipboard.** The tools write to your clipboard only when you select a copy button. They do not read your clipboard.
- **Downloads.** PNG and SVG files are created in the browser and saved by your browser's download feature.

Remembered in your browser

These preferences are stored in your browser's local storage on this device. They are not sent anywhere by the tools, and clearing the site's data in your browser removes them.

What	Where it is used
The view you used last (Workspace, Catalogue or Panels)	Basic Color Tools
Level (Basic, Advanced or Both) and category	Basic Color Tools Catalogue
Favourite tools	Basic Color Tools Catalogue
Whether you dismissed the quick start note	Basic Color Tools Catalogue
Each Colour Tool's panel size, position and pin state	Colour Tools
The position of the Colour Tools control bar	Colour Tools

Open panels and their contents are not stored. They are gone when you reload or leave the page.

Uses the network

- **Loading the page and tools.** The page, its scripts and each Colour Tool are downloaded from auricartisan.com when you open them.
- **Colour Tools data.** Some Colour Tools download reference data from the site the first time you open a tab that needs it, such as the Color Inspector's **Names** and **Psychology** tabs.
- **Save to Library.** Items you save go to your Auric Artisan Library. How the Library stores and syncs items is described in the Library Kit documentation.
- **The rest of the site.** Site-wide features such as analytics and advertising follow your cookie choices. See the Help centre and Company and legal.

Related

- Limits and accuracy
- Import, export and the Library
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Related tools and reading

Tools, colour libraries, kits and Learn articles that go well with Basic Color Tools and the Colour Tools.

In the colour workspace

Tool	Why you might use it next
Color Science Lab	Deeper colour-space, conversion and colour-difference work.
Personalization Generator	Turn colours into a personalised design system, palettes and gradients.
Design Engine	Generate a full design system (colour, type, spacing, tokens) and open its colours in the Colour Tools.
Font Library	Choose type to go with your palette.

Tool	Why you might use it next
Icon Library	Find icons and colour them with your palette.
ASCII Art Generator	Turn images and text into ASCII art.

Other tool groups

- Accessibility and vision: the Contrast Checker, Vision Simulation, the URL Analyzer and more, for full accessibility checks beyond colour pairs.
- Image and file tools: the full Collage Maker page, Image Picker (palettes from images), SVG Recolour, LUT Lab and more.
- Colorimetry and rendering: gamut, illuminant and tone-mapping tools.
- Perception and spectral: spectral, dye and material tools.

Colour libraries

Browse large collections of palettes, shades, gradients, harmonies and named colours, and save them to your Library: Colour libraries.

Kits and site features

- Library Kit: where items you save with **Save to Library** live.
- Site features: the right-click menu that opens the Colour Tools, settings, themes and search.

Learn articles

These articles on the site explain the ideas the tools use:

- Color Theory and Harmony
- Oklab and Oklch: Modern Perceptual Color Spaces
- CIELAB and LCH Explained
- Accessible Color Design and WCAG Contrast
- Color Blindness and Color Vision Deficiency
- Color Gradients and Interpolation
- Color in Typography and Text: Readable, Accessible, Beautiful
- Color Order Systems: Munsell, NCS, Pantone, and RAL
- CMYK and the Four-Color Printing Process
- Gamut Mapping
- Simultaneous Contrast: Why a Color Depends on Its Neighbors

More articles: Learn.

Related

- Basic Color Tools overview
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Color Science Lab

What the Color Science Lab is, who it is for, where to find it, and how to start with its Workbench, Compare, Gamut audit and Illuminants views and its 45 panel tools.

The Color Science Lab is a research workspace for colour science on the Auric Artisan website. It has five views. Four of them work on colours you enter and compute their results with published formulas: the **Workbench** resolves one colour into CIE XYZ, xyY, Lab, LCh, Oklab, Oklch and more under the illuminant and standard observer you choose; **Compare** measures the difference between two colours with ΔE 2000, ΔE 1994, ΔE 1976 and Oklab distance; **Gamut audit** checks a palette against sRGB, Display P3, Adobe RGB and Rec. 2020; and **Illuminants** shows how one colour shifts under seven standard lights.

The fifth view, the **Catalogue**, opens 45 tools in eight domains (colorimetry, chromatic adaptation, spectral analysis, appearance models, perceptual spaces, colour spaces, theory and history, and gamut and rendering) as floating panels. Each panel has the same controls (sample and compare colours, model, colour space, theory, white points, strength and colour temperature) and six tabs, and it reads out coordinates in any of 63 colour spaces. Up to five panels can be open, linked so that shared controls move together, arranged with a floating control centre, and exported as one image.

Everything is calculated in your browser. The coordinates are computed from standard formulas; some panel previews are illustrations of a model's effect rather than full implementations, which Limits and accuracy explains.

Who it is for

- Colour scientists, imaging engineers and students who want to see conversions, white points and observers side by side.
- Designers and developers who need Lab, Oklch, xy or ΔE values, or want to know whether a palette survives a wide-gamut conversion.
- Print and product teams checking how a colour holds up under different lights.

Where to find it

- Page: https://auricartisan.com/tool/color-science-lab/ (the short address https://auricartisan.com/model also opens it).
- No sign-in is needed. Saving panel presets uses your Auric Artisan Library.
- A related REST endpoint exists: the page notes **Automate this — 5 API credits per call to POST /v1/science/dossier**. See the Public API.

Quick start

1. Open https://auricartisan.com/tool/color-science-lab/. The **Workbench** view opens first.
2. Type a HEX code such as `#4F7FE0` in the **Specimen** box, or pick a preset swatch.
3. Read **Resolved coordinates**: XYZ, xyY, Lab, LCh, Oklab, Oklch, sRGB, HSL and CMYK.
4. Select illuminant **A** and watch the Lab values and the **Under other light** list change.
5. Open **Compare**, set a **Candidate** colour, and read ΔE 2000.
6. Open **Catalogue**, select a tool card such as **Bradford Adaptation**, and it opens as a floating panel.
7. Open a second tool, select the link button on both panels, and change the sample colour: both follow.

What you can do

- Resolve a colour into nine notations under seven illuminants, two observers and three adaptation transforms.
- Measure colour difference four ways and see whether the gap is lightness, chroma or hue.

- Audit a palette of up to 16 colours for gamut headroom in four RGB working spaces.
- See a colour adapted to seven CIE illuminants with ΔE 2000 shifts.
- Open 45 colour-science tools as panels, each with an experiment canvas, metrics, chart, response graph, model diagram and a per-space dossier.
- Compare up to eight tools at once on a six-colour test palette, and explore a heat map, mind map, statistics and a catalogue of 63 colour spaces and 24 theories, exportable as JSON or CSV.
- Link, tile, split, cascade, stack, pin, dock and export panels.

In this folder

- docs/ — how to use the lab.
  - Getting started
  - The Workbench
  - Compare two colours
  - Gamut audit
  - Illuminants
  - The Catalogue and its panels
  - The 45 tools
  - Catalogue Compare and Analytics
  - Arrange, link, save and export panels
  - Reference
  - Troubleshooting
  - FAQ
- others/ — supporting material.
  - Glossary
  - Shortcuts
  - Limits and accuracy
  - Related
  - Privacy

WEBSITE › TOOLS › COLOUR WORKSPACE

Color Science Lab documentation

The list of guides for the Color Science Lab.

These guides cover the lab's five views, the 45 panel tools, and how to arrange, link, save and export panels.

File	What it covers
getting-started.md	Open the lab, tour the views, make a first measurement.
workbench.md	Resolve one colour under any illuminant, observer and adaptation transform.
compare-two-colours.md	ΔE 2000, 1994, 1976 and Oklab distance, and where the difference lies.
gamut-audit.md	A palette checked against four RGB working spaces.
illuminants.md	One colour adapted to seven standard illuminants.
catalogue-and-panels.md	Finding tools, and every control and tab in a tool panel.
tool-list.md	All 45 tools by domain, with their level and default model.
catalogue-compare-and-analytics.md	Tool comparison on a test palette; heat map, mind map, stats and catalogue exports.
arrange-link-save-export.md	Panel windows, linking, the Color Control centre, the dock, image export and the Library.
reference.md	Every control, option, default and output.
troubleshooting.md	Symptoms, causes and fixes.
faq.md	Short answers to common questions.

Related

- Color Science Lab overview
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Getting started with the Color Science Lab

Open the Color Science Lab, learn its five views, and make your first measurements.

Open the lab

1. Go to https://auricartisan.com/tool/color-science-lab/ in a current desktop browser.
2. Under the title **Color Science Lab** you see a short description and labels: **8 Domains**, **45 Tools**, **60+ Color Spaces**, **Von Kries · Newton · Fairchild** and **In-page Floating Panels**, then a note about the matching API call, which you can dismiss with ×.
3. A row of five view buttons follows: **Workbench**, **Catalogue**, **Compare**, **Gamut audit** and **Illuminants**. The lab opens on **Workbench** the first time and on the view you used last after that.

The five views

View	Use it to
Workbench	See one colour in every CIE and perceptual notation, under the illuminant, observer and adaptation you choose.

View	Use it to
Catalogue	Browse 45 tools and open them as floating panels.
Compare	Measure the difference between a reference and a candidate colour.
Gamut audit	Check how much room a palette has in sRGB, Display P3, Adobe RGB and Rec. 2020.
Illuminants	See one colour adapted to seven standard lights.

The Workbench and Illuminants views share one colour, the *specimen*; Compare's **Reference** starts from it. The lab remembers the specimen in your browser.

First measurement: one colour, two lights

1. On **Workbench**, type `#D3AF37` in the specimen HEX box (include the `#`).
2. Read **Resolved coordinates**. For example, the CIE Lab row reads about `72.75 0.95 62.79` under D65.
3. Under **Illuminant**, select **A** (incandescent). The Lab row changes, the note reads *A · 2°*, and the table header shows **Bradford**.
4. Read **Under other light** on the right: each illuminant with the ΔE 2000 shift from the current one.

Result: you can see how the same surface colour reads under daylight and under tungsten light.

First panel

1. Select **Catalogue**. The tool cards appear, grouped by domain.
2. Select **Delta E Comparator**. It opens as a floating panel over the page.
3. Set **Sample color** and **Compare color** in the panel's left column.
4. Select the **Metrics** tab and read **dE76 to compare**.

Result: a panel you can move, resize and keep open while you try other tools. A floating **Color Control** box also appears, for arranging panels.

Next steps

- The Workbench
- The Catalogue and its panels
- Arrange, link, save and export panels

Related

- Color Science Lab overview
- Shortcuts

WEBSITE › TOOLS › COLOUR WORKSPACE

Arrange, link, save and export panels

Move, resize, pin, dock and link Color Science Lab panels, arrange them with the Color Control centre, export a combined image, and save panels to your Library.

Panel windows

Lab panels float over the whole page, whichever view you are on. Each opens at 1120 by 720 pixels (or smaller to fit your window), offset slightly from the last one.

Title-bar buttons

Button	Tooltip	What it does
Link	**Link shared controls**	Links this panel. Linked panels copy each other's control changes (see below).
Duplicate	**Open another instance**	Opens another copy of the tool with its own settings.
Pin	**Keep panel open**	Protects the panel from being closed automatically or by **Close unpinned** and **Close all**.
–	**Minimize**	Hides the panel and adds it to the dock.
□	**Maximize or restore**	Fills the window; select again to restore.
×	**Close**	Closes the panel.

- Drag the title bar to move; double-click it to maximise or restore.
- Drag the bottom-right corner to resize. The minimum is 520 by 420 pixels, or the window size if smaller.
- Select a panel to bring it to the front.
- With the keyboard focus inside a panel, `Escape` restores a maximised panel, or closes the panel otherwise.
- Selecting a tool that is already open brings its panel forward instead of opening a second one.

The five-panel limit

Up to five panels can be open. Opening a sixth closes the oldest panel that is not pinned. If all five are pinned, the new tool does not open until you unpin or close one.

The dock

Minimised panels appear as items in a dock. Select an item to restore the panel; select its × to close it.

Linking panels

Link two or more panels with their link buttons (or **Link all** / **Sync all**). When you change a control in a linked panel, the same control in every other linked panel changes too: sample and compare colours, model, colour space, theory, white points, strength, colour temperature and the two check boxes. Use it to run one colour through several tools at once, for example Bradford Adaptation, CAT Matrix Comparison and White Point Reference Atlas.

The Open Panels list

Choose **Open Panels** in the Catalogue's domain list. A toolbar offers ⊞ **Tile**, ⩤ **Cascade**, ▤ **Stack**, 🔗 **Link all** (or **Unlink all**), 📤 **Export image** and ✕ **Close unpinned**. Below it, a card per panel shows its domain, name, whether it is active or *Minimized in dock*, its level and **Linked** or **Pinned** tags. Select a card to bring the panel forward.

The Color Control centre

When at least one panel is open, a floating **Color Control** box appears, with the number of open panels. Drag it by its header; collapse or expand it with the arrow button.

Button	What it does
Grid	Tiles the panels: one fills the window, two to four in two columns, five in three columns.
Split H	Places the panels side by side at full height.
Split V	Stacks the panels at full width.
Cascade	Gives each panel the default size and staggers them.
Stack	Centres every panel in the same place.
Sync all	Links every panel, or unlinks them if all are linked.
Minimize / **Restore**	Docks or restores every panel.
Export	Downloads the combined image (below).
Close all	Closes every panel that is not pinned.

The layouts apply to panels that are neither minimised nor maximised. Below the buttons, each panel is listed with its domain and **Sync**, **Keep** (pin), **Min** or **Open**, **Max**, **Copy** (duplicate) and **Close**. Select a panel's name to bring it forward.

Export a combined image

Export image (Open Panels) or **Export** (Color Control) downloads one PNG, `color-science-lab-` followed by the date and time, with each open panel's experiment canvas in a grid, labelled with its name.

Save to and restore from the Library

With a panel focused, select the floating **Save to Library** button. The panel is saved as a preset with a picture of its canvas, its colours and every control value. With no panel open, it saves a *Color Science Lab — workspace* bookmark instead.

Opening a saved preset from your Library opens a fresh copy of that tool (closing any open copy) and re-applies the saved controls once the panel has drawn. See the Library Kit.

Panels themselves are not kept when you reload the page.

Related

- The Catalogue and its panels
- Shortcuts
- Troubleshooting

WEBSITE › TOOLS › COLOUR WORKSPACE

The Catalogue and its panels

Find tools in the Catalogue, and use every control and tab of a Color Science Lab tool panel.

The **Catalogue** view lists the lab's 45 tools as cards. Selecting a card opens that tool as a floating panel over the page. All 45 panels share one layout, so once you know one, you know them all; the tool you choose sets the defaults (model, colour space and theory) and what the canvas draws.

The filter bar

Control	What it does
Color Lab	Reloads the page.
Search box	*Search tools, models, spaces, illuminants…*. Matches tool names, summaries, tags, domains, default models and spaces, and, for colour-space and theory tools, the names and details of every space and theory. Results update a moment after you stop typing.
Clear	Empties the search.
Domain list	**All**, the eight domains, **Favorites**, **Compare**, **Analytics** and **Open Panels**.
Level	**Basic** (20 Foundation tools), **Advanced** (25 Research tools) or **Both** (all 45, the default).
Count	Cards shown out of the total, for example **20 of 45**.
Copy link	Copies the page address.
Quick start button	Shows or hides the quick start note.

The **Level** setting is shared with the site-wide complexity mode (the Advanced Kit), so changing it in one place changes it in the other.

The domains are **Colorimetry**, **Chromatic Adaptation**, **Spectral Analysis**, **Appearance Models**, **Perceptual Spaces**, **Color Spaces**, **Theory and History** and **Gamut and Rendering**. **Compare** and **Analytics** are described in Catalogue Compare and Analytics; **Open Panels** in Arrange, link, save and export.

Tool cards

Each card shows the domain, the tool name, a strip of colours computed for that domain, a summary, and tags: the domain, **Basic / Foundation** or **Advanced / Research**, up to three topic tags, and ● **Open** when it has a panel.

- Select a card to open the tool, or to bring its panel forward.
- Hold `Shift`, `Ctrl` or `Cmd` while selecting to open another copy.
- Select the star to add or remove a favourite.

The full list is in The 45 tools.

The panel

Controls (left column)

Control	What it does	Values or range	Default
Sample color	The colour under study: colour well, HEX box, **Random**.	Any sRGB colour	#D3AF37
Compare color	A second colour for difference metrics.	Any sRGB colour	#2FA7D8
Model	The model for the tool's domain.	Depends on the domain (see below)	The tool's default
Color space / standard	The space used for coordinate readouts and the dossier.	63 colour spaces	The tool's default
Theory / observer	The theory or observer framework.	24 theories and models	The tool's default

Control	What it does	Values or range	Default
White points: Source	Reference white for the sample.	A, B, C, D50, D55, D60 / ACES, D65, D75, DCI, E, F2, F7, F11	D65
White points: Target	Reference white for the result.	Same list	D50
Experiment strength	How strongly the model is applied to the result preview.	0 to 100 %	72 %
Color temperature	A light temperature used by spectral tools and the temperature metric.	1800 to 12000 K, step 100	6500 K
Show reference guides	Draws guides on the canvases.	On or off	On
Apply adaptation preview	Turns the result preview on or off (off shows the sample unchanged).	On or off	On

The HEX boxes accept a partial value while you type and apply it when it is complete, when you press `Enter`, or when you leave the box.

Models by domain

Domain	Models
Colorimetry	XYZ + Lab, xyY, u'v'Y, Delta E, CIE 1931 observer, CIE 1964 observer
Chromatic adaptation	XYZ scaling, Von Kries, Bradford, CAT02, CAT16, CMCCAT97, CMCCAT2000, Sharp CAT, Hunt-Pointer-Estevez, Bianco-Schettini, Bianco-Schettini PC
Spectral	Visible spectrum, Blackbody, Emission lines, Metamerism, CIE daylight locus, Planckian locus, Standard observer
Appearance	Fairchild appearance, Hunt, Nayatani, RLAB, LLAB, CIECAM02, CIECAM16, CAM16, CAM02-UCS, HDR appearance, Surround study, iCAM image appearance
Perceptual	CIELAB, CIELCh(ab), CIELUV, DIN99, Jzazbz, Oklab, Oklch, IPT, ICtCp, CAM02-UCS, CAM16-UCS
Color spaces	Full space atlas, RGB matrix profiles, Wide gamut RGB, Video component spaces, CIE family, Perceptual family, Print/process family, HDR family
Theory	Newton prism and wheel, Goethe, Young-Helmholtz, Hering opponent, Maxwell triangle, Munsell, Ostwald, Von Kries, Retinex, Fairchild, CIECAM02, CAM16
Rendering	sRGB, Display P3, Rec.2020, ACES, ICC rendering intent, Gamut mapping, Tone map, PQ HDR, HLG HDR

Tabs (right column)

Tab	What it shows
Experiment	The tool's main canvas, and three swatches: **Reference** (the sample), **Result** (the model's output) and **Compare**, each with HEX, RGB, xy and its white point.
Metrics	Twelve cards: **dE76 to compare**, **dE76 after model**, **Luma drift**, **Chroma drift**, **Lab result**, **CCT color dE**, **Source Lab**, **Active space**, **Sample coordinates** (with an in-gamut or out-of-gamut note), **Result coordinates**, **Theory model** and **Target white**.
Chart	A channel and Lab chart for the sample, result, compare and temperature colours.
Graph	A response graph for the domain, such as an adaptation response, spectral sweep, appearance curve, uniformity check, space preview, theory transform or rendering compression.
Diagram	The model path: input sample, reference white, and the domain's steps through to the output.
Audit	The tool summary; a dossier of the selected colour space; the current model, space, theory, whites and catalogue size; and tables of related spaces and theories.

The colour-space dossier

On the **Audit** tab, the dossier describes the space chosen in **Color space / standard** for the current sample: its family and category, a summary, the sample's coordinates in that space, an **In gamut** or **Out of … gamut** pill, key facts (primaries, white, transfer function, and for many spaces the year, standard, typical encoding and gamut area relative to sRGB), the **Conversion chain**, typical uses and notes. **This sample across 21 spaces** expands to a table of the sample's coordinates in 21 representative spaces.

Tasks

Adapt a colour from D65 to D50

1. Open **Bradford Adaptation** (Chromatic Adaptation).
2. Set **Sample color**. Leave **Source** on D65 and set **Target** to D50.
3. Set **Experiment strength** to 100 %.
4. Read the **Result** swatch and, on **Metrics**, **Lab result** and **dE76 after model**.

Result: the adapted colour and how far it moved.

Read a colour in an unfamiliar space

1. Open **40+ Color Space Atlas** (Color Spaces).
2. Set **Color space / standard**, for example to *ICtCp* or *Munsell HVC*.
3. Open **Audit** and read the dossier and the conversion chain.

Result: the sample's coordinates in that space, with what the space is for.

Related

- The 45 tools
- Arrange, link, save and export panels
- Limits and accuracy

WEBSITE › TOOLS › COLOUR WORKSPACE

Catalogue Compare and Analytics

Preview up to eight tools on a six-colour test palette, and explore the lab's heat map, mind map, statistics and catalogue, with JSON and CSV export.

Two entries in the Catalogue's domain list are not domains. **Compare** previews several tools side by side before you open any, and **Analytics** maps the whole lab. (This Compare is different from the top-level **Compare** view, which measures two colours.)

Compare (in the Catalogue)

Choose **Compare** in the domain list, or press `C` when the cursor is not in a text box.

Select tools

The left column, **Select tools**, lists the tools that match the current level and search, each marked **F** (Basic / Foundation) or **R** (Advanced / Research). Tick the ones to compare. Buttons: **Basic** (select all Foundation tools), **Advanced** (all Research tools),

Filtered (everything listed) and **Clear**. The first four tools selected at the start are CIE XYZ and Lab Workbench, Bradford Adaptation, Blackbody Radiation and 40+ Color Space Atlas.

The matrix

The matrix shows up to eight selected tools as columns and six test colours as rows: **Auric gold** (#D3AF37), **Signal red** (#D84444), **Cyan blue** (#2FA7D8), **Forest green** (#3C8D5A), **Violet** (#7C5CE2) and **Skin tone** (#C98E69). The **Reference** column shows the original colour. Each tool cell shows that tool's preview of the colour, its HEX code and its ΔE 76 from the reference. Each column header has **Open** to open that tool as a panel.

With nothing selected, it shows **No tools selected** and **Quick Start: Core Lab**, which selects six tools covering colorimetry, adaptation, spectral, colour spaces, theory and gamut.

The matrix previews are quick approximations of each domain's typical effect, for scanning; open a tool for its full readout.

Analytics

Choose **Analytics** in the domain list, or press `A`. The header, **Color Lab Intelligence**, shows the number of visible tools, domains and open panels, with four buttons:

Button	What it does
Random tool	Opens a random tool from those currently visible (also `R`).
JSON	Downloads the catalogue as JSON.
CSV	Downloads the catalogue as CSV.
?	Shows the keyboard shortcuts.

Four tabs:

- **Heat Map**: one row per domain; each tool is a cell coloured by an *analysis depth* score from 1 to 10 (green to orange), from Basic / Foundation to Advanced / Research. The score is the lab's own rating of model complexity and breadth, not a measurement. Select a cell to open the tool.
- **Mind Map**: *Unified model map*: the lab at the centre, the eight domains around it, and up to four tools per domain as small nodes. Select a small node to open that tool.
- **Stats**: **Total**, **Categories**, **Basic**, **Advanced**, **Avg Score**, **Favorites** and **Open**; then **Highest Analysis Depth** (top eight tools), **Domain Balance** (average score and count per domain) and **Recently Opened** (your last tools, up to ten remembered).
- **Catalog**: three figures (**Color spaces** 63, **Theory models** 24, **White points** 13), the **Color Space Registry** table (space, family, white, transfer, standard, gamut with approximate sRGB coverage) and the **Theory and Model Registry** (name, family, author, era, summary). The search box filters both.

Exports

- **JSON**: `color-science-lab-catalog-YYYY-MM-DD.json` with counts, every colour space, theory and white point, and every tool with its domain, level, summary, tags, defaults and score.
- **CSV**: `color-science-lab-catalog-YYYY-MM-DD.csv` with one row per tool, colour space and theory, and the columns `type, slug, name, category, depth_or_family, score_or_white, model_or_transfer, space_or_gamut, theory_or_author, tags_or_summary, summary_or_era`.

Related

- The Catalogue and its panels
- Shortcuts

WEBSITE › TOOLS › COLOUR WORKSPACE

Compare two colours

Measure the difference between a reference and a candidate colour with ΔE 2000, ΔE 1994, ΔE 1976 and Oklab distance, and see whether the gap is lightness, chroma or hue.

The **Compare** view measures how different two colours are. It is useful for checking a match (a print against a screen colour, a paint against a brand colour, a new token against an old one) and for understanding why two colours look different.

ΔE (delta E) is a number for colour difference. Several formulas exist; newer ones follow human perception more closely.

Layout

Two cards side by side:

- **Reference** (**A**) and **Candidate** (**B**), each with a large swatch, a HEX box, a colour well, and a table of **CIE XYZ**, **CIE Lab**, **CIE LCh** and **Oklab** coordinates.

The Reference starts as the Workbench specimen; the Candidate starts as #5A86D8.

Below them, four verdicts:

Verdict	Meaning
ΔE 2000	The current CIE standard (CIEDE2000). Its note says whether the difference is below the just-noticeable threshold (under 1), noticeable only side by side (1 to 2), perceptible at a glance but acceptable side by side (2 to 5), or two different colours (5 and above).
ΔE 1976	The original straight-line Lab distance. It overstates differences in blues, which is why it was replaced.
ΔE 1994	The step between the two: it weights chroma by its own size.
Oklab distance	Plain distance in Oklab, a space designed to make that distance meaningful.

Each figure is green (small), amber or red (large).

Where the difference sits splits the gap into **Δ lightness**, **Δ chroma** and **Δ hue angle**, and a sentence says which dominates: lightness (match L first), chroma (same hue, different saturation) or hue (genuinely different colours).

Tasks

Check whether a match is close enough

1. Enter the target in **Reference** and the sample in **Candidate** (HEX with `#`, or the colour well).
2. Read **ΔE 2000**. Under 1 is effectively identical; under 2 is usually acceptable for close work; above 5 is a clear mismatch.
3. Read **Where the difference sits** to know what to adjust.

Result: a measured, explained difference.

Controls

Control	Values	Default
Reference HEX box and colour well	`#RRGGBB`	The Workbench specimen
Candidate HEX box and colour well	`#RRGGBB`	#5A86D8

Lab values here use D65 and the 2° observer.

Related

- The Workbench
- Quick Converter's Compare tab
- Limits and accuracy

WEBSITE › TOOLS › COLOUR WORKSPACE

Gamut audit

Check a palette against sRGB, Display P3, Adobe RGB and Rec. 2020, see how much headroom its tightest colour has, and what clipping would cost.

The **Gamut audit** view checks a palette against four RGB working spaces. A **gamut** is the range of colours a space can hold. A palette written in HEX is sRGB by definition, so it always fits sRGB; the useful number is **headroom** (margin): how far the palette's tightest colour is from the edge of each space. At 0 % a colour already sits on a primary, and any adjustment clips it.

Layout

Palette under audit

The palette (*N colours · target sRGB*): each colour with its HEX code and a line saying its sRGB margin (for example *12% margin*) or, for a colour from a wider space, *clips ΔE N*. Each has a **×** to remove it. Below: a colour well and **Add to palette**.

The starting palette has eight colours: #4F7FE0, #D3AF37, #00C264, #C4562C, #2F6D3A, #00E07A, #8F3B22 and #2B6F8F. A palette holds from 1 to 16 colours.

Coverage by working space

A table with one row per space (**sRGB**, **Display P3**, **Adobe RGB**, **Rec. 2020**):

- **Tightest headroom**: a bar for the smallest margin in the palette.
- **In gamut**: how many colours fit, for example **8 / 8**.
- **Margin**: the tightest margin as a percentage. Red if any colour falls outside, amber if the margin is under 2 %, green otherwise.

The verdict and the cost of clipping

- A large figure such as **8/8** with *Every colour survives sRGB unchanged.*, or how many clip and whether they fit Display P3 instead.
- **What clipping costs** (*ΔE00 after clip*): for each colour that clips, its clipped HEX code and the ΔE 2000 lost. With nothing clipping, it names the colour closest to the edge and its margin.

Tasks

Find the palette colour most at risk in conversion

1. Replace the palette with yours: remove the starting colours with × and add yours with the colour well and **Add to palette**.
2. Read the **Margin** column. The space with the lowest margin, and the colour named under **What clipping costs**, is where trouble will start.

Result: you know which colour to watch when converting or adjusting the palette.

Controls

Control	Values	Default
× on a colour	Removes it	At least 1 colour remains
Colour well + **Add to palette**	Adds a colour	Up to 16 colours

Related

- The Workbench for one colour's gamut
- Colorimetry and rendering tools for gamut mapping

WEBSITE › TOOLS › COLOUR WORKSPACE

Illuminants

See one colour adapted to seven CIE illuminants with the Bradford transform, and how far it shifts under each.

The **Illuminants** view shows the Workbench specimen as it would appear to an observer adapted to each of seven standard lights, and measures how far it moves from its D65 appearance.

Layout

- **The same colour, adapted** (*Bradford · from D65*): one patch per illuminant with its name, colour temperature and **ΔE00** from D65. The explanation under it: each patch is the specimen re-rendered for an observer adapted to that illuminant, using the Bradford transform.
- The verdict: the largest shift and the illuminant that causes it, with *A match made under D65 will not hold there.* when the shift is above 5, or *The colour holds reasonably across the set.*
- **Shift by illuminant** (*ΔE00 from D65*): each illuminant other than D65 with its description and shift, coloured green below 3, amber 3 to 8 and red above 8.

The illuminants

Illuminant	Description	Temperature
D65	Daylight · reference	6504 K
D50	Print viewing	5003 K
D55	Mid daylight	5503 K

Illuminant	Description	Temperature
C	Average daylight (obsolete)	6774 K
A	Incandescent	2856 K
F2	Cool white fluorescent	4230 K
F11	Narrow tri-band fluorescent	4000 K

Tasks

See which light is most risky for a colour

1. Set the specimen on the **Workbench** (or keep the current one).
2. Open **Illuminants** and read the verdict and **Shift by illuminant**.

Result: the light under which the colour changes most.

Notes

- This view always uses the Bradford transform and the 2° observer, whatever you chose on the Workbench.
- It models adaptation of an sRGB colour, not the spectral reflectance of a real surface. Two surfaces that match under D65 can still separate under another light (metamerism) in ways only spectral data can predict.

Related

- The Workbench
- Illuminants tool and other colorimetry tools
- Limits and accuracy

WEBSITE › TOOLS › COLOUR WORKSPACE

The 45 tools

Every Color Science Lab panel tool by domain, with its level, what it is for and its default model.

Level **F** means Basic / Foundation (shown when the level is Basic or Both); **R** means Advanced / Research (shown when the level is Advanced or Both). Every tool opens the same panel described in The Catalogue and its panels; the default model, space and theory differ.

Colorimetry (5)

Tool	Level	What it is for
CIE XYZ and Lab Workbench	F	Convert samples between sRGB, CIE XYZ, xyY and CIE Lab with live white-point context.
Chromaticity Diagram	F	Plot samples, white points and approximate sRGB gamut positions on the CIE xy diagram.
Delta E Comparator	F	Compare two colours by Lab distance, luma drift, chroma drift and hue direction.
Standard Observer Lab	R	Compare 2° and 10° observers, xy chromaticity and u'v' uniform chromaticity. Default model: CIE 1931 observer.

Tool	Level	What it is for
Color Difference Suite	R	Audit Delta E 76 distance, chroma drift, luma drift and perceptual proxy spaces side by side. Default space: DIN99.

Chromatic Adaptation (6)

Tool	Level	What it is for
Bradford Adaptation	F	Move a colour from one reference white to another with a Bradford transform.
CAT02 and CAT16 Bench	R	Compare modern adaptation transforms with source and target whites.
Illuminant Match Bench	R	Study a colour under A, D50, D65, D75 and equal-energy whites.
Von Kries Adaptation Lab	F	Compare classic von Kries scaling with Bradford-style transforms.
CAT Matrix Comparison	R	Switch between XYZ scaling, von Kries, Bradford, CMCCAT, Sharp, CAT02, CAT16 and Bianco-Schettini. Default: CAT16.
White Point Reference Atlas	R	Inspect A, B, C, D-series, equal-energy, fluorescent, DCI and ACES D60 whites.

Spectral Analysis (6)

Tool	Level	What it is for
Visible Spectrum Explorer	F	Sweep wavelengths from violet to red and inspect approximate sRGB mapping.
Blackbody Radiation	F	Explore colour temperature from incandescent to daylight and blue sky.
Emission Line Mixer	R	Mix narrow spectral lines and compare against a broadband reference.
Metamerism Workspace	R	Compare similar colours under several illuminants and estimate metameric instability.
Daylight and Planckian Locus	R	Compare CIE daylight and Planckian blackbody behaviour and chromaticity drift.
Observer Sensitivity Curves	R	Explore wavelength bands through cone-like, CIE-like and opponent sensitivity proxies.

Appearance Models (6)

Tool	Level	What it is for
CIECAM16 Sandbox	R	Adjust luminance, surround and adaptation strength to see appearance behaviour.
HDR Appearance Lab	R	Push samples through high-dynamic-range viewing assumptions.
Surround Adaptation Study	F	Place a sample on dark, dim and average surrounds.
Fairchild Appearance Lab	R	Fairchild-style appearance ideas: media white, adapting luminance, colourfulness, context.
Hunt and Nayatani Models	R	Compare the Hunt and Nayatani models with modern CAM-style controls.
CIECAM02 and CAM16 Comparison	R	Compare CIECAM02, CAM16, CAT02 and CAT16 assumptions.

Perceptual Spaces (6)

Tool	Level	What it is for
Jzazbz Slice Visualizer	R	Inspect a perceptual slice for HDR and high-chroma relationships.
CAM02-UCS Distance	R	Compare uniformity estimates across Lab-like and CAM-derived coordinates.
Uniformity Stress Test	F	Generate hue and chroma ramps to see where equal numeric steps stop looking equal.
Oklab and Oklch Workbench	F	Oklab and Oklch coordinates for UI gradients, gamut checks and hue paths.
CIELAB, CIELUV, and LCh Suite	F	Compare Lab, Luv, LCh(ab), LCh(uv), Hunter Lab, DIN99 and related coordinates.
HDR Uniform Space Bench	R	Compare Jzazbz, JzCzhz, ICtCp, IPT, CAM02-UCS and CAM16-UCS.

Color Spaces (7)

Tool	Level	What it is for
40+ Color Space Atlas	F	Browse RGB, CIE, perceptual, print, video, HDR and colour-order spaces from one panel.
RGB Working Space Matrix Lab	F	Compare sRGB, linear sRGB, Display P3, Adobe RGB, ProPhoto RGB, Rec.2020, ACES and classic RGB profiles.
Wide Gamut and HDR Space Lab	R	Rec.2020, ACES2065-1, ACEScg, ICtCp, Jzazbz, PQ, HLG and scene-linear workflows.
Video Component Space Lab	F	YCbCr, YPbPr, YUV, YIQ, Rec.601, Rec.709, Rec.2020 and HDR encodings.
Artist Cylindrical Space Lab	F	HSL, HSV, HSI, HWB, HSLuv, HPLuv, LCh, Oklch and Munsell-like hue systems.
Print and Process Space Lab	F	CMY, CMYK, ECI RGB v2, ColorMatch RGB, Best RGB, Beta RGB, Don RGB and D50 print workflows.
CIE and Perceptual Space Map	R	XYZ, xyY, u'v'Y, LMS, Lab, Luv, Hunter Lab, DIN99, Oklab, IPT, CAM02-UCS and CAM16-UCS.

Theory and History (4)

Tool	Level	What it is for
Newton Color Theory Studio	F	Newton's prism spectrum, colour circle and hue ordering, linked to modern wavelength and RGB views.
Classic Theory Comparison	F	Compare Newton, Goethe, Young-Helmholtz, Hering, Maxwell, Munsell, Ostwald and Grassmann.
Opponent and Color Constancy Lab	R	Hering opponent channels, von Kries adaptation, Retinex, NCS notation and colour constancy.
Fairchild and CAM Theory Lab	R	Trace Fairchild, Hunt, Nayatani, RLAB, LLAB, CIECAM02, CAM16, iCAM, IPT and Oklab ideas.

Gamut and Rendering (5)

Tool	Level	What it is for
Gamut Map	F	Compare samples against sRGB, Display P3 and Rec.2020 gamut boundaries.
Tone Curve Compression	R	Apply tone and chroma compression to see rendering trade-offs.
ACES Rendering Pipeline	R	ACES scene-linear, AP0/AP1, look, output transform and display rendering.

Tool	Level	What it is for
HDR Tone Mapping Lab	R	Compare PQ, HLG, Rec.2020, ICtCp and Jzazbz-style HDR compression.
ICC Rendering Intent Lab	F	Compare perceptual, relative colorimetric, absolute colorimetric and saturation intents.

The colour-space registry

Color space / standard lists 63 spaces: 30 RGB working spaces (sRGB, linear sRGB, scRGB, Display P3, DCI-P3, P3-D65, Adobe RGB (1998), ProPhoto RGB / ROMM, Rec.2020, Rec.2100 PQ and HLG, Rec.709, Rec.601 525 and 625, NTSC (1953), SMPTE-240M, Apple RGB, ColorMatch RGB, Adobe Wide Gamut RGB, CIE RGB, Best RGB, Beta RGB, Bruce RGB, Don RGB 4, ECI RGB v2, Ekta Space PS5, ACES2065-1, ACEScg, ACEScc, ACEScct), the CIE spaces (XYZ, xyY, u'v'Y), LMS, perceptual spaces (CIELAB, CIELCh(ab), CIELUV, CIELCh(uv), Hunter Lab, DIN99, DIN99o, Oklab, Oklch, HSLuv, HPLuv, IPT), HDR spaces (ICtCp, Jzazbz, JzCzhz), appearance spaces (CAM16-UCS, CAM02-UCS), cylindrical RGB (HSL, HSV / HSB, HSI, HWB), print (CMY, CMYK), video (Y'CbCr (709), Y'PbPr, Y'UV, Y'IQ) and colour-order systems (Munsell HVC, Natural Color System).

Related

- The Catalogue and its panels
- Catalogue Compare and Analytics

WEBSITE › TOOLS › COLOUR WORKSPACE

The Workbench

Resolve one colour into CIE XYZ, xyY, Lab, LCh, Oklab, Oklch, sRGB, HSL and CMYK under a chosen illuminant, standard observer and adaptation transform, and see its gamut and how it shifts under other lights.

The Workbench takes one colour, the **specimen**, and computes it in nine notations at once, all from the same linear-light values. You can change the viewing conditions (the illuminant, the standard observer and the chromatic adaptation transform) and every row moves together. The page's own note under the board says: *Computed, not looked up.*

A few terms:

- **Illuminant**: a standard light defined by its white point, such as D65 (average daylight) or A (incandescent).
- **Standard observer**: the CIE's model of human colour matching over a 2° field (1931) or a 10° field (1964). Lab values depend on which one you use.
- **Chromatic adaptation transform**: a formula that predicts how a colour appears after the eye adapts to a different white. Bradford, CAT02 and CAT16 are three versions of the von Kries method.

Layout

Left: the controls

- **Specimen** (*sRGB in*): a large swatch, a HEX box, a colour well and eight preset swatches (#4F7FE0, #D3AF37, #2F6D3A, #C4562C, #8F3B22, #2B6F8F, #B0169A, #F0E6D2).
- **Observer** (*CIE standard*): **2° 1931** or **10° 1964**.
- **Illuminant** (*white point*): **D65**, **D50**, **D55**, **C**, **A**, **F2** and **F11**.
- **Adaptation** (*von Kries family*): **Bradford**, **CAT02** or **CAT16**. While the illuminant is D65, a note says no adaptation is applied: the transform only works once you move the white point.

Centre: the results

Resolved coordinates, with a note naming the illuminant, observer and transform:

Space	Coordinates	Note
CIE XYZ	X, Y, Z	reference
CIE xyY	x, y, Y	chromaticity
`CIE L*a*b*`	L, a, b	illuminant and observer
CIE LCh	L, C, h	polar Lab
Oklab	L, a, b	perceptual
Oklch	L, C, h	polar Oklab
sRGB	R, G, B	display
HSL	H, S, L	cylindrical
CMYK	C, M, Y, K	naive, no profile

CIE 1931 chromaticity plots the specimen on the xy diagram and gives x and y, with three figures: `CIE L*`, `C* ab` and **hue angle**.

Right: gamut and other lights

- **Gamut** (*this colour*): **sRGB** (IEC 61966-2-1), **Display P3** (SMPTE RP 431-2), **Adobe RGB** (1998) and **Rec. 2020** (ITU-R BT.2020), each marked **inside** or with the ΔE of clipping. A HEX colour is always inside sRGB.
- **Under other light** (*ΔE00 vs the current illuminant*): every other illuminant with the ΔE 2000 shift the colour would undergo. Green below 3, amber 3 to 8, red above 8. Select a row to switch to that illuminant.

How the numbers are computed

1. The specimen's sRGB values are decoded to linear light and converted to XYZ for D65 and the 1931 observer, where sRGB is defined.
2. If you choose another illuminant or the 10° observer, XYZ is adapted from D65 to that white with the chosen transform.
3. Lab and LCh are computed against the chosen white. Oklab and Oklch come straight from sRGB, as the Oklab definition specifies.

Tasks

Get print-condition Lab values

1. Enter your colour.
2. Select **D50** and **2° 1931**.
3. Choose the transform your workflow uses (Bradford is the common choice in ICC workflows).
4. Read the `CIE L*a*b*` row.

Result: Lab values relative to D50, as used in print.

Check a colour's stability under shop lighting

1. Enter the colour.

2. Read **Under other light**. Large values under **A**, **F2** or **F11** mean the colour will look noticeably different under incandescent or fluorescent light.

Result: a quick risk check before choosing a surface or product colour.

Controls

Control	Values	Default
Specimen HEX box	`#RRGGBB` (with `#`)	#4F7FE0, or your last specimen
Colour well	Any sRGB colour	Same
Preset swatches	8 colours	—
Observer	2° 1931, 10° 1964	2° 1931
Illuminant	D65, D50, D55, C, A, F2, F11	D65
Adaptation	Bradford, CAT02, CAT16	Bradford
Under other light row	Switches the illuminant	—

Related

- Illuminants
- Compare two colours
- Limits and accuracy

WEBSITE › TOOLS › COLOUR WORKSPACE

Color Science Lab reference

Every view, control, option, default, output and file of the Color Science Lab in one place.

Page

Item	Value
Address	https://auricartisan.com/tool/color-science-lab/ (also /model)
Views	Workbench, Catalogue, Compare, Gamut audit, Illuminants
Tools	45 in 8 domains (20 Foundation, 25 Research)
Colour spaces	63
Theories and models	24
White points (panels)	13
Maximum open panels	5
API equivalent	POST /v1/science/dossier, 5 API credits per call

Workbench

Control	Values	Default
Specimen HEX / colour well	`#RRGGBB`	#4F7FE0 (then remembered)
Presets	#4F7FE0, #D3AF37, #2F6D3A, #C4562C, #8F3B22, #2B6F8F, #B0169A, #F0E6D2	—
Observer	2° 1931, 10° 1964	2° 1931
Illuminant	D65, D50, D55, C, A, F2, F11	D65
Adaptation	Bradford, CAT02, CAT16	Bradford

Outputs: CIE XYZ, CIE xyY, `CIE L*a*b*`, CIE LCh, Oklab, Oklch, sRGB, HSL, CMYK; xy plot with L, C and hue angle; gamut in sRGB, Display P3, Adobe RGB, Rec. 2020; ΔE 2000 under each other illuminant.

Compare (view)

Control	Default
Reference (A)	The specimen
Candidate (B)	#5A86D8

Outputs: coordinates in CIE XYZ, Lab, LCh and Oklab for each; ΔE 2000, ΔE 1976, ΔE 1994, Oklab distance; Δ lightness, Δ chroma, Δ hue angle with a verdict sentence.

Gamut audit

Control	Values	Default
Remove (×)	—	At least 1 colour
Add to palette	Up to 16 colours	8 starting colours

Outputs: per-colour sRGB margin or clip ΔE; per-space tightest headroom, in-gamut count and margin; verdict; ΔE 2000 cost of clipping.

Illuminants

Seven illuminants (D65, D50, D55, C, A, F2, F11) with temperature and ΔE 2000 from D65, Bradford transform, 2° observer.

Catalogue filter bar

Control	Values	Default
Search	Free text, applied after a short pause	Empty
Domain list	All, Colorimetry, Chromatic Adaptation, Spectral Analysis, Appearance Models, Perceptual Spaces, Color Spaces, Theory and History, Gamut and Rendering, Favorites, Compare, Analytics, Open Panels	All (then remembered)
Level	Basic, Advanced, Both	Both (shared with the site's complexity mode)

Control	Values	Default
Copy link, Quick start	—	—

Panel controls

Control	Values or range	Default
Sample color	Colour well, HEX, Random	#D3AF37
Compare color	Colour well, HEX	#2FA7D8
Model	Per domain (see catalogue-and-panels.md)	Tool default
Color space / standard	63 spaces	Tool default
Theory / observer	24 entries	Tool default
Source white / Target white	A, B, C, D50, D55, D60 / ACES, D65, D75, DCI, E (Equal Energy), F2, F7, F11	D65 / D50
Experiment strength	0–100 %	72 %
Color temperature	1800–12000 K, step 100	6500 K
Show reference guides	On/off	On
Apply adaptation preview	On/off	On

Tabs: Experiment, Metrics, Chart, Graph, Diagram, Audit.

Theories and models: Newton prism and color circle, Goethe theory of colors, Young-Helmholtz trichromacy, Hering opponent process, Grassmann color laws, Maxwell color triangle, Munsell color order, Ostwald color system, CIE 1931 standard observer, MacAdam ellipses, Judd-Vos correction, Von Kries coefficient law, Land Retinex, Hunt appearance model, Nayatani model, RLAB, LLAB, Fairchild color appearance, CIECAM02, CAM16, Image color appearance / iCAM, IPT hue linearity, Oklab modern uniformity, Basic color terms.

Panel windows

Control	What it does
Link, Duplicate, Pin, Minimize, Maximize, Close	See arrange-link-save-export.md
Drag title bar / double-click	Move / maximise
Corner handle	Resize (minimum 520 × 420)
`Escape`	Restore if maximised, otherwise close

Open Panels toolbar and Color Control

Open Panels	Color Control
Tile, Cascade, Stack, Link all / Unlink all, Export image, Close unpinned	Grid, Split H, Split V, Cascade, Stack, Sync all, Minimize, Restore, Export, Close all; per panel Sync, Keep, Min / Open, Max, Copy, Close

Catalogue Compare and Analytics

Item	Values
Compare selection buttons	Basic, Advanced, Filtered, Clear; Quick Start: Core Lab
Compare columns	Up to 8 tools
Test palette	Auric gold, Signal red, Cyan blue, Forest green, Violet, Skin tone
Analytics tabs	Heat Map, Mind Map, Stats, Catalog
Analytics buttons	Random tool, JSON, CSV, ?

Files

File	From
`color-science-lab-<date-time>.png`	Export image / Export
`color-science-lab-catalog-YYYY-MM-DD.json`	Analytics → JSON
`color-science-lab-catalog-YYYY-MM-DD.csv`	Analytics → CSV

Remembered in your browser

Last view, Workbench specimen, domain, level (site-wide), favourites, recently opened tools (10), quick-start note dismissal.

Related

- Shortcuts
- Glossary

WEBSITE › TOOLS › COLOUR WORKSPACE

Troubleshooting the Color Science Lab

Symptoms you may meet in the Color Science Lab, why they happen, and what to do.

Typing a HEX code on the Workbench does nothing

- **Why:** the Workbench, Compare and the gamut tools accept a full `#RRGGBB` code, with the `#`.
- **What to do:** type all six digits with the `#`, or use the colour well or a preset.

A panel did not open

- **Why:** five panels are open and all are pinned.
- **What to do:** unpin or close one, then open the tool again.

A panel closed by itself when I opened another tool

- **Why:** at five panels, opening another closes the oldest unpinned panel.
- **What to do:** pin panels you want to keep.

A panel closed when I pressed Escape

- **Why:** `Escape` inside a panel closes it (or restores it first if it is maximised).
- **What to do:** select outside the panel before pressing `Escape` for other reasons.

Pressing a letter changed the Catalogue or tiled my panels

- **Why:** single-letter shortcuts (`/`, `?`, `F`, `A`, `C`, `R`, `T`, `1` to `5`) work anywhere on the page when the cursor is not in a text box.
- **What to do:** click into a text box before typing, or see Shortcuts.

Linked panels do not follow each other

- **Why:** only panels whose link button is on share changes.
- **What to do:** turn on linking in each panel, or use **Sync all**.

The search seems slow to respond

- **Why:** the search waits a moment after you stop typing.
- **What to do:** pause briefly after typing.

The Workbench and a panel give different Lab values for the same colour

- **Why:** the Workbench uses the illuminant and observer you selected with a chosen adaptation; a panel reports Lab against its source or target white.
- **What to do:** set the same white on both (for example D65 on the Workbench and D65 as the panel's source white).

A canvas looks blurry or too small after resizing

- **Why:** the canvases redraw at the panel's size.
- **What to do:** resize or maximise the panel once more to redraw.

The exported image is missing a panel's latest state

- **Why:** the export uses each panel's experiment canvas as last drawn; minimised panels are not redrawn.
- **What to do:** restore all panels and let them redraw before exporting.

A restored Library preset shows old values at first

- **Why:** saved controls are applied just after the panel draws.
- **What to do:** wait a moment; if values still look wrong, check that the preset was saved from a focused panel.

Everything disappeared after a reload

- **Why:** panels are not saved across reloads.
- **What to do:** save important panels to your Library, or export an image.

Related

- FAQ
- Limits and accuracy
- Help centre

WEBSITE › TOOLS › COLOUR WORKSPACE

Color Science Lab FAQ

Short answers to common questions about the Color Science Lab.

Are the numbers accurate enough for research?

The Workbench, Compare, Gamut audit and Illuminants views and the panels' coordinate readouts use published formulas (sRGB, CIE XYZ and Lab, Bradford, CAT02, CAT16, CIEDE2000, Oklab). They are exact within 8-bit sRGB input and floating-point rounding. Some panel **Result** previews are illustrations, not full model outputs. Read Limits and accuracy before citing a figure.

Can I enter spectral data or measured values?

No. Every input is an sRGB colour. The lab does not accept spectral reflectance or measured XYZ.

Why does a HEX colour always show "inside" sRGB?

A HEX code is an sRGB colour by definition. The useful figure is the headroom, which the Gamut audit reports.

What is the difference between the Catalogue's Compare and the Compare view?

The **Compare** view measures two colours. The Catalogue's **Compare** entry previews several tools on a test palette.

Do I need an account?

No. Saving panels uses your Auric Artisan Library.

Does the level switch affect other pages?

Yes. **Basic**, **Advanced** and **Both** follow the site-wide complexity mode, so changing it here changes it on other pages that use it.

Related

- Troubleshooting
- Getting started

WEBSITE › TOOLS › COLOUR WORKSPACE

Supporting material

Glossary, shortcuts, limits, related tools and privacy notes for the Color Science Lab.

Reference material that supports the Color Science Lab guides.

File	What it covers
glossary.md	Colour-science terms used in these docs.
shortcuts.md	Keyboard shortcuts and mouse gestures.
limits-and-accuracy.md	What is computed exactly, what is illustrative, and what the lab does not do.
related.md	Related tools and Learn articles.
privacy.md	What stays in your browser and what is stored.

Related

- Color Science Lab overview
- Documentation contents

WEBSITE › TOOLS › COLOUR WORKSPACE

Shortcuts

Keyboard shortcuts and mouse gestures in the Color Science Lab.

The single-key shortcuts work anywhere on the page when the cursor is not in a text box, list or text area, and when `Ctrl`, `Cmd` and `Alt` are not held. The letter shortcuts act on the Catalogue, even if another view is showing.

Keys	What it does
`/`	Focus the Catalogue search.
`Esc`	In the search box: clear the search. Also closes the shortcuts overlay.
`?`	Show or hide the keyboard shortcuts overlay.
`F`	Show Favorites in the Catalogue.
`A`	Show Analytics in the Catalogue.
`C`	Show Compare in the Catalogue.
`R`	Open a random tool from those visible.
`T`	Tile all panels.
`1` to `5`	Bring panel 1 to 5 (in opening order) to the front.
`Shift` or `Ctrl` + select a card	Open another copy of a tool.
`Esc` inside a panel	Restore a maximised panel; otherwise close the panel.
Double-click a title bar	Maximise or restore.
`Enter` in a panel HEX box	Apply the typed colour.
`Arrow Down`, `Enter` or `Space` on a panel list	Open the list; `Esc` closes it.

Related

- Supporting material
- Arrange, link, save and export

WEBSITE › TOOLS › COLOUR WORKSPACE

Glossary

Plain-language definitions of the colour-science terms used in the Color Science Lab documentation.

Term	Meaning
ACES	Academy Color Encoding System, a family of colour spaces for film and visual effects (AP0, AP1, ACEScg).
Appearance model	A model that predicts how a colour looks under given viewing conditions, such as CIECAM02 or CAM16.
Bradford transform	A widely used chromatic adaptation transform from the von Kries family.
CAT02, CAT16	Chromatic adaptation transforms from the CIECAM02 and CAM16 appearance models.
CCT	Correlated colour temperature, in kelvin.
Chromatic adaptation	The eye's adjustment to the colour of the light, and the formulas that predict it.
Chromaticity (x, y)	The colour of a stimulus apart from its brightness, plotted on the CIE 1931 diagram.
CIEDE2000 (ΔE 2000)	The CIE's current formula for perceived colour difference.
CIE XYZ	The tristimulus values that other colour spaces are defined from.
CIELAB / LCh	A perceptual space (lightness, a, b) and its polar form (lightness, chroma, hue).
Dossier	The panel's detailed description of a colour space for the current sample.
Gamut	The range of colours a space or device can hold.
Headroom (margin)	How far a colour sits from the edge of a gamut.
Illuminant	A standard light defined by its spectrum or white point, such as D65 or A.
Linked panels	Panels that copy each other's control changes.
Metamerism	Two colours that match under one light and differ under another.
Oklab / Oklch	A modern perceptual space and its polar form, used in CSS colour.
Planckian locus	The path of blackbody colours on the chromaticity diagram as temperature changes.
Standard observer	The CIE model of average human colour matching: 2° (1931) or 10° (1964).
Von Kries	The principle of scaling each cone response independently to model adaptation.
White point	The colour treated as white in a space or under a light.

Related

- Supporting material
- Basic Color Tools glossary

WEBSITE › TOOLS › COLOUR WORKSPACE

Limits and accuracy

What the Color Science Lab computes with published formulas, which previews are illustrative, and what it cannot do.

Computed with published formulas

- **Workbench**: sRGB decoding, CIE XYZ (D65, 2°), adaptation with the published Bradford, CAT02 and CAT16 matrices, Lab and LCh against the chosen white (with separate white points for the 2° and 10° observers), Oklab and Oklch, and gamut checks with the published matrices for sRGB, Display P3, Adobe RGB and Rec. 2020.
- **Compare**: CIEDE2000 written out in full, CIE94, CIE76 and Oklab distance.
- **Gamut audit** and **Illuminants**: the same matrices and CIEDE2000.
- **Panels**: the **Sample coordinates**, **Result coordinates** and the colour-space dossier come from a colour-space engine that implements each of the 63 spaces' conversions. The **Chromatic Adaptation** tools' Result colour uses real adaptation matrices, scaled by **Experiment strength**.

Illustrative, not full models

In the panels of the Spectral, Appearance, Perceptual, Color Spaces, Theory and Rendering domains, the **Result** swatch (and the metrics derived from it, such as **dE76 after model**) is an illustration of the direction and size of the named model's effect, controlled by **Experiment strength**. It is not the output of a full implementation of CIECAM16, ICtCp, ACES or the other models named. The same applies to the Catalogue Compare matrix and the Heat Map's analysis-depth score, which is the lab's own rating.

Use the coordinate readouts and the four computed views for figures; use the illustrative previews to explore ideas.

Inputs and precision

- Every input is an 8-bit sRGB colour. The lab cannot take spectral data, measured XYZ or out-of-sRGB values.
- Displayed values are rounded (two to four decimals).
- The CMYK row is a naive formula with no press profile.
- Colours drawn on screen depend on your display; they are not calibrated.

What the lab is not

- Not a spectrophotometer or colour-management system.
- Not a substitute for spectral analysis when predicting metamerism: adaptation of an RGB colour does not capture how two real surfaces with different spectra behave.

Related

- The Workbench
- Privacy

WEBSITE › TOOLS › COLOUR WORKSPACE

Privacy

What the Color Science Lab keeps in your browser and what uses the network.

Stays in your browser

All calculations run in the page. Colours you enter are not sent anywhere to be computed. Exports (PNG, JSON, CSV) are created in your browser.

Remembered in your browser

In your browser's local storage on this device: the last view, the Workbench specimen colour, the Catalogue domain, favourites, the last ten tools opened, whether you dismissed the quick start note, and the level, which is shared with the site-wide complexity setting. Open panels are not stored.

Uses the network

- Loading the page and its scripts.
- **Save to Library** stores presets in your Auric Artisan Library; see the Library Kit.
- Site-wide analytics and advertising follow your cookie choices; see the Help centre.

Related

- Limits and accuracy

WEBSITE › TOOLS › COLOUR WORKSPACE

Related tools and reading

Tools and Learn articles that go with the Color Science Lab.

Tools

- Colorimetry and rendering tools: chromatic adaptation, gamut maps, illuminants, HDR plots, tonal steps and tone mapping as full pages.
- Perception and spectral tools: spectral, perception and material tools.
- Basic Color Tools and its Color Space Converter for quick conversions.
- Design System Generator, which can open the lab with your palette.

Services

- Public API: the colour dossier as a REST call.

Learn articles

- CIELAB and LCh Explained
- Oklab and Oklch: Modern Perceptual Color Spaces
- Gamut Mapping
- Gamut Volume: How Much Color a Wide Gamut Actually Adds
- HDR, Wide Gamut, PQ, HLG, and Modern Displays
- Color Order Systems: Munsell, NCS, Pantone, and RAL

Related

- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Design System Generator

What the Design System Generator (Design Engine) is, who it is for, where to find it, and how to generate, check and export your first design system.

The Design System Generator, also called the Design Engine, turns three choices into a complete design system. You pick an **industry**, a **style** and a **contrast target**, and the tool composes colour ramps in OKLCH, light and dark colour roles that meet your contrast target, semantic colours, a font pairing with a modular type scale, an icon family, and spacing, radius, shadow, motion, z-index and breakpoint scales. That is 177 design tokens in the default light-and-dark mode.

There is no Generate button. Every change regenerates the whole system at once, and a contrast audit on the right of the screen re-runs with it, showing a score out of 100, the twelve pairings it checks and whether each passes. When you are happy, you export the tokens to any of twelve targets, from CSS and Tailwind to Figma, SwiftUI and Jetpack Compose, or share a link that rebuilds the exact same system.

Everything is calculated in your browser, and the same inputs always give the same system. The tool does not know your brand, your competitors or your product: the page itself says the result is *generated, not designed*. Treat it as a well-built, well-checked starting point.

Who it is for

- Product designers who want a coherent token set and the evidence that it passes contrast.
- Front-end developers who want ready-to-use tokens in their stack's format and naming convention.
- Accessibility leads who want to see every colour pairing's WCAG ratio and APCA value in one place.
- Founders and small teams starting a product without a design system.

Where to find it

- Page: https://auricartisan.com/tool/design-engine/
- No sign-in is needed to generate, share or export. Saving systems uses your Auric Artisan Library.
- The same job is available through the public API: the page shows the endpoint and its cost in API credits. See the Public API.

Quick start

1. Open https://auricartisan.com/tool/design-engine/. A system is generated straight away.
2. In **The brief** on the left, choose an **Industry**, for example **Healthcare**.
3. Choose a **Style**, for example **Modern**.
4. Under **Contrast target**, choose **AA** or **AAA**.
5. Watch **The audit** on the right: the score, and whether all twelve pairings pass.
6. Open the **Colour**, **Type** and **Components** tabs to look at the result.
7. Open **Tokens**, pick an export format such as **CSS**, and select **Copy** or **Download**.
8. Select **Share** at the top to copy a link that rebuilds this system.

What you can do

- Generate a design system from 15 industry settings (Auto plus 14 industries), 10 styles and 3 contrast targets.
- Override the brand seed colour, harmony, colour mode, density, corner style, type ratio and ramp shape.

- Rename every token with a prefix and a naming case (kebab, camel, snake or Pascal) before export.
- Read colour ramps, semantic colours, surface roles, gradients, the type scale and roles, the icon family and every scale.
- Try the tokens on a live, clickable component gallery in light and dark.
- Audit twelve contrast pairings for both themes with WCAG 2.1 ratios and APCA values, with suggested fixes.
- Compare the same brief across all ten styles or all industries.
- Export to CSS, modern CSS, SCSS, Tailwind v3 and v4, Figma / W3C tokens, JSON, TypeScript, Android XML, SwiftUI, Flutter and Compose.
- Save systems to your Library, restore them, and send the palette to other colour tools.

In this folder

- docs/ — how to use the tool.
  - Getting started
  - Set the brief
  - Read the system
  - Use the component playground
  - Check contrast with the audit
  - Compare styles and industries
  - Export tokens
  - Share, save and connect
  - Reference
  - Troubleshooting
  - FAQ
- others/ — supporting material.
  - Glossary
  - Limits and accuracy
  - Related
  - Privacy

WEBSITE › TOOLS › COLOUR WORKSPACE

Design System Generator documentation

The list of guides for the Design System Generator.

These guides explain how to use the Design System Generator: setting the brief, reading each tab, using the component playground and the contrast audit, comparing systems, exporting tokens, and saving and sharing. The reference lists every control.

File	What it covers
getting-started.md	Open the tool, tour the screen, generate and export a first system.
set-the-brief.md	Industry, style, contrast target, density, seed, harmony, colour mode, corners, type ratio, ramp shape and token naming.
read-the-system.md	The Overview, Colour, Type, Icons and Scales tabs.
component-playground.md	The live component gallery and its preview controls.
contrast-audit.md	The audit rail, the score, the full audit view and suggested fixes.
compare.md	Running the same brief through every style or every industry.
export-tokens.md	The Tokens tab, the twelve export formats and what each contains.
share-save-and-connect.md	Share links, saving to the Library, the Workspace view and sending the palette to other tools.
reference.md	Every control, option, default and output.
troubleshooting.md	Symptoms, causes and fixes.
faq.md	Short answers to common questions.

Related

- Design System Generator overview
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Getting started with the Design System Generator

Open the Design System Generator, learn the layout, and generate, check and export your first design system.

This guide takes about ten minutes. By the end you will have a design system that meets a contrast target, and its tokens in a file.

Open the tool

1. Go to https://auricartisan.com/tool/design-engine/ in a current desktop browser. The tool works on smaller screens, but the three-column layout is easiest on a wide one.
2. A system appears straight away. Which one depends on how you arrived:
   - from a shared link, the link's settings are applied first;
   - otherwise, on a return visit, the settings you used last in this browser;
   - otherwise the defaults: **Auto** industry, **Modern** style, **AAA** contrast.

A tour of the screen

The masthead

The top of the page reads *Three choices, one whole system.* with a short description, the number of tokens, **Targets 12** and *OKLCH · WCAG 2.1 · APCA*. On the right are five buttons and a link:

Control	What it does
Surprise	Picks a random industry, style and contrast target, resets the seed to auto, and sets harmony to auto or a random harmony.
Share	Copies a link that rebuilds this system.
Save	Saves the system to your Library.
Help	Opens **How this works**: what the defaults assume and how the score is worked out.
Workspace	Opens your saved systems and the options for sending the palette to other tools.
Read the API reference →	Opens the public API documentation.

Under the masthead a short note shows what the same job costs through the REST API (**Automate this — 4 API credits per call to POST /v1/tokens/generate**). You can dismiss it with its ×.

The tab row

Seven tabs: **Overview**, **Colour**, **Type**, **Icons**, **Scales**, **Components** and **Tokens**. The **Tokens** tab shows the token count. At the right end, **Expand** hides the two side rails so the tab you are reading gets the full width; select it again (**Collapse**) to bring them back. The page remembers this choice.

The three columns

- **The brief** (left): every input. Changing any of them regenerates the system.
- **The stage** (middle): the tab you selected.
- **The audit** (right): the score, where the points went, and the twelve contrast pairings. It is always visible.

Below the columns a note reads *Generated, not designed.*

Your first system

1. In **The brief**, open **Industry** and choose **Education**.
2. Open **Style** and choose **Playful**.
3. Under **Contrast target**, select **AA**.
4. Look at **The audit**. The large number is the score out of 100. Under it, a line says how it changed (*+3 on that change*) and a pill says whether all twelve pairings pass. A line below says how many tokens moved.
5. Open the **Colour** tab. Read the ramps, the semantic colours and the **Surfaces & roles** card. Select any colour to copy it.
6. Open **Components**. Try the buttons, tabs and switches, and switch **Theme** to **Dark**.
7. Open **Tokens**. Scroll to **Export tokens**, select the **CSS** pill, then select **Download**. The file `tokens.css` downloads.

Result: a complete token file for an education product with a playful style, whose text colours meet WCAG AA.

Next steps

- Anchor the system to your own brand colour: Set the brief.
- Understand the score and the pairings: Check contrast with the audit.
- Choose the right export for your stack: Export tokens.
- Keep the system: Share, save and connect.

Related

- Design System Generator overview
- Reference

WEBSITE › TOOLS › COLOUR WORKSPACE

Compare styles and industries

Run your brief through every style or every industry at once, sort the results, and adopt the one you prefer.

Because a system is generated almost instantly, the tool can build many at once. The **Compare** view keeps your brief and varies one thing, the style or the industry, so you can see every option side by side.

Open Compare

Select **Compare every style** → at the bottom of **The brief**. ← **Back to the system** returns to the tab you were on.

The heading says what is being compared, for example *Healthcare at AAA, through all ten styles.*, and the range of scores.

The controls

Control	Options	What it does
Vary	**Style**, **Industry**	Varies the style (ten cards) or the industry (one card per industry setting, including Auto).
Order by	**Score**, **Chroma**, **Listed**	Sorts by score, by how colourful the primary is, or in the list order.

Compare uses your contrast target, density and other overrides, but ignores your seed colour so that each style or industry shows its own hue.

Each card

- The name, **on screen** if it matches your current system, and the score (colour-coded).
- The primary ramp, and chips for primary, accent, success, warning and danger.
- The heading font drawn in its weight, with the body font, type ratio and icon family.
- A sample **Continue** button and a tile showing the corner and shadow.
- A footer with the corner style and harmony, **audit clean** or *N failing* (light and dark together), and **Use**.

When sorted by score, the highest-scoring card is highlighted.

Adopt a result

Select **Use** on a card. Your brief switches to that style or industry, the system regenerates, and you return to your tab with a message such as **Using …**.

Things to know

- The score measures contrast headroom, surviving chroma and text hierarchy, not taste; the page says as much under the cards.
- Compare is built only while the view is open, so it does not slow the rest of the tool.

Related

- Set the brief
- Check contrast with the audit

WEBSITE › TOOLS › COLOUR WORKSPACE

Use the component playground

Try your generated tokens on a live, clickable component gallery, and change the preview's theme, density, corners, elevation, button shape and viewport.

The **Components** tab draws a complete interface gallery with your generated tokens, and it works: tabs switch, menus open, switches toggle, modals and drawers open and close, toasts appear, steppers count and ratings set. It is the fastest way to notice that muted text is too light, that the primary is too close to the info colour, or that the corners feel wrong.

What is in the gallery

Choose a category with the chips above the stage. ✦ **All** shows every category in this order:

Category	What it contains
Navigation	A navigation bar, **Tabs**, **Breadcrumbs**, **Pagination**, **Steps / wizard**
Buttons	**Variants**, **Sizes & icon buttons**, **Loading, block & FAB**
Forms	**Text & select**, **Choice & toggles**, **Slider, stepper & rating**, **Tags, upload & OTP**
Feedback	**Alerts**, **Toast, banner & tooltip**, **Progress & loading**, **Validation & empty**
Data	**Stat tiles**, **Data table**, **List & avatars**, **Timeline**
Overlays	**Modal & drawer**, **Dropdown menu**, **Command palette**, **Consent & popover**
Commerce	**Product card**, **Pricing**, **Cart & checkout**
Marketing	A hero, **Feature grid**, **Testimonial**, **Newsletter & FAQ** and a closing section
Industry	An **Industry showcase**: a small branded app screen written for one industry

The Industry showcase

The showcase is a short app screen (app name, navigation, a headline figure, a call to action, data rows and a trust chip) written for one of the fourteen industries. A row of chips lets you switch the scene independently of the industry you

generated with, so you can check that a fintech palette still reads on a wellness screen. It starts on your industry, or on SaaS when the industry is **Auto**.

The preview controls

The bar above the chips changes how the preview looks without changing the generated system. These settings are not saved, shared or exported.

Control	Options	Effect
Theme	**Light**, **Dark**	Draws the preview with the light or dark roles. Only the themes your colour mode includes are offered.
Density	**Compact**, **Cozy**, **Spacious**	Padding, gaps, font size and control height inside the preview. Separate from the brief's Density.
Corners	**Auto**, **Sharp**, **Soft**, **Round**, **Pill**	Tries another corner family without regenerating.
Elevation	**None**, **Flat**, **Auto**, **Deep**	Removes, flattens, keeps or deepens the shadows. **Deep** moves every surface up one shadow level.
Buttons	**Auto**, **Square**, **Pill**	Button corner radius only.
Viewport	**Fluid**, Desktop, Tablet, Mobile	Frames the gallery at a fixed width (the desktop, tablet and mobile buttons are icons with tooltips), and the narrow frames stack the gallery in one column.
↺ **Reset**	—	Returns density, corners, elevation, buttons and viewport to their defaults. It keeps your theme and category.

Tasks

Check the system in dark mode

1. Make sure **Colour mode** in the brief is **Both** or **Dark**.
2. On **Components**, set **Theme** to **Dark**.
3. Look at **Feedback** and **Data**: muted text, borders and alert backgrounds are where dark themes usually struggle.

Result: the gallery drawn with the dark roles. To see the numbers, open the full audit and choose **Dark**; see Check contrast with the audit.

Decide on a corner style

1. Change **Corners** between **Sharp**, **Soft**, **Round** and **Pill** and watch the cards, inputs and buttons.
2. When you like one, set **Corner style** in the brief to the matching option so it reaches the tokens.

Result: a corner family chosen by eye, then made real in the system.

Check a phone layout

Set **Viewport** to Mobile. The gallery is framed at phone width and stacks into one column.

Things to know

- Changing any input in the brief rebuilds this tab from scratch. An open modal, a typed value or a toggled switch is reset.
- Avatars, product images and figures in the gallery are placeholders.

- Switches respond to `Enter` and `Space` as well as clicks. Clicking outside an open menu closes it; clicking a modal's backdrop closes the modal.

Related

- Read the system
- Set the brief
- Reference

WEBSITE › TOOLS › COLOUR WORKSPACE

Check contrast with the audit

Read the always-visible audit rail, understand the score, open the full audit for both themes, and use the suggested fixes.

The audit is the evidence that a generated system meets its contrast target. It re-runs on every change, it covers both the light and the dark roles, and it reports two measures:

- the **WCAG 2.1 contrast ratio**, from 1:1 to 21:1, the measure most accessibility requirements use;
- the **APCA Lc** value, the Accessible Perceptual Contrast Algorithm proposed for WCAG 3, which handles light-on-dark and mid-tone pairs better. The tool shows its size (without a sign) as context, not as a pass or fail gate.

The audit rail

The right-hand column, **The audit** (*live*), has four sections.

1. **The score**: a large number out of 100, *out of 100*, a line saying how the last change moved it (*first run*, *no change* or, for example, *+3 on that change*), and a pill: **all 12 pass** or **N failing**. Select the score to open the full audit. A line underneath says how many tokens changed on the last change, for example *14 of 177 tokens moved on that change*.
2. **Where the points went**: the four parts of the score as bars, and a sentence naming the weakest part.
3. **Light** (*wcag · apca*): the twelve pairings for the light roles. Each row shows a sample, the pairing, the kind of content and its requirement (*normal text · needs 7*), the ratio (green for pass, red for fail) and the Lc value.
4. **Dark**: *clean* or *N failing* for the dark roles, and **See the whole audit** →.

The twelve pairings

Each pairing is held to the requirement that fits its content, not one number for everything.

Pairing	Requirement	Why
Body text (on `bg`)	Your target	Normal text
Muted text (on `bg`)	4.5:1	Still body copy, held to AA
Subtle text (on `bg`)	3:1	Large or secondary text
Text on surface	Your target	Normal text on a card
Text on muted bg	Your target	Normal text on a wash
On-primary label	4.5:1	Button label
Primary vs bg	3:1	A user-interface element
Strong border (on `bg`)	1.4:1	Decorative; it only has to be seen

Pairing	Requirement	Why
Success solid	3:1	Labelled component
Warning solid	3:1	Labelled component
Danger solid	3:1	Labelled component
Info solid	3:1	Labelled component

For the semantic solids the pairing is the solid colour with its automatic black or white label. Translucent tokens (the dark theme's borders, for example) are blended over the surface beneath them before they are measured, and the full audit marks them **composited**.

The score

The score is a weighted total, kept between 40 and 100:

Part	Up to	What it rewards
Accessibility	55	The share of the twelve light pairings that pass.
Vibrancy	18	How much colour (OKLCH chroma) the primary step 500 kept after gamut mapping.
Headroom	15	How far body text clears your target.
Hierarchy	12	Full marks when body, muted and subtle text are three distinct tiers; 5 otherwise.

The rail colours the score green from 85, amber from 70, red below. The Overview ring uses green from 85 and amber from 65.

Things to know about the score:

- It is computed from the light roles. The dark theme is audited, but it does not change the score.
- Loosening the contrast target rarely raises it: lighter text tiers are more likely to collide, and the headroom shrinks. For a higher score, try a more saturated style or a seed with more chroma.
- It measures contrast, colour and hierarchy, not taste. Two very different systems can score the same.

The full audit

Select the score, or **See the whole audit** →, to open **The audit** view. ← **Back to the system** returns to the tab you were on.

- **Theme audited**: **Light** or **Dark**, with a pill for each theme's result.
- **The engine's own requirements**: the twelve pairings for the chosen theme, each with ratio, Lc, requirement and a **PASS** or **CHECK** badge.
- **What would fix it** (only when something fails): for each failing pairing, the nearest step in the generated ramp that would clear the requirement, with its token name and new ratio. If nothing in the ramp can reach it, it says that the seed hue or the requirement has to move.
- **Every foreground on every surface**: a matrix of `text`, `text-muted`, `text-subtle`, `primary`, `border-strong` and the four semantic solids against `bg`, `bg-subtle`, `bg-muted`, `surface` and `elevated`. Gold meets your body-text target, green is 4.5:1 and above, amber is 3:1 and above (large text and UI only), red is below 3:1.
- **Color-vision preview**: primary, secondary, accent, success, warning and danger drawn four times: **Normal**, **Deuteranopia**, **Protanopia** and **Tritanopia**. Look for colours that stay distinct in the first row but merge in another.

Tasks

Get to all twelve passing

1. Look for red ratios in the rail, or open the full audit.
2. Read **What would fix it**. Often the fastest fixes are: raise or lower the **Contrast target** by one step, choose a less saturated **Style**, or try another **Seed**.
3. Watch the pill turn to **all 12 pass**.

Result: a system whose defined pairings all meet their requirements.

Check the dark theme

1. Open the full audit and select **Dark** under **Theme audited**.
2. Read the pairings and the matrix.

Result: the same checks for the dark roles.

What the audit does not cover

- Your layout: text over images, disabled states, focus rings on coloured buttons and hover states.
- Font size and weight, which change what a ratio means in practice.
- Other accessibility requirements such as focus order, target size, motion preferences and semantics.

For those, see the Accessibility and vision tools.

Related

- Set the brief
- Component playground
- Limits and accuracy

WEBSITE › TOOLS › COLOUR WORKSPACE

Export tokens

Browse and search every token, then copy or download the system in any of twelve formats, and know what each format contains.

A **design token** is a named design decision stored as data, such as `--color-primary-500: #1f6feb` or `--space-4: 16px`. Tokens let designers and developers share one source of truth. The **Tokens** tab lists every token the system generates and exports them in twelve formats.

The token list

- **Find token** filters the list by name or value as you type, for example `primary`, `radius` or `#1f`. **Clear** shows everything again.
- **Design tokens** shows the total and the tokens in groups, each with a count: Color · Primary, Color · Secondary, Color · Accent, Color · Neutral, Color · Semantic, Color · Roles (light), Color · Roles (dark), Color · Gradients, Typography, Typography · Size, Typography · Leading, Typography · Tracking, Spacing, Radius, Shadow, Motion, Z-index and Breakpoints.

- Colour tokens have a chip. Select any token to copy its value.

The default light-and-dark system has 177 tokens; with **Colour mode** set to **Light**, the fourteen dark role tokens are left out (163).

The names follow **Token naming** in the brief; see Set the brief.

Export

Scroll to **Export tokens**. Select a format pill; the code below it changes, and its bar shows the file name and language. Three buttons:

- **Copy** copies the code shown.
- **Download** saves it under the format's file name.
- **Download all (.zip-less bundle)** saves every format, one after another with a comment line between them, in one text file, `design-system-bundle.txt`. It is not a zip archive; split it by hand or download formats one at a time.

Pill	File	What you get
CSS	`tokens.css`	A `:root` block with every token, and a dark block that applies under `[data-theme="dark"]` or `.dark`.
CSS (modern)	`tokens.modern.css`	Colour roles paired with `light-dark()` under `color-scheme: light dark`, and every colour restated in `oklch()` inside a `@media (color-gamut: p3)` block for wide-gamut screens.
SCSS	`_tokens.scss`	One `$variable` per token and a `$tokens` map.
Tailwind v3	`tailwind.config.js`	A `theme.extend` block: colour ramps, semantic solids, font families, font sizes with line heights, spacing, radius and shadows.
Tailwind v4	`theme.css`	An `@theme` block.
Figma / W3C	`design-tokens.json`	The W3C Design Tokens format for Figma Variables or Tokens Studio.
JSON	`design-system.json`	The whole system with its settings. The most complete export.
TypeScript	`tokens.ts`	The JSON as a typed constant with a `Tokens` type.
Android XML	`colors.xml`	Colour and dimension resources.
SwiftUI	`DesignTokens.swift`	A `Color` extension and dimension constants.
Flutter	`app_tokens.dart`	A class of `Color` and number constants.
Compose	`Tokens.kt`	Jetpack Compose Kotlin values.

The CSS, SCSS, Tailwind and TypeScript files start with a comment naming the industry, style, target, seed and generation time; the JSON export carries the same details as data, and the Figma / W3C file has a one-line description.

What each format covers

Not every format carries every token. Check before you choose one.

Content	CSS, CSS (modern), Tailwind v4	SCSS	Tailwind v3	Figma / W3C	JSON, TypeScript	Android, SwiftUI, Flutter, Compose
Colour ramps and semantic colours	Yes	Yes	Ramps and semantic solids	Ramps and semantic solids	Ramps and semantic solids	Yes

Content	CSS, CSS (modern), Tailwind v4	SCSS	Tailwind v3	Figma / W3C	JSON, TypeScript	Android, SwiftUI, Flutter, Compose
Light roles	Yes	Yes	No	No	Yes	Yes
Dark roles	CSS and CSS (modern) only	No	No	No	Yes	Yes
Gradients	Yes	Yes	No	No	Yes	No
Font families	Yes	Yes	Yes	Names only	Names only	No
Font sizes, spacing, radius	Yes	Yes	Yes	Yes	Yes	Yes
Shadows	Yes	Yes	Yes	Yes	Yes	No
Leading and tracking	Yes	Yes	Line heights with sizes	No	Yes	No
Motion, z-index	Yes	Yes	No	No	Yes	No
Breakpoints	Yes	Yes	No	No	Yes	Yes

Notes:

- **Native formats** (Android, SwiftUI, Flutter, Compose) keep only plain HEX colours and pixel dimensions (spacing, radius, font sizes and breakpoints). Translucent colours, such as the focus ring and the dark theme's borders, are left out; define them by hand on those platforms.
- The roles `primary-active` and `primary-subtle` appear on the **Colour** tab and in the JSON and TypeScript exports, but not in the other formats.
- If you set a token prefix or a case other than kebab, the native formats may leave out the dimension tokens; export them with the default naming if you need them.
- For a production system, pick one complete format (CSS or JSON) as your source of truth and generate the rest from it, rather than mixing formats with different coverage.

Example

The start of a **CSS** export:

```
/* Auric Artisan · Design System */
/* Education · Playful · AA · seed #… */
/* Generated 2026-09-24T10:00:00.000Z */
:root {
  --color-primary-50: #…;
  --color-primary-100: #…;
  /* … */
}

/* Dark theme — apply [data-theme="dark"] or .dark on a wrapper */
[data-theme="dark"], .dark {
  --color-bg: #…;
  /* … */
}
```

Related

- Set the brief for token naming

- Share, save and connect
- Reference

WEBSITE › TOOLS › COLOUR WORKSPACE

Read the system

What the Overview, Colour, Type, Icons and Scales tabs show, and how to copy any value from them.

The first five tabs show the generated system from different angles. Every tab is made of cards; select a card's heading to fold or unfold it. Some cards start folded. Almost every colour, gradient and token can be selected to copy its value, and a short message confirms the copy.

Overview

Four cards:

- **System summary**: a score ring (green at 85 and above, amber from 65, red below), the industry and style, the mood and target, the harmony, and badges for the seed, **All pairings pass** or **N to review**, the font pairing and the icon family. Under it, eight headline swatches: primary, secondary, accent, success, warning, danger, info and text. The primary swatch also shows its contrast against white.
- **At a glance**: **Color ramps** (4 × 11), **Type sizes**, **Base size**, **Scale ratio**, **Spacing unit**, **Radius**, **Shadows** (6 levels) and **Total tokens**.
- **Generation recipe**: every input written out, including the resolved primary hue and the secondary and accent hue angles, the colour mode, the font pairing and the icon family with its stroke width. Use it to explain how a system was made.
- **Jump to**: shortcuts to **Color**, **Type**, **Components** and **Export**.

Colour

- **Color ramps**: primary, secondary and accent at eleven steps (50, 100, 200 … 900, 950), and a neutral ramp at thirteen steps (0, 50, 100 … 950, 1000). The card's note lists the three hues. Steps sit on even perceptual lightness in OKLCH, with chroma highest in the middle steps and lower at the ends. Neutral step 0 is pure white; the other neutral steps carry a trace of the brand hue.
- **Semantic colors**: success, warning, danger and info at fixed hues (green, amber, red and blue), each shown as a solid with a label and four small chips for the background wash, border, text and solid colours. Select a chip to copy it.
- **Surfaces & roles**: **Light surfaces & roles** and, unless the colour mode is Light, **Dark surfaces & roles**. These are the tokens you build with: `bg`, `bg-subtle`, `bg-muted`, `surface`, `elevated`, `border`, `border-strong`, `text`, `text-muted`, `text-subtle`, `primary`, `primary-hover`, `primary-active`, `primary-subtle`, `on-primary` and `ring`.
- **Gradients** (folded): five gradients named **brand**, **brand-soft**, **sunrise**, **mesh** and **subtle**. Select one to copy its CSS.

How roles are chosen:

- Light text tiers are the lightest neutral steps that still meet their requirement: your target for `text`, 4.5:1 for `text-muted` and 3:1 for `text-subtle`. This gives three visibly different tiers rather than three near-blacks.
- The primary fill is the first of steps 600, 700, 500, 800, 400, 900, 300, 950 and 200 that reaches 3:1 against the background and takes a black or white label at your target (at least 4.5:1).
- The dark set is built on neutral step 950, a tinted near-black. Its text colour is chosen against the lightest dark surface, so it passes on cards as well as on the page. Dark borders are translucent white.

Colours that would fall outside sRGB have their chroma reduced until they fit, so a very vivid seed can come back calmer: what you see is what a screen can show.

Type

- **Font pairing**: specimens of the heading, body and mono families, the pairing's note, *Served from fonts.auricartisan.com*, and **Browse in Font Library ↗**, which opens the Font Library searching for the heading family.
- **Type scale**: thirteen sizes named `2xs`, `xs`, `sm`, `base`, `md`, `lg`, `xl`, `2xl`, `3xl`, `4xl`, `5xl`, `6xl` and `7xl`, largest first, each with pixels, rem and a specimen. The note shows the ratio and base size.
- **Roles**: twelve text styles (**display**, **h1** to **h5**, **body-lg**, **body**, **small**, **caption**, **overline** and **code**), each drawn at its size, weight, line height and letter spacing, with the family it uses.
- **Available weights** (folded): the weights of the heading family.

The pairing is chosen from sixteen curated pairings by how many tags it shares with your industry and style. The families include Inter, Space Grotesk, Sora, Manrope, Plus Jakarta Sans, Outfit, DM Serif Display, Fraunces, Playfair Display with Source Sans 3, Cormorant Garamond with Jost, Bricolage Grotesque, Lexend, Epilogue, Archivo Expanded with Archivo, Spline Sans, and IBM Plex Sans with IBM Plex Mono; the mono family is JetBrains Mono unless the pairing has its own.

Each size also has its own line height, tighter for large sizes; the Tailwind v3 export carries it with each font size.

Icons

- **Recommended icon family**: **Recommended set**, **Why**, **Stroke width**, **Grid size** (24 px), **Corner join**, **Suggested sizes** (16 · 20 · 24 · 32 px), **Source** (icons.auricartisan.com) and a button to browse the set in the Icon Library.
- **Preview set**: twenty common icons at the recommended stroke.
- **On brand color** (folded): eight icons in the primary colour.
- **Size ramp** (folded): one icon at 16, 20, 24, 32 and 40 pixels.

The family is one of seven: **Feather**, **Lucide**, **Heroicons**, **Phosphor**, **Tabler**, **Remix Icon** and **Material Symbols**, chosen by tag overlap. The stroke is made slightly thinner for Minimal and Elegant and slightly thicker for Bold and Brutalist. When the Auric icon CDN is reachable, the previews show the real family; otherwise they show built-in line icons of the same weight.

Scales

Four columns in one tab:

- **Spacing scale**: nineteen steps (`0`, `px`, `0.5`, `1`, `1.5`, `2`, `2.5`, `3`, `4`, `5`, `6`, `8`, `10`, `12`, `16`, `20`, `24`, `32`, `40`), each drawn as a bar. `0` is 0 px and `px` is 1 px; every other step is its number × the base unit × the density multiplier.
- **Border radius scale**: `none`, `sm`, `md`, `lg`, `xl`, `2xl`, `3xl` and `pill`, each drawn as a box, with the corner style named.
- **Elevation / shadow scale**: six levels (`xs` to `2xl`) plus `primary` and `inner`, drawn on a surface, with the brand-tinted shadow colour. **Shadow values** (folded) lists the CSS for each. The **Hard** preset (Brutalist) gives offset shadows with no blur; the **Glow** preset (Futuristic) adds a brand-coloured glow to the two largest levels.
- **Durations** (fast, base, slow) and **Easing curves** (standard, spring, linear): hover a row to replay the movement.

Z-index (`base` 0, `dropdown` 1000, `sticky` 1100, `banner` 1200, `overlay` 1300, `modal` 1400, `popover` 1500, `toast` 1600, `tooltip` 1700) and breakpoints (`sm` 640, `md` 768, `lg` 1024, `xl` 1280, `2xl` 1536 px) have no tab; they are in the Tokens list and the exports.

Copying values

- Select a ramp cell, swatch, gradient or token to copy its value. The item flashes and a message such as **Copied #1f6feb** appears.
- Changing any input rebuilds every tab, so folded or unfolded cards return to their starting state.

Related

- Set the brief
- Export tokens
- Reference

WEBSITE › TOOLS › COLOUR WORKSPACE

Set the brief

What every input in the brief rail does: industry, style, contrast target, density, seed, harmony, colour mode, corner style, type ratio, ramp shape and token naming.

The brief is the left-hand column. It holds every input the system is built from. Each section shows a short note on the right that says what the choice resolved to, for example *auto → analogous* or *4.5:1 body text*. Any change regenerates the whole system immediately, and the audit shows what the change did.

The first three inputs (industry, style and contrast target) decide most of the system. The rest are overrides.

Industry

Industry sets the brand hue and how much colour (chroma) the system uses, plus a mood and a set of tags used to pick the font pairing and icon family. A coloured dot beside the list shows the industry's hue.

Industry	Hue	Mood
Auto (default)	250°	No vertical assumed; your seed decides.
Fintech	258°	trustworthy, precise, secure
Healthcare	192°	calm, clean, reassuring
SaaS / B2B	264°	efficient, modern, focused
E-commerce	22°	energetic, inviting, urgent
Education	234°	approachable, clear, friendly
Creative / Agency	320°	expressive, bold, original
Gaming	282°	high-energy, immersive, sharp
Luxury	44°	refined, exclusive, timeless
Food & Drink	18°	warm, appetizing, organic
Travel	200°	open, adventurous, fresh
Real Estate	168°	stable, premium, grounded
Non-profit	148°	hopeful, human, honest
Crypto / Web3	270°	futuristic, bold, technical
Wellness	138°	serene, natural, balanced

Auto gives a balanced hue and a neutral tag set, so the font and icon choice is not nudged towards any vertical. When you set a seed colour, the seed decides the hue and chroma whichever industry you choose; the industry then only influences the font and icon choice.

Style

Style multiplies the industry's saturation and sets the shape and movement of the system.

Style	Saturation	Corners	Shadows	Heading weight	Type ratio	Motion
Modern (default)	× 1.00	Soft	Balanced	700	1.25	Smooth
Minimal	× 0.62	Sharp	Subtle	600	1.2	Subtle
Classic	× 0.82	Soft	Balanced	700	1.25	Smooth
Bold	× 1.28	Rounded	Deep	800	1.333	Snappy
Playful	× 1.22	Rounded	Deep	800	1.333	Bouncy
Elegant	× 0.72	Soft	Subtle	600	1.414	Smooth
Corporate	× 0.88	Soft	Balanced	700	1.2	Subtle
Brutalist	× 1.15	Sharp	Hard	900	1.5	Snappy
Organic	× 0.90	Rounded	Soft	700	1.25	Smooth
Futuristic	× 1.20	Soft	Glow	700	1.333	Snappy

The note beside **Style** shows the corner, shadow and motion character it applies.

Contrast target

The minimum contrast that text colours are chosen to meet. It changes which colour steps become text and fills; it is not a check applied afterwards.

Option	Body text must reach	Effect
AA Large	3:1	Lightest text tiers, most brand colour kept.
AA	4.5:1	The usual production choice.
AAA (default)	7:1	Darkest text; safest, least colourful.

Whatever you choose, the primary fill is always picked so that a black or white button label reaches at least 4.5:1 on it.

Density

Option	Body size	Spacing unit	Spacing multiplier
Compact	15 px	4 px	× 0.9
Comfort (default)	16 px	4 px	× 1.0
Spacious	17 px	5 px	× 1.15

Seed

The seed is the brand colour the whole palette is built around.

- The colour chip shows the current seed.

- The colour well and the HEX box set a seed. The well regenerates as you pick; the HEX box applies when you press `Enter` or leave it.
- **Auto** returns to a seed derived from the industry (the HEX box then reads *auto*).
- Twelve hue buttons under the seed set a seed at 0°, 30°, 60° and so on round the hue circle, at the current system's chroma.

The note reads *from the industry* or *overridden*. A seed's chroma is kept within a workable range, so a near-grey or an extremely vivid colour is adjusted to something a ramp can be built from.

Harmony

Where the secondary and accent hues sit relative to the primary.

Option	Secondary	Accent
Auto (default)	Chosen by the style (see below)	
Complementary	+ 180°	+ 150°
Analogous	+ 32°	− 32°
Triadic	+ 120°	+ 240°
Split-complementary	+ 150°	+ 210°
Tetradic	+ 90°	+ 180°
Monochrome	same hue, less chroma	same hue

With **Auto**: Minimal uses monochrome; Bold and Brutalist use complementary; Playful uses triadic; Futuristic uses split-complementary; Modern, Classic, Elegant, Corporate and Organic use analogous. The note shows the result, for example *auto → analogous*; a chosen harmony shows *locked*.

Colour mode

Light, **Dark** or **Both** (default). With **Both**, dark roles are generated alongside the light ones (*two blocks*). With **Light**, the dark role tokens are left out of the tokens and exports and the playground offers only a light theme; with **Dark**, the playground offers only a dark theme.

Corner style

Auto (default, from the style), **Sharp**, **Soft**, **Rounded** or **Pill**. It replaces the whole radius family:

Corner style	sm	md	lg	xl	2xl	3xl
Sharp	1	2	3	4	6	8
Soft	3	6	8	12	16	22
Rounded	6	10	14	20	28	36
Pill	8	14	22	32	full	full

Values are pixels. Every family also has **none** (0) and **pill** (fully round).

Type ratio

The ratio between neighbouring font sizes. **Auto** (default) uses the style's ratio. The options are **1.125 · minor second**, **1.2 · minor third**, **1.25 · major third**, **1.333 · perfect fourth**, **1.414 · augmented fourth**, **1.5 · perfect fifth** and **1.618 · golden ratio**.

Ramp shape

Two rows of three buttons that reshape the primary, secondary and accent ramps. The semantic colours (success, warning, danger, info) ignore them on purpose.

- **Span**: **Tight**, **Even** (default) or **Wide**: how far each ramp travels from light to dark.
- **Peak**: **Flat**, **Natural** (default) or **Vivid**: whether chroma stays level across the ramp or bulges in the middle steps.

Token naming

Renames every token in every export, so the file matches your codebase.

- **Prefix**: free text, for example `brand`, which turns `--color-primary-500` into `--brand-color-primary-500`. It applies after a short pause in typing.
- **Case**: **kebab** (default), **camel**, **snake** or **Pascal**.
- A line under the controls shows two real token names in your chosen style, for example `--brandColorPrimary500 · --brandSpace4`.

Compare every style

The button at the bottom of the brief, **Compare every style** →, opens the Compare view. See Compare styles and industries.

Tasks

Build a system around your brand colour

1. Type your brand HEX code in the **Seed** box and press `Enter`.
2. Choose the **Industry** that best fits your product, so the font and icon choice suits it.
3. Choose a **Style** and a **Contrast target**.
4. If the brand colour must appear as the primary button, check the **Colour** tab's **Surfaces & roles** card: the primary fill is the ramp step that passes contrast, which may be lighter or darker than your seed.

Result: ramps, roles and every scale built around your colour.

Match your codebase's naming

1. Type your prefix under **Token naming**, for example `acme`.
2. Choose the case your code uses.
3. Check the example line, then export.

Result: exported tokens that need no renaming.

Related

- Read the system
- Check contrast with the audit
- Reference

WEBSITE › TOOLS › COLOUR WORKSPACE

Share, save and connect

Share a link that rebuilds a system, save systems to your Library and restore them, and send the palette, fonts and icons to other Auric Artisan tools.

The generator is deterministic: the same inputs always produce the same system. That makes a system easy to share and keep, because only its inputs need to travel.

Share a link

1. Select **Share** in the masthead.
2. The link is copied, **Shareable link copied** ✓ appears, and the address bar updates to the same link.

The link lists your inputs as address parameters. Settings left on **Auto** are left out. For example:

```
https://auricartisan.com/tool/design-engine/?
industry=wellness&rampSpan=default&rampPeak=default&tokenPrefix=&tokenCase=kebab&style=elegant&a11y=AA&mode=both&density=spacious
```

When someone opens the link, the page applies these parameters before generating, so they see the same system. You can also write a link by hand with only the parameters you need, for example `?industry=wellness&style=elegant&a11y=AA`.

Parameter	Values
`industry`	`fintech`, `healthcare`, `saas`, `ecommerce`, `education`, `creative`, `gaming`, `luxury`, `food`, `travel`, `realestate`, `nonprofit`, `crypto`, `wellness`
`style`	`modern`, `minimal`, `classic`, `bold`, `playful`, `elegant`, `corporate`, `brutalist`, `organic`, `futuristic`
`a11y`	`AA-large`, `AA`, `AAA`
`mode`	`light`, `dark`, `both`
`density`	`compact`, `comfortable`, `spacious`
`harmony`	`complementary`, `analogous`, `triadic`, `split`, `tetradic`, `monochrome`
`ratio`	`1.125`, `1.2`, `1.25`, `1.333`, `1.414`, `1.5`, `1.618`
`corner`	`sharp`, `soft`, `rounded`, `pill`
`seedHex`	A HEX colour, for example `%231f6feb` or `1f6feb`

The link also lists the ramp shape and token naming (`rampSpan`, `rampPeak`, `tokenPrefix`, `tokenCase`), but opening a link does not apply them: those settings open at their defaults (**Even**, **Natural**, no prefix, **kebab**). Tell the recipient if you changed them.

Your last settings

The page remembers the inputs you used last in this browser and restores them on your next visit, unless you arrive through a shared link. Like shared links, this restores the industry, style, target, colour mode, density, harmony, type ratio, corner style and seed, but not the ramp shape or token naming. The **Expand** choice is also remembered.

Save to your Library

Two buttons save the current system to your Auric Artisan Library:

- **Save** in the masthead (message: **Saved to workspace** ⭐);
- the site's floating save button, labelled **Save design system** on this page, which opens the Library's save window so you can name the item and file it.

A saved system stores its name (industry and style), a colour strip, a summary with the target, fonts and score, and its full set of inputs. If the Library is not available on the page, a message says so instead.

The Workspace view

Select **Workspace** in the masthead. ← **Back to the system** returns you to your tab. The view has two parts.

Saved systems

Each saved system shows its colour strip, name and summary, with three buttons:

- **Restore** applies the saved inputs, regenerates and returns to **Overview** (**Restored from workspace** ✓). As with shared links, the ramp shape and token naming are not re-applied.
- **Open** shows the item in your Library.
- **Delete** removes it (**Removed from workspace**).

With nothing saved it shows **No saved systems yet**. If the Library did not load, it shows **Workspace unavailable** with an **Open workspace** button.

Connect

Four cards send the system to other tools:

- **Send the brand color into a tool**: **Harmony Studio**, **Gradient Library**, **Accessibility Lab**, **Contrast System**, **Color Spaces** and **Color Psychology**. Each opens that Colour Tool as a floating panel on this page, seeded with your primary colour. See the Colour Tools.
- **Open a full generator with this palette**: a strip of the twelve-colour palette, and links to nine tools that receive your seed and palette in the address: **Palette Generator**, **Harmony Generator**, **Gradient Generator**, **Shade Generator**, **Accessibility Generator**, **Personalization Gen**, **Color Science Lab**, **Vision Simulation** and **URL Analyzer**.
- **Libraries & workspace**: **Font Library** (opens the Font Library searching for your heading family), **Icon Library** (opens the Icon Library on your recommended set), **Save to Workspace** and **Open Workspace** (opens your Library).
- **Copy palette for anywhere** (folded): **Copy palette (HEX)** (comma-separated), **Copy as JSON array** and **Copy as URL slug** (codes without `#`, joined by hyphens).

> **Note:** At the time of writing, the Palette, Harmony, Gradient, Shade and Accessibility Generator links lead to pages that are not available on the site. Their collections are in the library area instead; see Colour libraries.

The twelve-colour palette is: primary, secondary 500, accent 500, neutral 700, the four semantic solids, primary 200, primary 700, secondary 300 and accent 700.

Related

- Export tokens
- Library Kit
- Font Library and Icon Library
- Privacy

WEBSITE › TOOLS › COLOUR WORKSPACE

Design System Generator reference

Every control, option, default, view and output of the Design System Generator in one place.

Page

Item	Value
Address	https://auricartisan.com/tool/design-engine/
Tabs	Overview, Colour, Type, Icons, Scales, Components, Tokens
Views	Compare, Workspace, Audit, Help
Tokens	177 (light + dark), 163 (light only)
Export targets	12
Contrast measures	WCAG 2.1 ratio, APCA Lc
API equivalent	POST /v1/tokens/generate, 4 API credits per call

Masthead

Control	What it does
Surprise	Random industry, style and target; seed reset to auto; harmony auto or random. Keeps other overrides.
Share	Copies a link with the current inputs and updates the address bar.
Save	Saves the system to your Library.
Help	Opens the **How this works** view.
Workspace	Opens saved systems and the Connect cards.
Read the API reference →	Opens the API documentation.
API note ×	Hides the "Automate this" note; remembered in your browser.
Expand / **Collapse**	Hides or shows both side rails; remembered in your browser.

The brief

Control	What it does	Values	Default
Industry	Brand hue, chroma, mood and font/icon tags	Auto, Fintech, Healthcare, SaaS / B2B, E-commerce, Education, Creative / Agency, Gaming, Luxury, Food & Drink, Travel, Real Estate, Non-profit, Crypto / Web3, Wellness	Auto
Style	Saturation, corners, shadows, weights, ratio, motion	Modern, Minimal, Classic, Bold, Playful, Elegant, Corporate, Brutalist, Organic, Futuristic	Modern
Contrast target	Body-text requirement	AA Large (3:1), AA (4.5:1), AAA (7:1)	AAA
Density	Body size and spacing	Compact (15 px, 4 px × 0.9), Comfort (16 px, 4 px), Spacious (17 px, 5 px × 1.15)	Comfort
Seed colour well and HEX box	Brand colour	Any HEX; **Auto**	Auto (from the industry)
Seed hue buttons	Set a seed at a fixed hue	0° to 330° in 30° steps	—
Harmony	Secondary and accent hue positions	Auto, Complementary, Analogous, Triadic, Split-complementary, Tetradic, Monochrome	Auto
Colour mode	Which role sets are built into the tokens	Light, Dark, Both	Both
Corner style	Radius family	Auto, Sharp, Soft, Rounded, Pill	Auto
Type ratio	Modular scale ratio	Auto, 1.125, 1.2, 1.25, 1.333, 1.414, 1.5, 1.618	Auto
Ramp shape span	Lightness travel of brand ramps	Tight, Even, Wide	Even
Ramp shape peak	Chroma curve of brand ramps	Flat, Natural, Vivid	Natural
Token naming prefix	Prefix for every token	Free text	Empty
Token naming case	Name style	kebab, camel, snake, Pascal	kebab
Compare every style →	Opens Compare	—	—

Generated system

Part	Contents
Ramps	Primary, secondary, accent: 50, 100, 200, 300, 400, 500, 600, 700, 800, 900, 950. Neutral: 0, 50, 100 … 950, 1000.
Semantic	Success, warning, danger, info: solid, bg, border, text, and a label colour.
Roles	bg, bg-subtle, bg-muted, surface, elevated, border, border-strong, text, text-muted, text-subtle, primary, primary-hover, primary-active, primary-subtle, on-primary, ring (light, and dark unless mode is Light).
Gradients	brand, brand-soft, sunrise, mesh, subtle.
Type	Heading, body and mono families; 13 sizes (2xs to 7xl); 12 roles; leading (none, tight, snug, normal, relaxed, loose); tracking (tighter, tight, normal, wide, wider, widest).
Icons	One of Feather, Lucide, Heroicons, Phosphor, Tabler, Remix Icon, Material Symbols, with stroke width.

Part	Contents
Spacing	19 steps: 0, px, 0.5, 1, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 12, 16, 20, 24, 32, 40.
Radius	none, sm, md, lg, xl, 2xl, 3xl, pill.
Shadows	xs, sm, md, lg, xl, 2xl, inner, primary.
Motion	Durations fast, base, slow; easings standard, spring.
Z-index	base 0, dropdown 1000, sticky 1100, banner 1200, overlay 1300, modal 1400, popover 1500, toast 1600, tooltip 1700.
Breakpoints	sm 640, md 768, lg 1024, xl 1280, 2xl 1536 (px).

Motion by style personality:

Personality	fast / base / slow (ms)	Used by
Subtle	120 / 200 / 320	Minimal, Corporate
Smooth	150 / 250 / 400	Modern, Classic, Elegant, Organic
Snappy	110 / 180 / 280	Bold, Brutalist, Futuristic
Bouncy	160 / 280 / 460	Playful

Tabs

Tab	Cards
Overview	System summary, At a glance, Generation recipe, Jump to
Colour	Color ramps, Semantic colors, Surfaces & roles, Gradients (folded)
Type	Font pairing, Type scale, Roles, Available weights (folded)
Icons	Recommended icon family, Preview set, On brand color (folded), Size ramp (folded)
Scales	Spacing scale, Border radius scale, Elevation / shadow scale, Shadow values (folded), Durations, Easing curves
Components	Component playground
Tokens	Find token, Design tokens, Export tokens

Component playground controls

Control	Values	Default
Theme	Light, Dark (as allowed by colour mode)	Light
Density	Compact, Cozy, Spacious	Cozy
Corners	Auto, Sharp, Soft, Round, Pill	Auto
Elevation	None, Flat, Auto, Deep	Auto
Buttons	Auto, Square, Pill	Auto
Viewport	Fluid, Desktop, Tablet, Mobile	Fluid
↺ **Reset**	Resets density, corners, elevation, buttons, viewport	—
Category chips	✦ All, Navigation, Buttons, Forms, Feedback, Data, Overlays, Commerce, Marketing, Industry	✦ All

Control	Values	Default
Scene chips (Industry)	The 14 industries	Your industry, or SaaS

Audit

Item	Value
Pairings	12 per theme (8 roles + 4 semantic solids)
Requirements	Target (body, text on surface, text on muted), 4.5 (muted text, on-primary), 3 (subtle text, primary vs bg, semantic solids), 1.4 (strong border)
Score	Accessibility 55, Vibrancy 18, Headroom 15, Hierarchy 12; range 40–100
Rail score colours	Green 85+, amber 70+, red below
Overview ring colours	Green 85+, amber 65+, red below
Full audit	Theme audited (Light, Dark), requirements, What would fix it, Every foreground on every surface, Color-vision preview
Matrix colours	Gold: meets target; green: 4.5+; amber: 3+; red: below 3
APCA rating labels	Lc 90 fine print, Lc 75 body text, Lc 60 fluent text, Lc 45 large/bold, Lc 30 headlines, Lc 15 non-text, insufficient

Compare

Control	Values	Default
Vary	Style, Industry	Style
Order by	Score, Chroma, Listed	Score
Use	Adopts that style or industry	—

Export formats

Pill	File	Language
CSS	tokens.css	css
CSS (modern)	tokens.modern.css	css
SCSS	_tokens.scss	scss
Tailwind v3	tailwind.config.js	javascript
Tailwind v4	theme.css	css
Figma / W3C	design-tokens.json	json
JSON	design-system.json	json
TypeScript	tokens.ts	typescript
Android XML	colors.xml	xml
SwiftUI	DesignTokens.swift	swift
Flutter	app_tokens.dart	dart

Pill	File	Language
Compose	Tokens.kt	kotlin

Buttons: **Copy**, **Download**, **Download all (.zip-less bundle)** (`design-system-bundle.txt`). Coverage per format: see Export tokens.

Share link parameters

`industry`, `style`, `a11y`, `mode`, `density`, `harmony`, `ratio`, `corner` and `seedHex` are applied when a link is opened. `rampSpan`, `rampPeak`, `tokenPrefix` and `tokenCase` are written into shared links but not applied. See Share, save and connect.

Messages

Message	When
Copied …	A value was copied.
Shareable link copied ✓	After **Share**.
Saved to workspace ⭐	After **Save**.
Workspace library isn't available on this page	Save could not reach the Library.
Restored from workspace ✓	After **Restore**.
Couldn't read that saved system	A saved item had no inputs.
Removed from workspace	After **Delete**.
Export copied ✓, **Downloaded …**, **Downloaded full bundle ✓**	After export actions.
Using …	After **Use** in Compare.
Generation failed — see console	Generation hit an error.

Related

- Getting started
- Glossary

WEBSITE › TOOLS › COLOUR WORKSPACE

Troubleshooting the Design System Generator

Symptoms you may meet in the Design System Generator, why they happen, and what to do.

Generation and inputs

My open modal, typed text or folded cards reset when I changed an input

- **Why:** every change regenerates the whole system and rebuilds every tab.
- **What to do:** settle the brief first, then explore the tabs and the playground.

My seed colour looks duller than the colour I entered

- **Why:** the seed's chroma is kept within a workable range, and ramp steps outside sRGB have their chroma reduced until they fit.
- **What to do:** this is expected. For more colour, try a more saturated style (Bold, Playful, Futuristic) or a **Vivid** ramp peak.

My brand colour is not the primary button colour

- **Why:** the primary fill is the ramp step that is visible on the background (3:1) and takes a readable label at your target. Your exact seed may not meet that.
- **What to do:** keep the generated fill for accessibility, or lower the **Contrast target** if your requirements allow.

Changing Industry does nothing to the colours

- **Why:** when a seed is set, the seed decides hue and chroma; the industry then only affects fonts and icons.
- **What to do:** select **Auto** beside the seed to go back to the industry's hue.

Generation failed — see console

- **Why:** an unexpected error while generating.
- **What to do:** reload the page. If a shared link caused it, check its parameter values against Share, save and connect.

Audit

Some pairings say CHECK or the pill says N failing

- **Why:** a role colour does not meet its requirement.
- **What to do:** open the full audit and read **What would fix it**. Try the next contrast target, a less saturated style, or a different seed.

The score went down when I relaxed the contrast target

- **Why:** lighter text tiers can collide (losing hierarchy points) and headroom shrinks.
- **What to do:** keep the stricter target and raise the score through a more saturated style or seed instead.

The dark theme is not reflected in the score

- **Why:** the score uses the light roles. The dark roles are audited separately.
- **What to do:** open the full audit and choose **Dark**.

Fonts and icons

The type specimens look like system fonts

- **Why:** the fonts load from the Auric font CDN, with Google Fonts as a fallback. If both are blocked (by an extension, a firewall or no connection), the page uses system fonts. The names, sizes and weights shown are still correct.
- **What to do:** allow fonts.auricartisan.com and fonts.googleapis.com, or check your connection, then reload.

The icons look generic, not like the recommended family

- **Why:** the real icons come from the Auric icon CDN. If it cannot be reached, the tool shows built-in line icons of the same stroke weight.

- **What to do:** allow icons.auricartisan.com and reload.

Share, save and export

A shared link opens with different ramp shape or token names

- **Why:** shared links, saved systems and your last visit do not apply the ramp shape or token naming.
- **What to do:** set **Ramp shape** and **Token naming** again after opening the link, or share the exported file.

Save shows "Workspace library isn't available on this page"

- **Why:** the Library did not load on the page.
- **What to do:** reload. If it persists, use **Share** or an export to keep your system.

A generator link in Connect shows a missing page

- **Why:** the Palette, Harmony, Gradient, Shade and Accessibility Generator pages are not available on the site at the time of writing.
- **What to do:** use the collections in the library area instead: Colour libraries.

Download all gives a .txt file, not a zip

- **Why:** the bundle is one text file with every format separated by comment lines.
- **What to do:** download the formats you need one at a time, or split the bundle by hand.

My Tailwind v3 or Figma export has no background or text roles

- **Why:** those formats do not carry role tokens.
- **What to do:** use CSS, Tailwind v4, JSON or a native export for roles. See the coverage table in Export tokens.

Native exports are missing spacing and radius after I set a token prefix

- **Why:** with a prefix or a non-kebab case, the native exporters may not recognise dimension tokens.
- **What to do:** export native formats with no prefix and kebab case, then rename in your project.

Copy does nothing

- **Why:** the browser blocked clipboard access.
- **What to do:** select **Download** instead, or select the code in the export block and copy it by hand.

The "Automate this" API note has gone from every tool

- **Why:** dismissing it with × hides it on all tool pages in this browser.
- **What to do:** clear the site's data for auricartisan.com in your browser to see it again.

Related

- FAQ
- Limits and accuracy
- Help centre

WEBSITE › TOOLS › COLOUR WORKSPACE

Design System Generator FAQ

Short answers to common questions about the Design System Generator.

Is it the same as "Design Engine"?

Yes. The page's title is the Design System Generator; its address and some labels use Design Engine.

Do I need an account?

No, to generate, share and export. Saving systems uses your Auric Artisan Library; see the Library Kit.

Is anything generated on a server?

No. The system is calculated in your browser. The page downloads fonts and icons for the previews; see Privacy. The same job is also available through the public REST API, which is billed in API credits.

Will the same inputs always give the same system?

Yes. That is why a share link reproduces a system. The only exceptions are the ramp shape and token naming, which links and saved systems do not carry; see Share, save and connect.

Where is the Generate button?

There is none. Every change regenerates the system at once.

Why is there no dark-theme score?

The score is calculated from the light roles. The dark roles are fully audited in the audit view under **Dark**.

Which export should I use?

CSS or JSON if you want everything. Tailwind v4 for Tailwind projects with roles. Figma / W3C to import into Figma Variables or Tokens Studio (colours, type, spacing, radius and shadow only). The native formats for Android, iOS, Flutter or Compose (colours and dimensions). See Export tokens.

Can I use my own font or icon set?

The tool recommends from its own list. After export, replace the font families and icon set in your project as you like.

Are the fonts free to use?

The recommended families are served through the Auric font CDN and Google Fonts. Check each family's licence in the Font Library before using it in a product.

Does passing the audit make my product accessible?

No. It shows that the defined colour pairings meet their contrast requirements. Layout, font size, focus states and many other requirements are up to you.

Related

- Troubleshooting
- Getting started

WEBSITE › TOOLS › COLOUR WORKSPACE

Supporting material

Glossary, limits, related tools and privacy notes for the Design System Generator.

Reference material that supports the Design System Generator guides. The tool has no keyboard shortcuts of its own beyond the standard ones, so there is no shortcuts page.

File	What it covers
glossary.md	Design-system, colour and contrast terms used in these docs.
limits-and-accuracy.md	What the generator and audit calculate, and what they do not.
related.md	Related tools, services and Learn articles.
privacy.md	What stays in your browser and what uses the network.

Related

- Design System Generator overview
- Documentation contents

WEBSITE › TOOLS › COLOUR WORKSPACE

Glossary

Plain-language definitions of the terms used in the Design System Generator documentation.

Term	Meaning
APCA / Lc	Accessible Perceptual Contrast Algorithm, a contrast method proposed for WCAG 3; Lc is its lightness-contrast value, shown here without its sign.
Brief	The set of inputs (industry, style, target and overrides) a system is generated from.
Chroma	How colourful a colour is; in OKLCH, the C value.
Colour mode	Whether the system builds light roles, dark roles or both.
Contrast target	The minimum WCAG contrast body text must reach: 3:1, 4.5:1 or 7:1.
Density	How compact the type and spacing are.
Design system	A shared set of design decisions (colours, type, spacing and more) and the components built from them.
Design token	One named design decision stored as data, such as `--space-4: 16px`.
Deterministic	Always producing the same output from the same inputs.
Gamut mapping	Reducing a colour's chroma until it can be shown in sRGB.
Harmony	The positions of the secondary and accent hues relative to the primary.
Headroom	How far body text's contrast clears the target.
Hierarchy	Body, muted and subtle text reading as three distinct levels.
Modular scale	Font sizes that grow by a fixed ratio, such as 1.25.
Neutral ramp	A ramp of greys, here tinted slightly towards the brand hue.
OKLCH	A perceptual colour space with lightness (L), chroma (C) and hue (H), where equal steps look roughly equal.

Term	Meaning
Ramp	A series of steps of one hue from light to dark, such as 50 to 950.
Ramp shape	How far a ramp travels in lightness (span) and where its chroma peaks (peak).
Role	A token named for its job, such as `text-muted` or `surface`, rather than its colour.
Seed	The brand colour a system's hue and chroma come from.
Semantic colour	A colour with a fixed meaning: success, warning, danger or info.
Token naming	The prefix and case (kebab, camel, snake, Pascal) applied to every token name.
W3C Design Tokens	A standard JSON format for tokens, readable by Figma Variables and Tokens Studio.
WCAG contrast ratio	The luminance ratio between two colours, from 1:1 to 21:1, used by WCAG 2.x.

Related

- Supporting material
- Basic Color Tools glossary for general colour terms

WEBSITE › TOOLS › COLOUR WORKSPACE

Limits and accuracy

What the Design System Generator calculates, how reliable the audit and score are, and where the output stops.

What is calculated

- Colours are built in OKLCH and converted to sRGB HEX codes. Steps outside sRGB have their chroma reduced by search until they fit, so no channel is clipped.
- Contrast ratios use the WCAG 2.x relative-luminance formula. Translucent tokens are blended over the surface beneath them before measuring.
- APCA values come from a compact implementation of the APCA formula. APCA is still a draft for WCAG 3, and its values are shown as context, not as a gate.
- Colour-vision previews use simple matrix simulations of deuteranopia, protanopia and tritanopia.

What the audit covers

- Twelve pairings per theme: eight role pairings and the four semantic solids with their labels, each against the requirement for its content.
- A matrix of every foreground role against every surface role.

It does not cover your layout, font sizes and weights in use, text over images, focus rings on coloured backgrounds, hover and disabled states, or any accessibility requirement beyond colour contrast.

What the score means

The score (40 to 100) blends accessibility (55), vibrancy (18), headroom (15) and hierarchy (12), from the light roles only. It is a heuristic for comparing systems, not a quality grade or a compliance result.

What the generator does not do

- It does not know your brand, your audience or your competitors. The page's own note: *Generated, not designed.*
- Font pairings and icon families are chosen from fixed lists by tag matching.
- Token names are structural (for example `primary-600`, `space-4`). Rename them to your product's vocabulary if you prefer.
- Preview settings in the component playground are never saved, shared or exported.
- Export formats differ in coverage; see Export tokens.
- Shared links, saved systems and your last visit do not restore the ramp shape or token naming.

Previews depend on the network

Font specimens need the Auric font CDN or Google Fonts, and the real icon families need the Auric icon CDN. Without them the previews use system fonts and built-in icons. The tokens are unaffected.

Related

- Check contrast with the audit
- Privacy

WEBSITE › TOOLS › COLOUR WORKSPACE

Privacy

What the Design System Generator keeps in your browser, what it saves, and which services it contacts.

Stays in your browser

- The whole system is generated in the page. Your inputs are not sent to a server to generate it.
- Copies go to your clipboard only when you select a copy control.
- Downloads are created in the browser.

Remembered in your browser

Stored in your browser's local storage on this device:

- your last set of inputs, used to restore the system on your next visit;
- whether the side rails are expanded;
- whether you dismissed the API note.

Uses the network

What	Where	Why
Fonts for the type previews	fonts.auricartisan.com (Auric font CDN) and fonts.googleapis.com (fallback)	To draw the specimens in the recommended families. The requests name the font families.
Icons for the previews	icons.auricartisan.com (Auric icon CDN)	To show the recommended icon family.
Saving	Your Auric Artisan Library	When you select **Save** or the save button.

What	Where	Why
Share links	Your clipboard and address bar	The link contains your inputs; anyone you send it to can see them.

Opening another tool from **Connect** passes your seed colour and palette to that tool in the address, and keeps a copy for that tool in your browser's session storage.

For the CDNs as services, see the Font CDN and Icon CDN. Site-wide analytics and advertising follow your cookie choices; see the Help centre.

Related

- Limits and accuracy
- Share, save and connect

WEBSITE › TOOLS › COLOUR WORKSPACE

Related tools and reading

Tools, services and Learn articles that go with the Design System Generator.

Tools

Tool	Why
Personalization Generator	Another way to generate a personalised palette and tokens from a brief.
Basic Color Tools	Palettes, gradients and conversions, and the Colour Tools the Connect view opens.
Font Library	Inspect, compare and embed the recommended font families, and check their licences.
Icon Library	Browse and export the recommended icon family.
Color Science Lab	Deeper colour analysis of your palette.
Accessibility and vision tools	Contrast Checker, Vision Simulation and the URL Analyzer for checks beyond tokens.

Services

- Font CDN: the service that serves the fonts.
- Icon CDN: the service that serves the icons.
- Public API: generate design tokens from your own code.

Library and kits

- Library Kit: where saved systems live.
- Colour libraries: palette, shade, gradient and harmony collections.

Learn articles

- Oklab and Oklch: Modern Perceptual Color Spaces
- Accessible Color Design and WCAG Contrast

- Color Theory and Harmony
- Color in Typography and Text: Readable, Accessible, Beautiful
- Color Blindness and Color Vision Deficiency

Related

- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Font Library

What the Font Library is, who it is for, where to find it, and how to find, test, compare and take away open-licence fonts from the Auric Artisan catalogue.

The Font Library is the browsing and testing page for the Auric Artisan font catalogue: 2,213 families and 4,883 font files, 550 of them variable fonts, covering 60 scripts and Unicode blocks. Every card is set in its own typeface, so you judge fonts by looking at them. Type your own words and every card shows them.

Open a family and a panel docks beside the grid with eleven tabs: a specimen you can type into and style, variable-axis sliders and OpenType switches, computed suitability scores, suggested pairings, the full glyph set, every style and file, the licence and its permissions, the character coverage, developer exports (font stacks, Tailwind, SCSS, tokens and a modular type scale) and ready-to-paste embed code.

Star the families you like and they gather in **Your kit**, kept in your browser. From there you can compare up to four side by side, copy one combined embed snippet, or download a self-hosting kit: a ZIP with the font files, a stylesheet, a demo page and the licence. A **Recommend** panel suggests families for an industry, a brand style or even a colour palette.

The fonts are served by the Auric Artisan Font CDN, so previews and embeds make no request to any third party. The CDN itself (hosts, loader script, self-hosting and licences) is documented under Font CDN.

Who it is for

- Web designers and developers choosing type for a site or product.
- Brand designers looking for a typeface and a partner for it.
- Anyone who needs fonts with a clear open licence to self-host.
- People checking whether a font covers a language or has a feature such as tabular figures.

Where to find it

- Page: https://auricartisan.com/tool/font/
- No sign-in is needed. Your selection is kept in your browser.

Quick start

1. Open https://auricartisan.com/tool/font/.
2. Type a sentence in *Type to preview every family…* above the grid.
3. On the left, pick a category such as **Serif**, or type a name in *Family, designer or foundry*.
4. Select a card. The family opens in the panel on the right, on its **Specimen** tab.
5. Try the **Suitability** and **Pairings** tabs.
6. Select + **Select** to add the family to **Your kit**.
7. Open **Use & Embed** and select **Copy**, or select **Download kit** for a self-hosting ZIP.

What you can do

- Search, sort and filter 2,213 families by category, variable or italic, weights, OpenType features, scripts, licence and source.
- Browse six curated collections, or ask the recommender for a shortlist.
- Test a family with your own text, sizes, weights, colours, variable axes and OpenType features.

- Read computed scores for accessibility, readability, UI, code and editorial use, and see suggested pairings.
- Copy any glyph, check character coverage, and read the licence and its permissions.
- Copy a font stack, Tailwind, SCSS, design-token or type-scale code, and embed code for one family or a whole selection.
- Compare two to four families side by side, and download self-hosting kits.

In this folder

- docs/ — how to use the Font Library.
  - Getting started
  - Browse, search and filter
  - Collections and the recommender
  - The specimen panel
  - Family details
  - Develop and embed
  - Your kit, compare and download
  - Reference
  - Troubleshooting
  - FAQ
- others/ — supporting material.
  - Glossary
  - Shortcuts
  - Limits and accuracy
  - Related
  - Privacy

WEBSITE › TOOLS › COLOUR WORKSPACE

Font Library documentation

The list of guides for the Font Library.

File	What it covers
getting-started.md	Open the page, learn the layout, find and keep a first family.
browse-and-filter.md	Preview text and size, search, sort, categories, facets, applied filters, grid and list.
collections-and-recommender.md	The six collections, the recommender and recently viewed families.
specimen-panel.md	The docked panel, and the Specimen tab's editor, axes and OpenType switches.
family-details.md	Suitability, Pairings, Glyphs, Features, Styles, About, License and Charset tabs.
develop-and-embed.md	Font stacks, code exports, the type scale, and embed code.
kit-compare-and-download.md	Selecting families, Your kit, the Selected view, Compare, combined embed and download kits.
reference.md	Every control, option and message.
troubleshooting.md	Symptoms, causes and fixes.
faq.md	Short answers to common questions.

Related

- Font Library overview
- Supporting material
- Font CDN

WEBSITE › TOOLS › COLOUR WORKSPACE

Getting started with the Font Library

Open the Font Library, learn its layout, and find, test and keep your first family.

Open the page

Go to https://auricartisan.com/tool/font/. The masthead reads *Two thousand faces, set in their own.* Its labels give the live counts of families, font files, variable fonts and scripts.

The catalogue loads from the Font CDN. If it cannot be reached, a notice explains that the library is temporarily unavailable; see Troubleshooting.

The layout

Three views sit in a row at the top: **Specimens** (with the number of families shown), **Collections** (6) and **Selected** (the number you have starred).

The **Specimens** view has two columns:

- **The rail** on the left, which stays in view as you scroll:
  - **Find**: the search box, the sort menu, **Filters**, **Recommend**, and the number of matching families;
  - **Applied**: what is switched on, when anything is;

  - **Category**: **All**, **Serif**, **Sans Serif**, **Handwriting**, **Monospace** and **Display**, with counts;
  - **Narrow it**: six groups of filter chips;
  - at the foot, **Reset all filters** and, once you have starred something, **Download selected**.
- **The stage** on the right: the preview text box with **Shuffle**, **Grid** and **List**, the **Size** slider, and the grid of cards, which loads more as you scroll.

Once you have opened a family, a **Recently viewed** strip appears under the three views.

On screens narrower than about 1,180 pixels the rail sits above the grid, and its filter groups start folded.

A card

Each card shows the family name, the designer and the number of styles, a star to select it, a specimen in the family's own face, and badges: the category, ⚡ **Variable** if it is a variable font, the computed accessibility score, the licence, and **Installed** for families that are listed but not served (see Limits and accuracy).

Select a card, or focus it and press `Enter`, to open the family.

Your first family

1. In *Type to preview every family…*, type the headline you want to set, for example *Harbour Street Bakery*.
2. Under **Category**, select **Serif**.
3. Open **Filters** if the chips are hidden, and under **Properties** select ⚡ **Variable**.
4. Select a card you like. The panel opens on the right on the **Specimen** tab, with your text.
5. Drag the **Weight** axis slider, then open **Pairings** to see body fonts that suit it.
6. Select **+ Select** in the panel. The **Your kit** button appears in the bottom-right corner with a count of 1.
7. Open **Use & Embed** and select **Copy** to copy a stylesheet link.

Result: a chosen family, a suggested partner, and the code to use it on a web page.

Next steps

- Browse, search and filter
- The specimen panel
- Your kit, compare and download

Related

- Font Library overview
- Shortcuts

WEBSITE › TOOLS › COLOUR WORKSPACE

Browse, search and filter

Preview your own text, set the specimen size, search, sort, pick a category, narrow with filter chips, and switch between grid and list.

Everything here happens on the **Specimens** view. The grid shows 48 families at first and loads more as you scroll.

Preview your own text

Type in *Type to preview every family…* above the grid. Every card shows your text in its own face. Clear the box to go back to each family's name.

Your text also starts the specimen in the family panel and appears in the **Styles** tab and in **Compare**.

Specimen size

The **Size** row sets how large the card specimens are. Drag the slider, use **−** and **+** (2 pixels a step), or select **Reset to 40**. The slider snaps near 16, 24, 48, 72 and 96.

Cards keep their specimens between 22 and 48 pixels, so settings below 22 or above 48 make no further difference to the grid. For a bigger view, open the family: the specimen editor goes up to 400 pixels.

Grid or list

Grid shows cards in columns; **List** shows one family per row, which suits long preview text.

Search

Type in *Family, designer or foundry*. Results update as you type. The search looks for what you type anywhere in the family name, the designer's name or the category (for example *mono* or *serif*), ignoring capitals. Foundry names are not searched, despite the placeholder.

Sort

Option	Order
Alphabetical	A to Z (default)
Accessibility score	Intended to put the highest accessibility score first; see Troubleshooting
UI score	Highest UI score first
Readability score	Intended to put the highest readability score first; see Troubleshooting
Most weights	Families with the most weights first
Most glyphs	Families with the most characters first
Variable first	Variable fonts first, then A to Z
Surprise me	Random order

Shuffle, beside the preview box, switches the sort to **Surprise me** and deals a new random order each time.

Category

The **Category** chips pick one category at a time: **All** (2,213), **Serif** (1,172), **Sans Serif** (780), **Handwriting** (97), **Monospace** (83) or **Display** (81). Categories are detected automatically from the font files; see Limits and accuracy.

Narrow it: the filter chips

Filters shows or hides the chip groups. Select a chip to turn it on; select it again to turn it off.

Group	Chips	How several chips combine
Properties	**Variable**, **Italic**, **Monospace**	A family must match all
Weights available	100 to 900 (hover for the name, such as SemiBold)	A family must have every weight chosen
Features	Ligatures, Small Caps, Oldstyle Figures, Tabular Figures, Fractions, Slashed Zero, Stylistic Sets, Swashes	A family must have every feature chosen
Scripts	The 18 most common scripts and Unicode blocks, such as Basic Latin, Cyrillic or Greek	A family must cover every one chosen
Licence	Open Font License, Unspecified, Apache 2.0, Creative Commons, MIT, Custom, with counts	Any of those chosen
Source	**Library** (served fonts) and **System** (listed, not served)	Either

Tips:

- **Filter these options** at the top of the chip groups finds a chip by name across all groups, for example *tabular* or *Cyrillic*. It shows **No option matches that.** if nothing fits.
- Groups with more than ten chips show **+ n More**; select it to see the rest, and **Less** to fold them again.
- Select a group's heading to fold it; a number on the heading says how many of its chips are on. The page remembers which groups you folded.
- Chips can be reached with `Tab` and switched with `Enter` or `Space`.

See and clear what is applied

When anything is switched on, **Applied** appears under the search box with a chip for each filter, the search and the category. Select **×** on a chip to remove just that one, or **Clear all** to remove everything.

Reset all filters at the foot of the rail also clears every chip, the category and the search.

The number of matching families appears beside the search tools and on the **Specimens** view button. If nothing matches, the grid reads **No fonts match these filters.**

Related

- Collections and the recommender
- The specimen panel
- Shortcuts

WEBSITE › TOOLS › COLOUR WORKSPACE

Collections and the recommender

The six curated collections, the Find your font recommender, and the Recently viewed strip.

Three shortcuts past the alphabet: curated collections, a recommender that ranks families for a brief, and a strip of families you opened recently.

Collections

Open the **Collections** view. Under **Featured collections** (*curated starting points, ranked by our computed scores*) there are six rows of up to 14 families each:

Collection	Description on the page
Popular Fonts	Widely-used, battle-tested families
New Fonts	Most recently published in the collection
Best UI Fonts	Interface & product type — clean, high-legibility sans
Best Accessibility Fonts	Highest legibility & character distinction
Best Developer Fonts	Monospaced, distinct, code-ready
Best Editorial Fonts	For headlines & long-form reading

Each mini card shows the family in its own face with its accessibility score. Select one to open the family.

The collections are hidden while a search, a filter or the recommender is active on the **Specimens** view. If the **Collections** view looks empty, clear those first; see Troubleshooting.

Find your font: the recommender

Select **Recommend** in the rail (or choose **Find your font** from the command palette) to open ✨ **Find your font** above the grid: *Describe the brand or drop a colour palette — we rank the catalogue by computed fit.*

1. **Industry**: choose one of 13, or **Any / not sure**. The options are SaaS / Software, Developer / DevTools, Startup / Tech, E-commerce / Retail, Luxury, Fashion / Beauty, Editorial / Publishing, Finance / Legal, Healthcare, Education / Kids, Food / Hospitality, Corporate / B2B and Creative / Agency.
2. **Brand style**: select any of **Modern**, **Classic**, **Elegant**, **Playful**, **Bold**, **Minimal**, **Technical** and **Friendly**.
3. **Accessibility priority**: **Essential**, **Balanced** (default) or **Any**. The higher the priority, the more the accessibility score counts.
4. **Colour palette** (optional): paste HEX colours such as `#0F172A #F8FAFC #3B82F6`. Swatches appear as you type.
5. Select **Recommend fonts**.

The results are twelve ranked cards, each set in its own face, with a short reason (the family's two strongest style traits, and its accessibility score if it is 80 or more) and a **match** score. Select a card to open the family.

A note beside the buttons explains the ranking. With a palette it reads, for example, *Palette reads cool, high-contrast → leaning modern / minimal.*; without one, *12 matches ranked by fit.*

Reset clears your choices; ✕ closes the panel.

How it ranks

- Each industry, brand style and palette mood adds to a target style fingerprint across eight traits: modern, classic, elegant, playful, technical, friendly, bold and minimal.
- A family scores on how closely its own fingerprint matches, how strong it is in the role the industry needs (UI, code, editorial, readability or accessibility), and its accessibility score, weighted by your priority.
- Families outside the categories that suit the industry are marked down.
- The palette is read for warmth, saturation, lightness and contrast. A cool, high-contrast palette leans modern and minimal; warm, vivid colours lean playful and friendly; dark, muted colours lean elegant and classic.

The recommender ranks within a shortlist of about 130 well-known, versatile families, not the whole catalogue, so its suggestions are families you can rely on. The match score shows how well a family fits your brief compared with a perfect fit, from 40 to 99.

Recently viewed

Every family you open is added to **Recently viewed**, which appears under the three views; it keeps the last twelve. Selecting a name types it into the search box, so the family's card comes up in the grid. **Clear** empties the list.

Related

- Browse, search and filter
- Family details
- Limits and accuracy

WEBSITE › TOOLS › COLOUR WORKSPACE

Develop and embed

Copy a font stack, CSS variables, Tailwind, SCSS, design tokens or a modular type scale from the Develop tab, and embed code from the Use & Embed tab.

Two tabs in the family panel give you code. **Develop** is for your stylesheets and design system; **Use & Embed** is for loading the font on a page. For how the Font CDN serves these files, and for self-hosting, see the Font CDN documentation.

Develop

Font-family stack

A `font-family` line with the family first and safe fallbacks for its category, for example for a serif:

```
font-family: 'Lora', ui-serif, Georgia, Cambria, 'Times New Roman', Times, serif;
```

Copy copies it; the message reads **Copied stack**.

Export

Five formats, each with **Copy**:

Format	What you get
CSS variables	A `:root` variable such as `--font-serif` and a `body` rule using it
Tailwind	A `tailwind.config.js` fragment that adds the stack to `fontFamily`
SCSS	A `$font-…` variable and a `body` rule
Design tokens	A JSON token such as `font.family.serif` with type `fontFamily`
CSS @import	An `@import` of the family's CDN stylesheet and the variable

The variable and token names use **mono** for monospace families, **serif** for serif families and **sans** for everything else.

Modular type scale

A scale of nine sizes, **xs**, **sm**, **base**, **md**, **lg**, **xl**, **2xl**, **3xl** and **4xl**, each shown in the family with its size in pixels and rem.

Control	Options	Default
Base	10 to 24 px	16
Ratio	1.125 Major Second, 1.200 Minor Third, 1.250 Major Third, 1.333 Perfect Fourth, 1.414 Augmented Fourth, 1.500 Perfect Fifth, 1.618 Golden Ratio	1.200 Minor Third

base is the base size; each step up multiplies by the ratio and each step down divides by it. **Copy CSS** copies a `:root` block of `--text-xs` to `--text-4xl` in rem (16 pixels = 1 rem). The preview caps very large steps at 60 pixels so they fit.

Use & Embed

Snippets

Six tabs of code for this family, with **Copy**:

Tab	What it is
Link	A `<link>` to the family's stylesheet on the Font CDN
Import	The same stylesheet as a CSS `@import`
Script	The Font CDN loader script with a `data-families` attribute naming the family
JS	A call that loads the family from JavaScript and then applies it
@font-face	`@font-face` rules for up to the first four font files
Usage	A CSS rule applying the family, with its weight range for variable fonts

A note under the code says the files are self-hosted, with no third-party request, and names the family's licence.

The **Selected weights** chips above the tabs can be switched on and off, but they do not change the code: every snippet always covers the whole family. To load fewer weights, edit the `@font-face` rules or self-host only the files you need.

Families that are listed but not served

Families marked **Installed** exist only as fonts installed on a computer. Their code refers to the font by its local name, so it works only on devices where that font is installed. See Limits and accuracy.

Embed several families at once

To get one block of code for several families, select them and use **Embed** in **Your kit**. See Your kit, compare and download.

Related

- Font CDN: embed a family
- Font CDN: use the loader
- Family details

WEBSITE › TOOLS › COLOUR WORKSPACE

Family details

The Suitability, Pairings, Glyphs, Features, Styles, About, License and Charset tabs of the family panel.

These eight tabs in the family panel tell you what a family is good for, what goes with it, what it contains and what you may do with it.

Suitability

The page explains: *Scores are computed from this font's own outlines and metrics — x-height, character distinction, proportions, weight range and screen rendering — not editorial opinion.*

Five scores from 0 to 100, sorted best first:

Score	Meaning on the page
Accessibility	Legibility & character distinction for the widest audience
Readability	Comfort for long-form body text
UI / Product	Clarity at small sizes in interfaces
Developer / Code	Monospacing & glyph disambiguation for code
Editorial	Headlines & long-form print character

Why these scores shows the measurements behind them, each 0 to 100:

- **x-height**, with the measured value in em;
- **Character distinction (I l 1 · 0 O)**, noting *distinct zero* where the zero differs from the letter O;
- **Proportion / width**;
- **Weight flexibility**, with the number of weights;
- **Screen rendering**;
- **Low stroke-contrast**.

Best for lists uses the family qualifies for: *UI & product* (UI 78 or more), *Body text* (readability 78 or more), *Code* (developer 75 or more), *Headlines* (editorial 78 or more) and *Accessibility-critical* (accessibility 85 or more). A family that reaches none shows *Display & accents*.

Style fingerprint rates eight traits from 0 to 100: Modern, Classic, Elegant, Playful, Technical, Friendly, Bold and Minimal. The recommender and the pairings use these.

Colours on scores run from low to great. For what the scores can and cannot tell you, see Limits and accuracy.

Pairings

Pairings suggest a partner that contrasts in role but matches in proportion. The page says whether the open family is treated as the **heading** or the **body**: serif and display families, and any family with an editorial score of 72 or more, are headings; others are bodies.

- A preview sets *Designing with type* in the heading face and a paragraph in the body face, each labelled with its family.
- **Best body partners** (or **Best heading partners**) lists eight partners with a score from 60 to 99. A **Super-family** badge marks a partner from the same family of designs, such as a sans and a serif sharing a name.
- Select a partner to preview it. **+ Select both** adds both families to **Your kit**; **Open** *family* opens the partner in the panel.

The score blends how well the two categories pair, how close their x-heights are, how strong the partner is in its role, and its accessibility and UI quality. Partners come from the same shortlist of about 130 well-known families the recommender uses, without handwriting faces.

Glyphs

Every character the family maps, in its own face, with the total at the top (a **+** after it means the count stopped at the limit). Hover a glyph to see its code point, such as *U+0041*.

- Select a glyph to copy the character; the message reads, for example, **Copied U+00E9**.
- **Size** (18 to 80 pixels) sets the glyph size.
- Glyphs appear 600 at a time; **Show more glyphs** adds the next batch. Up to 8,000 are shown.

Features

Each OpenType feature the family supports, shown *default → enabled* on a sample chosen for it, for example fractions on *1/2 3/4 5/8 11/16* or tabular figures on *1111 2222 3333*. Each row shows the feature's name and four-letter tag.

Select a row to copy its CSS, for example `font-feature-settings: "tnum" 1;`. The message reads **Copied "tnum" CSS**.

If the family has none, the tab reads *This family exposes no toggleable OpenType features.*

Styles

- **All weights & styles**: a line of your preview text (or the family name) in every weight, and every italic if the family has them.
- A download card: **Download self-hosting kit**, with the number of font files, and **Download kit**. For a family that is listed but not served, it reads **Installed system family** and the button is disabled.
- **Font files**: a table of every file with **Style**, **Weight**, **Width**, **Format** (or **Variable**), **Size**, **Glyphs**, and a **Download** link for that single file (or an **Installed** badge).

About

The family's description, if it has one, then **Details**: **Designer**, **Foundry / Publisher**, **Category**, **Classification** (serif or sans-serif, detected from the letter shapes), **Version**, **Released / Created**, **Last modified**, **Outline format** (TrueType or PostScript/CFF), **Styles**, **Weights**, **Variable axes**, **Glyph count**, **Embedding** and **Source**. Designer and foundry names link to their websites where the font gives one. **Copyright** and **Trademark** notices follow when present.

Everything here is read from the font files themselves, so it is only as complete as the files.

License

- A badge with the licence name, and buttons for the **License URL** ↗ and the **Original file** the licence came from, when known.
- Six permissions: **Commercial use**, **Modification**, **Redistribution**, **Private / personal use**, **Web embedding** and **Sell font on its own**. Each shows a tick, a cross, or *Not stated in this file* when the licence could not be read for that point. Unknowns are listed last.
- **Copyright**, then **Full license**: the complete licence text, or *No bundled license text was found for this family. Refer to the foundry or the license URL above.*

The permissions summarise the licence for convenience. The full licence text is what applies. See also Font CDN licences.

Charset

- **Supported scripts**, as chips.
- **Character coverage**: **Characters mapped**, **Glyphs in font** and **First / last codepoint**.
- **Unicode ranges (first 40)**.
- **Sample**: a sample sentence in the family's face.

Use this tab to check a language before choosing a family; the **Scripts** filter only offers the 18 most common scripts and blocks.

Related

- The specimen panel
- Develop and embed
- Glossary

WEBSITE › TOOLS › COLOUR WORKSPACE

Your kit, compare and download

Select families into Your kit, see them in the Selected view, compare up to four side by side, copy combined embed code, and download self-hosting kits.

Families you select gather in **Your kit**. The kit is kept in your browser on this device, so it is still there next time, and a second tab of the page sees the same kit.

Select families

Any of these adds a family to your kit:

- the star (☆) on a card, which turns to ★;
- + **Select** in the family panel, which turns to ✓ **Selected**;
- + **Select both** on the **Pairings** tab, which adds the open family and its partner.

Select the star or the button again to remove the family.

The Your kit tray

Once something is selected, a **Your kit** button with a count appears in the bottom-right corner. Select it to open the tray: **Your kit** with the count and *Kept in this browser*, a list of families (each with ✕ to remove it), and four buttons:

Button	What it does	Needs
Embed	Shows combined embed code for the whole kit	1 family
Download	Downloads one ZIP of self-hosting kits	1 family
Compare	Opens a side-by-side comparison	2 families
Clear	Empties the kit	always available

Hide closes the tray.

Download selected at the foot of the rail does the same as **Download**, and shows the count, for example **Download selected (3)**.

The Selected view

The **Selected** view at the top lists your kit as cards with each family's name and category. With nothing selected it reads *Nothing selected yet. Star a family on the Specimens view and it lands here, ready to download as one kit.*

Compare

Select **Compare** in the tray (or press `C` with two or more families selected). The panel shows up to four families, the first four in your kit, side by side:

Row	Shows
Specimen	Your preview text, or *Quick brown fox 0123*
Category, **Designer**, **Weights**	From the catalogue
Variable	The axis tags, or No
x-height	In em
Glyphs	Number of characters
OpenType	Number of features
Accessibility, **UI / Product**, **Readability**, **Developer**, **Editorial**	The five scores, colour-coded
License	The licence name

Select a family's name at the top of its column to open it. With fewer than two families, the message reads **Select at least 2 families to compare**.

Combined embed code

Embed in the tray shows **Your selection**, *n families ready to embed*, with one block of code:

- a `<link>` to each family's stylesheet;
- or, instead, a single loader `<script>` whose `data-families` lists every family, separated by `|`;
- a `<style>` block with a CSS variable for each family, such as `--font-lora`.

Copy all copies the lot; the message reads **Copied combined embed**.

Download a self-hosting kit

Download kit in the family panel (or on the **Styles** tab) downloads one family; **Download** in the tray, or **Download selected**, downloads your whole kit. While it builds, the message reads **Preparing download kit…**, then **Downloaded** and the file name.

Download	File name
One family	*Family-name*-font-kit.zip

Download	File name
Several families	auric-font-selection-kit.zip

Inside, each family has its own folder with:

- `fonts/`: the font files;
- `css/`: a stylesheet of `@font-face` rules pointing at those files, and a CSS variable for the family;
- `demo.html`: a page showing the family, which works straight from the folder;
- `README.txt`: the family, category, designer, licence, file list and three steps to use it;
- `LICENSE.txt`: the licence text, when the catalogue has it.

If a file could not be fetched, a `DOWNLOAD_NOTES.txt` at the top of the ZIP says which. Families marked **Installed** are left out, because they are not served; if every family in the request is one of them, the message reads **Only bundled library fonts can be downloaded**.

The ZIP is assembled in your browser from files on the Font CDN. To self-host, keep the folders together and link the stylesheet from your page. More detail is in Font CDN: self-host a family.

Related

- Develop and embed
- Privacy
- Troubleshooting

WEBSITE › TOOLS › COLOUR WORKSPACE

The specimen panel

How the family panel docks, resizes and closes, and how to use the Specimen tab's editor, variable axes, named instances, OpenType switches and presets.

Selecting a card opens that family in a panel. It is where you test a family and find everything about it.

Where the panel sits

- **On a wide screen** (wider than about 1,180 pixels), the panel docks to the right edge as a column of the page. The page moves aside, nothing is covered, and you can keep selecting cards: the panel simply shows the next family.
  - Drag the grip on the panel's left edge to make it wider or narrower (380 to 720 pixels; 460 at first). Double-click the grip to go back to 460.
  - With the grip focused, `←` and `→` resize it by 16 pixels, or 48 with `Shift`.
  - The width you choose is remembered.
- **On a narrower screen**, the panel is a sheet from the bottom, covering about 62% of the screen so the card you tapped stays visible.
  - Drag the grip up to open it to 94%, or down to close it.
  - With the grip focused, `↑` and `↓` move it in steps.
  - Tapping the dimmed page above closes it.

Close the panel with ✕ or `Esc`. Focus returns to the card you opened it from.

The panel header

- The family name, and a line with its category, *by* designer, number of styles, ϟ number of variable axes (for variable fonts) and number of glyphs.
- ＋ **Select** adds the family to **Your kit**; it then reads ✓ **Selected**, and selecting it again removes the family.
- **Download kit** downloads a self-hosting ZIP for this family. For families that are listed but not served it reads **Installed only** and is disabled. See Your kit, compare and download.

Eleven tabs follow: **Specimen**, **Suitability**, **Pairings**, **Glyphs**, **Features**, **Styles**, **About**, **License**, **Charset**, **Develop** and **Use & Embed**. The Specimen tab is described here; the others are in Family details and Develop and embed.

The Specimen tab

A large editable specimen, set at 64 pixels in the family's regular weight. It starts with your preview text, or *The quick brown fox jumps over the lazy dog*. Click into it and type anything; the placeholder reads *Type here to test the typeface…* when it is empty.

The toolbar

Control	What it does
B, **I**, **U**, **S**	Bold, italic, underline and strikethrough for the selected text.
Align left, centre, right	Aligns the specimen.
Weight menu	The family's weights, such as *400 Regular* or *700 Bold*.
Size	8 to 400 pixels (64 at first).
Line height	0.6 to 3 (1.2 at first).
Letter spacing	−10 to 40 pixels (0 at first).
Text colour	Colours the selected text, or the whole specimen if nothing is selected.
↺	Resets text, size, weight, spacing and axes.

Bold and italic only look right if the family has those styles; otherwise the browser imitates them.

Variable axes

For a variable font, a slider appears for each axis under the specimen, labelled with its name, tag and range, for example *Weight · wght · 100–900*. Moving a slider changes the specimen at once. Choosing a weight from the menu also moves the weight axis.

If the font has named instances (such as *Light* or *Condensed Bold*), they appear as chips; select one to set every axis to that instance.

OpenType features

If the family has features you can switch, an **OpenType features** box lists them, grouped as **Ligatures**, **Letterforms**, **Figures & numbers**, **Spacing**, **Stylistic sets** and **Character variants**, and says how many there are. Select a chip to switch a feature on in the specimen; select it again to switch it off. The **Features** tab shows each one before and after.

Presets

Chips under the specimen replace the text with a standard sample:

Preset	Text
Pangram	The quick brown fox jumps over the lazy dog
Paragraph	A paragraph about typography, for testing body text
Alphabet	a to z
Uppercase	A to Z
Numbers	Digits, currency signs, fractions and other number signs
Symbols	Ampersand, at sign, copyright and other punctuation
Code	A short program with operators, look-alike characters and a hexadecimal number

Code also caps the size at 24 pixels, keeps the line breaks, and switches on contextual alternates and ligatures if the family has them, so you can see coding ligatures such as `=>` and `!=`.

Related

- Family details
- Develop and embed
- Shortcuts

WEBSITE › TOOLS › COLOUR WORKSPACE

Font Library reference

Every view, control, option, file and message in the Font Library in one place.

Address

https://auricartisan.com/tool/font/

Catalogue

Figure	Value
Families	2,213
Font files	4,883
Variable families	550
Families with italics	401
Served (Library)	2,023
Listed only (System, marked Installed)	190
Scripts and Unicode blocks	60
Licences	Open Font License 1,929; Unspecified 191; Apache 2.0 45; Creative Commons 40; MIT 7; Custom 1

Views

View	Contents
Specimens	Rail and grid; the count of matching families
Collections	Six curated rows
Selected	Your kit as cards; the count

Rail

Control	Values	Default
Search	Family name, designer or category	empty
Sort	Alphabetical, Accessibility score, UI score, Readability score, Most weights, Most glyphs, Variable first, Surprise me	Alphabetical
Filters	Shows or hides the chip groups	shown
Recommend	Opens Find your font	closed
Category	All, Serif, Sans Serif, Handwriting, Monospace, Display	All
Filter these options	Text	empty
Properties	Variable, *I* Italic, ⌗ Monospace	off
Weights available	100, 200, 300, 400, 500, 600, 700, 800, 900	off
Features	Ligatures, Small Caps, Oldstyle Figures, Tabular Figures, Fractions, Slashed Zero, Stylistic Sets, Swashes	off
Scripts	The 18 most common	off
Licence	Six licence types	off
Source	Library, System	off
Reset all filters	Clears chips, category and search	button
Download selected	Downloads your kit	shown when the kit has families

Stage

Control	Values	Default
Preview text	Any text	the family name
Shuffle	Random order	button
Grid, **List**	Layout	Grid
Size	12 to 120 px, − and + in 2 px steps, **Reset to 40**; cards use 22 to 48 px	40

Family panel

Item	Values
Width (wide screens)	380 to 720 px, 460 at first, remembered
Height (narrow screens)	62% at rest, 94% open
Header buttons	+ **Select** / ✓ **Selected**, **Download kit** / **Installed only**, ✕
Tabs	Specimen, Suitability, Pairings, Glyphs, Features, Styles, About, License, Charset, Develop, Use & Embed

Specimen tab

Control	Values	Default
Size	8 to 400 px	64
Line height	0.6 to 3	1.2
Letter spacing	−10 to 40 px	0
Weight	The family's weights	400, or the nearest available
Text colour	Any colour	dark ink
Presets	Pangram, Paragraph, Alphabet, Uppercase, Numbers, Symbols, Code	the preview text

Scores

Score	Range
Accessibility, Readability, UI / Product, Developer / Code, Editorial	0 to 100
Style fingerprint: Modern, Classic, Elegant, Playful, Technical, Friendly, Bold, Minimal	0 to 100
Pairing score	60 to 99
Recommender match	40 to 99

Develop tab

Item	Values
Exports	CSS variables, Tailwind, SCSS, Design tokens, CSS @import
Type scale steps	xs, sm, base, md, lg, xl, 2xl, 3xl, 4xl
Base	10 to 24 px, 16 at first
Ratio	1.125, 1.200, 1.250, 1.333, 1.414, 1.500, 1.618; 1.200 at first

Use & Embed tab

Link, Import, Script, JS, @font-face, Usage.

Recommender

Control	Values	Default
Industry	Any / not sure and 13 industries	Any
Brand style	Modern, Classic, Elegant, Playful, Bold, Minimal, Technical, Friendly	none
Accessibility priority	Essential, Balanced, Any	Balanced
Colour palette	HEX colours	empty
Results	12 cards	always 12

Files

File	From
Family-name-font-kit.zip	**Download kit**, or a kit of one family
auric-font-selection-kit.zip	**Download** or **Download selected** with several families
Single font file	**Download** in the Styles tab's file table

Messages

Message	Meaning
No fonts match these filters.	Nothing fits the search and filters.
No option matches that.	No filter chip matches the text in Filter these options.
Copied …	Code, a stack, a glyph or a feature was copied.
Added both to selection	+ Select both worked.
Select some families first	Embed or Download with an empty kit.
Select at least 2 families to compare	Compare needs two.
Preparing download kit…	A ZIP is being built.
Downloaded …	The ZIP is ready.
Only bundled library fonts can be downloaded	Every family asked for is Installed only.
Could not prepare this download	The family's details could not be loaded.
Font library is temporarily unavailable	The catalogue could not be reached.
The live font engine didn’t load	The page's font engine did not start; **Retry** reloads.

Related

- Shortcuts
- Glossary
- Browse, search and filter

WEBSITE › TOOLS › COLOUR WORKSPACE

Troubleshooting

Fixes for common problems in the Font Library, from an empty page to downloads and shortcuts that do not behave.

The page stays empty, or says the library is unavailable

The catalogue and the fonts load from the Font CDN. If it cannot be reached, after a few seconds the grid shows **Font library is temporarily unavailable** or **The live font engine didn't load**, with the likely causes.

- Check that you are online.
- Turn off extensions that block scripts or requests for this site, and allow requests to fonts.auricartisan.com.
- Select **Retry**, or reload the page, once the connection is back.

Some cards show a plain font instead of the family

The specimen appears once the family's font file has loaded; slow connections show a stand-in first. Cards marked **Installed** only appear in their own face if that font is installed on your device; see Limits and accuracy.

Sorting by Accessibility score or Readability score shows A to Z

At the time of writing these two sort options do not take effect, and the grid stays in alphabetical order. **UI score** works. For accessible families, open **Collections** and use **Best Accessibility Fonts**, or open **Recommend** with **Accessibility priority** set to **Essential**. Each card's badge also shows its accessibility score.

The Collections view is empty

The collections are hidden while a search, a filter chip, a category or the recommender is active on **Specimens**. Select **Clear all** under **Applied** (or **Reset all filters**) and close the recommender, then open **Collections** again.

The Size slider stops changing the cards

Card specimens stay between 22 and 48 pixels. Settings outside that range make no further difference. Open a family for larger sizes; its specimen goes up to 400 pixels.

Searching for a foundry finds nothing

The search covers family names, designers and categories, not foundries. Open a family's **About** tab to see its foundry.

I cannot filter for my language

The **Scripts** filter offers the 18 most common scripts and blocks. For others, such as Arabic, Devanagari or Thai, search for the script name (many families carry it in their name, such as *Noto Sans Devanagari*), then check the family's **Charset** tab.

The F and R keys do nothing

At the time of writing, pressing `F` (filters) or `R` (recommender) switches the panel twice, so nothing appears to change. Use the **Filters** and **Recommend** buttons, or press `?` and choose **Toggle filters** or **Find your font** from the command palette.

Other shortcuts work normally; see Shortcuts.

The Selected view shows names in a plain font

The **Selected** view lists names and categories; its sample word may appear in a stand-in font. Open a family, or use **Compare**, to see the families in their own faces.

The weight chips in Use & Embed do not change the code

They are not connected to the snippets. Every snippet covers the whole family. Edit the code, or download the kit and keep only the files you need.

Download kit is disabled or says Installed only

That family is listed from installed system fonts and is not served, so it cannot be downloaded or embedded from the CDN. Choose a family from the **Library** source.

The ZIP is missing a file

If a font file could not be fetched, the ZIP holds `DOWNLOAD_NOTES.txt` naming it. Try again later. Large kits take a while to build; wait for the **Downloaded** message.

My kit disappeared

The kit is kept in this browser on this device. It does not follow you to another browser or device, and clearing site data or using a private window empties it. Keep embed code or the ZIP for anything you need to keep.

The panel covers the page on my tablet

On narrower screens the panel is a sheet from the bottom. Drag its grip down or tap the dimmed page to close it, or turn the device to landscape if it is wide enough for the docked layout.

Related

- FAQ
- Limits and accuracy
- Font CDN troubleshooting

WEBSITE › TOOLS › COLOUR WORKSPACE

Frequently asked questions

Short answers to common questions about the Font Library.

Do I need an account?

No. Browsing, testing, embed code and downloads all work without signing in. Your kit is kept in your browser.

Are the fonts free to use?

Most families are under the SIL Open Font License, and others under Apache 2.0, MIT, Creative Commons or other terms; 191 families have no licence the catalogue could identify. Check each family's **License** tab and its full licence text before you use it, especially commercially. The Font CDN licences page explains the licence types.

What does Installed mean?

The family is listed for reference but not served: it came from fonts installed on a computer, whose licences do not allow redistribution. The page can show it only if the same font is installed on your device, and it cannot be downloaded or embedded from the CDN.

Does embedding a font send my visitors' data to a third party?

The embed code points at the Auric Artisan Font CDN, not at a third-party font service. To avoid any external request, download the kit and self-host the files.

How are the suitability scores worked out?

They are computed from each font's own outlines and metrics, such as x-height, how distinct look-alike characters are (I, l, 1; 0, O), proportions, weight range and how well it renders on screens. They are not editorial ratings. See Limits and accuracy.

Why does the recommender keep suggesting well-known fonts?

It ranks within a shortlist of about 130 versatile, widely used families, so its suggestions are dependable. To explore beyond them, use the filters, the sort and the collections.

Can I test variable fonts?

Yes. The **Specimen** tab has a slider for each axis and chips for the font's named instances. Filter for them with **Variable** under **Properties**, or sort by **Variable first**.

Can I see a font in my own layout?

The specimen editor lets you type, style and colour text. For your own pages, copy the embed code or download the kit, which includes a demo page.

Can I upload my own fonts?

No. The library shows the Auric Artisan catalogue only.

Is my kit shared between devices?

No. It is kept in the browser on the device where you made it.

Related

- Troubleshooting
- Getting started
- Font CDN FAQ

WEBSITE › TOOLS › COLOUR WORKSPACE

Supporting material

The glossary, shortcuts, limits, related tools and privacy notes for the Font Library.

File	What it covers
glossary.md	Type and font terms used on the page.
shortcuts.md	Keyboard shortcuts and the command palette.
limits-and-accuracy.md	What the catalogue, scores and previews can and cannot tell you.
related.md	Tools, services and reading that go with the library.
privacy.md	What is kept in your browser and what uses the network.

Related

- Font Library overview
- Documentation contents

WEBSITE › TOOLS › COLOUR WORKSPACE

Shortcuts

Keyboard shortcuts and the command palette in the Font Library.

Shortcuts work when you are not typing in a field. They do not use `Ctrl`, `Cmd` or `Alt`: `Ctrl` + `K` belongs to the site-wide search.

Key	What it does
`?`	Opens the library command palette
`/`	Focuses the search box
`S`	Shuffles the results into a random order
`G`	Grid view
`L`	List view
`T`	Opens the **Your kit** tray (close it with **Hide**)
`[`	Makes card specimens smaller
`]`	Makes card specimens larger
`C`	Compares your kit (two or more families)
`Esc`	Closes the command palette or the family panel
`F`	Meant to show or hide filters; see below
`R`	Meant to open the recommender; see below

At the time of writing, `F` and `R` switch their panel twice, so nothing appears to change. Use the **Filters** and **Recommend** buttons, or the command palette.

The command palette

Press `?` to open a search box of commands. Type to narrow the list (for example *shuffle*, *filters* or *size*), move with `↑` and `↓`, run a command with `Enter`, and close with `Esc`.

Commands include **Focus search**, **Shuffle results**, **Grid view**, **List view**, **Toggle filters**, **Find your font**, **Open selected tray**, **Download selected**, **Increase preview size**, **Decrease preview size**, **Scroll to top**, **Toggle light / dark theme** and **Focus preview text**, plus a **Category ·** command for each category. Each shows its key, if it has one.

In the family panel

Key	Where	What it does
`Enter`	A focused card	Opens the family
`←`, `→`	The panel grip, wide screens	Makes the panel wider or narrower; add `Shift` for bigger steps
`↑`, `↓`	The panel grip, narrow screens	Raises or lowers the sheet
`Tab`, then `Enter` or `Space`	Filter chips	Switches a chip

Related

- Getting started
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Glossary

Type and font terms used in the Font Library and its documentation.

Term	Meaning
Axis	One dimension a variable font can change along, such as weight (wght) or width (wdth).
Character variant	An OpenType feature (cv01 to cv99) that swaps the design of one particular character.
Code point	The number Unicode gives a character, written like U+0041.
Family	All the styles of one typeface, such as Regular, Bold and Italic.
Fallback stack	The fonts listed after the one you want, used if it is missing.
@font-face	The CSS rule that tells a browser where a font file is and which style it provides.
Foundry	The company or person that publishes a typeface.
Glyph	A single drawn shape in a font, usually a character.
Heading and body	The two roles in a type pairing: a display face for titles and a text face for reading.
Installed	A family listed from system fonts, shown only if you have it installed, and not served.
Instance	A named point on a variable font's axes, such as Light or Condensed Bold.
Ligature	Two or more characters drawn as one glyph, such as fi, or => in coding fonts.
Modular type scale	A set of sizes where each step is the previous one times a fixed ratio.
OpenType feature	A switchable behaviour inside a font, named by a four-letter tag such as liga or tnum.

Term	Meaning
Pangram	A sentence using every letter of the alphabet.
Self-hosting	Serving font files from your own website instead of a font service.
Specimen	A sample of text set in a typeface, used to judge it.
Stroke contrast	The difference between thick and thin parts of letters.
Stylistic set	An OpenType feature (ss01 to ss20) that swaps a group of alternative designs.
Super-family	Related typefaces designed to work together, such as a sans and a serif sharing a name.
Tabular figures	Digits of equal width, so numbers line up in columns.
Variable font	One font file that holds a continuous range of styles along one or more axes.
Weight	How heavy the strokes are, from 100 (Thin) to 900 (Black); 400 is Regular and 700 Bold.
x-height	The height of lowercase letters such as x, as a share of the font's size (em).

Related

- Reference
- Font CDN glossary

WEBSITE › TOOLS › COLOUR WORKSPACE

Limits and accuracy

What the Font Library's catalogue, categories, scores, recommendations and previews can and cannot tell you.

The catalogue is read from the font files

Names, designers, foundries, versions, dates, weights, scripts and licences come from the data inside each font file. Where a file leaves something out, the page leaves it out too, or shows it as unknown. 191 families have no licence the catalogue could identify; they appear under **Unspecified**.

Categories are detected automatically

Each family's category (serif, sans serif, display, handwriting or monospace) and its serif or sans-serif classification are worked out from the font's data and letter shapes, not assigned by hand. Some families are misclassified; rounded sans-serifs in particular can be listed as serif. If the category filter hides a family you expect, search for it by name.

Scores are computed, not judged

The suitability scores and style fingerprints are calculated from measurements of each font: x-height, how distinct look-alike characters are, proportions, weight range, stroke contrast and how it renders on screen. They are useful for sorting and comparing, but:

- they do not know how a font was designed to be used, so a display face can score well in a category it was not made for;
- a high accessibility score is a good sign, not a guarantee. Test your text at real sizes with real readers, and check colour contrast separately.

This is why the collections and the recommender draw on curated lists rather than the raw top scores.

Recommendations and pairings use a shortlist

The recommender and the pairings rank within about 130 well-known, versatile families. Other families never appear there, however well they would fit. Pairings leave out handwriting faces.

Installed families

190 families marked **Installed** are listed from installed system fonts and are not served. Their cards and specimens use your own copy of the font if you have one; otherwise you see a stand-in. They cannot be downloaded, and their embed code only works on devices with the font installed.

Previews are your browser's rendering

Specimens are drawn by your browser from the served font files. Rendering differs a little between operating systems and browsers, and bold or italic buttons in the specimen editor are imitated by the browser if the family lacks those styles.

Page limits

Limit	Value
Card specimen size	22 to 48 pixels, whatever the slider shows
Script filter chips	The 18 most common scripts and blocks
Glyphs shown in the Glyphs tab	Up to 8,000
Unicode ranges listed in the Charset tab	The first 40
Families compared at once	4 (the first four in your kit)
Collection rows	Up to 14 families each
Recently viewed	The last 12 families
Recommender results	12
Pairing suggestions	8

Known issues

At the time of writing:

- sorting by **Accessibility score** or **Readability score** leaves the grid in alphabetical order;
- the `F` and `R` keys appear to do nothing;
- the **Selected weights** chips in **Use & Embed** do not change the code;
- the search box mentions foundries, but foundry names are not searched.

See Troubleshooting for workarounds.

Related

- Font CDN limits and accuracy
- Family details

- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Privacy

What the Font Library keeps in your browser and what uses the network.

Stays in your browser

Your preview text, specimen edits, recommender choices and palettes are used only in the page. They are not sent anywhere. Download kits are assembled in your browser.

Remembered in your browser

In your browser's local storage on this device:

- **Your kit**: the families you selected;
- which filter groups you folded;
- the width of the family panel;
- the last twelve families you opened (**Recently viewed**).

Like other pages on the site, the page also keeps the values of some fields, such as your preview text, in the page address after `#p=`, so a reload restores them. If you share the address, the other person sees those values. See Shareable links.

Uses the network

- Loading the page and its scripts.
- The catalogue, the specimens and every font file load from the Auric Artisan Font CDN at fonts.auricartisan.com. No third-party font service is involved.
- Downloads fetch the font files from the same CDN.
- Site-wide analytics and advertising follow your cookie choices. See the Help centre.

For what the CDN itself records, see Font CDN privacy.

Related

- Limits and accuracy
- Your kit, compare and download

WEBSITE › TOOLS › COLOUR WORKSPACE

Related tools and reading

Tools, services and reading that go with the Font Library.

Services

- Font CDN: the service that serves these fonts, with its loader, self-hosting guide and licences.
- Icon CDN: the matching service for icons.

Tools

- Icon Library: the same kind of browser for 12,896 icons.
- Design System Generator: build a colour and type system from a brief.
- Personalization Generator: palettes, type pairings and a brand Studio from your colours.
- Contrast Checker: check the colour contrast of your text.

Learn articles

- Color in Typography and Text
- Building a Color System

Related

- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Icon Library

What the Icon Library is, who it is for, where to find it, and how to find, restyle and take away icons from seven open-licence sets.

The Icon Library puts seven open-licence icon sets under one search: Phosphor, Tabler, Lucide, Remix, Heroicons, Feather and Material. That is 12,896 icons in 32,803 variants across 21 styles, such as outline, fill, duotone or 16-pixel solid.

Resize, recolour and change the stroke of every icon at once, then open one to see it large, switch its style, and copy it as SVG, a React (JSX) component, an image data URI or a one-line CDN tag. Select the icons you need into **Your kit** and take them away as a ZIP of SVG files or a single SVG sprite.

The icons are served by the Auric Artisan Icon CDN, so nothing loads from a third party. The CDN itself (the loader script, embedding, self-hosting and licences) is documented under Icon CDN.

Who it is for

- Developers who need an icon as SVG, JSX or an embed tag.
- Designers comparing how an idea is drawn across several sets.
- Anyone building a small, consistent icon kit for a project.

Where to find it

- Page: https://auricartisan.com/tool/icon/
- No sign-in is needed. Your kit and view settings are kept in your browser.

Quick start

1. Open https://auricartisan.com/tool/icon/.
2. Type what you need in *Name, tag or set*, for example *arrow*.
3. Pick a set on the left, such as **Lucide**, and, if it has several, a style.
4. Set **Size**, **Stroke** and **Colour** above the grid to preview every icon that way.
5. Select an icon. It opens in the panel on the right.
6. Select **Copy SVG** or **Copy JSX**, or **+ Select** to add it to **Your kit**.
7. Open **Your kit** and select **Download ZIP** or **Copy sprite**.

What you can do

- Search 12,896 icons by name, keyword or set; filter by set, style, category and licence.
- Preview every icon at your size, stroke width and colour.
- Browse the seven sets and popular categories, or use **Find your icon** for a shortlist.
- Copy an icon as SVG, JSX, an `<img>` data URI or a CDN tag, or download it as an SVG file.
- Collect icons in **Your kit** and download them as a ZIP or copy them as an SVG sprite.

In this folder

- docs/ — how to use the Icon Library.

    - Getting started
    - Browse, search and filter
    - Sets, collections and Find your icon
    - The icon panel
    - Your kit and export
    - Reference
    - Troubleshooting
    - FAQ
- others/ — supporting material.
    - Glossary
    - Shortcuts
    - Limits and accuracy
    - Related
    - Privacy

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Icon Library documentation

The list of guides for the Icon Library.

File	What it covers
getting-started.md	Open the page, learn the layout, copy a first icon.
browse-and-filter.md	Search, sort, sets and styles, category and licence filters, size, stroke and colour.
sets-and-finder.md	The Sets view, featured collections, Find your icon and Recently viewed.
icon-panel.md	The docked panel: preview controls, copy and download, code blocks, details, Material icons.
kit-and-export.md	Selecting icons, Your kit, the ZIP and the SVG sprite.
reference.md	Every control, set, style and message.
troubleshooting.md	Symptoms, causes and fixes.
faq.md	Short answers to common questions.

Related

- Icon Library overview
- Supporting material
- Icon CDN

WEBSITE › TOOLS › COLOUR WORKSPACE

Getting started with the Icon Library

Open the Icon Library, learn its layout, and find, restyle and copy your first icon.

Open the page

Go to https://auricartisan.com/tool/icon/. The masthead reads *Twelve thousand icons, one line to serve them* and its labels give the live counts of icons, variants, sets and styles.

The catalogue loads from the Icon CDN. If it cannot be reached, a notice explains that the library is temporarily unavailable; see Troubleshooting.

The layout

Two views sit in a row at the top: **Icons** (with the number of icons shown) and **Sets** (7).

The **Icons** view has two columns:

- **The rail** on the left, which stays in view as you scroll:
  - **Find**: the search box, the sort menu, **Recommend**, **Shuffle**, **Filters** and the number of matching icons;
  - **Applied**: what is switched on, when anything is;
  - **Set**: **All** and the seven sets with their counts, and a style switch when the chosen set has more than one style;
  - **Narrow it**: the **Category** and **License** chips;
  - at the foot, **Reset filters** and **Download selected**.

- **The stage** on the right: **Size**, **Stroke** and **Colour** controls, then the grid of icons with their names, which loads more as you scroll.

Once you have opened an icon, a **Recently viewed** strip appears under the two views.

On screens narrower than about 1,180 pixels the rail sits above the grid and its filter groups start folded.

Your first icon

1. Type *bell* in *Name, tag or set*.
2. Under **Set**, select **Phosphor**, then the **Duotone** style.
3. Drag **Size** to 64 and pick a colour with **Colour**. Every icon in the grid changes.
4. Select an icon. The panel opens on the right with a large preview.
5. Select **Copy SVG**. The message **SVG copied** confirms it; paste it into your HTML or design tool.
6. Select **+ Select** to keep it in **Your kit** for later.

Result: an SVG in your chosen style and colour, ready to use.

Next steps

- Browse, search and filter
- The icon panel
- Your kit and export

Related

- Icon Library overview
- Shortcuts

WEBSITE › TOOLS › COLOUR WORKSPACE

Browse, search and filter

Search, sort, choose a set and style, filter by category and licence, and preview every icon at your size, stroke and colour.

Everything here happens on the **Icons** view. The grid shows 120 icons at first and loads more as you scroll. Each tile shows the icon and its name; hover it to see the name in full. Select a tile to open the icon.

Search

Type in *Name, tag or set*. Results update as you type. The search looks for your text in the icon's name, in its identifier (which begins with the set's name, so *lucide* finds every Lucide icon), and in its keywords. Only Tabler and Remix icons carry keywords, so a word such as *notification* finds more in those two sets.

Sort

Option	Order
Alphabetical	A to Z by name (default)
Group by set	Set by set, A to Z within each

Option	Order
Surprise me	Random order

Shuffle switches the sort to **Surprise me** and deals a new random order.

Sets and styles

The **Set** chips pick one set at a time: **All** (12,896), **Phosphor** (1,512), **Tabler** (5,093), **Lucide** (1,714), **Remix** (1,690), **Heroicons** (324), **Feather** (287) or **Material** (2,276).

When the chosen set has more than one style, a style switch appears under the set chips:

Set	Styles, in the order shown (default in bold)
Phosphor	Thin, Light, **Regular**, Bold, Fill, Duotone
Tabler	**Outline**, Filled
Lucide	**Regular** only
Remix	**Line**, Fill
Heroicons	**24 · Outline**, 24 · Solid, 20 · Solid, 16 · Solid
Feather	**Regular** only
Material	**Outlined**, Round, Sharp, Regular, Twotone

With **All** chosen, each icon shows in its set's default style. Choosing a set clears the category filter.

Narrow it: category and licence

Filters shows or hides the chip groups.

- **Category** lists up to 40 categories for the current set, largest first, each with its count, such as **System**, **Arrows** or **Devices**. Selecting several shows icons in any of them. Only Tabler and Remix icons have categories; with Phosphor, Lucide, Heroicons, Feather or Material chosen, the group reads *No categories for this set.*
- **License** lists the licences in the current set with counts: **MIT**, **ISC** (Lucide), **Remix v1.0** (Remix) and **Apache-2.0** (Material). Selecting several shows icons under any of them.

Tips:

- **Filter these options** finds a chip by name, for example *weather*. It shows **No option matches that.** if nothing fits.
- **Category** shows ten chips at first; **+ n More** shows the rest and **Less** folds them again.
- Select a group's heading to fold it; a number on the heading says how many of its chips are on. The page remembers which groups you folded.

See and clear what is applied

When anything is switched on, **Applied** appears under the search box with a chip for the search, the set and each filter. Select × on a chip to remove just that one, or **Clear all** to remove everything and go back to **All**.

Reset filters at the foot of the rail clears the category and licence chips and the search, but keeps the chosen set.

If nothing matches, the grid reads *No icons match. Try another set or clear filters.*

Size, stroke and colour

The controls above the grid restyle every icon at once, and the panel starts from them when you open an icon.

Control	Range	Default	Notes
Size	16 to 128 px	48	All icons
Stroke	0.5 to 3, in steps of 0.25	**auto**	Shows **auto** until you move it; **Reset** goes back to each icon's own stroke. Affects outline icons only, such as Tabler, Lucide, Feather and outline Heroicons.
Colour	Any colour	the page's text colour	**currentColor** goes back to following the text colour.

These settings, the chosen set and style, and the sort are remembered in your browser.

Related

- Sets, collections and Find your icon
- The icon panel
- Shortcuts

WEBSITE › TOOLS › COLOUR WORKSPACE

The icon panel

Preview an icon large, switch its style, size, colour and stroke, copy it as SVG, JSX, a data URI or a CDN tag, download it, and read its details.

Selecting an icon opens it in a panel.

Where the panel sits

- **On a wide screen** (wider than about 1,180 pixels), the panel docks to the right edge as a column of the page. Nothing is covered, and selecting another icon shows that one instead.
  - Drag the grip on the panel's left edge to resize it (380 to 720 pixels; 460 at first). Double-click the grip to go back to 460. With the grip focused, `←` and `→` resize it, faster with `Shift`. The width is remembered.
- **On a narrower screen**, the panel is a sheet from the bottom at about 62% of the screen. Drag the grip up to open it to 94%, or down to close it. Tapping the dimmed page closes it.

Close the panel with ✕ or `Esc`.

The header

- The icon's name, and a line with its set, licence, category (Tabler and Remix only) and number of styles.
- **+ Select** adds the icon to **Your kit**; it then reads **✓ Selected**. See Your kit and export.
- **Download .svg** downloads the icon in the style the grid is showing, as it comes from the set, without your panel colour or stroke. Use **Download .svg** among the copy buttons for a restyled file.

Preview controls

A large preview sits beside the controls. Changing these affects the preview; colour and stroke also change the code below.

Control	Range	Notes
Style	The icon's styles	Only when it has more than one. Switches the preview, code and copy buttons to that style.
Size	16 to 160 px	Starts at the grid size. Changes the preview only.
Color	Any colour	The picker starts at the grid colour, or near-black; until you pick, the preview follows the text colour. **currentColor** goes back to that.
Stroke	0.5 to 3	Only for outline sets (Tabler, Lucide, Heroicons, Feather). Starts at the grid stroke.

Copy and download

Button	What you get	Message
Copy SVG	The SVG markup with your colour and stroke	**SVG copied**
Copy JSX	A React component named after the icon, passing props to the SVG	**JSX copied**
Copy data-URI	An `<img>` tag whose source is the SVG as a data URI, sized to the grid size	**data-URI copied**
Copy CDN tag	An `<i data-icon="set:name">` tag and the Icon CDN script that fills it	**CDN tag copied**
Download .svg	An SVG file named after the icon and style, with your colour and stroke	none

The icons use `currentColor`, so without a chosen colour they take the colour of the text around them. Picking a colour replaces `currentColor` with that colour in the copied code.

The code blocks

Under the preview, four blocks show the code in full, each with its own **Copy** (message **Copied**):

- **SVG**;
- **JSX / React**: `export const IconName = (props) => (…);`;
- `<img>` **data-URI**;
- **CDN (auto-inject)**: the tag with a `data-style` attribute when the style is not the set's default, and the loader script.

The blocks update when you change the colour or stroke.

For how the CDN tag and loader work, see Icon CDN: embed an icon.

Details

At the foot of the panel: **Set** (linked to the set's own website), **License**, **Author**, **Category** and **Keywords** when the icon has them, and **Path**, which says which set and style the icon comes from.

Material icons

Material icons are served as a webfont, so the panel works differently for them:

- the copy buttons are **Copy glyph**, which copies the icon's HTML character reference such as `` (message **Glyph entity copied**), and **Copy CDN tag**;
- the code blocks are **HTML (CDN)**, a tag and the loader script, and **Webfont @font-face**, a rule for the Material font of that style with a class to apply it;

- **Download .svg** shows **Material is a webfont — use the embed code** instead of downloading;
- there is no stroke control.

Related

- Your kit and export
- Icon CDN: style, size and colour
- Browse, search and filter

WEBSITE › TOOLS › COLOUR WORKSPACE

Your kit and export

Select icons into Your kit, and take them away as a ZIP of SVG files or a single SVG sprite.

Icons you select gather in **Your kit**. The kit is kept in your browser on this device, so it is still there next time.

Select icons

Open an icon and select **+ Select** in its panel; it turns to **✓ Selected**, and the icon's tile in the grid is marked as selected. Select the button again to remove it.

Each icon is kept in the style the grid was showing when you selected it. To keep a different style, switch the style in the rail first.

The Your kit tray

Once something is selected, a **Your kit** button with a count appears. Select it to open or close the tray: **Your kit** with the count and *Kept in this browser*, a list of your icons with a small preview, name and set (each with ✕ to remove it), and three buttons:

Button	What it does	Message
Download ZIP	Downloads all your icons as SVG files	**Downloaded *n* icons**
Copy sprite	Copies one SVG sprite containing all your icons	**SVG sprite copied**
Clear	Empties the kit	none

Hide closes the tray. **Download selected** at the foot of the rail does the same as **Download ZIP** and shows the count, for example **Download selected (5)**; it is disabled while the kit is empty. With an empty kit, the messages read **Select some icons first**.

The ZIP

The file is `auric-icons.zip`. Inside:

- a folder per set, holding one SVG per icon named after the icon and style, for example `lucide/bell-regular.svg`;
- `README.txt`, noting the number of icons and that each uses `currentColor` for fill or stroke, so you set the colour with CSS and the size with `width` and `height` or CSS.

The files are the icons as they come from their sets, without the size, stroke or colour you previewed. For a restyled single icon, use **Download .svg** or **Copy SVG** in the icon panel.

Material icons are a webfont, so each one appears in the ZIP as a short text note under `material/` giving its character reference, not as an SVG.

The SVG sprite

Copy sprite copies a hidden `<svg>` block with one `<symbol>` per icon, each with the icon's identifier as its `id`, followed by a comment showing how to use it:

```
<svg><use href="#lucide-bell"/></svg>
```

Paste the sprite once near the top of your page, then show any icon with a `<use>` that points at its identifier. Material icons are left out of the sprite.

Related

- The icon panel
- Icon CDN: export and self-host
- Privacy

WEBSITE › TOOLS › COLOUR WORKSPACE

Sets, collections and Find your icon

The Sets view, the featured collections, the Find your icon panel and the Recently viewed strip.

The Sets view

Open **Sets** at the top. *Seven sets, each a shortcut past the alphabet.* A card for each set shows its name, **Open set** and its number of icons. Select a card to choose that set and go back to the **Icons** view.

Set	Icons	Styles	Licence	Author
Phosphor	1,512	6	MIT	Phosphor Icons
Tabler	5,093	2	MIT	Paweł Kuna
Lucide	1,714	1	ISC	Lucide Contributors
Remix	1,690	2	Remix v1.0	Remix Design
Heroicons	324	4	MIT	Tailwind Labs
Feather	287	1	MIT	Cole Bemis
Material	2,276	5	Apache-2.0	Google

Material icons are a webfont rather than SVG files, which changes what you can copy; see The icon panel.

Featured collections

Under the set cards, **Featured collections** (*Fast starts, just like the font library.*) has two rows:

- **Icon sets** (*Browse complete families*): a card per set with its count and licence. Selecting one chooses the set, clears the search and category, and scrolls to the icons.

- **Popular categories** (*Jump straight into a use case*): the ten largest categories, such as **System**, **Design**, **Arrows**, **Brand** and **Devices**, with their counts. Selecting one switches to **All** sets, filters to that category, and scrolls to the icons.

Featured collections are hidden while a search, a set other than **All**, a filter or **Find your icon** is active. The set cards above them are always there.

Find your icon

Select **Recommend** in the rail (or choose **Find your icon** from the command palette) to open **Find your icon** above the grid: *Filter the catalogue by set, category, and words from your brief.*

1. **Set**: **Any set** or one of the seven.
2. **Category**: **Any category** or a category from the list.
3. **Brief** (*name or keyword*): a word such as *arrow*, *brand*, *user* or *chart*.
4. Select **Recommend icons**.

Up to twelve results appear, each with its rank, name, set and category. Select one to open the icon. The note beside the buttons gives the number found, for example *12 strong starting points*, or *No matches yet* with the suggestion *Try a broader set or simpler keyword.*

Reset clears the fields; **×** closes the panel.

Find your icon filters rather than ranks: it lists the first twelve icons, in catalogue order, that match all three fields. The **match** figure on each card only reflects its place in the list, from 96 for the first down to 72, not a measured fit. Keywords and categories exist only for Tabler and Remix icons, so a category or brief finds most in those two sets.

Recently viewed

Every icon you open is added to **Recently viewed**, under the two views; it keeps the last twelve. Selecting a name types it into the search box, so icons with that name, from every set shown, come up in the grid. **Clear** empties the list.

Related

- Browse, search and filter
- The icon panel
- Limits and accuracy

WEBSITE › TOOLS › COLOUR WORKSPACE

Icon Library reference

Every view, control, set, style, file and message in the Icon Library in one place.

Address

https://auricartisan.com/tool/icon/

Catalogue

Figure	Value
Icons	12,896
Variants (icon and style combinations)	32,803
Sets	7
Styles	21
Icons with categories and keywords	Tabler and Remix only (51 categories)

Sets

Set	Icons	Styles	Default style	Licence	Kind
Phosphor	1,512	Thin, Light, Regular, Bold, Fill, Duotone	Regular	MIT	SVG, filled shapes
Tabler	5,093	Outline, Filled	Outline	MIT	SVG, stroked
Lucide	1,714	Regular	Regular	ISC	SVG, stroked
Remix	1,690	Line, Fill	Line	Remix v1.0	SVG, filled shapes
Heroicons	324	24 · Outline, 24 · Solid, 20 · Solid, 16 · Solid	24 · Outline	MIT	SVG, mixed
Feather	287	Regular	Regular	MIT	SVG, stroked
Material	2,276	Outlined, Round, Sharp, Regular, Twotone	Outlined	Apache-2.0	Webfont

Views

View	Contents
Icons	Rail, controls and grid; the count of matching icons
Sets	Seven set cards and Featured collections

Rail

Control	Values	Default
Search	Name, identifier or keyword	empty
Sort	Alphabetical, Group by set, Surprise me	Alphabetical
Recommend	Opens Find your icon	closed
Shuffle	Random order	button
Filters	Shows or hides the chip groups	shown
Set	All and seven sets	All
Style	The chosen set's styles	The set's default
Category	Up to 40 for the current set	off
License	MIT, ISC, Remix v1.0, Apache-2.0	off

Control	Values	Default
Reset filters	Clears category, licence and search	button
Download selected	Downloads Your kit as a ZIP	disabled when empty

Grid controls

Control	Range	Default
Size	16 to 128 px	48
Stroke	0.5 to 3, steps of 0.25	auto
Colour	Any colour	currentColor

Icon panel

Item	Values
Width (wide screens)	380 to 720 px, 460 at first, remembered
Height (narrow screens)	62% at rest, 94% open
Preview size	16 to 160 px
Preview stroke	0.5 to 3 (outline sets)
Copy buttons	Copy SVG, Copy JSX, Copy data-URI, Copy CDN tag, Download .svg; Material: Copy glyph, Copy CDN tag
Code blocks	SVG, JSX / React, `<img>` data-URI, CDN (auto-inject); Material: HTML (CDN), Webfont @font-face

Find your icon

Field	Values
Set	Any set, or one of seven
Category	Any category, or one of the categories
Brief	A name or keyword
Results	Up to 12

Files

File	From
icon-id-style.svg	**Download .svg** in the icon panel
auric-icons.zip	**Download ZIP** or **Download selected**

Messages

Message	Meaning
SVG copied, **JSX copied**, **data-URI copied**, **CDN tag copied**, **Copied**	Code was copied.

Message	Meaning
Glyph entity copied	A Material icon's character reference was copied.
Material is a webfont — use the embed code	Material icons cannot be downloaded as SVG.
Select some icons first	Download ZIP or Copy sprite with an empty kit.
Downloaded *n* icons	The ZIP is ready.
SVG sprite copied	The sprite is on the clipboard.
No categories for this set.	The chosen set has no categories.
No option matches that.	No chip matches the text in Filter these options.
Icon library is temporarily unavailable	The catalogue could not be reached.
The live icon engine didn’t load	The page's icon engine did not start; **Retry** reloads.

Related

- Shortcuts
- Glossary
- Browse, search and filter

WEBSITE › TOOLS › COLOUR WORKSPACE

Troubleshooting

Fixes for common problems in the Icon Library, from an empty page to filters and exports that do not behave as expected.

The page stays empty, or says the library is unavailable

The catalogue and the icons load from the Icon CDN. If it cannot be reached, after a few seconds the grid shows **Icon library is temporarily unavailable** or **The live icon engine didn’t load**, with the likely causes.

- Check that you are online.
- Turn off extensions that block scripts or requests for this site, and allow requests to icons.auricartisan.com.
- Select **Retry**, or reload the page, once the connection is back.

Some tiles show a plain rounded square

That square stands in for an icon whose drawing could not be loaded. Scroll away and back, or reload the page. If it persists for a particular set and style, try another style.

The Category filter says No categories for this set

Only Tabler and Remix icons have categories. Choose **All**, **Tabler** or **Remix**, or search by name instead.

A keyword search finds only Tabler and Remix icons

Only those two sets carry keywords. For the other sets, search by the words in the icon's name, such as *bell*, *arrow* or *user*.

Moving Stroke changes nothing

Stroke affects outline icons only: Tabler (Outline), Lucide, Feather and outline Heroicons. Filled sets such as Phosphor, Remix and Material have no stroke to change.

The Featured collections are missing from the Sets view

They are hidden while a search, a set other than **All**, a filter or **Find your icon** is active. Select **Clear all** under **Applied**, close **Find your icon**, and open **Sets** again. The seven set cards are always shown.

Removing a licence filter with × turns on a different licence

At the time of writing, the **×** on a licence chip under **Applied** can switch the wrong licence when more than one is listed. Select the chip itself in the **License** group instead, or use **Clear all**.

Find your icon shows high match numbers for poor matches

The match figure only shows each result's place in the list, not how well it fits. Refine the **Brief**, or pick a **Set** or **Category**.

The ZIP icons are not the size or colour I previewed

The ZIP holds the icons as they come from their sets. They use `currentColor`, so set the colour and size with CSS where you use them, or download single restyled icons with **Download .svg** in the icon panel.

The header Download .svg ignores my colour

The **Download .svg** button at the top of the panel saves the icon in the grid's style as it comes from the set. Use **Download .svg** among the copy buttons for a file with your colour and stroke.

A Material icon will not download as SVG

Material icons are served as a webfont. Use **Copy CDN tag** or the **Webfont @font-face** code, or choose a similar icon from another set.

My kit disappeared

The kit is kept in this browser on this device. It does not follow you to another browser or device, and clearing site data or using a private window empties it.

Related

- FAQ
- Limits and accuracy
- Icon CDN troubleshooting

WEBSITE › TOOLS › COLOUR WORKSPACE

Frequently asked questions

Short answers to common questions about the Icon Library.

Do I need an account?

No. Browsing, copying and downloading all work without signing in.

Can I use these icons in commercial projects?

Each set keeps its own open licence: MIT (Phosphor, Tabler, Heroicons, Feather), ISC (Lucide), the Remix Icon licence (Remix) and Apache 2.0 (Material). The panel shows each icon's licence and author, and links to the set's website where the full terms are. See also Icon CDN licences.

Why do icons change colour with my text?

The icons are drawn in `currentColor`, which means "the colour of the surrounding text". Pick a colour in the panel before copying to fix it, or set `color` in your CSS.

Which format should I copy?

- **SVG** to paste into HTML or a design tool.
- **JSX** for a React component.
- **data-URI** for an `<img>` tag or a CSS background without a separate file.
- **CDN tag** to let the Icon CDN script insert the icon for you.

Can I mix icons from different sets?

Yes, but sets differ in line weight, corner style and grid size. For a consistent look, keep to one set, or compare the same idea across sets by searching its name with **All** chosen.

Can I upload my own icons?

No. The library shows the seven bundled sets only.

Is my kit shared between devices?

No. It is kept in the browser on the device where you made it.

Related

- Troubleshooting
- Getting started
- Icon CDN FAQ

WEBSITE › TOOLS › COLOUR WORKSPACE

Supporting material

The glossary, shortcuts, limits, related tools and privacy notes for the Icon Library.

File	What it covers
glossary.md	Icon and SVG terms used on the page.
shortcuts.md	Keyboard shortcuts and the command palette.
limits-and-accuracy.md	What the catalogue, search and exports can and cannot do.
related.md	Tools and services that go with the library.
privacy.md	What is kept in your browser and what uses the network.

Related

- Icon Library overview
- Documentation contents

WEBSITE › TOOLS › COLOUR WORKSPACE

Shortcuts

Keyboard shortcuts and the command palette in the Icon Library.

Shortcuts work when you are not typing in a field. They do not use `Ctrl`, `Cmd` or `Alt`: `Ctrl` + `K` belongs to the site-wide search.

Key	What it does
`?`	Opens the library command palette
`/`	Focuses the search box
`S`	Shuffles the icons into a random order
`F`	Shows or hides the filter chips
`R`	Opens or closes **Find your icon**
`T`	Opens or closes the **Your kit** tray
`[`	Makes the icons smaller
`]`	Makes the icons larger
`Esc`	Closes the command palette or the icon panel

The command palette

Press `?` to open a search box of commands. Type to narrow the list, move with `↑` and `↓`, run a command with `Enter`, and close with `Esc`.

Commands include **Focus search**, **Shuffle results**, **Toggle filters**, **Find your icon**, **Open selected tray**, **Download selected**, **Increase preview size**, **Decrease preview size**, **Scroll to top**, **Toggle light / dark theme**, **Pick icon color**, **Thicker stroke** and **Thinner stroke**, plus a **Category ·** command for each set. Each shows its key, if it has one.

In the icon panel

Key	Where	What it does
←, →	The panel grip, wide screens	Makes the panel wider or narrower; add Shift for bigger steps
↑, ↓	The panel grip, narrow screens	Raises or lowers the sheet

Related

- Getting started
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Glossary

Icon and SVG terms used in the Icon Library and its documentation.

Term	Meaning
currentColor	A CSS keyword meaning the current text colour. Icons drawn with it take the colour of the text around them.
Data URI	A way to put a whole file inside a link or src attribute, so no separate file is needed.
Duotone	A style with two tones: a solid shape and a lighter fill behind it.
Filled and outline	Two ways of drawing icons: solid shapes, or lines of a set width.
Identifier	An icon's unique name, beginning with its set, such as lucide-bell.
JSX	The HTML-like syntax used to write React components.
Set	A family of icons drawn to one design, such as Lucide or Material.
Sprite	One SVG file holding many icons as symbols, each shown with a <use> reference.
Stroke width	The thickness of the lines in an outline icon.
Style	One drawing of a set, such as Outline, Fill or 16 · Solid.
SVG	Scalable Vector Graphics: an image format made of shapes, sharp at any size.
Symbol	A reusable piece of an SVG sprite, identified by an id.
Variant	One icon in one style. 12,896 icons come in 32,803 variants.
Webfont	A font whose characters are icons, used for the Material set.

Related

- Reference
- Icon CDN glossary

WEBSITE › TOOLS › COLOUR WORKSPACE

Limits and accuracy

What the Icon Library's catalogue, search, recommendations and exports can and cannot do.

Categories and keywords cover two sets

Only Tabler and Remix icons (6,783 of 12,896) have categories and keywords. For Phosphor, Lucide, Heroicons, Feather and Material, the category filter, the popular categories and keyword search find nothing, and you search by the words in the icon's name.

Find your icon filters, it does not rank

Find your icon lists the first twelve icons that match your set, category and brief, in catalogue order. Its match figures follow the list order and do not measure fit.

Names are the sets' own

Icon names come from each set, so the same idea can have different names in different sets (for example *trash*, *delete* or *bin*). Try more than one word.

Previews and exports

- The grid and panel controls change the preview. Only the panel's colour and stroke carry into the copied code and the restyled **Download .svg**.
- The **Download .svg** button in the panel header, the ZIP and the sprite hold icons as they come from their sets, in the style the grid showed when you selected them.
- The data URI is sized to the grid's **Size**, not the panel's preview size.
- Stroke only affects outline icons.
- Material icons are a webfont: they cannot be downloaded as SVG, are left out of the sprite, and appear in the ZIP as text notes.

Page limits

Limit	Value
Icons shown before scrolling loads more	120
Category chips per set	40
Find your icon results	12
Recently viewed	The last 12 icons
Panel preview size	16 to 160 pixels

Known issues

At the time of writing, the × on a licence chip under **Applied** can switch the wrong licence when more than one is listed. Use the chip itself in the **License** group.

Related

- Icon CDN limits and accuracy

- Troubleshooting
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Privacy

What the Icon Library keeps in your browser and what uses the network.

Stays in your browser

Your searches, colour and stroke choices and copied code are used only in the page. ZIP files and sprites are assembled in your browser.

Remembered in your browser

In your browser's local storage on this device:

- **Your kit**: the icons you selected;
- the chosen set, style, sort, size, stroke and colour;
- which filter groups you folded;
- the width of the icon panel;
- the last twelve icons you opened (**Recently viewed**).

Like other pages on the site, the page address may also record how far you have scrolled, after `#p=`, so a reload returns you there. See Shareable links.

Uses the network

- Loading the page and its scripts.
- The catalogue and the icon drawings load from the Auric Artisan Icon CDN at icons.auricartisan.com. No third-party icon service is involved.
- Site-wide analytics and advertising follow your cookie choices. See the Help centre.

For what the CDN itself records, see Icon CDN privacy.

Related

- Limits and accuracy
- Your kit and export

WEBSITE › TOOLS › COLOUR WORKSPACE

Related tools and reading

Tools and services that go with the Icon Library.

Services

- Icon CDN: the service that serves these icons, with its loader, embedding, export and licences.
- Font CDN: the matching service for fonts.

Tools

- Font Library: the same kind of browser for 2,213 font families.
- Basic Color Tools: pick and test colours for your icons.
- Contrast Checker: check that an icon colour stands out from its background.
- Design System Generator: build the colours and type your icons sit among.

Related

- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Personalization Generator

What the Personalization Generator is, who it is for, where to find it, and how to turn a colour pool and a short brief into palettes, gradients, type, UI pairs, posters, design tokens and a live brand Studio.

The Personalization Generator turns the colours you like and a short brief into a whole visual system. You give it a **pool** of colours (your brand colours, colours you admire, a preset), pick a few style words and a use case, and describe what you want, such as *calm premium dashboard for creators*. It then runs a **sweep**: it samples hundreds or thousands of candidate palettes, scores each one for closeness to your pool, colour harmony, fit with your styles, variety, contrast and colour-vision safety, and keeps the best.

From the kept palettes it derives gradients in twelve styles, typography pairings, UI colour pairs, posters in twelve layouts, a full design-token system (semantic roles, tonal ramps, light, dark and high-contrast themes, a WCAG and APCA contrast audit and data-visualisation scales) and a **Brand Studio** that paints any palette across sixteen live mockups, from a dashboard to an email. Everything can be copied as CSS, downloaded as JSON or saved to your Library.

The sweep runs in your browser in small chunks, so the page stays responsive, and you can stop it at any time without losing what it has found.

Who it is for

- Brand and product designers exploring colour directions from a starting set.
- Founders and small teams who want a coherent system without a designer on hand.
- Front-end developers who want tokens and CSS from a palette.
- Anyone who wants to see a palette on real interfaces before committing to it.

Where to find it

- Page: https://auricartisan.com/tool/personalization-gen/ (the short address https://auricartisan.com/personalization also opens it).
- No sign-in is needed. Saving uses your Auric Artisan Library.
- The page notes the matching REST call: **Automate this — 14 API credits per call to POST /v1/personalize/system**. See the Public API.

Quick start

1. Open https://auricartisan.com/tool/personalization-gen/. A quick first sweep runs by itself, so there are results on screen straight away.
2. Open **Pool** and add one or two of your own colours with **Add**.
3. In **The brief**, type what you want, for example *warm editorial brand for a bookshop*.
4. Select **Generate** (or press `Enter` in the brief). Watch **The run** on the right.
5. Open **Palettes** to see the ranked results with their scores.
6. Select **Open in Studio** on a palette you like and look at the sixteen mockups.
7. Open **Export** and select **Copy CSS**, or **Download JSON** for everything.

What you can do

- Build a colour pool by hand, from quick seed sets, from 17 presets, by expanding it automatically or by adding random related colours.
- Steer the search with 18 style chips, 5 intents, a preference note and scoring weights for diversity, accessibility and warmth.
- Run sweeps of 40 to 4,000 candidates, stop them, or run them endlessly.
- Get ranked palettes with score breakdowns, and gradients, type combinations, UI pairs and posters derived from them.
- Get a design-token system from the top palette, with a contrast audit and chart scales.
- Try any palette on sixteen Studio mockups, with theme, density, token inspection and live token swaps.
- Copy CSS for any result, export all tokens as CSS, download a JSON snapshot, and save results to your Library.

In this folder

- docs/ — how to use the generator.
  - Getting started
  - Build your colour pool
  - Tune the sweep
  - Run a sweep and read the results
  - Palettes, gradients, typography, UI pairs and posters
  - Design tokens
  - Brand Studio
  - Presets
  - Export, save and share
  - Reference
  - Troubleshooting
  - FAQ
- others/ — supporting material.
  - Glossary
  - Limits and accuracy
  - Related
  - Privacy

WEBSITE › TOOLS › COLOUR WORKSPACE

Personalization Generator documentation

The list of guides for the Personalization Generator.

These guides follow the generator's own order: pool, tune, generate, Studio, export.

File	What it covers
getting-started.md	Open the page, tour the layout, run a first sweep.
build-the-pool.md	Adding, building, randomising, jittering and removing pool colours; pool statistics.
tune-the-sweep.md	Style chips, intent, preference note, engine settings, scoring weights and filters; Simple and Advanced modes.
run-a-sweep.md	Generate, Stop, Endless, progress, the run panel and the Overview tab.
result-tabs.md	Palettes, Gradients, Typography, UI Pairs and Posters, and their actions.
design-tokens.md	Semantic roles, ramps, themes, contrast audit and data-visualisation scales.
brand-studio.md	The sixteen mockups and their controls.
presets.md	The seventeen presets.
export-and-save.md	CSS, JSON, copying and saving to the Library.
reference.md	Every control, option, range and default.
troubleshooting.md	Symptoms, causes and fixes.
faq.md	Short answers to common questions.

Related

- Personalization Generator overview
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Getting started with the Personalization Generator

Open the Personalization Generator, learn its layout, and run your first sweep.

Open the page

Go to https://auricartisan.com/tool/personalization-gen/. The masthead reads *Describe it once, keep what survives.*, with a short description, a row of labels and **Read the API reference** →.

Each time the page opens, a quick sweep of up to 120 candidates runs by itself, so there are results on screen straight away. Results are not kept between visits; your pool and settings are. See Privacy.

The layout

On a wide screen the tool has a row of tabs and three columns under it.

- **Tabs**: **Overview**, **Pool**, **Tune**, **Palettes**, **Gradients**, **Typography**, **UI Pairs**, **Posters**, **Tokens**, **Studio**, **Presets** and **Export**. **Pool** and the five result tabs show a count.

- **The brief** (left column):
  - a text box, *Describe the system you want… e.g. calm premium dashboard for creators*, with **Generate**, **Stop** and **Endless** under it;
  - **Search**: the **Simple** and **Advanced** switch. In Advanced mode it also shows the **Depth**, **Pool**, **Size** and **Keep** boxes, the **CVD-safe**, **Contrast** and **Unique** check boxes, and an **Advanced Kit** note;
  - **This run**: what the next sweep will do (depth, pool target, palette size, top kept, and which of the three filters are on). It shows in both modes.
- **The stage** (centre): the **Search results** box with **Clear**, a status line (**Idle.**, **Generating (320)…**, **Done · 24 kept · 812ms** and so on), and the open tab.
- **The run** (right column): the progress bar while a sweep runs, then **Where the sweep went**, **Score distribution** and **The leader, checked**. See Run a sweep.

On medium screens **The run** moves below the other two columns. On narrow screens all three stack, and on a phone the status line is hidden.

Your first sweep

1. Open **Pool**. The starting pool has eight colours: #D3AF37, #161616, #F8F7F1, #7D8A75, #2D6CDF, #C97B84, #0F172A and #14B8A6.
2. Type one of your own colours in the HEX box, for example `#3C6E71`, and select **Add**.
3. Open **Tune**. Under **Style chips**, select **Nordic**. Set **Use case** to **Brand**.
4. In the brief box, type *nordic studio* and press `Enter`.
5. Watch **The run**. When the status reads **Done**, open **Palettes**.
6. On the first card, select **Open in Studio**.

Result: a ranked set of palettes near your colours, in a Nordic direction, and the leader painted across sixteen mockups.

Next steps

- Build your colour pool
- Tune the sweep, including how the brief is actually used
- Brand Studio

Related

- Personalization Generator overview
- Reference

WEBSITE › TOOLS › COLOUR WORKSPACE

Brand Studio

Paint any generated palette across sixteen live mockups, switch theme and density, inspect tokens, swap colours live and save the brand system.

The **Studio** tab applies one palette to sixteen live mockups at once, so you can judge it on real layouts rather than as swatches. It opens on the leader of the last sweep; **Open in Studio** on any palette card, or a row in **Recent palettes** on **Overview**, opens that palette instead.

How the palette is turned into a brand system

From the palette the Studio picks:

- a **background** and **text** pair: the most readable pair in the palette;
- an **accent**: the most colourful remaining colour, and an **accent 2**, the next most colourful;
- a **surface**: the background mixed a little towards the accent;
- quieter tones mixed between background and text for secondary surfaces, muted text and borders;
- **success**, **warning**, **danger** and **info** colours: a palette colour in the green, amber, red or blue range where there is one, otherwise the accent or accent 2;
- a type system chosen from your use case and style chips.

Choose the palette

Available Palettes lists up to 24 of the kept palettes with their names and scores. Select one to apply it to every mockup. **Shuffle** in the toolbar picks another at random from the same 24.

Choosing a palette clears any colours you swapped in with **Test**.

The toolbar

Control	What it does
Auto, **Light**, **Dark**	**Auto** uses the palette's readable pair as it is. **Light** swaps background and text if the background is dark; **Dark** swaps them if the background is light.
Cozy, **Compact**	**Compact** tightens the grid and makes the mockup text slightly smaller.
Test	Opens the test panel, below.
Inspect	Outlines the parts of each mockup that use a token, and shows the token name and HEX when you hover one.
Shuffle	Opens a random palette from the first 24.
Save brand system	Saves the current palette and its brand system to your Library.

The mockups

Sixteen cards, each with a title and a small label:

Mockup	Label
Brand identity card	Identity
Product · Dashboard	UI
Marketing · Landing	Web
Component library	Tokens
Type scale	the display font's name
Charts	Data
Pricing	Marketing

Mockup	Label
Auth · Sign in	Onboarding
Messages	Realtime
Mobile · App	iOS
Calendar	Schedule
Music · Player	Media
Editorial · Magazine spread	Print
Marketing banner	CTA
Transactional · Email	Comms
Commerce · Product card	Shop

The mockups are pictures of interfaces built from the tokens. Their buttons and fields are for looks only and do nothing. The text in them is sample copy.

Test: swap colours live

Tick **Test** to open **Test · live swap**. It lists five tokens, **Background**, **Text**, **Accent**, **Accent 2** and **Surface**, each with its current value and a row of the palette's colours. Select a colour in a row to assign it to that token; every mockup updates straight away, and the token shows **override**.

The panel header shows the current **BG/FG** contrast ratio, so you can see at once whether a swap keeps text readable. **Reset** removes all overrides.

Overrides stay while you work, but choosing another palette or selecting **Shuffle** clears them.

Save the brand system

Save brand system saves an item named after the palette, for example *Balanced Analogous · Brand system*, to your Library. It holds the palette, its score, the derived brand system, your overrides, and the theme and density you chose. See Export, save and share.

Tips

- Use **Dark** and **Light** to check one palette works both ways before choosing it.
- If text is hard to read in several mockups, open **Test** and watch the **BG/FG** figure while you try other background and text colours.
- The Studio uses the same font stacks as **Typography**, so faces you do not have installed fall back to others. See Limits and accuracy.

Related

- Palettes, gradients, typography, UI pairs and posters
- Design tokens
- Export, save and share

WEBSITE › TOOLS › COLOUR WORKSPACE

Build your colour pool

Add, generate, randomise, nudge and remove the colours the Personalization Generator builds palettes around, and read the pool statistics.

The pool is the set of colours the generator treats as *yours*. Every palette it keeps is scored partly on how close its colours sit to the nearest pool colour, so the pool is the strongest steer you have. It holds up to 240 colours.

Open the **Pool** tab. The heading reads **Personal Color Pool**, with the count, for example **Pool: 8/240**.

Add one colour

1. Pick a colour with the colour well, or type a HEX value such as `#3C6E71` in the box next to it. Three-digit values such as `#3a7` also work.
2. Select **Add**.

The colour goes to the front of the pool. If it is already in the pool, nothing changes. An invalid value shows the message **Use a valid HEX color**.

Remove a colour

Select any swatch in the grid. Each swatch shows an **×** and removes itself when selected. **Clear** empties the whole pool; there is no undo.

If you generate with an empty pool, the generator puts the eight starting colours back first.

Grow the pool automatically

Button	What it does
Build 64	Fills the pool up to your **Pool target** (64 by default; the label shows the current target). New colours are variations of your pool and of built-in reference palettes that match your styles, use case and brief. If the pool is already larger than the target, it is cut down to the target.
+24 random	Adds 24 looser variations drawn from the matching reference palettes.

Set the pool target in **Tune** (**Pool target**, 5 to 240) or in the **Pool** box of the brief in Advanced mode.

Quick seeds

The **Quick seeds** card adds small fixed sets to the end of the pool:

Button	Colours added
+ 4 UI	#0F172A, #F8FAFC, #D3AF37, #1F2937
+ 4 Earthy	#F4EFE4, #473B2F, #C46D3B, #8AA17E
+ 4 Neon	#10002B, #7B2CBF, #E0AAFF, #00F5D4
+ 4 Pop	#FFFFFF, #000000, #E63946, #2D6CDF
+ Grayscale	seven greys from #0A0A0A to #F5F5F5

Jitter nudges every pool colour a little at random in OKLCH: up to about 6 degrees of hue, 0.02 of chroma and 0.03 of lightness either way. Use it to loosen a pool that is producing results too close to your inputs.

Other ways to fill the pool

- **Add to pool** on any result card (palette, gradient, typography, UI pair or poster) puts that result's colours at the front of the pool.
- Loading a preset replaces the pool. See Presets.
- From your Library, a palette, gradient, colour or shade saved in another tool can be opened here. Its colours are added to the front of the pool and the **Pool** tab opens. See Export, save and share.

Read the pool statistics

The side cards describe the pool as it stands:

- **Pool statistics**: **Count**, **Target**, **Avg chroma**, **Avg lightness**, **Light range** and **Neutrals** (colours with less than 8% saturation). Chroma and lightness here are HSL saturation and lightness, in per cent.
- **Hue distribution**: twelve 30-degree bands around the hue wheel. A fuller band holds more colours; hover a band to see its count.
- **Lightness ladder**: one tick per colour. An even spread leaves room for readable text and background pairs.

Tips

- Five to fifteen colours you really mean usually steer better than a pool of 200 automatic variations.
- Include at least one very light and one very dark colour. The generator looks for readable pairs, and a pool with no dark colours has fewer to offer.
- To lock a direction you like, select **Add to pool** on a good palette, then generate again.

Related

- Tune the sweep
- Presets
- Reference

WEBSITE › TOOLS › COLOUR WORKSPACE

Design tokens

The token system the Personalization Generator builds from the top palette — semantic roles, tonal ramps, light, dark and high-contrast themes, a WCAG and APCA audit, and data-visualisation scales — and how to copy it.

The **Tokens** tab turns the leader (the top palette of the last sweep) into a complete set of design tokens. It always uses the leader, whichever palette is open in the Studio, and it changes whenever a new sweep produces a new leader.

The heading reads, for example, **Semantic roles, 88 tonal stops, light/dark/high-contrast themes, APCA + WCAG audit, and chart scales — from your top palette.** Four figures sit beside it: **Roles**, **Tones**, **WCAG floor** and **APCA floor**. The two floors are the weakest of the first four contrast checks in the light theme (body text, muted text, text on surface and the primary button).

Semantic roles

Eight role chips. Select one to copy its HEX.

Role	How it is chosen
primary	The most chromatic palette colour with a usable mid lightness.
secondary	The most chromatic remaining colour.
accent	A strongly coloured palette colour whose hue is furthest from the primary.
neutral	The least chromatic palette colour.
success	A green, pulled a little towards the nearest palette hue.
warning	An amber, pulled the same way.
danger	A red, pulled the same way.
info	A blue, pulled the same way.

If the palette is missing something (for example, it has only one strongly coloured colour), the generator makes a suitable colour instead.

Copy CSS and **Copy JSON** sit beside the heading. See Copying the tokens, below.

Theme preview

Light and **Dark** switch a small console preview (**Acme Console**, a revenue card, **Get started** and **Learn more** buttons, and **Active**, **Pending**, **Failed** and **Info** chips) and list its background, surface, text, primary and accent values. The switch also changes which theme the contrast audit checks.

The generator builds three themes. The high-contrast theme is not previewed; it is included in the CSS and JSON.

Theme	Character
Light	Near-white background and white surfaces with a faint tint of the neutral's hue, dark text, the primary and accent as they are.
Dark	Dark tinted background and surfaces, near-white text, a lighter primary with near-black text on it.
High contrast	White background and surfaces, near-black text, a darker primary and accent with white text on them.

Each theme has fourteen values: background, surface, surface alt, border, text, text muted, primary, primary text, accent, accent text, success, warning, danger and info.

Tonal ramps

One ramp for each of the eight roles, eleven steps each: 50, 100, 200, 300, 400, 500, 600, 700, 800, 900 and 950. Lightness falls on a fixed OKLCH ladder from the light end to the dark end, and chroma is eased at both ends so the palest and darkest tones stay believable. The neutral ramp is kept almost grey.

Select any step to copy its HEX. Hover to see the role, the step and the value.

Contrast audit

A table for the theme you are previewing, with four columns: **Pair**, **WCAG**, **APCA** and **Use for**.

Pair	Checks
Body text	text on background
Muted text	muted text on background
Text on surface	text on surface
Primary button	primary text on primary
Accent on bg	accent on background
Primary on bg	primary on background

WCAG is the WCAG 2.1 contrast ratio. **APCA** is the APCA Lc value, and **Use for** names what that value supports:

Lc	Use for
75 or more	Body text
60 to 74	Fluent text
45 to 59	Large / bold
30 to 44	Headlines
15 to 29	Non-text only
under 15	Insufficient

Data-visualisation scales

Three strips, each with **Copy** (which copies the HEX values separated by spaces):

- **Categorical**: eight colours for separate series, chosen from the palette's colourful members to be as far apart in hue as possible, with extra hues made up if the palette has fewer than eight.
- **Sequential**: seven steps from light to dark in the hue of the most chromatic colour.
- **Diverging**: nine steps between two clearly different hues, meeting at a light neutral in the middle.

Copying the tokens

- **Copy CSS** (in the Tokens tab) copies the three themes as CSS custom properties named `--aa-background`, `--aa-surface`, `--aa-surface-alt`, `--aa-border`, `--aa-text`, `--aa-text-muted`, `--aa-primary`, `--aa-primary-text`, `--aa-accent`, `--aa-accent-text`, `--aa-success`, `--aa-warning`, `--aa-danger` and `--aa-info`:
  - the light theme under `:root`;
  - the dark theme under `:root[data-theme="dark"], .dark`;
  - the high-contrast theme under `:root` inside `@media (prefers-contrast: more)`.
- **Copy JSON** copies everything: roles, themes, ramps, the audit for both previewed themes, the data-visualisation scales, the CSS and the summary figures.
- The **Export** tab adds these tokens to its CSS as well. See Export, save and share.

The ramps and data-visualisation scales are in the JSON only, not in the CSS.

Related

- Brand Studio

- Limits and accuracy
- Glossary

WEBSITE › TOOLS › COLOUR WORKSPACE

Export, save and share

Copy CSS, download the JSON snapshot, save results to your Library, bring Library colours in, and automate the generator through the Public API.

Results live only in the open page; a reload starts a fresh sweep. Copy, download or save anything you want to keep.

The Export tab

The **Export** tab has two buttons at the top and two cards.

Control	What it does
Copy CSS (top)	Copies the CSS shown in the **CSS tokens** card.
Download JSON	Downloads the full snapshot as `auric-personalization-generator.json`.
Copy (CSS tokens card)	The same as **Copy CSS**.
Copy JSON (JSON snapshot card)	Copies the full snapshot as text.

The **JSON snapshot** card previews only the start of a long snapshot and marks it *(truncated, full file in download)*. The copy and the download are always complete.

What the CSS contains

A `:root` block with:

- `--aa-personal-1`, `--aa-personal-2` and so on: the leader's colours, in order (if there are no results yet, the first six pool colours);
- `--aa-personal-gradient`: the first gradient;
- `--aa-font-display`, `--aa-font-body` and `--aa-font-mono`: the first typography card's three font stacks;

followed by the design tokens for the light, dark and high-contrast themes, as described in Design tokens.

What the JSON contains

- `preferences`: your pool, styles, use case, brief, and the search settings (palette size, depth, keep, the three weights and the three filters);
- `results`: every palette (with its relationship, source and score breakdown), gradient, typography card, UI pair and poster, and the full token system;
- the time it was made.

There is no way to load a JSON snapshot back into the generator. Keep it as a record, or use it in your own code.

Copy single results

Every result card has its own copy buttons. See Palettes, gradients, typography, UI pairs and posters for exactly what each one copies. In short:

Where	Button	Copies
Palette card	**Copy hex**	HEX values separated by spaces
Overview	**Copy hex list**	the leader's HEX values
Any swatch or role chip	select it	that HEX
Gradient card	**Copy CSS**, **Copy stops**	a `background:` line, or the stop colours
Typography card	**Copy CSS**	five custom properties
UI pair card	**Copy CSS**	five custom properties
Poster card	**Copy CSS**	a short `.poster` rule
Tokens tab	**Copy CSS**, **Copy JSON**, scale **Copy**	the token system, or one scale

A short message confirms each copy.

Save to your Library

Save on a palette, gradient, typography, UI pair or poster card, and **Save brand system** in the Studio, store the item in your Auric Artisan Library with a preview image.

- The message reads, for example, **Saved "Balanced Analogous" to Library**.
- Saving the same card twice from the same sweep shows **Already saved** instead of making a copy.
- If the Library is not available, the message reads **Save failed**.

Saved from	Saved as	Holds
Palette card	palette	colours, relationship, source and score
Gradient card	gradient	CSS, stops, style and score
Typography card	typography	fonts, colours and score
UI pair card	preset	the four colours, accent text colour, contrast and colour-vision score
Poster card	preset	layout, colours, fonts and score
Studio	preset	palette, brand system, overrides, theme and density

The Library's own **Save to Library** captures the leader palette (or, before any sweep, your pool) and the first gradient. See the Library Kit for managing saved items.

Bring colours in from your Library

A palette, gradient, colour or shade saved in any Auric Artisan tool can be opened in the Personalization Generator from your Library. Its colours are added to the front of your pool and the **Pool** tab opens, ready for a sweep. See the Library Kit.

Automate it

The page notes the matching Public API call: **POST /v1/personalize/system**, at 14 API credits per call. API credits are a separate balance from your tool tokens. See the Public API and the API reference at https://auricartisan.com/library/documentation/api/.

Related

- Design tokens
- Brand Studio
- Privacy

WEBSITE › TOOLS › COLOUR WORKSPACE

Presets

The seventeen starting points in the Personalization Generator and what loading one changes.

The **Presets** tab holds seventeen starting points. Each has a name, five colours, two style chips and a use case.

Load a preset

1. Open **Presets**.
2. Select a preset card. A message confirms it, for example **Creator SaaS loaded**.
3. Select **Generate**.

Loading a preset:

- replaces your pool with the preset's five colours, then builds the pool up to your **Pool target** with variations of them;
- replaces your style chips with the preset's two;
- sets **Use case** to the preset's use case.

It does not change your brief, the engine settings, the weights or the filters, and it does not start a sweep.

Loading a preset discards your current pool. If you want to keep your own colours, note them first, or add them back with **Add** afterwards.

The presets

Preset	Styles	Use case
Creator SaaS	Tech, Minimal	Ui
Premium Studio	Luxe, Editorial	Brand
Festival Poster	Neon, Playful	Poster
Earth Brand	Earthy, Heritage	Brand
Calm Editorial	Calm, Editorial	Editorial
Data Console	Tech, Calm	Dashboard
Pastel Atlas	Pastel, Calm	Brand
Cyber Console	Cyber, Tech	Ui
Vintage Press	Heritage, Editorial	Editorial
Monochrome Lab	Monochrome, Minimal	Ui
Botanical Brand	Botanical, Earthy	Brand
Nordic Calm	Nordic, Minimal	Ui

Preset	Styles	Use case
Sunset Studio	Sunset, Playful	Poster
Midnight Tech	Midnight, Tech	Ui
Coastal Brand	Coastal, Calm	Brand
Editorial Plum	Editorial, Luxe	Editorial
Solstice Type	Sun, Editorial	Poster

Each card shows the preset's five colours as a strip.

Related

- Build your colour pool
- Tune the sweep

WEBSITE › TOOLS › COLOUR WORKSPACE

Palettes, gradients, typography, UI pairs and posters

What each result tab of the Personalization Generator shows, how the cards are scored, and what every card button copies or saves.

Five tabs hold the results of the last sweep. Palettes come straight from the sweep; the other four are built from the kept palettes, in rank order.

Tab	Built from	Most cards
Palettes	the sweep	up to 96
Gradients	the top 28 palettes	28
Typography	the top 24 palettes	24
UI Pairs	the top 28 palettes	28
Posters	the top 28 palettes	28

Every card has **Save**, which saves it to your Library, and **Add to pool**, which puts its colours at the front of your pool. See Export, save and share.

Palettes

The heading summarises the set, for example **24 kept · best 81 (Balanced Analogous) · avg 74 · most common Analogous**, with **Top**, **Average**, **Kept** and **Sweeps** figures.

Each card shows:

- the palette name, such as **Balanced Analogous**: a mood word (**Vivid**, **Deep**, **Soft** or **Balanced**, from the average saturation and lightness) and the colour relationship;
- a line such as **Analogous · from Soft Product**. The first part is the colour relationship the palette was built on (19 relationships, from **Analogous** to **Brand trio**; see the Reference). The second is the built-in reference palette it drew on;
- the score, from 0 to 100 (highlighted from 60 and again from 80);
- the swatches with their HEX values. Select a swatch to copy its HEX;

- tags: the number of stops, the relationship and **BG/FG** with the best contrast ratio in the palette;
- four scores: **Preference**, **Harmony**, **Contrast** and **CVD / UI**;
- three contrast bars, **AAA text**, **AA UI** and **Large text**.

The three contrast bars all use the same number, the highest WCAG ratio between any two colours in the palette, measured against 7:1, 4.5:1 and 3:1 respectively. They tell you whether the palette has a pair that clears each bar, not how every pair performs.

Buttons: **Copy hex** (the HEX values separated by spaces), **Add to pool**, **Open in Studio** and **Save**.

How a palette is scored

The total blends six parts. The card shows four of them.

Part	What it measures
Preference	How close each colour is to the nearest pool colour (CIEDE2000), with a small reward for not copying the pool exactly.
Harmony	How well the hues fit the palette's relationship, and how evenly lightness and chroma are spread.
Style	How well the colours fall inside the ranges of your style chips. Not shown on the card.
Diversity	How clearly the colours differ from one another. Weighted by the **Diversity** slider. Not shown on the card.
Contrast	The best contrast pair and the overall light-to-dark range. Weighted by the **Accessibility** slider.
CVD / UI	How distinct the colours stay under three colour-vision simulations, plus whether a pair reaches 4.5:1 (or 3:1 with **Contrast** off). Weighted by the **Accessibility** slider.

The score ranks palettes against your settings. It is not a measure of quality in general. See Limits and accuracy.

Gradients

Each kept palette in the top 28 becomes one gradient, in one of twelve styles taken in turn: **linear**, **radial**, **conic**, **mesh**, **layered**, **sharp**, **soft-mesh**, **diagonal**, **spot**, **duotone**, **prism** and **halo**. Stops are the palette's colours sorted by hue, six at most.

Filter chips above the grid (**All · 28**, **linear · 3** and so on) show one style at a time. Each card has a large preview with the style and score, the stops, the CSS, and:

- **Copy CSS**: a line such as `background: linear-gradient(…);`
- **Copy stops**: the stop colours separated by spaces
- **Add to pool** and **Save**

A gradient's score is its palette's score.

Typography

Each card pairs one palette with a type system: a display face, a body face and an accent face. The preview shows **H1**, **H2**, **Body** and **Mono** lines on the palette's most readable background and text pair, with the palette's most colourful colour as the accent, and lists the three faces. The header line gives the palette name and the **BG/FG** ratio.

The type systems offered are the ones whose tags best match your use case and style chips; when several match equally, the cards rotate through them. There are 17 systems in all, such as **Product Grotesk**, **Editorial Serif**, **Luxury Contrast** and **Console Stack**.

The fonts are named as CSS font stacks and are not downloaded by the page. If a named font is not installed on your device, the preview uses the next font in the stack. Check the faces on your own system before relying on the preview.

Buttons: **Copy CSS** copies five custom properties, `--font-display`, `--font-body`, `--color-bg`, `--color-text` and `--color-accent`; **Add to pool** adds the background, text and accent; **Save**.

UI Pairs

Each card is a small interface in the palette's colours: a title bar, **Background**, **Text**, **Surface** and **Accent** with their HEX values, a search field and three buttons (**Primary action**, **Surface button**, **Ghost button**), and a contrast meter.

- The meter and the first tag grade the text on the background: **AAA** (7:1 or more), **AA** (4.5:1), **Large** (3:1) or **Fail**.
- The header line gives **WCAG** ratio, **APCA Lc** and **CVD**, the palette's colour-vision score out of 100.
- The APCA tag also names the use the value supports, from **Body text** (Lc 75 or more) down to **Insufficient**.

The heading counts how many pairs pass WCAG AA, AAA and APCA 60 or more.

Buttons: **Copy CSS** copies `--ui-bg`, `--ui-text`, `--ui-surface`, `--ui-accent` and `--ui-accent-ink`; **Add to pool** adds the four colours; **Save**.

Posters

Each card is a poster in one of twelve layouts, taken in turn: **bauhaus**, **type-dominant**, **asymmetric**, **magazine**, **minimal-centered**, **brutalist**, **festival**, **swiss**, **collage**, **kinetic**, **archive** and **gradient-poster**. Each uses the palette's background, text and two accent colours, and a type system.

Filter chips show one layout at a time. The text on the posters is fixed sample copy.

Buttons: **Copy CSS** copies a short `.poster` rule with background, colour and font, plus `.poster-accent` and `.poster-accent-2` colours; **Add to pool** adds the text and accent colours; **Save**. There is no image download for posters. **Save** stores the poster's colours, layout and fonts in your Library, with a swatch strip of its colours as the preview, not a picture of the poster.

Related

- Design tokens
- Brand Studio
- Export, save and share

WEBSITE › TOOLS › COLOUR WORKSPACE

Run a sweep and read the results

Start, stop and repeat sweeps, follow their progress, and read The run panel and the Overview tab.

A sweep samples a set number of candidate palettes (the **Search depth**), scores every one, and keeps the best (**Keep top**). From the kept palettes the generator then builds the gradients, typography, UI pairs, posters, tokens and Studio.

Start a sweep

Do any one of these:

- select **Generate** under the brief;

- press `Enter` in the brief box;
- select **Run sweep** in **Tune**.

While it runs, **Generate** reads **Generating…**, the status line reads, for example, **Generating (320)…**, and the progress bar appears at the top of **The run** with a label such as **Sweep #3 — 320 candidates**, then **Sampling… 45%** and a line like **144/320 · 139 kept · ETA 0.4s**. The count of kept candidates there is everything that has survived so far, before the final cut.

The sweep works in small batches and lets the page redraw between them, so you can scroll, switch tabs and read results while it runs.

A new sweep replaces the previous results. To add to them instead, use **Endless**.

Stop a sweep

Select **Stop** under the brief, or **Cancel** beside the progress bar. The sweep stops at the end of the current batch and keeps the best of what it had already found. The status reads, for example, **Cancelled after 192/320 (140ms).**

Endless mode

Select **Endless** to keep sweeping. The button reads **Endless ✓** while it is on. A new sweep starts about two seconds after each one finishes, and each sweep's best palettes are merged into the existing results rather than replacing them. Duplicates are removed, and the list is capped at the 96 highest-scoring palettes.

Select **Endless** again, or **Stop**, to end it.

When it finishes

The status reads, for example, **Done · 24 kept · 812ms**. The tab counts update, and the leader (the highest-scoring palette) becomes the first card in **Palettes**, the source of the **Tokens** tab, and the palette shown in **Studio**.

The run panel

The right-hand column explains the last sweep.

Where the sweep went

Four bars:

- **sampled**: the candidates generated (the search depth);
- **repeats**: candidates thrown out because they repeated one already seen (only when **Unique** is on);
- **ranked**: the candidates that were scored;
- **kept**: those that made the cut.

A note under the bars explains that contrast and colour-vision are weights in the score, not filters: a candidate that reads badly is ranked down, never discarded.

Score distribution

A histogram of every ranked candidate's score, from the lowest on the left to the highest on the right, with the kept bars highlighted and the cut-off score marked. The line under it reads, for example, **24 of 318 ranked candidates cleared the cut.**, and says so if the sweep was cancelled.

The leader, checked

The leader's swatches, its name and score, and three WCAG contrast checks between its own colours, each against the 4.5:1 bar for body text:

- **Lightest on darkest**
- **Mid on darkest**
- **Darkest on lightest**

Each shows an *Ag* sample and the ratio. This is measured directly from the colours, so it is a quick check of whether the leader can carry text at all.

The Overview tab

Overview summarises the current results:

- five tiles: **Top score** (with the number of sweeps run), **Pool** (up to 14 swatches, the pool target and the selected styles), **Search space** (depth, keep, last sweep time and the total number of candidates sampled since the page opened), **Output** (counts of palettes, gradients, posters and type combos, and the use case) and **Avg score** (with the peak);
- **Top palette preview**: the leader's swatches (select one to copy its HEX), its relationship and source, **Copy hex list**, **Open in Studio**, **Add to pool** and **Save**, and a **Score breakdown** of **Preference**, **Harmony**, **Contrast** and **CVD / UI**;
- **Sweep diagnostics**: the average of those four scores across all kept palettes, plus **Intent**, **Mode**, **CVD-safe** and **Unique**;
- **Recent palettes**: the top six by name and score. Select one to open it in the Studio.

Filter the results

Type in **Search results** above the tabs to show only the cards in the open tab that match, for example a palette name, a HEX value, a relationship such as *Triadic*, a gradient style or a font name. **Clear** empties the box. The filter applies to the tab you are on; switching tabs applies it to the new tab.

Related

- Palettes, gradients, typography, UI pairs and posters
- Tune the sweep
- Limits and accuracy

WEBSITE › TOOLS › COLOUR WORKSPACE

Tune the sweep

Style chips, use case, the brief, engine settings, scoring weights and filters, and what each one changes in the Personalization Generator.

Open the **Tune** tab to shape the next sweep. It has four cards (**Style chips**, **Intent & note**, **Engine** and **Scoring weights**) and a **Current configuration** summary. Changes apply to the next sweep; they do not change results you already have.

Style chips

Eighteen chips: **Calm**, **Luxe**, **Playful**, **Tech**, **Earthy**, **Editorial**, **Neon**, **Minimal**, **Heritage**, **Pastel**, **Cyber**, **Sun**, **Monochrome**, **Botanical**, **Nordic**, **Sunset**, **Midnight** and **Coastal**. **Calm** and **Editorial** are on at first.

Select a chip to turn it on or off. You can combine several; at least one is always on, and turning the last one off switches **Calm** back on.

Each style describes a range of saturation, lightness and chroma, a warmth and a contrast level. With several chips on, the generator averages them. Styles affect the sweep in three ways:

- new colours are drawn towards the style's ranges;
- each palette gets a **style** score for how well its colours fall inside those ranges;
- built-in reference palettes that share your styles are drawn on more often.

Use case

Use case has five options: **Ui** (the default), **Brand**, **Poster**, **Editorial** and **Dashboard**. It favours the colour relationships and reference palettes suited to that use, and, together with your styles, chooses the type systems offered in **Typography** and the Studio.

The brief and the preference note

The brief box in the left column and **Preference note** in Tune are the same setting: typing in either one sets it.

The generator does not read the brief as a sentence. It splits it into words and compares each word with the names of about 80 built-in reference palettes, such as *Clay Garden*, *Quiet Finance*, *Night Market* or *Coast Studio*. A reference palette whose name contains one of your words is drawn on more often. So *garden*, *coast*, *console*, *finance* or *studio* have an effect; words that match no name have none.

Short words such as *a* or *for* also match parts of names, so a brief of a few distinctive words steers better than a long sentence. For a real change of direction, use the pool, the style chips and the use case.

Engine settings

Setting	Range	Default	What it does
Pool target	5 to 240	64	How many colours **Build** fills the pool to.
Palette size	5 to 12	6	Colours in each generated palette.
Search depth	40 to 4,000, steps of 20	320	Candidate palettes sampled per sweep. More finds better results but takes longer.
Keep top	6 to 96	24	How many of the best palettes survive the sweep.

The same four settings appear in the left column in Advanced mode as **Depth**, **Pool**, **Size** and **Keep**. Moving a Tune slider updates those boxes. The **Keep** box accepts up to 120, but the generator never keeps more than 96.

Scoring weights

Weight	Range	Default	Effect
Diversity	0 to 100	68	How much the score rewards palettes whose colours are clearly different from one another.
Accessibility	0 to 100	72	How much the score rewards contrast and colour-vision safety.
Warmth	0 to 100	52	Pulls new colours towards warmer (higher) or cooler (lower) hues. It is averaged with your styles' own warmth.

Filters

Three check boxes, all on by default. They are in the **Scoring weights** card and, in Advanced mode, in the left column.

Tune label	Left-column label	When on
CVD-safe penalty	**CVD-safe**	Palettes whose colours become hard to tell apart under simulated protanopia, deuteranopia or tritanopia score lower. When off, every palette gets the same middling colour-vision score.
UI contrast bonus (4.5:1)	**Contrast**	Palettes with at least one colour pair of 4.5:1 or more score higher (the bar is 3:1 when off), and new candidates are built with a wider light-to-dark spread.
Strict perceptual dedupe	**Unique**	Candidates that repeat one already seen in the sweep are thrown out. With **Endless**, repeats of palettes you already have are thrown out too.

Contrast and colour-vision safety are weights, not hard filters: a palette that reads badly is ranked lower, but it can still be kept if everything else about it scores well. Only repeats are discarded.

The two sets of check boxes set the same values, but the left-column boxes only pick up a change made in Tune after a reload. Use one place consistently.

Simple and Advanced

The **Simple** and **Advanced** switch in the left column only shows or hides the search settings there. Every setting keeps working in both modes, and all of them are always in **Tune**.

Current configuration

The right-hand card in Tune summarises the next sweep: up to ten pool colours, the selected styles, **Intent**, the four engine settings, the three weights and **Search space**, a rough count of possible palettes from your pool size and palette size. It has two buttons:

- **Run sweep** starts a sweep, the same as **Generate**.
- **Reset defaults** puts the engine settings, weights and filters back to their defaults. It does not change your pool, styles, use case or brief.

Related

- Build your colour pool
- Run a sweep and read the results
- Reference

WEBSITE › TOOLS › COLOUR WORKSPACE

Personalization Generator reference

Every control, setting, range, default and list in the Personalization Generator in one place.

Address

- https://auricartisan.com/tool/personalization-gen/

- https://auricartisan.com/personalization opens the same page.

The brief column

Control	Type	Notes
Brief box	text	Same setting as **Preference note** in Tune. `Enter` starts a sweep.
Generate	button	Starts a sweep. Reads **Generating…** while one runs.
Stop	button	Stops the running sweep and keeps what it found. Also ends **Endless**.
Endless	toggle	Keeps sweeping and merging results until stopped. Reads **Endless** ✓ when on.
Simple, **Advanced**	switch	Advanced shows the search settings below. Default **Simple**.
Depth	number, 40 to 4,000	Advanced only. Same as **Search depth**.
Pool	number, 5 to 240	Advanced only. Same as **Pool target**.
Size	number, 5 to 12	Advanced only. Same as **Palette size**.
Keep	number, 6 to 120	Advanced only. Values above 96 act as 96.
CVD-safe, **Contrast**, **Unique**	check boxes	Advanced only. All on by default.
This run	summary	Depth, pool target, palette size, top kept and the filters.

Tabs

Tab	Purpose
Overview	Summary tiles, the leader with its score breakdown, sweep diagnostics, recent palettes.
Pool	Add, remove, build, randomise and nudge pool colours; pool statistics; quick seeds.
Tune	Style chips, use case, preference note, engine settings, weights, filters.
Palettes	Kept palettes, ranked.
Gradients	One gradient per top palette, twelve styles.
Typography	Type systems paired with palettes.
UI Pairs	Background, text, surface and accent sets with contrast checks.
Posters	Poster layouts in the palettes' colours.
Tokens	Roles, themes, ramps, contrast audit and chart scales from the leader.
Studio	Sixteen mockups of one palette.
Presets	Seventeen starting points.
Export	CSS and JSON.

Pool

Item	Value
Maximum pool	240 colours
Starting pool	#D3AF37, #161616, #F8F7F1, #7D8A75, #2D6CDF, #C97B84, #0F172A, #14B8A6

Item	Value
Accepted input	HEX with 3, 4, 6 or 8 digits, with or without `#`
Build	Fills or trims the pool to the pool target
+24 random	Adds 24 variations
Jitter	Nudges every colour by up to about 6° hue, 0.02 chroma and 0.03 lightness (OKLCH)
Quick seeds	**+ 4 UI**, **+ 4 Earthy**, **+ 4 Neon**, **+ 4 Pop**, **+ Grayscale** (7 greys)

Tune settings

Setting	Range	Default
Style chips	18, at least one on	Calm, Editorial
Use case	Ui, Brand, Poster, Editorial, Dashboard	Ui
Preference note	text	empty
Pool target	5 to 240	64
Palette size	5 to 12	6
Search depth	40 to 4,000, steps of 20	320
Keep top	6 to 96	24
Diversity	0 to 100	68
Accessibility	0 to 100	72
Warmth	0 to 100	52
CVD-safe penalty	on or off	on
UI contrast bonus (4.5:1)	on or off	on
Strict perceptual dedupe	on or off	on

Reset defaults restores the last ten rows. It leaves the pool, style chips, use case and note alone.

Style chips

Calm, Luxe, Playful, Tech, Earthy, Editorial, Neon, Minimal, Heritage, Pastel, Cyber, Sun, Monochrome, Botanical, Nordic, Sunset, Midnight, Coastal.

Colour relationships

Palettes are built on one of nineteen hue relationships: Analogous, Split complement, Triadic, Tetradic, Mono accent, Editorial pop, UI system, Poster clash, Natural, Aurora, Duo (warm/cool), Monochrome tonal, Complementary, Square, Hexadic, Nature spectrum, Sunset burn, Compound, Brand trio.

The **Ui** and **Dashboard** use cases favour UI system; **Poster** favours Poster clash; **Editorial** favours Editorial pop.

Palette names

A mood word and the relationship, for example *Deep Triadic*:

Mood	When
Vivid	average saturation above 70%
Deep	average lightness under 38%
Soft	average lightness above 72%
Balanced	anything else

Score parts

Part	Base weight	Grows with
Preference	26%	fixed
Harmony	20%	fixed
Style	16%	fixed
Diversity	11%	**Diversity**, up to a further 13%
Contrast	10%	**Accessibility**, up to a further 10%
CVD / UI	5%	**Accessibility**, up to a further 12%

The total is scaled to 0 to 100 and capped at 100.

Gradient styles

linear, radial, conic, mesh, layered, sharp, soft-mesh, diagonal, spot, duotone, prism, halo. Assigned in turn down the ranked palettes.

Type systems

Product Grotesk, Editorial Serif, Quiet Humanist, Poster Condensed, Luxury Contrast, Data Mono, Pastel Round, Solar Slab, Modern Sans, Magazine Serif, Geometric Sans, Console Stack, Workshop Slab, Botanic Type, Nordic Sans, Vapor Display, Coast Casual.

Poster layouts

bauhaus, type-dominant, asymmetric, magazine, minimal-centered, brutalist, festival, swiss, collage, kinetic, archive, gradient-poster. Assigned in turn down the ranked palettes.

Studio

Control	Options	Default
Theme	Auto, Light, Dark	Auto

Control	Options	Default
Density	Cozy, Compact	Cozy
Test	on or off	off
Inspect	on or off	off
Palette switcher	up to 24 palettes	the leader

Mockups: Brand identity card, Product · Dashboard, Marketing · Landing, Component library, Type scale, Charts, Pricing, Auth · Sign in, Messages, Mobile · App, Calendar, Music · Player, Editorial · Magazine spread, Marketing banner, Transactional · Email, Commerce · Product card.

Tokens

Item	Value
Roles	primary, secondary, accent, neutral, success, warning, danger, info
Ramp steps	50, 100, 200, 300, 400, 500, 600, 700, 800, 900, 950 (88 tones in all)
Themes	light, dark, high contrast (14 values each)
Audit pairs	Body text, Muted text, Text on surface, Primary button, Accent on bg, Primary on bg
Chart scales	Categorical (8), Sequential (7), Diverging (9)
CSS prefix	`--aa-`

Result limits

Result	Most
Palettes kept	96
Gradients, UI pairs, posters	28
Typography cards	24
Studio palette switcher	24

Messages

Message	Meaning
Idle.	No sweep has run since the page opened.
Generating (320)…	A sweep of 320 candidates is running.
Done · 24 kept · 812ms	The sweep finished.
Cancelled after 192/320 (140ms).	The sweep was stopped; results so far are kept.
Use a valid HEX color	The HEX box could not be read.
Pool is empty	**Jitter** was selected with no pool colours.
Defaults restored	**Reset defaults** was selected.
Nothing to save yet	**Save brand system** was selected with no palettes.

Message	Meaning
Already saved "…"	That item from this sweep is already in your Library.
Save failed	The Library could not store the item.

Related

- Tune the sweep
- Glossary
- Limits and accuracy

WEBSITE › TOOLS › COLOUR WORKSPACE

Troubleshooting

Fixes for common problems in the Personalization Generator, from lost results to unreadable palettes.

My results disappeared after a reload

Results are kept only while the page is open. On every visit a quick new sweep runs, using your saved pool and settings. Before you leave, copy what you need, select **Download JSON** in **Export**, or **Save** the results you want to your Library. See Export, save and share.

The brief box is empty after a reload, but the results still follow my old brief

The brief is remembered, but the box in the left column starts empty. The remembered text is still used, and it shows in **Tune** under **Preference note**. Edit or clear it there, or type a new brief.

Changing the brief does not change much

The brief is matched word by word against the names of built-in reference palettes; it is not read as a sentence. Words that appear in no name have no effect. Use a few distinctive words, and steer with the pool, style chips and **Use case** for bigger changes. See Tune the sweep.

Every palette looks like my pool

Preference, closeness to your pool, carries the largest share of the score.

- Select **Jitter** once or twice in **Pool**.
- Raise **Diversity** in **Tune**.
- Remove near-duplicates from the pool.
- Use **+24 random** or **Build** to widen the pool.

The palettes have no readable text pair

- Make sure **Contrast** (**UI contrast bonus (4.5:1)**) is on.
- Raise **Accessibility** in **Tune**.
- Add a very dark and a very light colour, or select **+ Grayscale** in **Quick seeds**.

Contrast is a weight, not a filter, so a palette with weak contrast can still be kept if the rest of its score is high. Check **The leader, checked** in the right column after each sweep.

Generate does nothing

A sweep is already running; **Generate** reads **Generating…** until it ends. Wait, or select **Stop**. If **Endless** is on, each new sweep starts on its own; select **Endless** again to turn it off.

A big sweep is slow

Sweeps of several thousand candidates take longer, especially on older devices. Lower **Search depth**, or select **Stop** at any time; the best of what the sweep has already found is kept.

Endless mode shows the same kind of palette over and over

Endless merges each sweep into the list and keeps only the 96 best, so strong early palettes stay at the top. Change the pool or styles and run a normal sweep with **Generate** to start afresh.

The Studio shows colours that are not in the palette

The **Test** overrides from an earlier palette are still set: a new sweep or a reload keeps them. Open **Test** and select **Reset**, or choose a palette from **Available Palettes**, which clears them.

The check boxes in the left column disagree with Tune

The left-column boxes pick up changes made in **Tune** only after a reload. The Tune settings are the ones in force. Use one place for the filters, or reload the page to bring the two into line.

This run shows old numbers

This run refreshes when you change a setting in the left column. After moving a slider in **Tune**, it can show the previous value until you next touch the left column. The Tune slider value is the one in force.

The fonts in Typography or the Studio look wrong

The page does not download the named fonts. If a font is not installed on your device, the browser uses the next one in the stack, often a system sans or serif. Install the font, or read the three faces listed on each card.

Save says Already saved

That card from the same sweep is already in your Library. Run a new sweep to save a new copy, or open your Library to find the saved item.

Save says Save failed

The Library could not store the item. Reload the page and try again. If it keeps failing, see the Library Kit.

Loading a preset lost my colours

Loading a preset replaces the pool. There is no undo. Add your colours back with **Add**, or from a palette you saved to your Library.

Related

- FAQ
- Limits and accuracy
- Help centre

WEBSITE › TOOLS › COLOUR WORKSPACE

Frequently asked questions

Short answers to common questions about the Personalization Generator.

Do I need an account?

No. The generator runs in your browser without signing in. Saving to your Library uses your Auric Artisan Library.

Is the brief read by an AI?

No. The brief is split into words, and each word is compared with the names of built-in reference palettes. A match makes that reference palette more likely to be drawn on. Your pool, style chips and use case steer the sweep far more. See Tune the sweep.

Where do the generated colours come from?

From your pool, from built-in reference palettes that match your styles, use case and brief, and from small variations of both, arranged on a colour relationship such as Analogous or Triadic. Every candidate is then scored and only the best are kept.

Will I get the same results if I run the same settings again?

Usually not. Each sweep starts from a different random seed, so results vary from run to run. Save or download the results you like.

Are my results saved?

Your pool and settings are remembered in your browser. The results are not: each visit starts with a fresh quick sweep. See Export, save and share.

Can I load a JSON snapshot back in?

No. The JSON is a record of your settings and results for your own use. To carry a palette back in, save it to your Library and open it in the generator, or add its HEX values to the pool.

What does the score mean?

It is how well a palette fits your settings: closeness to your pool, harmony, fit with your styles, variety, contrast and colour-vision safety, weighted by your sliders. A score of 85 under one set of settings is not comparable with 85 under another. See Palettes, gradients, typography, UI pairs and posters.

Is a high-scoring palette guaranteed to pass WCAG?

No. Contrast is part of the score, not a requirement. Check the contrast bars on each palette card, **The leader, checked**, the **UI Pairs** grades and the **Tokens** audit, and test your final colour pairs in your own designs.

Are the design tokens ready to use in code?

They are ordinary CSS custom properties and JSON, and you can paste them into a project. Review them first: the roles are chosen automatically, and the status colours (success, warning, danger, info) are made up when the palette has no suitable colour.

Can I use my own fonts in the Typography tab or the Studio?

No. The type systems are fixed. The page shows them with the fonts installed on your device.

Can I export the posters or mockups as images?

No. You can copy their CSS and save their settings to your Library.

Can I automate the generator?

Yes. The Public API offers the same generation as **POST /v1/personalize/system**, at 14 API credits per call. See the Public API.

Does it work on a phone?

Yes. The columns stack on narrow screens. Large sweeps take longer on slower devices; lower **Search depth** if it feels slow.

Related

- Troubleshooting
- Getting started

WEBSITE › TOOLS › COLOUR WORKSPACE

Supporting material

The glossary, limits, related tools and privacy notes for the Personalization Generator.

File	What it covers
glossary.md	Terms used by the generator.
limits-and-accuracy.md	What the scores, checks and previews can and cannot tell you.
related.md	Tools, services and reading that go with the generator.
privacy.md	What stays in your browser and what is saved.

The only keyboard shortcut specific to the generator is `Enter` in the brief box, which starts a sweep.

Related

- Personalization Generator overview
- Documentation contents

WEBSITE › TOOLS › COLOUR WORKSPACE

Glossary

Terms used in the Personalization Generator and its documentation.

Term	Meaning
Accent	The most colourful colour in a palette once the background and text are chosen. Accent 2 is the next most colourful.
APCA Lc	A contrast value from the Accessible Perceptual Contrast Algorithm. Higher means easier to read. The generator shows its size without the sign.
Brief	The text box at the top of the left column; the same setting as Preference note in Tune. Its words are matched against reference palette names.
Candidate	One palette generated during a sweep, before it is scored.
CIEDE2000	A formula for the perceived difference between two colours.
CVD	Colour vision deficiency. The generator simulates protanopia, deuteranopia and tritanopia.
Endless	A mode that runs sweep after sweep and merges the results until stopped.
Leader	The highest-scoring palette of the last sweep. It feeds the Tokens tab and opens first in the Studio.
OKLCH	A colour model of lightness, chroma and hue in which equal steps look roughly equal.
Pool	Your own set of colours, up to 240. Palettes are scored on how close they stay to it.
Pool target	The size Build fills the pool to.
Preset	A starting point of five colours, two style chips and a use case.
Reference palette	One of about 80 built-in named palettes the generator draws on, such as Clay Garden.
Relationship	The hue pattern a palette is built on, such as Analogous or Triadic.
Semantic role	A named job for a colour: primary, secondary, accent, neutral, success, warning, danger or info.
Style chip	A mood word, such as Calm or Neon, with preferred ranges of saturation, lightness, chroma, warmth and contrast.

Term	Meaning
Sweep	One run: sample candidates, score them, keep the best.
Tonal ramp	One colour from very light (50) to very dark (950) in eleven steps.
Use case	What the system is for: Ui, Brand, Poster, Editorial or Dashboard.
WCAG contrast ratio	The Web Content Accessibility Guidelines contrast measure, from 1:1 to 21:1. Body text needs 4.5:1 for AA and 7:1 for AAA; large text 3:1 for AA.

Related

- Reference
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Limits and accuracy

What the Personalization Generator's scores, contrast checks, colour-vision checks and previews can and cannot tell you.

The score is a ranking, not a verdict

A palette's score says how well it fits your settings: your pool, style chips, use case and weights. It is useful for sorting one sweep's candidates. It does not measure whether a palette is good in general, and scores from sweeps with different settings are not comparable.

The brief is keyword matching

The brief is not understood as language. Its words are compared with the names of about 80 built-in reference palettes, and even short words such as *a* can match part of a name. The pool, style chips and use case have a much larger effect.

Contrast and colour-vision are weights, not guarantees

- A palette with weak contrast or poor colour-vision separation is ranked lower, not removed. It can still be kept.
- The three contrast bars on a palette card all show the single best pair in the palette. A palette can have one strong pair and many weak ones.
- The colour-vision check simulates protanopia, deuteranopia and tritanopia. It does not model partial forms (such as deuteranomaly) or achromatopsia, and simulations only approximate what any one person sees.
- With **CVD-safe** off, every palette gets the same fixed colour-vision score.

Always check your final text and background pairs in context, with the WCAG and APCA figures in **UI Pairs** and **Tokens**, and with tools built for accessibility review.

WCAG and APCA

WCAG ratios follow WCAG 2.1. APCA values are shown as the size of Lc, and the **Use for** labels are general guidance, not a formal conformance result. APCA is not part of WCAG 2.

Tokens are generated, not designed

Roles are picked by rules (most colourful, least colourful, furthest hue). Status colours are made up near fixed green, amber, red and blue hues when the palette has none. Review every role before using the tokens in a product.

Fonts are not downloaded

Typography cards, posters and the Studio name font stacks such as *'Playfair Display', Garamond, Georgia, serif*. The page does not load those fonts, so you see whichever font in the stack is installed on your device. Other people may see something different.

Results are not reproducible

Each sweep uses a new random seed, so the same settings give different palettes each time. Save or download results you want to keep.

Size and speed limits

Limit	Value
Pool	240 colours
Candidates per sweep	40 to 4,000
Palettes kept	96 at most, even if **Keep** is set higher
Colours per palette	5 to 12
Derived gradients, UI pairs, posters	top 28 palettes
Derived typography	top 24 palettes

Large sweeps run in small batches so the page stays usable, but they take longer on slower devices.

Mockups and posters are illustrations

Studio mockups and posters show how colours sit together on typical layouts. They are not working components, and their text is sample copy.

Related

- Tune the sweep
- Design tokens
- Supporting material

WEBSITE › TOOLS › COLOUR WORKSPACE

Privacy

What the Personalization Generator keeps in your browser and what uses the network.

Stays in your browser

Every sweep, score, token system and mockup is computed in the page. Your colours and your brief are not sent anywhere to generate results. The JSON download is created in your browser.

Remembered in your browser

In your browser's local storage on this device: your pool, style chips, use case, brief, engine settings, weights and filters, the open tab, Simple or Advanced mode, and the Studio's palette choice, theme, density, **Test** and **Inspect** settings and overrides.

Results are not stored. Clearing your browser's site data resets the generator to its starting pool and defaults.

Like other pages on the site, the page may also keep the values of some on-page controls in the page address after `#p=`, so a reload or a shared link restores them. See Shareable links.

Uses the network

- Loading the page and its scripts.
- **Save** and **Save brand system** store items in your Auric Artisan Library. See the Library Kit.
- The Public API, if you choose to use it, runs on Auric Artisan's servers. See the Public API.
- Site-wide analytics and advertising follow your cookie choices. See the Help centre.

Related

- Limits and accuracy
- Export, save and share

WEBSITE › TOOLS › COLOUR WORKSPACE

Related tools and reading

Tools, services and Learn articles that go with the Personalization Generator.

Tools

- Design System Generator: build a full design system from a brief, with more control over ramps and token export.
- Basic Color Tools: palettes, gradients, a colour picker and the Colour Tools workbench.
- Color Science Lab: look closely at any colour the generator produces.
- Contrast Checker: test a final text and background pair.
- Vision Lab: see a palette through colour-vision and other vision conditions.

Library

- Colour libraries: browse palettes, gradients and shades, save them, and open them here to seed your pool.
- Library Kit: find and manage the items you saved from the generator.

Services

- Public API: the same generation as a REST call.

Learn articles

- Color Theory and Harmony
- Building a Color System
- Color in Branding and Identity
- Accessible Color Design and WCAG Contrast
- Color Blindness and Color Vision Deficiency
- Color in Typography and Text
- Color in Data Visualization

Related

- Supporting material

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Colorimetry and rendering tools

What the colorimetry, gamut and rendering tools on auricartisan.com are for, where to find them, and which one to open for your question.

This area documents seven colour-science tools on auricartisan.com. They answer questions that sit underneath everyday colour work: what "white" means under a given light, how a colour changes when the light changes, which colours a screen or file format can hold, what happens to a colour that does not fit, and how a high-dynamic-range signal becomes light on a display.

The tools are split into two groups on the site:

- **Colorimetry** — measuring and converting colour with the standard CIE methods: light sources (illuminants), white points, chromatic adaptation and gamut mapping.
- **Gamut and rendering** — the limits of colour spaces and displays, HDR transfer curves, tone mapping, and ramps of tints and shades.

Every tool runs entirely in your browser. Each one has a **Data** tab that lists the published tables and formulas it uses, with a status for each (verbatim, computed, stand-in or absent), so you can see which figures are standard values and which are illustrations. None of them is a certified measurement instrument.

Where to find them

The tools are listed in the **Colorimetry** and **Gamut & Rendering** categories of the site's tool collection at https://auricartisan.com/collections/.

The three hub addresses below are kept for old links. Each one now sends you to the collection page:

Address	Where it goes
https://auricartisan.com/tool/general/	The tool collection
https://auricartisan.com/tool/general/colorimetry/	The tool collection
https://auricartisan.com/tool/general/gamut-and-rendering/	The tool collection

The tools

Tool	Address	Use it to
Illuminants	https://auricartisan.com/tool/general/colorimetry/illuminants/	Inspect standard light sources: spectra, white points, CCT and Duv, chromaticity, adaptation and measurement uncertainty
Chromatic Adaptation	https://auricartisan.com/tool/general/colorimetry/chromatic-adaptation-dynamic/	Apply eight adaptation transforms to an image and study how the eye adapts over time
Gamut Mapping	https://auricartisan.com/tool/general/colorimetry/gamut-map/	See what a colour becomes when the target colour space cannot hold it
Gamut Lab	https://auricartisan.com/tool/general/gamut-and-rendering/gamut/	Draw and measure 13 RGB colour spaces (or your own) and their transfer curves
HDR Gamut Plot	https://auricartisan.com/tool/general/gamut-and-rendering/hdr-gamut-plot/	Compare Rec.709, Display P3 and Rec.2020 and explore PQ, HLG, ICtCp and Jzazbz

Tool	Address	Use it to
Tonal Steps	https://auricartisan.com/tool/general/gamut-and-rendering/tonal-steps-tints-shades/	Build a lightness ramp, tints, shades and tones from one colour and check contrast
Tone Mapping	https://auricartisan.com/tool/general/gamut-and-rendering/tone-mapping/	Compare tone-mapping operators that turn HDR values into displayable ones

Which tool do I need?

- "What colour is this light, and is it close to daylight?" — **Illuminants**.
- "My photo was lit by tungsten; what does it look like adapted to daylight?" — **Chromatic Adaptation**.
- "This brand colour is outside sRGB. What will it look like after conversion?" — **Gamut Mapping**.
- "How much bigger is Display P3 than sRGB, and what curve does Rec.2020 use?" — **Gamut Lab**.
- "What does a PQ signal of 0.5 mean in nits?" — **HDR Gamut Plot**.
- "I need an 11-step ramp with known contrast against white." — **Tonal Steps**.
- "Which tone curve keeps my highlights?" — **Tone Mapping**.

Concepts in one page

- **Illuminant** — a light source defined by its spectral power distribution (how much power it emits at each wavelength). CIE D65 stands for average daylight; CIE A for a tungsten lamp.
- **White point** — the colour of an illuminant, usually written as chromaticity coordinates x, y. A colour space is defined against one (D65 for sRGB, D50 for ProPhoto RGB).
- **Chromaticity diagram** — a flat map of colour without brightness. The CIE 1931 xy diagram is the classic one; the CIE 1976 u′v′ diagram spaces colours more evenly.
- **Gamut** — the set of colours a device or colour space can represent. On a chromaticity diagram an RGB gamut is a triangle.
- **Chromatic adaptation** — the eye's adjustment to the colour of the light. A chromatic adaptation transform (CAT) predicts it numerically.
- **Transfer function** — the curve that converts between stored code values and light (for example the sRGB curve, PQ or HLG).
- **Tone mapping** — compressing a wide range of scene brightness into what a display can show.

Each tool folder has its own glossary with fuller definitions.

In this folder

Folder	What it covers
illuminants/	The Standard Illuminants lab
chromatic-adaptation/	The Chromatic Adaptation lab
gamut-map/	The Gamut Mapping lab
gamut-lab/	The Gamut Lab
hdr-gamut-plot/	The HDR Gamut Plot
tonal-steps/	Tonal Steps, Tints and Shades
tone-mapping/	The Tone Mapping lab

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Chromatic Adaptation

What the Chromatic Adaptation lab does, who it is for, where to find it and how to adapt your first image.

When you walk from daylight into a room lit by a tungsten lamp, white paper looks orange for a moment and then white again. Your eyes have rebalanced to the new light. This is **chromatic adaptation**, and colour science models it with **chromatic adaptation transforms (CATs)** that convert a colour seen under one white point into the colour that would look the same under another.

The Chromatic Adaptation lab applies eight published transforms (Bradford, Von Kries, CAT02, CAT16, Sharp, CMCCAT2000, HPE and plain XYZ scaling) to a photograph of your own or to a built-in test sweep, between any two of 12 standard illuminants. It also models the time the eye takes to adapt, using the two-phase time course measured by Fairchild and Reniff (1995): about half the adaptation happens within a second and the rest over tens of seconds. You can scrub or animate that timeline, and set how complete the adaptation becomes.

Supporting views show the matrices behind each transform (with copy-ready code), compare all eight on your illuminant pair, list every dataset the lab uses and its status, and export the result as an image, JSON or a shareable link. Everything runs on your device; images are never uploaded.

Who it is for

- Photographers and designers who want to see how a white-balance change affects colours.
- Developers implementing colour management who need a CAT matrix in NumPy, JSON, CSS or GLSL form.
- Students of colour science learning von Kries adaptation and the differences between transforms.
- Researchers who want a quick visual demonstration of the adaptation time course.

Where to find it

- Address: https://auricartisan.com/tool/general/colorimetry/chromatic-adaptation-dynamic/
- In the tool collection (https://auricartisan.com/collections/) under **Colorimetry**, as **Chromatic Adaptation**.
- No account is needed. It works in current desktop and mobile browsers.

Quick start

1. Open https://auricartisan.com/tool/general/colorimetry/chromatic-adaptation-dynamic/.
2. Select **Generate the sample sweep** (or **Choose an image** to use your own photo).
3. Leave **Source illuminant** on D65 and set **Destination illuminant** to **CIE A (2856 K) — tungsten**.
4. Keep **CAT Method** on **Bradford (Lam 1985)**.
5. Drag **Time (scrubber)** from 0 to 5 s and watch the adapted image warm up as adaptation progresses.
6. Select **Animate** to play the five seconds in real time.
7. Open the **Matrix** tab to see the adaptation matrix, then **Export** and select **Export Frame (PNG)**.

What you can do

- Adapt an image between 12 illuminants: D65, D50, D55, D60, D75, A, B, C, E, F2, F7 and F11.
- Choose among eight chromatic adaptation transforms.
- Set the final degree of adaptation D from 0 (none) to 1 (complete).
- Scrub or animate a five-second adaptation timeline, and explore the full two-minute time course on the Dynamics tab.

- Inspect the RGB histogram, the adaptation curve, cone (LMS) channel scaling and the two illuminant spectra.
- View and copy the adaptation matrix in NumPy, JSON, CSS matrix3d or GLSL form, with a round-trip check.
- Compare what all eight transforms do to your illuminant pair.
- Adapt a list of up to 50 HEX colours and read their ΔE_{00} and LCh°.
- Export JSON, a PNG frame, or a share link.

In this folder

- docs/ — how to use the lab
  - Getting started
  - Adapt an image to a new light
  - Explore the adaptation time course
  - Compare transforms and copy a matrix
  - Adapt a list of colours
  - Reference
  - Troubleshooting
  - FAQ
- others/ — background and supporting pages
  - Glossary
  - Shortcuts
  - Limits and accuracy
  - Related tools
  - Release notes
  - Privacy

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Chromatic Adaptation documentation

The guides, reference and help pages for the Chromatic Adaptation lab.

This folder explains how to use the Chromatic Adaptation lab: a screen tour, task guides for images, the time course, matrices and colour lists, a complete reference and help pages.

File	What it covers
getting-started.md	Opening the lab, the six tabs and your first adapted image
adapt-an-image.md	Adapting a photo between two illuminants and exporting it
explore-the-time-course.md	The Dynamics tab and how adaptation unfolds over time
compare-transforms-and-copy-matrices.md	The Matrix tab, the eight-transform comparison and copying code
adapt-a-colour-list.md	Batch Colour Analysis for HEX lists
reference.md	Every control, readout and export
troubleshooting.md	Symptoms, causes and fixes
faq.md	Short answers

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Getting started with Chromatic Adaptation

Open the Chromatic Adaptation lab, learn its tabs and controls, and adapt your first image.

Open the lab

Go to https://auricartisan.com/tool/general/colorimetry/chromatic-adaptation-dynamic/. The page heading is **Chromatic Adaptation**. You do not need an account. Images you load are processed in your browser and are never uploaded.

Key ideas

- A **white point** is the colour of the light a scene is viewed under, such as D65 (average daylight) or A (tungsten).
- A **chromatic adaptation transform (CAT)** predicts what a colour seen under the **source** white point would have to become to look the same under the **destination** white point. It works by converting XYZ into a cone-like space (L, M, S), scaling each channel by the ratio of the two whites, and converting back. The transforms differ in the cone-like matrix they use.
- The **degree of adaptation D** says how complete the adaptation is: 0 means none, 1 means the source white maps exactly onto the destination white.
- Adaptation takes **time**. In this lab, the adaptation at a moment t is the starting state moved towards D along a measured two-phase curve (fast phase with a 1 s time constant carrying half the change, slow phase with a 30 s half-life carrying the rest).

A tour of the screen

A note at the top says every figure names its source: which table a white point came from, which paper a matrix is published in, and whether a spectrum is tabulated or reconstructed.

Tab	What it is for
Lab	Load an image, choose the illuminants, transform and timing, and see the result
Dynamics	The adaptation time course over two minutes, with adjustable fast and slow phases
Matrix	The matrices behind the chosen transform, a round-trip check, code to copy, and all eight transforms compared
Data	The register of the 11 datasets the lab uses, with their status (the tab shows the count)
Export	JSON and PNG export, links, the all-methods comparison and batch colour analysis
Reference	Standards, formulas, citations and research notes with a search filter

The Lab tab

The left panel holds the controls: **Image Source**, **Illuminants**, **CAT Method**, **Display Gamma**, **Degree of adaptation D**, **Per-Pixel Modulation**, **Spatial Adaptation**, **Timeline** and **Options**, then the **Run**, **Animate**, **Pause** and **Reset** buttons.

The right side shows **Adaptation Analysis** with two status chips (what the lab is doing, and the model with the current time and D), then:

- **Original Image** and **Adapted Image**;
- **RGB Histogram (Adapted)**;
- **Adaptation Curve A(t)** — adaptation over the 0–5 s timeline with the current time marked;
- **Per-Pixel Adaptation State Map** — white is fully adapted, black is unadapted;
- **LMS Cone-Channel Analysis** — the L, M and S responses for source (light) and destination (dark);
- **SPD Spectral Overlay (Source vs Dest)** — the two illuminants' spectra, where the lab holds them.

Until you load an image, an empty-state panel covers the image area and offers **Generate the sample sweep** and **Choose an image**.

Adapt your first image

1. Select **Generate the sample sweep**. A 360 × 240 test image of hues and lightnesses appears, and the status chip reads "Sample loaded (360×240)".
2. Under **Illuminants**, keep **Source illuminant** as **CIE D65 (6504 K) — sRGB daylight** and set **Destination illuminant** to **CIE A (2856 K) — tungsten**.
3. Keep **CAT Method** on **Bradford (Lam 1985)** and **Degree of adaptation D** at 1.00.
4. Drag **Time (scrubber)** to 5 s. With **Live** ticked, the adapted image updates as you drag. It becomes warmer as the model adapts towards tungsten.
5. Select **Reset** to return to 0 s, then **Animate** to play the five seconds. Select **Pause** to stop at any moment.
6. Look at the **LMS Cone-Channel Analysis**: the short-wavelength (S) channel changes the most between D65 and A.
7. Look at **SPD Spectral Overlay** to see the D65 and A spectra together.

Why the image is not fully adapted after five seconds

The Lab's timeline runs from 0 to 5 s and starts from a small initial adaptation (0.1). With the measured two-phase time course, the fast phase is nearly complete by 5 s but the slow phase has barely begun, so the applied D reaches roughly 0.6 of the way to the value you set. The status chip always shows the D in use. To see the whole curve over two minutes, open the **Dynamics** tab (Explore the adaptation time course).

Next steps

- Adapt an image to a new light
- Compare transforms and copy a matrix
- Adapt a list of colours

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Adapt a list of colours

Paste up to 50 HEX colours and get their adapted values, colour shift and L*C*h° at the current adaptation state.

Batch Colour Analysis on the **Export** tab applies the current adaptation to a list of colours — useful for checking how a brand palette shifts between D65 and D50, or under tungsten light.

Steps

1. On the **Lab** tab, set the source and destination illuminants, the CAT method, the degree of adaptation D and the time. The batch uses the adaptation applied at the current moment of the timeline, exactly as the image does.
2. Open the **Export** tab and find **Batch Colour Analysis**.
3. Paste HEX colours into the box, one per line. Each must have six digits; the `#` is optional. Up to 50 lines are used; any more are ignored.
4. Select **Run Batch**. A message confirms how many colours were analysed.
5. Read the table:

Column	Meaning
Color	Your input colour and its swatch
Adapted	The adapted colour as an sRGB HEX, clipped to the displayable range
ΔE_{00}	The CIEDE2000 difference between the colour's CIELAB value under the source white and the adapted colour's CIELAB value under the destination white. Green below 1, amber below 3, red from 3 up
L*	Lightness of the adapted colour
C*	Chroma of the adapted colour
h°	Hue angle of the adapted colour

L*, C* and h° are computed against the destination white point.

Tips

- Lines that are not six-digit HEX values (for example three-digit shorthand such as `#abc`) are skipped. If nothing valid is found, a message asks for valid HEX values.
- A low ΔE_{00} means the transform keeps the colour's appearance almost unchanged relative to each white; a high one means the adapted value moved a lot.
- The table is not exported as a file. Select and copy it from the page if you need it elsewhere, or use **Export JSON** for the matrix that produced it.

Result

You have each colour's adapted HEX, its shift in ΔE_{00} and its lightness, chroma and hue under the destination light.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Adapt an image to a new light

Load a photo, adapt it between two illuminants with a chosen transform and degree of adaptation, and export the frame.

Use this to see how the colours of a picture shift when it is re-expressed for a different light: for example a D65 photo prepared for a D50 print booth, or a daylight scene shown as it would appear once the eye has adapted to tungsten.

Steps

1. Open the **Lab** tab.
2. Under **Image Source**, select **Choose image** and pick a PNG, JPEG or other image file your browser can open. The file name appears beside the button. Large images are scaled down to fit 480 × 320 pixels for processing; small images are not enlarged. The status chip shows the size used, for example "Image loaded (480×320)".
3. Under **Illuminants**, set **Source illuminant** to the light the picture was taken or balanced for. Most camera and web images are D65.
4. Set **Destination illuminant** to the light you want to adapt to. Use **Swap** to exchange the two.
5. Choose a **CAT Method**. Bradford is the common choice in colour management; CAT16 is the most recent model.
6. Set **Degree of adaptation D**. 1.00 means complete adaptation; lower values leave part of the source cast in place, as happens when the eye does not fully adapt.
7. Set the moment in time with **Time (scrubber)** (0 to 5 s), or select **Animate** to play it.
8. If bright colours clip harshly after a large shift (for example D65 to A), tick **HDR Tonemap**. It compresses values above white with a Reinhard curve (c ÷ (1 + c) on each channel in linear light) before the image is clipped to the display range.
9. With **Live** ticked, every change re-renders immediately. If you untick Live, select **Run** after each change.
10. Open **Export** and select **Export Frame (PNG)** to download the adapted image.

What each result shows

Result	Read it as
Adapted Image	The picture with the adaptation at the current time applied
RGB Histogram (Adapted)	How the red, green and blue values of the adapted image are distributed
Adaptation Curve A(t)	The applied adaptation over 0–5 s, with the current time and value marked
Per-Pixel Adaptation State Map	A grey strip whose brightness is the adaptation in use (white = fully adapted)
LMS Cone-Channel Analysis	The L, M and S response to the source white (light bars) and the destination white (dark bars), with the scale factor under each
SPD Spectral Overlay	The spectra of the two illuminants, with a line saying how each was obtained

For B, C, F2, F7 and F11 the lab holds the white point but not the spectrum. The overlay then says "No spectrum held for …" and explains that the adaptation is unaffected, because only the white point is needed to adapt.

Choosing D and time

- The adaptation applied at time t starts from an initial state of 0.1 and moves towards your D along the measured time course. After 5 s it has covered about 55% of that distance.
- To see the fully adapted result, you would need a later time than the Lab timeline offers. Use the **Dynamics** tab to read when adaptation reaches 50% and 90%, and use the Matrix tab, which shows the matrix at the D you set.

Tips

- The share link on the **Export** tab stores the illuminants, method, D and time, but not the image. Send the image separately.
- **Copy Link** copies the page's current address, which only includes settings if the page was opened from a share link.
- Keyboard: `Space` plays or pauses and `R` resets the time, when the focus is not in a text field. See Shortcuts.

Result

You have an adapted version of your image at a chosen moment of adaptation, plus a PNG of it.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Compare transforms and copy a matrix

Read the adaptation matrix, check its round trip, copy it as code, and compare all eight transforms on your illuminant pair.

The maths in one line

Every transform in the lab has the same shape:

```
X′ = M⁻¹ · diag(d) · M · X
d = 1 + D × (ρ_destination ÷ ρ_source − 1)   for each of L, M, S
```

M converts XYZ into a cone-like space. diag(d) scales each cone channel by the ratio of the destination white to the source white, softened by the degree of adaptation D. M^{-1} converts back. The transforms differ only in M.

The eight transforms

Transform	Source of the matrix	Notes
Bradford	Lam 1985	The usual choice in ICC colour management; the reference the comparison is measured against
Von Kries	Hunt–Pointer–Estévez (Estévez 1979), normalised to D65	Reads the same as HPE, because normalising rows cancels out in the formula
CAT02	CIE 159:2004 (CIECAM02)	—
CAT16	Li et al. 2017 (CAM16)	—
Sharp	Finlayson and Süsstrunk 2000	A spectrally sharpened matrix
CMCCAT2000	Li, Luo, Rigg and Hunt 2002	—
HPE	Hunt–Pointer–Estévez	—

Transform	Source of the matrix	Notes
XYZ scaling	Identity matrix	The null case: scales X, Y and Z directly

Read and copy the matrix

1. Choose the illuminants and transform on the **Lab** tab, and set **Degree of adaptation D**.
2. Open the **Matrix** tab. The title names the transform, the pair and the D, for example "Bradford, D65 → D50 at D = 1.00".
3. Read the three blocks:
   - **M — XYZ to cone-like**: the transform's own matrix;
   - **diag(d)**: the three cone gains, with the formula beneath;
   - **Full adaptation matrix**: the single 3 × 3 matrix that takes XYZ under the source white to XYZ under the destination white.
4. Under **Decimals**, choose **4**, **6** (default) or **9**. Published matrices carry four to six figures; nine shows the round-off residue, which is useful when checking an inverse.
5. Check **Round trip**: the largest error of $M^{-1} \times M$ against the identity matrix, recomputed every time.
6. Under **Copy as**, choose **NumPy**, **JSON**, **CSS matrix3d** or **GLSL mat3**. The transform's M matrix is copied to your clipboard in that form.

What each format contains:

Format	Content
NumPy	`import numpy as np` and `M = np.array([...])` with the three rows
JSON	An object with the transform name, its citation and M as three rows
CSS matrix3d	A `matrix3d(...)` value with M in the upper-left 3 × 3 and the rest of a 4 × 4 identity
GLSL mat3	`const mat3 M = mat3(...)` in column-major order, as GLSL expects

The copy buttons copy M, not the full adaptation matrix. The full matrix for the current moment is in the JSON export on the **Export** tab.

Compare all eight on your pair

The **All eight on this pair** list on the Matrix tab shows what each transform does, not which is more accurate. For each transform:

Column	Meaning
Transform	The name; the one selected in the Lab is highlighted
ΔE vs Bradford	Mean CIEDE2000 difference between this transform's result and Bradford's over a fixed 12-colour probe
cond(M)	The matrix's condition number: how much it can stretch a difference. A description, not a quality score
Source	The paper the matrix comes from
Status	**Published**, or **Null case** for XYZ scaling

Rows are sorted by ΔE vs Bradford, smallest first. Ranking transforms by accuracy would need the published corresponding-colour data sets, which the lab does not hold (see the Data tab).

The Compare All Methods panel

On the **Export** tab, **Compare All Methods** runs all eight transforms at the current moment of the Lab timeline over eight test colours (#735244, #C29682, #627A9D, #D3AF37, #FF6347, #3BB273, #FFFFFF and #000000). For each it reports the **Mean ΔE_{00}**, **Median**, **Min** and **Max** colour shift between the colour under the source white and the adapted colour under the destination white, plus **det(M)** and **||M||F** of the full adaptation matrix. A bar chart shows the means, and a star marks the transform with the smallest mean shift. That is the smallest change on these eight colours, not proof of the most accurate transform.

Result

You have the exact matrix for your adaptation, verified by its round trip, in the code format you need, and a picture of how the eight transforms differ on your pair.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Explore the adaptation time course

Use the Dynamics tab to see how quickly the eye adapts to a new light, with the two-phase model and its single-exponential predecessor side by side.

Adaptation is not instant. Fairchild and Reniff (1995) measured it for colour-appearance judgements and found two components: a **fast phase**, completing about half the adaptation within roughly a second, and a **slow phase** with a half-life near 30 seconds. The **Dynamics** tab plots that model over two minutes.

The model

The fraction of adaptation completed t seconds after the light changes is:

```
fraction(t) = w1 × (1 − e^(−t / τ1)) + (1 − w1) × (1 − e^(−t / τ2))
τ2 = t½ ÷ ln 2
```

- **τ1** is the fast time constant (default 1.0 s).
- **w1** is the share of adaptation in the fast phase (default 0.50).
- **t½** is the half-life of the slow phase (default 30 s, which is τ2 ≈ 43 s).

The defaults come from the paper. For comparison the chart also draws the single exponential the page used before (time constant 0.6 s after a 0.4 s delay), which reaches 90% within two seconds.

Steps

1. Choose your illuminants and transform on the **Lab** tab. The Dynamics heading shows the pair, for example "D65 → A".
2. Open the **Dynamics** tab.
3. Under **Model**, keep **Two components** selected. (**Single exponential** hides the fast and slow controls and reads the playhead value from the older curve; the half-adapted and 90% figures still describe the two-component model.)
4. Adjust **τ1** (0.2 to 4 s), **w1** (0 to 1) and **t½** (5 to 120 s).
5. Move the **Playhead** (0 to 120 s) to a moment of interest.
6. Read the statistics under the chart:
   - **D at t = …** — the fraction adapted at the playhead;
   - **Half-adapted at** — when the fraction reaches 0.5;

- **90% at** — when it reaches 0.9;
- **Legacy 90% at** — the same point for the old single exponential, for contrast.

Reading the chart

- The solid gold curve is the two-component model with your settings. The dashed red curve is the single exponential, always drawn so the difference stays visible.
- The vertical line and dot mark the playhead.
- With the defaults, the curve rises steeply for the first few seconds and then creeps up over the next minute or two.

How this relates to the Lab tab

The Dynamics controls are for exploring the time course. The Lab tab's image timeline (0 to 5 s) always uses the published defaults (τ1 = 1 s, w1 = 0.5, t½ = 30 s), starts from an initial adaptation of 0.1, and scales the result towards the **Degree of adaptation D** you set.

Result

You can see how long adaptation to a new light takes under the measured model, and how different that is from a one-second exponential.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Chromatic Adaptation reference

Every control, readout, option and export in the Chromatic Adaptation lab.

Address: https://auricartisan.com/tool/general/colorimetry/chromatic-adaptation-dynamic/

Lab tab

Image Source

Control	What it does	Values	Default
Choose image	Opens a file picker for an image	Any image type the browser can decode	—
File name	Shows the chosen file	—	"No file chosen"
Sample	Generates a synthetic HSL sweep (a test target, not a photograph)	360 × 240 pixels	—
Generate the sample sweep, **Choose an image**	The same two actions, offered in the empty state before any image is loaded	—	—

Loaded images are scaled to fit within 480 × 320 pixels (never enlarged).

Illuminants and transform

Control	What it does	Values	Default
Source illuminant	The original viewing white point	12 illuminants, below	CIE D65 (6504 K) — sRGB daylight
Swap	Exchanges source and destination	Button	—
Destination illuminant	The target white point	12 illuminants	CIE D50 (5003 K) — ICC PCS / printing
CAT Method	The chromatic adaptation transform	Bradford (Lam 1985); Von Kries (Hunt–Pointer–Estévez); CAT02 (CIE 159:2004); CAT16 (Li et al. 2017); Sharp (Finlayson & Süsstrunk 2000); CMCCAT2000 (Li et al. 2002); HPE (Hunt–Pointer–Estévez); XYZ scaling (null case)	Bradford

The 12 illuminants and their white points (CIE xy):

Choice	x	y	Spectrum drawn?
CIE D65 (6504 K) — sRGB daylight	0.3127	0.3290	Yes, reconstructed from the daylight basis
CIE D50 (5003 K) — ICC PCS / printing	0.3457	0.3585	Yes
CIE D55 (5503 K)	0.3324	0.3474	Yes
CIE D60 (6000 K) — ACES	0.32168	0.33767	Yes
CIE D75 (7504 K) — north sky	0.2990	0.3149	Yes
CIE A (2856 K) — tungsten	0.4476	0.4074	Yes, Planck at 2848 K with the 1931 constant
CIE B (4874 K) — noon sun	0.3484	0.3516	No, table not held
CIE C (6774 K) — average daylight	0.3101	0.3162	No, table not held
Equal-energy E	0.3333	0.3333	Yes, flat
FL F2 (4230 K) — cool white	0.3721	0.3751	No, table not held
FL F7 (6500 K) — broadband	0.3129	0.3292	No, table not held
FL F11 (4000 K) — narrow tri	0.3805	0.3769	No, table not held

Adaptation needs only the white point, so all 12 adapt correctly whether or not a spectrum is drawn.

Adaptation and timing controls

Control	What it does	Range	Default
Degree of adaptation D	The final degree of adaptation the timeline moves towards	0 to 1, step 0.01	1.00
Time (scrubber)	The moment after the light changes	0 to 5 s, step 0.01	0 s
Run	Re-renders the image now	Button	—
Animate	Plays the timeline from 0 to 5 s in real time, then stops	Button	—
Pause	Stops the animation at the current time	Button	—
Reset	Stops and returns to 0 s	Button	—
Live	Re-renders after every change	On or off	On
HDR Tonemap	Compresses values above white with a Reinhard curve before clipping	On or off	Off

The adaptation applied at time t is 0.1 + (D − 0.1) × fraction(t), where fraction(t) follows the two-phase time course with the published defaults (see Explore the adaptation time course). The starting value 0.1 can only be changed through the `init` value of a share link.

Controls with no visible effect in the current version

These controls are on the page and their values are recorded in the JSON export, but they do not change the rendered result:

Control	Range	Default	Current behaviour
Display Gamma	1.8 to 2.6	2.2	The image is always decoded and encoded with the sRGB transfer function
τ luminance scale	0 to 1	0.0	Intended to slow adaptation in dark pixels; the current time course does not use it
Delay luminance scale	0 to 1	0.0	As above
Local radius	0 to 256 px	64 px	Not applied
Local strength	0 to 1	0.60	Not applied
Mix (global ↔ local)	0 to 1	0.50	Has no effect while the per-pixel scales have none
Tile factor	1× to 8×	2×	Not applied
Target FPS	8 to 120	60	The animation runs at the browser's frame rate
GPU	On or off	Off	Processing always runs on the CPU

Because the per-pixel settings have no effect, the **Per-Pixel Adaptation State Map** is a uniform grey whose brightness is the adaptation in use.

Readouts

Readout	What it shows
Status chip	Ready, Image loaded (size), Sample loaded (360×240), Animating..., Paused at t = ..., Reset
Model chip	"two components · t = … s · D = …", the adaptation in use
Original Image	The loaded or generated image
Adapted Image	The result at the current time
RGB Histogram (Adapted)	Red, green and blue distributions of the result
Adaptation Curve A(t)	The applied adaptation over 0 to 5 s, with the current point labelled
Per-Pixel Adaptation State Map	Adaptation per pixel as grey (white = adapted)
LMS Cone-Channel Analysis	Source white (light bars) and destination white (dark bars) in L, M and S, with each channel's scale factor
SPD Spectral Overlay (Source vs Dest)	The two spectra from 380 to 780 nm with a line naming how each was obtained, or a message when a spectrum is not held

Every chart can be opened full screen with its **Fullscreen** button or by double-clicking it.

Dynamics tab

Control	What it does	Range	Default
Model	**Two components** (measured) or **Single exponential** (the page's former curve)	—	Two components
τ1	Fast-phase time constant	0.2 to 4 s, step 0.1	1.0 s

Control	What it does	Range	Default
w1	Share of adaptation in the fast phase	0 to 1, step 0.05	0.50
t½	Slow-phase half-life	5 to 120 s, step 1	30 s
Playhead	Time to read	0 to 120 s, step 0.5	12.0 s

Readouts: the chart (both curves, 0–120 s), **D at t = …**, **Half-adapted at**, **90% at**, **Legacy 90% at**, and a note explaining the two curves. These settings do not change the Lab tab.

Matrix tab

Control or readout	What it does	Values	Default
Decimals	Precision of the matrices shown	4, 6, 9	6
Copy as	Copies the transform's M matrix	NumPy, JSON, CSS matrix3d, GLSL mat3	—
Round trip	Largest error of $M^{-1} \times M$ against the identity	—	—
Matrices	**M — XYZ to cone-like**, **diag(d)**, **Full adaptation matrix**, at the D set on the Lab tab	—	—
All eight on this pair	Transform, ΔE vs Bradford, cond(M), Source, Status	—	—

Data tab

Control	What it does
Status chips	Filter the register: **All**, **Verbatim**, **Computed**, **Absent** (a **Stand-in** chip appears only when a stand-in exists)
Rows	Expand to read the note and, for absent data, **What it would take**
Install a table	A drop area labelled "Drop a CSV or JSON here". It does not load files in the current version

Dataset	Status
Illuminant white points, CIE xy	Verbatim
Chromatic adaptation matrices, 8	Verbatim
sRGB primaries and transfer function	Verbatim
Daylight eigenvectors S0 S1 S2	Computed
Planckian radiators	Computed
Adaptation time constants	Computed
CIEDE2000 colour difference	Computed
Illuminants B and C spectra	Absent
Fluorescent F2 F7 F11 spectra	Absent
Corresponding-colour data sets	Absent
Colour matching functions, 10°	Absent

Export tab

Control	What it does	Output
Export JSON	Downloads the settings and the adaptation at the current time	`cad-<method>-<source>-<destination>.json`
Export Frame (PNG)	Downloads the adapted image	`cad-frame-t<time>.png`
Copy Link	Copies the page's current address	Text
Share URL field and **Generate**	Builds a link to the current settings, shows it and copies it	A URL
Copy	Copies the link in the field	Text
Compare All Methods	Runs all eight transforms on eight test colours at the current adaptation	Bar chart and table
Batch Colour Analysis, **Run Batch**	Adapts up to 50 HEX colours	Table

The JSON file contains: `srcIllum`, `dstIllum`, `catMethod`, `gamma`, `delay`, `tau`, `strength` (the D setting), `initAdapt`, `tauLocal`, `delayLocal`, `radius`, `localStrength`, `mix`, `tile`, `time`, `fps`, `hdr`, `matrix` (the full 3 × 3 adaptation matrix as nine numbers, row by row), `srcLMS`, `dstLMS`, `scale` (the three cone gains) and `det` (its determinant). `delay` and `tau` belong to the former single-exponential model.

The share link carries `src`, `dst`, `cat`, `tau`, `delay`, `str` (D), `init` (initial adaptation) and `t` (time). It does not carry the image.

Reference tab

Control	What it does
Standards, **Formulas**, **Citations**, **Research notes**	Switch sections; each shows a count
Filter	Searches all four sections; `Esc` clears it
Where it is used chips (Citations)	**All**, **Runs here**, **Corrected**, **Context only**

Standards covers CIE 160:2004, CIE 159:2004 (CIECAM02), CAM16/CAT16, ICC.1:2022, Fairchild and Reniff (1995), Brainard and Wandell (1992), Foster (2011) and CIE 15:2018. Formulas covers the adaptation models, von Kries diagonal adaptation, the eight matrices, sRGB conversion, CIELAB, ΔE, spectrum generation and the spatial model. Some background prose predates the current time-course model; the figures the lab uses are those on the Dynamics tab.

Save to Library

The floating **Save to Library** button stores a preset of the current form settings in your Library.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Troubleshooting Chromatic Adaptation

Fixes for common problems in the Chromatic Adaptation lab.

Symptom	Why it happens	What to do
The image area is dark and says "No image yet"	No image has been loaded	Select **Generate the sample sweep** or **Choose an image**

Symptom	Why it happens	What to do
Nothing changes when I move a slider	**Live** is off, or no image is loaded	Tick **Live**, or select **Run** after changes
The image hardly changes at 0 s	Adaptation starts at 0.1 and grows with time	Move **Time (scrubber)** towards 5 s
The image is only partly adapted even at 5 s and D = 1	The measured slow phase has a 30 s half-life, so at 5 s about 55% of the change has happened	Expected. The **Matrix** tab shows the matrix at the D you set; the **Dynamics** tab shows when 90% is reached
Display Gamma, the per-pixel scales, Local radius, Local strength, Mix, Tile factor or Target FPS change nothing	These controls are not applied in the current version; they are only recorded in the JSON export	See Reference
The GPU box (or `G`) changes nothing	Processing always runs on the CPU	Expected
The state map is a flat grey	The per-pixel model has no effect in the current version, so every pixel has the same adaptation	Expected. Its brightness equals the adaptation in use
The SPD overlay says "No spectrum held"	B, C, F2, F7 and F11 spectra are not held	The adaptation is still correct, because it uses the tabulated white point
Bright colours clip after a large shift such as D65 to A	Adapted values exceed the sRGB range and are clipped	Tick **HDR Tonemap** to compress highlights first, or choose a smaller shift
My photo looks smaller or softer	Images are scaled to fit 480 × 320 for processing	Expected. The exported PNG is at the processing size
Run Batch says "Enter valid hex values"	No six-digit HEX lines were found	Enter values such as `#735244` or `735244`, one per line
Only 50 of my colours appear	The batch is capped at 50	Split the list and run it in parts
The share link opens without my image	Links carry settings only	Send the image separately
Copy Link gave a link without my settings	Copy Link copies the page's current address	Use **Generate** under Share URL instead
Copying fails with "Copy failed"	The browser blocked clipboard access	Allow clipboard access, or copy the text from the field manually
Dropping a file on **Install a table** does nothing	The drop area is not active in the current version	No workaround; the lab uses its built-in tables
Space, R, C or G does nothing	The focus is in a text box, slider or list	Click an empty part of the page first
A chart is too small to read	—	Double-click it, or select its **Fullscreen** button; `Esc` closes

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Chromatic Adaptation FAQ

Short answers to common questions about the Chromatic Adaptation lab.

Which transform should I use?

For ICC colour management and D65 ↔ D50 conversions, Bradford is the common convention. CAT16 is the transform of the most recent CIE-lineage appearance model (CAM16). The lab does not rank them for accuracy, because that needs corresponding-colour data sets it does not hold; it shows what each one does on your pair.

Why do Von Kries and HPE give the same result?

The lab's Von Kries matrix is the Hunt–Pointer–Estévez matrix normalised to D65. Normalising scales each row, and that scaling cancels out in $M^{-1} \cdot \mathrm{diag}(d) \cdot M$, so the two produce identical adaptations.

What is XYZ scaling for?

It is the simplest possible adaptation, scaling X, Y and Z directly. It is not a model of the eye and is kept as a baseline to compare the others against.

What does "degree of adaptation" mean?

How complete the adaptation is. At D = 1 the source white maps exactly onto the destination white. Real viewers often adapt incompletely, especially in dim or complex scenes, so values below 1 are realistic in some situations.

How long does adaptation really take?

The lab uses Fairchild and Reniff's (1995) measurements: about half the adaptation within a second or so, and the rest with a half-life of about 30 seconds. With the default settings, 90% adaptation takes about 70 seconds. The Dynamics tab computes the exact times.

Why does the time-course prose in Reference quote different time constants?

Some background text in the Reference section predates the current model. The time course the lab actually uses is the one on the Dynamics tab.

Is my image uploaded?

No. The image is read and processed in your browser, and the exported PNG is created there too.

Can I adapt a video?

No. The lab adapts one still image (or the sample sweep) at a time.

Can I get the full adaptation matrix as numbers?

Yes. The Matrix tab shows it at the D you set, and **Export JSON** includes it for the current moment of the timeline. The **Copy as** buttons copy the transform's cone matrix M.

Does the lab model colour appearance (CIECAM02 or CAM16)?

No. It applies the chromatic adaptation step only. The Reference notes explain how CIECAM02 would compute D from the adapting luminance and surround.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Chromatic Adaptation: supporting pages

Glossary, shortcuts, limits, related tools, release notes and privacy for the Chromatic Adaptation lab.

These pages support the guides in docs/: the terms used, the keyboard shortcuts, how far the results can be trusted, related tools, what has changed and what stays on your device.

File	What it covers
glossary.md	Terms used in the lab
shortcuts.md	Keyboard and mouse shortcuts
limits-and-accuracy.md	What the results are and are not
related.md	Related tools and Learn articles
release-notes.md	What has changed for users
privacy.md	What stays in your browser

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Chromatic Adaptation shortcuts

Keyboard and mouse shortcuts in the Chromatic Adaptation lab.

These keys work when the focus is not in a text box, slider or list. Click an empty part of the page first if a key does nothing.

Key	What it does
`Space`	Plays or pauses the 0–5 s animation
`R`	Resets the time to 0 s
`C`	Runs **Compare All Methods** (results on the Export tab)
`G`	Toggles the **GPU** checkbox (which has no effect on processing)
`Esc`	Clears the Reference filter; closes a full-screen chart

Mouse	What it does
Double-click a chart or image	Opens it full screen

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Chromatic Adaptation glossary

Plain-language definitions of the terms used in the Chromatic Adaptation lab.

Adaptation time course — how the degree of adaptation grows after the light changes. The lab uses a fast and a slow exponential phase.

Bradford transform — a CAT whose cone-like matrix comes from Lam (1985). The usual choice in ICC colour management.

CAT02 — the CAT inside the CIECAM02 colour appearance model (CIE 159:2004).

CAT16 — the CAT inside the CAM16 colour appearance model (Li et al., 2017).

Chromatic adaptation — the visual system's adjustment to the colour of the light, keeping surface colours roughly stable.

Chromatic adaptation transform (CAT) — a formula that converts a colour under one white point into the corresponding colour under another: $X' = M^{-1} \cdot \mathrm{diag}(d) \cdot M \cdot X$.

CIEDE2000 (ΔE_{00}) — the CIE colour-difference formula (CIE 142:2001). Around 1 is a just-noticeable difference for many viewers.

CIELAB (L*a*b*) and L*C*h° — a colour space with lightness L*, and either a*, b* or chroma C* and hue angle h°. Values depend on the reference white.

CMCCAT2000 — a CAT published by Li, Luo, Rigg and Hunt (2002).

Condition number, cond(M) — how much a matrix can stretch a small difference. The lab shows it as a description, not a quality score.

Cone-like space (LMS) — the space a CAT matrix M maps XYZ into, roughly representing the long-, medium- and short-wavelength cone responses.

Corresponding colours — pairs of colours that look the same under two different lights, measured in experiments. They are needed to rank CATs for accuracy; the lab does not hold them.

Degree of adaptation (D) — how complete adaptation is, from 0 (none) to 1 (complete).

Destination illuminant — the light you adapt to.

Determinant, det(M) — a single number describing how a matrix scales volume. Shown in the comparison as a diagnostic.

diag(d) — the three cone gains, d = 1 + D × (destination ÷ source − 1) for each channel.

Frobenius norm, ||M||F — the overall size of a matrix's entries. Shown as a diagnostic.

HPE (Hunt–Pointer–Estévez) — a classic cone-response matrix. The lab's Von Kries option is HPE normalised to D65.

Reinhard tone mapping — the curve c ÷ (1 + c), which compresses values above 1 smoothly. Used by the **HDR Tonemap** option.

Round trip — multiplying a matrix by its inverse should give the identity; the largest error shows how exact the inverse is.

Sharp transform — a CAT with a spectrally sharpened matrix (Finlayson and Süsstrunk, 2000).

Source illuminant — the light the colours are originally expressed under.

Spectral power distribution (SPD) — the relative power of a light at each wavelength.

von Kries adaptation — the principle that each cone type adjusts its gain independently. All eight transforms use it.

White point — the chromaticity of the reference white, such as D65 (x 0.3127, y 0.3290).

XYZ scaling — the simplest adaptation, scaling X, Y and Z directly. A baseline, not a model of the eye.

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Chromatic Adaptation limits and accuracy

What the Chromatic Adaptation lab computes exactly, what it approximates, and what it does not do.

The lab is an educational and research simulator that runs in your browser. It is not a substitute for measured device profiles, calibrated displays, spectrophotometer data or a validated ICC workflow.

What is exact

- **White points** of all 12 illuminants are tabulated CIE xy values, held verbatim. Adaptation needs only these, so it is correct even where a spectrum is not held.
- **Transform matrices** are the published values; each inverse is recomputed, and the Matrix tab shows the round-trip error.
- **sRGB conversion** uses the IEC 61966-2-1 primaries and transfer function. Adaptation happens in linear light; the transfer function is applied only at the ends.
- **ΔE_{00}** is the CIEDE2000 formula (CIE 142:2001).

What is modelled or simplified

- **Time course.** The two-phase model uses constants from Fairchild and Reniff (1995). It describes average behaviour in their experiments, not any particular viewer or scene.
- **Degree of adaptation** is yours to set. CIECAM02 would compute it from the adapting luminance and the surround; the lab does not.
- **Clipping.** Adapted colours outside sRGB are clipped per channel (or compressed first with HDR Tonemap). This is not perceptual gamut mapping. For that, see Gamut Mapping.
- **Spectra.** The daylight illuminants are reconstructed from the CIE daylight basis; A is computed from Planck's law. B, C, F2, F7 and F11 are not drawn because their tables are not held.

What is not done

- **Ranking transforms by accuracy.** That needs corresponding-colour data sets, which are not held. The comparisons report what each transform does, measured against Bradford or as a colour shift, and say so.
- **Spatial or per-pixel adaptation.** The Spatial Adaptation and Per-Pixel Modulation controls do not affect the result in the current version, nor do Display Gamma, Target FPS or GPU.
- **Colour appearance modelling.** Only the adaptation step is applied; lightness, colourfulness and hue correlates of CIECAM02 or CAM16 are not computed.
- **The 10° observer.** Not held.

Precision

- Images are processed at up to 480 × 320 pixels, in 8-bit sRGB.
- Batch results round ΔE_{00} to two decimals and L*, C* and h° to one.
- The Lab timeline covers 0 to 5 seconds only; the Dynamics tab covers 0 to 120 seconds.

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Chromatic Adaptation privacy

What the Chromatic Adaptation lab does with your images and settings.

- **Your image never leaves your device.** When you choose an image, the browser reads it locally, draws it onto the page and adapts it there. Nothing is uploaded.
- **Exports are created locally.** The PNG frame and JSON file are generated in the browser and saved through your normal download.
- **Share links carry settings only** — illuminants, transform, D, initial adaptation and time. They never include your image.
- **Clipboard.** Copy Link, Generate, Copy and the Copy as buttons write to your clipboard only when you use them; your browser may ask for permission.

- **Batch colours** you paste stay in the page and are gone when you close it.
- **Save to Library** stores a preset of your settings in your Auric Artisan Library; it does not store the image.
- The page loads the site's shared features in the same way as every other page on auricartisan.com; see the site's privacy policy.

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Chromatic Adaptation release notes

User-facing changes to the Chromatic Adaptation lab.

September 2026 — the workbench rebuild

- The tabs are now **Lab**, **Dynamics**, **Matrix**, **Data**, **Export** and **Reference**. Standards, Formulas, References and Research became one Reference view with a search filter.
- The adaptation timeline now follows the two-phase time course measured by Fairchild and Reniff (1995). The former single exponential (0.6 s with a 0.4 s delay) is drawn on the new **Dynamics** tab for comparison. The delay and τ sliders were removed from the Lab.
- The Lab gained a **Degree of adaptation D** control, and every figure states the D it used.
- Illuminant spectra now come from the same checked tables as the Illuminants lab. Where a spectrum is not held (B, C, F2, F7, F11), the plot says so instead of drawing a daylight curve under that name.
- The new **Matrix** tab shows M, diag(d) and the full matrix, with a round-trip check, copy-as-code buttons and a comparison of all eight transforms against Bradford.
- The new **Data** tab lists the 11 datasets and their status.
- Three citations were corrected (Sharp, CAM16 and CMCCAT2000); corrected entries are marked in the Reference view.

May 2026

The Chromatic Adaptation Dynamic User Guide was published in the site's Documentation library, describing the earlier version.

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Related to Chromatic Adaptation

Tools and Learn articles that go well with the Chromatic Adaptation lab.

Tools

Tool	Why
Illuminants	Spectra, white points and CCT of the illuminants, plus a 24-patch adaptation preview with six transforms
Gamut Mapping	Includes an "adapt white, then clip" mapping between spaces with different white points
Gamut Lab	White points and primaries of 13 RGB spaces
Colour Science Lab	Adaptation and appearance models in a wider laboratory

Learn articles

- White Points and Chromatic Adaptation: von Kries, Bradford, and CAT02 — https://auricartisan.com/library/learn/articles/2026-07-11-white-points-chromatic-adaptation-transforms
- Color Constancy, Adaptation, and Why Colors Change with Context — https://auricartisan.com/library/learn/articles/2026-05-24-color-constancy-adaptation
- Color Appearance Models — https://auricartisan.com/library/learn/articles/2026-06-21-color-appearance-models-ciecam
- ΔE Metrics from CIE76 to CIEDE2000 and ΔEITP — https://auricartisan.com/library/learn/articles/2026-05-26-delta-e-metrics-cie76-ciede2000-itp
- CIELAB and LCH Explained — https://auricartisan.com/library/learn/articles/2026-05-24-cielab-and-lch-explained

Documentation on the site

The site's Documentation library (https://auricartisan.com/library/documentation/) has an earlier **Chromatic Adaptation Dynamic User Guide**. It describes delay and τ sliders that the current page no longer has; this documentation follows the current page.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Lab

What the Gamut Lab does, who it is for, where to find it and how to compare your first colour spaces.

The Gamut Lab shows where an RGB colour space ends. Every RGB space is defined by three **primaries** (the reddest red, greenest green and bluest blue it can make), a **white point**, and a **transfer function** that turns stored values into light. Plot the primaries on a chromaticity diagram and they form a triangle: the space's **gamut**. The lab draws those triangles for 13 standard spaces — from sRGB and Display P3 to Rec. 2020, the HDR Rec. 2100 encodings, Adobe RGB, ProPhoto RGB, the ACES spaces and legacy television standards — plus any space you define yourself.

Beyond the diagram, it derives the RGB-to-XYZ matrices from the primaries, draws each space's transfer curve (and tells you whether that curve is exactly the one its standard defines), slices gamuts at constant lightness in CIELAB, and measures gamut volume in CIELAB and overlap in u′v′ with quasi-random sampling. You can keep your own spaces in a library in your browser, import and export them as JSON, and export the diagram as SVG or PNG.

The Gamut Lab defines and measures spaces. It does not map colours from one space into another; that is the job of Gamut Mapping.

Who it is for

- Designers and developers choosing a working or output colour space.
- Video, VFX and HDR practitioners comparing Rec. 709, P3, Rec. 2020 and ACES.
- Display and camera engineers checking primaries and matrices for a custom space.
- Students learning chromaticity diagrams, transfer functions and gamut volume.

Where to find it

- Address: https://auricartisan.com/tool/general/gamut-and-rendering/gamut/
- In the tool collection (https://auricartisan.com/collections/) under **Gamut & Rendering**, as **Gamut Explorer**. The page itself is titled **Gamut Lab**.

Quick start

1. Open https://auricartisan.com/tool/general/gamut-and-rendering/gamut/. The diagram shows sRGB, Display P3 and Rec. 2020 on the CIE 1931 xy diagram.
2. Under **Working color space**, choose **Adobe RGB**. Its triangle is added and its primaries, white point and matrices appear in the panel.
3. Click inside the diagram. **Vertex readout** shows the x, y you clicked, a colour chip and which spaces contain that chromaticity.
4. Change **Diagram mode** to **CIE 1976 u′v′** to see a more evenly spaced diagram.
5. Open the **Compare** tab and select **Measure them all**.
6. Open **Transfer** to see every transfer curve and whether it is exact.
7. Open **Export**, choose **The diagram** and **SVG**, and select the download button.

What you can do

- Draw 13 standard RGB spaces and your own on CIE 1931 xy or CIE 1976 u′v′ diagrams, or as 3-D wireframes in CIELAB.
- Read a space's primaries, white point and both RGB ↔ XYZ matrices, and solve the matrices for custom primaries.

- Probe any chromaticity for its colour, CIELAB value and gamut membership.
- Plot transfer curves (EOTF/OETF) and see which are exact, partial or corrected.
- Slice gamuts at constant CIELAB L* or at OKLab L.
- Measure gamut volume in CIELAB and overlap in u′v′ between two spaces, or measure all spaces at once.
- Keep, import and export custom spaces as JSON.
- Export the diagram (SVG, PNG), a space (JSON), all spaces (JSON), the comparison (CSV, JSON) or a link.

In this folder

- docs/ — how to use the lab
  - Getting started
  - Compare colour spaces
  - Define a custom colour space
  - Inspect transfer functions
  - Reference
  - Troubleshooting
  - FAQ
- others/ — background and supporting pages
  - Glossary
  - Shortcuts
  - Limits and accuracy
  - Related tools
  - Release notes
  - Privacy

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Lab documentation

The guides, reference and help pages for the Gamut Lab.

This folder explains how to use the Gamut Lab: a screen tour, task guides for comparing spaces, defining your own and reading transfer curves, a complete reference and help pages.

File	What it covers
getting-started.md	Opening the lab, the six tabs and the chromaticity diagram
compare-colour-spaces.md	Diagram, slices, volume, overlap and the Compare tab
define-a-custom-space.md	Entering primaries, solving matrices, importing, exporting and the library
inspect-transfer-functions.md	The Lab's transfer curve panel and the Transfer tab
reference.md	Every control, readout and export
troubleshooting.md	Symptoms, causes and fixes
faq.md	Short answers

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Getting started with the Gamut Lab

Open the Gamut Lab, learn its tabs and panels, and read the chromaticity diagram.

Open the lab

Go to https://auricartisan.com/tool/general/gamut-and-rendering/gamut/. The heading is **Gamut Lab**. All calculations run in your browser.

Key ideas

- The **chromaticity diagram** maps colour without brightness. Its curved edge, the **spectral locus**, is made of single wavelengths of light; the straight line closing it at the bottom joins deep violet and deep red (the purples). Every colour a person can see lies inside.
- An RGB space's **gamut** on the diagram is the triangle between its three primaries. A larger triangle holds more saturated colours.
- The **CIE 1931 xy** diagram is the classic one, but equal distances on it are not equal visual differences (it stretches greens). The **CIE 1976 u'v'** diagram is more even, so the lab quotes shares of the locus in u'v'.
- A triangle shows chromaticity only. **Gamut volume** in CIELAB also accounts for lightness, which is why the lab measures it separately.

A tour of the screen

Tab	What it is for
Lab	The diagram, the working space's primaries and matrices, a transfer curve, volume and slice panels
Transfer	Every distinct transfer curve, plotted, with its fidelity to its standard
Compare	Every space measured the same way: volume, ratio to sRGB and share of the locus

Tab	What it is for
Data	The eight datasets the lab uses and their status
Export	Choose what to take and in which format, preview it, download or copy; import spaces; the library
Reference	Standards, formulas, citations and research notes with a filter

The Lab tab

The left panel holds: **Working color space**, **Diagram mode**, **Standard observer**, **Overlays**, **Vertex readout**, **Actions** (Reset view, Export PNG, Export SVG), **Primaries (xy)**, **White point XYZ**, **RGB ↔ XYZ matrices** (with **Solve from primaries** and the triangle area), **Transfer function**, **Volume comparison** and **Gamut slice**.

The right side shows four canvases: **Chromaticity diagram**, **EOTF / OETF curve**, **Gamut slice — Lab / OKLab** and **3-D gamut wireframe**.

Read the diagram

1. The diagram opens with sRGB, Display P3 and Rec. 2020 drawn on CIE 1931 xy.
2. Choose another space under **Working color space**, for example **ACES 2065-1 (AP0)**. It is added to the diagram (up to five spaces are drawn; the oldest is dropped when you add a sixth). Its primaries and matrices fill the panel.
3. Notice AP0's blue primary lies below the diagram's edge: it is an imaginary colour, which lets the triangle enclose every real colour.
4. Click or drag on the diagram. A marker follows the pointer and **Vertex readout** shows x, y, a colour chip with its sRGB HEX, a CIELAB value, and a tick or cross for each drawn space.
5. Tick **Show all library spaces** to draw every space at once, including your own.
6. Choose **CIE 1976 u'v'** under **Diagram mode** to redraw everything on the more uniform diagram, or **3-D Wireframe (Lab)** to see each drawn space as a wireframe solid in CIELAB.
7. Select **Reset view** to return to CIE 1931 xy with the default overlays and no probe marker.

Next steps

- Compare colour spaces
- Define a custom colour space
- Inspect transfer functions

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Compare colour spaces

Compare RGB spaces by triangle, slice, CIELAB volume, u'v' overlap and share of the spectral locus.

The Gamut Lab gives four ways to compare spaces. Each answers a different question.

Measure	Question it answers	Where
Triangle area and share of the locus	How much of the range of visible chromaticities does the space cover?	Lab panel, Compare tab
Constant-lightness slice	At a given lightness, which hues can each space make vivid?	Lab › Gamut slice
CIELAB volume	How many distinguishable colours, including lightness, does the space hold?	Lab › Volume comparison, Compare tab

Measure	Question it answers	Where
Overlap in u′v′	How much of one space's triangle lies inside another's?	Lab › Volume comparison

Compare two spaces in the Lab

1. On the **Lab** tab, find **Volume comparison**.
2. In the first list, choose the reference: **sRGB** (default), **Display P3**, **Rec. 2020** or **ACEScg**.
3. In the second list, choose **Active space** (the working space selected at the top) or one of the same four spaces.
4. Select **Compute volume**. The panel shows "Computing…" and then:
   - each space's CIELAB volume in Lab³, with a ± spread against a run at half the samples and the half-width of the sampling box;
   - **Overlap in u′v′** — the share of space A inside B ("of A") and of B inside A ("of B");
   - **Jaccard** — the area both share divided by the area either covers.

These figures use 10,000 quasi-random (Halton) samples for volume and 5,000 for overlap.

Slice at constant lightness

1. Under **Gamut slice**, choose **Constant L* (CIELAB)**.
2. Move **Level** from 0 to 100 (default 50). The slice canvas outlines up to five drawn spaces at that lightness.
3. Choose **Constant L (OKLab)** to see the slice in OKLab, which keeps hue lines straighter and makes boundaries easier to read. In the current version the OKLab slice is always drawn at L = 0.5; the Level slider changes its label but not the slice.

Measure every space at once

1. Open the **Compare** tab.
2. Under **Measure in**, choose **u′v′ share** (default) or **xy share (legacy)** for the share column. The xy figure is kept because other tools quote it; it overstates green.
3. Set **Samples** (**n**) from 5,000 to 200,000 (default 40k).
4. Select **Measure them all**. The button reads "Measuring…" while it works.
5. Read the table, sorted from smallest to largest volume:

Column	Meaning
Space	Name and colour swatch
White	D65, D50, ACES, DCI, C, or the coordinates for others
Lab volume ± spread	CIELAB volume, and how much it changed against a half-size run
vs sRGB	Volume as a multiple of sRGB's
u′v′ share or xy share	The triangle's area as a percentage of the spectral locus in that plane; "> 100%" when the primaries lie outside the locus

The note above the table records the sample count and that each space is measured against its own white.

Why "> 100%" appears

ACES AP0 and ProPhoto RGB use imaginary primaries outside the spectral locus. Part of their triangle is colour no light can make, so their share cannot be read as coverage. The lab prints "> 100%" instead of a misleading percentage.

Why the figures differ from published percentages

The locus area here is computed from the CIE 1931 2° observer data the lab holds, from 380 to 715 nm, in the plane named. Shares quoted elsewhere are often against a slightly different locus area. For example, the lab measures Rec. 2020 at 63.4% of the xy locus rather than the 75.8% often quoted.

Result

You can say how two or more spaces differ in chromaticity coverage, in vivid hues at a given lightness, and in total CIELAB volume.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Define a custom colour space

Enter your own primaries and white point, derive the RGB ↔ XYZ matrices, and keep, import or export the space.

Use this to check a display's measured primaries, a camera's native gamut, or a proposed working space.

How the matrices are derived

Given the xy chromaticities of the red, green and blue primaries and of the white point, the lab:

1. turns each primary into XYZ with Y = 1;
2. builds a matrix with those three columns;
3. solves for the scale of each column that makes RGB (1, 1, 1) equal the white point;
4. applies the scales to get the RGB → XYZ matrix;
5. inverts it to get the XYZ → RGB matrix.

This is the standard construction (as described by Bruce Lindbloom), so a custom space gets exactly the same treatment as a published one.

Enter and solve

1. On the **Lab** tab, set **Working color space** to **Custom**.
2. Under **Primaries (xy)**, type the **R**, **G** and **B** x and y values, and the **W** (white point) x and y. The fields accept four decimal places.
3. Select **Solve from primaries**.
4. Read **RGB ↔ XYZ matrices**: the RGB → XYZ and XYZ → RGB matrices, then the triangle's **Area (xy)** and **Area % sRGB**.
5. **White point XYZ** shows the white as X, Y, Z with Y = 1, for reference.
6. Your triangle is drawn on the diagram as **Custom**.

A custom space made this way uses the sRGB transfer curve. It lasts until you reload the page. To keep it, export it with **This space** (below) while **Custom** is the working space, then import the JSON into the library.

Keep it in the library

The **Export** tab has a library table listing every standard space and any you have added. Standard spaces show **Load**; your spaces show **Load** and **Del**. Your spaces are kept in this browser's storage, so they are there next time you open the lab on the same device and browser.

To add a space to the library, import it:

1. Open **Export**.
2. Paste a colour space definition into **Bring one in**. The format is:

```
{
  "name": "MyDisplay",
  "label": "My display",
  "primaries": { "r": [0.680, 0.320], "g": [0.265, 0.690], "b": [0.150, 0.060] },
  "whitePoint": [0.3127, 0.3290],
  "gamma": "sRGB"
}
```

`name` and `primaries` are required; `label` defaults to the name and `gamma` to `sRGB`. 3. Select **Add to the library**. The status line says "Imported: MyDisplay", or "Invalid JSON" or "Parse error: …" if something is wrong. 4. Select **Load** in the library table to draw the space and make it the working space.

The `gamma` value names the transfer curve: `sRGB`, `BT.2020`, `ROMM`, `AdobeRGB`, `PQ`, `HLG`, `linear`, or a number such as `2.2` for a pure power law.

Export a space

1. On **Export**, under **What to take**, choose **This space** (the working space) or **Every space** (all of them, including imports).
2. The format is **JSON**. **Exactly what you will get** previews the file and its size.
3. Select **Download** (the button names the file, for example `gamut-lab-srgb.json`) or **Copy to clipboard**.

An exported space contains `name`, `label`, `primaries`, `whitePoint`, `gamma`, `rgb2xyz`, `xyz2rgb` and `area` (the xy triangle area). It can be pasted back into **Bring one in**.

Result

You have a custom space with verified matrices, drawn beside the standards, saved in your browser and portable as JSON.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Inspect transfer functions

Read each space's transfer curve, understand EOTF and OETF, and check which curves are exact to their standard.

Key ideas

- A **transfer function** links stored values to light. The **OETF** (opto-electronic) encodes light into a signal; the **EOTF** (electro-optical) turns a signal back into light on a display.
- Most spaces use a **relative** curve: signal 0 to 1 becomes light 0 to 1 of the display's peak. **PQ** (SMPTE ST 2084) is **absolute**: a signal value means a specific luminance in cd/m² (nits), up to 10,000.
- **HLG** (hybrid log-gamma) is a relative HDR curve. Its full EOTF needs a display peak luminance and a surround; this lab draws the inverse OETF only, and marks it **Partial**.

The ten distinct curves

Curve	Used by	Fidelity
IEC 61966-2-1 (sRGB piecewise)	sRGB, Display P3, Custom	Exact
ITU-R BT.2020-2	Rec. 2020	Exact (corrected)
ROMM / ISO 22028-2, γ 1.8 plus toe	ProPhoto RGB	Exact (corrected)
Adobe RGB (1998), γ 563/256	Adobe RGB	Exact (corrected)
SMPTE ST 2084 (PQ)	Rec. 2100 PQ	Exact
ARIB STD-B67 — inverse OETF	Rec. 2100 HLG	Partial
Linear	ACES 2065-1 (AP0), ACEScg (AP1)	Exact
Gamma 2.6	DCI-P3	Exact
Gamma 2.2	Adobe Wide Gamut, NTSC 1953	Exact
Gamma 2.8	PAL/SECAM	Exact

"Corrected" marks a curve that an earlier version of the page did not run as its standard defines.

The Transfer tab

1. Open the **Transfer** tab. The note gives the number of distinct curves across the spaces.
2. Under **Plot**, choose **Relative, 0 to 1** for the display-relative curves or **Absolute, cd/m²** for PQ, which has its own log-scale axis from 1 to 10,000 cd/m².
3. Under **Fidelity**, filter the rows: **All**, **Exact**, **Partial** or **Corrected** (each chip shows its count).
4. Read the legend under the plot; a dashed line is the linear reference.
5. Open a row to see the defining document, what the curve used to be (for corrected rows) and, for HLG, why it is partial.

The Lab tab's transfer panel

1. On the **Lab** tab, find **Transfer function**.
2. Choose a preset: **sRGB**, **Display P3**, **Rec. 2020**, **PQ (ST 2084)**, **HLG**, **Adobe RGB (γ 563/256)** or **ProPhoto (ROMM)**.
3. The **EOTF / OETF curve** canvas plots it, the summary names the defining document, and an equation box shows the formula for the sRGB, PQ, HLG and linear curves. For the other presets the equation box shows a simplified power-law form; the Transfer tab and Reference › Formulas give the exact definitions.

Result

You know which curve each space uses, whether it is relative or absolute, and whether the lab runs it exactly as its standard defines.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Lab reference

Every control, readout, option and export in the Gamut Lab.

Address: https://auricartisan.com/tool/general/gamut-and-rendering/gamut/

The 13 standard spaces

Space	Red (x, y)	Green (x, y)	Blue (x, y)	White (x, y)	Transfer
sRGB / BT.709	0.64, 0.33	0.30, 0.60	0.15, 0.06	0.3127, 0.329	sRGB
Display P3	0.68, 0.32	0.265, 0.69	0.15, 0.06	0.3127, 0.329	sRGB
DCI-P3	0.68, 0.32	0.265, 0.69	0.15, 0.06	0.314, 0.351	Gamma 2.6
Rec. 2020	0.708, 0.292	0.17, 0.797	0.131, 0.046	0.3127, 0.329	BT.2020
Rec. 2100 PQ	as Rec. 2020	as Rec. 2020	as Rec. 2020	0.3127, 0.329	PQ
Rec. 2100 HLG	as Rec. 2020	as Rec. 2020	as Rec. 2020	0.3127, 0.329	HLG (inverse OETF)
Adobe RGB	0.64, 0.33	0.21, 0.71	0.15, 0.06	0.3127, 0.329	γ 563/256
ProPhoto RGB	0.7347, 0.2653	0.1596, 0.8404	0.0366, 0.0001	0.3457, 0.3585	ROMM
Adobe Wide Gamut	0.7347, 0.2653	0.1152, 0.8264	0.1566, 0.0177	0.3457, 0.3585	Gamma 2.2
ACES 2065-1 (AP0)	0.7347, 0.2653	0.0, 1.0	0.0001, −0.077	0.32168, 0.33767	Linear
ACEScg (AP1)	0.713, 0.293	0.165, 0.83	0.128, 0.044	0.32168, 0.33767	Linear
NTSC 1953	0.67, 0.33	0.21, 0.71	0.14, 0.08	0.31, 0.316	Gamma 2.2
PAL/SECAM	0.64, 0.33	0.29, 0.60	0.15, 0.06	0.3127, 0.329	Gamma 2.8

Custom is a 14th option whose primaries you enter.

Lab tab

Controls

Control	What it does	Values	Default
Working color space	Chooses the space shown in the panel and adds it to the diagram (up to five drawn)	The 13 spaces and Custom	sRGB / BT.709
Diagram mode	How the main canvas draws	CIE 1931 xy; CIE 1976 u′v′; 3-D Wireframe (Lab)	CIE 1931 xy
Standard observer	Shown with two options, but only the CIE 1931 2° observer is used	CIE 1931 2°; CIE 1964 10°	CIE 1931 2°
Spectral locus	Draws the locus	On or off	On
Grid	Draws the grid	On or off	On
MacAdam ellipses	Draws 24 illustrative ellipses (a stand-in, see Limits)	On or off	Off
λ labels	Labels wavelengths on the locus	On or off	On
Show all library spaces	Draws every space, including imported ones	On or off	Off
WebGL chromaticity fill	Fills the locus with approximate colours using WebGL 2	On or off	Off
Reset view	Returns to CIE 1931 xy, default overlays, no probe	Button	—
Export PNG	Downloads the diagram as `gamut-<mode>.png`	Button	—

Control	What it does	Values	Default
Export SVG	Downloads the diagram as `gamut-<mode>.svg`	Button	—
Primaries (xy) R, G, B, W	The working space's primaries and white; editable for Custom	Numbers, 4 decimals	From the space
White point XYZ	The white as X, Y, Z with Y = 1	—	From the space
Solve from primaries	Builds the Custom space from the fields and derives its matrices	Button	—
Transfer function	Curve shown on the EOTF / OETF canvas	sRGB, Display P3, Rec. 2020, PQ (ST 2084), HLG, Adobe RGB (γ 563/256), ProPhoto (ROMM)	sRGB
Volume comparison first list	Reference space	sRGB, Display P3, Rec. 2020, ACEScg	sRGB
Volume comparison second list	Space to compare	Active space, sRGB, Display P3, Rec. 2020, ACEScg	Active space
Compute volume	Measures both volumes and their overlap	Button	—
Gamut slice mode	Slice space	Constant L* (CIELAB); Constant L (OKLab)	Constant L*
Level	Slice lightness	0 to 100	50

Readouts

Readout	What it shows
Chromaticity diagram	Drawn spaces as triangles, the locus, overlays and the probe marker with its coordinates. Click or drag to probe
Vertex readout	x, y, Y, a chip with sRGB HEX, CIELAB (L*, a*, b*) and a tick or cross per drawn space
RGB ↔ XYZ matrices	RGB → XYZ, XYZ → RGB, Area (xy) and Area % sRGB
EOTF / OETF curve	The chosen transfer curve, its defining document and an equation
Volume metrics	Both volumes (Lab^3) with spread and box, overlap of A and of B in u′v′, and Jaccard
Gamut slice — Lab / OKLab	Up to five drawn spaces outlined at the slice level
3-D gamut wireframe	Wireframes of the drawn spaces in CIELAB, at a fixed viewing angle

Transfer tab

Control	What it does	Values	Default
Plot	Which curves and axis	Relative, 0 to 1; Absolute, cd/m^2 (PQ, log scale 1 to 10,000)	Relative
Fidelity	Filters the rows	All, Exact, Partial, Corrected (with counts)	All

Rows give each distinct curve, the spaces that use it, a fidelity tag, the defining document and, where relevant, what it used to be or why it is partial.

Compare tab

Control	What it does	Values	Default
Measure in	The share column's plane	u′v′ share; xy share (legacy)	u′v′ share

Control	What it does	Values	Default
Samples (n)	Halton samples per space	5,000 to 200,000, step 5,000	40k
Measure them all	Measures every space	Button	—

Table columns: Space, White, Lab volume ± spread, vs sRGB, share.

Data tab

Dataset	Status
Primaries and white points, 13 spaces	Verbatim
Transfer functions, 10 distinct curves	Computed
HLG opto-electronic transfer	Synthesised (stand-in): inverse OETF only
Colour matching functions, 2°	Verbatim
Spectral locus	Computed, 380–715 nm
RGB ↔ XYZ matrices	Computed
Gamut volume in CIELAB	Computed, Halton sampling
MacAdam ellipses	Synthesised (stand-in)

Status chips filter the list. **Install a table** ("Drop a CSV or JSON") does not load files in the current version.

Export tab

Control	What it does
What to take	**This space**, **Every space**, **The comparison**, **The diagram**, **A link to this view**
As	Formats that fit the choice: JSON; JSON; CSV or JSON; SVG or PNG; Link
Download (names the file) or **Copy the link**	Saves the file, or copies the link
Copy to clipboard	Copies text payloads (hidden for PNG and link)
Exactly what you will get	Previews the payload and its size
Bring one in, **Add to the library**	Imports a space from JSON
Library table	Name, Type (Standard or Custom), White, Transfer, and **Load** / **Del**

Choice	File
This space, JSON	`gamut-lab-<space>.json`
Every space, JSON	`gamut-lab-spaces.json`
The comparison, CSV	`gamut-comparison.csv` with Name, Area (xy), Area % sRGB, Volume (Lab^3), Transfer, White Point
The comparison, JSON	`gamut-comparison.json` with area, area %, volume, spread, box, u′v′ share, transfer and white per drawn space
The diagram, SVG	`gamut-diagram.svg`
The diagram, PNG	`gamut-diagram.png`, at the canvas size
A link to this view	A URL encoding the drawn spaces, diagram mode, overlays and other view settings

The comparison covers only the spaces currently drawn on the diagram, not the whole library. A link longer than about 1,800 characters shows a warning that some clients truncate long links.

Reference tab

Control	What it does
Standards, **Formulas**, **Citations**, **Research notes**	Switch sections
Filter	Searches all four; `Esc` clears it

Standards are grouped as display and broadcast, cinema and VFX, photography and print, legacy, and colorimetry foundations. Formulas cover xy and u′v′, matrix derivation, the sRGB, PQ and HLG curves, shoelace area, CIELAB volume and overlap. Citations list 22 references. Research notes include a computed table of each space's area in xy and u′v′, % of sRGB and % of the locus.

Save to Library

The floating **Save to Library** button stores a snapshot of the diagram in your Auric Artisan Library.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Troubleshooting the Gamut Lab

Fixes for common problems in the Gamut Lab.

Symptom	Why it happens	What to do
A space I picked earlier disappeared from the diagram	The diagram draws up to five spaces; adding a sixth drops the oldest	Tick **Show all library spaces**, or pick the space again
Choosing **Custom** shows nothing	The custom space does not exist until you solve it	Enter primaries and white, then select **Solve from primaries**
My custom space vanished after reloading	A solved Custom space is not saved	Export it (**This space**, JSON) and import it with **Add to the library**
Import says "Invalid JSON"	The pasted object has no `name` or no `primaries`	Use the format in Define a custom colour space
Import says "Parse error: …"	The text is not valid JSON	Check quotes, commas and brackets
Choosing the 10° observer changes nothing	Only the CIE 1931 2° observer is used	Expected
The OKLab slice does not move with **Level**	In the current version the OKLab slice is fixed at L = 0.5; only the label changes	Use **Constant L* (CIELAB)** to move through lightness
In the u′v′ diagram, the vertex readout colour and membership look wrong	The readout treats the clicked coordinates as CIE 1931 x, y	Switch **Diagram mode** to **CIE 1931 xy** before probing
The vertex readout always shows L* 100	The probe takes the chromaticity at Y = 1, the brightest colour of that chromaticity	Expected; it describes chromaticity, not a particular lightness
The WebGL fill does nothing	The browser or device does not support WebGL 2, or it is disabled	Leave the fill off; the diagram works without it
The wireframe cannot be rotated	It is drawn at a fixed viewing angle	Expected
A share of "> 100%" appears	The space's primaries lie outside the spectral locus	Expected for ACES AP0 and ProPhoto RGB; read the volume instead

Symptom	Why it happens	What to do
Percentages differ from other tools	Other tools often divide by a different locus area, and often in xy	See Compare colour spaces
Volume figures change between runs	They are sampled estimates	Raise **Samples** on the Compare tab; the ± spread shrinks
The equation box shows an odd power-law form for Rec. 2020, Adobe RGB or ProPhoto	The box simplifies those curves	Use the Transfer tab or Reference › Formulas
The comparison export has fewer rows than spaces	It covers only the spaces drawn on the diagram	Draw the spaces you want first
A long link warning appears	Links over about 1,800 characters may be cut off by some apps	Send the link as a file or shorten the view (fewer spaces)
Copy to clipboard is missing	It is hidden for PNG and link choices	Use **Download**, or **Copy the link** for links

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Lab FAQ

Short answers to common questions about the Gamut Lab.

What is the difference between Display P3 and DCI-P3?

They share primaries. Display P3 uses the D65 white and the sRGB transfer curve; DCI-P3 (the cinema reference projector) uses a greener white at x 0.314, y 0.351 and a pure 2.6 gamma.

Why do Rec. 2020 and Rec. 2100 have the same triangle?

Rec. 2100 (HDR television) uses the Rec. 2020 primaries and white. It differs in the transfer curve: PQ or HLG instead of the BT.2020 curve.

Why does the ACES AP0 blue primary sit outside the diagram?

AP0 uses an imaginary blue primary so its triangle contains every visible colour. That makes it suitable for archiving, and it is why its share of the locus reads "> 100%".

Why is the u′v′ share quoted rather than xy?

Equal distances on the xy diagram are not equal visual differences; it exaggerates greens. u′v′ is more uniform, so areas there compare more fairly.

Is a bigger triangle always a bigger gamut?

Not exactly. The triangle ignores lightness. CIELAB volume counts how much colour a space holds at every lightness, which is why the lab measures it separately.

Why does the spectral locus stop at 715 nm?

The lab builds the locus from the published colour matching functions at four decimals. Beyond about 715 nm the values are too small to give a reliable chromaticity, so those points are left out rather than distorting the areas.

Are the MacAdam ellipses real data?

No. The overlay is an illustrative approximation of MacAdam's (1942) figure, useful for showing that ellipses vary in size and orientation. It is not a basis for reading discrimination thresholds.

Can the lab convert or map colours between spaces?

No. It defines and measures spaces. Use Gamut Mapping to map colours.

Where are my imported spaces stored?

In your browser's local storage for this site. They stay on this device and browser until you delete them with **Del** or clear site data.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Lab: supporting pages

Glossary, shortcuts, limits, related tools, release notes and privacy for the Gamut Lab.

These pages support the guides in docs/.

File	What it covers
glossary.md	Terms used in the lab
shortcuts.md	Keyboard and mouse shortcuts
limits-and-accuracy.md	What the figures are and are not
related.md	Related tools and Learn articles
release-notes.md	What has changed for users
privacy.md	What is stored and where

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Lab shortcuts

Keyboard and mouse shortcuts in the Gamut Lab.

The Gamut Lab has no single-key shortcuts of its own.

Shortcut	Where	What it does
Click or drag	Chromaticity diagram	Moves the probe and updates the vertex readout
Double-click	A chart	Opens it full screen
`Esc`	Full-screen view	Closes it
`Esc`	Reference › Filter	Clears the filter

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Lab glossary

Plain-language definitions of the terms used in the Gamut Lab.

ACES (AP0, AP1) — the Academy Color Encoding System. AP0 (ACES 2065-1) is an archival space with imaginary primaries; AP1 (ACEScg) is a working space with primaries inside the locus. Both are linear.

Chromaticity diagram — a map of colour without brightness. CIE 1931 xy is the classic form; CIE 1976 u′v′ is more uniform.

CIELAB — a colour space with lightness L* and axes a* (green–red) and b* (blue–yellow), used here to measure gamut volume and slices.

EOTF — electro-optical transfer function: turns a signal into light on a display.

Gamma — shorthand for a power-law transfer curve, such as 2.2 or 2.6.

Gamut — the colours a space can represent. On a chromaticity diagram, an RGB gamut is a triangle.

Gamut volume — the size of a gamut in CIELAB, in Lab^3 (units cubed).

Halton sampling — quasi-random sampling that covers space evenly; its error shrinks faster than random sampling.

HLG (hybrid log-gamma) — a relative HDR transfer curve (ARIB STD-B67, ITU-R BT.2100).

Imaginary primary — a primary outside the spectral locus: a mathematical colour no light can produce.

Jaccard index — the overlap of two areas divided by their combined area.

Line of purples — the straight edge of the chromaticity diagram between violet and red.

MacAdam ellipses — regions on the chromaticity diagram within which colours look the same to an observer, measured by MacAdam in 1942.

OETF — opto-electronic transfer function: encodes light as a signal.

OKLab — a perceptual colour space (Ottosson, 2020) with straighter hue lines than CIELAB.

PQ (perceptual quantiser) — the absolute HDR transfer curve of SMPTE ST 2084, mapping signal to 0–10,000 cd/m².

Primaries — the three reddest, greenest and bluest colours an RGB space mixes from.

RGB ↔ XYZ matrix — the 3 × 3 matrices that convert linear RGB of a space to CIE XYZ and back.

ROMM RGB — the ISO 22028-2 encoding behind ProPhoto RGB: a 1.8 power with a linear toe.

Shoelace formula — the method used to compute a polygon's area from its corners.

Spectral locus — the curved edge of the chromaticity diagram, made by single wavelengths.

Spread — here, how much a volume estimate changed against a run at half the samples. Not a confidence interval.

Transfer function — the curve between stored values and light.

u’v’ share — a triangle's area as a percentage of the spectral locus area in the u’v’ diagram.

White point — the chromaticity a space treats as white, such as D65 or D50.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Lab limits and accuracy

How far the Gamut Lab's diagrams, curves, areas and volumes can be trusted.

Exact

- **Primaries and white points** of the 13 spaces are the published chromaticities.
- **Matrices** are derived from the primaries and white by the standard construction.
- **Transfer curves** run as their standards define them, except HLG (see below). The BT.2020, ROMM and Adobe RGB curves were corrected in the current version and are marked so on the Transfer tab.
- **Areas** are computed by the shoelace formula in the named plane.

Computed with stated limits

- **Spectral locus** — built from the CIE 1931 2° colour matching functions at 5 nm and four decimals, from 380 to 715 nm. Beyond 715 nm the published values are too small to give a reliable chromaticity.
- **Share of the locus** — divided by the lab's own locus area in the same plane. Published shares divided by other locus areas differ (for Rec. 2020: 63.4% of the xy locus here against the often-quoted 75.8%).
- **Volume** — Halton quasi-random sampling in a box fitted to each space. The ± is the spread against a half-size run, not a confidence interval.

- **Overlap** — sampled in u′v′ with point-in-triangle tests.

Stand-ins

- **HLG** — the lab draws the inverse OETF only. The full HLG EOTF needs a display peak luminance and a surround the lab does not ask for.
- **MacAdam ellipses** — 24 hand-rounded ellipses approximating the published figure, drawn at a fixed exaggeration. MacAdam published 25 as coefficient triples; those are not held. Do not read thresholds from the overlay.

Not done

- The CIE 1964 10° observer (the option has no effect).
- The CIE 170-1 cone-fundamental chromaticity diagram.
- Colour conversion or gamut mapping between spaces.
- Loading tables through the Data tab's install area.

Current-version caveats

- The OKLab slice stays at L = 0.5 whatever the Level slider shows.
- The vertex readout treats clicks as CIE 1931 x, y, so its colour and membership are only correct in the xy diagram mode.
- The 3-D wireframe is a fixed projection.
- The WebGL fill's colours are an approximate sRGB rendering of each chromaticity.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Lab privacy

What the Gamut Lab stores, where, and what leaves your device.

- **All calculations run in your browser.** Diagrams, matrices, volumes and overlaps are computed on your device.
- **Imported colour spaces are stored in this browser.** They are kept in the site's local storage on your device, not on a server, and are removed with **Del** or by clearing site data. They do not follow you to another browser or device.
- **Exports are created locally** and saved through your browser's download, or copied to your clipboard when you choose.
- **The link** encodes your view settings (drawn spaces, diagram mode, overlays and similar). Anyone with the link sees them. If you defined a custom space, its primaries may be part of the encoded settings.
- **Save to Library** stores a snapshot in your Auric Artisan Library.
- The page loads the site's shared features like every other page on auricartisan.com; see the site's privacy policy.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Lab release notes

User-facing changes to the Gamut Lab.

Current version (2026 workbench rebuild)

- Tabs are now **Lab**, **Transfer**, **Compare**, **Data**, **Export** and **Reference**.
- Three transfer curves now match their standards: Rec. 2020 runs the BT.2020 curve (it previously ran the sRGB curve), ProPhoto runs ROMM with its linear toe, and Adobe RGB uses 563/256 rather than 2.2. HLG is labelled as the inverse OETF.

- The new **Transfer** tab plots every distinct curve, with PQ on its own absolute axis, and tags each as Exact, Partial or Corrected.
- Volume is measured with Halton sampling in a box fitted to each space, and quoted with a spread against a half-size run. Overlap is measured in u′v′.
- Shares of the spectral locus are quoted in u′v′, with xy kept as a legacy option, and "> 100%" is shown for spaces with imaginary primaries.
- The spectral locus is computed from the observer data and stops at 715 nm.
- The MacAdam ellipse overlay is labelled as a stand-in.
- Export now previews the exact payload and names the file before you download it; the library, import and export sit together on the Export tab.
- Citations were checked and several corrected; corrected entries are marked in Reference.

May 2026

The Gamut Lab User Guide was published in the site's Documentation library, describing the earlier version.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Related to the Gamut Lab

Tools and Learn articles that go well with the Gamut Lab.

Tools

Tool	Why
Gamut Mapping	Map colours from one space into another
HDR Gamut Plot	Rec.709, P3 and Rec.2020 with PQ, HLG, ICtCp and Jzazbz
Illuminants	The white points behind each space, and the Planckian and daylight loci
ICC Profile Parser	Read the primaries and curves stored in an ICC profile
LUT Lab	Build lookup tables that move colour between spaces

Learn articles

- Chromaticity Diagrams Without the Intimidation — https://auricartisan.com/library/learn/articles/2026-05-24-chromaticity-diagrams
- Gamut Volume: How Much Color a Wide Gamut Actually Adds — https://auricartisan.com/library/learn/articles/2026-08-26-gamut-volume-how-much-color
- HDR, Wide Gamut, PQ, HLG, and Modern Displays — https://auricartisan.com/library/learn/articles/2026-05-25-hdr-wide-gamut-pq-hlg-modern-displays
- Oklab and Oklch: Modern Perceptual Color Spaces — https://auricartisan.com/library/learn/articles/2026-06-02-oklab-oklch-perceptual-color-spaces
- CIE XYZ Explained — https://auricartisan.com/library/learn/articles/2026-05-24-cie-xyz-explained

Documentation on the site

The site's Documentation library (https://auricartisan.com/library/documentation/) has an earlier **Gamut Lab User Guide**. This documentation follows the current page.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Mapping

What the Gamut Mapping lab does, who it is for, where to find it and how to map your first colour.

A **gamut** is the set of colours a colour space or device can hold. When a colour moves from a wide space (such as Display P3 or ProPhoto RGB) into a narrower one (such as sRGB), some colours will not fit. **Gamut mapping** decides what they become.

The Gamut Mapping lab shows this as a horizontal slice through CIELAB at a lightness you choose: the source gamut on one side, the colours mapped into the target on the other, with each space's boundary outlined. It supports five RGB spaces (sRGB, Display P3, Rec. 2020, Adobe RGB and ProPhoto RGB), each decoded with its own transfer curve and white point, and four mapping methods named for what they do: chroma compression, clipping in the target RGB, uniform RGB scaling, and adapting the white point before clipping.

Around the slice you can probe any point, map a base colour or a list of colours, measure every space's volume in CIELAB with a stated standard error, compare the four methods on a fixed set of probe colours, and compare colour differences with three ΔE formulas. The lab is explicit that none of its methods is an ICC rendering intent: ICC perceptual and saturation intents are lookup tables built into profiles, not algorithms.

Who it is for

- Designers and brand teams checking what a wide-gamut colour becomes on standard screens or in sRGB files.
- Developers working with CSS Color 4 (Display P3, Rec. 2020) who want to see what clipping does.
- Photographers comparing ProPhoto, Adobe RGB and sRGB.
- Students learning how gamut boundaries change shape with lightness.

Where to find it

- Address: https://auricartisan.com/tool/general/colorimetry/gamut-map/
- In the tool collection (https://auricartisan.com/collections/) under **Colorimetry**, as **Gamut Mapping**.

Quick start

1. Open https://auricartisan.com/tool/general/colorimetry/gamut-map/.
2. Under **Colour Spaces**, set **Source** to **Display P3** and **Target** to **sRGB / Rec.709** (or select **Swap Source & Target** from the defaults).
3. Type a vivid P3 colour such as `#00FF00` in the base colour HEX field. The **Base Colour Info** panel says whether it is inside each space.
4. Move **Lightness Slice L*** to find the lightness where the source and target boundaries differ most.
5. Under **Method**, try each of the four options and watch the **Before / After (12 Test Colours)** strip change.
6. Click inside the **Source Slice** to probe a point.
7. Open **Export** and paste a list of HEX colours into **Map a list of colours at once**, then select **Run the list**.

What you can do

- Draw CIELAB *ab* slices of five RGB spaces at any L* from 5 to 95.
- Map colours with four methods and a compression strength.
- Probe any slice point for L*ab*, C*h°, sRGB, HEX and gamut membership.
- Compare ΔE_{00}, ΔE*94 and ΔE*76 between two colours.

- See the boundary of up to five spaces at five lightness levels on one scale.
- Measure each space's CIELAB volume by Monte Carlo sampling, with a standard error.
- Compare the four methods on the base colour or on 12 fixed probe colours.
- Export the slices as PNG, the settings as JSON, a link to the exact state, and a mapped colour list.

In this folder

- docs/ — how to use the lab
  - Getting started
  - Map a colour into a smaller gamut
  - Compare spaces and methods
  - Map a list of colours
  - Reference
  - Troubleshooting
  - FAQ
- others/ — background and supporting pages
  - Glossary
  - Shortcuts
  - Limits and accuracy
  - Related tools
  - Release notes
  - Privacy

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Mapping documentation

The guides, reference and help pages for the Gamut Mapping lab.

This folder explains how to use the Gamut Mapping lab: a screen tour, task guides for mapping a colour, comparing spaces and methods and mapping a list, a complete reference and help pages.

File	What it covers
getting-started.md	Opening the lab, the six tabs, and reading a slice
map-a-colour.md	Mapping one colour and choosing a method and strength
compare-spaces-and-methods.md	The Boundary and Methods tabs, volumes and the ΔE playground
map-a-colour-list.md	Mapping up to 50 colours and exporting
reference.md	Every control, readout and export
troubleshooting.md	Symptoms, causes and fixes
faq.md	Short answers

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Getting started with Gamut Mapping

Open the Gamut Mapping lab, learn its tabs, and read a CIELAB gamut slice.

Open the lab

Go to https://auricartisan.com/tool/general/colorimetry/gamut-map/. The heading is **Gamut Mapping**. Everything is computed in your browser.

Key ideas

- **CIELAB** describes a colour by lightness **L*** (0 black to 100 white) and two opponent axes, **a*** (green to red) and **b*** (blue to yellow). **Chroma C*** is the distance from the grey axis and **hue h°** the angle around it.
- A **gamut slice** is a cut through CIELAB at one L*. Seen from above, each RGB space's gamut at that lightness is an irregular shape around the grey centre. Colours outside the shape cannot be represented in that space.
- A colour is **out of gamut (OOG)** for a space when converting it to that space's linear RGB gives a channel below 0 or above 1.
- Each space has its own **transfer function** (the curve between stored values and light) and **white point**. The lab decodes a HEX value with the source space's own curve, and draws the slice on the source space's white.

A tour of the screen

Tab	What it is for
Lab	Choose the colour, spaces, method and slice, and see the mapping
Boundary	Boundaries of up to five spaces at five lightness levels, and volumes in CIELAB
Methods	The four mapping methods explained, with figures on the base colour or 12 probe colours
Data	The nine datasets the lab uses and their status

Tab	What it is for
Export	PNG, JSON, a link to the exact state, and batch mapping
Reference	Standards, formulas, citations and research notes with a filter

The Lab tab

The left panel groups the controls: **Base Colour**, **Lightness Slice L***, **Colour Spaces**, **Method**, **Compression Strength**, **Sampling**, **Overlays** and **ΔE Playground**.

The right side, **Gamut Mapping Results**, shows:

- **Source Slice** — every colour inside the source gamut at the chosen L*, drawn as an a*b* disc (a* across, b* up, both from −128 to +128);
- **Mapped to Target** — the same colours after mapping into the target;
- a statistics line with the slice areas and their ratio;
- a line naming the transfer function, the Lab white and the method;
- **Probe Point** — details of the point you last clicked;
- **Before / After (12 Test Colours)** — twelve vivid colours before and after mapping.

Read your first slice

1. Keep the defaults: base colour `#247DEB`, L* 65, source **sRGB / Rec.709**, target **Display P3**.
2. Look at the **Source Slice**. The filled region is sRGB at L* 65. The thicker outline is the source boundary; the thinner outline (with **Mark out-of-gamut in source** on) is the target's boundary, so you can see where P3 extends beyond sRGB.
3. Select **Swap Source & Target**. Now the source is Display P3 and the target sRGB. The filled region grows, and the sRGB outline sits inside it.
4. Look at **Mapped to Target**. Colours outside sRGB have been pulled inside its boundary.
5. Click near the edge of the source slice. **Probe Point** shows L*, a*, b*, C*, h°, an sRGB value and HEX, and whether the point is in each space ("Yes" or "OOG").
6. Move **Lightness Slice L*** from 20 to 90 and watch the boundaries change shape.

The slices are painted in sRGB so your screen can show them. Colours outside sRGB are displayed clipped, even though the geometry is correct.

Next steps

- Map a colour into a smaller gamut
- Compare spaces and methods
- Map a list of colours

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Compare spaces and methods

Use the Boundary and Methods tabs and the ΔE Playground to compare colour spaces, gamut volumes, mapping methods and colour differences.

Compare boundaries up the lightness axis

1. Open the **Boundary** tab.
2. Under **Spaces on the plot**, select the spaces to draw. Each chip toggles one space; sRGB and Display P3 are on at first.
3. Read **The boundary up the lightness axis**: five tiles at L* 20, 40, 65, 80 and 90. All tiles share one scale (shown in the note, for example "one scale, ±150"), so sizes can be compared directly. The source space from the Lab tab is drawn with a thicker line.

What to look for: at low and high lightness all gamuts shrink towards the grey axis; the biggest differences are in the middle; yellows keep chroma at high lightness and blues at low lightness.

Measure gamut volumes

1. On the **Boundary** tab, set **Volume samples** (**n**) from 5,000 to 200,000 in steps of 5,000. The default is 20k.
2. Select **Measure the five**. The button reads "Measuring…" while it works.
3. Read **The five spaces, measured**, sorted from smallest to largest:

Column	Meaning
Space	The space and its colour swatch
White	The white point the space is defined against (D65, or D50 for ProPhoto)
Lab volume ± 1 SE	The volume in CIELAB units cubed, and its standard error
vs sRGB	The volume as a multiple of sRGB's
Sample box	The a*, b* half-width of the sampling box, fitted to each space

The volume is estimated by random (Monte Carlo) sampling in a box fitted to each space. Raising the sample count shrinks the standard error. The results change slightly on each run.

Volume in CIELAB is a more meaningful size than the area of a triangle on the xy chromaticity diagram, which ignores lightness and overstates green. The lab no longer shows the xy area percentage.

Compare the four methods

1. Open the **Methods** tab.
2. Under **Compare on**, choose **The base colour** or **12 probe colours**. The probe set is fixed (#FF0000, #00FF00, #0000FF, #FFFF00, #FF00FF, #00FFFF, #FF7F00, #7F00FF, #00FF7F, #BF4040, #4080BF and #808080), so the comparison means the same thing whatever base colour you pick.
3. Open each method's row to read its description and three figures:
   - **mean ΔE_{00}** — average colour change caused by mapping;
   - **max ΔE_{00}** — the largest change;
   - **out of gamut** — how many of the colours were outside the target.
4. **Provenance** in the rail counts published intents (0) and this tool's methods (4).

The figures use the source, target and strength set on the Lab tab. The method selected on the Lab tab opens by default.

Compare two colours with the ΔE Playground

1. On the **Lab** tab, find **ΔE Playground**.

2. Set colour **A** and colour **B** with the pickers, or type a six-digit HEX in each field and press `Enter` or move focus away.
3. Read the result line: **ΔE_{00}** with a verbal band (Imperceptible, Barely perceptible, Noticeable, Significant or Very different), **ΔE^*_{94}** and **ΔE^*_{76}**.

The formula menu beside the result does not change the readout in the current version; all three figures are always shown.

Result

You can see how the five spaces differ in shape and size, and how differently the four methods treat the same colours.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Map a list of colours

Map up to 50 HEX colours at once and export images, settings and a link.

Map the list

1. On the **Lab** tab, set the **Source** and **Target** spaces, the **Method** and, for chroma compression, the **Compression Strength**. The list goes through the same settings.
2. Open the **Export** tab and find **Map a list of colours at once**.
3. Enter HEX values, one per line (commas also separate them). Three-digit (`#F0A`) and six-digit (`#FF00AA`) forms are accepted, with or without `#`. Up to 50 are used.
4. Select **Run the list**.
5. Read the table:

Column	Meaning
Input	Your colour and its swatch
Output	The mapped colour, converted to sRGB HEX for display on screen
OOG	**Yes** if the input was outside the target gamut, otherwise **No**
ΔE_{00}	The colour change caused by mapping
Lab	The input's L*, a*, b*

A summary line gives the number processed, how many were out of gamut and the average ΔE_{00}. If no valid HEX is found, the panel says "No valid hex values found."

The **Output** HEX is an sRGB value for your screen. When the target is wider than sRGB (for example Rec. 2020), it is not the target-space code value.

Export

Button	What you get
Source slice, PNG	`gamut-map-gm-canvas-src.png`: the source slice at its drawn size, on the dark background, with outlines
Mapped slice, PNG	`gamut-map-gm-canvas-dst.png`: the mapped slice
Everything as JSON	`gamut-map-config.json`: source space, target space, method, strength, L*, resolution, sampler and a timestamp
Link, then **Copy**	A link that reopens this state

The link records the source and target spaces, the method, L*, resolution, sampler, strength and base colour. Overlays and the ΔE Playground colours are not included. The JSON file holds settings only, not the figures on screen.

Result

You have every colour in your list mapped with its out-of-gamut flag and ΔE_{00}, and files or a link to share the setup.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Map a colour into a smaller gamut

Check whether a colour fits a target space, choose a mapping method and strength, and read the result.

The four methods

All four are this tool's own methods. None is an ICC rendering intent.

Method	What it does	Keeps	Gives up
Chroma compression	Holds L* and hue, and reduces chroma until the colour fits (found by bisection). **Compression Strength** sets how far: 1 lands exactly on the boundary, 0 keeps the original chroma and leaves the final clip to deal with it	Lightness and hue	Vividness
Clip in target RGB	Converts to the target's linear RGB and clamps each channel to 0–1	Everything already inside the gamut	Detail and hue near the boundary; different out-of-gamut colours can collapse onto one
Uniform RGB scale	Divides all three channels by the largest, so the brightest channel becomes 1; if a channel is negative it falls back to clipping	The ratio between channels	Lightness: everything darkens together
Adapt white, then clip	Adapts from the source white to the target white with the Bradford transform, then clips. This is the shape of ICC relative colorimetric	The relationship to each space's white	Accuracy of out-of-gamut colours, which are clipped

The last method only differs from **Clip in target RGB** when the two spaces have different white points. Of the five spaces, ProPhoto RGB uses D50; the others use D65.

Steps

1. On the **Lab** tab, enter your colour under **Base Colour**: use the colour picker or type a HEX value such as `#00C853` in the field. **Random** picks a random colour.
2. Under **Colour Spaces**, set **Source** to the space the colour is defined in and **Target** to where it must go. The HEX value is interpreted in the source space, decoded with that space's own transfer curve.
3. Read **Base Colour Info**: L*, a*, b*, C* and h°, then whether the colour is inside the source space and inside the target space, each shown as **Yes** or **OOG**.
4. Set **Lightness Slice L*** to the colour's L* (shown in Base Colour Info) so the white ring marking the base colour sits on the slice you are looking at.
5. Choose a **Method**.
6. For **Chroma compression**, set **Compression Strength** (0 to 1, default 0.90). An in-gamut colour is never changed, whatever the strength.
7. Read the effect in **Before / After (12 Test Colours)**: each row shows a vivid test colour on the left and its mapped result on the right.

8. To get exact numbers for your colour, open **Methods**, choose **The base colour** under **Compare on**, and read mean ΔE_{00}, max ΔE_{00} and whether it was out of gamut for each method. Or add the colour to the batch list on the **Export** tab (Map a list of colours).

About the Mapped to Target slice

In the current version the **Mapped to Target** slice is always drawn with chroma compression all the way to the boundary, whichever method and strength you choose. The **Before / After** strip, the batch list and the **Methods** tab use your chosen method and strength. Use those to judge the method; use the slice to see the shape of the target gamut.

Reading ΔE00

ΔE_{00} (CIEDE2000) measures how different the mapped colour is from the original. As a rough guide, which the lab also uses in its playground: below 1 imperceptible, 1–2 barely perceptible, 2–3.5 noticeable, 3.5–5 significant, and above 5 very different.

Result

You know whether the colour fits the target, what each method turns it into, and how large the change is.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Mapping reference

Every control, readout, option and export in the Gamut Mapping lab.

Address: https://auricartisan.com/tool/general/colorimetry/gamut-map/

The five colour spaces

Space	White	Transfer curve used to decode HEX
sRGB / Rec.709	D65	sRGB piecewise curve (IEC 61966-2-1)
Display P3	D65	sRGB piecewise curve
Rec. 2020	D65	BT.2020 curve
Adobe RGB	D65	Power law 563/256 (about 2.2)
ProPhoto RGB	D50	ROMM: power 1.8 with a linear toe

The slice is drawn on the source space's white. A space with a different white is adapted onto it with the Bradford transform before its boundary is tested.

Lab tab

Controls

Control	What it does	Values or range	Default
Base colour picker and HEX field	The colour to analyse, interpreted in the source space	Any six-digit HEX	#247DEB
Random	Picks a random base colour	Button	—

Control	What it does	Values or range	Default
Lightness Slice L*	The lightness of the *ab* slice	5 to 95, step 1	65
Source	The space colours come from	The five spaces	sRGB / Rec.709
Target	The space colours go to	The five spaces	Display P3
Swap Source & Target	Exchanges the two	Button	—
Method	The mapping method	Chroma compression, Clip in target RGB, Uniform RGB scale, Adapt white, then clip	Chroma compression
Compression Strength	How far chroma compression goes (used by chroma compression only)	0 to 1, step 0.01	0.90
Resolution	Sample grid for the slices	60 to 512, step 20	120 × 120
Sampler	How sample points are placed	Uniform grid; Stratified (2×2), randomly jittered; Halton (base 2/3), a low-discrepancy sequence	Uniform grid
Adaptive edge refinement	Shown but not applied in the current version	On or off	Off
Mark out-of-gamut in source	Draws the target's boundary on the source slice	On or off	On
Show base colour on slice	Rings the base colour on the source slice	On or off	On
Hue tick lines	Draws radial lines every 30° of hue	On or off	Off
Gamut boundary outlines	Outlines the source boundary on the source slice and the target boundary on the mapped slice	On or off	On
Axes & labels	Shown but not applied; axes are always drawn	On or off	On
ΔE Playground **A**	First colour (picker and HEX field)	Six-digit HEX	#735244
ΔE Playground **B**	Second colour	Six-digit HEX	#3BB273
ΔE formula menu	Shown but does not change the readout; all three formulas are always listed	ΔE_{00}, ΔE_{94}, ΔE_{76}	ΔE_{00}

Readouts

Readout	What it shows
Base Colour Info	L*, a*, b*, C*, h°, and whether the colour is in the source and in the target (Yes or OOG)
Source Slice	Colours inside the source gamut at the chosen L*, painted in sRGB, a* and b* from −128 to +128. Click to probe
Mapped to Target	The source-gamut colours mapped into the target. In the current version this slice always uses chroma compression to the boundary, whatever the method and strength
Statistics line	L*, the source and target slice areas (in *ab* units), their ratio as a percentage, the method and the sampler with its resolution
Provenance line	The transfer curve, the white the Lab values are referenced to, and the method
Probe Point	L*, a*, b*, C*, h°, sRGB, HEX, and whether the point is in the source and the target
Before / After (12 Test Colours)	#FF0000, #00FF00, #0000FF, #FFFF00, #FF00FF, #00FFFF, #FF8800, #88FF00, #0088FF, #FF0088, #8800FF and #00FF88, before (left) and after (right) mapping with your method and strength
ΔE Playground result	ΔE_{00} with its verbal band, ΔE*94 and ΔE*76

ΔE_{00} verbal bands: below 1 Imperceptible; below 2 Barely perceptible; below 3.5 Noticeable; below 5 Significant; 5 and above Very different.

Boundary tab

Control or readout	What it does	Values	Default
Spaces on the plot	Toggles each space on the boundary tiles	Five chips	sRGB and Display P3 on
Boundary tiles	a*b* boundaries at L* 20, 40, 65, 80 and 90 on one shared scale	—	—
Volume samples (n)	Monte Carlo samples per space	5,000 to 200,000, step 5,000	20k
Measure the five	Estimates every space's CIELAB volume	Button	—
Volume table	Space, White, Lab volume ± 1 SE, vs sRGB, Sample box	—	—

Methods tab

Control or readout	What it does	Values	Default
Compare on	The colours the figures are computed on	The base colour; 12 probe colours	The base colour
Method rows	Name, what it does, tag "this tool", explanation, and mean ΔE_{00}, max ΔE_{00} and out-of-gamut count	—	The Lab's method open
Provenance	Count of published intents (0) and this tool's methods (4)	—	—

Data tab

Dataset	Status
Primaries, 5 RGB spaces	Verbatim
White points D65 and D50	Verbatim
Transfer functions, 5 spaces	Computed
Bradford adaptation matrix	Verbatim
Gamut boundary, a*b* slice	Computed (360 hues per L*, 24 bisection steps)
ΔE_{00}, ΔE94, ΔE76	Computed
Gamut volume in CIELAB	Computed (uniform Monte Carlo in a fitted box)
ICC perceptual and saturation tables	Absent
Reference gamut-mapping results	Absent

Status chips filter the list (**All**, **Verbatim**, **Computed**, **Absent**). The **Install a table** area ("Drop a CSV, JSON or ICC profile") does not load files in the current version.

Export tab

Control	Output
Source slice, PNG	`gamut-map-gm-canvas-src.png`
Mapped slice, PNG	`gamut-map-gm-canvas-dst.png`
Everything as JSON	`gamut-map-config.json` with `sourceSpace`, `targetSpace`, `intent`, `strength`, `lStar`, `resolution`, `sampler`, `timestamp`
Link field, **Link**, **Copy**	A URL with `src`, `dst`, `int` (method), `l` (L*), `res`, `sam`, `str` and `hex`
Map a list of colours at once, **Run the list**	Table of Input, Output, OOG, ΔE_{00} and Lab for up to 50 colours, with a summary

Method values in the link: `compress`, `clip`, `scale`, `adapt-clip`. Sampler values: `grid`, `stratified`, `halton`. Space values: `sRGB`, `P3`, `Rec2020`, `AdobeRGB`, `ProPhoto`.

Reference tab

Control	What it does
Standards, **Formulas**, **Citations**, **Research notes**	Switch sections; each shows a count
Filter	Searches all four; `Esc` clears it

Standards: IEC 61966-2-1 (sRGB), SMPTE ST 432-1 (Display P3), ITU-R BT.2020-2, Adobe RGB (1998), ProPhoto RGB (ROMM), ICC.1:2022 rendering intents, ISO 11664-6 (CIEDE2000) and CIE 15:2018, each with a badge read from the data register. Formulas: hard clip, chroma compression, uniform scaling, Bradford adaptation plus clip, boundary bisection, RGB–XYZ matrix derivation, ΔE formulas, and the Halton and stratified samplers. Some research notes use the older intent names and mention area comparisons that have moved to the Boundary tab; the Methods tab reflects the current methods.

Save to Library

The floating **Save to Library** button stores a snapshot of the first slice image in your Library.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Troubleshooting Gamut Mapping

Fixes for common problems in the Gamut Mapping lab.

Symptom	Why it happens	What to do
The Mapped to Target slice does not change when I change method or strength	In the current version that slice is always drawn with chroma compression to the boundary	Judge methods with **Before / After**, the **Methods** tab or the batch list, which use your method and strength
Changing Compression Strength has no effect	Strength only applies to **Chroma compression**, and never changes a colour that is already in gamut	Choose Chroma compression and use an out-of-gamut colour
Adapt white, then clip gives the same result as **Clip in target RGB**	Both spaces share the D65 white, so there is nothing to adapt	Use ProPhoto RGB (D50) as one of the spaces to see a difference
The slice looks dull or clipped at the edges	Slices are painted in sRGB so your screen can show them; wider colours are displayed clipped	Expected. Use the probe and figures, not the painted colour, for wide-gamut values

Symptom	Why it happens	What to do
The slice looks grainy or speckled	A low **Resolution**, or the **Stratified** sampler, which places points at random	Raise Resolution, or choose **Uniform grid** or **Halton**
The page is slow after raising Resolution	512 × 512 samples per slice is a lot of work	Lower Resolution
My base colour's ring is not on the coloured area	The slice is at a different lightness from the colour	Set L* to the colour's L* shown in Base Colour Info
Adaptive edge refinement or **Axes & labels** does nothing	These controls are not applied in the current version	No workaround needed
The ΔE formula menu does not change the result	All three formulas are always shown	Read the one you need from the result line
A HEX typed into the ΔE Playground is ignored	The field needs a full `#RRGGBB` value, applied when you press Enter or leave the field	Type a six-digit value with `#`
Volume figures change each time	They are Monte Carlo estimates	Raise **Volume samples** to shrink the standard error
The batch says "No valid hex values found."	No line was a 3- or 6-digit HEX	Put one colour per line, for example `#FF00AA`
Batch Output for Rec. 2020 looks the same as for P3	Output HEX is converted to sRGB for display	Use OOG and ΔE_{00} to compare; the HEX is a screen preview
The JSON file has no numbers from the screen	**Everything as JSON** saves the settings only	Use the batch table or PNG images for figures
The link field stays empty	The link is built only when you select **Link**	Select **Link**, then **Copy**
Dropping a file on **Install a table** does nothing	The area is not active in the current version	No workaround

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Mapping FAQ

Short answers to common questions about the Gamut Mapping lab.

Are these the ICC rendering intents?

No. ICC perceptual and saturation intents are lookup tables that a profile author builds for a particular device, so no general algorithm can reproduce them. The lab's four methods are named for what they do. **Adapt white, then clip** has the shape of ICC relative colorimetric; **Clip in target RGB** (no white adaptation) resembles absolute colorimetric clipping.

Which method should I use?

For photographs, chroma compression usually looks most natural because it keeps lightness and hue. For logos and spot colours that must match exactly where they can, clipping keeps in-gamut colours untouched. The lab does not rank methods, because there is no agreed reference set of correct mappings; it reports what each does.

Why is the slice painted in sRGB?

Most screens and browsers show sRGB reliably. The geometry (which colours are in each gamut) is computed exactly; only the painting is limited to what your screen can show.

Why does ProPhoto RGB look lopsided?

ProPhoto RGB is defined against D50 and includes colours outside human vision. The lab adapts it onto the Lab plane's white with Bradford before testing its boundary, and its chroma can run past the ±128 range the slice draws.

Why do the areas in the statistics line differ from percentages I see elsewhere?

They are areas of the a*b* slice at one lightness, not the "percentage of the spectral locus" often quoted from the xy diagram. For total size, use the volume on the Boundary tab.

Is a ΔE00 of 2 acceptable?

For critical work an average below 2 is a common target, and the lab's research notes suggest the same. It depends on your application.

Can I load an ICC profile?

No. The Data tab shows an **Install a table** area, but it does not load files in the current version.

Is anything uploaded?

No. All mapping runs in your browser.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Mapping: supporting pages

Glossary, shortcuts, limits, related tools, release notes and privacy for the Gamut Mapping lab.

These pages support the guides in docs/.

File	What it covers
glossary.md	Terms used in the lab
shortcuts.md	Keyboard and mouse shortcuts
limits-and-accuracy.md	What the figures are and are not
related.md	Related tools and Learn articles
release-notes.md	What has changed for users
privacy.md	What stays in your browser

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Mapping shortcuts

Keyboard and mouse shortcuts in the Gamut Mapping lab.

The lab has no single-key shortcuts of its own.

Shortcut	Where	What it does
Click	Source Slice or Mapped to Target	Probes that *ab* point at the current L*
Double-click	A slice or the Before / After strip	Opens it full screen
`Esc`	Full-screen view	Closes it
`Esc`	Reference › Filter	Clears the filter
`Enter`	A ΔE Playground HEX field	Applies the typed colour

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Mapping glossary

Plain-language definitions of the terms used in the Gamut Mapping lab.

a*, b* — the two opponent axes of CIELAB: a* runs green to red, b* blue to yellow.

Adobe RGB (1998) — a D65 space with a greener green primary than sRGB, encoded with a power of 563/256.

Bisection — finding a boundary by repeatedly halving an interval. The lab finds each gamut boundary this way along 360 hue directions.

Bradford transform — a chromatic adaptation transform used to move colours between white points.

C* (chroma) — distance from the grey axis in CIELAB: how vivid a colour is.

CIELAB — a colour space with lightness L* and axes a* and b*, designed so equal distances are roughly equal perceived differences.

Clipping — forcing each out-of-range RGB channel to 0 or 1.

Display P3 — a D65 space with the DCI-P3 primaries and the sRGB curve, used on many modern devices and in CSS Color 4.

ΔE (delta E) — a colour difference. ΔE*76 is plain distance in CIELAB; ΔE*94 and ΔE_{00} (CIEDE2000) weight lightness, chroma and hue to match perception better.

Gamut — the set of colours a space or device can represent.

Gamut boundary — the edge of a gamut. On a slice it is a closed outline around the grey centre.

Gamut mapping — choosing what an out-of-gamut colour becomes in the target space.

Gamut volume — the size of a gamut in CIELAB, in units cubed.

Halton sequence — a low-discrepancy sequence of points that covers an area more evenly than random points.

h° (hue angle) — the angle around the grey axis in CIELAB.

ICC rendering intent — one of four ways an ICC colour-management system converts colours: perceptual, relative colorimetric, saturation and absolute colorimetric.

L* (lightness) — CIELAB lightness, from 0 (black) to 100 (white).

Monte Carlo estimate — an estimate made from random samples, reported here with its standard error.

OOG (out of gamut) — outside the gamut of a given space.

ProPhoto RGB (ROMM RGB) — a very wide D50 space whose primaries include imaginary colours; encoded with a power of 1.8 and a linear toe.

Rec. 2020 (BT.2020) — the ultra-high-definition television space with near-spectral primaries.

Slice — a cut through CIELAB at one lightness.

sRGB — the standard web and display space (IEC 61966-2-1), with D65 white and a piecewise transfer curve.

Standard error (SE) — the typical size of the error in an estimate.

Stratified sampling — dividing the area into cells and placing one random point in each.

Transfer function — the curve between stored code values and linear light.

White point — the reference white a space is defined against: D65 or D50 here.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Mapping limits and accuracy

What the Gamut Mapping lab computes exactly, what it approximates and what it does not do.

The lab is a visual and numerical explorer that runs in your browser. It is not an ICC colour-management engine and does not reproduce any vendor's conversion.

Exact

- **Primaries and white points** of the five spaces are published values, held verbatim.
- **Transfer curves** are each space's own: the sRGB curve (sRGB and Display P3), the BT.2020 curve, Adobe RGB's 563/256 power law and ProPhoto's ROMM curve with its linear toe.
- **Boundaries** are traced by 24 bisection steps along 360 hue directions, accurate to about 0.00001 in chroma — far below what the plot can show.

- **ΔE formulas** follow CIE definitions; ΔE_{00} follows the Sharma, Wu and Dalal implementation notes.

Estimated

- **Gamut volume** is a Monte Carlo estimate with a standard error. It changes slightly each run and narrows as you raise the sample count.
- **Out-of-gamut tests** allow a tiny tolerance so that colours on the boundary count as inside.

Not done

- **ICC perceptual and saturation intents** — these are profile lookup tables, not algorithms, and no profile is loaded.
- **Ranking methods** — there is no published reference set of correct mappings, so the lab reports what each method does.
- **Loading profiles or tables** — the Data tab's install area is not active.
- **Showing wide-gamut colours as they really are** — all painting is in sRGB.

Current-version caveats

- The **Mapped to Target** slice always uses chroma compression to the boundary; the other mapping outputs follow your method and strength.
- **Adaptive edge refinement**, **Axes & labels** and the ΔE formula menu do not change the output.
- The slices show a* and b* from −128 to +128; ProPhoto RGB's chroma can exceed that range, so its outer boundary may run off the slice.
- **Output** HEX values in the batch list are sRGB display values, clipped to sRGB.
- The JSON export records settings, not results.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Mapping privacy

What the Gamut Mapping lab does with your colours and settings.

- **All mapping runs in your browser.** Colours, slices, volumes and batch lists are computed on your device and not sent anywhere.
- **Exports are created locally** and saved through your browser's download.
- **The share link** contains the spaces, method, L*, resolution, sampler, strength and base colour. Anyone with the link sees those settings.
- **Clipboard. Copy** writes the link to your clipboard only when you select it.
- **Save to Library** stores a snapshot in your Auric Artisan Library.
- The page loads the site's shared features like every other page on auricartisan.com; see the site's privacy policy.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Gamut Mapping release notes

User-facing changes to the Gamut Mapping lab.

Current version (2026 workbench rebuild)

- Tabs are now **Lab**, **Boundary**, **Methods**, **Data**, **Export** and **Reference**.

- The four methods are renamed for what they do: **Chroma compression**, **Clip in target RGB**, **Uniform RGB scale** and **Adapt white, then clip**. They were previously labelled with ICC intent names, two of which were on the wrong method.
- Every space now decodes HEX values with its own transfer curve, and ProPhoto RGB uses its own D50 white, adapted onto the Lab plane.
- **Compression Strength** now affects the result of chroma compression.
- Gamut volume moved to the new **Boundary** tab, measured in a box fitted to each space and reported with a standard error. The xy-area percentage was removed.
- The new **Methods** tab compares the four methods on the base colour or 12 fixed probe colours.
- The new **Data** tab lists the nine datasets and their status.
- The Export tab offers PNG slices, a settings JSON, a link to the exact state and a 50-colour batch.

May 2026

The Gamut Map User Guide was published in the site's Documentation library, describing the earlier version.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Related to Gamut Mapping

Tools and Learn articles that go well with the Gamut Mapping lab.

Tools

Tool	Why
Gamut Lab	13 RGB spaces on the chromaticity diagram, their transfer curves and volumes
HDR Gamut Plot	Rec.709, P3 and Rec.2020 with HDR transfer curves and perceptual planes
Chromatic Adaptation	Moving colours between white points with eight transforms
ICC Profile Parser	Inspect the ICC profiles whose lookup tables implement rendering intents
Contrast Checker	Check that a mapped colour still has enough contrast

Learn articles

- Gamut Mapping — https://auricartisan.com/library/learn/articles/2026-06-17-gamut-mapping-clipping-and-rendering-intents
- Gamut Volume: How Much Color a Wide Gamut Actually Adds — https://auricartisan.com/library/learn/articles/2026-08-26-gamut-volume-how-much-color
- CIELAB and LCH Explained — https://auricartisan.com/library/learn/articles/2026-05-24-cielab-and-lch-explained
- ΔE Metrics from CIE76 to CIEDE2000 and ΔEITP — https://auricartisan.com/library/learn/articles/2026-05-26-delta-e-metrics-cie76-ciede2000-itp
- HDR, Wide Gamut, PQ, HLG, and Modern Displays — https://auricartisan.com/library/learn/articles/2026-05-25-hdr-wide-gamut-pq-hlg-modern-displays

Documentation on the site

The site's Documentation library (https://auricartisan.com/library/documentation/) has an earlier **Gamut Map User Guide** that uses the old intent names. This documentation follows the current page.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

HDR Gamut Plot

What the HDR Gamut Plot does, who it is for, where to find it and how to take your first reading.

High dynamic range (HDR) video changes two things at once: it widens the **gamut** (more saturated colours, typically Rec. 2020 primaries) and extends the **luminance range** (from about 100 nits for standard video to as much as 10,000 nits). The HDR Gamut Plot puts both on screen.

It draws the three broadcast gamuts — Rec. 709/sRGB, Display P3 and Rec. 2020 — on a CIE 1931 xy or CIE 1976 u′v′ chromaticity diagram, and measures each as a share of the spectral locus. It plots the HDR transfer curves (PQ from SMPTE ST 2084 and HLG from BT.2100, with its system gamma) against standard gamma 2.2 and 2.4, on an axis in real cd/m^2. It draws the gamuts in two HDR perceptual spaces, ICtCp and Jzazbz, at a luminance you choose, and stacks their boundaries at eight luminance levels from 0.1 to 10,000 nits. You can also drop in an image to measure how much of each gamut its colours occupy.

The page is careful about what a nit is. A PNG or JPEG does not record absolute luminance, so the lab does not claim one from an image; where a luminance comes from the peak-brightness control rather than from a signal, it says so.

Who it is for

- Video, streaming and games developers working with HDR10, HLG or wide-gamut content.
- Designers preparing assets for P3 and HDR displays.
- Students learning PQ, HLG, ICtCp and the difference between absolute and display-referred curves.

Where to find it

- Address: https://auricartisan.com/tool/general/gamut-and-rendering/hdr-gamut-plot/
- In the tool collection (https://auricartisan.com/collections/) under **Gamut & Rendering**, as **HDR Gamut Plot**.

Quick start

1. Open https://auricartisan.com/tool/general/gamut-and-rendering/hdr-gamut-plot/. The diagram shows the three gamuts on CIE 1931 xy.
2. Read **Gamut metrics**: each space's primaries, white, share of the locus and area.
3. Move the pointer over the diagram to read x, y, XYZ, L* and ICtCp values under it.
4. Press `A` to switch to the u′v′ diagram, then `A` again to switch back.
5. Open **Curves**, set **Peak brightness** to 1000 nits, and press `3` (PQ) and `4` (HLG) to compare the curves.
6. Open **Planes** to see the gamuts in ICtCp at the luminance set by **Signal level**.
7. Open **Export**, choose **The measurements** as **CSV**, and select the download button.

What you can do

- Compare Rec. 709/sRGB, Display P3 and Rec. 2020 on xy or u′v′, with the spectral locus and line of purples.
- Read each gamut's area and share of the spectral locus in the plane on screen.
- Plot PQ, HLG, gamma 2.4 (BT.1886) and gamma 2.2 as absolute luminance against signal.
- Set peak brightness (100 to 10,000 nits) and HLG surround, and read HLG's system gamma.
- View the gamuts in ICtCp or Jzazbz at a chosen luminance, and as boundary rings at eight luminance levels.
- Measure how much of each gamut an image's colours occupy.

- Convert a list of HEX colours to CIELAB, ICtCp and Jzazbz.
- Export measurements (CSV, JSON, CSS), the curve (CSV), the diagram, curve plot and plane (PNG), or a link.

In this folder

- docs/ — how to use the tool
  - Getting started
  - Compare broadcast gamuts
  - Read HDR transfer curves
  - Explore ICtCp and Jzazbz planes
  - Reference
  - Troubleshooting
  - FAQ
- others/ — background and supporting pages
  - Glossary
  - Shortcuts
  - Limits and accuracy
  - Related tools
  - Release notes
  - Privacy

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HDR Gamut Plot documentation

The guides, reference and help pages for the HDR Gamut Plot.

This folder explains how to use the HDR Gamut Plot: a screen tour, task guides for gamuts, transfer curves and perceptual planes, a complete reference and help pages.

File	What it covers
getting-started.md	Opening the tool, the six tabs and the chromaticity diagram
compare-broadcast-gamuts.md	The three gamuts, locus shares and plotting an image
read-hdr-curves.md	PQ, HLG and gamma curves, peak brightness and surround
explore-perceptual-planes.md	ICtCp and Jzazbz planes, boundary rings and HEX conversion
reference.md	Every control, readout and export
troubleshooting.md	Symptoms, causes and fixes
faq.md	Short answers

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Getting started with the HDR Gamut Plot

Open the HDR Gamut Plot, learn its tabs, and read the chromaticity diagram.

Open the tool

Go to https://auricartisan.com/tool/general/gamut-and-rendering/hdr-gamut-plot/. The heading is **HDR Gamut Plot**. Everything runs in your browser.

Key ideas

- **Gamut** — the range of chromaticities a space can make, drawn as a triangle on a chromaticity diagram. HDR video usually uses the wide **Rec. 2020** gamut.
- **Luminance** — how much light, in candelas per square metre (cd/m², also called nits). Standard video is mastered around 100 nits; HDR can reach 1,000 to 10,000.
- **Transfer function** — the curve from signal (0 to 1) to light. **PQ** gives an absolute luminance for every signal value. **HLG** and the gamma curves are display-referred: their output depends on the display's peak.
- **ICtCp** and **Jzazbz** — colour spaces designed for HDR, in which equal steps are closer to equal visible differences across a wide luminance range.

A tour of the screen

A note at the top reminds you that a nit is a measurement, not a setting.

Tab	What it is for
Lab	The chromaticity diagram, overlays, gamut metrics and plotting an image
Curves	Tone curves from signal to cd/m², peak brightness, HLG surround and signal level

Tab	What it is for
Planes	The gamuts in ICtCp or Jzazbz, and boundary rings at eight luminances
Data	The ten datasets the tool uses and their status
Export	Measurements, curves and images to download or copy, HEX conversion, and a side-by-side comparison of the three spaces
Reference	Standards, formulas, citations and research notes with a filter

The Lab tab

The left panel holds **Working colour space**, **Overlay gamuts**, **Advanced overlays**, **Chromaticity axes**, **Standard observer**, **Plot an image on it** and a shortcut hint. The right side shows the **Gamut chromaticity diagram**, a hover readout, and the **Gamut metrics** table.

Read the diagram

1. The diagram shows three triangles: Rec.709/sRGB, Display-P3 and Rec.2020, with R, G and B labels at the corners and the D65 white point marked.
2. Move the pointer over it. The line under the diagram shows the coordinates (x, y or u′, v′), the XYZ of that chromaticity at a fixed Y of 0.35, its L*, and its I, Ct and Cp at the luminance that the Curves tab's **Signal level** represents on the PQ curve.
3. Untick a gamut under **Overlay gamuts** to hide it.
4. Under **Advanced overlays**, untick **Spectral locus** or **Line of purples** to hide the edge of the diagram.
5. Change **Chromaticity axes** to **CIE 1976 u′v′** (or press `A`). The label over the diagram changes and the metrics are recomputed in u′v′.
6. Read **Gamut metrics**: primaries, white, **Share of locus** (the triangle's area as a percentage of the spectral locus area in the same plane) and **Area**.

Next steps

- Compare broadcast gamuts
- Read HDR transfer curves
- Explore ICtCp and Jzazbz planes

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Compare broadcast gamuts

Compare Rec.709/sRGB, Display P3 and Rec.2020 by area and share of the spectral locus, and measure how much of each an image occupies.

The three spaces

Space	Red (x, y)	Green (x, y)	Blue (x, y)	White
Rec.709 / sRGB	0.640, 0.330	0.300, 0.600	0.150, 0.060	D65
Display-P3	0.680, 0.320	0.265, 0.690	0.150, 0.060	D65
Rec.2020	0.708, 0.292	0.170, 0.797	0.131, 0.046	D65

All three share the D65 white. The P3 here is Display P3, not the DCI cinema version with its greener white.

Read the share of the locus

1. On the **Lab** tab, read **Gamut metrics**.
2. **Share of locus** is each triangle's area divided by the area of the spectral locus in the same plane. In CIE 1931 xy the tool measures 33.6% for Rec.709, 45.6% for Display-P3 and 63.6% for Rec.2020.
3. Press `A` or change **Chromaticity axes** to **CIE 1976 u'v'**. The shares change because u'v' spaces colours more evenly; the u'v' figures are the fairer comparison.

The often-quoted 35.9%, 53.6% and 75.8% are against a smaller locus area than the one the tool draws. Neither set is wrong; the tool divides by the locus it shows. Its locus is the CIE 1931 2° observer's, tabulated at 65 points from 380 to 700 nm.

Measure an image's colours

1. Under **Plot an image on it**, select **Drop a PNG or JPEG** and choose an image. Images larger than 1,024 pixels on a side are reduced for analysis.
2. Read the results under the button:
   - **Pixels on the diagram** — the analysed size, how many pixels were sampled and how many distinct chromaticities they contain;
   - **How much of each gamut it occupies** — for each space, the area of its triangle covered by the image's chromaticities (measured in xy, in cells 0.002 wide), as a percentage.
3. A note explains the result. For an ordinary untagged image every pixel falls inside all three gamuts, because such a file is read as sRGB and sRGB sits inside the other two.

Occupancy answers "how much of this gamut does the image use?". A grey ramp occupies 0%; a full sweep of the sRGB cube occupies about 99.9% of sRGB, 75.0% of P3 and 53.8% of Rec.2020.

The tool reports no luminance for the image. An 8-bit PNG or JPEG does not record what light its values stood for.

Compare side by side

Open **Export**. Under **The three spaces, side by side**, a bar chart and table list each space's **Area**, **Share of locus** and **White** in the plane currently on the Lab tab.

Result

You can state how much of the visible chromaticity range each broadcast gamut covers, in the plane you choose, and how much of each an image actually uses.

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Explore ICtCp and Jzazbz planes

View the three gamuts in the ICtCp or Jzazbz perceptual planes, read the boundary rings from 0.1 to 10,000 nits, and convert HEX colours.

Key ideas

- **ICtCp** (ITU-R BT.2100, from Dolby) splits colour into **I** (intensity), **Ct** (tritan, blue–yellow) and **Cp** (protan, red–green). It converts linear Rec. 2020 RGB to cone responses, encodes each with the PQ curve, then rotates to I, Ct, Cp. It is the basis of the ΔE ITP colour difference.

- **Jzazbz** (Safdar et al., 2017) gives lightness **Jz** and opponent axes **az** and **bz**. It starts from absolute XYZ and uses its own PQ-like quantiser with the exponent raised by a factor of 1.7 (p = 134.034375).
- Both are built to keep equal steps roughly equal from very dark to very bright, which CIELAB does not.

View the plane

1. Set the luminance first: on the **Curves** tab, choose a tone curve, peak brightness and **Signal level**. Together they define the absolute luminance the plane is drawn at.
2. Open the **Planes** tab.
3. Under **Perceptual plane**, choose **ICtCp (Ct–Cp)** (default) or **JzAzBz (Az–Bz)**. The title and note change to name the space and its quantiser.
4. Read the plane: each gamut switched on in the Lab's **Overlay gamuts** is outlined, showing how far its chroma extends at that luminance.
5. Read **Boundary rings by luminance**: the same boundaries at eight luminances — 0.1, 1, 10, 100, 500, 1,000, 4,000 and 10,000 nits — listed under **Boundary rings** in the rail with a matching key. A ring is a slice at one luminance, so the stack shows the boundary changing with brightness.

The **Tone mapping operator** menu (Reinhard, Hable (Filmic), ACES Filmic, Uchimura (Gran Turismo)) is shown on this tab, but in the current version it does not change any plot. For tone mapping, use the Tone Mapping lab.

Convert a list of colours

1. Set the tone curve, peak and **Signal level** on the Curves tab; the conversion uses the resulting luminance.
2. Open **Export** and find **Convert a list**.
3. Enter six-digit HEX colours, one per line or separated by commas.
4. Select **Convert to ICtCp**. The table lists **Hex**, **L***, **a***, **b***, **I**, **Ct**, **Cp**, **Jz**, **Az** and **Bz** for each colour.
5. Select **Download the result** to copy the table as CSV to your clipboard (the button copies rather than downloads).

The ICtCp and Jzazbz values depend on the luminance you set. Treat them as illustrations of how the spaces behave; see Limits and accuracy.

Export the plane

On **Export**, choose **The plane** (PNG) to download the ICtCp or Jzazbz plane at the current luminance.

Result

You can see how much chroma each broadcast gamut offers at a given HDR luminance, in a space designed for HDR, and how that changes from 0.1 to 10,000 nits.

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Read HDR transfer curves

Compare PQ, HLG and gamma curves in cd/m², set peak brightness and HLG surround, and read what a signal value means in nits.

The four curves

Curve	Defined by	Kind	What the output means
PQ (ST.2084)	SMPTE ST 2084	Absolute	Signal 0 to 1 maps to 0 to 10,000 cd/m² on any display that can reach it. Signal 0.5 is about 92 cd/m²
HLG (BT.2100)	ARIB STD-B67, with the BT.2100 OOTF	Display-referred	Scene light is shaped by a system gamma that depends on the display's peak, then scaled to that peak
Gamma 2.4 (BT.1886, zero black)	ITU-R BT.1886	Display-referred	signal^2.4 × peak
Gamma 2.2 (SDR)	Pure power law	Display-referred	signal^2.2 × peak

Absolute means the curve states a luminance outright. **Display-referred** means it reaches a luminance only by way of the display's peak, which the **Peak brightness** control supplies.

HLG system gamma

HLG applies an opto-optical transfer function (OOTF) with a system gamma that depends on the display peak L_W:

```
γ = 1.2 + 0.42 × log10(L_W ÷ 1000)
```

At 1,000 nits γ is 1.2. The tool also offers two surround adjustments: **Dim surround** multiplies γ by 0.95 and **Bright surround** by 0.90. These two factors are this tool's own; they are not in BT.2100 or BT.2390. Choosing one marks the HLG reading **Adjusted**. **Reference dark** uses BT.2100's value unchanged. The system gamma is never set below 1.0.

Steps

1. Open the **Curves** tab.
2. Set **Peak brightness** (100 to 10,000 nits in steps of 50; default 1,000). It scales the display-referred curves and sets HLG's system gamma. It does not scale PQ.
3. Choose a **Tone curve**: **Gamma 2.2 (SDR)**, **Gamma 2.4 (BT.1886, zero black)**, **PQ (ST.2084)** (default) or **HLG (BT.2100)**. Keys `1` to `4` select them in that order.
4. For HLG, choose an **HLG environment**: **Reference dark** (default), **Dim surround** or **Bright surround**. The line under it shows γsys, and a note says whether the reading is standard or adjusted.
5. Read **Tone curve output**: signal on the horizontal axis, cd/m² on a log vertical axis that extends to whatever the drawn curves need (up to 10,000 for PQ). The selected curve is gold; gamma 2.2 is drawn faintly for reference, and PQ too when another curve is selected.
6. Open the rows under the plot. Each is tagged **Absolute**, **Display-referred** or **Adjusted**. The PQ row lists its constants as the standard's exact fractions; the HLG row gives the system gamma at the current peak.
7. Set **Signal level** (0 to 1, default 0.5). It is used with the tone curve and peak to compute an absolute luminance for the Planes tab and the HEX conversion. The **I =** readout shows that luminance as an ICtCp intensity.

The EOTF comparison chart

EOTF Comparison — PQ vs HLG vs sRGB plots all three on one log-nits axis: PQ over 0 to 10,000 cd/m², HLG at the 1,000-nit reference, and sRGB peaking near 100 nits. It is drawn when the page loads and does not follow the peak control.

Export the curve

On the **Export** tab, choose **The curve** (CSV) to download the drawn curve sampled at 257 points as `signal,nits`, with a footer recording the curve, whether it is absolute, the peak (or that it was ignored for PQ) and, for HLG, the system gamma and surround. Choose **The curve plot** (PNG) for an image of the plot.

Result

You can say exactly what luminance a signal value produces under each curve, and when that luminance depends on the display rather than on the signal.

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HDR Gamut Plot reference

Every control, readout, option and export in the HDR Gamut Plot.

Address: https://auricartisan.com/tool/general/gamut-and-rendering/hdr-gamut-plot/

Lab tab

Control	What it does	Values	Default
Working colour space	Shown on the page, but in the current version it does not change the diagram or figures	sRGB / Rec.709 (BT.709); Display-P3 (D65); Rec.2020 (BT.2020)	Rec.2020
Overlay gamuts	Shows or hides each triangle (also used by the Planes tab)	Rec.709 / sRGB, Display-P3, Rec.2020	All on
Spectral locus	Draws the locus	On or off	On
Line of purples	Draws the straight edge of the locus	On or off	On
ICtCp marker	Shown, but draws nothing in the current version	On or off	Off
Chromaticity axes	The diagram's plane	CIE 1931 xy; CIE 1976 u′v′	CIE 1931 xy
Standard observer	Shown, but the locus is always the CIE 1931 2° observer's	CIE 1931 2° (Standard); CIE 2015 10° (Revised)	CIE 1931 2°
Drop a PNG or JPEG	Opens a file picker and measures the image's gamut occupancy	Image files; reduced to at most 1,024 pixels a side	—

Readout	What it shows
Gamut chromaticity diagram	The triangles, locus, purples line and D65 marker; the axis name is shown over it
Hover line	Coordinates, XYZ at Y = 0.35, L*, and I, Ct, Cp at the PQ luminance of the Signal level
Gamut metrics	Space, Primaries, White, Share of locus, Area, in the plane on screen
Image results	Size analysed, pixels sampled, distinct chromaticities, occupancy of each gamut, and a note

Curves tab

Control	What it does	Values	Default
Peak brightness	The display peak L_W: scales display-referred curves and sets HLG's system gamma; does not scale PQ	100 to 10,000 nits, step 50	1000
Tone curve	The curve drawn and used for the Signal level	Gamma 2.2 (SDR); Gamma 2.4 (BT.1886, zero black); PQ (ST.2084); HLG (BT.2100)	PQ
HLG environment	Surround factor for HLG's system gamma	Reference dark (×1); Dim surround (×0.95); Bright surround (×0.90)	Reference dark
Signal level	A signal value to convert to absolute luminance	0 to 1, step 0.01	0.5

Readout	What it shows
γsys	HLG system gamma at the current peak and surround
Surround note	Whether the HLG reading is BT.2100's own or adjusted
I =	The ICtCp intensity of the signal's absolute luminance
Tone curve output	Signal against cd/m² (log), with a peak label
Curve rows	One per curve, tagged Absolute, Display-referred or Adjusted, with its defining document and details
EOTF Comparison — PQ vs HLG vs sRGB	All three on one log-nits axis, drawn at page load

Planes tab

Control	What it does	Values	Default
Perceptual plane	The space for the plane and rings	ICtCp (Ct–Cp); JzAzBz (Az–Bz)	ICtCp
Tone mapping operator	Shown, but does not change any plot in the current version	Reinhard; Hable (Filmic); ACES Filmic; Uchimura (Gran Turismo)	Reinhard
Boundary rings	Key to the eight ring levels	0.1, 1, 10, 100, 500, 1k, 4k, 10k nits	—

Readout	What it shows
Plane	Gamut boundaries at the Signal level's absolute luminance
Boundary rings by luminance	Boundaries at the eight levels for each gamut switched on in the Lab

Data tab

Dataset	Status
Primaries and white, 3 spaces	Verbatim
Spectral locus (65 xy pairs, 380–700 nm)	Verbatim
PQ transfer (ST 2084 constants as rationals)	Verbatim
ICtCp matrices (BT.2100)	Verbatim
HLG transfer (with the OOTF)	Computed
JzAzBz (Safdar et al. 2017, own quantiser)	Computed

Dataset	Status
Gamut area and share	Computed
Tone-mapping operators (four curves)	Computed
HLG surround factor	Synthesised (this tool's own)
Image luminance	Absent: 8-bit PNG and JPEG carry none

Status chips filter the list. **Install a table** ("Drop a CSV or JSON") does not load files in the current version.

Export tab

Control	What it does
What to take	**The measurements**, **The curve**, **The diagram**, **The curve plot**, **The plane**, **A link to this view**
As	Only the formats that fit (below)
Download (names the file)	Saves it; for the link, copies it
Copy to clipboard	Copies text payloads
Exactly what you will get	Preview of the payload and its size
Convert a list, **Convert to ICtCp**, **Download the result**	HEX conversion; the last button copies the result as CSV
The three spaces, side by side	Bar chart and table of Area, Share of locus and White

Choice	Format	File
The measurements	CSV	`hdr-gamut.csv`: Space, Plane, Red x, Red y, Green x, Green y, Blue x, Blue y, White x, White y, Area, Locus share %, plus a footer naming the locus area, plane and observer
The measurements	JSON	`hdr-gamut.json`: per space label, primaries, white point, plane, area and locus share, and a `_meta` block (axes, peak, curve, locus area, observer, locus points)
The measurements	CSS	`hdr-gamut.css`: custom properties such as `--gamut-rec2020-locus-share`, `--gamut-rec2020-area`, the R, G and B primaries, `--gamut-plane`, `--gamut-locus-area`, `--hdr-peak-nits` and `--hdr-curve-type`
The curve	CSV	`hdr-curve-<curve>.csv`: 257 rows of signal and nits, with a footer
The diagram	PNG	`hdr-gamut-diagram.png`
The curve plot	PNG	`hdr-tone-curve.png`
The plane	PNG	`hdr-perceptual-plane.png`
A link to this view	Link	A URL with `axes`, `peak`, `curve` and `mode` (the perceptual plane)

Reference tab

Control	What it does
Standards, **Formulas**, **Citations**, **Research notes**	Switch sections
Filter	Searches all four; `Esc` clears it

Standards: ITU-R BT.2020, BT.2100, SMPTE ST 2084, ARIB STD-B67, ICtCp, Jzazbz, BT.709 and sRGB, and Display P3. Formulas: PQ, HLG OETF and OOTF, ICtCp, Jzazbz, gamut area and locus share, four tone-mapping operators, and xy/u′v′ conversion. Citations: 17 references. Research notes explain why a file has no nits, occupancy against coverage, the single locus denominator, absolute against display-referred curves, the surround factor, the two quantisers, and what "ACES" means here.

Save to Library

The floating **Save to Library** button stores a snapshot of the diagram in your Auric Artisan Library.

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Troubleshooting the HDR Gamut Plot

Fixes for common problems in the HDR Gamut Plot.

Symptom	Why it happens	What to do
Changing **Working colour space** changes nothing	The control is not applied in the current version	Use **Overlay gamuts** to choose which spaces are drawn
Choosing the 2015 10° observer changes nothing	The locus is always the CIE 1931 2° observer's	Expected
ICtCp marker draws nothing	Not applied in the current version	No workaround
The **Tone mapping operator** menu changes nothing	Not applied in the current version	Use the Tone Mapping lab
PQ does not change when I move **Peak brightness**	PQ is absolute; the peak does not scale it	Expected. Choose HLG or a gamma curve to see the peak's effect
The HLG reading is marked **Adjusted**	Dim or bright surround applies this tool's own factor	Choose **Reference dark** for BT.2100's value
The EOTF comparison chart does not follow my settings	It is drawn once when the page loads	Use **Tone curve output** for your settings
My image does not appear on the diagram	The current version reports the image's occupancy as text under the drop area; it does not scatter the pixels on the diagram	Read the results under **Plot an image on it**
The image readout shows no nits	A PNG or JPEG does not record absolute luminance	Expected; the tool will not invent one
Every image shows 100% of pixels inside all three gamuts	Untagged images are read as sRGB, which sits inside P3 and Rec.2020	Read the occupancy figures, which do vary
Large images look squashed in the analysis	The image is reduced to at most 1,024 pixels on each side, which can change its proportions	Proportions do not affect chromaticity occupancy
Convert to ICtCp says "No valid hex colours found."	Only six-digit HEX values are read	Use values such as `#FF0000`
Download the result did not download a file	It copies the CSV to your clipboard	Paste it into a spreadsheet or text file
Copying failed	The browser blocked clipboard access	Allow clipboard access, or select the table and copy manually
Pressing 1–4 or A does nothing	The focus is in a text box, slider or list	Click an empty area of the page first
My shared link opens with different overlays	Links carry axes, peak, curve and plane only	Set the other controls again
Shares differ from published figures	The tool divides by its own drawn locus area	See Compare broadcast gamuts

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HDR Gamut Plot FAQ

Short answers to common questions about the HDR Gamut Plot.

What is the difference between PQ and HLG?

PQ (SMPTE ST 2084) is absolute: each signal value means a specific luminance, up to 10,000 nits, and displays that cannot reach it must tone-map. HLG (ARIB STD-B67 / BT.2100) is relative: the display applies a system gamma based on its own peak, which makes HLG backward-compatible with standard displays.

What does a PQ signal of 0.5 mean?

About 92 cd/m^2, on any display that can show it.

Why is HLG's system gamma 1.2?

That is its value for a 1,000-nit display under BT.2100: $\gamma = 1.2 + 0.42 \times \log10(L_W \div 1000)$. It rises for brighter displays and falls for dimmer ones.

Are the dim and bright surround settings standard?

No. The multipliers 0.95 and 0.90 are this tool's own. Readings made with them are marked Adjusted.

Why can't the tool tell me the peak brightness of my image?

An ordinary 8-bit PNG or JPEG stores code values with no record of the light they stood for. Absolute luminance needs an HDR format with PQ or HLG and mastering metadata, or a format such as OpenEXR, which the tool does not read.

Why do ICtCp and Jzazbz use different quantisers?

ICtCp uses ST 2084 as written. Jzazbz's authors raised the curve's exponent by a factor of 1.7 ($p = 134.034375$). The tool runs each with its own.

Is the "ACES Filmic" operator the Academy's transform?

No. It is Krzysztof Narkowicz's 2016 curve fit that approximates the look of the ACES output transforms. The Reference section says so.

Why 33.6%, 45.6% and 63.6% rather than the usual figures?

They are shares of the spectral locus the tool draws (the CIE 1931 2° locus, 65 points from 380 to 700 nm). Commonly quoted figures use a smaller locus area.

Does it work with HDR displays?

The tool draws its charts in standard sRGB, so it does not need an HDR display. It explains HDR numerically rather than showing HDR images.

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HDR Gamut Plot: supporting pages

Glossary, shortcuts, limits, related tools, release notes and privacy for the HDR Gamut Plot.

These pages support the guides in docs/.

File	What it covers
glossary.md	HDR and colour terms used in the tool
shortcuts.md	Keyboard and mouse shortcuts
limits-and-accuracy.md	What the figures are and are not
related.md	Related tools and Learn articles
release-notes.md	What has changed for users
privacy.md	What stays in your browser

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HDR Gamut Plot shortcuts

Keyboard and mouse shortcuts in the HDR Gamut Plot.

These keys work when the focus is not in a text box, slider or list. The Lab tab's rail lists them as "1-4 cycle curves" and "A toggle axes".

Key	What it does
`1`	Tone curve: Gamma 2.2 (SDR)
`2`	Tone curve: Gamma 2.4 (BT.1886, zero black)
`3`	Tone curve: PQ (ST.2084)
`4`	Tone curve: HLG (BT.2100)
`A`	Switches the chromaticity axes between CIE 1931 xy and CIE 1976 u′v′
`Esc`	Closes a full-screen chart; clears the Reference filter

Mouse	What it does
Move over the diagram	Updates the hover readout
Double-click a chart	Opens it full screen

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HDR Gamut Plot glossary

Plain-language definitions of the HDR and colour terms used in the HDR Gamut Plot.

Absolute (curve) — a transfer curve whose output is a stated luminance, independent of the display. PQ is absolute.

BT.1886 — the ITU-R reference EOTF for standard-dynamic-range flat-panel displays; with zero black level it is a 2.4 power law.

BT.2020 (Rec. 2020) — the ultra-high-definition television standard, whose wide primaries HDR video usually uses.

BT.2100 — the ITU-R standard for HDR television. It defines PQ and HLG, the HLG system gamma and the ICtCp colour space.

cd/m² (nit) — the unit of luminance.

Display-referred (curve) — a curve whose output depends on the display's peak luminance. HLG and gamma curves are display-referred.

EOTF — electro-optical transfer function: signal to light.

HDR (high dynamic range) — images and displays with a much wider luminance range than standard video.

HLG (hybrid log-gamma) — a relative HDR transfer function (ARIB STD-B67), combining a square-root segment for darks with a logarithmic segment for highlights.

ICtCp — an HDR colour space from BT.2100: I (intensity), Ct (blue–yellow) and Cp (red–green), built on PQ-encoded cone responses.

Jzazbz — a perceptually uniform colour space for HDR and wide gamut (Safdar et al., 2017), with its own PQ-like quantiser.

Line of purples — the straight edge of the chromaticity diagram joining violet and red.

Locus share — a gamut triangle's area as a percentage of the spectral locus area, in the same plane.

Occupancy — how much of a gamut's area an image's chromaticities cover.

OETF — opto-electronic transfer function: light to signal.

OOTF — opto-optical transfer function: the rendering from scene light to display light, including HLG's system gamma.

Peak brightness (L_W) — the maximum luminance of the display the curves are rendered for.

PQ (perceptual quantiser) — the SMPTE ST 2084 transfer function, mapping signal 0–1 to 0–10,000 cd/m² with steps close to the threshold of visibility.

Signal level — a code value from 0 to 1, as carried in the video signal.

Spectral locus — the curved edge of the chromaticity diagram, made by single wavelengths.

System gamma (γsys) — the exponent of HLG's OOTF, 1.2 at 1,000 nits.

Tone mapping — compressing a wide luminance range into what a display can show.

u′v′ — the CIE 1976 uniform chromaticity coordinates.

xy — the CIE 1931 chromaticity coordinates.

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HDR Gamut Plot limits and accuracy

How far the HDR Gamut Plot's curves, planes, shares and image readings can be trusted.

Exact to the standard

- **Primaries and white** of Rec.709/sRGB, Display P3 and Rec.2020.
- **PQ** with the ST 2084 constants written as the standard's exact fractions.
- **ICtCp** matrices, matching BT.2100.
- **HLG** with the BT.2100 system gamma (reference surround).

Computed

- **Jzazbz** from Safdar et al. (2017), with its own quantiser.
- **Areas and locus shares** by the shoelace formula, divided by the tabulated CIE 1931 2° locus (65 points, 380–700 nm; area 0.33323 in xy against a published 0.334).
- **Tone-mapping operators** with published constants. "ACES Filmic" is Narkowicz's approximation, not the Academy's transform.
- **Image occupancy**, binned at 0.002 in xy.

This tool's own

- The **dim and bright surround** multipliers for HLG (0.95 and 0.90). They are not in BT.2100 or BT.2390, and readings made with them are marked Adjusted.

Not available

- **Image luminance.** 8-bit PNG and JPEG files carry no absolute luminance. HDR formats (OpenEXR, or PNG with PQ or HLG and mastering metadata) are not read.
- **The 10° observer.** The option has no effect.

Current-version caveats

- **Working colour space**, **ICtCp marker** and **Tone mapping operator** do not change the output.
- The image's pixels are measured but not drawn on the diagram.
- The **EOTF comparison** chart is drawn at page load and does not follow the controls.
- The **Convert a list** feature scales each HEX value's channel numbers directly to the chosen luminance when computing ICtCp, without first converting sRGB to linear Rec. 2020 light. Treat its ICtCp figures as illustrative; its CIELAB figures are standard sRGB conversions.
- For the Planes tab, the gamma 2.4 curve's luminance is computed with a 2.2 exponent.
- Charts are drawn in sRGB; the tool does not display HDR images.

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HDR Gamut Plot privacy

What the HDR Gamut Plot does with your images, colours and settings.

- **Images stay on your device.** An image you drop in is read and analysed by your browser; it is not uploaded, and the tool keeps only the resulting figures on the page.
- **All curves, planes and measurements are computed in your browser.**
- **Exports are created locally** and saved through your browser's download, or copied to your clipboard when you choose.
- **The link** carries the axes, peak brightness, tone curve and perceptual plane. Anyone with the link sees those settings.
- **Save to Library** stores a snapshot in your Auric Artisan Library.
- The page loads the site's shared features like every other page on auricartisan.com; see the site's privacy policy.

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HDR Gamut Plot release notes

User-facing changes to the HDR Gamut Plot.

Current version (2026 workbench rebuild)

- Tabs are now **Lab**, **Curves**, **Planes**, **Data**, **Export** and **Reference**.
- Gamut shares are quoted against the spectral locus in the plane on screen (33.6%, 45.6% and 63.6% in xy). They were previously divided by Rec.2020's area, so Rec.2020 always read 100%.
- The image readout no longer prints a "peak luminance" or 100% coverage for every image. It measures the image's occupancy of each gamut instead.
- The tone-curve axis now covers whatever is drawn, so PQ is shown in full to 10,000 cd/m^2. Peak brightness is documented as not scaling PQ.
- Jzazbz now runs its own quantiser (p = 134.034375) instead of ICtCp's.
- HLG surround adjustments are marked as this tool's own, and readings using them are labelled Adjusted.
- The boundary rings are now drawn automatically with a key to their eight luminance levels.
- Export previews the exact payload and names the file; the CSV column formerly headed "Coverage%" is now "Locus share %".
- The "ACES" operator is identified as Narkowicz's 2016 approximation.

May 2026

The HDR Gamut Plot User Guide was published in the site's Documentation library, describing the earlier version.

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Related to the HDR Gamut Plot

Tools and Learn articles that go well with the HDR Gamut Plot.

Tools

Tool	Why
Tone Mapping	Apply and compare tone-mapping operators
Gamut Lab	13 RGB spaces, including Rec. 2100 PQ and HLG, with transfer curves and volumes
Gamut Mapping	What happens to colours outside a smaller gamut
LUT Lab	Build and preview lookup tables for grading

Learn articles

- HDR, Wide Gamut, PQ, HLG, and Modern Displays — https://auricartisan.com/library/learn/articles/2026-05-25-hdr-wide-gamut-pq-hlg-modern-displays
- ΔE Metrics from CIE76 to CIEDE2000 and ΔEITP — https://auricartisan.com/library/learn/articles/2026-05-26-delta-e-metrics-cie76-ciede2000-itp
- Chromaticity Diagrams Without the Intimidation — https://auricartisan.com/library/learn/articles/2026-05-24-chromaticity-diagrams
- Gamut Volume: How Much Color a Wide Gamut Actually Adds — https://auricartisan.com/library/learn/articles/2026-08-26-gamut-volume-how-much-color
- Color in Real-Time 3D: Albedo, PBR, and Tone Mapping — https://auricartisan.com/library/learn/articles/2026-08-02-color-in-real-time-3d-albedo-pbr-tone-mapping

Documentation on the site

The site's Documentation library (https://auricartisan.com/library/documentation/) has an earlier **HDR Gamut Plot User Guide**. This documentation follows the current page.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Illuminants

What the Standard Illuminants lab does, who it is for, where to find it and how to take your first reading.

The Standard Illuminants lab shows you what a light source is made of and what that does to colour. You pick one of 18 CIE standard illuminants (A, D50, D55, D65, D75, equal-energy E and the fluorescent series F1 to F12) or a blackbody radiator at any colour temperature from 1,500 K to 25,000 K. The lab draws its spectral power distribution, integrates it against the CIE 1931 2° standard observer, and reports its white point, correlated colour temperature (CCT), Duv, colour rendering, how surfaces look under it, and how a set of colours adapts from one white point to another.

Around that core it adds views for the chromaticity diagram (in three planes), a Monte Carlo estimate of how measurement error propagates into x, y, CCT and Duv, a comparison of calculation methods, a small query console for comparing many illuminants at once, and a data register that says where every table comes from. Every reading carries a line naming the dataset, observer and interval behind it.

The lab is honest about its limits. The CIE fluorescent tables and the CIE 13.3 test colour samples are not held, so fluorescent spectra and colour rendering figures are built from stand-ins and labelled as such. IES TM-30 is not computed at all. See Limits and accuracy.

Who it is for

- Designers and photographers who want to know why a colour looks different under office, shop or home lighting.
- Developers and colour-management practitioners working with D50 and D65 white points (ICC workflows, sRGB, print).
- Students learning how illuminants, CCT, Duv and chromatic adaptation fit together.
- Lighting and measurement people who want a quick, traceable reference calculation — not a certified one.

Where to find it

- Address: https://auricartisan.com/tool/general/colorimetry/illuminants/
- In the tool collection (https://auricartisan.com/collections/) under **Colorimetry**, as **Illuminants**.
- It works in any current desktop or mobile browser. No account is needed.

Quick start

1. Open https://auricartisan.com/tool/general/colorimetry/illuminants/. The **Lab** tab opens with CIE D65 selected.
2. Read the **Illuminant Summary** at the top of the results: CCT, Duv, CIE x and y, CRI Ra and XYZ.
3. Open the **Illuminant** list and choose **CIE A (2856 K) — Incandescent**. Watch the spectrum tilt towards red.
4. Tick **Enable comparison** and leave the comparison illuminant on D65 (choose **CIE D65 (6504 K)** in the list). The chart overlays both curves and a comparison table appears at the bottom.
5. Look at **Surface appearance** to see how skin, paint and ink samples shift under the new light.
6. Open the **Chromaticity** tab to see where the light sits against the Planckian locus.
7. Open **Export** and select **Export SPD CSV** to download the spectrum.

What you can do

- Plot the spectrum of any of 18 standard illuminants or a blackbody at a custom CCT.
- Read white point (x, y, XYZ), CCT and Duv computed by Ohno's method, with a validity check.
- Compare two illuminants and get a metamerism index.

- See how eight sample surfaces render under the light, with CIEDE2000 differences.
- Adapt a 24-colour chart between white points with six chromatic adaptation transforms (or none) and inspect the matrix.
- View chromaticity in CIE 1931 xy, CIE 1976 u′v′ or CIE 1960 uv, with the Planckian locus, isotemperature lines, the daylight locus and RGB primaries.
- Read the dominant wavelength and purity.
- Propagate an instrument uncertainty budget into x, y, CCT and Duv by Monte Carlo simulation.
- Compare interval and interpolation choices, and the Ohno and McCamy CCT methods.
- Query many illuminants at once and export the results as CSV or JSON.
- Export the spectrum as CSV, JSON or PNG, copy the white as HEX, and share a link to your settings.

In this folder

- docs/ — how to use the lab
  - Getting started
  - Compare two light sources
  - Adapt colours between white points
  - Estimate measurement uncertainty
  - Query many illuminants at once
  - Reference
  - Troubleshooting
  - FAQ
- others/ — background and supporting pages
  - Glossary
  - Shortcuts
  - Limits and accuracy
  - Related tools
  - Release notes
  - Privacy

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Illuminants documentation

The guides, reference and help pages for the Standard Illuminants lab.

This folder holds the working guides for the Standard Illuminants lab: a screen tour, step-by-step tasks, a full reference to every control and output, and help for when a reading looks wrong.

File	What it covers
getting-started.md	Opening the lab, the eight tabs, and your first reading
compare-light-sources.md	Comparing two illuminants: spectra, CRI, metamerism and surface shifts
adapt-between-white-points.md	Adapting a colour chart from one white point to another, and reading the matrix
estimate-uncertainty.md	Running the Monte Carlo uncertainty propagation
query-console.md	Using the query console, its grammar and its exports
reference.md	Every control, readout, option and export, tab by tab
troubleshooting.md	Symptoms, causes and fixes
faq.md	Short answers to common questions

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Getting started with Illuminants

Open the Standard Illuminants lab, learn its eight tabs and take your first reading of a light source.

Open the lab

Go to https://auricartisan.com/tool/general/colorimetry/illuminants/. The page title is **Standard Illuminants**. Nothing needs installing and you do not need to sign in. All calculations run in your browser.

Key ideas before you start

- An **illuminant** is a light source described by its **spectral power distribution (SPD)**: the relative power it emits at each wavelength from 380 nm (violet) to 780 nm (deep red).
- The lab turns an SPD into colour by weighting it with the **CIE 1931 2° standard observer** — three curves that describe how an average observer matches colours in a small (2°) field of view. The result is the tristimulus values **X, Y, Z**, and from them the chromaticity **x, y** — the illuminant's **white point**.
- **Correlated colour temperature (CCT)** is the temperature, in kelvin, of the blackbody (Planckian) radiator whose colour is closest to the light. Low CCT is warm (candle, tungsten); high CCT is cool (north sky).
- **Duv** is how far the light's chromaticity sits from the blackbody curve, in the CIE 1960 uv plane. Positive Duv is greenish; negative is pinkish. CCT only means something when Duv is small.

A tour of the screen

A note at the top of the tool explains that every figure is traceable: the line under each result names the dataset, observer and interval behind it.

The tabs across the top are:

Tab	What it is for
Lab	Choose the light, compare it, see surfaces and chromatic adaptation. The main working view.
Chromaticity	The chromaticity diagram in three planes, with loci, isotemperature lines and the dominant wavelength.
Uncertainty	Monte Carlo propagation of an instrument uncertainty budget into x, y, CCT and Duv.
Methods	How much the interval and interpolation choices move the result, and how two CCT methods compare.
Console	A small query language for comparing many illuminants in one table and plot.
Data	The register of every dataset the lab uses, with its status. The tab label shows the count (15).
Export	Downloads, the share link and the all-illuminant comparison table.
Reference	Standards, formulas, citations and research notes, with a search filter.

The Lab tab

The left rail holds the controls, grouped as **Source mode**, **Illuminant**, **CCT**, **Comparison illuminant**, **Surface reflectance samples** and **Chromatic adaptation**. The results column on the right shows, from top to bottom:

1. **Illuminant Summary** — a swatch of the white, its name, type and HEX, then CCT, Duv, CIE x, CIE y, CRI Ra and XYZ.
2. **SPD — Spectral Power Distribution (380–780 nm)** — the spectrum. Gold is the selected light; blue is the comparison light when comparison is on.
3. **CRI — Colour Rendering Index (R1–R14)** — one bar per test sample and the Ra figure.
4. **Surface appearance — D65 reference vs test illuminant** — one row per ticked surface.
5. **Chromatic adaptation preview (24 patches)** — original colours on top, adapted colours below.
6. **Adaptation matrix details** — the transform matrix and cone scaling.
7. **Illuminant comparison** — metrics side by side when comparison is on.

Take your first reading

1. On the **Lab** tab, make sure **Standard** is selected under **Source mode**.
2. Open the **Illuminant** list and choose **CIE D50 (5003 K) — Daylight, ICC PCS**.
3. Read the **Illuminant Summary**. CIE x and y should be close to 0.3457 and 0.3585, the published D50 white point.
4. Click anywhere on the SPD chart. The line under the chart shows the nearest wavelength and its relative power, with a colour chip for that wavelength.
5. Select the **Fullscreen** button on the chart (or double-click the chart) to see it larger. Press `Esc` or select the close button to return.
6. Drag the **CCT** slider to 3000 K. The lab switches to **Blackbody** mode and the spectrum becomes a smooth Planckian curve rising towards red.
7. Select **Reset to D65** to go back to standard D65 at 6500 K.

You should now see the D65 spectrum, a CCT of about 6504 K and a Duv close to zero.

Where the numbers come from

Under several results you will see a provenance line such as "CIE 15:2018 Table 5, verbatim · CIE 1931 2° · 5 nm · Ohno 2014, parabolic · ∑ 380–780 nm". It tells you the dataset and its status, the observer, the wavelength interval, the method and the range summed over. When a reading rests on a stand-in, the line says "not a standard value". The Data tab explains each dataset.

Next steps

- Compare two light sources
- Adapt colours between white points
- Estimate measurement uncertainty
- Query many illuminants at once

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Adapt colours between white points

Use the chromatic adaptation panel to move a 24-colour chart from one white point to another and read the transform matrix.

Colour-managed work often needs a colour expressed under one white point to be re-expressed under another. The classic case is ICC colour management, whose profile connection space uses D50, while sRGB and most screens use D65. A **chromatic adaptation transform (CAT)** does this by imitating how the eye rebalances its cone sensitivities to a new light.

How a CAT works, briefly

1. Convert the colour's XYZ values into a cone-like space (L, M, S) with a 3 × 3 matrix M.
2. Scale each of L, M and S by the ratio of the destination white to the source white. This is the **von Kries** idea: each cone type adjusts its gain independently.
3. Convert back to XYZ with the inverse of M.

The transforms differ only in the matrix M. The lab offers:

Choice	What it is
Bradford	The matrix from Lam (1985), the usual choice in ICC workflows. Default.
CAT16	The matrix from the CAM16 colour appearance model (Li et al., 2017).
CAT02	The matrix from CIECAM02 (CIE 159:2004).
Von Kries	The Hunt–Pointer–Estévez cone matrix, normalised to D65.
Sharp	A spectrally sharpened matrix (Finlayson and Süsstrunk, 2000).
XYZ scaling	Scales X, Y and Z directly. Kept as the simplest case to compare against; it is not a model of the eye.
None	No adaptation. The control turned off, so you can see the unadapted colours.

Steps

1. Open the **Lab** tab and find **Chromatic adaptation** in the left rail.
2. Under **Adapt from**, choose the white point the colours are currently expressed in. The choices are D65, D50, D55, D75, A and E. The default is D65.
3. Under **Adapt to**, choose the destination white point. The default is D50.
4. Select a transform chip: **Bradford**, **CAT16**, **CAT02**, **Von Kries**, **Sharp**, **XYZ scaling** or **None**.
5. Look at **Chromatic adaptation preview (24 patches)**. Each column has the original colour on top and the adapted colour below. The caption names the source, destination and method.
6. Read **Adaptation matrix details**:
   - the transform's **M matrix**;

  - **LMS scaling (dst/src)** — the three gains applied to L, M and S;
  - **Source white LMS** and **Dest white LMS** — the two whites in the cone-like space.
7. To reverse the direction, select **Swap** between the two lists.

Reading the result

- Adapting D65 to D50 makes colours look slightly warmer on a D65 screen, because the preview is still displayed as sRGB. That is expected: the preview shows the adapted values, not what an adapted eye would see.
- Choosing A as the destination produces a strong orange cast for the same reason.
- If you choose **None**, the matrix panel says that no adaptation is applied, and the bottom row matches the top row.
- The 24 patches are a fixed set of 18 chromatic colours and a six-step grey scale, similar to a colour-checker chart. They are defined as sRGB values.

Tips

- For ICC work, D65 to D50 with Bradford matches common practice.
- To see how much the choice of transform matters for a real image, use the Chromatic Adaptation lab, which applies eight transforms to a photograph and compares them.
- The adaptation settings (from, to and method) are saved in the share link on the **Export** tab.

Result

You have a chart of 24 colours re-expressed from one white point to another, and the exact matrix and gains that did it.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Compare two light sources

Overlay two illuminants, read their CCT, CRI and white points side by side, and judge metamerism risk with the surface samples.

Use this when you need to explain why a product, print or swatch looks different in two places — for example daylight at a window against tungsten at home, or a D50 viewing booth against a D65 screen.

Before you start

- **Metamerism** is when two surfaces match under one light and differ under another. The lab estimates the risk between two lights with a **metamerism index**: the average CIEDE2000 colour difference of 14 test samples rendered under each light. Bigger means more risk.
- **CRI (colour rendering index)** scores how faithfully a light renders test colours compared with a reference light of the same CCT (a blackbody below 5,000 K, CIE daylight at or above). **Ra** is the mean of the first eight scores; R9 is saturated red.
- In this lab the test samples are approximations of the CIE 13.3 set, not the published reflectances. CRI and the metamerism index are therefore indicative. See Limits and accuracy.

Steps

1. Open the **Lab** tab.
2. Under **Illuminant**, choose the first light, for example **CIE D65 (6504 K) — Daylight, sRGB**.

3. Under **Comparison illuminant**, tick **Enable comparison**.
4. Open the comparison list and choose the second light. The choices are CIE A, D50, D55, D65, D75, equal-energy E, F2, F7 and F11. The default is **CIE A (2856 K)**.
5. Look at the SPD chart. The gold curve is your first light and the blue curve is the comparison light. Both are drawn on one relative scale.
6. Scroll to **Illuminant comparison**. The table shows, for each light:
   - **CCT** in kelvin;
   - **CRI Ra**;
   - **CIE x, y**;
   - and, across both, **Metamerism Index (avg ΔE_{00})**.
7. Under **Surface reflectance samples**, turn on the surfaces you care about. The choices are White, Gray, Red, Green, Blue, Skin, Yellow and Cyan. White, Gray, Red, Green and Skin are on by default.
8. Read **Surface appearance — D65 reference vs test illuminant**. Each row shows two swatches (under D65, then under your first light after adaptation), both HEX values and a ΔE_{00} figure.
9. Check the **CRI** chart for the first light. Look at R9 separately; Ra can hide weak red rendering.

Reading the result

- A metamerism index below about 0.5 suggests very low risk; 0.5 to 2 is moderate; 2 to 4 is high; above 4 means a visible shift is almost certain. These bands come from the lab's own research notes (Reference › Research notes) and are a rule of thumb, not a standard.
- The surface rows answer a slightly different question: after your eye has adapted to the light, how far does each surface's colour move compared with how it looks in daylight?
- A light can have a similar white point to D65 and still render surfaces differently. That is the spectrum at work, and it is why the lab shows the spectrum and not only the white.

Tips

- To study a fluorescent source, choose **F2**, **F7** or **F11** as the first light and D65 as the comparison. Remember that the fluorescent spectra are stand-ins; the chart labels them so.
- To see the full list of 18 illuminants side by side, open **Export** and select **Generate comparison table**. It lists CCT, x, y, CRI Ra and a white swatch for each.
- To compare many lights by x, y, CCT, Duv, efficacy or spectral spread, use the query console.

Result

You have a side-by-side table of two lights, a metamerism index, and a row of surface swatches showing how much each material shifts between them.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Estimate measurement uncertainty

Propagate a spectroradiometer uncertainty budget into x, y, CCT and Duv with the Monte Carlo view.

A measured spectrum is never exact. The wavelength scale may be slightly off, the instrument's slit blurs sharp peaks, and the detector adds noise. The **Uncertainty** tab asks how much those errors move the numbers you would report: x, y, CCT and Duv.

It uses the **Monte Carlo method** described in JCGM 101:2008 (Supplement 1 to the Guide to the Expression of Uncertainty in Measurement): it perturbs the spectrum thousands of times at random, according to the error sizes you give, and measures the spread of the results.

The input budget

The budget is preset for a typical array spectroradiometer. It is a starting point, not your instrument: every value is editable. Hover over a label to see what the term models and whether it is treated as a normal or a rectangular distribution.

Term	Unit	Preset	Distribution	What it models
Wavelength scale	nm	0.20	Normal	A shift of the whole wavelength axis. Dominates on a spiky source.
Bandwidth (FWHM)	nm	5.0	Rectangular	The finite slit width, modelled as a triangular blur.
Stray light	relative	0.0001	Rectangular	A flat pedestal added across the range.
Non-linearity	%	0.10	Rectangular	A gain error that depends on signal level.
Detector noise	%	0.05	Normal	Independent noise at each wavelength.
Reference lamp	%	0.80	Normal	The calibration scale. Largely cancels in chromaticity.

For a normal term the value is one standard uncertainty. For a rectangular term the value is the half-width of the range, so its standard uncertainty is the value divided by √3.

Steps

1. On the **Lab** tab, choose the illuminant you want to study. The Uncertainty tab always works on the light currently selected in the Lab.
2. Open the **Uncertainty** tab.
3. Edit any budget value. Negative or non-numeric entries are reset to 0.
4. Under **Trials**, choose **1k**, **4k** (the default) or **20k**. More trials give a steadier estimate and take longer.
5. Select **Run the propagation**. The button shows "Running…" and then returns.
6. Read **Expanded uncertainty, k = 2**. For each of x, y, CCT and Duv the table shows:
   - **Value** — the reading itself;
   - **u** — the standard uncertainty;
   - **U (k=2)** — the expanded uncertainty, twice u, which covers about 95% of cases for a normal distribution;
   - **Relative** — U as a percentage of the value. Figures over 5% are highlighted.
7. Read **Contribution to u(CCT)**. Each bar shows the share of the CCT uncertainty caused by one budget term. Bars above 40% are highlighted.
8. Look at **95% coverage region in u'v'**. The solid ellipse is the region that holds 95% of the simulated chromaticities; the dashed ellipse is one standard deviation. The centre coordinates are printed in the corner.
9. The line under the results records the number of trials, how long the run took, the observer and the interval.

Reading the result

- For a smooth source such as D65 or a blackbody, the wavelength and bandwidth terms usually matter little. For a spiky source they can dominate.
- The reference lamp term mostly changes absolute level, which cancels when you compute chromaticity, so its share of the CCT uncertainty is small.
- The inputs are treated as uncorrelated. The page says so under the results.

- The contribution bars come from a shorter run (about a sixth of the trials, never fewer than 200), so they can move slightly between runs.

Tips

- The interval and interpolation set on the **Methods** tab also apply here.
- Monte Carlo results vary a little each time. If two runs differ more than you are comfortable with, use **20k** trials.

Result

You have expanded uncertainties for x, y, CCT and Duv, a ranking of which error source matters most, and a picture of the 95% region in u'v'.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Query many illuminants at once

Use the Console tab's query language to tabulate, sort, plot and export readings for many illuminants.

The **Console** tab answers questions such as "which of these lights is furthest from the Planckian locus?" in one table. You type a short query; the lab computes every requested column with the same engine the other tabs use, so a console row and a Lab reading cannot disagree. The query is parsed, never run as code.

Steps

1. Open the **Console** tab. The **Query** box already holds an example.
2. Edit the query, or select one of the **Recipes** to load a ready-made one.
3. Select **Run**, or press `Ctrl` + `Enter` in the query box.
4. Read the **Results** table. Its heading shows the number of rows. The status line beside the buttons shows rows, time taken and the interval and interpolation used.
5. Under **Any two columns, plotted**, choose a **Horizontal axis** and a **Vertical axis**. The scatter plot labels each point with the illuminant name.
6. Select **CSV** or **JSON** to download the results.

The grammar

A query is a few lines, each starting with a keyword. Anything after `//` on a line is a comment.

Line	What it does	Example
`select`	The illuminants to include, separated by commas or spaces. Names are case-insensitive. A range such as `F1..F12` expands to every name in between.	`select A, D50, D65, F1..F12`
`with`	Pipeline settings as `name=value` pairs.	`with interval=1nm, interpolation=sprague`
`report`	The columns to show, in order.	`report x, y, CCT, Duv`
`order by`	Sort by one reported column, `asc` (default) or `desc`.	`order by Duv desc`
`limit`	Keep only the first n rows after sorting.	`limit 8`

Names you can select

A, D50, D55, D65, D75, E and F1 to F12.

Settings for `with`

Setting	Values	Notes
`interval`	`1nm`, `5nm`, `10nm`	Any other value falls back to the interval set on the Methods tab.
`interpolation`	`linear`, `sprague`, `pchip`	Any other value falls back to the Methods tab setting.
`observer`	`2deg`	Accepted for readability. Only the CIE 1931 2° observer is held.

Columns for `report`

Column	Shown as	Meaning	Decimals
`x`, `y`	x, y	CIE 1931 chromaticity	5
`u`, `v`	u′, v′	CIE 1976 chromaticity	5
`cct`	CCT	Correlated colour temperature (Ohno); blank when not valid	0
`duv` (also `d_uv`)	Duv	Signed distance from the Planckian locus	5
`efficacy` (also `eff`)	lm/W	Luminous efficacy of radiation	1
`entropy`	Entropy	A measure of how spread out the spectrum is	3
`peak`	Peak nm	Wavelength of the highest power	0

If you leave out `report`, the table shows x, y, CCT and Duv.

The recipes

Recipe	What it asks
The whole catalogue by CCT	All 18 illuminants at 5 nm: x, y, CCT, Duv and efficacy, sorted by CCT
Which sources are furthest off the locus	CCT, Duv and entropy, sorted by Duv descending, top 8
Luminous efficacy against spectral spread	Efficacy, entropy and peak, sorted by efficacy descending
The D series at 1 nm	D50, D55, D65 and D75 at 1 nm with Sprague interpolation

Stand-in rows

Fluorescent illuminants F1 to F12 are built from stand-in spectra, not the CIE tables. Their rows carry a **stand-in** tag in the table, "(stand-in)" after the name in the CSV, and `"synthetic": true` in the JSON. Do not quote them as CIE values.

The export files

- **CSV** downloads `illuminants.csv`. The first line is a comment recording the observer, interval and interpolation; then a header row (`source` and your columns) and one row per illuminant.
- **JSON** downloads `illuminants.json` with the generation time, the pipeline settings, the column list and one entry per row (name, synthetic flag and values).

The buttons do nothing until a query has returned rows.

Result

You have a sorted table of readings for the lights you chose, a scatter plot of any two columns, and a file you can open in a spreadsheet.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Illuminants reference

Every control, readout, option and export in the Standard Illuminants lab, tab by tab.

Address: https://auricartisan.com/tool/general/colorimetry/illuminants/

Lab tab

Controls

Control	What it does	Values or range	Default
Source mode	Chooses between a tabulated standard illuminant and a Planckian (blackbody) spectrum	**Standard**, **Blackbody**	Standard
Illuminant	The standard illuminant to analyse	18 choices, listed below	CIE D65 (6504 K) — Daylight, sRGB
CCT slider and number box	Colour temperature of the blackbody source. Moving either switches to Blackbody mode	1,500 to 25,000 K in 50 K steps	6500
Link to select	When ticked, choosing a standard illuminant also sets the CCT field to that illuminant's nominal CCT	On or off	On
Reset to D65	Returns to Standard mode, D65 and 6500 K	Button	—
Enable comparison	Overlays a second illuminant on the SPD chart and fills the comparison table	On or off	Off
Comparison illuminant list	The second illuminant	CIE A, D50, D55, D65, D75, Equal-energy E, F2, F7, F11	CIE A (2856 K)
Surface reflectance samples	Which surfaces appear under Surface appearance	White, Gray, Red, Green, Blue, Skin, Yellow, Cyan	White, Gray, Red, Green, Skin on
Adapt from	Source white point for the adaptation preview	D65, D50, D55, D75, A, E	D65
Swap	Exchanges Adapt from and Adapt to	Button	—
Adapt to	Destination white point	D65, D50, D55, D75, A, E	D50
Transform chips	The chromatic adaptation transform	Bradford, CAT16, CAT02, Von Kries, Sharp, XYZ scaling, None	Bradford

The 18 standard illuminants

Choice in the list	Kind
CIE A (2856 K) — Incandescent	Tungsten, computed from Planck's law
CIE D50 (5003 K) — Daylight, ICC PCS	Daylight, reconstructed from the CIE daylight basis

Choice in the list	Kind
CIE D55 (5503 K) — Daylight	Daylight
CIE D65 (6504 K) — Daylight, sRGB	Daylight
CIE D75 (7504 K) — North sky	Daylight
Equal-energy E (5455 K)	Equal power at every wavelength
F1 — Daylight fluorescent (6430 K)	Fluorescent, stand-in spectrum
F2 — Cool white fluorescent (4230 K)	Fluorescent, stand-in
F3 — White fluorescent (3450 K)	Fluorescent, stand-in
F4 — Warm white fluorescent (2940 K)	Fluorescent, stand-in
F5 — Daylight fluorescent (6350 K)	Fluorescent, stand-in
F6 — Lite white fluorescent (4150 K)	Fluorescent, stand-in
F7 — D65 simulator fluorescent (6500 K)	Broadband fluorescent, stand-in
F8 — D50 simulator fluorescent (5000 K)	Broadband fluorescent, stand-in
F9 — Cool white fluorescent (4150 K)	Broadband fluorescent, stand-in
F10 — Narrow-band 3-phosphor (5000 K)	Three-band fluorescent, stand-in
F11 — TL84 3-phosphor (4000 K)	Three-band fluorescent, stand-in
F12 — TL83 3-phosphor (3000 K)	Three-band fluorescent, stand-in

The kelvin figures in the list are nominal labels. The lab computes its own CCT from each spectrum.

Readouts

Readout	What it shows
Illuminant Summary	A swatch of the white as sRGB, the name, the type and HEX, then **CCT**, **Duv**, **CIE x**, **CIE y**, **CRI Ra** and **XYZ** (Y normalised to 100)
SPD chart	Relative power from 380 to 780 nm at 5 nm. Gold is the selected light, blue the comparison. A faint rainbow fill marks the wavelength regions. Click to probe the nearest wavelength and its power
CRI chart	Bars for R1 to R14 and the Ra value (mean of R1 to R8). Reference: blackbody below 5,000 K, CIE daylight at or above. Computed on stand-in test samples
Surface appearance	For each ticked surface: its colour under D65, its colour under the selected light after adaptation to D65, both HEX values and the ΔE_{00} between them
Chromatic adaptation preview	24 fixed colours (top) and their adapted versions (bottom), with a caption naming source, destination and method
Adaptation matrix details	The transform's M matrix, the LMS scaling (destination ÷ source), and the source and destination white in LMS. With None selected it says no adaptation is applied
Illuminant comparison	With comparison on: CCT, CRI Ra and CIE x, y for both lights, and the **Metamerism Index (avg ΔE_{00})** over the 14 test samples

Surface samples: White Paint, Neutral Gray, Saturated Red, Leaf Green, Sky Blue, Skin-like, Yellow Paint and Cyan Ink. They are simple educational reflectance curves, not measured materials.

Each chart has a **Fullscreen** button; double-clicking the chart does the same.

Chromaticity tab

Control or readout	What it does	Values	Default
Plane	The chromaticity diagram to draw	**1931 xy**, **1976 u′v′**, **1960 uv**	1931 xy
Planckian locus	Draws the blackbody curve from 1,500 to 25,000 K	On or off	On
Isotemperature lines	Short lines across the locus at 2000, 2856, 4000, 5000, 6500, 10000 and 20000 K (needs the Planckian locus on)	On or off	On
Daylight locus	Draws CIE daylight chromaticities from 4,000 to 25,000 K	On or off	Off
sRGB / P3 / Rec.2020	Draws the three RGB primary triangles	On or off	Off
Marks	A, D50, D65 and E in grey; the current light in gold, labelled	—	—
Dominant wavelength	**λ d** (a minus sign means complementary wavelength), **Purity**, **D uv** and **CCT**	—	—

The dominant wavelength is measured from the D65 reference white against the spectral locus. When the light is almost the reference white (purity below 0.2%), λ d shows a dash because it has no defined direction. CCT shows "n/a" when it is not valid (see Methods).

MacAdam ellipses are not drawn. A note under the overlays explains that the measured ellipse parameters are not held.

Uncertainty tab

Control or readout	What it does	Values	Default
Input budget	Six editable error terms	See Estimate measurement uncertainty	Preset spectroradiometer budget
Trials	Number of Monte Carlo trials	**1k**, **4k**, **20k**	4k
Run the propagation	Runs the simulation on the light selected in the Lab	Button	—
Expanded uncertainty, k = 2	Table of Quantity, Value, u, U (k=2) and Relative for x, y, CCT and Duv	—	"Not run yet."
Contribution to u(CCT)	Share of the CCT uncertainty from each term	—	—
95% coverage region in u′v′	Ellipse holding 95% of results; dashed ellipse at one standard deviation	—	—

Methods tab

Control	What it does	Values	Default
Observer	The standard observer	**1931 2°**; **1964 10°** is shown but switched off because its tables are not held	1931 2°
Interval	Wavelength step used for integration	**1 nm**, **5 nm**, **10 nm**	5 nm
Interpolation	How the source table is resampled onto the working grid	**Sprague**, **Linear**, **PCHIP**	Sprague

The interval and interpolation apply to the Chromaticity, Uncertainty and Console readings and to the tables on this tab. The Lab tab's own readouts are always computed at 5 nm.

Readouts:

- **What the pipeline is worth** — a table with the current pipeline's x, y, CCT and Duv, then the change caused by switching to 1 nm, to 5 nm, to linear and to PCHIP. A dash means no change. The note beneath explains that a 5 nm table resampled onto a finer grid adds no information.
- **CCT method** — the CCT and Duv by Ohno's method (named "Ohno 2014, triangular" or "Ohno 2014, parabolic" depending on how close the light is to the locus), with a validity note, and the older McCamy (1992) cubic for comparison, marked as having no validity test.
- **Rendition metrics** — three cards, all switched off: TM-30-18, CIE 13.3 and CIE S 026. Each lists the dataset it would need.

CCT validity notes: "in range" (Duv within ±0.02), "far from the locus" (Duv between ±0.02 and ±0.05), a note that CCT is not defined when Duv is beyond ±0.05, or "outside 1000–100000 K". A CCT is reported as valid only between 1,000 and 100,000 K and with Duv within ±0.05.

Console tab

See Query many illuminants at once for the grammar.

Control	What it does
Query box	The query text. `Ctrl` + `Enter` runs it
Run	Runs the query
CSV	Downloads `illuminants.csv`
JSON	Downloads `illuminants.json`
Recipes	Four ready-made queries; selecting one loads and runs it
Results table	One row per illuminant
Horizontal axis, **Vertical axis**	Columns to plot in the scatter chart

Data tab

The register lists every dataset the lab touches, with its source, the interval it is tabulated at, its status and why it matters. The summary counts datasets, verbatim, computed and not held.

Status	Meaning
Verbatim	The published table, copied
Computed	Derived exactly from a defining formula or a verbatim table
Synthesised	A stand-in that is not the published data; never reported as a standard value
Absent	Not held; features that need it stay switched off

Dataset	Status
Colour matching functions, 2° (CIE 1931)	Verbatim
Colour matching functions, 10° (CIE 1964)	Absent
Daylight eigenvectors S0 S1 S2	Verbatim

Dataset	Status
Illuminant A	Computed
Illuminants D50 D55 D65 D75	Computed
Illuminant E	Computed
Planckian radiators	Computed
Fluorescent F1–F12	Synthesised
Fluorescent FL3.1–FL3.15	Absent
High pressure discharge HP1–HP5	Absent
LED-B1–B5, BH1, RGB1, V1–V2	Absent
Test colour samples 1–14	Synthesised
Colour evaluation samples 1–99	Absent
α-opic action spectra	Absent
MacAdam ellipses	Absent

Switched off, and why lists the features that stay off: TM-30-18 Rf / Rg and the colour vector graphic; CIE 13.3 Ra and R1–R14 as standard values; melanopic EDI and the α-opic ratios; the CIE 1964 10° observer; F1–F12 as published illuminants; MacAdam ellipse overlays.

Export tab

Control	What it does	Output
Export SPD CSV	Downloads the selected light's spectrum	`<name>-spd.csv` with columns `wavelength_nm` and `relative_power`, 380–780 nm at 5 nm
Export JSON	Downloads the light's data	`<name>.json` with the illuminant label, CCT, xy, XYZ, the CRI object (Ra and R1–R14) and the SPD
Export SPD PNG	Saves the SPD chart as an image	`<name>-spd.png`
Copy white HEX	Copies the sRGB HEX of the light's white to the clipboard	Text
Share URL field, **Generate**, **Copy**	Builds and copies a link to the current settings	A URL
Generate comparison table	Tables all 18 standard illuminants	Illum, CCT, x, y, CRI Ra and a white swatch

`<name>` is the illuminant (for example `D65`) or, in Blackbody mode, `BB_<CCT>K`.

The share link stores the source mode, the illuminant (Standard mode) or the CCT (Blackbody mode), and the adaptation source, destination and method. It does not store the surface selection or the comparison settings.

Reference tab

Control	What it does
Standards, **Formulas**, **Citations**, **Research notes**	Switch between the four sections. Each shows a count

Control	What it does
Filter	Searches all four sections at once and jumps to one with matches. `Esc` clears it
Status chips (Standards)	**All**, **Implemented**, **Stand-in**, **Cited only**
Expand, **Collapse** (Standards)	Open or close every row
Where it is used chips (Citations)	**All**, **Runs here**, **Data missing**, **Context only**
Open in the data register (inside a standard)	Jumps to the Data tab

Each standard shows a badge read from the data register, the datasets it rests on, and any datasets it specifies that the lab does not hold. **Formulas** has eight sub-tabs: Planck Blackbody, CIE Daylight, SPD to XYZ, CRI Formula, TM-30 Overview, Bradford CAT, CCT Estimation and ΔE Formulas. These explain the background mathematics; where a formula (such as McCamy's CCT) is not what produces the readings, the Methods tab says which method does. **Citations** lists 22 sources, each marked Runs here, Data missing or Context only.

Save to Library

The floating **Save to Library** button stores a preset of the current settings in your Auric Artisan Library, so you can reopen it later.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Troubleshooting Illuminants

What to do when a reading, chart, export or shortcut in the Standard Illuminants lab does not behave as expected.

Symptom	Why it happens	What to do
Moving the CCT slider changed my illuminant to a blackbody	The CCT controls drive Blackbody mode, so touching them switches the source	Choose **Standard** under Source mode and pick the illuminant again, or select **Reset to D65**
The CCT box jumps back when I pick a standard illuminant	**Link to select** copies the illuminant's nominal CCT into the box	Untick **Link to select** to keep your own CCT value
The Illuminant comparison panel is empty	Comparison is off	Tick **Enable comparison**
CCT shows "n/a" or a validity warning	The light is too far from the blackbody curve (Duv beyond ±0.05) or outside 1,000–100,000 K, where CCT is not defined	Read Duv and x, y instead. See Limits and accuracy
The dominant wavelength shows a dash	The light is almost the same colour as the D65 reference white, so there is no direction to measure	Expected. Purity will read close to 0%
The CRI of a fluorescent light looks wrong compared with a datasheet	The lab's test colour samples and fluorescent spectra are stand-ins, not the CIE tables	Treat CRI as indicative. Use a published or measured value for specifications
I want TM-30 Rf and Rg	The 99 colour evaluation samples are not held, so TM-30 is not computed	The Methods tab shows the card as switched off. Use IES software for TM-30
The 1964 10° observer button is greyed out	Its colour matching functions are not held	Only the 1931 2° observer is available
The equal-energy E spectrum is a flat line	E has equal power at every wavelength by definition	Expected

Symptom	Why it happens	What to do
Changing interval or interpolation does not change the Lab summary	Those settings apply to the Chromaticity, Uncertainty, Console and Methods views; the Lab tab always works at 5 nm	Read the changed figures on those tabs
Changing interval or interpolation changes nothing at all	The source tables are tabulated at 5 nm, so a finer grid adds no information; the Methods note says so	Expected. The deltas show as dashes
The Uncertainty results change slightly each run	Monte Carlo sampling is random	Use **20k** trials for a steadier result
Console CSV or JSON does nothing	No query has returned rows yet	Select **Run** first, then export
Console says "Nothing matched"	A name was misspelt, or a range used an unknown prefix	Use names such as `D65` or `F1..F12`
Pressing S, R, C or E does nothing	The Lab rail shows these keys as a hint, but they are not active in the current version	Use **Swap**, **Reset to D65**, **Enable comparison** and **Export SPD CSV** instead
Copy white HEX or **Copy** gives no toast and nothing is copied	The browser blocked clipboard access (permissions, or a page not opened over https)	Allow clipboard access, or select the share field and copy it manually
The share link opened without my surfaces or comparison	The link stores source, illuminant or CCT, and adaptation settings only	Re-tick the surfaces and comparison after opening
The full-screen chart will not close	—	Press `Esc`, select the close button, or click outside the chart
Charts look blurred after resizing the window	Some charts redraw only when their view changes	Switch tabs and back, or reload the page

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Illuminants FAQ

Short answers to common questions about the Standard Illuminants lab.

Why does CIE A show 2856 K when the standard defines it at 2848 K?

Illuminant A is defined as a Planckian radiator at 2848 K using the 1931 value of the second radiation constant. Measured on a modern (ITS-90) Planckian locus, its CCT is 2856 K. The lab computes A the way it is defined and reports the CCT you would measure, which is why the list says 2856 K. The Data tab explains this under Illuminant A.

Are the D65 and D50 spectra the published CIE tables?

They are reconstructed from the published CIE daylight basis functions (S0, S1, S2), which the lab holds verbatim. The reconstructions land on their published chromaticities to better than 0.0003, which is why they are marked "computed" rather than "verbatim".

Why are the fluorescent illuminants marked as stand-ins?

The lab does not hold the CIE 15 table of fluorescent spectra. F1 to F12 are built from smooth peaks as illustrations, so their chromaticity, CCT and rendering figures describe the stand-in, not the real CIE illuminant. They are flagged wherever they appear.

Is the CRI value a real CIE 13.3 result?

No. The 14 test colour samples are approximations, so Ra and R1–R14 are indicative only. The Methods tab shows the CIE 13.3 card as switched off for that reason.

Why is there no TM-30 figure?

TM-30 is defined on 99 colour evaluation samples that the lab does not hold. Rather than invent a figure, the lab leaves TM-30 switched off. The Reference tab still explains what TM-30 measures.

What is the difference between the Ohno and McCamy CCT?

McCamy's (1992) formula is a quick cubic fit that returns a number for any input, even where CCT is meaningless. Ohno's method (2014) searches a finely computed Planckian locus, returns a signed Duv, and says when the result is not valid. The lab uses Ohno for every reading and shows McCamy only on the Methods tab for comparison.

What does a negative Duv mean?

The light sits below the Planckian locus in the uv plane, which looks slightly pink or magenta. Positive Duv is above the locus, slightly green.

Can I use the 10° observer?

Not yet. The CIE 1964 10° colour matching functions are not held, so the option is switched off rather than silently returning 2° results.

Can I load my own measured spectrum?

The page has no control for importing a spectrum. You can choose any standard illuminant or a blackbody at any CCT from 1,500 to 25,000 K.

Does it work offline?

Once the page has loaded, all calculations run in your browser without a network connection. Loading the page needs one.

Can I save my settings?

Yes. Use **Export › Share URL** for a link, or the floating **Save to Library** button to keep a preset in your Library.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Illuminants: supporting pages

Glossary, shortcuts, limits, related tools, release notes and privacy for the Standard Illuminants lab.

These pages support the guides in docs/. They define the terms the lab uses, say how far its numbers can be trusted, and point to related tools and articles.

File	What it covers
glossary.md	Terms used in the lab, from SPD to Duv
shortcuts.md	Keyboard and mouse shortcuts
limits-and-accuracy.md	What is standard, what is a stand-in, and what is not computed
related.md	Related tools and Learn articles
release-notes.md	What has changed for users
privacy.md	What stays in your browser

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Illuminants shortcuts

Keyboard and mouse shortcuts in the Standard Illuminants lab.

Shortcut	Where	What it does
`Ctrl` + `Enter`	Console, in the Query box	Runs the query
`Esc`	Reference, in the Filter box	Clears the filter
`Esc`	A full-screen chart	Closes the full-screen view
Double-click	SPD or CRI chart	Opens the chart full screen
Click	SPD chart	Probes the nearest wavelength and shows its relative power

On macOS, `Cmd` + `Enter` also runs a console query.

About the S, R, C and E hint

The Lab rail shows a **Keyboard shortcuts** hint: S swap adaptation, R reset D65, C toggle compare, E export CSV. These single-key shortcuts are not active in the current version of the page. Use the buttons instead:

Hint	Button to use
S	**Swap** under Chromatic adaptation
R	**Reset to D65**
C	**Enable comparison**
E	**Export SPD CSV** on the Export tab

The site's own shortcuts (such as the search palette) also work on this page; see the site features documentation.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Illuminants glossary

Plain-language definitions of the colour-science terms used in the Standard Illuminants lab.

Blackbody (Planckian radiator) — an idealised object whose emitted light depends only on its temperature, described by Planck's law. Tungsten lamps are close to one.

Chromatic adaptation — the eye's adjustment to the colour of the prevailing light, so that a white sheet still looks white under warm or cool light.

Chromatic adaptation transform (CAT) — a calculation that predicts chromatic adaptation. It maps XYZ into a cone-like space, scales each channel by the ratio of two white points, and maps back. Bradford, CAT02, CAT16, Von Kries (HPE) and Sharp differ in the cone-like matrix.

Chromaticity — the colour of a light or surface without its brightness, written as two coordinates.

Chromaticity diagram — a plot of chromaticity. The lab offers CIE 1931 xy, CIE 1960 uv (where Duv is defined) and CIE 1976 u′v′ (more evenly spaced).

CIE — the International Commission on Illumination, which publishes the standard observers, illuminants and formulas used here.

Colour matching functions (CMFs) — three curves describing how a standard observer matches colours. Weighting a spectrum by them gives X, Y and Z.

Colour rendering index (CRI) — a score of how faithfully a light renders test colours compared with a reference light at the same CCT. **Ra** is the mean of the first eight scores (R1–R8); R9 is saturated red.

Complementary wavelength — for a colour that has no dominant wavelength (a purple), the wavelength opposite it through the white point. The lab shows it with a minus sign.

Correlated colour temperature (CCT) — the temperature, in kelvin, of the blackbody whose colour is closest to the light. Only meaningful when Duv is small.

Daylight locus — the curve of CIE daylight chromaticities, slightly above the Planckian locus.

D-series illuminants — CIE daylight illuminants D50, D55, D65 and D75, built from three daylight basis functions (S0, S1, S2).

Dominant wavelength — the single wavelength that, mixed with white, matches the colour's hue.

Duv — the signed distance of a chromaticity from the Planckian locus in the CIE 1960 uv plane. Positive is greenish, negative is pinkish.

Efficacy (luminous efficacy of radiation) — lumens per watt of radiant power: how much visible light a spectrum delivers for its power.

Equal-energy illuminant (E) — a theoretical light with the same power at every wavelength. Its chromaticity is x = y = 1/3.

Expanded uncertainty (U) — the standard uncertainty multiplied by a coverage factor k. With k = 2 it covers about 95% of cases for a normal distribution.

Illuminant — a light source defined by a table of relative spectral power.

Isotemperature line — a short line across the Planckian locus joining chromaticities that share the same CCT.

Metamerism — two surfaces that match under one light but not under another. The **metamerism index** in the lab is the average CIEDE2000 difference of 14 test samples between two lights.

Monte Carlo method — estimating uncertainty by repeating a calculation many times with randomly perturbed inputs and measuring the spread.

Observer (standard observer) — the CIE's average human colour vision for a 2° (1931) or 10° (1964) field. The lab holds only the 2° observer.

PCHIP — piecewise cubic Hermite interpolation; a resampling method that does not overshoot near narrow peaks.

Planckian locus — the curve of blackbody chromaticities as temperature changes.

Purity (excitation purity) — how far a colour lies from white towards its dominant wavelength, as a percentage.

Spectral locus — the horseshoe-shaped edge of the chromaticity diagram, made by single wavelengths.

Spectral power distribution (SPD) — the relative power a light emits at each wavelength.

Sprague interpolation — a smooth resampling method recommended by CIE 167:2005 for evenly spaced spectral data.

Stand-in — a dataset built to illustrate a shape, not the published table. Readings from a stand-in are never reported as standard values.

Test colour samples (TCS) — the 14 reflectance samples CIE 13.3 uses to compute CRI. The lab approximates them.

TM-30 — the IES method for colour rendition using 99 samples and two indices, Rf (fidelity) and Rg (gamut). Not computed by the lab.

Tristimulus values (X, Y, Z) — the three numbers that describe a colour for the standard observer. Y is related to brightness.

White point — the chromaticity of an illuminant, used as the reference white for colour calculations.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Illuminants limits and accuracy

Which Illuminants readings are standard values, which rest on stand-ins, and what the lab does not compute.

The lab is a reference calculator that runs in your browser. It is not calibrated measurement software and not a substitute for official CIE or IES tools. This page says what each kind of reading rests on.

Standard values you can rely on

Reading	Rests on	Notes
x, y, XYZ of A, D50, D55, D65, D75, E and blackbody sources	CIE 1931 2° colour matching functions (verbatim, 5 nm, 380–780 nm)	The D series is reconstructed from the verbatim daylight basis and lands on the published chromaticities to better than 0.0003
CCT and Duv	Ohno's method over a Planckian locus computed through the same observer	Reported as valid only for 1,000–100,000 K and Duv within ±0.05
Chromaticity in xy, u′v′ and uv	Exact conversions	—
Chromatic adaptation matrices	Published Bradford, CAT02, CAT16, HPE and Sharp matrices	XYZ scaling is included as the simplest case, not as a model of vision
Dominant wavelength and purity	The spectral locus of the 2° observer, from the D65 white	Undefined (shown as a dash) when the light is almost white
Monte Carlo uncertainty	JCGM 101:2008 method	Only as good as the budget you enter; inputs are treated as uncorrelated

Readings that rest on stand-ins

Reading	Why it is not a standard value
Anything for F1 to F12 (spectrum, x, y, CCT, CRI)	The CIE fluorescent tables are not held; the spectra are smooth stand-ins and the mercury lines are not in the right places
CRI Ra and R1 to R14	The 14 CIE 13.3 test colour samples are approximations
Metamerism index	Uses the same approximate test samples
Surface appearance	Uses eight simple educational reflectance curves, not measured materials

The lab labels these wherever they appear ("stand-in", "not a standard value").

Not computed at all

- **IES TM-30** (Rf, Rg and the colour vector graphic): the 99 colour evaluation samples are not held.
- **CIE S 026 α-opic quantities** (melanopic EDI and related ratios): the action spectra are not held.
- **The CIE 1964 10° observer**: its colour matching functions are not held.
- **MacAdam ellipses**: the measured ellipse parameters are not held.
- **CIE FL3, HP and LED illuminant families**: their tables are not held.

Precision notes

- The Lab tab integrates at 5 nm over 380–780 nm. The source tables are tabulated at 5 nm (the daylight basis at 10 nm), so choosing 1 nm on the Methods tab does not add information; the Methods tab shows the deltas so you can see this.
- The sRGB swatches and HEX values are display previews. A white point shown as a swatch is not a calibrated target.
- Summary figures are rounded for display (CCT to the nearest kelvin, x and y to five decimals, Duv to five decimals).
- The kelvin values in the illuminant list are nominal labels; the computed CCT can differ slightly.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Illuminants privacy

What the Standard Illuminants lab keeps on your device and what, if anything, leaves it.

- **All calculations run in your browser.** Spectra, readings, uncertainty runs and console queries are computed on your device. The lab does not send them anywhere.
- **Nothing you type is uploaded.** Budget values and console queries stay in the page and are gone when you close it.
- **Downloads are made on your device.** CSV, JSON and PNG exports are generated in the browser and saved through your browser's normal download.
- **Clipboard. Copy white HEX** and the share **Copy** button write to your clipboard only when you select them. Your browser may ask for permission.
- **Share links** contain only the settings listed in the reference (mode, illuminant or CCT, and adaptation settings). Anyone you send the link to sees those settings.
- **Save to Library** stores a preset in your Auric Artisan Library. See the Library documentation for where that is kept.
- The page itself loads the site's shared features in the same way as every other page on auricartisan.com. See the site's privacy policy for those.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Illuminants release notes

User-facing changes to the Standard Illuminants lab.

September 2026 — the workbench rebuild

The lab was rebuilt around a new colorimetric core and a new layout.

- New tabs: **Chromaticity**, **Uncertainty**, **Methods**, **Console** and **Data** join Lab, Export and Reference.
- CCT and Duv are now computed with Ohno's method, with a signed Duv and a validity check. The older McCamy figure is shown only on the Methods tab for comparison.
- The daylight illuminants are rebuilt from the published daylight basis at its tabulated interval, and Illuminant A is computed as defined (2848 K with the 1931 constant, measuring 2856 K).
- Chromatic adaptation now offers Bradford, CAT16, CAT02, Von Kries, Sharp, XYZ scaling and None, with the matrix shown.
- A provenance line under results names the dataset, observer and interval.
- A data register marks each dataset verbatim, computed, stand-in or absent. Fluorescent spectra and CRI test samples are labelled as stand-ins; TM-30 is shown as not computed rather than estimated.
- The Standards, Formulas, References and Research tabs became one **Reference** view with a search filter and status chips; the citation list grew to 22 entries, each marked Runs here, Data missing or Context only.

May 2026

The Illuminants User Guide was published in the site's Documentation library, describing the earlier version of the lab (Lab, Actions, Standards, Formulas, References and Research tabs).

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Related to Illuminants

Tools and Learn articles that go well with the Standard Illuminants lab.

Tools

Tool	Why
Chromatic Adaptation	Applies eight adaptation transforms to a real image, with a time course of adaptation
Gamut Mapping	What happens to a colour moved between RGB spaces with different white points
Gamut Lab	RGB primaries and white points on the chromaticity diagram
Material Lighting	Surface reflectance rendered under different illuminants
Colour Science Lab	A broader laboratory for adaptation, appearance models and spectra

Learn articles

- Spectral Power Distributions and Why RGB Is Not Enough — https://auricartisan.com/library/learn/articles/2026-05-24-spectral-power-distributions
- CIE XYZ Explained — https://auricartisan.com/library/learn/articles/2026-05-24-cie-xyz-explained
- Chromaticity Diagrams Without the Intimidation — https://auricartisan.com/library/learn/articles/2026-05-24-chromaticity-diagrams

- Metamerism Explained — https://auricartisan.com/library/learn/articles/2026-05-25-metamerism
- White Points and Chromatic Adaptation: von Kries, Bradford, and CAT02 — https://auricartisan.com/library/learn/articles/2026-07-11-white-points-chromatic-adaptation-transforms
- Dominant Wavelength and Purity: Naming a Color by Its Spectrum — https://auricartisan.com/library/learn/articles/2026-08-11-dominant-wavelength-and-purity
- Spectrophotometers, Colorimeters, and Spectroradiometers — https://auricartisan.com/library/learn/articles/2026-05-25-spectrophotometers-colorimeters-spectroradiometers

Documentation on the site

The site's Documentation library (https://auricartisan.com/library/documentation/) has an earlier **Illuminants User Guide**. Where it differs from the current page, this documentation follows the page.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tonal Steps

What the Tonal Steps, Tints and Shades tool does, who it is for, where to find it and how to build your first ramp.

Tonal Steps builds a lightness **ramp** from one colour: a sequence of steps running from white, through versions of your colour, to black. Alongside it the tool builds **tints** (your colour mixed towards white), **shades** (towards black), **tones** (towards grey), a **neutral ladder** of greys, and a 50–950 or 50–900 **key set** of the kind design systems use.

Two ramp engines are offered. **HSL lightness** spaces the steps evenly in HSL lightness. **Equal luminance** spaces them evenly in relative luminance, the quantity WCAG contrast is built on, with an easing exponent to bunch steps where you want them. Tints, shades, tones and key sets are mixed in the space you choose: CIELAB, OKLab, HSL or linear sRGB.

Every step is measured: OKLab lightness, WCAG 2.2 contrast ratio and APCA-W3 0.1.9 Lc against white, black or your base colour. A Ramps view compares the engines and reports how even each ramp is in three measures. Exports give CSS custom properties, an SCSS map, JSON or CSV, each with a header recording how the ramp was made.

Who it is for

- Designers building colour scales for a design system or brand.
- Front-end developers who need ramp tokens with known contrast against white and black.
- Accessibility specialists checking which steps meet WCAG AA or AAA, or an APCA Lc level.
- Students comparing HSL, CIELAB and OKLab mixing.

Where to find it

- Address: https://auricartisan.com/tool/general/gamut-and-rendering/tonal-steps-tints-shades/
- In the tool collection (https://auricartisan.com/collections/) under **Gamut & Rendering**, as **Tonal Steps, Tints & Shades**.

Quick start

1. Open https://auricartisan.com/tool/general/gamut-and-rendering/tonal-steps-tints-shades/.
2. Type your colour in **Base colour**, for example `#D3AF37`, or use the picker.
3. Choose a **Ramp engine**: **HSL lightness** or **Equal luminance**.
4. Set **Steps** (3 to 21; default 11).
5. Read each step's HEX, OKLab L and contrast against white.
6. Under **Also show**, tick **Tints — toward white** and **Shades — toward black**.
7. Open **Contrast** to see every step's WCAG rating and APCA Lc.
8. Open **Export**, choose **The ramp** as **CSS**, and select the download button.

What you can do

- Build an HSL-lightness or equal-luminance ramp of 3 to 21 steps, with easing.
- Build tints, shades, tones and a neutral ladder, mixed in CIELAB, OKLab, HSL or linear sRGB.
- Read each step's OKLab L and WCAG ratio, and the base colour's HEX, RGB, HSL, CIELAB, LCh, OKLab L and luminance.
- Compare the engines side by side and measure evenness in OKLab L, CIELAB L* or relative luminance.
- Build 50–950 and 50–900 key sets.
- Check every step's WCAG 2.2 ratio and rating and APCA-W3 Lc against white, black or the base.

- Export CSS, SCSS, JSON, CSV or a link; analyse a list of colours.

In this folder

- docs/ — how to use the tool
  - Getting started
  - Build a ramp
  - Make tints, shades, tones and key sets
  - Check contrast
  - Export tokens
  - Reference
  - Troubleshooting
  - FAQ
- others/ — background and supporting pages
  - Glossary
  - Shortcuts
  - Limits and accuracy
  - Related tools
  - Release notes
  - Privacy

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tonal Steps documentation

The guides, reference and help pages for Tonal Steps, Tints and Shades.

This folder explains how to use Tonal Steps: a screen tour, task guides for ramps, tints and shades, contrast and exports, a complete reference and help pages.

File	What it covers
getting-started.md	Opening the tool, the six tabs and your first ramp
build-a-ramp.md	Choosing an engine, steps and easing, and reading evenness
tints-shades-tones.md	Tints, shades, tones, the neutral ladder, mixing spaces and key sets
check-contrast.md	WCAG 2.2 and APCA against white, black or the base
export-tokens.md	CSS, SCSS, JSON, CSV, links and list analysis
reference.md	Every control, readout and export
troubleshooting.md	Symptoms, causes and fixes
faq.md	Short answers

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Getting started with Tonal Steps

Open Tonal Steps, learn its tabs, and build and read your first ramp.

Open the tool

Go to https://auricartisan.com/tool/general/gamut-and-rendering/tonal-steps-tints-shades/. The heading is **Tonal Steps**. Everything runs in your browser.

Key ideas

- A **ramp** is a sequence of colours from light to dark built from one base colour.
- A **tint** is a colour mixed with white, a **shade** mixed with black, and a **tone** mixed with grey.
- **Relative luminance (Y)** is the physical brightness of a colour on a 0 (black) to 1 (white) scale. WCAG contrast ratios are calculated from it.
- **OKLab L** is a perceptual lightness from 0 to 1 (Ottosson, 2020): equal steps in it look roughly equally spaced. The tool uses it to show where your steps fall.
- **WCAG 2.2 contrast** is a ratio from 1:1 to 21:1 with thresholds of 3, 4.5 and 7. **APCA Lc** is a signed contrast value from the APCA-W3 0.1.9 algorithm; its sign shows which colour is lighter.

A tour of the screen

A note at the top explains a correction the tool made: OKLab needs linear sRGB as input, and the page now provides it.

Tab	What it is for
Lab	Build the ramp and see its steps, lightness chart, base colour details and optional sets

Tab	What it is for
Ramps	Compare engines, measure evenness, see the ramp in chroma, and build key sets
Contrast	Every step's WCAG and APCA contrast against a colour you choose
Data	The eight datasets and formulas the tool rests on, with their status
Export	Download or copy the ramp, key set, contrast table or link; analyse a list
Reference	Standards, formulas, citations and notes, and a chromaticity diagram of the ramp

The Lab tab

The left panel holds **Base colour** (HEX field, picker and **Random base**), **Ramp engine**, **Mix in**, **Steps**, **Easing γ** and **Also show**.

The right side shows:

- the caption (steps, engine and mixing space) and a base note;
- the ramp as one strip;
- one row per step: swatch, HEX, OKLab L and contrast against white;
- four statistics: **Base OKLab L**, **Largest step**, **Smallest step** and **Evenness**;
- **Where the steps fall** — a chart of OKLab L across the ramp;
- **Base colour** — HEX, RGB, HSL, L*a*b*, LCh, OKLab L, Y (luminance), and contrast against white and black with ratings;
- **The other sets**, when you tick any under Also show.

Build your first ramp

1. Type `#2563EB` in **Base colour** (it is the default) and press `Enter` or move out of the field.
2. Keep **Ramp engine** on **HSL lightness** and **Steps** at 11.
3. Read the caption: "11 steps, HSL lightness, mixed in OKLab".
4. Read the base note. It says whether your base colour is one of the steps ("step 4") or "not a step". Usually it is not: HSL lightness steps land on 100, 90, 80 … 0, and the base is rarely exactly on one.
5. Read the step rows. The first step is white and the last black.
6. Switch the engine to **Equal luminance**. The steps now fall at equal intervals of relative luminance. In perceived lightness most of them sit in the light half, with a large final jump to black; the **Where the steps fall** chart shows it.
7. Move **Easing γ** above 1. The target luminances fall faster, so more steps land in the darker half and the ramp looks more evenly spaced; below 1 the steps crowd further towards white. Easing applies to the Equal luminance engine and the neutral ladder, not to HSL lightness.

Next steps

- Build a ramp
- Make tints, shades, tones and key sets
- Check contrast
- Export tokens

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Build a ramp

Choose a ramp engine, step count and easing, and judge how even the ramp is.

The engines

Engine	How it builds the steps	Holds even	Easing γ
HSL lightness	Keeps your colour's HSL hue and saturation and steps HSL lightness evenly from 100 (white) to 0 (black)	HSL L, a coordinate rather than a perceptual quantity	Not used
Equal luminance	For each step, finds the HSL lightness (at your colour's hue and saturation) that gives a target relative luminance Y = (1 − t)^γ, from 1 (white) to 0 (black)	Relative luminance Y, the quantity WCAG contrast is built on (at γ = 1)	Bends the targets

A third engine, **Equal contrast** (equal steps in WCAG ratio against black), is not offered on the Lab tab because it produces the same ramp as Equal luminance: equal steps in WCAG ratio are equal steps in luminance. The Ramps tab shows it with a "same ramp as equal luminance" badge, and it can still be reached with the `3` key or a saved link.

Both engines start at white and end at black. The **Mix in** setting does not affect the main ramp; it affects tints, shades, tones and key sets.

Steps

1. On the **Lab** tab, enter the **Base colour** as a six-digit HEX (or use the picker, or **Random base**).
2. Choose **HSL lightness** or **Equal luminance** under **Ramp engine**.
3. Set **Steps** from 3 to 21 (default 11).
4. For Equal luminance, set **Easing γ** from 0.4 to 2.6 (default 1.0). At 1.00 the steps are evenly spaced in luminance. Higher values move steps towards the dark end, which usually looks more even; lower values crowd them towards white.
5. Read the step rows and the **Where the steps fall** chart.
6. Read the statistics:
   - **Base OKLab L** — your colour's perceptual lightness;
   - **Largest step** and **Smallest step** — the biggest and smallest OKLab L gaps between neighbouring steps;
   - **Evenness** — how far the step sizes stray from their average, as ±%. Smaller is more even.

The note under the statistics gives the ratio of the largest to the smallest step.

Compare the engines

1. Open the **Ramps** tab.
2. Under **Compare**, keep **Every engine** to see all three, or choose **One at a time** for the Lab's engine only.
3. Under **Measure evenness in**, choose **OKLab L** (default), **CIELAB L*** or **Relative Y**. A ramp even in one is rarely even in another.
4. Under **Endpoints**, choose **Include black** (default) or **Interior only**. The last step into pure black usually dominates the figure; the interior figure drops the first and last gaps. Both figures are shown, and the one you choose is emphasised.
5. Read each engine's row: its name, what it holds constant, its strip, and its evenness figures.
6. Read **Step sizes, as the chosen measure sees them** for a chart of the gaps, and **Where the ramp sits in chroma** for the ramp (with tints, shades and tones) plotted on the OKLab a–b plane. A ramp that only changes lightness traces a line there; one mixed towards white and black in a non-linear space does not.

The comparison uses the same base, step count, easing and mixing space as the Lab.

Tips

- The equal-luminance engine is, perhaps surprisingly, the least even in OKLab L at γ = 1: equal steps in luminance are very unequal in perceived lightness, with the biggest jump at the dark end. The Ramps footer says so.
- For a ramp whose steps line up with WCAG thresholds, use Equal luminance and check the Contrast tab.
- Press `1` or `2` to switch engines, and `R` for a random base, when the focus is not in a field.

Result

You have a ramp built by the engine you chose, with its step sizes measured, and a side-by-side view of how the engines differ.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Check contrast

Read every ramp step's WCAG 2.2 ratio and rating and its APCA-W3 Lc against white, black or the base colour.

Two measures

- **WCAG 2.2 contrast ratio** — (L1 + 0.05) ÷ (L2 + 0.05), lighter over darker, where L is relative luminance. Thresholds: 3:1 for large text (AA large), 4.5:1 for normal text (AA), 7:1 for enhanced (AAA).
- **APCA Lc** (APCA-W3 0.1.9) — a perceptual contrast value that treats dark-on-light and light-on-dark differently. It is signed: the sign tells you which way round the pair is. APCA is a working draft of what WCAG 3 may adopt, not a ratified requirement.

Steps

1. Build your ramp on the **Lab** tab.
2. Open the **Contrast** tab.
3. Under **Against**, choose **White** (default), **Black** or **The base** (your base colour). The caption changes to "Every step against white", and so on.
4. Read the table:

Column	Meaning
Step	The step's swatch and HEX
WCAG	The ratio, for example 4.63:1
Rating	**AAA** (7 and above), **AA** (4.5 and above), **AA large** (3 and above) or **Fail**
APCA Lc	The signed Lc, to one decimal

5. Under **Superseded APCA**, choose **Show the difference** to add two columns: **Superseded** (the approximate figure an earlier version of the page computed) and **Difference**. Differences of 1 Lc or more are highlighted. Choose **Hide** to remove them (the default).
6. Look at **The ladder** for a chart of the steps' contrast.

The **Metric** choice (**WCAG 2.2** or **APCA**) does not change the table in the current version; both figures are always shown.

Reading the result

- Pick text and background steps whose WCAG rating meets your target (AA for body text is the usual minimum).
- For APCA, larger absolute Lc means more contrast. Common reference levels are Lc 45, 60, 75 and 90, used by APCA guidance for progressively smaller or thinner text.
- The difference column shows why the correction mattered: under 0.3 Lc through the mid-range, 8 at black against white, and up to 20 against black.

Tips

- Each step row on the Lab tab already shows its ratio against white; the Contrast tab adds black, the base and APCA.
- The contrast table export on the Export tab lists every step's WCAG ratio and APCA Lc against both white and black, whichever **Against** you chose here.
- For checking a single text and background pair in depth, use the Contrast Checker.

Result

You know which steps of your ramp pass WCAG AA or AAA, and their APCA Lc, against white, black or your base colour.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Export tokens

Export the ramp, key set or contrast table as CSS, SCSS, JSON or CSV, share a link, and analyse a list of colours.

Steps

1. Build the ramp you want on the **Lab** tab.
2. Open the **Export** tab.
3. Under **What to take**, choose:
   - **The ramp** — every step of the Lab's ramp;
   - **The key set** — the 50–950 or 50–900 ladder chosen on the Ramps tab;
   - **The contrast table** — the ramp with its contrast figures;
   - **A link to this view**.
4. Under **As**, choose a format. Only formats that fit are offered: CSS, SCSS, JSON or CSV for the ramp and key set; CSV or JSON for the contrast table; URL for the link.
5. Check **Exactly what you will get**: the preview and its size in bytes.
6. Select the download button, which names the file (for example **Download tonal-2563eb.css**), or **Copy to clipboard**. For a link the button reads **Copy the link**.

File names

Choice	File
The ramp	`tonal-<hex>.<ext>`
The key set	`tonal-<hex>-50-950.<ext>` or `tonal-<hex>-50-900.<ext>`
The contrast table	`tonal-<hex>-contrast.<ext>`

`<hex>` is your base colour without `#`, in lower case.

What each format contains

Every file records how the ramp was made: the base, engine, mixing space, step count and gamma, and a footer stating that OKLab was fed linear sRGB, WCAG 2.2, APCA-W3 0.1.9 and the D65 white.

CSS — custom properties in `:root`, one per step, with a comment giving OKLab L, WCAG ratio against white and APCA Lc against white. An abbreviated example:

```
:root {
  /* base #2563EB · engine HSL lightness · mixed in OKLab · 11 steps · gamma 1.00 */
  --tonal-0:  #ffffff;  /* OKLab L 1.000  WCAG vs white  1.00:1  APCA    0.0 */
  --tonal-1:  …
}
```

HEX values in the files are lower case.

SCSS — a map: `$tonal: (0: #ffffff, 1: …, …);` with the provenance as comments.

JSON — a `provenance` object and a `steps` array. Each step has `index`, `hex`, `rgb`, `oklab_l`, `wcag_vs_white`, `wcag_vs_black`, `apca_on_white` and `apca_on_black`.

CSV — two comment lines, then the columns `index, hex, r, g, b, oklab_l, wcag_vs_white, wcag_vs_black, apca_on_white, apca_on_black`.

The contrast table uses the same JSON and CSV layouts, so it lists contrast against both white and black. It does not use the **Against** choice from the Contrast tab. Key-set properties are numbered from 0 like the ramp, not by key.

The link

The link reopens the same ramp. It carries `hex`, `engine`, `steps`, `gamma` and `mode` (the mixing space). The read-only field on the Export tab always shows the current link.

Analyse a list of colours

1. Under **Analyse a list**, enter six-digit HEX colours, one per line or separated by commas (the `#` is optional).
2. Select **Analyse**. The table shows **Hex**, **L***, **a***, **b***, **OKL** (OKLab L), **Y** (luminance), and the number of **Tints** and **Shades** built for it at the current step count.
3. Select **Download the result** to copy the table as CSV (`hex, L*, a*, b*, okL, Y, tints, shades`) to your clipboard. Despite its label, the button copies rather than downloads.

If nothing valid is found, the panel says "No valid hex colours found."

Result

You have ramp or key-set tokens in the format your project uses, each carrying a record of how it was made, and a link that rebuilds the same ramp.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Make tints, shades, tones and key sets

Build tints, shades, tones, a neutral ladder and 50–950 or 50–900 key sets, and choose the space they are mixed in.

The sets

Set	How it is made	Number of swatches
Tints — toward white	Your colour mixed with white in equal steps, from 0% (your colour) to 100% (white)	Steps + 1
Shades — toward black	Your colour mixed with black, from your colour to black	Steps + 1
Tones — toward grey	Your colour mixed with mid grey (#808080), from your colour to that grey	Steps + 1
Neutral ladder	Greys from black to white; each value is (i ÷ (steps − 1))^γ	Steps
Key set 50–950	Keys 50, 100 … 900, 950. Below 500 your colour is mixed towards white by (1 − key ÷ 500); 500 is your colour; above 500 it is mixed towards black by ((key − 500) ÷ 500)	11
Key set 50–900	The same ladder with keys 50 to 900	10

The key sets borrow the Tailwind and Material key names only. They are not the Tailwind or Material palettes: Tailwind's colours are hand-tuned per hue, and Material's tonal palettes are specified in HCT at fixed tones.

Mixing spaces

The **Mix in** setting chooses the space in which tints, shades, tones and key sets are interpolated. It changes the middle of a mix, not its ends.

Choice	Effect
OKLab (default)	Perceptually even mixes; hues stay steady
CIELAB	Mixes in CIE L*a*b* against D65
HSL	Mixes in hue, saturation and lightness
sRGB linear	In the current version this produces the same mixes as OKLab

Steps

1. On the **Lab** tab, set the base colour, **Steps** and **Mix in**.
2. Under **Also show**, tick any of **Tints — toward white**, **Shades — toward black**, **Tones — toward grey** and **Neutral ladder**. **Full ramp** is on by default.
3. Read **The other sets**. Each set is titled with its swatch count. Each tile shows its HEX and, beneath it, its contrast ratio against black; hovering shows the HEX, luminance Y and that ratio.
4. For a key set, open the **Ramps** tab and choose **50–950** or **50–900** under **Key sets**. **Design-scale key sets** lists the ladder.
5. To export a key set, open **Export** and choose **The key set** (Export tokens).

Result

You have tints, shades, tones, greys and a keyed ladder built from one colour, mixed in the space you chose.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tonal Steps reference

Every control, readout, option and export in Tonal Steps, Tints and Shades.

Address: https://auricartisan.com/tool/general/gamut-and-rendering/tonal-steps-tints-shades/

Lab tab

Controls

Control	What it does	Values or range	Default
Base colour field and picker	The colour every step is built from	Six-digit HEX	#2563EB
Random base	Picks a random base colour	Button	—
Ramp engine	How the main ramp is built	HSL lightness; Equal luminance	HSL lightness
Mix in	Space for tints, shades, tones and key sets	sRGB linear; CIELAB; OKLab; HSL	OKLab
Steps	Number of ramp steps	3 to 21	11
Easing γ	Bends the Equal luminance targets and the neutral ladder	0.4 to 2.6, step 0.05	1.0
Full ramp	Adds the ramp to The other sets	On or off	On
Tints — toward white	Shows tints	On or off	Off
Shades — toward black	Shows shades	On or off	Off
Tones — toward grey	Shows tones	On or off	Off
Neutral ladder	Shows greys	On or off	Off

In the current version, **sRGB linear** mixes the same way as OKLab.

Readouts

Readout	What it shows
Caption	Steps, engine and mixing space
Base note	The base HEX, whether it is one of the steps ("step n" or "not a step") and γ
Strip and step rows	Each step's swatch, HEX, OKLab L and contrast against white
Base OKLab L, **Largest step**, **Smallest step**, **Evenness**	Lightness and step-size statistics in OKLab L; evenness as ±%
Where the steps fall	OKLab L of each step
Base colour	HEX, RGB, HSL, L*a*b*, LCh, OKLab L, Y (luminance), vs white and vs black with ratings
The other sets	The sets ticked under Also show, each titled with its count; tiles show HEX and contrast against black

Ramps tab

Control	What it does	Values	Default
Compare	Which engines to show	Every engine; One at a time	Every engine

Control	What it does	Values	Default
Measure evenness in	Measure for the evenness figure and step-size chart	OKLab L; CIELAB L*; Relative Y	OKLab L
Endpoints	Which evenness figure leads	Include black; Interior only	Include black
Key sets	The key list for the design-scale ladder	50–950; 50–900	50–950

Readout	What it shows
Engine rows	Name, what it holds even, a "same ramp as" badge where two engines produce the same colours, the strip, evenness with and without the end steps
Step sizes, as the chosen measure sees them	Chart and statistics of the gaps between steps
Where the ramp sits in chroma	The ramp, tints, shades and tones on the OKLab a–b plane
Design-scale key sets	The ladder for the chosen key list

Contrast tab

Control	What it does	Values	Default
Against	The colour every step is measured against	White; Black; The base	White
Metric	Shown, but the table always lists both measures in the current version	WCAG 2.2; APCA	WCAG 2.2
Superseded APCA	Adds the former approximate figure and the difference	Show the difference; Hide	Hide

Readout	What it shows
Caption and note	"Every step against …" and "WCAG 2.2 ratio · APCA-W3 0.1.9 Lc · n steps"
Table	Step, WCAG, Rating (AAA, AA, AA large, Fail), APCA Lc, and optionally Superseded and Difference
The ladder	A chart of the steps' contrast

Data tab

Dataset	Status
sRGB transfer and primaries (IEC 61966-2-1:1999)	Verbatim
WCAG 2.2 contrast	Verbatim
OKLab matrices (Ottosson 2020)	Verbatim
OKLab lightness	Computed from linear sRGB
CIELAB (CIE 015:2018, D65)	Computed
Ramp engines (two distinct)	Computed
APCA Lc (APCA-W3 0.1.9)	Computed
Tailwind and Material palettes	Absent: key names only

Status chips filter the list. **Install a table** ("Drop a CSV or JSON") does not load files in the current version.

Export tab

Control	What it does
What to take	The ramp; The key set; The contrast table; A link to this view
As	CSS, SCSS, JSON, CSV or URL, as fits the choice
Download button	Names and saves the file; for a link, **Copy the link**
Copy to clipboard	Copies the payload
Link field	Read-only; always shows the current link
Analyse a list, **Analyse**	Table of Hex, L*, a*, b*, OKL, Y, Tints, Shades
Download the result	Copies the analysis as CSV to the clipboard
Exactly what you will get	Preview, size in bytes and a footer explaining the provenance lines

Link parameters: `hex`, `engine` (`hsl`, `luminance`, or `contrast`), `steps`, `gamma`, `mode` (`srgb`, `lab`, `oklab`, `hsl`).

Reference tab

Control	What it does
Standards, **Formulas**, **Citations**, **Research notes**	Switch sections
Filter	Searches all four

Standards include IEC 61966-2-1:1999, ITU-R BT.709-6, CIE 015:2018, WCAG 2.2, APCA-W3 0.1.9, the WCAG 3.0 working draft, the Design Tokens Format Module and Ottosson (2020). Formulas list 11 formulas as written, from the sRGB curve to the evenness measure. Citations list 14 references, tagged runs, context or names only, with corrected ones marked. Research notes describe six fixes. **Where these colours sit on the 1931 diagram** plots the ramp on the CIE 1931 chromaticity diagram.

Save to Library

The floating **Save to Library** button saves the colours shown on the page as a palette in your Auric Artisan Library. A palette or colour sent back from the Library to this tool sets the base colour.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Troubleshooting Tonal Steps

Fixes for common problems in Tonal Steps, Tints and Shades.

Symptom	Why it happens	What to do
My base colour is not in the ramp	The engines step lightness or luminance evenly from white to black; the base rarely lands exactly on a step. The base note says "not a step"	Expected. Use the key sets, where 500 is always your colour, or tints and shades, which start from it
Changing **Mix in** does not change the ramp	The mixing space applies to tints, shades, tones and key sets, not to the main ramp	Tick a set under **Also show** to see the effect
sRGB linear gives the same result as OKLab	In the current version the sRGB linear option mixes in OKLab	Use CIELAB or HSL for a visibly different mix

Symptom	Why it happens	What to do
Easing γ does nothing	Easing applies only to Equal luminance and the neutral ladder	Switch the engine to **Equal luminance**
The **Equal contrast** engine is missing	It produces the same ramp as Equal luminance, so it is not offered on the Lab tab	Use Equal luminance; the Ramps tab shows both
Pressing `3` shows a ramp that looks like Equal luminance	`3` selects the Equal contrast engine, which gives the same colours	Expected
The base colour field ignores my value	It needs a six-digit HEX with `#`	Type a value such as `#D3AF37`
The **Metric** choice on Contrast changes nothing	Both WCAG and APCA figures are always shown	Read the column you need
The contrast export ignores my **Against** choice	Exports list contrast against both white and black	Use the Contrast tab for the base colour
Download the result did not save a file	It copies the analysis as CSV to your clipboard	Paste it where you need it
"No valid hex colours found."	The list has no six-digit HEX values	Put one value per line, for example `2563eb`
Copying fails	The browser blocked clipboard access	Allow clipboard access, or select the preview and copy manually
`1`, `2`, `3` or `R` do nothing	The focus is in a field or slider	Click an empty part of the page first
Colours differ from the Tailwind or Material palettes	The key sets borrow the key names only	Expected; they are this tool's own ladders
OKLab L values differ from an older export	Earlier versions fed OKLab gamma-encoded values; OKLab now receives linear sRGB	Re-export; new files note `oklab_input=linear_srgb`
APCA values differ from an older export	APCA-W3 0.1.9 now runs as published	Turn on **Show the difference** to compare

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tonal Steps FAQ

Short answers to common questions about Tonal Steps, Tints and Shades.

Which engine should I use?

Use **HSL lightness** for a quick, familiar ramp. Use **Equal luminance** when you want steps tied to WCAG contrast, and adjust easing so the steps look even. The Ramps tab measures how even each one is.

Why are there two engines, not three?

Equal steps in WCAG contrast ratio are, mathematically, equal steps in relative luminance. The third engine produced the same colours as Equal luminance, so it is not offered as a separate choice.

Why is OKLab used to measure the steps?

OKLab lightness tracks perceived lightness well, so gaps in OKLab L show whether a ramp looks evenly spaced. You can switch to CIELAB L* or relative Y on the Ramps tab.

What is the difference between tints, shades and tones?

A tint adds white, a shade adds black, and a tone adds grey. All three start from your base colour.

Are the 50–950 keys the Tailwind colours?

No. They are this tool's ladder with Tailwind's key names: mixed towards white below 500 and towards black above. Tailwind's own colours are tuned by hand per hue.

Is APCA a requirement?

No. APCA is a working draft of what WCAG 3 may adopt. WCAG 2.2 ratios remain the published requirement. The tool shows both.

Why does the APCA sign change?

APCA Lc is signed. Positive and negative values mark which colour is lighter, because APCA treats dark text on light backgrounds differently from light text on dark.

Can I use more than 21 steps?

No. The range is 3 to 21.

Does anything leave my browser?

No. Ramps, contrast and exports are computed and created on your device.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tonal Steps: supporting pages

Glossary, shortcuts, limits, related tools, release notes and privacy for Tonal Steps.

These pages support the guides in docs/.

File	What it covers
glossary.md	Terms used in the tool
shortcuts.md	Keyboard shortcuts
limits-and-accuracy.md	What the figures are and are not
related.md	Related tools and Learn articles
release-notes.md	What has changed for users
privacy.md	What stays in your browser

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tonal Steps shortcuts

Keyboard shortcuts in Tonal Steps, Tints and Shades.

These keys work when the focus is not in a text box, slider or list.

Key	What it does
`1`	Ramp engine: HSL lightness
`2`	Ramp engine: Equal luminance
`3`	Ramp engine: Equal contrast (gives the same colours as Equal luminance)
`R`	Random base colour
`Esc`	Closes a full-screen chart

Mouse	What it does
Double-click a chart	Opens it full screen
Hover a swatch in The other sets	Shows its HEX, luminance Y and contrast against black

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tonal Steps glossary

Plain-language definitions of the terms used in Tonal Steps, Tints and Shades.

AA, AAA, AA large — WCAG 2.2 contrast ratings: AAA at 7:1 or more, AA at 4.5:1 or more, AA large at 3:1 or more (for large text). Below 3:1 the tool shows **Fail**.

APCA (Accessible Perceptual Contrast Algorithm) — a contrast method by Myndex; the tool runs version APCA-W3 0.1.9. Its result, **Lc**, is signed.

CIELAB (L*a*b*) — a CIE colour space with lightness L* and opponent axes a* and b*, computed here against the D65 white.

Easing γ — an exponent that bends where steps fall in the Equal luminance engine and the neutral ladder.

Evenness — how much the gaps between neighbouring steps stray from their average, as ±%. Smaller is more even.

HSL — hue, saturation and lightness: a simple rearrangement of RGB. HSL lightness is a coordinate, not a measure of perceived lightness.

Key set — a ladder of colours named by keys such as 50, 100 … 900, as used in design-system palettes.

LCh — CIELAB expressed as lightness, chroma and hue angle.

Linear sRGB — sRGB values with the transfer curve removed, proportional to light. OKLab and luminance need it as input.

Mixing space — the colour space in which two colours are interpolated. It changes the middle of a mix.

Neutral ladder — a scale of greys from black to white.

OKLab — a perceptual colour space (Ottosson, 2020). **OKLab L** is its lightness, from 0 (black) to 1 (white).

Ramp — a sequence of steps from light to dark built from one colour.

Relative luminance (Y) — the brightness of a colour on a 0 to 1 scale, computed from linear sRGB with the BT.709 weights 0.2126, 0.7152 and 0.0722.

Shade — a colour mixed with black.

Tint — a colour mixed with white.

Tone — a colour mixed with grey (here #808080).

WCAG contrast ratio — (L1 + 0.05) ÷ (L2 + 0.05) for the lighter and darker luminances, from 1:1 to 21:1.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tonal Steps limits and accuracy

What the Tonal Steps figures rest on, how precise they are, and what the tool does not claim.

Standard values

- **WCAG 2.2 ratios and ratings** use relative luminance from linear sRGB with the BT.709 weights, exactly as WCAG defines.
- **APCA Lc** runs APCA-W3 0.1.9 as published: a plain 2.4 power, separate exponent pairs for each polarity, the soft clamp below Y 0.022 and the ±0.027 low-clip offset. APCA itself is a working draft, not a ratified requirement.
- **OKLab** uses Ottosson's matrices on linear sRGB.
- **CIELAB** follows CIE 015:2018 against D65.

This tool's own

- **The ramp engines** are the tool's own constructions. Equal luminance searches HSL lightness at the base's hue and saturation for each target luminance; the first and last steps are exactly white and black.
- **The key sets** borrow Tailwind and Material key names only. Nothing here is a Tailwind or Material colour.
- **Evenness** is the tool's own measure: the largest deviation of a step size from the mean step size, as a percentage.

Precision

- All colours are 8-bit sRGB; HEX values are rounded to the nearest code value, so luminance and contrast figures reflect the rounded colour.

- OKLab L is shown to three or four decimals, WCAG ratios to two, APCA Lc to one.

Current-version caveats

- **sRGB linear** mixing produces the same result as OKLab.
- The **Metric** choice on the Contrast tab does not change the table.
- The contrast table export lists contrast against white and black only, not the base.
- **Install a table** on the Data tab does not load files.

Not a substitute for

Testing real text at real sizes on real screens. Contrast figures describe colour pairs, not legibility of a particular font, weight or size.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tonal Steps privacy

What Tonal Steps does with your colours and settings.

- **Everything runs in your browser.** Ramps, sets, contrast figures and list analyses are computed on your device.
- **Exports are created locally** and saved through your browser's download, or copied to your clipboard when you choose.
- **The link** carries the base colour, engine, step count, gamma and mixing space. Anyone with the link sees them.
- **Colours you paste** into Analyse a list stay in the page.
- **Save to Library** stores a palette in your Auric Artisan Library.
- The page loads the site's shared features like every other page on auricartisan.com; see the site's privacy policy.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tonal Steps release notes

User-facing changes to Tonal Steps, Tints and Shades.

Current version (2026 workbench rebuild)

- Tabs are now **Lab**, **Ramps**, **Contrast**, **Data**, **Export** and **Reference**.
- OKLab now receives linear sRGB, as its definition requires. OKLab lightness values, the lightness chart and the OKLab mixing mode changed for every tone except black, white and the primaries.
- The Lab offers two ramp engines instead of three: the equal-contrast engine produced the same colours as equal luminance (and ran in reverse). The Ramps tab shows all three side by side.
- The equal-luminance engine's end steps are exactly white and black.
- The APCA column now runs APCA-W3 0.1.9 as published. The Contrast tab can show the former approximate figure and the difference.
- The "Tailwind 50–950" and "Material 50–900" scales are now presented as key sets that borrow the names only.
- New evenness measures in OKLab L, CIELAB L* and relative luminance.
- Exports now carry a header recording the base, engine, mixing space, steps and gamma, and a footer recording the OKLab input, WCAG and APCA versions.
- The share link control works and the link field always shows the current link.

May 2026

The Tonal Steps, Tints and Shades User Guide was published in the site's Documentation library, describing the earlier version.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Related to Tonal Steps

Tools and Learn articles that go well with Tonal Steps, Tints and Shades.

Tools

Tool	Why
Contrast Checker	Check one text and background pair in depth
M-Gradient Palette	Multi-stop gradients and ramps across colour spaces
Design Engine	Generate a whole design system with accessible OKLCH ramps
Basic Colour Tools	Quick palettes, gradients and conversions
Gamut Mapping	See whether ramp colours fit a target space

Learn articles

- The Vocabulary of Color: Hue, Saturation, Value, Tint, Tone, and Shade — https://auricartisan.com/library/learn/articles/2026-06-29-vocabulary-of-color-hue-saturation-value
- Oklab and Oklch: Modern Perceptual Color Spaces — https://auricartisan.com/library/learn/articles/2026-06-02-oklab-oklch-perceptual-color-spaces
- Accessible Color Design and WCAG Contrast — https://auricartisan.com/library/learn/articles/2026-05-25-wcag-contrast-accessible-color
- CIELAB and LCH Explained — https://auricartisan.com/library/learn/articles/2026-05-24-cielab-and-lch-explained
- Black, White, and Gray: The Achromatic Colors — https://auricartisan.com/library/learn/articles/2026-07-15-black-white-and-gray-achromatic-colors

Documentation on the site

The site's Documentation library (https://auricartisan.com/library/documentation/) has an earlier **Tonal Steps, Tints and Shades User Guide**, which describes three engines. This documentation follows the current page.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tone Mapping

What the Tone Mapping lab does, who it is for, where to find it and how to map your first frame.

A real scene can hold far more light than any display can show: a sunlit window may be thousands of times brighter than the shadows in the room. **Tone mapping** is the curve that squeezes that range into the display's range while deciding what to keep — shadow detail, mid-tones, or colour in the highlights.

The Tone Mapping lab runs a frame through an eleven-step pipeline you control: decode the source, convert it to a working space with white-point adaptation, set exposure, apply an optional low-level operator, apply the tone curve, grade it, convert to the target display space with gamut mapping, and encode it for output. It offers seven working tone curves — clipping, Reinhard, Uchimura (Gran Turismo), Hable filmic, the Narkowicz fit of the ACES look, a soft log curve and your own three-point curve — and shows two retired operators on the Operators tab for comparison.

You can work on built-in test scenes or your own image, read the result on a histogram, waveform, vectorscope and false-colour map, compare two versions side by side, and export the tone curve as a 1D `.cube` or CSV, or the whole pipeline as a 3D `.cube` LUT.

Who it is for

- Game, 3D and VFX artists choosing a tone curve for renders.
- Colourists and video editors who want a LUT that matches a preview.
- Developers implementing tone mapping in a renderer.
- Students learning scene-referred and display-referred imaging.

Where to find it

- Address: https://auricartisan.com/tool/general/gamut-and-rendering/tone-mapping/
- In the tool collection (https://auricartisan.com/collections/) under **Gamut & Rendering**, as **Tone Mapping**.

Quick start

1. Open https://auricartisan.com/tool/general/gamut-and-rendering/tone-mapping/. The HDR test scene is mapped with the Hable filmic curve.
2. Move **Exposure** up and down to see which part of the curve the scene lands on.
3. Under **Tone curve**, try **Reinhard**, **Narkowicz fit** and **None (clip)**. Watch the highlights.
4. Read the statistics: where mid-grey 0.18 lands, and the input at which the scene reaches display white.
5. Open **Scopes** to read the histogram, waveform and vectorscope.
6. Open **Operators** to compare every curve on one set of inputs.
7. Open **Export**, choose **The tone curve** as **.cube**, and select the download button.

What you can do

- Tone-map three built-in test scenes or your own image.
- Declare the source's encoding (linear, sRGB, PQ, HLG, gamma 2.2 or 2.4), primaries and white.
- Set exposure in stops, choose one of seven tone curves or shape your own.
- Apply a low-level operator and a grade (contrast, saturation, lift, gamma, gain, temperature, tint, hue, vibrance).
- Choose working and target spaces, target white, gamut mapping and output encoding; mark out-of-gamut pixels; dither.

- Compare operators numerically and on one chart.
- Read histogram, waveform, vectorscope and a false-colour exposure map; store two frames and split the view.
- Export a 1D `.cube` or CSV of the curve, a 3D `.cube` of the pipeline, or a link.
- Map a list of HEX colours through the pipeline.

In this folder

- docs/ — how to use the lab
  - Getting started
  - Map a frame
  - Compare operators
  - Read the scopes
  - Export a LUT
  - Reference
  - Troubleshooting
  - FAQ
- others/ — background and supporting pages
  - Glossary
  - Shortcuts
  - Limits and accuracy
  - Related tools
  - Release notes
  - Privacy

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tone Mapping documentation

The guides, reference and help pages for the Tone Mapping lab.

This folder explains how to use the Tone Mapping lab: a screen tour, task guides for mapping, comparing, measuring and exporting, a complete reference and help pages.

File	What it covers
getting-started.md	Opening the lab, the six tabs and the pipeline
map-a-frame.md	Source, exposure, curve, grade and output for one frame
compare-operators.md	The Operators tab and what each curve does
read-the-scopes.md	Histogram, waveform, vectorscope, false colour and A/B
export-a-lut.md	1D and 3D `.cube`, CSV, links and colour lists
reference.md	Every control, readout and export
troubleshooting.md	Symptoms, causes and fixes
faq.md	Short answers

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Getting started with Tone Mapping

Open the Tone Mapping lab, learn its tabs and the eleven-step pipeline, and read your first mapped frame.

Open the lab

Go to https://auricartisan.com/tool/general/gamut-and-rendering/tone-mapping/. The heading is **Tone Mapping**. All processing happens in your browser.

Key ideas

- **Scene-referred** values describe light in the scene. They start at 0 and can go far above 1; 0.18 is the conventional **mid-grey**, and 1.0 is often treated as diffuse white.
- **Display-referred** values run from 0 to 1, where 1 is the brightest the display shows.
- A **tone curve** (tone-mapping operator) maps scene values to display values. A good curve keeps mid-grey where you expect it and rolls highlights off gently instead of clipping them.
- **Exposure** is measured in **stops (EV)**. One stop doubles or halves the light, and it is applied before the curve, so it changes which part of the curve the scene lands on.

A tour of the screen

A note at the top warns that two of the nine operators the page once offered were one-line stand-ins; see the Data tab.

Tab	What it is for
Lab	The pipeline controls, the mapped frame and the curve
Operators	All nine operators on one table and one chart

Tab	What it is for
Scopes	Histogram, waveform, vectorscope, false colour, A/B split and timing
Data	The eight datasets and transforms the lab rests on, with their status
Export	Curve and pipeline LUTs, CSV, a link, and HEX list mapping
Reference	Standards, formulas, citations and research notes with a filter

The pipeline

Every pixel goes through these steps in order:

1. Linearise (decode the source encoding)
2. Source → working, with white-point adaptation
3. Exposure
4. Low-level operator
5. Tone curve
6. Contrast, saturation, lift / gamma / gain
7. Temperature and tint
8. Hue and vibrance
9. Working → target, gamut mapping
10. Output OETF (encode for the display)
11. Dither

The Export tab refers to these step numbers.

The Lab tab

The left panel groups the controls: source (**Source**, encoding, primaries, white), **Exposure**, **Tone curve** (with a custom curve option), **Low-level operator**, **Grade**, **Working → target** (working space, target primaries and white, gamut mapping, output OETF, out-of-gamut mark, dither), **Display peak**, **Source peak** and **Presets**.

The right side shows **The mapped frame** with its resolution and status chips, then **The curve, and where the scene sits on it** (the tone curve drawn over a histogram of the scene), statistics and a note.

Read your first frame

1. The lab opens on **HDR test scene**, **Linear (scene)** encoding, Rec.709 primaries, D65, the **Hable filmic** curve and the **ACEScg AP1 (D60)** working space.
2. Look at the chips over the frame: the working and target spaces, the curve and the processing time; **Scene-referred** (because the source is linear) and **Photographic**; and **out of gamut** if any pixel left the target gamut.
3. Read the statistics:
   - **Mid-grey, curve** — what 0.18 becomes;
   - **Scene white at** — the input at which the curve reaches display white ("never" if it only approaches it);
   - **Working luma** — the working space's luminance weights;
   - **Display peak** — in nits.
4. Move **Exposure** to +2.0 EV. The frame brightens, and the histogram behind the curve shifts to the right.
5. Choose **None (clip)** under **Tone curve**. Highlights now clip hard at white.

Next steps

- Map a frame
- Compare operators
- Read the scopes
- Export a LUT

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Compare operators

Compare all nine operators by what they return for chosen inputs, where they reach white, and how their curves look.

The nine operators

Operator	Source	Formula or shape	Offered in the Lab
None (clip)	—	Clamps to 0–1	Yes
Reinhard	Reinhard et al. 2002	L ÷ (1 + L); approaches 1 but never reaches it	Yes
Uchimura / Gran Turismo	Uchimura 2017	A toe, a linear section and a shoulder blended together	Yes
Hable filmic	Hable 2010 (Uncharted 2)	A filmic curve with six constants, exposure bias 2.0, white point 11.2	Yes
Narkowicz fit	Narkowicz 2016	(x(2.51x + 0.03)) ÷ (x(2.43x + 0.59) + 0.14), clamped. Approximates the look of ACES; it is not the Academy's transform	Yes
Log (soft)	This tool	log2(1 + v) ÷ log2(11)	Yes
Custom 1D curve	This tool	Knee, mid and shoulder	Yes
Gamma-domain scale	This tool; formerly labelled "BT.2446 Method A"	Y × k^2.4, a constant multiplication (about 0.004 at the default peaks) that turns mid-grey nearly black	No
Scaled Reinhard	This tool; formerly labelled "BT.2446 Method C"	reinhard(Y × r) ÷ r, which can never exceed 1 ÷ r (0.1 at the default ratio of 10)	No

The last two are not tone curves and do not implement ITU-R BT.2446. They are shown so that old links still open and so you can see what they did.

Steps

1. Set the **Display peak** and **Source peak** on the Lab tab. They only affect the two retired operators.
2. Open the **Operators** tab.
3. Under **Compare**, choose **All nine** or **The seven that work**.
4. Under **Read out at**, choose the inputs:
   - **Mid-grey 0.18** — 0.18, 1, 4 and 12;
   - **Scene white** — 1, 4, 12 and 32;
   - **Both** — 0.18, 1, 2, 4, 8 and 16.
5. Read the table:

Column	Meaning
Operator	The name, and its source (or "was: …" for a retired operator)
One column per input	The output value; values above 1 are highlighted
White at	The input where the curve reaches display white, or "never"
Output	A grey strip of the operator's output across the domain, as a display would show it
Tag	**runs**, **exceeds 1** (the curve passes 1.0, so it clips at the end of the pipeline) or **not a curve**

6. Under **Curve domain**, choose **0 – 1** or **0 – 12** (default) for the chart. Scene values run past 1, so the wider domain shows how each curve treats highlights.
7. Read the chart: every operator on one pair of axes, with a dashed line at display white and a line at mid-grey 0.18. Retired operators are dashed and labelled in red. Select **Redraw the curves** if the chart looks out of date.

Reading the result

- Mid-grey is where a tone map is judged. An operator that moves 0.18 a long way has changed the picture's exposure, not just its highlights.
- A curve that reaches white at a modest input (Narkowicz fit, None) clips everything above it. A curve that never reaches white (Reinhard) keeps compressing highlights but never shows pure white.
- The table's notes under the chart explain what the two retired operators computed and what the real BT.2446 methods involve.

Result

You can see, for the same inputs, how each curve treats mid-grey, white and highlights, and which ones are not really tone curves.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Export a LUT

Export the tone curve as a 1D .cube or CSV, the whole pipeline as a 3D .cube, a link to the view, and map a list of colours.

A **LUT (lookup table)** stores a colour transform as a table so another program — a grading application, a game engine, a video tool — can apply it. A **1D LUT** maps each channel on its own; a **3D LUT** maps every RGB combination and can carry cross-channel steps such as saturation, hue and gamut conversion. The `.cube` format is the Adobe Cube LUT specification.

Steps

1. Set up the frame on the **Lab** tab.
2. Open **Export**.
3. Under **What to take**, choose:
   - **The tone curve** — step 5 of the pipeline only;
   - **The whole pipeline** — all eleven steps;
   - **A link to this view**.
4. Under **As**, choose a format: **.cube** or **CSV** for the curve; **.cube** for the pipeline; **URL** for the link.
5. For the tone curve, choose the **Input domain**: **0 – 1** (default) or **0 – 12**. It is written into the file as DOMAIN_MIN and DOMAIN_MAX (or, in the CSV, as the input range).

6. Check **Exactly what you will get**: a preview (long files show the start and end), the size in bytes and the number of lines.
7. Read **What this file carries, of the eleven steps the viewport ran**: a tick or cross for each step. A curve export carries only step 5.
8. Select the download button (it names the file) or **Copy to clipboard**. For a link, the button reads **Copy the link**.

The files

Choice	File	Contents
Tone curve, .cube	`tone-<operator>-curve.cube`	`TITLE "Auric Artisan tone curve"`, comments naming the operator, spaces and exposure and listing the steps not included, `LUT_1D_SIZE 33`, `DOMAIN_MIN 0.0 0.0 0.0`, `DOMAIN_MAX` 1.0 or 12.0
Tone curve, CSV	`tone-<operator>-curve.csv`	A comment line, `input,output`, then 1,024 rows across the domain
Whole pipeline, .cube	`tone-<operator>-pipeline.cube`	`TITLE "Auric Artisan tone map"`, comments, `LUT_3D_SIZE 17`, domain 0 to 1, then 4,913 entries each run through all eleven steps

`<operator>` is the operator's short name, for example `hable`, `reinhard`, `aces` (the Narkowicz fit), `uchimura`, `log`, `none` or `custom`.

Use the pipeline cube when you want the file to reproduce the preview: it includes the colour-space conversion, white-point adaptation, gamut mapping and grade. Use the curve when you only want the tone shape.

The link

The page keeps its address up to date as you work. **A link to this view** copies that address, which records the operator, exposure, source encoding, display peak, output encoding and target primaries. Other settings (the grade, working space, custom curve and source peak) are not in the link.

Map a list of colours

1. Under **Analyse a list**, enter six-digit HEX colours, one per line or separated by commas. (The box's example shows numbers, but only HEX colours are read.)
2. Select **Analyse**. Each colour is treated as a source pixel, decoded with the source encoding you set, and run through the whole pipeline. The table lists **#**, **Source**, **Mapped** and **OOG** (a warning sign if the colour left the target gamut, a tick if not).
3. Select **Download the result** to save `tone-map-batch.csv` with Index, Source, Mapped and OOG. Run the analysis first, or a message asks you to.

If no HEX values are found, the panel says "No valid hex colours found."

Result

You have a `.cube` or CSV file that another application can load, clearly labelled with which pipeline steps it contains.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Map a frame

Choose a source, set exposure and a tone curve, grade the result and set the output for your display.

1. Choose the source

1. Under **Source**, choose **HDR test scene**, **Exposure gradient**, **ColorChecker patches** or **Uploaded image**.
2. For **Uploaded image**, select **Choose a file** and pick an image. The file name replaces "no file" and a message confirms the load.
3. Tell the lab how the source was written:
   - encoding: **Linear (scene)**, **sRGB**, **PQ (ST 2084)**, **HLG (inverse OETF)**, **Gamma 2.2** or **Gamma 2.4**;
   - primaries: **Rec.709**, **Display P3**, **Rec.2020** or **ACEScg AP1**;
   - white point: **D65**, **D60** or **D50**.

An ordinary PNG or JPEG is 8-bit sRGB: choose **sRGB** encoding and **Rec.709** primaries for it. Its values stop at 1.0, so there are no true highlights above white for the curve to shape; use Exposure to push it into the curve's shoulder. The built-in scenes are generated with values above 1.

The chip over the frame says **Scene-referred** when the encoding is linear and **Display-referred** otherwise.

2. Set exposure

Move **Exposure** from −6 to +6 EV (steps of 0.1). Each stop doubles or halves the values before the curve. Watch the histogram behind the curve slide along it.

3. Choose a tone curve

Select one of the seven curves under **Tone curve**: **None (clip)**, **Reinhard**, **Uchimura / Gran Turismo**, **Hable filmic** (default), **Narkowicz fit**, **Log (soft)** or **Custom 1D curve**. See Compare operators for what each does.

To shape your own curve, tick **Use the custom curve instead** and set the three fields, in order **knee**, **mid** and **shoulder** (each 0 to 1; defaults 0.15, 0.5, 0.85). The curve is three straight segments: from 0 up to (knee, mid × knee), then to (shoulder, mid), then to (1, 1). Inputs above 1 give 1.

The three buttons under the curve chart — **Linear axis**, **Log axis** and **Film stops** — fill the custom curve's fields with presets (0.33, 0.5, 0.66; 0.05, 0.35, 0.7; and 0.15, 0.5, 0.85). They do not change the chart's axis, and they only affect the frame when **Use the custom curve instead** is ticked.

4. Optional low-level operator

Low-level operator applies a simple function to each channel before the tone curve, using parameters **a** and **b**:

Choice	Function
None	No change (default)
Add	v + a ÷ 10,000
Multiply	v × a
Soft clip	v ÷ (v + a)
Power	v^a
Sigmoid	1 ÷ (1 + e^(−(v − b) × a × 10))
Exp	1 − e^(−v × a)
Log	log2(1 + v × a) ÷ log2(1 + a)

Choice	Function
Tanh	tanh(v × a)

Both parameters start at 0; most functions need a positive **a** to do anything useful.

5. Grade

The **Grade** row holds nine number fields, applied after the curve in the working space, in this order:

Field	Range	Default	Effect
Contrast	0 to 3	1	Expands or compresses values around 0.5
Saturation	0 to 3	1	Mixes towards or away from the working space's own luminance
Lift	—	0	Adds to every channel
Gamma	0.1 to 4	1	Raises each channel to 1 ÷ gamma
Gain	0 to 4	1	Multiplies every channel
Temperature	—	0	Positive warms (more red, less blue)
Tint	—	0	Positive adds green
Hue rotation	−180 to 180°	0	Rotates hue around the luminance axis
Vibrance	0 to 3	1	Changes saturation more for muted colours than for vivid ones

6. Output

1. Choose the **Working space**: **ACEScg AP1 (D60)** (default), **Rec.709 (D65)**, **Display P3 (D65)** or **Rec.2020 (D65)**.
2. Choose the target primaries (**sRGB / Rec.709**, **Display P3**, **Rec.2020**) and target white (**D65**, **D60**, **D50**). Different white points are bridged with the Bradford transform.
3. Choose the gamut mapping: **Soft compress**, **Clip** or **Preserve hue** (scales the colour down by its largest channel, keeping channel ratios). In the current version Soft compress gives the same result as Clip.
4. Choose the output encoding: **sRGB**, **Gamma 2.2**, **Gamma 2.4**, **PQ** or **Linear**. With PQ, a display value of 1.0 is encoded as 10,000 cd/m².
5. Tick **Mark out-of-gamut pixels** to paint pixels that left the target gamut in magenta.
6. Tick **Dither the 8-bit output** to add a tiny amount of noise that hides banding.

Presets

Preset	Curve	Target	Display peak	Output	Source encoding
SDR 100	Hable filmic	sRGB / Rec.709	100 nits	sRGB	sRGB
P3 600	Hable filmic	Display P3	600 nits	sRGB	sRGB
HDR 1000	Narkowicz fit	Rec.2020	1,000 nits	PQ	Linear (scene)

Reset reloads the lab with its default settings.

Result

You have a frame mapped with the curve, grade and output you chose, with the chips and statistics telling you how it was made.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Read the scopes

Use the histogram, waveform, vectorscope, false-colour map and A/B split to judge a mapped frame.

The **Scopes** tab measures the mapped output — the end the question is about. The note at the top names the working and target spaces and the operator.

The three scopes

Scope	What it shows	How to read it
Histogram — output, 256 bins	How many pixels sit at each output level	A pile-up at the right edge means clipped highlights; at the left, crushed shadows
Waveform — output luma by column	Brightness of each column of the image, left to right	The dashed line is display white. A trace pressed flat against it is clipping
Vectorscope — output chroma	Colour direction and strength, after gamut mapping	Distance from the centre is saturation; angle is hue. It shows only what the target can carry

The false-colour exposure map

1. Tick **Show the exposure map** under **False colour**. The frame on the Lab tab is replaced by bands of colour.
2. Read the key under **False colour, and the peak it is counted against**. Pixels are coloured by estimated luminance:

Up to (nits)	Colour
0.01	Black
0.1	Deep violet
1	Blue
10	Teal
100	Green
203	Yellow (203 nits is the HDR diffuse white of ITU-R BT.2408)
1,000	Orange
4,000	Red
10,000	Magenta
Above	White

3. The note reports the source's brightest pixel, using the **Source peak**, and greys out bands the source cannot reach.

The map estimates luminance from the source pixel's luma multiplied by a peak. In the current version the map itself uses the **Display peak** for that multiplication while the note uses the **Source peak**, so they can disagree. Untick the box to see the frame again.

Compare two versions (A / B)

1. Set up the first version and select **Store A**.
2. Change a setting and select **Store B**.
3. Tick **Split the viewport**. The frame shows A on the left and B on the right.
4. Move the split slider (0 to 1; default 0.5) to move the dividing line.

Both versions must be stored, and the image must stay the same size, for the split to show.

Timing

- **Time a pass** runs the whole image through the pipeline 20 times and reports the average in milliseconds per frame.
- **Time the curve** does the same 5 times.

The chip over the frame also shows the time the last update took.

Controls that were removed

Controls that were read and never used lists three controls from an earlier version — Scene reference white, Highlight stress, and Zebra low / high — each tagged "did nothing", with an explanation. They are not on the page any more.

Result

You can tell whether highlights or shadows are clipping, how saturated the output is, where the scene's luminance falls, and how two versions differ.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tone Mapping reference

Every control, readout, option and export in the Tone Mapping lab.

Address: https://auricartisan.com/tool/general/gamut-and-rendering/tone-mapping/

Lab tab

Source

Control	What it does	Values	Default
Source	The frame to map	HDR test scene; Exposure gradient; ColorChecker patches; Uploaded image	HDR test scene
Choose a file	Picks an image (shown for Uploaded image)	Image files	"no file"
Source encoding	How the source values are decoded (step 1)	Linear (scene); sRGB; PQ (ST 2084); HLG (inverse OETF); Gamma 2.2; Gamma 2.4	Linear (scene)
Source primaries	The source's RGB space	Rec.709; Display P3; Rec.2020; ACEScg AP1	Rec.709
Source white	The source's white point	D65; D60; D50	D65

Exposure and curve

Control	What it does	Values	Default
Exposure	Multiplies by 2^EV before the curve (step 3)	−6 to +6 EV, step 0.1	0.0 EV
Tone curve	The operator (step 5)	None (clip); Reinhard; Uchimura / Gran Turismo; Hable filmic; Narkowicz fit; Log (soft); Custom 1D curve	Hable filmic
Use the custom curve instead	Uses the knee / mid / shoulder curve whatever is selected	On or off	Off
Custom curve knee, mid, shoulder	The three points of the custom curve	0 to 1 each	0.15, 0.5, 0.85
Low-level operator	A per-channel function before the curve (step 4)	None; Add; Multiply; Soft clip; Power; Sigmoid; Exp; Log; Tanh	None
Parameters a, b	Values for the low-level operator	Numbers, step 0.1	0, 0

Grade (steps 6–8)

Field	Range	Default
Contrast	0 to 3	1
Saturation	0 to 3	1
Lift	Any	0
Gamma	0.1 to 4	1
Gain	0 to 4	1
Temperature	Any	0
Tint	Any	0
Hue rotation	−180 to 180°	0
Vibrance	0 to 3	1

The fields appear as one row in this order. See Map a frame for what each does.

Working → target (steps 2, 9–11)

Control	What it does	Values	Default
Working space	The space the curve and grade run in	ACEScg AP1 (D60); Rec.709 (D65); Display P3 (D65); Rec.2020 (D65)	ACEScg AP1 (D60)
Target primaries	The display's RGB space	sRGB / Rec.709; Display P3; Rec.2020	sRGB / Rec.709
Target white	The display's white; Bradford adapts between whites	D65; D60; D50	D65
Gamut mapping	What happens to out-of-gamut colours	Soft compress; Clip; Preserve hue	Soft compress
Output OETF	How the result is encoded	sRGB; Gamma 2.2; Gamma 2.4; PQ; Linear	sRGB
Mark out-of-gamut pixels	Paints pixels that left the target gamut in magenta	On or off	Off
Dither the 8-bit output	Adds noise of up to half a code value	On or off	Off

Control	What it does	Values	Default
Display peak	Display peak luminance	80 to 4,000 nits, step 10	100
Source peak	Source peak luminance	100 to 10,000 nits, step 50	1,000

The two peaks change only the two retired operators, the false-colour map (display peak) and the false-colour note (source peak). The seven working curves do not depend on them.

Presets

Button	Sets
SDR 100	Hable filmic, sRGB / Rec.709 target, 100 nits, sRGB output, sRGB source
P3 600	Hable filmic, Display P3 target, 600 nits, sRGB output, sRGB source
HDR 1000	Narkowicz fit, Rec.2020 target, 1,000 nits, PQ output, linear source
Reset	Reloads the lab with defaults

Readouts

Readout	What it shows
The mapped frame	The result, with its resolution
Frame chips	Working → target, curve and processing time; Scene-referred or Display-referred; Photographic or Display; "out of gamut" when any pixel left the target
The curve, and where the scene sits on it	The tone curve over a histogram of the scene
Linear axis, **Log axis**, **Film stops**	Fill the custom curve with presets (0.33 / 0.5 / 0.66; 0.05 / 0.35 / 0.7; 0.15 / 0.5 / 0.85)
Statistics	Mid-grey, curve; Scene white at; Working luma; Display peak
Behaviour label	A short description of the curve, such as Photographic or Filmic shoulder
Note	What mid-grey becomes, where white is reached, or why a curve clips

Operators tab

Control	What it does	Values	Default
Compare	Rows shown	All nine; The seven that work	All nine
Read out at	Input columns	Mid-grey 0.18 (0.18, 1, 4, 12); Scene white (1, 4, 12, 32); Both (0.18, 1, 2, 4, 8, 16)	Mid-grey 0.18
Curve domain	Chart range	0 – 1; 0 – 12	0 – 12
Redraw the curves	Redraws the chart	Button	—

Table columns: Operator, one column per input, White at, Output strip, tag (runs, exceeds 1, not a curve).

Scopes tab

Control	What it does	Values	Default
Show the exposure map	Replaces the frame with false colour	On or off	Off

Control	What it does	Values	Default
Store A, **Store B**	Stores the current frame	Buttons	—
Split the viewport	Shows A left and B right	On or off	Off
Split position	Where the split falls	0 to 1, step 0.01	0.5
Time a pass	Average of 20 full passes, in ms per frame	Button	—
Time the curve	Average of 5 full passes	Button	—

Readouts: **Histogram — output, 256 bins**; **Waveform — output luma by column**; **Vectorscope — output chroma**; the false-colour key and note; **Controls that were read and never used**.

Data tab

Dataset	Status
sRGB, PQ and HLG transfer	Verbatim
Primaries and white points, 4 spaces	Verbatim
Bradford adaptation	Verbatim
Tone-mapping operators	Computed: seven of nine do what they are named for
Working-space luminance	Computed from each space's own matrix
Hue rotation	Computed
BT.2446 Method A and Method C	Absent: the names were held, the methods were not
Absolute scene luminance	Absent: the generated scenes carry none

Status chips filter the list. **Install a table** ("Drop a CSV or JSON") does not load files in the current version.

Export tab

Control	What it does
What to take	The tone curve; The whole pipeline; A link to this view
As	.cube or CSV (curve); .cube (pipeline); URL (link)
Input domain	0 – 1 or 0 – 12 for the curve export (default 0 – 1)
Download button	Names and saves the file; **Copy the link** for links
Copy to clipboard	Copies the payload
Exactly what you will get	Preview, bytes and lines
What this file carries	Which of the eleven steps the file includes
Analyse a list, **Analyse**, **Download the result**	Maps HEX colours; saves `tone-map-batch.csv`

Link parameters: `op`, `exp`, `src` (source encoding), `peak` (display peak), `oetf`, `tp` (target primaries).

Reference tab

Control	What it does
Standards, **Formulas**, **Citations**, **Research notes**	Switch sections
Filter	Searches all four

Standards include IEC 61966-2-1, SMPTE ST 2084, ARIB STD-B67 / BT.2100, BT.709, BT.2020, Academy S-2014-004 (ACEScg), BT.2446-1 (not implemented), BT.2408 and the Adobe Cube LUT Specification 1.0. Formulas give each operator, the transfer curves, working luminance, hue rotation, Bradford adaptation and the out-of-gamut test.

Save to Library

The floating **Save to Library** button stores an image of the mapped frame in your Auric Artisan Library.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Troubleshooting Tone Mapping

Fixes for common problems in the Tone Mapping lab.

Symptom	Why it happens	What to do
My photo looks flat or unchanged by the curve	An 8-bit PNG or JPEG stops at 1.0, so there are no highlights above white for the curve to compress	Raise **Exposure**, or use the built-in HDR test scene to compare curves
My photo looks too dark or washed out	The source encoding does not match the file	For ordinary images choose **sRGB** encoding and **Rec.709** primaries
Changing **Display peak** or **Source peak** changes nothing	The seven working curves do not use the peaks	Expected. The peaks affect the retired operators and the false-colour map
Soft compress looks exactly like **Clip**	In the current version the two give the same result	Use **Preserve hue** to keep channel ratios in out-of-gamut colours
The HDR 1000 preset looks pale and washed out on my screen	PQ output is an HDR encoding (a display value of 1.0 is encoded as 10,000 nits), and an ordinary screen shows PQ code values as if they were sRGB	Use sRGB output for viewing on a standard display; use PQ only when exporting for HDR
Linear axis, **Log axis** or **Film stops** does not change the chart	They set the custom curve's knee, mid and shoulder	Tick **Use the custom curve instead** to see their effect
The custom curve fields do nothing	The custom curve is used only when its box is ticked or **Custom 1D curve** is selected	Tick **Use the custom curve instead**
The low-level operator does nothing	Parameters a and b start at 0	Enter a positive **a**
Gamma-domain scale or Scaled Reinhard are not in the list	They are not tone curves and are no longer offered	See them on the **Operators** tab
The false-colour note and the map disagree	The note uses the Source peak; the map uses the Display peak	Set both peaks to the same value when reading the map
The split view shows only one version	Both A and B must be stored	Select **Store A** and **Store B**, then tick **Split the viewport**
Analyse says "No valid hex colours found."	Only six-digit HEX colours are read, although the box's example shows numbers	Enter values such as `#FF8800`

Symptom	Why it happens	What to do
Download the result says "Run batch first"	No analysis has been run	Select **Analyse** first
The link does not restore my grade	The link records operator, exposure, source encoding, display peak, output and target primaries only	Re-enter the grade, or export the pipeline `.cube`
A LUT looks different from the preview in another program	A curve export carries only step 5	Export **The whole pipeline** as `.cube`
Copy fails	The browser blocked clipboard access	The text is in the preview; select and copy it
Reset cleared everything	It reloads the lab with defaults	Use a stored link to return to a setup

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tone Mapping FAQ

Short answers to common questions about the Tone Mapping lab.

Which curve should I use?

For a photographic look with gentle highlight roll-off, Hable filmic or the Narkowicz fit are common choices. Reinhard is simple and never clips, but never reaches pure white. Compare them on the Operators tab at mid-grey and scene white before deciding.

Is "Narkowicz fit" the ACES tone mapper?

No. It is Krzysztof Narkowicz's 2016 curve fit that approximates the look of the ACES output transforms. The Academy's own transforms are not implemented.

What happened to the BT.2446 operators?

The two operators once labelled BT.2446 Method A and Method C did not implement ITU-R BT.2446. One multiplied by a small constant and turned the frame nearly black; the other could never exceed a 10% linear grey. They are renamed Gamma-domain scale and Scaled Reinhard, shown on the Operators tab, and no longer offered in the Lab.

What is the difference between scene-referred and display-referred?

Scene-referred values describe light in the scene and can exceed 1. Display-referred values are 0 to 1 of what the display can show. The lab treats a linear source as scene-referred and anything encoded (sRGB, PQ, HLG, gamma) as display-referred.

Why ACEScg as the default working space?

ACEScg is a wide, linear working space designed for rendering and compositing. Grading in it and converting to the display at the end is common practice. You can choose Rec.709, Display P3 or Rec.2020 instead.

Which export reproduces the preview?

The whole-pipeline 3D `.cube`. The curve export holds only the tone curve.

Can I load an EXR or HDR file?

The lab reads images your browser can decode (such as PNG and JPEG), which are 8-bit. It does not read OpenEXR or Radiance HDR files.

Can I save and reload my settings as a file?

Not in the current version. Use the link, or the floating **Save to Library** button, which stores an image of the frame.

Is my image uploaded?

No. It is decoded and processed in your browser.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tone Mapping: supporting pages

Glossary, shortcuts, limits, related tools, release notes and privacy for the Tone Mapping lab.

These pages support the guides in docs/.

File	What it covers
glossary.md	Terms used in the lab
shortcuts.md	Keyboard and mouse shortcuts
limits-and-accuracy.md	What the results are and are not
related.md	Related tools and Learn articles
release-notes.md	What has changed for users
privacy.md	What stays in your browser

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tone Mapping shortcuts

Keyboard and mouse shortcuts in the Tone Mapping lab.

The lab has no single-key shortcuts of its own.

Shortcut	Where	What it does
Double-click	A chart or scope	Opens it full screen
`Esc`	Full-screen view	Closes it

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tone Mapping glossary

Plain-language definitions of the terms used in the Tone Mapping lab.

ACEScg (AP1) — a linear, wide-gamut working space from the Academy Color Encoding System, with a D60 white.

Bradford transform — a chromatic adaptation method used to move colours between white points.

.cube — the Adobe Cube LUT file format, used by many grading and video applications.

Display-referred — values relative to what a display can show, from 0 (black) to 1 (peak white).

Dither — tiny random noise added before quantising to 8 bits, to hide banding.

EV (exposure value, stop) — a factor of two in light.

False colour — an overlay that paints each pixel by its brightness band instead of its colour.

Gamut mapping — handling colours outside the target space: clipping, compressing or scaling them.

Grade — creative adjustments after the curve: contrast, saturation, lift, gamma, gain, temperature, tint, hue and vibrance.

Hable filmic — a filmic tone curve published by John Hable (2010).

HLG — hybrid log-gamma, an HDR transfer function (ARIB STD-B67).

Histogram — a count of pixels at each level.

Lift, gamma, gain — adjustments that mainly move the shadows, the mid-tones and the highlights.

LUT (lookup table) — a table that stores a colour transform. A 1D LUT maps each channel alone; a 3D LUT maps every RGB combination.

Mid-grey — scene value 0.18, the usual reference for exposure.

Narkowicz fit — a five-constant curve approximating the ACES look (2016).

OETF — the encoding curve applied to the output, such as sRGB or PQ.

Out of gamut (OOG) — a colour the target space cannot represent: a channel below 0 or above 1.

PQ — the perceptual quantiser, SMPTE ST 2084, an absolute HDR encoding up to 10,000 cd/m^2.

Reinhard — the tone curve L ÷ (1 + L) (Reinhard et al., 2002).

Scene-referred — values describing light in the scene, which can exceed 1.

Tone mapping — compressing scene luminance into the display's range.

Uchimura (Gran Turismo) — a tone curve with a toe, a linear section and a shoulder (Uchimura, 2017).

Vectorscope — a plot of colour direction and strength.

Vibrance — a saturation change weighted towards less saturated colours.

Waveform — a plot of brightness across the image, column by column.

Working space — the colour space in which the curve and grade are applied.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tone Mapping limits and accuracy

What the Tone Mapping lab implements exactly, what is its own, and what it does not do.

Exact to the source

- **Transfer functions**: the sRGB curve, PQ (SMPTE ST 2084, constants as the standard gives them) and HLG (the OETF and its inverse; the OOTF is not applied).
- **Primaries and white points** of Rec.709, Display P3, Rec.2020 and ACEScg.
- **Bradford** adaptation between white points.
- **Operators** with their published constants: Reinhard (2002), Hable (2010), Narkowicz (2016) and Uchimura (2017).

This tool's own

- **Log (soft)** and the **custom curve**.
- The **grade** controls, the **low-level operators** and the **gamut mapping** options.
- **Working luminance** is read from each working space's own matrix, so saturation and vibrance mix towards the right grey.

Not implemented

- **ITU-R BT.2446** Methods A and C. The two operators once named after them are kept as "Gamma-domain scale" and "Scaled Reinhard" and are not tone curves.
- **The Academy's ACES transforms** (RRT and ODT). The Narkowicz fit approximates their look.
- **HDR image files** such as OpenEXR. Uploaded images are 8-bit and carry no absolute luminance.

Current-version caveats

- The operators work per channel, not on luminance, so strongly saturated highlights can shift hue.
- **Soft compress** gives the same result as **Clip**.
- **Display peak** and **Source peak** affect only the two retired operators and the false-colour display; they do not scale the working curves or the PQ output.
- With **PQ** output, display value 1.0 is encoded as 10,000 cd/m².
- The false-colour map multiplies source luma by the display peak, while its note uses the source peak.
- The link records only operator, exposure, source encoding, display peak, output encoding and target primaries.
- The **Linear axis**, **Log axis** and **Film stops** buttons set custom-curve presets rather than changing the chart's axis.
- **Install a table** on the Data tab does not load files.

Precision

- The curve `.cube` has 33 entries; the pipeline `.cube` is 17 × 17 × 17. Programs interpolate between entries, so very steep curves lose some detail.
- Output is 8-bit per channel.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tone Mapping privacy

What the Tone Mapping lab does with your images and settings.

- **Your image stays on your device.** An uploaded image is decoded and processed in your browser, and is never sent to a server.
- **Exports are created locally** and saved through your browser's download, or copied to your clipboard when you choose.
- **The page address** is updated as you work to record the operator, exposure, source encoding, display peak, output encoding and target primaries. Copying or sharing the address shares those settings, never your image.
- **Stored A and B frames** exist only in the open page.
- **Save to Library** stores an image of the mapped frame in your Auric Artisan Library.
- The page loads the site's shared features like every other page on auricartisan.com; see the site's privacy policy.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Tone Mapping release notes

User-facing changes to the Tone Mapping lab.

Current version (2026 workbench rebuild)

- Tabs are now **Lab**, **Operators**, **Scopes**, **Data**, **Export** and **Reference**.

- The Lab offers seven tone curves instead of nine. The two labelled BT.2446 Method A and C were renamed Gamma-domain scale and Scaled Reinhard and moved to the Operators tab, because neither implements BT.2446.
- The "ACES" operator is named Narkowicz fit.
- Display peak and source peak are now two separate controls.
- The grade mixes towards the working space's own luminance, and the hue control rotates hue.
- **Mark out-of-gamut pixels** now paints an overlay.
- The false-colour scale was rebuilt with a key, including the 203-nit HDR diffuse white.
- Scene reference white, Highlight stress and the zebra thresholds were removed; the Scopes tab lists them as controls that did nothing.
- Export now offers a 1D curve `.cube` or CSV with a stated domain, and a 3D `.cube` that runs the whole pipeline, with a preview and a list of the steps each file carries.

May 2026

The Tone Mapping User Guide was published in the site's Documentation library, describing the earlier version.

WEBSITE › TOOLS › COLORIMETRY AND RENDERING

Related to Tone Mapping

Tools and Learn articles that go well with the Tone Mapping lab.

Tools

Tool	Why
HDR Gamut Plot	PQ and HLG curves in cd/m², and gamuts in ICtCp and Jzazbz
LUT Lab	Build, read and preview `.cube` LUTs
Gamut Mapping	Compare gamut-mapping methods between RGB spaces
Gamut Lab	The working and target spaces, their primaries and curves

Learn articles

- Color in Real-Time 3D: Albedo, PBR, and Tone Mapping — https://auricartisan.com/library/learn/articles/2026-08-02-color-in-real-time-3d-albedo-pbr-tone-mapping
- HDR, Wide Gamut, PQ, HLG, and Modern Displays — https://auricartisan.com/library/learn/articles/2026-05-25-hdr-wide-gamut-pq-hlg-modern-displays
- Spectral Rendering: Why Three Numbers Aren't Enough — https://auricartisan.com/library/learn/articles/2026-08-22-spectral-rendering-beyond-rgb

Documentation on the site

The site's Documentation library (https://auricartisan.com/library/documentation/) has an earlier **Tone Mapping User Guide**. This documentation follows the current page.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL

Perception and spectral tools

What the three perception and neuroscience tools and the two spectral tools on auricartisan.com are for, where to find them, and which one to open for each job.

This group holds five workbenches for questions that sit behind a colour value rather than in it. Three of them are about the observer: how a colour travels through the early visual system, how a scene looks to a different visual system, and how to measure the faintest contrast a person can see. Two of them are about light and matter: what colour a dye bath makes under a named light, and how a material looks under a light source at a given angle.

All five are teaching and exploration tools that show their working. Each page says, in a strip at the top that cannot be closed, what its pictures and numbers rest on. Each has a **Data** tab that lists every table, formula and model the tool uses and marks it **Verbatim** (taken unchanged from a publication or standard), **Computed** (derived from verbatim data by stated arithmetic), **Stand-in** (this tool's own values, chosen to be plausible) or **Absent** (not held, with a note on what it would take). A figure that rests on a stand-in or an absent entry is never presented as a published value. None of the five is a clinical, physiological or laboratory instrument.

Everything runs in your browser. Nothing you load, type or measure is uploaded, and nothing leaves the page until you press an export or copy button.

Where to find them

The group has two hub addresses:

- https://auricartisan.com/tool/general/perception-and-neuroscience/ (the site menu calls it **Perception and Neuroscience Hub**)
- https://auricartisan.com/tool/general/spectral-tools/ (the site menu calls it **Spectral Tools Hub**)

Both addresses currently forward you to the Collections page at https://auricartisan.com/collections/, which lists every tool on the site. Find the tools there, or open them directly from the table below. The Collections page is described in Collections.

The tools

Tool	Page title	Address	Use it to
Neuroscience	Visual Pathway	https://auricartisan.com/tool/general/perception-and-neuroscience/neuroscience/	Follow a grating from cone signals through opponent channels, retina, V1, V2 and V4
Perception and Vision	Perception Vision	https://auricartisan.com/tool/general/perception-and-neuroscience/perception-vision/	Simulate colour vision deficiency, night vision, lens ageing, cataract, central loss and acuity over an image
Psychophysical Experiment Engine	Psychophysics Bench	https://auricartisan.com/tool/general/perception-and-neuroscience/psychophysical-experiment-engine/	Measure a contrast detection threshold with an adaptive staircase
Industrial Dye Chemistry	Dye in Solution	https://auricartisan.com/tool/general/spectral-tools/industrial-dye-chemistry/	Model a dye bath with pH, concentration and path length, and see its colour under four lights
Material Lighting	Material Under Light	https://auricartisan.com/tool/general/spectral-tools/material-lighting/	Render a material's reflectance under an illuminant and a surface model, and compare it with its reference light

The names differ between places. The Collections page lists the tools as **Neuroscience**, **Perception & Vision**, **Psychophysics Engine**, **Industrial Dye Chemistry** and **Material Lighting**; the site menu uses **Neuroscience of Color**, **Perception and**

Vision, **Psychophysical Experiment Engine**, **Industrial Dye Chemistry** and **Material Lighting**.

> **Important:** On the Psychophysical Experiment Engine, at the time of writing, only the **Lab** tab opens. Its other tabs, including **Export**, do not respond. See its troubleshooting page.

Which tool for which job

- You want to see what each stage of the early visual system does to a colour or a pattern: use Neuroscience.
- You want to check a design, a screenshot or a palette for colour vision deficiency, low light or an ageing eye: use Perception and Vision.
- You want to measure your own contrast threshold for a Gabor patch, or learn how staircase procedures work: use the Psychophysical Experiment Engine.
- You want to see how a pH indicator changes colour, what an isosbestic point is, or how far a bath's colour moves between daylight and tungsten light: use Industrial Dye Chemistry.
- You want to see why the same material looks different under two light sources, or how a surface's roughness and metalness change its colour: use Material Lighting.

What the five share

- **Six tabs.** A working tab (**Lab**) followed by explanation tabs, a **Data** tab (the register described above), an **Export** tab and a **Reference** tab of sources. Links inside the page text, such as **Data** or **Method** in a note, jump to that tab.
- **Stated inputs.** Where a number depends on a condition, the condition is a control: viewing distance and pixel pitch for anything in cycles per degree, the illuminant for anything that is a colour of a physical sample.
- **Files that carry their caveats.** Every CSV and JSON the tools write begins with a header (or contains fields) that name the method, the units, the light, the data sources and whether the result may be reported as a published value.
- **Links that reproduce a setup.** Each tool can copy a link that restores its main controls when opened.

All five tools are free to use, like every tool on the site.

In this folder

Folder	What it covers
neuroscience/	The Visual Pathway: stimulus, viewing geometry, the five stages and their maps, the cone table, the colour list export
perception-and-vision/	The eight simulations, sources, comparison views, the grey-ramp check, presets, the colour list export
psychophysical-experiment-engine/	The two tasks, the stimulus, the five staircases, running a session, the threshold readouts and the exports
industrial-dye-chemistry/	The bath, the two-form dye model, pH and titration, Beer–Lambert, lights and colour inconstancy, the tolerance sweep, measured spectra
material-lighting/	Illuminants, the twenty materials, the five lobes, geometry, the reference comparison, the sensitivity sweep, measured reflectance

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Industrial Dye Chemistry (Dye in Solution)

What the Dye in Solution tool does, who it is for, where to find it and how to watch a first pH indicator change colour.

The Industrial Dye Chemistry tool, whose page is titled **Dye in solution**, models a dye bath one wavelength at a time. You choose up to four dyes, their concentrations and the bath's pH and path length. The tool works out how much light the bath absorbs at each of 81 wavelengths from 380 to 780 nm, what colour that makes under a light source you name, and how far that colour moves under the other lights it holds.

Each dye is modelled as two forms, an acid form and a base form, each with its own absorption band. The pH sets how much of the dye is in each form, so a pH indicator changes colour as the two bands trade places, and the curves for every pH cross at one wavelength, the isosbestic point. Absorbance follows the Beer–Lambert law, and the page flags when the bath is too strongly coloured for that law to hold. Colour is computed against the CIE 1931 standard observer under daylight D65 or D50, tungsten illuminant A, or equal-energy light E, and colour differences use CIEDE2000.

The model is real; the dyes are not. The six preset dyes stand for classes of dye (for example "the methyl-orange family") with constants chosen by the tool, and their bands are drawn as Gaussians. You can replace a band with a measured UV–Vis spectrum. The page holds no safety or toxicological data, does not perform colour-fastness tests and does not offer fluorescent illuminants such as F11. Everything runs in your browser.

Who it is for

- Students of chemistry and colour science learning absorbance, indicators and isosbestic points.
- Textile and dye professionals who want to see, in principle, how concentration, path length, pH and the light source move a colour, and how far a colour shifts between lights.
- Colour enthusiasts curious about why a solution's colour depends on the lamp.

Where to find it

Open https://auricartisan.com/tool/general/spectral-tools/industrial-dye-chemistry/ in a current browser. It is listed as **Industrial Dye Chemistry** on the Collections page (https://auricartisan.com/collections/) and in the site menu.

Quick start

1. Open the tool. The bath holds one **Weak-acid indicator** at pH 3.50, in a 1.0 cm path, under D65.
2. Look at the spectrum: two dashed bands (acid and base form), the filled curve you would measure, and a green circle at the isosbestic point.
3. Select **Run the titration** and watch the pH sweep from 0 to 14 while the swatch and curves change.
4. Drag **pH** back to 3.5 and raise **Concentration** until **Beer–Lambert holds** reads **no**.
5. Change the light to **A — a blackbody at 2856 K** and compare the four swatches under **The same bath under every light this tool holds**.
6. Add a second dye: pick one from the list under **The components** and select **Add**.
7. Open **Export** and select **Download the spectrum** to save the 81-row CSV.

What you can do

- Build a bath of up to four dyes from six presets, and set each dye's pKa and concentration.
- Set pH (0–14) and path length (0.1–10 cm), and animate a titration.

- See both forms of the selected dye, their weighted sum and the isosbestic point.
- Read the colour as sRGB and CIE L*ab*, the peak absorbance and its wavelength, the Beer–Lambert check, the dominant wavelength and excitation purity.
- Compare the bath under D65, D50, A and E as CIEDE2000 colour differences.
- Run a tolerance sweep over concentration, path length and molar absorptivity.
- Replace a dye band with your own measured spectrum.
- Export the spectrum as CSV, the whole bath as JSON, or a link.

In this folder

- docs/ — how to use the tool.
  - Getting started — open it, the screen tour, the first bath.
  - Build a bath — dyes, concentration, path length and the Beer–Lambert law.
  - Run a titration — pH, the two forms, the base fraction and the isosbestic point.
  - Compare lights — illuminants, L*ab*, colour inconstancy, metamerism and dominant wavelength.
  - Run a tolerance sweep — how far the colour moves when the recipe varies.
  - Use a measured spectrum — replace a drawn band with your own data.
  - Export and share — the spectrum CSV, the bath JSON and links.
  - Reference — every control, readout, tab and file format.
  - Troubleshooting — symptoms, causes and fixes.
  - FAQ — short answers to common questions.
- others/ — background material.
  - Glossary — terms used in this tool.
  - Shortcuts — keyboard and pointer actions.
  - Limits and accuracy — what is published, what is the tool's own, what is not held, and safety.
  - Related — related tools and Learn articles.
  - Release notes — user-facing history.
  - Privacy — what stays in your browser.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Using Dye in Solution

Contents of the how-to documentation for the Dye in Solution tool.

These pages explain how to build and analyse a dye bath, from the first titration to exporting the spectrum. Start with getting started; the task guides can be read in any order; the reference lists every control.

File	What it covers
getting-started.md	Opening the tool, the six tabs, the Lab screen tour and a first bath
build-a-bath.md	The six dyes, adding, muting and removing components, concentration, path length and Beer–Lambert
run-a-titration.md	pH, pKa, the base fraction, the two forms, the titration animation and the isosbestic point
compare-lights.md	The four illuminants, the colour readouts, colour inconstancy against metamerism, dominant wavelength and purity
run-a-tolerance-sweep.md	The Method tab's sweep over concentration, path length and molar absorptivity
use-a-measured-spectrum.md	Loading a UV–Vis spectrum into a dye's acid or base form
export-and-share.md	The spectrum CSV, the bath JSON and links
reference.md	Every tab, control, readout, message and file format, with ranges and defaults
troubleshooting.md	Refused files, flags, limits on components, links
faq.md	Short answers to common questions

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Getting started with Dye in Solution

Open Dye in Solution, learn the layout of the Lab tab and read the default bath.

Open the tool

Go to https://auricartisan.com/tool/general/spectral-tools/industrial-dye-chemistry/. The page opens on the **Lab** tab with a default bath: one **Weak-acid indicator** (standing for the methyl-orange family) at 34.0 µM, pKa 3.5, at pH 3.50, in a 1.0 cm path, lit by D65 daylight.

A note strip above the tabs, which cannot be closed, says what an earlier version of the page got wrong and links to the **Method** and **Safety** tabs.

The six tabs

Tab	What it is for
Lab	Build the bath, see the spectrum, read the colour and compare the lights
Method	How the titration, Beer–Lambert law and light are modelled; the tolerance sweep
Safety	What the page does and does not know about hazards
Data	The register of 18 tables and formulas, with a status for each; load a measured spectrum here
Export	The spectrum CSV, the bath JSON and a link

Tab	What it is for
Reference	Thirteen sources, filterable by whether they run

A tour of the Lab tab

The rail (left), top to bottom:

- **The bath** — **pH**, **Path length** and **Run the titration**.
- **The light** — the illuminant menu.
- **The components** — the list of dyes in the bath, a menu of dyes to add, and **Add**, **Mute** and **Remove**.
- **The selected dye** — its name, what it stands for, and its **pKa** and **Concentration** sliders.
- **Reset the bath**.

The stage (right):

- **The two forms, and what you measure** — the spectrum plot for the selected dye, 380–780 nm, with a legend: **Acid form** `HA` (dashed), **Base form** `A⁻` (dashed), **What you measure** (the filled curve, their weighted sum) and **Isosbestic** (a green circle and label where the two forms cross).
- Three rows of readouts: **sRGB**, **CIE L***, **a*, b***, **Base fraction**; **Peak absorbance**, **λmax**, **Beer–Lambert holds**, **Isosbestic point**; **Dominant λ**, **Excitation purity**, **Inside sRGB**, **Observer**.
- **The same bath under every light this tool holds** — one swatch per illuminant, with the colour difference from the current light, and a card for F11 marked **not held**.

Read the default bath

1. Look at the plot. The acid form peaks near 508 nm and the base form near 462 nm. At pH 3.50, equal to the dye's pKa, half the dye is in each form, so **Base fraction** reads 0.500 and the filled curve sits halfway between them.
2. Find the green circle: **Isosbestic point** reads about 480 nm, where the two forms absorb equally.
3. Read the colour: **sRGB** `#FFBF9F`, a pale orange-pink, with **CIE L*** 84.8 and **a*, b*** 28.2, 28.9.
4. Read **Peak absorbance** 0.630 at **λmax** 485 nm, and **Beer–Lambert holds yes, A 0.63**: the bath is inside the range where absorbance grows in proportion to concentration.
5. Look at the lights row: the same bath differs by about 1.60 ΔE_{00} under D50, 6.57 under A and 0.74 under E.
6. Select **Run the titration**. The pH climbs from 0 to 14 in about six seconds and the swatch moves from pink (`#FF96C7` at pH 0) to yellow (`#FFF17A` at pH 14).

You should now see a bath whose colour, spectrum and readouts respond to every control.

Next steps

- Build a bath — add dyes and understand absorbance.
- Run a titration — pH and the isosbestic point.
- Compare lights — what the lights row means.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Build a bath

Add, select, mute and remove dyes, set concentration and path length, and understand absorbance and the Beer–Lambert law.

Absorbance in brief

A dye solution looks coloured because it absorbs some wavelengths of light more than others. **Transmittance** T is the fraction of light at a wavelength that gets through. **Absorbance** A is its negative base-10 logarithm, $A = -\log_{10} T$: an absorbance of 1 lets 10% through, 2 lets 1% through.

The **Beer–Lambert law** says absorbance is proportional to how much dye the light meets:

$A = \varepsilon \times c \times l$

- **ε** (epsilon), the **molar absorptivity**, is the dye's own strength at that wavelength, in litres per mole per centimetre (M^{-1} cm^{-1}).
- **c** is the concentration in moles per litre. The page shows it in micromolar (μM, millionths of a mole per litre).
- **l** is the **path length**, the distance the light travels through the solution, in centimetres.

Absorbances of several dyes, and of the two forms of one dye, simply add. So doubling the concentration or the path length doubles the absorbance at every wavelength.

The law holds well only up to about A = 1. Above that, real solutions fall below the straight line, because dye molecules clump together and spectrophotometers have finite bandwidth. The page flags a bath whose absorbance exceeds 1 anywhere.

The six preset dyes

Every preset stands for a **class** of dye and says which; none is a named commercial substance, and all their constants are the tool's own. Each has an acid form and a base form, each drawn as a Gaussian band.

Dye	Stands for	pKa	Default concentration	Acid form: λmax, ε	Base form: λmax, ε
Weak-acid indicator	The methyl-orange family: a large shift, low in the pH range	3.5	34 μM	508 nm, 26,000	462 nm, 22,000
Mid-range indicator	The bromothymol-blue family: the widest shift, near neutral	7.1	20 μM	430 nm, 15,000	615 nm, 34,000
High-range indicator	The phenolphthalein family: one form barely absorbs	9.4	25 μM	400 nm, 900	553 nm, 31,000
Azo red, pH-stable	A textile azo red whose colour does not move over the working range	12.5	50 μM	505 nm, 30,000	512 nm, 29,000
Anthraquinone blue, pH-stable	A textile blue with a narrow band and little pH response	11.8	40 μM	600 nm, 16,000	606 nm, 16,500
Yellow, pH-stable	A short-wavelength dye for mixing, flat across the range	12.0	60 μM	428 nm, 21,000	431 nm, 21,000

λmax is the wavelength at which a band absorbs most.

Add, select, mute and remove components

The list under **The components** shows each dye in the bath as a button with a swatch (that dye alone in the current bath), its name, its pKa, concentration and the two band peaks, and a tag: **Stand-in** for drawn bands or **Measured** after you load a spectrum.

1. Pick a dye in the menu under the list and select **Add**. It joins the bath and becomes the selected dye. A bath holds at most four; a fifth shows **Four components is the ceiling here**.

2. Select a dye's button to make it the selected dye. The spectrum plot and **The selected dye** group switch to it.
3. Select **Mute** to take the selected dye out of the bath without deleting it; its button dims. Select **Mute** again to bring it back.
4. Select **Remove** to delete the selected dye. The bath keeps at least one; removing the last shows **The bath needs at least one dye**.

Reset the bath returns to the default single weak-acid indicator at pH 3.50, 1.0 cm and D65, and clears any sweep and measured spectra.

Set concentration and path length

- **Concentration** (under **The selected dye**, 0.5–200 µM in steps of 0.5) sets the selected dye's concentration.
- **pKa** (0–14 in steps of 0.1) sets the selected dye's pKa; see Run a titration.
- **Path length** (under **The bath**, 0.1–10 cm in steps of 0.1, default 1.0 cm) applies to the whole bath. A standard spectrophotometer cell is 1 cm.

Read the absorbance readouts

These describe the whole bath (every component that is not muted):

Readout	Meaning
Peak absorbance	The highest absorbance at any of the 81 wavelengths, three decimals
λmax	The wavelength of that peak
Beer–Lambert holds	**yes, A 0.63** when the peak is at or below 1; **no — A 1.85 is past 1** when above, highlighted

With the default dye at pH 3.5, raising **Concentration** above about 54 µM, or **Path length** above about 1.6 cm, pushes the peak past 1.

The spectrum plot is different: it shows the **selected** dye's two forms and their weighted sum as molar absorptivity (the vertical axis is in thousands of M^{-1} cm^{-1}), not scaled by concentration or path. Use it to see the shape of a dye; use the readouts for the bath.

Try it

1. Add the **Anthraquinone blue, pH-stable** to the default bath.
2. Watch the swatch move to a muted grey (`#ACA1A6`) as the blue dye absorbs the orange-red light the indicator lets through; the two dyes together absorb across most of the spectrum.
3. Select the blue dye's button and lower its **Concentration** until you like the shade.
4. Select **Mute** on it to compare the bath with and without it.

You should now have a mixture whose colour and absorbance you can adjust dye by dye.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Compare lights

Choose the illuminant, read the colour readouts, compare the bath under every light, and understand colour inconstancy, metamerism, dominant wavelength and purity.

Why the light matters

A dye does not have a colour on its own. The colour you see is the light that the bath lets through, and that depends on what light went in. The page computes it wavelength by wavelength: the illuminant's power × the bath's transmittance × the CIE 1931 2° standard observer's colour-matching functions, summed over 81 samples from 380 to 780 nm at 5 nm. The observer is used at the 5 nm spacing it is published at, so no sample is interpolated.

Choose the illuminant

Under **The light**:

Option	What it is
D65 — daylight, 6504 K (default)	CIE standard average daylight, reconstructed from the published CIE daylight basis
D50 — daylight, 5003 K	Warmer daylight, the usual reference in graphic arts
A — a blackbody at 2856 K	Incandescent (tungsten) light, from Planck's law
E — equal energy	Equal power at every wavelength; a theoretical reference

F11 is not offered. Fluorescent lamps such as F11, the light most often used in shops and the one a dyer most wants, have narrow mercury emission lines. The CIE publishes them as measured tables that cannot be computed from a formula, and the page does not hold them. It shows F11 as a card marked **not held** rather than offering a smooth imitation under that name.

Read the colour readouts

Readout	Meaning
sRGB	The hex colour of the light leaving the bath under the chosen illuminant, not adapted to it, so a bath under A looks warmer
CIE L*	Lightness from 0 to 100
a*, b*	Red–green and yellow–blue coordinates
Inside sRGB	Whether the colour fits the sRGB range; **no** (highlighted) means the swatch is the nearest colour a screen can show
Observer	CIE 1931 2°

L*, a* and b* are CIE L*a*b* taken against the white of the **chosen illuminant itself**, not against D65. That way, when you switch lights, a change in L*a*b* measures the dye's colour moving relative to the light, not simply the fact that the lamp is a different colour.

The swatch is un-adapted on purpose: it shows how the bath would look if the light changed while your eyes stayed adapted to the screen. The two answer different questions, and the page labels both.

Compare the bath under every light

The same bath under every light this tool holds shows one swatch per illuminant. The current light is labelled **— this bath** with its hex; the others show **ΔE_{00}**, the colour difference from the current light.

ΔE_{00} (CIEDE2000, CIE 142-2001) is the CIE's recommended measure of how different two colours look; the page uses the full formula including its rotation term for blues. A difference around 1 is at the edge of what most people notice side by side under good conditions; several units is plainly visible.

With the default bath under D65, the differences are about 1.60 to D50, 6.57 to A and 0.74 to E.

Colour inconstancy is not metamerism

What the lights row measures is **colour inconstancy**: how far one bath's colour moves between two lights.

Metamerism is a property of a **pair**: two samples with different spectra that match under one light and stop matching under another. It is the everyday failure in industrial dyeing, where a dyed batch matches the standard in the dyehouse and not in the shop. It needs two baths. The **Method** tab's **The light** view explains the difference. The current interface compares one bath across lights and has no control for comparing two baths, so it does not report a metamerism index.

Dominant wavelength and excitation purity

Readout	Meaning
Dominant λ	The wavelength of the spectral colour that, mixed with the illuminant's white, would match the bath; found by drawing a line on the chromaticity diagram from the white point through the bath's colour to the edge of the diagram
Excitation purity	How far the bath's colour lies along that line, from 0% at the white point to 100% at the spectral edge

For purples, the line meets the straight "purple line" at the bottom of the diagram instead, where there is no spectral colour. The page then gives the **complementary** wavelength with a `c` prefix (for example `c540 nm`), shows purity as **—**, and explains in a note under the readouts. A bath whose colour sits exactly on the white point has no dominant wavelength, and a note says so.

The default bath reads **Dominant λ** about 592 nm and **Excitation purity** about 36.7%.

Try it

1. With the default bath, switch **The light** to **A — a blackbody at 2856 K**.
2. The swatch and **sRGB** turn a strong orange (`#FFB535`). **CIE L*** and **a*, b*** move less, to about 89.3 and 27.3, 39.7, because they are taken against illuminant A's own white; what remains is the dye's own colour inconstancy.
3. In the lights row, the D65 swatch now shows about 6.57 ΔE_{00}: the same size of shift, seen from the other side.
4. Add the **Anthraquinone blue, pH-stable** and compare the differences again: mixtures can be more or less constant than their components.

You should now be able to say how far a bath's colour moves between lights, and why that number is not a metamerism index.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Export and share

Download the 81-row spectrum CSV and the whole-bath JSON, and copy a link that rebuilds the bath.

Open the **Export** tab. **What to take** switches the preview between **The spectrum** (the first lines of the CSV), **The whole bath** (the first part of the JSON) and **A link to this**; the note above it gives the row count. Nothing leaves the page until you press a button. Exports are never gated on a tick-box.

Button	Result
Download the spectrum	Saves `dye-in-solution-spectrum.csv`
Download the bath	Saves `dye-in-solution-bath.json`
Copy a link to this bath	Copies a link; the message **Link copied** confirms it, or the link is shown if the clipboard is unavailable

The spectrum CSV

The file starts with `#` header lines that state everything a reader needs to compare it with another spectrum:

- the time; the sampling (380–780 nm at 5 nm, 81 samples); the observer (CIE 1931 2°, not interpolated); the illuminant;
- the bath's pH and path length;
- for each component: name (and "(disabled)" if muted), concentration, pKa, both forms' λmax and ε, whether the band shape is MEASURED or SYNTHESISED, the base fraction at this pH and the isosbestic point;
- the peak absorbance and whether it is inside the Beer–Lambert linear range;
- which white point the swatch and the L*a*b* use;
- the colour inconstancy (ΔE_{00}) to each other illuminant;
- statements that the file is not a metamerism index, that F11 is not held, that it is not a fastness test, and that it holds no hazard data;
- the tolerance sweep's median and 95th percentile, if you ran one.

Then 81 rows with these columns:

Column	Meaning
`wavelength_nm`	380 to 780 in steps of 5
`absorbance`	The whole bath's absorbance
`transmittance`	10 to the power of minus the absorbance
`eps_acid`, `eps_base`	The selected dye's two forms, molar absorptivity
`observer_x`, `observer_y`, `observer_z`	The CIE 1931 2° colour-matching functions

The bath JSON

Field	Contents
`tool`, `generated`	The tool name and time
`sampling`	Range, step and number of samples, with a note that nothing is interpolated
`illuminant`, `illuminant_not_held`	The light used, and F11 with why it is not held and what it would take
`bath`	`ph` and `path_cm`
`components`	For each dye: name, what it stands for, provenance, enabled, pKa, concentration in mol/L, base fraction, both forms' λmax and ε, isosbestic point
`beer_lambert`	Peak absorbance, its wavelength, whether it is inside the linear range, and the limit
`colour`	sRGB hex (un-adapted), L*a*b*, the white it was taken against, gamut flag, dominant wavelength, purity and note
`colour_inconstancy`	ΔE_{00} to each other illuminant, each marked `is_metamerism: false`
`sensitivity`	The sweep result, or `null`
`not_held`	Every register entry marked absent, with why and what it would take
`register`	Every register entry with its source and status
`spectrum`	Wavelengths, absorbance and transmittance arrays

Links

A link rebuilds the bath: `ph`, `path`, `ill` (`d65`, `d50`, `a` or `e`) and `dyes`, a comma-separated list of `id:pKa:concentration-in-µM`, for example:

```
https://auricartisan.com/tool/general/spectral-tools/industrial-dye-chemistry/?ph=3.50&path=1.00&ill=d65&dyes=weak-acid:3.50:34.00,anthraquinone-blue:11.80:40.00
```

Dye identifiers: `weak-acid`, `mid-range`, `high-range`, `azo-red`, `anthraquinone-blue`, `yellow`. A link does not carry muted state, measured spectra or sweep tolerances.

You should now have a file that anyone can read without the page, with its light, observer and data sources stated.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Run a titration

Change the pH, understand pKa, the base fraction and the two forms of a dye, animate a titration and find the isosbestic point.

How pH changes a dye's colour

Many dyes, and all pH indicators, exist in two chemical forms: an **acid form**, written HA, which holds an extra hydrogen ion, and a **base form**, written A^-, which has given it up. Each form has its own absorption band, so each has its own colour.

The **pKa** is the pH at which half the dye is in each form. The **Henderson–Hasselbalch** relation gives the share in the base form, the **base fraction** f:

f = 1 ÷ (1 + 10^(pKa − pH))

Two pH units below the pKa, about 1% of the dye is in the base form; at the pKa, 50%; two units above, about 99%.

What a spectrophotometer measures is the weighted sum of the two bands:

A(λ) = c × l × [(1 − f) × ε_HA(λ) + f × ε_A(λ)]

Because the total amount of dye does not change, there is one wavelength where the two forms absorb equally. At that wavelength the absorbance is the same at every pH, so every curve of a titration passes through one point: the **isosbestic point**. It is the signature of a clean two-form equilibrium.

Change the pH

1. Select the dye you want to study in **The components** list.
2. Drag **pH** (0–14, step 0.05, default 3.50).
3. Watch **Base fraction** (for the selected dye, three decimals) and the plot. As the pH rises past the pKa, the filled curve moves from the acid band to the base band. Where the two forms absorb comparably, the peak dips at the pKa; where one form is much stronger, the peak shifts to it.
4. Watch the swatch, **sRGB**, **CIE L*** and **a*, b*** change with it.

pKa under **The selected dye** (0–14, step 0.1) moves the midpoint of the change for that dye.

Animate a titration

Select **Run the titration**. The pH steps from 0 to 14 in steps of 0.2, about 11 steps a second, and every readout, the swatch and the plot follow. The button reads **Stop** while it runs; select it to stop early. At the end a message says **Titration complete** and the pH is left at 14.

With the default weak-acid indicator the bath runs from pink (`#FF96C7`) at pH 0 to yellow (`#FFF17A`) at pH 14, changing fastest around pH 3.5.

Find the isosbestic point

The **Isosbestic point** readout gives the wavelength where the selected dye's two forms cross, located precisely rather than at the nearest 5 nm sample:

- **480 nm** (for the default dye): a clear crossing, marked on the plot with a green circle and the label **isosbestic 480 nm**.
- **515 nm (weak)**: the forms cross, but so deep in their tails (less than 5% of the peak) that the point is invisible on a plot. It is highlighted and not marked.
- **none in range**: the two forms do not cross between 380 and 780 nm.

Dye	Isosbestic point
Weak-acid indicator	About 480 nm, clear
Mid-range indicator	About 515 nm, weak
High-range indicator	About 461 nm, weak
Azo red, pH-stable	About 515 nm, clear
Anthraquinone blue, pH-stable	About 600 nm, clear
Yellow, pH-stable	About 430 nm, clear

The pH-stable dyes have clear crossings because their two forms are almost the same band; their colour barely changes with pH.

Compare with the old model on the Method tab

Open **Method** and select **The titration**. It shows the chain from **pH** through **Henderson–Hasselbalch**, **Two spectra** and **Their weighted sum** to **Transmittance**, then two tables for the selected dye:

- **Do the curves ever agree?** — compares a single sliding band (the model an earlier version used, labelled **One sliding band — what it did**) with the two-form model (**Two species — what it does now**): the wavelength of tightest agreement, how far apart the curves are there, and the isosbestic point.
- **The peak across the titration** — the base fraction, the old and new peak absorbance and the peak wavelength at pKa − 3, − 1, 0, + 1 and + 3 (for a 1 cm path).

Try it

1. Select **Reset the bath**.
2. Add the **Mid-range indicator** and select **Remove** on the weak-acid indicator (select it first).
3. Run the titration. The bath goes from pale yellow at low pH, through pale green-cyan near pH 7, to a stronger cyan.
4. Note that **Isosbestic point** reads about 515 nm and is marked weak: this dye's two bands barely overlap.

You should now be able to predict a dye's colour at any pH from its pKa and its two bands.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Run a tolerance sweep

Estimate how far a bath's colour moves when concentration, path length and molar absorptivity vary within stated tolerances.

A real bath never matches its recipe exactly. The balance and flask set the concentration to within a tolerance, the cell's specification sets the path length, and a published molar absorptivity comes with its own uncertainty. The sweep on the **Method** tab asks: if each of those varies within its tolerance, how far does the colour move, in ΔE_{00}?

Run it

1. Set up the bath on the **Lab** tab.
2. Open the **Method** tab. The sweep controls are under **The sweep** on the rail:

Control	Range and step	Default
Concentration	±0–20%, step 0.5	±2.0%
Path length	±0–2 mm, step 0.01	±0.10 mm
Molar absorptivity	±0–30%, step 0.5	±5.0%
Draws	100–2000, step 100	500

3. Replace the defaults with your own tolerances if you know them. The defaults are the tool's guess at a hand-weighed bath in a 10 mm cell, not a measurement of your laboratory.
4. Select **Run the sweep**. A message reports how many draws took how long.

Read the result

Under **How far the colour moves when the tolerances do**, three figures appear, each a ΔE_{00} from the unperturbed bath under the current light:

Figure	Meaning
Median	Half the draws moved less than this
95th percentile	95% of draws moved less than this
Worst draw	The largest movement seen

The note beside the heading gives the number of draws and the time taken, and a line under the figures restates the tolerances used.

For the default bath with the default tolerances, the median is typically around 0.3 and the 95th percentile around 0.8; because the draws are random, each run differs slightly.

How the draws are made

Each draw changes, independently and uniformly within ± the tolerance:

- each dye's concentration (as a percentage of its value);

- the path length (in millimetres, the same for the whole bath);
- each dye's molar absorptivity (as a percentage, the same factor for both of its forms).

The pH and the illuminant are not varied. The result is written into the export: the CSV header gains a **sensitivity** line and the JSON a `sensitivity` object. It is cleared by **Reset the bath**, but it is not rerun when you change the bath, so run the sweep again after any change before you export.

What this is not

This is a tolerance sweep, not a statistical confidence interval from repeated measurements, and not a bootstrap. The **Method** tab's **What was resampled** view shows what an earlier version reported instead and explains why that interval was meaningless.

You should now have a colour tolerance in ΔE_{00} that follows from tolerances you can defend.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Use a measured spectrum

Replace a drawn dye band with a measured UV–Vis molar absorptivity spectrum, and go back to the drawn bands.

The preset dyes' bands are drawn as Gaussian curves around a stated λmax. Real absorption bands are asymmetric, with a longer tail towards the blue, and the tails matter most where dyes overlap in a mixture. If you have a UV–Vis scan of a real dye, you can load it into one form of a dye in the bath.

Prepare the file

- A plain-text file: `.csv`, `.tsv` or `.txt`.
- Two columns per line: **wavelength in nm** and **molar absorptivity** (ε, in M^{-1} cm^{-1}). Columns may be separated by commas, semicolons, tabs or spaces.
- At least eight numeric rows, covering at least **420 to 680 nm**. Lines that are not two numbers (headers, comments) are skipped. Negative values are read as 0.

If your instrument exported absorbance rather than molar absorptivity, divide each absorbance by the concentration (in mol/L) and path length (in cm) of the measured sample first.

Load it

1. On the **Lab** tab, select the dye whose band you want to replace in **The components** list.
2. Open the **Data** tab. Under **Measure your own dye**, choose **Into the acid form** or **Into the base form**.
3. Drop the file on **Drop a spectrum**, or select the area (or press `Enter` or `Space` on it) to choose the file. You can also drop the file directly on the spectrum plot on the Lab tab, or click the plot to choose a file; it goes into the form chosen on the Data tab.
4. Read the message: for example **74 rows into the acid form — peak 506 nm, ε 25300**, or **Not installed:** with a reason.

What changes

- The file is resampled onto the 5 nm grid from 380 to 780 nm by straight-line interpolation between your rows. Beyond the ends of your file, the first and last values are held flat.
- That form's λmax and ε become the peak of your curve.

- The dye's tag in the components list changes from **Stand-in** to **Measured**.
- On the **Data** tab, **The absorption band shape** entry changes to **Verbatim**, with your file name and the span it covered.
- The CSV header and JSON mark the band shape as **MEASURED** and the component's provenance as `measured`.
- **Back to the drawn bands** appears on the Data tab.

The dye's pKa and concentration are unchanged; the pH still sets how much of it is in each form. If you have scans of both forms, load both.

Go back

Select **Back to the drawn bands**. The selected dye's bands return to its preset, and every measured curve in the bath is cleared. A message confirms **Back to the drawn bands**. **Reset the bath** also clears measured curves.

Measured spectra are not carried by links and are lost when you reload the page.

You should now have a dye in the bath whose band is your own measurement.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Dye in Solution reference

Every tab, control, readout, message and file format in Dye in Solution, with ranges and defaults.

Address: https://auricartisan.com/tool/general/spectral-tools/industrial-dye-chemistry/

Tabs

Tab	Contents
Lab	Bath, light and component controls; the spectrum plot; colour, absorbance and wavelength readouts; the lights row
Method	Four explanation views and the tolerance sweep
Safety	The withdrawn hazard gate and what stays
Data	The register of 18 tables and formulas, filterable by status; the tab label shows the count; measured-spectrum loader
Export	Spectrum CSV, bath JSON and link
Reference	Thirteen sources, filterable

Links in the page text (for example **Method** or **Safety** in the note strip) switch to that tab.

Lab tab — The bath

Control	Range and step	Default
pH	0–14, step 0.05	3.50
Path length	0.1–10 cm, step 0.1	1.0 cm
Run the titration	Steps pH from 0 to 14 by 0.2 every 90 ms; reads **Stop** while running	—

Lab tab — The light

Control	Values	Default
Illuminant	**D65 — daylight, 6504 K**, **D50 — daylight, 5003 K**, **A — a blackbody at 2856 K**, **E — equal energy**	D65

Lab tab — The components and the selected dye

Control	What it does	Values or range	Default
Component buttons	Select a dye; show its swatch, pKa, concentration, band peaks and **Stand-in** or **Measured**	1 to 4 dyes	Weak-acid indicator
Dye menu	The dye **Add** will add	Six presets	Weak-acid indicator
Add	Adds the dye in the menu and selects it	Up to 4 components	—
Mute	Toggles the selected dye out of and back into the bath	—	—
Remove	Deletes the selected dye	At least 1 remains	—
pKa	The selected dye's pKa	0–14, step 0.1	The preset's
Concentration	The selected dye's concentration	0.5–200 µM, step 0.5	The preset's
Reset the bath	Default bath, clears sweep and measured spectra	—	—

Preset dyes, with pKa, default concentration, and acid and base λmax, ε and band width (Gaussian standard deviation):

Dye	pKa	µM	Acid form	Base form
Weak-acid indicator	3.5	34	508 nm, 26,000, 34 nm	462 nm, 22,000, 30 nm
Mid-range indicator	7.1	20	430 nm, 15,000, 30 nm	615 nm, 34,000, 32 nm
High-range indicator	9.4	25	400 nm, 900, 40 nm	553 nm, 31,000, 30 nm
Azo red, pH-stable	12.5	50	505 nm, 30,000, 36 nm	512 nm, 29,000, 36 nm
Anthraquinone blue, pH-stable	11.8	40	600 nm, 16,000, 26 nm	606 nm, 16,500, 26 nm
Yellow, pH-stable	12.0	60	428 nm, 21,000, 28 nm	431 nm, 21,000, 28 nm

Lab tab — Spectrum plot

Element	Description
Acid form `HA`	The selected dye's acid-form molar absorptivity, dashed
Base form `A⁻`	The base-form molar absorptivity, dashed
What you measure	Their weighted sum at the current pH, filled
Isosbestic	Green circle and label where the forms cross, when the crossing is clear
Axes	380–780 nm; molar absorptivity in thousands (k)
Drop and click	Drop a spectrum file on the plot, or click it to choose one

Lab tab — Readouts

Readout	Meaning	Format
sRGB	Un-adapted colour under the chosen light	Hex
CIE L*	Lightness against the light's own white	One decimal
a*, b*	Opponent coordinates against the light's own white	One decimal each
Base fraction	Share of the selected dye in the base form	Three decimals
Peak absorbance	Highest bath absorbance	Three decimals
λmax	Wavelength of that peak	nm
Beer–Lambert holds	**yes, A x.xx** or **no — A x.xx is past 1** (highlighted)	—
Isosbestic point	**N nm**, **N nm (weak)** or **none in range** (last two highlighted)	—
Dominant λ	Dominant wavelength, or complementary with a `c` prefix	nm
Excitation purity	Distance from white towards the spectral edge	%, one decimal, or **—**
Inside sRGB	**yes** or **no** (highlighted)	—
Observer	Fixed	CIE 1931 2°

Lab tab — The same bath under every light

Swatches for **D65**, **D50**, **A** and **E**; the current light is labelled **— this bath** with its hex, the others show **ΔE_{00}** from it. An **F11** card reads **not held**.

Method tab

Control	Values or range	Default
Method	**The titration**, **Beer–Lambert**, **The light**, **What was resampled**	The titration
Concentration (sweep)	±0–20%, step 0.5	±2.0%
Path length (sweep)	±0–2 mm, step 0.01	±0.10 mm
Molar absorptivity (sweep)	±0–30%, step 0.5	±5.0%
Draws	100–2000, step 100	500
Run the sweep	Runs the sweep; shows **Median**, **95th percentile** and **Worst draw** in ΔE_{00}	—

Safety tab

Shows **GHS hazard assessment and PPE compliance gate**, marked **Withdrawn**, with a table of what an earlier version did and what GHS classification actually is, and **What stays, because it is true and needs no calculation**: gloves, eye protection and a lab coat for any dye work; the supplier's safety data sheet (section 2 for the GHS classification, section 8 for exposure controls and PPE); and disposing of dye liquor through your institution's waste stream.

Data tab

Control	What it does
Status	**All** (18), **Verbatim** (6), **Computed** (5), **Stand-in** (4), **Absent** (3)
Register row	Select, or press `Enter` or `Space`, to expand its note and **what it would take**
Form menu	**Into the acid form** or **Into the base form**
Drop a spectrum	Loads a measured spectrum; click or `Enter` / `Space` opens a file chooser
Back to the drawn bands	Shown after loading; restores drawn bands

Entries: **CIE 1931 2° standard observer**, **CIE D-series eigenvector basis**, **sRGB primaries and transfer curve**, **CIEDE2000 colour difference**, **The Beer–Lambert law**, **Henderson–Hasselbalch** (Verbatim); **The two-species absorbance**, **The isosbestic point**, **Dominant wavelength and excitation purity**, **Colour inconstancy index**, **Special metamerism index, change of illuminant** (Computed); **The absorption band shape**, **The six dye presets**, **The default sweep tolerances**, **Kubelka-Munk on an opaque substrate** (Stand-in); **The CIE fluorescent illuminants (F-series)**, **Colourfastness and content testing**, **GHS classification and PPE compliance** (Absent). The metamerism index and Kubelka–Munk entries describe methods that have no control in the current interface.

Export tab

Control	What it does
What to take	Preview **The spectrum**, **The whole bath** or **A link to this**
Download the spectrum	Saves `dye-in-solution-spectrum.csv` (81 rows)
Download the bath	Saves `dye-in-solution-bath.json`
Copy a link to this bath	Copies a link

Reference tab

Control	Values
Reference	**All** (13), **Runs** (6), **Partly** (1), **Not held** (6)

Messages

Message	Meaning
Titration complete	The titration reached pH 14
Four components is the ceiling here	A fifth dye was refused
The bath needs at least one dye	The last dye cannot be removed
Bath reset	Defaults restored
Swept N draws in M ms	The sweep finished
N rows into the acid form — peak N nm, ε N	A spectrum was loaded
No two-column numeric rows in that file	The file had no usable rows

Message	Meaning
Not installed: fewer than eight rows / fewer than eight usable rows	Too few rows
Not installed: covers X to Y nm; at least 420 to 680 is needed …	The file does not span 420–680 nm
Back to the drawn bands	Measured spectra cleared
Link copied	The link is on the clipboard

File formats

File	Format	Contents
`dye-in-solution-spectrum.csv`	CSV with `#` header lines	`wavelength_nm,absorbance,transmittance,eps_acid,eps_base,observer_x,observer_y,observer_z`
`dye-in-solution-bath.json`	JSON	See Export and share
Measured spectrum (input)	`.csv`, `.tsv` or `.txt`	Two columns: wavelength (nm), molar absorptivity (M^{-1} cm^{-1}); at least 8 rows spanning 420–680 nm

Link parameters

`ph`, `path` (cm), `ill` (`d65`, `d50`, `a`, `e`), `dyes` (`id:pKa:µM`, comma-separated; ids `weak-acid`, `mid-range`, `high-range`, `azo-red`, `anthraquinone-blue`, `yellow`).

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Troubleshooting Dye in Solution

Symptoms, causes and fixes for Dye in Solution: flags, refused spectra, component limits, the plot and links.

"Beer–Lambert holds" reads "no"

Why: the bath's absorbance is above 1 at some wavelength. Past that point real solutions depart from the law and spectrophotometers become unreliable, so the computed colour is less trustworthy.

What to do: lower **Concentration** or **Path length**. For the default dye at pH 3.5, the limit is reached at about 54 µM in a 1 cm path.

Moving pH changes almost nothing

Why: the selected dye is one of the pH-stable presets (azo red, anthraquinone blue, yellow), whose two forms have nearly the same band, or the pH is far from the dye's pKa, where almost all of it is already in one form.

What to do: use an indicator preset, or move the pH to within about two units of the pKa (watch **Base fraction**).

"Isosbestic point" is highlighted or reads "(weak)" or "none in range"

Why: (weak) means the two forms cross so deep in their tails (below 5% of the peak) that the point cannot be seen on the plot. **none in range** means they do not cross between 380 and 780 nm.

What to do: nothing is wrong; it describes the dye. The mid-range and high-range indicators have weak crossings.

The plot does not change when I change concentration or path length

Why: the plot shows the selected dye's molar absorptivity (a property of the dye), not the bath's absorbance.

What to do: read **Peak absorbance** and **λmax** for the bath, or export the spectrum CSV, whose `absorbance` column is the whole bath.

The plot shows a different dye from the one I am editing

Why: the plot follows the selected dye. Adding a dye selects the new one.

What to do: select the dye's button in **The components** list.

A file chooser opens when I click the spectrum

Why: the plot is also a drop zone for measured spectra; clicking it opens the file chooser.

What to do: cancel the chooser if you did not mean to load a file.

My spectrum file is refused

Message	Fix
No two-column numeric rows in that file	Make sure each data line has two numbers: wavelength and molar absorptivity
Not installed: fewer than eight rows	Supply at least eight rows
Not installed: covers X to Y nm; at least 420 to 680 is needed …	Extend the scan to cover 420–680 nm

The loader expects molar absorptivity. A file of raw absorbance loads but gives the wrong strength; convert it first (see Use a measured spectrum).

I cannot add a fifth dye or remove the last one

Why: a bath holds one to four components.

What to do: remove or mute a dye first; to empty the bath, use **Reset the bath** and build again.

"Inside sRGB" reads "no"

Why: the bath's colour under this light is more saturated, or lighter, than sRGB can show. The swatch is the nearest colour a screen can display. Pale, strongly tinted baths such as the default often fall just outside.

What to do: nothing; the L*a*b* values are still exact for the model.

Dominant wavelength starts with "c" and purity reads "—"

Why: the colour is a purple, for which no single spectral wavelength exists; the page gives the complementary wavelength instead.

What to do: nothing; the note under the readouts explains it.

The sweep result looks out of date

Why: the sweep is not rerun when you change the bath.

What to do: select **Run the sweep** again on the **Method** tab before exporting.

A link did not restore a muted dye or my measured spectrum

Why: links carry pH, path length, the light and each dye's preset, pKa and concentration only.

What to do: mute the dye and load the spectrum again after opening the link.

I need F11, a fastness result or hazard information

Why: the page does not hold fluorescent illuminant tables, cannot perform physical fastness tests, and holds no toxicological data.

What to do: see Limits and accuracy. For hazards, read the supplier's safety data sheet.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Dye in Solution FAQ

Short answers to common questions about Dye in Solution.

Are the preset dyes real products?

No. Each stands for a class of dye, such as "the methyl-orange family", with constants chosen by the tool to be plausible. None is a named commercial substance, and every preset is marked **Stand-in**.

Can I use it to formulate a real recipe?

Not as it stands. The model (two forms, Beer–Lambert, a stated light and observer) is sound, but the dye data are the tool's own and the bands are Gaussian. With measured spectra for both forms of your dyes it becomes much closer, but it still models dye in solution, not on cloth.

Does it model dye on fabric?

No. There is no substrate or reflectance mode in the current interface. The Data register describes a Kubelka–Munk model for opaque substrates as a stand-in, and the Reference tab explains why it is not offered.

Why is there no F11 or other fluorescent light?

Fluorescent illuminants are measured spectra with narrow emission lines, published as tables, and the page does not hold them. It will not substitute a smooth imitation under that name.

Is the "ΔE00" in the lights row a metamerism index?

No. It is colour inconstancy: one bath under two lights. Metamerism concerns two baths that match under one light and not under another. See Compare lights.

Which colour difference formula does it use?

CIEDE2000 (CIE 142-2001), the full formula including its rotation term, checked against published test pairs.

Why do the swatch and the L*a*b* disagree when I change the light?

On purpose. The swatch shows the light leaving the bath without adaptation, so a bath under tungsten looks warm. L*a*b* is taken against the chosen light's own white, so it measures the dye's colour relative to that light.

Can the tool tell me if a dye is hazardous?

No. The page holds no toxicological data. Hazard classification belongs to each substance and is on its supplier's safety data sheet (section 2; section 8 covers exposure controls and protective equipment). See the **Safety** tab.

Does it do lightfastness or wash-fastness tests?

No. ISO 105-B02, ISO 105-C06, AATCC TM 61 and ISO 14184-1 are laboratory procedures on physical cloth and cannot be performed in a browser.

Is my data uploaded?

No. Everything, including loaded spectra, stays in the page. See Privacy.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Dye in Solution background

Contents of the background material for the Dye in Solution tool.

This folder supports the how-to pages with the chemistry and colour terms the tool uses, its keyboard and pointer actions, a statement of what is published, what is the tool's own and what is not held (including safety), related tools and articles, its history and what stays on your device.

File	What it covers
glossary.md	Chemistry and colour terms used in Dye in Solution
shortcuts.md	Drop zones, `Enter` and `Space`, and tab links
limits-and-accuracy.md	The data register entry by entry, precision, what is not held, and safety
related.md	Related tools on the site and Learn articles
release-notes.md	User-facing history of the tool
privacy.md	What stays in your browser

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Dye in Solution shortcuts

Keyboard and pointer actions in the Dye in Solution tool.

The tool has no keyboard shortcuts of its own beyond these. Site-wide shortcuts are described in Site features.

Action	Where	Result
Drop a file	The spectrum plot on the Lab tab, or **Drop a spectrum** on the Data tab	Loads a measured spectrum into the form chosen on the Data tab
Click	The spectrum plot or **Drop a spectrum**	Opens a file chooser
`Enter` or `Space`	Focused **Drop a spectrum** area	Opens a file chooser
`Enter` or `Space`	A focused row on the **Data** tab	Expands or collapses the row
Click a tab name in the page text	Note strip and hints	Switches to that tab
`Tab`	Anywhere	Moves between controls; sliders then respond to the arrow keys

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Dye in Solution glossary

Chemistry and colour terms used in the Dye in Solution tool, each explained briefly.

Term	Meaning
Absorbance (A)	$-\log_{10}$ of the fraction of light transmitted; A = 1 lets 10% through
Acid form (HA)	The protonated form of a dye, holding an extra hydrogen ion
Base form (A^-)	The deprotonated form of a dye
Base fraction (f)	The share of a dye in its base form at a given pH, from Henderson–Hasselbalch

Term	Meaning
Beer–Lambert law	A = ε × c × l: absorbance is proportional to molar absorptivity, concentration and path length
CIE 1931 2° observer	The standard set of colour-matching functions describing average human colour vision for small fields
CIE L*a*b*	A colour space with lightness L* and opponent axes a* (red–green) and b* (yellow–blue)
CIEDE2000 (ΔE_{00})	The CIE's recommended formula for how different two colours look
Colour inconstancy	How far one sample's colour changes between two lights
Complementary wavelength	For purples, the wavelength on the opposite side of the white point, given instead of a dominant wavelength
Concentration (c)	Amount of dye per volume; shown in micromoles per litre (µM)
Dominant wavelength	The spectral colour that, mixed with the light's white, matches the sample
Excitation purity	How far a colour lies from the white point towards the spectral edge, as a percentage
Gaussian band	A bell-shaped absorption curve; the tool draws preset bands this way
GHS	The Globally Harmonized System for classifying and labelling chemical hazards, per substance
Henderson–Hasselbalch	The relation f = 1 ÷ (1 + 10^(pKa − pH)) between pH, pKa and the base fraction
Illuminant	A standard light defined by its spectral power: D65, D50 (daylights), A (tungsten), E (equal energy)
Isosbestic point	The wavelength where both forms of a dye absorb equally, so absorbance there does not change with pH
Kubelka–Munk	A model of colour in opaque, light-scattering layers such as dyed cloth; not offered in the current interface
λmax	The wavelength at which a band absorbs most
Metamerism	Two samples with different spectra that match under one light but not under another
Molar absorptivity (ε)	How strongly one mole of dye absorbs at a wavelength, in M^{-1} cm^{-1}
Path length (l)	The distance light travels through the solution, in cm
pKa	The pH at which half of a dye is in each form
Safety data sheet (SDS)	The supplier's document for a substance, giving its hazards (section 2) and controls (section 8)
sRGB	The standard colour space of screens and the web
Transmittance (T)	The fraction of light at a wavelength that passes through the solution
UV–Vis spectrum	A measurement of absorbance across ultraviolet and visible wavelengths

Limits and accuracy

What Dye in Solution takes from published sources, what is its own, how precise it is, what it does not hold, and what it cannot tell you about safety.

What the tool is

A spectral model of a dye bath for learning and exploration. The physics and colorimetry follow published standards; the dyes are the tool's own inventions unless you load measured spectra. It is not a formulation system, a spectrophotometer, a fastness tester or a safety tool.

The data register, in plain terms

Entry	Status	What it means for you
CIE 1931 2° standard observer	Verbatim	Colours use the published 5 nm observer table, with no interpolated samples
CIE D-series eigenvector basis	Verbatim	D65 and D50 are reconstructed from the published daylight basis and land within 0.0003 of their published chromaticities
sRGB primaries and transfer curve	Verbatim	The swatch and gamut flag use the published sRGB definition
CIEDE2000 colour difference	Verbatim	Every ΔE_{00} uses the full CIE 142-2001 formula
The Beer–Lambert law	Verbatim	Absorbance is ε × c × l, additive over dyes and forms, flagged above A = 1
Henderson–Hasselbalch	Verbatim	The base fraction follows the standard relation
The two-species absorbance	Computed	Each dye is two forms weighted by pH, which is what produces an isosbestic point
The isosbestic point	Computed	Located precisely between samples, with a strength
Dominant wavelength and excitation purity	Computed	A proper chromaticity construction from the light's white point, with purples flagged
Colour inconstancy index	Computed	ΔE_{00} for one bath between two lights; not a metamerism index
Special metamerism index	Computed	Described in the register; no control uses it in the current interface
The absorption band shape	Stand-in	Preset bands are Gaussians; real bands are asymmetric; loading a measured spectrum makes this Verbatim
The six dye presets	Stand-in	Plausible constants for classes of dye, never a named substance
The default sweep tolerances	Stand-in	±2% concentration, ±0.1 mm path, ±5% ε are the tool's guess; replace them with yours
Kubelka–Munk on an opaque substrate	Stand-in	Described in the register; no cloth mode is offered in the current interface
The CIE fluorescent illuminants	Absent	F11 and the other F-series lights are not held and not imitated
Colourfastness and content testing	Absent	ISO 105-B02, ISO 105-C06, AATCC TM 61 and ISO 14184-1 are physical laboratory tests
GHS classification and PPE compliance	Absent	The page holds no hazard data; see Safety below

Precision

- The spectrum has 81 samples, 380–780 nm at 5 nm. Features narrower than about 10 nm are not resolved, and light below 380 nm or above 780 nm is ignored.
- Beer–Lambert is flagged, not corrected: above A = 1 the numbers are the law's, and a real solution would absorb less.
- Gaussian bands put the peak position and height about right and the wings wrong. Mixtures, whose colour depends on where bands overlap, are affected most.
- The swatch is limited to what sRGB can show; **Inside sRGB** says when it is clipped. L*a*b* values are not clipped.
- Your own screen is assumed to be sRGB.

What the tool does not do

- It models dye in solution, not on cloth, paper or other substrates.
- It does not compare two baths for metamerism in the current interface.
- It offers no fluorescent illuminants.

- It performs no fastness, wash or formaldehyde testing.
- It does not account for temperature, ionic strength, solvent, dye aggregation or fluorescence.

Safety

This page models what a dye does to light. It holds no toxicological data, and a λmax and a molar absorptivity say nothing about toxicity. GHS hazard classification belongs to each substance, from its own data, and is published on the supplier's safety data sheet (section 2 for the classification, section 8 for exposure controls and personal protective equipment). The page's **Safety** tab also lists what always applies: gloves, eye protection and a lab coat for any dye work; reading the SDS before opening a container; and disposing of dye liquor through the waste stream your institution specifies. An earlier version of the page assigned hazard levels by chemical class and blocked exports until boxes were ticked; that gate was withdrawn.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Privacy

What Dye in Solution keeps in your browser and what uses the network.

What stays in your browser

- Every calculation: spectra, colours, the lights row and the tolerance sweep are computed in the page.
- Spectrum files you load are read from your disk by the page. They are not uploaded and are forgotten when you reload.

What leaves the page, and only when you ask

- **Download the spectrum** and **Download the bath** create files through your browser's normal download. The bath file includes the name of any spectrum file you loaded.
- **Copy a link to this bath** writes a link to your clipboard. It contains the pH, path length, light and each dye's preset, pKa and concentration, never your spectrum files.

The rest of the page

The tool page loads the site's usual pages, styles and scripts, and, depending on your cookie choices, advertising and consent-based analytics, like the rest of auricartisan.com. See the Privacy Policy at https://auricartisan.com/legal/privacy/ and the Cookie Policy at https://auricartisan.com/legal/cookies/.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Release notes

User-facing history of the Dye in Solution tool.

The site-wide history is at https://auricartisan.com/library/changelog/. The entries below cover what changed for users of this tool.

September 2026 — rebuilt as Dye in Solution

The page was rebuilt with six tabs (**Lab**, **Method**, **Safety**, **Data**, **Export**, **Reference**).

Changed:

- Each dye is now two forms (acid and base) weighted by pH, so the pH control changes every dye's colour through its pKa, and a titration shows an isosbestic point. Earlier, most presets did not respond to pH at all.
- The colour is computed under a named light source, and L*a*b* is taken against that light's own white. Earlier there was no light source.
- The spectrum is sampled at 5 nm from 380 to 780 nm, the grid the CIE observer is published on.
- The preset list is now six dye classes with the tool's own constants, instead of named commercial dyes with unsourced values.
- **Dominant λ** is now a true dominant wavelength with excitation purity, rather than the wavelength of peak transmittance.

Added: the illuminant menu (D65, D50, A, E) and the lights row with colour inconstancy; the **Beer–Lambert holds** check; **Base fraction** and **Isosbestic point**; the tolerance sweep; loading measured spectra; the **Data** register; header lines stating the method in every export.

Removed:

- The confidence interval that resampled the recipe, replaced by the tolerance sweep.
- The GHS hazard levels and the tick-box gate on exports; the **Safety** tab explains why.

Earlier

- An Industrial Dye Chemistry User Guide for the previous version was published in the site documentation in May 2026; it is now marked as superseded.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › INDUSTRIAL DYE CHEMISTRY

Related tools and articles

Tools and Learn articles related to Dye in Solution.

Tools in this group

- Material Lighting — the same spectral approach for opaque materials: reflectance, illuminant and surface model.
- Perception and Vision — check how a dyed colour looks to viewers with colour vision deficiency.

Other tools on the site

- Illuminants — the standard light sources in detail.
- Chromatic adaptation — how colours are predicted to shift when the light changes.

Learn articles

The Learn section is described in Learn.

- Color on Textiles: Dyeing, Printing, and Why Fabric Color Is Hard — https://auricartisan.com/library/learn/articles/2026-07-10-color-on-textiles-dyeing-and-printing
- Metamerism Explained — https://auricartisan.com/library/learn/articles/2026-05-25-metamerism
- Spectral Power Distributions and Why RGB Is Not Enough — https://auricartisan.com/library/learn/articles/2026-05-24-spectral-power-distributions
- ΔE Metrics from CIE76 to CIEDE2000 and ΔEITP — https://auricartisan.com/library/learn/articles/2026-05-26-delta-e-metrics-cie76-ciede2000-itp

Documentation on the site

- Dye in Solution Developer Reference — https://auricartisan.com/library/documentation/articles/2026-06-02-dye-in-solution-developer-reference (for how the model works)
- The earlier Industrial Dye Chemistry User Guide describes a previous version of the tool and is marked as superseded on the site.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Material Lighting (Material Under Light)

What the Material Under Light tool does, who it is for, where to find it and how to render a first material under a chosen light.

The Material Lighting tool, whose page is titled **Material under light**, computes the colour of a surface the way light actually produces it: one wavelength at a time. It multiplies three things at each of 81 wavelengths from 380 to 780 nm: the power of a light source (the illuminant), the share of that light the material reflects (its reflectance spectrum), and how much of the reflected light travels towards your eye at the angles you set (the reflectance lobe). It then turns the result into a colour with the CIE 1931 standard observer and shows it as a shaded sphere, a swatch and a set of numbers.

Because the calculation is spectral, the tool can show things a three-channel RGB renderer cannot: why the same material looks different under daylight and tungsten, and why two materials that match under one light can part under another (metamerism). It also compares your material under your chosen light with the same material under the reference light the CIE rule names for that light's colour temperature, and reports the difference as a CIEDE2000 colour difference.

The twenty material presets are the tool's own curves, built from a flat level plus bell-shaped bumps; none is a measurement. You can load a measured reflectance file instead. The two colour rendering scores an earlier version printed (CRI and TM-30) are withdrawn, and the **Metrics** tab explains why. Everything runs in your browser.

Who it is for

- Designers, 3D artists and lighting designers who want to see how a light source changes a material's colour.
- Students of colour science learning spectral power distributions, reflectance, colour temperature and reference illuminants.
- Anyone curious why gold, skin or a leaf look different under a warm lamp.

Where to find it

Open https://auricartisan.com/tool/general/spectral-tools/material-lighting/ in a current browser. It is listed as **Material Lighting** on the Collections page (https://auricartisan.com/collections/) and in the site menu.

Quick start

1. Open the tool. The scene is **Gold** under **D65 — daylight, 6504 K**, with the **GGX** lobe, roughness 0.40, light from 30° and view from 20°.
2. Look at the spectrum plot: the illuminant (dashed), the reflectance, and the light leaving the surface (filled).
3. Switch **The light** to **A — a blackbody at 2856 K** and watch the sphere and swatch warm up.
4. Read **Measured temperature** and **Reference for it**, and the ΔE_{00} between **Under your source** and **Under its reference**.
5. Choose **Skin** under **The material**, then select **Conductor** and back to **Dielectric**.
6. Pick a different lobe card, such as **Oren-Nayar**, and move **Roughness**.
7. Select **Save the spectrum** to download the 81-row CSV.

What you can do

- Light a material with D65, D50, D55, D75, A, E, or a blackbody or daylight at a temperature you choose.
- Choose from twenty materials, or load a measured reflectance file.
- Set the surface as a dielectric with a refractive index, or a conductor (metal), and its roughness.
- Choose one of five published reflectance lobes: GGX, Beckmann, Ward, Ashikhmin-Shirley or Oren-Nayar.
- Set the light angle, the view angle and the azimuth between them.

- Read the colour as sRGB, CIE L*a*b* and Jz, with the gamut flag.
- Read the source's measured colour temperature with its error band, and compare the material with its CIE reference light.
- Run a sensitivity sweep that perturbs the reflectance curve.
- Export the spectrum as CSV, the whole scene as JSON, or a link.

In this folder

- docs/ — how to use the tool.
  - Getting started — open it, the screen tour, the first scene.
  - Set up a scene — the light, the material, the surface, the lobes and the geometry.
  - Compare with the reference light — colour temperature, the reference rule, the colour difference and the two white points.
  - Run a sensitivity sweep — the Metrics tab and the withdrawn rendering indices.
  - Use a measured reflectance — load your own reflectance file.
  - Export and share — the spectrum CSV, the scene JSON and links.
  - Reference — every control, readout, tab and file format.
  - Troubleshooting — symptoms, causes and fixes.
  - FAQ — short answers to common questions.
- others/ — background material.
  - Glossary — terms used in this tool.
  - Shortcuts — keyboard and pointer actions.
  - Limits and accuracy — what is published, what is the tool's own, what is not held.
  - Related — related tools and Learn articles.
  - Release notes — user-facing history.
  - Privacy — what stays in your browser.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Using Material Under Light

Contents of the how-to documentation for the Material Under Light tool.

These pages explain how to set up a scene, compare it with its reference light, test its sensitivity, load your own data and export. Start with getting started; the reference lists every control.

File	What it covers
getting-started.md	Opening the tool, the six tabs, the Lab screen tour and a first scene
set-up-a-scene.md	Illuminants and colour temperature, the twenty materials, dielectric and conductor, roughness, the five lobes, the three angles
compare-with-the-reference.md	Measured colour temperature, the reference rule, the rendering shift, the two white points, metamerism
run-a-sensitivity-sweep.md	The Metrics tab: the sweep, and why CRI and TM-30 are withdrawn
use-a-measured-reflectance.md	Loading a reflectance file and going back to the presets
export-and-share.md	The spectrum CSV, the scene JSON and links
reference.md	Every tab, control, readout, message and file format, with ranges and defaults
troubleshooting.md	Black spheres, disabled sliders, clipping flags, refused files, links
faq.md	Short answers to common questions

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Getting started with Material Under Light

Open Material Under Light, learn the layout of the Lab tab and read the default gold-under-daylight scene.

Open the tool

Go to https://auricartisan.com/tool/general/spectral-tools/material-lighting/. The page opens on the **Lab** tab with a default scene: **Gold**, as a dielectric with refractive index 1.50, under **D65 — daylight, 6504 K**, with the **GGX** lobe at roughness 0.40, the light 30° and the eye 20° from the surface normal, and 0° between them.

A note strip above the tabs, which cannot be closed, says that two rendering figures from an earlier version were withdrawn and links to the **Metrics** and **Method** tabs.

The six tabs

Tab	What it is for
Lab	Set up the scene, see the spectrum and the sample, read the colour, compare with the reference light
Method	How a spectrum becomes a colour, how the reference is chosen, the five lobes, the two white points
Metrics	The sensitivity sweep, and the two withdrawn rendering indices
Data	The register of 17 tables and formulas, with a status for each; load a measured reflectance here
Export	The spectrum CSV, the scene JSON and a link
Reference	Fourteen sources, filterable

A tour of the Lab tab

The rail (left), top to bottom:

- **The light** — the illuminant menu and **Temperature** (used only by the last two options).
- **The material** — the material menu, **Dielectric** / **Conductor**, and **Refractive index**.
- **The surface** — **Roughness**.
- **The geometry** — **Light from**, **Seen from** and **Between them**.
- **Save the spectrum** and **Reset the scene**.

The stage (right):

- **The spectrum** — three curves against wavelength: **Illuminant** (dashed), **Reflectance** (solid) and **What leaves the surface** (filled), each scaled to its own maximum. The legend names the illuminant, the material and whether it is synthesised or measured.
- **The sample** — a sphere of the material lit from your light angle, with the note **scaled so a white card here would read Y = 1** and, when colours had to be clipped, a **clipped to sRGB** badge.
- Five lobe cards: **GGX**, **Beckmann**, **Ward**, **Ashikhmin-Shirley** and **Oren-Nayar**, each with a one-line description and its source.
- Two rows of readouts: **sRGB**, **CIE L***, **a*, b***, **Jz**; **Measured temperature**, **Reference for it**, **Reflectance**, **Inside sRGB**.
- **Against the reference the standard names** — two swatches, **Under your source** and **Under its reference**, with the ΔE_{00} between them.

Read the default scene

1. In the plot, the dashed D65 curve is fairly flat, the gold reflectance rises from low values in the blue to a plateau in the orange and red, and the filled product follows the reflectance.
2. The swatch reads **sRGB** `#FFD026`, with **CIE L*** 87.2 and **a*, b*** 9.6, 82.9: a strong yellow.
3. **Inside sRGB** reads **no** and the sphere shows **clipped to sRGB**: this gold is more saturated than sRGB can show, so the screen shows the nearest colour it can.
4. **Measured temperature** reads **6500 K +/- 11 K**: the page measured the D65 spectrum's colour temperature back out of it. **Reference for it** reads **daylight 6500 K**.
5. **Under your source** and **Under its reference** look the same, and the difference reads about 0.01: D65 is its own reference.
6. **Reflectance** reads **synthesised**: the gold curve is the tool's own.

Now switch **The light** to **A — a blackbody at 2856 K**. The sphere and swatch turn orange (`#FFC500`), **Measured temperature** reads about 2860 K, and **Reference for it** becomes **blackbody 2860 K**.

You should now see a scene whose spectrum, sample and numbers follow every control.

Next steps

- Set up a scene covers every control.
- Compare with the reference light explains the colour temperature and the reference swatches.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Compare with the reference light

Read the measured colour temperature and its error band, understand how the reference light is chosen, and interpret the colour difference between a material under its source and under that reference.

Colour temperature

The **correlated colour temperature** (CCT) of a light is the temperature of the blackbody whose colour is closest to it, in kelvin. Low values (2700 K) are warm and orange; high values (6500 K and above) are cool and bluish.

Measured temperature gives the CCT the page computes back from the illuminant spectrum you chose, using McCamy's formula, rounded to 10 K, with its error band. The error bands were measured against blackbodies of known temperature:

Temperature	Worst measured error
1800–2500 K	±18 K
2500–4000 K	±13 K
4000–7000 K	±11 K
7000–10000 K	±139 K
10000–15000 K	±1069 K

Above 7000 K the readout is highlighted and adds **McCamy's fit is out of its range up here**. The reference chosen is a daylight either way.

The reference rule

To judge how a light renders colours, the CIE compares it with a reference light of the same CCT (CIE 13.3, section 5.1):

- below 5000 K, a **blackbody** at that temperature;
- at or above 5000 K, a **CIE daylight** at that temperature.

Reference for it shows the choice (for example **daylight 6500 K** or **blackbody 2860 K**), and the note above the swatches gives the reason, for example **below 5000 K the reference is a Planckian radiator**. A light that is its own reference, such as D65 or a blackbody, compares with itself.

The colour difference

Against the reference the standard names shows the material twice: **Under your source** (captioned with your illuminant) and **Under its reference** (captioned **D** or **BB** and the temperature). Between them is ΔE_{00}, the CIEDE2000 colour difference.

Each colour is taken against its own light's white, so what remains is how the **material** moved, not the fact that the two lights differ. That is why a source that is its own reference reads about zero: D65 about 0.01, a 2700 K blackbody about 0.002, illuminant A about 0.05. The small remainders come from the error in measuring the temperature.

Some examples with the preset materials and default surface:

Scene	Reference	ΔE_{00}
Gold under D65	Daylight 6500 K	About 0.01
Gold under E	Daylight 5460 K	About 2.0
Gold under a 10000 K blackbody	Daylight 9860 K	About 3.1
Skin under 4000 K daylight	Blackbody 4010 K	About 1.6
Leaf under illuminant A	Blackbody 2860 K	About 0.05

This is a colour difference for **this one material**. It is not a rendering index and does not characterise the lamp: a rendering index needs a specified set of many measured samples, which the tool does not hold. The line under the swatches says so, and adds, for a synthesised material, that the difference is a property of the model rather than of a real pigment.

Two white points

The page uses two white points on purpose, and the **Method** tab's **Two white points** view explains them:

Readout	White point	Question it answers
The swatch and its hex	None: un-adapted	What does this material look like under this light?
L*, a*, b* and every ΔE	The illuminant's own	How far did the material move?

Metamerism

Two materials with different reflectance spectra can produce the same colour under one light and different colours under another. This is **metamerism**, and it is why a spectral tool is needed to predict it: RGB values alone cannot tell you whether a match will hold. The tool shows one material at a time; to explore metamerism, note a material's L*a*b* under two lights, then compare with another material (for example a measured file) under the same two lights.

The Method tab's reference view

Method › **Choosing the reference** shows the rule as a table of D65, D50, illuminant A and a 2700 K blackbody against the reference each should get, the published and computed chromaticities of D50, D55, D65 and D75, and the colour-temperature error bands.

You should now be able to say how far a material's colour moves between a light and its reference, and what that number does and does not mean.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Export and share

Download the 81-row spectrum CSV and the whole-scene JSON, and copy a link that rebuilds the scene.

Nothing leaves the page until you press a button. **Save the spectrum** on the Lab tab and **Download the spectrum** on the **Export** tab save the same file.

On the Export tab, **What to take** switches the preview between **The spectrum** (the first lines of the CSV), **The whole scene** (the first part of the JSON) and **A link to this**; the note above it gives the row count.

Button	Result
Download the spectrum	Saves `material-under-light-spectrum.csv`
Download the scene	Saves `material-under-light-scene.json`
Copy a link to this scene	Copies a link; **Link copied** confirms it, or the link is shown if the clipboard is unavailable

The spectrum CSV

The file starts with `#` header lines:

Line	Contents
`sampling`	380–780 nm at 5 nm, 81 samples
`observer`	CIE 1931 2°, published at 10 nm, interpolated to 5
`illuminant`	The light and its definition
`measured_cct`	The measured colour temperature and its error band
`reference`	The reference light's kind, temperature and the reason
`material`, `reflectance`	The material, and whether its curve is MEASURED (with the file name) or SYNTHESISED
`surface`	Conductor, or dielectric with its refractive index; the lobe; the roughness
`geometry`	Light, view and azimuth angles
`exposure`	The scale that makes a white diffuser read Y = 1
`swatch_white`, `lab_white`	That the swatch is un-adapted and L*a*b* uses the light's own white
`rendering_shift`	ΔE_{00} for this sample between the source and its reference
`NOT_A_RENDERING_INDEX`	That this is one sample, and that CRI and TM-30 are withdrawn
`sensitivity`	The sweep's median and 95th percentile, if you ran one

Then 81 rows:

Column	Meaning
`wavelength_nm`	380 to 780 in steps of 5
`illuminant_relative`	The illuminant's power, scaled to peak at 1
`reflectance`	The material's reflectance
`radiance_relative`	The light leaving the surface at your geometry, scaled by the exposure
`observer_xbar`, `observer_ybar`, `observer_zbar`	The CIE 1931 2° colour-matching functions

The scene JSON

Field	Contents
`tool`, `generated`	The tool name and time
`sampling`, `observer`	The grid, and the observer's published and used steps with a note on interpolation
`illuminant`, `measured_cct`, `reference`	The light, its measured temperature and band, and the reference chosen
`material`	Its id, reflectance status and source, conductor flag and refractive index
`surface`, `geometry`, `exposure`	Lobe and roughness; the three angles; the exposure scale
`colour`	sRGB hex (un-adapted), L*a*b*, the white used, JzAzBz, gamut flag
`rendering_shift`	ΔE_{00}, marked `is_rendering_index: false`
`withdrawn`	CRI and TM-30, each with why and what it would take
`sensitivity`	The sweep result, or `null`
`spectrum`	Wavelength, illuminant, reflectance and radiance arrays
`register`	Every Data register entry with its source and status

Links

A link rebuilds the scene with these parameters: `illuminant` (`d65`, `d50`, `d55`, `d75`, `a`, `e`, `planck`, `daylight`), `cct`, `material` (for example `gold`), `ior`, `roughness`, `thetaL`, `thetaV`, `phi`, `lobe` (`ggx`, `beckmann`, `ward`, `ashikhmin-shirley`, `oren-nayar`) and `metal` (`1` for conductor, `0` for dielectric). For example:

```
https://auricartisan.com/tool/general/spectral-tools/material-lighting/?
illuminant=a&cct=2700&material=skin&ior=1.5&roughness=0.4&thetaL=30&thetaV=20&phi=0&lobe=ggx&metal=0
```

A link does not carry a measured reflectance file or the sweep settings.

You should now have a file that records the light, the material's provenance and the geometry along with the numbers.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Run a sensitivity sweep

Use the Metrics tab to see how far the colour moves when the reflectance curve is perturbed, and read why the CRI and TM-30 figures were withdrawn.

A reflectance curve is never known exactly: a spectrophotometer's reading carries error, and the presets are inventions. The sweep asks how much that matters for the colour.

Run it

1. Set up the scene on the **Lab** tab.
2. Open the **Metrics** tab. Under **Sensitivity sweep**:

Control	Range and step	Default	Meaning
Perturbation	±0.005–0.1, step 0.005	±0.02	The size of the random change to the reflectance, in reflectance units
Correlation length	0–120 nm, step 5	40 nm	How far along the spectrum neighbouring changes move together; 0 shows **independent**
Draws	100–2000, step 100	500	How many perturbed curves to try

3. Select **Run the sweep**. A message reports how many draws took how long.

Read the result

Under **How far the colour moves when the curve does** appear **Median**, **95th percentile** and **Worst draw**, each a ΔE_{00} between the unperturbed colour and a perturbed one. The note gives the number of draws and the time, and the line under the figures restates the perturbation and correlation, adding, for a synthesised material, that this is the sensitivity of the model, not of any pigment.

For gold under D65 with the defaults, the median is typically about 0.5 and the 95th percentile about 1.2. The draws are random, so each run differs slightly.

Why the perturbation is correlated

A real instrument's error does not jump independently from one wavelength to the next; it drifts over tens of nanometres. The sweep smooths its random noise over the **Correlation length** so neighbouring wavelengths move together. Independent noise (correlation length 0) largely cancels when the spectrum is summed into a colour, and gives a misleadingly small spread: for the default scene, roughly half the correlated value.

This is a sensitivity sweep, not a bootstrap: there is no observed data to resample, so the page perturbs the curve and says by how much.

The result is written into the exports until you run another sweep or reset the scene; it is not rerun when the scene changes.

What used to print here

Below the sweep, **What used to print here** (marked **2 withdrawn**) explains two rendering figures an earlier version printed.

A **colour rendering index** (CRI, CIE 13.3) scores how faithfully a light source renders a specified set of measured test colours, compared with its reference light, on a scale up to 100. **TM-30** (IES TM-30-20) is a newer method with a fidelity score, Rf, and a gamut score, Rg, over 99 samples.

- **CIE colour rendering index** — marked **Withdrawn**. The earlier version compared every source with D65 regardless of its temperature, used invented test colours and the wrong colour space. A table reruns that old formula for D65, D50, D55, illuminant A and blackbodies at 2700 K and 2000 K, beside the value each ought to have (100, since each is its own reference) and the tool's current colour difference for skin.
- **TM-30 Rf and Rg** — marked **Withdrawn**. Both old numbers were simple functions of the same fourteen values, so they could never disagree. A table shows this over several sources.

Each panel ends with **What it would take** to hold the index properly. The tool now holds the reference-selection rule, but not the measured sample sets or the colour spaces the indices need, so it prints neither.

You should now know how sensitive your material's colour is to errors in its curve.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Set up a scene

Choose the light, the material, the surface model and the viewing geometry, and understand what each does to the spectrum and the colour.

How the colour is made

At each of 81 wavelengths, 380 to 780 nm in 5 nm steps, the tool multiplies:

1. **The illuminant's spectral power** — how much light the source emits at that wavelength (its **spectral power distribution**, SPD).
2. **The material's reflectance** — the fraction of that light the material sends back, from 0 to 1.
3. **The lobe's weight** — how much of the reflected light travels towards your eye at the angles you set.

The products are summed against the CIE 1931 2° observer's colour-matching functions to give CIE XYZ, and converted to sRGB for the screen. The **Method** tab's **From spectrum to colour** view shows these steps. The published observer table is at 10 nm; the tool interpolates it to 5 nm, and says so in its exports.

1. Choose the light

Option	What it is
D65 — daylight, 6504 K (default)	CIE average daylight
D50 — daylight, 5003 K	Warmer daylight, the graphic arts reference
D55 — daylight, 5503 K	Mid-morning or mid-afternoon daylight
D75 — daylight, 7504 K	Cool north-sky daylight
A — a blackbody at 2856 K	Incandescent (tungsten) light
E — equal energy	Equal power at every wavelength
Blackbody at a temperature you choose	Planck's law at **Temperature**
Daylight at a temperature you choose	The CIE daylight model at **Temperature**

The daylights are reconstructed from the CIE D-series basis, which is published and lands on the published chromaticities. A **blackbody** (Planckian radiator) is an ideal hot object whose light depends only on its temperature; daylight has a slightly different, greener-or-bluer balance at the same temperature.

Temperature (1800–10000 K in steps of 50, default 2700 K) is used only by the last two options, and is disabled otherwise; its hint says which. The CIE daylight model is defined from 4000 K, so **Daylight at a temperature you choose** raises lower values to 4000 K and says so in the exports.

2. Choose the material

The menu offers twenty materials: **White**, **Gray**, **Black**, **Red**, **Green**, **Blue**, **Yellow**, **Cyan**, **Magenta**, **Skin**, **Leaf**, **Ocean**, **Concrete**, **Wood**, **Sky**, **Gold** (default), **Copper**, **Silver**, **Ceramic** and **Rubber**.

Every one is a curve the tool invented to look right: a flat level plus one or two bell-shaped bumps. For example, silver is a flat 0.75 across the spectrum and gold is a low level plus bumps near 590 and 620 nm. None is a measurement, and **Reflectance** reads **synthesised**. This matters most for metals, whose reflectance spectrum is their colour. To use real data, load a measured file; see Use a measured reflectance.

3. Set the surface

Dielectric or **Conductor** (default Dielectric):

- A **dielectric** (paint, plastic, ceramic, skin) reflects a small, nearly colourless highlight from its surface and gets its colour from light that enters and scatters back out (the diffuse term). **Refractive index** (1.00–3.00, step 0.01, default 1.50) sets how strong the surface highlight is: the reflectance straight on, F0, is $((n - 1) \div (n + 1))^2$, 4% at 1.50.
- A **conductor** (metal) has no diffuse term: all its colour comes from the specular reflection, tinted by its reflectance spectrum. **Refractive index** is disabled.

Roughness (0.02–1.00, step 0.01, default 0.40) sets how spread out the reflection is: low values give a tight, bright highlight; high values a broad, dim one.

4. Choose a lobe

A **lobe** (a bidirectional reflectance distribution function, BRDF) describes how a surface spreads reflected light over directions. Most here are **microfacet** models, which treat the surface as tiny mirror facets whose tilts are spread according to the

roughness. Select a card under the sample:

Lobe	Description on the card	Source
GGX (default)	Trowbridge-Reitz, with Smith height-correlated masking	Walter et al. 2007
Beckmann	A Gaussian slope distribution; narrower tails than GGX	Beckmann & Spizzichino 1963
Ward	Fitted to measured gonioreflectometer data	Ward 1992
Ashikhmin-Shirley	An anisotropic Phong lobe with a coupled diffuse term (used here in its isotropic form)	Ashikhmin & Shirley 2000
Oren-Nayar	Rough diffuse, no specular: a surface of Lambertian facets	Oren & Nayar 1994

The **Method** tab's **The five lobes** view gives each lobe's specular and diffuse values at your current geometry and roughness.

Oren-Nayar has no specular term, and a **Conductor** has no diffuse term, so together they reflect nothing: the sphere and swatch go black. Use a specular lobe for metals.

5. Set the geometry

Control	Range	Default	Meaning
Light from	0–89°	30°	Angle of the light from the surface normal (straight up)
Seen from	0–89°	20°	Angle of your eye from the normal
Between them	0–180°	0°	Azimuth between the light and the eye around the normal

At 0° and 0° the eye is in the mirror direction and the highlight is strongest.

The three angles set the geometry for the readouts, the swatches and the spectrum's product curve: they describe one point on a flat surface. The **sphere** is always drawn seen head-on, lit from the **Light from** angle, so it shows the whole range of surface angles at once; **Seen from** and **Between them** do not change it.

6. Read the sample

- The sphere and swatch are scaled so that a perfect white diffuser in the same light and at the same angles would read Y = 1 (the note **scaled so a white card here would read Y = 1**). This makes brightness comparable between scenes.
- The swatch and **sRGB** are not adapted to the light: under a 2700 K lamp even a white card looks warm, as it would to a camera.
- **CIE L***, **a***, **b*** are taken against the light's own white, so they describe the material relative to the light; see Compare with the reference light.
- **Jz** is the lightness of JzAzBz (Safdar et al. 2017), a colour space designed for high dynamic range, given as a second opinion on lightness.
- **Inside sRGB** reads **yes**, **no**, or **the highlight is not** when only the sphere's highlight had to be clipped. **clipped to sRGB** appears on the sphere in the last two cases.

Save the spectrum downloads the CSV; **Reset the scene** returns to the defaults.

You should now be able to build a scene of any preset material under any of the eight light options.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Use a measured reflectance

Load a spectrophotometer reflectance file in place of the synthesised presets, and go back to the presets.

The twenty presets are invented curves. If you have a measured reflectance spectrum, from a spectrophotometer or a published dataset, you can render it instead.

Prepare the file

- A plain-text file: `.csv`, `.tsv` or `.txt`.
- Two columns per line: **wavelength in nm** and **reflectance**. Columns may be separated by commas, semicolons, tabs or spaces. Lines that are not two numbers are skipped.
- Reflectance from 0 to 1. A file in percent is detected automatically: if any value is above 1, every value is divided by 100. Values outside 0–1 after that are clipped.
- At least eight rows, covering at least **420 to 680 nm**.

Load it

1. Drop the file on the spectrum plot on the **Lab** tab, or on **Drop a reflectance file** under **Measure the material** on the **Data** tab. You can also click either area (or press `Enter` or `Space` on the Data tab's area) to choose the file.
2. Read the message: for example **41 rows accepted, read as percent — the register now says verbatim**, or **Not installed:** with a reason.

What changes

- The file is resampled onto the 5 nm grid from 380 to 780 nm by straight-line interpolation between your rows; beyond the ends of your file, the first and last values are held flat.
- The material menu now holds a single entry, named after your file with **— measured**, and it is selected. The twenty presets are hidden.
- **Reflectance** on the Lab tab reads **measured —** and your file name, and the plot legend names your file.
- On the **Data** tab, **The twenty pigment reflectances** entry becomes **Verbatim**, with your file name, the number of rows and the wavelength span.
- Exports mark the reflectance as **MEASURED, from** your file.
- **Go back to the presets** appears on the Data tab.

Every other control, including dielectric or conductor, roughness, lobe and geometry, applies as before.

Go back

Select **Go back to the presets** on the Data tab. The message **Back to the twenty synthesised presets** confirms it. The loaded curve is also lost when you reload the page; links do not carry it.

You should now see your own material rendered under any light and lobe.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Material Under Light reference

Every tab, control, readout, message and file format in Material Under Light, with ranges and defaults.

Address: https://auricartisan.com/tool/general/spectral-tools/material-lighting/

Tabs

Tab	Contents
Lab	Light, material, surface and geometry controls; the spectrum plot; the sample sphere and lobe cards; readouts; the reference comparison
Method	Four explanation views
Metrics	The sensitivity sweep and the two withdrawn rendering indices
Data	The register of 17 tables and formulas, filterable by status; the tab label shows the count; reflectance loader
Export	Spectrum CSV, scene JSON and link
Reference	Fourteen sources, filterable

Links in the page text (for example **Metrics** or **Data**) switch to that tab.

Lab tab — The light

Control	Values or range	Default
Illuminant	**D65 — daylight, 6504 K**, **D50 — daylight, 5003 K**, **D55 — daylight, 5503 K**, **D75 — daylight, 7504 K**, **A — a blackbody at 2856 K**, **E — equal energy**, **Blackbody at a temperature you choose**, **Daylight at a temperature you choose**	D65
Temperature	1800–10000 K, step 50; enabled only for the last two options; daylight is raised to at least 4000 K	2700 K

Lab tab — The material and surface

Control	Values or range	Default
Material	**White**, **Gray**, **Black**, **Red**, **Green**, **Blue**, **Yellow**, **Cyan**, **Magenta**, **Skin**, **Leaf**, **Ocean**, **Concrete**, **Wood**, **Sky**, **Gold**, **Copper**, **Silver**, **Ceramic**, **Rubber**; or one measured file	Gold
Dielectric / **Conductor**	Surface type	Dielectric
Refractive index	1.00–3.00, step 0.01; disabled for Conductor	1.50
Roughness	0.02–1.00, step 0.01	0.40

Lab tab — The geometry

Control	Range and step	Default
Light from	0–89°, step 1	30°
Seen from	0–89°, step 1	20°
Between them	0–180°, step 1	0°

Lab tab — Buttons

Button	What it does
Save the spectrum	Saves `material-under-light-spectrum.csv`
Reset the scene	Restores D65, 2700 K, Gold (if the presets are loaded), Dielectric, 1.50, GGX, 0.40, 30°, 20°, 0°, and clears the sweep

Lab tab — Lobe cards

Card	Source
GGX (default)	Walter et al. 2007
Beckmann	Beckmann & Spizzichino 1963
Ward	Ward 1992
Ashikhmin-Shirley	Ashikhmin & Shirley 2000
Oren-Nayar	Oren & Nayar 1994

Lab tab — Spectrum and sample

Element	Description
Spectrum plot	**Illuminant** (dashed), **Reflectance** and **What leaves the surface** (filled), each scaled to its own maximum, 380–780 nm
Legend	The illuminant's name; the material with "synthesised" or the file name with "measured"; "their product, times the lobe"
Sample	A 232 × 232 sphere, seen head-on, lit from the **Light from** angle
Exposure note	**scaled so a white card here would read Y = 1**
clipped to sRGB	Shown when the swatch or the sphere had to be clipped
Drop and click	Drop a reflectance file on the plot, or click it to choose one

Lab tab — Readouts

Readout	Meaning	Format
sRGB	Un-adapted colour at the chosen geometry	Hex
CIE L*	Lightness against the light's own white	One decimal
a*, b*	Opponent coordinates against the light's own white	One decimal each
Jz	JzAzBz lightness	Four decimals
Measured temperature	CCT of the illuminant with its error band; highlighted above 7000 K	Nearest 10 K
Reference for it	**daylight N K** or **blackbody N K**	Nearest 10 K
Reflectance	**synthesised** (highlighted) or **measured —** and the file name	—
Inside sRGB	**yes**, **no** or **the highlight is not** (last two highlighted)	—

Lab tab — Against the reference the standard names

Element	Description
Under your source	Swatch, captioned with the illuminant
ΔE_{00}	CIEDE2000 difference between the two, two decimals
Under its reference	Swatch, captioned **D** or **BB** and the temperature
Note	The reason for the reference choice

Method tab

Control	Values	Default
Method	**From spectrum to colour**, **Choosing the reference**, **The five lobes**, **Two white points**	From spectrum to colour

Metrics tab

Control	Range and step	Default
Perturbation	±0.005–0.1, step 0.005	±0.02
Correlation length	0–120 nm, step 5 (0 is **independent**)	40 nm
Draws	100–2000, step 100	500
Run the sweep	Shows **Median**, **95th percentile** and **Worst draw** in ΔE_{00}	—

Below: **CIE colour rendering index** and **TM-30 Rf and Rg**, each marked **Withdrawn**, with tables and **What it would take**.

Data tab

Control	What it does
Status	**All** (17), **Verbatim** (8), **Computed** (5), **Stand-in** (2), **Absent** (2)
Register row	Select, or press `Enter` or `Space`, to expand its note and **what it would take**
Drop a reflectance file	Loads a measured reflectance; click or `Enter` / `Space` opens a file chooser
Go back to the presets	Shown after loading; restores the twenty presets

Entries: **CIE 1931 2° standard observer**, **CIE D-series eigenvector basis**, **sRGB primaries and transfer curve**, **CIEDE2000 colour difference**, **JzAzBz**, **Planck's radiation law**, **GGX / Trowbridge-Reitz microfacet distribution**, **The other four reflectance lobes** (Verbatim); **Correlated colour temperature**, **The reference illuminant for a source**, **Rendering shift**, **Exposure normalisation**, **Illumination and view geometry** (Computed); **The twenty pigment reflectances**, **The sensitivity sweep** (Stand-in); **CIE colour rendering index (Ra)**, **IES TM-30 fidelity and gamut (Rf, Rg)** (Absent).

Export tab

Control	What it does
What to take	Preview **The spectrum**, **The whole scene** or **A link to this**

Control	What it does
Download the spectrum	Saves `material-under-light-spectrum.csv` (81 rows)
Download the scene	Saves `material-under-light-scene.json`
Copy a link to this scene	Copies a link

Reference tab

Control	Values
Reference	**All** (14), **Colorimetry** (6), **Reflectance** (6), **Not held** (2)

Messages

Message	Meaning
Scene reset	Defaults restored
Swept N perturbations in M ms	The sweep finished
N rows accepted — the register now says verbatim (with **, read as percent** when applicable)	A reflectance file was loaded
No two-column numeric rows in that file	The file had no usable rows
Not installed: fewer than eight rows / fewer than eight usable rows	Too few rows
Not installed: covers X to Y nm; at least 420 to 680 is needed …	The file does not span 420–680 nm
Back to the twenty synthesised presets	The loaded file was removed
Link copied	The link is on the clipboard

File formats

File	Format	Contents
`material-under-light-spectrum.csv`	CSV with `#` header lines	`wavelength_nm,illuminant_relative,reflectance,radiance_relative,observer_xbar,observer_ybar,observer_zbar`
`material-under-light-scene.json`	JSON	See Export and share
Reflectance (input)	`.csv`, `.tsv` or `.txt`	Two columns: wavelength (nm), reflectance (0–1 or percent); at least 8 rows spanning 420–680 nm

Link parameters

`illuminant`, `cct`, `material`, `ior`, `roughness`, `thetaL`, `thetaV`, `phi`, `lobe`, `metal`. Values are listed in Export and share.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Troubleshooting Material Under Light

Symptoms, causes and fixes for Material Under Light: black samples, disabled controls, clipping, refused files and links.

The sphere and swatch are black

Why: you chose **Conductor** with the **Oren-Nayar** lobe. A conductor has no diffuse term, and Oren-Nayar has no specular term, so nothing is reflected.

What to do: choose **Dielectric**, or a specular lobe such as **GGX**, **Beckmann**, **Ward** or **Ashikhmin-Shirley**.

The Temperature slider does nothing or is greyed out

Why: it is used only by **Blackbody at a temperature you choose** and **Daylight at a temperature you choose**. Every other light has its own fixed definition.

What to do: choose one of those two options.

I set a daylight below 4000 K but the readout says 4000 K

Why: the CIE daylight model is defined from 4000 K upwards, so lower values are raised to 4000 K. The exports record this.

What to do: use **Blackbody at a temperature you choose** for warmer sources.

"Measured temperature" is highlighted and mentions McCamy

Why: above 7000 K the colour-temperature formula is outside its accurate range; the error band grows to ±139 K and beyond.

What to do: nothing; the reference chosen above 5000 K is a daylight either way. Treat the temperature as approximate.

"clipped to sRGB" appears, or "Inside sRGB" reads "no" or "the highlight is not"

Why: the material's colour, or its highlight, is outside what sRGB can show; the swatch and sphere show the nearest displayable colour. Saturated materials such as the default gold, and bright highlights on shiny surfaces, often do this.

What to do: nothing is wrong; the L*a*b* values are computed before clipping. Raise **Roughness** or change the angles to dim the highlight.

Moving "Seen from" or "Between them" does not change the sphere

Why: the sphere is always drawn seen head-on, lit from **Light from**. The view angles set the geometry of the single point the readouts and swatches describe.

What to do: read the swatch and the readouts, which do change.

The Refractive index slider is greyed out

Why: it applies only to dielectrics.

What to do: select **Dielectric**.

Every material looks similar under E or a high-temperature light

Why: the presets are smooth invented curves, and the lobe mixes in a colourless highlight on dielectrics.

What to do: compare L*a*b* values rather than swatches, or load a measured reflectance.

My reflectance file is refused

Message	Fix
No two-column numeric rows in that file	Make sure each data line has two numbers: wavelength and reflectance
Not installed: fewer than eight rows	Supply at least eight rows
Not installed: covers X to Y nm; at least 420 to 680 is needed ...	Extend the data to cover 420–680 nm

A file in percent is converted automatically. A file mixing fractions and percentages is read as percent throughout.

The material menu shows only my file

Why: a loaded reflectance replaces the twenty presets until you remove it.

What to do: select **Go back to the presets** on the Data tab.

A file chooser opens when I click the spectrum

Why: the plot is also a drop zone for reflectance files; clicking it opens a chooser.

What to do: cancel it if you did not mean to load a file.

The sweep result is out of date

Why: the sweep is not rerun when the scene changes.

What to do: select **Run the sweep** again on the Metrics tab before exporting.

A link did not restore my measured material

Why: links carry the light, preset material, surface, lobe and angles only.

What to do: load the file again after opening the link.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Material Under Light FAQ

Short answers to common questions about Material Under Light.

Are the materials measured?

No. All twenty presets are invented curves (a flat level plus bell-shaped bumps) and are marked **synthesised** everywhere. Load a measured file on the Data tab to use real data.

Why is there no CRI or TM-30 score?

Both need specified sets of measured test colours and specific colour spaces that the tool does not hold. The figures an earlier version printed were wrong in ways the **Metrics** tab demonstrates, so they are withdrawn.

So what does the ΔE00 under the swatches mean?

How far this one material's colour moves between your light and the CIE reference light for its colour temperature, each taken against its own light's white. It describes the material in that light, not the light in general.

Why does the swatch look warm under tungsten when L*a*b* barely changes?

The swatch is un-adapted: it shows the light that actually leaves the surface, as a camera would record it. L*a*b* is taken against the light's own white, so it measures the material relative to the light.

Is the rendering physically accurate?

The spectral pipeline, observer, daylights, colour difference and the five lobes follow their published forms. The materials are invented, the scene is a single light and a single surface with no inter-reflection, and the sphere is a simple shaded preview. It is a teaching and exploration tool, not a renderer for production images.

Can it show metamerism?

It shows one material at a time. Because it is spectral, you can compare two materials (for example a preset and a measured file) under two lights one after the other and see whether a match holds; it has no side-by-side metamerism index.

What does Jz add?

The lightness of JzAzBz, a colour space designed for high-dynamic-range and wide-gamut signals, as a second view of lightness beside CIE L*.

Why does a metal go black with Oren-Nayar?

Oren-Nayar is a diffuse-only model and a conductor has no diffuse reflection. Choose a specular lobe for metals.

How accurate is the measured colour temperature?

Within about ±11 to ±18 K below 7000 K, and much less accurate above; the readout shows the band. See Compare with the reference light.

Is my reflectance file uploaded?

No. It is read and used in the page only. See Privacy.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Material Under Light background

Contents of the background material for the Material Under Light tool.

This folder supports the how-to pages with the colour and rendering terms the tool uses, its keyboard and pointer actions, a statement of what is published, what is the tool's own and what is not held, related tools and articles, its history and what stays on your device.

File	What it covers
glossary.md	Colour, light and reflectance terms used in Material Under Light
shortcuts.md	Lobe cards, drop zones, `Enter` and `Space`, and tab links
limits-and-accuracy.md	The data register entry by entry, precision, and what the tool is not
related.md	Related tools on the site and Learn articles
release-notes.md	User-facing history of the tool
privacy.md	What stays in your browser

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Material Under Light shortcuts

Keyboard and pointer actions in the Material Under Light tool.

The tool has no keyboard shortcuts of its own beyond these. Site-wide shortcuts are described in Site features.

Action	Where	Result
Select a card	The five lobe cards under the sample	Chooses that lobe
Drop a file	The spectrum plot on the Lab tab, or **Drop a reflectance file** on the Data tab	Loads a measured reflectance
Click	The spectrum plot or **Drop a reflectance file**	Opens a file chooser
`Enter` or `Space`	Focused **Drop a reflectance file** area	Opens a file chooser
`Enter` or `Space`	A focused row on the **Data** tab	Expands or collapses the row
Click a tab name in the page text	Note strip and hints	Switches to that tab
`Tab`	Anywhere	Moves between controls; sliders then respond to the arrow keys

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Material Under Light glossary

Colour, light and reflectance terms used in the Material Under Light tool, each explained briefly.

Term	Meaning
Azimuth	The angle between the light and the eye measured around the surface normal (**Between them**)
Blackbody (Planckian radiator)	An ideal hot object whose light depends only on its temperature, given by Planck's law
BRDF (lobe)	A model of how a surface spreads reflected light over directions

Term	Meaning
CIE 1931 2° observer	The standard colour-matching functions describing average human colour vision
CIE L*a*b*	A colour space with lightness L* and opponent axes a* (red–green) and b* (yellow–blue)
CIEDE2000 (ΔE_{00})	The CIE's recommended formula for how different two colours look
Conductor	A metal; its colour comes entirely from specular reflection tinted by its reflectance
Correlated colour temperature (CCT)	The temperature of the blackbody whose colour is closest to a light's, in kelvin
CRI (colour rendering index)	A CIE score of how faithfully a light renders a set of test colours; withdrawn from this tool
D-series daylight (D50, D55, D65, D75)	CIE standard daylights reconstructed from a published basis, named for their temperature
Dielectric	A non-metal; a faint colourless highlight plus coloured diffuse reflection
Diffuse reflection	Light that enters a material, scatters and leaves in many directions, carrying the material's colour
Exposure normalisation	Scaling so a perfect white diffuser in the same light and geometry reads Y = 1
F0	The reflectance of a surface seen straight on, set here by the refractive index
Illuminant A	The CIE standard for incandescent light: a blackbody at 2856 K
Illuminant E	A theoretical light with equal power at every wavelength
JzAzBz (Jz)	A colour space for high-dynamic-range signals; Jz is its lightness
McCamy's formula	A published approximation for computing CCT from chromaticity
Metamerism	Two materials with different spectra matching under one light but not under another
Microfacet model	A lobe that treats a surface as tiny mirror facets with a spread of tilts
Reference illuminant	The light a source is compared with: a blackbody below 5000 K, a CIE daylight at or above
Reflectance spectrum	The fraction of light a material reflects at each wavelength, from 0 to 1
Refractive index (n)	How strongly a material bends light; sets a dielectric's F0
Rendering shift	The ΔE_{00} of one material between a source and its reference light
Roughness	How spread out a surface's reflection is; low is glossy, high is matte
Specular reflection	Mirror-like reflection from the surface itself
Spectral power distribution (SPD)	A light's power at each wavelength
sRGB	The standard colour space of screens and the web
TM-30 (Rf, Rg)	An IES method scoring colour fidelity (Rf) and gamut (Rg); withdrawn from this tool
Un-adapted swatch	A colour shown as the light actually leaving the surface, without compensating for the light's colour

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Limits and accuracy

What Material Under Light takes from published sources, what is its own, how precise it is, and what it is not.

What the tool is

A spectral appearance bench for learning and exploration: one light, one flat material, one geometry, computed wavelength by wavelength. It is not a production renderer, a lighting-design certification tool or a colour-rendering meter.

The data register, in plain terms

Entry	Status	What it means for you
CIE 1931 2° standard observer	Verbatim	The published 10 nm table, unaltered; every other 5 nm sample is interpolated
CIE D-series eigenvector basis	Verbatim	D50, D55, D65, D75 and chosen daylights land within 0.0003 of their published chromaticities
sRGB primaries and transfer curve	Verbatim	The swatch uses the published sRGB definition, un-adapted
CIEDE2000 colour difference	Verbatim	Every ΔE_{00} uses the full formula, against the light's own white
JzAzBz	Verbatim	The published constants
Planck's radiation law	Verbatim	Every blackbody is computed at the 81 samples
GGX microfacet distribution	Verbatim	The published form, including Smith masking
The other four reflectance lobes	Verbatim	Beckmann, Ward, Ashikhmin-Shirley and Oren-Nayar in their published forms
Correlated colour temperature	Computed	McCamy's formula, with error bands measured for the whole chain (±11 to ±18 K below 7000 K)
The reference illuminant for a source	Computed	The CIE 13.3 selection rule only, not the index built on it
Rendering shift	Computed	A colour difference for one material; never a property of the lamp
Exposure normalisation	Computed	A white diffuser in the same light and geometry reads Y = 1
Illumination and view geometry	Computed	From the three angle controls, with the normal fixed vertical
The twenty pigment reflectances	Stand-in	Invented curves; loading a measured file makes this Verbatim
The sensitivity sweep	Stand-in	A perturbation the tool chooses, not repeat measurements
CIE colour rendering index	Absent	Withdrawn; needs the fourteen measured test colours and the specified colour space
TM-30 Rf and Rg	Absent	Withdrawn; needs 99 samples, CAM02-UCS and a gamut-area calculation

Precision

- 81 samples, 380–780 nm at 5 nm. Narrow emission lines (such as fluorescent or LED spikes) can fall between samples, and light outside 380–780 nm is ignored.
- The observer is interpolated from 10 nm to 5 nm.
- The CCT error bands are those in Compare with the reference light.
- Clipped colours are flagged; L*a*b*, Jz and ΔE_{00} are computed before clipping.
- Your own screen is assumed to be sRGB.

Model limits

- One light, one surface, no shadows, inter-reflections, subsurface depth, transparency or fluorescence.
- Refractive index is a single number for all wavelengths, and a conductor's Fresnel term is fixed at full reflectance, with its colour coming from the reflectance curve.
- The Ashikhmin-Shirley lobe is used in its isotropic form; Oren-Nayar in its qualitative approximation.
- The sphere is a preview seen head-on; the view angles apply to the readouts, not to the sphere.
- There are no fluorescent or LED illuminants.

What the tool does not do

- It does not score light sources (no CRI, no TM-30).
- It does not compare two materials side by side for metamerism.
- It does not measure real materials; it uses invented curves unless you load a measured file.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Privacy

What Material Under Light keeps in your browser and what uses the network.

What stays in your browser

- Every calculation: spectra, the sphere, the readouts, the reference comparison and the sweep are computed in the page.
- Reflectance files you load are read from your disk by the page. They are not uploaded and are forgotten when you reload.

What leaves the page, and only when you ask

- **Save the spectrum**, **Download the spectrum** and **Download the scene** create files through your browser's normal download. They include the name of any reflectance file you loaded.
- **Copy a link to this scene** writes a link to your clipboard. It contains your scene settings, never your files.

The rest of the page

The tool page loads the site's usual pages, styles and scripts, and, depending on your cookie choices, advertising and consent-based analytics, like the rest of auricartisan.com. See the Privacy Policy at https://auricartisan.com/legal/privacy/ and the Cookie Policy at https://auricartisan.com/legal/cookies/.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Release notes

User-facing history of the Material Under Light tool.

The site-wide history is at https://auricartisan.com/library/changelog/. The entries below cover what changed for users of this tool.

September 2026 — rebuilt as Material Under Light

The page was rebuilt with six tabs (**Lab**, **Method**, **Metrics**, **Data**, **Export**, **Reference**).

Changed:

- The D-series daylights (D50, D55, D65, D75 and chosen daylights) are now true CIE daylights. Earlier they were blackbodies at the same temperature, which shifted colours slightly (for example about 1.2 ΔE_{00} for skin under D65).
- The twenty material curves are unchanged in shape but now labelled synthesised everywhere, including exports.
- The earlier "bootstrap" is now a sensitivity sweep with a stated, correlated perturbation you control.

Added:

- **Measured temperature** with its error band, **Reference for it**, and the comparison of the material under its source and under the CIE reference light.
- The two-white-point treatment: an un-adapted swatch, and L*a*b* against the light's own white.
- Loading a measured reflectance file.
- The **Data** register and header lines stating the method in every export.

Removed:

- The CIE colour rendering index and the TM-30 Rf and Rg figures. The **Metrics** tab shows why and what they would need.

What stayed the same: the spectral pipeline, the observer, sRGB, CIEDE2000, JzAzBz, all five lobes and every control.

Earlier

- A Material & Lighting Lab user guide for the previous version was published in the site documentation in April 2026; it is now marked as superseded.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › MATERIAL LIGHTING

Related tools and articles

Tools and Learn articles related to Material Under Light.

Tools in this group

- Industrial Dye Chemistry — the same spectral approach for transparent dye baths, with colour inconstancy between lights.
- Perception and Vision — how the viewer's visual system changes what a colour looks like.

Other tools on the site

- Illuminants — the standard light sources and their spectra in detail.
- Chromatic adaptation — predicting how colours appear when the eye adapts to a new light.
- Colour Science Lab — colour spaces and conversions.

Learn articles

The Learn section is described in Learn.

- Spectral Power Distributions and Why RGB Is Not Enough — https://auricartisan.com/library/learn/articles/2026-05-24-spectral-power-distributions
- Spectral Rendering: Why Three Numbers Aren't Enough — https://auricartisan.com/library/learn/articles/2026-08-22-spectral-rendering-beyond-rgb
- Color Rendering: Why Two "White" Lights Reveal Colors Differently — https://auricartisan.com/library/learn/articles/2026-08-07-color-rendering-and-cri
- Metamerism Explained — https://auricartisan.com/library/learn/articles/2026-05-25-metamerism
- ΔE Metrics from CIE76 to CIEDE2000 and ΔEITP — https://auricartisan.com/library/learn/articles/2026-05-26-delta-e-metrics-cie76-ciede2000-itp

Documentation on the site

- Material Under Light Developer Reference — https://auricartisan.com/library/documentation/articles/2026-05-31-material-under-light-developer-reference (for how the model works)
- The earlier Material & Lighting Lab user guide describes a previous version of the tool and is marked as superseded on the site.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › NEUROSCIENCE

Neuroscience (Visual Pathway)

What the Visual Pathway tool does, who it is for, where to find it and how to follow a first grating from the cones to V4.

The Neuroscience tool, whose page is titled **Visual Pathway**, follows one stimulus through five stages of the early visual system, one stage at a time. It starts with a colour on your screen, turns it into the signals of the three cone types, combines those into opponent channels, and then shows what a retinal centre-surround cell, a bank of V1 orientation filters, a V2 boundary signal and a V4 colour cell would each respond to. Every stage produces a map you can view beside the stimulus, with a scale bar that says what a bright pixel means.

The stimulus is a grating, a pattern of stripes whose spatial frequency you set in cycles per degree of visual angle. Because cycles per degree depends on how far you sit from the screen and how large its pixels are, the page asks for both and derives every frequency from them. It also warns you when the frequency you ask for is finer than your pixel grid can draw.

The tool is an educational approximation, not a physiological measurement. The page says so, and its **Data** tab lists which of its numbers come from published sources and which are its own. In particular, the cone curves drawn in the figure are the tool's own and put the L and M cones at the same wavelength, while the cone signals that drive the stages come from a published transform. Everything runs in your browser.

Who it is for

- Students and teachers of colour vision and visual neuroscience who want to see what each stage computes.
- Designers and developers curious about why some colour pairs are hard to see as patterns, and what "opponent colour" means.
- Anyone who wants a worked, traceable example of cone signals and opponent channels for a list of colours.

Where to find it

Open https://auricartisan.com/tool/general/perception-and-neuroscience/neuroscience/ in a current desktop browser. It is listed as **Neuroscience** on the Collections page (https://auricartisan.com/collections/) and as **Neuroscience of Color** in the site menu. No installation or account is needed.

Quick start

1. Open the tool. The **Lab** tab is selected and a blue grating is shown at 4.0 c/deg and 25% contrast.
2. Under **Geometry**, set **Viewing distance** to how far your eyes are from the screen and **Pixel pitch** to your screen's pixel size (screen width in millimetres divided by its width in pixels).
3. Under **Show**, choose **Retina, on-centre**. The right-hand canvas shows the retinal map and the scale bar under it gives its range.
4. Switch **Show** to **LGN, L − M**, then **LGN, S − (L+M)/2**, and watch the chromatic channels respond to the grating.
5. Choose **V1, preferred orientation** to see the orientation map as a colour wheel.
6. Change the **Carrier** from **Chromatic** to **Luminance** and compare the maps again.
7. Open the **Export** tab, paste your own hex colours into **Colours** and select **Download the CSV** to take away their cone signals and opponent values.

What you can do

- Show a chromatic or luminance grating, or a built-in test card, and set its base colour, frequency and contrast.

- State your viewing distance and pixel pitch, and read the degrees per pixel and the Nyquist limit of your screen.
- Switch each of the five stages on or off, and view any of nine maps in one of three colourmaps.
- Tune the retinal centre-surround, the V1 Gabor bank and the V4 preferred direction.
- Simulate protanopia, deuteranopia or tritanopia before the stages see the stimulus.
- Compare the cone-curve figure with the published peak wavelengths, and install a cone table of your own.
- See every stage side by side on the **Cells** tab, and the colour chain step by step on the **Pipeline** tab.
- Export a colour list as CSV or to the clipboard, the active map as PNG, the settings as JSON, or a link to the view.

In this folder

- docs/ — how to use the tool.
  - Getting started — open it, the screen tour, the first grating.
  - Explore the five stages — the stimulus, the stages, the maps and how to read them.
  - Cones, opponent channels and the cone table — the Pipeline tab, the cone figure and installing your own table.
  - Export and share — the colour list, the map, the settings file and links.
  - Reference — every control, readout, tab and file format.
  - Troubleshooting — symptoms, causes and fixes.
  - FAQ — short answers to common questions.
- others/ — background material.
  - Glossary — terms used in this tool.
  - Shortcuts — keyboard and pointer actions.
  - Limits and accuracy — what is published, what is the tool's own, what is not held.
  - Related — related tools and Learn articles.
  - Release notes — user-facing history.
  - Privacy — what stays in your browser.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › NEUROSCIENCE

Using the Visual Pathway

Contents of the how-to documentation for the Visual Pathway tool.

These pages explain how to use the Visual Pathway, from the first grating to exporting cone signals for your own colours. Start with getting started, then read the task guides in any order. The reference lists every control.

File	What it covers
getting-started.md	Opening the tool, the six tabs, the Lab screen tour and a first walk through the stages
explore-the-stages.md	Stimulus and geometry, each of the five stages, the nine maps, the scale bar and the Cells tab
cones-and-opponent-channels.md	Cone signals, opponent axes, the Pipeline tab, the cone figure, colour vision simulation and installing a cone table
export-and-share.md	The colour list, the CSV and clipboard formats, saving the map, the settings JSON and view links
reference.md	Every tab, control, readout, message and file format, with ranges and defaults
troubleshooting.md	Flat maps, alias and contrast warnings, refused cone tables, links and copying
faq.md	Short answers to common questions

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › NEUROSCIENCE

Getting started with the Visual Pathway

Open the Visual Pathway, learn the layout of the Lab tab and follow a first grating through all five stages.

Open the tool

Go to https://auricartisan.com/tool/general/perception-and-neuroscience/neuroscience/. The page opens on the **Lab** tab with a default stimulus already drawn: a chromatic grating based on the colour `#3A8CFF`, at 4.0 cycles per degree and 25% contrast, viewed from 57 cm on a screen with 0.248 mm pixels.

A note strip above the tabs says what the pictures rest on: the L and M cones peak 25 nm apart in the published data, but the cone table this page draws peaks them at the same wavelength. The strip cannot be closed. Its **Data** link opens the Data tab, where the reason is explained.

The six tabs

Tab	What it is for
Lab	The workbench: controls on the left, the stimulus and the chosen stage map on the right, the cone figure and the readouts below
Pipeline	Where a colour goes before any cortex sees it, step by step, with the opponent values of seven test colours
Cells	All the stages side by side over the same small stimulus
Data	The register of every dataset and transform the tool rests on, with a status for each; you can install a cone table here
Export	A list of colours through the front end, and the download, copy and link buttons
Reference	Standards, formulas, citations and research notes, with a search box

A tour of the Lab tab

The rail (left). From top to bottom: **Stimulus**, **Geometry**, the frequency, contrast and pupil sliders, **Stages**, **Show**, then the settings of the stages that are switched on (**Retina**, **V1**, **V4 preferred direction**) and **Colour vision**. Each group has a short hint under it. Sliders show their current value beside their label.

The stage (right).

- Two square canvases, 256 × 256 pixels each: **Stimulus** on the left and the selected map on the right, named under it (for example **Retina, on-centre**). The header gives the size and the angle one pixel covers at your stated geometry.
- The scale bar: the map's minimum on the left, the colour ramp, the maximum on the right, and what the numbers are.
- **The front end: where the two cone curves actually sit** — a figure of the L, M and S curves with dashed lines at the published peak wavelengths.
- Two rows of readouts: **L − M peak gap**, **Published gap**, **Neutral off-axis**, **Contrast achieved**, then **Sensitivity at this frequency**, **The curve's peak**, **Frame RMS contrast** and **Nyquist limit of the grid**.
- Notes about what the page does not report, and two buttons: **Save the map** and **Copy a link to this view**.

Warnings appear between these parts only when needed: one when the frequency is above your screen's Nyquist limit, one when the contrast you asked for does not fit in sRGB.

Your first walk through the stages

1. Set **Viewing distance** and **Pixel pitch** for your own screen. For a pixel pitch, divide the screen's visible width in millimetres by its horizontal resolution. The header above the canvases updates with the degrees per pixel, and **Nyquist limit of the grid** shows the finest frequency your screen can draw.
2. Leave **Spatial frequency** at 4.0 c/deg and **Contrast** at 25%.
3. In **Show**, pick **Retina, on-centre**. The map shows where the centre-surround cells respond. Read the numbers at the ends of the scale bar: every map is stretched to its own range, so those numbers are what the colours mean.
4. Pick **Retina, off-centre**. With the default chromatic grating this map can come out flat, and the scale bar then says **flat — this stage found no variation in this stimulus**. That is a result, not a fault: see Explore the five stages.
5. Pick **LGN, L − M** and **LGN, S − (L+M)/2**. These are the red-green and blue-yellow opponent channels, and the grating drives them strongly.
6. Pick **V1, complex-cell energy**, then **V1, preferred orientation**. The second is drawn as a colour wheel: hue is the preferred orientation from 0° to 180°, and paler pixels are less selective.
7. Change **Carrier** to **Luminance**. The grating is now grey stripes, lighter and darker than a grey with the base colour's luminance. The chromatic LGN channels flatten, and the retinal and V1 maps respond to the light-dark swing.
8. Double-click either canvas to see it enlarged. Press `Esc` to close.

You should now see the stimulus on the left and a map on the right that changes with every control you move.

Next steps

- Explore the five stages explains each stage and every map.
- Cones, opponent channels and the cone table explains the colour side of the pipeline.
- Export and share shows how to take numbers away.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › NEUROSCIENCE

Cones, opponent channels and the cone table

How the Visual Pathway turns a screen colour into cone signals and opponent channels, how to read the Pipeline tab and the cone figure, and how to install your own cone table.

The idea in brief

Your retina has three kinds of cone. The **L** (long-wavelength) cones respond most to yellowish-red light, the **M** (medium) cones to green light and the **S** (short) cones to violet-blue light. Their sensitivities overlap heavily: in the published CIE 2006 fundamentals the L and M cones peak at 566.8 nm and 541.2 nm, only about 25 nm apart, and the S cones at 441.8 nm. A colour you see is summarised by three numbers, the **cone signals**, one per type.

Early in the visual pathway these three signals are recombined into **opponent channels**: one that compares L with M (red against green), one that compares S with the sum of L and M (blue against yellow), and one that adds L and M (roughly, lightness). A neutral grey drives neither chromatic channel, which is why you never see a "reddish green" or a "bluish yellow".

How the tool gets from a hex colour to opponent values

The **Pipeline** tab draws the chain as a row of steps:

Step	What happens
sRGB	The 8-bit, gamma-encoded screen value
linear	Decoded through the sRGB transfer curve (IEC 61966-2-1)
XYZ	Converted to CIE XYZ with the sRGB primaries at D65
LMS	Converted to cone signals with the Hunt-Pointer-Estévez matrix
cone contrast	Marked **not held**: the tool has no adapting white to express cone contrast against
opponent	L − M, S − (L+M)/2 and L + M
stages	The retina-to-V4 stages read these values

The Hunt-Pointer-Estévez matrix is published for XYZ and is scaled so that the D65 white point gives equal L, M and S signals. That is what puts every neutral grey at zero on both chromatic channels. The **Neutral off-axis** readout on the Lab tab checks it: the largest departure of a nine-step grey ramp from zero is about 2.2×10^{-4}, which is the rounding of the published four-decimal matrix.

These axes are not DKL

The opponent space of Derrington, Krauskopf and Lennie (DKL), used widely in vision science, is built on cone **contrast** relative to a stated adapting white, with its chromatic axes scaled so they carry no luminance change. The tool's axes are plain differences of absolute cone signals, and its luminance channel gives L and M equal weight, where their published contribution to luminance is closer to two to one. They are useful axes and they are the tool's own. Nothing on the page is reported in cone-contrast units.

Use the Pipeline tab

The rail has two groups of buttons.

Show

- **The chain** — the step row above and a table of seven colours (`#000000`, `#404040`, `#808080`, `#C0C0C0`, `#FFFFFF`, `#3A8CFF`, `#E03131`) with their **L − M** and **S − (L+M)/2** values to four decimals. Greys are highlighted when they sit at zero.

- **The opponent axes** — the same table, with a note giving the worst neutral departure.
- **The sensitivity curve** — a table of pupil sizes (2, 3, 5 and 8 mm) against the peak of the tool's sensitivity curve (always 4.0 c/deg) and the published photopic peak range (2–6 c/deg).

Against

- **What it was** (default) — adds columns showing the values an earlier version of this page produced, marked **as shipped**, so you can see how large each correction was.
- **Hide** — shows the current values only.

Try it: with **The chain** and **What it was** selected, compare the white row. The current columns read 0.0000; the "as shipped" columns do not.

Read the cone figure

On the Lab tab, **The front end: where the two cone curves actually sit** plots three curves from 390 to 730 nm: L in red, M in green and S in blue, each scaled to peak at 1. Dashed vertical lines mark the published peaks, labelled 442, 541 and 567 nm. The note under the figure explains what you are seeing.

The built-in curves are the tool's own. They are smooth and plausibly shaped, but both the L and M curves peak at 590 nm, on top of each other, where the published curves peak 25 nm apart. That is why **L − M peak gap** reads 0 nm and is highlighted. These curves draw the figure and nothing else: every cone signal the stages use comes from the matrix described above, not from this table.

Install your own cone table

You can replace the figure's curves with a published table, for example the CIE 170-1:2006 2° cone fundamentals, which the page does not include.

1. Prepare a CSV or JSON file with four columns in this order: wavelength in nm, L, M, S.
   - CSV: one row per line, values separated by commas, semicolons, tabs or spaces. Lines that begin with `#` or a letter (such as a header row) are skipped.
   - JSON: an array of four-number rows, or an object with a `rows` array.
2. Open the **Data** tab.
3. Drop the file on **Drop a CSV or JSON** under **Install a cone table**, or select that area to choose the file.
4. Read the message at the top of the page: **Cone table installed — L peaks at N nm**, or **Not installed:** followed by the reason.

The table is refused when it has fewer than eight rows, when a row has fewer than four columns or a non-numeric value, when the wavelengths do not increase, or when its L and M curves peak less than 5 nm apart.

Once installed, the cone figure, the curve labels, **L − M peak gap** and the note under the figure change to the new table, and the settings file and CSV record the file name as the cone table with `reportable` set to `true`. The stages do not change, because they never used the table. The installed table lasts until you reload the page.

Simulate a colour vision deficiency

Colour vision on the Lab rail applies a dichromacy simulation to the stimulus before the stages: **Protanopia** (no L cones), **Deuteranopia** (no M cones) or **Tritanopia** (no S cones). The tool uses the three projection matrices of Viénot, Brettel and Mollon (1999), applied in cone space as they are defined, at full strength. Try **Protanopia** with the default blue grating, then with a red base colour, and watch the **LGN, L − M** map. For a fuller set of simulations over your own images, use Perception and Vision.

Take the numbers away

The **Export** tab runs any list of hex colours through the same front end and gives L, M, S and the three opponent values; see Export and share. Put a grey in the list: its two chromatic values should read zero.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › NEUROSCIENCE

Explore the five stages

Set up the stimulus and viewing geometry, then read what the retina, LGN, V1, V2 and V4 stages do to it, map by map.

This guide covers the stimulus, the viewing geometry, each stage of the pathway, the nine maps and the Cells tab. All controls named here are on the **Lab** tab unless stated.

1. Choose a stimulus

Under **Stimulus**, choose one of two sources.

- **Grating** — stripes whose brightness or colour varies as a sine wave across the frame. It is the only source with a defined spatial frequency, so it is the one the stages can be read against. Its controls are a colour picker (**Base colour**, default `#3A8CFF`) and a **Carrier** menu:
  - **Chromatic** — the stripes swing between the base colour's hue and its opposite, around a grey with the same L + M cone signal. Both phases have the same L + M, so the pattern changes colour with little change in lightness.
  - **Luminance** — grey stripes that swing lighter and darker around a grey with the base colour's relative luminance. The pattern changes lightness only.
- **Test card** — six colour bars (`#E03131`, `#2F9E44`, `#1971C2`, `#F59F00`, `#C2255C`, `#0CA678`) across the top half, a nine-step grey ramp below them, and a red-to-green gradient along the bottom. The base colour, carrier, frequency and contrast do not apply to it, and the grating controls are hidden.

Contrast (0 to 100%, default 25%) sets how far the grating swings either side of its mean. A saturated base colour's opposite is often outside the sRGB range a screen can show. When the swing you ask for does not fit, the page shrinks it to the largest swing both phases can survive, shows the note **The contrast you asked for does not fit in sRGB**, and reports what it drew under **Contrast achieved**. It does not clip one phase, because that would quietly change the contrast.

2. State the viewing geometry

A spatial frequency in **cycles per degree** (c/deg) counts how many light-dark cycles fit into one degree of your field of view. It is not a property of an image alone; it depends on how far away the image is and how large its pixels are. So the page asks for both:

- **Viewing distance** — 20 to 200 cm, default 57 cm. At 57 cm one degree covers very nearly one centimetre, which is why vision laboratories use it.
- **Pixel pitch** — 0.05 to 0.6 mm, default 0.248 mm. Your screen's value is its visible width in millimetres divided by its width in pixels.

From these the page works out the degrees per pixel (shown above the canvases; 0.0249° at the defaults) and the **Nyquist limit of the grid**, the finest frequency the pixel grid can carry: half a cycle per pixel, 20.1 c/deg at the defaults. If **Spatial frequency** (0.2 to 30 c/deg, default 4.0) is above that limit, the note **Above the pixel grid's Nyquist limit** appears: the screen cannot draw the frequency you asked for, and what you see is an alias, a coarser false pattern. Ask for a lower frequency, or check your geometry: the limit rises as the viewing distance grows and as the pixel pitch shrinks. (The warning's own text suggests moving closer, which in fact lowers the limit.)

Pupil (2 to 8 mm, default 3.0 mm) is shown on the rail but, in the current version, does not change any map or readout. The sensitivity curve peaks at 4 c/deg whatever the pupil size; the **Pipeline** tab explains why.

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3. Switch stages on and off

Under **Stages** there is one checkbox per stage, all ticked by default:

- **Retina — centre-surround**
- **LGN — opponent channels**
- **V1 — Gabor bank**
- **V2 — boundary signal**
- **V4 — colour cell**

Each one really switches its stage. When a stage is off, its settings group is hidden, and a map that belongs to it is drawn black with **— stage off** after its name. **V2** is computed from the V1 energy map, so it needs **V1** on as well.

---

4. What each stage computes

All five are standard forms from the vision literature, implemented with constants chosen by this tool. They show the shape of what each area does; they are not measurements of real neurons.

Retina: centre-surround

A retinal ganglion cell with a centre-surround receptive field responds to a spot that differs from its surroundings. The tool models this as a **difference of Gaussians**: the stimulus's relative luminance blurred by a small Gaussian (the centre), minus a weight times the same image blurred by a larger Gaussian (the surround). Positive responses form the **on-centre** map and negative ones the **off-centre** map.

Control	Range	Default
Centre σ	0.5 to 6, in pixels	1.5
Surround σ	1 to 14, in pixels	4.0
Surround weight	0 to 1.5	0.75

The sigmas are in pixels, not degrees. With a surround weight below 1 the centre and surround do not cancel over a uniform area, so the response carries a constant proportional to the mean luminance. That is why the off-centre map can be empty while the on-centre one is not. Raise the weight towards 1 to balance them.

LGN: opponent channels

The lateral geniculate nucleus carries colour as differences between cone signals. The tool computes, for every pixel, three channels from the cone signals L, M and S:

- **L − M** — the red-green channel.
- **S − (L+M)/2** — the blue-yellow channel.
- **L + M** — the tool's luminance channel.

A neutral grey sits at zero on both chromatic channels. Cones, opponent channels and the cone table explains how the cone signals are obtained and how these axes differ from the published DKL space.

V1: Gabor bank

Simple cells in primary visual cortex respond to stripes of a particular orientation and spacing. The tool filters the stimulus's luminance with a bank of **Gabor** filters — a sine-wave stripe pattern inside a Gaussian envelope — at evenly spaced orientations from 0° to 180°. For each orientation it uses a pair of filters a quarter-cycle apart and takes the length of the pair's response, which is the classic **complex-cell energy** model: it responds to a stripe pattern wherever the stripes' phase falls. The energies of all orientations are summed into one map. The bank runs at half the image resolution and the result is enlarged back to 256 × 256.

Control	Range	Default
Gabor frequency	0.2 to 30 c/deg	4.0 c/deg
Orientations	2 to 16	8
Kernel size	9 to 51 pixels, odd	31
Envelope σ	1 to 16	6.0

Gabor frequency is in cycles per degree, the same unit as the stimulus, so a filter set to the grating's frequency is tuned to it.

The **preferred orientation** map (the hypercolumn view) shows, for every pixel, which orientation in the bank responded most. It is drawn as a colour wheel: hue runs through the orientations from 0° to 180°, and saturation shows how selective the pixel is (the winning energy as a share of the total). Grey or pale pixels have no clear preference.

V2: boundary signal

The tool's V2 stage takes the V1 energy map, measures how steeply it changes from pixel to pixel, and divides the result by a blurred copy of itself (divisive normalisation). It marks where texture regions meet. Real V2 does more than this, and the Data tab says so.

V4: colour cell

A V4 colour cell prefers a direction in colour space. The tool projects each pixel's two chromatic values onto the direction you pick under **V4 preferred direction**, keeps only positive responses, and normalises. The directions are **Red** (default), **Green**, **Blue**, **Yellow**, **Orange** and **Purple**. Red points along +L − M, Green along −(L − M), Blue along +S − (L+M)/2, Yellow along its negative, and Orange and Purple between them. V4 reads the same cone signals and opponent axes as the LGN stage.

Colour vision

Colour vision changes the stimulus before any stage sees it: **Typical** (default), **Protanopia**, **Deuteranopia** or **Tritanopia**. The simulation uses the Viénot, Brettel and Mollon (1999) projection at full strength: the missing cone type's signal is replaced by a combination of the other two, so colours that differed only along the lost axis become the same. The stimulus canvas shows the simulated frame. The same setting applies to the colour list on the **Export** tab.

---

5. Choose and read a map

Under **Show**, the first menu picks the map for the right-hand canvas:

Map	Stage	What bright means
Retina, on-centre	Retina	Brighter than the surround
Retina, off-centre	Retina	Darker than the surround
LGN, L − M	LGN	More L than M signal

Map	Stage	What bright means
LGN, S − (L+M)/2	LGN	More S than the L and M average
LGN, L + M	LGN	More combined L and M signal
V1, complex-cell energy	V1	Strong oriented stripe energy at the Gabor frequency
V1, preferred orientation	V1	Colour wheel of the winning orientation
V2, boundary signal	V2	A change in V1 energy
V4, colour cell	V4	Colour close to the preferred direction

The second menu picks the colourmap: **Viridis** (default), **Inferno** or **Grayscale**. Grayscale is the one to use for printing.

Show the scale (ticked by default) shows the scale bar. Every map is stretched between its own minimum and maximum, so the numbers at the ends are the only thing that says what a bright pixel means: without them a bright pixel could be 0.2 or 20. The unit label reads **response, this map's own range**. For the orientation map the ends read 0° and 180°, the bar shows the hue wheel and the label reads **preferred orientation**. When a map has no variation at all, the label reads **flat — this stage found no variation in this stimulus**; for example, the off-centre cells find little to answer to in a chromatic grating.

Double-click either canvas to enlarge it; see Shortcuts.

6. Read the readouts

Readout	Meaning
L − M peak gap	How far apart the L and M curves peak in the cone table the page holds; highlighted when under 5 nm
Published gap	25 nm, the gap in the published cone fundamentals
Neutral off-axis	The largest distance from zero that a nine-step grey ramp reaches on either chromatic channel
Contrast achieved	The contrast actually drawn, after any cap
Sensitivity at this frequency	The tool's bandpass weighting curve at the stimulus frequency, from 0 to 1
The curve's peak	Where that curve peaks: 4.0 c/deg
Frame RMS contrast	The root-mean-square spread of relative luminance across the stimulus frame
Nyquist limit of the grid	The finest frequency your stated screen can draw

Neutral off-axis reads about 2.2×10^{-4} rather than zero because the published cone matrix is given to four decimals. The sensitivity reading is a weighting, not a probability of seeing the grating: the page gives no detection figure, because that would need a threshold criterion and a calibrated model it does not hold.

7. See every stage at once on the Cells tab

The **Cells** tab draws small (96 × 96) previews of every stage over the same grating, each with a name, what it computes, a short explanation and a tag.

- **Show**: **Everything** adds two cards for panels that were withdrawn from the page (**Spike raster** and **Detection %**); **The five that run** shows only the working stages.
- **Contrast**: **10%**, **25%** (default) or **100%**. At 100% each stage is at full drive, where a stage that does nothing is easy to spot.

The previews use your base colour, carrier, frequency, geometry, retinal settings, Gabor frequency and V4 direction, with a fixed V1 bank of 8 orientations, and the Viridis colourmap. They are drawn when you first open the tab and redrawn when you press one of its buttons, so after changing the Lab, press a **Show** or **Contrast** button to refresh them.

Below the grid, **Controls that were read and never used** lists six controls from an earlier version of the page and whether each was removed or connected up.

You should now be able to predict which maps a given stimulus will drive, and read their scale bars.

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Export and share

Run a list of colours through the Visual Pathway's front end, download it as CSV or copy it, save the active map as PNG, download the settings and copy a link to the view.

Nothing leaves the page until you press one of the buttons described here. Downloads are saved by your browser; copies go to your clipboard.

Run a list of colours

1. Open the **Export** tab.
2. In **Colours**, enter hex colours, one per line. Commas and semicolons also separate entries. Three-digit (`#F80`) and six-digit (`#FF8800`) forms work, with or without `#`. Anything else is ignored.
3. The table updates as you type; **Run them** redraws it.

The default list is `#E03131`, `#2F9E44`, `#1971C2`, `#F59F00` and `#808080`.

The table has one row per colour: **Input** (a swatch and the hex), **L**, **M**, **S**, **L − M**, **S − (L+M)/2** and **L + M**, each to four decimals. The header note gives the number of rows. The current **Colour vision** setting on the Lab tab is applied first, at full strength, so you can see what a dichromat's cone signals would be.

Read the grey row first. Because the cone matrix gives equal L, M and S for the D65 white, a neutral has both chromatic values at zero, and the page highlights them. If a grey row shows non-zero chromatic values, a colour vision simulation is on.

Download or copy the table

- **Download the CSV** saves `cones-opponent.csv`. It starts with comment lines, then one row per colour:

  ```
  # visual-pathway — cone signals and opponent channels
  # chain: sRGB → linear → XYZ → LMS → opponent
  # cone table: this tool's own (L peak 590 nm, M peak 590 nm)
  # opponent axes: L-M, S-(L+M)/2, L+M — not DKL
  # geometry: 57 cm, 0.248 mm pixel pitch
  # reportable against a published table: false
  # educational approximation, not a physiological measurement
  Input,L,M,S,RG,BY,Lum
  ```

  `RG` is L − M, `BY` is S − (L+M)/2 and `Lum` is L + M. Values have six decimals.

- **Copy to clipboard** copies the same table as tab-separated text with a header row (`Input`, `L`, `M`, `S`, `RG`, `BY`, `Lum`) and four decimals, ready to paste into a spreadsheet. A message confirms **Copied**, or says **Copy blocked** if the browser refused.

If the list is empty, the page says **No colours to write** or **No colours to copy**.

Save the active map

Save the map (on the Lab tab, under the readouts) saves the right-hand canvas as a 256 × 256 PNG named after the map, for example `visual-pathway-retina-on.png` or `visual-pathway-v1-orient.png`. The scale bar is not part of the image, so note its range if you need it.

Download the settings

Download the settings saves `visual-pathway-settings.json`. The Export tab shows its current content under **The settings file, and what it carries**. It records:

Field	Meaning
`tool`	`visual-pathway`
`stimulus`	`grating` or `testcard`
`spatial_frequency_cpd`, `viewing_distance_cm`, `pixel_pitch_mm`, `degrees_per_pixel`	The stimulus frequency and the geometry
`chain`	The colour chain used
`cone_table`, `cone_peaks_nm`	Whose cone curves drew the figure, and where L, M and S peak in them
`opponent_axes`	The three axes, marked "not DKL"
`stages`	The stages that ran for the current frame
`colour_vision`	`none`, `protan`, `deutan` or `tritan`
`reportable`	`true` only after you install a cone table
`clinical`	Always `false`

Copy a link to the view

Copy a link to this view copies an address that reopens the tool with the same stimulus, geometry, stage settings, colour vision setting, colourmap and map on screen. The message **Link copied** confirms it. The link carries these parameters: `source`, `baseHex`, `carrier`, `cpd`, `contrast`, `distanceCm`, `pitchMm`, `nOrient`, `rfSize`, `gaborCpd`, `gaborSigma`, `dogCenter`, `dogSurround`, `dogWeight`, `v4Preset`, `cvdType`, `colormap` and `viewMap`. It does not carry the pupil size, the stage checkboxes, the scale-bar setting, an installed cone table or your colour list.

The What to take buttons

The **What to take** buttons at the top of the Export rail are shortcuts: **A list of colours** keeps you on the list, **The active map** saves the map, **The settings** downloads the JSON and **A link to this view** copies the link.

You should now have your colours' cone and opponent values in a file, with the chain and the cone table named in its header.

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Visual Pathway reference

Every tab, control, readout, message and file format in the Visual Pathway, with ranges and defaults.

Address: https://auricartisan.com/tool/general/perception-and-neuroscience/neuroscience/

Tabs

Tab	Contents
Lab	Stimulus, geometry and stage controls; stimulus and map canvases; scale bar; cone figure; readouts; save and link buttons
Pipeline	The colour chain, the opponent-axis table of seven colours and the sensitivity table, with an earlier-version comparison
Cells	Previews of every stage over one stimulus, and the list of controls from an earlier version
Data	The register of nine datasets and transforms, filterable by status; the tab label shows the count (9); cone-table installer
Export	Colour list, CSV, clipboard, settings JSON and link
Reference	Standards, formulas, citations and research notes, with a search box

Links in the page text (for example **Data** or **Cells** in a hint) switch to that tab.

Lab tab — Stimulus

Control	What it does	Values or range	Default
Stimulus	Chooses the source	**Grating**, **Test card**	Grating
Base colour	The colour the grating is built from (grating only)	Any colour from the picker	`#3A8CFF`
Carrier	What the grating modulates (grating only)	**Chromatic**, **Luminance**	Chromatic

Lab tab — Geometry, frequency and contrast

Control	What it does	Range and step	Default
Viewing distance	Eye-to-screen distance used for every frequency	20–200 cm, step 1	57 cm
Pixel pitch	Physical size of one pixel	0.05–0.6 mm, step 0.001	0.248 mm
Spatial frequency	Grating frequency	0.2–30 c/deg, step 0.1	4.0 c/deg
Contrast	Grating swing either side of its mean	0–100%, step 1	25%
Pupil	Shown only; does not change any output in this version	2–8 mm, step 0.1	3.0 mm

Lab tab — Stages

Checkbox	Stage	Default
Retina — centre-surround	Difference-of-Gaussians on and off maps	On
LGN — opponent channels	L − M, S − (L+M)/2 and L + M maps	On
V1 — Gabor bank	Complex-cell energy and preferred orientation	On
V2 — boundary signal	Normalised gradient of V1 energy; needs V1	On
V4 — colour cell	Projection onto a preferred colour direction	On

Lab tab — Show

Control	Values	Default
Map menu	**Retina, on-centre**; **Retina, off-centre**; **LGN, L − M**; **LGN, S − (L+M)/2**; **LGN, L + M**; **V1, complex-cell energy**; **V1, preferred orientation**; **V2, boundary signal**; **V4, colour cell**	Retina, on-centre
Colourmap menu	**Viridis**, **Inferno**, **Grayscale**	Viridis
Show the scale	Shows or hides the scale bar	On

Lab tab — Stage settings

Shown only while the stage is on.

Control	Range and step	Default	Unit
Centre σ (Retina)	0.5–6, step 0.1	1.5	pixels
Surround σ (Retina)	1–14, step 0.1	4.0	pixels
Surround weight (Retina)	0–1.5, step 0.01	0.75	none
Gabor frequency (V1)	0.2–30, step 0.1	4.0	c/deg
Orientations (V1)	2–16, step 1	8	count
Kernel size (V1)	9–51, step 2	31	pixels
Envelope σ (V1)	1–16, step 0.1	6.0	pixels
V4 preferred direction	**Red**, **Green**, **Blue**, **Yellow**, **Orange**, **Purple**	Red	—

Lab tab — Colour vision

Control	Values	Default
Colour vision	**Typical**, **Protanopia**, **Deuteranopia**, **Tritanopia** (Viénot, Brettel and Mollon 1999, full strength)	Typical

Lab tab — Canvases, scale bar and figure

Element	Description
Stimulus canvas	256 × 256; the frame the stages see, after any colour vision simulation
Map canvas	256 × 256; the selected map, named underneath, with **— stage off** when its stage is off
Header note	`256 × 256 · N° per pixel` at the stated geometry
Scale bar	Minimum, colour ramp, maximum and label: **response, this map's own range**; **flat — this stage found no variation in this stimulus**; or 0°–180° **preferred orientation**
Cone figure	L, M and S curves 390–730 nm with dashed lines at the published peaks (442, 541, 567 nm) and a note

Lab tab — Readouts

Readout	Meaning	Format
L − M peak gap	Distance between the L and M peaks of the held cone table; highlighted below 5 nm	nm
Published gap	Fixed reference value	25 nm
Neutral off-axis	Worst chromatic departure of a nine-step grey ramp; highlighted above 0.001	Scientific notation
Contrast achieved	Contrast drawn after any sRGB cap; highlighted when capped	%
Sensitivity at this frequency	The tool's log-Gaussian weighting (peak 4 c/deg, 1.6 octaves) at the stimulus frequency	0–1, three decimals
The curve's peak	Peak of that weighting	4.0 c/deg
Frame RMS contrast	Root-mean-square spread of relative luminance over the stimulus frame	Four decimals
Nyquist limit of the grid	Half a cycle per pixel at the stated geometry	c/deg, one decimal

Lab tab — Notes and buttons

Element	When it appears or what it does
Above the pixel grid's Nyquist limit.	Spatial frequency is above the Nyquist limit; the drawn grating is an alias
The contrast you asked for does not fit in sRGB.	The requested swing does not fit both phases; the contrast was reduced
Save the map	Saves the map canvas as `visual-pathway-<map>.png`
Copy a link to this view	Copies a link with the current settings

Pipeline tab

Control	Values	Default
Show	**The chain**, **The opponent axes**, **The sensitivity curve**	The chain
Against	**What it was**, **Hide**	What it was

Outputs: the chain of seven steps; a table of `#000000`, `#404040`, `#808080`, `#C0C0C0`, `#FFFFFF`, `#3A8CFF` and `#E03131` with **L − M** and **S − (L+M)/2** (plus **as shipped** columns with **What it was**); or a table of **Pupil** (2, 3, 5, 8 mm), **Peak, as shipped**, **Peak now** and **Published photopic peak**.

Cells tab

Control	Values	Default
Show	**Everything** (eight stage cards plus two withdrawn panels), **The five that run**	Everything
Contrast	**10%**, **25%**, **100%**	25%

Cards: **Stimulus**, **Retina — centre-surround**, **LGN — L − M**, **LGN — S − (L+M)/2**, **V1 — Gabor energy**, **V1 — hypercolumn**, **V2 — boundary signal**, **V4 — colour cell**; withdrawn: **Spike raster**, **Detection %**. Each card is 96 × 96 with a tag: **pass-through**, **units noted**, **space fixed**, **units fixed**, **runs**, **stand-in**, **inherits** or **withdrawn**. The list **Controls that were read and never used** has six rows marked **removed** or **wired up**.

Data tab

Control	What it does
Status	Filters the register: **All** (9), **Verbatim** (2), **Computed** (2), **Stand-in** (4), **Absent** (1)
Register row	Select (or press `Enter` or `Space`) to expand its note and **What it would take**
Drop a CSV or JSON	Installs a cone table: wavelength, L, M, S; see Cones, opponent channels and the cone table

Register entries: **sRGB transfer and primaries** (Verbatim), **Dichromacy projection** (Verbatim), **Cone transform** (Computed), **Spectral sensitivity table** (Stand-in), **Opponent axes** (Stand-in), **Spatial sensitivity** (Stand-in), **Cortical stage models** (Stand-in), **Viewing geometry** (Computed), **Spike raster** (Absent).

Export tab

Control	What it does
What to take	**A list of colours**, **The active map** (saves the PNG), **The settings** (downloads the JSON), **A link to this view** (copies the link)
Colours	Hex colours, one per line; commas and semicolons also separate
Run them	Redraws the table (it also updates as you type)
Download the CSV	Saves `cones-opponent.csv`
Copy to clipboard	Copies a tab-separated table
Download the settings	Saves `visual-pathway-settings.json`

Table columns: **Input**, **L**, **M**, **S**, **L − M**, **S − (L+M)/2**, **L + M**.

Reference tab

Control	Values
Reference	**Standards** (5), **Formulas** (12), **Citations** (13, default), **Research notes** (7)
Filter	Searches all four groups at once; shows **Nothing in the reference matches that.** when empty

A ● before a citation marks one the current version corrected. Each entry carries a tag such as **runs**, **not held**, **re-named**, **form only**, **stand-in**, **context** or **measured**.

Messages

Message	Meaning
Cone table installed — L peaks at N nm	A dropped table was accepted
Not installed: too few rows	Fewer than eight rows
Not installed: row N has fewer than 4 columns / **row N is not numeric**	A malformed row
Not installed: wavelengths must increase	Rows out of order
Not installed: L and M peak N nm apart, which is not a cone table	L and M peaks closer than 5 nm

Message	Meaning
CSV saved, **PNG saved**, **Settings exported**, **Copied**, **Link copied**	The action succeeded
Copy blocked	The browser refused clipboard access
No colours to write / **No colours to copy**	The colour list is empty

File formats

File	Format	Contents
`cones-opponent.csv`	CSV with `#` comment lines	Chain, cone table and peaks, axes, geometry, reportable flag; columns `Input,L,M,S,RG,BY,Lum`
Clipboard table	Tab-separated text	`Input`, `L`, `M`, `S`, `RG`, `BY`, `Lum`, four decimals
`visual-pathway-<map>.png`	PNG, 256 × 256	The selected map without its scale bar
`visual-pathway-settings.json`	JSON	See Export and share
Cone table (input)	CSV or JSON	Four columns: wavelength (nm), L, M, S; at least eight rows, increasing wavelengths

Link parameters

`source`, `baseHex`, `carrier`, `cpd`, `contrast`, `distanceCm`, `pitchMm`, `nOrient`, `rfSize`, `gaborCpd`, `gaborSigma`, `dogCenter`, `dogSurround`, `dogWeight`, `v4Preset` (`red`, `green`, `blue`, `yellow`, `orange`, `purple`), `cvdType` (`none`, `protan`, `deutan`, `tritan`), `colormap` (`viridis`, `inferno`, `grayscale`), `viewMap` (`retina-on`, `retina-off`, `lgn-rg`, `lgn-by`, `lgn-lum`, `v1`, `v1-orient`, `v2`, `v4`).

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Troubleshooting the Visual Pathway

Symptoms, causes and fixes for flat maps, warnings, refused cone tables, links and copying in the Visual Pathway.

A map is completely black and its name ends in "— stage off"

Why: the stage that produces that map is switched off under **Stages**, or it is **V2** and **V1** is off.

What to do: tick the stage's checkbox. For **V2, boundary signal**, tick both **V1 — Gabor bank** and **V2 — boundary signal**.

The scale bar says "flat — this stage found no variation in this stimulus"

Why: the stage gave the same response everywhere. This is a real result. Common cases: the off-centre retinal map with a chromatic grating (the retinal stage reads luminance, and a surround weight below 1 adds a positive constant), the chromatic LGN maps with a luminance grating (a grey pattern has no colour to oppose), or any map at **Contrast** 0%.

What to do: change the **Carrier**, raise **Contrast**, or set **Surround weight** closer to 1 for the off-centre map.

A warning says "Above the pixel grid's Nyquist limit"

Why: at your viewing distance and pixel pitch, the screen cannot draw the spatial frequency you asked for. The pattern you see is an alias, a coarser false pattern.

What to do: lower **Spatial frequency** below the **Nyquist limit of the grid** readout. Also check that **Viewing distance** and **Pixel pitch** match your real setup. The limit rises when you sit further away and when the pixels are smaller; the warning's own text suggests moving closer, but moving closer lowers the limit.

A warning says "The contrast you asked for does not fit in sRGB"

Why: the colour opposite your base colour, at the swing you asked for, is outside what an sRGB screen can show. The page reduced the swing so both phases fit, rather than clipping one of them.

What to do: read **Contrast achieved** for the value actually used. To reach a higher contrast, choose a less saturated base colour.

Moving the Pupil slider changes nothing

Why: in the current version the pupil size is not used by any map or readout. The tool's sensitivity curve peaks at 4 c/deg whatever the pupil.

What to do: nothing; the **Pipeline** tab's **The sensitivity curve** view shows the reason.

L – M peak gap reads 0 nm and is highlighted

Why: the built-in cone table is the tool's own and puts the L and M peaks together at 590 nm. It affects only the cone figure.

What to do: install a published table on the **Data** tab; see Cones, opponent channels and the cone table.

My cone table is refused

Message	Fix
Not installed: too few rows	Supply at least eight rows
Not installed: row N has fewer than 4 columns	Every row needs wavelength, L, M and S
Not installed: row N is not numeric	Remove text from data rows (in CSV, a header line that starts with a letter is skipped automatically)
Not installed: wavelengths must increase	Sort the rows by wavelength, lowest first, with no repeats
Not installed: L and M peak N nm apart, which is not a cone table	Check the column order is wavelength, L, M, S

A JSON file must be an array of rows or an object with a `rows` array. A JSON file in any other shape is read as CSV text and usually fails.

My installed cone table disappeared

Why: an installed table lasts only for the current page visit.

What to do: install it again after reloading. Links do not carry it.

The Cells previews do not match my new settings

Why: the previews are drawn when you first open the **Cells** tab and when you press one of its buttons, not every time a Lab control moves.

What to do: press any **Show** or **Contrast** button on the Cells tab.

Nothing is copied, or the page says "Copy blocked"

Why: the browser refused clipboard access, which can happen in private windows, in embedded views or when the page is not focused.

What to do: click inside the page and try again, or use **Download the CSV** instead of **Copy to clipboard**.

A link I opened does not restore the stage checkboxes or pupil

Why: links carry the stimulus, geometry, stage settings, colour vision, colourmap and map, but not the stage checkboxes, the pupil or the scale-bar setting.

What to do: set those by hand after opening the link. See Export and share for the full list.

Sliders feel slow on an older device

Why: each change recomputes all five stages over a 256 × 256 frame.

What to do: switch off stages you are not looking at, or lower **Orientations** and **Kernel size**, which set most of the V1 work.

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Visual Pathway FAQ

Short answers to common questions about the Visual Pathway.

Is this a model of real neurons?

No. Each stage uses a standard form from the literature (a difference of Gaussians, a Gabor energy model, divisive normalisation, a projection in an opponent plane), but every constant is the tool's own and sizes are in pixels rather than degrees of retina. The maps show the shape of what each area does. The page marks itself "Educational, not physiological".

Why are the tool's L and M cone curves on top of each other?

The built-in spectral table is the tool's own and puts both peaks at 590 nm. It only draws the figure. The cone signals the stages use come from a published matrix (Hunt-Pointer-Estévez) applied to CIE XYZ. You can install a published table on the **Data** tab to fix the figure.

Are the opponent channels the DKL space?

No. They are differences of absolute cone signals (L − M, S − (L+M)/2) and a luminance channel of L + M. DKL uses cone contrast against a stated white and scaled isoluminant axes. See Cones, opponent channels and the cone table.

Why does a grey not read exactly zero on the chromatic channels?

It reads about 0.0002 at worst, because the published cone matrix is given to four decimals. That is the **Neutral off-axis** readout.

What does "Sensitivity at this frequency" mean? Is it the chance I can see the grating?

It is a weighting curve that peaks at 1.0 at 4 c/deg and falls off either side. It is not a probability of detection; the page deliberately gives no detection figure. To measure your own threshold, use the Psychophysical Experiment Engine.

Why do I need to enter a viewing distance and pixel pitch?

Cycles per degree depends on both. Without them, the same image has a different spatial frequency at every distance. Defaults are 57 cm and 0.248 mm.

Can I load my own image?

No. The sources are the grating and the built-in test card. To run simulations over your own images, use Perception and Vision.

Can I simulate a milder colour vision deficiency?

Not on this page: **Colour vision** applies full dichromacy. Perception and Vision has a **Severity** control.

Does the pupil control do anything?

Not in the current version. It is shown on the rail, but no map or readout uses it.

Is there a spike raster or firing rate?

No. An earlier version drew a panel called a spike raster; it had no time axis and was withdrawn. Nothing on the page may be reported as a spike train or a firing rate.

Does anything I enter leave my computer?

No. All processing is in the page. Files and copies are only created when you press an export or copy button. See Privacy.

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Visual Pathway background

Contents of the background material for the Visual Pathway tool.

This folder holds material that supports the how-to pages: the terms the tool uses, its keyboard and pointer actions, an honest statement of what is published and what is the tool's own, related tools and articles, its history and what stays on your device.

File	What it covers
glossary.md	Terms used in the Visual Pathway, each in a sentence or two
shortcuts.md	Double-click to enlarge, `Esc`, and keyboard use of the register
limits-and-accuracy.md	The data register entry by entry, precision, and what the tool is not
related.md	Related tools on the site and Learn articles
release-notes.md	User-facing history of the tool
privacy.md	What stays in your browser and what uses the network

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Visual Pathway shortcuts

Keyboard and pointer actions in the Visual Pathway tool.

The tool has no keyboard shortcuts of its own beyond these. Site-wide shortcuts, such as the search palette, are described in Site features.

Action	Where	Result
Double-click	**Stimulus** or map canvas on the Lab tab	Opens the canvas enlarged, filling the window
`Esc`	Enlarged view	Closes it
Select the × button or the dark area around the image	Enlarged view	Closes it
`Enter` or `Space`	A focused row on the **Data** tab	Expands or collapses the row
`Tab`	Anywhere	Moves between controls; sliders then respond to the arrow keys
Drag a file	**Drop a CSV or JSON** on the Data tab	Installs a cone table

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Visual Pathway glossary

Terms used in the Visual Pathway tool, each explained briefly.

Term	Meaning
Alias	A false, coarser pattern that appears when a screen is asked to draw stripes finer than its pixels can carry
Carrier	What a grating varies: colour at constant L + M (**Chromatic**) or lightness in grey (**Luminance**)
Centre-surround	A receptive field that compares a small central region with a larger surrounding one

Term	Meaning
Colourmap	The ramp of colours used to draw a map's values, here Viridis, Inferno or Grayscale
Complex-cell energy	The combined response of two filters a quarter-cycle apart; it responds to stripes regardless of their exact position
Cone signals (L, M, S)	The responses of the long-, medium- and short-wavelength-sensitive cones to a colour
Cone table	A table of cone spectral sensitivities by wavelength; the page's built-in one is its own, and you can install another
Cycles per degree (c/deg)	How many light-dark cycles of a pattern fall within one degree of visual angle
Degrees per pixel	The angle one screen pixel covers at your viewing distance: atan(pixel pitch ÷ distance)
Deuteranopia	Colour vision without M cones
Dichromacy	Colour vision with only two of the three cone types working
Difference of Gaussians (DoG)	A centre blur minus a weighted surround blur, the standard model of a retinal centre-surround cell
Divisive normalisation	Dividing a response by the local average response, so each part of a map is read relative to its neighbourhood
DKL space	The Derrington–Krauskopf–Lennie opponent space, built on cone contrast against an adapting white; the tool's axes are not DKL
Gabor filter	A sine-wave stripe pattern inside a Gaussian envelope, the standard model of a V1 simple cell
Hunt-Pointer-Estévez matrix	A published 3 × 3 matrix that converts CIE XYZ to cone signals, scaled so D65 white gives equal L, M and S
Hypercolumn map	A map of the preferred orientation at each point, drawn here as a colour wheel
LGN	The lateral geniculate nucleus, the relay between retina and cortex where colour is carried in opponent channels
Luminance (relative)	The lightness-related quantity Y of CIE XYZ, from 0 for black to 1 for white; the retina and V1 stages read it
Nyquist limit	The finest frequency a sampled grid can carry: half a cycle per pixel, converted here to c/deg
Opponent channel	A signal formed by comparing cone signals, such as L − M (red against green)
Pixel pitch	The physical width of one screen pixel, in millimetres
Protanopia	Colour vision without L cones
RMS contrast	The root-mean-square spread of luminance values across an image
Scale bar	The bar under the map giving the minimum and maximum the colours stand for
Spatial frequency	How finely a pattern repeats across space, here in cycles per degree
sRGB	The standard colour space of web content, with a defined transfer curve and primaries (IEC 61966-2-1)
Stand-in	A register status: values chosen by the tool to be plausible, not taken from a publication
Tritanopia	Colour vision without S cones
V1, V2, V4	Areas of visual cortex: V1 analyses orientation and spatial frequency, V2 boundaries and texture, V4 colour and form
Verbatim, Computed, Absent	Register statuses: taken unchanged from a source; derived from verbatim data by stated arithmetic; not held
Viewing distance	The distance from your eyes to the screen

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › NEUROSCIENCE

Limits and accuracy

What the Visual Pathway takes from published sources, what is its own, how precise its numbers are and what it is not.

What the tool is

An educational simulation of how early vision transforms a stimulus. It is not a physiological model fitted to measurements, not a diagnostic tool and not a vision test. Nothing it shows is a measurement of any real visual system.

The data register, in plain terms

The **Data** tab lists nine entries. This is what each means for the numbers you see.

Entry	Status	What it means for you
sRGB transfer and primaries	Verbatim	Screen colours are decoded and converted with the published IEC 61966-2-1 curve and matrices
Dichromacy projection	Verbatim	The three colour-vision simulations use the published Viénot, Brettel and Mollon (1999) matrices, applied in cone space
Cone transform	Computed	Cone signals come from the published Hunt-Pointer-Estévez matrix applied to XYZ; greys sit within about 2.2×10^{-4} of zero on the chromatic axes
Spectral sensitivity table	Stand-in	The cone curves in the figure are the tool's own, with L and M both peaking at 590 nm; they draw the figure and nothing else
Opponent axes	Stand-in	L − M, S − (L+M)/2 and L + M are useful axes but are not DKL; nothing is in cone-contrast units
Spatial sensitivity	Stand-in	The weighting curve is the tool's own log-Gaussian peaking at 4 c/deg; it has no threshold criterion and gives no detection probability
Cortical stage models	Stand-in	Retina, V1, V2 and V4 use standard forms with constants chosen by the tool, with sizes in pixels rather than degrees
Viewing geometry	Computed	Degrees per pixel and the Nyquist limit follow exactly from the distance and pitch you enter
Spike raster	Absent	The tool has no time axis and no firing model, so nothing may be reported as a spike train or a firing rate

A figure that depends on a stand-in or absent entry is never labelled as a published value. The settings file and CSV carry `reportable: false` (or the line "reportable against a published table: false") until you install a published cone table.

Precision

- Colours are handled as 8-bit sRGB values and computed in double precision.
- The maps are 256 × 256 pixels. The V1 bank runs at half resolution (128 × 128) and is enlarged back, so V1 and V2 maps are blockier than the retinal and LGN maps.
- Every map is stretched to its own range. Compare maps by their scale-bar numbers, not by their brightness.
- The grating's contrast is capped where sRGB cannot show it; **Contrast achieved** gives the value used.
- The geometry is only as accurate as the viewing distance and pixel pitch you enter. The page cannot measure your screen or your seat.
- Your screen is assumed to follow the sRGB standard. An uncalibrated or wide-gamut screen shows the stimulus differently.

What it does not do

- It does not model the eye's optics, eccentricity, eye movements or time: every stage is a single static frame.
- It does not simulate anomalous trichromacy or a partial deficiency: **Colour vision** is full dichromacy.
- It does not load your own images.
- It does not report detection probabilities, thresholds, spike trains or firing rates.

- The **Pupil** control does not change any output in the current version.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › NEUROSCIENCE

Privacy

What the Visual Pathway keeps in your browser and what, if anything, uses the network.

What stays in your browser

- Every calculation: the stimulus, the five stages, the readouts and the colour list are computed in the page.
- A cone table you install is read from your disk by the page. It is not uploaded and is forgotten when you reload.
- The colours you type in the Export tab stay in the page.

What leaves the page, and only when you ask

- **Download the CSV**, **Save the map** and **Download the settings** create files through your browser's normal download.
- **Copy to clipboard** and **Copy a link to this view** write to your clipboard.
- A copied link contains your settings in its address (for example the base colour and viewing distance). Anyone you send it to can read them.

The rest of the page

The tool page loads the site's usual pages, styles and scripts, and, depending on your cookie choices, advertising and consent-based analytics, like the rest of auricartisan.com. See the Privacy Policy at https://auricartisan.com/legal/privacy/ and the Cookie Policy at https://auricartisan.com/legal/cookies/.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › NEUROSCIENCE

Release notes

User-facing history of the Visual Pathway tool.

The site-wide history is at https://auricartisan.com/library/changelog/. The entries below cover what changed for users of this tool.

September 2026 — rebuilt as the Visual Pathway

The page was rebuilt with six tabs (**Lab**, **Pipeline**, **Cells**, **Data**, **Export**, **Reference**). The **Pipeline** and **Cells** tabs can show values from the earlier version beside the current ones.

Added:

- **Viewing distance** and **Pixel pitch** controls; every frequency is now in cycles per degree, including the V1 **Gabor frequency** (previously cycles per image).
- The **Nyquist limit of the grid** readout and a warning when the grating is finer than the screen can draw.
- **Contrast achieved** and a note when the requested contrast does not fit in sRGB.
- A working scale bar under every map, including a hue wheel for the orientation map and a "flat" label for maps with no variation.
- The **Test card** as a stimulus source.

- The cone figure with the published peak wavelengths marked, the **L − M peak gap**, **Published gap** and **Neutral off-axis** readouts, and a cone-table installer on the **Data** tab.
- The **Data** register with a status for every dataset and model.

Changed:

- Cone signals are now computed from CIE XYZ, so greys sit at zero on both chromatic channels. This changes all LGN and V4 values and the Export table.
- The colour-vision simulation is now applied in cone space and credited to Viénot, Brettel and Mollon (1999).
- The opponent axes are no longer called DKL, and the sensitivity curve is no longer credited to Watson and Ahumada; it now peaks at 4 c/deg regardless of pupil size.
- Maps are computed in the background, so sliders respond faster.

Removed:

- The **Detection** percentage, the spike raster panel and its Poisson option, the dominant-wavelength readout, and two checkboxes that switched nothing.

May 2026

- The Neuroscience User Guide was published in the site documentation.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › NEUROSCIENCE

Related tools and articles

Tools and Learn articles related to the Visual Pathway.

Tools in this group

- Perception and Vision — the same dichromacy simulation, plus night vision, lens ageing, cataract, central loss and acuity, over your own images and with a severity control.
- Psychophysical Experiment Engine — measure a real contrast threshold for a Gabor patch, using the same viewing-geometry idea.
- Material Lighting and Industrial Dye Chemistry — where the colour a cone receives comes from: light and matter, wavelength by wavelength.

Other tools on the site

- Vision simulation — the Vision Lab for checking designs under different visual conditions.
- Colour Science Lab — colour spaces and conversions in a workspace.

Learn articles

The Learn section is described in Learn.

- Human Color Vision: Cones, Opponent Signals, and the Brain — https://auricartisan.com/library/learn/articles/2026-05-24-human-color-vision-cones-opponent
- Afterimages and the Opponent Process: Why Colors Leave Ghosts — https://auricartisan.com/library/learn/articles/2026-07-02-afterimages-and-the-opponent-process

- Contrast Sensitivity: The Spatial Frequencies You Can and Can't See — https://auricartisan.com/library/learn/articles/2026-08-24-contrast-sensitivity-spatial-frequency
- Color Blindness and Color Vision Deficiency — https://auricartisan.com/library/learn/articles/2026-05-24-color-blindness-cvd

Documentation on the site

- Neuroscience User Guide — https://auricartisan.com/library/documentation/articles/2026-05-24-neuroscience-user-guide

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PERCEPTION AND VISION

Perception and Vision

What the Perception Vision tool does, who it is for, where to find it and how to run a first colour vision simulation over your own image.

Perception and Vision (the page is titled **Perception Vision**) runs eight simulations of how a scene reaches a different visual system: colour vision deficiency (protan, deutan and tritan), rod-only night vision, twilight vision, a yellowing lens, cataract, loss of central vision, light adaptation and reduced acuity. You run them over a built-in test card, a grey ramp, a Snellen chart or an image of your own, and compare the result with the original in a split view.

The colour vision simulation follows the published method of Viénot, Brettel and Mollon (1999): each colour goes from sRGB through CIE XYZ into cone signals, the missing cone type is projected away, and the result comes back to sRGB. The page checks itself as it goes. A grey ramp shows whether the simulation tints neutrals (a correct dichromacy simulation never does), a counter reports how many of the frame's colours merged into one, and the graphics-card and processor renderers are compared on 216 colours after every change.

The tool is an educational and design-review aid. Only the colour vision projection comes from a publication; the other seven simulations use the tool's own constants and show that an effect exists rather than how strong it is for any person. It is not a clinical instrument, and everything runs in your browser.

Who it is for

- Designers and front-end developers checking whether a screen, chart or palette still works for people with colour vision deficiency, in dim light or with ageing eyes.
- Accessibility specialists who want a traceable simulation with a stated method.
- Students learning what dichromacy, rod vision and lens yellowing do to colour.

Where to find it

Open https://auricartisan.com/tool/general/perception-and-neuroscience/perception-vision/ in a current desktop browser. It is listed as **Perception & Vision** on the Collections page (https://auricartisan.com/collections/) and as **Perception and Vision** in the site menu. WebGL2 is used when available; without it the page runs the same simulations on the processor.

Quick start

1. Open the tool. The **Lab** tab shows the test card, split down the middle: original on the left, protanopia on the right.
2. Under **Source**, choose **Uploaded image**, select **Choose a file** and pick a screenshot or picture.
3. Under **Simulation**, keep **Colour vision** and try **Protan**, **Deutan** and **Tritan** under **Type**.
4. Drag on the image to move the divider, or set **Divider** on the rail.
5. Read **Colours merged** and **Distinct colours left** to see how many of your image's colours became indistinguishable.
6. Try **Scotopic**, **Lens yellowing** or **Acuity**, or pick a **Presets** entry.
7. Select **Save the frame** to download the view as a PNG, or open **Export** to check a list of hex colours.

What you can do

- Simulate eight visual conditions over a test card, a grey ramp, a Snellen chart or your own image.
- Set severity, adaptation level, age, scotoma radius, cataract level and Snellen acuity.
- Daltonise instead of simulate, to see a colour correction for dichromats.

- Relight the scene with illuminant D50 or A before it reaches the simulated eye.
- Add contrast, gamma, blur, glare and noise adjustments to both halves of the comparison.
- Compare as a split view, simulated only, or an amplified difference image.
- Read neutral drift, merged colours, out-of-gamut share, renderer agreement and frame luminance.
- Run a list of hex colours through the simulation and export CSV, clipboard text, settings JSON, a PNG of the frame or a link.

In this folder

- docs/ — how to use the tool.
  - Getting started — open it, the screen tour, the first simulation.
  - Simulate a design or image — sources, the eight simulations, their controls, presets and comparison views.
  - Read the checks — the grey ramp, the readouts, the Simulation and Modes tabs.
  - Check a palette and export — the colour list, CSV, clipboard, frame PNG, settings and links.
  - Reference — every control, readout, tab and file format.
  - Troubleshooting — symptoms, causes and fixes.
  - FAQ — short answers to common questions.
- others/ — background material.
  - Glossary — terms used in this tool.
  - Shortcuts — keyboard and pointer actions.
  - Limits and accuracy — what is published, what is the tool's own, what is not held.
  - Related — related tools and Learn articles.
  - Release notes — user-facing history.
  - Privacy — what stays in your browser.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PERCEPTION AND VISION

Using Perception and Vision

Contents of the how-to documentation for the Perception Vision tool.

These pages explain how to run the simulations over your own material, how to read the page's built-in checks and how to take results away. Start with getting started; the reference lists every control.

File	What it covers
getting-started.md	Opening the tool, the six tabs, the Lab screen tour and a first simulation
simulate-a-design.md	Sources, the eight simulations and their settings, daltonising, the illuminant, adjustments, presets and comparison views
read-the-checks.md	The grey ramp, every readout, renderer agreement, and the Simulation and Modes tabs
check-a-palette-and-export.md	Running hex colours through the simulation, CSV and clipboard formats, the frame PNG, settings JSON and links
reference.md	Every tab, control, readout, message and file format, with ranges and defaults
troubleshooting.md	Camera, WebGL2, uploads, readouts that say n/a, links and copying
faq.md	Short answers to common questions

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PERCEPTION AND VISION

Getting started with Perception and Vision

Open Perception Vision, learn the layout of the Lab tab and run a first colour vision simulation over the test card and your own image.

Open the tool

Go to https://auricartisan.com/tool/general/perception-and-neuroscience/perception-vision/. The page opens on the **Lab** tab in its default state: the built-in test card, the **Colour vision** simulation at **Protan**, **Severity** 1.00, a **Split** comparison with the divider in the middle, illuminant D65 (the scene as authored) and the **GPU** renderer.

A note strip above the tabs, which cannot be closed, states the check this page is built around: a colour-blind viewer sees white as white. Its **Data** link opens the register that explains the method.

The six tabs

Tab	What it is for
Lab	The workbench: controls on the left, the simulated frame on the right, the grey-ramp check and the readouts below
Simulation	One colour's route through the colour vision simulation, the matrices, and a table of test colours
Modes	Every simulation side by side over the same source
Data	The register of every dataset and transform the tool rests on, with a status for each
Export	A list of colours through the simulation, and the download, copy, load and link buttons
Reference	Standards, formulas, citations and research notes, with a search box

A tour of the Lab tab

The rail (left), from top to bottom:

- **Source** — test card, grey ramp, Snellen chart, your image, or camera.
- **Simulation** — nine buttons: the eight simulations and **None**.
- Settings for the chosen simulation. For **Colour vision** these are **Type** (**Protan**, **Deutan**, **Tritan**), **Daltonise instead of simulate** and **Severity**. Other simulations show their own slider (adaptation level, age, scotoma radius, cataract or acuity) under **Severity**.
- **Compare** — **Split**, **Simulated only** or **Difference**, and the **Divider** position for a split.
- **Renderer** — **GPU** or **CPU**, and **Fixation cross**.
- **Before the eye** — the illuminant and its **Adaptation** strength.
- **Adjustments** — contrast, gamma, blur, glare and noise.
- **Presets** and **Reset**.

The stage (right):

- A caption naming what is shown (for example **Colour vision, protanopia at severity 1.00**) and, on the right, a coloured dot, the renderer in use and the frame size (for example **GPU · 800 × 600**).
- The frame. In a split, the left half is labelled **Original** and the right half with the simulation's name. The line under the frame names the method's source.
- **The grey ramp, which is the test that needs no reference**: nine grey steps with a number under each, the channel spread the simulation adds to that step.
- Two rows of readouts: **Neutral drift**, **Colours merged**, **Out of sRGB**, **GPU and CPU agree**, then **Distinct colours left**, **Largest channel shift**, **Frame luminance, mean**, and **min and max**.
- Two buttons: **Save the frame** and **Copy a link to this view**.

Your first simulation

1. Look at the test card. Its top band has eight colour patches, the middle band a nine-step grey ramp, and the bottom three single-channel ramps (red, green and blue). The line printed on the card reads "a simulation that tints the ramp below is wrong".
2. With **Colour vision** and **Protan** selected, compare the halves. Red and green patches move towards similar yellowish browns; the grey ramp stays grey.
3. Check the numbers under the grey-ramp strip below the frame: all nine should be `0`, and **Neutral drift** should read `0`.
4. Read **Colours merged**: the share of the card's colours that the simulation mapped onto the same colour as another.
5. Select **Deutan**, then **Tritan**, and watch the patches and the merged figure change.
6. Under **Source**, choose **Uploaded image**, select **Choose a file** and open an image. It appears in the frame, scaled to fit within 1200 × 900 pixels.
7. Drag across the frame to move the divider over the part you care about.
8. Select **Save the frame** to download the view.

You should now have a PNG of your image, half original and half as a protanope would see it, and a figure for how many of its colours merged.

Next steps

- Simulate a design or image covers every simulation and control.
- Read the checks explains the grey ramp and the readouts.

- Check a palette and export shows how to test a list of brand colours.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PERCEPTION AND VISION

Check a palette and export

Run a list of hex colours through the active simulation, find colours that become indistinguishable, and export CSV, clipboard text, the frame, the settings and links.

Nothing leaves the page until you press one of the buttons described here.

Check a palette

1. On the **Lab** tab, set up the simulation you want to test, for example **Colour vision**, **Deutan**, **Severity** 1.00.
2. Open the **Export** tab.
3. In **Colours**, enter your hex colours, one per line. Commas and semicolons also separate entries; three- and six-digit forms work, with or without `#`; anything else is ignored. Include a grey.
4. The table updates as you type; **Run them** redraws it.

The default list is `#E03131`, `#2F9E44`, `#1971C2`, `#F59F00` and `#808080`. The caption names the simulation, for example **The colours, through colour vision (protan) at severity 1.00**.

Column	Meaning
Input	Your colour, as a swatch and hex
Simulated	The colour after the simulation, as a swatch and hex
L, **M**, **S**	The cone signals of the input colour, four decimals
ΔE00	The CIEDE2000 colour difference between input and simulated colour, one decimal
In sRGB	**yes**, or **clamped** when the simulated colour fell outside sRGB and was clipped

How to read it:

- **Two rows with the same or nearly the same Simulated hex** are two colours the simulated viewer cannot tell apart. Change one of them.
- **The grey row** should read ΔE00 0.0 and an identical simulated hex under Colour vision; a dichromacy projection leaves greys alone.
- **ΔE00** measures how far each colour moved. A large value means the colour looks very different to the simulated viewer, not necessarily that it is confused with another; compare the simulated hexes for that.
- **clamped** means the hex you would paste is not exactly what the simulation produced.

The colour list uses the per-colour part of the simulation, including the illuminant under **Before the eye**. Central loss and Acuity depend on position, so colours pass through them unchanged; Cataract applies only its colour cast and contrast loss.

ΔE00 (CIEDE2000) is the CIE's current formula for how different two colours look. Around 1 is the smallest difference most people notice side by side; differences of 10 or more are obvious.

Download or copy the table

- **Download the CSV** saves a file named after the simulation and severity, for example `perception-cvd-100.csv`. It starts with comment lines naming the simulation, the chain, the matrix, the illuminant and adaptation, the renderer and whether

the renderers agreed, and the line "educational approximation, not a clinical measurement". The columns are `Input,Simulated,L,M,S,RG,BY,DeltaE00,InSRGB`, where `RG` is L − M and `BY` is S − (L+M)/2 of the input.
- **Copy to clipboard** copies a tab-separated table with a header row: `Input`, `Simulated`, `L`, `M`, `S`, `DeltaE00`, `InSRGB`.

If the list is empty the page says **No colours to write** or **No colours to copy**.

Save the frame

Save the frame (Lab tab) downloads the frame as a PNG, for example `perception-scotopic-70.png`, including the divider and fixation cross if they are showing. In the Export tab, **What to take** › **The frame** does the same.

Download and load settings

Download the settings saves `perception-vision-settings.json`. The Export tab shows the current content under **The settings file, and what it carries**:

Field	Meaning
`tool`	`perception-vision`
`simulation`, `label`, `type`, `severity`	The simulation, its name, the colour vision type (or `null`) and the severity
`space`	The chain used, for example `sRGB → linear → XYZ → LMS (HPE) → project → XYZ → linear → sRGB`
`matrix`	The published matrix or the tool's own model used
`illuminant`, `adaptation`	The illuminant and the Bradford strength, or "none (source white)"
`source`, `compare`	The source and comparison mode
`adjustments`	`contrast`, `gamma`, `blur`, `noise` and `bloom` (glare)
`renderer`	`gpu` or `cpu`
`agrees_with_cpu`, `agreement_worst_code_values`	The renderer check result, or `null` when it could not run
`clinical`	Always `false`

Load settings opens a settings file and applies its simulation, type, severity, illuminant, source, comparison mode and adjustments. It does not load an image: if the file's source is **Uploaded image**, choose the file again. The message **Settings imported** confirms it; **Import failed:** followed by a reason means the file was not valid JSON.

Copy a link to the view

Copy a link to this view copies an address that reopens the tool with the same simulation, type, severity, simulation settings, adjustments, illuminant and strength, comparison mode, source and renderer. The link does not include the divider position, the daltonise and fixation-cross checkboxes, or an uploaded image.

The What to take buttons

A list of colours keeps you on the list; **The frame** saves the PNG; **The settings** downloads the JSON; **A link to this view** copies the link.

You should now have a table showing which of your colours survive the simulation, and a file that records exactly how it was produced.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PERCEPTION AND VISION

Read the checks

Understand the grey-ramp test, every readout under the frame, the renderer agreement check, and the Simulation and Modes tabs.

The page does not ask you to trust its simulation. It shows checks you can read yourself. This guide explains each one.

The grey ramp: the test that needs no reference

Under the frame, **The grey ramp, which is the test that needs no reference** shows nine grey steps from black to white, and under each step a number: how far apart the red, green and blue channels of that grey end up after the simulation, in 8-bit code values (0 to 255).

- For **Colour vision** the numbers must all be `0`. A dichromacy projection removes one colour axis and leaves the grey axis alone, so a person with a colour vision deficiency sees a grey as grey and white as white. A non-zero step is a defect and is shown in red. The note under the strip also shows what an earlier version of the page produced.
- For **Scotopic**, **Mesopic**, **Lens yellowing**, **Cataract** and **Light adaptation**, non-zero numbers are the effect itself: these simulations are meant to tint or shift neutrals. The note says so.
- For **Central loss** and **Acuity**, which depend on position rather than colour, the ramp measures only their colour part, which is none.
- For **None**, every number is zero.

The ramp leaves out the illuminant, because relighting a scene with illuminant A tints greys on purpose; including it would hide the signal the ramp exists to show.

The readouts

Readout	What it tells you
Neutral drift	The largest of the nine grey-ramp numbers; highlighted when it is above zero under Colour vision
Colours merged	The share of the frame's distinct colours that the simulation mapped onto a colour already taken by another, as a percentage
Out of sRGB	The share of the frame whose simulated colour fell outside what sRGB can show and had to be clipped
GPU and CPU agree	Whether the two renderers give the same answer on a grid of 216 colours: **yes** (within 2 code values), **N off**, **n/a — spatial** or **not run**
Distinct colours left	How many different colours remain after the simulation, out of how many before, for example **812 of 1540**
Largest channel shift	The biggest change in any channel of any pixel, in code values
Frame luminance, mean	The average relative luminance of what is on screen, from 0 to 1
min and max	The darkest and lightest relative luminance on screen

Colours merged is the figure a colour vision simulation is for: colours that were distinct and arrive at the same place are colours the simulated viewer cannot tell apart. The merged, out-of-sRGB, distinct and largest-shift figures are measured on a reduced copy of the source (64 × 48 pixels), so they describe the image's main colours rather than every pixel. They read **n/a** or **spatial mode** for **Cataract**, **Central loss** and **Acuity**, which change pixels according to their neighbours or position.

The luminance figures are read from the frame actually on screen, including the split and any adjustments.

The readouts refresh shortly after each change.

Renderer agreement

The colour part of every simulation is one calculation that both renderers use. After each change the page draws 216 colours (six levels each of red, green and blue) on the GPU, reads them back, and compares them with the processor's answer. A gap of up to 2 code values counts as agreement and reads **yes**. The check covers colour only, so it reads **n/a — spatial** for Cataract, Central loss and Acuity, and **not run** when the GPU is unavailable. The settings file records the result.

The Simulation tab

This tab follows a colour through the colour vision simulation.

Show

- **The chain** — the steps: **sRGB** (8-bit, encoded) → **linear** (the transfer curve) → **XYZ** → **LMS** (Hunt-Pointer-Estévez) → **project** (the 3×3 for the type) → **XYZ** → **linear** (and back to sRGB). The two XYZ steps are marked as ones an earlier version skipped.
- **The matrices** — four matrices to four decimals: **sRGB** → **XYZ**, **XYZ** → **LMS**, **project** for the type, and **the whole chain, in linear sRGB**, the single matrix both renderers use.

Against

- **What it was** (default) — the table adds **On screen** and **In the file** columns, the results an earlier version produced on screen and in its export, and **Apart**, the largest channel gap to the current result.
- **Hide** — current results only.

Type — **Protan**, **Deutan**, **Tritan** or **All three**.

The table runs seven colours (`#F59F00`, `#E03131`, `#C2255C`, `#1971C2`, `#2F9E44`, `#808080`, `#FFFFFF`) at severity 1.00 and shows each input and its result **Through XYZ** as swatches and hex. Under it, **The tell: a neutral must stay neutral** repeats the grey-ramp check for the chosen type.

The Modes tab

This tab draws every simulation over the same source, as small previews computed on the processor.

- **Show**: **Everything** (all nine entries plus a card for the withdrawn **CSF overlay**) or **The eight that simulate**.
- **On**: **Test card**, **Grey ramp** or **Snellen chart**.
- **Severity**: **0.50** or **1.00** (default).

Each card gives the simulation's name, its source, a short explanation and a tag (**pass-through**, **space fixed**, **weights noted** or **runs**). The previews use fixed settings: adaptation level 0.35, age 70, cataract 0.60, scotoma radius 0.18 and acuity 20/90; the colour vision type follows the Lab. The CSF overlay card explains why a contrast-sensitivity mode was withdrawn. Below the grid, **Controls that were read and never used** lists six controls from an earlier version and whether each was removed or connected.

The previews are drawn when you first open the tab and when you press one of its buttons.

The Data and Reference tabs

The **Data** tab lists nine entries with a status each: **Verbatim** (3), **Computed** (3), **Stand-in** (2) and **Absent** (1). Select a row, or press `Enter` or `Space` on it, to read its note and what it would take to hold it properly. The **Install a table** area on this tab (**Drop a CSV or JSON**) does not act on files in the current version.

The **Reference** tab groups its entries as **Standards** (7), **Formulas** (11), **Citations** (13, the default) and **Research notes** (9). **Filter** searches all four. A ● marks a citation the current version corrected.

You should now be able to tell, for any simulation, whether the page's own checks pass.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PERCEPTION AND VISION

Simulate a design or image

Choose a source, run any of the eight simulations with its settings, relight the scene, apply adjustments and presets, and compare the result.

All controls in this guide are on the **Lab** tab.

1. Choose a source

Source	What it is
Test card (default)	Eight colour patches, a nine-step grey ramp and red, green and blue ramps, 800 × 600
Grey ramp only	The nine grey steps across the whole frame, for reading neutral drift at a glance
Snellen chart	Letter rows labelled 20/200, 20/100, 20/70, 20/50, 20/40, 20/30 and 20/20, so the **Acuity** simulation has something to be judged on
Uploaded image	A picture from your device
Camera	Live frames from a camera; see the note below

Uploaded image. Choose it, then select **Choose a file** and pick any image your browser can open (PNG, JPEG, WebP and so on). The file name appears beside the button. Large images are scaled down to fit within 1200 × 900 pixels; small ones are not enlarged. The image is read by the page and never uploaded.

Camera. Choose it, then **Start or stop** switches the camera on and off; the status reads **off**, **active** or **unavailable**. The camera needs your browser's permission, and auricartisan.com currently does not allow its pages to use the camera, so in most browsers the page shows a message beginning **Camera:**, sets the status to **unavailable** and returns to the test card. Use a screenshot or photo with **Uploaded image** instead.

2. Pick a simulation

Select one of the buttons under **Simulation**. Every simulation has a **Severity** slider (0 to 1, default 1.00): 0 leaves the image alone and 1 applies the simulation in full; values between blend the two.

Colour vision

Simulates dichromacy, colour vision with one cone type missing. Choose the **Type**:

- **Protan** — no L (long-wavelength) cones; reds darken and reds and greens converge.
- **Deutan** — no M (medium-wavelength) cones; reds and greens converge without the darkening.

- **Tritan** — no S (short-wavelength) cones; blues and greens converge, and yellows move towards pink.

Each colour is converted from sRGB to linear light, to CIE XYZ, to cone signals (Hunt-Pointer-Estévez), projected with the Viénot, Brettel and Mollon (1999) matrix for the type, and converted back. The projection leaves greys unchanged.

At a **Severity** below 1 the page blends between the original and full dichromacy. That is a convention of this tool, not a model of anomalous trichromacy (a weakened rather than missing cone type).

Daltonise instead of simulate shows a correction instead: the colour difference the projection would lose is pushed into channels a dichromat can still see (after Fidaner, Lin and Ozguven, 2005). Use it to preview how a daltonising filter would recolour the image.

Scotopic

Night vision with rods only. Colour collapses to a single rod signal, which is darkened and drawn with a blue cast. The three weights used to form the rod signal from red, green and blue (0.10, 0.59 and 0.31) are the tool's own and are not the published scotopic curve, so no scotopic luminance is reported.

Mesopic

Twilight vision, between rods and cones. **Adaptation level** (0 to 1, default 0.50) crossfades between the rod image at 0 and the original colours at 1. This is a crossfade, not the CIE mesopic photometry system.

Lens yellowing

The eye's lens absorbs more blue light with age. **Age** (20 to 80 years, default 40) sets a blue-absorbing filter that is zero at 20 and grows to its maximum at 80. The direction follows the published ageing literature; the strength is the tool's own.

Cataract

A clouded lens scatters light forward. **Cataract** (0 to 1, default 0.50) sets a blur, a slight yellow cast and a loss of contrast towards grey together. This simulation reads neighbouring pixels.

Central loss

A soft-edged disc of lost vision at the centre of the frame, drawn as a darkened grey. **Scotoma radius** (0.02 to 0.45, default 0.15) is the disc's radius as a fraction of the frame. It is named for what it draws, not for a particular disease. Tick **Fixation cross** to mark the centre.

Light adaptation

The eye adjusting to a brighter or dimmer scene. **Adaptation level** (0 to 1, default 0.50) sets a gain and a saturation together: low levels brighten and desaturate the image, high levels darken it slightly with full saturation. The constants are the tool's own.

Acuity

Blur from reduced sharpness of vision. **Acuity** is the denominator of a Snellen fraction, from 20/20 (no blur) to 20/200 in steps of 5; the blur grows with the denominator. The mapping from Snellen value to blur is the tool's own and does not use a viewing distance, so it shows the kind of loss, not how a given person with that acuity would see your screen. Use the **Snellen chart** source to judge it.

None

Shows the source unchanged. It is what the other half of a split is compared against, and it is not counted as a simulation.

3. Relight the scene (Before the eye)

The illuminant menu changes the light falling on the scene before the simulated eye sees it:

- **D65 — as authored** (default) — no change; screen images are authored under D65.
- **D50** — a slightly warmer daylight.
- **A — tungsten** — incandescent light, much warmer.

The change uses the Bradford chromatic adaptation transform. **Adaptation** (0 to 1, default 1.00) sets how far it goes. Under illuminant A many colours leave the sRGB range; that is the correct result, and **Out of sRGB** reports how much.

4. Apply adjustments

These change the source for both halves of a comparison, so the only difference across the divider is the simulation. They are not simulations.

Adjustment	Range	Default	Effect
Contrast	−0.9 to 0.9	0.00	Stretches or flattens values around mid-grey
Gamma	0.4 to 2.6	1.00	Raises linear values to this power; above 1 darkens mid-tones
Blur	0 to 8	0.0	Blurs the source
Glare	0 to 1	0.00	Adds a halo around bright areas
Noise	0 to 0.08	0.000	Adds random speckle, redrawn each frame

5. Use a preset

Presets sets several controls at once:

Preset	Sets
Protanopia	Colour vision, Protan, Severity 1.00
Mild deuteranomaly	Colour vision, Deutan, Severity 0.40
Tritanopia	Colour vision, Tritan, Severity 1.00
Low light with glare	Scotopic, Severity 0.70, Glare 0.30, Noise 0.010, Contrast −0.20, Gamma 1.30
Lens at 70	Lens yellowing, Severity 1.00, Age 70
Cataract	Cataract, Severity 0.80, Cataract 0.60
Central loss	Central loss, Severity 0.90, Scotoma radius 0.18
Acuity 20/80	Acuity, Severity 1.00, Acuity 20/80

A preset changes only the values listed; other adjustments stay as they were. **Reset** returns everything to the defaults: test card, Colour vision Protan at 1.00, split at 50%, D65 at full adaptation, all adjustments neutral, daltonise and fixation cross off, and the GPU renderer if available. It also stops the camera.

6. Compare

Under **Compare**:

- **Split** (default) — original on the left of the divider, simulated on the right. Drag anywhere on the frame to move the divider, or use the **Divider** slider (0 to 100%, default 50%). The side labels read **Original** and the simulation's name.
- **Simulated only** — the whole frame simulated.
- **Difference** — the absolute difference between original and simulated, amplified four times, so black means "unchanged".

Renderer chooses where the drawing is computed: **GPU** (WebGL2, the default) or **CPU**. The colour arithmetic is the same single calculation in both; the **GPU and CPU agree** readout checks it. If your browser has no WebGL2, the page uses the CPU automatically, disables the GPU button and shows **No WebGL2 here.**

Double-click the frame to see it enlarged; press `Esc` to close.

7. Save the result

Save the frame downloads the current view as a PNG named after the simulation and severity, for example `perception-cvd-100.png`. The divider line and fixation cross, if shown, are included. See Check a palette and export for the other exports.

You should now be able to produce a side-by-side view of any image under any of the eight conditions.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PERCEPTION AND VISION

Perception and Vision reference

Every tab, control, readout, message and file format in Perception Vision, with ranges and defaults.

Address: https://auricartisan.com/tool/general/perception-and-neuroscience/perception-vision/

Tabs

Tab	Contents
Lab	Source, simulation, comparison, renderer, illuminant, adjustments and presets; the frame, grey-ramp check and readouts
Simulation	The colour vision chain, its matrices and a seven-colour table, with an earlier-version comparison
Modes	Previews of every simulation over one source, and the list of controls from an earlier version
Data	The register of nine datasets and transforms, filterable by status; the tab label shows the count (9)
Export	Colour list, CSV, clipboard, settings download and load, link
Reference	Standards, formulas, citations and research notes, with a search box

Lab tab — Source

Control	Values	Default
Source	**Test card**, **Grey ramp only**, **Snellen chart**, **Uploaded image**, **Camera**	Test card
Choose a file	Opens a file chooser for images (shown for Uploaded image); images are fitted within 1200 × 900	—
Start or stop	Switches the camera (shown for Camera); status **off**, **active** or **unavailable**	off

Lab tab — Simulation

Control	Values or range	Default	Shown for
Simulation	**Colour vision**, **Scotopic**, **Mesopic**, **Lens yellowing**, **Cataract**, **Central loss**, **Light adaptation**, **Acuity**, **None**	Colour vision	Always
Type	**Protan**, **Deutan**, **Tritan**	Protan	Colour vision
Daltonise instead of simulate	On or off	Off	Colour vision
Severity	0–1, step 0.01	1.00	Always
Adaptation level	0–1, step 0.01	0.50	Mesopic, Light adaptation
Age	20–80, step 1	40	Lens yellowing
Scotoma radius	0.02–0.45, step 0.01	0.15	Central loss
Cataract	0–1, step 0.01	0.50	Cataract
Acuity	20/20–20/200, step 5	20/20	Acuity

Lab tab — Compare and renderer

Control	Values or range	Default
Compare	**Split**, **Simulated only**, **Difference** (absolute difference × 4)	Split
Divider	0–100%, step 1%; also set by dragging on the frame (Split only)	50%
Renderer	**GPU**, **CPU**; GPU is disabled without WebGL2	GPU
Fixation cross	Draws a cross and ring at the frame centre	Off

Lab tab — Before the eye

Control	Values or range	Default
Illuminant	**D65 — as authored**, **D50**, **A — tungsten**	D65
Adaptation	0–1, step 0.01 (Bradford strength)	1.00

Lab tab — Adjustments

Control	Range and step	Default
Contrast	−0.9 to 0.9, step 0.01	0.00
Gamma	0.4 to 2.6, step 0.01	1.00
Blur	0 to 8, step 0.1	0.0
Glare	0 to 1, step 0.01	0.00
Noise	0 to 0.08, step 0.001	0.000

Lab tab — Presets

Control	Values
Presets	**—**, **Protanopia**, **Mild deuteranomaly**, **Tritanopia**, **Low light with glare**, **Lens at 70**, **Cataract**, **Central loss**, **Acuity 20/80** (values in Simulate a design or image)
Reset	Returns every control to its default and stops the camera

Lab tab — Stage elements

Element	Description
Caption	The simulation, type and severity, or **The frame, unchanged**
Renderer note	A dot (green when WebGL2 is available), **GPU** or **CPU**, and the frame size
Frame	The simulated view; in Split the halves are labelled **Original** and the simulation's name; the line under it names the method's source
No WebGL2 here.	Shown when the browser has no WebGL2; the CPU is used
Grey-ramp strip	Nine steps and the channel spread added to each, in code values
Save the frame	Downloads `perception-<simulation>-<severity × 100>.png`
Copy a link to this view	Copies a link to the current settings

Lab tab — Readouts

Readout	Format
Neutral drift	Code values, 0–255
Colours merged	Percentage, one decimal; **n/a** for spatial simulations
Out of sRGB	Percentage, one decimal; **n/a** for spatial simulations
GPU and CPU agree	**yes**, **N off**, **n/a — spatial** or **not run**
Distinct colours left	**N of M**, or **spatial mode**
Largest channel shift	Code values; **n/a** for spatial simulations
Frame luminance, mean	Relative luminance, four decimals
min and max	Relative luminance, four decimals

Simulation identifiers used in file names and links: `none`, `cvd`, `scotopic`, `mesopic`, `age-yellowing`, `cataracts`, `macular-degeneration` (Central loss), `photopic-adaptation` (Light adaptation), `acuity`.

Simulation tab

Control	Values	Default
Show	**The chain**, **The matrices**	The chain
Against	**What it was**, **Hide**	What it was

Control	Values	Default
Type	**Protan**, **Deutan**, **Tritan**, **All three**	Protan

Table columns: **Input**, **Type**, **Through XYZ**, and with **What it was** also **On screen**, **In the file** and **Apart**.

Modes tab

Control	Values	Default
Show	**Everything**, **The eight that simulate**	Everything
On	**Test card**, **Grey ramp**, **Snellen chart**	Test card
Severity	**0.50**, **1.00**	1.00

Data tab

Control	What it does
Status	**All** (9), **Verbatim** (3), **Computed** (3), **Stand-in** (2), **Absent** (1)
Register row	Select, or press `Enter` or `Space`, to expand the note and **What it would take**
Drop a CSV or JSON	Shown under **Install a table**; not active in the current version

Entries: **sRGB transfer and primaries**, **Hunt-Pointer-Estévez cone fundamentals**, **Bradford chromatic adaptation** (Verbatim); **Dichromacy projection**, **One implementation, two renderers**, **The shader's matrix literals** (Computed); **Rod spectral weighting**, **Lens, cataract, acuity and adaptation models** (Stand-in); **Contrast sensitivity function** (Absent).

Export tab

Control	What it does
What to take	**A list of colours**, **The frame**, **The settings**, **A link to this view**
Colours	Hex colours, one per line; commas and semicolons also separate
Run them	Redraws the table (it also updates as you type)
Download the CSV	Saves `perception-<simulation>-<severity × 100>.csv`
Copy to clipboard	Copies a tab-separated table
Download the settings	Saves `perception-vision-settings.json`
Load settings	Applies a settings JSON file

Reference tab

Control	Values
Reference	**Standards** (7), **Formulas** (11), **Citations** (13, default), **Research notes** (9)
Filter	Searches all four groups; shows **Nothing in the reference matches that.** when empty

Messages

Message	Meaning
Preset: …	A preset was applied
Reset	All controls returned to defaults
Camera: …	The camera could not start; the source returned to the test card
PNG saved, **CSV saved**, **Settings exported**, **Settings imported**, **Copied**, **Link copied**	The action succeeded
Import failed: …	The settings file could not be read
Copy blocked	The browser refused clipboard access
No colours to write / **No colours to copy**	The colour list is empty

File formats

File	Format	Contents
`perception-<simulation>-<severity>.png`	PNG	The frame at its own size, with the divider and fixation cross if shown
`perception-<simulation>-<severity>.csv`	CSV with `#` comment lines	`Input,Simulated,L,M,S,RG,BY,DeltaE00,InSRGB`
Clipboard table	Tab-separated text	`Input`, `Simulated`, `L`, `M`, `S`, `DeltaE00`, `InSRGB`
`perception-vision-settings.json`	JSON	See Check a palette and export

Link parameters

`simMode`, `cvdType` (`protan`, `deutan`, `tritan`), `severity`, `contrast`, `gamma`, `blur`, `noise`, `bloom` (glare), `adaptLevel`, `age`, `scotomaR`, `cataractLvl`, `acuity`, `illum` (`D65`, `D50`, `A`), `adaptStrength`, `compare` (`split`, `sim`, `diff`), `source` (`sample`, `ramp`, `snellen`, `upload`, `camera`), `renderer` (`gpu`, `cpu`).

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PERCEPTION AND VISION

Troubleshooting Perception and Vision

Symptoms, causes and fixes for the camera, WebGL2, uploads, readouts that say n/a, links and copying in Perception Vision.

The camera does not start

Symptom: after choosing **Camera** or selecting **Start or stop**, a message beginning **Camera:** appears, the status reads **unavailable** and the source goes back to **Test card**.

Why: the camera needs permission, and auricartisan.com currently does not allow its pages to use the camera. Browsers also block cameras on pages that are not secure, and a camera already in use by another application may refuse.

What to do: take a screenshot or photo and load it with **Uploaded image**.

The page says "No WebGL2 here."

Why: your browser or device does not provide WebGL2, or it is switched off.

What to do: nothing is required: every simulation runs on the processor (**CPU**) and gives the same colours, more slowly. The **GPU** button is disabled and **GPU and CPU agree** reads **not run**. To use the GPU, update the browser or enable hardware acceleration in its settings.

My uploaded image looks smaller than the original

Why: images larger than 1200 × 900 pixels are scaled down to fit. Smaller images are shown at their own size.

What to do: nothing; to inspect detail, crop the image before loading it. Double-click the frame to enlarge the view.

The grey ramp shows non-zero numbers

Why: under **Scotopic**, **Mesopic**, **Lens yellowing**, **Cataract** and **Light adaptation**, shifting greys is the effect being simulated. Under **Colour vision** the numbers should be zero; the strip shows them in red if not.

What to do: check which simulation is selected. Adjustments (contrast, gamma, glare, noise) and the illuminant do not affect the strip.

Readouts show "n/a", "spatial mode" or "n/a — spatial"

Why: Cataract, **Central loss** and **Acuity** change pixels according to their neighbours or their position, so per-colour measurements do not apply.

What to do: use **Difference** under **Compare** to see where these simulations change the image.

"GPU and CPU agree" shows a number followed by "off"

Why: the GPU's answer differs from the processor's by more than 2 code values on at least one of the 216 test colours. This can happen with unusual graphics drivers.

What to do: switch **Renderer** to **CPU** for figures you intend to rely on. The settings file records the check.

The divider moves when I click the frame

Why: in **Split** mode, pressing anywhere on the frame moves the divider there.

What to do: use **Simulated only** if you do not want a split, or set **Divider** with its slider.

The Install a table area on the Data tab does nothing

Why: the area is shown but not active in the current version.

What to do: nothing; the tool's tables cannot be replaced from a file at present.

A preset did not reset my adjustments

Why: a preset only sets the values it lists. Earlier adjustments stay.

What to do: select **Reset** first, then the preset.

A link or loaded settings file did not bring back my image

Why: links and settings files carry settings, not images.

What to do: after opening the link or loading the settings, choose **Uploaded image** and the file again. Links also omit the divider position, **Daltonise instead of simulate** and **Fixation cross**.

Nothing is copied, or the page says "Copy blocked"

Why: the browser refused clipboard access.

What to do: click inside the page and try again, or use **Download the CSV** or **Download the settings**.

Sliders are slow with a large image

Why: the readouts and the renderer check rerun after every change, and the CPU renderer processes every pixel.

What to do: use the **GPU** renderer if available, or load a smaller image.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PERCEPTION AND VISION

Perception and Vision FAQ

Short answers to common questions about Perception Vision.

Can I use this to diagnose colour blindness?

No. It simulates how an image might appear to someone with a condition; it does not test anyone's vision. The settings file and CSV carry `clinical: false` and "not a clinical measurement".

Which colour vision simulation does it use?

The Viénot, Brettel and Mollon (1999) projection, applied in cone space: sRGB is decoded, converted to CIE XYZ and then to cone signals with the Hunt-Pointer-Estévez matrix, projected, and converted back. The **Simulation** tab shows every matrix.

Does a severity of 0.5 simulate anomalous trichromacy?

Not as a physiological model. Between 0 and 1 the page blends linearly between normal vision and full dichromacy. The Machado, Oliveira and Fernandes (2009) model would be needed for anomalous trichromacy, and the page does not implement it. The **Mild deuteranomaly** preset uses this blend at 0.40.

Why does white stay white under protanopia?

Because it should: people with a colour vision deficiency see neutrals as neutral. The grey ramp under the frame checks this for every change.

What is daltonising?

A recolouring that moves information from a colour axis a dichromat cannot see into one they can. Tick **Daltonise instead of simulate** to preview it. It changes colours for everyone, so it is a preview of a correction filter, not a design fix.

How accurate are the night-vision, lens, cataract and acuity simulations?

They show that an effect exists and its direction, not its size for any person. Their constants are the tool's own; see Limits and accuracy.

Why is there no contrast sensitivity mode?

An earlier "CSF overlay" mode applied a contrast sensitivity curve to the distance from the image centre instead of to spatial frequency, and at its own defaults produced a black frame. It was withdrawn. The **Modes** tab shows what it did and what a real one would need.

Can I check a whole palette at once?

Yes. Paste the hex colours into the **Export** tab. Rows whose **Simulated** hex values match or nearly match are colours the simulated viewer cannot tell apart. See Check a palette and export.

Does my image get uploaded?

No. The image is read and processed in your browser. See Privacy.

Why does the camera option not work?

The site does not currently allow its pages to use the camera. Use a screenshot or photo instead.

GPU or CPU: which should I use?

Either. The colour calculation is the same, and the page checks the two against each other. The GPU is faster; the CPU is used automatically without WebGL2.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PERCEPTION AND VISION

Perception and Vision background

Contents of the background material for the Perception Vision tool.

This folder supports the how-to pages with the terms the tool uses, its keyboard and pointer actions, a statement of what is published and what is the tool's own, related tools and articles, its history and what stays on your device.

File	What it covers
glossary.md	Terms used in Perception Vision
shortcuts.md	Dragging the divider, enlarging the frame, `Esc` and keyboard use of the register
limits-and-accuracy.md	The data register entry by entry, precision, and what the tool is not
related.md	Related tools on the site and Learn articles
release-notes.md	User-facing history of the tool
privacy.md	What stays in your browser, the camera, and what uses the network

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PERCEPTION AND VISION

Perception and Vision shortcuts

Keyboard and pointer actions in the Perception Vision tool.

The tool has no keyboard shortcuts of its own beyond these. Site-wide shortcuts are described in Site features.

Action	Where	Result
Press or drag	The frame, with **Compare** set to **Split**	Moves the divider to the pointer
Double-click	The frame	Opens the frame enlarged, filling the window
`Esc`	Enlarged view	Closes it
Select the × button or the dark area around the image	Enlarged view	Closes it
`Enter` or `Space`	A focused row on the **Data** tab	Expands or collapses the row
`Tab`	Anywhere	Moves between controls; sliders then respond to the arrow keys

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PERCEPTION AND VISION

Perception and Vision glossary

Terms used in the Perception Vision tool, each explained briefly.

Term	Meaning
Acuity (Snellen)	Sharpness of vision written as a fraction such as 20/40: the viewer sees at 20 feet what a typical eye sees at 40
Anomalous trichromacy	Colour vision with all three cone types but one of them shifted, giving weaker colour discrimination than normal
Bradford transform	A published method for predicting how colours shift when the light source changes, used here to relight the scene
Cataract	Clouding of the eye's lens, which scatters light and lowers contrast
CIEDE2000 (ΔE00)	The CIE's current formula for how different two colours look; larger is more different

Term	Meaning
Clamped	A colour pushed back inside the sRGB range because the simulation produced a value a screen cannot show
Cone signals (L, M, S)	The responses of the long-, medium- and short-wavelength-sensitive cones
Daltonising	Recolouring an image to move information into colour differences a dichromat can see
Deutan	Relating to the M (medium-wavelength) cones; deuteranopia is the absence of M cones
Dichromacy	Colour vision with only two working cone types
Illuminant (D65, D50, A)	A standard light source: D65 and D50 are daylights, A is tungsten (incandescent) light
Mesopic vision	Vision in twilight, when both rods and cones contribute
Neutral drift	How far the simulation pulls a grey away from grey, in code values
Projection	The operation that removes one colour axis, replacing the missing cone's signal with a combination of the others
Protan	Relating to the L (long-wavelength) cones; protanopia is the absence of L cones
Relative luminance	The lightness-related quantity Y of CIE XYZ, from 0 (black) to 1 (white)
Rods	Photoreceptors that work in dim light and do not distinguish colour
Scotoma	An area of lost vision in the visual field
Scotopic vision	Night vision with rods only
Severity	How strongly a simulation is applied, from 0 (none) to 1 (full)
Spatial simulation	A simulation that depends on neighbouring pixels or position (Cataract, Central loss, Acuity)
sRGB	The standard colour space of web content, with a defined transfer curve and primaries
Tritan	Relating to the S (short-wavelength) cones; tritanopia is the absence of S cones
WebGL2	A browser graphics interface that lets the page compute on the graphics card

Limits and accuracy

What Perception Vision takes from published sources, what is its own, how precise it is and what it is not.

What the tool is

An educational and design-review simulator. It shows how an image may reach a viewer with a different visual system. It is not a clinical or diagnostic instrument and nothing it prints is a measurement of anyone's vision.

The data register, in plain terms

Entry	Status	What it means for you
sRGB transfer and primaries	Verbatim	Colours are decoded and re-encoded with the published IEC 61966-2-1 curve and matrices
Hunt-Pointer-Estévez cone fundamentals	Verbatim	Cone signals come from the published matrix, applied to CIE XYZ as it is defined
Bradford chromatic adaptation	Verbatim	The D50 and A options relight the scene with the published Bradford transform and CIE chromaticities

Entry	Status	What it means for you
Dichromacy projection	Computed	Protan, deutan and tritan use the published Viénot, Brettel and Mollon (1999) matrices; the severity blend is the tool's convention
One implementation, two renderers	Computed	GPU and CPU use one combined matrix and are checked against each other on 216 colours
The shader's matrix literals	Computed	The GPU receives its matrices in the correct orientation; the grey ramp confirms it
Rod spectral weighting	Stand-in	Scotopic and Mesopic use the tool's own rod weights (0.10, 0.59, 0.31), not the CIE scotopic curve; no scotopic luminance is reported
Lens, cataract, acuity and adaptation models	Stand-in	Their constants are the tool's own; they show the direction of an effect, not its clinical size
Contrast sensitivity function	Absent	There is no contrast-sensitivity simulation; nothing may be reported as a contrast-sensitivity result

What is and is not modelled

- **Colour vision deficiency.** Full dichromacy follows a published method. Partial severities are a linear blend towards it, not a model of anomalous trichromacy. The matrices describe a typical dichromat and are not individualised.
- **Night and twilight vision.** A rod signal from the tool's own weights; the mesopic setting is a crossfade, not CIE mesopic photometry.
- **Lens yellowing and cataract.** Direction from the ageing literature (Pokorny, Smith and Lutze 1987 for the lens); strengths are the tool's own.
- **Central loss.** A disc of reduced luminance, not a model of any particular disease.
- **Acuity.** A blur sized from the Snellen denominator with no viewing distance; it does not show what a person with that acuity sees at your screen distance.
- **Illuminant.** A chromatic adaptation of the image as authored under D65; it is not a spectral relighting.

Precision

- Colours are 8-bit sRGB in and out and are computed in floating point.
- The GPU and CPU colour results agree within 2 code values on the 216-colour check when **GPU and CPU agree** reads **yes**. Blur-based effects use slightly different filters in the two renderers, so spatial simulations can look a little different between them.
- **Colours merged**, **Out of sRGB**, **Distinct colours left** and **Largest channel shift** are measured on a 64 × 48 reduced copy of the source.
- Your screen is assumed to be sRGB. What you see also depends on your own display and your own eyes.

What the tool does not do

- It does not test vision, diagnose or measure a person.
- It does not model anomalous trichromacy physiologically, contrast sensitivity, eye movements, or the time course of adaptation.
- It does not use the camera on auricartisan.com at present.
- The **Install a table** area on the Data tab is not active.

Privacy

What Perception Vision keeps in your browser, how it treats images and the camera, and what uses the network.

What stays in your browser

- Images you load are read by the page, drawn on a canvas and processed on your own graphics card or processor. They are not uploaded, and they are forgotten when you leave or reload the page.
- Every simulation, readout and the colour list are computed in the page.
- Settings files you load are read locally.

The camera

The page has a **Camera** source. It would process frames locally in the same way as images, but auricartisan.com currently does not allow its pages to use the camera, so the page reports it as unavailable. No camera frames are captured or sent anywhere.

What leaves the page, and only when you ask

- **Save the frame**, **Download the CSV** and **Download the settings** create files through your browser's normal download. A saved frame contains your image as simulated.
- **Copy to clipboard** and **Copy a link to this view** write to your clipboard. A link contains your settings in its address, but never your image.

The rest of the page

The tool page loads the site's usual pages, styles and scripts, and, depending on your cookie choices, advertising and consent-based analytics, like the rest of auricartisan.com. See the Privacy Policy at https://auricartisan.com/legal/privacy/ and the Cookie Policy at https://auricartisan.com/legal/cookies/.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PERCEPTION AND VISION

Release notes

User-facing history of the Perception Vision tool.

The site-wide history is at https://auricartisan.com/library/changelog/. The entries below cover what changed for users of this tool.

September 2026 — rebuilt

The page was rebuilt with six tabs (**Lab**, **Simulation**, **Modes**, **Data**, **Export**, **Reference**). The **Simulation** tab can show the results an earlier version produced on screen and in its export beside the current ones.

Changed:

- The colour vision simulation now converts through CIE XYZ into cone signals before projecting, as the published method defines, and the screen and the export use the same calculation. Colours in earlier exports and screenshots can differ widely from current results; white now stays white under every type.
- Uploaded images are no longer drawn upside down.

- The readouts refresh after every change.
- The hero now counts eight simulations; **None** is listed but not counted.

Added:

- A CPU renderer, used automatically when WebGL2 is not available, and a check that the GPU and CPU agree.
- The grey-ramp strip, **Neutral drift**, **Colours merged**, **Distinct colours left**, **Out of sRGB** and **Largest channel shift**.
- The **Grey ramp only** and **Snellen chart** sources.
- **Load settings**, and the chain, matrix, illuminant and renderer check in the settings file and CSV header.

Connected (these controls previously did nothing): **Blur**, **Glare**, the illuminant with its **Adaptation** strength, and **Fixation cross**.

Removed: the CSF overlay simulation and the test overlay checkbox.

May 2026

- The Perception & Vision User Guide was published in the site documentation.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PERCEPTION AND VISION

Related tools and articles

Tools and Learn articles related to Perception Vision.

Tools in this group

- Neuroscience — the same cone and dichromacy maths, followed through the retina, LGN, V1, V2 and V4.
- Psychophysical Experiment Engine — measure a contrast threshold, the quantity a contrast sensitivity simulation would need.
- Material Lighting — how an illuminant changes a material's colour, spectrally rather than by adaptation.

Other tools on the site

- Vision simulation — the Vision Lab, for checking designs under many visual conditions.
- Contrast Checker — check the contrast of a colour pair against accessibility guidelines.
- Ishihara plate generator — generate pseudo-isochromatic plates.

Learn articles

The Learn section is described in Learn.

- Color Blindness and Color Vision Deficiency — https://auricartisan.com/library/learn/articles/2026-05-24-color-blindness-cvd
- Human Color Vision: Cones, Opponent Signals, and the Brain — https://auricartisan.com/library/learn/articles/2026-05-24-human-color-vision-cones-opponent
- Color Vision Across the Lifespan: Development, Aging, and the Yellowing Lens — https://auricartisan.com/library/learn/articles/2026-07-07-color-vision-across-the-lifespan
- Tetrachromacy, Anomalous Trichromacy, and the Range of Human Color Vision — https://auricartisan.com/library/learn/articles/2026-07-14-tetrachromacy-and-the-range-of-human-color-vision

- ΔE Metrics from CIE76 to CIEDE2000 and ΔEITP — https://auricartisan.com/library/learn/articles/2026-05-26-delta-e-metrics-cie76-ciede2000-itp

Documentation on the site

- Perception & Vision User Guide — https://auricartisan.com/library/documentation/articles/2026-05-24-perception-vision-user-guide

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Psychophysical Experiment Engine

What the Psychophysics Bench does, who it is for, where to find it and how to measure a first contrast threshold.

The Psychophysical Experiment Engine, whose page is titled **Psychophysics bench**, measures a contrast detection threshold: the faintest pattern you can reliably see. It shows a Gabor patch (a soft-edged patch of stripes) or a grating in a circular aperture, asks you to say where or whether you saw it, and uses an adaptive staircase to move the contrast up after errors and down after correct answers, so that most trials fall near your threshold. At the end it reports the threshold as a Michelson contrast, together with the percentage correct that the threshold belongs to.

The bench states its units. Contrast is Michelson contrast in relative luminance, about a background grey you choose, passed through the sRGB transfer curve; spatial frequency is in cycles per degree of visual angle, worked out from a viewing distance and pixel pitch you enter. When your screen cannot show the contrast or the frequency you asked for, the page says so instead of silently drawing something else.

It is an educational instrument that runs in your browser, not a calibrated laboratory display. It assumes your screen follows the sRGB standard rather than measuring it. Two of the five staircase procedures are published methods (Levitt 1971); the other three are the tool's own and are named as such.

> **Important:** At the time of writing (25 September 2026), only the **Lab** tab of this page opens: selecting **Method**, **Paradigms**, **Data**, **Export** or **Reference** does not change the view. You can design and run experiments and read the results on the Lab tab, but the download and link buttons, which are on the **Export** tab, cannot be reached. These pages document every tab so they remain useful when the tabs work; see Troubleshooting for workarounds.

Who it is for

- Students and teachers of psychophysics and vision science who want to run a real staircase and see how it converges.
- Designers and developers curious about how faint a pattern people can actually see on a screen.
- Anyone who wants to understand what a threshold "at 70.7% correct" means.

Where to find it

Open https://auricartisan.com/tool/general/perception-and-neuroscience/psychophysical-experiment-engine/ on a desktop or laptop with a keyboard. It is listed as **Psychophysics Engine** on the Collections page (https://auricartisan.com/collections/) and as **Psychophysical Experiment Engine** in the site menu.

Quick start

1. Open the tool. The **Lab** tab shows a Gabor patch at the starting contrast.
2. Under **Geometry**, set **Viewing distance** to your eye-to-screen distance and **Pixel pitch** to your screen's pixel size.
3. Leave **Task** at **Two-interval forced choice** and **Staircase** at **1-up / 2-down — 70.7%**.
4. Dim the room, sit at the stated distance and look at the centre of the stage.
5. Select **Practice** and answer a few trials: press `←` if the patch was in the first interval and `→` if it was in the second. Guess when unsure.
6. Select **Run** and answer until the run stops by itself (after 12 reversals or 120 trials by default).
7. Read **Threshold, Michelson** and **At what percentage**, and look at the staircase trace.

What you can do

- Choose between two tasks: two-interval forced choice and yes/no detection.
- Set the stimulus: Gabor or grating, spatial frequency, orientation, envelope size and background grey.
- State the viewing geometry and read the degrees per pixel and your screen's Nyquist limit.
- Choose one of five procedures: two transformed up-down staircases, a Bayesian tracker, an accelerated staircase or constant stimuli, and set the step, starting contrast, stopping rule and trial ceiling.
- Set stimulus duration and the pause between trials.
- Run practice and measured sessions with the keyboard or on-screen buttons, and stop at any time.
- Read the threshold, its target percentage, the reversals used, the delivered contrast, trials run and proportion correct, with a live staircase trace.
- Export every trial as CSV, the session as JSON, or a link to the setup (from the Export tab, when it opens).

In this folder

- docs/ — how to use the tool.
  - Getting started — open it, the screen tour, a first threshold.
  - Design an experiment — tasks, stimulus, geometry, staircases and timing.
  - Run a session — practice, the trial sequence, responding and stopping.
  - Read the results — the threshold, the readouts, the trace and the explanation tabs.
  - Export your data — the trials CSV, the session JSON and setup links.
  - Reference — every control, readout, tab and file format.
  - Troubleshooting — symptoms, causes, fixes and workarounds.
  - FAQ — short answers to common questions.
- others/ — background material.
  - Glossary — psychophysics terms used in this tool.
  - Shortcuts — the response keys and other actions.
  - Limits and accuracy — what is published, what is the tool's own, what is not shown.
  - Related — related tools and Learn articles.
  - Release notes — user-facing history.
  - Privacy — what stays in your browser.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Using the Psychophysics Bench

Contents of the how-to documentation for the Psychophysics Bench.

These pages cover designing an experiment, running it, reading the results and taking the data away. Start with getting started; the reference lists every control.

File	What it covers
getting-started.md	Opening the tool, the tabs, the Lab screen tour and a first threshold
design-an-experiment.md	The two tasks, the stimulus, viewing geometry, the five procedures and their parameters, timing
run-a-session.md	Practice and measured runs, the trial sequence, responding, stopping rules and good practice
read-the-results.md	Threshold, target percentage, reversals, delivered contrast, the trace, and the Method, Paradigms, Data and Reference tabs
export-your-data.md	The trials CSV, the session JSON and setup links, and what to do while the Export tab does not open
reference.md	Every tab, control, readout, message, file format and link parameter
troubleshooting.md	Tabs that do not open, warnings, stuck staircases, keys that do nothing
faq.md	Short answers to common questions

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Getting started with the Psychophysics Bench

Open the Psychophysics Bench, learn the Lab tab and measure a first contrast threshold with the default two-interval task.

Open the tool

Go to https://auricartisan.com/tool/general/perception-and-neuroscience/psychophysical-experiment-engine/ on a computer with a keyboard. The page opens on the **Lab** tab. A note strip above the tabs, which cannot be closed, says what the numbers are of: contrast in light, not in screen code values, and frequency in cycles per degree.

The six tabs

Tab	What it is for
Lab	Design and run an experiment and read the results
Method	What a contrast is on this page, how the frequency axis is built, and what each staircase estimates
Paradigms	The two tasks that can be scored and three that were withdrawn
Data	The register of every method and constant a threshold rests on (13 entries)
Export	Downloads of the trials and the session, and a link to the setup
Reference	Eleven sources and what each is used for

Note: At the time of writing, only the **Lab** tab opens. See Troubleshooting.

A tour of the Lab tab

The rail (left), from top to bottom:

- **Task** — **Two-interval forced choice** or **Yes / no detection**.
- **Stimulus** — **Gabor** or **Grating in an aperture**, then **Spatial frequency**, **Orientation**, **Envelope σ** and **Background**.
- **Geometry** — **Viewing distance** and **Pixel pitch**.
- **Staircase** — the procedure, then **Step**, **Starting contrast**, **Stop after** (reversals) and **Trial ceiling**.
- **Timing** — **Stimulus** duration and the pause **Between trials**.
- **Run**, **Practice** and, while a run is going, **Stop**.

The stage (right):

- **The trial**, with the run state on the right (**not running**, **practice**, **running**, **stopped** or **done — N trials**). Between runs the square stage shows the stimulus at its current contrast, so you can see what you are about to measure.
- **Interval 1** and **Interval 2** markers, lit in turn during a two-interval trial.
- Two response buttons: **First interval** ← and **Second interval** →, which become **Yes, I saw it** ← and **No, I did not** → for the yes/no task.
- **The staircase**: a plot of contrast against trial number, with a key for **reversals used**, **warm-up, discarded** and **threshold**.
- Two rows of readouts: **Threshold, Michelson**, **At what percentage**, **Reversals used**, **Delivered / asked**, then **Trials run**, **Proportion correct**, **Degrees per pixel** and **Nyquist limit of the grid**.
- Warnings, when needed, about the Nyquist limit and about contrast that does not fit around the background.

Measure your first threshold

1. Set **Viewing distance** to the distance from your eyes to the screen, and **Pixel pitch** to your screen's width in millimetres divided by its width in pixels. The defaults are 57 cm and 0.248 mm.
2. Keep the defaults for the rest: a Gabor at 4.0 c/deg, 45°, envelope 0.50°, background code 128; **1-up / 2-down — 70.7%** with a 2.0 dB step from a starting contrast of 0.300; 12 reversals or 120 trials; 200 ms stimulus and 600 ms between trials.
3. Dim the room, keep the screen brightness fixed, and sit at the distance you entered.
4. Select **Practice**. A fixation cross appears, then **Interval 1** lights with a patch or a blank, then **Interval 2**. The patch is in exactly one of them. Press ← for the first or → for the second. If you did not see it, guess: chance is 50%.
5. After a few practice trials, select **Stop** or let practice finish (it ends after 12 trials).
6. Select **Run** and keep answering. The patch will get fainter as you answer correctly and stronger after mistakes. The run stops by itself at 12 reversals or 120 trials, whichever comes first, and the state reads **done — N trials**.
7. Read **Threshold, Michelson**: the contrast at which you were right 70.7% of the time, as **At what percentage** says. **Reversals used** shows how many turning points went into it.

You should see a staircase trace that drops from 0.3, zig-zags, and settles in a band, with a dashed gold line at your threshold.

Next steps

- Design an experiment explains every setting.
- Run a session covers stopping rules and good practice.
- Read the results explains the numbers and the trace.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Design an experiment

Choose a task, build the stimulus, state the viewing geometry, pick a staircase procedure and set its parameters and the timing.

Every setting in this guide is on the **Lab** rail. Settings take effect at the next **Run** or **Practice**; the stage preview updates as you change them.

What a threshold is

Psychophysics measures the relationship between a physical stimulus and what a person reports. A **detection threshold** is the stimulus strength at which an observer reports the stimulus a stated proportion of the time. The proportion matters: "your threshold is 0.01" means nothing until it says "at 70.7% correct" or "at 79.4% correct", because the chance of a correct answer rises smoothly with contrast. That rising curve is the **psychometric function**. This bench always reports a threshold together with the percentage it belongs to, and says when a procedure has no stated percentage.

1. Choose the task

Task	What happens on a trial	Chance level
Two-interval forced choice (default)	Two intervals are shown one after the other, marked **Interval 1** and **Interval 2**. The patch is in exactly one, chosen at random. You say which.	50%
Yes / no detection	One interval. The patch is present on a random half of trials and absent on the rest. You say whether you saw it.	No guess rate of this kind; your answers depend on how willing you are to say yes

Forced choice is the usual method for thresholds because guessing is built in: you must answer, and the known chance level (50%) lets the analysis separate seeing from guessing. **Yes/no** is the task that has stimulus-absent trials, so it is the only one that can separate sensitivity from willingness to say yes (signal detection theory). In yes/no, a correct "no" on a blank trial counts as correct and moves the staircase just as a correct "yes" does.

Three more tasks from an earlier version (m-alternative forced choice, same/different and odd-one-out) are withdrawn because they could never score a trial correct; the **Paradigms** tab explains each.

2. Build the stimulus

Control	Range	Default	What it sets
Stimulus type	**Gabor**, **Grating in an aperture**	Gabor	A **Gabor** is a sine-wave grating that fades out under a Gaussian envelope; a grating in an aperture has the same stripes inside a hard-edged circle
Spatial frequency	0.25–20 c/deg, step 0.25	4.0 c/deg	How fine the stripes are
Orientation	0–179°, step 1	45°	The angle of the stripes
Envelope σ	0.10–2.00°, step 0.05	0.50°	The size of the Gabor's Gaussian envelope, in degrees of visual angle (used by the Gabor only)
Background	Code 40–220, step 1	Code 128	The grey the contrast swings about, as an 8-bit screen value

Human contrast sensitivity is highest at a few cycles per degree, so thresholds are lowest near the default 4 c/deg.

How contrast is defined

Michelson contrast is (Lmax − Lmin) ÷ (Lmax + Lmin), where L is luminance. The bench converts the **Background** code value to relative luminance through the sRGB transfer curve, places the stripes' peaks and troughs symmetrically around that

luminance, and only then converts back to code values. So a contrast of 0.02 on this page is 0.02 in light, not in screen numbers.

The background limits how much contrast fits. A trough cannot be darker than black and a peak cannot be brighter than white, so the largest contrast possible about a background of relative luminance L is 1 ÷ L − 1, capped at 1. Around the default code 128 (relative luminance about 0.216) the full range up to 1.0 fits; around code 200 (about 0.578) the most is about 0.73, and around code 220 (about 0.716) about 0.40. When a staircase asks for more than fits, the bench shows the largest contrast that fits, reports **Delivered / asked** below 1.00, and shows the note **The contrast asked for does not fit about this background.** It never clips one side of the stripes, because that would be a different stimulus.

3. State the viewing geometry

Control	Range	Default
Viewing distance	20–200 cm, step 1	57 cm
Pixel pitch	0.05–0.6 mm, step 0.001	0.248 mm

Cycles per degree and an envelope in degrees are undefined without these. At 57 cm one degree covers very nearly 1 cm of screen. Your pixel pitch is your screen's visible width in millimetres divided by its horizontal resolution.

The readouts **Degrees per pixel** (0.0249° at the defaults) and **Nyquist limit of the grid** (20.1 c/deg at the defaults) update as you change them. The Nyquist limit is the finest frequency the pixel grid can show: half a cycle per pixel. Above it the note **Above the pixel grid's Nyquist limit** appears, because what reaches your eye would be an alias at a different frequency and contrast. Lower the frequency, or check the geometry: the limit rises as you sit further away and as pixels get smaller.

4. Choose the procedure

An **adaptive staircase** chooses each trial's contrast from your previous answers, lowering it after correct answers and raising it after errors, so trials cluster near your threshold instead of being wasted where the patch is obviously visible or invisible. A **reversal** is a trial where the staircase changes direction.

Procedure	How it moves	Converges at	Source
1-up / 2-down — 70.7% (default)	Down one step after two correct in a row, up one step after any error	70.7% correct (the p at which $p^2 = 0.5$)	Levitt 1971
3-down / 1-up — 79.4%	Down after three correct in a row, up after any error	79.4% correct ($p^3 = 0.5$)	Levitt 1971
Bayesian tracker — this tool's own	Updates a probability distribution over the threshold after every trial and tests at its mean	About 80.3% correct in two-interval, 61.9% in yes/no	The tool's own
Accelerated — this tool's own	Moves every trial; halves the step at each reversal and doubles it after three moves in the same direction	No stated percentage	The tool's own
Constant stimuli — no adaptation	Every trial at the starting contrast	Not a threshold procedure here	—

Staircase parameters

Control	Range	Default	Applies to
Step	0.5–12 dB, step 0.5	2.0 dB	1-up / 2-down, 3-down / 1-up, Accelerated
Starting contrast	0.01–1, step 0.01	0.300	All
Stop after	4–24 reversals, step 2	12 reversals	1-up / 2-down, 3-down / 1-up, Accelerated

Control	Range	Default	Applies to
Trial ceiling	20–400 trials, step 10	120	All

Step is a ratio in decibels of contrast: each step multiplies or divides the contrast by 10^(dB ÷ 20), so 2 dB is a factor of about 1.26 at any contrast. A ratio step resolves a threshold of 0.005 as finely as one of 0.5, which a fixed step cannot. For the two up-down staircases the step halves after the second reversal and halves again after the fourth. The accelerated staircase starts at the set step, halves it at each reversal (down to one eighth) and doubles it after three moves in one direction (up to four times the set step). Contrast is kept between 0.0001 and 1.

The Bayesian tracker keeps a probability for 240 candidate thresholds spread evenly on a logarithmic scale from 0.0001 to 1. It starts from a Gaussian prior centred on a quarter of the starting contrast with a width of one decade (1 dex), assumes a Weibull psychometric function with a slope of 3.5, a lapse rate of 0.02 and the task's guess rate, and after each answer tests at the mean of the updated distribution. Its slope, lapse rate and prior are the tool's choices, not estimated from your data, which is why it is not called QUEST.

Constant stimuli here shows every trial at the starting contrast, so it measures your proportion correct at one contrast. The classical method of constant stimuli uses several fixed levels in random order; the bench does not do that.

5. Set the timing

Control	Range	Default	What it sets
Stimulus	20–1000 ms, step 10	200 ms	How long each interval shows the patch or blank
Between trials	100–2000 ms, step 50	600 ms	The pause after your answer before the next trial

A fixation cross shows for 240 ms before each trial, and in the two-interval task the intervals are separated by a 300 ms blank. There is no time limit on answering. Browser timing follows your screen's refresh, so very short durations are rounded to whole frames by the display.

6. Save your design

When the **Export** tab opens, **Copy a link to this setup** copies every rail setting as a link. Until then, see Export your data for how to build that link by hand.

You should now have a stimulus, geometry and procedure that match the question you want to answer.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Export your data

The trials CSV, the session JSON and setup links on the Export tab, their exact contents, and what to do while the Export tab does not open.

> **Important:** At the time of writing, the **Export** tab does not open, so the three buttons below cannot be reached. The last section of this page describes what you can do meanwhile.

The Export tab

What to take switches the preview on the right between **Every trial** (the CSV), **The session** (the JSON, first part) and **A link to this**. The note above the preview gives the number of rows. Nothing leaves the page until you press a button:

Button	Result
Download the trials	Saves `psychophysics-trials.csv`
Download the session	Saves `psychophysics-session.json`
Copy a link to this setup	Copies a link with every rail setting; shows **Link copied**, or the link itself if the clipboard is unavailable

With no trials yet, the download buttons show **No trials yet — run the experiment first**.

The trials CSV

The file starts with header lines that record what the numbers mean, then one row per trial:

```
# Auric Artisan · psychophysics bench · 2026-09-25T10:00:00.000Z
# contrast_unit    = michelson, relative luminance, sRGB EOTF
# mean_luminance   = 0.2159 (code 128)
# frequency_unit   = cycles per degree
# viewing_distance = 570 mm ; pixel_pitch = 0.248 mm ; deg_per_px = 0.0249
# display_gamma    = sRGB standard curve, not a measurement of this screen
# paradigm         = Two-interval forced choice
# staircase        = 1-up / 2-down, 2.0 dB step, target 70.7%
# reportable       = false   (the fit is this tool's own; see the Data register)
trial,contrast,delivered,response,correct,rt_ms,interval,present,cpd
```

Column	Meaning
`trial`	Trial number
`contrast`	The Michelson contrast the procedure asked for
`delivered`	The contrast the screen could show; lower than `contrast` when capped
`response`	`0` (first interval) or `1` (second) in two-interval; `yes` or `no` in yes/no
`correct`	`1` or `0`
`rt_ms`	Response time from the end of the stimulus, in milliseconds
`interval`	The interval that held the patch (`0` or `1`); `-1` in yes/no
`present`	`1` if the patch was shown, `0` if not (always `1` in two-interval)
`cpd`	Spatial frequency in cycles per degree

The header carries the background luminance and the geometry because a Michelson contrast cannot be reconstructed without its mean, and a frequency in cycles per degree means nothing without a viewing distance.

The session JSON

`psychophysics-session.json` contains:

Field	Meaning
`tool`, `version`, `recorded`	`psychophysics-bench`, `2` and the time of export
`units`	Contrast unit, `mean_luminance`, `mean_code`, frequency unit and `deg_per_px`
`display_gamma`	"sRGB standard curve, not a measurement of this screen"
`settings`	Every rail setting

Field	Meaning
`paradigm`, `staircase`	The task and a description of the procedure and step
`threshold`, `threshold_at`	The current threshold and its proportion correct, or `null`
`reversals_used`	The same text as the readout
`trials`	Number of trials
`reportable`	Whether the result may be reported as a published quantity (currently `false`)
`data`	Every trial, with the fields of the CSV

Setup links

A setup link opens the bench with the same rail settings. It uses these parameters:

Parameter	Control	Values
`paradigm`	Task	`2afc`, `yesno`
`stimType`	Stimulus type	`gabor`, `grating`
`cpd`	Spatial frequency	0.25–20
`orientation`	Orientation	0–179
`sigmaDeg`	Envelope σ	0.1–2
`meanCode`	Background	40–220
`distanceCm`	Viewing distance	20–200
`pitchMm`	Pixel pitch	0.05–0.6
`stair`	Staircase	`1u2d`, `3d1u`, `bayes`, `accel`, `constant`
`stepDb`	Step	0.5–12
`start`	Starting contrast	0.01–1
`nReversals`	Stop after	4–24
`trials`	Trial ceiling	20–400
`duration`	Stimulus (ms)	20–1000
`iti`	Between trials (ms)	100–2000

A link carries settings, not results.

While the Export tab does not open

- **Share or save a setup:** build the link by hand. Start from the tool address and add the parameters you want to set, for example:

  ```
  https://auricartisan.com/tool/general/perception-and-neuroscience/psychophysical-experiment-engine/?
  paradigm=yesno&cpd=2&stair=3d1u&trials=160
  ```

  Parameters you leave out keep their defaults.

- **Keep a result:** write down **Threshold, Michelson**, **At what percentage**, **Reversals used**, **Delivered / asked**, **Trials run** and **Proportion correct**, together with your settings, before starting another run. A screenshot of the Lab tab records the trace as well. Double-clicking the trace enlarges it first.
- **Per-trial data** cannot be taken away until the tab opens.

Read the results

Understand the threshold readouts, the staircase trace and the explanation tabs of the Psychophysics Bench.

The readouts

Readout	What it tells you
Threshold, Michelson	Your threshold contrast, four decimals; highlighted when there is not yet an estimate
At what percentage	The proportion correct the threshold belongs to
Reversals used	How many reversals went into the estimate, out of how many occurred
Delivered / asked	The contrast the screen could show divided by the contrast the staircase asked for; 1.00 means nothing was capped
Trials run	Trials completed and scored
Proportion correct	Correct trials divided by trials run, three decimals
Degrees per pixel	The angle one pixel covers at your geometry
Nyquist limit of the grid	The finest frequency your screen can show at your geometry; highlighted when the stimulus is above it

How the threshold is worked out

1-up / 2-down, 3-down / 1-up and Accelerated. The first two reversals are a warm-up and are discarded. Of the rest, an even number is used (if an odd number remain, the earliest is dropped too), because up-runs and down-runs of a transformed staircase are not symmetric around the threshold. The threshold is the **geometric mean** of those reversal contrasts, because the staircase steps by ratios. **Reversals used** reads, for example, **10 of 12**; before there are two usable reversals it reads **fewer than two reversals after the warm-up** and the threshold shows **—**.

Bayesian tracker. The threshold is the tracker's current estimate (the mean of its distribution on a log scale), and **Reversals used** reads, for example, **80 trials, posterior mean**.

Constant stimuli. There is no threshold estimate; **Threshold, Michelson** and **Reversals used** show **—**. Read **Proportion correct**: it is your score at the starting contrast.

What the percentage means

Procedure	At what percentage
1-up / 2-down	70.7%
3-down / 1-up	79.4%
Bayesian tracker	80.3% (two-interval) or 61.9% (yes/no)
Accelerated, **Constant stimuli**	**not defined for this ladder**

A threshold from the accelerated staircase is a contrast near your threshold, but not a threshold at a stated percentage, and should not be compared directly with one.

What is a normal value?

For a Gabor at about 4 c/deg in a dim room, detection thresholds typically fall roughly between 0.005 and 0.02 Michelson contrast on a well-behaved display. A bright room, an uncalibrated screen, glare or inattention all raise it. At such low contrasts the stripes differ from the background by only a few screen code values, so screen quality matters.

The staircase trace

The staircase plots the contrast on every trial (horizontal axis: trial number; vertical axis: contrast on a logarithmic scale, labelled on the left). The note above it names the procedure.

- **Grey dots** — the first two reversals, the warm-up, discarded.
- **Red dots** — the reversals the estimate uses.
- **Dashed gold line** — the threshold, with its value on the right.

A good run falls from the starting contrast, then zig-zags in a narrowing band. The trace is drawn for the three staircases that have reversals (**1-up / 2-down**, **3-down / 1-up**, **Accelerated**). For the **Bayesian tracker** and **Constant stimuli** it stays empty with the text **The trace appears here once the run starts**; read the numbers instead.

Double-click the trace or the stimulus to see it enlarged; press `Esc` to close.

Warnings under the readouts

- **Above the pixel grid's Nyquist limit.** — the frequency is too fine for your screen at your geometry; what you see is an alias.
- **The contrast asked for does not fit about this background.** — the staircase asked for more contrast than fits; **Delivered / asked** is below 1.00. A threshold measured while this shows is a threshold for a different stimulus. Choose a darker background.

What the page does not show

The bench computes more than it displays. A psychometric function fit (a logistic with the task's guess rate and a fitted lapse rate), d′ and criterion for yes/no sessions, and a bootstrap interval are described in the **Data** register, but no readout on the page shows them in the current version. The session file records every trial, so you can fit them in your own analysis software.

The explanation tabs

When the tabs open (see Troubleshooting), they hold the following.

Method has three views:

- **The contrast chain** — the steps from **Contrast asked** through **Mean luminance**, **L × (1 ± m)** and the **sRGB OETF** to **Code values**, and a table of what an earlier version delivered for contrasts of 0.02 to 1 (**Contrast asked**, **Codes sent**, **Luminance reached**, **Contrast delivered**, **Error**).
- **The frequency axis** — a table of your current viewing distance, pixel pitch, degrees per pixel, cycles per pixel at your frequency, and the Nyquist limit, with where each comes from.
- **What each ladder estimates** — each procedure and the percentage it converges at, and notes on the multiplicative step, the even reversal count and why the fit carries a guess rate.

Paradigms lists the five tasks the menu once offered: the two that score (tagged **Scores**) and three withdrawn ones (tagged **Never correct**), each with what it would take to bring back.

Data lists 13 entries with a status each: **Verbatim** (3: sRGB transfer, Levitt's rules, the 1/2N extreme-proportion correction), **Computed** (5: Michelson contrast in luminance, viewing geometry, signal-detection formulae, inverse normal, bootstrap), **Stand-in** (3: Bayesian tracker, psychometric fit, accelerated staircase) and **Absent** (2: Weibull fit, ROC curve). Its **Calibrate the display** area (**Drop a measured gamma table**) is not active in the current version.

Reference lists eleven sources, from Levitt (1971) to Abramowitz and Stegun (1964), each with what it is used for and a tag: **Runs**, **Stand-in** or **Withdrawn**.

You should now be able to say what your threshold is a threshold of, and how much to trust it.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Run a session

Practise, run a measured session, respond with keys or buttons, and understand when and how a run stops.

Before you start

- Sit at the **Viewing distance** you entered and keep your head still.
- Dim the room and turn off automatic brightness, night-light or colour-shifting modes on your screen. Keep brightness fixed for the whole session.
- Look at the centre of the stage throughout.
- Close other busy windows; a slow computer can delay intervals.

Practise

Select **Practice**. A practice run shows up to 12 trials (fewer if **Trial ceiling** is lower) and does not stop on reversals. Its answers appear in the readouts while you practise, and are cleared when you start a measured run. A message confirms **Practice — arrow keys respond, Esc stops**, and the run state reads **practice**.

Run

Select **Run**. The message **Running — arrow keys respond, Esc stops** appears, the run state reads **running**, **Run** and **Practice** are disabled, and **Stop** appears.

What one trial looks like

Two-interval forced choice

1. A fixation cross on the background grey, for 240 ms.
2. **Interval 1** lights and the stage shows either the patch or a blank patch of the background, for the **Stimulus** duration.
3. A 300 ms blank.
4. **Interval 2** lights and shows the other one.
5. The stage returns to the background and waits for your answer.

Yes / no detection

1. A fixation cross for 240 ms.
2. One interval: the patch on about half of trials, a blank on the rest, for the **Stimulus** duration.
3. The stage returns to the background and waits for your answer.

Answering

Task	Left	Right
Two-interval forced choice	← or **First interval**	→ or **Second interval**
Yes / no detection	← or **Yes, I saw it**	→ or **No, I did not**

Answers count only after the intervals have finished. There is no time limit, but answer promptly: the page records your response time. When you are unsure in the two-interval task, guess. The method assumes you will be guessing near threshold, and waiting to be certain biases the result upwards.

After each answer the page scores the trial, updates the staircase and the readouts, waits for the **Between trials** pause, and starts the next trial.

When a run stops

Procedure	Stops at
1-up / 2-down, **3-down / 1-up**, **Accelerated**	The **Stop after** reversal count, or the **Trial ceiling**, whichever comes first
Bayesian tracker, **Constant stimuli**	The **Trial ceiling**

The run state then reads **done — N trials** and the stage shows the background.

To stop early, press `Esc` while the page is waiting for your answer, or select **Stop**. **Stop** takes effect at the end of the current interval; if the page is already waiting for your answer, the run ends when you next press `Esc`, a response key or a response button, and that last answer is not recorded. The state then reads **stopped**, and the trials completed so far remain in the readouts.

A normal session is 60 to 120 trials. Starting a new run clears the previous one's trials.

Good practice

- Run **Practice** once before every measured session.
- Measure more than once and compare; single runs vary.
- Keep the geometry, background and room lighting the same between runs you want to compare.
- If the trace climbs to the top and stays there, you could not see the patch at any contrast: check the frequency against the Nyquist limit, the background, and the **Delivered / asked** readout.

You should now have a completed run with a threshold on the Lab tab; Read the results explains it.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Psychophysics Bench reference

Every tab, control, readout, message, file format and link parameter in the Psychophysics Bench, with ranges and defaults.

Address: https://auricartisan.com/tool/general/perception-and-neuroscience/psychophysical-experiment-engine/

Tabs

Tab	Contents	Opens at the time of writing
Lab	All experiment controls, the trial stage, response buttons, the staircase trace, readouts and warnings	Yes
Method	The contrast chain, the frequency axis, what each ladder estimates	No
Paradigms	Two scoring tasks and three withdrawn ones	No
Data	The register of 13 methods and constants; the tab label shows the count	No
Export	Trials CSV, session JSON, setup link	No
Reference	Eleven sources and their use	No

Lab tab — Task

Control	Values	Default
Task	**Two-interval forced choice**, **Yes / no detection**	Two-interval forced choice

Lab tab — Stimulus

Control	Range and step	Default
Stimulus type	**Gabor**, **Grating in an aperture**	Gabor
Spatial frequency	0.25–20 c/deg, step 0.25	4.0 c/deg
Orientation	0–179°, step 1	45°
Envelope σ	0.10–2.00°, step 0.05 (Gabor only)	0.50°
Background	Code 40–220, step 1	Code 128

Lab tab — Geometry

Control	Range and step	Default
Viewing distance	20–200 cm, step 1	57 cm
Pixel pitch	0.05–0.6 mm, step 0.001	0.248 mm

Lab tab — Staircase

Control	Values or range	Default
Procedure	**1-up / 2-down — 70.7%**, **3-down / 1-up — 79.4%**, **Bayesian tracker — this tool's own**, **Accelerated — this tool's own**, **Constant stimuli — no adaptation**	1-up / 2-down
Step	0.5–12 dB, step 0.5	2.0 dB
Starting contrast	0.01–1, step 0.01	0.300

Control	Values or range	Default
Stop after	4–24 reversals, step 2	12 reversals
Trial ceiling	20–400, step 10	120

Lab tab — Timing

Control	Range and step	Default
Stimulus	20–1000 ms, step 10	200 ms
Between trials	100–2000 ms, step 50	600 ms

Fixed timings: fixation cross 240 ms before each trial; 300 ms blank between the two intervals.

Lab tab — Buttons

Button	What it does
Run	Starts a measured run; disabled while a run is going
Practice	Starts a practice run of up to 12 trials with no reversal stop
Stop	Shown during a run; ends it (see Run a session)
First interval ← / **Yes, I saw it** ←	Answers first interval (two-interval) or yes (yes/no)
Second interval → / **No, I did not** →	Answers second interval or no

Lab tab — Stage and trace

Element	Description
Run state	**not running**, **practice**, **running**, **stopped**, **done — N trials**
Stimulus stage	256 × 256; shows the stimulus at the current contrast between runs
Interval 1, **Interval 2**	Light in turn during two-interval trials
The staircase	Contrast (log axis) against trial; grey warm-up reversals, red reversals used, gold dashed threshold; drawn for the up-down and accelerated staircases

Lab tab — Readouts

Readout	Format
Threshold, Michelson	Four decimals, or **—**
At what percentage	70.7%, 79.4%, 80.3% or 61.9%, or **not defined for this ladder**
Reversals used	**N of M**, **fewer than two reversals after the warm-up**, **N trials, posterior mean**, or **—**
Delivered / asked	Ratio, two decimals; highlighted when capped
Trials run	Count
Proportion correct	Three decimals

Readout	Format
Degrees per pixel	Degrees, four decimals
Nyquist limit of the grid	c/deg, one decimal; highlighted when exceeded

Lab tab — Warnings

Warning	When
Above the pixel grid's Nyquist limit.	Spatial frequency above the Nyquist limit
The contrast asked for does not fit about this background.	The contrast asked for exceeds what fits around the background

Method tab

Control	Values
Show	**The contrast chain**, **The frequency axis**, **What each ladder estimates**

Data tab

Control	What it does
Status	**All** (13), **Verbatim** (3), **Computed** (5), **Stand-in** (3), **Absent** (2)
Register row	Select, or press `Enter` or `Space`, to expand its note and **what it would take**
Drop a measured gamma table	Shown under **Calibrate the display**; not active in the current version

Entries: **sRGB transfer and primaries**, **Transformed up-down rules**, **Extreme-proportion correction** (Verbatim); **Michelson contrast in luminance**, **Viewing geometry**, **Signal-detection formulae**, **Inverse normal CDF**, **Nonparametric bootstrap** (Computed); **Bayesian adaptive staircase**, **Psychometric function fit**, **Accelerated staircase** (Stand-in); **Weibull fit**, **ROC curve and its area** (Absent).

Export tab

Control	What it does
What to take	Preview **Every trial**, **The session** or **A link to this**
Download the trials	Saves `psychophysics-trials.csv`
Download the session	Saves `psychophysics-session.json`
Copy a link to this setup	Copies a link with every rail setting

Messages

Message	Meaning
Practice — arrow keys respond, Esc stops	A practice run started
Running — arrow keys respond, Esc stops	A measured run started

Message	Meaning
No trials yet — run the experiment first	A download was requested with no trials
Link copied	The setup link is on the clipboard
A trial could not be scored; the run stopped rather than record it	A safety stop that should not occur with the two shipped tasks

File formats

File	Format	Contents
`psychophysics-trials.csv`	CSV with `#` header lines	`trial,contrast,delivered,response,correct,rt_ms,interval,present,cpd`
`psychophysics-session.json`	JSON	Units, settings, procedure, threshold and its percentage, every trial

Details are in Export your data.

Link parameters

`paradigm`, `stimType`, `cpd`, `orientation`, `sigmaDeg`, `meanCode`, `distanceCm`, `pitchMm`, `stair`, `stepDb`, `start`, `nReversals`, `trials`, `duration`, `iti`. Values are listed in Export your data.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Troubleshooting the Psychophysics Bench

Symptoms, causes, fixes and workarounds for the Psychophysics Bench, including tabs that do not open.

Only the Lab tab opens; Method, Paradigms, Data, Export and Reference do nothing

Why: at the time of writing (checked on the live site in September 2026), the page's tab buttons do not switch the view. Links in the page text that point to another tab (for example **Data** in the note strip, or **Paradigms** under **Task**) do nothing for the same reason.

What this affects: everything on the Lab tab works: designing, running and reading results. The explanation tabs and the **Export** tab's downloads and link button cannot be reached.

What to do:

- Read the content of those tabs in these docs: Read the results summarises Method, Paradigms, Data and Reference, and Export your data describes the files.
- Record results from the Lab readouts, or take a screenshot, before starting another run.
- Share a setup by building a link by hand, as shown in Export your data.

The arrow keys do nothing

Why: answers are accepted only after the intervals of the current trial have finished and the page is waiting; between runs they do nothing. The keys also do nothing when focus is in another window or the browser's address bar.

What to do: click an empty part of the page, then use the keys, or use the on-screen response buttons.

Stop does not end the run immediately

Why: Stop takes effect at the end of the current interval. If the page is already waiting for your answer, it ends the run at your next key or button press, without recording that answer.

What to do: press `Esc` while the page is waiting for an answer; it stops at once.

The staircase climbs to the top and stays there

Why: you could not see the patch at any contrast offered. Common causes: a spatial frequency above the Nyquist limit, a very small **Envelope σ**, a background that caps the contrast, or looking away from the centre.

What to do: check the **Nyquist limit of the grid** readout and warning, check **Delivered / asked**, try a lower frequency or larger envelope, and run **Practice** first.

"The contrast asked for does not fit about this background" appears

Why: the procedure asked for more contrast than a symmetric swing around the background can hold. Bright backgrounds leave little room: around code 220 the maximum is about 0.40.

What to do: lower **Background** (code 128 allows up to 1.0), or lower **Starting contrast**. A threshold measured while the warning shows is for a capped stimulus.

"Above the pixel grid's Nyquist limit" appears

Why: at your viewing distance and pixel pitch, the screen cannot draw the frequency you asked for.

What to do: lower **Spatial frequency**, and check the geometry. The limit rises as the viewing distance grows and as the pixel pitch shrinks. The warning's own text suggests moving closer or raising the pitch; both in fact lower the limit.

The threshold reads "—"

Why: for the up-down and accelerated staircases, there are not yet two reversals after the two warm-up reversals (**Reversals used** says so). For **Constant stimuli** there is no threshold estimate at all.

What to do: run longer (raise **Trial ceiling** or **Stop after**), or read **Proportion correct** for constant stimuli.

"At what percentage" says "not defined for this ladder"

Why: the **Accelerated** staircase and **Constant stimuli** do not converge on a stated proportion correct.

What to do: use **1-up / 2-down** or **3-down / 1-up** for a threshold at a stated percentage.

The trace stays empty during a run

Why: the trace is drawn only for procedures with reversals. The **Bayesian tracker** and **Constant stimuli** show **The trace appears here once the run starts** throughout.

What to do: read the numeric readouts for those procedures.

Practice results disappeared

Why: starting a measured run clears the trials of the previous run, practice included.

What to do: this is expected; practice is not meant to be kept.

Intervals look uneven or flicker

Why: the timing runs in your browser. A busy computer, a background tab or power saving can delay drawing.

What to do: close other applications and tabs, keep the page in the foreground, plug in a laptop, and avoid very short **Stimulus** durations.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Psychophysics Bench FAQ

Short answers to common questions about the Psychophysics Bench.

Is this a vision test?

No. It measures a contrast threshold on your screen under your conditions, as a teaching and exploration instrument. It is not calibrated, not clinical and not a diagnosis.

Which staircase should I use?

1-up / 2-down — 70.7% for most purposes: it is the standard transformed up-down rule and converges at a stated percentage. **3-down / 1-up — 79.4%** converges higher and takes longer. The other three are the tool's own or not adaptive.

Why does the threshold come with a percentage?

Because the probability of a correct answer rises gradually with contrast. A threshold is the contrast at a chosen point on that curve, so it means nothing without the point.

Is the Bayesian tracker QUEST?

No. It works on a log scale with a stated prior, like QUEST, but its slope, lapse rate and prior width are the tool's choices rather than estimated or taken from previous measurements, so it carries its own name.

Is the accelerated staircase PEST?

No. PEST is a sequential procedure driven by a statistical test; the accelerated staircase simply halves its step at reversals and doubles it after three moves in one direction.

Does it fit a psychometric function, compute d’ or give a confidence interval?

Not on screen in the current version. The Data register describes a logistic fit with guess and lapse rates, d’ and criterion for yes/no sessions, and a trial bootstrap, but no readout shows them. Export the session (when the Export tab opens) and fit the data in your own software.

Why is my threshold different every run?

Staircase estimates vary from run to run; attention, fatigue, lighting and viewing distance add more. Average several runs under the same conditions.

Can I use a different number of alternatives, same/different or odd-one-out?

No. Those tasks were withdrawn because they could not score trials correctly. The Paradigms tab says what each would need.

Why does the page ask for my viewing distance and pixel pitch?

Spatial frequency in cycles per degree and an envelope size in degrees depend on both. Without them the frequency would change with every screen and seat.

Does it measure my screen's gamma?

No. It assumes the sRGB standard curve, and every export says so. The **Drop a measured gamma table** area on the Data tab is not active in the current version.

Why can I not download my results?

At the time of writing the **Export** tab does not open. See Export your data for workarounds.

Is anything I do sent anywhere?

No. Trials are run and stored in the page until you leave or start another run. See Privacy.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Psychophysics Bench background

Contents of the background material for the Psychophysics Bench.

This folder supports the how-to pages with the psychophysics terms the bench uses, its response keys, a statement of what is published and what is the tool's own, related tools and articles, its history and what stays on your device.

File	What it covers
glossary.md	Psychophysics and display terms used in the bench
shortcuts.md	Response keys, `Esc`, enlarging the stage and the trace
limits-and-accuracy.md	The data register entry by entry, display limits and what is not shown
related.md	Related tools on the site and Learn articles
release-notes.md	User-facing history of the bench
privacy.md	What stays in your browser

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Psychophysics Bench shortcuts

Response keys and other keyboard and pointer actions in the Psychophysics Bench.

Keys work while the page is waiting for an answer. Site-wide shortcuts are described in Site features.

Key or action	Two-interval forced choice	Yes / no detection
`←`	First interval	Yes, I saw it
`→`	Second interval	No, I did not
`Esc`	Stops the run	Stops the run

Action	Where	Result
Double-click	The stimulus stage or the staircase trace	Opens it enlarged, filling the window
`Esc`	Enlarged view	Closes it
Select the × button or the dark area around the image	Enlarged view	Closes it
`Tab`	Anywhere	Moves between controls; sliders then respond to the arrow keys

Between runs the arrow keys are not used by the bench, so a focused slider responds to them instead. Click an empty part of the page before a run so the response keys are not also moving a slider.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Psychophysics Bench glossary

Psychophysics and display terms used in the Psychophysics Bench, each explained briefly.

Term	Meaning
2AFC (two-interval forced choice)	A task with two intervals, one holding the stimulus; the observer must pick one, so chance is 50%
Adaptive staircase	A procedure that sets each trial's stimulus from the previous answers so trials cluster near threshold
Alias	A false pattern that appears when a screen is asked to draw a frequency above its Nyquist limit
Background (mean luminance)	The grey the contrast swings around; given as an 8-bit code and converted to relative luminance
Bayesian tracker	A procedure that keeps a probability distribution over the threshold and updates it after each answer
Bootstrap	Estimating the uncertainty of a result by refitting many resampled copies of the data
Constant stimuli	A method that tests at fixed contrasts rather than adapting; here, at the starting contrast only
Criterion (c) and β	Signal-detection measures of an observer's willingness to say "yes"
d′ (d-prime)	A signal-detection measure of sensitivity that separates seeing from willingness to say "yes"; needs stimulus-absent trials
Decibel step	A step size given as a ratio: $20 \times \log_{10}(\text{ratio})$; 2 dB multiplies or divides contrast by about 1.26
Degrees per pixel	The visual angle one pixel covers: atan(pixel pitch ÷ viewing distance)
Envelope σ	The width of a Gabor's Gaussian envelope, in degrees of visual angle
Gabor patch	A sine-wave grating multiplied by a Gaussian, fading smoothly into the background
Geometric mean	The average of values on a logarithmic scale; used because the staircase steps by ratios
Guess rate	The proportion correct expected by chance: 50% in two-interval forced choice
Lapse rate	The small proportion of errors an observer makes even on easy trials
Michelson contrast	(Lmax − Lmin) ÷ (Lmax + Lmin), where L is luminance
Nyquist limit	The finest frequency a pixel grid can show: half a cycle per pixel
Orientation	The angle of the grating's stripes, in degrees
Psychometric function	The curve relating stimulus strength to the probability of a correct or "yes" response
Relative luminance	The lightness-related quantity Y from 0 (black) to 1 (white), obtained from a screen value through the sRGB curve
Reversal	A trial at which the staircase changes direction
Spatial frequency	How many stripe cycles fall within one degree of visual angle (c/deg)
sRGB transfer curve (EOTF)	The standard curve relating a screen code value to light output
Threshold	The stimulus strength at which the observer is correct a stated proportion of the time
Transformed up-down (Levitt)	Staircases that step down after n correct in a row and up after an error, converging at p where $p^n = 0.5$
Warm-up reversals	The first two reversals, discarded because the staircase is still approaching threshold
Yes / no detection	A task with one interval, stimulus present on some trials, where the observer says whether it was seen

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Limits and accuracy

What the Psychophysics Bench takes from published sources, what is its own, the limits of an uncalibrated screen, and what it does not show.

What the bench is

An educational instrument for measuring a contrast detection threshold in a browser. It is not a calibrated laboratory system and not a clinical test. Results depend on your screen, your room and your attention as much as on your vision.

The data register, in plain terms

Entry	Status	What it means for you
sRGB transfer and primaries	Verbatim	Contrast is converted to screen values with the standard sRGB curve, which is a standard, not a measurement of your screen
Transformed up-down rules	Verbatim	1-up / 2-down and 3-down / 1-up follow Levitt (1971), including averaging an even number of reversals
Extreme-proportion correction	Verbatim	The 1/2N rule of Macmillan and Kaplan (1985) for hit rates of 1 or false-alarm rates of 0
Michelson contrast in luminance	Computed	Every contrast is placed symmetrically around the background's luminance, so the contrast asked for is the contrast drawn, unless capped
Viewing geometry	Computed	Degrees per pixel and the Nyquist limit follow exactly from the distance and pitch you enter
Signal-detection formulae	Computed	d′, criterion and β as in Macmillan and Creelman (2005), for yes/no only
Inverse normal CDF	Computed	An approximation accurate to 4.5×10^{-4}; it caps any d′ at about 6.18
Nonparametric bootstrap	Computed	A trial bootstrap; it gives intervals that are too narrow for adaptively sampled data
Bayesian adaptive staircase	Stand-in	The tool's own tracker, not QUEST; its slope and lapse rate are assumed
Psychometric function fit	Stand-in	A logistic with the task's guess rate and a fitted lapse rate; search and binning are the tool's own
Accelerated staircase	Stand-in	The tool's own; not PEST, and no stated target percentage
Weibull fit	Absent	No Weibull function is ever fitted to data
ROC curve and its area	Absent	A single yes/no criterion gives one point, not a curve

Because the fit is a stand-in, exports carry `reportable: false`.

Limits of a screen

- **No calibration.** The page assumes the sRGB curve. Real screens differ, and brightness, contrast and colour settings change the light that reaches you. A measured display calibration cannot be loaded in the current version.
- **8-bit steps.** Near threshold, a stripe pattern may differ from the background by only one or two screen code values, so the smallest contrasts the page can draw are coarse. The page does not dither.
- **Capped contrast.** A bright background cannot hold high contrast; the page caps it and says so.
- **Geometry you enter.** The frequency and envelope in degrees are only as accurate as the distance and pixel pitch you give, and assume you stay put.
- **Timing.** Durations are timed by the browser and shown on the screen's refresh; background load can delay them.

What the bench does not show

In the current version no readout displays the psychometric fit, d′, criterion, β or bootstrap interval, although the register describes them. The threshold shown comes from the staircase itself: the reversal average or the Bayesian tracker's estimate.

What it does not do

- Only two tasks run: two-interval forced choice and yes/no. m-alternative, same/different and odd-one-out are withdrawn.
- Constant stimuli tests only the starting contrast.
- At the time of writing, only the Lab tab opens, so exports cannot be made.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Privacy

What the Psychophysics Bench keeps in your browser and what uses the network.

What stays in your browser

- Your trials, answers and response times are held by the page only for the current run. They are cleared when you start another run or leave the page.
- All stimulus drawing and every calculation happen in the page.

What leaves the page, and only when you ask

- **Download the trials** and **Download the session** create files through your browser's normal download. They contain your answers and response times, so treat them as personal data if you collect them from other people.
- **Copy a link to this setup** writes a link to your clipboard. It contains settings only, never results.

The rest of the page

The tool page loads the site's usual pages, styles and scripts, and, depending on your cookie choices, advertising and consent-based analytics, like the rest of auricartisan.com. See the Privacy Policy at https://auricartisan.com/legal/privacy/ and the Cookie Policy at https://auricartisan.com/legal/cookies/.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Release notes

User-facing history of the Psychophysics Bench.

The site-wide history is at https://auricartisan.com/library/changelog/. The entries below cover what changed for users of this tool.

September 2026 — known issue

- Only the **Lab** tab opens; the other five tabs, including **Export**, do not respond. See Troubleshooting.

September 2026 — rebuilt as the Psychophysics Bench

Changed:

- Contrast is now Michelson contrast in relative luminance about a stated background, through the sRGB curve. Earlier versions built the grating in screen code values, so a requested 0.02 was drawn at about 0.051. Sessions recorded before this change cannot be converted to the new axis, because their exports did not record the background.

- Spatial frequency is now in cycles per degree, from a viewing distance and pixel pitch you enter. Earlier versions used cycles across the patch.
- Staircase steps are ratios in decibels instead of a fixed 0.05, and reversals are averaged geometrically in an even number.
- The Bayesian and accelerated procedures are named as the tool's own rather than QUEST and PEST.
- The stage shows the stimulus at the starting contrast between runs.

Added: **Background**, **Viewing distance**, **Pixel pitch**, **Envelope σ** in degrees, **Delivered / asked**, **At what percentage**, the Nyquist and contrast-cap warnings, and the header lines in the trials CSV.

Removed: the m-alternative forced choice, same/different and odd-one-out tasks, which could not score a trial correctly.

May 2026

- The Psychophysical Experiment Engine User Guide was published in the site documentation.

WEBSITE › TOOLS › PERCEPTION AND SPECTRAL › PSYCHOPHYSICAL EXPERIMENT ENGINE

Related tools and articles

Tools and Learn articles related to the Psychophysics Bench.

Tools in this group

- Neuroscience — the same viewing-geometry controls, and a V1 Gabor bank that models the cells a Gabor patch is designed to drive.
- Perception and Vision — simulate acuity loss, cataract and low light, which all raise contrast thresholds.

Other tools on the site

- Contrast Checker — contrast between two colours for text accessibility, a different measure from the Michelson contrast of a grating.

Learn articles

The Learn section is described in Learn.

- Contrast Sensitivity: The Spatial Frequencies You Can and Can't See — https://auricartisan.com/library/learn/articles/2026-08-24-contrast-sensitivity-spatial-frequency
- Human Color Vision: Cones, Opponent Signals, and the Brain — https://auricartisan.com/library/learn/articles/2026-05-24-human-color-vision-cones-opponent
- Color Illusions and the Limits of Perception — https://auricartisan.com/library/learn/articles/2026-06-10-color-illusions-and-perception

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Documentation on the site

- Psychophysical Experiment Engine User Guide — https://auricartisan.com/library/documentation/articles/2026-05-25-psychophysical-experiment-engine-user-guide

WEBSITE › TOOLS › IMAGE AND FILE TOOLS

Image and file tools

What the six image and file tools on auricartisan.com are for, where to find them, and which one to open for each job.

These six tools work on files you already have: photographs, SVG artwork, colour profiles and colour lookup tables. Each one reads your file in the browser, shows you what is in it, lets you change or measure it, and gives you a file or a report to take away. Nothing you load is uploaded; the reading and the arithmetic happen on the page.

The tools share a layout. A row of tabs runs across the top of each tool. The first tab is the working area, and later tabs explain the method, list where every number comes from (the **Data** tab), offer the exports (the **Export** or **Actions** tab) and cite the published sources (the **Reference** tab). The Image Picker is the exception: it has three tabs, **Workbench**, **Analysis** and **Actions**.

Where to find them

The group's own address is https://auricartisan.com/tool/general/tools/. Opening it takes you to the Collections page at https://auricartisan.com/collections/, which lists every tool on the site under **All tools**. Use the category filter or the **Sort** menu there, or open a tool directly from the table below. The Collections page is described in Collections.

The tools

Tool	Address	Use it to
Collage Maker	https://auricartisan.com/tool/general/tools/collage-maker/	Place pictures on a canvas, arrange them, measure coverage and export PNG or JPEG
ICC Profile Parser	https://auricartisan.com/tool/general/tools/icc-profile-parser/	Read an .icc or .icm profile: header, tags, primaries, tone curves and conformance
Image Picker	https://auricartisan.com/tool/general/tools/image-picker/	Extract a palette from a picture, pin exact pixels and export the colours
LUT Lab	https://auricartisan.com/tool/general/tools/lut-lab/	Apply a look or a .cube file, see what it moved, and write a .cube
M-Gradient Palette	https://auricartisan.com/tool/general/tools/m-gradient-palette/	Build a colour ramp in OKLCH, sRGB, linear RGB or HSL and check its steps and contrast
SVG Recolour	https://auricartisan.com/tool/general/tools/svg-recolor/	Remap every colour in an SVG, check contrast and colour vision, and export the file

The page titles differ slightly from the names in the Collections list: Collage Maker opens as **Collage Studio**, the ICC Profile Parser as **ICC Profile Reading**, the Image Picker as **Image Color Picker & Palette Tool**, the M-Gradient Palette as **Gradient Ramp**, and SVG Recolor as **SVG Recolour**.

Which tool for which job

- You have photographs and want one image made of several: use the Collage Maker.
- You have a colour profile from a monitor, printer or camera and want to know what it says: use the ICC Profile Parser.
- You have a picture and want its colours as a palette: use the Image Picker.
- You have a .cube grading file, or want to make one: use the LUT Lab.
- You want a smooth, evenly stepped colour scale between two colours: use the M-Gradient Palette.
- You have an icon or illustration as SVG and want it in different colours: use SVG Recolour.

All six tools are free to use, like every tool on the site.

In this folder

Folder	What it covers
collage-maker/	Canvas shapes, pictures, arrangements, measurements, undo, and PNG, JPEG and session exports
icc-profile-parser/	Loading profiles, every header field and tag shown, identification, gamut, conformance rules and exports
image-picker/	Palette extraction, the eyedropper and pins, contrast pairs, harmony, charts, exports and batch analysis
lut-lab/	LUT concepts, the looks, loading and writing .cube files, the before and after readouts and round trips
m-gradient-palette/	Ramps, interpolation spaces and hue routes, evenness, gamut mapping, contrast and colour-vision checks, exports
svg-recolour/	Loading SVGs, mapping colours, harmonies, previews, contrast and colour-vision checks, exports

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Collage Maker

What the Collage Maker does, who it is for, where to find it and how to make a first collage.

The Collage Maker (the page is titled **Collage Studio**) puts your pictures on a canvas so you can move, scale, rotate, flip and stack them, then export the result as a PNG or JPEG. You choose the shape of the canvas from six presets, round its corners, and decide whether the background is transparent or white. A grid arranger lays the pictures out for you and compares every possible grid before you commit to one.

Beside the canvas, the tool measures what you have made. It reports how much of the canvas the pictures cover, how much of it is painted twice where pictures overlap, and how much picture area hangs off the edge and will not be exported. Coverage and overlap are read from the pixels of the same image the export produces, so the figures describe the file you download.

Undo reaches back to where you started, including pictures you removed and a canvas you cleared. Pictures are read in the page and drawn straight to the canvas; nothing is uploaded.

Who it is for

- Designers and social media editors assembling a set of photographs into one image.
- Anyone who needs a transparent PNG composite with rounded corners.
- People who want to know, in numbers, how much of the frame their pictures fill before they export.

Where to find it

Open https://auricartisan.com/tool/general/tools/collage-maker/ in a current desktop or mobile browser. It is listed as **Collage Maker** on the Collections page at https://auricartisan.com/collections/. No installation is needed.

Quick start

1. Open the tool. The **Studio** tab is selected.
2. Drop your pictures on the **Drop pictures** box, or select it to choose files (JPG, PNG, WebP or GIF).
3. Pick a **Shape** for the canvas, for example **1 : 1 — 1200 × 1200**.
4. Select **Arrange in a grid** to lay the pictures out in the arrangement that covers the most canvas.
5. Click a picture to select it, drag it to move it, and drag a corner handle to resize it.
6. Open the **Export** tab and select **Save the PNG**. The file downloads as `collage.png`.

What you can do

- Choose from six canvas shapes, all 1200 pixels wide, and set corner radius, padding and gap.
- Add any number of pictures, and move, scale (2% to 400% by slider), rotate, fade, flip, hide, duplicate, reorder and remove each one.
- Snap moves to a 10 px grid and show centre and rule-of-thirds guides while you work.
- Compare every grid arrangement, fitted or filled, and apply one in a single step.
- Undo and redo up to 60 steps, including removals and clearing the canvas.
- Read measured coverage, overlap and off-canvas area as you work.
- Export PNG (with transparency) or JPEG at 0.25× to 4× scale, copy a PNG to the clipboard, and save or load the layout as a session file.
- Save a short JSON reading of what was measured.

In this folder

- docs/ — how to use the tool.
  - Getting started — open it, the screen tour, the first collage.
  - Build and edit a collage — pictures, the canvas, selecting, editing, layers and undo.
  - Arrange pictures in a grid — the Layout tab, fit and fill, and applying an arrangement.
  - Read the measurements — coverage, overlap, off-canvas area and the Measure and Data tabs.
  - Export, copy and save sessions — PNG, JPEG, clipboard, session files and the reading.
  - Reference — every control, option, output and file format.
  - Troubleshooting — symptoms, causes and fixes.
  - FAQ — short answers to common questions.
- others/ — background material.
  - Glossary — terms used in this tool.
  - Shortcuts — keyboard and pointer actions.
  - Limits and accuracy — what is measured, how precisely, and what is not held.
  - Related — related tools and Learn articles.
  - Release notes — user-facing history.
  - Privacy — what stays in your browser.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Collage Maker documentation

The list of how-to guides, the reference, troubleshooting and FAQ for the Collage Maker.

This folder holds the task guides for the Collage Maker, the full reference of its controls and outputs, and the troubleshooting and FAQ pages. Start with Getting started if you have not used the tool before; go to the Reference when you need the exact range or default of a control.

File	What it covers
getting-started.md	Opening the tool, a tour of the six tabs, and a first collage from start to export
build-a-collage.md	Adding pictures, the canvas settings, selecting and editing pictures, the layers list, undo and redo
arrange-in-a-grid.md	The Layout tab: fit or fill, enlarging small pictures, the arrangement table and applying a grid
read-the-measurements.md	The five readout cards, what each figure measures, and the Measure and Data tabs
export-and-sessions.md	Saving PNG and JPEG, export scale and quality, the clipboard, session files and the reading
reference.md	Every tab, control, button, message and file format, with ranges and defaults
troubleshooting.md	What to do when pictures will not load, exports look wrong, or undo behaves unexpectedly
faq.md	Short answers to common questions

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Getting started with the Collage Maker

Open the Collage Maker, find your way around its six tabs, and make and export a first collage.

Open the tool

Go to https://auricartisan.com/tool/general/tools/collage-maker/. The page opens on the **Studio** tab with an empty 1200 × 900 canvas. You do not need an account, and nothing needs installing.

A note under the page heading explains what the numbers on the page are measurements of. It has no close button on purpose.

The screen at a glance

The tool has six tabs.

Tab	What it is for
Studio	The working area: pictures, canvas settings, history, the canvas, layers and readout cards
Layout	Every grid arrangement for your pictures, laid out and measured, with a button to apply each one
Measure	How the coverage, overlap and off-canvas figures are measured, with worked examples
Data	A list of every figure on the page and where it comes from; the tab shows the number of entries (12)
Export	Export scale and JPEG quality, the save, copy and session buttons, file sizes and the reading
Reference	The published sources the tool relies on, and the arithmetic it runs

The Studio tab

The **Studio** tab has two parts.

- **The left-hand panel** holds, from top to bottom: **Pictures** (the drop box), **The canvas** (shape, padding, gap, corner radius and three switches), **History** (undo and redo), **The selected picture** (sliders and buttons for the picture you have selected), and two buttons: **Arrange in a grid** and **Clear the canvas**.
- **The right-hand panel** holds **The canvas** itself, with its size above it, the **Layers · top first** list beside it, and a row of cards under **What is on the canvas**.

A chequerboard behind the canvas shows where it is transparent. The chequerboard, the guides, the selection outline and the corner handles are only on screen; none of them is ever written into an exported file.

Make a first collage

1. On the **Studio** tab, drop three or four pictures onto the **Drop pictures** box. You can also select the box, or press `Enter` or `Space` while it has keyboard focus, to choose files. The box accepts JPG, PNG, WebP and GIF.
2. Each picture appears in the middle of the canvas, shrunk if necessary so that it fits within 42% of the canvas width and height. The most recently added picture is selected, and all of them appear in the **Layers · top first** list.
3. Open the **Shape** menu under **The canvas** and choose a shape, for example **1 : 1 — 1200 × 1200**. The canvas changes size; the pictures stay where they were.
4. Select **Arrange in a grid**. The tool tries every number of columns, measures how much of the canvas each arrangement covers, and applies the best one. A short message says which grid it chose and how much it covers.
5. Click a picture on the canvas to select it. Drag it to move it. Drag one of its four corner handles to make it larger or smaller around its centre.
6. With a picture selected, use the **Rotation** slider to turn it, or **Flip across** to mirror it.
7. Look at the cards under **What is on the canvas**. **canvas covered** is the share of the canvas the pictures paint; **outside the canvas** turns into a warning when part of a picture hangs off the edge.
8. Open the **Export** tab and select **Save the PNG**. Your browser downloads `collage.png`, and a message confirms the file size.

You should now have a PNG of your collage, at 1200 pixels wide, with a transparent background and rounded corners (the default corner radius is 18 px).

Next steps

- To change the background to white or remove the rounded corners, see Build and edit a collage.
- To choose a grid yourself rather than letting the tool pick, see Arrange pictures in a grid.
- To export larger, as JPEG, or to save the layout for later, see Export, copy and save sessions.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Arrange pictures in a grid

Use the Layout tab to compare every grid arrangement for your pictures and apply the one you want.

The **Layout** tab answers one question: which grid gives these pictures the most canvas? Every arrangement is laid out on the real canvas and measured, rather than estimated from the size of the cells, because a cell is only filled by a picture that shares its shape.

How a grid is built

A grid is a set of equal cells, filled row by row in the order the pictures are painted (the bottom of the stack goes in the first cell). Two settings from the **Studio** tab decide the size of the cells:

- **Padding** is the margin between the canvas edge and the block of cells.
- **Gap** is the space between neighbouring cells.

Both come out of the same canvas. The cell width is the canvas width, minus padding on both sides, minus one gap between each pair of columns, divided by the number of columns; the height works the same way. With a large gap on a small canvas this can come out as zero or less. The tool then refuses that arrangement and says why, rather than drawing it inside out.

Only visible pictures are arranged. Hidden pictures keep their place and are not counted.

Choose how a picture fits its cell

The **Fitting** group on the left of the **Layout** tab has two buttons and a switch.

Control	What it does
Fit inside the cell (default)	Scales each picture so all of it sits inside its cell; a different shape leaves part of the cell empty
Fill the cell, crop	Scales each picture so it covers its whole cell; the excess is not cut away (see below)
Enlarge a small picture to fill its cell	Off by default. Allows scaling above 100% of a picture's own pixels

With **Fill the cell, crop**, the part of a picture beyond its cell is not trimmed off. It overlaps the neighbouring pictures or hangs off the canvas, and the **Overlap** and **Off-canvas** columns of the table show how much.

With enlarging off, a picture smaller than its cell stays at its own resolution and leaves the cell short. That keeps small pictures sharp. It is a preference, not a rule; switch it on when you would rather fill the frame than keep every pixel crisp.

Read the arrangement table

The table under **Which arrangement gives these pictures the most canvas** has one row for every number of columns, from one column up to one column per picture.

Column	Meaning
Arrangement	Columns × rows, for example `3 × 2`
Cell	The size of each cell in pixels
Covered	The share of the canvas the pictures cover in this arrangement
Overlap	The share of the canvas where pictures sit on top of one another (only possible with **Fill the cell, crop**)
Off-canvas	The share of picture area outside the canvas (only possible with **Fill the cell, crop**)
What happened	**the most canvas of these**, a count of cells left empty, or the reason a row was refused

A refused row shows **refused** in the **Covered** column. A grid with spare cells says so in the last column, for example **1 cell left empty**.

A row is only marked as the winner when it truly covers more than every other row. If every arrangement covers the same amount, no row is marked, and the note under the table explains why: none of the pictures is larger than its cell, so each keeps its own size wherever it goes. Switch on enlarging, or use larger pictures, and the rows separate.

Under the table, **The grid, and what it leaves** draws the three best arrangements with their coverage and the number of cells used, for example **cells used 4 of 6**.

Apply an arrangement

1. Open the **Layout** tab.
2. Choose **Fit inside the cell** or **Fill the cell, crop**, and set **Enlarge a small picture to fill its cell** as you want it.
3. Under **Apply**, select the button for the arrangement you want, for example **3 × 2**. Only arrangements that fit are offered.
4. A message confirms: **Arranged. Undo puts every picture back.**

Every visible picture is moved to the centre of its cell, scaled as the fitting choice says, and turned back to 0° rotation. The whole arrangement is one history step, so one **Undo** returns every picture to where it was.

To let the tool choose for you, select **Arrange in a grid** on the **Studio** tab instead. It applies the arrangement with the highest coverage; where several tie, it takes the one with the fewest columns.

Tips

- If every arrangement shows the same coverage, your pictures are all smaller than their cells. Reduce **Padding** or **Gap**, switch on enlarging, or accept the tie.
- If an arrangement is refused, the message gives the padding, gap, grid and canvas that caused it. Reduce the padding or gap and look again.
- Mixed portrait and landscape pictures rarely fill a grid well with **Fit inside the cell**. Compare the rows, or use **Fill the cell, crop** and check the **Overlap** column.
- The arrangement only sets position, scale and rotation. You can still move and resize any picture afterwards on the **Studio** tab.

What you should see

After you apply an arrangement, the **Studio** canvas shows the pictures in their cells, and the **canvas covered** card is close to the **Covered** figure the table gave for that row. The two are measured differently: the table works from where the pictures are placed, while the card reads the pixels. A picture with transparent areas, such as a PNG cut-out, therefore covers less on the card than in the table.

The **Covered** column shows a dash when three or more pictures overlap at one point, because the placement arithmetic cannot then give an exact figure. The card on the **Studio** tab is still exact in that case.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Build and edit a collage

Add pictures, set up the canvas, select and edit pictures, use the layers list, and undo or redo any step.

Everything in this guide happens on the **Studio** tab.

Add pictures

1. Drop one or more image files onto the **Drop pictures** box in the left-hand panel. You can also drop them straight onto the canvas, or select the box (or press `Enter` or `Space` while it has focus) to open a file chooser that accepts several files at once.

2. Wait for the message at the bottom of the screen. It reads **Added** followed by the file name, or **Added 4 pictures** for a batch.

What happens to each picture:

- It is centred on the canvas.
- It keeps its own pixels. If it is larger than 42% of the canvas width or height, it is scaled down to fit inside that box; a smaller picture stays at 100%.
- It is added on top of the others and becomes the selected picture.
- It appears at the top of the **Layers · top first** list, with a thumbnail, its file name, its size in pixels and its current scale.

Any image type your browser can open is accepted; the box names JPG, PNG, WebP and GIF. Files that are not images are ignored. If a file cannot be decoded, a message reads **Could not read** followed by its name, and the other files in the batch are still added. Adding a batch is one history step.

Set up the canvas

The controls under **The canvas** apply to the whole collage.

Control	What it does	Range	Default
Shape	Sets the canvas size. Every shape is 1200 pixels wide	4 : 3, 1 : 1, 3 : 2, 16 : 9, 9 : 16, 2 : 3	4 : 3 — 1200 × 900
Padding	The margin between the canvas edge and the block of grid cells. Used only when arranging in a grid	0 to 120 px	24 px
Gap	The space between grid cells. Used only when arranging in a grid	0 to 80 px	8 px
Corner radius	Rounds the corners of the canvas. The export is clipped to the rounded shape	0 to 80 px	18 px
Transparent background	When ticked, empty areas are transparent. When cleared, the background is white	On or off	On
Centre and rule-of-thirds guides	Shows thirds lines on the canvas, and dashed lines through the centre of the selected picture	On or off	On
Snap to a 10 px grid	While dragging, the picture's centre lands on a multiple of 10 px	On or off	On

Above these controls, **size** and **background** repeat the current canvas size and background (transparent, or the colour in use).

Changing the shape does not move or resize any picture. Pictures that were near the edge of a larger canvas can end up partly outside a smaller one; the **outside the canvas** card tells you when that happens.

The guides are drawn only on screen. They never appear in an export.

Select a picture

- Click a picture on the canvas. Where pictures overlap, the click selects the one on top.
- Or select a row in the **Layers · top first** list. This is the only way to select a hidden picture, because hidden pictures cannot be clicked on the canvas.
- Click an empty part of the canvas to clear the selection.

The selected picture has a gold outline and four square corner handles. Its controls appear under **The selected picture**; when nothing is selected that area reads **nothing selected**.

Move, resize and nudge

Action	How
Move	Drag the picture. With snapping on, its centre lands on a multiple of 10 px
Resize	Drag a corner handle; the picture scales around its centre, keeping its proportions, from 2% to 800%
Nudge 1 px	Click the canvas so it has keyboard focus, then press an arrow key
Nudge 10 px	Hold `Shift` and press an arrow key
Remove	Press `Delete` or `Backspace` while the canvas has focus, or select **Remove**

Each drag is one history step, recorded when you let go. Each arrow-key press is one step. Arrow-key nudges are exact and are not snapped to the grid.

Change the selected picture

The controls under **The selected picture** change only the selected picture.

Control	What it does	Range
Scale	Size relative to the picture's own pixels	2% to 400%
Rotation	Turns the picture about its centre	−180° to 180°
Opacity	How see-through the picture is	0% to 100%
Bring to front	Moves it to the top of the stack	—
Send to back	Moves it to the bottom of the stack	—
Centre	Moves its centre to the centre of the canvas	—
Duplicate	Adds a copy 24 px to the right and 24 px down, and selects the copy	—
Flip across	Mirrors it left to right; select again to undo the flip	—
Flip down	Mirrors it top to bottom	—
Hide	Hides it from the canvas, the measurements and the export. The button then reads **Show**	—
Remove	Takes it off the canvas. Undo brings it back	—

Sliders show the value as you drag. The history step is recorded when you release the slider.

The layers list

Layers · top first lists every picture in the order it is painted, with the top of the stack first. Each row shows:

- a thumbnail;
- the file name (hover to see the full name);
- the picture's own size and its current scale, for example `3024×4032 · 12%`;
- a dot, or **hidden** for a hidden picture.

Select a row to select that picture.

Arrange in a grid, or start again

- **Arrange in a grid** tries every number of columns for the visible pictures, measures each arrangement, and applies the one that covers the most canvas. It uses the fitting choices on the **Layout** tab. A message reports the grid and its coverage, for example **Arranged 2 × 2 — 88.1% covered. Undo puts them back.** See Arrange pictures in a grid to choose an arrangement yourself.
- **Clear the canvas** removes every picture. It is one history step, so **Undo** brings them all back.

Undo and redo

The **History** group has ← **Undo** and **Redo** →. The note under them says how many steps you can go back, for example **5 steps back, to the start.**

- Every change is recorded: adding pictures, moving, resizing, each slider release, every button, changing the shape, padding, gap, corner radius or background, arranging, clearing, removing and loading a session.
- A removed picture comes back with the step that removed it.
- The history holds up to 60 steps. Older steps are dropped once you pass that.
- Making a new change after undoing clears the redo steps.
- Keyboard: `Ctrl` + `Z` undoes; `Ctrl` + `Shift` + `Z` or `Ctrl` + `Y` redoes. On a Mac, `Cmd` works in place of `Ctrl`. The shortcuts do nothing while a slider, menu or text box has focus.

> **Tip:** When the canvas itself has keyboard focus, one press of the undo or redo shortcut can move two steps. If you want exactly one step, use the **Undo** and **Redo** buttons, or click outside the canvas before using the keyboard.

Switching **Centre and rule-of-thirds guides** and **Snap to a 10 px grid** is not recorded, because neither changes the collage.

What you should see

After these steps your pictures sit where you want them, the layers list shows their order, and the **canvas covered**, **overlapping** and **outside the canvas** cards describe the collage exactly as it will export. To export it, see Export, copy and save sessions.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Export, copy and save sessions

Save your collage as PNG or JPEG at the size you need, copy it to the clipboard, and save or reload the layout as a session file.

All exports are on the **Export** tab.

What an export contains

The exported image is the canvas drawn again at the scale you choose. It contains:

- every visible picture, at its position, scale, rotation, opacity and flip;
- the background: transparent, or white when **Transparent background** is cleared;
- the rounded corners set by **Corner radius**, scaled with the export.

It does not contain:

- the guides, the selection outline, the corner handles or the chequerboard, which are only on screen;
- hidden pictures;
- any part of a picture outside the canvas. The **outside the canvas** card on the **Studio** tab tells you how much that is.

Set the size and quality

Under **What to take**:

Control	What it does	Range	Default
Scale	Multiplies the canvas size for the export	0.25× to 4×, in steps of 0.25	1.00×
JPEG quality	Compression quality for JPEG only	0.10 to 1.00	0.92

The **size** line shows the export size in pixels, for example `2400 × 1800` at 2×, and **background** shows transparent or the colour in use. Pictures are drawn from their own pixels, so exporting larger than 1× is sharp for pictures that were scaled down, and soft for any picture scaled above its own size.

Save an image

1. Set **Scale** and, for JPEG, **JPEG quality**.
2. Select **Save the PNG** or **Save a JPEG**.
3. Your browser downloads `collage.png` or `collage.jpg`. A message confirms the format and file size, for example **Saved PNG · 2.4 MB**.

If there is nothing on the canvas, the message reads **Nothing on the canvas yet.** and nothing is saved.

PNG or JPEG

The table under **What leaves this page** shows the real size of each file for your collage, measured by encoding it before you save.

Format	Carries	Good for
PNG	Lossless pixels with transparency	A transparent background, text and hard edges
JPEG	Lossy pixels, no transparency	Photographs, where a smaller file matters
Session JSON	The layout, not the pictures	Coming back to the collage later

> **Important:** JPEG cannot store transparency. Clear **Transparent background** on the **Studio** tab before saving a JPEG, so empty areas and rounded corners are white rather than left to your browser's encoder.

The same quality setting gives slightly different files in different browsers, because each browser has its own JPEG encoder.

Copy to the clipboard

Select **Copy to the clipboard** to place a PNG of the collage, at the current export scale, on your clipboard. Paste it into another application.

- If your browser cannot put images on the clipboard, the message reads **This browser will not take an image from the clipboard.**

- If the browser refuses permission, it reads **The browser refused the clipboard.**

Save and load a session

A session file records the layout so you can come back to it. It saves the canvas shape and size, padding, gap, corner radius and background, and for every picture its file name, position, own size, scale, rotation, opacity, flips and whether it is visible.

> **Important:** A session file does not contain the pictures. Keep the original files; you need them to reopen the collage.

Save a session

1. Select **Save the session**.
2. Your browser downloads `collage-session.json`. The message reminds you: **Session saved. It describes the layout, not the pictures.**

Load a session

1. On the **Studio** tab, add the same pictures, in the same order you originally added them. The session is applied to pictures by their position in the stack, bottom first.
2. On the **Export** tab, select **Load a session** and choose the `.json` file.
3. The canvas settings are applied, and each loaded picture takes the position, scale, rotation, opacity, flips and visibility stored for it. The message reads, for example, **Session applied to the 4 pictures you have loaded.**

Loading a session is one history step, so **Undo** returns to the layout you had before.

If you load a session before adding any pictures, only the canvas settings are applied. Add the pictures, then load the session file again. A file that the tool did not write produces the message **That is not a session file this page wrote.**

Save the reading

Save the reading downloads `collage-reading.json`, a short record of what was measured, to keep beside the image. The same text is shown under **The reading that goes with it**. It holds:

- the canvas width, height and background;
- the number of pictures;
- under `measured`: `covered`, `overlapping` and `outsideCanvas` as fractions (0 to 1, four decimal places), and `at`, which says whether they were measured at export resolution or at a reduced size;
- under `choices`: whether enlarging past 100% was off, the snap step, and the JPEG quality;
- under `notHeld`: the things the reading does not know, physical print size and output density.

```
{
  "tool": "Auric Artisan — collage",
  "canvas": { "w": 1200, "h": 900, "background": "transparent" },
  "pictures": 4,
  "measured": {
    "covered": 0.8123,
    "overlapping": 0,
    "outsideCanvas": 0,
    "at": "export resolution"
  },
  "choices": { "neverEnlargeBeyond": 1, "snapStep": 10, "jpegQuality": 0.92 },
  "notHeld": ["physical print size", "output density"]
}
```


The figures in this example are illustrative; your reading contains the values measured for your collage.

What you should see

After saving, you have an image file at the size shown on the **Export** tab, without any on-screen guides or outlines, and optionally a session file and a reading to go with it.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Read the measurements

What the canvas covered, overlapping, outside the canvas and history cards measure, and how to use the Measure and Data tabs.

Under the canvas on the **Studio** tab, a row of cards headed **What is on the canvas** describes the collage. The figures update after every change.

The five cards

Card	What it reports	How it is found
canvas covered	The share of the canvas that pictures paint	Measured from the pixels
overlapping	The share of the canvas painted by two or more pictures	Measured from the pixels
outside the canvas	The share of total picture area that lies beyond the edges and will not be exported	Computed exactly from positions
history	How many steps you can undo, back to the start	Counted
the old readout said	The sum of every picture's area divided by the canvas area	Computed, for comparison only

The **outside the canvas** card changes colour as a warning when more than 0.1% of picture area is off the canvas.

canvas covered

The tool draws the canvas twice, once without the pictures and once with them, exactly as the export would. Every pixel that differs between the two drawings is a pixel a picture has put something on. The figure is that count divided by the number of pixels on the canvas.

Because it reads pixels, this figure notices transparency. A PNG cut-out with a transparent background covers only its visible part. A picture at 0% opacity covers nothing.

overlapping

Each visible picture is drawn on its own, and the tool counts the pixels where two or more pictures have painted. The figure is that count divided by the pixels on the canvas.

outside the canvas

The tool works out the four corners of each picture after its scale and rotation, clips that shape to the canvas rectangle, and compares what is left with the whole picture. The figure is the share of total picture area that falls outside. It needs no pixels, so it is exact. Hidden pictures are not counted.

A picture hanging off the edge may be exactly what you intended, for example a bleed. The figure is shown on its own so you can decide.

the old readout said

This card keeps the figure an earlier version of the tool reported: the sum of all picture areas divided by the canvas area. It does not look at where the pictures are, so stacked pictures are counted twice and pictures dragged off the canvas are still counted in full. It is shown beside the measured figures so you can see the difference; do not rely on it.

Measurement resolution

The measurement is made at the canvas size. For canvases larger than 1,440,000 pixels (for example **9 : 16 — 1200 × 2133**), it is made on a proportionally smaller copy to keep dragging smooth. The label above the canvas says which: **measured at full size**, or **measured at** followed by the size used. At the reduced size the figures agree with a full-size measurement to well under a tenth of a percentage point.

The Measure tab

The **Measure** tab explains the method with worked examples that are laid out and measured live by the same code the **Studio** uses.

- **One number for three different collages** draws four pictures three ways, **Stacked**, **Spread out** and **Dragged off the canvas**, and shows what the old sum-of-areas figure says (**it said**) beside the measured coverage (**really**) for each. The sum gives the same answer for all three; only the spread-out arrangement is actually covered to that extent.
- Further panels describe how the undo history works, what is and is not written into an export, and two features that were withdrawn because they did not measure the collage.
- The left-hand panel lists **What is measured** (covered and overlapping by pixels, off-canvas by geometry, arrangements by laying them out) and has two buttons, **Bootstrap CI** and **Batch coverage**, that scroll to the notes on the withdrawn features.

The Data tab

The **Data** tab lists every figure on the page and where it comes from. Each entry has a status:

Status	Meaning
Measured	Read from the canvas the export will use
Computed	Worked out from the layout by arithmetic the entry states
This tool's choice	A default this tool picked, where someone else might reasonably choose differently

Status	Meaning
Absent	Named on the page but not held, with what it would take to hold it

Use the **Status** buttons on the left to filter the list; each shows how many entries it holds. Select an entry to expand its explanation. The twelve entries cover canvas coverage, overlap, the off-canvas area, the arrangement comparison, the transform arithmetic, the grid cell size, the fitted scale, the export file size, the choice not to enlarge past 100%, the 10 px snap step, the 0.92 JPEG quality default, and physical print size, which the tool does not hold.

Below the list, **Two entries that are gone rather than reclassified** names the two withdrawn features.

The Reference tab

The **Reference** tab lists the sources behind the tool: the HTML Canvas 2D standard, PNG (ISO/IEC 15948), JPEG (ISO/IEC 10918-1), affine transforms in the plane, and WCAG 2.2 target size and keyboard operation, each marked **Runs**, and one withdrawn citation. Under **The arithmetic**, a table gives the formulas for a picture's corners, the grid cell size, the fitted scale, and each of the three measurements.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Collage Maker reference

Every tab, control, button, readout, message and file format in the Collage Maker, with ranges and defaults.

Address: https://auricartisan.com/tool/general/tools/collage-maker/

Tabs

Tab	Contents
Studio	Pictures, canvas settings, history, selected-picture controls, the canvas, the layers list, the readout cards
Layout	Fitting options, apply buttons, the arrangement table and drawings of the three best arrangements
Measure	Worked examples of how coverage, overlap and off-canvas area are measured
Data	The list of every figure and its source, filterable by status; the tab label shows the count (12)
Export	Export scale, JPEG quality, save, copy and session buttons, file sizes, the reading
Reference	Sources, each marked **Runs** or **Withdrawn**, and the arithmetic table

Links in the page text (for example **Measure** or **Data** in a note) switch to that tab.

Studio tab — Pictures

Control	What it does	Accepts
Drop pictures box	Adds pictures by drag and drop, or opens a file chooser when selected or when `Enter` or `Space` is pressed	Any image type the browser opens; JPG, PNG, WebP, GIF named
The canvas	Also accepts dropped image files	As above

Several files can be added at once. New pictures are centred, scaled down to fit within 42% of the canvas width and height (never enlarged), placed on top and selected.

Studio tab — The canvas

Control	Values or range	Default
Shape	4 : 3 — 1200 × 900; 1 : 1 — 1200 × 1200; 3 : 2 — 1200 × 800; 16 : 9 — 1200 × 675; 9 : 16 — 1200 × 2133; 2 : 3 — 1200 × 1800	4 : 3
Padding	0 to 120 px, step 1	24 px
Gap	0 to 80 px, step 1	8 px
Corner radius	0 to 80 px, step 1	18 px
Transparent background	On (transparent) or off (white)	On
Centre and rule-of-thirds guides	On or off; screen only	On
Snap to a 10 px grid	On or off; applies to dragging	On
size (readout)	Current canvas size	1200 × 900
background (readout)	transparent, or the colour in use	transparent

Padding and gap affect only grid arrangements. The corner radius clips the canvas and every export.

Studio tab — History

Control	What it does
← **Undo**	Steps back one change; disabled when there is nothing to undo
Redo →	Steps forward one undone change; disabled when there is nothing to redo
Note	**Nothing to undo yet**, or the number of steps back to the start

Up to 60 steps are kept. A new change clears the redo steps.

Studio tab — The selected picture

Shown when a picture is selected; otherwise the area reads **nothing selected**.

Control	Range or effect	Default for a new picture
Scale	2% to 400%, step 1%	Whatever fits 42% of the canvas, at most 100%
Rotation	−180° to 180°, step 1°	0°
Opacity	0% to 100%, step 1%	100%
Bring to front	Moves the picture to the top of the stack	—
Send to back	Moves the picture to the bottom	—
Centre	Centres the picture on the canvas	—
Duplicate	Adds a copy offset 24 px right and 24 px down, and selects it	—
Flip across	Mirrors left to right (toggles)	Not flipped
Flip down	Mirrors top to bottom (toggles)	Not flipped
Hide / **Show**	Hides or shows the picture	Shown

Control	Range or effect	Default for a new picture
Remove	Removes the picture	—

Studio tab — Other buttons

Button	What it does
Arrange in a grid	Applies the arrangement with the highest measured coverage, using the **Layout** tab's fitting choices
Clear the canvas	Removes every picture; undoable

Studio tab — Canvas and layers

Element	What it shows or does
Canvas heading	**Canvas ·** width × height, and **measured at full size** or **measured at** the reduced size
Note beside **The canvas**	**no pictures**, or the number of pictures and the canvas size
Canvas	Click to select, drag to move, drag a corner handle to resize, click empty space to deselect
Layers · top first	One row per picture, top of the stack first: thumbnail, file name, own size and scale, visibility

Pointer resizing by a corner handle ranges from 2% to 800%.

Studio tab — Readout cards

Card	Meaning
canvas covered	Share of canvas pixels painted by pictures (measured from pixels)
overlapping	Share of canvas pixels painted by two or more pictures
outside the canvas	Share of picture area beyond the edges; warns above 0.1%
history	Undo steps available
the old readout said	Sum of picture areas divided by canvas area; for comparison only

Percentages are shown to one decimal place.

Layout tab

Control	What it does	Default
Fit inside the cell	Scales pictures so all of each is inside its cell	Selected
Fill the cell, crop	Scales pictures to cover their cells; the excess is not trimmed	—
Enlarge a small picture to fill its cell	Allows scaling above 100% of a picture's own pixels	Off
Apply buttons	One per valid arrangement, labelled columns × rows	—

Arrangement table columns: **Arrangement**, **Cell**, **Covered**, **Overlap**, **Off-canvas**, **What happened**. One row per column count from 1 to the number of visible pictures. The three highest-covering arrangements are drawn under **The grid, and what it leaves**, each with **covered** and **cells used**.

Applying an arrangement moves each visible picture to the centre of its cell, sets its scale, and sets its rotation to 0°. It is one history step.

Measure tab

Element	Contents
What is measured	covered: pixels; overlapping: pixels; off-canvas: geometry; arrangements: laid out
Withdrawn buttons	**Bootstrap CI** and **Batch coverage** scroll to the notes on two withdrawn features
One number for three different collages	Three live panels, **Stacked**, **Spread out**, **Dragged off the canvas**, with **it said** and **really**
Further panels	How undo records history, what an export contains, withdrawn features, and what stayed unchanged

Data tab

Status filter buttons: **All**, **Measured**, **Computed**, **This tool's choice**, **Absent**, each with a count. Each entry expands to an explanation; stand-in and absent entries add **what it would take**.

Export tab

Control	Range or effect	Default
Scale	0.25× to 4×, step 0.25	1.00×
JPEG quality	0.10 to 1.00, step 0.01	0.92
size (readout)	Export size in pixels	1200 × 900
background (readout)	transparent or the colour in use	transparent
Save the PNG	Downloads `collage.png`	—
Save a JPEG	Downloads `collage.jpg` at the chosen quality	—
Copy to the clipboard	Copies a PNG at the export scale	—
Save the session	Downloads `collage-session.json`	—
Load a session	Opens a chooser for a `.json` session file	—
Save the reading	Downloads `collage-reading.json`	—

The table under **What leaves this page** lists **PNG**, **JPEG** and **Session JSON** with what each carries, its measured size for your collage, and what it is good for.

File formats

collage.png and collage.jpg

Property	Value
Size	Canvas size × **Scale**, rounded to whole pixels
Colour	8 bits per channel, as the browser draws the canvas
Transparency	PNG keeps it; JPEG cannot store it

Property	Value
Contents	Visible pictures, background, rounded corners; no guides, outlines or handles

collage-session.json

```
{
  "tool": "Auric Artisan — collage",
  "canvas": {
    "w": 1200, "h": 900, "aspect": "4:3",
    "pad": 24, "gap": 8, "radius": 18,
    "background": "transparent"
  },
  "pictures": [
    {
      "name": "harbour.jpg", "cx": 300, "cy": 225, "bw": 4032, "bh": 3024,
      "scale": 0.1339, "rot": 0, "opa": 1,
      "flipX": false, "flipY": false, "visible": true
    }
  ]
}
```

Field	Meaning
`canvas.w`, `canvas.h`	Canvas size in pixels
`canvas.aspect`	Shape preset, for example `4:3`
`canvas.pad`, `canvas.gap`, `canvas.radius`	Padding, gap and corner radius in pixels
`canvas.background`	`transparent`, or a colour
`pictures[].name`	The file name the picture had
`pictures[].cx`, `cy`	Position of the picture's centre on the canvas, in pixels
`pictures[].bw`, `bh`	The picture's own width and height in pixels
`pictures[].scale`	Scale factor (1 is 100%)
`pictures[].rot`	Rotation in degrees
`pictures[].opa`	Opacity from 0 to 1
`pictures[].flipX`, `flipY`	Mirrored across or down
`pictures[].visible`	Shown or hidden

The file holds no image data. On loading, stored pictures are matched to loaded pictures by order in the stack, bottom first. The values in the example are illustrative.

collage-reading.json

Fields: `tool`, `canvas` (`w`, `h`, `background`), `pictures` (a count), `measured` (`covered`, `overlapping`, `outsideCanvas` as fractions to four decimals, and `at`), `choices` (`neverEnlargeBeyond`, `snapStep`, `jpegQuality`), and `notHeld`. See Export, copy and save sessions for an example.

Messages

Message	When it appears
Added name / **Added N pictures**	Pictures were added
Could not read name	A file could not be decoded as an image
Removed. Undo brings it back.	A picture was removed
Cleared. Undo brings every picture back.	The canvas was cleared
Arranged. Undo puts every picture back.	An arrangement was applied from the **Layout** tab
Arranged C × R **—** N% **covered. Undo puts them back.**	**Arrange in a grid** was used
No arrangement fits at this padding and gap.	Every arrangement was refused
Nothing on the canvas yet.	Export, copy or arrange with no pictures
Saved PNG or **Saved JPG**, with a size	An image was saved
The browser could not encode that.	The browser failed to produce the image file
Copied with a size	The PNG was copied to the clipboard
This browser will not take an image from the clipboard.	The browser does not support copying images
The browser refused the clipboard.	Clipboard permission was refused
Session saved. It describes the layout, not the pictures.	A session file was saved
Session applied to the N **pictures you have loaded.**	A session was loaded onto loaded pictures
Session read. Load the same pictures and it will be applied to them.	A session was loaded with no pictures on the canvas
That is not a session file this page wrote.	The chosen file is not a valid session
Reading saved.	The reading was saved
A picture in that step is no longer loaded.	An undo or redo step refers to a picture the page no longer holds

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Troubleshooting the Collage Maker

Symptoms you may meet in the Collage Maker, why they happen, and what to do.

Pictures will not load

Symptom	Why	What to do
Nothing happens after dropping a file	The file is not an image type, so it is ignored	Convert it to JPG, PNG, WebP or GIF and try again
Could not read followed by a file name	Your browser could not decode that file	Save it again as JPG or PNG in another program; HEIC and camera RAW files are not opened by every browser
The page slows down with many pictures	Every change is measured from pixels, once per picture for the overlap figure	Use fewer or smaller pictures, or resize very large photographs before adding them

A picture is in the wrong place or missing

Symptom	Why	What to do
A picture cannot be clicked	It is hidden, or another picture is on top of it	Select it in the **Layers · top first** list; use **Show** or **Bring to front**
A picture moved partly off the canvas after changing **Shape**	Changing the shape does not move pictures	Use **Arrange in a grid**, **Centre**, or drag it back; watch the **outside the canvas** card
Dragging jumps in steps	**Snap to a 10 px grid** is on	Clear the switch, or use the arrow keys for 1 px nudges
Arrow keys scroll the page instead of nudging	The canvas does not have keyboard focus	Click the canvas once, then use the arrow keys

Grid arrangements

Symptom	Why	What to do
No arrangement fits at this padding and gap.	Padding and gap leave no room for any grid	Reduce **Padding** or **Gap** on the **Studio** tab
A row in the **Layout** table reads **refused**	That grid's cells come out at zero or less	Reduce padding or gap, or choose a grid with fewer columns or rows
Every row shows the same **Covered** figure	No picture is larger than its cell, so each keeps its own size	Switch on **Enlarge a small picture to fill its cell**, or use larger pictures
Pictures overlap after applying a grid	**Fill the cell, crop** scales pictures past their cells without trimming them	Use **Fit inside the cell**, or check the **Overlap** column first
Rotated pictures became straight	Applying an arrangement sets every picture's rotation to 0°	Rotate again afterwards, or undo

Undo and redo

Symptom	Why	What to do
One key press undid two steps	With the canvas focused, the shortcut can be handled twice	Use the **Undo** and **Redo** buttons, or click outside the canvas before pressing the shortcut
`Ctrl` + `Z` does nothing	A slider, menu or text box has focus	Click an empty part of the page, or use the **Undo** button
Early steps can no longer be undone	The history keeps 60 steps	Save a session file at points you may want to return to
A picture in that step is no longer loaded.	The step refers to a picture whose image the page no longer holds	Add the picture again

Exports

Symptom	Why	What to do
The JPEG has dark or unexpected empty areas or corners	JPEG cannot store transparency	Clear **Transparent background** before saving a JPEG, or save a PNG
Part of a picture is missing from the export	It lies outside the canvas, or the rounded corners clip it	Check **outside the canvas**; reduce **Corner radius**

Symptom	Why	What to do
A picture looks soft in a large export	It was scaled above its own size, or the export scale enlarges it	Use a larger original, or a lower **Scale**
The browser could not encode that.	The image was too large for the browser, or encoding failed	Lower **Scale** and try again
Nothing on the canvas yet.	There are no pictures	Add pictures first
This browser will not take an image from the clipboard.	The browser cannot copy images	Use **Save the PNG** instead
The browser refused the clipboard.	Clipboard permission was blocked	Allow clipboard access for the site, or save the file

Sessions

Symptom	Why	What to do
That is not a session file this page wrote.	The file is not JSON, or lacks the canvas and picture lists	Choose a `collage-session.json` saved by this tool
Loading a session changed only the canvas	No pictures were loaded when the session was read	Add the pictures, then load the session again
Pictures took each other's positions	They were added in a different order from the original	Clear the canvas, add the pictures in their original order, and load the session again
The session reopened without the pictures	A session never contains image data	Keep the original picture files with the session

Still stuck

Reload the page to start fresh; nothing you loaded was stored, so reloading clears the canvas and history. For help with the site in general, see the help pages at Help.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Collage Maker FAQ

Short answers to common questions about the Collage Maker.

Are my pictures uploaded?

No. Pictures are read in the page and drawn straight to a canvas in your browser. Nothing is sent to a server. See Privacy.

What file types can I add?

Any image your browser can open. The drop box names JPG, PNG, WebP and GIF. The export is a still image, so an animated GIF contributes a single frame.

How many pictures can I add?

The tool sets no fixed limit. Each picture is measured separately for the overlap figure, so very many large pictures make the page slower.

Can I change the canvas size to something other than the presets?

No. There are six shapes, all 1200 pixels wide. To get a larger file, use **Scale** on the **Export** tab: 4× gives a canvas 4800 pixels wide.

Can I choose a background colour other than white?

The switch offers transparent or white. A session file that carries another background colour will apply it when loaded, but the page has no colour control of its own.

Why do the rounded corners appear in my export?

Corner radius clips the canvas itself, and the default is 18 px. Set it to 0 px for square corners.

Why does the export not show the gold outline and the guides?

They are drawing aids shown only on screen. The export contains the pictures and the background, nothing else.

Does "Fill the cell, crop" actually crop my pictures?

No. It scales each picture to cover its cell, but the part beyond the cell is not cut away; it overlaps the neighbouring pictures or hangs off the canvas. The **Overlap** and **Off-canvas** columns on the **Layout** tab show how much.

Why is "canvas covered" lower than I expected for a PNG?

The figure reads pixels. Transparent parts of a PNG, and any picture set to 0% opacity, paint nothing and are not counted as covered.

What is "the old readout said"?

The figure an earlier version of the tool gave: the sum of picture areas divided by the canvas area. It ignores overlap and anything off the canvas, so it is kept only for comparison. Use **canvas covered** instead.

Does a session file contain my pictures?

No. It records positions, sizes, rotation, opacity, flips, visibility and the canvas settings. Keep your original files, add them in the same order, then load the session.

Can I undo after clearing the canvas or removing a picture?

Yes. Both are single history steps, and **Undo** brings the pictures back. The history keeps up to 60 steps.

Does the tool know how big my collage will print?

No. A canvas has pixels and no physical size. Print size depends on the output density you print at, which the tool does not ask for. The reading file lists physical print size and output density under `notHeld`.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Collage Maker background material

The contents of the Collage Maker's others folder: glossary, shortcuts, limits, related pages, release notes and privacy.

This folder holds reference material that supports the how-to guides: the terms the tool uses, its keyboard and pointer actions, what it measures and what it does not, related tools and reading, its history, and what happens to your files.

File	What it covers
glossary.md	Definitions of the terms used in the Collage Maker
shortcuts.md	Keyboard shortcuts and pointer actions
limits-and-accuracy.md	What each figure measures, how precisely, and what the tool does not hold
related.md	Related tools on the site and Learn articles
release-notes.md	User-facing history of the tool
privacy.md	What stays in your browser and what uses the network

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Collage Maker shortcuts

Keyboard shortcuts and pointer actions in the Collage Maker.

Keyboard

Keys	Action	Works when
`Ctrl` + `Z`	Undo	Focus is not in a slider, menu or text box
`Ctrl` + `Shift` + `Z`	Redo	As above
`Ctrl` + `Y`	Redo	As above
Arrow keys	Nudge the selected picture 1 px	The canvas has focus (click it first)
`Shift` + arrow keys	Nudge the selected picture 10 px	The canvas has focus
`Delete` or `Backspace`	Remove the selected picture	The canvas has focus
`Enter` or `Space`	Open the file chooser	The **Drop pictures** box has focus

On a Mac, `Cmd` works in place of `Ctrl`. With the canvas focused, one press of the undo or redo shortcut can move two steps; use the **Undo** and **Redo** buttons when you want exactly one.

Every control on the page can be reached with `Tab`. Each tab button, slider, switch and button responds to the keyboard in the usual way.

Pointer

Action	Result
Click a picture	Select it (the top one where pictures overlap)
Click empty canvas	Clear the selection
Drag a picture	Move it; snapped to 10 px when snapping is on

Action	Result
Drag a corner handle	Resize the selected picture around its centre, from 2% to 800%
Click a row in **Layers · top first**	Select that picture, including a hidden one
Drop image files on the canvas or the drop box	Add them as pictures

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Collage Maker glossary

Definitions of the terms used in the Collage Maker.

Term	Meaning
Arrangement	A grid of equal cells, given as columns × rows, into which the visible pictures are placed in stacking order
Canvas	The rectangle the collage is drawn on; always 1200 pixels wide, with a height set by the shape
Canvas covered	The share of canvas pixels that pictures paint, found by drawing with and without them and counting differences
Cell	One rectangle of a grid arrangement, sized from the canvas, padding, gap and grid
Chequerboard	The on-screen pattern behind the canvas that shows transparency; never exported
Corner radius	How far the canvas corners are rounded, in pixels; exports are clipped to the rounded shape
Export scale	The factor, 0.25× to 4×, by which the canvas size is multiplied when an image is saved or copied
Fill	Scaling a picture so it covers its whole cell; in this tool the excess is not trimmed away
Fit	Scaling a picture so all of it lies inside its cell, leaving part of the cell empty if the shapes differ
Gap	The space between neighbouring cells of a grid arrangement
Guides	The thirds lines across the canvas and the dashed lines through the selected picture's centre; screen only
History step	One recorded change that **Undo** can reverse; the history holds up to 60
JPEG	A lossy image format that makes small files from photographs and cannot store transparency
Layer	One picture in the stack; the layers list shows them top first
Off-canvas area	The share of total picture area outside the canvas, which will not appear in an export
Opacity	How solid a picture is drawn, from 0% (invisible) to 100% (solid)
Overlap	The share of the canvas painted by two or more pictures
Padding	The margin between the canvas edge and the block of grid cells
PNG	A lossless image format that can store transparency
Reading	A small JSON file recording what was measured about the collage
Rule of thirds	A composition guide dividing the canvas into three equal parts across and down
Session file	A JSON file with the canvas settings and each picture's placement, but no image data
Snap	Rounding a dragged picture's centre to the nearest multiple of 10 px

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Limits and accuracy

What the Collage Maker measures, how precisely, and what it does not know.

What is measured, and how

Figure	Method	Precision
canvas covered	The canvas drawn with and without the pictures; differing pixels counted	Exact at full size; see resolution below
overlapping	Each picture drawn alone; pixels painted by two or more counted	As above
outside the canvas	Each picture's corners after scale and rotation, clipped to the canvas	Exact geometry, no pixels involved
Covered on the **Layout** tab	Placed area minus pairwise overlap	Exact unless three pictures meet at one point, when it shows a dash
Export file sizes	The encoded file, measured before it is offered	Exact for your browser

Coverage and overlap are read from the same drawing the export uses, so they describe the file you download at 1× scale.

Measurement resolution

Canvases up to 1,440,000 pixels are measured at full size. Larger canvases (the **9 : 16** and **2 : 3** shapes) are measured on a proportionally smaller copy; the label above the canvas then reads **measured at** followed by the size used. The resulting percentages agree with a full-size measurement to well under a tenth of a percentage point.

What counts as covered

A pixel counts as covered when any picture changes it. Fully transparent pixels in a picture and pictures at 0% opacity change nothing and are not counted. A picture at low opacity still changes the pixels it sits on, so it counts as covering them.

Choices the tool makes for you

These are defaults, not rules. Each is listed on the **Data** tab as **This tool's choice**.

Choice	Default	How to change it
Never enlarge a picture past 100% when arranging	On	**Enlarge a small picture to fill its cell** on the **Layout** tab
Snap step	10 px	The step is fixed; switch snapping off with **Snap to a 10 px grid**
JPEG quality	0.92	**JPEG quality** on the **Export** tab

The same JPEG quality setting produces slightly different files in different browsers, because each has its own encoder. The size shown on the **Export** tab is the size your browser actually produced.

Limits

- **Canvas sizes.** Six shapes, all 1200 pixels wide. Export scale multiplies them from 0.25× to 4×.
- **Background.** Transparent or white. There is no background colour control.
- **Formats.** PNG and JPEG images, a JSON session and a JSON reading. There is no PDF, SVG or layered file export.
- **Colour.** The canvas works in the browser's standard colour handling (sRGB). It is not a colour-managed publishing workflow, and exported files are 8 bits per channel.
- **Physical size.** A canvas has pixels and no physical size. The tool does not convert to inches or millimetres, because that depends on an output density it does not ask for.

- **Session files** hold the layout only. The pictures must be added again, in the same order, before a session is applied.
- **History** keeps up to 60 steps, and is lost when you reload or leave the page.
- **Animated images** are exported as a still.
- **Performance.** There is no fixed limit on the number of pictures, but every change is measured from pixels, so many large pictures make the page slower.

What the tool is not

The measurements say how much of the frame your pictures fill and where they overlap. They are not a judgement of composition or quality, and the tool gives no score.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Privacy

What the Collage Maker does with your pictures, and what stays in your browser.

Your pictures stay in your browser

- Pictures are read by the page itself and drawn straight to a canvas in your browser. The tool does not upload them or send them anywhere.
- The measurements, the arrangement comparison and the exports are all worked out on your device.
- Exported files (PNG, JPEG, session and reading) are created in the browser and saved by your browser's normal download.
- The clipboard is used only when you select **Copy to the clipboard**, and only to place the PNG there.

Nothing is kept after you leave

The tool does not store your pictures, your layout or your history. Reloading or leaving the page clears them. To keep a layout, save a session file; to keep the picture, save a PNG or JPEG.

What a session file contains

A session file contains the canvas settings and, for each picture, its file name, position, size, scale, rotation, opacity, flips and visibility. It contains no image data. If you share a session file, the recipient sees your picture file names.

The page itself

The Collage Maker is a normal page on auricartisan.com. Site-wide features such as sign-in, preferences and any consent-based services follow the site's own privacy policy and your cookie choices; they are covered in the site's help pages at Help.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Release notes

User-facing history of the Collage Maker.

Current version — the measured studio

The tool was rebuilt as the Collage Studio workbench, with six tabs: **Studio**, **Layout**, **Measure**, **Data**, **Export** and **Reference**. The developer reference that describes this version in the site's documentation library is dated 5 June 2026.

What changed for you:

- **Undo reaches the start.** Every step records the state before the change, so undoing takes you all the way back. A removed picture, or a cleared canvas, comes back with the step that removed it.
- **Exports contain only the collage.** Guides, the selection outline, corner handles and the transparency chequerboard are drawn only on screen. A PNG saved with **Transparent background** ticked is transparent.
- **Coverage is measured.** The single coverage figure, which added up picture areas, is replaced by **canvas covered** and **overlapping** read from pixels, and **outside the canvas** computed from positions. The old figure stays on screen as **the old readout said** for comparison.
- **The Layout tab** lays out and measures every grid arrangement before you apply one, and marks a winner only when one truly covers more.
- **Two features were withdrawn:** a "95% confidence interval" for coverage and a batch table of hypothetical layouts. Neither measured the collage on the canvas. The question the batch table was aiming at, which grid gives these pictures the most canvas, is now the **Layout** tab.
- **Export sizes are measured.** The **Export** tab shows the real size of the PNG, JPEG and session file for your collage.
- **The reading.** A small JSON file records what was measured, to keep with the image.

Some features described in the earlier user guide of 25 May 2026 are not part of the current tool, including a background colour picker, a snap step setting, per-layer lock and rename, and a full-screen preview.

25 May 2026 — earlier version

The Collage Maker Lab user guide was published in the site's documentation library. It describes the earlier version of the tool.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › COLLAGE MAKER

Related tools and reading

Other Auric Artisan tools and Learn articles that work well with the Collage Maker.

Tools on the site

Tool	Why you might use it with the Collage Maker
Image Picker	Take a palette from a finished collage exported as PNG or JPEG
SVG Recolour	Recolour vector artwork before exporting it as a picture for a collage
LUT Lab	Check what a colour grade does to your photographs
Basic Tools	The Basic Tools workspace includes its own quick collage panel
Vision Simulation	See how the finished collage looks with different kinds of colour vision

Learn articles

- Color Theory and Harmony — https://auricartisan.com/library/learn/articles/2026-06-04-color-theory-and-harmony

- RGB, sRGB, Adobe RGB, ProPhoto, Display P3, and Rec.2020 — https://auricartisan.com/library/learn/articles/2026-05-25-rgb-working-spaces-compared
- Skin Tones and Inclusive Color — https://auricartisan.com/library/learn/articles/2026-07-18-skin-tones-and-inclusive-color

Other documentation

- The group overview: Image and file tools.
- The site's documentation library carries a developer reference for the collage measurements, "Collage Measurement Developer Reference", at https://auricartisan.com/library/documentation/articles/2026-06-05-collage-measurement-developer-reference. It is written for people maintaining the tool; this guide covers everything you need to use it.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

ICC Profile Parser

What the ICC Profile Parser does, who it is for, where to find it and how to read a first profile.

The ICC Profile Parser (the page is titled **ICC Profile Reading**) opens a colour profile file, `.icc` or `.icm`, and tells you what it says. An ICC profile is a small file, defined by the International Color Consortium, that describes how a device or colour space reproduces colour, so that software can convert colours between devices. The tool reads the profile's header at the byte positions the ICC specification (ICC.1:2010, also published as ISO 15076-1:2010) sets out, lists every tag in the file, and runs a conformance check in which every finding names the clause of the specification it enforces.

For RGB display-style profiles, it recovers the primaries behind the stored colorants, draws them on a CIE 1931 chromaticity diagram, plots the tone curves, and identifies which of six well-known colour spaces the profile matches most closely: sRGB, Rec. 709, Display P3, Adobe RGB (1998), Rec. 2020 or ProPhoto RGB. It reports gamut area and coverage against the space you choose, using the more even CIE 1976 u′v′ diagram. For press and other profiles with no colorants, it walks the profile's lookup table to draw the gamut in CIELAB instead.

The file is read in the page and never sent anywhere. You can save the whole reading as JSON, copy it, or save the tag table as CSV.

Who it is for

- Designers and photographers checking what a monitor, camera or printer profile actually contains.
- Colour-management and prepress staff triaging a profile that an application rejects.
- Developers who need the header fields, the tag table and a clause-by-clause verdict to paste into a ticket.
- Students learning how ICC profiles are put together.

Where to find it

Open https://auricartisan.com/tool/general/tools/icc-profile-parser/ in a current browser. It is listed as **ICC Parser** on the Collections page at https://auricartisan.com/collections/. Nothing needs installing.

Quick start

1. Open the tool. The **Lab** tab is selected.
2. Drop an `.icc` or `.icm` file on the **Drop a profile** box, or select the box to choose one.
3. Read the verdict under **What this profile is**, for example "An sRGB profile, matrix/TRC, ICC v2.1.0".
4. Check **What the header says** on the left for the version, class, data space, connection space and more.
5. Look at the **conformance** card and the **Findings** list for anything that breaks the specification.
6. Select **Save the reading** to download the full reading as JSON.

What you can do

- Read every header field: size, version, device class, data colour space and channel count, connection space, rendering intent, CMM, platform, creation date, tag count and profile ID.
- See every tag with its type, offset and size, and which tags share one data element.
- Recompute the profile ID and see whether it matches the bytes.
- Identify the closest of six reference colour spaces, with primaries distance, tone-curve fit and the adaptation the profile was built with.

- Compare gamut area and coverage against any of the six spaces in CIE 1976 u′v′, with the CIE 1931 xy figure alongside.
- Plot stored colorants, recovered primaries and the reference space on a chromaticity diagram, and the tone curves against the reference transfer function.
- Draw the CIELAB *ab* gamut of a CMYK or other LUT-based profile from its A2B table.
- Run 23 conformance rules, each graded error, warning or note and citing its clause.
- Export the reading as JSON and the tag table as CSV.

In this folder

- docs/ — how to use the tool.
  - Getting started — open it, the screen tour, the first profile.
  - Understand ICC profiles — classes, tags, the connection space, versions, intents and curves.
  - Identify a profile and read its gamut — the verdict, the ranking, the diagram, area and coverage, tone curves.
  - Check conformance — findings, severities, clauses and the rule list.
  - Export a reading — JSON, clipboard and the tag table CSV.
  - Reference — every field, tag, value, rule and file format.
  - Troubleshooting — symptoms, causes and fixes.
  - FAQ — short answers to common questions.
- others/ — background material.
  - Glossary — terms used in this tool.
  - Limits and accuracy — what it can and cannot read, and how precise it is.
  - Related — related tools and Learn articles.
  - Release notes — user-facing history.
  - Privacy — what stays in your browser.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

ICC Profile Parser documentation

The list of guides, the reference, troubleshooting and FAQ for the ICC Profile Parser.

This folder holds the task guides for the ICC Profile Parser, a concept page on how ICC profiles are organised, the full reference of every field and rule the tool shows, and the troubleshooting and FAQ pages. If ICC profiles are new to you, read Understand ICC profiles before the other guides.

File	What it covers
getting-started.md	Opening the tool, a tour of the six tabs, and reading a first profile
understand-icc-profiles.md	What a profile is: classes, colour spaces, the connection space, tags, versions, intents, curves and tables
identify-and-gamut.md	The verdict, the ranking table, the chromaticity diagram, gamut area and coverage, and the tone curves
check-conformance.md	Reading findings, what error, warning and note mean, and the Conformance tab
export-a-reading.md	Saving and copying the JSON reading and the tag table CSV
reference.md	Every header field, value table, tag type, rule, readout and export format
troubleshooting.md	What to do when a file will not read or a panel stays empty
faq.md	Short answers to common questions

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

Getting started with the ICC Profile Parser

Open the ICC Profile Parser, find your way around its tabs, and read a first profile.

Open the tool

Go to https://auricartisan.com/tool/general/tools/icc-profile-parser/. The page opens on the **Lab** tab. Until you load a file, the main panel reads **Nothing loaded yet** and explains what the tool will and will not tell you.

A note under the heading explains what the numbers on the page are measurements of. It has no close button on purpose.

Find a profile to read

An ICC profile is a file ending in `.icc` or `.icm` that describes how a device or colour space reproduces colour. Your computer already has some:

- **Windows:** `C:\Windows\System32\spool\drivers\color\`
- **macOS:** `/Library/ColorSync/Profiles/`, `~/Library/ColorSync/Profiles/` and `/System/Library/ColorSync/Profiles/`

Monitor calibration software, printer drivers and paper manufacturers also supply profiles.

The tool reads standalone profile files. A profile embedded inside a JPEG, PNG, TIFF or PDF must first be saved out as its own file by another application.

The screen at a glance

Tab	What it is for
Lab	Load a profile; read the header, verdict, diagrams, figures, findings and tag table
Method	How each figure is derived, in five sections
Conformance	The rules the check runs, filtered by severity, with the clause each one enforces
Data	Every figure and what it rests on; the tab shows the number of entries (20)
Export	What the JSON reading contains, and a preview of it
Reference	The published sources, marked by whether they run in the tool

The Lab tab

- **The left-hand panel** holds **The profile** (the drop box), **What the header says** (the header fields), **Compare against** (a menu of six reference spaces), and **Take it away** (four buttons: **Save the reading**, **Copy it**, **Tag table, CSV** and **Clear**).
- **The right-hand panel** shows, once a profile is loaded, **What this profile is**: the verdict, the ranking table **How close, and to what**, the chromaticity diagram and tone-curve plot, six figure cards, the **Findings** list and **The tag table**.

Read a first profile

1. Drop a profile on the **Drop a profile** box, or select the box (or press `Enter` or `Space` while it has focus) to open a file chooser. The chooser offers `.icc` and `.icm` files.
2. A message confirms **Read** followed by the file name.
3. Read the verdict at the top of the right-hand panel. For a standard sRGB profile it reads "An sRGB profile, matrix/TRC, ICC v2.1.0", followed by how far its colorants sit from sRGB's primaries, how well its tone curve fits, and whether it is conformant.
4. In **What the header says**, check **version**, **class**, **data space**, **PCS** and **intent**. The **profile ID** line says **not carried**, **matches** or **does NOT match**.
5. Look at the cards under the diagrams. **conformance** reads **clean** when no rule raised anything, **no errors** when only warnings or notes were raised, or the number of errors.
6. Scroll to **Findings** to see any rule the profile does not meet, each with its severity and clause.
7. Scroll to **The tag table** to see every tag the file contains.
8. Select **Save the reading** to download the full reading as a JSON file.

You should now know what kind of profile it is, which colour space it describes, and whether anything in it breaks the specification.

Next steps

- If the terms on the page are unfamiliar, read Understand ICC profiles.
- To interpret the ranking, the diagram and the area figures, read Identify a profile and read its gamut.
- To understand a finding, read Check conformance.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

Check conformance

Read the conformance verdict and findings, understand error, warning and note, and use the Conformance tab to see every rule and the clause it enforces.

Every profile you load is checked against 23 rules. Each rule either holds or raises a finding. Every rule cites the clause of ICC.1:2010 it enforces, except two that are marked as house preferences because the specification does not require them.

Read the result

1. Load a profile on the **Lab** tab.
2. Look at the **conformance** card under the diagrams:
   - **clean** — no rule raised anything;
   - **no errors** — only warnings or notes were raised; the profile is still conformant;
   - **1 error**, **2 errors** and so on — the profile breaks that many rules graded as errors. The card turns red.
3. Read the last sentence of the verdict. It says either "It is conformant: N of 23 rules pass", adding how many findings are below the level of an error, or how many rules the profile should satisfy do not hold.
4. Scroll to **Findings**. When every rule holds you see a single **Pass** line, **All 23 rules hold**. Otherwise there is one row per finding.

A profile is **conformant** when no rule graded as an error is broken. There is no score out of 100: a profile is conformant or it is not, and where it is not the clause says why.

Read a finding

Each finding row shows:

Part	Example	Meaning
Severity	**warning**	**error**, **warning** or **note**
Title	**The profile ID matches the bytes**	The rule that did not hold
Clause	**§ 7.2.18**	The clause of ICC.1:2010 it enforces, or **house preference**
Detail	"stored 0f3e…, recomputed 8a21…"	What the tool found in your file

What the severities mean

Severity	Meaning
error	The specification says this shall not happen. Software may refuse the profile or misread it
warning	Something the specification requires or expects is missing or wrong, but the profile can usually still be used
note	A detail worth knowing, such as a creation date that is not a real date

The grade given to each rule is the tool's own judgement; the specification says what shall be true but does not grade consequences. Because every finding names its clause, you can check it and disagree with the grade without disagreeing with the finding.

The rules

Rule	Severity	Clause
Profile file signature (bytes 36 to 39 read `acsp`)	error	7.2.5

Rule	Severity	Clause
Declared size matches the file	error	7.2.2
Major version is one this reader knows (2 or 4)	warning	7.2.4
Version reserved bytes are zero	note	7.2.4
Device class is one the specification defines	error	7.2.5
PCS is XYZ or Lab (DeviceLink profiles are exempt)	error	7.2.7
PCS illuminant is D50	warning	7.2.16
Rendering intent is 0-3	error	7.2.15
Creation date is a real date	note	7.2.8
Every tag payload lies inside the file	error	7.3.1
Tag payloads begin on a 4-byte boundary	warning	7.3.1
No tag payload partly overlaps another	error	7.3.4
Tag table fits in the file	error	7.3.1
No tag signature appears twice	error	7.3.1
Required tags for this device class are present	error	8.2
A v4 profile carries a profile ID	warning	7.2.18
The profile ID matches the bytes	warning	7.2.18
An RGB matrix profile carries all three colorants and curves	warning	8.3
Media white point is the PCS white (display profiles)	warning	9.2.36
Colorants sum to the media white point	note	house preference
Every tag payload could be decoded	warning	7.3.1
A LUT tag holds the grid its header declares	error	10.11
Padding between elements is zero	note	house preference

In total: 11 errors, 8 warnings and 4 notes.

The required tags

The **Required tags for this device class are present** rule checks these tags:

Profile class	Required tags
Display or ColorSpace, Grayscale data	`desc`, `cprt`, `wtpt`, `kTRC`
Display or ColorSpace, any other data space	`desc`, `cprt`, `wtpt`
Output or Input	`desc`, `cprt`, `wtpt`
DeviceLink	`desc`, `cprt`, `pseq`, `A2B0`
NamedColor	`desc`, `cprt`, `ncl2`, and `wtpt` in version 4
Abstract	`desc`, `cprt`, and `wtpt` in version 4

Shared tags are not an error

Two tags pointing at the same offset with the same size are sharing one data element, which clause 7.3.4 allows. The tag table marks them **shares this element with** followed by the other tag. Only a block that partly overlaps another, without being the same block, is an error.

Tables that are too short

A truncated file can leave a lookup table with fewer values than its header promises. Rather than failing, the tool reads what is there, raises **A LUT tag holds the grid its header declares** with how many values are missing, and reads the rest of the profile as normal. A tag it cannot decode at all raises **Every tag payload could be decoded**.

The Conformance tab

The **Conformance** tab lists the rules independently of any loaded profile.

- The **Rules** buttons filter by severity: **All**, **Errors**, **Warnings** and **Notes**, each with its count. The note under them says how many rules are shown and how many cite no clause.
- **What counts as a defect** explains two checks that are no longer made, because they fired on correct files: one that reported every profile's version as unusual, and one that treated shared tag data as corrupt.
- **The rules that stayed, and what each cites** lists every rule with its severity and clause.
- **What the old validator got right** lists the checks kept unchanged: the tag table walk, alignment and bounds; the required-tag table; the declared size against the file; and the classification as matrix/TRC, LUT or both.

What a pass means

A clean result means the file is well formed and says what the specification requires. It does not mean the device the profile describes still behaves that way. See Limits and accuracy.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

Export a reading

Save or copy the full JSON reading of a profile, save its tag table as CSV, and see what each export contains.

A reading is the tool's full account of a profile as a JSON file: the header with the clause behind each field, the tag table, the identification, the gamut figures and every conformance finding. Every derived figure carries the clause or the method it came from, so someone else can check it without opening this page.

Save the reading

1. Load a profile on the **Lab** tab.
2. Under **Take it away**, select **Save the reading**.
3. Your browser downloads a file named after the profile, for example `srgb-color-space-profile-reading.json` for `sRGB Color Space Profile.icm`. The name is the file name in lower case, with the `.icc` or `.icm` ending removed and anything other than letters and digits turned into hyphens.

If no profile is loaded, the message reads **Load a profile first** and nothing is saved.

Copy the reading

Select **Copy it** to place the same JSON on the clipboard, ready to paste into a ticket or a message. The message reads **Copied the reading**, or **The clipboard refused** if your browser blocks clipboard access.

Save the tag table as CSV

Select **Tag table, CSV** to download a spreadsheet-ready list of every tag, named like the reading but ending `-tags.csv`.

```
signature,type,offset,size,aligned,in_bounds,shares_with
desc,desc,240,108,yes,yes,
A2B0,mft2,576,41094,yes,yes,A2B2
```

Column	Meaning
`signature`	The tag's four-character signature
`type`	The four-character type at the start of its data
`offset`	Where the data starts, in bytes from the start of the file
`size`	The length of the data in bytes
`aligned`	`yes` if the offset is a multiple of 4
`in_bounds`	`yes` if the data lies inside the file
`shares_with`	Other tags pointing at the same data, separated by spaces

The rows in the example are illustrative.

Clear

Clear removes the loaded profile from the page and returns the **Lab** tab to **Nothing loaded yet**. The message reads **Cleared**.

Preview the reading on the Export tab

The **Export** tab shows what the reading will contain before you save it.

- On the left, **What to take** lists the parts: the reading as JSON, the header with clauses, the tag table, and the conformance rules and findings, followed by the **size** of the reading in KB and its **file name**.
- On the right, **What the reading says about itself** explains the provenance, and **The reading, as it will be written** shows the JSON. Long readings are cut after 58 lines, with a count of the lines not shown; the saved file is always complete.

What the reading contains

Section	Contents
`tool`, `specification`, `generated`	The tool name, the specification (ISO 15076-1:2010), and the date and time of the reading
`file`	The file name and its size in bytes
`header`	Version, device class, data space, PCS, rendering intent, PCS illuminant, creation date and profile ID, each with its clause

Section	Contents
`tags`	Every tag: signature, type, offset, size, alignment, whether it is in bounds, and any tags it shares data with
`identity`	For RGB profiles with colorants: the closest space, primaries distance, curve fit, adaptation, and the full ranking
`gamut`	For RGB profiles: the space compared against, stored colorants, recovered primaries, area ratios in u′v′ and xy, and coverage
`gamut` (no colorants)	For profiles walked from an A2B table: the table, grid points, *ab* hull area, lightness range and encoding
`conformance`	Number of rules, number passed, whether conformant, and every finding with severity, clause, title and detail
`register`, `registerCounts`	The status and source of every entry on the **Data** tab

For a file that could not be read, the reading holds the file details and the error message.

The `profileId` entry holds the stored ID, the recomputed ID, and `matches`: `true`, `false`, or `null` when the profile carries no ID. The `identity` entry records that the white point does not contribute to the comparison. The `gamut` and `conformance` entries each carry a short note explaining their figures.

The reading uses the space currently chosen under **Compare against** for the gamut figures. Choose the space you want before saving.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

Identify a profile and read its gamut

Read the verdict and ranking, choose a reference space, and interpret the chromaticity diagram, gamut area, coverage and tone curves.

After you load a profile on the **Lab** tab, the right-hand panel opens with **What this profile is**. This guide explains each part of it.

The verdict

The verdict is a heading and a few sentences.

Profile	Heading reads, for example
Has RGB colorants	"An *sRGB* profile, matrix/TRC, ICC v2.1.0"
No colorants, but a readable A2B table	"A *CMYK* output profile, ICC v2.0.0"
Anything else	"An ICC v4.3.0 display profile"

The sentences under the heading say:

- how far the colorants sit from the closest space's primaries, in CIE 1976 u′v′, and which of two adaptation methods, Bradford or XYZ scaling, fits the profile better;
- how closely the tone curve fits that space's published transfer function, as a peak difference;
- for a profile with no colorants, which table its gamut was read from, how many grid points it has and the range of CIELAB lightness (L*) it covers;
- whether it is conformant: "It is conformant: 21 of 23 rules pass, with 2 below the level of an error", or how many rules it should satisfy do not hold.

Two extra lines can appear:

- **"This profile cannot be told apart from Rec. 709 here"** appears when two reference spaces share the same primaries and white, and the profile has no tone curve to separate them. sRGB and Rec. 709 are the usual pair.
- **"Separated from Rec. 709 by the tone curve, not by the primaries"** appears when the primaries tie and the tone curve decides.

The ranking table

How close, and to what appears for profiles with RGB colorants. It ranks all six reference spaces, closest first; the first row is highlighted.

Column	Meaning
Space	The reference space
Primaries Δu′v′	The largest of the three distances between the profile's colorants and the space's primaries, in CIE 1976 u′v′
Curve fit	The largest difference between the profile's tone curve and the space's transfer function, on a 0 to 1 scale
White	Always **matched**, because the comparison is made at one white (see below)
Verdict	**this profile**, **same primaries, different curve**, **same curve, different primaries**, or a dash

Rows are ordered by the primaries distance, and by the curve fit when two spaces tie. The note above the table reads **adapted into the PCS ·** followed by **Bradford** or **XYZ scaling**, the adaptation that fits the profile better.

Why the white point is not a term

A conformant display profile stores its white as D50, the connection-space white, while five of the six reference spaces are defined at D65. Comparing whites directly would penalise every correct profile by the same amount. So the tool first adapts each reference space's primaries into the connection space, and only then compares them with the profile's colorants. Both sides are then at the same white, and the white cannot add a difference.

A distance in u′v′

The CIE 1976 u′v′ diagram is a version of the CIE 1931 chromaticity diagram stretched so that equal distances look more nearly equal. The smaller the **Primaries Δu′v′**, the closer the match; compare the first row with the second to see how clearly the winner stands out.

Choose the space to compare against

The **Compare against** menu on the left lists the six reference spaces: **sRGB**, **Rec. 709**, **Display P3**, **Adobe RGB (1998)**, **Rec. 2020** and **ProPhoto RGB**. The choice sets:

- the white dashed triangle on the chromaticity diagram;
- the dashed reference curve on the tone-response plot;
- the **area, u′v′**, **covers of it** and **area, xy** cards.

When a profile loads, the comparison switches to the closest space. The label of the menu may still show your previous choice. The first row of the ranking table names the closest space; to be certain which space the area figures refer to, choose it from the menu yourself.

Closest means closest among these six. A profile that describes a space outside the list is still ranked against them, and the distances in the table show how poor even the best match is.

The chromaticity diagram

For a profile with RGB colorants, the left plot is headed **Chromaticity · CIE 1931 xy** and marked **2° observer**. The horseshoe outline is the spectral locus, the chromaticities of pure spectral light from 380 to 700 nm. Inside it:

Mark	What it is
Blue dashed triangle	The colorants as stored in the profile, adapted to D50
Gold solid triangle, lightly filled	The primaries recovered from those colorants
White dashed triangle	The space you are comparing against, adapted into the connection space
White dot	The profile's media white point

The caption under the plots says how the primaries were recovered: with the profile's own `chad` tag, or with the Bradford transform from an assumed white because the profile carries no `chad` tag.

The tone-response plot

The right plot, **Tone response**, marked **device → linear**, draws each tone curve from input value 0 to 1 against linear light 0 to 1: red, green and blue curves, a gold grey curve for `kTRC`, and a dashed curve for the transfer function of the space you are comparing against. A curve that lies on the dashed line matches that space's published transfer function.

The reference curves are: the piecewise sRGB function for sRGB and Display P3; a 2.4 power for Rec. 709; the Adobe RGB power of 563/256 (about 2.2); the Rec. 2020 function; and the ProPhoto (ROMM) 1.8 power with a short straight section near black.

The plot is shown only for profiles with RGB colorants.

Profiles with no colorants

A CMYK press profile, and other LUT-based profiles, carry no colorants and no tone curves. For these, when the profile has an `A2B1` or `A2B0` table in the lut8 or lut16 format, the tool reads every grid point of that table, converts it to CIELAB, and draws the outline of the result.

- The plot is headed **Gamut · CIELAB a*b* hull**, with the a* axis across and the b* axis up, and rings every 20 units.
- The tone-response plot is hidden.
- The caption names the table and the number of grid points, for example 6,561 for a table of 9 × 9 × 9 × 9.
- The area cards show a dash, because area in u′v′ applies only to RGB triangles.

No gamut is drawn when the table has more than 200,000 grid points, is incomplete, does not have three output channels, or uses one of the version 4 table formats (`mAB` or `mpet`).

The figure cards

Card	Meaning
version	The ICC version, major.minor.bug-fix
conformance	**clean** (nothing raised), **no errors** (only warnings or notes), or the number of errors
tags	How many tags the tag table lists
area, u′v′	The area of the profile's primaries triangle as a percentage of the reference space's triangle, in CIE 1976 u′v′

Card	Meaning
covers of it	The share of the reference space's triangle that lies inside the profile's triangle
area, xy	The same area ratio computed in CIE 1931 xy, shown for comparison

Area is not coverage

area, u’v’ answers "how big is this gamut compared with that one?" It can exceed 100%: Adobe RGB is larger than sRGB. **covers of it** answers "how much of that space can this profile reproduce?" It is a real overlap and cannot exceed 100%. A figure such as "136% sRGB" in marketing material is an area ratio, not coverage.

Why u’v’ rather than xy

The CIE 1931 xy diagram stretches the greens and squeezes the blues and reds, so areas measured on it do not match what people see. The **area, xy** card is kept because it is the figure much of the industry quotes. Measured across the reference spaces, xy overstates Adobe RGB by 18.2 percentage points and Rec. 2020 by 16.8, and understates ProPhoto RGB by 14.4, compared with u’v’.

The Method tab

The **Method** tab explains each step in five sections, chosen with the buttons under **The chain**:

Section	What it shows
The header	Bytes 8 to 11 of the loaded profile, labelled, and how they make the version number
Colorant vs primary	How far sRGB's stored colorants sit from its primaries, and why both adaptation methods are tested
Identification	Why the white point is taken out of the comparison, and the order of the steps
Gamut area	Every reference space's area as a percentage of sRGB in xy and in u’v’, and its coverage of sRGB
A profile with no colorants	How the gamut is read from an A2B table; with such a profile loaded, its figures

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

Understand ICC profiles

The concepts you need to read an ICC profile: classes, colour spaces, the connection space, tags, versions, intents, colorants, curves and tables.

This page explains the parts of an ICC profile that the ICC Profile Parser shows you. It is a short primer, not a replacement for the specification, ICC.1:2010 (also published as ISO 15076-1:2010).

What a profile is for

A colour management system converts colours from one device to another, for example from a camera to a monitor, or from a monitor to a printing press. It does this in two halves. Each device has a profile that converts between the device's own colour values and a shared, device-independent space called the **profile connection space** (PCS). Converting from device A to device B means going from A into the PCS with A's profile, then out of the PCS with B's profile.

An ICC profile is the file that holds one of those halves. It has three parts:

1. a 128-byte **header** that says what kind of profile it is;
2. a **tag table** that lists the pieces of data inside;

3. the **tag data** itself: curves, matrices, tables and text.

The profile connection space

The PCS is either CIE XYZ or CIELAB, always relative to the D50 illuminant, a standard daylight white at about 5000 K. The header's **PCS** field says which. D50 is fixed by the specification, which is why the tool checks that the header's PCS illuminant is D50 and that a display profile's media white point is the PCS white.

Device classes

The header's class field says what the profile describes. The tool shows the four-character code and its name.

Code	Class	What it describes
`scnr`	Input	A capture device such as a scanner or camera
`mntr`	Display	A monitor or other display
`prtr`	Output	A printer or press
`link`	DeviceLink	A direct conversion from one device to another, with no PCS in between
`spac`	ColorSpace	Conversion between the PCS and a standard colour space
`abst`	Abstract	An effect applied within the PCS, such as a look
`nmcl`	NamedColor	A list of named colours, such as a spot-colour book

Data colour space and channels

The **data space** is the colour space of the device values: RGB, CMYK, Grayscale, CIELAB, or a multi-ink space such as `6CLR` (six colours). The tool shows the channel count beside it, for example **RGB · 3 ch** or **CMYK · 4 ch**.

Versions

Two major versions are in use: version 2 (for example 2.1.0 or 2.4.0) and version 4 (for example 4.3.0). The version is stored in bytes 8 and 9 of the header: byte 8 is the major version, and byte 9 holds the minor version and the bug-fix level. The tool shows it as major.minor.bug-fix. In version 4, every profile must carry the media white point tag and should carry a profile ID. In version 2, the white point is required for most classes but not all, and the profile ID is optional.

Rendering intents

A rendering intent is the strategy for handling colours one device can show and another cannot. The header records one of these four values.

Value	Intent	In short
0	Perceptual	Compresses the whole gamut so relationships between colours are kept
1	Media-relative colorimetric	Keeps in-gamut colours exact relative to the paper or display white; clips the rest
2	Saturation	Keeps colours vivid, for charts and graphics
3	ICC-absolute colorimetric	Keeps in-gamut colours exact, including the white of the medium

Tags

Every piece of data in a profile is a **tag**: a four-character signature, such as `wtpt` or `rTRC`, that points to a block of data at an offset in the file. The block starts with its own four-character **type**, such as `XYZ` or `curv`, which says how to read it. The tool's tag table shows the signature, type, offset and size of every tag.

The specification allows two tags to point to the same block of data. Press profiles use this to store an identical table once, for example when the perceptual and saturation tables are the same. The tool reports this as sharing, not as a fault. What is a fault is a block that partly overlaps another without being the same block.

Tags you will often see

Signature	Name	What it holds
`desc`	Profile description	The profile's name as software shows it
`cprt`	Copyright	Copyright text
`wtpt`	Media white point	The white of the medium, as XYZ
`rXYZ`, `gXYZ`, `bXYZ`	Red, green, blue colorants	The XYZ of each primary, adapted to D50
`rTRC`, `gTRC`, `bTRC`	Red, green, blue tone curves	How each channel's values map to light
`kTRC`	Grey tone curve	The tone curve of a greyscale profile
`chad`	Chromatic adaptation	The 3×3 matrix used to adapt the colorants to D50
`A2B0`, `A2B1`, `A2B2`	Device-to-PCS tables	Perceptual, colorimetric and saturation conversions into the PCS
`B2A0`, `B2A1`, `B2A2`	PCS-to-device tables	The same three intents, out of the PCS
`gamt`	Gamut	A table that says whether a PCS colour is in the device gamut
`pseq`	Profile sequence	For a DeviceLink, the profiles it was built from
`ncl2`	Named colour	The named colours of a NamedColor profile

Many display profiles made by calibration software also carry `vcgt`, a private tag that is not part of ICC.1:2010. The tool lists it in the tag table but does not read it.

Two ways to describe a device

Matrix/TRC profiles describe an RGB device with three colorants and three tone response curves (TRCs). Most display and RGB working-space profiles, including sRGB, are built this way. The tool's verdict calls these "matrix/TRC".

LUT-based profiles describe a device with lookup tables in the `A2B` and `B2A` tags. Every CMYK press profile is built this way, and some RGB profiles are too. The tool's verdict calls an RGB profile without colorants "LUT".

Colorants are not primaries

The colorants in `rXYZ`, `gXYZ` and `bXYZ` are stored after adaptation to the D50 connection space. They are therefore not the chromaticities of the colour space's primaries. For sRGB, the stored green colorant sits about 0.021 away in the CIE 1931 xy diagram from sRGB's published green primary.

To recover the real primaries, the tool reverses the adaptation. When the profile carries a `chad` tag, the tool uses the inverse of that matrix, which recovers them exactly. When it does not, the tool assumes the adaptation was the Bradford transform

from the white of the space being compared, and says so. Bradford is the chromatic adaptation method the `chad` tag conventionally carries; some older profiles instead scale XYZ directly, and the tool tests both and reports which fits better.

Tone response curves

A TRC maps a channel's device value (0 to 1) to linear light (0 to 1). The specification allows several forms, all of which the tool reads:

- **Identity**: output equals input.
- **Gamma**: a single power, for example 2.2.
- **Table**: a list of sampled values, read with straight-line interpolation between them.
- **Parametric**: one of five formulas with up to seven parameters, which can express a gamma with a linear segment near black, as sRGB uses.

The profile ID

Bytes 84 to 99 of the header may hold a profile ID: an MD5 checksum of the whole file, calculated with the flags, rendering intent and ID fields set to zero. If the ID is present and matches, the file has not been changed since the ID was written. Version 4 profiles should carry one; in version 2 the field is usually all zeros, which the tool reports as **not carried**.

What a profile cannot tell you

A profile describes a device as it was when the profile was made. It is not the device. A perfectly formed profile can describe a monitor that has drifted since it was calibrated. Checking the device itself needs a measuring instrument; see Limits and accuracy.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

ICC Profile Parser reference

Every tab, control, header field, value table, tag type, readout, rule and export format in the ICC Profile Parser.

Address: https://auricartisan.com/tool/general/tools/icc-profile-parser/

Tabs

Tab	Contents
Lab	The drop box, header fields, **Compare against**, export buttons, and the full reading of the loaded profile
Method	Five sections explaining how each figure is derived
Conformance	All 23 rules, filterable by severity, with clauses and the two withdrawn checks
Data	The 20 entries that every figure rests on, filterable by status
Export	The contents, size and file name of the reading, and a preview
Reference	The nine published sources, filterable by whether they run in the tool

Lab tab — controls

Control	What it does	Values
Drop a profile	Loads a profile by drag and drop, or opens a chooser when selected or when `Enter` or `Space` is pressed	`.icc`, `.icm` or `application/vnd.iccprofile` in the chooser
Compare against	Chooses the reference space for the diagram, the tone curve plot and the area cards	sRGB, Rec. 709, Display P3, Adobe RGB (1998), Rec. 2020, ProPhoto RGB
Save the reading	Downloads the reading as JSON	—
Copy it	Copies the reading to the clipboard	—
Tag table, CSV	Downloads the tag table as CSV	—
Clear	Removes the loaded profile	—

Only one profile is loaded at a time. Loading another replaces it.

Lab tab — What the header says

Field	Shows	Example
file	The file name	`display.icc`
size	The size declared in the header, in bytes	3,048 bytes
version	Major.minor.bug-fix, from bytes 8 and 9	2.1.0
class	Four-character code and class name	mntr · display
data space	Data colour space and channel count	RGB · 3 ch
PCS	Profile connection space	nCIEXYZ
intent	Rendering intent number and name	0 · perceptual
CMM	Preferred colour management module signature, or none declared	—
platform	Primary platform	Microsoft
created	Creation date and time, or **not a valid date**	2024-03-18 09:30:00
tags	Number of tags	17
profile ID	**not carried**, **matches** or **does NOT match**	not carried

If the file cannot be read, the panel shows the file name and **status: unreadable**. Example values are illustrative.

Value tables

Device classes

Code	Shown as
`scnr`	Input
`mntr`	Display

Code	Shown as
`prtr`	Output
`link`	DeviceLink
`spac`	ColorSpace
`abst`	Abstract
`nmcl`	NamedColor

Data colour spaces and the PCS

Code	Shown as	Channels
`XYZ`	nCIEXYZ	3
`Lab`	CIELAB	3
`Luv`	CIELUV	3
`YCbr`	YCbCr	3
`Yxy`	CIEYxy	3
`RGB`	RGB	3
`GRAY`	Grayscale	1
`HSV`	HSV	3
`HLS`	HLS	3
`CMYK`	CMYK	4
`CMY`	CMY	3
`2CLR` to `FCLR`	2 colour to 15 colour	2 to 15

Rendering intents

Value	Shown as
0	Perceptual
1	Media-relative colorimetric
2	Saturation
3	ICC-absolute colorimetric

Platforms

Code	Shown as
`APPL`	Apple
`MSFT`	Microsoft
`SGI`	Silicon Graphics
`SUNW`	Sun Microsystems
All zeros	none declared

Tag types the tool reads

Type	What it is	How the tool uses it
`text`	Plain text	Read
`desc`	Version 2 description text	Read
`mluc`	Multi-language text (version 4)	Read, every language record
`XYZ`	One or more XYZ values	Colorants and white point
`curv`	Tone curve: identity, gamma or table	Tone curves and curve fit
`para`	Parametric tone curve, five function types	Tone curves and curve fit
`sf32`	Signed fixed-point array	The `chad` matrix
`sig`	A signature	Read
`mft1`	lut8 table	Gamut hull for profiles without colorants; completeness check
`mft2`	lut16 table	As above
`mpet`	Multi-process elements (version 4)	Structure only: element count and types; not evaluated
`ncl2`	Named colour list	Structure only: count and up to 64 names; no colour resolved

Any other type is listed in the tag table with its type code and is not decoded. The page does not print the text of description or copyright tags.

Lab tab — the reading

Element	Shown when	Contents
Verdict	A profile is loaded	Heading, distances, gamut source, conformance sentence, optional aside
How close, and to what	The profile has RGB colorants	Ranking of six spaces: **Space**, **Primaries Δu'v'**, **Curve fit**, **White**, **Verdict**
Chromaticity diagram	The profile has RGB colorants	CIE 1931 xy, 2° observer: stored colorants, primaries, reference space, white point
Gamut hull	No colorants, with a readable A2B table	CIELAB *ab* hull, rings every 20
Tone response	The profile has RGB colorants	Red, green, blue and grey curves against the reference transfer function
Figure cards	A profile is loaded	**version**, **conformance**, **tags**, **area, u'v'**, **covers of it**, **area, xy**
Findings	A profile is loaded	One row per finding, or **All 23 rules hold**
The tag table	A profile is loaded	**Tag**, **Type**, **Offset**, **Size**, **Anything notable**

The note beside **Findings** gives the tag count and how many tags share an element. The **Anything notable** column shows one of: **outside the file**, **could not be decoded**, **shares this element with** another tag, **not 4-byte aligned**, or a dash.

Reference spaces

Space	Defined by	White	Transfer function used for the curve fit
sRGB	IEC 61966-2-1:1999	D65	Piecewise sRGB curve

Space	Defined by	White	Transfer function used for the curve fit
Rec. 709	ITU-R BT.709-6	D65	2.4 power
Display P3	SMPTE RP 431-2 primaries, sRGB transfer	D65	Piecewise sRGB curve
Adobe RGB (1998)	Adobe RGB (1998) Color Image Encoding	D65	Power of 563/256
Rec. 2020	ITU-R BT.2020-2	D65	BT.2020 curve
ProPhoto RGB	ISO 22028-2:2013 (ROMM RGB)	D50	1.8 power with a linear toe

sRGB and Rec. 709 share their primaries and white, so only the tone curve can tell them apart.

Conformance rules

The 23 rules, with their severities and clauses, are listed in Check conformance. Summary: 11 errors, 8 warnings, 4 notes; 2 rules are house preferences with no clause.

Method tab

Section	Contents
The header	Bytes 8 to 11 of the loaded profile (or an example), and how the version is read
Colorant vs primary	sRGB's stored colorants against its primaries, and the two adaptation hypotheses
Identification	Why a raw white-point term cannot be part of the score, and the order of the comparison
Gamut area	Each reference space as a percentage of sRGB in xy and u′v′, and its coverage of sRGB
A profile with no colorants	How an A2B table becomes a CIELAB hull, with figures when such a profile is loaded

Conformance tab

Element	Contents
Rules buttons	**All** (23), **Errors** (11), **Warnings** (8), **Notes** (4)
Rail note	How many rules are shown, and how many cite no clause
Withdrawn	Two checks removed because they fired on correct files
The rules that stayed, and what each cites	Every rule with severity and clause, or **house preference**
What the old validator got right	The checks carried across unchanged

Data tab

Status buttons: **All** (20), **Verbatim** (8), **Computed** (7), **Stand-in** (2), **Absent** (3). Select an entry to expand it; stand-in and absent entries add **what it would take**.

Status	Meaning	Entries
Verbatim	A published table, clause or definition carried unaltered	Header, tag table rules, required tags, CIE 1931 observer, sRGB, the other spaces, Bradford, MD5

Status	Meaning	Entries
Computed	Derived here by stated arithmetic	Primaries, adaptation method, identification, area and coverage, curve fit, profile ID, CLUT gamut
Stand-in	The tool's own choice in place of something published	The list of six spaces; the severity each rule carries
Absent	Named but not held	Evaluating `mpet` pipelines; resolving named colours; any measurement of the device

Reference tab

Status buttons: **All** (9), **Runs** (7), **Partly** (1), **Not held** (1).

Source	Status
ISO 15076-1:2010 / ICC.1:2010	Runs
IEC 61966-2-1:1999 (sRGB)	Runs
CIE 015:2018, Colorimetry	Runs
Lam (1985), the Bradford matrix	Runs
ITU-R BT.709-6, SMPTE RP 431-2:2011, Adobe RGB (1998), ITU-R BT.2020-2, ISO 22028-2:2013	Runs
CIE 142-2001 (CIEDE2000)	Runs
RFC 1321 (MD5)	Runs
ICC.1:2010 §10.14, multiProcessElementsType	Partly
ICC.1:2010 §10.17, namedColor2Type	Not held

Export formats

File	Name	Contents
Reading	*profile-name* `-reading.json`	Header with clauses, tags, identity, gamut, conformance, register
Tag table	*profile-name* `-tags.csv`	`signature,type,offset,size,aligned,in_bounds,shares_with`

See Export a reading for the fields.

Messages

Message	When
Read followed by the file name	A profile was read
A profile is at least 132 bytes; this file is N.	The file is too short to be a profile
Bytes 36-39 are "...", **not 'acsp' — this is not an ICC profile.**	The file lacks the profile signature
The reader could not get through this file: followed by a reason	Reading stopped unexpectedly
That file could not be read	The browser could not read the file
Load a profile first	An export was requested with nothing loaded

Message	When
Saved the reading / **Copied the reading** / **Saved the tag table**	An export succeeded
The clipboard refused	Clipboard access was blocked
Cleared	**Clear** was selected

When a file cannot be read, the verdict area turns red and reads **This file could not be read**, followed by the reason.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

Troubleshooting the ICC Profile Parser

Symptoms you may meet in the ICC Profile Parser, why they happen, and what to do.

The file will not read

Symptom	Why	What to do
A profile is at least 132 bytes; this file is N.	The file is too short to hold a header and tag count	Check you chose the right file; it may be empty or cut short by a failed download
Bytes 36-39 are … **not 'acsp' — this is not an ICC profile.**	The file does not carry the ICC signature	You may have dropped an image, a PDF or another format. Save the embedded profile out as a standalone `.icc` file first
My `.icc` file is greyed out in the file chooser	The chooser filters for `.icc` and `.icm`	Drag the file onto the **Drop a profile** box instead, or rename it with the right ending
The reader could not get through this file: …	Reading stopped part-way, usually because the file is damaged	Try the original source of the profile. The reason after the colon says where it failed
That file could not be read	The browser could not open the file	Check the file is not locked or still downloading, then try again

A panel is empty or missing

Symptom	Why	What to do
No ranking table, diagram or tone curves	The profile has no RGB colorants (for example a CMYK, greyscale or LUT-only RGB profile)	This is expected. A CMYK profile with an A2B table shows a CIELAB hull instead
No gamut hull for a CMYK profile	Its A2B table is in a version 4 format, is incomplete, has more than 200,000 grid points, or is missing	The header, tags and findings are still read; the gamut cannot be drawn
The area cards show a dash	Area and coverage apply only to profiles with RGB colorants	—
I cannot see the profile's name	The page does not print the text of the description tag	The tag table shows whether `desc` is present; open the profile in another viewer to read the name
A tag's row says **could not be decoded**	Its data is damaged or shorter than its type needs	See the **Every tag payload could be decoded** finding for the reason

The figures look wrong

Symptom	Why	What to do
The area figures seem to refer to another space	After loading, the comparison moves to the closest space, but the menu label may still show your earlier choice	Choose the space you want from **Compare against**
sRGB and Rec. 709 tie in the ranking	They share primaries and white; only the tone curve separates them	Read the aside under the verdict; a profile with no tone curve cannot be told apart
area, u'v' is over 100%	It is an area ratio; a larger gamut gives more than 100%	Use **covers of it** to see how much of the reference space is covered
area, xy differs from **area, u'v'**	The xy diagram is uneven; the u'v' figure is the more meaningful	Quote the u'v' figure, and name it
The verdict names a space that seems wrong	Closest means closest among six	Compare the first and second rows of the ranking; a large distance means none of the six is a good match
The caption says the primaries were recovered with an assumed white	The profile carries no `chad` tag, so the white it was adapted from is not recorded	Treat the gold triangle as a best estimate

Findings

Symptom	Why	What to do
The profile ID matches the bytes raised as a warning	The file was changed after its ID was written, or the ID was written incorrectly	Get a fresh copy from the source of the profile
A v4 profile carries a profile ID raised	A version 4 profile has an all-zero ID	It is graded a warning, not an error; regenerate the profile if your workflow needs the ID
Media white point is the PCS white raised on a display profile	The white point tag holds the display's measured white instead of D50	Common in older version 2 profiles; version 4 requires D50 here
Declared size matches the file raised as an error	The file was cut short or padded after writing	Get an undamaged copy
A LUT tag holds the grid its header declares	A table is shorter than its header says, often from a truncated download	Download the profile again

Clipboard and downloads

Symptom	Why	What to do
The clipboard refused	Your browser blocked clipboard access	Allow clipboard access for the site, or use **Save the reading**
Load a profile first	Nothing is loaded	Drop a profile on the **Lab** tab first

For help with the site in general, see Help.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

ICC Profile Parser FAQ

Short answers to common questions about the ICC Profile Parser.

Is my profile uploaded?

No. The file is read in the page and never sent anywhere. See Privacy.

Which files can I open?

Standalone ICC profiles, usually ending `.icc` or `.icm`, of version 2 or version 4. A profile embedded in an image or PDF must be saved out as its own file first. A version other than 2 or 4 is still read, with a warning.

Can I compare two profiles side by side?

Not on one page. The tool holds one profile at a time. To compare, save a reading of each and compare the JSON files, or open the tool in two browser tabs.

Why does my monitor profile say "sRGB" when I calibrated a wide-gamut screen?

The verdict names the closest of six reference spaces. Check the **Primaries Δu′v′** column: if the first row's distance is large, the match is loose. Also check whether the menu under **Compare against** shows the space you expect.

What does "matrix/TRC" mean?

The profile describes an RGB device with three colorants (a matrix) and three tone response curves (TRCs). See Understand ICC profiles.

Why are the stored colorants and the primaries different triangles?

Colorants are stored after adaptation to D50, the connection-space white. The primaries of the space the profile describes are recovered by reversing that adaptation. The diagram shows both so you can see the difference.

Is "136% sRGB" coverage?

No. That kind of figure is an area ratio. The tool gives both: **area, u′v′** is the area ratio, and **covers of it** is the true share of the reference space inside the profile's gamut, which cannot exceed 100%.

Does a clean result mean my colours are right?

It means the file is well formed. It says nothing about whether the device still behaves as the profile describes; that needs a measuring instrument and the device.

Why is there no score out of 100?

A profile either meets the specification or it does not, and where it does not, each finding names the clause. Averaging unlike rules into one number hides which rule mattered.

Can the tool fix or edit a profile?

No. It reads profiles and reports on them. It does not write or modify profile files.

Can it convert colours using the profile?

No. It reads tone curves and, for CMYK and similar profiles, walks the A2B table to draw the gamut, but it does not offer colour conversion through the profile.

Why is a named-colour profile nearly empty?

A named-colour profile is little more than its list of named colours, and the tool does not resolve those names to colours or list them on the page. For that class of profile you get the header, the tag table and the conformance findings: the shape of the data, not its content.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

ICC Profile Parser background material

The contents of the ICC Profile Parser's others folder: glossary, limits, related pages, release notes and privacy.

This folder holds reference material that supports the guides: the terms the tool uses, what it can and cannot read and how precisely, related tools and reading, its history, and what happens to your files. The tool has no keyboard shortcuts of its own beyond standard keyboard access, so there is no shortcuts page.

File	What it covers
glossary.md	Definitions of the terms used in the ICC Profile Parser
limits-and-accuracy.md	What the tool reads, how precisely, and what it does not hold
related.md	Related tools on the site and Learn articles
release-notes.md	User-facing history of the tool
privacy.md	What stays in your browser

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

ICC Profile Parser glossary

Definitions of the terms used in the ICC Profile Parser.

Term	Meaning
A2B / B2A table	Tables converting device values to the PCS (A2B) or back (B2A); 0, 1, 2 are perceptual, colorimetric, saturation
Area ratio	The area of one gamut triangle divided by another's; can exceed 100%
Bradford transform	The standard chromatic adaptation method, from Lam (1985), carried by the `chad` tag
`chad` tag	A 3×3 matrix recording how the colorants were adapted to D50
Chromatic adaptation	Converting colour values from one white point to another
Chromaticity diagram	A plot of colour without brightness; the tool uses CIE 1931 xy for drawing
CIE 1976 u′v′	A chromaticity diagram in which equal distances look more nearly equal than in xy
CIELAB	A colour space with lightness L* and two colour axes, a* (green to red) and b* (blue to yellow)
Clause	A numbered section of ICC.1:2010 that a conformance rule enforces
CLUT	Colour lookup table: a grid of output values sampled at points across the input
CMM	Colour management module, the software that performs conversions; named in the header
Colorant	The stored XYZ value of a primary, already adapted to D50
Conformant	Breaking no rule graded as an error
Coverage	The share of a reference gamut that lies inside the profile's gamut; at most 100%
D50 / D65	Standard daylight whites at about 5000 K and 6500 K; D50 is the ICC connection-space white
Data colour space	The colour space of the device values, such as RGB or CMYK
Device class	What the profile describes: Input, Display, Output, DeviceLink, ColorSpace, Abstract or NamedColor
Finding	A rule that did not hold for the loaded profile, with severity, clause and detail
Gamut	The range of colours a device or colour space can reproduce

Term	Meaning
Header	The first 128 bytes of a profile, describing its kind
House preference	A rule the tool applies that no clause requires
ICC profile	A file, defined by the International Color Consortium, describing how a device or space reproduces colour
Matrix/TRC	A profile built from three colorants and three tone response curves
Media white point	The white of the display or paper, in the `wtpt` tag
MD5	A checksum algorithm (RFC 1321) used for the profile ID
PCS	Profile connection space: CIE XYZ or CIELAB at D50, through which profiles are joined
Primaries	The red, green and blue chromaticities that define an RGB colour space
Profile ID	An MD5 checksum of the profile, stored in bytes 84 to 99 of the header
Rendering intent	The strategy for out-of-gamut colours: perceptual, media-relative, saturation or absolute
Spectral locus	The horseshoe outline of pure spectral colours on a chromaticity diagram
Tag	A four-character signature pointing to a block of data in the profile
Tag type	The four-character code at the start of a tag's data that says how to read it
TRC	Tone response curve: how a channel's values map to linear light
Transfer function	The published curve relating encoded values to light for a colour space
XYZ	The CIE 1931 tristimulus values, the basis of colorimetry
XYZ scaling	A simple adaptation that scales X, Y and Z directly; used by some older profiles

Limits and accuracy

What the ICC Profile Parser can and cannot read, how precise its figures are, and what it is not.

What it reads

- Standalone profile files of ICC version 2 and version 4, at the byte positions ICC.1:2010 (ISO 15076-1:2010) gives.
- The whole header, the full tag table, and the tag types listed in the Reference: text, descriptions, multi-language text, XYZ values, tone curves in every form, the `chad` matrix, signatures, and lut8 and lut16 tables.
- A tag that cannot be decoded is reported as a finding, and the rest of the profile is still read.

What it cannot read

Item	What happens
Profiles embedded in images or PDFs	Not read. Save the profile out as its own file first
Version 4 `mAB` , `mBA` tables	Listed in the tag table but not decoded; no gamut hull is drawn from them
`mpet` (multi-process elements)	Its structure is read; the pipeline is not run, so its behaviour is not measured
`ncl2` named colours	Not resolved to colours and not listed on the page
Private tags such as `vcgt`	Listed with their type, not decoded

Item	What happens
Description and copyright text	Read, but not printed on the page or in the reading

How the figures are found, and how precise they are

Figure	Method	Notes
Header fields	Read directly at the published offsets	Exact
Profile ID match	MD5 recomputed with the flags, intent and ID fields zeroed	Exact
Primaries	Colorants un-adapted with the profile's `chad` tag, or with Bradford from an assumed white	Exact with `chad`; an estimate without, and the caption says so
Primaries Δu′v′	Largest per-primary distance in CIE 1976 u′v′, after adapting the reference into the PCS	Two adaptation methods tried; the closer is reported
Curve fit	Peak difference from the reference transfer function over 257 samples	Uses the red curve, or the grey curve for greyscale
area, u′v′ and **area, xy**	Triangle area ratios	Area, not coverage
covers of it	Polygon intersection in u′v′	At most 100%
CIELAB hull	Every grid point of the A2B1 (or A2B0) table converted to CIELAB, then its outline	Up to 200,000 grid points
Spectral locus	CIE 1931 2° observer, 380 to 700 nm at 5 nm	Taken from CIE 015:2018 data

Choices the tool makes

These are listed as **Stand-in** on the **Data** tab.

- **The six reference spaces.** Identification means closest among sRGB, Rec. 709, Display P3, Adobe RGB (1998), Rec. 2020 and ProPhoto RGB. Each definition is taken from its own publication; the choice of list is the tool's.
- **Severity.** The grade of each rule (error, warning or note) is the tool's judgement. Each rule states its clause so you can apply your own.

What it is not

- **Not a device measurement.** Everything on the page is a reading of a file. Whether the display, printer or press still behaves as the profile says is a question for a measuring instrument and the device. A profile that is internally perfect can describe a monitor that has drifted since calibration.
- **Not an official validator.** The conformance check cites the specification clause by clause, but it is not the International Color Consortium's own validation tool.
- **Not a profile editor or converter.** It does not change profiles or convert colours through them.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

Privacy

What the ICC Profile Parser does with your profile files.

Your profile stays in your browser

- The profile file is read by the page itself in your browser. The tool does not upload it or send it anywhere.
- Every figure, diagram and finding is worked out on your device.
- **Save the reading** and **Tag table, CSV** create their files in the browser and save them through your browser's normal download.
- **Copy it** writes the reading to your clipboard only when you select it.

Nothing is kept after you leave

The tool does not store the profile or the reading. **Clear**, reloading, or leaving the page removes them.

What a reading contains

A reading contains the profile's file name and size, its header values, its tag table, the tool's identification, gamut figures and findings. It does not contain the profile file itself. The file name appears in the reading, so check that it says nothing you would rather not share.

The page itself

The ICC Profile Parser is a normal page on auricartisan.com. Site-wide features such as sign-in, preferences and any consent-based services follow the site's own privacy policy and your cookie choices; see Help.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

Release notes

User-facing history of the ICC Profile Parser.

4 June 2026 — ICC Profile Reading

The earlier engine was replaced by the current workbench, titled **ICC Profile Reading**, with six tabs: **Lab**, **Method**, **Conformance**, **Data**, **Export** and **Reference**. The site's earlier user guide carries a notice that it describes the engine replaced on this date.

What changed for you:

- **The version is read correctly.** The version now comes from the whole of byte 8 and the two halves of byte 9, as the specification says. Profiles are no longer reported as version 0.2 or 0.4, and no longer flagged for an "unusual" version.
- **Shared tag data is allowed.** Two tags pointing at one data block, which the specification permits, are shown as sharing rather than reported as corrupt. A partial overlap is still an error.
- **Identification works on conformant profiles.** Reference spaces are adapted into the connection space before comparing, so a correct D50 white no longer counts against a profile. The ranking shows primaries distance, curve fit and white separately instead of one blended score.
- **Colorants and primaries are drawn separately.** The diagram shows the stored colorants, the primaries recovered from them, and the reference space.
- **Area in u′v′, and coverage.** Gamut area is given in CIE 1976 u′v′ with the xy figure beside it, and true coverage is shown separately from the area ratio.

- **CMYK and other LUT profiles get a gamut.** The gamut of a profile without colorants is drawn from its A2B table as a CIELAB hull.
- **No score out of 100.** Each finding carries a severity and the clause it enforces; a profile is conformant or it is not.
- **Damaged tags no longer stop the reading.** A tag that cannot be decoded, or a truncated table, becomes a finding and the rest of the profile is read.
- **The profile ID is recomputed** and compared with the stored one.

25 May 2026 — earlier version

The ICC Profile Parser user guide was published in the site's documentation library. It describes the earlier engine and is now marked as superseded.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › ICC PROFILE PARSER

Related tools and reading

Other Auric Artisan tools and Learn articles that go with the ICC Profile Parser.

Tools on the site

Tool	Why you might use it with the ICC Profile Parser
Gamut Explorer	Inspect device and working-space gamuts in chromaticity and 3D views
Gamut Mapping	Explore how colours are mapped between gamuts, and rendering intents
Chromatic Adaptation	Work with Bradford and other adaptation transforms directly
Illuminants	Look up D50, D65 and other standard illuminants
LUT Lab	Read and write .cube lookup tables, a different kind of colour transform file

Learn articles

- ICC Profiles and How Color Management Actually Works — https://auricartisan.com/library/learn/articles/2026-05-25-icc-profiles-color-management
- How to Calibrate and Profile a Display — https://auricartisan.com/library/learn/articles/2026-05-25-how-to-calibrate-and-profile-a-display
- RGB, sRGB, Adobe RGB, ProPhoto, Display P3, and Rec.2020 — https://auricartisan.com/library/learn/articles/2026-05-25-rgb-working-spaces-compared
- Gamma, Linear Light, and Transfer Functions — https://auricartisan.com/library/learn/articles/2026-05-25-gamma-linear-light-transfer-functions
- Gamut Mapping — https://auricartisan.com/library/learn/articles/2026-06-17-gamut-mapping-clipping-and-rendering-intents
- Gamut Volume: How Much Color a Wide Gamut Actually Adds — https://auricartisan.com/library/learn/articles/2026-08-26-gamut-volume-how-much-color

Other documentation

- The group overview: Image and file tools.
- The site's documentation library carries "ICC Profile Reading Developer Reference" at https://auricartisan.com/library/documentation/articles/2026-06-04-icc-profile-reading-developer-reference, written for

people maintaining the tool.

- The ICC specification itself, ICC.1:2010, is published by the International Color Consortium and as ISO 15076-1:2010.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Image Picker

What the Image Picker does, who it is for, where to find it and how to take a first palette from a picture.

The Image Picker (the page is titled **Image Color Picker & Palette Tool**) turns a picture into a palette. Drop a photograph, a screenshot or a scan and the tool samples its pixels, groups them into clusters with the k-means++ method, and shows the resulting colours with the share of the picture each one takes up. For flat artwork such as logos and icons, where the picture has no more distinct colours than you asked for, it keeps the exact colours instead of averages.

Alongside the palette it sorts the colours into shadows, midtones and highlights, lists the pairs with the most contrast for text by the WCAG 2 ratio, and spots harmony relationships between the colours. A live eyedropper reads any pixel under the pointer, and a click pins it, so you can keep exact colours the clustering merged away. The **Analysis** tab charts the same palette four ways.

You can copy the palette as hex values, CSS custom properties or JSON, download it as SVG, JSON or CSV, and analyse a list of hex colours without any picture at all. Pictures are read in the page; nothing is uploaded.

Who it is for

- Designers building a palette from a photograph, moodboard or product shot.
- Brand and editorial teams auditing the colours in a set of images.
- Developers who want an image's colours as CSS custom properties or JSON.
- Anyone who needs the exact colour of one pixel.

Where to find it

Open https://auricartisan.com/tool/general/tools/image-picker/ in a current browser. It is listed as **Image Picker** on the Collections page at https://auricartisan.com/collections/. Nothing needs installing.

Quick start

1. Open the tool. The **Workbench** tab is selected.
2. Drop an image anywhere on the large panel in the middle, or select the panel to choose a file (PNG, JPEG, WEBP, GIF or BMP).
3. The palette appears under **Extracted palette**, largest share first.
4. Move the **Clusters** slider to ask for more or fewer colours; the palette updates when you release it.
5. Hover the preview to read any pixel, and click to pin one under **Pinned**.
6. Select **Hex**, **CSS** or **JSON** under **Take it with you** to copy the palette, or **SVG** to download it.

What you can do

- Extract 2 to 12 colour clusters (16 when the page first opens), at five sampling qualities, with optional merging of near-identical clusters.
- Leave out near-black or near-white pixels.
- Read any pixel as hex, RGB, HSL, relative luminance and position, and pin up to 24 exact colours.
- See each colour's share of the picture, its tone band, the eight strongest contrast pairs with WCAG ratings, and detected harmonies.
- Chart population, distribution, hue against saturation, and luminance, each viewable full screen.

- Copy hex, CSS custom properties or JSON; download SVG, JSON or CSV.
- Analyse a typed list of hex colours in RGB, HSL, CIELAB, LCH, luminance and tone band.
- Load a picture from a web address, where the server allows it.

In this folder

- docs/ — how to use the tool.
  - Getting started — open it, the screen tour, the first palette.
  - Extract a palette — loading pictures and every extraction setting.
  - Pick exact colours — the eyedropper and pins.
  - Read contrast pairs, tone bands and harmony — what each readout means.
  - Chart the palette — the four charts on the Analysis tab.
  - Export and share — every copy and download format.
  - Analyse a list of hex colours — batch colour analysis.
  - Reference — every control, output and format.
  - Troubleshooting — symptoms, causes and fixes.
  - FAQ — short answers to common questions.
- others/ — background material.
  - Glossary — terms used in this tool.
  - Shortcuts — keyboard shortcuts.
  - Limits and accuracy — how the palette is made and what it is not.
  - Related — related tools, the API and Learn articles.
  - Release notes — user-facing history.
  - Privacy — what stays in your browser and what uses the network.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Image Picker documentation

The list of guides, the reference, troubleshooting and FAQ for the Image Picker.

This folder holds the task guides for the Image Picker, the full reference of its controls and outputs, and the troubleshooting and FAQ pages. Start with Getting started, then go to the guide for the job you have.

File	What it covers
getting-started.md	Opening the tool, a tour of the three tabs, and a first palette
extract-a-palette.md	Loading a picture from a file or a web address, and every extraction setting
pick-exact-colours.md	The live eyedropper, pinning pixels, and managing pins
read-contrast-and-harmony.md	Swatches, tone bands, contrast pairs and harmony relationships
chart-the-palette.md	The four charts on the Analysis tab and the full-screen view
export-and-share.md	Copying hex, CSS and JSON; downloading SVG, JSON and CSV; the page link
batch-analysis.md	Analysing a typed list of hex colours on the Actions tab
reference.md	Every control, value, readout, file format and keyboard key
troubleshooting.md	What to do when a picture will not load or a control seems to do nothing
faq.md	Short answers to common questions

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Getting started with the Image Picker

Open the Image Picker, find your way around its three tabs, and take a first palette from a picture.

Open the tool

Go to https://auricartisan.com/tool/general/tools/image-picker/. The page opens on the **Workbench** tab with a large empty panel that reads **Drop an image anywhere in here**.

Under the page heading there may be a short line that begins **Automate this**. It says the same job is available through the site's REST API, what one call costs in API credits, and links to the API documentation. You can close it with ×; it stays closed in that browser. You do not need the API to use the tool.

The screen at a glance

Tab	What it is for
Workbench	Load a picture, set the extraction, see the preview, the palette, the tone bands and the readouts
Analysis	Four charts of the current palette; the tab label shows **4**
Actions	Download JSON or CSV, copy the page link, and analyse a list of hex colours

The Workbench tab

The **Workbench** has three columns.

- **Left: The image** (file name, size, and the **Replace** and **Load URL** buttons with a web address field), **Extraction** (the **Clusters**, **Quality** and **Merge threshold** sliders and the **Ignore dark** and **Ignore light** switches), **Run** (**Re-extract** and **Reset**), and **History** (how many extractions you have run).
- **Middle:** before a picture is loaded, the drop area. After, the **Preview** with the eyedropper readout and **Pin** button, the **Extracted palette** swatches, and the **Tone bands**.
- **Right:** before a picture is loaded, the keyboard keys and a note of what comes back. After, **Pairs that carry text**, **Harmony**, **Pinned** and **Take it with you**.

Take a first palette

1. Drop an image onto the middle panel, or select the panel to choose a file. PNG, JPEG, WEBP, GIF and BMP are all accepted.
2. The picture appears under **Preview**, and the file name and sizes appear on the left, for example `4032 × 3024 → 667 × 500 | harbour.jpg`. Large pictures are shrunk to fit an 800 × 500 working area; the first size is the original.
3. Under **Extracted palette**, the swatches appear largest share first. Each shows its share of the picture, its hex value, its HSL values and its relative luminance. The heading says how many clusters and samples were used.
4. Move **Clusters** to 6 and release it. The palette is extracted again with six colours.
5. Hover the preview. The readout under it shows the colour under the pointer. Click to pin that exact colour; it appears under **Pinned** on the right.
6. Under **Take it with you**, select **CSS**. The palette is copied as CSS custom properties. Paste it into your stylesheet:

```
:root {
  --palette-01: #2E4A5C;
  --palette-02: #D9C7A8;
  --palette-03: #8A6A4F;
}
```

The values above are an example; you get the colours of your picture.

Next steps

- To tune the palette, see Extract a palette.
- To keep exact pixels, see Pick exact colours.
- To check which colours work together as text and background, see Read contrast pairs, tone bands and harmony.
- For every export, see Export and share.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Analyse a list of hex colours

Use batch colour analysis on the Actions tab to get RGB, HSL, CIELAB, LCH, luminance and tone band for a list of hex colours.

Batch colour analysis gives the same figures the palette uses for any list of colours you type or paste. No picture is needed.

Run a batch

1. Open the **Actions** tab.
2. In the box under **Batch colour analysis**, type or paste hex colours. Separate them with new lines, commas, semicolons or spaces. The `#` is optional.

```
#D3AF37
#735244
#4A90D9
#E84855
#3BB273
```

3. Select **Run batch**.
4. A table appears under the box, one row per colour.

The results table

Column	Meaning	Example
#	Position in your list	1
Swatch	The colour	—
Hex	The colour as six-digit hex, in capitals	#D3AF37
RGB	Red, green, blue, 0 to 255	211,175,55
HSL	Hue, saturation, lightness	46° 64% 52%
Lab	CIELAB lightness, a*, b*, for a D65 white	72.8,0.9,62.8
LCH	CIELAB lightness, chroma, hue angle	72.8,62.8,89°
Lum	WCAG relative luminance, 0 to 1	0.4478
Tone	Shadow, Midtone or Highlight	Midtone

The example column shows the row for `#D3AF37`. The tone bands use the same limits as the **Workbench**: below 0.18 is a Shadow, below 0.65 a Midtone, and 0.65 or above a Highlight.

Input rules

- Three-digit hex values such as `#abc` are expanded to six digits (`#AABBCC`).
- Entries that are not hex values are skipped.
- Use three- or six-digit hex. The box accepts four- and eight-digit values (with an alpha part), but they are not read as the colour you mean, so remove the alpha digits first.

Messages

Message	When
Click Run batch to analyse colours.	Before the first run
Enter hex colours above.	The box is empty
No valid hex colours found.	Nothing in the box is a hex value

Uses

- Check a brand palette's luminance and tone spread without an image.
- Get CIELAB and LCH values for a list of colours to paste into a specification.

- Compare colours you pinned from a picture: copy the JSON, take the `picks`, and paste their hex values here.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Chart the palette

Read the four charts on the Analysis tab and open any of them full screen.

The **Analysis** tab draws the current palette four ways. Every chart is drawn in the palette's own colours, so a bar and a swatch of the same colour are the same cluster.

Open the charts

1. Extract a palette on the **Workbench** tab.
2. Select the **Analysis** tab.
3. The left-hand panel, **Charting**, names the picture and its working size and pixel count, for example `667 × 500 · 333,500 px`.

Until a picture is loaded, the tab reads **Nothing charted yet**.

The four charts

Chart	Subtitle	What it shows
Cluster population	share of the samples	One bar per cluster, largest first, with the share of the largest and smallest at the top
Colour distribution	every cluster	A ring divided by share, with the leading share, its hex value and the cluster count in the middle
Hue vs saturation	HSL scatter	Each cluster placed by hue (0° to 360° across) and saturation (0% to 100% up); larger dots are larger shares
Relative luminance	WCAG, 0 → 1	One bar per cluster, sorted from darkest to lightest, height set by relative luminance

Details:

- **Cluster population** labels each bar with its share when there is room; with many clusters the labels are left off, and the share is still on the swatch.
- **Colour distribution** labels slices larger than 8% with their share.
- **Hue vs saturation** has a faint rainbow behind it to show where each hue sits.
- **Relative luminance** labels each bar with its value when there is room, and marks **darkest** and **lightest** at the ends.

Full screen

To see a chart larger, double-click it, or select the ⛶ button in its corner. The chart opens over the page with its title, such as **Cluster Population Histogram**. Close it with the **×** button, the `Escape` key, or by clicking the dark area around it.

When the charts update

The charts redraw whenever the palette changes and the **Analysis** tab is open, and when you resize the window. Extractions you run while on the **Workbench** tab are charted the next time you open **Analysis**.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Export and share

Copy the palette as hex, CSS custom properties or JSON, download it as SVG, JSON or CSV, and copy the page link.

There are two places to take a palette away: **Take it with you** on the right of the **Workbench** tab, and **Download** on the **Actions** tab.

The page shows no confirmation after copying or downloading. Paste the text, or check your downloads, to confirm.

What each export contains

Export	Where	Contains	Pins included
Hex	Workbench	Palette hex values, comma separated	No
CSS	Workbench	Palette as CSS custom properties	No
JSON (copy)	Workbench	Full analysis	Yes
SVG	Workbench	A strip of palette swatches with labels, downloaded	No
JSON (download)	Actions	Full analysis, as `palette-analysis.json`	Yes
CSV	Actions	One row per palette colour, as `palette.csv`	No

The heading **Take it with you** says **palette + pins**, but only the JSON carries the pins.

Hex

Selecting **Hex**, or pressing `H`, copies the palette in order of share:

```
#2E4A5C, #D9C7A8, #8A6A4F, #C0392B
```

CSS

Selecting **CSS** copies a block of numbered custom properties:

```
:root {
  --palette-01: #2E4A5C;
  --palette-02: #D9C7A8;
  --palette-03: #8A6A4F;
  --palette-04: #C0392B;
}
```

JSON

Selecting **JSON** on the **Workbench**, or pressing `J`, copies the full analysis. **JSON** on the **Actions** tab downloads the same content as `palette-analysis.json`.

```
{
  "image": {
    "width": 4032, "height": 3024,
    "canvasWidth": 667, "canvasHeight": 500,
    "label": "harbour.jpg"
  },
  "settings": {
    "clusterCount": 6, "quality": 4, "mergeThreshold": 0,
    "ignoreDark": false, "ignoreLight": false
  },
  "sampleCount": 41750,
  "palette": [
    {
      "hex": "#2E4A5C", "r": 46, "g": 74, "b": 92,
      "hsl": { "h": 203.5, "s": 33.3, "l": 27.1 },
      "lab": { "L": 30.04, "a": -4.96, "b": -13.89 },
      "lch": { "L": 30.04, "C": 14.75, "h": 250.3 },
      "luminance": 0.06251,
      "population": "24.13%",
      "count": 10074
    }
  ],
  "picks": [ { "r": 192, "g": 57, "b": 43, "hex": "#C0392B" } ],
  "exportedAt": "2026-09-24T10:15:00.000Z"
}
```


Field	Meaning
`image`	Original width and height, working width and height, and the file name or address
`settings`	Clusters, quality (1 Fast to 5 Max detail), merge threshold, and the two ignore switches
`sampleCount`	Pixels sampled
`palette[].hex`, `r`, `g`, `b`	The colour
`palette[].hsl`	Hue in degrees, saturation and lightness in percent
`palette[].lab`	CIELAB, calculated for a D65 white
`palette[].lch`	CIELAB lightness, chroma and hue angle
`palette[].luminance`	WCAG relative luminance, 0 to 1
`palette[].population`, `count`	Share of the samples as a percentage, and the number of samples
`picks`	Your pinned colours
`exportedAt`	When the export was made

The values in the example are illustrative.

SVG

Selecting **SVG** downloads `palette.svg`: one 60 × 60 swatch per palette colour with rounded corners, 4 pixels apart, with the hex value written under each. Open it in any drawing program or browser.

CSV

On the **Actions** tab, **CSV** downloads `palette.csv`:

```
#,Hex,R,G,B,H,S,L,Luminance,Population%,Count
1,#2E4A5C,46,74,92,203.5,33.3,27.1,0.06251,24.13,10074
```

Column	Meaning
`#`	Position in the palette
`Hex`, `R`, `G`, `B`	The colour
`H`, `S`, `L`	HSL hue in degrees, saturation and lightness in percent
`Luminance`	Relative luminance, five decimal places
`Population%`	Share of the samples
`Count`	Number of samples in the cluster

Copy the page link

On the **Actions** tab, **Copy page link** copies the address of the page. It does not carry the picture, the settings or the palette; whoever opens it starts with an empty tool. To share a palette, send the JSON, CSS or SVG instead.

Automate it

The same job, extracting a palette from an image, is available through the site's REST API as `POST /v1/palette/from-image`. The line under the page heading gives the current cost in API credits, and **API docs** opens the documentation for that endpoint at https://auricartisan.com/library/documentation/api/#ep-post-palette-from-image. API credits are a separate balance from the tokens that tools use. See Public API.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Extract a palette

Load a picture from a file or a web address, and use every extraction setting to get the palette you want.

Load a picture

From a file

1. On the **Workbench** tab, drop an image onto the middle panel, or select the panel to open a file chooser.
2. To load a different picture later, select **Replace** under **The image** and choose a file.

The drop area is only there until a picture is loaded. After that, use **Replace**, or select **Reset** (or press `R`) to bring the drop area back.

Accepted: any image your browser can open. The drop area names PNG, JPEG, WEBP, GIF and BMP. A file that is not an image is ignored.

From a web address

1. Type or paste the image's address into the field under **The image**, for example `https://example.com/image.jpg`.
2. Select **Load URL**, or press `Enter` in the field.

This only works when the server hosting the image allows other websites to read its pixels (known as cross-origin access, or CORS). Many sites do not. If nothing appears, download the image and load it as a file instead.

What happens on loading

- The picture is shrunk, if necessary, to fit a working area of 800 × 500 pixels, keeping its proportions. It is never enlarged.
- The line under the file name shows the original size, the working size and the name, for example `4032 × 3024 → 667 × 500 | harbour.jpg`.
- The palette is extracted straight away with the current settings.

How the palette is made

1. **Sampling.** The tool reads pixels on a regular grid across the whole picture. The **Quality** setting decides how far apart the grid points are. To keep the work fast, the grid is widened when needed so no more than about 50,000 pixels are sampled; they are still spread across the whole picture.
2. **Filtering.** Nearly transparent pixels are skipped. **Ignore dark** and **Ignore light** skip pixels at the extremes.
3. **Exact colours or clusters.** If the sampled pixels contain no more distinct colours than the number of clusters you asked for, those exact colours are the palette. This is typical of logos, icons and flat illustrations. Otherwise the tool groups the pixels into clusters with k-means++, a method that picks well-spread starting colours and then moves each cluster to the average of the pixels nearest it.
4. **Merging.** Clusters closer together than the **Merge threshold** are combined, weighted by how many pixels each holds.
5. **Ordering.** The palette is sorted by share of the sampled pixels, largest first.

Because k-means++ picks its starting colours at random, extracting the same picture twice can give slightly different palettes. Select **Re-extract** to try again.

The extraction settings

Setting	What it does	Range	When the page opens
Clusters	The largest number of colours to find	2 to 12 on the slider	16
Quality	How densely pixels are sampled	Fast, Quick, Balanced, Detailed, Max detail	Detailed
Merge threshold	Combines clusters closer than this distance	0 to 100	0 (off)
Ignore dark	Skips pixels with relative luminance below 0.03	On or off	Off
Ignore light	Skips pixels with relative luminance above 0.97	On or off	Off

When the page opens, the tool asks for 16 clusters: the label beside **Clusters** reads 16 and the slider sits at its right-hand end. As soon as you move the slider, it sets a value from 2 to 12.

Clusters

Fewer clusters give a short summary palette; more give a detailed study. The palette can have fewer colours than you ask for when merging combines some, or when the picture has fewer distinct colours.

Quality

Level	Samples every
Fast	10th pixel across and down
Quick	6th pixel
Balanced	4th pixel
Detailed	2nd pixel

Level	Samples every
Max detail	Every pixel, subject to the 50,000-sample cap

Higher quality catches small areas of colour at the cost of a little time.

Merge threshold

The distance is measured in RGB values from 0 to 255 per channel. At 0, nothing merges. Raise it when the palette shows several near-identical shades of the same colour. At high values, quite different colours start to merge.

Ignore dark and Ignore light

Relative luminance is a measure of how bright a colour is, from 0 for black to 1 for white, as used by WCAG. **Ignore dark** leaves out near-black pixels, for example a black border or deep shadows. **Ignore light** leaves out near-white pixels, for example a white background or a blown-out sky.

Re-extract, reset and history

- Every setting re-runs the extraction when you release the slider or change the switch.
- **Re-extract** (or `E`) runs it again with the same settings.
- **Reset** (or `R`) clears the picture, the palette and the pins, and brings back the drop area. It does not change the settings or the history count.
- **History** counts the extractions you have run in this session.

What you should see

After loading and adjusting, the **Extracted palette** shows the colours you want, each labelled with its share of the picture, and the readouts on the right describe the same palette.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Pick exact colours

Use the live eyedropper to read any pixel, pin exact colours, and copy them.

The palette is made of averages. When you need the exact colour of one spot, such as a logo's red or a product's trim, use the eyedropper and pins.

Read a pixel

1. Load a picture on the **Workbench** tab.
2. Hover the **Preview**. The readout beside the swatch under it shows the colour under the pointer:

```
#C0392B
RGB(192,57,43) · HSL(6°,63%,46%) · L=0.1431 · px(212,88)
```

Part	Meaning
`#C0392B`	The hex value
`RGB(…)`	Red, green and blue, 0 to 255

Part	Meaning
`HSL(…)`	Hue in degrees, saturation and lightness in percent
`L=`	Relative luminance, 0 (black) to 1 (white), as used by WCAG
`px(…)`	The pixel's position in the working copy of the picture, not in the original

When the pointer leaves the preview, the readout returns to **Hover the image to preview**.

The eyedropper reads the working copy, which is shrunk to fit 800 × 500. On a large photograph each working pixel stands for several original pixels.

Pin a colour

1. Hover the spot you want.
2. Click. The exact colour is added to **Pinned** on the right.

Rule	Detail
Limit	Up to 24 pins. Adding a 25th removes the oldest
Duplicates	A colour already pinned is not added again
Transparent pixels	Nearly transparent pixels cannot be pinned
Counter	The heading reads **Pinned** with the count, for example **3 of 24**

Clicking directly on the preview is the reliable way to pin. The **Pin** button beside the readout pins the colour under the pointer, but moving the pointer onto the button clears the readout first, so it usually has nothing to pin.

Use your pins

- Select a pin to copy its hex value.
- Pins are included when you copy or download the palette as JSON, in a `picks` list. The **Hex**, **CSS**, **SVG** and **CSV** exports contain the extracted palette only.
- **Reset** clears the pins along with the picture.

Copy a palette colour

Select any swatch under **Extracted palette** to copy its hex value.

The page shows no confirmation when something is copied. Paste it to check.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Read contrast pairs, tone bands and harmony

What the swatches, tone bands, contrast pairs and harmony readouts tell you about a palette taken from a picture.

Once a palette is extracted, the **Workbench** tab describes it in four ways. All four use the extracted palette only; pinned colours are not included.

The swatches

Each swatch under **Extracted palette** shows:

Line	Example	Meaning
Inside the swatch	`24%`	Share of the sampled pixels in this cluster
First line	`#2E4A5C`	Hex value
Second line	`203° 33% 27%`	HSL: hue, saturation, lightness
Third line	`L=0.063`	Relative luminance, 0 (black) to 1 (white)

The heading reads, for example, **6 clusters · 41750 samples**: how many colours the palette has and how many pixels were sampled.

Tone bands

Tone bands sorts the palette by relative luminance, the brightness measure used by WCAG:

Band	Relative luminance
Shadows	Below 0.18
Midtones	0.18 to below 0.65
Highlights	0.65 and above

A band with no colours reads **None**. The bands show where the weight of the palette sits: a moody photograph piles up in **Shadows**, a high-key one in **Highlights**.

Pairs that carry text

Pairs that carry text, marked **WCAG 2.1**, lists the eight pairs of palette colours with the highest contrast ratio. The contrast ratio is the WCAG measure of how different two colours are in brightness, from 1:1 (identical) to 21:1 (black on white).

Each row shows the two swatches, the ratio, a rating, and the two hex values:

Rating	Ratio	Suitable for
AAA	7:1 or more	Body text at the enhanced level
AA	4.5:1 or more	Body text at the minimum level
AA-lg	3:1 or more	Large text only (at least 18 pt, or 14 pt bold)
Fail	Below 3:1	Not suitable for text

With fewer than two colours, the panel reads **Need 2+ colors**.

This is a first pass. Colours taken from a picture are often beautiful as accents and weak as text on background. Check the exact pair you intend to use in a dedicated contrast tool such as the Contrast Checker.

Harmony

Harmony looks at the hue of every pair of palette colours and reports pairs whose hues sit at a classic colour-wheel angle. Hue here is the HSL hue, 0° to 360° around the colour wheel.

Relationship	Hue difference
Analogous	25° to 45°
Tetradic	80° to 100°
Triadic	110° to 130°
Split-Complementary	140° to 165°
Complementary	165° to 180° (the largest possible difference)

Each row shows the relationship, the two colours and the measured difference, for example **Δh=172.4°**. Up to 12 rows are listed, grouped by relationship name. When no pair matches, the panel reads **No strong harmonic relationships detected**.

These angles are a convention from colour-wheel teaching, not a measurement of how pleasing a palette is. Greys have no real hue, so a palette with grey clusters can show relationships that mean little.

What you should see

Reading the four panels together tells you what the picture is made of: which colours dominate, how light or dark it is, which pairs could carry text, and whether the hues follow a familiar scheme.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Image Picker reference

Every tab, control, readout, export format and keyboard key in the Image Picker, with ranges and starting values.

Address: https://auricartisan.com/tool/general/tools/image-picker/

Above the tabs

Element	What it does
Read the API reference →	Opens the site's API documentation
Automate this line	The REST endpoint and its cost in API credits; **API docs** opens its documentation; **×** hides the line

Tabs

Tab	Contents
Workbench	Loading, extraction settings, preview and eyedropper, palette, tone bands, contrast pairs, harmony, pins, quick exports
Analysis	Four charts of the current palette; the label shows **4**
Actions	JSON and CSV downloads, **Copy page link**, batch colour analysis

Workbench — The image

Control	What it does	Accepts
Drop area (middle panel, before loading)	Loads a dropped image, or opens a file chooser when selected	Any image type; PNG, JPEG, WEBP, GIF, BMP named
Replace	Opens a file chooser to load a different picture	As above
Address field	The web address of an image	`https://…`
Load URL	Loads the address in the field; `Enter` in the field does the same	Images whose server allows cross-origin reading
File name and size line	The name, and original size → working size	—

Pictures are scaled down to fit 800 × 500 pixels, never up.

Workbench — Extraction

Control	Values	Starting value
Clusters	2 to 12 on the slider	16 when the page opens
Quality	Fast (every 10th pixel), Quick (6th), Balanced (4th), Detailed (2nd), Max detail (every pixel)	Detailed
Merge threshold	0 to 100, RGB distance	0
Ignore dark	Skips pixels with relative luminance below 0.03	Off
Ignore light	Skips pixels with relative luminance above 0.97	Off

Sampling is capped at about 50,000 pixels by widening the grid. Pixels that are nearly transparent are skipped. Each change re-runs the extraction when the slider is released or the switch changes.

Workbench — Run and history

Control	What it does	Key
Re-extract	Runs the extraction again	`E`
Reset	Clears the picture, palette and pins, and shows the drop area again; settings are kept	`R`
Extractions	The number of extractions in this session	—

Workbench — Preview and palette

Element	What it shows or does
Preview	The working copy of the picture. Hover to read a pixel; click to pin it
Eyedropper readout	Hex, RGB, HSL, relative luminance and pixel position under the pointer
Pin	Pins the colour under the pointer, if there is one
Extracted palette	Swatches largest share first: share, hex, HSL, relative luminance. Select to copy the hex
Palette heading	Number of clusters and number of samples
Tone bands	**Shadows** (below 0.18), **Midtones** (below 0.65), **Highlights** (0.65 and above)

Workbench — Readouts

Panel	Contents
Keyboard (before loading)	`E` Re-extract, `H` Copy hex list, `J` Copy JSON, `R` Reset
Pairs that carry text	The eight highest-contrast pairs: swatches, ratio, rating (AAA, AA, AA-lg, Fail), hex values
Harmony	Up to 12 pairs at classic hue angles: Analogous, Tetradic, Triadic, Split-Complementary, Complementary
Pinned	Up to 24 pinned colours; select one to copy its hex
Take it with you	**Hex**, **CSS**, **JSON** (copy) and **SVG** (download)

Contrast ratings

Rating	Contrast ratio
AAA	7:1 or more
AA	4.5:1 or more
AA-lg	3:1 or more
Fail	Below 3:1

Harmony angles

Relationship	HSL hue difference
Analogous	25° to 45°
Tetradic	80° to 100°
Triadic	110° to 130°
Split-Complementary	140° to 165°
Complementary	165° to 180°

Analysis tab

Element	Contents
Charting panel	File name, working size and pixel count
Cluster population	Bar chart of each cluster's share
Colour distribution	Ring chart of shares, leading share in the middle
Hue vs saturation	Scatter of hue (0° to 360°) against saturation (0% to 100%)
Relative luminance	Bars from darkest to lightest, 0 to 1
⛶ button or double-click	Opens the chart full screen; close with ×, `Escape` or a click outside

Actions tab

Control	What it does
JSON	Downloads `palette-analysis.json`

Control	What it does
CSV	Downloads `palette.csv`
Copy page link	Copies the page address; no picture or palette is included
Hex box	One hex colour per line, or separated by commas, semicolons or spaces
Run batch	Analyses the colours in the box

Export formats

Export	File or clipboard	Format
Hex	Clipboard	`#RRGGBB` values, comma and space separated
CSS	Clipboard	`:root { --palette-01: #RRGGBB; … }`
JSON	Clipboard, or `palette-analysis.json`	`image`, `settings`, `sampleCount`, `palette`, `picks`, `exportedAt`
SVG	`palette.svg`	60 × 60 swatches, 4 px apart, hex labels underneath
CSV	`palette.csv`	`#,Hex,R,G,B,H,S,L,Luminance,Population%,Count`

Hex values are written in capitals. Pins appear only in the JSON. For field details, see Export and share.

Batch results

Columns: **#**, **Swatch**, **Hex**, **RGB**, **HSL**, **Lab**, **LCH**, **Lum**, **Tone**. CIELAB is calculated for a D65 white. See Analyse a list of hex colours.

Keyboard

Key	Action
`E`	Re-extract
`H`	Copy the hex list
`J`	Copy the JSON
`R`	Reset
`Escape`	Close a full-screen chart

The letter keys work when focus is not in a text field, slider, switch or menu.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Troubleshooting the Image Picker

Symptoms you may meet in the Image Picker, why they happen, and what to do.

The Image Picker does not show pop-up messages, so when something fails, the usual sign is that nothing changes. This page lists the common causes.

Loading pictures

Symptom	Why	What to do
Nothing happens after dropping or choosing a file	The file is not an image, or your browser cannot decode it	Save it as PNG or JPEG and try again. Some formats, such as HEIC or camera RAW, are not opened by every browser
Dropping a second picture on the preview opens it in the browser instead	The drop area is only active before a picture is loaded	Use **Replace**, or select **Reset** first to bring the drop area back
Load URL does nothing	The address is not an image, cannot be reached, or the server does not allow other sites to read its pixels	Download the image and load it with **Replace**
After **Load URL** the middle panel goes blank	The picture loaded, but its server does not allow its pixels to be read (cross-origin protection)	Download the image and load it as a file
The size line shows a much smaller working size	Pictures are shrunk to fit 800 × 500 for analysis	This is expected; the palette still covers the whole picture

The palette

Symptom	Why	What to do
The **Clusters** label says 16 but the slider will not go past 12	The page starts at 16; the slider covers 2 to 12	Move the slider to choose 2 to 12, or leave it untouched to keep 16
Fewer colours than the cluster count	Merging combined some, or the picture has fewer distinct colours	Lower **Merge threshold**, or accept it: flat artwork keeps its exact colours
Several near-identical shades	Clusters that are close are kept apart when merging is off	Raise **Merge threshold**
A white background or black border dominates	Those pixels are part of the picture	Turn on **Ignore light** or **Ignore dark**
The palette changes slightly each time	k-means++ starts from random colours	Select **Re-extract** until you are happy, then export
A small but important colour is missing	It covers too few pixels to form a cluster	Raise **Quality** or **Clusters**, or pin it with the eyedropper

Eyedropper and pins

Symptom	Why	What to do
Pin does nothing	Moving the pointer onto the button clears the colour under it	Click directly on the preview to pin
A click does not add a pin	The colour is already pinned, or the pixel is transparent	Check **Pinned**; pick an opaque pixel
The oldest pin disappeared	Pins are capped at 24	Copy the JSON before pinning more
Pins are missing from my CSS or hex copy	Only the JSON export includes pins	Copy or download the JSON

Copying and exports

Symptom	Why	What to do
No sign that anything was copied	The page shows no confirmation	Paste to check
Nothing is copied	Your browser blocked clipboard access	Allow clipboard access for the site, or use the downloads on the **Actions** tab
The shortcut keys do nothing	Focus is in a text field, slider, switch or menu	Click an empty part of the page first
The page reset unexpectedly	`R` resets the workbench when focus is not in a control	Load the picture again; your settings are kept
The copied page link opens an empty tool	The link does not carry the picture or palette	Share the JSON, CSS or SVG instead

Batch analysis

Symptom	Why	What to do
No valid hex colours found.	Nothing in the box is a hex value	Enter values such as `#D3AF37` or `D3AF37`
A colour's figures look wrong	It was entered with an alpha part (four or eight digits)	Use three or six digits

Charts

Symptom	Why	What to do
Nothing charted yet	No picture is loaded	Load a picture on the **Workbench**
Bar labels are missing	There are too many clusters for the labels to fit	Open the chart full screen, or read the shares on the swatches

For help with the site in general, see Help.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Image Picker FAQ

Short answers to common questions about the Image Picker.

Is my picture uploaded?

No. A picture you load from a file is read in the page and never leaves the tab. When you use **Load URL**, your browser fetches the image from that address, as it would for any web page. See Privacy.

Why do I get different colours each time I extract?

The k-means++ method starts from randomly chosen colours, so two runs on the same picture can settle on slightly different averages. Flat artwork with few colours is not affected, because its exact colours are used.

Why is a colour I can see not in the palette?

The palette holds the colours that cover the most of the picture. A small detail may not make up enough pixels to become a cluster. Raise **Clusters** or **Quality**, or pin the colour with the eyedropper.

Are the swatches real pixel colours?

For flat artwork with no more distinct colours than the cluster count, yes. For photographs, each swatch is the average of a cluster of similar pixels, and may not match any single pixel exactly. Pins are always exact pixels of the working copy.

How many colours can I extract?

The slider goes from 2 to 12. When the page first opens it uses 16 until you move the slider.

Does the contrast check mean the pairs are accessible?

It gives the WCAG 2 contrast ratio and rating for the strongest eight pairs, as a first pass. Test the exact text and background pair you plan to use, at its real size, in a dedicated contrast tool.

What colour space are the Lab and LCH values in?

CIELAB and its LCH form, calculated from sRGB with a D65 white.

Can I compare palettes from two pictures?

The **History** count keeps track of how many extractions you have run, but the page does not show a side-by-side comparison. Export the JSON of each picture and compare the files.

Can I share my palette with a link?

Copy page link copies the page address only. Share the CSS, JSON or SVG instead.

Is there an API for this?

Yes. The line under the page heading gives the REST endpoint, `POST /v1/palette/from-image`, and its cost in API credits, with a link to its documentation. API credits are separate from tool tokens. See Public API.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Image Picker background material

The contents of the Image Picker's others folder: glossary, shortcuts, limits, related pages, release notes and privacy.

This folder holds reference material that supports the guides: the terms the tool uses, its keyboard keys, how the palette is made and what it is not, related tools and reading, its history, and what happens to your pictures.

File	What it covers
glossary.md	Definitions of the terms used in the Image Picker
shortcuts.md	Keyboard shortcuts and pointer actions
limits-and-accuracy.md	How the palette is made, how precise it is, and what it is not
related.md	Related tools, the API and Learn articles
release-notes.md	User-facing history of the tool
privacy.md	What stays in your browser and what uses the network

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Image Picker shortcuts

Keyboard shortcuts and pointer actions in the Image Picker.

Keyboard

Key	Action
`E`	Re-extract the palette with the current settings
`H`	Copy the palette's hex values
`J`	Copy the full analysis as JSON
`R`	Reset: clear the picture, palette and pins
`Enter` in the address field	Load the image at that address
`Escape`	Close a full-screen chart

The letter keys work anywhere on the page when focus is not in a text field, slider, switch or menu. Because a single `R` resets the workbench, click into a field before typing if you do not want to reset.

Pointer

Action	Result
Hover the preview	Read the pixel under the pointer
Click the preview	Pin that pixel's colour
Click a palette swatch	Copy its hex value
Click a pin	Copy its hex value
Double-click a chart	Open it full screen
Click outside a full-screen chart	Close it

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Image Picker glossary

Definitions of the terms used in the Image Picker.

Term	Meaning
CIELAB	A colour space with lightness L* and two colour axes, a* and b*, designed so distances track perceived difference
Cluster	A group of similar sampled pixels, represented in the palette by their average colour
Contrast ratio	The WCAG 2 measure of brightness difference between two colours, from 1:1 to 21:1
CORS	Cross-origin resource sharing: a server's permission for other sites to read its images' pixels
D65	A standard daylight white at about 6500 K, used here as the white for CIELAB
Eyedropper	The live readout of the pixel under the pointer on the preview
Harmony	A pair of colours whose hues sit at a classic colour-wheel angle
Hex	A colour written as `#RRGGBB`, two hexadecimal digits each for red, green and blue
HSL	Hue (0° to 360°), saturation and lightness (0% to 100%): a cylindrical form of RGB
k-means++	A clustering method that picks well-spread starting colours, then moves each cluster to the average of its nearest pixels
LCH	CIELAB written as lightness, chroma (colourfulness) and hue angle
Merge threshold	The RGB distance below which two clusters are combined
Palette	The extracted colours, sorted by their share of the picture
Pin	An exact pixel colour kept by clicking the preview; up to 24
Population	A cluster's share of the sampled pixels
Quality	How densely the picture is sampled, from Fast to Max detail
Relative luminance	The WCAG measure of a colour's brightness, from 0 for black to 1 for white
Sample	One pixel read from the working copy for the extraction
Tone band	Shadows, Midtones or Highlights, by relative luminance
WCAG	The Web Content Accessibility Guidelines, which define the contrast ratio and its ratings
Working copy	The picture shrunk to fit 800 × 500 pixels, which the tool samples and the eyedropper reads

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Limits and accuracy

How the Image Picker makes a palette, how precise its figures are, and what it is not.

How the palette is made

Step	Detail
Working copy	The picture is shrunk to fit 800 × 500 pixels; it is never enlarged
Sampling	A regular grid set by **Quality**, widened so that no more than about 50,000 pixels are sampled
Skipped pixels	Nearly transparent pixels; with the switches on, relative luminance below 0.03 or above 0.97
Exact colours	Used when the samples hold no more distinct colours than the cluster count

Step	Detail
Clustering	k-means++ in RGB, up to 20 refinement rounds, stopping early once fewer than 1 in 2,000 samples change cluster
Merging	Clusters closer than the merge threshold in RGB are combined, weighted by size
Shares	Each cluster's sample count divided by all samples

Precision

- **Swatch colours** are averages of their clusters, rounded to whole RGB values. The same picture can give slightly different averages on different runs, because k-means++ starts from random colours.
- **Shares** are shares of the sampled pixels, not of every pixel in the original picture. With widely spaced sampling, very small areas may not be sampled at all.
- **The eyedropper and pins** read the working copy. On a large photograph each working pixel stands for a block of original pixels, so a pin is exact to the working copy, not to a single original pixel.
- **Relative luminance and contrast ratios** follow the WCAG 2 definitions exactly, from the sRGB values.
- **CIELAB and LCH** are calculated from sRGB with a D65 white.
- **Clusters are formed in RGB.** RGB distance does not match perceived difference evenly, so two swatches can look closer or further apart than their separation suggests.

Limits

- One picture at a time.
- Images from other websites load only if their server allows cross-origin reading.
- The slider sets 2 to 12 clusters; 16 is used only before it is first moved.
- Up to 24 pins.
- Harmony uses HSL hue and fixed angle ranges. Greys have no meaningful hue.
- Contrast pairs cover the eight highest-contrast pairs of palette colours; pins are not included.
- Batch analysis reads three- and six-digit hex correctly; values with an alpha part are not.
- The page does not compare palettes side by side, even though it counts extractions.
- The page shows no confirmation messages.

What it is not

The palette describes which colours occupy the picture and how much of it. It is not a judgement of which colours are important to the picture's meaning, and the harmony readout is a convention, not a measure of quality. The contrast ratings are a first pass; confirm text and background pairs in a dedicated contrast tool.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Privacy

What the Image Picker does with your pictures, and when it uses the network.

Pictures from your device stay in your browser

- A picture you drop or choose is read by the page itself and drawn to a canvas in your browser. The tool does not upload it or send it anywhere.
- The extraction, the readouts, the charts and the batch analysis are all worked out on your device.

- Downloads are created in the browser and saved by your browser's normal download.
- The clipboard is written only when you select a copy button or press `H` or `J`, or when you click a swatch or pin.

Pictures from a web address

When you use **Load URL**, your browser requests the image from the address you typed, exactly as it would when you open that address. The site that hosts the image sees that request. The Image Picker does not send the image anywhere afterwards.

Nothing is kept after you leave

The picture, the palette and the pins are not stored. **Reset**, reloading or leaving the page clears them. The only thing remembered is whether you closed the **Automate this** line, which is kept in your browser.

What an export contains

The JSON export contains the picture's file name (or the last part of its address), its sizes, your settings, the palette and your pins. It does not contain the picture itself.

The page itself

The Image Picker is a normal page on auricartisan.com. Site-wide features such as sign-in, preferences and any consent-based services follow the site's own privacy policy and your cookie choices; see Help.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Release notes

User-facing history of the Image Picker.

No dated changelog entry for the Image Picker has been published on the site. This page records what can be confirmed.

Current version

- Three tabs: **Workbench**, **Analysis** and **Actions**.
- The whole middle panel is the drop target before a picture is loaded.
- The four charts live on their own **Analysis** tab, with full-screen viewing.
- The keyboard keys are listed on the **Workbench** before a picture is loaded.
- A line under the heading points to the equivalent REST endpoint and its cost in API credits.

25 May 2026 — user guide published

The Image Picker User Guide was published in the site's documentation library. It describes a layout with two tabs, **Lab** and **Actions**, with the charts on the **Lab** tab. The extraction method, eyedropper, pins, exports and batch analysis it describes are the same as today.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › IMAGE PICKER

Related tools and reading

Other Auric Artisan tools, the public API and Learn articles that go with the Image Picker.

Tools on the site

Tool	Why you might use it with the Image Picker
Contrast Checker	Test the exact text and background pair you chose from the palette
Vision Simulation	See how the picture and palette look with different kinds of colour vision
M-Gradient Palette	Build an evenly stepped ramp between two colours taken from the picture
SVG Recolour	Apply colours from the palette to SVG artwork
Collage Maker	Combine several pictures into one before extracting a palette from all of them
Basic Tools	Quick palette, gradient and conversion utilities

The public API

The same job is available as a REST call, `POST /v1/palette/from-image`, documented at https://auricartisan.com/library/documentation/api/#ep-post-palette-from-image. See Public API for keys, credits and limits.

Learn articles

- Color Theory and Harmony — https://auricartisan.com/library/learn/articles/2026-06-04-color-theory-and-harmony
- Accessible Color Design and WCAG Contrast — https://auricartisan.com/library/learn/articles/2026-05-25-wcag-contrast-accessible-color
- ΔE Metrics from CIE76 to CIEDE2000 and ΔEITP — https://auricartisan.com/library/learn/articles/2026-05-26-delta-e-metrics-cie76-ciede2000-itp
- The Vocabulary of Color: Hue, Saturation, Value, Tint, Tone, and Shade — https://auricartisan.com/library/learn/articles/2026-06-29-vocabulary-of-color-hue-saturation-value
- Skin Tones and Inclusive Color — https://auricartisan.com/library/learn/articles/2026-07-18-skin-tones-and-inclusive-color

Other documentation

- The group overview: Image and file tools.
- The site's documentation library carries an "Image Picker User Guide" and an "Image Picker Developer Reference", both dated 25 May 2026. They describe an earlier layout with two tabs; where they differ from this guide, this guide follows the current page.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

LUT Lab

What LUT Lab does, who it is for, where to find it and how to try a first look and write a .cube file.

LUT Lab reads, applies and writes colour lookup tables. A lookup table, or LUT, is a file that says what colour each input colour should become; colourists use LUTs to carry a grade or a conversion from one program to another. LUT Lab works with the `.cube` format, in its one-dimensional (1D) and three-dimensional (3D) forms.

The **Lab** tab shows a scene before and after a transform, side by side. The transform is either one of six built-in looks, such as **Teal and orange** or a **Reinhard tone-map**, or a `.cube` file you load. Under the pair, the tool measures what the transform moved: the average and largest colour difference in CIEDE2000, the share of pixels it clipped to pure white or black, and, on a ring of test colours, how far it turned hue and changed lightness and chroma. You can test on four drawn scenes or on your own picture.

The **Export** tab writes the transform in force as a `.cube`, at any size from 2 to 65, as a 3D cube or a 1D curve. The file's header states plainly that the tool works in 8 bits and that the sample order is red fastest, as the format requires. Everything runs in your browser; nothing is uploaded.

Who it is for

- Colourists and editors checking what a `.cube` does before using it in a grading or editing program.
- Designers and developers who want a simple look as a `.cube` file.
- Anyone who needs to resample a `.cube` to a different size, or turn it into a 1D curve.
- Students learning how LUTs, interpolation and clipping work.

Where to find it

Open https://auricartisan.com/tool/general/tools/lut-lab/ in a current browser. It is listed as **LUT Lab** on the Collections page at https://auricartisan.com/collections/. Nothing needs installing.

Quick start

1. Open the tool. The **Lab** tab shows the **Neutral wedge** scene with the **Teal and orange** look applied.
2. Select another scene, for example **Colour patches**, or drop your own picture on **Drop an image**.
3. Choose a look from the menu under **The look**, and move **Strength** to set how much of it to apply.
4. Read the cards under **What it moved** and the hue circle under **Where it moved things to**.
5. To test a file instead, drop a `.cube` on **Drop a .cube**.
6. Open the **Export** tab, choose **Kind** and **Size**, and select **Save the .cube**.

What you can do

- Apply six built-in looks: Teal and orange, Bleach bypass, Film soft, Mono contrast, Log-ish lift and Reinhard tone-map, with a strength control and, for the tone-map, an exposure control.
- Load any 1D or 3D `.cube` file, see its size, domain and any problems with it, and apply it.
- Test on four drawn scenes or on your own picture, scaled to fit 480 × 270.
- See the pixels a transform drove to white or black, highlighted on the result.
- Measure mean, 95th-percentile and largest CIEDE2000 difference, and the share of clipped pixels.
- See hue, lightness and chroma shifts on a 72-point ring of constant lightness in CIELAB.

- Write a 3D `.cube` (2^3 to 65^3) or a 1D `.cube` (2 to 65 entries) of the transform in force, and copy it to the clipboard.
- Save the hue sweep as CSV and a full report as JSON, and copy a link that restores your settings.

In this folder

- docs/ — how to use the tool.
  - Getting started — open it, the screen tour, a first look.
  - LUT concepts — 1D and 3D LUTs, sizes, domains, interpolation, clipping and bit depth.
  - Apply a look — scenes, your own picture, the six looks, strength and exposure.
  - Load a .cube file — what the reader accepts and what it reports.
  - Read what a transform moved — the clip overlay, the difference cards and the hue circle.
  - Write a .cube file — kinds, sizes, the header, round trips and the other exports.
  - Reference — every control, readout, format and message.
  - Troubleshooting — symptoms, causes and fixes.
  - FAQ — short answers to common questions.
- others/ — background material.
  - Glossary — terms used in this tool.
  - Limits and accuracy — the 8-bit path, what is measured and what is not held.
  - Related — related tools and Learn articles.
  - Release notes — user-facing history.
  - Privacy — what stays in your browser.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

LUT Lab documentation

The list of guides, the concept page, the reference, troubleshooting and FAQ for LUT Lab.

This folder holds the task guides for LUT Lab, a concept page on how lookup tables work, the full reference of controls and formats, and the troubleshooting and FAQ pages. If LUTs are new to you, read LUT concepts first.

File	What it covers
getting-started.md	Opening the tool, a tour of the six tabs, and a first look
lut-concepts.md	1D and 3D LUTs, grid size, domain, sample order, interpolation, clipping and 8-bit precision
apply-a-look.md	Choosing a scene or your own picture, the six looks, strength, exposure and the Looks tab
load-a-cube-file.md	Loading a .cube, the file card, warnings and errors, and switching back to the looks
read-what-it-moved.md	The clip overlay, the difference cards and the hue circle
write-a-cube-file.md	Exporting a .cube, resampling, the header, round-trip accuracy, CSV, JSON and the share link
reference.md	Every control, readout, keyword, file format and message
troubleshooting.md	What to do when a file will not load or a result looks wrong
faq.md	Short answers to common questions

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

Getting started with LUT Lab

Open LUT Lab, find your way around its six tabs, and try a first look.

Open the tool

Go to https://auricartisan.com/tool/general/tools/lut-lab/. The page opens on the **Lab** tab, showing the **Neutral wedge** scene before and after the **Teal and orange** look.

A note under the heading explains what the numbers on the page are measurements of. It has no close button on purpose.

The screen at a glance

Tab	What it is for
Lab	The scene, the transform, the before and after pair, and what the transform moved
Method	Five short explanations: sample order, interpolation, clipping, round trip and linear light
Looks	A card for each built-in look, with what it is and what it does to a test strip
Data	Every figure and what it rests on; the tab shows the number of entries (17)
Export	Kind and size of the table, the file preview, and the save and copy buttons
Reference	The published sources, marked by whether they run in the tool

The Lab tab

- **The left-hand panel** holds **The scene** (four scene buttons and a **Drop an image** box), **The look** (a menu, **Strength** and **Tone-map exposure**), **A .cube file** (a **Drop a .cube** box and **Clear the file**), and **Show** (the **What the transform crushed**

switch and **Copy a link to this setting**).

- **The right-hand panel** holds **Before and after**, a key to the clip overlay, the cards under **What it moved**, and the hue circle under **Where it moved things to** with its own cards.

Try a first look

1. On the **Lab** tab, select **Colour patches** under **The scene**. The pair shows 24 flat colour patches before and after.
2. Open the menu under **The look** and choose **Bleach bypass**. The right-hand image loses colour and gains contrast. The label above the pair reads **Bleach bypass**.
3. Move **Strength** to 50%. The look is blended halfway with the original.
4. Read **What it moved**. **mean ΔE_{00}** is the average colour difference between before and after; **clipped** is the share of pixels driven to pure white or black.
5. Look at the hue circle. Each hollow dot is a test colour before the look; the filled dot joined to it is where the look moved it.
6. Open the **Export** tab. Leave **Kind** on **3D cube** and **Size** on 33, and select **Save the .cube**. Your browser downloads `auric-bleach-bypass-33.cube`.

You now have a `.cube` of the look at half strength, ready to load in a program that reads `.cube` files.

Next steps

- To understand sizes, domains and why the file header mentions 8 bits, read LUT concepts.
- To test a `.cube` you already have, read Load a .cube file.
- To test on your own picture, see Apply a look.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

Apply a look

Choose a test scene or your own picture, apply one of the six built-in looks, and set its strength and exposure.

Choose what to test on

Under **The scene** on the **Lab** tab, select one of four drawn scenes. Each is 320 × 180 pixels.

Scene	What it contains	Good for spotting
Neutral wedge	A 16-step grey ramp above 16-step red, green and blue ramps	Banding, contrast changes and colour casts in greys
Colour patches	24 flat patches: six hues at four lightness levels	Hue and saturation shifts across the colour wheel
Skin tones	Six skin-like tones from dark to light on a mid-grey surround	What a look does to faces
Saturation field	Hue across, from grey at the bottom to strong colour at the top	Hue twists and clipping in saturated colours

The scenes are drawings, not photographs. They show the shape of a transform evenly, but they have no grain, lens flare or real-camera colour.

Use your own picture

1. Drop an image on **Drop an image**, or select the box to choose one.
2. The picture is scaled to fit 480 × 270, keeping its proportions (it is never enlarged), and becomes the scene. A message confirms the size, for example **Loaded beach.jpg at 480×270**.

3. A fifth button with your file name appears among the scenes, selected.
4. To go back, select **Back to the drawn scenes**, or any of the four scene buttons.

Every figure on the page is then computed from your picture, and the **Data** tab marks the scene as your own file.

Choose a look

Open the menu under **The look**.

Look	Kind	What it does
None	—	Leaves the image unchanged
Teal and orange	Stylisation	Warms the highlights and cools the shadows. The look shown when the page opens
Bleach bypass	Stylisation	Pulls colours towards grey and raises contrast. Named after a film process it does not simulate
Film soft	Stylisation	Lifts the black point and lowers the white point, with a slight warm cast. Not based on a measured film stock
Mono contrast	Stylisation	Black and white from Rec. 709 luminance in linear light, then an S-shaped contrast curve
Log-ish lift	Stylisation	A 0.45 power curve with a lift and a gain, which brightens the image, lifting the shadows most
Reinhard tone-map	Transform	The Reinhard photographic tone-mapping operator, applied in linear light

The label above the pair names the transform in force.

Set the strength

Strength, from 0% to 100% (default 100%), blends the look with the original. At 50%, each channel of each pixel is halfway between its original value and the look's value. **Strength** applies to the looks only, not to a loaded `.cube` file.

Set the tone-map exposure

Tone-map exposure, from 0.25× to 8× (default 2.00×), is active only for **Reinhard tone-map**; it is greyed out for every other look and for a loaded file.

The tone-map works in linear light: it decodes the sRGB value to light, multiplies it by the exposure, compresses it with Reinhard's formula L ÷ (1 + L), and encodes it back to sRGB. Higher exposure brightens the picture and compresses the highlights more; lower exposure darkens it. An 8-bit picture carries no record of how bright the original scene was, so the exposure is a choice rather than a measured value.

Compare looks on the Looks tab

The **Looks** tab has a card for each look except **None**. The heading note counts them: five stylisations and one transform, plus two withdrawn.

Each card shows:

- the look's name and a badge, **Stylisation** or **Transform**;
- a strip of 18 test colours, a 12-step grey wedge and the six pure primaries and secondaries, before (top) and after (bottom);
- a sentence on what the look is;
- **mean ΔE_{00}**, **max hue** and **clipped** on the 72-colour hue ring (see Read what a transform moved);

- **Show it on the pair**, which applies the look and returns you to the **Lab** tab.

Under **Withdrawn**, two former features are explained: an ACEScct to ACEScg conversion, which an 8-bit canvas cannot hold, and a 32-bit float export. A table shows what the ACEScct decode produces for several input codes, and how many of the 256 codes would come out pure white.

Show what the transform crushed

The **What the transform crushed** switch under **Show** (on by default) hatches the pixels the transform drove to white in pink and those driven to black in blue, on the right-hand image. See Read what a transform moved.

What you should see

The right-hand image shows your scene with the look at the strength you chose, and the figures under the pair describe exactly that combination.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

Load a .cube file

Load a 1D or 3D .cube file, read what the tool reports about it, and switch between the file and the built-in looks.

Load the file

1. On the **Lab** tab, drop a `.cube` file on **Drop a .cube**, or select the box to choose one. The chooser offers `.cube` and plain-text files.
2. A message confirms **Loaded** followed by the file name, and says how many notes there are if the tool found any.
3. The before and after pair now shows your file applied to the current scene. The label above the pair shows the file's title, or its name if it has none.

While a file is shown, the look menu and **Strength** are greyed out and the note under them reads **The pair is showing the loaded file. The looks are paused.** A loaded file is always applied at full strength.

The file card

A card under the drop box describes the file.

Line	Meaning	Example
Heading	The `TITLE` in the file, or the file name	Warm Film
kind	1D or 3D	3D
size	For 3D, N^3 and the number of rows; for 1D, the number of rows	33^3 (35937 rows)
domain	The declared input range for red, green and blue	[0.000, 0.000, 0.000] – [1.000, 1.000, 1.000]
reachable	The share of the table that an 8-bit sRGB input can reach	100.0%

Under these lines is a list of notes. When there is nothing to report it reads **no discrepancies between the header and the body**.

What the reader accepts

LUT Lab reads the `.cube` format as defined in the Adobe / IRIDAS Cube LUT Specification 1.0.

Item	Handling
`TITLE "…"`	Read and shown as the file's name
`LUT_3D_SIZE N`	A 3D table of N × N × N rows, red index changing fastest
`LUT_1D_SIZE N`	A 1D table of N rows
`DOMAIN_MIN r g b`, `DOMAIN_MAX r g b`	The input range; 0 to 1 when absent
Lines starting with `#`	Comments, ignored
Blank lines	Ignored
Rows of three numbers	Table data
Any other keyword	Ignored

Only `.cube` is supported. Other LUT formats, such as `.3dl`, need converting to `.cube` in another program first.

Notes the tool may add

Note	Meaning	What happens
The file carries N rows beyond the ones the header declares	Extra data rows	The extra rows are ignored
No `LUT_3D_SIZE` in the file, but the row count is a whole cube	The size keyword is missing	The file is read as a 3D table of that size
No size keyword, and the row count is not a cube	The size keyword is missing	The file is read as a 1D table with one row per line
The declared domain is wider than 0 to 1	Part of the table is for inputs above 1 or below 0	The reachable share is given; only that part can affect an 8-bit picture

The specification requires a size keyword, so a file that needed a guess is worth fixing at its source.

Files that cannot be read

Message	Why
empty file	The file has no content
`LUT_3D_SIZE` N needs M rows; the file has K	The file is shorter than its header says
`LUT_1D_SIZE` N needs N rows; the file has K	As above, for a 1D file
no `LUT_1D_SIZE` or `LUT_3D_SIZE` keyword and no usable sample rows	Nothing in the file looks like a table

When a file cannot be read, the card shows **status: not loaded** with the reason, and the pair keeps showing the look.

How the file is applied

Each pixel of the scene is taken as 8-bit sRGB values, scaled to 0 to 1, placed within the file's domain, and looked up: with trilinear interpolation for a 3D table, or straight-line interpolation per channel for a 1D table. The result is rounded back to 8

bits. See LUT concepts.

Switch between the file and the looks

- **Back to the looks** (shown while a file is applied) returns the pair to the built-in look and keeps the file loaded.
- Choosing a look from the menu, or **Show it on the pair** on the **Looks** tab, also switches back to the looks.
- **Clear the file** removes the file entirely and returns to the looks. The message reads **Cleared the loaded file**.

To apply the file again after going back to the looks, drop it again.

Resample or convert the file

While a file is applied, the **Export** tab writes that file's transform. You can use this to save the same LUT at a different size, or as a 1D curve. See Write a .cube file for what changes when you do.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

LUT concepts

What a lookup table is, 1D and 3D LUTs, grid sizes, domains, sample order, interpolation, clipping and 8-bit precision.

This page explains the ideas behind LUT Lab's controls and figures.

What a LUT is

A lookup table (LUT) is a list that says, for a set of input colours, what each should become. Software applies it to every pixel of an image or video. Because it is just a table of numbers, a LUT carries a look or a conversion from one program to another without carrying the program's settings.

A LUT does not know what the colours mean. It works on whatever numbers it is given, so a LUT made for one kind of footage gives the wrong result on another. LUT Lab applies LUTs to ordinary 8-bit sRGB values, the kind of colour a web page or a JPEG holds.

1D and 3D LUTs

	1D LUT	3D LUT
What it maps	Each channel on its own: red in to red out, green to green, blue to blue	Every combination of red, green and blue to a new colour
Can express	Brightness, contrast, gamma, per-channel curves, colour casts that depend only on each channel	Anything a 1D can, plus hue shifts, saturation changes and effects that mix channels
Size	N rows, each holding a red, green and blue output	N × N × N rows
Keyword in `.cube`	`LUT_1D_SIZE`	`LUT_3D_SIZE`

A 3D LUT of size N is a cube of N points along each axis. The number of rows grows quickly:

Size	Rows in a 3D LUT
17	4,913
33	35,937

Size	Rows in a 3D LUT
65	274,625

Bigger cubes follow a curved transform more closely and make larger files. 17, 33 and 65 are the customary sizes.

The domain

A `.cube` may declare the range of input values its table covers with `DOMAIN_MIN` and `DOMAIN_MAX`. Without them, the domain is 0 to 1 on every channel.

LUT Lab's input is always an 8-bit sRGB pixel, which spans exactly 0 to 1. If a file declares a wider domain, for example 0 to 4 for high-dynamic-range footage, only part of its table can be reached from here. LUT Lab tells you what share is reachable, for example **reachable 25.0%**, and adds a note to the file card.

Sample order

A 3D `.cube` lists its rows with the red index changing fastest, then green, then blue. The first rows are red 0, 1, 2… at green 0 and blue 0. A program that reads or writes the order the other way round swaps the red and blue axes, which gives a plausible-looking but wrong result. LUT Lab reads and writes red fastest, as the format requires, and the **Method** tab shows the first eight rows of a 2 × 2 × 2 table in order.

Interpolation

An input colour rarely lands exactly on a grid point. For a 3D LUT, LUT Lab finds the small cube of eight grid points around the input and blends them, first along red, then green, then blue. This is trilinear interpolation. For a 1D LUT, it blends the two neighbouring rows of each channel. An input exactly on a grid point reads that point.

A straight-line transform, such as an identity (a LUT that changes nothing), is reproduced exactly at any size. Curves are only approximated between grid points, which is why larger cubes are more faithful to curved looks.

Clipping

A pixel is **clipped** when the transform drives one of its channels to 0 or 255 that was not already there. Clipping loses detail: highlights driven to white can no longer be told apart, and neither can shadows driven to black.

This is different from asking whether an output channel is at 0 or 255. A pure red pixel has green and blue at 0 before any transform, so it is not a casualty when a transform leaves it alone. LUT Lab's clip figure counts only pixels the transform actually pushed to an end.

8-bit precision

LUT Lab reads pixels from, and draws them to, a canvas at 8 bits per channel: 256 levels, from 0 to 255. Every value it writes into a `.cube` is therefore one of 256 levels divided by 255, written to the six decimal places the format uses. The file's header says so, because a `.cube` has no field of its own for precision, and a table written from 8 bits otherwise looks the same as one written from 32.

When grid points land on 8-bit values

A 3D grid of size N has points at 0, 1/(N−1), 2/(N−1) … 1. These land exactly on 8-bit values only when N−1 divides 255. Within the sizes LUT Lab writes, that is true for 2, 4, 6, 16, 18 and 52, and not for the customary 17, 33 and 65. At the other

sizes, each grid point is written to the nearest 8-bit value, up to half a step away. An 8-bit image put through the file is unchanged by this; an application reading the file at higher precision sees the small gap.

Stylisations and transforms

LUT Lab separates two kinds of look:

- A **stylisation** is a curve someone designed because it looks good. It has no definition to check against. Five of the built-in looks are stylisations.
- A **transform** claims to convert between two defined things and can be checked against its definition. The Reinhard tone-map is the one built-in transform.

Measuring colour difference

LUT Lab measures how far a transform moved each colour with CIEDE2000 (ΔE_{00}), the CIE's standard formula for colour difference, computed in CIELAB with a D65 white. Zero means no change, and larger numbers mean larger visible changes.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

Read what a transform moved

Interpret the clip overlay, the colour difference cards and the hue circle on the Lab tab.

Everything below the before and after pair on the **Lab** tab describes the difference between the two images, for the transform currently shown: a look at its strength, or a loaded file.

The pair

- **Before** is the scene, with its pixel count beside the label, for example **57600** for a 320 × 180 scene.
- **After** is the scene through the transform, with the clipped share beside the label, for example **0.83% clipped**.

The clip overlay

With **What the transform crushed** switched on, the **After** image is hatched with diagonal stripes where the transform clipped a pixel:

Stripes	Meaning
Pink	A channel was driven to white (255) that was not at 255 before
Blue	A channel was driven to black (0) that was not at 0 before

Pixels that were already at 0 or 255 and stayed there are not marked. Switch the overlay off to see the clean result; the figures do not change.

What it moved

Card	Meaning
mean ΔE_{00}	The average CIEDE2000 colour difference between each pixel before and after
95th percentile	The difference that 95% of pixels are at or below; shows how big the typical large change is
largest	The single largest difference on any pixel

Card	Meaning
clipped	The share of pixels with at least one channel driven to 0 or 255 by the transform; the card turns red above 1%
the old test said	The share of output pixels with any channel at 0 or 255, whatever it was before

The last card is the test an earlier version of the tool used. It counts every saturated or black or white pixel in the result, even those the transform did not touch, so it is shown only for comparison. On the **Colour patches** or **Saturation field** scene, compare it with **clipped** to see the difference.

Where it moved things to

The hue circle tests the transform on 72 colours arranged round a ring in CIELAB, at a constant lightness (L* 60) and chroma (C* 40), one every 5° of hue.

- The angle round the circle is the CIELAB hue; the distance from the centre is the chroma. Faint rings mark every 20 units of chroma.
- Each hollow dot is a test colour before the transform. The filled dot, joined to it by a thin line, is where the transform moved it.
- A pink ring around a filled dot marks a test colour the transform clipped.

Not every colour at L* 60, C* 40 fits inside sRGB. Test colours that do not fit are pulled in towards the centre, keeping their hue, until they do. The caption says how many: **13 of 72**, whatever the transform, because the ring is the same every time. This is done before the transform runs, so the shift you see belongs to the transform, not to the gamut.

The circle cards

Card	Meaning
largest hue shift	The largest turn of hue on any test colour, in degrees, with its direction (+ or −)
mean hue shift	The average size of the hue turn across all 72, in degrees
*largest L shift**	The largest change in CIELAB lightness, with its direction
clipped on the circle	How many of the 72 test colours the transform clipped

Hue alone rarely tells the whole story: most looks move lightness and chroma at the same time, often by more. The CSV export of the hue sweep gives all three for every test colour; see Write a .cube file.

Reading the numbers together

- A look with a small **mean ΔE_{00}** but a large **largest** changes most colours a little and a few a lot. Find the few with the overlay or the hue circle.
- A high **clipped** figure on your own picture means detail is being lost in highlights or shadows. Lower **Strength**, or for the tone-map, change **Tone-map exposure**.
- A consistent twist on the hue circle, all dots turning the same way, is a hue rotation. Dots moving inwards or outwards are chroma changes.

The Method tab

The **Method** tab explains the method in five sections, chosen with the buttons under **The chain**:

Section	What it shows
Sample order	How a `.cube` indexes its rows, with the first eight rows of a 2 × 2 × 2 table
Interpolation	How trilinear interpolation works, and the worst error on an identity at sizes 2 to 65
Clipping	Seven test pixels judged by the old test and by the clip test used now
Round trip	Writing a table and reading it back, and whether an identity survives at various sizes
Linear light	The tone-map applied in linear light compared with applying it to encoded values

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

Write a .cube file

Export the transform in force as a 3D or 1D .cube at any size, understand the header and round-trip accuracy, and save the hue sweep, report and share link.

The **Export** tab writes the transform currently shown on the **Lab** tab: a look at its strength and exposure, or a loaded `.cube` file.

Save a .cube

1. On the **Lab** tab, set up the transform you want: choose a look and its **Strength**, or load a `.cube`.
2. Open the **Export** tab.
3. Choose **Kind**: **3D cube** (default) or **1D curve**.
4. Set **Size** from 2 to 65 (default 33). The label shows, for example, **33^3** for a 3D cube.
5. Check the lines under the slider: **rows**, **file size** and **file name**, and the note on whether the grid lands on 8-bit values.
6. Select **Save the .cube**. Your browser downloads the file, for example `auric-teal-and-orange-33.cube`, and a message confirms it.

To paste the table somewhere instead, select **Copy it**. The message reads **Copied the .cube**, or **The clipboard refused** if clipboard access is blocked.

The file name is `auric-`, the transform's name in lower case with spaces and symbols turned into hyphens, the size, and `-1d` for a 1D curve.

3D cube or 1D curve

Kind	What it captures	Rows
3D cube	The full transform: every mix of red, green and blue	Size³
1D curve	Only what the transform does to neutral greys, written as a curve per channel	Size

A 1D curve is made by sending greys through the transform and recording each output channel. Any effect that depends on how channels mix, such as a hue shift or a change in saturation, is lost. **Mono contrast**, for example, cannot be expressed as a 1D curve: the curve would leave colours coloured. Use **1D curve** only for transforms that act on each channel separately, such as **Film soft** or **Log-ish lift**.

Choosing a size

- **17**, **33** and **65** are the sizes most programs expect.

- Larger sizes follow curved transforms more closely and make larger files; a 65^3 cube has 274,625 rows.
- Sizes **2**, **4**, **6**, **16**, **18** and **52** place every grid point exactly on an 8-bit value. At other sizes, grid points are written to the nearest 8-bit value, up to half a step away. The note under the slider says which applies.

The file preview

The file, as it will be written shows the first 26 lines of the file, a count of the lines not shown, and the last row. Above it, **What the file will say about itself** explains why the header states the precision.

A saved file looks like this:

```
# Auric Artisan — LUT lab
# https://auricartisan.com/tool/general/tools/lut-lab/
#
# Transform: Teal and orange
# Kind: stylisation
#
# Sample order: red varies fastest, then green, then blue.
# Precision: this pipeline is 8-bit end to end. Every value below
# is one of 256 levels written to six decimals;
# the decimals are the format's, not this tool's resolution.
# (size-1) does not divide 255, so the grid nodes fall between code values and
# are written to the nearest one - up to half a code out. An 8-bit result is
# unaffected; anything reading this file at higher precision is not.
TITLE "Auric Artisan — Teal and orange"
LUT_3D_SIZE 33
DOMAIN_MIN 0.0 0.0 0.0
DOMAIN_MAX 1.0 1.0 1.0

0.000000 0.000000 0.121569
…
```

Part	Meaning
Lines starting with `#`	Comments: the tool, the transform, its kind, strength and exposure when they apply, sample order and precision
`TITLE`	"Auric Artisan — " and the transform's name
`LUT_3D_SIZE` or `LUT_1D_SIZE`	The size you chose
`DOMAIN_MIN`, `DOMAIN_MAX`	Always 0 to 1
Data rows	Red, green and blue output, six decimals, red index changing fastest

A **Strength** below 100% adds a `Strength:` line, and the tone-map adds an `Exposure:` line. For a loaded file, the `Transform:` line gives the file's title or name. A 1D curve, or a 3D cube at a size whose grid lands on 8-bit values, leaves out the three lines about grid nodes. The first data row shown is what **Teal and orange** does to black: it lifts the blue.

Round trips: what survives

Two different questions are worth separating.

1. **Do the numbers survive the file?** Yes. A table written here and read back here reproduces every value to the six decimals the format writes.
2. **Does an identity come back as an identity?** For an 8-bit picture, yes, at every size: an identity table written here and applied back to an 8-bit image leaves every pixel unchanged. But at sizes where the grid does not land on 8-bit values

(including 17, 33 and 65), the values in the file sit up to half an 8-bit step off the exact grid. An application reading the file at higher precision sees an identity that is very nearly, but not exactly, one.

The **Round trip** section of the **Method** tab shows both measurements.

Resampling a loaded file

While a loaded `.cube` is applied, the export resamples it: every grid point of the new table is an 8-bit input looked up in your file with trilinear interpolation, and the result is rounded to 8 bits. The new file follows the original closely but is not a bit-for-bit copy, and a 1D export keeps only what the file does to greys. Keep your original file.

Other exports

Button	File	Contents
Hue sweep, CSV	`auric-lut-lab-hue-sweep.csv`	The 72-colour hue ring: shift in hue, lightness and chroma, ΔE_{00}, and whether clipped
Report, JSON	`auric-lut-lab-report.json`	The transform, the scene, the figures under the pair, the pipeline and the Data register

The hue sweep CSV

```
# Auric Artisan — LUT lab, hue sweep
# transform,Teal and orange
# constant L*,60
# requested C*,40
hue_deg,hue_shift_deg,lightness_shift,chroma_shift,dE00,clipped
0,1.9855,-0.2300,1.3764,0.9428,no
```

One row per test colour, every 5° from 0 to 355. Shifts and differences have four decimal places. The row shown is the first row for **Teal and orange**.

The report JSON

Field	Contents
`tool`, `url`, `generated`	The tool, its address, and when the report was made
`transform`	Name, whether it is a look or a file, the look, strength and exposure
`scene`	A drawn scene's id, or your picture's file name
`image`	Pixel count, mean, 95th-percentile and largest ΔE_{00}, clipped share, and the old test's share
`pipeline`	256 levels per channel, the sample order, and a note on precision
`register`	The status, source and table of every **Data** tab entry

Share your settings

Copy a link to this setting on the **Lab** tab copies an address that restores the scene, look, strength, exposure, and the export size and kind. For example:

```
https://auricartisan.com/tool/general/tools/lut-lab/?scene=patches&look=film-soft&s=0.5&e=2&n=33&k=3D
```

The link does not carry a loaded `.cube` or your own picture. Whoever opens it must load those again.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

LUT Lab reference

Every tab, control, readout, .cube keyword, export format, link parameter and message in LUT Lab.

Address: https://auricartisan.com/tool/general/tools/lut-lab/

Tabs

Tab	Contents
Lab	Scene, look, .cube file, overlay switch, share link; the before and after pair and what the transform moved
Method	Five explanations: **Sample order**, **Interpolation**, **Clipping**, **Round trip**, **Linear light**
Looks	A card per look with a test strip and figures; the two withdrawn features
Data	The 17 entries every figure rests on, filterable by status
Export	Kind, size, file details, save and copy buttons, the file preview
Reference	The ten sources, filterable by whether they run

Lab tab — The scene

Control	What it does	Default
Neutral wedge	16-step grey ramp over red, green and blue ramps, 320 × 180	Selected
Colour patches	24 flat patches, six hues at four lightness levels	—
Skin tones	Six skin-like tones on a mid-grey surround	—
Saturation field	Hue across, chroma from none at the bottom to strong at the top	—
Drop an image	Loads your picture, scaled to fit 480 × 270, as the scene	—
Back to the drawn scenes	Removes your picture and returns to the drawn scenes	Shown with your picture

Lab tab — The look

Control	Values	Default
Look menu	None, Teal and orange, Bleach bypass, Film soft, Mono contrast, Log-ish lift, Reinhard tone-map	Teal and orange
Strength	0% to 100%, step 1%	100%
Tone-map exposure	0.25× to 8×, step 0.05; active only for Reinhard tone-map	2.00×
Note	**The pair is showing a look this page computes.** or **The pair is showing the loaded file. The looks are paused.**	—

The look menu and **Strength** are disabled while a loaded file is shown.

Lab tab — A .cube file

Control	What it does
Drop a .cube	Loads a dropped `.cube`, or opens a chooser (`.cube`, plain text) when selected or on `Enter` or `Space`
File card	Title, **kind**, **size**, **domain**, **reachable**, and notes or errors
Back to the looks	Shows the look again, keeping the file loaded; visible while the file is shown
Clear the file	Removes the file and shows the look

Lab tab — Show

Control	What it does	Default
What the transform crushed	Hatches clipped pixels on the **After** image: pink for driven to white, blue for driven to black	On
Copy a link to this setting	Copies an address that restores the settings	—

Lab tab — Readouts

Readout	Meaning
Label above the pair	The transform shown
Before caption	Pixel count of the scene
After caption	Share of pixels clipped
mean ΔE_{00}	Average CIEDE2000 difference, before to after
95th percentile	Difference that 95% of pixels are at or below
largest	Largest difference on any pixel
clipped	Share of pixels with a channel driven to 0 or 255; red above 1%
the old test said	Share of output pixels with any channel at 0 or 255
Hue circle	72 test colours at L* 60, C* 40: hollow before, filled after, pink ring if clipped
largest hue shift	Largest hue turn, degrees, signed
mean hue shift	Average absolute hue turn, degrees
*largest L shift**	Largest lightness change, signed
clipped on the circle	Test colours clipped, out of 72
Caption count	Test colours pulled inside sRGB before the test: 13 of 72

ΔE figures have two decimal places, percentages two, hue and lightness shifts one.

Looks tab

Each card: name, **Stylisation** or **Transform** badge, an 18-colour before and after strip, a description, **mean ΔE_{00}**, **max hue** and **clipped** on the hue ring, and **Show it on the pair**.

Look	Kind	mean ΔE₀₀ on the ring	max hue	clipped
Teal and orange	Stylisation	1.2	+3.5°	0/72
Bleach bypass	Stylisation	7.9	−12.9°	0/72
Film soft	Stylisation	1.9	−7.7°	0/72
Mono contrast	Stylisation	22.6	+177.8°	0/72
Log-ish lift	Stylisation	16.9	−25.2°	0/72
Reinhard tone-map	Transform	6.6	−25.1°	0/72

These are the figures at full strength, and for the tone-map at the default exposure of 2×. **Mono contrast** removes colour entirely, so its hue figure has no useful meaning.

Withdrawn: ACEScct to ACEScg and **Export 32-bit float samples**, each with why and what it would take, and a table of the ACEScct decode for nine input codes.

Method tab

Section	Contents
Sample order	The index order, and the eight rows of a 2 × 2 × 2 identity
Interpolation	The seven blends of trilinear interpolation; worst identity error at sizes 2, 5, 9, 17, 33 and 65
Clipping	Seven test pixels judged by the old test and the clip test
Round trip	Write-and-read-back difference at 17 and 33 of the current transform; identity accuracy at 16, 17, 18, 33, 52 and 65
Linear light	The tone-map at six input codes, in linear light and in the old placement

Data tab

Status buttons: **All** (17), **Verbatim** (5), **Computed** (5), **Stand-in** (4), **Absent** (3). Select an entry to expand it.

Status	Entries
Verbatim	The .cube format, sRGB, CIE 1931 observer and CIELAB, CIEDE2000, Reinhard tone reproduction
Computed	Trilinear interpolation, clipping detection, the hue sweep, what a transform moves, the export round trip
Stand-in	The five stylisations, the 8-bit path, the tone-map exposure, the drawn scenes
Absent	ACEScct to ACEScg, 32-bit float export, PQ, BT.2020 and the ACES primaries

Loading your own picture changes the scenes entry to **Verbatim**, naming your file.

Export tab

Control	Values	Default
Kind	3D cube, 1D curve	3D cube
Size	2 to 65, step 1	33
rows	Size³ for 3D, size for 1D	35937

Control	Values	Default
file size	Size of the text, in KB	—
file name	`auric-` name `-` size `.cube`, with `-1d` for a curve	—
Grid note	Whether grid points land on 8-bit values	—
Save the .cube	Downloads the file	—
Copy it	Copies the file text	—
Hue sweep, CSV	Downloads `auric-lut-lab-hue-sweep.csv`	—
Report, JSON	Downloads `auric-lut-lab-report.json`	—

.cube keywords

Keyword	Read	Written
`TITLE`	Yes	`"Auric Artisan —` name `"`
`LUT_3D_SIZE`	Yes	For 3D
`LUT_1D_SIZE`	Yes	For 1D
`DOMAIN_MIN`	Yes; default 0 0 0	`0.0 0.0 0.0`
`DOMAIN_MAX`	Yes; default 1 1 1	`1.0 1.0 1.0`
`#` comments	Ignored	Header of about 12 lines
Other keywords	Ignored	—

Data rows are written with six decimal places, red index changing fastest.

Share link parameters

Parameter	Meaning	Values
`scene`	Drawn scene	`wedge`, `patches`, `skin`, `sweep`
`look`	Look	`none`, `teal-orange`, `bleach-bypass`, `film-soft`, `mono-contrast`, `log-lift`, `reinhard`
`s`	Strength	0 to 1
`e`	Tone-map exposure	0.25 to 8
`n`	Export size	2 to 65
`k`	Export kind	`3D`, `1D`

Reference tab

Status buttons: **All** (10), **Runs** (6), **Partly** (1), **Not held** (3).

Source	Status
Adobe / IRIDAS Cube LUT Specification 1.0	Runs
IEC 61966-2-1:1999 (sRGB)	Runs

Source	Status
CIE 015:2018, Colorimetry	Runs
CIE 142-2001 (CIEDE2000)	Runs
Sharma, Wu and Dalal (2005), CIEDE2000 test data	Runs
Reinhard et al. (2002), photographic tone reproduction	Runs
ACES S-2016-001, ACEScct	Partly
SMPTE ST 2084 (PQ) and ITU-R BT.2100	Not held
ITU-R BT.2020-2	Not held
SMPTE ST 2065-1 (ACES AP0)	Not held

Messages

Message	When
Loaded name, with a note count if any	A `.cube` was read
Loaded name **at** width×height	Your picture was loaded
That image could not be read	The picture could not be decoded
A reading error, such as **empty file**	The `.cube` could not be read
Cleared the loaded file	**Clear the file** was selected
Back to the drawn scenes	Your picture was removed
Saved file name	A `.cube` was saved
Copied the .cube / **Link copied**	A copy succeeded
The clipboard refused	Clipboard access was blocked
Saved the hue sweep / **Saved the report**	A CSV or JSON was saved

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

Troubleshooting LUT Lab

Symptoms you may meet in LUT Lab, why they happen, and what to do.

Loading a .cube

Symptom	Why	What to do
LUT_3D_SIZE N needs M rows; the file has K	The file is shorter than its header says, often from a failed download	Download the file again
no LUT_1D_SIZE or LUT_3D_SIZE keyword and no usable sample rows	The file is not a `.cube`, or is empty of data	Check the file; convert other LUT formats to `.cube` first
empty file	The file has no content	Choose the right file
My `.3dl` or other LUT file is greyed out or will not load	Only `.cube` is supported	Convert it to `.cube` in another program

Symptom	Why	What to do
A note says rows beyond the declared number were ignored	The file has extra data rows	Check the file at its source; the declared table is used
A note says the file was read as a 1D or 3D table because no size keyword was found	The size keyword is missing	The guess is stated; fix the file at its source if it is wrong
reachable is below 100%	The file's domain is wider than 0 to 1, for example for HDR or log footage	Only part of the table affects 8-bit sRGB pictures; the file was made for different input
The result looks badly wrong, with red and blue swapped	The file was written by a program that orders rows blue fastest	Re-export it from its source with the standard order; LUT Lab reads red fastest, as the format requires
The result is far too flat or too contrasty	The LUT was made for a different input, such as log camera footage	Apply it only to the kind of input it was made for

Looks and controls

Symptom	Why	What to do
The look menu and **Strength** are greyed out	A loaded file is being shown	Select **Back to the looks** or **Clear the file**
Tone-map exposure is greyed out	It applies only to **Reinhard tone-map**	Choose that look
Strength has no effect on my file	Loaded files are always applied at full strength	Blend in the program where you use the LUT
I went back to the looks and cannot show my file again	**Back to the looks** keeps the file but offers no way back	Drop the file again

Your picture

Symptom	Why	What to do
That image could not be read	The browser could not decode the picture	Save it as PNG or JPEG and try again
My picture looks small	Pictures are scaled to fit 480 × 270 for the analysis	This is expected; the figures cover the whole picture
My picture disappeared	Selecting a drawn scene removes your picture	Drop it again

Exports

Symptom	Why	What to do
A 1D export has lost the look's colour	A 1D curve keeps only what the transform does to greys	Export as **3D cube**
A resampled file differs slightly from the original	Resampling looks up each new grid point with interpolation and rounds to 8 bits	Keep the original; resample only when you need a different size
Another program reports banding or small errors	LUT Lab works in 8 bits, and the values in the file are one of 256 levels	Use the file for 8-bit work, or make high-precision LUTs in a tool with a floating-point pipeline
The clipboard refused	Clipboard access was blocked	Allow clipboard access for the site, or use **Save the .cube**

Share links

Symptom	Why	What to do
The link opened without my `.cube` or picture	Links carry only the scene, look, strength, exposure, size and kind	Send the files separately
After opening a link, the **Kind** menu or **Size** slider looks different from the file preview	The export uses the linked kind and size; the controls may not move to match	Trust the size label and the file preview, or set **Kind** and **Size** again

For help with the site in general, see Help.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

LUT Lab FAQ

Short answers to common questions about LUT Lab.

Are my files uploaded?

No. Your `.cube` and your picture are read in the page and never sent anywhere. See Privacy.

Which LUT formats can I use?

Only `.cube`, 1D or 3D, as defined in the Adobe / IRIDAS Cube LUT Specification 1.0. Convert other formats first.

Will a .cube from LUT Lab work in my editing program?

It is written to the `.cube` specification: `TITLE`, `LUT_3D_SIZE` or `LUT_1D_SIZE`, `DOMAIN_MIN` and `DOMAIN_MAX`, and rows with red changing fastest. Programs that read standard `.cube` files should read it. The looks are designed for ordinary sRGB images, not log camera footage.

What size should I choose?

33 is a common default. 17 is smaller and coarser, 65 larger and more faithful to curved looks. Sizes 2, 4, 6, 16, 18 and 52 put every grid point exactly on an 8-bit value.

Why does the file say it is 8-bit when it has six decimals?

The tool reads and writes pixels at 8 bits, so every value it can produce is one of 256 levels. Six decimals is simply how the format writes numbers. The header says this because the format has no other way to record precision.

Can I make a LUT with more precision than 8 bits?

Not in LUT Lab. A 32-bit float export was withdrawn because the tool has no floating-point path to fill it.

Why is there no ACEScct or log conversion?

Such conversions need values far above the 0 to 1 range an 8-bit canvas holds. The ACEScct look was withdrawn for that reason; the **Looks** tab shows why.

Is "Bleach bypass" a real film process simulation?

No. It is a stylisation named after the process, not a simulation of it. All five stylisations are hand-tuned curves; only the Reinhard tone-map is a defined transform.

What does "clipped" count?

Pixels where the transform drove a channel to 0 or 255 that was not already there. Pixels that were already pure black, white or fully saturated and stayed that way are not counted.

Can I adjust the strength of a loaded .cube?

No. **Strength** applies to the built-in looks. A loaded file is applied at full strength.

Can I use LUT Lab to convert a 3D LUT to a different size?

Yes. Load the file, then export at the size you want. The new table is resampled with interpolation and rounded to 8 bits, so keep the original. Exporting a 3D file as a 1D curve keeps only what it does to greys.

Does the tool show the LUT as a 3D cube?

No. It shows the effect on a scene, on 18 test colours on the **Looks** tab, and on a ring of 72 test colours.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

LUT Lab background material

The contents of LUT Lab's others folder: glossary, limits, related pages, release notes and privacy.

This folder holds reference material that supports the guides: the terms the tool uses, the limits of its 8-bit path and what it does not hold, related tools and reading, its history, and what happens to your files. LUT Lab has no keyboard shortcuts of its own beyond standard keyboard access, so there is no shortcuts page.

File	What it covers
glossary.md	Definitions of the terms used in LUT Lab
limits-and-accuracy.md	The 8-bit path, how figures are measured, and what the tool does not hold
related.md	Related tools on the site and Learn articles
release-notes.md	User-facing history of the tool
privacy.md	What stays in your browser

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

LUT Lab glossary

Definitions of the terms used in LUT Lab.

Term	Meaning
1D LUT	A table mapping each channel on its own, one row per input level
3D LUT	A cube of grid points mapping every combination of red, green and blue to a new colour
8-bit	256 levels per channel, 0 to 255; the precision LUT Lab works in
ACEScct / ACEScg	Encodings from the Academy Color Encoding System for film work; not offered, because they need values above 1
.cube	A plain-text LUT format defined by the Adobe / IRIDAS Cube LUT Specification 1.0
CIEDE2000 (ΔE_{00})	The CIE's standard formula for the size of a colour difference
CIELAB	A colour space of lightness L* and colour axes a* and b*, used here with a D65 white
Chroma (C*)	How colourful a colour is in CIELAB, its distance from grey
Clipped	A pixel with a channel driven by the transform to 0 or 255 that was not there before
Domain	The input range a `.cube` table covers, set by `DOMAIN_MIN` and `DOMAIN_MAX`; 0 to 1 by default
Exposure	In the tone-map, the factor the linear scene light is multiplied by before compression
Grid point	One of the sampled inputs of a LUT, at which the output is stored
Hue sweep	The ring of 72 test colours at L* 60, C* 40 used to measure hue, lightness and chroma shifts
Identity	A LUT that changes nothing
Interpolation	Estimating the output between grid points by blending the neighbouring points
Linear light	Values proportional to physical light, before the sRGB curve is applied
Look	A built-in transform in LUT Lab: a stylisation or a defined transform
LUT	Lookup table: a list of what each input colour should become
Reachable	The share of a table's domain that an 8-bit sRGB input can reach

Term	Meaning
Reinhard tone-map	A tone-mapping operator, L ÷ (1 + L), that compresses bright light smoothly
Round trip	Writing a table to a file and reading it back, to check nothing was lost
Sample order	The order of rows in a 3D `.cube`: red changes fastest, then green, then blue
Size	The number of grid points along each axis of a LUT
sRGB	The standard colour space for web and consumer images
Strength	How much of a look is blended with the original, 0% to 100%
Stylisation	A look designed because it looks good, with no definition to check against
Transform	A conversion between two defined things, which can be checked against its definition
Trilinear interpolation	Blending the eight grid points around an input along red, then green, then blue

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

Limits and accuracy

LUT Lab's 8-bit path, how its figures are measured, and what it does not hold.

The 8-bit path

Every value in LUT Lab travels as an 8-bit integer, because it comes from and returns to a canvas. This is a real limit:

- A loaded LUT's output is rounded to 8 bits when applied.
- An exported `.cube` holds only the 256 values k ÷ 255 per channel, written to six decimals. The header says so.
- A 1D export longer than 256 entries would only repeat values; the tool writes at most 65.
- At sizes where N − 1 does not divide 255 (including 17, 33 and 65), grid points in an exported file sit up to half an 8-bit step from the exact grid. An 8-bit image is unaffected; a program reading at higher precision is not.

What is measured, and how

Figure	Method
mean ΔE_{00}, 95th percentile, largest	CIEDE2000 between each pixel before and after, in CIELAB with a D65 white
clipped	Pixels with a channel moved to 0 or 255 from somewhere else
the old test said	Pixels with any output channel at 0 or 255; kept for comparison only
Hue circle	72 colours at L* 60, C* 40 in CIELAB, every 5°; 13 are pulled inside sRGB first
Hue shift	Change of CIELAB hue angle, not HSL hue
Round trip	Values written and read back agree to the six decimals of the format

The CIEDE2000 formula is checked against a second transcription of the standard and against published test data.

Choices the tool makes

These are listed as **Stand-in** on the **Data** tab.

- **The five stylisations** are hand-tuned curves. None imitates a measured film stock or a photochemical process, whatever its name.
- **The tone-map exposure** defaults to 2×, a one-stop lift. An 8-bit sRGB picture has no record of the scene's true brightness.
- **The drawn scenes** exercise a transform evenly but are not photographs: no grain, flare or real sensor colour. Load your own picture for a real test.
- **The 8-bit path** itself.

What the tool does not hold

Not held	Why
ACEScct to ACEScg	Its output runs far above 1, which an 8-bit canvas cannot hold
32-bit float export	There is no floating-point path to fill it
PQ (SMPTE ST 2084), BT.2020, ACES primaries	Not implemented; the tool works in sRGB only
LUT formats other than `.cube`	Only `.cube` is read and written
Input beyond 0 to 1	Pictures are 8-bit sRGB; wider domains are only partly reachable

Limits

- Your picture is scaled to fit 480 × 270 for the analysis.
- Export sizes are 2 to 65, for 3D and 1D alike.
- A loaded file is applied at full strength only.
- Share links do not carry a loaded file or your picture.
- The tool does not store anything between visits.

What it is not

LUT Lab is a place to read, test and write `.cube` files for 8-bit sRGB work. It is not a grading application, and it does not handle log, HDR or wide-gamut footage.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

Privacy

What LUT Lab does with your .cube files and pictures.

Your files stay in your browser

- A `.cube` file you load is read as text by the page itself, in your browser. It is not uploaded.
- A picture you load is read in the page and scaled to fit 480 × 270 for the analysis. It is not uploaded.
- The looks, the measurements and the exports are all worked out on your device.
- Exported files are created in the browser and saved by your browser's normal download.
- The clipboard is written only when you select **Copy it** or **Copy a link to this setting**.

Nothing is kept after you leave

LUT Lab does not store your `.cube`, your picture or your settings. Reloading or leaving the page clears them.

What leaves in an export or link

- An exported `.cube` contains the transform's name; for a loaded file, that is the file's title or name.
- The JSON report contains your picture's file name, if you used one, and the figures.
- A share link contains only the scene, look, strength, exposure, size and kind. It does not contain your file or picture.

The page itself

LUT Lab is a normal page on auricartisan.com. Site-wide features such as sign-in, preferences and any consent-based services follow the site's own privacy policy and your cookie choices; see Help.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

Release notes

User-facing history of LUT Lab.

3 June 2026 — the measured workbench

The earlier engine was replaced by the current workbench, with six tabs: **Lab**, **Method**, **Looks**, **Data**, **Export** and **Reference**. The site's earlier user guide carries a notice that it describes the engine replaced on this date.

What changed for you:

- **Exported .cube files use the standard sample order.** Files written by the earlier version had their red and blue axes swapped, so they looked right when read back by the same tool and wrong everywhere else. Re-export any `.cube` you made with the earlier version.
- **The overlay shows what the transform crushed.** The earlier out-of-gamut overlay flagged pure colours, black and white even when nothing had changed them. The new overlay marks only pixels the transform drove to white or black, in two colours.
- **"Cine Log to Rec.709" is now "Log-ish lift".** The arithmetic is unchanged; the old name claimed a log encoding and a conversion that were never applied.
- **The Reinhard tone-map runs in linear light,** with an exposure control. Mid grey now comes out where the operator puts it.
- **Two features were withdrawn:** the ACEScct to ACEScg look, which turned 114 of 256 input codes pure white, and a 32-bit float export that could not produce anything but 8-bit values.
- **New readouts:** CIEDE2000 mean, 95th percentile and largest difference; clipped share; and a hue circle in CIELAB with hue, lightness and chroma shifts.
- **Your own picture** can be the test scene.
- **The export header** states the sample order and the 8-bit precision, and the **Export** tab says whether a size's grid lands on 8-bit values.
- **New exports:** the hue sweep as CSV and a full report as JSON, and a link that restores your settings.

25 May 2026 — earlier version

The LUT Lab user guide was published in the site's documentation library. It describes the earlier engine and is now marked as superseded.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › LUT LAB

Related tools and reading

Other Auric Artisan tools and Learn articles that go with LUT Lab.

Tools on the site

Tool	Why you might use it with LUT Lab
Tone Mapping	Experiment with tone-mapping operators for HDR to standard range
ICC Profile Parser	Read ICC profiles, the other common kind of colour transform file
Image Picker	Take a palette from a picture before and after a look
Collage Maker	Combine test pictures into one scene to load into LUT Lab
Vision Simulation	Check a graded picture for different kinds of colour vision

Learn articles

- Color Grading and LUTs — https://auricartisan.com/library/learn/articles/2026-07-19-color-grading-and-luts
- Gamma, Linear Light, and Transfer Functions — https://auricartisan.com/library/learn/articles/2026-05-25-gamma-linear-light-transfer-functions
- ΔE Metrics from CIE76 to CIEDE2000 and ΔEITP — https://auricartisan.com/library/learn/articles/2026-05-26-delta-e-metrics-cie76-ciede2000-itp
- HDR, Wide Gamut, PQ, HLG, and Modern Displays — https://auricartisan.com/library/learn/articles/2026-05-25-hdr-wide-gamut-pq-hlg-modern-displays
- Color in Real-Time 3D: Albedo, PBR, and Tone Mapping — https://auricartisan.com/library/learn/articles/2026-08-02-color-in-real-time-3d-albedo-pbr-tone-mapping

Other documentation

- The group overview: Image and file tools.
- The site's documentation library carries "LUT Lab Developer Reference" at https://auricartisan.com/library/documentation/articles/2026-06-03-lut-lab-developer-reference, written for people maintaining the tool.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

M-Gradient Palette

What the M-Gradient Palette does, who it is for, where to find it and how to build a first ramp.

The M-Gradient Palette (the page is titled **Gradient Ramp**) builds a colour ramp: a row of evenly spaced steps between two colours, of the kind used for data-visualisation scales, tint and shade series, and design tokens. You choose the two ends, the number of steps, and the space the colours are blended in: OKLCH, sRGB, linear RGB or HSL, with a short or long way round the hue circle for the two polar spaces.

The tool then measures the ramp. It gives the CIEDE2000 colour difference between every pair of neighbouring steps, their average, and how even they are, and counts the steps whose colourfulness had to be reduced to fit sRGB. A second tab builds all thirty combinations of route and easing between your two colours and marks the evenest. A third checks every step as a background for a text colour you choose, with both the WCAG 2.2 ratio and the APCA lightness contrast, and the whole ramp can be viewed as people with protanopia, deuteranopia or tritanopia see it.

You can copy the ramp as a CSS gradient, as CSS custom properties, as a list of hex values or as SVG, and save a JSON reading of what was measured. Everything runs in your browser.

Who it is for

- Designers building tint and shade scales, and tokens for a design system.
- Data-visualisation designers who need sequential scales with even, visible steps.
- Front-end developers choosing between `linear-gradient` in sRGB and in OKLCH.
- Accessibility specialists checking which steps of a scale can carry text.

Where to find it

Open https://auricartisan.com/tool/general/tools/m-gradient-palette/ in a current browser. It is listed as **M-Gradient Palette** on the Collections page at https://auricartisan.com/collections/. Nothing needs installing.

Quick start

1. Open the tool. The **Ramp** tab shows a 12-step ramp from `#0B3D5C` to `#F2B134`, blended in **OKLCH, short hue**.
2. Set the two ends with the swatches or the hex boxes under **The two ends**, or choose a pair from **A starting point**.
3. Set **Steps**, and choose a route under **Interpolate in**.
4. Read **mean ΔE_{00}** and **evenness** under **What the ramp measures**; flatter bars mean more even steps.
5. Open the **Contrast** tab, choose a **Text colour**, and see which steps can carry text.
6. Open the **Export** tab and select **Copy the custom properties** to copy the steps as CSS.

What you can do

- Blend between two colours in four spaces by six routes, at 2 to 32 steps, with five easing curves.
- Start from 25 named preset pairs, and swap the ends.
- Measure the CIEDE2000 difference between every pair of neighbouring steps, their mean, spread and largest jump.
- See which steps were brought inside sRGB, and where the ramp travels in OKLCH hue and chroma.
- Compare all 30 route and easing combinations, built and measured, with the evenest marked.
- Check every step against a text colour with WCAG 2.2 and APCA, side by side.
- View the ramp as seen with protanopia, deuteranopia or tritanopia.

- Copy a CSS gradient, custom properties, a hex list or SVG; save SVG and a JSON reading.

In this folder

- docs/ — how to use the tool.
  - Getting started — open it, the screen tour, a first ramp.
  - Interpolation and gamut — OKLab, OKLCH, sRGB, linear RGB and HSL; hue routes; easing; gamut mapping.
  - Build a ramp — ends, presets, routes, steps, easing and the measurements.
  - Compare routes and easings — the Spaces tab.
  - Check contrast and colour vision — WCAG 2.2, APCA and the colour-vision views.
  - Export a ramp — every copy and save format.
  - Reference — every control, value, readout and format.
  - Troubleshooting — symptoms, causes and fixes.
  - FAQ — short answers to common questions.
- others/ — background material.
  - Glossary — terms used in this tool.
  - Limits and accuracy — what is measured, how precisely, and what is not held.
  - Related — related tools and Learn articles.
  - Release notes — user-facing history.
  - Privacy — what stays in your browser.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

M-Gradient Palette documentation

The list of guides, the concept page, the reference, troubleshooting and FAQ for the M-Gradient Palette.

This folder holds the task guides for the M-Gradient Palette, a concept page on colour spaces, interpolation and gamut mapping, the full reference, and the troubleshooting and FAQ pages. Read Interpolation and gamut if you want to know why the same two colours give different ramps.

File	What it covers
getting-started.md	Opening the tool, a tour of the six tabs, and a first ramp
interpolation-and-gamut.md	The four spaces, short and long hue routes, easing, CIEDE2000, evenness and gamut mapping
build-a-ramp.md	Setting the ends, presets, routes, steps and easing, and reading the Ramp tab
compare-routes.md	The Spaces tab: 30 combinations measured, and the desaturated middle
check-contrast-and-colour-vision.md	WCAG 2.2 and APCA against a text colour, and the colour-vision views
export-a-ramp.md	CSS gradient, custom properties, hex list, SVG and the JSON reading
reference.md	Every control, preset, readout, format and message
troubleshooting.md	What to do when a ramp or an export does not look right
faq.md	Short answers to common questions

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

Getting started with the M-Gradient Palette

Open the M-Gradient Palette, find your way around its six tabs, and build a first ramp.

Open the tool

Go to https://auricartisan.com/tool/general/tools/m-gradient-palette/. The page, titled **Gradient Ramp**, opens on the **Ramp** tab with a 12-step ramp from a deep blue, `#0B3D5C`, to a gold, `#F2B134`.

A note under the heading explains what the numbers on the page are measurements of. It has no close button on purpose.

The screen at a glance

Tab	What it is for
Ramp	The controls, the ramp, the step-by-step differences, the figures and the hue circle
Spaces	All six routes by five easings, built and measured, and three ramps drawn side by side
Contrast	A text colour, and every step checked as a background with WCAG 2.2 and APCA
Data	Every figure and what it rests on; the tab shows the number of entries (15)
Export	Copy and save buttons, previews and the JSON reading
Reference	The published sources and the arithmetic

The Ramp tab

- **The left-hand panel** holds **The two ends** (a swatch and a hex box for each, **Swap the ends**, and **A starting point**), **Interpolate in** (six route buttons), **Shape** (**Steps** and the **Easing** buttons), and **Show it as seen by** (four vision buttons).
- **The right-hand panel** holds **The ramp** (one button per step), the bars of **ΔE_{00} between adjacent steps**, the cards under **What the ramp measures**, and the circle under **Where the ramp goes in hue and chroma**.

Build a first ramp

1. Under **A starting point**, choose **Ocean**. The ends become `#0B3D5C` and `#69D2E7`, and the ramp redraws.
2. Move **Steps** to 9. The ramp now has nine steps.
3. Under **Interpolate in**, select **sRGB**, then **OKLCH, short hue**. Watch the bars under **ΔE_{00} between adjacent steps**: the flatter they are, the more evenly the steps are spaced to the eye.
4. Read **evenness** under **What the ramp measures**. Lower is more even.
5. Select a step in the ramp to copy its hex value.
6. Open the **Export** tab and select **Copy the custom properties**. Paste the result into your stylesheet:

```
--ramp-01: #0B3D5C;
--ramp-02: #164E6D;
--ramp-03: #215F7F;
--ramp-04: #2B7190;
--ramp-05: #3784A2;
--ramp-06: #4397B3;
--ramp-07: #4FAAC5;
--ramp-08: #5CBED6;
--ramp-09: #69D2E7;
```

These are the nine steps of **Ocean** in **OKLCH, short hue** with **linear** easing. Here **evenness** reads 0.61, against 0.80 for the same ramp in **sRGB**.

Next steps

- To understand why routes differ, read Interpolation and gamut.
- To find the evenest combination for your two colours, see Compare routes and easings.
- To check text on your ramp, see Check contrast and colour vision.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

Build a ramp

Set the two ends, use a preset, choose a route, steps and easing, and read everything on the Ramp tab.

Everything in this guide happens on the **Ramp** tab. The ramp and its figures update as soon as you change a control.

Set the two ends

Under **The two ends**, the first row is the start colour and the second the end colour.

1. Select a swatch to open the colour picker and choose a colour, or
2. type a hex value in the box beside it and press `Enter` or move focus away. Three- or six-digit hex is accepted, with or without `#`, for example `F2B134` or `#abc`.

If the box holds something that is not a colour, it returns to the previous value and a message reads **That is not a colour.**

Swap the ends exchanges the start and end colours.

Start from a preset

A starting point offers 25 named pairs. Choosing one sets both ends; the steps, route and easing stay as they are. The pairs are a starting point only and make no claim to be good ramps. The full list is in the Reference.

Choose a route

Under **Interpolate in**, select one of six routes:

Route	Space	Hue direction
OKLCH, short hue (default)	OKLCH	The shorter way round
OKLCH, long hue	OKLCH	The longer way round
sRGB	sRGB	—
Linear RGB	Linear RGB	—
HSL, short hue	HSL	The shorter way round
HSL, long hue	HSL	The longer way round

See Interpolation and gamut for how the spaces differ.

Set the shape

Control	What it does	Range	Default
Steps	The number of colours in the ramp, including both ends	2 to 32	12
Easing	Where along the route the steps land	linear, ease-in, ease-out, ease-in-out, step	linear

The **step** easing gives only five distinct colours, so with more than five steps, neighbouring steps repeat.

Read the ramp

The note above the ramp repeats your settings, for example **12 steps · OKLCH, short hue · linear**.

- **The ramp** shows one button per step. The first and last are labelled with their hex values underneath. Hover a step to see its hex value; select it to copy the value, and a message confirms, for example **#3E925C copied**.
- A **gold underline** under a step means its chroma was reduced to fit sRGB. The colour you get has the lightness and hue you asked for, on the edge of what sRGB can show.

Read the differences between steps

ΔE_{00} between adjacent steps draws one bar per gap: 11 bars for a 12-step ramp. Each is the CIEDE2000 difference between two neighbouring colours in the ramp. A line marks the mean, labelled with its value.

- Flat bars mean each step looks as far from the next as every other.

- A tall bar is a visible seam, where the ramp jumps.
- Bars are highlighted in two stronger colours when a difference reaches 2 and 5.

Hover a bar to see its value, for example **step 3 to 4: ΔE_{00} 8.12**.

Read the figures

Card	Meaning
mean ΔE_{00}	The average difference between neighbouring steps
evenness	The standard deviation of those differences; lower is more even. The card is highlighted when it is under 15% of the mean
largest jump	The largest difference, with the steps it lies between, for example **step 10 → 11**
moved to fit sRGB	How many steps had their chroma reduced; the card is highlighted when any were

With the default ends, 12 steps, **OKLCH, short hue** and **linear**, the figures are a mean of 8.26, an evenness of 0.69, a largest jump of 9.36 between steps 10 and 11, and 4 steps moved to fit sRGB.

Read the hue circle

Where the ramp goes in hue and chroma plots every step in OKLCH: the angle round the circle is hue, and the distance from the centre is chroma. Spokes mark every 30°, the four quarters are labelled 0°, 90°, 180° and 270°, and the label on the outer ring gives the largest chroma in the ramp, which sets the scale. A gold line joins the steps in order.

- A route that crosses from, say, 350° to 10° draws a short arc across the 0° spoke, not a line across the whole plot.
- Steps whose chroma was reduced are outlined in amber.
- A ramp through grey passes close to the centre.

View it with a colour-vision deficiency

Under **Show it as seen by**, select **Protan**, **Deutan** or **Tritan** to redraw the ramp, the hue circle dots and the contrast samples as they would look with protanopia, deuteranopia or tritanopia. **Normal vision** returns to the true colours. The figures, the copied values and the exports always use the true colours. See Check contrast and colour vision.

What you should see

A ramp whose bars are as flat as you need, a low **evenness**, and a hue path that goes where you intend, ready to export.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

Check contrast and colour vision

Check every step of a ramp as a background for a text colour with WCAG 2.2 and APCA, and view the ramp as seen with three colour-vision deficiencies.

Check text on every step

1. Build your ramp on the **Ramp** tab.
2. Open the **Contrast** tab.
3. Under **Text colour**, select the swatch and choose the colour your text will be. The default is white, `#FFFFFF`.

4. Read **Your ramp, checked both ways**. The note beside the heading names the text colour, for example **against #FFFFFF**.

Every step is treated as a background for text in the chosen colour, and measured by two methods at once.

Column	Meaning
Step	The step's swatch and hex value
WCAG 2	The WCAG 2.2 contrast ratio, for example **7.90:1**
Rating	**AAA**, **AA**, **AA large** or **fail**
APCA Lc	The APCA lightness contrast, signed
Good for	What APCA considers the pair suitable for
The old figure	The Lc an earlier version of this tool printed, struck through; highlighted where it differs by more than 20

WCAG 2.2

The WCAG 2.2 contrast ratio compares the relative luminance of the two colours, from 1:1 (identical) to 21:1 (black on white).

Rating	Ratio	Meaning
AAA	7:1 or more	Body text at the enhanced level
AA	4.5:1 or more	Body text at the minimum level
AA large	3:1 or more	Large text only
fail	Below 3:1	Not enough for text

APCA

APCA, the Advanced Perceptual Contrast Algorithm (version APCA-W3 0.1.9 here), measures lightness contrast in units called Lc. It treats dark text on a light background and light text on a dark background differently, so it is signed:

- a **positive** Lc means dark text on a lighter background;
- a **negative** Lc means light text on a darker background.

With white text, a dark step gives a negative figure, for example **−99.8** for `#0B3D5C`.

Absolute Lc	Good for
90 or more	any text
75 or more	body text
60 or more	large text
45 or more	headlines only
30 or more	non-text only
Below 30	not for text

Black text on white gives Lc 106.04; white text on black gives −107.88.

Why two figures

WCAG 2.2 and APCA disagree by design: they measure different things on different scales. A step can pass one and fail the other. Both are always shown so you can see the disagreement rather than being handed whichever one passes.

The samples

Under the table, three boxes show the sentence "The quick brown fox" in your text colour on the first, middle and last steps, so you can judge by eye.

What the evidence table shows

At the top of the **Contrast** tab, **The contrast figure passed text its own algorithm fails** compares, for five fixed pairs, what an earlier version of this tool printed (**It printed**) with what APCA gives (**APCA gives**) and the difference (**Out by**). The earlier version used the wrong scale and was missing two parts of the algorithm, so it reported contrast about 1.4 times too high: 161.80 for black on white, where APCA gives 106.04. The footnote under your own table says how far the old figure is off on your ramp, and how many steps it would wrongly have passed at Lc 60.

WCAG 2.2 was already right confirms the WCAG figures: black on white gives exactly 21.00, `#767676` on white gives 4.54, and `#595959` gives 7.00.

View the ramp with a colour-vision deficiency

On the **Ramp** tab, under **Show it as seen by**, choose:

Button	Simulates
Normal vision	No simulation
Protan	Protanopia: no working long-wavelength (red) cones
Deutan	Deuteranopia: no working medium-wavelength (green) cones
Tritan	Tritanopia: no working short-wavelength (blue) cones

The simulation redraws the ramp, the dots on the hue circle, the three ramps on the **Spaces** tab and the samples on the **Contrast** tab. It uses the model of Machado, Oliveira and Fernandes (2009) at full severity, applied in linear light.

The figures, the contrast table, the copied values and the exports always use the true colours.

What to look for

- Two neighbouring steps that become hard to tell apart under a simulation are a problem for a data scale.
- A ramp that runs from red to green can collapse to a single-hue ramp under **Protan** or **Deutan**; one that changes lightness steadily stays readable.

What the simulation does not cover

It simulates complete loss of one cone type (dichromacy). It has no setting for milder, partial forms (anomalous trichromacy), which are more common, so the case the tool cannot show is the more likely one.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

Compare routes and easings

Use the Spaces tab to build all thirty route and easing combinations for your two colours and find the evenest.

The **Spaces** tab answers: which route and easing give the evenest ramp between these two colours? Every row is built at your two ends and your step count and then measured; nothing is predicted from a model of the space.

Read the table

1. Set the ends and **Steps** on the **Ramp** tab.
2. Open the **Spaces** tab. The note beside the heading repeats your ends and step count, for example **#0B3D5C → #F2B134 · 12 steps**.

The table has thirty rows: six routes by five easings.

Column	Meaning
Route	One of the six routes
Easing	One of the five easings
Mean ΔE_{00}	Average difference between neighbouring steps
Min	Smallest difference between neighbouring steps
Max	Largest difference between neighbouring steps
Evenness	Standard deviation of the differences; lower is more even
What happens	**the evenest of the thirty**, or how many steps were moved to fit sRGB

The evenest row is highlighted and marked **the evenest of the thirty**. It is marked only when one row is strictly evenest; if two or more tie, none is marked, and the note under the table says so.

For the default ends at 12 steps, the evenest row is **OKLCH, short hue** with **linear** easing, at an evenness of 0.69.

What to look for

The note under the table points out two patterns you will usually see:

- **The mean barely changes between routes.** At linear easing, a ramp between the same two colours covers much the same perceptual distance whichever way it goes, so the mean alone cannot choose a route. The spread between rows in **Evenness** is what separates them.
- **Easing changes the evenness, not the mean.** The five rows for one route are the same colours spaced differently, so their means are close while their evenness varies.

A route marked as moving steps to fit sRGB is still a valid ramp, but those steps sit on the edge of sRGB rather than at the colourfulness the route asked for.

The desaturated middle, drawn

Under **The desaturated middle, drawn**, three ramps are drawn at your two ends with linear easing: **sRGB**, **Linear RGB** and **OKLCH, short hue**. Each heading gives its evenness, and each card gives the chroma of its **least chromatic step**, measured in OKLCH so all three are on one scale.

A much lower figure for sRGB or Linear RGB than for OKLCH means the straight-line blend passes near grey, the "muddy middle" that polar spaces avoid. For the default ends, the least chromatic step has a chroma of 0.030 in sRGB, 0.013 in Linear RGB and 0.074 in OKLCH.

The drawn ramps follow the **Show it as seen by** setting.

Use the result

The table does not change your ramp. To use the evenest row, go back to the **Ramp** tab and select that route and easing.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

Export a ramp

Copy the ramp as a CSS gradient, custom properties, a hex list or SVG, save an SVG, and save the JSON reading.

All exports are on the **Export** tab, under **Take it away**. They always use the true colours, whatever **Show it as seen by** is set to.

Button	What you get
Copy the CSS gradient	A CSS `linear-gradient` from the two ends, in the space of your route
Copy the custom properties	One CSS custom property per measured step
Copy the hex list	The steps' hex values, one per line
Copy the SVG	An SVG gradient built from the measured steps
Save the SVG	The same SVG, downloaded as `ramp.svg`
Save the reading	A JSON record of the ramp and its measurements, downloaded as `ramp-reading.json`

A message confirms each copy, for example **Custom properties copied** or **12 colours copied**. If the browser blocks the clipboard, it reads **The browser refused the clipboard.**

The table under **What leaves this page** gives the size of each form for your ramp and what each is good for.

The CSS gradient

The CSS gradient asks the browser to do the blending, from your two ends:

Route	Copied text
OKLCH, short hue	`background: linear-gradient(in oklch shorter hue, #0B3D5C 0%, #F2B134 100%);`
OKLCH, long hue	`background: linear-gradient(in oklch longer hue, #0B3D5C 0%, #F2B134 100%);`
HSL, short or long hue	As above, with `in hsl`
Linear RGB	`background: linear-gradient(in srgb-linear, #0B3D5C 0%, #F2B134 100%);`
sRGB	`background: linear-gradient(#0B3D5C 0%, #F2B134 100%);`

The copied text is laid out over two lines, as shown in the preview.

> **Important:** The CSS gradient carries only the two ends and the space. It does not carry your step count or easing, and it is smooth rather than stepped. Browsers that do not support interpolation in a named space fall back to sRGB, which is no longer the ramp you chose. A browser also brings out-of-gamut colours back into range in its own way, which may differ from this tool's chroma reduction.

The custom properties

The custom properties hand over the exact steps the tool measured:

```
--ramp-01: #0B3D5C;
--ramp-02: #004B63;
--ramp-03: #005A67;
--ramp-04: #00686A;
--ramp-05: #00776B;
--ramp-06: #0E8666;
--ramp-07: #3E925C;
--ramp-08: #649C50;
--ramp-09: #88A441;
--ramp-10: #ACAA32;
--ramp-11: #D0AE2A;
--ramp-12: #F2B134;
```

This is the default ramp. Paste the lines inside a rule such as `:root { … }`. Use them for design tokens, and anywhere a browser might not support the space.

The hex list

Copy the hex list copies the same steps as plain hex values, one per line, in capitals.

The SVG

The SVG is 600 × 80 pixels: a rectangle filled with a linear gradient whose stops are the measured steps, spaced evenly from 0% to 100%. Use it to hand the ramp to a drawing program.

```
<svg xmlns="http://www.w3.org/2000/svg" width="600" height="80">
  <defs><linearGradient id="ramp" x1="0" y1="0" x2="1" y2="0">
    <stop offset="0.00%" stop-color="#0B3D5C"/>
    …
    <stop offset="100.00%" stop-color="#F2B134"/>
  </linearGradient></defs>
  <rect width="600" height="80" fill="url(#ramp)"/>
</svg>
```

Between its stops, the SVG is blended by the program that displays it, usually in sRGB.

The reading

Save the reading downloads `ramp-reading.json`, shown under **The reading that goes with it**:

Field	Contents
`tool`	The tool's name
`ends`	The two end colours
`route`, `easing`, `steps`	Your settings; the route as an id such as `oklch-short`
`colors`	Every step's hex value
`measured`	`meanDeltaE00`, `evenness`, `largestJump` (three decimals) and `gamutMapped` (a count)

Field	Contents
`contrast`	The algorithm (`APCA-W3 0.1.9`), the text colour, and each step's APCA Lc and WCAG 2 ratio
`notHeld`	What the reading does not cover: colour-vision deficiency below full severity

The route ids are `oklch-short`, `oklch-long`, `srgb`, `linear`, `hsl-short` and `hsl-long`.

Which to use

- For a design system or a data scale, use the **custom properties** or the **hex list**: they are exactly the measured steps.
- For a smooth background, use the **CSS gradient**, and check it in the browsers you support.
- For a drawing program, use the **SVG**.
- To record what was measured, or to hand the ramp to someone else to check, keep the **reading**.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

Interpolation and gamut

How the four colour spaces blend two colours differently, what short and long hue routes and easing do, how evenness is measured, and how out-of-gamut steps are brought back.

A ramp is made by **interpolation**: choosing colours part of the way between the two ends. What "part of the way" means depends on the colour space the blending happens in, so the same two colours give different ramps in different spaces.

The four spaces

Space	What is blended	Character of the ramp
sRGB	The red, green and blue values as stored in a web colour	The browser's default. Middles between very different colours can turn dull or dark
Linear RGB	Red, green and blue as amounts of light, before the sRGB curve is applied	Physically like mixing light. Middles tend to look too light, and the dark end changes quickly
HSL	Hue, saturation and lightness, a cylinder built from sRGB	Keeps colours saturated, but HSL lightness is not perceived lightness, so brightness can bulge in the middle
OKLCH	Lightness, chroma and hue in OKLab	Designed so equal steps look equal. Keeps colourfulness through the middle. The default

OKLab and OKLCH

OKLab is a perceptual colour space published by Björn Ottosson in 2020. It has a lightness axis, L, and two colour axes, a and b, arranged so that equal distances look roughly equally different. OKLCH is the same space in polar form: **L** for lightness, **C** for chroma (how colourful, the distance from grey), and **H** for hue (an angle, 0° to 360°). Blending lightness, chroma and hue separately gives ramps whose steps look even and whose middle stays colourful.

sRGB and linear RGB

sRGB values are encoded with a curve that spends more of the 0 to 255 range on dark tones, roughly matching how eyes see. Blending the encoded values is quick and familiar; it is what CSS gradients do by default. Linear RGB undoes that curve first, so the blend is proportional to light. Neither is designed to make equal steps look equal.

HSL

HSL rearranges sRGB into hue, saturation and lightness. It is easy to reason about, but its lightness is a simple average of the largest and smallest channel, not a perceptual measure: fully saturated yellow and fully saturated blue have the same HSL lightness and look very different.

Short and long hue routes

In the two polar spaces, OKLCH and HSL, hue is an angle, and there are two ways round the circle from one hue to another.

- **Short hue** takes the smaller arc, never more than 180°.
- **Long hue** goes the other way round, through the hues the short route skips.

This is a choice of route, not a different space, which is why the tool offers four spaces but six routes. The long route covers much more distance when the two hues are close together; when they are nearly opposite, the two routes are similar in length.

sRGB and linear RGB have no hue angle, so they have one route each.

Easing

Easing decides where along the way each step lands. The tool uses the familiar CSS timing curves:

Easing	Where the steps land
linear	Evenly spaced along the route
ease-in	Bunched near the start, spreading out towards the end
ease-out	Spread out at the start, bunched near the end
ease-in-out	Bunched at both ends, spread in the middle
step	Only five distinct positions (0, ¼, ½, ¾ and the end), so neighbouring steps repeat colours

Easing changes where the steps land, not the path the ramp takes. That is why it barely changes the average difference between steps but can change how even they are a great deal.

Measuring a ramp

- **ΔE_{00}** is the CIEDE2000 colour difference, the CIE's standard formula for how different two colours look. The tool measures it between each step and the next.
- **mean ΔE_{00}** is the average of those differences: how far apart the steps are.
- **evenness** is the standard deviation of those differences: how much they vary. Zero would mean every step is exactly as far from the next. Lower is more even.
- **largest jump** is the biggest single difference, and where it is.

The mean and the evenness answer different questions and move independently. A ramp is usually being asked the evenness question.

Gamut and gamut mapping

A **gamut** is the range of colours a device or colour space can show. sRGB, the gamut of ordinary screens and web colours, cannot hold every colour OKLCH can describe. Blending in OKLCH between two saturated colours often asks for a middle step more colourful than sRGB allows.

There are two ways to bring such a colour back:

- **Clipping** each channel to 0 to 255 is simple, but it changes the hue and lightness as well as the colourfulness.
- **Gamut mapping by chroma**, which this tool uses, keeps the lightness and hue you asked for and reduces only the chroma, until the colour just fits. The result sits on the edge of sRGB at the right lightness and hue.

The tool finds the edge by halving the search twenty times, which is far finer than one step of an 8-bit channel. Steps that were brought back this way are marked with a gold underline in the ramp and outlined in amber on the hue circle, and **moved to fit sRGB** counts them.

Only the OKLCH routes can leave sRGB. Blends in sRGB, linear RGB and HSL always stay inside it.

The desaturated middle

A ramp between two colours on opposite sides of the colour wheel, blended in sRGB or linear RGB, passes close to grey in the middle, because the straight line between them crosses the neutral axis. Blending in OKLCH goes round the hue circle instead and stays colourful. The **Spaces** tab draws the same two ends in **sRGB**, **Linear RGB** and **OKLCH, short hue** and gives the chroma of the least colourful step of each, measured in OKLCH, so the three can be compared on one scale.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

M-Gradient Palette reference

Every tab, control, preset, readout, export format and message in the M-Gradient Palette.

Address: https://auricartisan.com/tool/general/tools/m-gradient-palette/

Tabs

Tab	Contents
Ramp	Ends, presets, routes, steps, easing, colour-vision view; the ramp, the step differences, figures and hue circle
Spaces	Thirty route and easing combinations measured; three ramps drawn with their least chromatic step
Contrast	Text colour; evidence table; every step checked by WCAG 2.2 and APCA; samples
Data	The 15 entries every figure rests on, filterable by status; the corrections list
Export	Six copy and save buttons, previews, the size table and the reading
Reference	Eight sources and the arithmetic table

Ramp tab — controls

Control	What it does	Values	Default
Start swatch and hex box	The first colour	Any colour; hex as 3 or 6 digits, `#` optional	`#0B3D5C`

Control	What it does	Values	Default
End swatch and hex box	The last colour	As above	`#F2B134`
Swap the ends	Exchanges start and end	—	—
A starting point	Sets both ends to a preset pair	25 presets, listed below	Choose a pair…
Interpolate in	The route	OKLCH, short hue; OKLCH, long hue; sRGB; Linear RGB; HSL, short hue; HSL, long hue	OKLCH, short hue
Steps	Number of colours, ends included	2 to 32	12
Easing	Where the steps land	linear, ease-in, ease-out, ease-in-out, step	linear
Show it as seen by	Colour-vision view for the display	Normal vision, Protan, Deutan, Tritan	Normal vision

Presets

Name	Start	End
Auric	`#3A2F10`	`#D3AF37`
Sunset	`#F56217`	`#0B486B`
Ocean	`#0B3D5C`	`#69D2E7`
Purple	`#41295A`	`#2F0743`
Cherry	`#EB3349`	`#F45C43`
Emerald	`#0F3443`	`#34E89E`
Lemon	`#CAC531`	`#F3F9A7`
Mango	`#FFE259`	`#FFA751`
Rosewater	`#E55D87`	`#5FC3E4`
Midnight	`#232526`	`#414345`
Cosmic	`#FF00CC`	`#333399`
Frozen	`#403B4A`	`#E7E9BB`
Mojito	`#1D976C`	`#93F9B9`
Sublime	`#FC5C7D`	`#6A82FB`
Peach	`#ED4264`	`#FFEDBC`
Aqua	`#136A8A`	`#267871`
Noon	`#00C9FF`	`#92FE9D`
Royal	`#141E30`	`#243B55`
Fire	`#CB2D3E`	`#EF473A`
Lavender	`#7F7FD5`	`#E8E8E8`
Deep space	`#000000`	`#434343`

Name	Start	End
Warm flame	`#FF9A9E`	`#FAD0C4`
Dusty	`#6D6027`	`#D3CBB8`
Heavy rain	`#CFD9DF`	`#E2EBF0`
Plum	`#DD5E89`	`#F7BB97`

Easing curves

Easing	Position of step at fraction t along the ramp
linear	t
ease-in	t^2
ease-out	t × (2 − t)
ease-in-out	$2t^2$ for the first half; −1 + (4 − 2t) × t for the second
step	t rounded to the nearest quarter

Ramp tab — readouts

Readout	Meaning
Note above the ramp	Steps, route and easing
The ramp	One button per step; select to copy the hex; gold underline when chroma was reduced
End labels	Hex of the first and last steps
ΔE_{00} between adjacent steps	One bar per gap, with a mean line; stronger colours at 2 and 5
mean ΔE_{00}	Average step difference
evenness	Standard deviation of the step differences; highlighted below 15% of the mean
largest jump	Largest step difference and where it is
moved to fit sRGB	Steps with reduced chroma; highlighted when above 0
Hue circle	Steps in OKLCH hue (angle) and chroma (radius); spokes every 30°; amber outline for reduced steps

Figures have two decimal places.

Spaces tab

Element	Contents
Note	Ends and step count
Route table	**Route**, **Easing**, **Mean ΔE_{00}**, **Min**, **Max**, **Evenness**, **What happens**; 30 rows
Footnote	The evenest row, and how much the mean and evenness vary
The desaturated middle, drawn	sRGB, Linear RGB and OKLCH, short hue at linear easing, each with evenness and **least chromatic step**

Contrast tab

Element	Contents
Text colour	The colour checked against every step; default `#FFFFFF`
Why two figures	WCAG 2.2 is a ratio; APCA is perceptual and signed
Evidence table	Five fixed pairs: **Text on background**, **It printed**, **APCA gives**, **Out by**
Your ramp, checked both ways	**Step**, **WCAG 2**, **Rating**, **APCA Lc**, **Good for**, **The old figure**
Footnote	How far the old figure is off on this ramp, and steps it wrongly passed at Lc 60
Samples	"The quick brown fox" on the first, middle and last steps
WCAG 2.2 was already right	Reference values: 21.00, 4.54, 7.00

WCAG ratings: **AAA** 7:1 and above, **AA** 4.5:1, **AA large** 3:1, **fail** below. APCA bands by absolute Lc: **any text** 90, **body text** 75, **large text** 60, **headlines only** 45, **non-text only** 30, **not for text** below.

Data tab

Status buttons: **All** (15), **Measured** (4), **Computed** (6), **This tool's choice** (4), **Absent** (1). Select an entry to expand it.

Status	Entries
Measured	ΔE_{00} between adjacent steps, evenness, the route × easing table, steps moved to fit sRGB
Computed	OKLab and OKLCH, the sRGB transfer function, relative luminance, the WCAG 2 ratio, APCA Lc, CIELAB
This tool's choice	How out-of-gamut steps are brought back, the five easings, calling a ramp even, the 25 presets
Absent	Colour-vision deficiency below full severity

What was corrected, and what it printed lists four corrections, each with **was** and **now**: the APCA scale factor, the colour-blindness citation, "six interpolation spaces", and "chromaticity paths".

Export tab

Button	Output
Copy the CSS gradient	`linear-gradient` in the route's space, from the two ends
Copy the custom properties	`--ramp-01` to `--ramp-NN`, one per step
Copy the hex list	One hex per line
Copy the SVG	600 × 80 SVG with one stop per step
Save the SVG	`ramp.svg`
Save the reading	`ramp-reading.json`

Reference tab

Source	Status
Ottosson (2020), OKLab	Runs

Source	Status
ISO/CIE 11664-6:2014, CIEDE2000	Runs
IEC 61966-2-1:1999, sRGB	Runs
W3C WCAG 2.2, contrast minimum	Runs
APCA-W3 0.1.9	Rebuilt
Machado, Oliveira and Fernandes (2009)	Runs
Brettel, Viénot and Mollon (1997)	Withdrawn
W3C CSS Color 4 and Images 4, interpolation in a named space	Runs

The arithmetic gives the formulas for the hue route, gamut mapping, evenness, the WCAG 2 ratio, and APCA Lc for both polarities.

Messages

Message	When
That is not a colour.	A hex box holds an invalid value
Hex value followed by **copied**	A step was copied
CSS gradient copied / **Custom properties copied** / **SVG copied**	An export was copied
N **colours copied**	The hex list was copied
Reading saved.	The reading was saved
The browser refused the clipboard.	Clipboard access was blocked
This browser will not take text from the clipboard.	The browser cannot copy text

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

Troubleshooting the M-Gradient Palette

Symptoms you may meet in the M-Gradient Palette, why they happen, and what to do.

Building the ramp

Symptom	Why	What to do
That is not a colour. and the hex box reverts	The value is not three or six hex digits	Type a value such as `F2B134` or `#abc`; eight-digit hex with alpha is not accepted
Some steps have a gold underline	Their chroma was reduced to fit sRGB	Expected with saturated ends in OKLCH. Choose less saturated ends, or accept the boundary colours
Neighbouring steps are identical	The **step** easing has only five positions	Choose another easing, or five steps
The middle of the ramp is grey or muddy	sRGB and linear RGB pass near grey between very different hues	Use **OKLCH, short hue**; see the **Spaces** tab
The ramp passes through unexpected hues	A long hue route, or HSL, goes round the colour wheel	Use a short hue route
One bar is much taller than the rest	A visible jump at that step	Try another route or easing; check **largest jump**

Symptom	Why	What to do
A preset gave a poor ramp	Presets make no claim to be good ramps	Adjust the ends, route or easing

The Spaces tab

Symptom	Why	What to do
No row is marked the evenest	Two or more rows tie	Choose any of the tied rows
Choosing a row does nothing	The table is a comparison, not a control	Select that route and easing on the **Ramp** tab

Contrast

Symptom	Why	What to do
APCA Lc is negative	The text is lighter than the step	Expected: the sign shows polarity; the **Good for** column uses the size
WCAG passes but APCA says **not for text**, or the reverse	The two measure different things	Treat both as evidence; choose steps that satisfy the standard you work to
The simulation changed the colours but not the contrast figures	Figures always use the true colours	Expected

Exports

Symptom	Why	What to do
The CSS gradient looks different from the ramp	It carries only the two ends and the space, not steps or easing	Use the custom properties for exact steps
The CSS gradient looks like an sRGB blend in one browser	That browser does not support interpolation in a named space	Use the custom properties, or an SVG
The SVG's middle differs from the ramp	Between stops, the program showing the SVG blends in its own space	Add more steps so the stops are closer
The browser refused the clipboard.	Clipboard access was blocked	Allow clipboard access for the site, or use **Save the SVG** and **Save the reading**
This browser will not take text from the clipboard.	The browser cannot copy text	Use a current browser, or the save buttons
Pasted custom properties have no effect	They were pasted outside a CSS rule	Put them inside a rule such as `:root { … }`

For help with the site in general, see Help.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

M-Gradient Palette FAQ

Short answers to common questions about the M-Gradient Palette.

Which route should I use?

For even-looking steps, start with **OKLCH, short hue**, and confirm on the **Spaces** tab, which measures all six routes and five easings for your two colours.

Why are there four spaces but six routes?

OKLCH and HSL have a hue angle, and each can go the short or the long way round the colour wheel. That is two routes in the same space, not two spaces.

What does "evenness" mean?

The standard deviation of the colour differences between neighbouring steps. Lower means the steps look more equally spaced. It is a ranking, not a pass mark: the tool marks the evenest of the thirty combinations but sets no threshold for "even enough".

Why did some steps get a gold underline?

Blending in OKLCH asked for a colour more colourful than sRGB can show. The tool reduced its chroma, keeping its lightness and hue, until it fitted.

Can I have more than two colours in a ramp?

No. A ramp runs between two ends. For a multi-stop scale, build one ramp per section, or join exported steps yourself.

Can I have more than 32 steps?

No. The **Steps** slider runs from 2 to 32.

Is the CSS gradient the same as the steps?

No. The CSS gradient asks the browser to blend smoothly between the two ends in the chosen space. The custom properties and the hex list are the exact steps the tool measured.

Why does the tool show two contrast figures?

WCAG 2.2 and APCA measure contrast differently and often disagree. Both are shown so you can see the disagreement.

Why is the APCA figure negative?

APCA is signed. Negative means light text on a darker background; positive means dark text on a lighter background.

Does the colour-vision view change the exports?

No. It changes only what is drawn on screen. All figures and exports use the true colours.

Does it simulate mild colour blindness?

No. It simulates full protanopia, deuteranopia and tritanopia only.

Is anything saved between visits?

No. The ramp and settings are not stored. Save the reading or copy the custom properties to keep them.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

M-Gradient Palette background material

The contents of the M-Gradient Palette's others folder: glossary, limits, related pages, release notes and privacy.

This folder holds reference material that supports the guides: the terms the tool uses, what it measures and does not hold, related tools and reading, its history, and what happens to your data. The tool has no keyboard shortcuts of its own beyond standard keyboard access, so there is no shortcuts page.

File	What it covers
glossary.md	Definitions of the terms used in the M-Gradient Palette
limits-and-accuracy.md	What is measured, how precisely, choices the tool makes and what it does not hold
related.md	Related tools on the site and Learn articles
release-notes.md	User-facing history of the tool
privacy.md	What stays in your browser

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

M-Gradient Palette glossary

Definitions of the terms used in the M-Gradient Palette.

Term	Meaning
APCA	Advanced Perceptual Contrast Algorithm; a lightness contrast measure in Lc units, signed by polarity
Chroma	How colourful a colour is, its distance from grey; in OKLCH, the C value
CIEDE2000 (ΔE_{00})	The CIE's standard formula for how different two colours look
CIELAB	A colour space of lightness and two colour axes; used here only to compute ΔE_{00}
Clipping	Forcing each channel into 0 to 255 separately, which also shifts hue; not used by this tool
Colour-vision deficiency	Reduced ability to tell some colours apart; simulated here as full protanopia, deuteranopia or tritanopia
Contrast ratio	The WCAG 2 measure of luminance difference, from 1:1 to 21:1
Custom property	A CSS variable such as `--ramp-01`, holding one step's colour
Deutan	Deuteranopia: no working medium-wavelength (green) cones
Easing	A curve that decides where along the route the steps land
Evenness	The standard deviation of the differences between neighbouring steps; lower is more even
Gamut	The range of colours a space or device can show
Gamut mapping	Bringing an out-of-range colour back inside, here by reducing chroma at fixed lightness and hue
HSL	Hue, saturation and lightness: a cylinder built from sRGB
Hue route	Short or long way round the hue circle between the two ends
Interpolation	Choosing colours part of the way between two others
Lc	The unit of APCA lightness contrast
Linear RGB	Red, green and blue as amounts of light, without the sRGB curve
Long hue	The route round the hue circle that goes the longer way

Term	Meaning
Mean ΔE_{00}	The average difference between neighbouring steps
OKLab	A perceptual colour space published by Björn Ottosson in 2020
OKLCH	OKLab in polar form: lightness L, chroma C and hue H
Protan	Protanopia: no working long-wavelength (red) cones
Ramp	A row of steps between two colours
Relative luminance	The WCAG measure of brightness, from 0 for black to 1 for white
Short hue	The route round the hue circle that goes the shorter way, at most 180°
sRGB	The standard colour space for web and consumer screens
Step	One colour of the ramp
Tritan	Tritanopia: no working short-wavelength (blue) cones
WCAG	The Web Content Accessibility Guidelines; version 2.2 defines the contrast ratio used here

Limits and accuracy

What the M-Gradient Palette measures, how precisely, the choices it makes, and what it does not hold.

What is measured, and how precisely

Figure	Method	Accuracy
ΔE_{00} per step	CIEDE2000 (ISO/CIE 11664-6:2014) between neighbouring steps, via CIELAB with a D65 white	Agrees with the 34 published test pairs of Sharma, Wu and Dalal (2005) to within 0.000042
Evenness	Standard deviation of the step differences	Exact from the above
OKLab and OKLCH	Ottosson's published matrices	Checked against the article's own reference values
sRGB conversions	The piecewise sRGB transfer function, both directions	Exact
WCAG 2 ratio	$(L_1 + 0.05) \div (L_2 + 0.05)$ from relative luminance	Exact: 21.00, 4.54 for `#767676`, 7.00 for `#595959` on white
APCA Lc	APCA-W3 0.1.9 with all its constants	Black on white 106.04
Colour-vision view	Machado, Oliveira and Fernandes (2009), severity 1.0, in linear light	Full dichromacy only
Gamut mapping	Chroma halved towards the boundary 20 times, lightness and hue held	Far finer than one 8-bit step

Every step is rounded to an 8-bit hex colour before it is measured, so the figures describe the colours you export.

Choices the tool makes

These are listed as **This tool's choice** on the **Data** tab.

- **How out-of-gamut steps are brought back:** by reducing chroma, not by clipping channels. A different method gives a visibly different ramp.

- **The five easings** are CSS animation curves, not a colour standard.
- **Calling a ramp even:** there is no published threshold for evenness, so the tool ranks rows and marks the evenest rather than passing or failing them.
- **The 25 presets** are pairs someone liked; they carry no claim to be good ramps.

What the tool does not hold

- **Colour-vision deficiency below full severity.** The simulation shows complete loss of one cone type only. Milder forms are more common and are not shown.

Limits

- Two ends only; 2 to 32 steps.
- Only the OKLCH routes can go outside sRGB; the other routes always stay inside it.
- The CSS gradient export carries the ends and the space only, and depends on browser support.
- The APCA figure is a guide to legibility, and the Lc bands are APCA's own guidance, not a conformance level.
- Nothing is stored between visits.

What it is not

The tool measures how evenly spaced and how legible a ramp is. It does not judge whether the colours suit your brand, and it does not test your final layout; check text in context at its real size.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

Privacy

What the M-Gradient Palette does with your colours and settings.

Everything runs in your browser

- The ramp, the measurements, the route comparison, the contrast checks and the colour-vision views are all worked out on your device.
- The tool sends nothing to a server.
- **Save the SVG** and **Save the reading** create their files in the browser and save them through your browser's normal download.
- The clipboard is written only when you select a copy button or a step of the ramp.

Nothing is kept after you leave

Your colours and settings are not stored. Reloading or leaving the page returns to the default ramp. To keep a ramp, copy the custom properties or save the reading.

The page itself

The M-Gradient Palette is a normal page on auricartisan.com. Site-wide features such as sign-in, preferences and any consent-based services follow the site's own privacy policy and your cookie choices; see Help.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

Release notes

User-facing history of the M-Gradient Palette.

Current version — Gradient Ramp

The tool was rebuilt as the **Gradient Ramp** workbench, with six tabs: **Ramp**, **Spaces**, **Contrast**, **Data**, **Export** and **Reference**. The developer reference that describes this version in the site's documentation library is dated 6 June 2026.

What changed for you:

- **APCA figures are now correct.** The earlier contrast readout used the wrong scale and left out two parts of the APCA algorithm, so it reported contrast about 1.4 times too high: 161.80 for black on white, where APCA gives 106.04. Across 2,704 grey-on-grey pairs, 520 cleared its Lc 60 line that real APCA fails. If you chose text colours with the earlier version, check them again.
- **Both contrast figures side by side.** Each step shows the WCAG 2.2 ratio and the APCA Lc together, and the old figure for comparison.
- **The colour-vision simulation is correctly attributed** to Machado, Oliveira and Fernandes (2009); the simulation itself is unchanged.
- **"Six spaces" is now four spaces and six routes.** Short and long hue are routes in the same space.
- **The hue plot is a circle.** A ramp crossing 0° no longer draws a line across the whole plot.
- **The route table marks the measured evenest row,** and none when rows tie, instead of starring every row under a fixed threshold.
- **Steps brought inside sRGB are counted and marked.**
- **The reading** records the ramp, its measurements and both contrast figures, naming the APCA version.

WCAG 2 contrast, CIEDE2000, OKLab and the route comparison were already correct and are unchanged.

25 May 2026 — earlier version

The M-Gradient Palette user guide was published in the site's documentation library. It describes the earlier version of the tool.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › M-GRADIENT PALETTE

Related tools and reading

Other Auric Artisan tools and Learn articles that go with the M-Gradient Palette.

Tools on the site

Tool	Why you might use it with the M-Gradient Palette
Contrast Checker	Test a single text and background pair in more depth
Vision Simulation	Simulate more kinds of colour vision on a finished design
Tonal Steps, Tints and Shades	Generate tonal step charts from one colour
Gamut Mapping	Explore how out-of-gamut colours are brought back
Design Engine	Generate a whole design system of OKLCH colour ramps

Tool	Why you might use it with the M-Gradient Palette
Image Picker	Take the two end colours from a picture
SVG Recolour	Apply ramp colours to SVG artwork

Learn articles

- Color Gradients and Interpolation — https://auricartisan.com/library/learn/articles/2026-06-07-color-gradients-and-interpolation
- Oklab and Oklch: Modern Perceptual Color Spaces — https://auricartisan.com/library/learn/articles/2026-06-02-oklab-oklch-perceptual-color-spaces
- ΔE Metrics from CIE76 to CIEDE2000 and ΔEITP — https://auricartisan.com/library/learn/articles/2026-05-26-delta-e-metrics-cie76-ciede2000-itp
- Accessible Color Design and WCAG Contrast — https://auricartisan.com/library/learn/articles/2026-05-25-wcag-contrast-accessible-color
- Color Blindness and Color Vision Deficiency — https://auricartisan.com/library/learn/articles/2026-05-24-color-blindness-cvd
- Gamut Mapping — https://auricartisan.com/library/learn/articles/2026-06-17-gamut-mapping-clipping-and-rendering-intents
- Gamma, Linear Light, and Transfer Functions — https://auricartisan.com/library/learn/articles/2026-05-25-gamma-linear-light-transfer-functions

Other documentation

- The group overview: Image and file tools.
- The site's documentation library carries "Gradient Ramp Developer Reference" at https://auricartisan.com/library/documentation/articles/2026-06-06-gradient-ramp-developer-reference, written for people maintaining the tool.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

SVG Recolour

What SVG Recolour does, who it is for, where to find it and how to recolour a first SVG.

SVG Recolour (listed as **SVG Recolor** in Collections) changes the colours of an SVG file. It finds every colour in the file, wherever and however it is written, lists each distinct colour once with the number of places it is used, and lets you choose what each should become. The new file is written from your original in one pass, so two colours can trade places, a colour written as a name such as `red` is changed like any other, and the count beside each colour is the number of places actually rewritten.

Before and after previews sit side by side on a chequerboard. You can recolour by hand, one colour at a time, or from a harmony built on the file's main colour: complementary, analogous, triadic, tetradic or split complementary. The **Check** tab measures every colour as a background for a text colour of your choice, with both WCAG 2.2 and APCA, and finds the two colours in the file that are hardest to tell apart, with normal vision and with each of three colour-vision deficiencies.

You can save the recoloured SVG, copy its markup, copy the new palette as CSS custom properties, and save a JSON report of every change. The file is parsed in the page and never uploaded, and scripts and other active content are removed from the previews.

Who it is for

- Designers adapting icons and illustrations to a new brand palette.
- Developers who need an SVG's colours to match CSS tokens.
- Accessibility specialists checking whether the colours in an icon set can be told apart.
- Anyone who has been sent an SVG and wants to change its colours without a drawing program.

Where to find it

Open https://auricartisan.com/tool/general/tools/svg-recolor/ in a current browser. It is listed as **SVG Recolor** on the Collections page at https://auricartisan.com/collections/. Nothing needs installing.

Quick start

1. Open the tool. A small sample SVG is already loaded on the **Recolour** tab.
2. Drop your own SVG on the **Drop an SVG** box, or paste its markup and select **Read the markup**.
3. Under **Every colour in the file**, select the right-hand swatch in a row and choose the new colour.
4. Watch the **After** preview change, and the row report how many places were written.
5. Open the **Check** tab to see contrast and the closest pair of colours.
6. Open the **Export** tab and select **Save the recoloured SVG**. Your browser downloads `recoloured.svg`.

What you can do

- Load an SVG by dropping it, choosing it, or pasting its markup; or load the built-in sample.
- Find colours in eight paint properties, in attributes, style attributes and style sheets, written as hex, `rgb()`, `hsl()` or any of the 148 CSS colour names.
- Remap each distinct colour, and swap colours with each other.
- Write new colours as hex, or keep the form each colour was written in.
- Recolour from five harmonies built in OKLCH, and put every colour back in one step.

- Preview before and after, also as seen with protanopia, deuteranopia or tritanopia.
- Check every colour as a background for text with WCAG 2.2 and APCA, and find the closest pair under four kinds of vision.
- Save or copy the recoloured SVG, copy the palette as custom properties, and save a JSON report.

In this folder

- docs/ — how to use the tool.
  - Getting started — open it, the screen tour, a first recolour.
  - Recolour an SVG — loading, what is found, mapping colours, swaps and write styles.
  - Recolour from a harmony — the five harmonies and how they are built.
  - Check contrast and colour vision — the Check tab and the vision previews.
  - Export the result — the SVG, the markup, custom properties and the report.
  - Reference — every control, readout, format and message.
  - Troubleshooting — symptoms, causes and fixes.
  - FAQ — short answers to common questions.
- others/ — background material.
  - Glossary — terms used in this tool.
  - Limits and accuracy — what is found, what is not, and how precise it is.
  - Related — related tools and Learn articles.
  - Release notes — user-facing history.
  - Privacy — what stays in your browser, and active content.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

SVG Recolour documentation

The list of guides, the reference, troubleshooting and FAQ for SVG Recolour.

This folder holds the task guides for SVG Recolour, the full reference of its controls and formats, and the troubleshooting and FAQ pages. Start with Getting started, then go to the guide for the job you have.

File	What it covers
getting-started.md	Opening the tool, a tour of the six tabs, and a first recolour
recolour-an-svg.md	Loading an SVG, what the tool finds, mapping and swapping colours, write styles, the Mapping tab
use-a-harmony.md	Recolouring the whole file from a harmony, and putting colours back
check-contrast-and-colour-vision.md	Contrast against a text colour, the closest pair, and the colour-vision previews
export-the-result.md	Saving and copying the SVG, custom properties and the JSON report
reference.md	Every control, property, colour form, readout, format and message
troubleshooting.md	What to do when a file will not parse or a colour does not change
faq.md	Short answers to common questions

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

Getting started with SVG Recolour

Open SVG Recolour, find your way around its six tabs, and recolour a first SVG.

Open the tool

Go to https://auricartisan.com/tool/general/tools/svg-recolor/. The page, titled **SVG Recolour**, opens on the **Recolour** tab with a small sample SVG already loaded, so you can try every control straight away.

A note under the heading explains what the numbers on the page are measurements of. It has no close button on purpose.

The screen at a glance

Tab	What it is for
Recolour	Load the file, choose how colours are written, preview before and after, and remap each colour
Mapping	Every colour, where it is used, what it becomes, and how many places were written
Check	Contrast of every colour against a text colour, and the colours hardest to tell apart
Data	Every figure and what it rests on; the tab shows the number of entries (15)
Export	Save and copy buttons, the custom properties and the report
Reference	The published sources, and how the tool works step by step

The Recolour tab

- **The left-hand panel** holds **The file** (the **Drop an SVG** box, **Load the sample**, a box to paste markup, and **Read the markup**), **Write colours as**, **Show it as seen by**, and **Recolour from a harmony** with **Put every colour back**.

- **The right-hand panel** holds **Before, and after** (two previews on a chequerboard), **Every colour in the file** (one row per colour), and the cards under **What the file holds**.

Recolour a first SVG

The sample contains six colour uses: a hex fill, an `rgb()` fill, the name `red`, a three-digit shorthand, a colour in a style sheet, and `currentColor`.

1. Under **Every colour in the file**, find the row for `#FF0000`, labelled **red · 1 place**.
2. Select the right-hand swatch in that row and choose a green. The **After** preview changes, the row's right-hand side shows the new hex value with **written in 1 place**, and the count on the far right reads **1 written**.
3. Find the row for `#E5484D` and set it to the colour that `#3E63DD` has, and set `#3E63DD` to the colour `#E5484D` had. The two rectangles trade colours.
4. The row for **currentcolor** is greyed out and reads **left alone**, because that colour comes from outside the file.
5. Read the heading of the **After** preview: for example **3 of 5 written**. Five of the six uses can be written; the sixth is `currentColor`.
6. Open the **Export** tab and select **Save the recoloured SVG**. Your browser downloads `recoloured.svg`, and a message confirms how many occurrences were written.

Next steps

- To load your own file and understand what is found, see Recolour an SVG.
- To recolour everything at once, see Recolour from a harmony.
- To check the result for legibility and colour vision, see Check contrast and colour vision.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

Check contrast and colour vision

Measure every colour in the SVG as a background for text with WCAG 2.2 and APCA, find the colours hardest to tell apart, and preview the artwork as seen with colour-vision deficiencies.

The **Check** tab measures the colours the file will have after your changes: the targets, not the originals.

Contrast against a text colour

1. Open the **Check** tab.
2. Under **Text laid over the artwork**, select the swatch and choose the colour of any text that will sit on the artwork. The default is white, `#FFFFFF`.
3. Read **What can be read on it**. The note beside the heading gives the number of colours checked.

Each colour is treated as a background for text in the chosen colour:

Column	Meaning
Colour	The colour's swatch and hex value
WCAG	The WCAG 2.2 contrast ratio, for example **4.54:1**
Rating	**AAA** (7:1 and above), **AA** (4.5:1), **AA large** (3:1) or **fail**
APCA Lc	The APCA-W3 0.1.9 lightness contrast; negative when the text is lighter than the colour

Column	Meaning
Good for	APCA's guidance by size of Lc, from **any text** to **not for text** (see the table below)
It used to say	The Lc an earlier version of this tool printed, struck through

The two methods measure different things, so a row can pass one and fail the other. Neither is a verdict on the artwork; both describe text laid on that colour.

Absolute Lc	Good for
90 or more	any text
75 or more	body text
60 or more	large text
45 or more	headlines only
30 or more	non-text only
Below 30	not for text

Under **The Lc that was always zero**, a small table shows what the earlier version printed for four pairs against what APCA gives: the earlier version printed zero for every pair.

The colours hardest to tell apart

The two colours hardest to tell apart compares every pair of colours in the file and reports the closest pair, measured with CIEDE2000, under four kinds of vision:

Column	Meaning
Seen as	**normal vision**, **protanopia**, **deuteranopia** or **tritanopia**
The two closest colours	Swatches as seen under that vision, and the two original hex values
ΔE_{00} between them	Their CIEDE2000 difference under that vision
What that means	**close to indistinguishable** (below 5), **tells apart with effort** (below 12), or **clearly different**

The row with the smallest difference among the three deficiencies is highlighted. You need at least two colours for this check.

The same figure appears on the **Recolour** tab as the **closest pair** card, highlighted when it is below 8.

The artwork under three dichromacies

The artwork under three dichromacies draws the recoloured file four times, as seen with **Normal vision**, **Protanopia**, **Deuteranopia** and **Tritanopia**, with **closest two** under each: the closest pair's difference, highlighted when below 8.

Preview with a colour-vision deficiency on the Recolour tab

On the **Recolour** tab, under **Show it as seen by**, choose **Normal vision**, **Protan**, **Deutan** or **Tritan**. Both previews are redrawn as seen with that deficiency. The exports always use the true colours.

Button	Simulates
Protan	Protanopia: no working long-wavelength (red) cones

Button	Simulates
Deutan	Deuteranopia: no working medium-wavelength (green) cones
Tritan	Tritanopia: no working short-wavelength (blue) cones

The simulation follows Machado, Oliveira and Fernandes (2009) at full severity, in linear light. It shows complete loss of one cone type only; milder, more common forms are not shown.

Using the results

- If two colours are **close to indistinguishable** under a deficiency, change one of them, usually in lightness, and check again.
- If text must sit on a colour, choose a text colour and target colour that pass the standard you work to, under both methods if you can.
- Colours the file does not hold, such as `currentColor`, are not checked, because their value comes from outside the file.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

Export the result

Save or copy the recoloured SVG, copy the new palette as CSS custom properties, and save a JSON report of every change.

All exports are on the **Export** tab, under **Take it away**. They always use the true colours, whatever **Show it as seen by** is set to.

Button	What you get
Save the recoloured SVG	Your file with the colour values replaced, downloaded as `recoloured.svg`
Copy the markup	The same SVG markup, copied to the clipboard
Copy the custom properties	The file's colours as CSS custom properties
Save the report	A JSON record of every mapping, downloaded as `recolour-report.json`

The table under **What leaves this page** gives the size of each for your file and what it is good for.

The recoloured SVG

The exported file is your file with the colour values replaced, and nothing else changed on purpose:

- Structure, ids, gradients, filters, titles, comments and any markup the tool does not understand are carried across.
- Only occurrences you changed are rewritten, in the style chosen under **Write colours as**.
- In style sheets, only the changed declarations are rewritten.

The message confirms, for example **Saved · 3 occurrences written**. If nothing is loaded, it reads **Nothing loaded.**

Things to know about the file:

- It is rewritten from your original, not from the preview, so any scripts, event handlers or `javascript:` links in the original are still there. The previews on this page have them removed; the file does not.
- The browser writes the markup back out, so small formatting details, such as the quote marks around attribute values, may differ from your original.
- Only the `<svg>` element and its contents are written. An XML declaration, a DOCTYPE or a comment placed before the `<svg>` tag is not carried across.

The custom properties

Copy the custom properties copies one line per colour the tool can change, in the order of the colour list, with the colour each ends up as and its CSS name or nearest name:

```
--colour-01: #E5484D;  /* near indianred */
--colour-02: #3E63DD;  /* near royalblue */
--colour-03: #00AA00;  /* near limegreen */
--colour-04: #AABBCC;  /* near lightsteelblue */
--colour-05: #30A46C;  /* near mediumseagreen */
```

This is the sample file with `red` changed to `#00AA00`. The preview under **The palette as custom properties** shows the same text. Paste the lines inside a rule such as `:root { … }` to keep a stylesheet in step with the artwork. An exact CSS name is given as itself; any other colour gets the nearest name, prefixed "near".

The report

Save the report downloads `recolour-report.json`, also shown under **The report that goes with it**:

```
{
  "tool": "Auric Artisan — SVG recolour",
  "source": { "elements": 9, "occurrences": 6, "colours": 5 },
  "writeStyle": "hex",
  "mapped": [
    {
      "from": "#FF0000", "to": "#00AA00",
      "occurrences": 1, "written": 1, "writtenAs": ["named"]
    }
  ],
  "notWritten": [
    { "value": "currentcolor", "occurrences": 1, "why": "resolves from CSS outside this file" }
  ],
  "measured": {
    "deltaE00Method": "ISO/CIE 11664-6",
    "contrast": { "wcag": "2.2", "apca": "APCA-W3 0.1.9" },
    "colourVision": "Machado, Oliveira & Fernandes (2009), severity 1.0"
  },
  "notHeld": ["what currentColor resolves to", "colour inside embedded raster art"],
  "bytes": 512
}
```

Field	Contents
`source`	Elements, occurrences and distinct colours in the original
`writeStyle`	`hex` or `keep`
`mapped`	Each changed colour: from, to, how many places use it, how many were written, and the forms it was written in
`notWritten`	Each value left alone, with its count and the reason
`measured`	The methods behind the figures
`notHeld`	What the tool cannot know
`bytes`	The size of the recoloured SVG

The written count in the report is the same number the colour list and the **Mapping** tab show, and the same number of changes the file carries. The example shows the sample with `red` changed to `#00AA00`; the `bytes` value is illustrative.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

Recolour an SVG

Load an SVG, understand what colours the tool finds and where, remap and swap colours, choose how they are written, and read the Mapping tab.

Load a file

On the **Recolour** tab, under **The file**, use any of these:

Method	How
Drop	Drop an `.svg` file on **Drop an SVG**
Choose	Select **Drop an SVG** (or press `Enter` or `Space` while it has focus) and choose an `.svg` file
Paste	Paste SVG markup into the box under **Or paste the markup**, then select **Read the markup**
Sample	Select **Load the sample**; its markup also appears in the paste box

A message confirms, for example **Loaded logo.svg** or **Loaded the pasted markup.** Loading a new file replaces the previous one and resets every mapping.

If the file will not parse

The file must be well-formed XML. If it is not, nothing is changed, the message reads **That file will not parse as SVG, so nothing was changed.**, and a warning under the paste box gives the parser's own complaint. A file is never half-recoloured.

Active content in the preview

An SVG can carry scripts, event handlers and `javascript:` links. Before drawing the previews, the tool removes `script`, `foreignObject` and `handler` elements, every attribute starting with `on`, and any `href` or `xlink:href` that starts with `javascript:`. If anything was removed, a notice above the previews lists it, for example `1 <script>, 2 onclick handlers`.

This applies to the previews only. The exported file is rewritten from your original file and still contains whatever it contained. See Privacy.

What the tool finds

The tool looks for colours in eight paint properties:

`fill`, `stroke`, `stop-color`, `flood-color`, `lighting-color`, `color`, `solid-color`, `text-decoration-color`

It looks in three places:

- **presentation attributes**, such as `fill="#E5484D"`;
- **style attributes**, one declaration at a time, such as `style="fill: red; stroke: #000"`;
- **style sheets** inside `<style>` elements, one colour-valued declaration at a time.

Each place a colour appears is one **occurrence**. The tool reads these colour forms:

Form	Examples
Six-digit hex	`#E5484D`
Three-digit hex	`#ABC`

Form	Examples
Hex with alpha	`#E5484D80`, `#ABC8`
`rgb()` and `rgba()`	`rgb(62, 99, 221)`, `rgba(62 99 221 / 0.5)`
`hsl()` and `hsla()`	`hsl(220, 70%, 55%)`
CSS colour names	`red`, `rebeccapurple` and all 148 names

Some values are paint but not a colour the tool can change. They are listed, counted and left alone:

Value	Shown as
`none`, `transparent`	No paint
`currentColor`	Resolves from CSS outside this file
`inherit`, `initial`, `unset`, `revert`, `context-fill`, `context-stroke`	Resolves from outside this file
`url(#gradient)`	A reference, not a colour

Colours written in other forms, such as `oklch()`, `lab()` or `var(--brand)`, are not recognised, and are neither counted nor changed.

The colour list

Every colour in the file has one row per distinct colour, however many ways it is written, most-used first. `#F00`, `#ff0000`, `rgb(255,0,0)` and `red` are the same row.

Part of the row	Meaning
Left swatch	The colour as it is now
Hex and note	The colour, with its CSS name or the nearest name prefixed "near", and how many places use it
Right swatch	What it should become; select it to choose
Right-hand hex and note	The target colour, and **unchanged** or **written in** N **places**
Count	How many places were actually written, or **—** when unchanged

Rows greyed out are values with no colour to change: they read **left alone**, with the reason, and **not held**.

Remap a colour

1. Select the right-hand swatch in the row.
2. Choose the new colour in the colour picker.
3. The **After** preview and the row update at once. Setting a colour back to its original value marks the row **unchanged**.

Every occurrence of that colour, in every form and every place, is rewritten. The heading of the **After** preview reads, for example, **4 of 5 written**: occurrences written out of those that can be written.

Swap two colours

Set colour A's target to colour B, and colour B's target to colour A. The tool plans every change from the original file before writing anything, so the two colours trade places cleanly.

Undo your changes

Put every colour back, under **Recolour from a harmony**, returns every target to its original colour. The message reads **Every colour back to what it was.**

Choose how colours are written

Under **Write colours as**:

Option	What is written
Write hex (default)	Every changed colour as six-digit hex in lower case, for example `#00aa00`
Keep the source form	Each changed colour in the form it was found in, as far as possible

With **Keep the source form**:

- a colour written as a name is written as a name if the new colour has an exact CSS name, and as hex if it does not;
- `rgb()` stays `rgb()`, and `rgba()` keeps its alpha;
- `hsl()` stays `hsl()`, and `hsla()` keeps its alpha;
- hex written in capitals stays in capitals;
- a three-digit shorthand is written as full six-digit hex, because very few colours have a three-digit form.

In both styles, a hex colour with an alpha part, such as `#E5484D80`, is written as six-digit hex, so its transparency is lost when you change it. Only changed occurrences are rewritten; everything else in the file is left exactly as it was.

What the file holds

Card	Meaning
elements	Elements in the parsed document, including the root
occurrences	Places a colour value is written
colours	Distinct colours the tool can change
left alone	Values with no colour to change
written	Occurrences the export will carry as changed
closest pair	The smallest CIEDE2000 difference between any two colours under the three colour-vision simulations; highlighted below 8

The note above the previews repeats elements, occurrences and colours.

The Mapping tab

The **Mapping** tab lists every colour with what will happen to it:

Column	Meaning
Swatch	The original colour
In the file	The hex value, or the keyword for a value left alone
Used	Number of occurrences
Becomes	The target, or **unchanged** or **left alone**

Column	Meaning
Written	Occurrences written; highlighted when it equals **Used**
Where, and what happened	How many are attributes, in a style attribute or in a style sheet, and the forms they were written in

The note under the table adds up the file: how many occurrences were written, how many left because nothing was asked of them, and how many could not be written, by keyword. The three always add up to the total.

Below it, **Two colours trading places** and **What the count used to mean** show, with measured examples, why an earlier version of the tool could not swap colours and miscounted them.

What you should see

The **After** preview shows your new colours, each row reports **written in** N **places**, and the **written** count matches the number you expect.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

Recolour from a harmony

Recolour every colour in an SVG at once from a colour harmony built on the file's main colour, and put the colours back.

A harmony is a set of colours at fixed angles round the colour wheel, such as opposite (complementary) or evenly spaced in threes (triadic). **Recolour from a harmony** builds one from your file's main colour and assigns it to every colour in the file in one step.

Apply a harmony

1. Load an SVG on the **Recolour** tab.
2. Under **Recolour from a harmony**, select one of the five buttons: **complementary**, **analogous**, **triadic**, **tetradic** or **split complementary**.
3. Every colour in the file gets a new target, the **After** preview updates, and a message names the harmony and the base colour, for example **A triadic palette from #E5484D.**

If no file is loaded, the message reads **Load an SVG first.**

How the harmony is built

- **The base colour** is the first row of **Every colour in the file**: the colour used in the most places. It keeps its own colour.
- **The angles** are applied to the base colour's hue in OKLCH, a perceptual colour space, keeping its lightness and chroma. Rotating hue in OKLCH keeps colours looking equally light and colourful, which rotating in HSL does not.
- **Assignment** goes down the colour list in order: the second row gets the second angle, and so on.

Harmony	Hue angles from the base
complementary	0°, 180°
analogous	0°, −30°, +30°
triadic	0°, 120°, 240°
tetradic	0°, 90°, 180°, 270°
split complementary	0°, 150°, 210°

When the file has more colours than the harmony has angles, the pattern repeats, turned a further 12° each time round, so later colours are close to, but not the same as, earlier ones.

If a new colour would fall outside sRGB, its chroma is reduced, keeping its lightness and hue, until it fits.

Colours left alone, such as `currentColor`, are not affected.

After applying

- Adjust any row by hand afterwards; the harmony only sets the starting targets.
- **Put every colour back** returns every target to the original colour.
- Applying a harmony replaces any targets you set by hand.

Things to know

- The five harmonies are a convention from colour-wheel teaching. They do not guarantee a pleasing or accessible result, and the **Data** tab lists them as this tool's choice.
- Because every generated colour shares the base colour's lightness, a harmony can make colours that differ in hue but are hard to tell apart for someone with a colour-vision deficiency, or too similar in lightness for text. Check the result on the **Check** tab; see Check contrast and colour vision.
- A file whose main colour is a grey or near-grey has little chroma, so the harmony produces greys or near-greys too.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

SVG Recolour reference

Every tab, control, paint property, colour form, readout, export format and message in SVG Recolour.

Address: https://auricartisan.com/tool/general/tools/svg-recolor/

Tabs

Tab	Contents
Recolour	Loading, write style, vision preview, harmonies; before and after previews, the colour list and figures
Mapping	Every colour's use, target and written count; worked examples of swaps and counts
Check	Text colour; WCAG 2.2 and APCA per colour; closest pairs; the artwork under three dichromacies
Data	The 15 entries every figure rests on, filterable by status; six corrections
Export	Four save and copy buttons, the size table, custom properties and the report
Reference	Nine sources and a six-step description of how the tool works

Recolour tab — The file

Control	What it does	Accepts
Drop an SVG	Loads a dropped file, or opens a chooser when selected or when `Enter` or `Space` is pressed	`.svg`, `image/svg+xml`
Load the sample	Loads a built-in sample and puts its markup in the paste box	—
Paste box	SVG markup	Text

Control	What it does	Accepts
Read the markup	Loads the markup in the paste box	Well-formed SVG

The sample loads automatically when the page opens.

Recolour tab — Options

Control	Values	Default
Write colours as	**Write hex**, **Keep the source form**	Write hex
Show it as seen by	**Normal vision**, **Protan**, **Deutan**, **Tritan**	Normal vision
Recolour from a harmony	complementary, analogous, triadic, tetradic, split complementary	—
Put every colour back	Resets every target to its original	—

Paint properties searched

`fill`, `stroke`, `stop-color`, `flood-color`, `lighting-color`, `color`, `solid-color`, `text-decoration-color`, in presentation attributes, style attributes and `<style>` elements.

Colour forms

Form	Read	Written with Keep the source form
`#RRGGBB`	Yes	Hex, keeping capitals if the original used them
`#RGB`	Yes	Six-digit hex
`#RRGGBBAA`, `#RGBA`	Yes	Six-digit hex; the alpha is lost
`rgb()`, `rgba()`	Yes, comma or space separated	`rgb(r, g, b)`, or `rgba(r, g, b, a)` keeping the alpha
`hsl()`, `hsla()`	Yes	`hsl(h, s%, l%)`, or `hsla(…)` keeping the alpha
148 CSS names	Yes	The CSS name if the new colour has one, otherwise hex
`none`, `transparent`	Listed, left alone	—
`currentColor`, `inherit`, `initial`, `unset`, `revert`, `context-fill`, `context-stroke`	Listed, left alone	—
`url(…)`	Listed, left alone	—
Other functions, `var()`	Not recognised	—

With **Write hex**, every changed colour is written as lower-case six-digit hex.

Active content removed from the previews

`script`, `foreignObject` and `handler` elements; attributes whose names begin with `on`; `href` and `xlink:href` values beginning with `javascript:`, including when whitespace or control characters are hidden inside the word. The exported file is

not stripped.

Recolour tab — Readouts

Readout	Meaning
Stripped notice	What active content was removed from the previews
Note above the previews	Elements, occurrences and colours, or **nothing loaded**
As it came	The original, on a chequerboard
After heading	Occurrences written out of those that can be written
Colour list row	Current swatch, hex, name, places; target swatch, hex, **unchanged** or **written in** N **places**; written count
Greyed row	Keyword, **left alone**, reason, **not held**
elements	Elements in the document, root included
occurrences	Places a colour value is written
colours	Distinct changeable colours
left alone	Values with no colour to change
written	Occurrences the export carries as changed
closest pair	Smallest CIEDE2000 difference under the three simulations; highlighted below 8

Reasons for greyed rows: **resolves from outside this file** (`currentColor`), **no paint** (`none`), **a reference, not a colour** (others).

Harmonies

Harmony	Angles from the base colour's OKLCH hue
complementary	0°, 180°
analogous	0°, −30°, +30°
triadic	0°, 120°, 240°
tetradic	0°, 90°, 180°, 270°
split complementary	0°, 150°, 210°

The base is the first colour in the list. Lightness and chroma are kept; the pattern repeats with a further 12° turn when there are more colours than angles.

Mapping tab

Element	Contents
Note	Number of occurrences
Table	Swatch, **In the file**, **Used**, **Becomes**, **Written**, **Where, and what happened**
Footnote	Written, unchanged and skipped counts, which add up to the total
Two colours trading places	Three measured cases: a swap, a named colour, a shorthand

Element	Contents
What the count used to mean	Six measured cases of the earlier count against the real one

Check tab

Element	Contents
Text laid over the artwork	The text colour; default `#FFFFFF`
What can be read on it	**Colour**, **WCAG**, **Rating**, **APCA Lc**, **Good for**, **It used to say**
The Lc that was always zero	Four pairs: **It printed** against **APCA gives**
The two colours hardest to tell apart	**Seen as**, **The two closest colours**, **ΔE_{00} between them**, **What that means**
The artwork under three dichromacies	Four previews with **closest two**

WCAG ratings: **AAA** 7:1, **AA** 4.5:1, **AA large** 3:1, **fail** below. APCA bands by absolute Lc: **any text** 90, **body text** 75, **large text** 60, **headlines only** 45, **non-text only** 30, **not for text** below. Closeness: **close to indistinguishable** below 5, **tells apart with effort** below 12, **clearly different** otherwise.

Data tab

Status buttons: **All** (15), **Measured** (3), **Computed** (7), **This tool's choice** (3), **Absent** (2). Select an entry to expand it.

Status	Entries
Measured	Where each colour is used, occurrences written back, colours that cannot be written
Computed	Reading a colour value, ΔE_{00}, WCAG ratio, APCA Lc, OKLab and OKLCH, the sRGB transfer function, colour-vision simulation
This tool's choice	When two colours count as the same, the harmony angles, the colour names
Absent	What `currentColor` resolves to; colour inside embedded raster images

What was corrected, and what it printed lists six corrections with **was** and **now**. Colours are grouped only when they are exactly the same colour; there is no control for merging near-identical colours.

Export tab

Button	Output
Save the recoloured SVG	`recoloured.svg`
Copy the markup	The recoloured SVG markup
Copy the custom properties	`--colour-01: #RRGGBB; /* name */` lines
Save the report	`recolour-report.json`

Reference tab

Source	Status
W3C SVG 1.1 and SVG 2, paint and presentation attributes	Runs

Source	Status
W3C CSS Color 4, colour syntax and named colours	Runs
ISO/CIE 11664-6:2014, CIEDE2000	Runs
IEC 61966-2-1:1999, sRGB	Runs
W3C WCAG 2.2, contrast minimum	Runs
APCA-W3 0.1.9	Rebuilt
Ottosson (2020), OKLab	Runs
Machado, Oliveira and Fernandes (2009)	Runs
Brettel, Viénot and Mollon (1997)	Withdrawn

How it works lists six steps: Parse, Strip, Walk, Resolve, Plan and Write.

Messages

Message	When
Loaded file name / **Loaded the pasted markup.**	A file or markup was loaded
A sample with a hex, an rgb(), a name, a shorthand, a stylesheet and a currentColor.	The sample was loaded
That file will not parse as SVG, so nothing was changed.	The markup is not well-formed
Paste some SVG markup first.	**Read the markup** with an empty box
Load an SVG first.	A harmony with nothing loaded
A harmony **palette from** hex.	A harmony was applied
Every colour back to what it was.	**Put every colour back**
Saved · N **occurrences written**	The SVG was saved
Nothing loaded.	Save or copy with nothing loaded
Markup copied / **Custom properties copied**	A copy succeeded
Report saved.	The report was saved
The browser refused the clipboard.	Clipboard access was blocked
This browser will not take text from the clipboard.	The browser cannot copy text

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

Troubleshooting SVG Recolour

Symptoms you may meet in SVG Recolour, why they happen, and what to do.

Loading

Symptom	Why	What to do
That file will not parse as SVG, so nothing was changed.	The markup is not well-formed XML	Read the parser's message under the paste box; fix the markup (often an unclosed tag or a stray `&`) and try again
Paste some SVG markup first.	The paste box is empty	Paste the markup, then select **Read the markup**

Symptom	Why	What to do
My file is greyed out in the file chooser	The chooser offers `.svg` files	Drag the file onto **Drop an SVG**, or rename it with the `.svg` ending
A notice says active content was removed	The file contains scripts, event handlers or `javascript:` links	The previews are safe; remember the exported file still contains them
My mappings disappeared	Loading a file, markup or the sample resets every mapping	Set the targets again, or apply a harmony

Colours not found or not changed

Symptom	Why	What to do
A colour I can see is not in the list	It is written in a form the tool does not read, such as `oklch()` or `var(--brand)`	Rewrite it as hex, `rgb()`, `hsl()` or a CSS name before loading
A colour inside a picture did not change	It is part of an embedded raster image, not markup	Recolour the raster image in an image editor
A row reads **left alone**	The value is `currentColor`, `none`, `transparent`, a keyword such as `inherit`, or a `url()` reference	Expected. `currentColor` takes its colour from the page the SVG is placed in
A gradient did not change	Its colours are in `stop-color` on the gradient's stops, listed as their own rows; the shape's `fill` is a `url()` reference	Change the rows for the stop colours
Few colours for a large file	The file is mostly embedded images, or uses `currentColor`	Check **left alone** and **occurrences** under **What the file holds**
A semi-transparent colour became solid	It was written as hex with an alpha part, which is rewritten as six-digit hex	Re-add the alpha afterwards, or write the original colour as `rgba()` and use **Keep the source form**
Colours I wrote as names came back as hex	**Write hex** is selected, or the new colour has no exact CSS name	Choose **Keep the source form**; names are kept only when the new colour has one

Previews and checks

Symptom	Why	What to do
The previews are tinted	A colour-vision view is selected under **Show it as seen by**	Select **Normal vision**
The **Check** tab says two colours or more are needed	The file has fewer than two changeable colours	Nothing to compare
APCA Lc is negative	The text colour is lighter than the background colour	Expected: the sign shows polarity; **Good for** uses the size
WCAG and APCA disagree	They measure different things	Treat both as evidence

Exports

Symptom	Why	What to do
Nothing loaded.	No file is loaded	Load a file first
The exported file still contains a script	Only the previews are stripped	Remove active content in an editor if you do not trust the file

Symptom	Why	What to do
The XML declaration or DOCTYPE is missing	Only the `<svg>` element and its contents are written	Add them back if your workflow needs them
Attribute quotes changed	The browser writes the markup back out	This does not change the drawing
The browser refused the clipboard.	Clipboard access was blocked	Allow clipboard access for the site, or use **Save the recoloured SVG**
This browser will not take text from the clipboard.	The browser cannot copy text	Use the save buttons

For help with the site in general, see Help.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

SVG Recolour FAQ

Short answers to common questions about SVG Recolour.

Is my SVG uploaded?

No. The file is parsed in the page and never sent anywhere. See Privacy.

Can I swap two colours?

Yes. Set each colour's target to the other. Every change is planned from the original file before anything is written, so the swap works cleanly.

Does it change colours written as names, like `red`?

Yes. Names, hex, `rgb()` and `hsl()` forms are all found and rewritten. The same colour written different ways appears as one row.

Why is `currentColor` left alone?

It takes its value from the `color` of the page the SVG is placed in, so the file does not contain the colour. The tool counts it and leaves it as it is, which keeps icons that rely on it working.

Does it recolour gradients?

Yes, through the `stop-color` of each gradient stop, which appear as their own rows. The shape that uses the gradient refers to it with `url()`, which is left alone.

Does it recolour embedded images?

No. Pictures embedded in an SVG are pixels, not markup, and are not found or changed.

Can I recolour several files at once?

No. The tool works on one file at a time.

Are near-identical colours merged into one row?

No. Only exactly identical colours share a row, however they are written.

Will the exported file look exactly like the After preview?

Yes, in colour: the preview is drawn from the same rewrite. The preview has active content removed, but the file keeps it.

What does "near royalblue" mean?

It is the nearest CSS colour name, found by colour difference, to make hex values easier to read. An exact CSS name is shown without "near". Nothing depends on the name.

Is the harmony a guarantee of good colours?

No. The harmonies are a convention from colour-wheel teaching. Check the result for contrast and colour vision on the **Check** tab.

Does the colour-vision preview change the exported file?

No. It changes only what is drawn on screen.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

SVG Recolour background material

The contents of SVG Recolour's others folder: glossary, limits, related pages, release notes and privacy.

This folder holds reference material that supports the guides: the terms the tool uses, what it finds and does not, related tools and reading, its history, and what happens to your files, including active content. The tool has no keyboard shortcuts of its own beyond standard keyboard access, so there is no shortcuts page.

File	What it covers
glossary.md	Definitions of the terms used in SVG Recolour
limits-and-accuracy.md	What the tool finds and writes, how precisely, and what it does not hold
related.md	Related tools on the site and Learn articles
release-notes.md	User-facing history of the tool
privacy.md	What stays in your browser, and how active content is handled

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

SVG Recolour glossary

Definitions of the terms used in SVG Recolour.

Term	Meaning
Active content	Parts of an SVG that run code: scripts, event handlers and `javascript:` links
APCA	Advanced Perceptual Contrast Algorithm; lightness contrast in Lc units, signed by polarity
CIEDE2000 (ΔE_{00})	The CIE's standard formula for how different two colours look
Closest pair	The two colours in the file with the smallest CIEDE2000 difference
Contrast ratio	The WCAG 2 measure of luminance difference, from 1:1 to 21:1
CSS colour name	One of the 148 named colours of CSS, such as `red` or `rebeccapurple`
`currentColor`	A value that takes the colour from the page the SVG is placed in
Custom property	A CSS variable, such as `--colour-01`, holding a colour
Dichromacy	Colour vision with one of the three cone types missing
Harmony	A set of colours at fixed angles round the colour wheel
Left alone	A value with no colour the tool can change, listed but not written
OKLCH	A perceptual colour space of lightness, chroma and hue, used for the harmonies
Occurrence	One place in the file where a colour value is written
Paint property	A property that sets colour, such as `fill` or `stroke`
Presentation attribute	An SVG attribute that sets a style directly, such as `fill="#E5484D"`
Protanopia, deuteranopia, tritanopia	Colour vision missing the long (red), medium (green) or short (blue) wavelength cones
Source form	The way a colour was written: hex, shorthand, `rgb()`, `hsl()` or a name
Style attribute	An SVG `style="…"` attribute holding CSS declarations
Style sheet	CSS inside a `<style>` element in the SVG

Term	Meaning
SVG	Scalable Vector Graphics: an XML format for drawings
Target	The colour an original colour will become
WCAG	The Web Content Accessibility Guidelines; version 2.2 defines the contrast ratio used here
Well-formed	XML that follows the syntax rules, so it can be parsed
Written	An occurrence the tool actually rewrote

Limits and accuracy

What SVG Recolour finds and writes, how precise its figures are, and what it does not hold.

What is found and written

- Eight paint properties: `fill`, `stroke`, `stop-color`, `flood-color`, `lighting-color`, `color`, `solid-color` and `text-decoration-color`.
- In presentation attributes, style attributes and `<style>` elements.
- Colours written as hex (with or without alpha), shorthand hex, `rgb()`, `rgba()`, `hsl()`, `hsla()` and all 148 CSS names.
- The count beside each colour is the number of occurrences, and the written count is the number actually rewritten. The colour list, the **Mapping** tab, the report and the exported file all use the same list, so they always agree.
- Every change is planned from the original file before anything is written, so no change can be caught by another.

What is not found or not changed

Item	What happens
`currentColor` and other inherited keywords	Counted and left alone; the colour comes from outside the file
`none`, `transparent`, `url()` references	Counted and left alone
Colours in other CSS functions or variables (`oklch()`, `lab()`, `var()`)	Not recognised, not counted, not changed
Embedded raster images	Not read; their pixels are not changed
Colours set by CSS outside the file	Not visible to the tool

Precision and choices

Item	Detail
Colour values	Parsed to 8-bit sRGB; an unrecognised value is never guessed as black
Alpha	Kept for `rgba()` and `hsla()` with **Keep the source form**; lost for hex with alpha when changed
Three-digit hex	Written back as six digits
Grouping	Only identical colours share a row; near-identical colours are kept apart
Colour names	The nearest CSS name by CIEDE2000, prefixed "near"; a label, not a classification
Harmony angles	A colour-wheel convention, applied in OKLCH with lightness and chroma kept
Contrast	WCAG 2.2 ratio, exact; APCA-W3 0.1.9 with all its constants

Item	Detail
CIEDE2000	ISO/CIE 11664-6:2014, agreeing with published test data to within 0.00004
Colour-vision simulation	Machado, Oliveira and Fernandes (2009), full severity, in linear light; no milder forms

Limits

- One file at a time; there is no batch mode.
- The previews have active content removed; the exported file does not.
- Only the `<svg>` element is exported: anything before it, such as an XML declaration or DOCTYPE, is not carried across, and the browser may normalise small formatting details.
- Nothing is stored between visits.

What it is not

SVG Recolour changes colour values in the markup. It does not edit shapes, text or structure, does not sanitise files for publishing, and does not judge whether a palette is attractive.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

Privacy

What SVG Recolour does with your SVG files, and how it handles active content.

Your file stays in your browser

- An SVG you drop, choose or paste is read and parsed by the page itself, in your browser. It is not uploaded.
- Finding colours, rewriting them, the checks and the previews all happen on your device.
- Exported files are created in the browser and saved by your browser's normal download.
- The clipboard is written only when you select a copy button.

Active content

An SVG is a document, not just a picture, and can contain scripts, event handlers and `javascript:` links. Because this tool is often used on files other people send, the previews on the page are drawn from a copy with that content removed, and a notice lists what was removed. The file's scripts are not run by the page.

The exported file is rewritten from your original, so it still contains any active content the original had. If you do not trust the file, remove that content in an editor before using the export elsewhere.

Nothing is kept after you leave

The tool does not store your file, your mappings or your settings. Loading another file, reloading or leaving the page clears them.

What the report contains

The report contains counts of elements and colours, every colour mapping and the values left alone. It does not contain the SVG markup.

The page itself

SVG Recolour is a normal page on auricartisan.com. Site-wide features such as sign-in, preferences and any consent-based services follow the site's own privacy policy and your cookie choices; see Help.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

Release notes

User-facing history of SVG Recolour.

7 June 2026 — rewrites planned from the original

The earlier engine was replaced by the current workbench, titled **SVG Recolour**, with six tabs: **Recolour**, **Mapping**, **Check**, **Data**, **Export** and **Reference**. The site's earlier user guide carries a notice that it describes the engine replaced on this date.

What changed for you:

- **Two colours can trade places.** The earlier version replaced colours one after another as text, so swapping red and blue left both red. Every change is now planned from the original file and written once.
- **Colours written as names are changed.** A `fill="red"` was listed and offered for remapping but never rewritten. It is now rewritten like any other form.
- **Three-digit colours stay the colour you chose.** The earlier version could write a new colour as a three-digit shorthand that browsers read as a different colour. New colours are written as six digits.
- **The count is the number of places.** The earlier count doubled or quadrupled depending on how the file was written. The count is now the number of occurrences, and the written count is what was actually changed.
- **The APCA column works.** It printed zero for every pair; it now gives real APCA-W3 0.1.9 figures beside the WCAG 2.2 ratio.
- **Grouping uses a real colour difference** (CIEDE2000) instead of a sum of hue, saturation and lightness.
- **New:** the **Check** tab's closest-pair check under three colour-vision deficiencies, **Keep the source form**, active content stripped from the previews, and the JSON report.

If you exported files with the earlier version, check swapped colours and colours written as names.

25 May 2026 — earlier version

The SVG Recolor user guide was published in the site's documentation library. It describes the earlier engine and is now marked as superseded.

WEBSITE › TOOLS › IMAGE AND FILE TOOLS › SVG RECOLOUR

Related tools and reading

Other Auric Artisan tools and Learn articles that go with SVG Recolour.

Tools on the site

Tool	Why you might use it with SVG Recolour
Icon Library	Find open-source SVG icons to recolour
M-Gradient Palette	Build an evenly stepped palette to apply to artwork
Image Picker	Take target colours from a photograph
Contrast Checker	Test one text and background pair in more depth
Vision Simulation	Simulate more kinds of colour vision on a finished design
Collage Maker	Place exported artwork, as PNG, into a collage

Learn articles

- Color Theory and Harmony — https://auricartisan.com/library/learn/articles/2026-06-04-color-theory-and-harmony
- Color Blindness and Color Vision Deficiency — https://auricartisan.com/library/learn/articles/2026-05-24-color-blindness-cvd
- Accessible Color Design and WCAG Contrast — https://auricartisan.com/library/learn/articles/2026-05-25-wcag-contrast-accessible-color
- Oklab and Oklch: Modern Perceptual Color Spaces — https://auricartisan.com/library/learn/articles/2026-06-02-oklab-oklch-perceptual-color-spaces
- ΔE Metrics from CIE76 to CIEDE2000 and ΔEITP — https://auricartisan.com/library/learn/articles/2026-05-26-delta-e-metrics-cie76-ciede2000-itp

Other documentation

- The group overview: Image and file tools.
- The site's documentation library carries "SVG Recolour Developer Reference" at https://auricartisan.com/library/documentation/articles/2026-06-07-svg-recolour-developer-reference, written for people maintaining the tool.